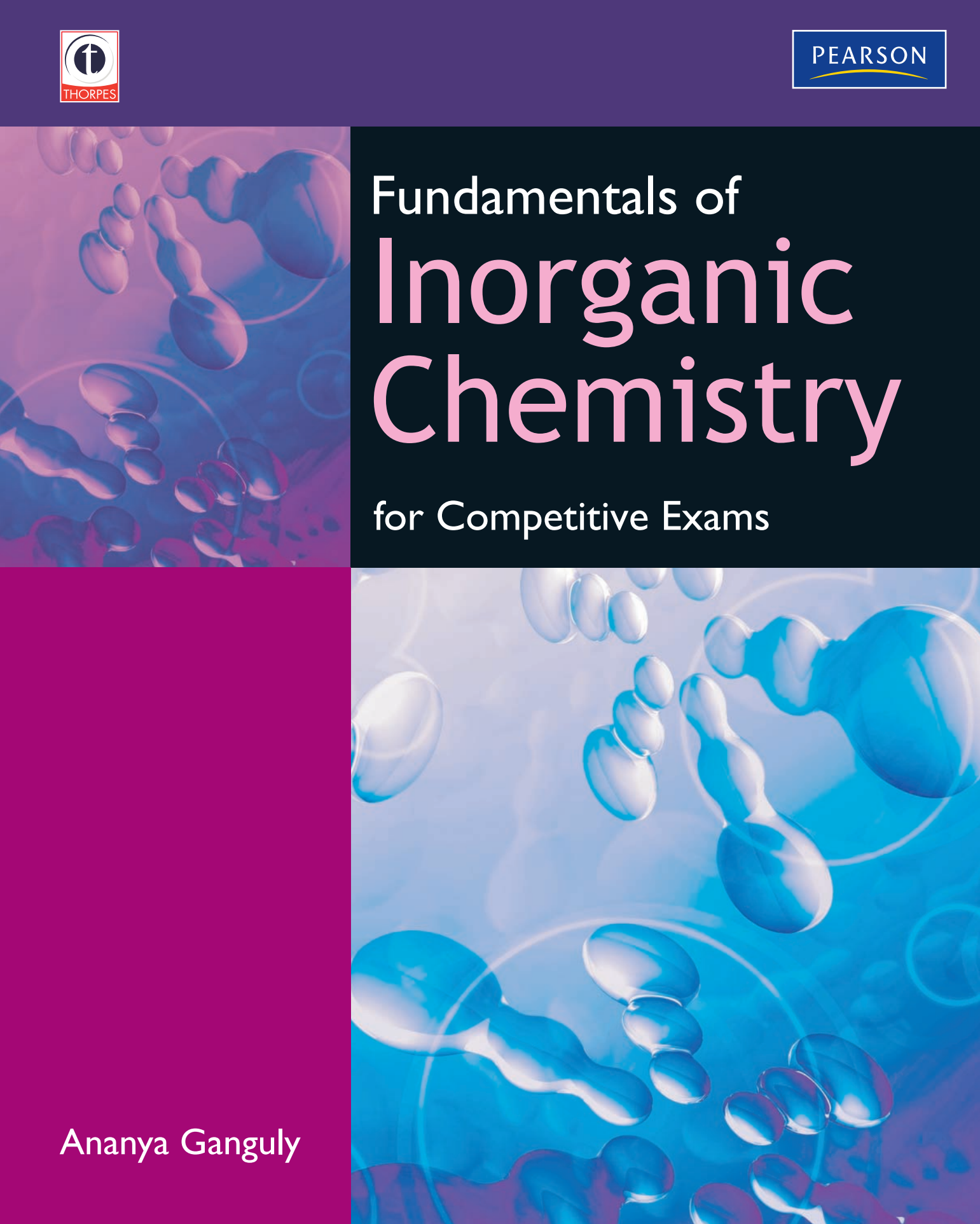




Fundamentals of Inorganic Chemistry

for Competitive Exams

Ananya Ganguly

Fundamentals of
**INORGANIC
CHEMISTRY**

Ananya Ganguly

PEARSON

Chandigarh • Delhi • Chennai

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Preface

In recent years, the question format of non-medical examinations like IIT-JEE and AIEEE has been restructured with greater emphasis on the theoretical and conceptual intricacies and the application of the underlying basic concepts and principles.

The poor performance of students in these examinations is partly due to non-availability of a comprehensive text book which lays adequate stress on the authentic logical theoretical concept building, numericals and related problems.

The present book grew out of my experience of classroom teaching through lectures, notes and assignments. Effort has been made to maintain lucid style and simplicity of expression. Apart from this, the book has numerous carefully selected examples and solved illustrations which include almost all the previous years' questions asked in IIT-JEE and AIEEE examinations.

Selected questions of different formats keeping in mind the recent pattern of examinations have been listed at the end of each chapter. Solutions to almost all advanced questions have been incorporated.

I would like to thank Showick Thorpe and Sanjay Sharma for their constant encouragement during the publication of this book. I extend my thanks to Nitkiran Bedi for her contribution in bringing out this book in record time. I am also thankful to the staff members of my academy for their sincere help.

I will appreciate comments, suggestions and criticism from the readers and will incorporate them in the subsequent editions.

Ananya Ganguly

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Periodic Properties and Chemical Bonding

INTRODUCTION

Chemists have long tried to find patterns in the properties of the elements. Some were discovered fairly readily; for example, elements were classified as metals, non-metals, and many of their compounds as acids, alkalis, or salts.

Table 1.1 Typical Properties of Metals and Non-metals

	<i>Metals</i>	<i>Non-metals</i>
Appearance and properties	Solids, some with high melting points; lustrous, malleable and ductile	Gases, or solids with low melting points
Conduction of heat and electricity	Very good	Poor
Compounds	Ionic compounds with non-metals; alloys with other metals	Ionic compounds with metals; covalent compounds with other non-metals
Charge on ions	Positive	Negative
Chemical nature	Reducing agents	Oxidizing agents
Electronegativity	Low	High

However, it was widely believed that there had to be an underlying reason for the patterns.

1. Prout's Hypothesis

One of the first suggestions was due to Prout. Prout's hypothesis was that all elements were made from a whole number of hydrogen atoms. (Be careful here: an atom' in Prout's time was a very different thing to our understanding of the word.) According to him the atomic masses of the elements should be a whole number of times that of hydrogen, i.e., they should be integers. Unfortunately, from Prout's point of view, the results of experiments showed that the atomic masses of many elements were not integers.

2. Dobereiner's Triads

According to Dobereiner, when elements of same properties are kept in the increasing order of their atomic weights, the atomic weight of middle element is equal to the mean atomic weight of remaining two elements. Such a group of elements is called **Dobereiner's triad**.

1.2 Periodic Properties and Chemical Bonding

Triad of atoms		Mean of first and last element	
Li	Na	K	
7	23	39	$\frac{7+39}{2} = 23$
Be	Mg	Ca	$\frac{8+40}{2} = 24$
8	24	40	

Dobereiner could arrange only a few elements as triads and there are some such elements present in a triad, whose atomic weights are approximately equal, e.g.,

Fe	Co	Ni
Ru	Rh	Pd

Therefore, this hypothesis was not acceptable for all elements.

3. The Telluric Helix

It was in 1862, that a periodic classification of the elements was developed that approached the idea we have today. At that time A.E. de Chancourtois, a professor of Geology at the Ecole des Mines in Paris presented an account of his telluric helix in which he indicated the relative properties of elements and their atomic weights.

He used a vertical cylinder with 16 equidistant lines on its surface, the lines lying parallel to the axes. Then he drew a helix at 45 degree to the axis and arranged the elements on the spiral in the order of their increasing atomic weights. In this manner, elements that differed from each other in atomic weight by 16 or multiples of 16 fell very nearly on the same vertical line. In addition to the 16 vertical lines, de Chancourtois felt that other connecting lines could be drawn, and that all elements lying on such lines were related in some manner. His arrangement resulted in the proposal by de Chancourtois that the properties of the elements are the properties of numbers.

4. Newland's Rule of Octave

A major attempt at making a link was made by new lands in 1864 (Table 1.2). He grouped elements into sets of eight and claimed that every eighth element in the pattern was chemically similar. Newlands' law of octaves was largely ignored, or at best treated with mild amusement.

The first thorough attempt at relating chemical properties to atomic masses was made by the Russian.

Table 1.2 Examples of Newlands' Octaves*

No.		No.		No.		No.		No.	
H	1	F	8	Cl	15	Co and Ni	22	Br	29
Li	2	Na	9	K	16	Cu	23	Rb	30
G	3	Mg	10	Ca	17	Zn	24	Sr	31
Bo	4	Al	11	Cr	18	Y	25	Ce and Le	32
C	5	Si	12	Ti	19	In	26	Zr	33
N	6	P	13	Mn	20	As	27	Di and Mo	33
O	7	S	14	Fe	21	Se	28	Ro and Ru	35

*This is part of a table that John Newlands presented in a talk he gave to the Chemical Society on 9 March 1866. The talk was entitled 'The Law of Octaves, and the Causes of Numerical Relations among the Atomic Weights'. (You might like to work out which elements G, Bo, etc., stand for.) Here is part of an account of the talk:

The author claims the discovery of a law according to which the elements analogous in their properties exhibit peculiar relationships, similar to those subsisting in music between a note and its octave.... Professor G. F. Foster humorously enquired of Mr. Newlands whether he had ever examined the elements according to the order of their initial letters?

Newlands was not at all happy about the credit that went to Mendeleeff over the discovery of the periodic law. In 1884 Newlands wrote:

Having been the first to publish the existence of the periodic law more than nineteen years ago, I feel, under existing circumstances, compelled to assert my priority in this matter.... As a matter of simple justice, and in the interest of all true workers in science, both theoretical and practical, it is right that the originator of any proposal or discovery should have the credit of his labour.

5. Lothar Meyer's Volume Curves

The graphs of atomic volumes against atomic weights are known as Lothar Meyer volume curves.

- (i) Lothar Meyer plotted a graph between atomic weight and atomic volume (i.e., atomic weight in solid state/density).
- (ii) Elements with similar properties occupied the similar positions on the graph.
- (iii) Strong electropositive elements of IA except Li, all others Na, K, Rb, Cs etc., occupied the top position on the graph.

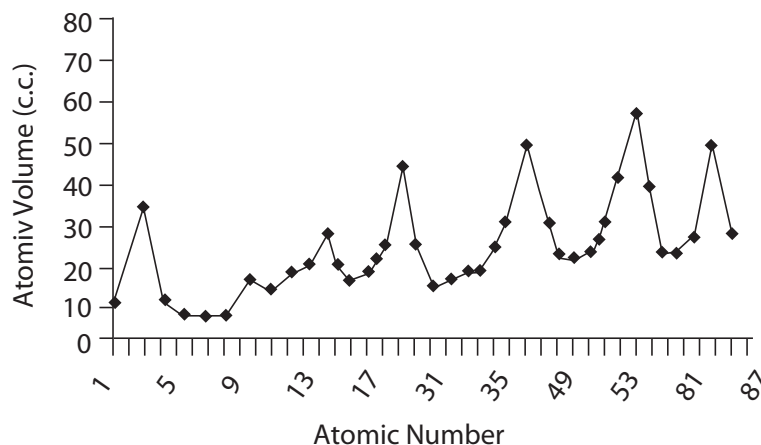


Fig. 1.1 Atomic volume versus atomic number curve

- (iv) IIA group elements Be, Mg, Ca, Sr, Ba etc., occupied the positions on the ascending part of the graph.
- (v) Inert gases except He occupied the positions on the descending part of the graph.
- (vi) Halogens also occupied the descending part of the graph.

- Transition elements have very small volumes and therefore these are present at the bottoms of the curve metals have highest atomic volumes.

6.(a) Mendeleef's Periodic Table

- Periodic table is based on atomic weight.
- In the periodic table, the horizontal lines are called periods and the vertical lines are called groups.
- The periodic table consists of seven periods and nine groups (The earlier periodic table had only 8 groups. The noble gases were added later in the zero group because these were not discovered when Mendeleef put forward his periodic table.
- All the groups (except VIII and Zero groups) are divided into subgroups A and B.
- 2, 8, 18 and 32 are called magic numbers.

(b) Merits of Mendeleef's Periodic Table

- Classification of elements then known, was done for the first time and the elements having similar properties were kept in the same group.
- It encouraged research and led to discovery of newer elements.
- Mendeleef had even predicted the properties of many elements not discovered at that time. This helped in the discovery of these elements. For example, Mendeleef predicted the properties of the following elements:

- (a) ka-boron—This was later called scandium (Sc)
- (b) Eka-aluminium—This was later called gallium (Ga)
- (c) Eka-silicon—This was later called germanium (Ge)

- Atomic weight of elements were corrected. Atomic weight of Be was calculated to be $3 \times 4.5 = 13.5$ by considering its valency 3. Mendeleef calculated it $2 \times 4.5 = 9$.

(c) Defects of Mendeleef's Periodic Table

- (1) **Position of Hydrogen:** Hydrogen resembles alkali metals and halogens in its properties. Hence, its position was not sure.
- (2) **Position of Isotopes:** The isotopes have different atomic weights and the periodic table is based on atomic weights. Therefore, isotopes should get different places in the periodic table on the basis of atomic weights.
- (3) The periodic table is not fully based on increasing order of atomic weights.
- (4) It is not proper to place together the elements having differing properties, such as coinage metals (Cu, Ag and Au) with alkali metals; Zn, Cd and Hg with alkaline earth metals and metal like Mn with halogens. Similarly, Pt and Au having similar properties have been placed in different groups.
- (5) There is no indication whether lanthanides and actinides are associated with group IIIA or group IIIB.
- (6) **Position of Isobars:** These elements have different groups when mass remains same.
- (7) Lot of stress was given to valency of elements.

MODERN PERIODIC LAW AND MODERN PERIODIC TABLE

- **Mosley** proved that the square root of frequency of the rays, which are obtained from a metal on showering high velocity electrons is proportional to the nuclear charge of the atom. This can be represented by the following expression. $\sqrt{\nu} = a(Z-b)$ where Z is nuclear charge on the atom and a and b are constants.
- The nuclear charge on an atom is equal to the atomic number.
- According to modern periodic law, "The properties of elements are the periodic functions of their atomic numbers"

Modern Periodic Table

- On the basis of the modern periodic law, a scientist named **Bohr** proposed a long form of periodic table that was prepared by Rang and Warner.
- In the periodic table, the horizontal lines are periods and the vertical lines are groups.
- The periodic table has a total of seven periods and 18 groups. *But according to CAS system*, the number of groups is 16, because the eighth group has been divided into three groups.
- There are two elements in the first period eight elements in each of the second and third periods, eighteen elements in each of the fourth and fifth period thirty two elements in the sixth period and only nineteen elements till now in the seventh period. Total 105 elements have been discovered so far.
- The first period is very short period, second and third are short periods fourth and fifth are long periods sixth is very long period, while the seventh is incomplete period.
- The lanthanides (Elements from atomic numbers 58 to 71) and actinides (elements from atomic numbers 90 to 103) are included in the sixth and seventh periods through these have been kept outside the periodic table.
- Period—The details about the seven periods are as follows.
 - (i) Each period starts with an alkali metal whose outermost electronic configuration is ns^1 .
 - (ii) Each period ends with a noble gas of outermost electronic configuration ns^2np^6 except He. The electronic configuration of He is $1s^2$.
 - (iii) The number of elements in a period is equal to the number of necessary electrons to acquire ns^2np^6 configuration in the outermost shell of first element (alkali metal) of the period. First period contains two elements.
 - (v) The number of elements in each period may be determined by the number of electrons in a stable configuration as under

Periods	Stable electronic configuration	Number of electrons	From	To
First	$1s^2$	2	H (1)	He (2)
Second	$2s^22p^6$	8	Li (3)	Ne (10)
Third	$3s^23p^6$	8	Na (11)	Ar (18)
Fourth	$4s^23d^{10}4p^6$	18	K (19)	Kr (36)
Fifth	$5s^24d^{10}5p^6$	18	Rb (37)	Xe (54)
Sixth	$6s^24f^{14}5d^{10}6p^6$	32(including lanthanoids)	Cs (55)	Rn (86)
Seventh		(including Actinoids)	Fr (87)	Ha (105)

Electronic Basis for the Periodic Classification

With a better understanding of the part that the electron plays great role in the properties of the elements, a corresponding understanding of the periodic system came about.

On the basis of electronic configuration, the elements may be divided into four groups:

s-block elements

- The elements of the periodic table in which the last electron enters in *s*-orbital, are called *s*-block elements.
- *s*-orbital can accommodate a maximum of two electrons.
- Their general formulae are ns^1 and ns^2 respectively, where, $n = (1 \text{ to } 7)$.
- I A group elements are known as alkali metals because they react with water to form alkali. II A group elements are known as alkaline earth metals because their oxides react with water to form alkali and these are found in the soil or earth.
- The total number of *s* block elements are 14.
- Fr^{87} and Ra^{88} are radioactive elements while H and He are gaseous elements.
- Cs and Fr are liquid elements belonging to *s*-block.

p-block elements

- The elements of the periodic table in which the last electron gets filled up in the *p*-orbital, called *p*-block elements.
- A *p*-orbital can accommodate a maximum of six electrons. Therefore, *p*-block elements are divided into six groups which are III A, IV A, V A, VI A, VII A and zero groups.
- The general formulae of *p* block elements are $ns^2 p^{1-6}$ (where, $n = 2 \text{ to } 6$)
- The zero group elements having general formula $ns^2 p^6$ are inert, because their energy levels are fully filled.
- The total number of *p* block elements in the periodic table is 30 (excluding He).
- There are nine gaseous elements (Ne, Ar, Kr, Xe, Rn, F_2 , Cl_2 , O_2 and N_2) belonging to *p*-block. Gallium (Ga) and bromine (Br) are liquids.
- The step-like thick lines drawn in the periodic table in the *p*-block divides elements into metals nonmetals and metalloids.

d-block elements

- The elements of the periodic table in which the last electron gets filled up in the *d* orbital, called *d* block elements.
- The *d* block elements are placed in groups named IIIB, IV B, V B, VI B, VII B, VIII, I B and II B.
- In *d* block elements, the electron gets filled up in the *d* orbital of the penultimate shell. That is why, these elements are known as transition elements.
- Though the total number of *d* block elements is 33 in the periodic table but there are only 30 transition elements. Because only those elements are transition in which *d* orbital is partially filled.
- The general formula of these elements is $(n-1)s^2, p^6, d^{1-10} ns^{1-2}$ where $n = 4 \text{ to } 7$.
- All of these elements are metals.
- Out of all the *d* block elements mercury is the only liquid element.

f-block elements

- The elements of the periodic table in which the last electron gets filled up in the *f* orbital, called *f* block elements.
- The *f* block elements are from atomic number 58 to 71 and from 90 to 103.
- The lanthanides occur in nature in low abundance and therefore, these are called rare earth elements.
- There are 28 *f* block elements in the periodic table.
- The elements from atomic number 58 to 71 are called lanthanides because they come after lanthanum (57). The elements from 90 to 103 are called actinides because they come after actinium (89).
- All the actinide elements are radioactive.
- All the elements after atomic number 92 (i.e., U^{92}) are transuranic elements.
- The general formula of these elements is $(n-2)s^2 p^6 d^{10} f^{1-14} (n-1)s^2 p^6 d^{0-1} ns^2$ where $n = 6 \text{ and } 7$.

1.6 Periodic Properties and Chemical Bonding

Representative elements												Representative elements						Noble gases
GROUP NUMBER												GROUP NUMBER						18
1 2																		0
I A II A																		
1 H 1s ¹																		
d-Transition elements																		
GROUP NUMBER																		
</																		

Table 1.3 Nomenclature of Elements with Atomic Number Above 100

<i>Atomic number</i>	<i>Name</i>	<i>Symbol</i>	<i>IUPAC official name</i>	<i>IUPAC symbol</i>
101	Unnilunium	Unu	Mendelevium	Md
102	Unnibium	Unb	Nobelium	No
103	Unniltrium	Unt	Lawrencium	Lr
104	Unnilquadium	Unq	Rutherfordium	Rf
105	Unnilpentium	Unp	Dubnium	Db
106	Unnihexium	Unh	Seaborgium	Sg
107	Unnilseptium	Uns	Bohrium	Bh
108	Unniloctium	Uno	Hassium	Hs
109	Unnilennium	Une	Meitnerium	Mt
110	Unnillium	Uun	Darmstadtium	Ds
111	Unununnium	Uuu	Röntgenium*	Rg*
112	Ununbium	Uub	*	*
113	Ununtrium	Uut	+	
114	Ununquadium	Uuq	*	*
115	Ununpentium	Uup	+	
116	Ununhexium	Uuh	*	*
117	Ununseptium	Uus	+	
118	Ununoctium	Uuo	+	

*Official IUPAC name yet to be announced + Elements yet to be discovered

Thus, the new element first gets a temporary name, with symbol consisting of three letters. Later permanent name and symbol are given by a vote of IUPAC representatives from each country. The permanent name might reflect the country (or state of country) in which the element was discovered or pay tribute to a notable scientist. As of now, elements with atomic numbers up to 112, 114 and have been discovered. Elements with atomic numbers 113, 115, 117, and 118 are not yet known.

Worked-out Example

1. What would be the IUPAC name and symbol for the element with atomic number 120?

Solution: From Table, the roots for 1, 2 and 0 are un, bi and nil, respectively. Hence, the symbol and the name respectively are Ubn and unbinilium.

The types of elements

Using electronic configuration as the criterion, we ordinarily recognize four general type of elements; the inert gas elements, the representative elements, the transition elements, and the inner transition elements. The classification of the elements into these groups is dependent on the extent to which the s, p, d and f orbitals are filled.

(i) Inert Gases

- s and p-orbitals of the outer most shell of these elements are completely filled. The outermost electronic configuration is ns^2np^6 .
- Helium is also inert gas but its electronic configuration is $1s^2$.

(ii) Representative or Normal Elements

- Outermost shell of these elements is incomplete. The number of electrons in the outermost shell is less than eight.

1.8 Periodic Properties and Chemical Bonding

- (b) Inner shells are complete.
- (c) s- and p-block elements except inert gases are called normal or representative elements.

(iii) Transition Elements

- (a) Last two shells of these elements namely outermost and penultimate shells are incomplete.
- (b) The last shell contains one or two electrons and the penultimate shell may contain more than eight up to eighteen electrons.
- (c) The outermost electronic configuration is similar to d-block elements, i.e., $(n-1)d^{1-10} ns^{1-2}$.
- (d) According to latest definition of transition elements those elements which have partly filled d-orbitals in neutral state or in any stable oxidation state are called transition elements. According to this definition, **Zn, Cd and Hg (IIB group) are d-block elements but not transition elements because these elements have d^0 configuration in neutral as well as in stable +2 oxidation state.**
- (e) Because of the extra stability which is associated with empty, half-filled, and filled subshells, there are some apparent anomalies in electronic arrangements in the transition series. This empirical rule is illustrated by the chromium and copper configuration in the first d series of elements:

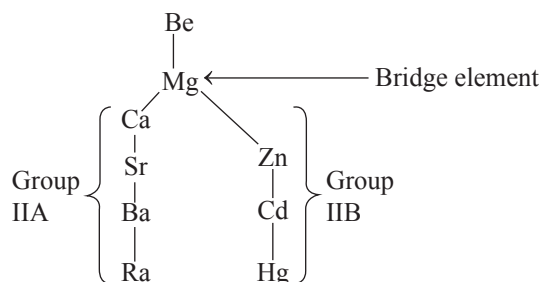
	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
3d	1	2	3	5	5	6	7	8	10	10
4s	2	2	2	1	2	2	2	2	1	2

(iv) Inner Transition Elements

- (a) In these elements last three shells i.e., last, penultimate and prepenultimate shells are incomplete.
- (b) These are related to IIIB i.e., group 3.
- (c) The last shell contains two electrons. Penultimate shell may contain eight or nine electrons and pre-penultimate shell contains more than 18 up to 32 electrons.
- (d) Their outermost electronic configuration is similar to f-block element i.e., $(n-2)f^{1-14}(n-1)s^2 (n-1)p^6 (n-1)d^{0-1}ns^2$.
- (v) **Typical Elements:** The elements of second and third periods are called typical elements as they summarize the properties of their respective groups.

Group	I	II	III	IV	V	VI	VII
2 nd period	Li	Be	B	C	N	O	F
3 rd period	Na	Mg	Al	Si	P	S	Cl

- (vi) Elements of the seventh period after atomic number 93 (i.e., actinides) are synthetic elements and are called transuranium elements.
- (i) **Bridge elements:** The typical elements of third period are also called bridge elements as the division between two subgroups A and B starts from these elements. In second group, Mg acts as a bridge element. The properties of bridge element are somewhat mixed properties of the elements of two subgroups as magnesium shows similarities with alkaline earth metals (IIA) on one hand and with zinc metals (IIB) on the other.



- (vii) **Diagonal relationship:** The first three members of second period (Li, Be and B) not only show similarities with the members of their own groups but show similarities with the elements diagonally placed in the higher groups. This resemblance is termed diagonal relationship.

Group	I	II	III	IV
2nd period	Li	Be	B	C
3rd period	Na	Mg	Al	Si

Lithium shows similarities with magnesium, beryllium with aluminium and boron with silicon.

- (viii) The elements belonging to the same subgroup exhibit either same properties or a regular gradation.
- (ix) **Coinage metals (Cu, Ag, Au) and Alkali metals:** There appears hardly any resemblance between the members of these two groups except that all show monovalency.

Table 1.4 Some Properties are Given Below to Justify this Statement

<i>IA (alkali metals)</i>	<i>IB (coinage metals)</i>
(i) Soft metals	(i) Hard metals
(ii) Low densities	(ii) High densities
(iii) Low melting points	(iii) High melting points
(iv) Highly reactive	(iv) Less reactive, noble metals
(v) Never found free in nature	(v) Found mainly free in nature
(vi) Decompose water readily	(vi) Do not decompose water

The Screening Effect or Shielding Effect

In a multielectron atom, the electrons of the valency shell (outermost shell) are attracted towards the nucleus and also these electrons are repelled by the electrons present in the inner shells. On account of this, the actual force of attraction between the nucleus and the valency electrons is somewhat decreased by the repulsive forces acting in opposite direction. ***This decrease in the force of attraction exerted by the nucleus on the valency electrons due to the presence of electrons in the inner shells, is called screening effect or shielding effect.*** The magnitude of the screening effect depends upon the number of inner electrons, i.e., higher the number of inner electrons, greater shall be the value of screening effect. The screening effect constant is represented by the symbol σ . The magnitude of ' σ ' is determined by the **Slater's rules**. The contribution of inner electrons to the magnitude of ' σ ' is calculated in the following ways:

For ns or np orbital electrons

- Write the electronic configuration of the element in the following order and group them as, (Is), (2s,2p), (3s,3p), (3d), (4s, 4p), (4d,4f), (5s, 5p), (5d, 5f), (6s, 6p), etc.
 - Electrons to the right of the (ns, np) group are not effective in shielding the ns or np electrons and contribute nothing to σ .
 - All other electrons in the (ns, np) group contribute to the extent of 0.35 each to the screening constant (except for Is for which the value is 0.30).
 - All the electrons in the (n – 1) th shell contribute 0.85 each to the screening constant.
 - All the electrons in the (n – 2) th shell or lower contribute 1.0 each to the screening constant.
- For d- or f-electron,** rules (i) to (iii) remain the same but rules (iv) and (v) get replaced by the rule (vi).
- All the electrons in the groups lying left to (nd, nf) group contribute 1.0 each to the screening effect.

Example 1. Calculation of screening constants of alkali metals for valency electrons.

$$\text{Li} \quad 2,1 = 2 \times 0.85 = 1.7$$

$$\text{Na} \quad 2, 8, 1 = 8 \times 0.85 + 2 \times 1 = 8.8$$

$$\text{K} \quad 2,8,8,1 = 8 \times 0.85 + 10 \times 1 = 16.8$$

$$\text{Rb} \quad 2,8,18,8,1 = 8 \times 0.85 + 28 \times 1 = 34.8$$

$$\text{Cs} \quad 2,8, 18, 18,8, 1 = 8 \times 0.85 + 46 \times 1 = 52.8$$

1.10 Periodic Properties and Chemical Bonding

Example 2. Calculation of screening constants of members of second period for valency electrons.

Li	2, 1	$= 2 \times 0.85$	$= 1.7$
Be	2, 2	$= 0.35 + 2 \times 0.85$	$= 2.05$
B	2, 3	$= 2 \times 0.35 + 2 \times 0.85$	$= 2.40$
C	2, 4	$= 3 \times 0.35 + 2 \times 0.85$	$= 2.75$
N	2, 5	$= 4 \times 0.35 + 2 \times 0.85$	$= 3.10$
O	2, 6	$= 5 \times 0.35 + 2 \times 0.85$	$= 3.45$
F	2, 7	$= 6 \times 0.35 + 2 \times 0.85$	$= 3.80$
Ne	2, 8	$= 7 \times 0.35 + 2 \times 0.85$	$= 4.15$

Example 3. Calculation of screening constant in zinc (a) for a 4s-electron (b) for 3d electron.

- (a) The electronic configuration of zinc (30) is,
 $(1s)^2 (2s2p)^8 (3s3p)^8 (3d)^{10} (4s)^2$
 $\sigma = 10 \times 1.0 + 18 \times 0.85 + 1 \times 0.35 = 25.65$

- (b) For 3d-electron,
 $\sigma = 18 \times 1.0 + 9 \times 0.35 = 21.15$

It is clear from the examples (1) and (2) that *the magnitude of screening constant in the case of s- and p-block elements increases in a period as well as in a group as the atomic number increases.*

From the example (3), it is evident that the *electrons in different orbitals are differently affected by the same nuclear charge depending upon their proximity to the nucleus.*

Effective Atomic Number

Due to screening effect the valency electron experiences less attraction towards nucleus. This brings decrease in the nuclear charge (Z) actually present on the nucleus. The reduced nuclear charge is termed effective nuclear charge and is represented by Z^* . It is related to actual nuclear charge (Z) by the following formula:

$$Z^* = (Z - \sigma) \text{ where } \sigma \text{ is screening constant.}$$

It is observed that *magnitude of effective nuclear charge or effective atomic number increases in a period when we move from left to right.*

II period	Li	Be	B	C	N	O	F	Ne
Z	3	4	5	6	7	8	9	10
σ	1.7	2.05	2.40	2.75	3.10	3.45	3.80	4.15
$Z^* = (Z - \sigma)$	1.3	1.95	2.60	3.25	3.90	4.55	5.20	5.85

In a subgroup of normal elements the magnitude of effective atomic number remains almost the same.

Alkali group	Li	Na	K	Rb	Cs
Z	3	11	19	37	55
σ	1.7	8.8	16.8	34.8	52.8
$Z^* = (Z - \sigma)$	1.3	2.2	2.2	2.2	2.2
IIA group	Be	Mg	Ca	Sr	Ba
Z	4	12	20	38	56
σ	2.05	9.15	17.15	35.15	53.15
$Z^* = (Z - \sigma)$	1.95	2.85	2.85	2.85	2.85

Atomic radius

- (i) The radius of an atom may be taken as the distance between atomic nucleus and the outermost shell containing electrons of the atom.
- (ii) According to the Heisenberg's uncertainty principle the position of a moving electron can not be accurately determined. So the distance between the nucleus and the outermost electron is uncertain.
- (iii) Atomic radius can be determined indirectly from the inter nuclear distance between the two atoms in a gaseous diatomic molecule. This internuclear distance between the two atoms is called bond length.
- (iv) The inter nuclear distance between the two atoms can be measured by $\times$ – ray diffraction or spectroscopic studies.
- (v) **Covalent radius:** One half of the distance between the nuclei (internuclear distance) of two covalently bonded atoms in a homodiatom molecule is called the covalent radius of that atom. The covalent bond must be single covalent bond. The covalent radius (r_A) of atom A in a molecule A_2 may be given as:

$$r_A = \frac{d_{A-A}}{2}$$

i.e., the distance between nuclei of two single covalently bonded atoms in a homodiatom molecule is equal to the sum of covalent radii of both the atoms.

$$d_{A-A} = r_A + r_A$$

In a heterodiatom molecule AB where the electronegativity of atoms A and B are different, the experimental values of internuclear distance d_{A-B} is less than the theoretical values ($r_A + r_B$). According to Schomaker and Stevenson (1941),

$$D_{A-B} = r_A + r_B - 0.09 \Delta_x$$

Where Δ_x is the difference of electronegativities of the atoms A and B.

According to Pauling – If the electronegativities of the two atoms A and B are x_A and x_B respectively then,

$$D_{A-B} = r_A + r_B - (C_1 x_A - C_2 x_B)$$

C_1 and C_2 are the Stevenson's coefficients for atoms A and B respectively.

Metallic Radius

Metal atoms are assumed to be closely packed spheres in the metallic crystal. These metal atom spheres are considered to touch one another in the crystal. One half of the internuclear distance between the two closest metal atoms in the metallic crystal is called metallic radius.

Metallic radius > Covalent radius

For example – Metallic radius and covalent radius of potassium are 2.3 Å and 2.03 Å respectively.

Van der waal's radius or collision radius

The molecules of non metal atoms are generally gases. On cooling, the gaseous state changes to solid state.

In the solid state, the non metallic elements usually exist as aggregations of molecules are held together by Van der Waal forces. **One half of the distance between the nuclei of two adjacent atoms belonging to two neighbouring molecules of a compound in the solid state is called Van der Waal's radius.**

It may also be defined as half of the inter nuclear distance of two non bonded neighbouring atoms of two adjacent molecules.

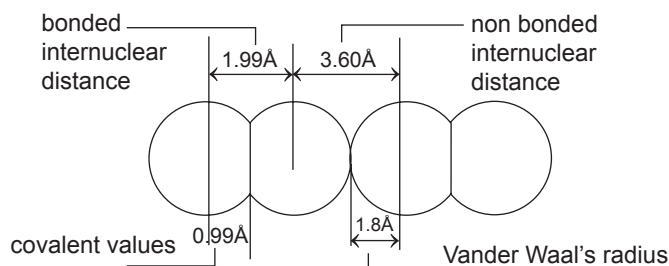


Fig. 1.3

Van der Waal's radius > metallic radius > covalent radius

The Van der Waal's radius and Covalent radius of Chlorine atom are $1.80\text{\AA}$ and $0.99\text{\AA}$ respectively.

Ionic Radius

A neutral atom changes to a cation by the loss of one or more electrons and to an anion by the gain of one or more electrons. The number of charge on cation and anion is equal to the number of electrons lost or gained respectively. The ionic radii of the ions present in an ionic crystal may be calculated from the internuclear distance between the two ions.

It is defined as the distance between the nucleus and outermost shell of an ion or it is the distance between the nucleus and the point where the nucleus exerts its influence on the electron cloud.

- (i) **Radius of a cation:** A cation is formed by removing one or more electrons from a neutral atom. Hence, **the ionic radii of a cation is always less than the atomic radii of that very atom.** This is due to the fact that nuclear charge in the case of a cation is acting on lesser number of electrons and pulls them closer.

	Na	Na ⁺
Number of e ⁻ =	11	10
Number of p =	11	11
	1s ² 2s ² 2p ⁶ 3s ¹	1s ² 2s ² 2p ⁶

Reasons

- (a) The effective nuclear charge increases. For example, in Na atom, 11 electrons are attracted by 11 protons and in Na⁺, 10 electrons are attracted by 11 protons. Thus in the formation of cation, number of electrons decreases and nuclear charge remains the same.
- (b) Generally, the formation of cation results in the removal of the whole outer shell.
- (c) Interelectronic repulsion decreases. The interelectronic repulsion in Na is among 11e⁻ and in Na⁺ among 10e⁻.
- (ii) **Radius of an anion:** In the case of negative ion or an anion, as electron or electrons are added to the neutral atom, the nuclear charge acts on more electrons so that each electron is held less tightly and the electron cloud expands.

Thus, the size of an anion is more than its parent atom.

Na	Na ⁺	Mg	Mg ²⁺	F	F ⁻	Cl	Cl ⁻
1.54 Å	0.95 Å	1.36 Å	0.65 Å	0.72 Å	1.36 Å	0.99 Å	1.81 Å
			Cl		Cl ⁻		
	Number of e ⁻ =	17		18			
	Number of p =	17		17			

Reasons

- (a) The effective nuclear charge decrease in the formation of anion. Thus the electrostatic force of attraction between the nucleus and the outer electrons decreases and the size of the anion increases.
- (b) Interelectronic repulsion increases.
- (iii) **Isoelectronic series:** A series of atoms, ions and molecules in which each species contains same number of electrons but different nuclear charge is called isoelectronic series.

	N ³⁻	O ²⁻	F ⁻	Ne	Na ⁺	Mg ²⁺
Number of e ⁻	10	10	10	10	10	10
Number of p	7	8	9	10	11	12

—————→

- (a) Number of electrons is same.
- (b) Number of protons is increasing.
- (c) So the effective nuclear charge is increasing and atomic size is decreasing. In an isoelectronic series atomic size decreases with the increase of charge.

Some of the examples of isoelectronic series are as under:

- (i) S²⁻, Cl⁻, K⁺, Ca²⁺, Sc³⁺
- (ii) SO₂, NO₃⁻, CO₃²⁻

(iii) N_2 , CO , CN^- (iv) NH_3 , H_3O^+

These can be explained on the basis of z/e ratio.

$\left(\frac{\text{Nuclear charge}}{\text{No. of electron}} \right)$ Whenever z/e ratio increases, the size decreases and when z/e ratio decreases, the size increases.

$$\begin{array}{cccccc} \text{Na} & \text{Na}^+ & \text{Cl} & \text{Cl}^- & \text{Fe}^{2+} & \text{Fe}^{3+} \\ z/e & \frac{11}{11} = 1.0; & \frac{11}{10} = 1.1; & \frac{17}{17} = 1.0; & \frac{17}{18} = 0.95; & \frac{26}{24} = 1.08; & \frac{26}{23} = 1.13 \end{array}$$

Isoelectronic species (ions or atoms) are those which have same number of electrons. For such species the size decreases with an increase of atomic number. This is illustrated in the following table:

Atom or Ion	Atomic number z	No. of electrons e	z/e ratio	Size in Å
C^{4-}	6	10	0.6	2.60
N^{3-}	7	10	0.7	1.71
O^{2-}	8	10	0.8	1.40
F^-	9	10	0.9	1.30
Ne	10	10	1.0	1.12
Na^+	11	10	1.1	0.95
Mg^{2+}	12	10	1.2	0.65
Al^{3+}	13	10	1.3	0.50

H^+ and Cs^+ are the smallest and largest cations respectively.

H^- and I^- are the smallest and largest anions respectively.

Periodicity in atomic radius and ionic radius

1. For normal elements

- (a) In a period from left to right effective nuclear charge increases because the next electron fills in the same shell. So the atomic size decreases. For example, the covalent radii of second period elements in Å are as follows:

Li	Be	B	C	N	O	F
1.23	0.89	0.80	0.77	0.74	0.73	0.72

- (b) In a group moving from top to bottom the number of shells increases. So the atomic size increases. Although the effective nuclear charge increases but its effect is negligible in comparison to the effect of increasing number of shells. For example the covalent radii of IA group elements in Å are as follows:

Li	Na	K	Rb	Cs
1.23	1.57	2.03	2.16	2.35

2. For inert gases

The atomic radius of inert gas (zero group) is shown largest in a period because of its Van der Waal's radius which is generally larger than the covalent radius. The Van der Waal's radius of inert gases also increases in moving from top to bottom in a group.

3. For transition elements

There are three series of transition elements:

3d – Sc (21) to Zn (30)

4d – Y (39) to Cd (48)

5d – La (57), Hf (72) to Hg (80)

1.14 Periodic Properties and Chemical Bonding

(a) From left to right in a period

- (i) The atomic size decreases due to the increase in effective nuclear charge.
- (ii) In transition elements, electrons are filled in the $(n-1)d$ orbitals. These $(n-1)d$ electrons screen the ns electrons from the nucleus. So the force of attraction between the ns electrons and the nucleus decreases.

This effect of $(n-1)d$ electrons over ns electrons is called shielding effect or screening effect. The atomic size increases due to shielding effect and balance the decrease in size due to increase in nuclear charge to about 80%.

- (iii) Thus moving from left to right in a period, there is a very small decrease in size and it may be considered that size almost remains the same.
- (iv) In the first transition series, the atomic size slightly decreases from Sc to Mn because effect of effective nuclear charge is stronger than the shielding effect. The atomic size from the Fe to Ni almost remains the same because both the effects balance each other. The atomic size from Cu to Zn slightly increases because shielding effect is more than effective nuclear charge due to d^{10} structure of Cu and Zn. The atomic radii of the elements of 3d transition series are as under.

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
1.44	1.32	1.22	1.18	1.17	1.17	1.16	1.15	1.17	1.25

4. Inner transition elements

As we move along the lanthanide series, there is a decrease in atomic as well as ionic radius. The decrease in size is regular in ions but not so regular in atoms. This is called lanthanide contraction. The atomic radii in Å are as under:

La	Ce	Pr	Nd	Pm	Sm	Eu	Gd
1.88	1.82	1.83	1.82	1.81	1.80	2.04	1.80
Tb	Dy	Ho	Er	Yb	Lu		
1.78	1.77	1.76	1.75	1.94	1.73		

There are two peaks one at Eu (63) and other at Yb (70). This is due to the difference in metallic bonding. Except Eu and Yb other lanthanides contribute three electrons in metallic bond formation. These two atoms contribute two electrons in the bond formation leaving behind half-filled and completely filled $4f$ -orbitals respectively.

Cause of lanthanide contraction

In lanthanides the additional electron enters into $(n-2)f$ orbital. The mutual shielding effect of $(n-2)f$ electrons is very little because the shape of f -subshell is very much diffused. Thus the effective nuclear charge increases in comparison to the mutual shielding effect of $(n-2)f$ electrons. The outer electrons are attracted more by the nucleus. Consequently the atomic and ionic radii decreases from La (57) to Lu (71).

This type of contraction also occurs in actinides. The jump in contraction between the consecutive elements in the actinides is greater than lanthanides. This is due to the lesser shielding of $5f$ -electrons which are therefore pulled more strongly by the nucleus.

In a group

- (i) The atomic radius of elements increases moving from first transition series (3d) to second transition series (4d). This is due to the increase in number of shells with the increase in atomic number.
- (ii) The atomic radii of second (4d) and third (5d) transition series in a group is almost same except Y(39) and La (57)

In third transition series, there are fourteen lanthanides in between La (57) of III B and Hf (72) of IV B groups, so the atomic radius of Hf(72) decreases much due to lanthanide contraction in lanthanides. The difference in the nuclear charge in the elements of a group in first and second transition series is + 18 units while this difference in second and third transition series is + 32 units except Y (39) $\rightarrow$ La(57). Due to the increase of + 32 units in the nuclear charge there is a sizable decrease in the atomic radius which balances the increase in size due to the increase in number of shells.

So in a group moving from second to third transition series, the atomic radii of the elements almost remain the same except IIIB. The difference is about 0.02Å.

Worked-out Examples

2. The correct order of radii is (IIT 2000)
 (a) $N < Be < B$ (b) $F^- < O^{2-} < N^{3-}$ (c) $Na < Li < K$ (d) $Fe^{3+} < Fe^{2+} < Fe^{4+}$

Ans. As we go from F to O to N, the positive charge in the nucleus decreases. This causes decrease in nucleus-electron attraction causing the electron cloud to spread more in space.

Also, larger negative charge involves larger electron-electron repulsion causing spreading of electron cloud in space.

3. According to the Periodic law of elements, the variation in properties of elements is related to their (AIEEE 2003)
 (a) Nuclear masses (b) Atomic numbers
 (c) Nuclear neutron-proton number ratio (d) Atomic masses

Ans. (b) According to modern periodic law, the properties of the elements are repeated after certain regular intervals when these elements are arranged in order of their increasing atomic Numbers.

4. The reduction in atomic size with increase in atomic number is a characteristic of elements of (AIEEE 2003)
 (a) d-block (b) f-block
 (c) radioactive series (d) high atomic masses.

Ans. (a)

5. The set representing the correct order of ionic radius is:
 (a) $Li^+ > Be^{2+} > Na^+ > Mg^{2+}$ (b) $Na^+ > Li^+ > Mg^{2+} > Be^{2+}$
 (c) $Li^+ > Na^+ > Mg^{2+} > Be^{2+}$ (d) $Na^+ > Mg^{2+} > Li^+ > Be^{2+}$

Ans. (d) Follow the periodic trends

6. The radii of Ar is greater than the radii of chlorine. Why?

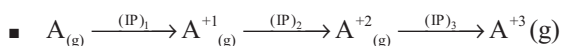
Solution: In chlorine, the radii means the atomic or covalent radii which is actually half the intermolecular distance between 2 atoms whereas in Argon the radii means the Van der Waal's radii as Argon is not a diatomic molecule. Van der Waal's radii is actually half the distance between adjacent molecule. So Van der Waal's radii being larger than atomic radii, Argon, has got a larger radii than chlorine

7. Berilium and Al are placed in different periods and groups but they show the similar properties. Explain

Solution: On moving across a period, the charge on the ions increases and the size decreases, causing the polarising power to increase. On moving down a group, the size increases and polarising power decreases. On moving diagonally i.e., from Be to Al these two effects partly cancel each other and so there is no marked change in properties.

Ionization Potential

- The energy required to remove the most loosely bound electron from the outermost orbit of one mole of isolated gaseous atoms or ion of an element, is called **ionization energy (IE)**. This ionization is an endoergic or energy-absorbing process. The corresponding potential difference to be applied to provide ionization energy to an electron is known as **Ionization Potential (IP)**. Hence, the ionization Energy of hydrogen is 13.6 eV and the ionization Potential is 13.6 V.
- An electron cannot be removed directly from an atom in solid state. For this purpose, the solid state is converted to gaseous state and the energy required for this is called **sublimation energy**.



1.16 Periodic Properties and Chemical Bonding

- The energy required to remove one electron from a neutral gaseous atom to convert it to monovalent cation, is called first ionization potential (I IP). The energy required to convert a monovalent cation to a divalent cation is called second ionization potential (II IP)
- $I \text{ IP} < II \text{ IP} < III \text{ IP}$ because as the electrons go out of the atom, the effective nuclear charge increases and the ionic size goes on decreasing. Thus the forces of attraction on valence shell electrons increases and hence the order.

Factors affecting ionization potential

- Number of shells:** With the increase in number of shells the atomic radius increases i.e., the distance of outer most shell electron from the nucleus increases and hence the ionization potential decreases.
- Effective nuclear charge:** Atomic size decreases with increase in effective nuclear charge because, higher the effective nuclear charge stronger will be the attraction of the nucleus towards the electron of the outermost orbit and higher will be the ionization potential.
- Shielding effect:** The electrons of internal orbits repel the electron of the outermost orbit due to which the attraction of the nucleus towards the electron of the outermost orbit decreases and thus atomic size increases and the value of ionization potential decreases.
- Stability of half-filled and fully filled orbitals:** The atoms whose orbitals are half-filled (p^3 , d^5 , f^7) or fully-filled (s^2 , p^6 , d^{10} , f^{14}) have greater stability than the others. Therefore, they require greater energy for removing an electron. However, stability of fully filled orbitals is greater than that of the half-filled orbitals.
- Penetration power:** In any atom, the s orbital is nearer to the nucleus in comparison to p , d and f orbitals. Therefore, greater energy is required to remove an electron from s orbital than from p , d and f orbitals. Thus the decreasing order of ionization potential of s , p , d and f orbitals is as follows:

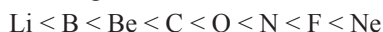
$$s > p > d > f$$

Periodic trends in ionization potential

- In a Period:** The value of ionization potential normally increase on going from left to right in a period, because effective nuclear charge increases and atomic size decreases.

Exceptions

- In second period, ionization, potential of Be is greater than that of B, and in the third period ionization potential of Mg is greater than that of Al due to high stability of fully filled orbitals.
- In second period, ionization, potential of N is greater than O and in the third period ionization potential of P is greater than that of S, due to stability of half-filled orbitals.
- The increasing order of the values of ionization potential of the second period elements is



The increasing order of the values of ionization potential of the third period elements is



Ionization Potential of Transition Elements

- In transition elements, the value of ionization potential has very little increase on going from left to right in a period because the outermost orbit remains the same but electrons get filled up in the $(n-1)d$ orbitals resulting in very little increase in the values of ionization potential.
- In transition element series, the first ionization potential normally increases with increase in atomic number on going from left to right, but this periodicity is not uniform. The value of ionization potential of transition elements depends on the following two important factors.
 - (a) The value of ionization potential increases with increase in effective nuclear charge.
 - (b) The value of ionization potential decreases with increase in shielding effect when the number of electrons increases in $(n-1)d$ orbitals
- In the first transition element series, the first ionization potential normally increases on going from left to right from Sc to Cr because shielding effect is much weaker in comparison to effective nuclear charge. The value of first ionization potential of Fe, Co and Ni remains constant, because shielding effect and effective nuclear charge balance one another.

The value of ionization potential shows slight increase from Cu to Zn because they have fully filled s and d orbitals. The value of first ionization potential of Mn is maximum because it has maximum stability due to fully filled s and half filled orbitals.

Ionization Potential of Inner Transition Elements

The size of inner transition elements is greater than that of d block elements. Therefore the value of ionization potential of f block elements is smaller than that of d block elements and due to almost constant atomic size of f block elements in a period the value of their ionization potential remains more constant than that of d block elements.

In a group

- The value of ionization potential normally decreases on going from top to bottom in a group because both atomic size and shielding effect increase.

Exception:

- The value of ionization potential remains almost constant from Al to Ga in the 13th A group. ($B > Al$, $Ga > In$)
- In 4th group i.e., Ti, Zr and Hf the I.P. of Hf is higher than that of Zr due to Lanthanide contraction. Thus the I.P. of 4th group varies as $Ti > Zr < Hf$.

Some Noteworthy Points

- In the periodic table the element having highest value of ionization potential is He.
- The values of ionization potential of noble gases are extremely high, because the orbitals of outermost orbit are fully-filled (ns^2 , np^6) and provide great stability.
- In a period, the element having least value of ionization potential is an alkali metal (group 1st) and that having highest value is inert gas (Group 18th)

Applications of ionization potential

- The elements having high values of ionization potential have low reactivity, e.g., inert gases.
- The value of ionization potential decreases more on going from top to bottom in a group in comparison to a period. Therefore, reactivity increases and the atom forms a cation by loss of electron.
- The elements having low value of ionization potential readily lose electron and thus behave as strong reducing agents.
- The elements having low value of ionization potential readily lose electron and thus exhibit greater metallic property.
- The elements having low value of ionization potential readily lose electron and thus oxide and hydroxides of these elements have basic property.

Worked-out Examples

8. The atomic numbers of vanadium (V), chromium (Cr), manganese (Mn) and iron (Fe) are respectively 23, 24, 25 and 26. Which one of these may be expected to have the highest second ionization enthalpy? (AIEEE 2003)
- (a) Cr (b) Mn (c) Fe (d) V.

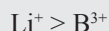
Ans. (a) Cr has $3d^5, 4s^1$ configuration and thus second electron is removed from 3d subshell. In rest, from 4s subshell.

9. Which of the following ions has the highest value of ionic radius? (AIEEE 2004)
- (a) O^{2-} (b) B^{3+} (c) Li^+ (d) F^-

Ans. (a) O^- and F^- are isoelectronic. Hence have same number of shells, therefore greater the nuclear charge smaller will be the size i.e.,



further Li^+ and B^{3+} are isoelectronic. Therefore



Hence, the correct order of atomic size is.



10. The increasing order of the first ionization enthalpies of the elements B, P, S and F (lowest first) is (AIEEE 2006)

(a) $\text{F} < \text{S} < \text{P} < \text{B}$ (b) $\text{P} < \text{S} < \text{B} < \text{F}$ (c) $\text{B} < \text{P} < \text{S} < \text{F}$ (d) $\text{B} < \text{S} < \text{P} < \text{F}$

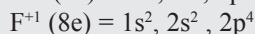
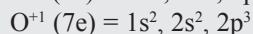
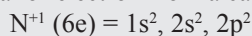
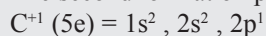
Ans. (b)

11. Which of the following should be the order of increasing values of second ionization potential of C_6 , N_7 , O_8 and F_9 ?

(1) $\text{C} > \text{N} > \text{F} > \text{O}$ (2) $\text{C} < \text{F} < \text{N} < \text{O}$ (3) $\text{C} < \text{F} < \text{N} < \text{O}$ (4) $\text{C} < \text{N} < \text{F} < \text{O}$

Ans. (4)

Solution: The second ionization potential means removal of electron from a cation



Therefore, $\text{C} < \text{N} < \text{F} < \text{O}$

12. Which of the following should be the correct order of the second ionization potential of Li, Be, B and C?

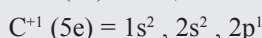
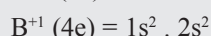
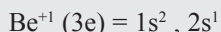
(1) $\text{Li} < \text{Be} > \text{B} < \text{C}$

(2) $\text{Be} < \text{B} < \text{C} < \text{Li}$

(3) $\text{Be} < \text{C} < \text{B} < \text{Li}$

(4) $\text{Li} < \text{C} < \text{B} < \text{Be}$

Ans. (3)



Therefore, $\text{Be} < \text{C} < \text{B} < \text{Li}$

Here, Li^{+1} has inert configuration and B^{+1} has fully filled s orbital. Hence these will have high value of second ionization potential.

13. Which of the following should have least value of ionization potential?

(1) Be^{+3}

(2) H

(3) Li^{+2}

(4) He^{+1}

Ans. (2)

Solution: All the above four species have one electron each, H has least number of proton. Therefore, H will have least value of ionization potential.

14. Which of the following should have less than 11.0 eV difference between the values of first and second ionization potentials?

(1) Mg

(2) Na

(3) K

(4) All of the above three

Ans. (1)

Solution: Mg exhibits higher oxidation state (Mg^{+2}) than Na (Na^{+1}) and K (K^{+1})

15. The first I.P. of nitrogen is greater than oxygen while the reverse is true for their second I.P. values. Explain

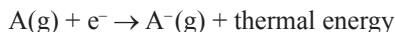
Solution: The first I.P. corresponds to the removal of first electron. Since nitrogen is already half-filled. So more energy is required to remove the electron. But once the electron is removed from oxygen it gains half-filled stability and therefore the 2nd I.P. becomes high.

16. The ionization energy of the coinage metals fall in the order $\text{Cu} > \text{Ag} < \text{Au}$. Explain

Solution: In all the 3 cases an s-electron in the unpaired state is to be removed. In the case of Cu, a 4s electron is to be removed which is closer to the nucleus than the 5s electron of Ag. So I.P. decreases from Cu to Ag. However, from Ag to Au the 14 f electrons are added which provide very poor shielding effect. The nuclear charge is thus enhanced and therefore the outer electron of Au is more tightly held and so the IP is high.

ELECTRON AFFINITY

Atoms not only lose electrons to form positive ions but also gain electrons to form negative ions. **Electron affinity**, as the name indicates, is a measure of an atom's tendency to gain an electron. The higher an atom's electron affinity the more likely it is to gain an electron. Quantitatively, electron affinity is the negative of the enthalpy change associated with the addition of an electron to a mole of gaseous atoms to form gaseous ions with a 1^- charge. Thermal energy is released when an electron is added to most atoms to form a 1^- ion



where A represents an atom of an element. The change shown in equation is exothermic. The energy change associated with an exothermic change always has a sign. The higher an element's electron affinity, the more thermal energy is given off when an electron is added to an atom of the element.

Period																	
1	IA	IIA											IIIA	IVA	VA	VIA	VIIA
2	3 Li 60	4 Be −19											5 B 27	6 C 122	7 N −7	8 O 141	9 F 328
3	11 Na 53	12 Mg −19							VIII								
4	19 K 48	20 Ca −10	21 Sc 18	22 Ti 8	23 V 51	24 Cr 64	25 Mn	26 Fe 16	27 Co 64	28 Ni 112	29 Cu 118	30 Zn −47	31 Ga 29	32 Ge 116	33 As 78	34 Se 195	35 Br 325
5	37 Rb 47	38 Sr	39 Y 30	40 Zr 41	41 Nb 86	42 Mo 72	43 Tc 53	44 Ru 101	45 Rh 110	46 Pd 54	47 Ag 126	48 Cd −32	49 In 29	50 Sn 116	51 Sb 103	52 Te 190	53 I 295
6	55 Cs 45	56 Ba	* Lu	71 Hf	72 Ta 31	73 W 79	74 Re 14	75 Os 106	76 Ir 151	77 Pt 205	78 Au 223	79 Hg −61	80 Tl 20	81 Pb 35	82 Bi 91	83 Po 183	84 At 270
7	87 Fr 44	88 Ra	◆ Lr	104 Unq	105 Unp	106 Unh	107 Uns	108 Uno	109 Uue								

Fig. 1.4 Electron affinities of the elements, kJ/mol

The most recent figure shows that electron affinity is a rather irregular periodic function of atomic number. In general, electron affinity increases going from left to right across a row of the periodic table, reaching a maximum with the halogens in Group VIIA. The halogens have a much greater tendency to gain an electron than the alkali metals in Group IA. Both actual and effective nuclear charges increase across a period. Atomic radius decreases and an electron added to the outer shell is closer to a larger positive charge in atoms on the right side of the periodic table. Thus, more energy is released when an electron is added.

As you can see from figure, only nitrogen and the elements of Groups 2, 12, and 18 have negative electron affinities. Energy is required to force electron into atoms of nitrogen, the alkaline earth metals (Group 2), the elements of Group 12, and the noble gases (Group 18). Addition of an electron to nitrogen and to atoms of the elements of the groups listed is an endothermic process. Because the noble gases have both high ionization energies and low electron affinities, the noble gases are unreactive.

In atoms of elements of Groups 2, 12, and 18, all subshells that contain: electrons are filled; the next electron must go into a higher energy subshell. The irregularities of the Group 15 elements, such as N and P, are also explained by electron configuration. An electron added to an atom of a Group VA element must go into a p orbital that already contains an electron. The electron already present in the orbital repels the second electron. As a result, less energy than expected is released when an electron is added to a Group VA element.

In general, electron affinity decreases going down groups in the periodic table, for example, $\text{Cl} > \text{Br} > \text{I} > \text{At}$ (see figure below). Going down a group, the radii of the shells increases because the principal quantum number is higher. The added electron is further from the positive charge in the nucleus. Therefore, the quantity of energy released when an electron is added is smaller.

Fluorine appears out of line compared to the other members of Group VIIA. The electron affinities of oxygen and nitrogen also seem low. These three atoms are small and have high effective nuclear charges. It is surprising that so little energy is given off when an electron is added to nitrogen, oxygen, and fluorine atoms. However, the second shell is small and repulsions are large between the electrons already present in the second shell and the electron that is entering. In third (and higher) period elements, the electrons already present in the outer shell are spread out through a larger volume. Repulsions between them and the electron that is entering are smaller.

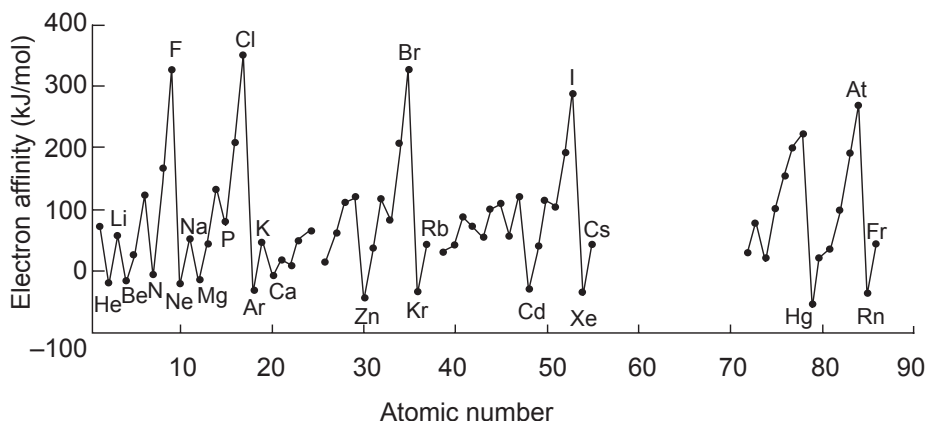


Fig. 1.5 Variation of electron affinities of the elements with atomic number. Unfortunately, the electron affinities of the lanthanides and of a few other elements have not yet been measured

Many properties of the first members of groups of representative elements are out of line compared with the properties of the larger members of the group. In several cases, the properties of elements in the second period are more like the properties of the third period element in the next-higher-numbered group than they are like the properties of the other members of the same group. Thus, lithium is similar to magnesium, beryllium to aluminum, and boron to silicon.

Worked-out Examples

17. The formation of the oxide ion $O^{2-}(g)$ requires first an exothermic and then an endothermic step as shown below:
 $O(g) + e^- = O^-(g), \Delta H^\circ = -142 \text{ kJ mol}^{-1}$
 $O(g) + e^- = O^{2-}(g), \Delta H^\circ = 844 \text{ kJ mol}^{-1}$

This is because

(AIEEE 2004)

- O^- ion will tend to resist the addition of another electron
- Oxygen has high electron affinity
- Oxygen is more electronegative
- O^- ion has comparatively larger size than oxygen atom.

Ans. (a) O^- ion exerts a force of repulsion on the incoming electron. The energy is required to overcome it.

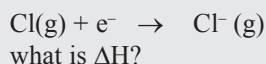
18. In which of the following arrangements the order is not according to the property indicated against it?

(AIEEE 2005)

- $Al^{3+} < Mg^{2+} < Na^{2+} < F^-$ increasing ionic size
- $B < C < N < O^-$ increasing first ionization enthalpy
- $I < Br < F < Cl^-$ increasing electron gain enthalpy (with negative sign)
- $Li < Na < K < Rb^-$ increasing metallic radius.

Ans. (b) In a period the value of ionization potential increases from left to right with breaks where the atoms have some what stable configuration. In this case, N has half-filled stable orbital. Hence has highest ionization energy. Thus the correct order is $B < C < O < N$.

19. For the addition of one mole of electrons to one mole of gaseous chlorine atoms



Solution: The electron affinity of Cl is 349 kJ/mol. Electron affinity is defined as the negative of the enthalpy change associated with the addition of an electron to a mole of gaseous atoms to form gaseous ions:

$$\text{electron affinity} = -\Delta H$$

$$349 \text{ kJ/mol} = -\Delta H$$

$$-349 \text{ kJ/mol} = \Delta H$$

The idea of electron affinity as a measure of tendency to gain an electron can be extended to ions. Electron affinities of anions are negative. An electron is repelled by the negative charge of an anion and energy must be added to force an electron onto a negative ion. Addition of electrons to negatively charged ions is always endothermic. To add a second electron to an O^- ion to form an O^{2-} ion requires 708 kJ/mol of energy:



The energy needed to force an electron into an O^- ion is greater than the energy given off when an electron is added to an oxygen atom to form the O^- ion:



Addition of above equations and gives:
$$\frac{[\text{O(g)} + \text{e}^- \rightarrow \text{O}^-(\text{g}) + 141 \text{ kJ}] + [\text{O}^-(\text{g}) + \text{e}^- + 708 \text{ kJ} \rightarrow \text{O}^{2-}(\text{g})]}{\text{O(g)} + 2\text{e}^- + 567 \text{ kJ} \rightarrow \text{O}^{2-}(\text{g})}$$

Thus, the formation of an oxide ion, O^{2-} , in the gas phase is endothermic. In addition, entropy (disorder) decreases when one oxygen atom and two electrons combine to form an oxide ion. Because of the combined unfavourable effects of both energy and entropy changes, oxide ions, O^{2-} , are not formed spontaneously in the gas phase.

However, oxide ions are common in solids. In solids, the negatively charged oxide ions are close to positively charged ions. The attraction between oppositely charged ions is large and the combination of anions and cations is very exothermic.

Periodicity in electron affinity:

- (i) In general electron affinity value increases on moving from left to right in a period because effective nuclear charge increases.

Exceptions

- The electron affinity value of alkaline earth metals of IIA group is zero.
 - Electron affinity value of alkali metals of IA group is also approximately zero because these elements have the tendency of losing the electron instead of gaining the electron.
 - Electron affinity values of nitrogen and phosphorous (15) are lesser than the electron affinity values of carbon and silicon respectively. It is due to the comparatively stable half-filled configuration (np^3) of nitrogen and phosphorus and the tendency to acquire the stable np^3 configuration by the gain of one electron in carbon and silicon (np^2).
 - The theoretical value of the electron affinity of zero group inert gas elements is zero due to stable s^2p^6 configuration.
- (ii) In a group moving from top to bottom the electron affinity value of elements decreases because the atomic size increases.

Exceptions

- Electron affinity values of second period elements are smaller than the electron affinity values of third period elements of group 15 to 17. This unexpected behaviour can be explained by the very much high value of charge

densities, of second period elements due to much smaller size. The electron being added experiences comparatively more repulsion and the electron affinity value decreases.

- (b) The electron affinity of fluorine (Second period) is less than the electron affinity of chlorine (third period). 2p-orbitals in fluorine are much more compact than 3p-orbitals of chlorine. So the electron being added in 2p-orbitals experiences comparatively more repulsion and the electron affinity value decreases.

Worked-out Example

20. Electron affinity of SF_5 is among the highest known but that of SF_6 is quite modest.

Solution: Sulphur in SF_6 is saturated in terms of maximum covalency and maximum co-ordination number and so it has very little tendency to attract electron. In SF_5 which is actually a free radical so it has got a very strong tendency to attract an electron to fulfill its covalency and co-ordination number. Therefore the electron affinity of SF_5 is high.

Electrovalent or covalent—the concept of electronegativity

Electronegativity is a measure of the power of an atom to attract electrons; it involves the two terms, ionization energy and electron affinity. Mulliken has defined electronegativity as the arithmetical mean of the first ionization energy and the electron affinity of an atom. Pauling's definition is somewhat different and it is his electronegativity values that are given in Table. Fluorine, the most electronegative element, is given an arbitrary value of 4.0 and the electronegativities of the atoms of other elements are related to it.

Table 1.5 The Pauling Electronegativity Values of Some Elements

			H 2.1			
Li 1.0	Be 1.5	B 2.0	C 2.5	N 3.0	O 3.5	F 4.0
Na 0.9	Mg 1.2	Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0
K 0.8	Ca 1.0	—	Ge 1.7	As 2.0	Se 2.4	Br 2.8
Rb 0.8	Sr 1.0	—	Sn 1.7	Sb 1.8	Te 2.1	I 2.4
Cs 0.7	Ba 0.9					

As the table shows, electronegativity decreases down a particular group. This is because the atom becomes progressively larger and the inner electron shells decrease the attraction between the positive nucleus and the peripheral electrons. Electronegativity increases across a particular period from alkali metal to halogen, since the progressive increase in nuclear charge exerts a contracting effect on the electron shells (in the same period, additional electrons reside in the same shell).

A bond formed between two atoms of similar electronegativity will be essentially covalent. Increase in electronegativity of one atom will result in that atom having greater control over the bonding pair of electrons, i.e., the covalent bond will be polarized. Further increase in electronegativity will result in increased polarity of the bond until eventually the electron pair can be considered to reside almost entirely on one atom, i.e., an electrovalent bond will be established, thus:



electronegativities of the atoms increase in the order $\text{X} < \text{Y} < \text{Z}$.

It now becomes clear why Group 1 and 2 elements react with Group 16th and 17th elements to give essentially electrovalent compounds, caesium fluoride, Cs^+F^- , being the most 'ionic' compound. On the other hand, non-metals react by forming essentially covalent bonds.

The reason why aluminium fluoride is an electrovalent compound, $\text{Al}^{3+}(\text{F}^-)_3$, whereas aluminium chloride, Al_2Cl_6 , is covalent is probably because the fluorine atom is more electronegative than the chlorine atom.

An explanation in terms of lattice energy would run as follows:

Since the fluoride ion is smaller than the chloride ion, the lattice energy of the $\text{Al}^{3+}(\text{F}^-)_3$ crystal structure would be greater than that of a similar $\text{Al}^{3+}(\text{Cl}^-)_3$ structure by virtue of the closer approach of the oppositely charged ions (other things being equal). In this instance, the two explanations amount to practically the same thing, since the fluorine atom is more electronegative than the chlorine atom because it is smaller and consequently gives rise to a smaller anion.

Electronegativity scale

Some arbitrary scales for the quantitative measurement of electronegativities are as under:

- (i) **Pauling's scale:** Pauling related the resonance energy (Δ_{AB}) of a molecule AB with the electronegativities of the atoms A and B. If x_{A} and x_{B} are the electronegativities of atoms A and B respectively then,

$$0.208 \sqrt{\Delta_{\text{AB}}} = x_{\text{A}} - x_{\text{B}} \text{ if } x_{\text{A}} > x_{\text{B}}$$

$$\text{or } \Delta_{\text{AB}} = 23.06 (x_{\text{A}} - x_{\text{B}})^2$$

$\Delta_{\text{AB}} = E_{\text{A-B(experimental)}} - E_{\text{A-B(theoretical)}}$ where $E_{\text{A-B}}$ is the energy of A–B bond. In a purely covalent molecule, AB, the experimental and theoretical values of bond energy A–B are equal.

$$\text{So } \Delta_{\text{AB}} = 0$$

$$\text{or } 0 = 23.06 (x_{\text{A}} - x_{\text{B}})^2$$

$$\text{or } x_{\text{A}} = x_{\text{B}}$$

In an ionic molecule AB, $E_{\text{A-B(experimental)}}$ is more than $E_{\text{A-B(Theoretical)}}$.

Pauling assumed the electronegativity value of fluorine 4 and calculated the electronegativity values of other elements from this value.

- (ii) **Mulliken's electronegativity:** In 1934, Mulliken suggested an alternative approach to electronegativity based on the ionization energy and electron affinity of an atom. Consider two atoms A and B. If an electron is transferred from A to B, forming ions A^+ and B^- , then the energy change is the ionization energy of atom A (I_{A}) minus the electron affinity of atom B (E_{B}), that is, $I_{\text{A}} - E_{\text{B}}$. Alternatively, if the electron was transferred the other way to give B^+ and A^- ions, then the energy change would be $I_{\text{B}} - I_{\text{A}}$. If A^+ and B^- are actually formed, then this process requires less energy, and

$$(I_{\text{A}} - E_{\text{B}}) < (I_{\text{B}} - E_{\text{A}})$$

Rearranging,

$$(I_{\text{A}} + E_{\text{A}}) < (I_{\text{B}} + E_{\text{B}})$$

Thus Mulliken suggested that electronegativity could be regarded as the average of the ionization energy and the electron affinity of an atom.

$$\text{Electronegativity} = \frac{(I + E)}{2}$$

Mulliken used I and E values measured in electron volts, and the values were about 2.8 times larger than the Pauling values. We now measure I and E in kJ mol^{-1} . The energy 1 eV/molecule = $96.48 \text{ kJ mol}^{-1}$, so the commonly accepted Pauling values are more nearly obtained by performing this calculation $(I + E)/(2 \times 2.8 \times 96.48)$ or $(I - E)/540$.

This method has a simple theoretical basis, and also has the advantage that different values can be obtained for different oxidation states of the same element. It suffers from the limitation that only a few electron affinities are known. It is more usual to use the approach based on bond energies.

ALLRED AND ROCHOW SCALE

In 1958, Allred and Rochow considered electronegativity in a different way, and worked out values for 69 elements. They defined electronegativity as the attractive force between a nucleus and an electron at a distance equal to the covalent radius. This force F is electrostatic, and is given by:

$$F = \frac{e^2 \cdot Z_{\text{effective}}}{r^2}$$

where e is the charge on an electron, r is the covalent radius, and $Z_{\text{effective}}$ is the effective nuclear charge. The latter is the nuclear charge modified by screening factors for the orbital electrons. The screening factors vary depending on the principal quantum number (the shell that the electron occupies), and the type of electron, *s*, *p*, *d* or *f*. Screening factors have been worked out by Slater, so this provides a convenient method of calculating electronegativity values. These F values may be converted to electronegativity values on the Pauling scale of values using an empirical relationship:

$$\chi = 0.744 + \frac{0.359 Z_{\text{effective}}}{r^2}$$

The electronegativity values so obtained agree quite closely with those obtained by Pauling and Mulliken.

As the oxidation number of an atom increases, the attraction for the electrons increases, so the electronegativity should also increase. Allred and Rochow's method gives slightly different values:

Mo(II)	2.18	Fe(II)	1.83	Tl(I)	1.62	Sn(II)	1.80
Mo(III)	2.19	Fe(III)	1.96	Tl(III)	2.04	Fe(IV)	1.96
Mo(IV)	2.24						
Mo(V)	2.27						
Mo(VI)	2.35						

Allred and Rochow's method depends on measuring covalent radii (and these are obtained with great accuracy by X-ray crystallography) so it might be expected to yield very accurate electronegativity values. This is not so, because although the interatomic distances can be measured very precisely, covalent radii are much less well known because the multiplicity of the bond is not known for certain, that is the bond may possess

Z = The actual number of charge present in the nucleus i.e., number of protons and σ = Shielding constant

Factors Affecting the Magnitude of Electronegativity

- (i) **Atomic radius:** As the atomic radius of the element increases the electronegativity value decrease.

$$\text{Electronegativity} \propto \frac{1}{\text{Atomic radius}}$$

- (iii) **Effective nuclear charge:** The electronegativity value increases as the effective nuclear charge on the atomic nucleus increases.

$$\text{Electronegativity} \propto \text{Effective nuclear charge } (Z_{\text{eff}})$$

- (iii) **Oxidation state of the atom:** The electronegativity value increases as the oxidation state (i.e., the number of positive charge) of the atom increases.

- (iv) **Hybridization state of an atom in a molecule:** If the *s*-character in the hybridization state of the atom increases electronegativity increases because *s*-electrons are comparatively nearer to the nucleus. For example, the electronegativity values of C-atom in various hybridization states are as under:

Hybridization states	sp^3	sp^2	sp
<i>s</i> -Character	25%	33.33%	50%
Electronegativity	2.48	2.75	3.25

→
s-character is increasing

So the electronegativity value is increasing

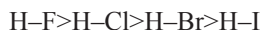
Periodicity in electronegativity

- (i) In a period, moving from left to right, the electronegativity increases due to the increase in effective nuclear charge.
- (ii) In a period, the electronegativity value of gp 1 alkali metal is minimum and that of gp 17 halogen is maximum.
- (iii) In a group moving from top to bottom, the electronegativity decreases because atomic radius increases.
- (iv) The electronegativity value of F is maximum and that of Cs is minimum in the periodic table.
- (v) The electronegativity of Cs(55) should be more than Fr(87) but it is less. This is due to the increase of +32 units in nuclear charge of Fr which makes the effective nuclear charge comparatively high.
- (vi) On moving from second to third transition series in a group [except Y(39) $\rightarrow$ La (57)], electronegativity increases due to the increase of +18 units in nuclear charge.
- (vii) The electronegativity of inert gas elements of zero group is zero due to stable s^2p^6 configuration. Inert gases are monoatomic molecules and the electronegativity is of bonded atoms.

Applications of electronegativity

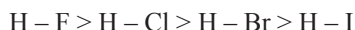
- (i) **Partial ionic character in covalent bond:** The ionic character of a covalent bond increases as the electronegativity difference of bonded atoms increases. According to Haney and Smith, if the electronegativity difference of bonded atoms is Δx then percentage ionic character of the bond $= 16\Delta x + 3.5\Delta x^2$

If the value of Δx is 2.1 then ionic character percentage is about 50. For example the order of ionic character in H-X bond is as follows:

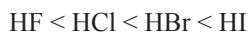


Because the electronegativity difference of bonded atoms (Δx) decreases.

- (ii) **Bond strength:** If the electronegativity difference of covalently bonded atoms (Δx) increases, the bond energy of the covalent bond also increases. For example, the order of the H-X bond strength is,

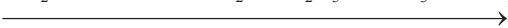


As the bond strength is decreasing the acid strength is increasing. So order of increasing acid strength is,



- (iii) **Acidic and basic nature of oxides of normal elements in a period:** The acidic nature of the oxides of normal elements increases as we move from left to right in a period. In a period from left to right the electronegativity of the elements increases. So the difference of the electronegativities of oxygen and the elements ($x_O - x_E$) decreases. If the ($x_O - x_E$) values is about 2.3 or more then oxide will be basic. If ($x_O - x_E$) values is less than 2.3 the oxide will be acidic. The oxides of the IIIA elements are amphoteric.

The order of acidic or basic nature of the oxides of third period elements may be given as under:



The value of $x_O - x_E$ is decreasing

Basic nature is decreasing

Acidic nature is increasing

- (iv) **Metallic and non metallic properties of elements:**

- (a) The metallic character decreases as the electronegativity of the element increases.
- (b) On moving from left to right in a periods, the electronegativity of the elements increases. So the metallic character decreases.
- (c) On moving down a group, the electronegativity of the elements decreases. So the metallic character increases.

- (v) **Basic nature of the hydroxides of elements:** A hydroxide MOH of an element M may ionize in two ways in water.



If the ionization is according to eqn (1) then it is acidic. It is possible when ionic character of O-H bonds is more than the ionic character of M-O bond i.e., $(x_O - x_H) > (x_O - x_M)$ where x_O , x_H and x_M are the electronegativities of oxygen, hydrogen and element respectively.

If the ionization is according to eqn. (2) then it is basic. This is only possible when ionic character of O–H bond is less than M–O bond i.e. $(x_{\text{O}} - x_{\text{H}}) < (x_{\text{O}} - x_{\text{M}})$

Chemical Bond

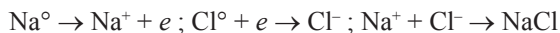
A chemical bond is defined as the force holding together two atoms or groups of atoms forming an aggregate of ions, or molecular species such that there occurs a lowering of energy. Such definition permits us to discuss chemical bonds under the following three classes:

1. The ionic bond (also called electrovalent or electrostatic bond),
2. The covalent bond and
3. The metallic bond.

It will be proper to warn at the outset that these three bonds represent only three extreme types of forces between atoms. There will be many chemical compounds whose properties would indicate bonds of intermediate types. The discussion on the above bond types will be followed by a short resume on the various intermolecular forces, often expressed as van der Waals forces.

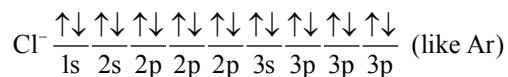
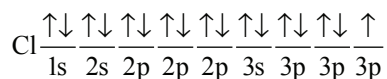
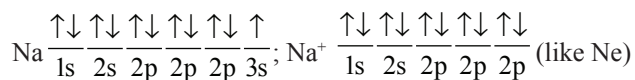
The ionic bond

It is wellknown that sodium and chlorine react vigorously to yield crystals of sodium chloride. Crystallographic studies have shown that there is no discrete NaCl molecule in the crystal lattice; instead each sodium ion (Na^+) is surrounded by six chloride ions (Cl^-) and vice versa. This packing of two ions is repeated throughout in a non-ending pattern. The two oppositely charged ions are held together by means of electrostatic forces of attraction. Each neutral sodium atom Na° loses one electron to form a monpositive sodium ion (Na^+). Similarly, a neutral chlorine atom (Cl°) gains one electron to form a mononegative chloride ion (Cl^-). The two ions Na^+ and Cl^- then build up a close packed type ionic crystal structure. Conceptually, we find a transfer of electron from sodium atom to the chlorine atom:



Ionic bonds are thus electrostatic forces that bind together oppositely charged ions formed by the transfer of electron(s) from an electropositive metal to an electronegative non-metal atom.

Let us now examine the motive of the above two elements behind the reaction. An examination of the electronic configuration of the elements and ions shows that both the ions have attained noble gas configuration.



The noble gas elements have filled subshells, and their chemical inertness points to the stability of their electronic configuration. The truth of this is strikingly shown by their high ionization energy/potentials and very poor electron affinities. *Attainment of stable noble gas electronic configuration is thus the clue to chemical reactions.* Such conclusion helps us to predict which elements are most likely to form ionic compounds. *The elements which have low ionization energy/potentials are the best candidates for forming positive ions (cations). Similarly, the elements which have high electron affinity are most likely to form negative ions (anions).* Factors that influence ionization potential, electron affinity and electronegativity also influence the formation of ionic compounds. It is possible to draw the following conclusions:

- (a) The lower the charge on the cation the better is the chance that an ionic compound will be formed. The lower the charge on the cation the lower is the ionization potential/energy.
- (b) The formation of ionic compounds is favoured by large size of the metal atom or ion since an increase in size is associated with a decrease in ionization potential/energy.

- (c) Small size and low charge on the anion will favour ionic compounds because these factors are associated with increasing electron affinity and electronegativity.

The above discussion gives us a rough guide as to when to expect an ionic compound. Unexpected results are sometimes observed. For instance, the second electron affinity of oxygen is negative, yet it forms stable, ionic compounds with alkalis or alkaline earths. Clearly, there must be other factors which play significant roles in the formation of ionic compounds.

Lattice Energy

Mere formation of oppositely charged ions does not lead to an ionic compound. In an idealised ionic compound there is considerable electrostatic force operating between the cations and anions considered to be hard spheres. When these cations and anions are brought together in large and equal (or equivalent) numbers they will arrange themselves in a regular pattern so as to attain a close packed type structure. For example the structure of sodium chloride reveals that each sodium ion is surrounded by as many as six chloride ions and vice versa. In caesium chloride each caesium ion has eight chloride ions as near neighbours and vice versa. While attaining such close packed structure energy is liberated and thus the system gets stabilized. *The energy liberated when one mole of the ionic crystal is formed from an assembly of free gaseous ions from infinite separation is called the lattice energy.*



Conversely, work has to be done on the ionic crystal to disperse the crystal into an assembly of widely separated ions. Formation of the lattice is an exothermic process while lattice dispersion is an endothermic process.

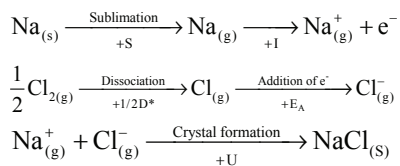
Table 1.6 Lattice Energies of Some Alkali Halides

Alkali halide	A–X distance in A (pm)	Calculated lattice energy kcal/ mole (kJ/mole)	Lattice energy from Born-Haber cycle kcal/mole (kJ/mole)
LiF	2.01 (201)	–254 (–1062)	–244 (–1021)
LiCl	2.57 (257)	–196 (–820)	–202 (–845)
LiBr	2.75 (275)	–184 (–770)	–192 (–803)
LiI	3.00 (300)	–169 (–707)	–80 (–753)
CsF	3.00 (300)	–175 (–732)	–173 (–724)
CsCl	3.56 (356)	–150 (–627)	–158 (–661)
CsBr	3.71 (371)	–143 (–598)	–150 (–627)
CsI	3.95 (395)	–135 (–565)	–145 (–606)

1 kcal/mole = 4.184 kJ/mole: 1 A = 100 pm

Energetics of Formation of Ionic Substances

The energy included in the formation of an ionic compound from its constituent elements may be considered as shown by the Born-Haber Cycle for the formation of one mole of sodium chloride from sodium and chlorine.



Where,

- S = heat of sublimation of sodium metal
- I = ionization energy of sodium
- D = heat of dissociation of molecular chlorine
- E_a = electron affinity of chlorine, and
- U = lattice energy of sodium chloride

1.28 Periodic Properties and Chemical Bonding

The amount of heat liberated in the overall reaction is the heat of formation of sodium chloride. From the above,

$$\Delta H_f = S + I + \frac{1}{2}D - E_a - U$$

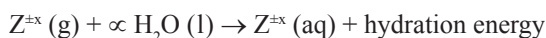
The most important of these energy terms are I , E_a and U , since these are considerably greater than the remaining terms S and D .

More the negative value of the heat of formation, greater would be the stability of the ionic compound produced. Thus on the basis of the above equation, formation of an ionic compound is favoured by,

- low ionization energy (I) of the metal.
- high electron affinity (E_a) of the other element.
- higher lattice energy (U) of the resulting compound.

Solvation Energy

When a cation or an anion in the gaseous state reacts with a solvent in the liquid state, energy is liberated. This liberated energy is termed solvation energy. When the solvent concerned is water the liberated energy is called hydration energy:



Water is a polar solvent (each hydrogen is δ^+ while each oxygen is δ^-). A gaseous cation is attracted by the negative oxygen end of the water molecule while the anion is attracted by the hydrogen end. The greater the *charge/surface area* i.e., the *charge density* of the ion the greater will be the amount of energy liberated. This explains why Li^+ has a higher solvation energy than either Na^+ or K^+ . Again Mg^{2+} has a higher hydration energy than Ba^{2+} or Li^+ . For the same reason hydration energy decreases from F^- to I^- ion.

Interestingly, there is a considerable difference in the hydration energies of F^- and K^+ ions (Table), although they have the same ionic radius ($\approx 1.36 \text{ \AA}$; 136 pm). This is due to the fact that F^- interacts with the small hydrogen atom (with δ^+ charge) of H_2O while K^+ has to interact with the much larger oxygen atom (with δ^-). More hydrogen atoms can get around a F^- ion compared to oxygen atoms around the K^+ ion. Thus, there is scope of strong hydrogen bonding in the case of F^- but not in the case of K^+ .

Table 1.7 Hydration Energy of Some Common ions

<i>Ion</i>	<i>Hydration energy kcal/mole (kJ/mole)</i>	<i>Ion</i>	<i>Hydration energy kcal/mole (kJ/mole)</i>
H^+	-260 (-1091)	F^-	-121 (-506)
Li^+	-124.4 (-521)	Cl^-	-87.0 (-364)
Na^+	-97.0 (-406)	Br^-	-80.0 (-335)
K^+	-77.0 (-322)	I^-	-70.0 (-293)
Mg^{2+}	-459.0 (-1920)	ClO_4^-	-57.0 (-238)
Ba^{2+}	-325.0 (-1360)		

The high charge density of H^+ leads to high hydration energy of the ion, Covalently bonded hydrogen chloride dissolves in water to form hydrated ions. In water there is an interaction between the dipoles of HCl and H_2O . The polar hydrogen end of HCl moves close to the negative oxygen end of water while the chlorine end orients itself towards the hydrogen end of water. Combined hydration energies of H^+ and Cl^- ($-1091 - 364 = -1455 \text{ kJ/mole}$) is so high that it leads to splitting of the covalent bond in HCl in favour of formation of the hydrated ions.

Properties Associated with Ionic Bond

- Ionic crystal lattice is built up of oppositely charged ions.
- Since electrostatic forces of attraction are omnidirectional i.e., extend in all directions, each ion tends to drag as many of opposite kind around itself. A non-ending array of alternate positive and negative ions exists. Hence no isolated discrete molecule exists in the crystal lattice; instead “giant molecules” are formed. Ionic bonds are non-directional.
- Ionic compounds are hard, and have high melting points and boiling points. This is because strong electrostatic forces have to be overcome in order to provide the necessary freedom of movement that characterises a liquid state.

So long as the ionic charge is not too large, hardness of ionic crystals increases with increasing ionic charge and decreasing ionic radius:

Oxide	SrO	CaO	MgO
M^{2+} radius: A (pm)	1.10(110)	0.99(99)	0.65(65)
Hardness (Mohs' scale)	3.5	4.5	6.5

(In Mohs scale hardness of diamond is taken as 10).

4. Ionic compounds are soluble in polar solvents and insoluble in nonpolar solvents. When an ionic compound is dissolved the crystal lattice is ruptured and the ions are held farther apart in solution than in the crystal. During the interaction of ionic compound with solvent, the lattice energy is lost but a solvation energy is gained due to the ions getting solvated. The negative end of a dipolar solvent molecule, attracts the cation and the positive end faces the anions. A polar solvent like water can penetrate the crystal lattice of an ionic compound and can effect dissolution. The solvent character is decided by the dielectric constant of the solvent, and the higher the dielectric constant the better it is as a solvent for ionic compounds. Conversely a non-polar solvent does not possess the necessary dipoles to initiate dissolution of an ionic compound. It has a low dielectric constant.
5. Ionic compounds are conductors in fused state or in solution. In solid crystalline state such compounds are only poor conductors. In fused state an ionic compound is composed of ions which are not confined to fixed positions as in the crystal lattice. The freely moving ions can therefore permit electric conductance.
6. The reactions of ionic compounds are reactions of their ions and these are instantaneous in solution. Common precipitation reactions used in analysis belong to the class of ionic reactions.

Some Noteworthy Points

- Ionic bond was proposed by Kossel.
- Elements of group 1 and group 2 on combining with halogens, oxygen and sulphur generally, form ionic bonds.
- Beryllium generally forms covalent/polar covalent bonds.
- Bonding in compounds of transition metals (in lower oxidation state) is ionic with partial covalent character.
- No bond is 100% ionic in character. It has some percentage of covalent character which is explained on the basis of Fajan rules (discussed later).
- The lattice energy of bi-bivalent ions > uni-bi, or bi-uni > uni- univalent ions.
- $PbSO_4$, AgCl, AgBr and AgI are ionic compounds but only sparingly soluble (almost insoluble) in water because of high value of lattice enthalpy.
- The polar solvents have high values of dielectric constant. Water is one of the best solvents as it has high value of dielectric constant. Due to high value of dielectric constant, the electrostatic force of attraction between the ions decreases and these ions get separated and ultimately get solvated by the solvent molecules. The non-polar solvents having very low value of dielectric constants are not capable of dissolving ionic compounds.
- The dielectric constant of D_2O is greater than H_2O .
- Dissolution of most of the ionic solids like NaCl, KCl, KBr, NH_4NO_3 etc., in water is endothermic i.e., ΔH_{soln} is positive. This clearly indicates that in all such cases;

$$\Delta H_{hydration\ energy} < \Delta H_{lattice\ energy}$$

Dissolution in such cases is possible due to large positive value of ΔS_{soln} which, makes the process spontaneous by making and ΔG_{soln} negative ($\Delta G = \Delta H - T\Delta S$).

Worked-out Examples

21. $BaSO_4$ being an electrovalent compound does not pass into solution state in water. Explain.

Ans. The hydration energy of $BaSO_4$ being lesser than the lattice energy and thus ions are not separated.

22. Why do electrovalent compounds exist as solid?

Ans. Strong non-directional coulombic forces between ions result in higher lattice energy.

23. Sodium conducts electricity in solid state while sodium chloride does not?

Ans. Sodium being metal involves metallic bonding between atoms in solid state having mobile electrons which conducts current. In NaCl, ions are held in lattice and not free to move.

The Covalent Bond

It is easy to have an understanding of an ionic bond which is formed as a result of electrostatic attraction between oppositely charged ions. Conceptually, it is, however, much difficult to gain an understanding of covalent bonding. It is unreasonable to suppose that the bond holding identical non-metal atoms to each other, as in H_2 or F_2 molecule, could be formed in the way sodium chloride is formed. Electron transfer seems more reasonable when two atoms with widely different electron affinity, such as hydrogen and fluorine, combine to form the compound HF. The properties of HF, however, differ widely from those characteristic of ionic compounds. Hydrofluoric acid is a non-conductor of electricity, is low boiling etc. Lewis suggested that such non-ionic molecular compounds (H_2 , F_2 , HF etc.), are formed as a result of sharing of electrons between atoms whereby each atom can attain a noble gas configuration (that is either two electrons as in helium, or an octet in the outermost shell like the other noble gases). *A covalent bond may be defined as a force holding together atoms through sharing of electrons.*

Hypervalent and Subvalent Compounds: There are many compounds where the number of electrons around an atom exceed an octet. Such compounds are called *hypervalent* compounds. Examples are SF_6 , $POCl_3$, $SOCl_2$, etc. On the other hand, there are compounds where the number of electrons around an atom falls short of an octet e.g., NO, NO e.g., etc. These latter are known as subvalent compounds.

Worked-out Example

24. Which properties of the elements depend on the electronic configuration of atoms and which do not?

Ans. Chemical and physical properties depends on electronic configuration; nuclear properties dot not.

Valence Bond (VB) Theory of Covalent Bond: We have so far not **attempted to** answer the logical question as to why at all sharing of electrons gives rise to the formation of a stable bond. The exact answer lies in quantum mechanics. We will endeavour to offer an explanation in popular language.

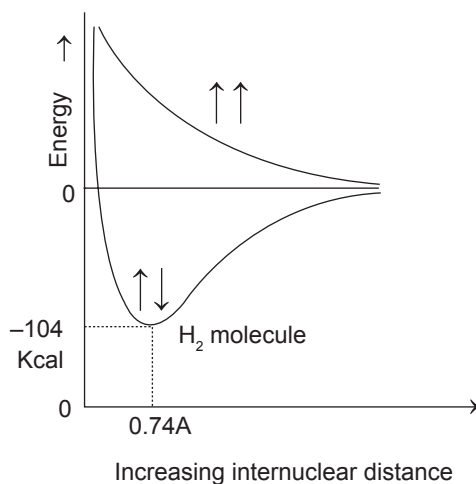


Fig. 1.6 Potential energy curve for hydrogen molecule

Let us consider the covalent bonding in the diatomic hydrogen molecule. Each hydrogen atom consists of a one-proton nucleus and a single 1s electron. We assume that nucleus A and electron A constitute the hydrogen atom H_A and that nucleus B and electron B make up the second hydrogen atom H_B . As the two atoms approach each other in addition to the Coulombic attraction between each nucleus and its own electron, new Coulombic attractive forces begin to operate between nucleus A and electron B and between nucleus B and electron A. Again Coulombic repulsive forces also exist between the two electrons, and between the two nuclei.

At large distances of separation, the system consists of two noninteracting hydrogen atoms. As the atoms approach each other the electrical potential energy is converted into kinetic energy. As the distance decreases, the conversion is more rapid. A critical internuclear distance is ultimately reached, at which the potential energy of the two-atom system reaches a minimum. At this distance (0.74 Å) (74 pm) the attractive forces balance the repulsive forces and a stable H_2 molecule results. When the internuclear distance falls below 0.74 Å (74 pm) the repulsive forces predominate and the molecule becomes unstable. There is a proton-proton attractive force only within the very short nuclear range of 10^{-13} cm. Below 0.74 Å (74 pm) the forces are still electrostatic in nature and because of the proximity of the two protons the proton-proton repulsion very much out-weighs the attractive forces of one proton over the electron of the other atom. The result is dissociation of H_2 molecule into atoms.

At the optimum internuclear distance of 0.74 Å (74 pm) each nucleus exerts an attractive forces on its own electron as also on the electron of the other atom. Consequently, the two electrons forming the covalent bond are shared by the two atoms. Quantum mechanically we say that there is a very high probability of finding the pair of electrons in between the two atoms. The higher the probability the stronger is the bond. That is, the greater is the energy necessary to break the bond. In terms of orbitals of the two electrons, there occurs a significant overlap of the two 1s orbitals of the two hydrogen atoms, such that one electron originally confined to a single orbital can now spread itself over the other orbital. Thus the two electrons of H_A and H_B now have a common orbital and allowing according to Pauli principle no more than two electrons per orbital, it follows that the electron spins of H_A and H_B have to be opposite if they are to share a common orbital. In the case of two with their electron spins parallel there is no minimum in the potential energy curve i.e., no H_2 molecule is formed. *Therefore in order that a covalent bond may be formed between two atoms a necessary condition is that each of the atoms must possess an unpaired spin.*

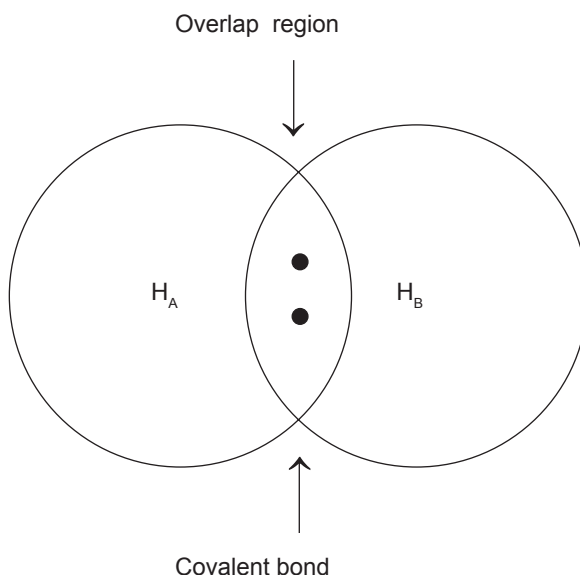


Fig. 1.7 Overlap of two 1s orbitals in H_2 molecule

This general approach which emphasizes overlap of atomic orbitals of individual atoms is referred to as Valence Bond (or Atomic Orbital) method. Covalent bond formation thus can occur between two hydrogen atoms, between two fluorine atoms, and between one hydrogen atom and a fluorine atom since each fulfils the criterion of having one unpaired electron in its outer orbital. But no such molecule formation can take place between two helium atoms since helium has a configuration of $1s^2$ that is, a filled is shell with no unpaired spin.

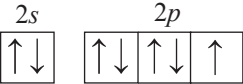
Covalency

It is defined as the number of electrons contributed by an atom of the element for sharing with other atoms as to achieve noble gas configuration. It can also be defined as the number of covalent bonds formed by the atom of the element with other atoms. The usual covalency of an element except hydrogen (which has covalency 1) is equal to (8-number of the group to which an element belongs). This is true for the elements belonging to 14, 15, 16 and 17 groups.

Element	Group	(8-Group number)	Covalency
C	14	$8 - 4 = 4$	4
Si	14	$8 - 4 = 4$	4
N	15	$8 - 5 = 3$	3
P	15	$8 - 5 = 3$	3
O	16	$8 - 6 = 2$	2
S	16	$8 - 6 = 2$	2
F	17	$8 - 7 = 1$	1
Cl	17	$8 - 7 = 1$	1

Generally, the covalency of an element is equal to the total number of unpaired electrons in s- and p-orbitals of the valency shell.

Hydrogen has one unpaired orbital  Covalency One

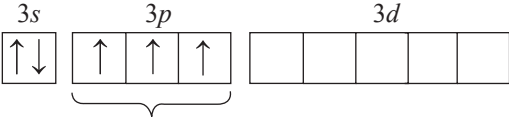
Fluorine has only one unpaired orbital  One

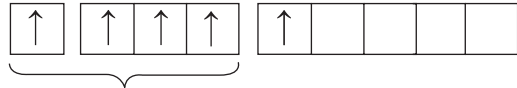
Oxygen has two unpaired p-orbitals  Two

Nitrogen has three unpaired p-electrons  Three

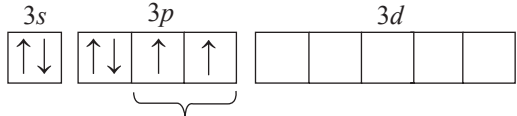
These four elements do not possess d-orbitals in their valency shell. However, the elements having vacant d-orbitals in their valency shell like P, S, Cl, Br, I, show **variable covalency** by increasing the number of unpaired electrons under excited conditions, i.e., unpairing the paired orbitals and shifting the electrons to vacant d-orbitals. **[Such a shifting is not possible in the case of H, N, O and F because d-orbitals are not present in their valency shell].**

Phosphorus shows 3 and 5 covalencies.

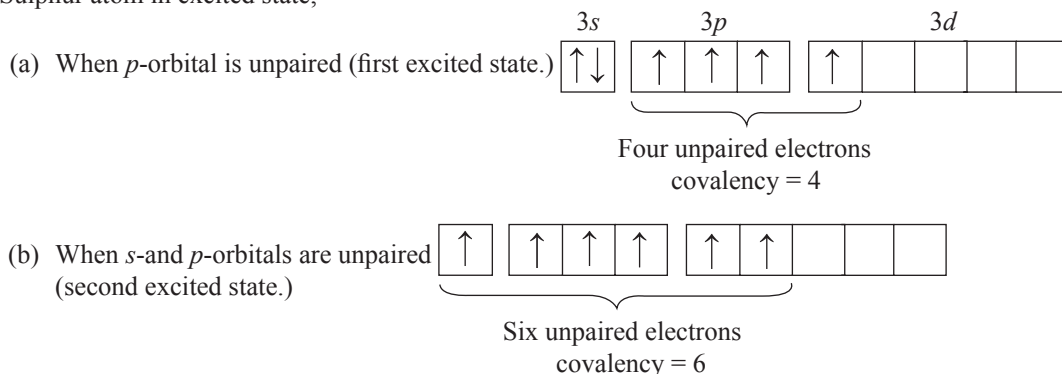
Phosphorus atom in ground state 
Three unpaired electrons
covalency = 3

Phosphorus atom in excited state 
Five unpaired electrons
covalency = 5

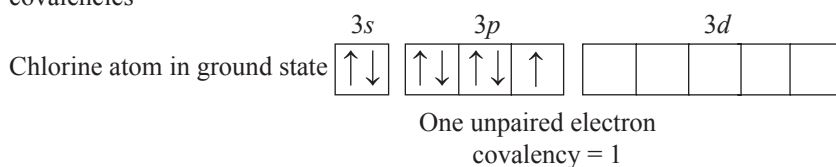
Sulphur atom shows 2, 4 and 6 covalencies

Sulphur atom in ground state 
Two unpaired electrons
covalency = 2

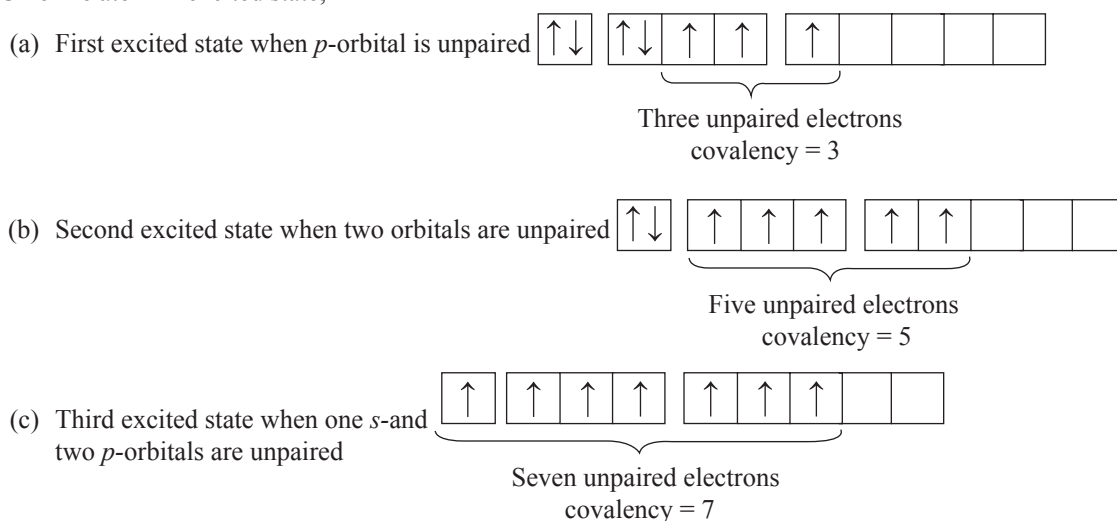
Sulphur atom in excited state,



Chlorine shows 1, 3, 5 and 7 covalencies
covalencies



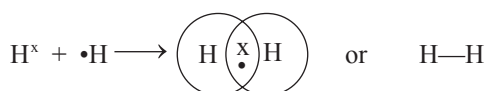
Chlorine atom in excited state,



Thus, variable covalency is shown by those elements whose atoms have vacant d -orbitals in their valency shell.

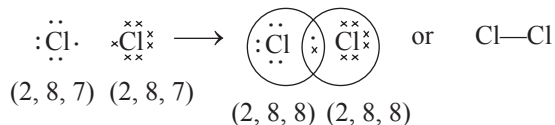
Covalent bond may be single, double or a triple bond. Double and triple covalent bonds are called **multiple covalent bonds**. Single covalent bond is formed by sharing of only one electron pair. This bond is represented by single dash ($—$). Double and triple covalent bonds are formed when atoms bonded together share two or three electron pairs, respectively. These bonds are represented by double dash ($=$) and triple dash ($\equiv$) respectively. Some examples of covalent bonding are given below:

- (i) **Formation of hydrogen molecule:** In the formation of hydrogen molecule, each hydrogen atom contributes one electron and then the pair is shared between two atoms. Both the atoms acquire stable configuration of helium. Thus, the molecule consists of one single covalent bond.

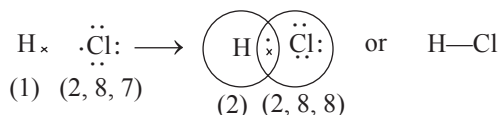


1.34 Periodic Properties and Chemical Bonding

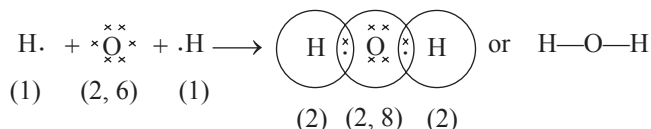
- (ii) **Formation of chlorine molecule:** Chlorine atom has seven electrons in the valency shell. In the formation of chlorine molecule, each chlorine atom contributes one electron and then the pair of electrons is shared between two atoms. Both the atoms acquire stable configuration of argon.



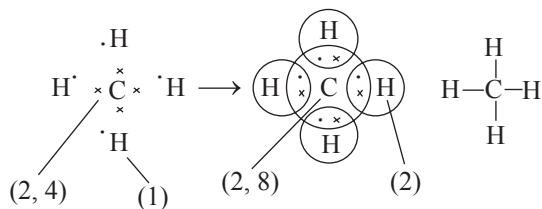
- (iii) **Formation of HCl molecule:** Both hydrogen and chlorine contribute one electron each and then the pair of electrons is equally shared. Hydrogen acquires the configuration of helium and chlorine acquires the configuration of argon.



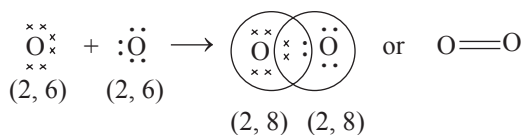
- (iv) **Formation of water molecule:** Oxygen atom has 6 valency electrons. It can achieve configuration of neon by sharing two electrons, one with each hydrogen atom.



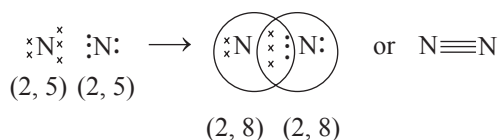
- (v) **Formation of methane:** Carbon has four electrons in the valency shell. It can achieve stable configuration of neon by sharing four electrons with four hydrogen atoms, one with each hydrogen atom. Each hydrogen atom acquires helium configuration.



- (vi) **Formation of O₂ molecule:** Each oxygen atom contributes two electrons and two pairs of electrons are then shared equally. Both the atoms acquire configuration of neon.

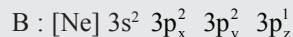


- (vii) **Formation of N₂ molecule:** Nitrogen atom has five valency electrons. Both nitrogen atoms achieve configuration of neon by sharing 3 pairs of electrons, i.e., each atom contributes 3 electrons.



Worked-out Example

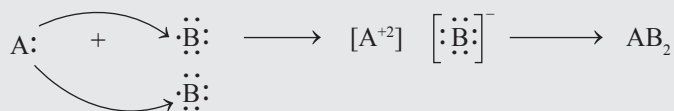
25. Two elements A and B have the following electronic configuration



The expected compound formed between A and B will have the formula

- (a) A_2B (b) AB (c) AB_2 (d) A_2B_3

Solution: (c)

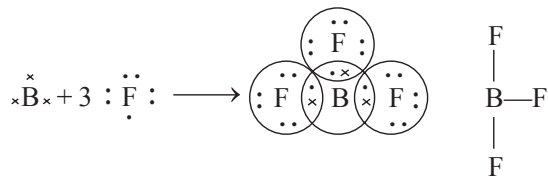
**Failure of Octet Rule**

There are several stable molecules known, in which the octet rule is violated, i.e., atoms in these molecules have number of electrons in the valency shell either short of octet or more than octet. Some important examples are:

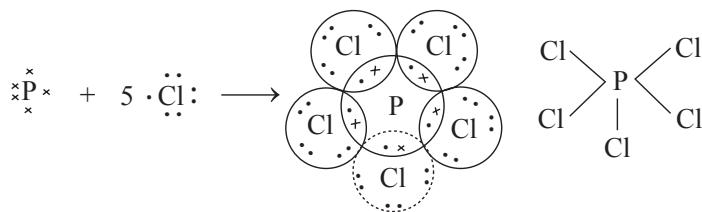
- (i) **BeCl_2 molecule:** BeCl_2 (beryllium chloride) is a stable molecule. Be atom forms two single covalent bonds with two chlorine atoms, i.e., it attains four electrons in the outer shell.



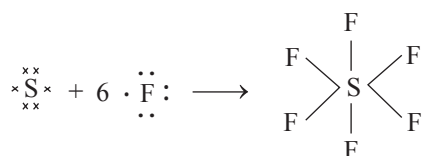
- (ii) **BF_3 molecule:** Boron atom forms three single covalent bonds with three fluorine atoms, i.e., it attains six electrons in the outer shell.



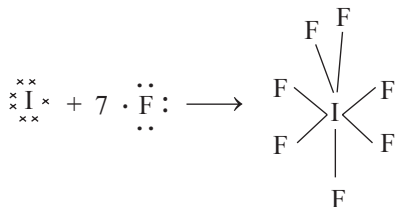
- (iii) **PCl_5 molecule:** Phosphorus atom have five electrons in valency shell. It forms five single covalent bonds with five chlorine atoms utilizing all the valency electrons and thereby attains 10 electrons in the outer shell.



- (iv) **SF_6 molecule:** Sulphur atom has six electrons in the valency shell. It forms six single covalent bonds with six fluorine atoms utilizing all the valency electrons and thereby attains 12 electrons in the outer shell.



- (v) **IF₇ molecule:** Iodine forms seven single covalent bonds the with seven fluorine atoms utilizing 7 valency electrons. The iodine atom attains 14 electrons in the outermost shell.



To explain the above abnormalities, the following two concepts were introduced:

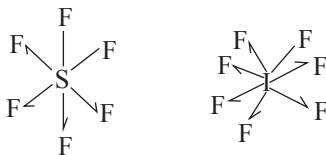
- Sugden's concept of singlet-linkage:** Sugden introduced the idea of singlet linkage in favour of octet rule. According to this concept, the maximum number of electrons in the outermost shell of any atom cannot exceed eight. In the molecules of PCl₅, SF₆, IF₇, etc., the central atom is linked with some of the combining atoms by single-electron bonds, called **singlet linkage** while the remaining atoms are linked by the normal two electrons bonds. The bond is represented by a half arrow ($\rightleftharpoons$) with the head pointing from donor towards the acceptor.

In PCl₅, three chlorine atoms are linked by normal covalent bonds and two chlorine atoms are linked by singlet linkages, thus, phosphorus achieves 8 electrons in the outermost shell.

This structure indicates that the nature of two chlorine atoms is different than the other three as singlet linkage is weaker than normal covalent bond. The above observation is confirmed by the fact that on heating, PCl₅ dissociates into PCl₃ and Cl₂.



Similarly, in SF₆, four singlet linkages are present while in IF₇, six singlet linkages are present.



Molecule can be represented by either of the following two diagrams.

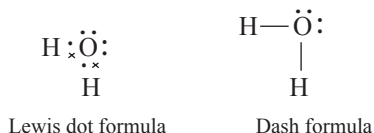
- Sidgwick's concept of maximum covalency:** This rule states that **the covalency of an element may exceed four and octet can be exceeded**. The maximum covalency of an element actually depends on the period of periodic table to which it belongs. The maximum covalency of the elements is tabulated below:

Period	Elements	Maximum covalency	No. of electrons in the outermost orbit
1st	H	2	4
2nd	Li to F	4	8
3rd	Na to Cl	6	12
4th	K to Br	6	12
5th } 6th }	Rb onwards } and rest of } the elements }	8	16

This rule explains the formation of PCl₅ and SF₆. This also explains, why nitrogen does not form NF₅ or NCl₅? because nitrogen belongs to second period and the maximum covalency of nitrogen is four.

Lewis Formulae for Molecules and Polyatomic Ions

Lewis dot formulae are used to represent atoms covalently bonded in a molecule or a polyatomic ion. For instance, a water molecule can be represented by either of the following two diagrams.



Lewis dot formulae show only the number of valency electron, the number and kinds of bonds, but do not depict the three dimensional shapes of molecules and polyatomic ions. Lewis formulae are based on the fact that the representative elements achieve a noble gas configuration in most of their compounds, i.e., 8 electrons in their outermost shell (except for H_2 , Li^+ and Be^{2+} ions which have 2 electrons).

The following steps are followed in constructing dot formulae for molecules and polyatomic ions.

(i) **Write a symmetrical ‘skeleton’ for the molecules and polyatomic ions.**

(a) The least electronegative element is usually taken as the central element except H.

Example: CO_2 has the skeleton OCO .

(b) Oxygen atoms do not bond to each other except in O_2 , O_3 , the peroxides and superoxides. The phosphate ion (PO_4^{3-}) has the skeleton $\text{O} \begin{array}{c} \text{O} \\ \text{P} \\ \text{O} \end{array} \text{O}$.

(c) Hydrogen actually bonds to an oxygen atom and not to the central atom in ternary acids (oxyacids).

Example: Nitrous acid HNO_2 , has the skeleton HONO . However, there are exceptions to this rule, such as for H_3PO_2 , the skeleton is $\text{H} \begin{array}{c} \text{O} \\ \text{P} \\ \text{H} \end{array} \text{OH}$.

(d) For polycentred species such as C_2H_4 , the most symmetrical skeleton is used. $\begin{array}{cc} \text{H} & \text{H} \\ & \diagdown \quad \diagup \\ & \text{C} = \text{C} \\ & \diagup \quad \diagdown \\ \text{H} & \text{H} \end{array}$ is the skeleton for C_2H_4 .

(ii) **Calculate the number of electrons available in the valency shell of all the atoms.**

For negatively charged ions add to the total number of electrons equal to the charge on the anion and for positively charged ions, subtract the number of electrons equal to the charge on the cations.

The total number of electrons calculated in this way is represented by symbol A.

A for H_2SO_4

$A = 2 \times 1$ (for hydrogen atoms) + 1×6 (for S atom) + 4×6 (for O atoms) = $2 + 6 + 24 = 32$ electrons.

A for PO_4^{3-} ion

$A = 1 \times 5$ (for P atom) + 4×6 (for O atoms) + 3 (for charge) = $5 + 24 + 3 = 32$ electrons.

A for NH_4^+ ion

$A = 1 \times 5$ (for nitrogen) + 4×1 (for H atoms) – 1 (for positive charge) = $5 + 4 - 1 = 8$ electrons.

(iii) **Calculate the total number of electrons needed by all atoms to achieve noble gas configuration. This number is represented by N.**

Example: N for H_2SO_4

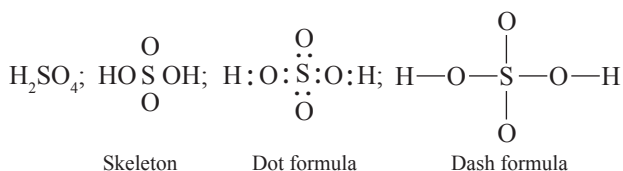
$W = 2 \times 2 + 8 \times 1 + 8 \times 4 = 4 + 8 + 32 = 44$ electrons.

(iv) **Calculate the total number of electrons shared. This is represented by S, which is equal to $N - A$.**

Example: S for H_2SO_4

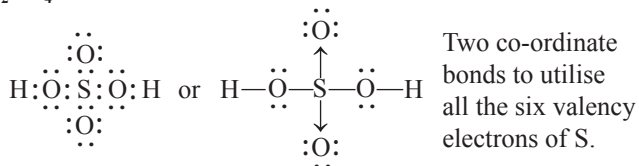
$S = N - A = 44 - 32 = 12$ electrons.

(v) **Place the shared pair of electrons into the skeleton, using double and triple bonds only when necessary. Example:**



- (vi) Place the additional unshared (lone) pairs of electrons to fill the octet to every atom except hydrogen which can have only 2 electrons as the total comes equal to A as calculated in step (ii).

For H_2SO_4



Worked-out Examples

26. Write the Lewis dot formula and dash formula for CO_2 .

Ans. Step (i) The skeleton is OCO.

Step (ii) $A = 1 \times 4$ (for C) + 2×6 (for O) = $4 + 12 = 16$ electrons

Step (iii) $N = 1 \times 8$ (for C) + 2×8 (for O) = $8 + 16 = 24$ electrons

Step (iv) $S = N - A = 24 - 16 = 8$ electrons shared

Step (v) $\text{O}:\text{C}:\text{O}$

Step (vi) $\text{:}\ddot{\text{O}}:\text{C}:\ddot{\text{O}}\text{:}$ or $\text{:}\ddot{\text{O}}=\text{C}=\ddot{\text{O}}\text{:}$

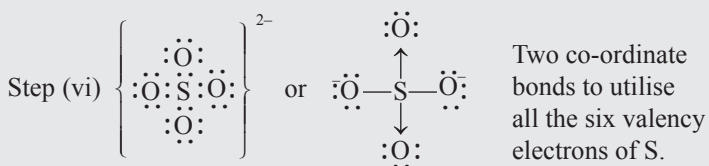
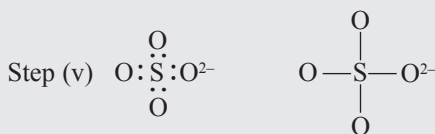
27. Write the Lewis dot formula and dash formula for the SO_4^{2-} ion.

Ans. Step (i) The skeleton $\left[\begin{array}{c} \text{O} \\ \text{O} \text{ S } \text{O} \\ \text{O} \end{array} \right]^{2-}$

Step (ii) $A = 1 \times 6$ (for S) + 4×6 (for O) + 2 (for negative charge) = $6 + 24 + 2 = 32$ electrons

Step (iii) $N = 1 \times 8$ (for S) + 4×8 (for O) = $8 + 32 = 40$ electrons

Step (iv) $S = N - A = 40 - 32 = 8$ electrons shared



28. Write the Lewis dot formula or dash formula for the CO_3^{2-} ion.

Ans. Step (i) Skeleton $\text{O} \begin{array}{c} \text{O} \\ \text{C} \text{ O} \end{array} \text{O}^{2-}$

Step (ii) $A = 1 \times 4$ (for C) + 3×6 (for O) + 2 (for negative charge) = $4 + 18 + 2 = 24$ electrons

Step (iii) $N = 1 \times 8$ (for C) + 3×8 (for O) = $8 + 24 = 32$ electrons.

Step (iv) $S = N - A = 32 - 24 = 8$ electrons shared (4 pairs shared)

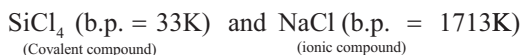
Step (v) $\text{O}:\text{C}::\text{O}^{2-}$
 O

Step (vi) $\text{:}\ddot{\text{O}}:\text{C}::\ddot{\text{O}}\text{:}^{2-}$ or $\begin{array}{c} \text{:}\ddot{\text{O}}-\text{C}=\ddot{\text{O}}\text{:}^{2-} \\ \text{:}\ddot{\text{O}}\text{:} \end{array}$

Characteristics of Covalent Compounds

Melting point and boiling point

In covalent compounds, except those consisting of giant molecules, the molecules are less powerfully attracted to each other, as a result of which their melting points and boiling points are relatively low compared to ionic compounds, e.g.,



Conductivity

Covalent substances (whether of the “molecular lattice” or “giant molecule” type) do not conduct electricity in the fused state since there are no free electrons or ions to carry the current. However, substances like graphite which consists of separate layers conduct electricity because the electrons have a passage in between the two flat layers.

Solubility

The characteristic solubility of covalent compounds in non-polar solvents such as benzene and carbon tetrachloride can be described to the similar covalent nature of the molecules of solute and solvent (i.e., like dissolves like). Covalent compounds in solution react more slowly as compared with the ionic compounds which react instantaneously in solution. The solubility of covalent compounds is, however, very much dependent upon the size of the molecule. Thus *covalent substances having giant molecules are insoluble in virtually all solvents due to the big size of the molecule unit.*

Fajan's Rules

When two oppositely charged ions approach each other closely, the positively charged cation attracts the outermost electrons of the anion and repel its positively charged nucleus. This results in the distortion or polarization of the anion followed by some sharing of electrons between the two ions, i.e., the bond becomes partly covalent in character.

- (i) **Charge on either of the ions:** As the charge on the cation increases, its tendency to polarize the anion increases. This brings more and more covalent nature in the electrovalent compound. Whereas with the increasing charge of anion, its ability to get polarized, by the cation, also increases.

For example, in the case of NaCl, MgCl₂ and AlCl₃ the polarization increases, thereby covalent character becomes more and more as the charge on the cation increases.

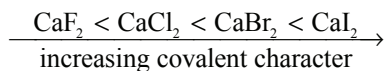
Similarly, lead forms two chlorides PbCl₂ and PbCl₄ having charges +2 and +4 respectively. PbCl₄ shows covalent nature. Similarly, among NaCl, Na₂S, Na₃P, the charge of the anions are increasing, therefore the increasing order of covalent character.



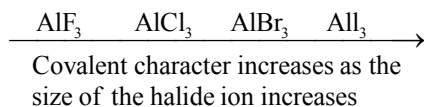
- (ii) **Size of the cation:** Polarization of the anion increases as the size of the cation decreases i.e., the electrovalent compounds having smaller cations show more of the covalent nature. For example, in the case of halides of alkaline earth metals, the covalent character decreases as we move down the group. Hence, melting point increases in the order of,



- (iii) **Size of anion:** The larger the size of the anion, more easily it will be polarized by the cation i.e., as the size of the anion increases for a given cation, the covalent character increases. For example, in the case of halides of calcium, the covalent character increases from F⁻ anion to I⁻ anion i.e.,



Similarly, in case of trihalides of aluminium, the covalent character increases with increase in size of halide anion i.e.,



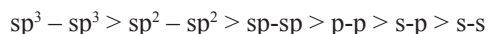
- (iv) **Nature of the cation:** Cations with 18 electrons ($s^2p^6d^{10}$) in outermost shell polarize an anion more strongly than cations of 8 electrons (s^2p^6) type. The d electrons of the 18 electron shell screen the nuclear charge of the cation less effectively than the s and p electrons of the 18-electron shell. Hence the 18-electron cations behave as if they had a greater charge. Copper (I) and Silver (I) halides are more covalent in nature compared with the corresponding sodium and potassium halides although charge on the ions is the same and the sizes of the corresponding ions are similar. This illustrates the effect of 18-electron configuration of Cu^+ ($3s^2, p^6, d^{10}$) and Ag^+ ($4s^2, p^6, d^{10}$) ions.

Some Noteworthy Points

- Isolated atoms combine to form molecules or ions because molecular state is a state of lower energy.
- A molecule results if the energy lowers by a minimum of 40 kJ mol^{-1} .
- More is the number of bonds between two atoms, the shorter is the bond length.
- The solubility of aluminium halides decreases from AlF_3 to AlI_3 due to increase in the covalent character in accordance with Fajan's rules.
- FeCl_2 is less covalent than FeCl_3 because polarizing power of Fe^{2+} is less than that of Fe^{3+} ion having smaller size and higher oxidation state.
- Head on overlapping is stronger than lateral or sidewise overlapping.

$$\underbrace{p-p > s-p > s-s}_{\text{Head on overlap}} > \underbrace{p-p}_{\text{Lateral overlap}}$$

- More directional is a bond, greater is the bond strength and vice versa. For example,



The hybrid orbitals with more p-character are more directional in character and hence the above order.

Worked-out Examples

29. Lattice energy of an ionic compound depends upon: (AIEEE 2005)

- (a) Charge on the ions only (b) Size of the ions only
(c) Packing of the ions only (d) Charge and size of the ions

Ans. (d) The value of lattice energy depends on the charges present on the two ions and the distance between them.

30. Which of the following does not contain iso-electronic species? (AIEEE 2005)

- (a) PO_4^{3-} , SO_4^{2-} , ClO_4^- (b) CN^- , N_2 , C_2^{2-}
(c) SO_3^{2-} , CO_3^{2-} , NO_3^- (d) BO_3^{3-} , CO_3^{2-} , NO_3^-

Ans. (c) SO_3^{2-} has 42 electrons, CO_3^{2-} has 32 electrons, NO_3^- has 32 electrons.

31. The charge/size ratio of a cation determines its polarizing power. Which one of the following sequences represents the increasing order of the polarizing power of the cationic species, K^+ , Ca^+ , Mg^{2+} , Be^{2+} ? (AIEEE 2007)

- (a) $\text{Mg}^{2+} < \text{Be}^{2+} < \text{K}^+ < \text{Ca}^{2+}$ (b) $\text{Be}^{2+} < \text{K}^+ < \text{Ca}^{2+} < \text{Mg}^{2+}$
(c) $\text{K}^+ < \text{Ca}^{2+} < \text{Mg}^{2+} < \text{Be}^{2+}$ (d) $\text{Ca}^{2+} < \text{Mg}^{2+} < \text{Be}^{2+} < \text{K}^+$

Ans. (c) Smaller the size and higher the charge more will be polarizing power of cation. Since the order of the size of cation is $\text{K}^+ > \text{Ca}^{++} > \text{Mg}^{++} > \text{Be}^{++}$. So the correct order of polarizing power is



32. Which one of the following constitutes a group of the isoelectronic species? (AIEEE 2008)

- (a) C_2^{2-} , O_2^- , CO , NO (b) NO^+ , C_2^{2-} , CN^- , N_2
(c) CN^- , N_2 , O_2^{2-} , CO_2^{2-} (d) N_2 , O_2^- , NO^+ , CO

Ans. (b)

33. A metal, M forms chloride in its +2 and +4 oxidation states. Which of the following statement about these chlorides is correct? (AIEEE 2006)

- (a) MCl_2 is more volatile than MCl_4
 (b) MCl_2 is more soluble in anhydrous ethanol than MCl_4
 (c) MCl_2 is more ionic than MCl_4
 (d) MCl_2 is more easily hydrolysed than MCl_4

Ans. (c) More is charge on cation (M^{4+}), more is deformation of anion, more is covalent nature.

34. Which of the following groupings represent a collection of isoelectronic species? (At. no. Cs = 55, Br = 35) (AIEEE 2003)

- (a) N^{3-} , F^- , Na^+ (b) Be, Al^{3+} , Cl^- (c) Ca^{2+} , Cs^+ , Br (d) Na^+ , Ca^{2+} , Mg^{2+}

Ans. (a)

35. Which one of the following sets of ions represents the collection of isoelectronic species? (AIEEE 2004)

- (a) K^+ , Cl^- , Mg^{2+} , Se^{3+} (b) Na^+ , Ca^{2+} , Se^{3+} , F^-
 (c) K^+ , Ca^{2+} , Se^{3+} , Cl^- (d) Na^+ , Mg^{2+} , Al^{3+} , Cl^-

Ans. (c)

36. The decomposition temperature of Li_2CO_3 is less than that of Na_2CO_3 . Explain.

Solution: As Li^+ ion is smaller than Na^+ ion, thus small cation (Li^+) will favour more covalent character in Li_2CO_3 and hence it has lower decomposition temperature than that of Na_2CO_3 .

37. Why is that Lithium salts have a greater degree of covalent character than other halides of the alkali metal?

Solution: The small size of Li^+ gives it a large polarizing power, hence covalent nature of its compound increases. It can be viewed in terms of Fajan's rule, smaller the cation, larger the anion, greater the charge density at the surface causes greater polarizing power of Li^+ . Hence, greater the covalent nature.

Sigma and Pi Bonds (σ and π Bonds)

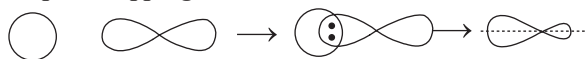
A covalent bond is formed by the overlapping of atomic orbitals. Covalent bonds formed are of two types depending upon the way the orbitals overlap each other:

1. **Sigma bond (σ bond):** The bond formed by the overlapping of two half-filled atomic orbitals along their axis is known as sigma bond. σ bond is a strong bond because overlapping in it takes place to large extent. The hybrid orbitals always from σ bond.

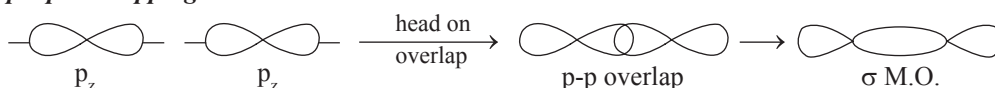
- (a) ***s-s overlapping***



- (b) ***s-p overlapping***

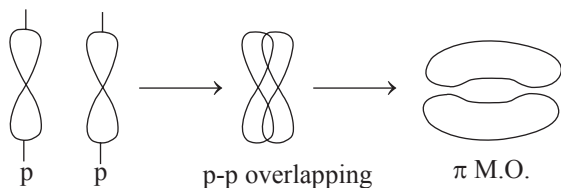


- (c) ***p-p overlapping***



2. **Pi bond (π bond):** The bond formed by the lateral overlapping of half-filled atomic orbitals is known as pi bond. The sidewise overlapping takes place to less extent. Therefore, π bond formed is a weak bond. π bond

overlapping takes place only at the sides of two lobes. A π bond is formed when a σ bond already exists between the combining atoms.



Example:

In A – B molecule, the bond formed is σ bond

In A $\begin{array}{c} \sigma \\ \hline \pi \end{array}$ B molecule, there are one σ and one π bonds

In A $\begin{array}{c} \sigma \\ \hline \pi \\ \hline \pi \end{array}$ B molecule, there are one σ and two π bonds

Thus, all the single bonds are σ bonds. Double bond has one σ and one π bond. Triple bond has one σ and two π bonds.

This simple pure atomic orbital overlap approach works well for very simple molecules but fails when we come to molecules like methane, boron trichloride etc. In order to cover these cases under the same overlap framework, hybridization of pure atomic orbitals has been invoked as a step preliminary to overlap of the orbitals. Hybridization of atomic orbitals is largely the work of Pauling and Slater.

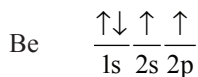
Hybridization of Atomic Orbitals

Simple valence bond theory requires the presence of unpaired electrons in an atom to enable it to participate in covalent bonding. This condition immediately limits the number of covalent bonds to the number of unpaired spins in an atom. The electronic configurations of Be, B, C are:

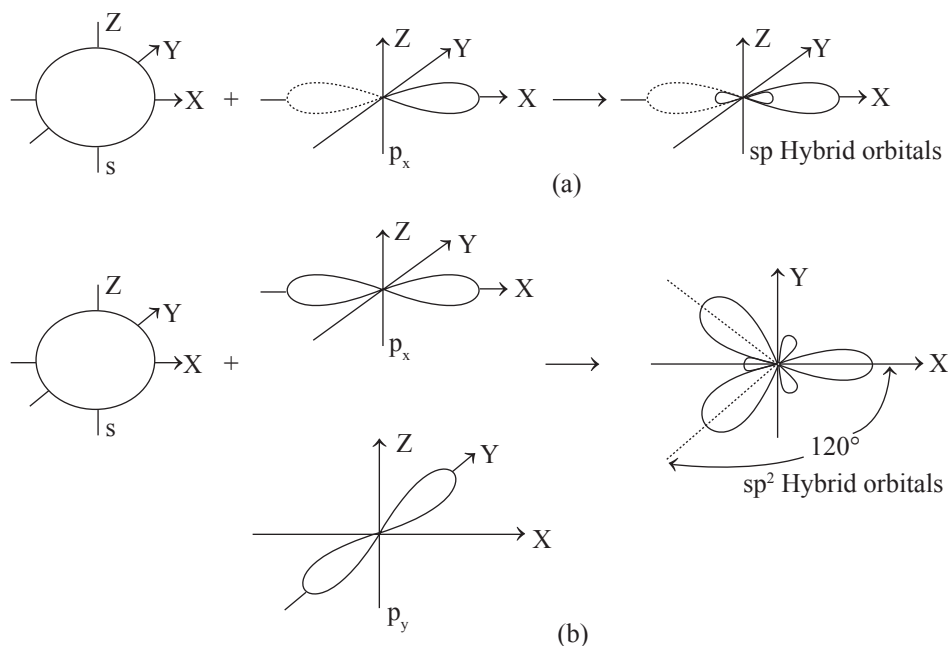
Be	$1s^2 2s^2$	or	$\begin{array}{cc} \uparrow\downarrow & \uparrow\downarrow \\ \hline 1s & 2s \end{array}$
B	$1s^2 2s^2 2p^1$	or	$\begin{array}{ccccc} \uparrow\downarrow & \uparrow\downarrow & \uparrow & _ & _ \\ \hline 1s & 2s & 2p & 2p & 2p \end{array}$
C	$1s^2 2s^2 2p^2$	or	$\begin{array}{ccccc} \uparrow\downarrow & \uparrow\downarrow & \uparrow & \uparrow & _ \\ \hline 1s & 2s & 2p & 2p & 2p \end{array}$

We are tempted to conclude that beryllium, boron and carbon are capable of forming 0, 1 and 2 single covalent bonds only. Yet experimental evidences show that beryllium, boron and carbon do form 2 (in BeCl_2), 3 (in BCl_3) and 4 (in CH_4) covalent bonds and further that the covalent bonds of the same element are all equivalent.

In order to explain the formation of two equivalent covalent bonds by beryllium, we assume that prior to reaction, one of the two $2s$ electrons of beryllium is promoted to the $2p$ level giving the excited state configuration:

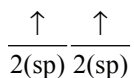


Now that the excited beryllium atom has two unpaired electrons, it is capable of forming two covalent bonds by overlap with other orbitals. But as it stands, the two bonds will not be equivalent since one will involve an s bond the other a p bond. Experiments show that the bonds in BeCl_2 are linear and completely equivalent. Quantum mechanical calculations have shown that one s orbital and a p orbital can be mixed to generate two equivalent orbitals at 180° to each other (figure). These orbitals are called hybrid orbitals and carry the symbol sp to indicate their origin. One of these two orbitals has high electron probability on one.

Formation of sp and sp^2 hybrid orbitals

One 's' and one 'p' orbital give two sp hybrid orbitals in (a). Note opposite each big lobe of a hybrid orbital there is a very small lobe. This indicates that the hybrid orbitals are very directional in character and the probability of finding the electron opposite the big lobe is very low.

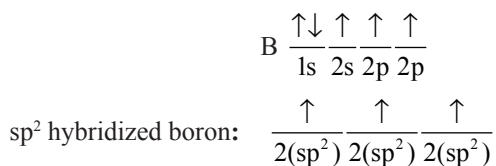
Similar is the case for other hybrid orbitals, e.g., sp^2 hybrid orbitals in (b). side of the nucleus while the other has high probability on the opposite side. The beryllium electrons are now placed in the two hybrid orbitals giving beryllium its *valence state*:



These two equivalent hybrid orbitals will overlap with say chlorine orbitals carrying unpaired electron to generate a linear BeCl_2 molecule. (It may be added that crystalline BeCl_2 , is polymeric and approximately tetrahedral but at $\sim 750^\circ\text{C}$ it is monomeric and linear),

One might object to the promotion of a $2s$ electron to a $2p$ orbital as being improbable on energy consideration. However, the energy released when the excited beryllium atom forms two stable bonds with another element (say chlorine) more than compensates the energy of promotion.

Coming to the case of boron, promotion of an electron from a $2s$ orbital to a $2p$ orbital gives the excited state configuration as:

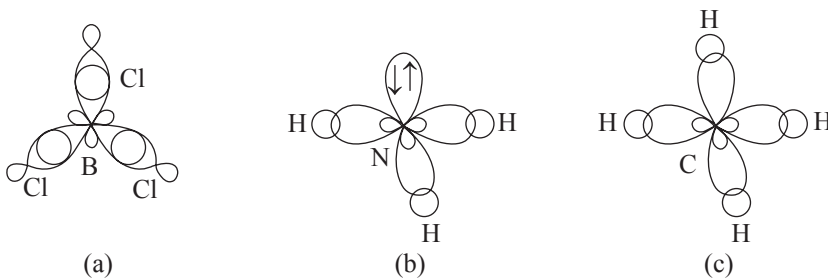


Mixing of one $2s$ orbital and two $2p$ orbitals provides three equivalent hybrid orbitals at 120° to each other. These are symbolized as sp^2 orbitals. Overlap of three such orbitals by chlorine gives the BCl_3 molecule with three equivalent bonds at 120° (figure). If the hybridization scheme were not invoked then overlap of the three atomic orbitals would have resulted in three covalent bonds—two of which formed from two p orbitals, would be at 90° to each other and the third

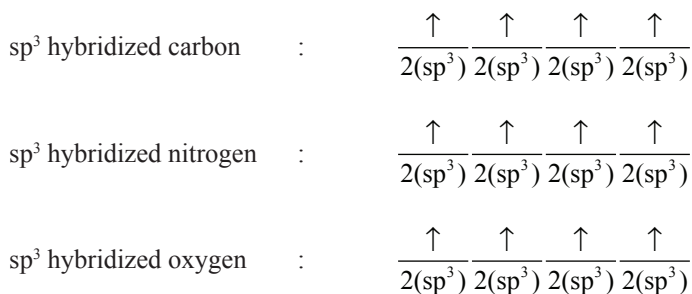
1.44 Periodic Properties and Chemical Bonding

obtained from the s orbital overlap would have stayed equidistant from the other two (minimum repulsion condition), either at 135° on the same plane or at 90° in a plane at right angles to the plane of the other two bonds.

Taking the case of carbon, promotion of a $2s$ electron to the third $2p$ and then mixing the s and the three p orbitals provides four equivalent sp^3 hybrid orbitals with one electron each. Overlap of these orbitals with $1s$ orbital of four hydrogen will give four equivalent carbon-hydrogen bonds. In fact in CH_4 we have all the carbon-hydrogen bond length the same: 1.09 Å (109 pm) and all the HCH bond angles the same: 109° . We have sp^3 hybrid orbitals of the central element in NH_3 and H_2O . Note in these two compounds there are one and two lone pairs respectively, being housed in the hybrid orbitals.



- (a) Overlap of boron sp^2 hybrid orbitals with p orbital of chlorine in BCl_3
 (b) Overlap of nitrogen sp^3 hybrid orbitals with $1s$ orbital of hydrogen in NH_3 . Note one sp^3 hybrid orbital houses the lone pair of nitrogen
 (c) Overlap of carbon sp^3 hybrid orbitals with $1s$ orbital of hydrogen in CH_4



Types of hybridization and spatial orientation of hybrid orbitals: The geometry and shapes of various species on the basis of VSEPR theory along with hybrid state of central atom is given below in tabular form.

	Types of atomic orbitals used	Hybridization	Orientation of hybrid orbitals	Examples
1.	one s + one p -orbital	sp	Linear	BeF_2 , BeCl_2 , C_2H_2 , HgCl_2
2.	one s + two p -orbitals	sp^2	Trigonal planar	BF_3 , C_2H_4 , NO_3^- , CO_3^{2-}
3.	one s + three p -orbitals	sp^3	Tetrahedral	CH_4 , CCl_4 , SiF_4 , NH_4^+ , SO_4^{2-} , ClO_4^-
4.	one s + three p + d	sp^3d	Trigonal bipyramidal	PF_5 , PCl_5
5.	one s + three p + two d	sp^3d^2	Octahedral	SF_6 , $[\text{CrF}_6]^{3-}$, IF_5
6.	one s + three p +three d	sp^3d^3	Pentagonal Bipyramidal	IF_7

Note:

- (i) Orbitals participating in hybridization must have only small difference in their energies.
- (ii) Both half-filled and completely filled orbitals can get involved in hybridization.
- (iii) The number of hybrid orbitals is equal to the number of orbitals participating in hybridization.
- (iv) Hybrid orbital form more stronger bonds than pure atomic orbitals.
- (v) Same atom can assume different hybrid states under different situations.
- (vi) Hybrid orbitals form sigma bonds.

Method of predicting the hybrid state of the central atom in covalent molecules of polyatomic ions:

The hybrid state of the central atom in similar covalent molecule or polyatomic ion can be predicted by using the generalized formula as described below:

<i>Simple molecule</i>	<i>Polyatomic anion</i>	<i>Poyatomic cation</i>
$X = \frac{1}{2}[V + G]$	$X = \frac{1}{2}[V + G + a]$	$X = \frac{1}{2}[V + G - c]$

In the above formulae,

V = Number of monovalent atoms or groups attached to the central atom

G = Number of outer shell electrons in ground state of the central atom

a = Magnitude of charge on anion

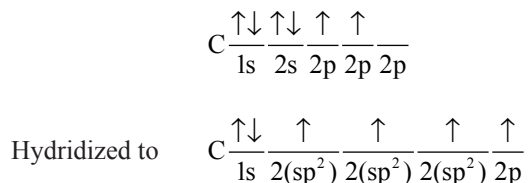
c = Magnitude of charge on cation

Calculate the value of X and decide the hybrid state of central atom as follows:

<i>X</i>	2	3	4	5	6	7
Hybrid state	sp	sp ²	sp ³	sp ³ d	sp ³ d ²	sp ³ d ³

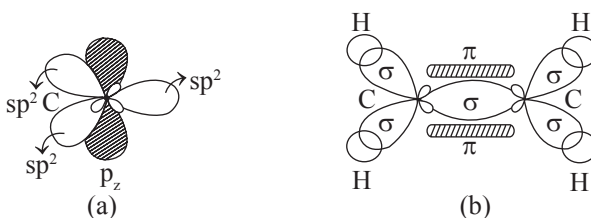
<i>PF₅</i>	<i>COCl₂</i>	<i>NH₄⁺</i>	<i>ClO₄⁻</i>
$X = \frac{1}{2}[5 + 5]$	$X = \frac{1}{2}[2 + 4]$	$X = \frac{1}{2}[4 + 5 - 1]$	$X = \frac{1}{2}[0 + 7 + 1]$
= 5	= 3	= 4	= 4
Hybrid state of P is sp ³ d	Hybrid state of C is sp ²	Hybrid state of N is sp ³	Hybrid state of Cl is sp ³
NO ₃ ⁻	IF ₅	CO ₂	XeF ₄
$X = \frac{1}{2}[0 + 5 + 1]$	$X = \frac{1}{2}[5 + 7]$	$X = \frac{1}{2}[0 + 4]$	$X = \frac{1}{2}[4 + 8]$
= 3	= 6	= 2	= 6
sp ²	sp ³ d ²	sp	sp ³ d ²
PCl ₆ ⁻	PH ₃	SF ₃ ⁺	SF ₄
$X = \frac{1}{2}[6 + 5 + 1]$	$X = \frac{1}{2}[3 + 5]$	$X = \frac{1}{2}[3 + 6 - 1]$	$X = \frac{1}{2}[4 + 6]$
= 6	= 4	= 4	= 5
Hybrid state sp ³ d ²	Hybrid state sp ³	Hybrid state sp ³	Hybrid state sp ³ d

Valence Bond Description of Multiple Bonds: Lewis formulation of C_2H_4 molecule involves a double bond between the two carbon atoms and two single bonds each between the hydrogens and carbon. Leaving aside the question of the second bond between two carbons, each carbon assumes three equivalent planar bonds: two of these are directed to two hydrogens and the third to the other carbon. These three equivalent bonds arise out of sp^2 hybrid orbitals of carbon, and a p orbital with an unpaired electron is left out of hybridization scheme.



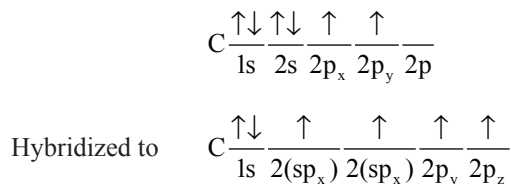
The valence bond model suggests that the overlap of the p orbitals of the two carbon atoms results in the fourth bond in C_2H_4 with the electron density concentrated in two sausage-shaped regions, one above the plane of the molecule and the other below the plane of the molecule.

The double bond area of C_2H_4 is composed therefore of two distinct parts having different electron distributions. The sp^2 hybrid orbitals overlap has symmetrical distribution of electron density about the carbon-carbon axis. *Such bond with cylindrically symmetrical distribution of electron density around bond axis is called sigma (σ) bond.* In C_2H_4 , there are five sigma bonds. Bond in which the electron density is not cylindrically symmetrical around the bond axis is called a pi (π) bond. A pi (π) bond has sideways or end-on overlap. In C_2H_4 there is one pi bond, the electron density of which is distributed in two particular regions above and below the carbon-carbon axis. For a π -bond the bond axis becomes a nodal plane. A nodal plane has zero electron density. Note that the two sausage-shaped regions together make one π -bond. The pictorial representation is (figure);



(a) sp^2 hybrid orbitals and p_z orbital of carbon, (b) valence bond model of C_2H_4 .

We can extend our discussion of multiple bonding to acetylene, C_2H_2 . In acetylene, each carbon is sigma bonded to one hydrogen and the second carbon. To satisfy the tetra valence of carbon there exists a triple bond between the two carbons. This triple bond is made up of one sigma bond and two pi bonds. Each carbon is sp hybridized to give a linear $H-C-C-H$ linkage.



One sp_x electron forms a sigma overlap with the $1s$ electron of hydrogen and the second sp electron makes the second sigma overlap with another sp electron of the second carbon. The two $2p_y$ orbitals overlap to give one pi bond. Similar overlap of the $2p_z$ orbitals of the two carbons makes the second pi bond. Thus acetylene is a linear molecule made up of three sigma bonds and two pi bonds.

As we know bond length decreases with increasing multiplicity of a bond. We can now state that addition of one (or two) π -bond(s) to a σ bond decreases the bond length.

Distinctions between sigma and pi bonds:

1. Because of a head-on overlap along the bond axis the sigma bond is stronger than a pi bond where the overlap is sideways and not along the bond axis. Note the energy of a C–C sigma bond is 83 kcal/mole (347 kJ/mole) while that of a C–C pi bond is 63 kcal/mole (263.6 kJ/mole).
2. The electrons of the weaker pi bond are more spread out in space and are therefore more exposed to any chemical reagent.
3. Due to cylindrical symmetry free rotation of atoms about a sigma bond is allowed. Similar rotation about a pi bond will lead to decrease in overlap because coplanarity of the pi orbitals overlap will be destroyed and hence rotation about a double or triple bond is hindered.
4. A sigma bond is cylindrically symmetrical about the bond axis but a pi bond is not so.
5. A sigma bond determines the direction between the bonding atoms but a pi bond has no primary effect in this respect.

Orbital Overlap and Bond Strength: For good covalent interaction there has to be a good overlap of the combining atomic orbitals. The greater the overlap (i.e., greater the electron density between the nuclei) the greater is the bond strength. In this respect the p orbitals with directional preference give better overlap than the spherically symmetrical non-directional s orbitals. The relative strengths of such interaction between *pure* atomic orbitals are:

Overlap	:	$s-s$	$p-p$
Relative strength	:	1	3

Strengths of $s-s$ and $p-p$ bonds are however considerably less than $s-p$ hybrid bonds. For example, the three molecules Li_2 ($2s-2s$), F_2 ($2p-2p$), C_2H_6 ($\text{C}-\text{C}$, $2s2p^3-2s2p^3$) have bond dissociation energies as 26, 38 and 80 kcal/mole (109, 159 and 335 kJ/mole) respectively. Then again highest overlap occurs with sp , then comes sp^2 and sp^3 in that order. In keeping with this overlap criterion, bond dissociation energy falls off from sp^2 to sp^3 and then to sp^3 .

Compare the following:

Hybridization	Example (A)	C–H bond Length (pm)		Bond Energy kcal/mole (kJ/mole)
sp	Acetylene	1.06	(106)	121 (506)
sp^2	Ethylene	1.069	(106.9)	106 (443)
sp^3	Methane	1.09	(109)	103 (431)

The stability of covalent hydrides of elements belonging to the same periodic group decreases with increasing atomic number. As the atomic number increases the size of the p orbital and hence of the hybrid orbitals (e.g., sp^3) increases so that the overlap of the $1s$ orbital of hydrogen with the sp^3 orbital of the (sp^3) hydride forming element (say, N, P, As, Sb) ($2p/3p/4p/5p$) becomes poorer and poorer leading to lower stability.

In 1957 Gillespie and Nyholm gave this theory to predict and explain molecular shapes and bond angles more exactly. The theory was developed extensively by Gillespie as the Valence Shell Electron Pair Repulsion (VSEPR) theory. This may be summarized as:

1. The shape of the molecule is determined by repulsions between all of the electron pairs present in the valence shell.
2. A lone pair of electrons takes up more space round the central atom than a bond pair, since the lone pair is attracted to one nucleus whilst the bond pair is shared by two nuclei. It follows that repulsion between two lone pairs is greater than repulsion between a lone pair and a bond pair, which in turn is greater than the repulsion between two bond pairs. Thus the presence of lone pairs on the central atom causes slight distortion of the bond angles from the ideal shape. If the angle between a lone pair, the central atom and a bond pair is increased, it follows that the actual bond angles between the atoms must be decreased. The order of repulsion between lone pairs and bond pairs of electrons follows the order as:
Lone pair–lone pair repulsion > lone pair – bond pair repulsion > bond pair – bond pair repulsion.
3. The magnitude of repulsions between bonding pairs of electrons depends on the electronegativity difference between the central atom and the other atoms.
4. Double bonds cause more repulsion than single bonds, and triple bonds cause more repulsion than a double bond.

Effect of Lone Pairs: Molecules with four electron pairs in their outer shell are based on a tetrahedron. In CH_4 there are four bonding pairs of electrons in the outer shell of the C atom, and the structure is a regular tetrahedron with bond angle $\text{H}-\text{C}-\text{H}$ of $109^\circ 28'$. In NH_3 and N atom has four electron pairs in the outer shell, made up of three bond pairs and one lone pair. Because of the lone pair, the bond angle $\text{H}-\text{N}-\text{H}$ is reduced from the theoretical tetrahedral angle of $109^\circ 28'$ to $107^\circ 28'$. In H_2O the O atom has four electron pairs in the outer shell. The shape of the H_2O molecule is based on a tetrahedron with two corners occupied by bond pairs and the other two corners occupied by lone pairs. The presence of two lone pairs reduces the bond angle further to $104^\circ 27'$.

In a similar way, SF_6 has six bond pairs in the outer shell and is a regular octahedron with bond angles of exactly 90° . In BrF_5 , the Br also has six outer pairs of electrons, made up of five bond pairs and one lone pair. The lone pair reduces the bond angles to $84^\circ 30'$. Whilst it might be expected that two lone pairs would distort the bond angles in an octahedral as in XeF_4 but it is not so. Actual bond angle is 90° , reason being that the lone pairs are trans to each other in the octahedron, and hence, the atoms have a regular square planar arrangement.

Molecules with five pairs of electrons are all based on a trigonal bipyramid. Lone pairs distort the structures as before. The lone pairs always occupy the *equatorial positions* (in an triangle), rather than the axial positions (up and down). Thus in I_3^- ion, the central I atom has five electron pairs in the outer shell, made of two bond pairs and three lone pairs. The lone pairs occupy all three equatorial positions and the three atoms occupy the top, middle, and bottom positions in the trigonal bipyramid, thus giving a linear arrangement with a bond angle of exactly 180° .

Effect of Electronegativity: NF_3 and NH_3 both have structures based on a tetrahedron with one corner occupied by a lone pair. The high electronegativity of F push the bonding electrons further away from N than in NH_3 . Hence the lone pair in NF_3 causes a greater distortion from tetrahedral and gives a $\text{F}-\text{N}-\text{F}$ bond angle of $102^\circ 30'$, compared with $107^\circ 48'$ in NH_3 . The same effect is found in H_2O (bond angle $104^\circ 27'$) and F_2O (bond angle 102°).

Worked-out Examples

38. In which of the following pairs bond angle is $109^\circ 28'$?

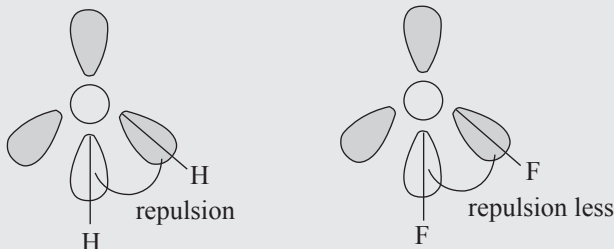
[AIEEE 2002]

- (a) $[\text{NH}_4]^+$, $[\text{BF}_4]^-$ (b) NH_4^+ , $[\text{BF}_3]$ (c) NH_3 , $[\text{BF}_4]^-$ (d) $[\text{NH}_3]$, $[\text{BF}_3]$

Ans. (c) In NH_3 and BF_4^- the hybridization is sp^3 and the bond angle is almost $109^\circ 28'$.

39. The bond angle of H_2O is 104° while that of F_2O is 102° . Explain

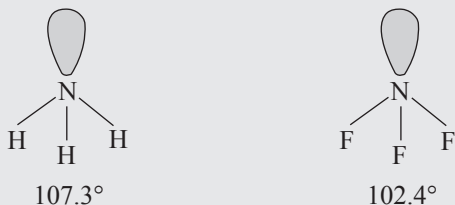
Solution: Both H_2O and F_2O have a lone pair of electrons. But fluorine being highly electronegative, the bond pair electrons are drawn more towards F in F_2O . The bond pairs being displaced away from the central atom has very little tendency to open up the angle. But in H_2O , this opening up is more as bond pair electrons are closer to each other. So bond angle of F_2O is less than H_2O .



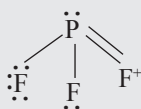
40. Explain why bond angle of NH_3 is greater than NF_3 while bond angle of PH_3 is less than that of PF_3 .

Solution: Both NH_3 and NF_3 are pyramidal in shape with one lone pair on N. However F has higher electronegativity than H, the electron pair is attracted more towards F in NF_3 i.e., the bond pairs of electrons are away from N or in other words distance between bond pairs ion is more. Hence repulsion between bond pairs in NF_3 is less

than NH_3 . Hence the lone pair repels the bond pairs of NF_3 more than it does in NH_3 . As a result, the bond angle decreases to 102.4° . Whereas, in NH_3 it decreases to 107.3° only.



PH_3 and PF_3 are also pyramidal in shape with one lone pair on P. But PF_3 has greater bond angle than PH_3 (opposite to NH_3 and NF_3). This is due to resonance in PF_3 , leading to partial double bond character as shown below:



As result repulsions between P – F bonds are large and hence the bond angle is large. There is no possibility of formation of double bonds in PH_3 .

Orbitals on central atom		Shape	Number of bond pairs	Number of lone pairs	Bond angle
BeCl ₂	2	Linear	2	0	180°
BF ₃	3	Plane triangle	3	0	120°
CH ₄	4	Tetrahedral	4	0	109°28'
NH ₃	4	Pyramidal	3	1	107°48'
NF ₃	4	Pyramidal	3	1	102°30'
H ₂ O	4	Bent (V-shape)	2	2	104°27'
F ₂ O	4	Bent (V-shape)	2	2	102°
PCl ₅	5	Trigonal bipyramid	5	0	120° and 90°
SF ₄	5	Trigonal bipyramid	4	1	101°36' and 86°33'
ClF ₃	5	T-shape	3	2	87°40'
XeF ₂	5	Linear	2	3	180°
SF ₆	6	Octahedral	6	0	90°
BrF ₅	6	Square pyramidal	5	1	84°30'
XeF ₄	6	Square planar	4	2	90°

Some Examples using the VSEPR Theory

Phosphorus pentachloride PCl_5

Gaseous PCl_5 is covalent. (The electronic structure P is $1s^2 2s^2 2p^6 3s^2 3p^3$). All five outer electrons are used to form bonds to the five Cl atoms. In the PCl_5 molecule the valence shell of the P atom contains five electron pairs: hence the structure is a trigonal bipyramid. There are no lone pairs, so the structure is not distorted. However, a trigonal bipyramid is not a completely regular structure, since some bond angles are 90° and others 120° . Symmetrical structures are usually more stable than asymmetrical ones.

Note: Thus PCl_5 is highly reactive, and in the solid state it splits into $[\text{PCl}_4]^+$ and $[\text{PCl}_6]^-$ ions, which have tetrahedral and octahedral structures respectively.

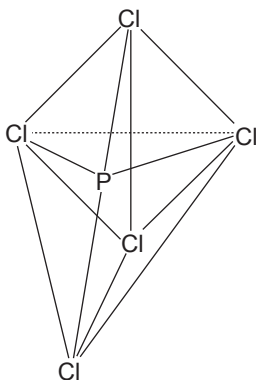


Fig. 1.8 Structure of PCl_5 molecule

Chlorine trifluoride ClF_3

The chlorine atom is at the centre of the molecule and determines its shape. The electronic configuration of Cl is $1s^2 2s^2 2p^6 3s^2 3p^5$. Three electrons form bonds to F, and four electrons do not take part in bonding. Thus in ClF_3 , the Cl atom has five electron pairs in the outer shell, hence the structure is a trigonal bipyramid. There are three bond pairs and two lone pairs.

It was noted previously that a trigonal bipyramid is not a regular shape since the bond angles are not all the same. It therefore follows that all the corners are not equivalent. Lone pair occupy two of the corners, and F atoms occupy the other three corners. Three different arrangements are theoretically possible, as shown in figure below.

The most stable structure will be the one of lowest energy, that is the one with the minimum repulsion between the five orbitals. The great repulsion occurs between two lone pairs. Lone pair bond pair repulsions are next strongest, and bond pair-bond pair repulsions are weakest. Groups at 90° repel each other strongly, whilst groups 120° apart repel each other much less.

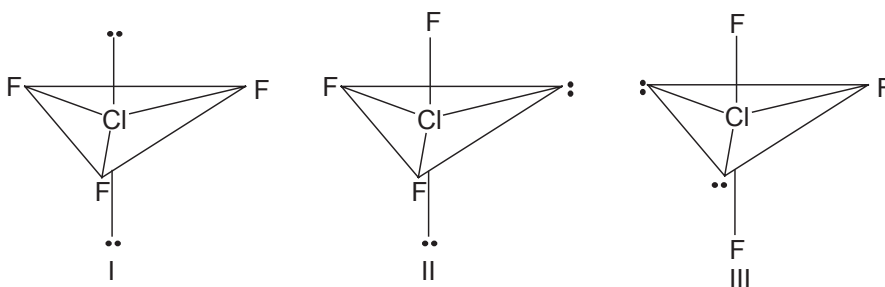


Fig. 1.9 Chlorine trifluoride molecule

Structure I is the most symmetrical, but has six 90° repulsions between lone pairs and atoms. Structure II has one 90° repulsion between two lone pairs, plus three 90° repulsions between lone pairs and atoms. These factors indicate that structure III is the most probable. The observed bond angles are $80^\circ 40'$, which is close to the theoretical 90° . This confirms that the correct structure is III, and the slight distortion from 90° is caused by the presence of the two lone pairs.

As a general rule, if lone pairs occur in a trigonal bipyramid they will be located in the equatorial position (round the middle) rather than the axial positions (top and bottom), since this arrangement minimizes repulsive forces.

Sulphur hexafluoride SF₆

The electronic structure of S is $1s^2 2s^2 2p^6 3s^2 3p^6$. All six of the outer electrons are used to form bonds with the F atoms. Thus in SF₆, the S has six electron pairs in the outer shell hence the structure is octahedral. There are no lone pairs, so the structure is completely regular with bond angles of 90°.

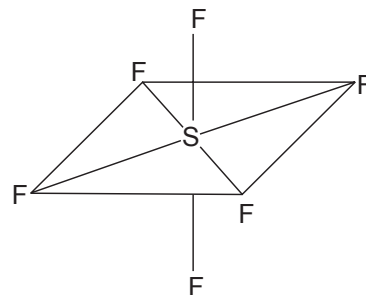
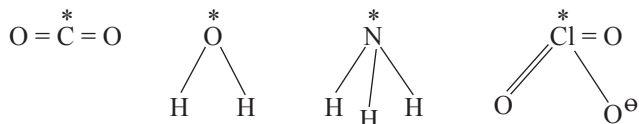


Fig. 1.10

How to find the real shape of a molecule

- The atoms to which maximum number of bonds are attached is called the central atom. For example 'C' atom in CO₂, oxygen atom in H₂O, nitrogen atom in NH₃, Cl atom in ClO₃[−] are the central atoms.



- Since maximum number of bonds are attached to central atom, **the state hybridization of central of central atom governs the geometry of molecule.**

Let us find out the real shape of NH₃ molecule

- First we determine the state of hybridization of 'N' atom in NH₃ we know

$$\text{Number of orbital of the atom hybridized} = \frac{1}{2} \left[\begin{array}{cccc} \text{Number of} & + \text{Number of} & + \text{unit} & - \text{unit} \\ \text{valence electrons} & \text{univalent atoms} & \text{negative} & \text{positive} \\ \text{of central atom} & \text{attached to it (bonded} & \text{charge} & \text{charge} \\ & \text{through single bond)} & & \end{array} \right]$$

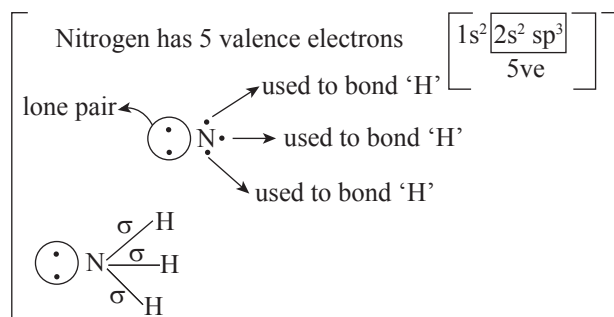
$$\frac{1}{2}[5 + 3] = 4 \text{ for 'N' atom in NH}_3$$

Four orbitals means one will be s and other will be 'p' orbitals. If the four hybrid orbitals will be sp³ orbitals.

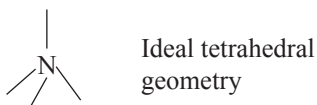
We can determine the state of hybridization by another as follows

$$\text{Number of the atom hybridized} = \left[\begin{array}{cc} \text{Number of s} & \text{Number of lone} \\ \text{bonds attached} & \text{pairs on it} \\ \text{to the atom} & \end{array} \right] = 3 + 1 = 4$$

Again we see it is sp³ hybridized

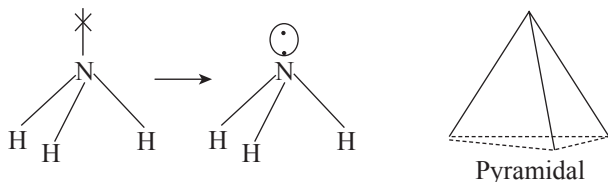


- Now we have to draw the ideal geometry as per the state of hybridization of central atom. Because 'N' in NH₃ is sp³ hybridized, the ideal geometry would be tetrahedral.



1.52 Periodic Properties and Chemical Bonding

- Now there are four stereochemical position to be occupied by atoms to generate the tetrahedral geometry. But we have only three 'H' atoms to fill up three stereochemical portions



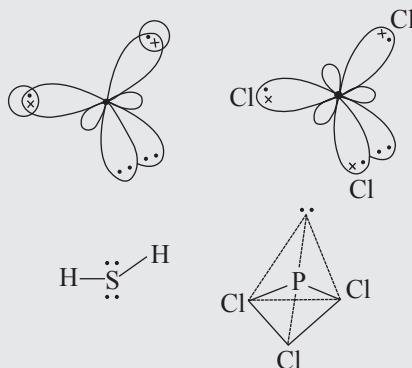
Now the bonds for the extra stereochemical positions should be erased and it should be replaced by a lone pair (non bonded electron pair).

Lone pair does not generate the geometry. If we join the nuclei of the bonded atoms by imaginary lines we get a pyramidal geometry. Hence the shape of NH_3 molecule is pyramidal.

Worked-out Examples

41. Interpret the non-linear shape of H_2S molecule and non-planar shape of PCl_3 using valence shell electron pair repulsion (VSEPR) theory.

[IIT 2001]



Ans. In H_2S , S is in sp^3 hybridized state having two lone pairs on it as shown in figure. The angle is contracted due to lone pair effect to produce V-shaped structure. On the other hand, in PCl_3 , P atom is also sp^3 hybridized having one lone pair on it. This time the shape from tetrahedron structure is distorted to produce pyramidal shape due to lone pair effect.

42. Using VSEPR theory, identify the type of hybridization and draw the structure of OF_2 . What are the oxidation state of O and F?

[IIT 1994]



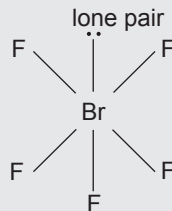
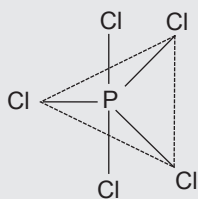
Ans. The structure of OF_2 (similar to H_2O) involves sp^3 hybridization on O-atom having V-shape.

Oxidation number of oxygen = +2

Oxidation number of fluorine = -1

43. Using VSEPR theory, draw the shape of PCl_5 and BrF_5 .

Ans. PCl_5 : sp^3d hybridization having trigonal bipyramidal shape with five 3sp^3 d-3p bonds.



BrF_5 : sp^3d^2 hybridization having distorted octahedral shape with one position occupied by lone pair (or square pyramidal) with five $4\text{sp}^3\text{d}^2$ -2p bonds.

44. Write the geometry of XeF_4 and OSF_4 using VSEPR theory and clearly indicates the position of lone pair of electrons and hybridization of He central atom. [IIT 2004]

Ans. In XeF_4 , Xe atom shows sp^3d^2 hybridization leading to octahedral structure with two positions occupied by lone pair giving rise to square planar shape. In OSF_4 , S atom shows sp^3d hybridization leading to trigonal bipyramidal geometry but distorted due to $\text{S}=\text{O}$ bond. The F atom are at axial and equatorial positions where as O being less electronegative occupies one of the three equatorial positions.

45. In XeF_2 , XeF_4 and XeF_6 , the number of lone pair of electrons on Xe are respectively:

[AIEEE 2002]

- (a) 2, 3, 1 (b) 1, 2, 3 (c) 4, 1, 2 (d) 3, 2, 1

Ans. (d) Xe in XeF_2 , XeF_4 and XeF_6 show sp^3d , sp^3d^2 and sp^3d^3 hybridization respectively.

46. In which of the following species, is the underlined carbon has sp^3 hybridization?

[AIEEE 2002]

- (a) $\text{CH}_3\text{C}\text{OOH}$ (b) $\text{CH}_3\text{C}\text{H}_2\text{OH}$ (c) $\text{CH}_3\text{C}\text{OCH}_3$ (d) $\text{CH}_2=\text{C}\text{H}-\text{CH}_3$

Ans. (b) $\text{CH}_3-\text{CH}_2-\text{OH}$
 $\text{sp}^3 \quad \quad \quad \text{sp}^3$

47. Molecular shape of SF_4 , CF_4 and XeF_4 are:

[IIT 2000; AIEEE 2005]

- (a) The same with 2, 0 and 1 lone pair of electron respectively
 (b) The same with 1, 1 and 1 lone pair of electron respectively
 (c) Different with 0, 1 and 2 lone pairs of electron respectively
 (d) Different with 1, 0 and 2 lone pairs of electron respectively

Ans. (d) SF_4 has sp^3d -hybridization with one lone pair, CF_4 has sp^3 -hybridization with no lone pair and XeF_4 has sp^3d^2 -hybridization with two lone pairs.

48. Which among the following has smallest bond angle?

[AIEEE 2003]

- (a) H_2S (b) NH_3 (c) SO_2 (d) H_2O

Ans. (a) In H_2S , due to low electronegativity of sulphur the L.P. – L.P. repulsion is more than B.P.-B.P. repulsion and hence the bond angle is minimum.

	SO_2	H_2O	H_2S	NH_3
Bond angle	119.5°	104.5°	92.5°	106.5°

49. Which pair of molecules will have permanent dipole moment for both members?

[AIEEE 2003]

- (a) NO_2 and O_3 (b) SiF_4 and CO_2 (c) SiF_4 and NO_2 (d) NO_2 and CO_2

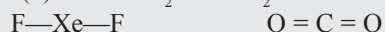
Ans. (a) Both NO_2 and O_3 have angular shape and hence will have net dipole moment.

50. The pair of species having identical shape of both species:

[AIEEE 2003]

- (a) BF_3 , PCl_3 (b) PF_5 , IF_5 (c) CF_4 , SF_4 (d) XeF_2 , CO_2

Ans. (d) Both XeF_2 and CO_2 have linear structure.

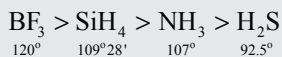


51. The correct order of bond angles is:

[AIEEE 2004]

- (a) $\text{H}_2\text{S} < \text{NH}_3 < \text{BF}_3 < \text{SiH}_4$ (b) $\text{NH}_3 < \text{H}_2\text{S} < \text{SiH}_4 < \text{BF}_3$
 (c) $\text{H}_2\text{S} < \text{NH}_3 < \text{SiH}_4 < \text{BF}_3$ (d) $\text{H}_2\text{S} < \text{SiH}_4 < \text{NH}_3 < \text{BF}_3$

Ans. (d) The order of bond angles

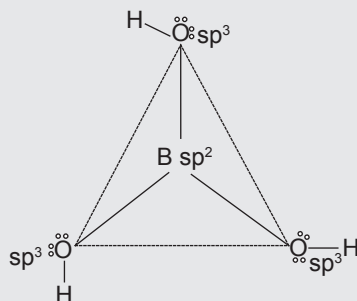


52. The states of hybridization of boron and oxygen atoms in boric acid (H_3BO_3) are respectively:

[AIEEE 2004]

- (a) sp^3 and sp^3 (b) sp^2 and sp^3 (c) sp^3 and sp^2 (d) sp^2 and sp^2

Ans. (c)



53. Which has regular tetrahedral geometry?

[AIEEE 2004]

- (a) SF_4 (b) BF_4^- (c) XeF_4 (d) $[\text{Ni}(\text{CN})_4]^{2-}$

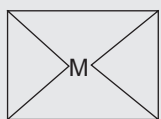
Ans. (b) XeF_4 (sp^3d^2 square planar), $[\text{Ni}(\text{CN})_4]^{2-}$ (dsp^2 square planar), BF_4^- (sp^3 tetrahedral), SF_4 (sp^3d see saw shaped)

54. The maximum number of 90° angles between bond pair-bond pair of electron is observed in:

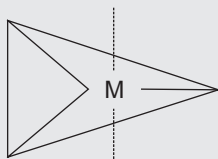
[AIEEE 2004]

- (a) sp^3d^2 -hybridization (b) sp^3d -hybridization
 (c) dsp^2 -hybridization (d) dsp^3 -hybridization

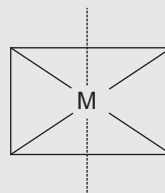
Ans. (a)



dsp^2 hybridization
 Number of 90° angle
 between bonds = 4



sp^3d or dsp^3 hybridization
 Number of 90° angle
 between bonds = 6



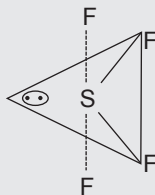
sp^3d^2 hybridization
 Number of 90° angle
 between bonds = 12

55. In which of the following molecules/ions, are all the bonds not equal?

[AIEEE 2006]

- (a) SF_4 (b) SiF_4 (c) XeF_4 (d) BF_4

Ans. (a) In SF_4 the hybridization is sp^3d and the shape of molecule is



- 56.** The decreasing values of bond angles from NH_3 (106°) to SbH_3 (101°) down the group 15 of the periodic table is due to: **[AIEEE 2006]**

- (a) Increase in bp-bp repulsion (b) Increase in p-orbital character in sp^3
(c) Decrease in lp-pp repulsion (d) Decrease in electronegativity

Ans. (d) The bond angle decreases on moving down the group due to decrease in bond pair-bond pair repulsion.

NH_3	PH_3	AsH_3	SbH_3	BiH_3
107°	94°	92°	91°	90°

This can also be explained by the fact that as the size of central atom increases sp^3 hybrid orbital becomes more distinct with increasing size of central atom i.e., pure p-orbitals are utilized in M-H bonding.

- 57.** The correct order of hybridization of the central atom in the following species NH_3 , $[\text{PtCl}_4]^{2-}$, PCl_5 and BCl_3 is
(a) dsp^2 , dsp^3 , sp^2 and sp^3 (b) sp^3 , dsp^2 , dsp^3 , sp^2
(c) dsp^2 , sp^2 , sp^3 , dsp^3 (d) dsp^2 , sp^3 , sp^2 , dsp^3 **(IIT 2001)**

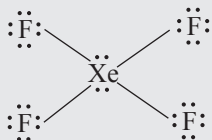
Ans. The structures of NH_3 , $[\text{PtCl}_4]^{2-}$, PCl_5 and BCl_3 are tetrahedral, square planar, triangle bipyramid and triangular planar, respectively. Hence, the central atoms respectively involve sp^3 , dsp^2 , dsp^3 and sp^2 hybrid orbitals.

- 58.** Specify the coordination geometry around and hybridization of N and B atoms in a 1 : 1 complex of BF_3 and NH_3 **(IIT 2002)**
(a) N : tetrahedral, sp^3 ; B : tetrahedral, sp^3 (b) N : pyramidal, sp^3 ; B : pyramidal, sp^3
(c) N : pyramidal, sp^3 ; B : planar, sp^2 (d) N : pyramidal, sp^3 ; B : tetrahedral, sp^3

Ans. Both N and B attain tetrahedral geometry

- 59.** Total number of lone pair of electrons in XeF_4 is **(IIT 2004)**
(a) 0 (b) 1 (c) 2 (d) 3

Ans. The total number of valence electrons in XeF_4 are 36 ($= 8 + 4 \times 7$). These are distributed as follows:



Thus, Xe contains six pairs of electrons around Xe out of which two are lone pairs. According to VSEPR, these six pairs of electrons are distributed in the octahedral geometry.

- 60.** The experimentally determined N—F bond length in NF_3 is greater than the sum of single bond covalent radii of N and F. **[IIT 1995]**

Ans. The experimentally determined N—F bond length in NF_3 is greater than the sum of single bond covalent radii of N and F. This is due to the partial ionic character of the bonds between unlike atoms and the difference is very much marked in the case of the most strongly electronegative elements F, O and N.

61. $(\text{SiH}_3)_3\text{N}$ is a weaker base than $(\text{CH}_3)_3\text{N}$.

[IIT 1995]

Ans. Lone pair of electrons on nitrogen in $(\text{SiH}_3)_3\text{N}$ are used up in *pπ-dπ* back bonding while in $(\text{CH}_3)_3\text{N}$ such *apn-dn* bonding is not possible due to absence of vacant *c/-* orbitals in carbon. Therefore $(\text{CH}_3)_3\text{N}$ is more basic than $(\text{SiH}_3)_3\text{N}$.

62. The central carbon-carbon bond in 1, 3-butadiene is shorter than that of n-butane.

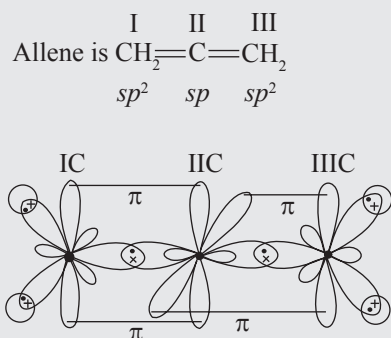
[IIT 1998]

Ans. Buta-1-3-diene, i.e., $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$ has $\text{sp}^2\text{-sp}^2$ (C—C) bond length; butane has $\text{sp}^3\text{-sp}^3$ (C—C) bond length. More is s-character in hybridization, lesser is bond length.

63. Discuss the hybridization of carbon atoms in allene (C_3H_4) and show the n-orbital overlaps.

[IIT 1999]

Ans.



64. Indicate the type of hybridization of each carbon atom in the following compounds:

[IIT 1998]

(a) CH_3CN ,

(b) $\text{CH}_3\text{CH}=\text{CH}_2$,

(c) $\text{H}_3\text{C}-\text{C}=\text{C}-\text{CH}_3$,

(d) $\text{HC}=\text{C}-\text{CH}=\text{CH}_2$

Ans. (a) sp^3 and sp

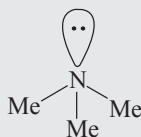
(b) sp^3 , sp^2 , sp^2

(c) sp^3 , sp , sp , sp^3

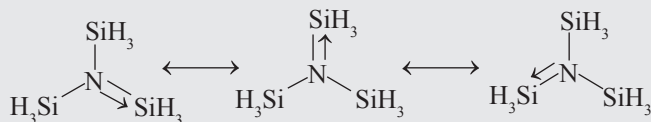
(d) sp , sp , sp^2 , sp^2

65. In trimethylamine, the nitrogen has a pyramidal geometry whereas in trisilylamine $\text{N}(\text{SiH}_3)_3$, it has a planar geometry. What is the reason behind this?

Solution: In $\text{N}(\text{CH}_3)_3$, there is sp^3 hybridization at nitrogen but due to lone pair-bond pair repulsion, shape become pyramidal.



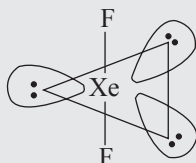
In trisilyl amine, there is vacant d-orbital at silicon, hence formation of $\text{p}\pi-\text{p}\pi$ back bonding takes place and geometry becomes planar.



66. Draw the molecular structures of XeF_2 and XeF_4 , indicating the location of lone pair(s) of electrons.

Solution: Structure of XeF_2
Hybridization – sp^3d

Shape – linear



Structure of XeF_4

Hybridization sp^3d^2

Shape – square planar



67. Which of the following statement is CORRECT?

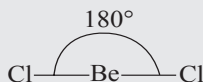
- (a) H_2O and OF_2 both are linear in shape (b) H_2O is linear whereas OF_2 is bent molecule
(c) both H_2O and OF_2 have a bent structure (d) H_2O is bent whereas OF_2 has a linear structure

Solution: (c) First statement is incorrect, both H_2O and OF_2 have a bent structure. Similarly, the second statement is also wrong, because H_2O is bent structure.

68. In which of the following molecules, the bond angle is maximum?

- (a) H_2O (b) BeCl_2 (c) H_2S (d) CH_4

Solution: (b) Of these molecules, BeCl_2 is linear. The bond angle is 180° .



69. Each of the following pair contains species with the same geometry EXCEPT

- (a) PO_4^{3-} , SO_4^{2-} (b) NH_4^+ , BF_4^- (c) CO_2 , C_2H_2 (d) N_2O , NO_2

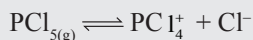
Solution: (d) In (a) PO_4^{3-} and SO_4^{2-} both are isoelectronic and tetrahedral in shape. Same is in (b) also. In (c) each C atom in C_2H_2 is sp hybridized same as C in CO_2 hence, both are linear. Only in (d) N_2O is linear whereas NO_2 is bent and therefore, the answer is (d).

70. Carbonate ion CO_3^{2-} , and nitrate ion NO_3^- , are iso-electronic, if NO_3^- has a trigonal planar geometry, then CO_3^{2-} will have a _____ shape.

- (a) bent (b) tetrahedral
(c) trigonal pyramidal (d) trigonal planar

Solution: (d) Isoelectronic species are species having the same number of electrons and the same geometry. Therefore, CO_3^{2-} will have the trigonal planar geometry and the correct answer is (d).

71. Phosphorus pentachloride, PCl_5 , dissociates in the vapour phase according to the equation



The hybridization of P in PCl_5 is sp^3d . What is the hybridization of P in PCl_4^+ ion?

- (a) sp^3 (b) sp^3d (c) sp^3d^2 (d) dsp^2

Solution: (a) PCl_4^+ is isostructural with NH_4^+ . The hybridization of P in PCl_4^+ is sp^3 . The correct answer is (a).

72. The correct order of increasing bond angle in NO_2 , NO_2^+ and NO_2^- is
 (a) $\text{NO}_2^- < \text{NO}_2 < \text{NO}_2^+$ (b) $\text{NO}_2 < \text{NO}_2^+ < \text{NO}_2^-$
 (c) $\text{NO}_2 < \text{NO}_2^- < \text{NO}_2^+$ (d) $\text{NO}_2^+ < \text{NO}_2^- < \text{NO}_2$

Solution: (a) In NO_2 , the bond angle is 134° , NO_2^+ is linear, bond angle is 180° . NO_2^- is bent, with bond angle slightly less than 120° . Therefore, the correct order of increasing bond angle is: $\text{NO}_2^- < \text{NO}_2 < \text{NO}_2^+$.

73. Amongst N_2O , SO_2 , I_3^+ and I_3^- , the linear species are _____ and _____.

(a) N_2O , SO_2 (b) N_2O , I_3^+ (c) SO_2 , I_3^+ (d) N_2O , I_3^-

Solution: The linear species are N_2O and I_3^- . The correct answer is (d). SO_2 is a bent molecule, therefore (a), (c) cannot be the answer. Whereas I_3^+ is bent. Therefore, (b) also cannot be the answer.

74. XeF_4 is planar whereas CCl_4 is tetrahedral because

(a) in XeF_4 , Xe is sp^2 hybridized and C is sp^3 hybridized in CCl_4 .
 (b) Xe is sp^3d^2 hybridized in XeF_4 but due to the presence of two lone pairs of electrons, the shape is planar whereas in CCl_4 , C is sp^3 hybridized.
 (c) in both XeF_4 and CCl_4 , the central atom is sp^3 hybridized.
 (d) Xe is a noble gas whereas C is a non-metal.

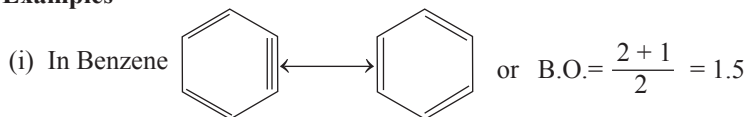
Solution: XeF_4 is planar because Xe is sp^3d^2 hybridized. Due to the presence of two lone pairs of electrons, the shape is reduced to a square planar. CCl_4 is tetrahedral because C is sp^3 hybridized. The correct answer is (b).

Some Noteworthy Points

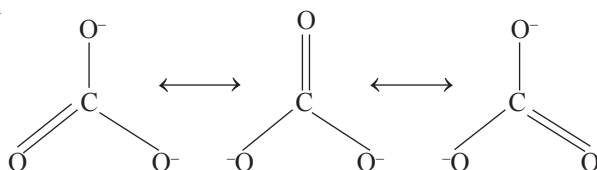
- VSEPR theory was proposed by Sedgwick and development by Gillespie and Nyholm.
- VBT was given by Heitler and London and was modified by Pauling and Slater
- Example of T-shaped molecules are ClF_3 , XeOF_2 , ICl_3 etc.
- Examples of pyramidal molecules are NH_3 , PH_3 , NCl_3 etc.
- SF_4 has see-saw shaped structure.
- Bond order can assume any value including zero.
- Higher the bond order, shorter is bond length and greater is bond energy.
- As a result of resonance the bond order changes in many molecules or ions.

$$\text{Bond order} = \frac{\text{Total number of bonds between two atoms in all the structure}}{\text{Total number of resonance structure}}$$

Examples



(ii) In carbonate ion

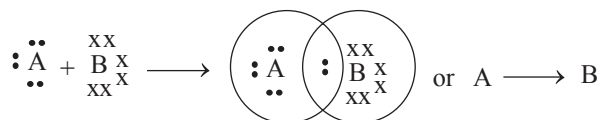


$$\text{B.O.} = \frac{2+1+1}{3} = \frac{4}{3} = 1.33$$

- The bond order of CO, CN⁻ and NO⁺ is 3.0; Bond order of CO⁺, CN and NO is 2.5 and Bond order of NO⁻ is 2.0
- In molecules like PCl₅ (sp³d-hybridization), the two axial bonds are slightly elongated and hence slightly weaker than equatorial bonds. This makes PCl₅ quite reactive.

COORDINATE BOND

It is a special type of covalent bond in which both the shared electrons are contributed by one atom only. It may be defined as **“a covalent bond in which both electrons of the shared pair are contributed by one of the two atoms”**. Such a bond is also called as dative bond. A coordinate or a dative bond is established between two such atoms, one of which has a complete octet and possesses a pair of valence electrons while the other is short of a pair of electrons.



This bond is represented by an arrow ($\rightarrow$) pointing towards acceptor atom.

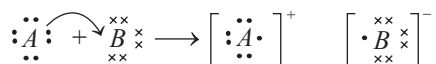
The atom which contributes electron pair is called the donor **while** the atom which accepts it is called acceptor.

The compound consisting of the coordinate bond is termed coordinate compound. Some examples of coordinate bond formation are given below:

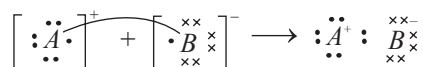
Note: Coordinate bond after formation is indistinguishable from a covalent bond.

The formation of a coordinate bond can be looked upon as a combination of electrovalent and covalent bonds. The formation may be assumed to have taken place in two steps:

- The donor atom loses one electron and transferred to acceptor atom. As a result donor atom acquires a positive charge and the acceptor atom acquires a negative charge.



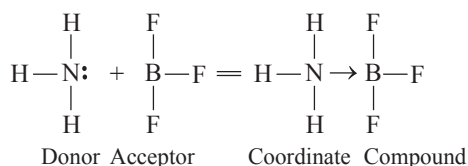
- These two charged particles now contribute one electron each and this pair is shared by both the atoms.



As the coordinate bond is a combination of one electrovalent bond and one covalent bond, it is also termed as **semi polar bond**.

The compound consisting of the coordinate bond is termed **coordinate compound**. Some examples of coordinate bond formation are given below:

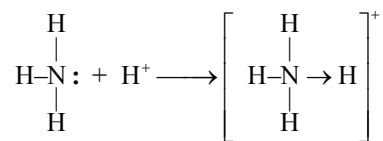
- Combination of ammonia and boron trifluoride:** Although the nitrogen atom has completed its octet in ammonia, it still has a lone pair of electrons in the valency shell which it can donate. The boron atom in boron trifluoride is short of two electrons which it accepts and completes its octet.



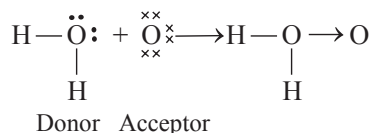
Any atom or ion or molecule which has one unshared electron pair which it can donate is termed as **Lewis base** while those which are capable of accepting the lone pair are termed as **Lewis acids**. In above example, ammonia is a Lewis base while boron trifluoride is a Lewis acid.

Note: H⁺ ion and cations of transition metals such as Cu²⁺, Co²⁺, Fe²⁺, Mn²⁺, Cr³⁺, Ni²⁺, etc., act as Lewis acids. The donors are also called as ligands.

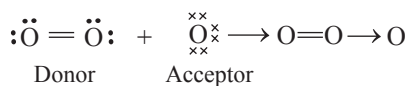
- (ii) **Formation of ammonium ion:** Hydrogen ion (H^+) has no electron and thus accepts a lone pair donated by nitrogen.



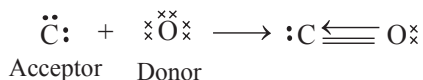
- (iii) **Formation of H_2O_2 :** Oxygen in water has two lone pairs of electrons, out of which one is donated to another oxygen atom forming hydrogen peroxide.



- (iv) **Formation of ozone:** Oxygen molecule consists of two oxygen atoms linked by a double covalent bond. Each oxygen atom has two lone pairs of electrons. When one lone pair of electrons is donated to a third oxygen atom which has six electrons, a coordinate bond is formed.



- (v) **Formation of CO:** Carbon has four valency electrons and oxygen has six. Two combine to form a double bond and a coordinate bond as to achieve their octet completed.



Characteristics of Coordinate Compounds: The properties of coordinate compounds are intermediate between the properties of electrovalent compounds and covalent compounds. The main properties are described below:

- (i) **Melting and Boiling Points:** Their melting and boiling points are higher than purely covalent compounds and lower than ionic compounds.
- (ii) **Solubility:** These are sparingly soluble in polar solvents like water but readily soluble in non-polar (organic) solvents.
- (iii) **Conductivity:** Like covalent compounds, these are also bad conductors of electricity. The solutions or fused mass do not allow the passage of electricity.

In the previous section, we discussed about those compound which deviate from fully ionic to some degree of covalency. A similar trend can also be observed with pure covalent molecules which can change to a partially ionic bond.

This happens when the electronegativities of the two atoms which form the covalent bond are not the same. The atom having higher electronegativity will draw the bonded electron pair more towards itself resulting in a partial charge separation. The distribution of the electron cloud in the bond does not remain uniform and shifts towards the more electronegative one. Such bonds are called polar covalent bonds. For example, the bond formed between hydrogen and chlorine or between hydrogen and oxygen in water is of this type.



Molecules like HCl, H_2O , NH_3 i.e., molecules of the type $H-X$ having two polar ends (positive and negative) are known as polar molecules. The extent of polar character or the degree of polarity in a compound is given by its dipole moment which is defined as the product of the net positive or negative charge and the distance of separation of the charges i.e., the bond length. The symbol of dipole moment is μ .

μ = electronic charge (e) $\times$ distance

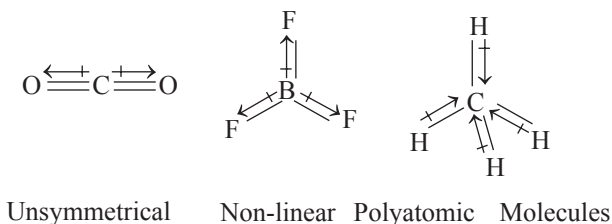
The unit of dipole moment is Debye (D)

1D = 3.33×10^{-30} cm = 10^{-18} esu cm

Dipole moment is indicated by an arrow having a symbol ($\rightarrow$) pointing towards the negative end. Dipole moment has both magnitude and direction and therefore it is a vector quantity.

To calculate the dipole moment of a molecule we should calculate the net dipole due to all bonds and for lone pair if any. Diatomic molecules like HCl, HF have the dipole moment of the bond (called bond dipole) equal to the molecular dipole as the structure has one bond only. But for poly atomic molecules the net dipole is the resultant of the individual bond dipoles. A compound having a zero dipole moment indicates that the compound is a symmetrical one.

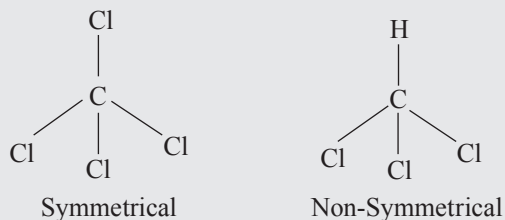
Symmetrical molecule is non-polar even though it contains polar bonds. For example, CO_2 , BF_3 , CH_4 , CCl_4 being symmetrical molecules have zero resultant dipole moments and hence are non-polar since dipole moments summation of all the bonds present in the molecule cancel each other. Dipole moment is usually indicated by an arrow having + on the tail ($\rightarrow$), above the polar bond and pointing towards the negative end.



Worked-out Examples

75. Dipole moment of CCl_4 is zero while that of CHCl_3 is non zero.

Solution: Both CCl_4 and CHCl_3 have tetrahedral structure but CCl_4 is symmetrical while CHCl_3 is non-symmetrical.



Due to the symmetrical structure of CCl_4 the resultant of bond dipoles comes out to be zero. But in case of CHCl_3 it is not possible as the presence of hydrogen introduces some dissymmetry.

76. Both CO_2 and N_2O are linear, but dipole moment of CO_2 is zero but N_2O is non-zero. Why?

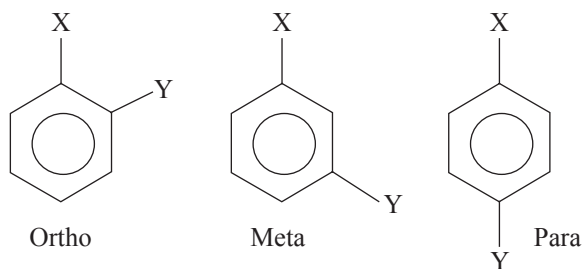
Solution: The answer lies in the structure of these molecules. While CO_2 is symmetrical, N_2O is not, for which the bond dipoles do not cancel each other leaving the molecule with a resultant dipole.



Exercise: Compare the dipole moment of NH_3 and NF_3 .

Dipole Moment in Aromatic Ring System

The dipole moments of the aromatic compounds present a very good illustration of dipole moment. We all know when a substituted benzene is treated with a reagent different products namely ortho, meta and para products are formed. The dipole moments of these products are different since the orientation of the groups are different at ortho, meta and para position. Let us take an example which will make it easily digestive for you. Suppose we have three isomers of o-nitrophenol, m-nitrophenol and p-nitrophenol. We have also the e.g., o-aminophenol, m-aminophenol and p-aminophenol.



In the case $X = Y$, the para isomer becomes symmetrical and have zero dipole.

Now the obvious question that is peeping through your mind is that which isomer in which case has got higher dipole moment. The answer lies in the nature of the groups linked to the benzene ring. In nitrophenol groups one group, is electron pushing and the other is electron withdrawing while in the second case both the groups attached are electron pushing. So depending on the nature of the groups attached one of the isomer, o, m or p has the largest dipole moment.

Case (i): Now when X and Y are both electron pushing or electron withdrawing.

Suppose, bond dipole of $C - X = \mu_1$
 And that of $C - Y = \mu_2$

Here, we have assumed a sign of (+) when groups are electron pushing and (–) when groups are electron withdrawing. The net dipole is the resultant of two bond dipoles at different orientations.

When both X and Y are electron pushing or electron withdrawing.

$$\mu_{ortho} = \sqrt{\mu_1^2 + \mu_2^2 + 2\mu_1\mu_2 \cos 60^\circ} = \sqrt{\mu_1^2 + \mu_2^2 + 2\mu_1\mu_2 \cdot \frac{1}{2}}$$

$$\therefore \mu_o = \sqrt{\mu_1^2 + \mu_2^2 + \mu_1\mu_2}$$

$$\mu_{meta} = \sqrt{\mu_1^2 + \mu_2^2 + 2\mu_1\mu_2 \cos 120^\circ}$$

$$\therefore \mu_m = \sqrt{\mu_1^2 + \mu_2^2 - \mu_1\mu_2}$$

$$\mu_{para} = \sqrt{\mu_1^2 + \mu_2^2 + 2\mu_1\mu_2 \cos 180^\circ} = \sqrt{\mu_1^2 + \mu_2^2 - 2\mu_1\mu_2}$$

$$\therefore \mu_p = \mu_1 - \mu_2$$

From the above expression of μ_o , μ_m and μ_p it is clear that when both X and Y are of the same nature i.e., both are electron withdrawing or both are electron pushing the para product has the least dipole moment and ortho product has the highest. Now, when $X = Y$, $\mu_p = \mu_1 - \mu_2 = \mu_1 - \mu_1 = 0$

Which we have already discussed.

Case (ii): When X is electron pushing and Y is electron withdrawing or vice versa.

Let $C - X$ dipole = μ_1 and $C - Y$ dipole = μ_2

$$\therefore \mu_o = \sqrt{\mu_1^2 + (-\mu_2)^2 + 2\mu_1(-\mu_2) \cos 60^\circ} = \sqrt{\mu_1^2 + \mu_2^2 - \mu_1\mu_2} = \sqrt{(\mu_1 + \mu_2)^2 - 3\mu_1\mu_2}$$

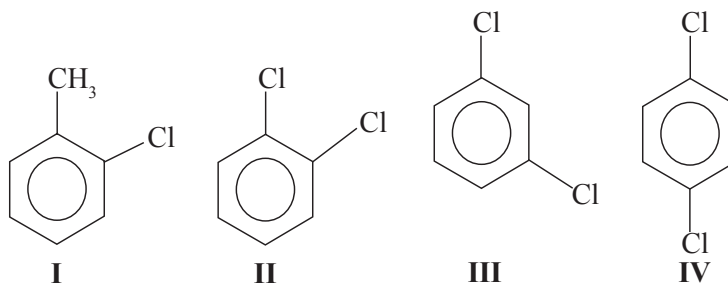
$$\mu_{meta} = \sqrt{\mu_1^2 + (-\mu_2)^2 + 2\mu_1(-\mu_2) \cos 120^\circ} = \sqrt{\mu_1^2 + \mu_2^2 + \mu_1\mu_2} = \sqrt{(\mu_1 + \mu_2)^2 - \mu_1\mu_2}$$

$$\mu_{para} = \sqrt{\mu_1^2 + (-\mu_2)^2 + 2\mu_1(-\mu_2) \cos 180^\circ} = \sqrt{\mu_1^2 + \mu_2^2 + 2\mu_1\mu_2} = \mu_1 + \mu_2$$

Now if you see the expressions, it is very clear that the para isomer has the highest dipole moment and ortho is the least.

So to calculate the dipole moments of disubstituted benzene one should consider about the nature of the groups linked and then only one can predict the dipole moment of the molecule.

Exercise: Arrange the following compounds in order of increasing dipole moments.



Percentage of Ionic Character

Every ionic compound having some percentage of covalent character according to Fajan's rule. The percentage of ionic character in a compound having some covalent character can be calculated by the following equation.

$$\text{The percent ionic character} = \frac{\text{Observed dipole moment}}{\text{Calculated dipole moment assuming 100\% ionic bond}} \times 100$$

Worked-out Examples

77. Dipole moment of KCl is 3.336×10^{-29} coulomb metre which indicates that it is highly polar molecule. The inter-atomic distance between K^+ and Cl^- is $2.6 \times 10^{-10}\text{m}$. Calculate the dipole moment of KCl molecule if there were opposite charges of one fundamental unit located at each nucleus. Calculate the percentage ionic character of KCl.

Ans. Dipole moment $\mu = e \times d$ coulomb metre

For KCl $d = 2.6 \times 10^{-10}\text{ m}$

For complete separation of unit charge

$$e = 1.602 \times 10^{-19}\text{ C}$$

Hence $\mu = 1.602 \times 10^{-19} \times 2.6 \times 10^{-10} = 4.1652 \times 10^{-29}\text{ cm}$

$$\mu_{\text{KCl}} = 3.336 \times 10^{-29}\text{ Cm}$$

$$\therefore \% \text{ ionic character of KCl} = \frac{3.336 \times 10^{-29}}{4.165 \times 10^{-29}} \times 100 = 80.09\%$$

78. Dipole moment of HX is 2.59×10^{-30} coulomb-metre. Bond length of HX is $1.39\text{\AA}$. Calculate percentage ionic character of molecule.

Ans. dipole moment $= q \times d$

$$\mu_{\text{cal}} = q \times 1.39 \times 10^{-10}\text{ coulomb - metre}$$

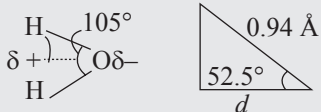
$$\mu_{\text{cal}} = 1.6 \times 10^{-19} \times 1.39 \times 10^{-10}$$

$$\therefore \% \text{ age ionic character} = \frac{\mu_{\text{exp}}}{\mu_{\text{cal}}} \times 100 = \frac{2.59 \times 10^{-30}}{1.6 \times 10^{-19} \times 1.36 \times 10^{-10}} \times 100 = 11.65\%$$

79. Calculate the percentage of covalent character of HI having bond length $= 1.62\text{\AA}$ and observed dipole moment $= 0.39\text{ D}$. 46 In water, the $\text{H}-\text{O}-\text{H}$ bond angle is 105° . Using the dipole moment of water and the covalent radii of the atoms, determine the magnitude of the charge on the oxygen atom in the water molecule.

Solution: $\mu = 1.85 \text{ D} = 1.85 \times 10^{-18} \text{ esu cm} = \delta d$

$$\cos 52.5^\circ = d/0.94 \text{ \AA}$$



$$d = (0.609)(0.94 \text{ \AA}) = 0.572 \text{ \AA}$$

$$\delta = \frac{\mu}{d} = \frac{1.85 \text{ D}}{0.572 \text{ \AA}} = \frac{1.85 \times 10^{-18} \text{ esu cm}}{0.572 \times 10^{-8} \text{ cm}} = 3.2 \times 10^{-10} \text{ esu}$$

or 0.67 times the electronic charge ($4.8 \times 10^{-10} \text{ esu}$).

Some Noteworthy Points

- The dipole moment helps to predict whether a molecule is polar or non polar.
- A molecule may contain polar covalent bonds but its dipole moment may be zero if it is a symmetrical molecule.
- The dipole moment helps to predict geometry of molecules.
- The percentage of ionic character can be calculated. For example,
 - (i) A diatomic molecule having a dipole moment of 1.92 Debye and bond length of 2.0 Å shall have 20% ionic character. This can be calculated as,

$$\mu = e \times d$$

$$\text{or } 1.92 \times 10^{-18} \text{ esu cm} = e \times 2 \times 10^{-8} \text{ cm}$$

$$\text{or } \text{change (e)} = \frac{1.92 \times 10^{-18}}{2 \times 10^{-8}} = 0.96 \times 10^{-10} \text{ esu}$$

Since charge on isolated ion = $4.8 \times 10^{-10} \text{ esu}$

$$\therefore \% \text{ age ionic character} = \frac{0.96 \times 10^{-10}}{4.8 \times 10^{-10}} \times 100 = 20\%$$

- (ii) The % age ionic character can also be calculate as follows;

$$\% \text{ age ionic character} = \frac{\text{Experimental value of dipole moment}}{\text{Theoretical value of dipole moment}} \times 100,$$

e.g., when the dipole moment of HX is 1.92 D and bond distance is 1.2 Å, the % age ionic character can be calculate as:

Theoretical value of dipole moment (μ) when ionic character is 100% is = $q \times d$

$$= 4.8 \times 10^{-10} \text{ esu} \times 1.2 \times 10^{-8} \text{ cm} \quad (\ominus 1 \text{ \AA} = 10^{-8} \text{ cm})$$

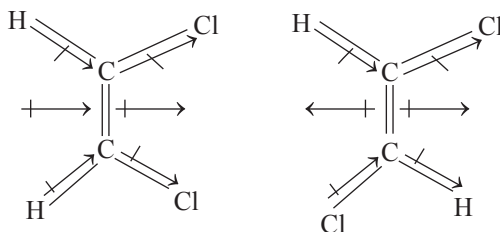
$$\mu = 5.76 \times 10^{-18} \text{ esu cm}$$

$$\mu = 5.76 \text{ D}$$

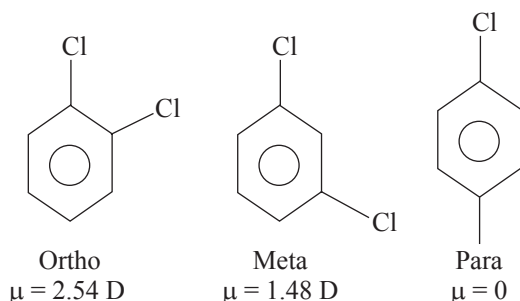
$$\% \text{ age ionic character} = \frac{\text{Experimental dipole moment}}{\text{Theoretical dipole moment}} \times 100 = \frac{1.92}{5.76} \times 100 = 33.3\%$$

- **Odd Electron bonds:** Luder (in 1916) postulated that there are a number of stable molecular in which the bonds are formed by sharing of an odd number of electrons (usually one or three). The bond of this type involving odd number of electrons is called odd electron bond. This type of bond is found in H_2^+ , He_2^+ , O_2 , NO, NO_2 , ClO_2 molecule etc.

- Dipole moment values can be used to distinguish between the cis and trans isomers. Usually, cis isomers have higher dipole moment and are more polar than the trans isomers, e.g.,



- Among the ortho, meta and para isomers, dipole moment is greatest for ortho isomer for para isomer (same substituents). In general, dipole moment of $o > m > p$

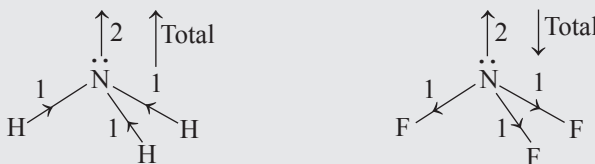


- If a molecule of the type MX_2 has zero dipole moment, the σ -bonding orbitals used by M ($Z < 21$) must be sp -hybridized (e.g., BeF_2).
- If a molecule of the type MX_3 has zero dipole moment, the σ -bonding orbitals used by M ($Z < 21$) must be sp^2 -hybridized (e.g., BF_3).
- If a molecule of the type MX_4 has zero dipole moment, the σ -bonding orbitals used by M ($Z < 21$) must be sp^3 -hybridized (e.g., CCl_4).

Worked-out Examples

80. The dipole moment of NH_3 is more than that of NF_3 . Explain

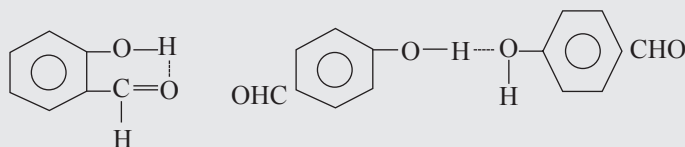
Ans. The dipole moment of NH_3 acts in the directions $\text{H} \longrightarrow \text{N}$ and thus moment due to unshared pair of electron will naturally increase the moment of the NH_3 molecule while in the case of NF_3 , the dipole moment acts in the direction $\text{N} \longrightarrow \text{F}$ and thus unshared electron pair will partially neutralize the dipole moment, causing a lower moment of NF_3 relative of NH_3 .



81. o-Hydroxy benzaldehyde is a liquid at room temperature while p-hydroxy benzaldehyde is a high melting solid.

[IIT 1999]

Ans. o-Hydroxy benzaldehyde show intramolecular H-bonding or chelation, a weaker one than intermolecular H-bonding in p-hydroxy benzaldehyde.



82. The dipole moment of KCl is 3.336×10^{-29} coulomb metre which indicates that it is a highly polar molecule. The interatomic distance between K^+ and Cl^- in this molecule is 2.6×10^{-10} m. Calculate the dipole moment of KCl molecule, if there were opposite charges of one fundamental unit located at each nucleus. Calculate percentage ionic character of KCl. [IIT 1993]

Ans. Dipole moment $\mu = \delta \times d$

$$\therefore 3.336 \times 10^{-29} = \delta \times 2.6 \times 10^{-10}$$

$$\therefore \delta = \frac{3.336 \times 10^{-29}}{2.6 \times 10^{-10}} = 1.283 \times 10^{-19} \text{ coulomb}$$

$$\therefore 1.602 \times 10^{-19} \text{ charge on each, \% character} = 100$$

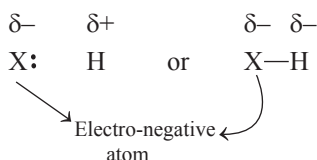
$$\therefore 1.283 \times 10^{-19} \text{ charge on each, \% character} = \frac{1.283 \times 10^{-19}}{1.602 \times 10^{-19}} \times 100 = 80.09\%$$

$$\therefore \text{If one unit charge, then } \delta = 1.602 \times 10^{-19} \text{ C}$$

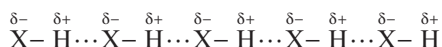
$$\therefore \mu = 1.602 \times 10^{-19} \times 2.6 \times 10^{-10} \\ = 4.1652 \times 10^{-29} \text{ coulomb metre}$$

HYDROGEN BONDING

In 1920, Latimer and Rodebush introduced the idea of “hydrogen bond” to explain the nature of association in liquid state of substance like water, hydrogen fluoride, ammonia, formic acid etc. In a hydrogen compound, when hydrogen is bonded to highly electronegative atom (such as F, O, N) by a covalent bond, the electron pair is attracted towards electronegative atom so strongly that a dipole results i.e., one end carries a positive charge (H-end) and other end carries a negative charge (X-end).



If a number of such molecules are brought nearer to each other, the positive end of one molecule and negative end of the other molecule will attract each other and weak electrostatic force will develop. Thus, these molecules will associate together to form a cluster of molecules.



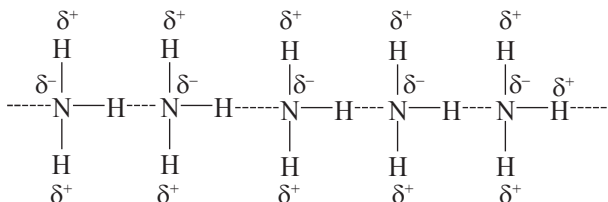
The attractive force that binds hydrogen atom of one molecule with electronegative atom of the other molecule of the same or different substance is known as hydrogen bond.

Hydrogen bonding is of two types:

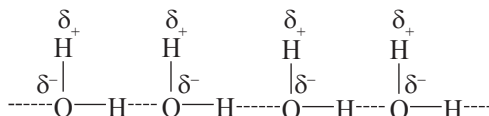
- (a) **Intermolecular hydrogen bonding:** This type of bonding results between the positive and negative ends of different molecules of the same or different substances.

Example

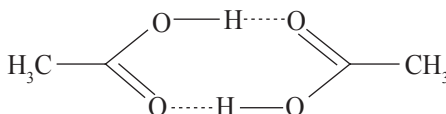
(i) *Ammonia*



(ii) *Water*

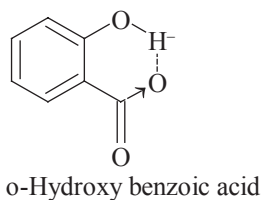
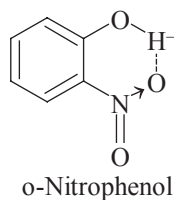


(iii) *Acetic acid*



This type of hydrogen bonding increases the boiling point of the compound and also its solubility in water. The increase in boiling point is due to association of several molecules of the compound.

- (b) **Intramolecular hydrogen bonding:** This type of bonding results between hydrogen and an electronegative element both present in the same molecule. This type of bonding is generally present in organic compounds. Examples are o-nitro-phenol, o-hydroxy benzoic acid, etc.



This type of bonding decreases the boiling point of the compound. The solubility of the compound also decreases. Hence, compound becomes more volatile.

Properties Explained by Hydrogen Bonding

- (a) Strength of certain acids and bases can be explained on the basis of hydrogen bonding.
- (b) **Solubility:** An organic substance is said to be insoluble in water if it does not form hydrogen bonding with water. The organic compound like alkanes, alkenes, ethers, etc., are insoluble in water as they do not form hydrogen bonding with water, while alcohols and acids are soluble because they readily form hydrogen bonds with water.
- (i) **Melting and boiling points of hydrides of N, O and F:** If the melting points and boiling points of the hydrides of the elements of IVA, VA, VIA and VIIA groups are plotted against the molecular weights of these hydrides, we shall get the plots as shown in figure (a) and (b).

From these plots it may be seen that although in case of SbH_3 , AsH_3 , PH_3 (VA group elements hydrides), H_2Te , H_2Se , H_2S (VIA group elements hydrides) and HI , HBr , HCl (VIIA group elements hydrides) there is a progressive decrease in their m.p's and b.p's with the decrease in their molecular weights, the m.p's and b.p's of NH_3 , H_2O and HF hydrides suddenly increase with a further decrease of their molecular weights. The sudden increase in m.p's and b.p's in these hydrides is due to the inter-molecular H-bonding in between H and F in case of HF, in between

H and O in case of H_2O and in between H and N in case of NH_3 respectively. The existence of H-bonding in these molecules gives polymerized molecules $(\text{NH}_3)_n$. Thus, m.p's and b.p's of these molecules are suddenly raised.

Having no power to form H-bonds, the simple carbon family hydrides (SnH_4 , GeH_4 , SiH_4 and CH_4) show a decrease in their bp's and mp's with the decrease in their molecular weights.

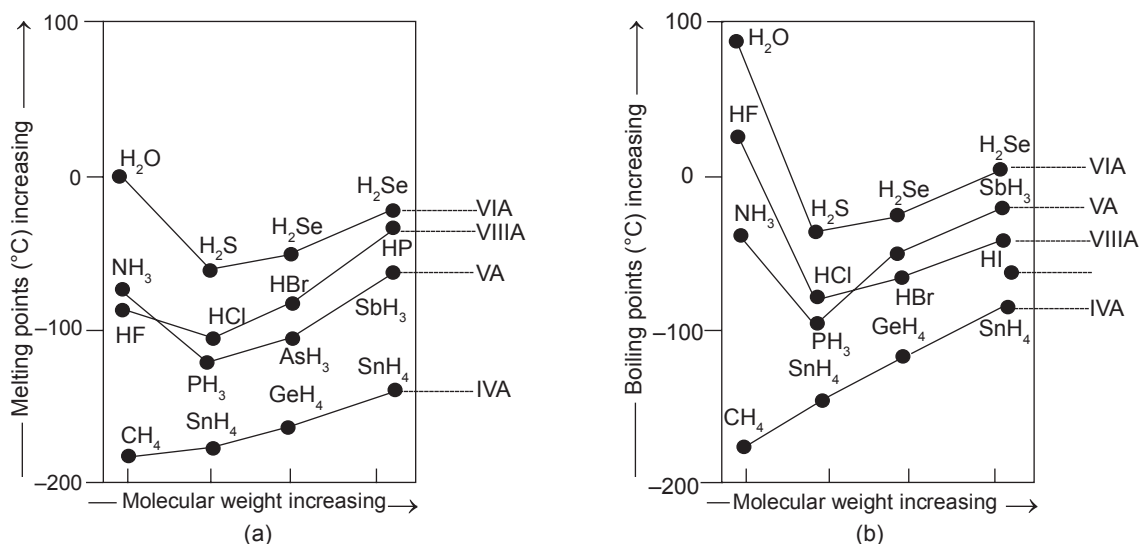
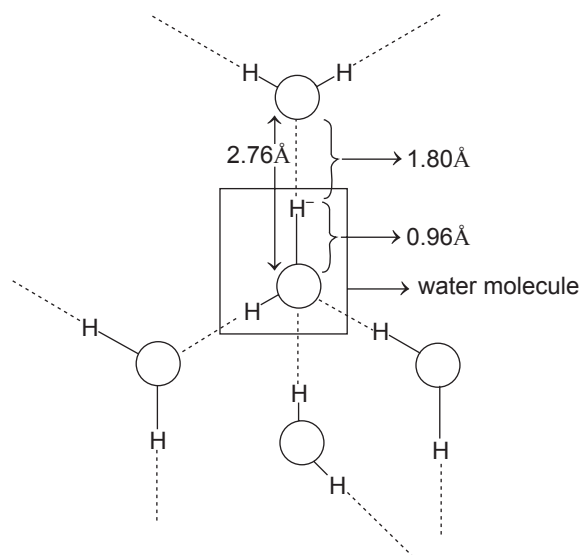


Fig. 1.11

- (ii) **Ice has less density than water:** The explanation of this fact is as follows: In the crystal structure of ice, the O-atom is surrounded by four H-atoms. Two H-atoms are linked to O-atom by covalent bonds as shown (by normal covalent bond) and the remaining two H-atoms are linked to O-atom by two H-bonds shown by dotted lines. Thus in ice every water molecule is associated with four other water molecules by H-bonding in a tetrahedral fashion. Ice has an open cage like structure with a large empty space due to the existence of H-bonds. As ice melts at 0°C , a number of H-bonds are broken down and the space between water molecules decreases so that water molecules move closer together. The density of water increases, from 0° to 4°C , and at 4°C it is maximum. Above 4°C the increase in kinetic energy of the molecules is sufficient to cause the molecules to begin to disperse and the result is that the density decrease with increasing temperature.



Open cage-like tetrahedral crystal structure of ice. Circles indicate oxygen atoms. Bonds represented by solid line are normal covalent bonds while those represented by dotted lines are hydrogen bonds.

Fig. 1.12

Worked-out Examples

83. Which of the following hydrogen bonds is the strongest? [AIEEE 2007]
 (a) O—H—N (b) F—H—F (c) O—H—O (d) O—H—F

Ans. (b) Greater the difference between electronegativity of bonded atoms, stronger will be bond. Since F is most electronegative hence $\text{F—H} \cdots \text{F}$ is the strongest bond.

84. An ether is more volatile than alcohol having same molecular formula. This is due to: [AIEEE 2003]
 (a) inter molecular H-bonding in ethers
 (b) inter molecular H-bonding in alcohols
 (c) dipolar character of ethers
 (d) resonance structure in alcohols

Ans. (b) In ether, there is no H-bonding while alcohols have intermolecular H-bonding
 (d) The C=O bond is weaker than the C—O bond

Rules for linear combination of atomic orbitals

In deciding which atomic orbitals may be combined to form molecular orbitals, three rules must be considered:

1. The atomic orbitals must be roughly of the same energy. This is important when considering overlap between two different types of atoms.
2. The orbitals must overlap one another as much as possible. This implies that the atoms must be close enough for effective overlap and that the radial distribution functions of the two atoms must be similar at this distance.
3. In order to produce bonding and anti-bonding MOs, either the symmetry of the two atomic orbitals must remain unchanged when rotated about the internuclear line, or both atomic orbitals must change symmetry in an identical manner.

In the same way that each atomic orbital has a particular energy, and may be defined by four quantum numbers, each molecular orbital has a definite energy, and is also defined by four quantum numbers.

1. The principal quantum number n has the same significance as in atomic orbitals.
2. The subsidiary quantum number l also has the same significance as in atomic orbitals.
3. The magnetic quantum number of atomic orbitals is replaced by a new quantum number λ . In a diatomic molecule, the line joining the nuclei is taken as a reference direction and λ , represents the quantization of angular momentum in $h/2\pi$ units with respect to this axis. λ takes the same values as m takes for atoms, i.e.,

$$\lambda = -l, \dots, -3, -2, -1, 0, +1, +2, +3, \dots, +l$$

When $\lambda = 0$, the orbitals are symmetrical around the axis and are called σ orbitals. When $\lambda = \pm 1$ they are called π orbitals and when $\lambda = \pm 2$ they are called δ orbitals.

4. The spin quantum number is the same as for atomic orbitals and may have values of $\pm \frac{1}{2}$.

The Pauli exclusion principle states that *in a given atom no two electrons can have all four quantum numbers the same*. The Pauli principle also applies to molecular orbitals: *No two electrons in the same molecule can have all four quantum numbers the same*.

The order of energy of molecular orbitals has been determined mainly from spectroscopic data. In simple homonuclear diatomic molecules, the order is:

$$\sigma 1s, \sigma^* 1s, \sigma 2s, \sigma^* 2s, \sigma 2p_x, \left\{ \begin{array}{l} \pi 2p_y, \pi^* 2p_y \\ \pi 2p_z, \pi^* 2p_z \end{array} \right\}, \sigma 2p_x \xrightarrow{\text{increasing energy}}$$

Note that the $2p_y$ atomic-orbital gives π bonding and π^* anti-bonding MOs and the $2p_z$ atomic orbital gives π bonding and, π^* anti-bonding MOs. The bonding $\pi 2p_y$ and $\pi 2p_z$ MOs have exactly the same energy and are said to be double degenerate. In a similar way, the anti-bonding $\pi^* 2p_y$ and $\pi^* 2p_z$ MOs have the same energy and are also doubly degenerate.

A similar arrangement of MOs exists from $\sigma 3s$ to $\sigma^* 3p_x$, but the energies are known with less certainty.

The energies of the $\sigma 2p$ and $\pi 2p$ MOs are very close together. The order of MOs shown above is correct for oxygen and heavier elements, but for the lighter elements boron, carbon and nitrogen the $\pi 2p_y$ and $\pi 2p_z$ are probably lower than $\sigma 2p_x$. For these atoms the order is:

$$\sigma 1s, \sigma^* 1s, \sigma 2s, \sigma^* 2s \quad \left\{ \begin{array}{l} \pi 2p_y, \sigma 2p_x, \sigma^* 2p_x \\ \pi 2p_z, \pi^* 2p_y, \pi^* 2p_z \end{array} \right\} \xrightarrow{\text{increasing energy}}$$

Examples of molecular orbital treatment for homonuclear diatomic molecules

In the build-up of atoms, electrons are fed into atomic orbitals. The *Aufbau* principle is used:

1. Orbitals of lowest energy are filled first.
2. Each orbital may hold up to two electrons, provided that they have opposite spins.

Hund's rule states that when several orbitals have the same energy (that is they are degenerate), electrons will be arranged so as to give the maximum number of unpaired spins.

In the molecular orbital method, we consider the whole molecule rather than the constituent atoms, and use molecular orbitals rather than atomic orbitals. In the build-up of the molecule, the total number of electrons from all the atoms in the molecule is fed into molecular orbitals. The *Aufbau* principle and Hund's rule are used as before.

For simplicity homonuclear diatomic molecules will be examined first. Homonuclear means that there is only one type of nucleus, that is one element present, and diatomic means that the molecule is composed of two atoms.

H_2^+ molecule ion

This may be considered as a combination of a H atom with a H^+ ion. This gives one electron in the molecular ion which occupies the lowest energy MO:

$$\sigma 1s^1$$

The electron occupies the $\sigma 1s$ bonding MO. The energy of this ion is thus lower than that of the constituent atom and ion, by an amount Δ , so there is some stabilization. This species exists but it is not common since H_2 is much more stable. However, H_2^+ can be detected spectroscopically when H_2 gas under reduced pressure is subjected to an electric discharge.

H_2 molecule

There is one electron from each atom, and hence there are two electrons in the molecule. These occupy the lowest energy MO:

$$\sigma 1s^2$$

This is shown in figure. The bonding $\sigma 1s$ MO is full, so the stabilization energy is 2Δ . A σ bond is formed, and the H_2 molecule exists and is well known.

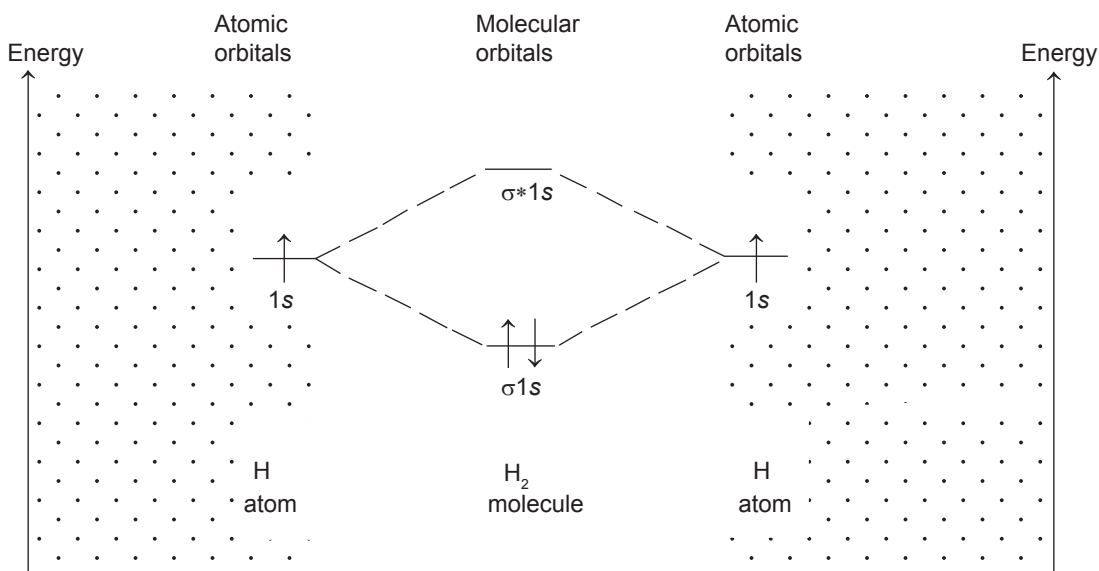
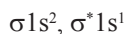


Fig. 1.13 Electronic configuration, atomic and molecular orbitals for hydrogen

He₂⁺ molecule ion

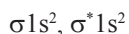
This may be considered as a combination of a He atom and a He⁺ ion. There are three electrons in the molecular ion, which are arranged in MOs:



The filled $\sigma 1s$ bonding MO gives 2Δ stabilization, while the half-filled $\sigma 1s^*$ gives Δ destabilization. Overall there is Δ stabilization. Thus the helium molecule ion can exist. It is not very stable, but it has been observed spectroscopically.

He₂ molecule

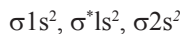
There are two electrons from each atom, and the four electrons are arranged in MOs:



The 2Δ stabilization energy from filling the $\sigma 2s$ MO is cancelled by the 2Δ destabilization energy from filling the $\sigma^* 1s$ MO. Thus a bond is not formed, and the molecule does not exist.

Li₂ molecule

Each Li atom has two electrons in its inner shell, and one in its outer shell, giving three electrons. Thus there is a total of six electrons in the molecule, and these are arranged in MOs:



This is shown in figure. The inner shell of filled $\sigma 1s$ and $\sigma^* 1s$ MOs do not contribute to the bonding in much the same way as in He₂. They are essentially the same as the atomic orbitals from which they were formed, and are sometimes written:



However, bonding occurs from the filling of the $\sigma 2s$ level, and Li₂ molecules do exist in the vapour state. However, in the solid it is energetically more favourable for lithium to form a metallic structure. Other Group 1 metals such as sodium behave in an analogous way:



Be₂ molecule

A beryllium atom has two electrons in the first shell plus two electrons in the second shell. Thus in the Be₂ molecule there are eight electrons. These are arranged in MOs:

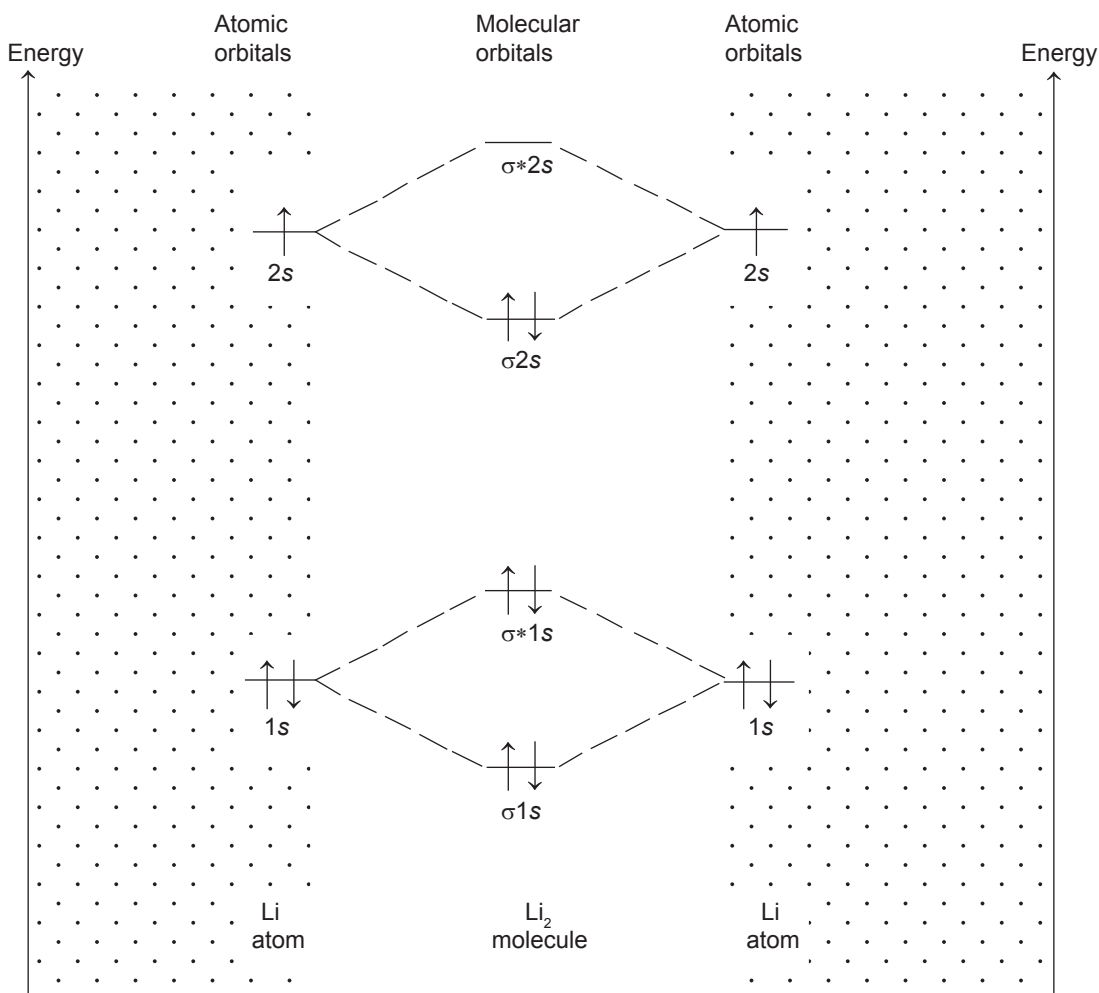


Fig. 1.14 Electronic configuration, atomic and molecular orbitals for lithium

$$\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2$$

or KK, $\sigma 2s^2, \sigma^* 2s^2$

Ignoring the inner shell as before, it is apparent that the effects of the bonding $\sigma 2s$ and anti-bonding $\sigma^* 2s$ levels cancel, so there is no stabilization and the molecule would not be expected to exist.

B₂ molecule

Each boron atom has 2 + 3 electrons. The B₂ molecule thus contains a total of ten electrons, which are arranged in MOs:

$$\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \begin{cases} \pi 2p_y^1 \\ \pi 2p_z^1 \end{cases}$$

This may be shown diagrammatically (figure). Note that B is a light atom and the order of energies of MOs is different from the 'usual' arrangement. Thus the $\pi 2p$ orbitals are lower in energy than the $\sigma 2p_x$. Since the $\pi 2p_y$ and $\pi 2p_z$ orbitals are

degenerate (identical in energy), Hund's rule applies, and each is singly occupied. The inner shell does not participate in bonding. The effects of bonding and anti-bonding $\sigma 2s$ orbitals cancel but stabilization occurs from the filling of the $\pi 2p$ orbitals, and hence a bond is formed and B_2 exists.

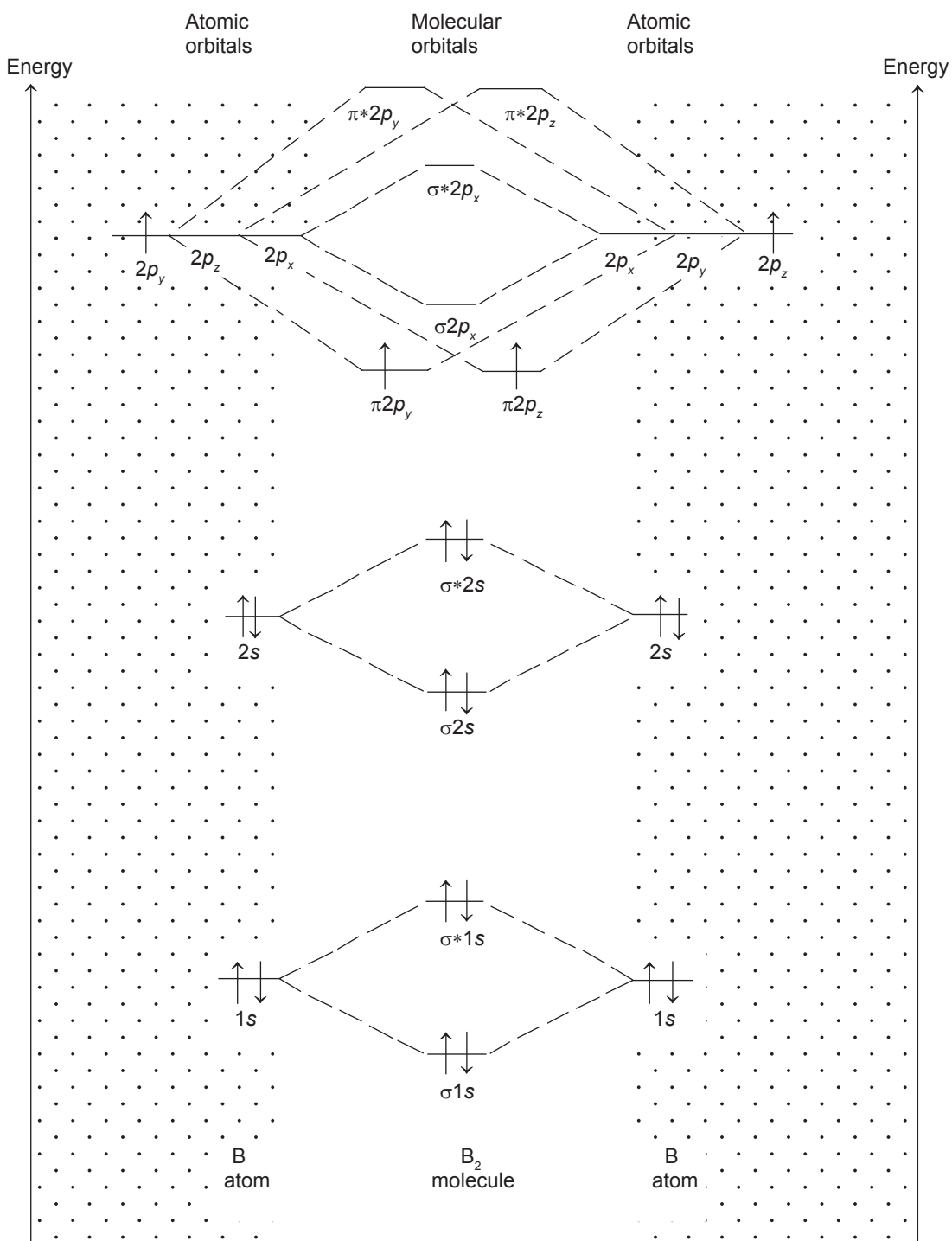


Fig. 1.15 Electronic configuration, atomic and molecular orbitals for boron

C₂ molecule

A carbon atom has 2 + 4 electrons. A C₂ molecule would contain a total of 12 electrons, and these would be arranged in MOs:

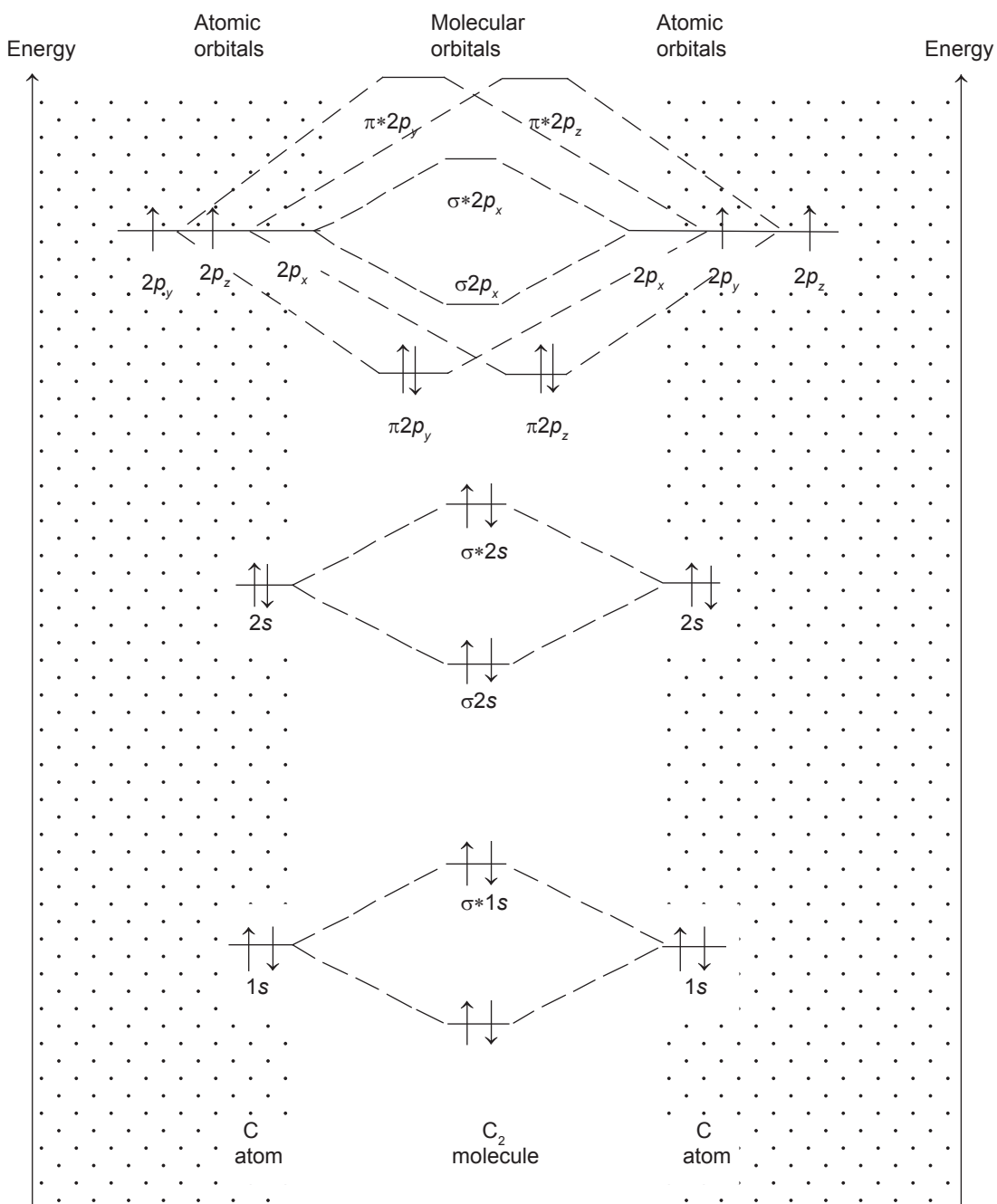


Fig. 1.16 Electronic configuration, atomic and molecular orbitals for carbon.

$$\sigma 1s^2, \sigma 1s^2, \sigma 2s^2, \sigma^* 2s^2, \begin{cases} \pi 2p_y^2 \\ \pi 2p_z^2 \end{cases}$$

The molecule should be stable, since the two $\pi 2p$ bonding orbitals provide 4Δ of stabilization energy, giving two bonds. In fact, carbon exists as a macromolecule in graphite and diamond, since these are an even more stable arrangement (where each carbon forms four bonds): hence diamond and graphite are formed in preference to C_2 .

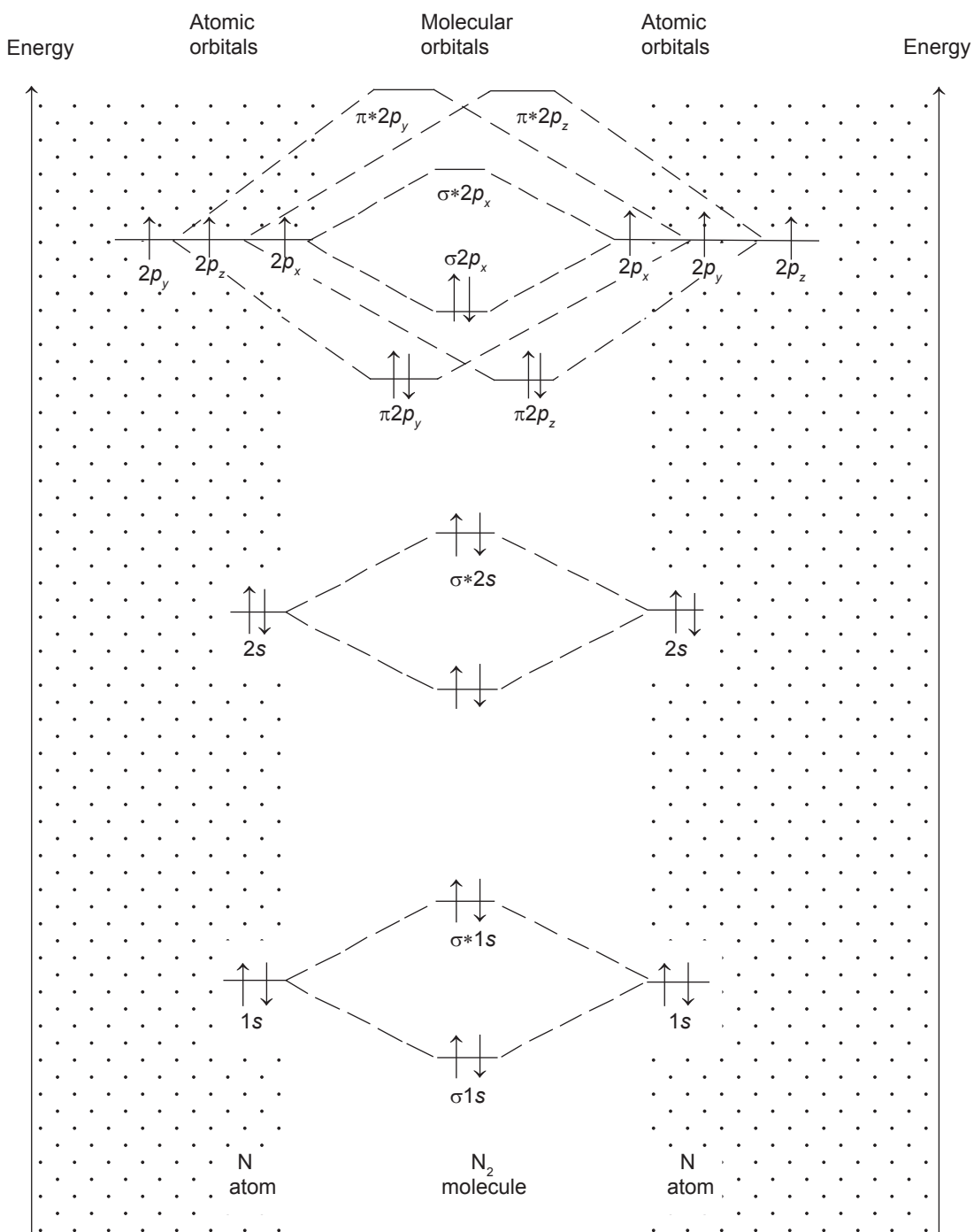


Fig. 1.17

N_2 molecule

A nitrogen atom has $2 + 5 = 7$ electrons. Thus the N_2 molecule contains 14 electrons. These are arranged in MOs:

$$\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \begin{cases} \pi 2p_y^2 \\ \pi 2p_z^2 \end{cases} \sigma 2p_x^2,$$

This is shown diagrammatically (figure).

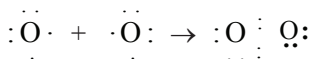
Assuming that the inner shell does not participate in bonding, and that the bonding and anti-bonding 2s levels cancel, one σ and two π bonding pairs remain, giving a total of three bonds. This is in agreement with the valence bond formulation as $N \rightarrow N$.

 O_2 molecule

Each oxygen atom has $2 + 6 = 8$ electrons. Thus the O_2 molecule contains a total of 16 electrons. These are arranged in MOs:

$$\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_x^2, \begin{cases} \pi 2p_y^2 \\ \pi 2p_z^2 \end{cases}, \begin{cases} \pi^* 2p_y^1 \\ \pi^* 2p_z^1 \end{cases}$$

The anti-bonding $\pi^* 2p_y$ and $\pi^* 2p_z$ orbitals are singly occupied in accordance with Hund's rule. Unpaired electrons give rise to paramagnetism. Since there are two unpaired electrons with parallel spins, this explains why dioxygen is paramagnetic. If this treatment is compared with the Lewis electron pair theory or the valence bond theory, these do not predict unpaired electrons or paramagnetism.



This was the first success of the molecular orbital theory in successfully predicting the paramagnetism of O_2 , a fact not even thought of with a valence bond representation of $O = O$.

As in the previous examples, the inner shell does not participate in bonding and the bonding and anti-bonding 2s orbitals cancel each other. A σ bond results from the filling of $\sigma 2p_x^2$. Since $\pi^* 2p_y^1$ is half-filled and therefore cancels half the effect of the completely filled $\pi 2p_y^2$ orbital, half of a π bond results. Similarly, another half of a π bond arises from $\pi 2p_z^2$ and $2p_z^1$, giving a total of $1 + \frac{1}{2} + \frac{1}{2} = 2$ bonds. The bond order is thus two.

Instead of working out the bond order by cancelling the effects of filled bonding and anti-bonding MOs, the bond order may be calculated as half the difference between the number of bonding and anti-bonding electrons:

$$\text{Bond order} = \frac{(\text{number of electrons occupying bonding orbitals}) - (\text{number of electrons in anti-bonding orbitals})}{2}$$

In the case of O_2 , the bond order calculates as $(10 - 6)/2 = 2$, which corresponds to a double bond.

 O_2^- ion

The compound potassium superoxide KO_2 contains the superoxide ion O_2^- . The O_2^- ion has 17 electrons, and has one more electron than the O_2 molecule.

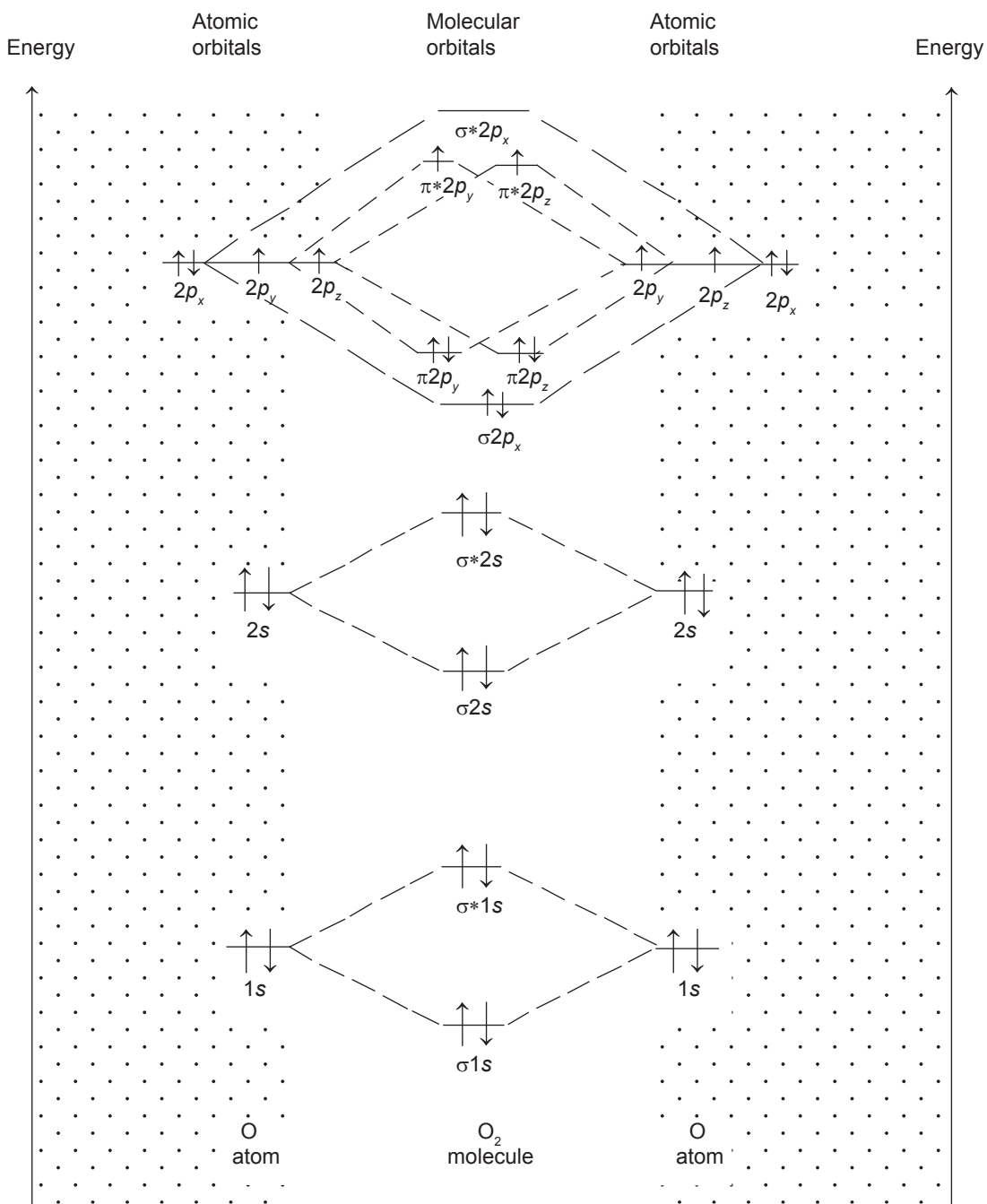


Fig. 1.18 Electronic configuration, atomic and molecular orbitals for oxygen

This extra electron occupies either the $\pi^* 2p_y$ or $\pi^* 2p_z$ orbital. It does not matter which it occupies since they are the same energy.

$$\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_x^2, \begin{cases} \pi 2p_y^2, \pi^* 2p_y^2 \\ \pi 2p_z^2, \pi^* 2p_z^2 \end{cases}$$

1.78 Periodic Properties and Chemical Bonding

The inner shell of electrons does not take part in bonding. The bonding $\sigma 2s^2$ and anti-bonding $\sigma^* 2s^2$ cancel. The $\sigma 2p^2$ orbital is filled and forms a σ bond. The effects of the bonding $\pi 2p_y^2$ and anti-bonding $\pi^* 2p_y^2$ orbitals cancel, and the completely filled bonding $\pi 2p_z^2$ is half cancelled by the half-filled anti-bonding $\pi^* 2p_z^1$, thus giving half a π bond. The order is thus $1 + \frac{1}{2} = \frac{1}{2}$. Alternatively, the bond order may be calculated like this: (bonding – anti-bonding)/2, that is $(10 - 7)/2 = 1\frac{1}{2}$. This corresponds to a bond that is intermediate in length between a single and a double bond. The superoxide ion has an unpaired electron is therefore paramagnetic. (A bond order of $1\frac{1}{2}$ is well accepted in benzene.)

O_2^{2-} ion

In a similar way, sodium peroxide Na_2O_2 contains the peroxide ion O_2^{2-} . This ion has 18 electrons, arranged:

$$\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_x^2, \begin{cases} \pi 2p_y^2, \\ \pi 2p_z^2, \end{cases} \begin{cases} \pi^* 2p_y^2 \\ \pi^* 2p_z^2 \end{cases}$$

Once again the inner shell takes no part in bonding. The bonding and anti-bonding 2s orbitals completely cancel each other. One σ bond forms from the filled $2p_x$ orbital. Both the bonding $2p_y$ and $2p_z$ orbitals are cancelled out by their corresponding anti-bonding orbitals. Thus the bond order is one, that is a single bond. Alternatively, the bond order may be calculated as (bonding – anti-bonding)/2, that is $(10 - 8)/2 = 1$.

F_2 molecules

Fluorine atoms have 2 + 7 electrons, so an F_2 molecule contains 18 electrons. These are arranged:

$$\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_x^2, \begin{cases} \pi 2p_y^2, \\ \pi 2p_z^2, \end{cases} \begin{cases} \pi^* 2p_y^2 \\ \pi^* 2p_z^2 \end{cases}$$

This is shown diagrammatically in figure.

The inner shell is non-bonding, and the filled bonding 2s, $2p_y$ and $2p_z$ are cancelled by the equivalent anti-bonding orbitals. This leaves a σ bond from the filled $\sigma 2p_x^2$ orbital, and thus a bond order of one. Alternatively the bond order may be calculated as (bonding – anti-bonding)/2, that is $(10 - 2)/2 = 1$.

It should be noted that Cl_2 and Br_2 have structure analogous to F_2 , except that additional inner shells of electrons are full.

The F–F bond is rather weak and this attributed to the small size of fluorine and repulsion between lone pairs of electrons on adjacent atoms.

Examples of molecular orbital treatment for heteronuclear diatomic molecules

The same principles apply when combining atomic orbitals from two different atoms as applied when the atoms were identical, that is,

1. Only atomic orbitals of about the same energy can combine effectively.
2. They should have the maximum overlap.
3. They must have the same symmetry.

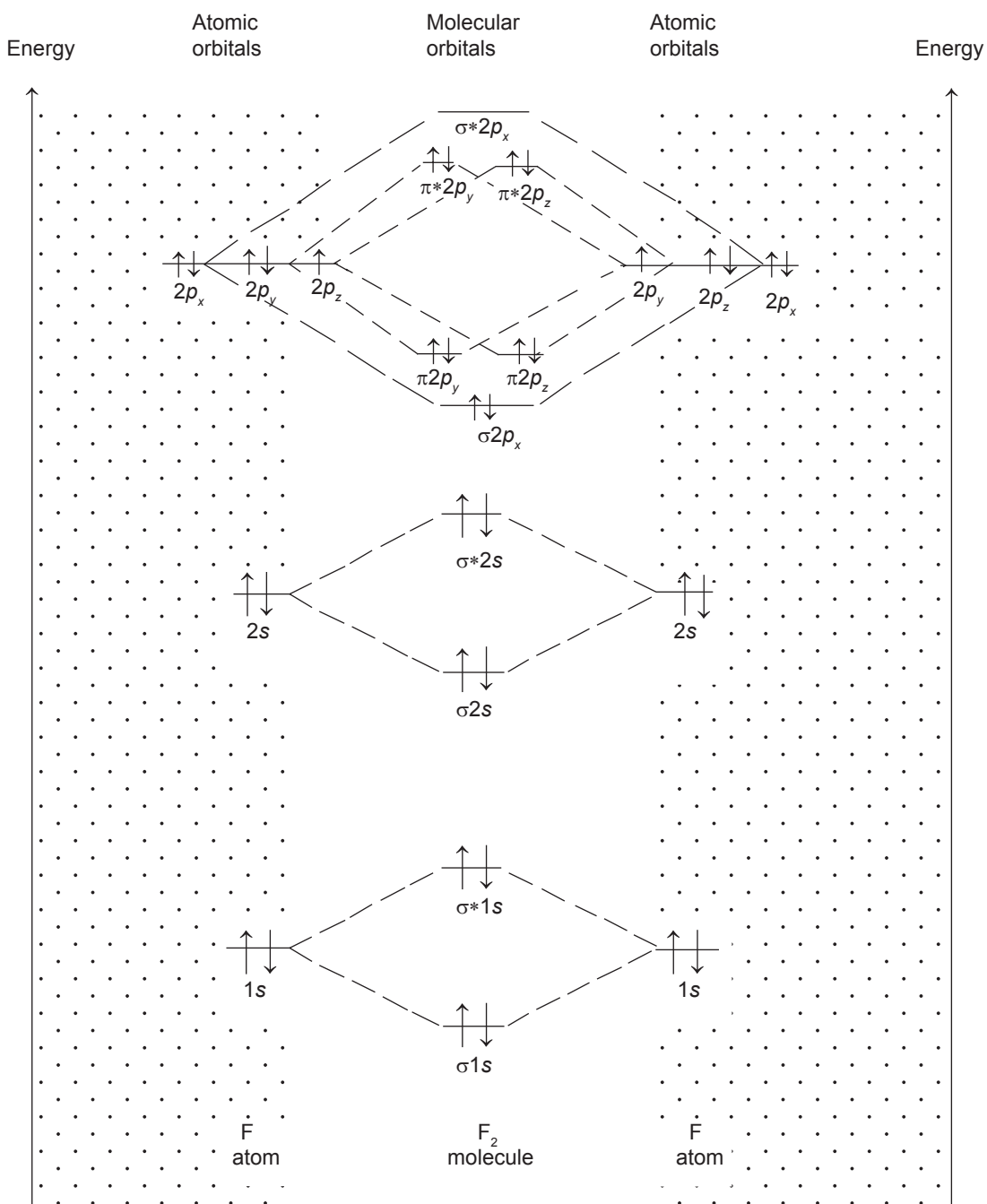


Fig. 1.19 Electronic configuration, atomic and molecular orbitals for fluorine

Since the two atoms are different, the energies of their atomic orbitals are slightly different. A diagram showing how they combine to molecular orbitals is given in figure

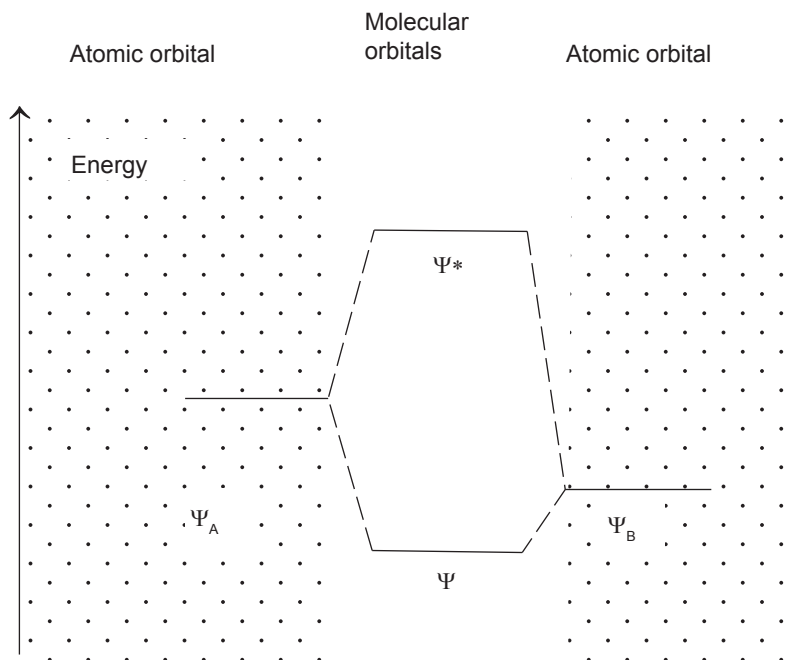


Fig. 1.20 The relative energy levels of atomic orbitals and molecular orbitals for a Heteronuclear diatomic molecules AB

The problem is that in many cases the order of MO energy levels is not known with certainty. Thus we will consider first some examples where the two different atoms are close to each other in the periodic table, and consequently it is reasonable to assume that the order of energies for the MOs are the same as for homonuclear molecules.

NO molecule

The nitrogen atom has $2 + 5 = 7$ electrons, and the oxygen atom has $2 + 6 = 8$ electrons, making 15 electrons in the molecule. The order of energy levels of the various MOs are the same as for homonuclear diatomic molecules heavier than C_2 , so the arrangement is:

$$\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_x^2, \begin{cases} \pi 2p_y^2, \\ \pi 2p_z^2 \end{cases}, \begin{cases} \pi^* 2p_y^1 \\ \pi^* 2p_z^0 \end{cases}$$

This is shown in figure.

The inner shell is non-bonding. The bonding and anti-bonding $2s$ orbitals cancel, and a σ bond is formed by the filled $\sigma 2p_x^2$ orbital. A π bond is formed by the filled $\pi 2p_z^2$ orbital. The half-filled $\pi^* 2p_y^1$ half cancels the filled $\pi 2p_y^2$ orbital, thus giving half a bond. The bond order is thus $2\frac{1}{2}$, that is in between a double and a triple bond. Alternatively the bond order may be worked out as (bonding-anti-bonding)/2, that is $(10 - 5)/2 = 2\frac{1}{2}$. The molecule is paramagnetic since it contains an unpaired electron. In NO there is a significant difference of about 250 kJ mol^{-1} in the energy of the AOs involved, so that combination of AOs to give MOs is less effective than in O_2 or N_2 . The bonds are therefore weaker than might be expected. Apart from this the molecular orbital pattern (figure).

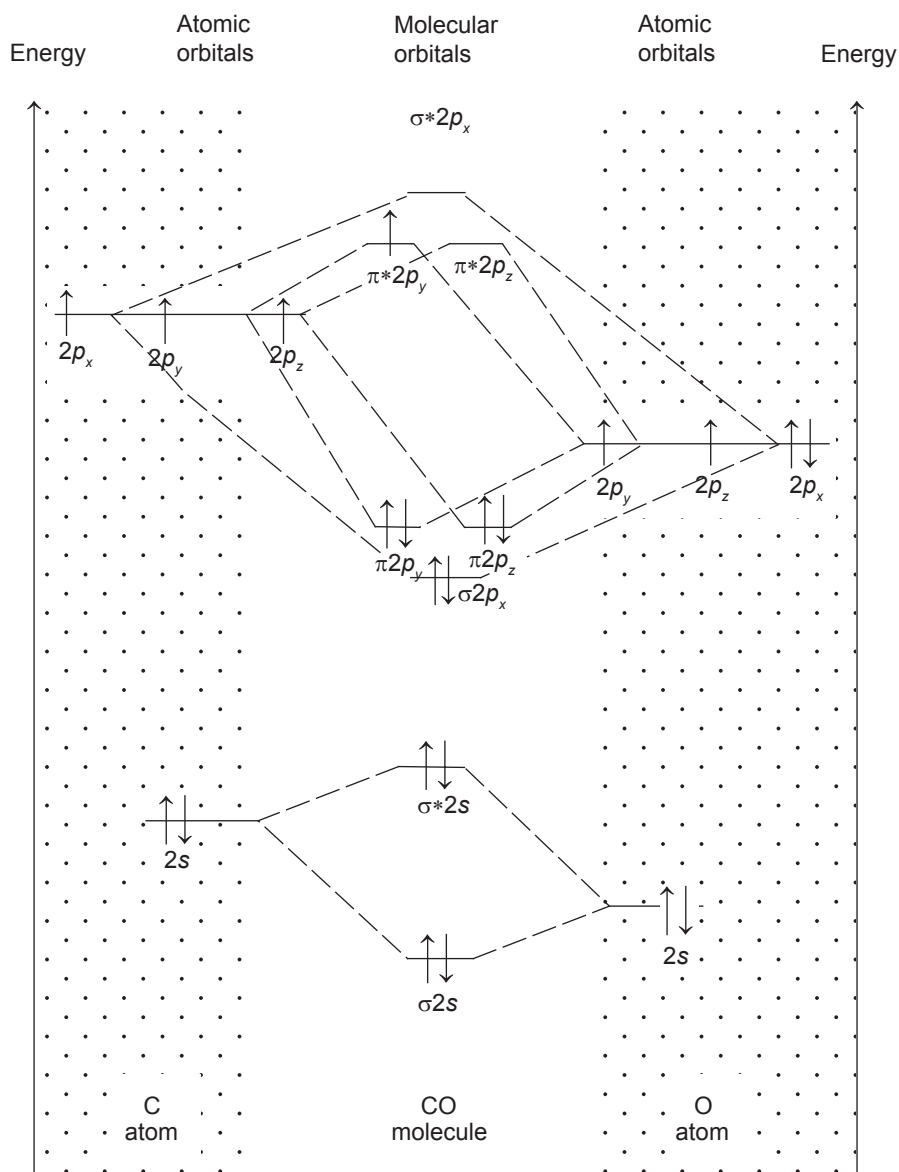


Fig. 1.21 Electronic configuration, atomic orbitals and molecular orbitals for nitric oxide. (This diagram is essentially the same as that for homonuclear diatomic molecules such as N_2 , O_2 or F_2 . The difference is that the atomic energy levels of N and O are not the same. The $\sigma 1s$ and $\sigma^* 1s$ MOs are omitted for simplicity.)

is similar to that for homonuclear diatomic molecules. Removal of one electron to make NO^+ results in a shorter and stronger bond because the electron is removed from an anti-bonding orbital, thus increasing the bond order to 3.

CO molecule

The carbon atom has $2 + 4 = 6$ electrons, and the O atom has $2 + 6 = 8$ electrons, so the CO molecule contains 14 electrons. In this case we are rather less certain of the order of energies of the MOs, since they are different for C and O. Assume the order is the same as for light atoms like C:

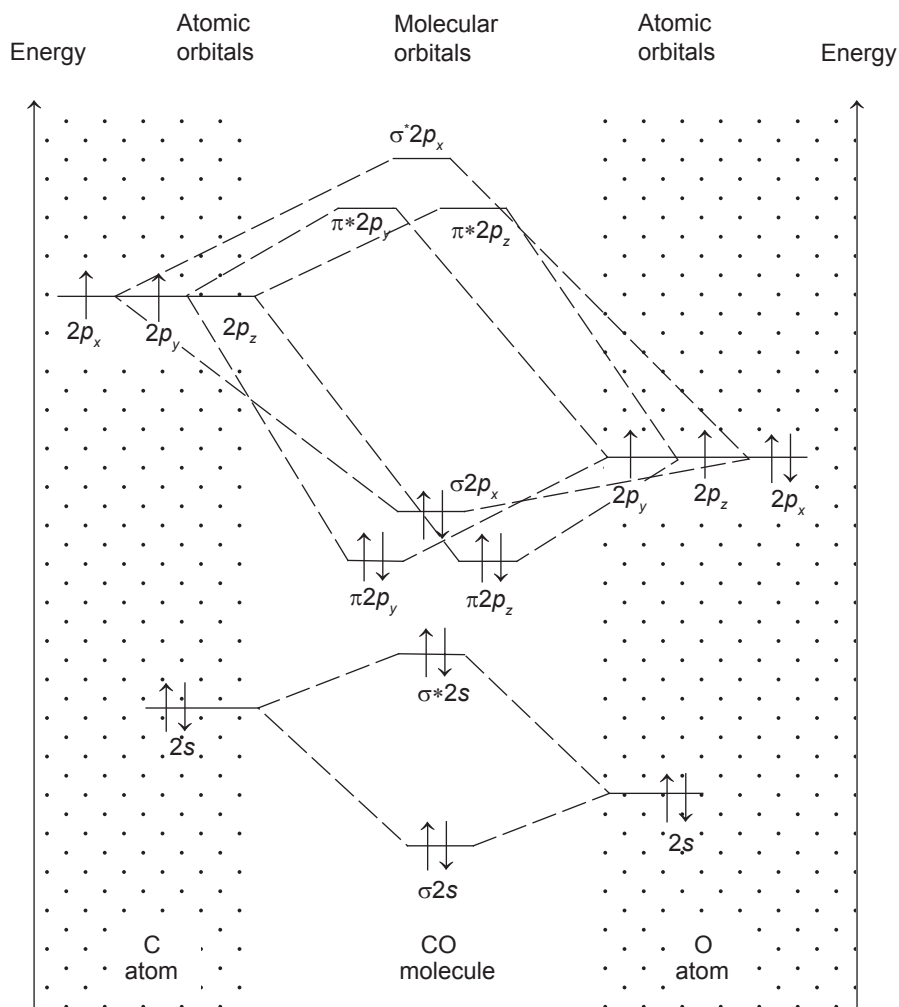


Fig. 1.22 Electronic configuration, atomic orbitals and molecular orbitals for carbon monoxide. (The $\sigma 1s$ and $\sigma^* 1s$ MOs are omitted for simplicity.)

The inner shell is non-bonding, and the bonding and anti-bonding $2s$ orbitals cancel, leaving one σ and two π bonds - and thus a bond order of 3. Alternatively the bond order may be calculated using the formula $(\text{bonding} - \text{anti-bonding})/2$, that is $(10 - 4)/2 = 3$. This simple picture is not adequate, since if CO is ionized to give CO^+ by removal of one electron from the $\sigma 2p_x$ orbital then the bond order should be reduced to $2\frac{1}{2}$ and the bond length increased. In fact the bond length in CO is $1.128 \text{ \AA}$ and in CO^+ it is $1.115 \text{ \AA}$. Thus the bond length decreases when we expected it to increase, and it indicates that the electron must have been removed from an anti-bonding orbital. The problem remains if we assume the order of energy for the MOs is the same as for atoms heavier than C, since this only reverses the position of the $\sigma 2p_x$ and the ($\pi 2p_y$ and $\pi 2p_z$) MOs. The most likely explanation of the bond shortening when CO is changed to CO^+ is that the $\sigma 2s$ and $\sigma^* 2s$ molecular orbitals differ in energy more than is shown in the figure. This means that they are wider apart, and the $\sigma 2s$ MO is higher in energy than the $\sigma 2p_x$, $\pi 2p_y$ and $\pi 2p_z$ MOs. This illustrates very plainly that the order of MO energy levels for simple homonuclear diatomic molecules used above is not automatically applicable when two different types of atoms are bonded together, and it is certainly incorrect in this particular heteronuclear case.

HCl molecule

With heteronuclear atoms, it is not obvious which AOs should be combined by the LCAO method to form MOs. In addition, because the energy levels of the AOs on the two atoms are not identical, some MOs will contain a bigger contribution from one AO than the other. This is equivalent to saying that the MO 'bulges' more towards one atom, or the electrons in the MO spend more time round one atom than the other. Thus some degree of charge separation δ^+ and δ^- occurs, resulting in a dipole. Thus partial ionic contributions may play a significant part in the bonding.

Consider the HCl molecule. Combination between the hydrogen 1s AO and the chlorine 1s, 2s, 2p and 3s orbitals can be ruled out because their energies are too low. If overlap occurred between the chlorine 3p_y and 3p_z orbitals it would be non-bonding because the positive lobe of hydrogen will overlap equally with the positive and negative lobes of the chlorine orbitals. Thus the only effective overlap is with the chlorine 3p_x orbital. The combination of H 1s¹ and Cl 3p_x¹ gives both bonding and anti-bonding orbitals, and the two electrons occupy the bonding MO, leaving the anti-bonding MO empty. It is assumed that all the chlorine AOs except 3p_x are localized on the chlorine atom and retain their original AO status, and the 3s, 3p_y and 3p_z orbitals are regarded as non-bonding lone pairs.

This over-simplification ignores any ionic contribution such as can be shown with the valence bond resonance structures



The former would be expected to contribute significantly, resulting in a stronger bond.

Worked-out Examples

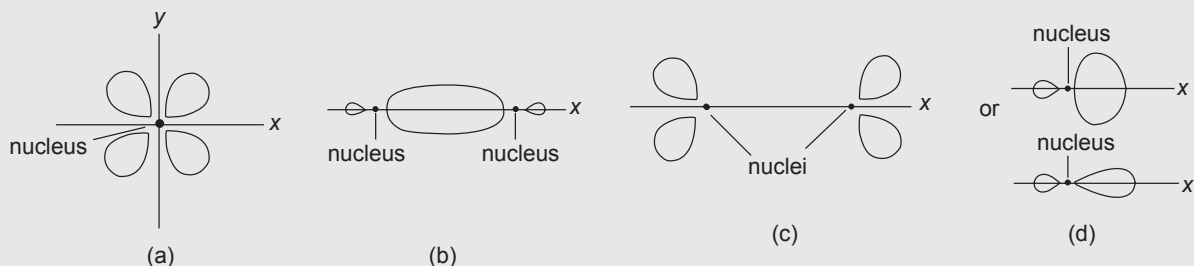
85. In which of the following pair, both the species have the same bond order?

- (a) $[\text{N}_2, \text{N}_2^+]$ (b) $[\text{O}_2, \text{F}_2]$ (c) $[\text{C}_2, \text{O}_2]$ (d) $[\text{C}_2, \text{N}_2]$

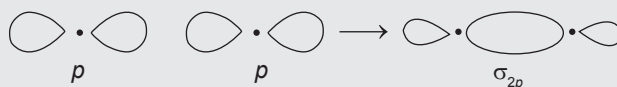
Solution: In the pair $(\text{N}_2, \text{N}_2^+)$, bond order of N_2 is 3 whereas N_2^+ has a bond order of 2.5. In (O_2, F_2) pair, bond order of O_2 and F_2 is 2 and 1 respectively whereas in the pair (C_2, N_2) , it is 2 and 3 respectively. Only in the pair (C_2, O_2) both the species have the same bond order of two hence the answer is (c).

86. Sketch the orbital configuration (boundary surface diagram) for the orbital types

- (a) $3d_{xy}$ (b) σ_{2p_x}
(c) $\sigma_{2p_x}^*$ (d) sp. Clearly label all axes and nuclei.



87. Draw a diagram showing the formation of σ_{2p} bonding orbital from the atomic orbitals.



88. (a) Draw and label an energy level diagram for molecular orbitals up to σ_{2p}^* for a molecule with 14 electrons.
(b) Use this diagram to give a molecular orbital description of the CO molecule,
(c) Briefly compare the CO molecule to the CN^- ion.

(b) OF is paramagnetic, since it has one unpaired electron. (With an odd number of electrons, it must be paramagnetic.)

(c) OF should be more stable, since it has one fewer anti-bonding electron.

91. Explain why NO^+ is more stable toward dissociation into its atoms than NO, whereas, CO^+ is less stable than CO?

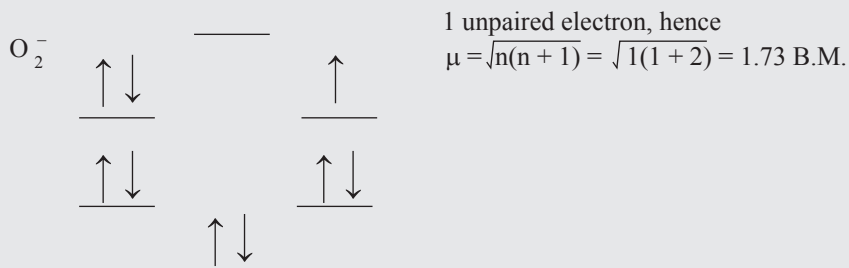
Ans. NO^+ has lost an anti-bonding electron; CO^+ has lost a bonding electron.

92. What would be the expected electronic arrangement and magnetic moment in

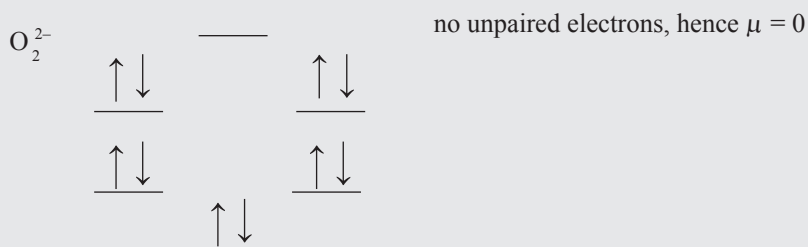
(a) the superoxide ion, O_2^- ?

(b) the peroxide ion, O_2^{2-} ?

Ans. (a)



(b)

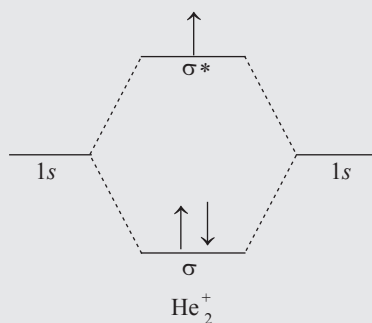


93. Has the peroxide ion, O_2^{2-} , a longer or shorter bond length than O_2 ? Explain.

Ans. O_2^{2-} has a bond order of 1; O_2 has a bond order of 2. Hence O_2^{2-} will have a longer and weaker bond.

94. Predict whether the He_2^+ ion in its electronic ground state is stable toward dissociation into He and He^+ .

Ans. It should be stable because it has one more bonding electron than anti-bonding.



95. Compare and contrast the concepts of hybrid orbitals and molecular orbitals with respect to (a) the number of atoms involved,
 (b) the number of orbitals produced from a given number of ground state orbitals, (c) the energies of the resulting orbitals with respect to one another.

<i>Hybrid</i>	<i>Molecular</i>
(a) One central atom	More than one atom
(b) Same number	Same number
(c) All the same	Bonding orbitals lower in energy than anti-bonding orbitals

96. State the bond order and indicate whether the species is paramagnetic:

(a) B_2 (b) C_2 (c) N_2 (d) O_2 (e) CN^- (f) Br_2 (g) Cl_2^+ (h) NO (i) NO^+ (j) CO

<i>Bond order</i>	<i>Magnetic behaviour</i>	<i>Bond order</i>	<i>Magnetic behaviour</i>
(a) 1	paramagnetic	(f) 1	diamagnetic
(b) 2	diamagnetic	(g) 1.5	paramagnetic
(c) 3	diamagnetic	(h) 2.5	paramagnetic
(d) 4	paramagnetic	(i) 3	diamagnetic
(e) 5	diamagnetic	(j) 3	diamagnetic

97. Which of the following molecules has the highest bond order?

(a) BN (b) CO (c) NO (d) Ne_2
 (e) F_2

Ans. (b) CO has a bond order of 3. [The others have bond orders of (a) 2 (c) 2.5 (d) 0 and (e) 1.]

98. Which of the following is paramagnetic?

(a) O_2^{2-} (b) BN

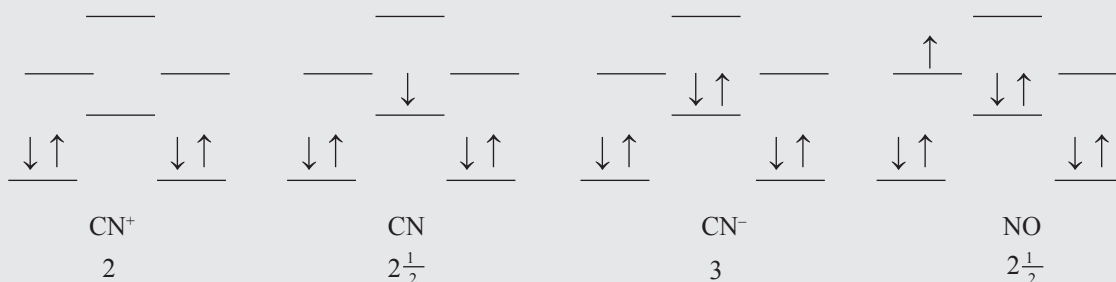
Ans. Neither (see Probs. 8.34 and 8.35)

99. Explain why N_2 has a greater dissociation energy than N_2^+ , whereas O_2 has a lower dissociation energy than O_2^+ .

Ans. N_2^+ has lost a bonding electron and so has been destabilized; O_2^+ has lost an anti-bonding electron and has been stabilized.

100. Deduce the bond order in each of the following: CN^+ , CN , CN^- , NO .

Ans.

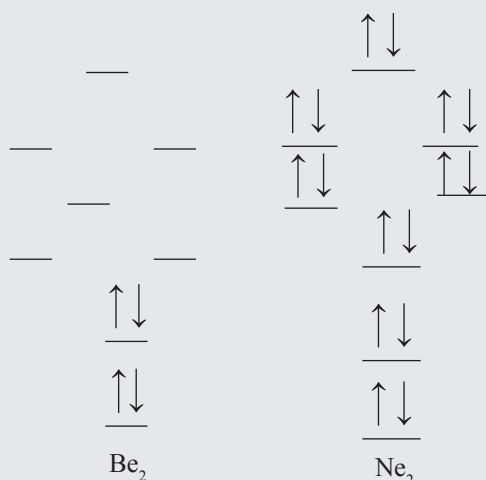


101. Make a table giving (i) number of orbitals with a given energy, (ii) maximum number of electrons per orbital, (iii) maximum number of electrons at a given energy for the following types of orbitals: (a) s (b) p (c) sp^2 (d) sp^3 (e) σ (f) σ^* (g) π^*

Ans.

	(i)	(ii)	(iii)		(i)	(ii)	(iii)
(a)	1	2	2	(e)	1	2	2
(b)	3	2	6	(f)	1	2	2
(c)	3	2	6	(g)	2	2	4
(d)	4	2	8				

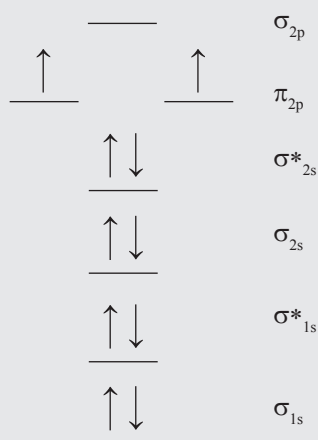
102. Which supposed homonuclear diatomic molecules of the second-period elements should have zero bond order?



Be₂ and Ne₂ would have as many antibonding as bonding electrons and thus would not be stable.

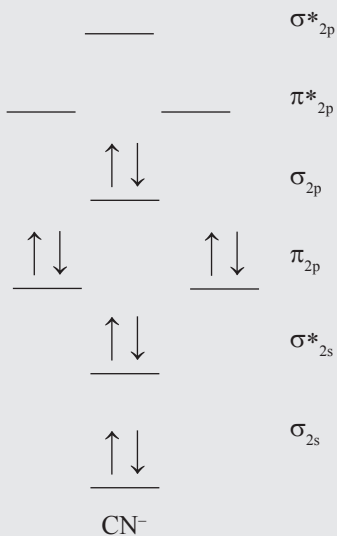
103. Which homonuclear diatomic molecule(s) of second-period elements, besides O₂, should be paramagnetic?

Ans.



B₂. Paramagnetism arises from unpaired electrons. Each diatomic molecule has an even number of electrons, and the only way any electrons can be unpaired is if they occupy π orbitals singly.

104. (a) What are the bond orders for CN^- , CN , and CN^+ ? (b) Which of these species should have the shortest bond length?



(a) CN^- has 4 bonding pairs and 1 antibonding pair, for a net bond order of 3. CN has one fewer bonding electron, for a bond order of $2\frac{1}{2}$, and CN^+ has still one fewer, for a bond order of 2.

(b) CN^- has the shortest bond because it has the highest bond order.

105. By the use of molecular orbital considerations, account for the fact that oxygen gas is paramagnetic. What is the bond order in O_2 ?

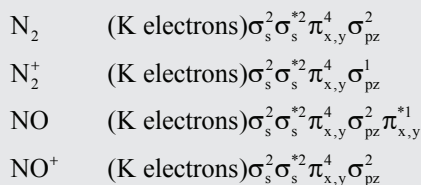
Ans. Each oxygen atom has the $1s^2 2s^2 2p^4$ configuration in the ground state. Aside from the K electrons of the two atoms in O_2 , which are so deeply imbedded in their respective atoms as not to overlap with other electrons, the remaining 12 electrons (6 from each of the atoms) will fill the lowest of the available molecular orbitals.

The last two electrons go into the anti-bonding equienergetic π^* orbitals, one into π_x^* and one into π_y^* , so as to maximize electron spin in accordance with Hund's rule. These two unpaired electrons confer paramagnetism upon the molecule.

$$\begin{aligned} \text{NBond order} &= (\text{number of electrons in bonding orbitals}) - (\text{number of electrons in anti bonding orbitals}) / 2 \\ &= 8 - 4 / 2 = 2 \end{aligned}$$

106. Explain the observations that the bond length in N_2^+ is $0.02 \text{ \AA}$ greater than in N_2 , while the bond length in NO^+ is $0.09 \text{ \AA}$ less than in NO .

Ans. The electron configurations should be written for the 4 molecules according to the buildup principle:



The computed bond orders for N_2 and N_2^+ are 3 and $2\frac{1}{2}$, respectively. N_2 therefore has the stronger bond and should have the shorter bond length. The computed bond orders for NO and NO^+ are 2.5 and 3, respectively. NO^+ has the stronger bond and should have the shorter bond length. As opposed to ionization of N_2 , which involves the loss of an electron in a bonding orbital, ionization of NO involves the loss of an electron in an anti-bonding orbital.

Some Noteworthy Points

- Valence bond theory fails to explain the paramagnetic behaviour of certain substances like O_2 and geometry of non-linear molecules.
- An atomic orbital is monocentric while a molecular orbital is polycentric.
- During the formation of M.Os. energy from A.Os. energy of the anti-bonding M.O. is raised above the energy of the combining A.Os. by an amount more than that by which the energy of the bonding M.O. has been lowered. The total energy of two M.Os., however, remains the same as that of the two original A.Os.
- In molecules such as B_2 , C_2 and N_2 the energy of σ_{2p_z} M.O. is higher than that of π_{2p_x} and π_{2p_y} M.Os. This is due to the mixing of 2s and 2p-orbitals in case B_2 , C_2 and N_2 .
- Through bond order for H_2^+ and H_2^- is same is same (i.e., 1/2), yet H_2 is slightly less stable than H_2^+ because H_2^- has one electron in the anti-bonding orbital resulting in repulsion (decreases in stability).
- Order of stability and bond dissociation energies are,

(a) $O_2^+ > O_2 > O_2^- > O_2^{2-}$

(b) $N_2 > N_2^+ = N_2^- > N_2^{2-}$
- Order of bond lengths

(a) $O_2^{2-} > O_2^- > O_2 > O_2^+$

(b) $N_2^{2-} > N_2^- = N_2^+ > N_2$

Worked-out Examples

107. Which of the following molecular species has unpaired electron(s)? [IIT 2002]

- (a) N_2 (b) F_2 (c) O_2^- (d) O_2^{2-}

Ans. The species O_2^- has one unpaired electron.

108. The species having bond order different from that in GO is [IIT 2007]

- (a) NO^- (b) NO^+ (c) CN^- (d) N_2

Ans. The diatomic species have the following scheme of molecular orbitals in the increasing order of energy.

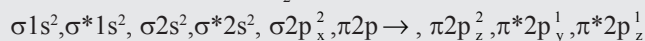


The species Co , NO^+ , CN^- and N_2 have the same number (=10) of valence electrons while NO^- has 12 electrons. Thus, NO^- will have different bond order.

109. Write the M.O. electron distribution of O_2 . Specify its bond order and magnetic property.

[IIT 2000]

Ans. M.O. configuration of O_2 is:



$$\text{Bond order} = \frac{1}{2} [10 - 6] = 2$$

Also O_2 is paramagnetic having two unpaired electrons.

110. Among the following compounds/species, write the order of O—O bond length in O_2 , $O_2[AsF_4]$, $K[O_2]$.

[IIT 2004]

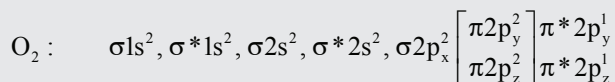
Ans. O_2



or



or



[AIEEE 2002]

111. The correct order of bond strength is:

- (a) $O_2^- < O_2 < O_2^+ < O_2^{2-}$ (b) $O_2^{2-} < O_2^- < O_2 < O_2^+$
 (c) $O_2^- < O_2^{2-} < O_2 < O_2^+$ (d) $O_2^+ < O_2 < O_2^- < O_2^{2-}$

Ans. (b) $O_2^+ (15) = KK(\sigma 2s)(\sigma^* 2s)^2(\sigma 2p_x)^2(\pi 2p_y)^2(\pi 2p_z)^2(\pi^* 2p_y)^1$

$$\text{Bond order} = \frac{1}{2}(8-3) = \frac{5}{2} = 2.5$$

$$O_2 (16) = KK(\sigma 2s)^2(\sigma^* 2s)^2(\sigma 2p)^2(\pi 2p)^2(\pi 2p_z)^2(\pi^* 2p_y)(\pi^* 2p_z)^1$$

$$\text{Bond order} = \frac{1}{2}(8-4) = 2$$

$$O_2^- (17) = KK(\sigma 2s)^2(\sigma^* 2s)^2(\sigma 2p_x)^2(\pi 2p_y)^2(\pi 2p_z)^2(\pi^* 2p_y)^2(\pi^* 2p_z)^1$$

$$\text{Bond order} = \frac{1}{2}(8-5) = 1.5$$

$$O_2^{2-} (18) = KK(\sigma 2s)^2(\sigma^* 2s)^2(\sigma 2p_x)^2(\pi 2p_y)^2(\pi^* 2p_y)^2(\pi^* 2p_z)^2$$

$$\text{Bond order} = \frac{1}{2}(8-6) = 1$$

[AIEEE 2007]

112. Which of the following species exhibits the diamagnetic behaviour?

- (a) O_2^{2-} (b) O_2^+ (c) O_2 (d) NO

Ans. (a) Diamagnetic species have n unpaired electrons

$$O_2^{2-} \rightarrow \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2, \pi 2p_x^2, \pi 2p_y^2, \pi^* 2p_x^2, \pi^* 2p_y^2$$

Whereas paramagnetic species has one or more unpaired electrons as in

$$O_2 \rightarrow \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2, \pi 2p_x^2, \pi 2p_y^2, \pi^* 2p_x^1, \pi^* 2p_y^1$$

$$\pi^* 2p_y^1 - 2 \text{ unpaired electrons}$$

$$O_2^+ \rightarrow \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2, \pi 2p_x^2, \pi 2p_y^2, \pi^* 2p_x^1, \pi^* 2p_y^1$$

$$\pi 2p_y^2, \pi^* 2p_x^1 - 1 \text{ unpaired electrons}$$

$$NO \rightarrow \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2, \pi 2p_x^2, \pi 2p_y^2, \pi^* 2p_x^1, \pi^* 2p_y^1$$

$$\pi 2p_y^2, \pi^* 2p_x^1 - 1 \text{ unpaired electrons}$$

[AIEEE 2007]

113. In which of the following ionization processes, the bond order has increased and the magnetic behaviour has changed:

- (a) $C_2 \longrightarrow C_2^+$ (b) $NO \longrightarrow NO^+$ (c) $O_2 \longrightarrow O_2^+$ (d) $N_2 \longrightarrow N_2^+$

Ans. (b) (a) N_2 : bond order 3, paramagnetic

$$N_2^-: \text{bond order, 2.5, paramagnetic}$$

(b) C_2 : bond order 2, diamagnetic

$$C_2^+: \text{bond order 1.5, paramagnetic}$$

(c) NO: bond order 2.5, paramagnetic

$$NO^+: \text{bond order 3, diamagnetic}$$

(d) O_2 : bond order 2, paramagnetic

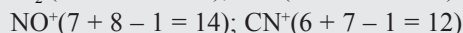
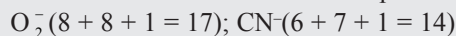
$$O_2^+: \text{bond order 2.5, paramagnetic}$$

[AIEEE 2008]

114. Which one of the following pairs of species have the same bond order?

- (a) CN^- and NO^+ (b) CN^- and CN^+ (c) O_2^- and CN^- (d) NO^+ and CN^+

Ans. (a) For any species to have same bond order we can expect them to have same number of electron. Calculating the number of electron in various species.



We find CN^- and NO^+ bond have 14 electrons so they have same bond order. Correct answer is (a).

115. Using MO theory predict which of the following species has the shortest bond length?

- (a) O_2^{2+} (b) O_2^+ (c) O_2^- (d) O_2^{2-}

Ans. (a) Bond length $\propto \frac{1}{\text{bond order}}$

$$\text{Bond order} = \frac{\text{no. of bonding } \bar{e} - \text{no. of antibonding } \bar{e}}{2}$$

Bond orders of O_2^+ , O_2^- , O_2^{2-} and O_2^{2+} are respectively

116. The bond order in NO is 2.5 while that in NO^+ is 3. Which statement is true?

[AIEEE 2004]

- (a) Bond length is unpredictable (b) Bond length in NO is greater than in NO^+
(c) Bond length in NO^+ is equal to that in NO (d) Bond length in NO^+ is greater than in NO

Ans. (b) Now since bond order of NO^+ given (3) is higher than that of NO (2.5). Thus bond length of NO^+ will be shorter.

117. Which one of the following species is diamagnetic in nature?

[AIEEE 2005]

- (a) He_2^+ (b) H_2 (c) H_2^+ (d) H_2^-

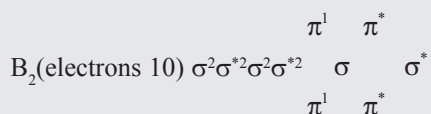
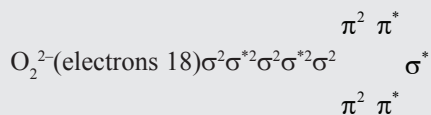
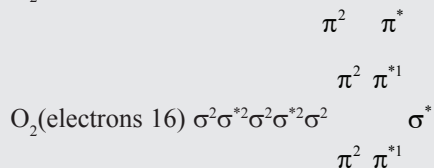
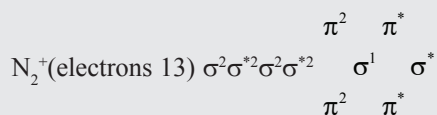
Ans. (b)

118. Which of the following molecules/ions does not contain unpaired electrons?

[AIEEE 2006]

- (a) O_2^{2-} (b) B_2 (c) N_2^+ (d) O_2

Ans. (a) The distribution of electrons in MOs is as follows:



As we know that as the bond order decreases, stability also decreases and hence the bond strength also decreases. Hence the correct order of their increasing bond strength is



ONE CORRECT OPTION TYPE QUESTIONS

PERIODIC PROPERTIES

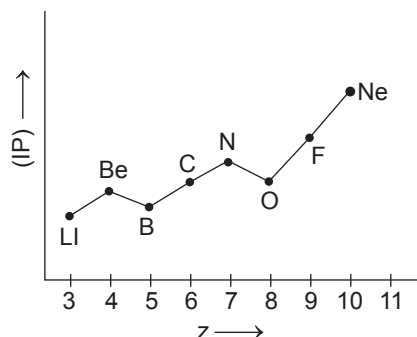
- According to modern periodic law, the properties of elements repeat at regular intervals when the elements are arranged in order of:
 - decreasing atomic number
 - increasing atomic weight
 - increasing atomic number
 - decreasing atomic weights
- In the fourth period of the periodic table, how many elements have one or more 4d electrons?
 - 2
 - 18
 - 0
 - 6
- The electronic configuration $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^{10}, 4s^2 4p^6 4d^{10}, 5s^2$ is for
 - f-block element
 - d-block element
 - p-block element
 - s-block element
- Give the symbol of the elements of lowest atomic number that has three 2p electrons:
 - Mg
 - P
 - N
 - Si
- Match list I (atomic number of elements) with list II (position in the Periodic Table) and select the correct answer using the codes given below the lists:

List I	List II
A. 52	1. s-block
B. 56	2. p-block
C. 57	3. d-block
D. 60	4. f-block

 - A 2 B 1 C 3 D 4
 - A 2 B 1 C 4 D 3
 - A 1 B 2 C 3 D 4
 - A 1 B 2 C 4 D 3
- Element Unq has atomic number of
 - 102
 - 103
 - 104
 - 112
- Element with electronic configuration $1s^2, 2s^2, 2p^6, 3s^2, 3p^6 3d^{10}, 4s^2 4p^6 4d^{10}, 5s^2 5p^3$ belongs to the group of the periodic table.
 - 13
 - 15
 - 17
 - 5
- If atomic number of an element is 17 it will be placed in the periodic table in:
 - group 1
 - group 5
 - group 8
 - group 17
- Which of the following sequence represents atomic number of only representative elements?
 - 55, 12, 48, 53
 - 13, 33, 54, 83
 - 3, 33, 53, 87
 - 22, 33, 55, 66
- Following triads have approximately equal size
 - $\text{Na}^+, \text{Mg}^{2+}, \text{Al}^{3+}$ (iso-electronic)
 - $\text{F}^-, \text{Ne}, \text{O}^{2-}$ (iso-electronic)
 - Fe, Co, Ni
 - $\text{Mn}^+, \text{Fe}^{2+}, \text{Cr}$ (iso-electronic)
- Which is correct order of size of $\text{O}^-, \text{O}^{2-}, \text{F}^-$ and F?
 - $\text{O}^{2-} > \text{O}^- > \text{F}^- > \text{F}$
 - $\text{O}^- > \text{O}^{2-} > \text{F}^- > \text{F}$
 - $\text{O}^{2-} > \text{F}^- > \text{F} > \text{O}^-$
 - $\text{O}^{2-} > \text{F}^- > \text{O}^- > \text{F}$
- Size of cation is smaller than that of the atom because of
 - the whole of the outer shell of electrons is removed
 - effective nuclear charge increases
 - due to gain of electrons
 - statement, that cation is smaller than atom, is wrong
- The ionic radii of $\text{O}^{2-}, \text{F}^-, \text{Na}^+$ and Mg^{2+} are 1.35, 1.34, 0.95 and $0.66\text{\AA}$ respectively. The radius of the Ne atom is
 - $1.39\text{\AA}$
 - $1.12\text{\AA}$
 - $0.85\text{\AA}$
 - $0.50\text{\AA}$
- Atomic radii of F and Ne in $\text{\AA}$ are given by
 - 0.72, 1.6
 - 1.6, 1.6
 - 0.72, 0.72
 - None of these
- If the ionic radii of K^+ and F^- are nearly the same (i.e., $1.34\text{\AA}$), then the atomic radii of K and F respectively are:
 - $1.34\text{\AA}, 1.34\text{\AA}$
 - $0.72\text{\AA}, 1.96\text{\AA}$
 - $1.96\text{\AA}, 0.72\text{\AA}$
 - $1.96\text{\AA}, 1.34\text{\AA}$
- Consider the isoelectronic ions: $\text{K}^+, \text{S}^{2-}, \text{Cl}^-$ and Ca^{2+}
The radii of these ionic species follow the order
 - $\text{Ca}^{2+} > \text{K}^+ > \text{Cl}^- > \text{S}^{2-}$
 - $\text{Cl}^- > \text{S}^{2-} > \text{K}^+ > \text{Ca}^{2+}$

- (c) $S^{2-} > Cl^- > K^+ > Ca^{2+}$
 (d) $K^+ > Ca^{2+} > S^{2-} > Cl^-$

17. First, second and third IP values are 100 eV, 150 eV and 1500 eV. Element can be
 (a) Be (b) B
 (c) F (d) Na
18. Following graph shows variation of ionization potential (IP) with atomic number in second period (Li — Ne). Value of ionization potential (IP) of Na(11) will be



- (a) above Ne
 (b) below Ne but above O
 (c) below Li
 (d) between N and O
19. The first ionization potential (in eV) of N, O atoms are
 (a) 14.6, 13.6 (b) 13.6, 14.6
 (c) 13.6, 13.6 (d) 14.6, 14.6
20. The first ionization potential of Al is smaller than that of Mg because
 (a) the atomic size of Al > Mg
 (b) the atomic size of Al < Mg
 (c) Al has one unpaired electron in p-orbital
 (d) the atomic number of Al > Mg
21. The highest first ionization energy of the following elements is that of
 (a) Cs (b) Cl
 (c) I (d) Li
22. The IP_1 , IP_2 , IP_3 , IP_4 and IP_5 of an element are 7.1, 14.3, 34.5, 46.8, 162.2 eV respectively. The element is likely to be
 (a) Na (b) Si
 (c) F (d) Ca
23. Which of the following isoelectronic ion has the lowest ionization energy?
 (a) K^+ (b) Cl^-
 (c) Ca^{2+} (d) S^{2-}
24. Which of the following statements is incorrect?
 (a) The second ionization energy of sulphur is greater than that of chlorine.
 (b) The third ionization energy of aluminium is greater than that of phosphorous.
 (c) The first ionization energy of aluminium is approximately the same as that of gallium.
 (d) The second ionization energy of boron is greater than that of carbon.
25. The third ionization energy is maximum for:
 (a) Nitrogen (b) Phosphorus
 (c) Aluminium (d) Boron
26. A molecule $H - \times$ will be 50% ionic if electronegativity difference of H and $\times$ is
 (a) 1.2 eV (b) 1.4 eV
 (c) 1.5 eV (d) 1.7 eV
27. The electronegativities of N, C, Si and P are such that
 (a) $P < Si < C < N$ (b) $Si < P < N < C$
 (c) $Si < P < C < N$ (d) $P < Si < N < C$
28. Which element has the highest electron gain enthalpy?
 (a) Br (b) Cl
 (c) I (d) F
29. Which of the following has zero electron gain enthalpy?
 (a) Radon (b) Nitrogen
 (c) Oxygen (d) Radium
30. The amount of energy released on the addition of an electron in outermost shell of an atom is called
 (a) lattice energy
 (b) hydration energy
 (c) electron gain enthalpy
 (d) electron negativity
31. The electron affinities of N, O, S and Cl are such that
 (a) $N < O < S < Cl$ (b) $O < N < Cl < S$
 (c) $O \approx Cl < N \approx S$ (d) $O < S < Cl < N$
32. Which of the following will have maximum electron affinity
 (a) $1s^2 2s^2 2p^5$ (b) $1s^2 2s^2 2p^6$
 (c) $1s^2 2s^2 2p^6 3s^2 3p^5$ (d) $1s^2 2s^2 2p^6 3s^2 3p^6$
33. Following are the values of the electron affinities of the formation of O^- and O^{2-} from O
 (a) -142, -702 (b) -142, 702
 (c) 142, 702 (d) -142, -142

34. EN of the element (A) is E_1 and IP is E_2 . Hence EA will be

- (a) $2E_1 - E_2$ (b) $E_1 - E_2$
(c) $E_1 - 2E_2$ (d) $(E_1 + E_2)/2$

35. The most negative electron of the following elements is that of

- (a) Br (b) Sn
(c) Ba (d) Li

36. Which one is incorrect statement?

- (a) IE_1 of He is maximum among all elements
(b) EA_1 for noble gases is zero
(c) Electronegativity is maximum for fluorine
(d) IE_1 for N < IE_1 for O

COMPREHENSION BASED QUESTIONS

COMPREHENSION-I

Nuclear charge actually experienced by an electron is termed as effective nuclear charge. The effective nuclear charge Z^* actually depends on type of shell and orbital in which electron is actually present. The relative extent to which the various orbitals penetrate the electron clouds of other orbitals is.

$$s > p > d > f \text{ (for the same value of } n\text{)}$$

The phenomenon in which penultimate shell electrons act as screen on shield in between nucleus and valence shell electrons and thereby reducing nuclear charge is known as shielding effect. The penultimate shell electrons repel the valence shell electron to keep them loosely held with nucleus. It is thus evident that more is the shielding effect, lesser is the effective nuclear charge and lesser is the ionization energy.

1. Which of the following valence electron experience maximum effective nuclear charge?

- (a) $4s^1$ (b) $4p^1$
(c) $3d^1$ (d) $2p^3$

2. Which of the following is not concerned to effective nuclear charge?

- (a) Higher ionization potential of carbon than boron
(b) Higher ionization potential of magnesium than aluminium
(c) Higher values of successive ionization energy
(d) Higher electronegativity of higher oxidation state

3. Ionization energy is not influenced by:

- (a) Size of atom
(b) Effective nuclear charge
(c) Electrons present in inner shell
(d) Change in entropy

COMPREHENSION-II

Ionization energies of five elements in kcal/mol are given below:

Atom	I	II	III
P	300	549	920
Q	99	734	1100
R	118	1091	1652
S	176	347	1848
T	497	947	1500

1. Which element is a noble gas?

- (a) P (b) T
(c) R (d) S

2. Which element form stable unipositive ion?

- (a) P (b) Q
(c) R (d) S

3. The element having most stable oxidation state +2 is?

- (a) Q (b) R
(c) S (d) P

4. Which is a non-metal (excluding noble gas)?

- (a) P (b) Q
(c) R (d) S

5. If Q reacts with fluorine and oxygen, the molecular formula of fluoride and oxide will be respectively:

- (a) QF_3, Q_2O_3
(b) QF, Q_2O
(c) QF_2, QO
(d) None of these

6. Which of the following pair represents elements of same group?

- (a) Q, R (b) P, Q
(c) P, S (d) Q, S

COMPREHENSION-III

Elements with their electronic configurations are given below:

Answer the following questions:

- I. $1s^2 2s^2$ II. $s^2 2s^2 2p^6$
 III. $1s^2 2s^2 2p^6 3s^2$ IV. $1s^2 2s^2 2p^3$
 V. $1s^2 2s^2 2p^5$

1. The element with highest I.e., is:

- (a) I (b) III
 (c) II (d) V

2. The element with lowest electron gain enthalpy is:

- (a) I (b) II
 (c) III (d) IV

3. The most ionic compound will be formed between:

- (a) I and IV (b) I and V
 (c) III and IV (d) III and V

4. Which of the following is the correct order of increasing size?

- (a) $I < III < IV < V$ (b) $V < IV < III < I$
 (c) $I < IV < V < III$ (d) $V < IV < I < III$

ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

1. Transition metals are characterised by the proprieties

- (a) variable valency
 (b) coloured compounds
 (c) high melting and boiling points
 (d) tendency to form complex

2. The factors that influence the ionization energies are

- (a) the size of the atom
 (b) the charge on the nucleus
 (c) how effectively the inner electron shell screen the nuclear charge
 (d) the type of electrons involved (s, p, d, f)

3. Which is/are correct statements:

- (a) I_1 of Be $>$ I_1 of B but I_2 of Be $<$ I_2 of B
 (b) I_1 of Be $<$ I_1 of B but I_2 of Be $<$ I_2 of B
 (c) I_2 of $\text{Be}^+ <$ I_1 of B^+
 (d) I_1 of Be^{2+} is abnormally high

4. Select equations having endothermic step:

- (a) $\text{S}^-(\text{g}) \rightarrow \text{S}^{2-}(\text{g})$
 (b) $\text{Na}^+(\text{g}) + \text{Cl}^-(\text{g}) \rightarrow \text{NaCl}(\text{s})$
 (c) $\text{N}(\text{g}) \rightarrow \text{N}^-(\text{g})$
 (d) $\text{Al}^{2+}(\text{g}) \rightarrow \text{Al}^{3+}(\text{g})$

5. Which of the following statements concerning elements with atomic number 10 is

- (a) It forms a covalent network solid
 (b) Element is monoatomic
 (c) It has a almost zero value of electron affinity
 (d) It has extremely high value of ionization energy

6. A change of Zn to Zn^{2+} is accompanied by a decrease in:

- (a) number of valence electrons
 (b) atomic mass

(c) atomic number

(d) number of shells

7. Which of the following statement regarding halogens is/are correct?

- (a) Ionization energy decreases with increase in atomic number.
 (b) Electronegativity decreases with increase in atomic number.
 (c) Electron affinity decreases with increase in atomic number.
 (d) Electron negativity increases with increase in atomic number.

8. Ionization energy of an element is:

- (a) equal in magnitude but opposite in sign to the electron gain enthalpy of the cation of the element.
 (b) same as electron affinity of the element.
 (c) energy required to remove one valence electron from an isolated gaseous atom in its ground state.
 (d) equal in magnitude but opposite in sign to the electron gain enthalpy of the anion of the element.

9. Which of the following statements are correct?

- (a) F is the most electronegative and Cs is the most electropositive element
 (b) The ionization energy of halogens decreases from F to I
 (c) The electron affinity of Cl is higher than that of F through their electronegativities are in the reverse order
 (d) The electron affinity of noble gases is almost zero

ASSERTION/REASON TYPE QUESTIONS

Each question contains Assertion and Reason. Examine the statements carefully and mark the correct answer according to the instructions given below:

- (a) If both the statement are TRUE and **Reason** is the correct explanation of **Assertion**
- (b) If both the statements are TRUE but **Reason** is NOT the correct explanation of **Assertion**
- (c) If **Assertion** is TRUE and **Reason** is FALSE
- (d) If **Assertion** is FALSE and **Reason** is TRUE

1. **Assertion:** F atom has less electron affinity than Cl atom.

Reason: Additional electrons are repelled more strongly by 3p electrons in Cl atom than by 2p electrons in F atom.

2. **Assertion:** Among the halogens bond energy of F_2 is minimum.

Reason: Among halogens F atom is small in size.

3. **Assertion:** The first ionization energy of Be is greater than that of B.

Reason: 2p-orbital is lower in energy than 2s-orbital.

4. **Assertion:** Noble gases have highest ionization enthalpies in their respective periods.

Reason: Noble gases have stable closed shell electronic configuration.

5. **Assertion:** Helium and beryllium have similar outer electronic configuration of the type ns^2 .

Reason: Both are chemically inert.

MATCH THE COLUMN TYPE QUESTIONS

Entries of Column-I are to be matched with entries of Column-II. Each entry of Column-I may have the matching with one or more than one entries of Column-II.

1.

Column-I (atomic number of element)	Column-II (IUPAC name)
(A) 105	(P) Uun
(B) 107	(Q) Uns
(C) 109	(R) Unp
(D) 110	(S) Une

2.

Column-I (atomic number)	Column-II (Position in the periodic table)
(A) 52	(P) s-block
(B) 56	(Q) p-block
(C) 57	(R) d-block
(D) 60	(S) f-block

3.

Column-I (Type of elements)	Column-II (Outer electronic configuration)
(A) Inert gas elements	(P) ns^{1-2} to ns^2np^5
(B) Representative elements	(Q) $1s^2$ and ns^2np^6
(C) Transition elements	(R) $(n-2)f^{1-14}(n-1)d^1$ or $2ns^2$
(D) Inner transition elements	(S) $(n-1)d^{1-10}ns^1$ or 2

4.

Column-I (elements)	Column-II (periodic properties)
(A) F	(P) Maximum ionization energy
(B) Cl	(Q) Maximum electronegativity
(C) Fe	(R) Maximum electronegativity
(D) He	(S) Variable valency

RECENT YEARS' QUESTIONS OF IIT-JEE AND AIEEE

- According to the Periodic law of elements, the variation in properties of elements is related to their (AIEEE 2003)
 - Nuclear masses
 - Atomic numbers
 - Nuclear neutron-proton number ratio
 - Atomic masses
- The reduction in atomic size with increase in atomic number is a characteristic of elements of
 - d-block
 - f-block
 - radioactive series
 - high atomic masses.
 (AIEEE 2003)
- Which of the following groupings represent a collection of isoelectronic species? (At. no. Cs = 55, Br = 35) (AIEEE 2003)
 - N^{3-} , F^- , Na^+
 - Be, Al^{3+} , Cl^-
 - Ca^{2+} , Cs^+ , Br
 - Na^+ , Ca^{2+} , Mg^{2+}
- The atomic numbers of vanadium (V), chromium (Cr), manganese (Mn) and iron (Fe) are respectively 23, 24, 25 and 26. Which one of these may be expected to have the highest second ionization enthalpy? (AIEEE 2003)
 - Cr
 - Mn
 - Fe
 - V.
- Which one of the following sets of ions represents the collection of isoelectronic species? (AIEEE 2004)
 - K^+ , Cl^- , Mg^{2+} , Sc^{3+}
 - Na^+ , Ca^{2+} , Sc^{3+} , F^-
 - K^+ , Ca^{2+} , Sc^{3+} , Cl^-
 - Na^+ , Mg^{2+} , Al^{3+} , Cl^-
- Which of the following ions has the highest value of ionic radius? (AIEEE 2004)
 - O^{2-}
 - V B^{3+}
 - Li^+
 - F^-
- The formation of the oxide ion $\text{O}^{2-}(\text{g})$ requires first an exothermic and then an endothermic step as shown below:

$$\text{O}(\text{g}) + \text{e}^- = \text{O}^-(\text{g}), \Delta H^\circ = -142 \text{ kJ mol}^{-1}$$

$$\text{O}^-(\text{g}) + \text{e}^- = \text{O}^{2-}(\text{g}), \Delta H^\circ = 844 \text{ kJ mol}^{-1}$$
 This is because
 - O^- ion will tend to resist the addition of another electron
 - Oxygen has high electron affinity
 - Oxygen is more electronegative
 - O^- ion has comparatively larger size than oxygen atom.
 (AIEEE 2004)
- In which of the following arrangements the order is NOT according to the property indicated against it?
 - $\text{Al}^{3+} < \text{Mg}^{2+} < \text{Na}^{2+} < \text{F}^-$ – increasing ionic size
 - $\text{B} < \text{C} < \text{N} < \text{O}$ – increasing first ionization enthalpy
 - $\text{I} < \text{Br} < \text{F} < \text{Cl}$ – increasing electron gain enthalpy (with negative sign)
 - $\text{Li} < \text{Na} < \text{K} < \text{Rb}$ – increasing metallic radius.
 (AIEEE 2005)
- The increasing order of the first ionization enthalpies of the elements B, P, S and F (lowest first) is (AIEEE 2006)
 - $\text{F} < \text{S} < \text{P} < \text{B}$
 - $\text{P} < \text{S} < \text{B} < \text{F}$
 - $\text{B} < \text{P} < \text{S} < \text{F}$
 - $\text{B} < \text{S} < \text{P} < \text{F}$
 - $\text{N} < \text{Be} < \text{B}$
 - $\text{F}^- < \text{O}^{2-} < \text{N}^{3-}$
 - $\text{Na} < \text{Li} < \text{K}$
 - $\text{Fe}^{3+} < \text{Fe}^{2+} < \text{Fe}^{4+}$
- The set representing the correct order of first ionization potential is: (2001 qualifying)
 - $\text{K} > \text{Na} > \text{Li}$
 - $\text{Be} > \text{Mg} > \text{Ca}$
 - $\text{B} > \text{C} > \text{N}$
 - $\text{Ge} > \text{Si} > \text{C}$

ANSWER KEYS

ONE CORRECT OPTION TYPE QUESTIONS

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (c) | 2. (c) | 3. (b) | 4. (c) | 5. (a) | 6. (c) | 7. (b) | 8. (d) | 9. (c) | 10. (c) |
| 11. (a) | 12. (b) | 13. (a) | 14. (a) | 15. (c) | 16. (c) | 17. (a) | 18. (c) | 19. (a) | 20. (c) |
| 21. (d) | 22. (b) | 23. (d) | 34. (b) | 25. (a) | 26. (d) | 27. (c) | 28. (b) | 29. (a) | 30. (c) |
| 31. (a) | 32. (c) | 33. (b) | 34. (a) | 35. (a) | 36. (b) | | | | |

COMPREHENSION BASED QUESTIONS

Comprehension-I

1. (d) 2. (b) 3. (d)

Comprehension-II

1. (b) 2. (b) 3. (c) 4. (a) 5. (b) 6. (a)

Comprehension-III

1. (c) 2. (b) 3. (d) 4. (d)

ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

1. (a, b, c, d) 2. (a, b, c, d) 3. (a, c, d) 4. (a, c, d) 5. (b, c, d) 6. (a, d) 7. (a, b) 8. (a, c)
9. (a, b, c, d)

ASSERTION/REASON TYPE QUESTIONS

1. (c) 2. (d) 3. (c) 4. (a) 5. (c)

MATCH THE COLUMN TYPE QUESTIONS

- | | | | | | | | |
|------------------------|---------------------|---------------------|-------------------|------------------------|---------------------|---------------------|-------------------|
| 1. $A \rightarrow R$; | $B \rightarrow Q$; | $C \rightarrow S$; | $D \rightarrow P$ | 2. $A \rightarrow Q$; | $B \rightarrow P$; | $C \rightarrow R$; | $D \rightarrow S$ |
| 3. $A \rightarrow Q$; | $B \rightarrow P$; | $C \rightarrow S$; | $D \rightarrow R$ | 4. $A \rightarrow Q$; | $B \rightarrow R$; | $C \rightarrow S$; | $D \rightarrow P$ |

RECENT YEARS' QUESTIONS OF IIT-JEE AND AIEEE

1. (b) 2. (b) 3. (a) 4. (a) 5. (c) 6. (a) 7. (a) 8. (d)
9. (b) 10. (b)

ONE CORRECT OPTION TYPE QUESTIONS

CHEMICAL BONDING

- Favourable conditions for electrovalency are:
 - Low charge on ions, large cation small anion
 - High charge on ions, small cation, large anion
 - High charge on ions, large cation small anion
 - Low charge on ions, small cation, large anion
- The electronic configurations of four elements L, P, Q and R are given in brackets
 L ($1s^2, 2s^2 2p^4$) Q ($1s^2, 2s^2 2p^6, 3s^2 3p^5$)
 P ($1s^2; 2s^2 2p^6, 3s^1$) R ($1s^2, 2s^2 2p^6, 3s^2$)
 The formulae of ionic compounds that can be formed between these elements are:
 - L_2P , RL, PQ and R_2Q
 - LP, RL, PQ and RQ
 - P_2L , RL, PQ and RQ_2
 - LP, R_2L , P_2Q and RQ
- Lattice energy (numerical value) of chloride of alkali metals is in order
 - $LiCl > NaCl > KCl > RbCl > CsCl$
 - $LiCl < NaCl < KCl < RbCl < CsCl$
 - $NaCl < KCl < LiCl < RbCl < CsCl$
 - $NaCl < KCl < RbCl < CsCl < LiCl$
- Which is maximum hydrated
 - NaCl
 - $MgCl_2$
 - $AlCl_3$
 - $BeCl_2$
- Least mobile ion is
 - $[Be(H_2O)_n]^{+2}$
 - $[Na(H_2O)_n]^+$
 - $[Mg(H_2O)_n]^{+2}$
 - $[Li(H_2O)_n]^+$
- The lesser solubility of silver halides in comparison to alkali metal halides is due to
 - low lattice energies of silver halides as compared to alkali metal halides.
 - lesser ionic character of silver halides because of greater polarization of Ag^+ ion.
 - tendency of Ag^+ ion to form complexes.
 - small size of Ag^+ ion.
- The solubility of LiF in water is less than that of KF because
 - Li^+ ions are not so heavily hydrated as K^+ ions
 - LiF is mainly covalent
 - LiF has larger lattice energy than KF
 - LiF has larger enthalpy of solution than KF

- Out of the following, which one is highly soluble in water
 - BeF_2
 - BaF_2
 - CaF_2
 - SrF_2
- Which of the following is least soluble?
 - BaF_2
 - SiF_2
 - CaF_2
 - MgF_2
- Solubility of NaCl, Na_2SO_4 and Na_3PO_4 in water in increasing order is:
 - $NaCl < Na_2SO_4 < Na_3PO_4$
 - $Na_3PO_4 < Na_2SO_4 < NaCl$
 - $NaCl < Na_3PO_4 < Na_2SO_4$
 - $Na_2SO_4 < NaCl < Na_3PO_4$
- Among LiCl, $BeCl_2$, BCl_3 , CCl_4 , the covalent character follows the order
 - $LiCl > BeCl_2 > BCl_3 > CCl_4$
 - $LiCl < BeCl_2 < BCl_3 < CCl_4$
 - $LiCl > BeCl_2 < BCl_3 < CCl_4$
 - $LiCl < BeCl_2 > BCl_3 > CCl_4$
- The correct order of decreasing polarisability of the ions is
 - F^-, I^-, Br^-, Cl^-
 - I^-, Br^-, Cl^-, F^-
 - Cl^-, Br^-, I^-, F^-
 - F^-, Cl^-, Br^-, I^-
- Of the following hydrides which one has the lowest boiling point?
 - AsH_3
 - SbH_3
 - PH_3
 - NH_3
- Which of the following is most ionic?
 - LiF
 - CsCl
 - CsF
 - LiCl
- Maximum polarizing power is of:
 - Li^+
 - Be^{++}
 - Na^+
 - K^+
- Which is the most covalent?
 - $C-O^{2-}$
 - $C-Br^-$
 - $C-S^{2-}$
 - $C-F^-$
- Which of the following is least ionic?
 - KCl
 - AgCl
 - $BaCl_2$
 - $CoCl_2$
- Amongst LiCl, RbCl, $BeCl_2$ and $MgCl_2$, the compounds with the greatest and the least ionic character respectively are:

- (a) LiCl, RbCl (b) RbCl and BeCl₂
 (c) RbCl, MgCl₂ (d) MgCl₂ and BeCl₂

19. Which one is the highest melting halide?

- (a) NaCl (b) NaBr
 (c) NaF (d) NaI

20. Which ion has a higher polarising power

- (a) Mg²⁺ (b) Al³⁺
 (c) Ca²⁺ (d) Na⁺

21. Which of the following bond is the most ionic?

- (a) Cs—Cl (b) Al—Cl
 (c) C—Cl (d) H—Cl

22. The following compounds are to be arranged in order of their increasing thermal stabilities. Identify the correct order:

K₂CO₃ (I), MgCO₃ (II), CaCO₃ (III), BeCO₃ (IV)

- (a) I < II < III < IV
 (b) IV < II < III < I
 (c) IV < II < I < III
 (d) II < IV < III < I

23. Which contains both ionic and covalent bonds?

- (a) CH₄ (b) NH₄Cl
 (c) CaCl₂ (d) H₂O

24. Covalent compounds have low m.p. because

- (a) Covalent molecules have definite shape
 (b) Covalent bond is weaker than ionic bond
 (c) Covalent bond is less exothermic
 (d) Covalent molecules are held by weak Van der Waal's forces of attraction

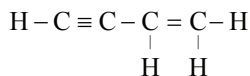
25. Carbon suboxide (C₃O₂) has recently been shown as a component of the atmosphere of Venus. Which of the following formulation represents the correct ground state Lewis structure for carbon suboxide?

- (a) :O:C::C:C:O:
 (b) :O::C:C::O:
 (c) :Ö::C::C::C::Ö
 (d) :O:C:C:C:O:.

26. The number of electron dots in Lewis structure indicates:

- (a) number of valence electrons in an atom
 (b) number of protons in the nucleus of the atom
 (c) total number of electrons in an atom
 (d) total number of electrons and protons in an atom

27. The number of σ (sigma) and π (pi) bonds in 1-buten-3-yne are:



- (a) 5σ and 5π (b) 7σ and 3π
 (c) 8σ and 2π (d) 6σ and 4π

28. Chlorine molecule is formed by the overlap of

- (a) s - s orbitals
 (b) s - p orbitals
 (c) p - p orbitals, end to end
 (d) p - p orbitals sideways

29. Ethyne molecule contains H - C ≡ C - H

- (a) 5 σ bonds
 (b) 4 σ bonds and 1 π bond
 (c) 3 σ bonds and 2 π bonds
 (d) 2 σ bonds and 3 π bonds

30. In which of the molecules does the central metal ion not employ sp³ hybrid orbital in bonding?

- (a) H₂O (b) NH₃
 (c) CO₂ (d) H₂S

31. In which of the molecules can the central atom be made to adopt sp² hybridization?

- (a) NH₃ (b) C₂H₂
 (c) BF₃ (d) BeF₂

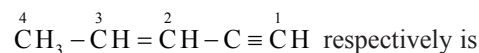
32. In which of the following the central atom does not use sp³ hybrid orbital:

- (a) BeF₃⁻ (b) OH₃⁺
 (c) NH₂⁻ (d) NF₃

33. Which of the following has a bond formed by the overlap of sp-sp³ hybrid orbitals?

- (a) CH₃—C≡C—H
 (b) CH₃—CH=CH—CH₃
 (c) CH₃=CH—CH=CH₂
 (d) CH≡CH

34. Hybridization of C-atoms numbered 1, 2, 3, 4 in



- (a) sp³, sp², sp², sp (b) sp, sp², sp², sp³
 (c) sp², sp², sp, sp³ (d) sp, sp², sp³, sp²

35. On hybridization of one s and p orbitals we get

- (a) Two mutually perpendicular orbitals
 (b) Two orbitals at 180°
 (c) Four orbitals directed tetrahedrally
 (d) Three orbitals in a plane

36. Hybridization in SO₂ is

- (a) sp (b) sp²
 (c) sp³ (d) sp³d

37. XeF_2 involves hybridization
 (a) sp^3 (b) sp^3d
 (c) sp^3d^2 (d) None of these
38. Hybridization involves
 (a) the mixing up of atomic orbitals centred on different atoms
 (b) the mixing up of atomic orbitals centred on the same atom
 (c) removal of an electron pair
 (d) addition of an electron pair
39. The hybridization of N in NO_2^+ , NO_3^- and NH_4^+ respectively are
 (a) sp , sp^2 , sp^3 (b) sp , sp^3 , sp^2
 (c) sp^2 , sp , sp^3 (d) sp^3 , sp^2 , sp
40. **Assertion:** Li^+ ion is more solvated than Na^+ in aqueous medium.
Reason: Solvation energy $\propto \frac{1}{\text{ionic radius}}$
 (a) When both assertion and reason are correct with respect to each other.
 (b) When assertion is correct but reason is wrong.
 (c) When both assertion and reason are wrong.
 (d) When both assertion and reason are correct but reason is not the explanation for assertion
41. Which of the following has tetrahedral shape?
 (a) NH_4^+ (b) NCl_3
 (c) BF_3 (d) H_2O
42. The molecule which has pyramidal shape is
 (a) PCl_3 (b) SO_3
 (c) CO_3^{2-} (d) NO_3^-
43. XeF_2 molecule is:
 (a) Linear (b) Triangular planar
 (c) Pyramidal (d) Square planar
44. Geometry of XeF_6 is:
 (a) Planar (b) Distorted octahedral
 (c) Octahedral (d) Plane triangular
45. Which of the following molecules has distorted geometry?
 (a) CH_4 (b) SF_6
 (c) NH_3 (d) BF_3
46. A molecule has two lone pairs and two σ bond pairs around the central atom. The molecule shape is expected to be
 (a) V-shaped (b) Triangular
 (c) Linear (d) Tetrahedral
47. The geometry of ICl_2^- is
 (a) Square planar
 (b) Trigonal
 (c) Trigonal bipyramidal
 (d) Octahedral
48. Which of the following molecules have trigonal planar geometry?
 (a) BF_3 (b) NH_3
 (c) PCl_3 (d) IF_3
49. The geometry of SF_2Cl_2 molecules is;
 (a) trigonal bipyramidal
 (b) square planar
 (c) sea-saw geometry
 (d) tetrahedral.
50. The linear structure is assumed by
 (i) SnCl_2 (ii) NCO^-
 (iii) NO_2^+ (iv) CS_2
 (a) (i), (ii) and (iii) (b) (ii), (iii) and (iv)
 (c) (i), (iii) and (iv) (d) None
51. The angle between covalent bonds is maximum in
 (a) CH_4 (b) BF_3
 (c) PF_3 (d) NH_3
52. Which has got the smallest bond angle?
 (a) NH_3 (b) AsH_3
 (c) PH_3 (d) SbH_3
53. Bond angle is minimum for
 (a) H_2O (b) H_2S
 (c) H_2Se (d) H_2Te
54. According to the VSEPR theory, the arrangement of lone pairs of an atom containing a total of three such pairs is
 (a) linear (b) triangular planar
 (c) tetrahedron (d) octahedron
55. Which of the following has/have maximum bond angle?
 (a) BeF_2 (b) H_2O
 (c) XeF_2 (d) NH_3 .
56. **Assertion:** The bond angle of H_2O is greater than H_2S .
Reason: These exist lesser bp-bp repulsion in H_2S
 (a) When both assertion and reason are correct with respect to each other.
 (b) When assertion is correct but reason is wrong.
 (c) When both assertion and reason are wrong.
 (d) When both assertion and reason are correct but reason is not the explanation for assertion

57. H_2O has a net dipole moment while BeF_2 has zero dipole moment because
 (a) 'F' is more electronegative than oxygen
 (b) Be is more electronegative than oxygen
 (c) H_2O molecule is linear and BeF_2 is bent
 (d) BeF_2 molecule is linear and H_2O is bent
58. Which of the following has maximum dipole moment.
 (a) NH_3 (b) NF_3
 (c) NCl_3 (d) NI_3
59. A molecule AB_3 exhibits a dipole moment = 0, then the hybrid state of A ($Z < 21$) should be
 (a) sp^2 (b) sp
 (c) sp^3 (d) sp^3d
60. Compound having a dipole moment is
 (a) CO_2 (b) CCl_4
 (c) C_6H_6 (d) H_2O
61. Pick up the molecule which has zero dipole moment
 (a) NH_3 (b) H_2O
 (c) BF_3 (d) SO_2
62. Which of the following has zero dipole moment?
 (a) ClF (b) PCl_3
 (c) SiF_4 (d) CFCl_3
63. If the electronegativity difference between the two atoms is 2.0, the percentage covalent character of the molecule is:
 (a) 46% (b) 72%
 (c) 54% (d) 10%
64. The dipole moment of HBr is 0.78×10^{-18} esu-cm. and the inter ionic spacing is $1.41 \text{ \AA}$. The percentage ionic character of HBr is:
 (a) 7.5 (b) 11.7
 (c) 15.0 (d) 27.0
65. % ionic character in a covalently bonded molecule =

$$\frac{\mu_{\text{obs}} - \mu_{\text{cal}}}{\mu_{\text{cal}}} \times 100$$

 (a) μ_{obs} (b) $\mu_{\text{obs}} - \mu_{\text{cal}}$
 (c) $\mu_{\text{cal}} - \mu_{\text{obs}}$ (d) $\mu_{\text{obs}} + \mu_{\text{cal}}$
66. The dipole moment of HCl is 1.03D. If H-Cl bond distance is $1.26 \text{ \AA}$, what is the percentage of ionic character in the H-Cl bond?
 (a) 60% (b) 39%
 (c) 29% (d) 17%
67. H-bonding is exhibited by
 (a) All the molecules containing H-atoms
 (b) Molecules in which H is covalently bonded to F, O, or N
 (c) Molecules in which two H atoms are present
 (d) Molecules in which H is bonded to atoms with electronegativity greater than 2.1.
68. The boiling point is not affected due to hydrogen bonding in
 (a) water (b) ammonia
 (c) methyl alcohol (d) hydrogen chloride
69. The correct order of the strength of H-bonds is
 (a) $\text{H} \cdots \text{F} > \text{H} \cdots \text{O} > \text{H} \cdots \text{N}$
 (b) $\text{H} \cdots \text{N} > \text{H} \cdots \text{O} > \text{H} \cdots \text{F}$
 (c) $\text{H} \cdots \text{O} > \text{H} \cdots \text{N} > \text{H} \cdots \text{F}$
 (d) $\text{H} \cdots \text{F} > \text{H} \cdots \text{N} > \text{H} \cdots \text{O}$
70. In ice, the length of H-bonds
 (a) is less than that of covalent bonds
 (b) is greater than that of covalent bonds
 (c) is same as that of covalent bonds
 (d) can be less, greater or same as that of covalent bonds
71. The crystal lattice of ice is mostly formed by
 (a) ionic forces
 (b) covalent bonds
 (c) intramolecular H-Bonds
 (d) covalent as well as H-bonds
72. The boiling point of water (H_2O) is 100°C whereas that of hydrogen sulphide (H_2S) is -42°C . This can be attributed to
 (a) larger bond angle in H_2O than in H_2S
 (b) smaller size of oxygen atom as compared to sulphur
 (c) larger ionization energy of oxygen than sulphur
 (d) larger tendency of H_2O to form hydrogen bonds than H_2S .
73. Which of the following compounds has the least tendency to form hydrogen bonds between molecules?
 (a) NH_3 (b) NH_2OH
 (c) HF (d) CH_3F .
74. The weakest bond among the following is
 (a) ionic (b) covalent
 (c) metallic (d) hydrogen bond
75. Incorrect order of decreasing boiling points is
 (a) $\text{HF} > \text{HI} > \text{HBr} > \text{HCl}$
 (b) $\text{H}_2\text{O} > \text{H}_2\text{Te} > \text{H}_2\text{Se} > \text{H}_2\text{S}$
 (c) $\text{Br}_2 > \text{Cl}_2 > \text{F}_2$
 (d) $\text{CH}_4 > \text{GeH}_4 > \text{SiH}_4$

76. The no. of unpaired electrons in O_2^{-1} are
 (a) 1 (b) 2
 (c) 3 (d) zero
77. Which of the following statements on peroxide ion is false?
 (a) bond order is one
 (b) has no unpaired electron
 (c) each oxygen has an oxidation of -1
 (d) stability is higher than O_2
78. On the basis of molecular orbital theory, the formation of which is not possible
 (a) H_2 (b) H_2^+
 (c) He_2^+ (d) He_2
79. The molecule having one unpaired electron is:
 (a) NO (b) CO
 (c) CN^- (d) O_2
80. Which is paramagnetic and has bond order $1/2$:
 (a) O_2 (b) N_2
 (c) F_2 (d) H_2^+
81. What is the correct sequence of bond order?
 (a) $O_2^+ > O_2^- > O_2$ (b) $O_2 > O_2^- > O_2^+$
 (c) $O_2^+ > O_2 > O_2^-$ (d) $O_2^- > O_2^+ > O_2$
82. Which of the following is the correct order of Bond strength
 (a) $N_2^+ > N_2 > N_2^-$ (b) $N_2^- > N_2 > N_2^+$
 (c) $N_2 > N_2^+ > N_2^-$ (d) $N_2^- > N_2^+ > N_2$
83. Which of the following species is paramagnetic?
 (a) CN^- (b) NO
 (c) O_2^{2-} (d) CO
84. Which of the following is paramagnetic?
 (a) H_2O (b) K_2O
 (c) Na_2O_2 (d) KO_2
85. Which of the following has minimum internuclear distance?
 (a) O_2 (b) O_2^+
 (c) O_2^- (d) O_2^{2-}
86. The number of anti-bonding electron pairs in the O_2^{2-} molecular ion on the basis of M.O. theory is
 (a) 2 (b) 3
 (c) 4 (d) 5
87. Which out of the following is not paramagnetic?
 (a) N_2^+ (b) CO
 (c) O_2^- (d) NO
88. According to the molecular orbital theory, which of the following statements about magnetic character and bond order is correct regarding O_2^{+} ?
 (a) Paramagnetic and bond order $< O_2$
 (b) Paramagnetic and bond order $> O_2$
 (c) Diamagnetic and bond order $< O_2$
 (d) Diamagnetic and bond order $> O_2$
89. Which of the following has least bond energy.
 (a) H_2 (b) Mg_2
 (c) F_2 (d) O_2^{-2}
90. **Assertion:** B_2 is paramagnetic in behaviour.
Reason: Because it contains 2 unpaired electrons.
 (a) When both assertion and reason are correct with respect to each other.
 (b) When assertion is correct but reason is wrong.
 (c) When both assertion and reason are wrong.
 (d) When both assertion and reason are correct but reason is not the explanation for assertion
91. Bond order of C_2 is
 (a) 2 (b) 3
 (c) 1 (d) 0

COMPREHENSION BASED QUESTIONS

COMPREHENSION-I

The rule governing the transition from ionic to covalent bonding, one called Fajan's rule. By this rule, the degree of covalency of a molecules are known, which one based on deformation of interacting ions in the bond. When an anion and cation each approach each other, then the electron cloud of anion is not only attracted by the nucleus, but also by the charge on cation.

At the same time, the cation also tends to repel, the positively charged nucleus of anions. The combined

effect of these two forces is that the electron symmetrical, but is elongated towards the cation. This is called distortion of the anion by the cation. This ability of cation to polarise anion is known polarisation of power of cation (ϕ).

$$\phi = \frac{\text{cationic charge}}{\text{cationic radius}}$$

The ability of anion to get deformed is known as polarisability of anion.

As we know that larger the anion, smaller the cation with larger cationic charge, larger will be extent of deformation and higher will be the covalent character. Effective nuclear charge of pseudo noble gas configuration is higher than that of noble gas configuration.

- Which of the following is the increasing order of covalent character of the compound?
 - $\text{KCl} < \text{NaCl} < \text{MgO} < \text{AlN}$
 - $\text{AlN} < \text{MgO} < \text{NaCl} < \text{KCl}$
 - $\text{MgO} < \text{AlN} < \text{KCl} < \text{NaCl}$
 - $\text{KCl} > \text{MgO} > \text{NaCl} > \text{AlN}$.
- Which of the Following compound is most covalent in character?
 - LiI
 - LiF
 - NaF
 - NaI .
- Which of the following anion has the highest polarisability?
 - I^-
 - Br^-
 - Cl^-
 - F^-

COMPREHENSION-II

The concept of hybridization has been introduced to explain the shapes of molecules. It involves the intermixing of two or more atomic orbitals belonging to same atom but in different sub-shells so as to intermix and redistribute energies to form equivalent orbitals called hybrid orbitals. Depending upon the number and nature of the orbitals involved the hybridization may be divided into sp (linear), sp^2 (trigonal), sp^3 (tetrahedral), sp^3d (trigonal bipyramidal), sp^3d^2 (octahedral) and sp^3d^3 (pentagonal bipyramidal) types. It may be noted that the orbitals of isolated atoms never hybridise and they do so at the time of bond formation.

- A hybrid orbital from s - and p -orbitals can contribute to
 - σ - bond only
 - π -bond only
 - either σ - or π -bond
 - cannot be predicted.
- Which carbon is maximum electronegative?
 - sp^3 – hybridized carbon
 - sp – hybridized carbon
 - sp^2 – hybridized carbon
 - the electron attracting power of carbon is always the same irrespective of its hybrid state.
- The hybridization of phosphorus in POCl_3 is the same as:
 - P in PCl_3
 - S in SF_4
 - Cl in ClF_3
 - B in BCl_3

COMPREHENSION -III

Gillespie and Nyholm proposed this theory which is based on the interactions of the electron pairs in the valence shell of the atoms. It explains the shape or the real geometry of the molecules having localized electron pairs, bonded or non-bonded.

The VSEPR's are regarded as occupying the localized orbitals oriented in space in such a way so that the Columbic repulsion be minimum between the electron pairs leading to stable configuration.

According to this theory, order of repulsion in between lp and lp , lp and bp , bp and bp is as under $lp - lp > lp - bp > bp - bp$. $lp - lp$ repulsion chosen the bond angle and $bp - bp$ repulsion opens the bond angle. Further, more electronegative substituents occupy less angular space, than that of less electronegative substituents and the lone pairs are assigned positions where $lp - lp$ repulsions are a minimum.

NF_3 and NH_3 both have tetrahedral geometry with one of the apices of the tetrahedron occupied by a lone-pair of electrons. The high electronegativity of F pulls the electron cloud towards itself in NF_3 while the electron cloud drifts toward N from H in NH_3 due to which repulsion between bond pairs is lesser in NF_3 than in NH_3 . Hence the lone pair in NF_3 causes a greater distortion from tetrahedral geometry giving $\text{F}-\text{N}-\text{F}$ bond angle of $102^\circ 30'$ compared to $107^\circ 48'$ and NH_3 .

- The expected bond angle of H_2O and F_2O respectively are
 - 102° and $104^\circ 27'$
 - $104^\circ 27'$ and 102°
 - 120° and $104^\circ 27'$
 - $104^\circ 27'$ and 120°
- Which among the following statements is true?
 - The dipole moments of NF_3 is greater than NH_3
 - The dipole moment of NF_3 is lesser than NH_3
 - The dipole moment of OF_2 is greater than H_2O
 - The dipole moment of OF_2 is equal to that of H_2O .
- The shape of XeO_2F_2 is expected to be
 - trigonal bipyramidal
 - trigonal pyramidal
 - sea-saw-shape
 - tetrahedral.
- The shape of ClO_2^- is expected to be
 - angular
 - tetrahedral
 - trigonal planar
 - square planar.

5. The shape of ClF_3 is T-shaped because
- the three bond pairs have maximum repulsion when placed at 120° angle
 - the lone pairs over Cl have greater repulsion with bond pairs when placed at 90° angle
 - the two lone pairs of electrons have minimum repulsion when placed at 120° angle
 - none of these.

COMPREHENSION-IV

L.C.A.O. principal is involved in the formation of the molecular orbitals according to molecular orbital theory. The energy of the bonding molecular orbital is less than that of the combining atomic orbitals while that of the anti-bonding molecular orbital is more. The bond order (B.O.) = $\frac{1}{2} (N_b - N_a)$ helps in predicting (i) formation of molecules/ molecular ions, bond dissociation energy, stability and bond length. Only the molecules or ions with positive B.O. can be formed. These will be diamagnetic if all molecular orbitals are filled and paramagnetic if one or more are half-filled. The atomic orbitals at the time of overlap must have the same symmetry as well.

- Which of the following combinations is not allowed. (assume z-axis as the internuclear axis)?
 - 2s and 2s
 - 2px and 2px
 - 2s and 2pz
 - 2px and 2py
- Bond order is:
 - directly related to bond length
 - inversely related to bond length
 - inversely related to bond strength
 - never fractional
- In the formation of N_2^+ from N_2 , the electron is removed from a
 - σ -orbital
 - π -orbital
 - σ^* -orbital
 - π^* -orbital
- The bond order (B.O.) in B_2 molecule is:
 - 0
 - 1
 - 2
 - 3
- In the homonuclear molecule which of the following sets of M.O. orbitals are degenerate?
 - σ_{1s} and σ_{1s}^*
 - π_{2px} and π_{2py}
 - π_{2px} and σ_{2pz}
 - σ_{2pz} and π_{2px}^*

ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

- Which of the following species is paramagnetic?
 - CN^-
 - NO
 - O_2^{2-}
 - O_2
- Ionic compounds in general do not possess:
 - high melting points and non-directional bonds
 - high melting points and low-boiling points
 - directional bonds and low-boiling points
 - high solubilities in polar and non-polar solvents
- Which of the following statements is incorrect?
 - A σ bond is weaker than a π bond.
 - There are four co-ordinate bonds in the NH_4^+ ions
 - The covalent bond is directional in nature
 - HF is less polar than HCl
- Correct order of decreasing boiling points is:
 - $\text{HF} > \text{HI} > \text{HBr} > \text{HCl}$
 - $\text{H}_2\text{O} > \text{H}_2\text{Te} > \text{H}_2\text{Se} > \text{H}_2\text{S}$
 - $\text{Br}_2 > \text{Cl}_2 > \text{F}_2$
 - $\text{CH}_4 > \text{GeH}_4 > \text{SiH}_4$
- Which of the following statements is/are correct?
 - All carbon to carbon bonds contain a sigma bond and one or more pi bonds
 - All carbon to hydrogen bonds are pi bonds
 - All oxygen to hydrogen bonds are hydrogen bonds
 - All carbon to hydrogen bonds are sigma bonds
- Non-polar molecules are:
 - CH_4
 - C_2F_2
 - C_2F_4
 - OF_2
- In which of the following there is intermolecular hydrogen bonding?
 - Water
 - Ethanol
 - Acetic acid
 - $\text{H}-\text{F}$
- In which species the hybrid state of central atom is sp^3d :
 - I_3^+
 - SF_4
 - PF_5
 - IF_5

9. Which of the following statement is incorrect?

- (a) O_2 is paramagnetic, O_3 is also paramagnetic
- (b) O_2 is paramagnetic, O_3 is diamagnetic
- (c) B_2 is paramagnetic, C_2 is also paramagnetic
- (d) Different observation is found in their bond length when $NO \rightarrow NO^+$ and $CO \rightarrow CO^+$

10. Which of the following is true for N_2O ?

- (a) Its molecule is linear
- (b) Symmetric $N-O-N$ is a favoured structure as compared to $N-N-O$ skeleton
- (c) Bond orders are fractional for $N-N$ and $N-O$ bonds
- (d) It is a neutral oxide

ASSERTION/REASON TYPE QUESTIONS

Each question contains Assertion and Reason. Examine the statements carefully and mark the correct answer according to the instructions given below:

- (a) If both the statement are TRUE and Reason is the correct explanation of Assertion
- (b) If both the statements are TRUE but Reason is NOT the correct explanation of Assertion
- (c) If Assertion is TRUE and Reason is FALSE
- (d) If Assertion is FALSE and Reason is TRUE

1. **Assertion:** Xe atom in XeF_2 assumes sp hybrid state.
Reason: XeF_2 molecule does not follow octet rule.

2. **Assertion:** The atoms in a covalent molecule are said to share electrons, yet some covalent molecules are polar.
Reason: In polar covalent molecule, the shared electrons spend more time on the average near one of the atoms.

3. **Assertion:** H bonding occurs in H_2O due to larger size of O atom.

Reason: The size of O atom is larger than H atom.

4. **Assertion:** CCl_4 is a non-polar molecule.

Reason: CCl_4 has polar bonds.

5. **Assertion:** Geometry of ICl_3 is tetrahedral.

Reason: Due to the presence of two lone pair its shape is T-shape.

MATCH THE COLUMN TYPE QUESTIONS

Column-I and Column-II contains four entries each. Entries of Column-I are to be matched with some entries of Column-II. One or more than one entries of Column-I may have the matching with the same entries of Column-II.

1.

Column-I

- (A) NH_4Cl
- (B) $CuSO_4 \cdot 5H_2O$
- (C) HNC
- (D) Liquid H_2O_2

Column-II

- (P) Hydrogen bond
- (Q) Co-ordinate bond
- (R) Ionic bond
- (S) Covalent bond

2.

Column-I

(Decreasing order)

- (A) NH_3 , SbH_3 , AsH_3 , PH_3
- (B) HI, HBr, HCl, HF

Column-II

(Physical properties)

- (P) Dipole moment
- (Q) Reducing property

(C) SnH_4 , GeH_4 , SiH_4 ,
 CH_4

(D) H_2O , H_2Te , H_2Se ,
 H_2S

(R) Enthalpy of
vaporization

(S) Boiling point

3.

Column-I

- (A) NH_2^-
- (B) $XeOF_2$
- (C) ICl_4^-
- (D) $[SbF_5]^{2-}$

Column-II

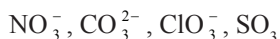
- (P) Square pyramidal
- (Q) V-shaped
- (R) T-shaped
- (S) Square planar

RECENT YEARS' QUESTIONS OF IIT-JEE AND AIEEE

- In XeF_2 , XeF_4 and XeF_6 , the number of lone pair of electrons on Xe are respectively: [AIEEE 2002]
 - 2, 3, 1
 - 1, 2, 3
 - 4, 1, 2
 - 3, 2, 1
- The correct order of bond strength is: [AIEEE 2002]
 - $\text{O}_2^- < \text{O}_2 < \text{O}_2^+ < \text{O}_2^{2-}$
 - $\text{O}_2^{2-} < \text{O}_2^- < \text{O}_2 < \text{O}_2^+$
 - $\text{O}_2^- < \text{O}_2^{2-} < \text{O}_2 < \text{O}_2^+$
 - $\text{O}_2^+ < \text{O}_2 < \text{O}_2^- < \text{O}_2^{2-}$
- In which of the following pairs, bond angle is $109^\circ 28'$? [AIEEE 2002]
 - $[\text{NH}_4]^+$, $[\text{BF}_4]^-$
 - NH_4^+ , $[\text{BF}_3]$
 - NH_3 , $[\text{BF}_4]^-$
 - $[\text{NH}_3]$, $[\text{BF}_3]$
- In which of the following species, is the underlined carbon has sp^3 hybridization: [AIEEE 2002]
 - $\text{CH}_3\text{C}\underline{\text{C}}\text{OOH}$
 - $\text{CH}_3\text{C}\underline{\text{H}}_2\text{OH}$
 - $\text{CH}_3\text{C}\underline{\text{O}}\text{CH}_3$
 - $\text{CH}_2=\text{C}\underline{\text{H}}-\text{CH}_3$
- Which of the following statement is true? [AIEEE 2002]
 - HF is less polar than HBr
 - Absolutely pure water does not contain any ions
 - Chemical bond formation takes place when forces of attraction overcome the forces of repulsion
 - In covalence transfer of electrons takes place
- An ether is more volatile than alcohol having same molecular formula. This is due to: [AIEEE 2003]
 - Inter molecular H-bonding in ethers
 - Inter molecular H-bonding in alcohols
 - Dipolar character of ethers
 - Resonance structure in alcohols
- Which among the following has smallest bond angle? [AIEEE 2003]
 - H_2S
 - NH_3
 - SO_2
 - H_2O
- Which pair of molecules will have permanent dipole moment for both members? [AIEEE 2003]
 - NO_2 and O_3
 - SiF_4 and CO_2
 - SiF_4 and NO_2
 - NO_2 and CO_2
- The pair of species having identical shape of both species: [AIEEE 2003]
 - BF_3 , PCl_3
 - PF_5 , IF_5
 - CF_4 , SF_4
 - XeF_2 , CO_2
- The correct order of bond angles is: [AIEEE 2004]
 - $\text{H}_2\text{S} < \text{NH}_3 < \text{BF}_3 < \text{SiH}_4$
 - $\text{NH}_3 < \text{H}_2\text{S} < \text{SiH}_4 < \text{BF}_3$
 - $\text{H}_2\text{S} < \text{NH}_3 < \text{SiH}_4 < \text{BF}_3$
 - $\text{H}_2\text{S} < \text{SiH}_4 < \text{NH}_3 < \text{BF}_3$
- Which has regular tetrahedral geometry? [AIEEE 2004]
 - SF_4
 - BF_4^-
 - XeF_4
 - $[\text{Ni}(\text{CN})_4]^{2-}$
- The bond order in NO is 2.5 while that in NO^+ is 3. Which statement is true? [AIEEE 2004]
 - Bond length is unpredictable
 - Bond length in NO is greater than in NO^+
 - Bond length in NO^+ is equal to that in NO
 - Bond length in NO^+ is greater than in NO
- Which one of the following species is diamagnetic in nature? [AIEEE 2005]
 - He_2^+
 - H_2
 - H_2^+
 - H_2^-
- Lattice energy of an ionic compound depends upon: [AIEEE 2005]
 - Charge on the ions only
 - Size of the ions only
 - Packing of the ions only
 - Charge and size of the ions
- Which of the following does not contain iso-electronic species? [AIEEE 2005]
 - PO_4^{3-} , SO_4^{2-} , ClO_4^-
 - CN^- , N_2 , C_2^{2-}
 - SO_3^{2-} , CO_3^{2-} , NO_3^-
 - BO_3^{3-} , CO_3^{2-} , NO_3^-
- Which of the following molecules/ions does not contain unpaired electrons? [AIEEE 2006]
 - O_2^{2-}
 - B_2
 - N_2^+
 - O_2
- In which of the following molecules/ions, are all the bonds not equal? [AIEEE 2006]
 - SF_4
 - SiF_4
 - XeF_4
 - BF_4
- The increasing order of the first ionization enthalpies of the elements B, P, S and F (lowest first) is: [AIEEE 2006]

- (a) $F < S < P < B$ (b) $P < S < B < F$
 (c) $B < P < S < F$ (d) $B < S < P < F$
19. The decreasing values of bond angles from NH_3 (106°) to SbH_3 (101°) down the group 15 of the periodic table is due to: [AIEEE 2006]
 (a) Increase in bp-bp repulsion
 (b) Increase in p-orbital character in sp^3
 (c) Decrease in lp-pp repulsion
 (d) Decrease in electronegativity
20. Which of the following species exhibits the diamagnetic behaviour? [AIEEE 2007]
 (a) O_2^{2-} (b) O_2^+
 (c) O_2 (d) NO
21. In which of the following ionization processes, the bond order has increased and the magnetic behaviour has changed? [AIEEE 2007]
 (a) $\text{C}_2 \rightarrow \text{C}_2^+$ (b) $\text{NO} \rightarrow \text{NO}^+$
 (c) $\text{O}_2 \rightarrow \text{O}_2^+$ (d) $\text{N}_2 \rightarrow \text{N}_2^+$
22. Which of the following hydrogen bonds is the strongest? [AIEEE 2007]
 (a) $\text{O}-\text{H}-\text{N}$ (b) $\text{F}-\text{H}-\text{F}$
 (c) $\text{O}-\text{H}-\text{O}$ (d) $\text{O}-\text{H}-\text{F}$
23. The charge/size ratio of a cation determines its polarizing power. Which one of the following sequences represents the increasing order of the polarizing power of the cationic species, K^+ , Ca^+ , Mg^{2+} , Be^{2+} ? [AIEEE 2007]
 (a) $\text{Mg}^{2+} < \text{Be}^{2+} < \text{K}^+ < \text{Ca}^{2+}$
 (b) $\text{Be}^{2+} < \text{K}^+ < \text{Ca}^{2+} < \text{Mg}^{2+}$
 (c) $\text{K}^+ < \text{Ca}^{2+} < \text{Mg}^{2+} < \text{Be}^{2+}$
 (d) $\text{Ca}^{2+} < \text{Mg}^{2+} < \text{Be}^{2+} < \text{K}^+$
24. Which one of the following constitutes a group of the isoelectronic species? [AIEEE 2008]
 (a) C_2^{2-} , O_2^- , CO , NO
 (b) NO^+ , C_2^{2-} , CN^- , N_2
 (c) CN^- , N_2 , O_2^{2-} , CO_2^{2-}
 (d) N_2 , O_2^- , NO^+ , CO
25. Which one of the following pairs of species have the same bond order? [AIEEE 2008]
 (a) CN^- and NO^+ (b) CN^- and CN^+
 (c) O_2^- and CN^- (d) NO^+ and CN^+
26. Molecular shape of SF_4 , CF_4 and XeF_4 are: [IIT 2000; AIEEE 2005]
 (a) The same with 2, 0 and 1 lone pair of electron respectively
 (b) The same with 1, 1 and 1 lone pair of electron respectively
 (c) Different with 0, 1 and 2 lone pairs of electron respectively
 (d) Different with 1, 0 and 2 lone pairs of electron respectively
27. The hybridization of atomic orbitals of nitrogen in NO_2^+ , NO_3^- and NH_4^+ are: [IIT 2000]
 (a) sp , sp^3 and sp^2 respectively
 (b) sp , sp^2 and sp^3 respectively
 (c) sp^2 , sp and sp^3 respectively
 (d) sp^2 , sp^3 and sp respectively
28. Amongst H_2O , H_2S , H_2Se and H_2Te , the one with the highest boiling point is: [IIT 2000]
 (a) H_2O because of H-bonding
 (b) H_2Te because of higher molecular weight
 (c) H_2S because of H-bonding
 (d) H_2Se because of lower molecular weight
29. Specify the co-ordination geometry around and hybridization of N and B complex of NH_3 and BF_3 : [IIT 2002]
 (a) N : tetrahedral, sp^3 ; B : tetrahedral, sp^3
 (b) N : Pyramidal, sp^3 ; B : Pyramidal, sp^3
 (c) N : Pyramidal, sp^3 ; B : Planar, sp^2
 (d) N : Pyramidal, sp^3 ; B : tetrahedral, sp^3
30. The least stable in amongst the following is: [IIT 2002]
 (a) Li^- (b) Be^-
 (c) B^- (d) C^-
31. Which of the following molecular species has unpaired electrons? [IIT 2002]
 (a) N_2 (b) F_2
 (c) O_2^- (d) O_2^{2-}
32. The nodal plane is the π -bond of ethene is located in: [IIT 2002]
 (a) The molecular plane
 (b) A plane parallel to molecular plane
 (c) A plane perpendicular to the molecular plane which bisects the carbon-carbon sigma bond at right angles
 (d) A plane perpendicular to the molecular plane which contains the carbon-carbon sigma bond
33. Among the following the molecule with the highest dipole moment is: [IIT 2003]
 (a) CH_3Cl (b) CH_2Cl_2
 (c) CHCl_3 (d) CCl_4

34. Which of the following are isoelectronics and iso-structural? [IIT 2003]



- (a) NO_3^- , CO_3^{2-} (b) SO_3 , NO_3^-
 (c) ClO_3^- , CO_3^{2-} (d) CO_3^{2-} , SO_3
35. Which of the following represents the given mode of hybridization sp^2 - sp^2 - sp - sp from left to right? [IIT 2003]
- (a) $\text{CH}=\text{CH}-\text{C}\equiv\text{CH}$
 (b) $\text{CH}\equiv\text{C}-\text{C}\equiv\text{CH}$
 (c) $\text{CH}_2=\text{C}=\text{C}=\text{CH}_2$
 (d) $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$
36. Total number of lone pair of electrons in XeOF_4 is: [IIT 2004]
- (a) 0 (b) 1
 (c) 2 (d) 3

37. Which statement is correct about O_2^+ ? [IIT 2004]

- (a) Paramagnetic and bond order $< \text{O}_2$
 (b) Paramagnetic and bond order $> \text{O}_2$
 (c) Diamagnetic and bond order $< \text{O}_2$
 (d) Diamagnetic and bond order $> \text{O}_2$

38. Which species has the maximum number of lone pair of electrons on the central atom? [IIT 2005]

- (a) $[\text{ClO}_3]^-$ (b) XeF_4
 (c) SF_4 (d) $[\text{I}_3]^-$

39. The species having bond order different from that in CO is: [IIT 2007]

- (a) NO^- (b) NO^+
 (c) CN^- (d) N_2

40. Among the following, the paramagnetic compound is: [IIT 2007]

- (a) Na_2O_2 (b) O_3
 (c) N_2O (d) KO_2

ANSWER KEYS

ONE CORRECT OPTION TYPE QUESTIONS

- | | | | | | | | | | |
|---------|---------|---------|---------|-----------|---------|---------|---------|---------|---------|
| 1. (c) | 2. (c) | 3. (a) | 4. (d) | 5. (a) | 6. (b) | 7. (c) | 8. (b) | 9. (b) | 10. (b) |
| 11. (b) | 12. (b) | 13. (c) | 14. (c) | 15. (b) | 16. (c) | 17. (b) | 18. (b) | 19. (c) | 20. (b) |
| 21. (a) | 22. (b) | 23. (b) | 24. (d) | 25. (c) | 26. (a) | 27. (b) | 28. (c) | 29. (c) | 30. (c) |
| 31. (c) | 32. (a) | 33. (a) | 34. (b) | 35. (b) | 36. (b) | 37. (b) | 38. (b) | 39. (a) | 40. (a) |
| 41. (a) | 42. (a) | 43. (a) | 44. (b) | 45. (c) | 46. (a) | 47. (c) | 48. (a) | 49. (c) | 50. (b) |
| 51. (b) | 52. (d) | 53. (d) | 54. (b) | 55. (a,c) | 56. (a) | 57. (d) | 58. (a) | 59. (a) | 60. (d) |
| 61. (c) | 62. (c) | 63. (a) | 64. (b) | 65. (a) | 66. (d) | 67. (b) | 68. (d) | 69. (a) | 70. (b) |
| 71. (d) | 72. (d) | 73. (d) | 74. (d) | 75. (d) | 76. (a) | 77. (d) | 78. (d) | 79. (a) | 80. (d) |
| 81. (c) | 82. (c) | 83. (b) | 84. (d) | 85. (b) | 86. (c) | 87. (b) | 88. (b) | 89. (b) | 90. (a) |
| 91. (a) | | | | | | | | | |

COMPREHENSION BASED QUESTIONS

Comprehension-I

1. (a) 2. (a) 3. (a)

Comprehension-II

1. (a) 2. (b) 3. (a)

Comprehension-III

1. (b) 2. (b) 3. (c) 4. (a) 5. (d)

Comprehension-IV

1. (d) 2. (b) 3. (a) 4. (b) 5. (b)

ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

1. (b, d) 2. (b, c, d) 3. (a, b, d) 4. (a, b, c) 5. (a, d) 6. (a, b, c) 7. (a, b, c, d) 8. (a, b, c)
 9. (a, c) 10. (a, c, d)

ASSERTION/REASON TYPE QUESTIONS

1. (d) 2. (b) 3. (d) 4. (b) 5. (d)

MATCH THE COLUMN TYPE QUESTIONS

1. $A \rightarrow Q, R, S;$ $B \rightarrow P, Q, R, S;$ $C \rightarrow Q, S;$ $D \rightarrow P, S$
2. $A \rightarrow P, R;$ $B \rightarrow Q;$ $C \rightarrow Q, R, S;$ $D \rightarrow S, R$
3. $A \rightarrow Q;$ $B \rightarrow R;$ $C \rightarrow S;$ $D \rightarrow P$

RECENT YEARS' QUESTIONS OF IIT-JEE AND AIEEE

1. (d) 2. (b) 3. (a) 4. (b) 5. (c) 6. (b) 7. (a) 8. (a)
9. (d) 10. (c) 11. (b) 12. (b) 13. (b) 14. (d) 15. (c) 16. (a)
17. (a) 18. (d) 19. (d) 20. (a) 21. (b) 22. (b) 23. (c) 24. (b)
25. (a) 26. (d) 27. (b) 28. (a) 29. (a) 30. (b) 31. (c) 32. (a)
33. (a) 34. (a) 35. (a) 36. (b) 37. (b) 38. (d) 39. (a) 40. (d)

CHAPTER 2

Alkali Metals (+2)

INTRODUCTION

Table 2.1 Physical Data of Group 1A Elements

	<i>Atomic number</i>	<i>Electronic configuration</i>	<i>First Ionization energy (kJ/mol)</i>	<i>Standard Electrode potential (volts)</i>	<i>Atomic radius (Å)</i>	<i>Ionic radius (Å)</i>	<i>M.p. °C</i>	<i>B.p. °C</i>
Li	3	2.1 1s ² 2s ¹	520	−3.04	1.33	0.60	179.1	1340
Na	11	2.8.1 ...2s ² 2p ⁶ 3s ¹	494	−2.71	1.57	0.95	97.5	885
K	19	2.8.8.1 ...3s ² 3p ⁶ 4s ¹	418	−2.92	2.03	1.33	63.5	775
Rb	37	2.8.18.8.1 ...4s ² 4p ⁶ 5s ¹	403	−2.99	2.16	1.48	39.0	690
Cs	55	2.8.18.18.8.1 ...5s ² 5p ⁶ 6s ¹	374	−3.02	2.35	1.69	28.5	670

Some general remarks about group 1

- The elements lithium, sodium, potassium, rubidium, caesium and francium are called the alkali metals. Not much is known about the last-named, since it is radioactive and all its isotopes are exceedingly shortlived; it is formed during the radioactive decay of actinium.
- All the alkali elements are silvery white solids. These are soft in nature and can be cut with the help of knife except the lithium. When freshly cut, they have a bright lustre which quickly fades due to surface oxidation and lose their silvery appearance in air, they are generally stored under oil. These are highly malleable and ductile. The silvery lustre of alkali metal is due to presence of highly mobile electrons of metallic lattice. There being only single electron per atom. So metallic bonding is not so strong. As a result the metals are soft in nature. Softness increases with increase in atomic number due to continuous decrease in metallic bond strength on account of an increase in atomic size.

2.2 Alkali Metals (+2)

- Group I is known as alkali group as the hydroxides of these metals are soluble in water and these solutions are highly alkaline in nature. The word alkali has been derived from Arabic word 'Alquili' meaning the ashes of plants from which compounds of sodium and potassium were first isolated.
- The metals are extremely reactive and electropositive (prone to lose electrons) and exist in combination with other elements or radicals as positive ions, e.g., Na^+Cl^- , $(\text{K}^+)_2\text{SO}_4^{2-}$; they are therefore never found in the free state in nature. They all adopt the body-centered cubic structure in which each atom is surrounded by eight nearest neighbors with six more atoms only slightly further distant.

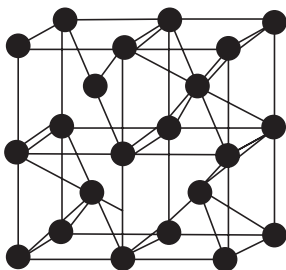


Fig. 2.1

- All alkali metal atoms have one electron in the outer shell preceded by a closed shell containing eight electrons—except for lithium which has a closed shell of two. In chemical combination, this single electron is very readily transferred, giving a unipositive metal ion with the stable electronic configuration of a noble gas, e.g., $\text{Na}^+(2.8)$ is isoelectronic with Ne (2.8). Compounds of the alkali metals are therefore generally predominantly ionic and exist as high melting-point solids in which as many ions of opposite charge surround each other as possible. These compounds are generally water soluble and white because in alkali metals ions all the electrons are paired. It may be mentioned here that some compounds of alkali metals are coloured. Here the colour of the compound is due to anion. For example, KMnO_4 (purple), $\text{K}_2\text{Cr}_2\text{O}_7$ (orange) and Na_2CrO_4 (yellow). In these cases, the colour comes from the anions $[\text{MnO}_4]^-$, $[\text{CrO}_4]^{2-}$, $[\text{Cr}_2\text{O}_7]^{2-}$ and not from the Group I metal ion.

Atomic and Ionic Radii

- The atoms of alkali metals have the largest size in their respective periods. The atomic radii increases on moving down the group among the alkali metals as a new shell is progressively added. Although the nuclear charge also increases down the group but the effect of addition of new shells is more predominant due to increasing screening effect of inner filled shell, on the valence-s- electrons. Hence the atomic size increases in a group. Alkali metals change into positively charged ions by losing valence electron. The size of cation is smaller than parent atom of alkali metals. However, within the group, the ionic radii increases with increases in atomic number. The alkali metal ions get extensively hydrated in aqueous solution follows the order:



- The charge density on Li^+ is higher in comparison to other alkali metals due to which it is extensively hydrated.

Density

- All are light metals. The densities are low. Lithium, sodium and potassium are lighter than water. Density gradually increases in moving down from Li to Cs. Due to large size, the atom of alkali metals are less closely packed and consequently have low density. On going down the group, both atomic size and atomic mass increases but the increase in atomic mass compensates the bigger atomic size. As a result, the density of alkali metals increases from Li to Cs.

	Li	Na	K	Rb	Cs
density (g/ml)	0.54	0.97	0.86	1.53	1.87

- Potassium is, however, lighter than sodium. The reason for the low values is that these metals have high atomic volumes. The abnormal value of potassium is due to unusual increase in atomic size, i.e., atomic volume.
- The density of potassium is lesser than that of sodium contrary to the expectations. This is probably because of the abnormal increase in atomic size on moving from Na (186 pm) to K (227 pm). Hence, potassium is lighter than sodium. Lithium is the lightest known metal (density = 0.534 g/cc).

Ionization Energies and Electropositive Character

- Due to their large size, the outermost electron is far from the nucleus and can easily be removed. Their ionization energies or ionisation potentials are relatively low. Thus, the metals have a great tendency to lose the ns^1 electron to change into M^+ ions. These metals are highly electropositive in nature. As the ionization potential decreases from Li to Cs, the electropositive character increases, i.e., metallic character increases. The reactivity of these metals increases from Li to Cs.

	Li	Na	K	Rb	Cs
Ionization potential (eV)	5.4	5.1	4.3	4.2	3.9
	$\xrightarrow{\text{Decreases}}$				
Electropositive or metallic nature	$\xrightarrow{\text{Increases}}$				
Reactivity	$\xrightarrow{\text{Increases}}$				

- The ns^1 electron is so loosely held that even the low energy photons (light) can eject this electron from the metal surface. This property is termed as **photoelectric effect** K and Cs are used in photoelectric cells which are sensitive to blue light.
- The alkali metals have lower ionization energies than any other group of similar elements; even so the removal of the first electron from the caesium atom requires the input of 374 kJ of energy per mol and for the other alkali metals the energy is somewhat larger still. The reason why the single electron is so readily transferred during chemical combination is because the resulting ions attract each other strongly, with the liberation of energy, and therefore the overall process is highly exothermic.
- The second ionization energies of all the alkali metals are very large because when one electron is lost from these elements, the resulting ions acquire noble gas configurations which are very stable and have high effective nuclear charge. Thus, large amounts of energies are required to remove the second electron.

Photoelectric Effect

Alkali metals (except Li) exhibit photoelectric effect (A phenomenon of emission of electrons from the metal surface when light of suitable frequency fall on them). The ability to exhibit photoelectric effect is due to low value of ionization energy of alkali metals. Li does not emit photoelectrons due to high value of ionization energy. Potassium and caesium are used in photoelectric cells.

Oxidation State

- The alkali metal shows +1 oxidation state. The alkali metals can easily lose their valence electron and change into unit positive ions



and gain stable noble gas configuration.

- But the alkali metals cannot form M^{2+} ions as the magnitude of 2nd ionization energy is very high.

Melting and Boiling Points

- The energy binding the atoms in the crystal lattices of these metals is relatively low on account of a single electron in the valency shell. Consequently, the metals have low melting and boiling points. These decrease in moving down from Li to Cs as the metallic bond strength decreases or cohesive force decreases.

2.4 Alkali Metals (+2)

- The specific heat values decrease from Li to Cs.
- Heats of atomization decreases from Li to Cs.

Reducing Character

- Alkali metals are very **good reducing agents** because of their great tendency to lose electrons. The reducing character increases from Na to Cs.
- Li is however, stronger reducing agent than Na due to greater hydration energy.

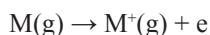
Explanation: An element, which acts as a reducing agent, must have low ionization energy. Alkali metals act as strong reducing agents as their ionization energy values are low. Since ionization energy decreases on moving down from Li to Cs, the reducing property increases in the same order. Thus, Li is the weakest reducing agent while Cs is the strongest reducing agent amongst alkali metals in free gaseous state.

The tendency of an element to lose an electron in solution is measured by its standard oxidation potential value (E_{ox}°). Since alkali metals have high E_{ox}° values, these are strong reducing agents. However, it is observed that Li is the strongest reducing agent amongst alkali metals in solution as E_{ox}° value of Li is maximum.

	Li	Na	K	Rb	Cs
Oxidation potential (V)	+3.05	+2.71	+2.93	+2.99	+2.99

It looks surprising at first sight that lithium having high value of ionization energy amongst alkali metals acts as strongest reducing agent in solution. This can be explained if we understand the fact that **ionization energy is the property of an isolated atom in gaseous state while oxidation potential is concerned when the metal atom goes into the solution.**

The ionization energy involves the change of gaseous atom to gaseous ion



while oxidation potential involves the following change:



The above change occurs in three steps:

- $\text{M(s)} \rightarrow \text{M(g)}$ – sublimation energy
- $\text{M(g)} \rightarrow \text{M}^{\text{+}}(\text{g}) + \text{e}$ – ionization energy
- $\text{M}^{\text{+}}(\text{g}) + \text{H}_2\text{O} \rightarrow \text{M}^{\text{+}}(\text{aq.}) + \text{Hydration energy}$

Sublimation energy is nearly same for all the alkali metals.

The energy required in (ii) step is the ionization energy which is highest for lithium. $\text{Li}^{\text{+}}$ ion is smallest in size, hence it has maximum degree of hydration. Thus, in (iii) step, maximum hydration energy is released in lithium.

	Li	Na	K
Step (i)	+122 kJ	+ 78 kJ	+ 61 kJ
Step (ii)	+520 kJ	+ 496 kJ	+ 419 kJ
Step (iii)	–480 kJ	–371 kJ	–299 kJ
overall	<u>162 kJ</u>	<u>203 kJ</u>	<u>181 kJ</u>

Thus, with greater ease the following overall change occurs in lithium and it acts as a strongest reducing agent.

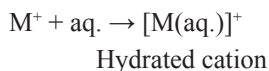


- **The compounds of alkali metals are diamagnetic.** Superoxides of alkali metals are, however, paramagnetic.

Hydration of ions, hydrated radii and hydration energy

Relative ionic radii	$\text{Cs}^+ > \text{Rb}^+ > \text{K}^+ > \text{Na}^+ > \text{Li}^+$
Relative hydrated ionic radii	$\text{Li}^+ > \text{Na}^+ > \text{K}^+ > \text{Rb}^+ > \text{Cs}^+$
Relative conducting power in water	$\text{Cs}^+ > \text{Rb}^+ > \text{K}^+ > \text{Na}^+ > \text{Li}^+$

The salts of alkali metals are ionic and soluble in water. The solubility is due to the fact that cations get hydrated by water molecules.



The smaller the cation, the greater is the degree of its hydration. Thus, the degree of hydration of M^+ ions decreases from Li^+ to Cs^+ . Consequently the radii of the hydrated ion decreases from Li^+ to Cs^+ .

	$[\text{Li}(\text{aq.})]^+$	$[\text{Na}(\text{aq.})]^+$	$[\text{K}(\text{aq.})]^+$	$[\text{Rb}(\text{aq.})]^+$	$[\text{Cs}(\text{aq.})]^+$
Hydrated radii (Å)	3.40	2.76	2.32	2.28	2.27

The ionic conductance of these hydrated ions increases from $[\text{Li}(\text{aq.})]^+$ to $[\text{Cs}(\text{aq.})]^+$.

Hydration of ions is an exothermic process. The energy released when one gram mole of an ion is dissolved in water to get it hydrated is called hydration energy. Since the degree of hydration decreases from Li^+ to Cs^+ , the hydration energy of alkali metal ion also decreases from Li^+ to Cs^+ .

- Among all the alkali metals, Li is the strongest and Na is the weakest reducing agent. Lithium, although, has the highest ionization energy (i.e., it holds its valence electrons most tightly), yet it is the strongest reducing agent.

Flame Colouration

In the alkali ions, transitions are possible from filled noble gas shell to the next higher quantum shell, for example, for rubidium $4p^6 \rightarrow 4p^5 5s^1$. These need much energy and such transitions are located deep into the ultraviolet region. Alkali metal salts on being held over a Bunsen flame mostly dissociate into M^+ and X^- ions which ions give no characteristic colour. But most alkali salts (e.g., NaCl) decompose to some extent into component atoms. These alkali atoms give coloured spectra. For example, in the sodium atom transition of an electron from 3s to several higher energy states occurs and when the excited electron returns from such an excited 3p level to 3s ground level it emits the characteristic golden yellow colour. As the atomic number increases, the energy gap between successive levels decreases so that 4p to 4s transition in potassium should occur at a lower energy. However, the characteristic violet colour of potassium is really due to a transition from 5p to 4s (and not due to 4p to 4s) occurring at a higher energy than in sodium, being at the violet end of the spectrum. It should be noted that the 5p to 4s transition is just one of the several possible transitions. It is also interesting to note that the higher energy transition in potassium is less realizable in a Bunsen flame. Hence the violet flame is less intense. The violet flame is masked by the presence of sodium. However, this can be viewed if the flame is observed through cobalt glass blue filter which is not transparent to the yellow sodium flame. Lithium, rubidium and caesium are best detected by spectroscopic techniques.

Elements	Li	Na	K	Rb	Cs
Emitted colour					
Wavelength (mμ or nm)	crimson 670.8	yellow 589	violet 766	red-violet 780	blue 455.5
of emitted colour					

- There is no definite trend of wavelengths of different lines emitted, when alkali metals or their salts are introduced into an oxidizing flame.

ANALYTICAL REACTIONS OF THE ALKALI ELEMENTS

Most alkali salts being quite soluble in aqueous medium have only a few useful precipitation reactions. Precipitation reactions are known only with large anions, and the larger the alkali cation the more insoluble salts it forms. Sodium forms an insoluble precipitate with zinc (or magnesium) uranyl acetate: $\text{Na}(\text{UO}_2)_3\text{Zn}(\text{CH}_3\text{COO})_9 \cdot 6\text{H}_2\text{O}$ in dilute acetic acid solution. Whereas NaClO_4 is highly soluble K, Rb, Cs perchlorates are rather insoluble and the latter alkalies can be estimated gravimetrically as perchlorates. Sodium cobaltinitrite $\text{Na}_3[\text{Co}(\text{NO}_2)_6]$ (sodium hexanitrito-kN-cobaltate (III)) and sodium tetraphenylborate (III) $\text{Na}[\text{B}(\text{C}_6\text{H}_5)_4]$ give quantitative precipitates suitable for gravimetric estimation of K, Rb and Cs.

1. Caesium is the most electropositive of all the alkali metals.
2. Since alkali metals are highly electropositive
 - (a) they (except Li) mostly form ionic compounds.
 - (b) their electronegativity values are low.
3. Due to their low ionization energies, alkali metals (except Li) show photoelectric effect. Cs can emit photoelectrons even with red light.
4. All the alkali metal are silvery white, soft and light metals.

Relative ionic radii : $\text{Cs}^+ > \text{Rb}^+ > \text{K}^+ > \text{Na}^+ > \text{Li}^+$

Relative ionic radii in water or

Relative degree of hydration : $\text{Li}^+ > \text{Na}^+ > \text{K}^+ > \text{Rb}^+ > \text{Cs}^+$

1. Alkali metals are light metals. Their densities are as follows:

	Li	Na	K	Rb
Density (g/cc)	0.54	0.97	0.86	1.53

Li is the lightest metal at room temperature. Potassium is lighter than sodium.

2. Lithium can't be stored in kerosene as it floats to the surface, due to its very low density. Therefore, lithium is generally kept wrapped in paraffin wax.
3. In fact, lithium is the lightest metal known. It is the least fusible, least dense and least soft of all the alkali metals.
4. Although Li has the highest ionization energy, yet Li is the strongest reducing agent, because of its large heat of hydration, which is sufficient to overcome its ionization energy.
5. Metallic nature of elements increases from Li to Cs because IE decreases in the same order.
6. The most metallic elements in the periodic table are caesium and francium.
7. KOH is a better absorbed of CO_2 than NaOH because potassium carbonate is more soluble and does not separate out.
8. Specific heat of alkali metals decreases from Li and Cs.
9. Na and K are 6th and 7th most abundant elements in the earth crust.
10. When a pellet of NaOH is exposed to moist air, a liquid layer is formed around it which is aqueous NaOH.
11. If a crystal of NaOH is kept exposed to moist air for longer periods a white powder is formed which is sodium carbonate.
12. Smaller the size and higher the charge on the ion, greater is its tendency to distort the electron cloud of anion so greater is its polarizing power (Fajan rule).
13. Alkali metals are the only elements which are capable of forming most stable solid bicarbonate because these metals are highly electropositive. Lithium does not form solid bicarbonate because it is least electropositive among the alkali metals.

Worked-out Examples

1. The following compounds have been arranged in order of their increasing thermal stabilities. Identify the correct order. K_2CO_3 (I) MgCO_3 (II) CaCO_3 (III) BeCO_3 (IV) (IITJEE 1996)
 (a) $\text{I} < \text{II} < \text{III} < \text{IV}$ (b) $\text{IV} < \text{II} < \text{III} < \text{I}$ (c) $\text{IV} < \text{II} < \text{I} < \text{III}$ (d) $\text{II} < \text{IV} < \text{III} < \text{I}$

Ans. (b)

2. Which is most reactive alkali metal and why?

Solution: Cs is most reactive alkali metal due to its lowest first ionization energy and lowest value of electronegativity

3. Li is the strongest reducing agent in the periodic table why?

Solution: The E^0 value depends on the three factors i.e., sublimation, ionization and hydration enthalpy. Due to small size as Li^+ ion it has highest hydration enthalpy. It has high negative E^0 value and is strongest reducing agent.

4. In aqueous solution Li^+ ion has the lowest mobility. Why?

Solution: Li^+ ions are highly hydrated in aqueous solution which result in decrease in its mobility.

5. Why Li is kept wrapped in paraffin wax and not stored in kerosene oil?

Solution: It is because lithium is a light metal and therefore it floats at the surface of kerosene oil. To prevent its exposure to air it is kept wrapped in paraffin wax.

6. Why Cs can be used in photoelectronic cell, while Li can not be?

Solution: Cs has the lowest while Li has the highest ionization energy among all the alkali metals. Hence, Cs can lose electron very easily while lithium cannot.

Chemical Nature

7. The metallic lustre exhibited by sodium is explained by: (IITJEE1987)
 (a) diffusion of sodium ions (b) oscillations of loose electrons
 (c) excitation of free proton (d) existence of body centered cubic lattice

Ans. (b)

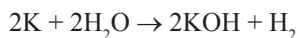
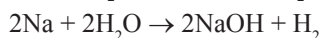
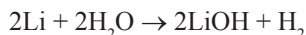
Chemical Properties

Reaction	Comment
$\text{M} + \text{H}_2\text{O} \rightarrow \text{MOH} + \text{H}_2$	The hydroxides are the strongest bases known
with excess oxygen $\text{Li} + \text{O}_2 \rightarrow \text{Li}_2\text{O}$	Monoxide is formed by Li and to a small extent by Na
$\text{Na} + \text{O}_2 \rightarrow \text{Na}_2\text{O}_2$	Peroxide formed by Na and to a small extent by Li
$\text{K} + \text{O}_2 \rightarrow \text{KO}_2$	Superoxide formed by K, Rb, Cs
$\text{M} + \text{H}_2 \rightarrow \text{MH}$	Ionic 'salt-like' hydrides
$\text{Li} + \text{N}_2 \rightarrow \text{Li}_3\text{N}$	Nitride formed only by Li
$\text{M} + \text{P} \rightarrow \text{M}_3\text{P}$	All the metals form phosphides
$\text{M} + \text{As} \rightarrow \text{M}_3\text{As}$	All the metals form phosphides
$\text{M} + \text{Sb} \rightarrow \text{M}_3\text{Sb}$	All the metals form phosphides
$\text{M} + \text{S} \rightarrow \text{M}_2\text{S}$	All the metals form sulphides
$\text{M} + \text{Se} \rightarrow \text{M}_2\text{Se}$	All the metals form selenides
$\text{M} + \text{Te} \rightarrow \text{M}_2\text{Te}$	All the metals form tellurides
$\text{M} + \text{F}_2 \rightarrow \text{MF}$	All the metals form fluorides
$\text{M} + \text{Cl}_2 \rightarrow \text{MCl}$	All the metals form chlorides
$\text{M} + \text{Br}_2 \rightarrow \text{MBr}$	All the metals form bromides
$\text{M} + \text{I}_2 \rightarrow \text{MI}$	All the metals form iodides
$\text{M} + \text{NH}_3 \rightarrow \text{MNH}_2$	All the metals form amides

2.8 Alkali Metals (+2)

Reaction with water

- Group I metals all react with water, liberating hydrogen and forming the hydroxides.
- The reaction becomes increasingly violent on descending the group. Thus lithium reacts gently, sodium melts on the surface of the water and the molten metal skates about vigorously and may catch fire (especially if localized), and potassium melts and always catches fire.



- The standard electrode potentials E° are $\text{Li}^+|\text{Li} = -3.05$ volts, $\text{Na}^+|\text{Na} = -2.71$, $\text{K}^+|\text{K} = -2.93$, $\text{Rb}^+|\text{Rb} = -2.92$, $\text{Cs}^+|\text{Cs} = -2.92$. Lithium has the most negative standard electrode potentials E° and Gibbs free energy? G are related by the equation:

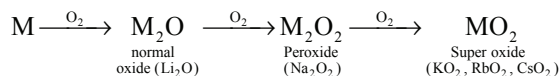
$$\Delta G = -nFE^\circ$$

where n is the number of electrons removed from the metal to produce the ion, and F is the Faraday constant.

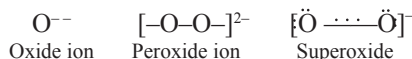
The reaction $\text{Li}^+ + e \rightarrow \text{Li}$ has the largest negative E° value, and hence the largest positive ΔG value. Thus the reaction does not occur. However, the reverse reaction $\text{Li} \rightarrow \text{Li}^+ + e$ has a large negative value of ΔG , so lithium liberates more energy than the other metals when it reacts with water. In view of this, it is at first sight rather surprising that lithium reacts gently with water, whereas potassium, which liberates less energy, reacts violently and catches fire. The explanation lies in the kinetics (that is the rate at which the reaction proceeds), rather than in the thermodynamics (that is the total amount of energy liberated). Potassium has a low melting point, and the heat of reaction is sufficient to make it melt, or even vaporize. The molten metal spreads out, and exposes a larger surface to the water, so it reacts even faster, gets even hotter and catches fire.

Action with oxygen (oxides and hydroxides)

- Affinity towards oxygen increases. When heated in atmosphere of oxygen, the alkali metals ignite and form oxides. On combustion Li forms Li_2O ; sodium gives the peroxide, Na_2O_2 ; and potassium, rubidium and caesium give super oxides, MO_2 .



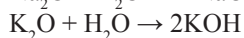
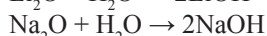
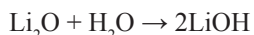
- The normal oxides contain O^{2-} ion, the peroxides contain O_2^{2-} ion and superoxides contain O_2^- ion. The peroxides and superoxides become more stable with increase in atomic number of the alkali metal. The formation and stability of these oxides can be explained on the basis of 'lattice energy effects'. Li^+ ion being a small ion has a strong positive field around it and can stabilize only a small anion, O^{2-} whereas Na^+ being a large cation can stabilize a large ion and so on.



- The monoxides are ionic, for example 2Li^+ and O^{2-} . Li_2O and Na_2O are pure white solids as expected, but surprisingly K_2O is pale yellow, Rb_2O is bright yellow and Cs_2O is orange. Metallic oxides are usually basic. The typical oxides M_2O are strongly basic oxides, and they react with water forming strong bases.



Example:



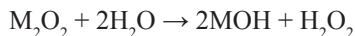
The crystal structures of Li_2O , Na_2O , K_2O and Rb_2O are anti-fluorite structures.

The basic nature of the oxides (M_2O) increases gradually on moving down in the group. The hydroxides (MOH) are colourless, strong alkaline and corrosive compounds. These are soluble in water and dissolve with the evolution of heat.

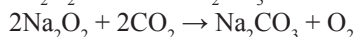
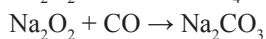
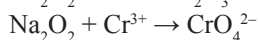
Solubility increases from LiOH to CsOH due to increasing ionic character. The hydroxides are thermally stable except LiOH. The relative strength of the hydroxides increases from LiOH to CsOH.



- All the peroxides contain the $[\text{—O—O—}]^{2-}$ ion. They are diamagnetic (all the electrons are paired), and are oxidizing agents. They may be regarded as salts of the dibasic acid H_2O_2 , and they react with water and acid, giving hydrogen peroxide H_2O_2 .



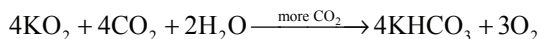
Na_2O_2 is pale yellow in colour. It is used industrially for bleaching wood pulp, paper and fabrics such as cotton and linen. It is a powerful oxidant and many of its reactions are dangerously violent, particularly with materials that are reducing agents such as aluminium powder, charcoal, sulphate and many organic liquids. Because it reacts with CO_2 in the air it has been used to purify the air in submarines and confined spaces, as it both moves CO_2 and produces O_2 .



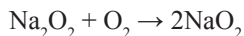
- Super-oxides are even stronger oxidizing agents than peroxides, and give both H_2O_2 and O_2 with either water or acids.



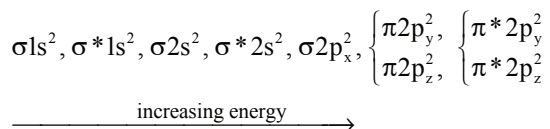
- KO_2 is used in space capsules, submarines, and breathing masks because it both produces oxygen and removes carbon dioxide. Both functions are important in life support systems.



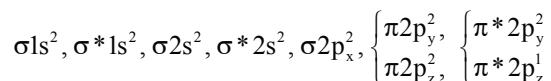
Sodium super-oxide cannot be prepared by burning the metal in oxygen (atmospheric pressure) but it is made commercially and in good yields by reacting sodium peroxide with oxygen at a high temperature and pressure (450°C and 300 atmospheres) in a stainless steel bomb.



The bonding in peroxides and super-oxides is described in the examples of molecular orbital treatment in Chapter 4. The peroxide ion $[\text{—O—O—}]^{2-}$ has 18 electrons, which occupy the molecular orbitals as shown:



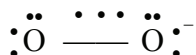
Thus the bond order is one, corresponding to a single bond. The superoxide ion $[\text{O}_2]^-$ has only 17 electrons, which give a bond order of 1.5.



- Generally, large atoms or ions form weaker bonds than small ones. The peroxide and superoxide ions are large, and it is noteworthy that the stability of the peroxides and super-oxides increases as the metal ions become larger. This shows that large cations can be stabilized by large anions, since if both ions are similar in size the coordination number will be high, and this gives a high lattice energy.

2.10 Alkali Metals (+2)

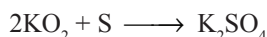
1. Superoxide ion O_2^- contains a three electron bond.



2. The peroxides and superoxides are strong oxidizing agents.
3. Peroxides react with water to give O_2 .



4. The peroxides are colourless and diamagnetic while the superoxides are paramagnetic and coloured. Superoxides are orange or yellow solids at room temperature.
5. The superoxides when heated with sulphur form sulphates.



Worked-out Examples

8. Among KO_2 , AlO_2^- , BaO_2 and NO_2^+ , unpaired electron is present in (IITJEE 1997)
(a) NO_2^+ and BaO_2 (b) KO_2 and AlO_2^- (c) KO_2 only (d) BaO_2 only

Ans. (c)

9. Why Li forms only oxides and not peroxide or superoxides?

Solution: Due to small size of Li, it has a strong positive field around it. On combination with the oxide anion (O^{2-}), the positive field of lithium ion restricts the spread of negative charge towards another oxygen atom and thus prevent the formation of higher oxide.

Carbonates, Bicarbonates

The carbonates are remarkably stable, and will melt before they eventually decompose into oxides at temperatures above 1000°C . As ionic nature of carbonates increases from Li to Cs (according to Fajan's rule because polarizing power of the cation decreases down the group), the stability of the carbonates as well as their solubility in water increase in the same order.



Li_2CO_3 decomposes on heating and insoluble in water. The abnormal behaviour of Li_2CO_3 towards heat can be explained in the following manner.

The Li^+ ion exerts a strong polarizing action and distorts the electron cloud of the nearby oxygen atom of the large CO_3^{2-} ion. This results in the weakening of the $\text{C}=\text{O}$ bond and strengthening of the $\text{Li}-\text{O}$ bond. This ultimately facilitates the decomposition of Li_2CO_3 into Li_2O and CO_2 . The lattice energy of Li_2O is higher than the lattice energy of carbonate. This also favours the decomposition of Li_2CO_3 .

Because Group I metals are so strongly basic, they also form solid bicarbonates (also called hydrogen carbonates). No other metals form solid bicarbonates, though NH_4HCO_3 also exists as a solid. Bicarbonates evolve carbon dioxide and turn into carbonates on gentle warming. This is one test for bicarbonates in qualitative analysis. The crystal structures of NaHCO_3 and KHCO_3 both show hydrogen bonding, but are different. In NaHCO_3 , the HCO_3^- ions are linked into an infinite chain, whilst in KHCO_3 a dimeric anion is formed.

Lithium is exceptional in that it does not form a solid bicarbonate, though LiHCO_3 can exist in solution. All the carbonates and bicarbonates are soluble in water.

Na_2CO_3 is used as washing soda to soften water in hard water areas, and NaHCO_3 is used as baking powder.

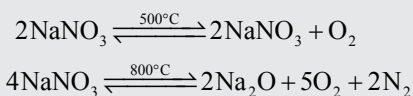
Worked-out Examples

10. Sodium nitrate decomposes above $\sim 800^\circ\text{C}$ to give

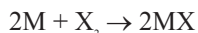
(IITJEE1998)

- (a) N_2 (b) O_2
 (c) NO_2 (d) Na_2O

Solution: (a), (b), (d) The nitrates of Group I decompose on heating into nitrites and to oxides on strongly heating temperature.



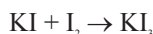
Reaction with Halogens: Alkali metals react with halogens to form metal halides. Metal halides are ionic in nature and have high melting and boiling points. All halides except lithium fluoride dissolve in water.



$\text{M} = \text{Li, Na, K, Rb or Cs}$

$\text{X} = \text{F, Cl, Br or I}$

Reactivity of alkali metals with particular halogen increases from Li to Cs. On other hand, reactivity of halogen decreases from F_2 to I_2 . Potassium, rubidium and caesium have the property of combining with extra halogen to form polyhalides.



Action with Halogens

Alkali metals react with X_2 directly and vigorously to form ionic halides MX ($\text{X} = \text{F, Cl, Br, I}$). These ionic substances are crystalline having high melting point and boiling point.

Except LiF , all other halides are water soluble. The insolubility of LiF is due to high lattice energy which results from the combination of small Li^+ ion with F^- anion.

Alkali metals react vigorously with halogens to form metal halides of the general formula MX where M is an alkali metal ion and X is the halide ion. Alkali metal halides are “ideal” ionic compounds, i.e., their physical and chemical properties can be interpreted with reasonable accuracy on the assumption that ions are present in solid state.

The structure and stability of alkali metal halides are determined in part by the lattice energies and radius ratio effects as discussed below:

(a) Lattice energy Effect

It is observed that lithium fluoride is insoluble in water whereas fluorides of other alkali metals are soluble. This is because the small lithium cation is in combination with small fluoride anion. This results in a high value for the lattice energy of LiF and thus it does not dissolve in water.

LiCl has exceptionally low melting point. LiCl lattice is very unstable because of the very small size of the Li^+ ion. The Li^+/Cl^- radius ratio is 0.33, which is much below the minimum (0.414) required for adopting NaCl type structure.

(b) Radius-ratio Effects

Alkali metal halides furnish a beautiful example of how coordination number increases with an increase in the size of the cation. The radius ratio in CsCl ($\text{Cs}^+/\text{Cl}^- = 0.93$) allows a coordination number 8 while radius ratio for NaCl ($\text{Na}^+/\text{Cl}^- = 0.52$) allows a coordination number of 6. Lithium, potassium and rubidium also exhibit coordination number 6 in their chlorides, though theoretically, lithium should show coordination number 4 only. This anomalous behaviour of lithium is thought to be due to more favourable lattice energy acquired by achieving a higher coordination number.

2.12 Alkali Metals (+2)

General Properties of Halides are given below:

(a) ΔH_f , Heat of formation (KJmol^{-1})

<i>Metal</i>	<i>MF</i>	<i>MCl</i>	<i>MBr</i>	<i>MI</i>
Li	612	398	350	271
Na	569	400	360	288
K	563	428	392	328
Rb	549	423	389	329
Cs	531	424	395	337

Observations on the above table:

(i) ΔH_f decreases as $\text{F}^- > \text{Cl}^- > \text{Br}^- > \text{I}^-$ for any given metal. Thus MF most and MI least stable.

(ii) ΔH_f of MF decreases down the group (lattice energy effect)

Stability, MF $\xrightarrow{\text{Li to Cs}}$ decreases

(iii) ΔH_f of MI(alkali iodides) increases down the group (stabilization of large cation by large anion).

Stability, MI $\xrightarrow{\text{Li to Cs}}$ increases

(iv) ΔH_f of MCl and MBr do not change regularly.

$\Delta H_f(\text{MCl})$ $\text{KCl} > \text{CsCl} > \text{RbCl} > \text{NaCl} > \text{LiCl}$

(MBr) $\text{CsBr} > \text{KBr} > \text{RbBr} > \text{NaBr} > \text{LiBr}$

(No concrete explanation is found for the above anomaly)

The stability follows the same order as ΔH_f .

(b) Lattice energies and mp (KCal mol^{-1} and $^{\circ}\text{C}$)

<i>Alkali metal</i>	<i>MF</i>		<i>MCl</i>		<i>MBr</i>		<i>MI</i>	
	LE	MP	LE	MP	LE	MP	IE	MP
Li	244	870	202	614	191	552	178	440
Na	215	992	184	800	175	747	163	662
K	190	857	167	700	160	742	149	682
Rb	182	833	161	717	154	677	145	638
Cs	172	715	155	642	149	636	140	621

Comments

(i) LE of LiX ($\text{X} = \text{F}, \text{Cl}, \text{Br}, \text{I}$) is highest.

(ii) Mp of LiX is lower (due to covalent nature in LiX) compared to NaX.

(iii) LE of MX ($\text{X} = \text{F}, \text{Cl}, \text{Br}, \text{I}$) decreases down the group.

(iv) Mp of NaX ($\text{X} = \text{F}, \text{Cl}, \text{Br}, \text{I}$) is greater than LiX (due to increase in ionic nature of NaX).

(v) Mp of KI is highest among alkali metal iodides.

(c) Solubility of CsX ($\text{X} = \text{F}, \text{Cl}, \text{Br}, \text{I}$)

<i>Property</i>	<i>CsF</i>	<i>CsCl</i>	<i>CsBr</i>	<i>CsI</i>
LE (KCalmol^{-1})	172	155	149	140
HE of CsX(KCalmol^{-1})	-181	-151	-143	-132
LE + HE	-9	+4	+6	+8
Solubility g/100 g water	370	186	108	79

Observations

- (i) CsF is most soluble (due to high HE of small F⁻ ion).
- (ii) Solubility decreases from CsF to CsI.
- (iii) CsI least soluble. It is due to low HE of large Cs⁺ and I⁻ ions.

(d) Solubility of some alkali metal halides.

<i>Property</i>	<i>CsF</i>	<i>CsCl</i>	<i>CsBr</i>	<i>CsI</i>
LE (KCalmol ⁻¹)	247	186	166	162
HE(KCalmol ⁻¹)	-240	-185	-168	-159
LE + HE	+7	+1	-2	+3
Solubility mol lit ⁻¹	0.11	6.1	11.0	7.5

Observations

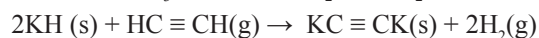
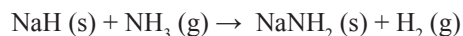
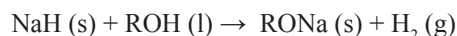
- (i) Solubility of LiF least (as LE + HE, most positive). LiF least soluble due to high lattice energy of LiF which results from combination of small F⁻ and Li⁺ ions.
- (ii) Solubility of NaI highest (as LE + HE, most negative).

Reaction with hydrogen: Alkali metals react with dry hydrogen at about 673 K to form colourless crystalline hydrides. All the alkali metal hydrides are ionic solids with high melting point.



Some important features of hydrides are:

- (i) Stability of hydrides decreases from Li to Cs. It is because of the fact that M – H bond becomes weaker due to increase in the size of alkali metals down the group.
- (ii) These hydrides react with water to form corresponding hydroxides and hydrogen gas.
- (iii) These hydrides are strong reducing agents and their reducing nature increases down the group. Alkali metals also form complex hydrides such as LiAlH₄ and NaBH₄ which are good reducing agents.
- (iv) All these hydrides react with proton donors such as water, alcohols, gaseous ammonia and alkynes liberating H₂ gas.



The order of reactivity of the alkali metals towards hydrogen decreases as we move down the group from Li to Cs. This is due to the reason that the lattice energies of these hydrides decreases progressively as the size of the metal cation increases and thus the stability of these hydrides decreases from LiH to CsH.

Worked-out Example

11. Assertion: The alkali metals can form ionic hydrides which contain the hydride ion H⁻.

Reason: The alkali metals have low electronegativity; their hydrides conduct electricity when fused and liberate hydrogen at the anode.

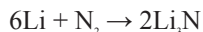
(IITJEE 1994)

- (a) Both S and E are true and E is the correct explanation of S,
- (b) Both S and E are true but E is not the correct explanation of S
- (c) S is true but E is false
- (d) S is false but E is true.

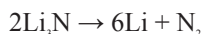
Ans. (a)

2.14 Alkali Metals (+2)

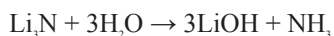
Reaction with nitrogen: Only lithium reacts in the nitrogen to form nitride.



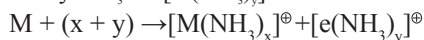
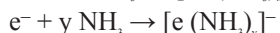
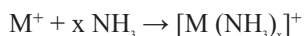
On heating to a high temperature, it decomposes to the constituent elements.



It reacts with water to form ammonia.

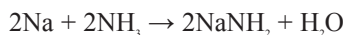


Reaction with ammonia: All alkali metals are soluble in liquid ammonia. Dilute alkali-metal–ammonia solutions are blue in colour. The blue colour solution of alkali metal ammonia shows certain characteristics which are explained on the basis of the formation of ammoniated metal cations and ammoniated electrons in the metal ammonia solution in following way:



The important characteristics of the alkali metal – ammonia are as follows:

- (i) **Conductivity:** It has high electrical conductivity due to presence of ammoniated electron in the cavities formed by electronic polarization between the electrons and ammonia molecule.
- (ii) **Paramagnetism:** The blue colour of the solution is paramagnetic due to presence of an unpaired electron in the cavities of ammonia solution.
- (iii) **Colour:** The blue colour of the solution is due to the excitation of free electrons to higher energy levels by the absorption of light in visible region.
- (iv) **Stability:** The solution is quite stable and can be considered as a dilute metal or an alloy in which electrons behave essentially as in a metal with the alkali metals slightly apart due to the presence of ammonia molecules. However, in the presence of a catalyst like iron oxide, the solution decomposes to form amide and hydrogen.



- (v) **Reducing property:** The free ammoniated electrons make the solution a very powerful reducing agent. The ammoniacal solution of an alkali metal is favoured as a reducing agent than its aqueous solution because in aqueous solution on the alkali metal being highly electropositive evolves hydrogen from water while its solution in ammonia is quite stable, provided no catalyst is present.

Worked-out Examples

12. Highly pure dilute solution of sodium in liquid ammonia:

(IITJEE 1998)

- (a) shows blue colour
- (b) exhibits electrical conductivity
- (c) produces sodium amide
- (d) produces hydrogen gas

Ans. (a, b)

13. **Assertion:** Alkali metals dissolve in liquid ammonia to give a blue solution.

Reason: Alkali metals in liquid ammonia give solvated species of the type $[\text{M}(\text{NH}_3)_n]^{n+}$ (M = alkali metal).

- (a) If both assertion and reason are correct and reason is the correct explanation of the assertion.
- (b) If both assertion and reason are correct but reason is not the correct answer of assertion.
- (c) If assertion is correct and reason is incorrect
- (d) If assertion is incorrect and reason is correct.

Ans. (b)

14. What happens when

- (i) Sodium hydrogen carbonate is heated?
- (ii) Sodium amalgam reacts with water?
- (iii) Fused sodium metal reacts with ammonia?

Ans.

- (i) $2\text{NaHCO}_3 \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O} + \text{CO}_2$
- (ii) $2\text{Na/Hg} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2 + 2\text{Hg}$
- (iii) $2\text{Na} + 2\text{NH}_3 \rightarrow 2\text{NaNH}_2 + \text{H}_2$

15. Sodium sulphate is soluble in water whereas barium sulphate is sparingly soluble because: **(IITJEE 1989)**

- (a) the hydration energy of sodium sulphate is more than its lattice energy
- (b) the lattice energy of barium sulphate is more than its hydration energy
- (c) the lattice energy has no role to play in solubility
- (d) the hydration energy of sodium sulphate is less than its lattice energy.

Ans. (a), (b)

Reaction with sulphur and phosphorous: All metals form sulphides and phosphides.

Reaction with mercury: Alkali metals combine with mercury to form amalgams. The reaction is highly exothermic in nature.

Anomalous Behaviour of Lithium

Difference between lithium and other alkali metals lithium shows the properties of alkali metals but at the same time it differs in many aspects. This is mainly due to the exceptionally smaller size of the Li atom. Li has the highest ionization energy and electronegativity as compared to other alkali metals.

Some differences:

1. Melting point and boiling point of lithium metal are much higher than those of other alkali metals.
2. Lithium is much harder than other alkali metals.
3. Lithium reacts least readily with oxygen forming oxide.
4. LiOH is less basic than other hydroxides therefore many of its salts are less stable Li_2CO_3 , LiNO_3 and LiOH , all form oxide on gentle heating, though the analogous compounds of the rest of the group elements are stable.
5. Lithium is the only alkali metal which directly reacts with nitrogen to form Li_3N .
6. LiHCO_3 is known in solution but not in solid state while the bicarbonates of other alkali metals are known in solid state.

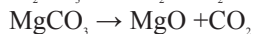
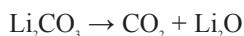
Diagonal relationship–Similarities with magnesium

Lithium resembles magnesium through diagonal relationship. The reason for resemblance of properties are:

1. On moving along a period from left to right, the charge on the ion increases while ionic size decreases hence polarizing power increases. On moving diagonally, ionic charge increases and ionic radius also increases, thus polarizing power remain nearly constant.
2. Electronegativity of Li and Mg are quite comparable.
3. Atomic radii and ionic radii of Li and Mg are not much different.

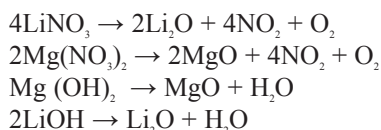
Lithium resembles magnesium in following properties:

- (i) Chlorides of both lithium and magnesium are deliquescent. These are soluble in alcohol as well as in pyridine.
- (ii) The carbonates, phosphates and fluorides of Li and Mg are partly or sparingly soluble in water.
- (iii) Carbonates of Li and Mg decompose on heating.



2.16 Alkali Metals (+2)

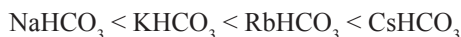
(iv) Hydroxide and nitrates of Li and Mg decompose on heating.



(v) Li and Mg react directly with nitrogen to form nitrides Li_3N and Mg_3N_2 .

With NH_3 , lithium forms imide Li_2NH while other alkali metals form amides MNH_2 .

1. Lithium does not form alum due to its small size.
2. Li metal is used as scavenger in metallurgy to remove O_2 and N_2 gases.
3. All alkali metals form body centred cubic lattices with coordination number 8.
4. Alkali metals dissolve in mercury forming amalgams. The dissolution is highly exothermic.
5. Lithium is strongest reducing agent due to highest positive oxidation potential (+ 3.05 V).
6. Potassium has three isotopes of mass numbers 39, 40 and 41 of which the first one predominates. However, due to the presence of traces (0.012%) of the radioactive isotope, K^{40} (half life = 10^9 years), it emits β -rays.
7. Alkali metal cations (Li^+ , Na^+ , K^+ , Rb^+ , Cs^+) have no unpaired electrons and are diamagnetic while alkali metal are paramagnetic due to the presence of unpaired ns^1 electrons.
8. Bicarbonates of alkali metals are crystalline solids. Their solubility in water and thermal stability increases in the order:



Complexes of alkali metals

Alkali metals are weak complex forming ions but polydentate ligands such as crown ethers and cryptates form highly stable complexes with alkali metal ions. Only the alkali metals which fit exactly into the available space (hole) of the polydentate ligand form stable complexes. The bonding between the metal ion and cyclic polyether is largely electrostatic. The name cryptate came from the fact that the metal ion is hidden in the structure.

SODIUM (Na)

Occurrence

Since alkali metals are highly reactive, they do not occur in free state in nature, Na and K are quite abundant in the earth's crust with 6th and 7th position in the order of abundance.

Ores of sodium

- (i) Chile Saltpeter, NaNO_3
- (ii) Glauber's Salt, $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$
- (iii) Trona, $\text{Na}_2\text{CO}_3 \cdot 2\text{NaHCO}_3 \cdot 3\text{H}_2\text{O}$
- (iv) Cryolite, Na_3AlF_6
- (v) Albite (soda feldspar), $\text{NaAlSi}_3\text{O}_8$
- (vi) Common salt or rock salt, NaCl
- (vii) Tincal (Borax), $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$

Extraction of alkali metal

Note:

- (i) The alkali metals are the strongest reducing agents known, therefore, they cannot be prepared by chemical reduction of their oxides.

- (ii) Since these metals are highly electropositive, they react with water. Therefore, attempted displacement of one element from solution by another higher in the electrochemical series will be unsuccessful.
- (iii) These elements cannot be extracted by the electrolysis of aqueous solutions because the metals liberated at once react with water.
- (iv) In order to overcome the above difficulties the **alkali metals are generally obtained by electrolysis of the fused halide with impurity added to lower the melting point.**

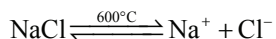
Down's Electrolysis Process for the Extraction of Sodium.

Now-a-days sodium metal is manufactured by this process.

Electrolyte—fused NaCl containing CaCl_2 and KF

Cathode—Fe

Anode—Graphite



At cathode: $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$

At anode: $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$

The electrolysis of pure NaCl presents the following difficulties:

- (a) The fusion temperature of NaCl is high, i.e., 803°C . At this temperature both sodium and chlorine are corrosive.
- (b) Sodium forms a metallic fog at this temperature.

To remove above difficulties, the fusion temperature is reduced to 600°C by adding CaCl_2 and KF. This is a cheaper method and chlorine is obtained as a byproduct.

Note: Na can also be obtained by electrolysis of fused NaOH at 330°C using Fe as cathode and Ni as anode (castner's process).

Worked-out Example

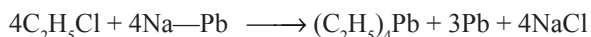
16. Of the following, the metals that cannot be obtained by electrolysis of the aqueous solution of their salts are: (IITJEE1990)

- | | |
|--------|--------|
| (a) Ag | (b) Mg |
| (c) Cu | (d) Al |

Ans. (b, d)

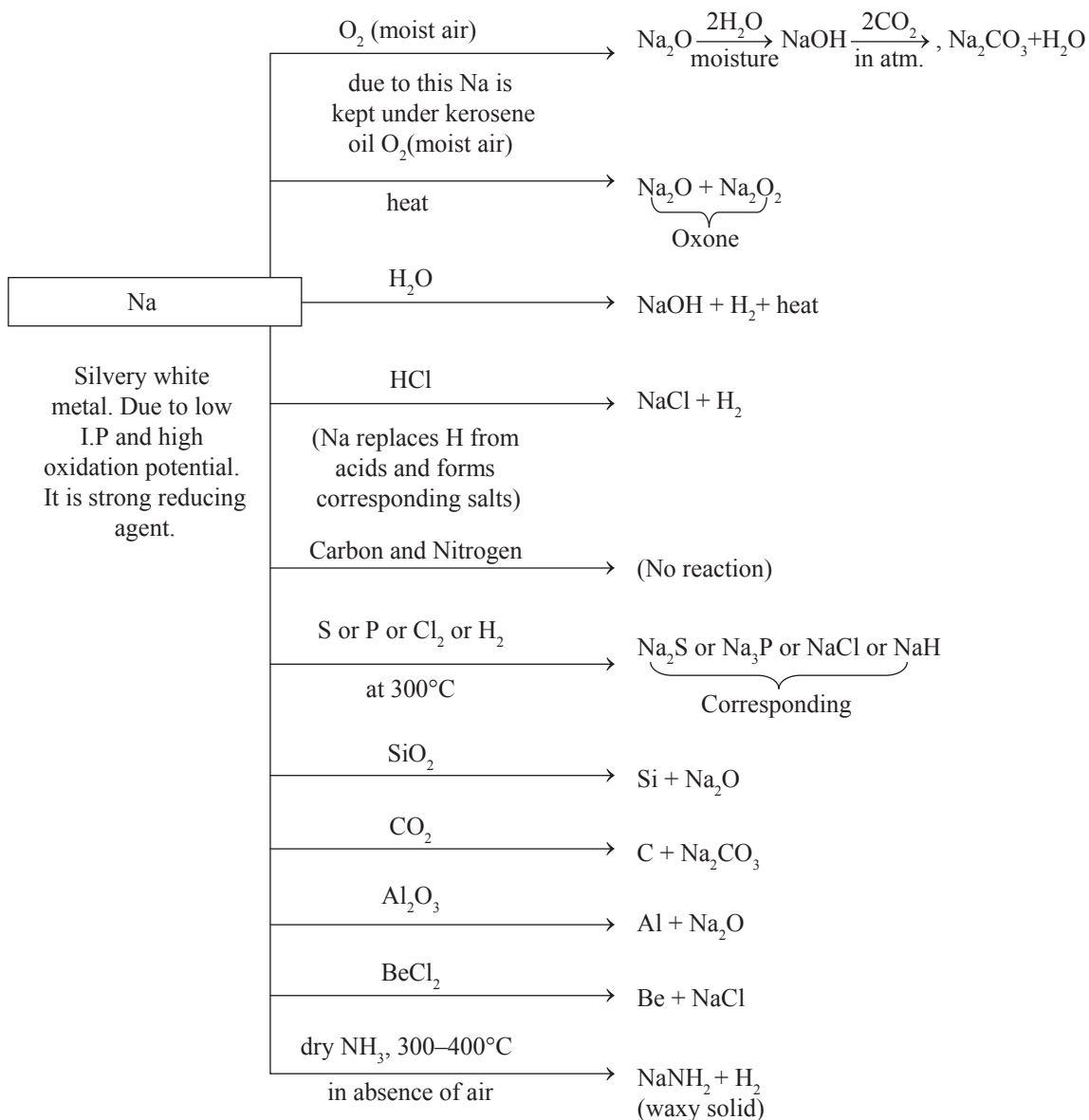
Uses: Sodium is used as

- (i) in the manufacture of various chemicals like sodium peroxide (Na_2O_2), sodamide (NaNH_2), sodium cyanide (NaCN) etc.
- (ii) for making lead tetraethyl (used as anti-knock in petrol).



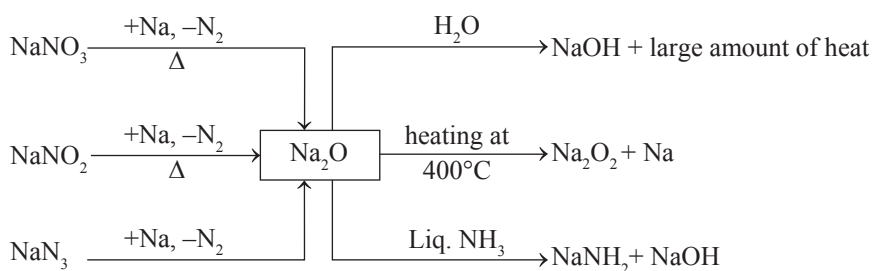
- (iii) in the preparation of sodium amalgam (used as a reducing agent).
- (iv) in molten state in nuclear reactors as heat transfer medium.
- (v) in sodium vapour lamps which emit yellow light.
- (vi) as a laboratory reagent for organic analysis.
- (vii) in high temperature thermometers in the form of sodium potassium alloy.
- (viii) for the extraction of C, Be, Mg and Si.

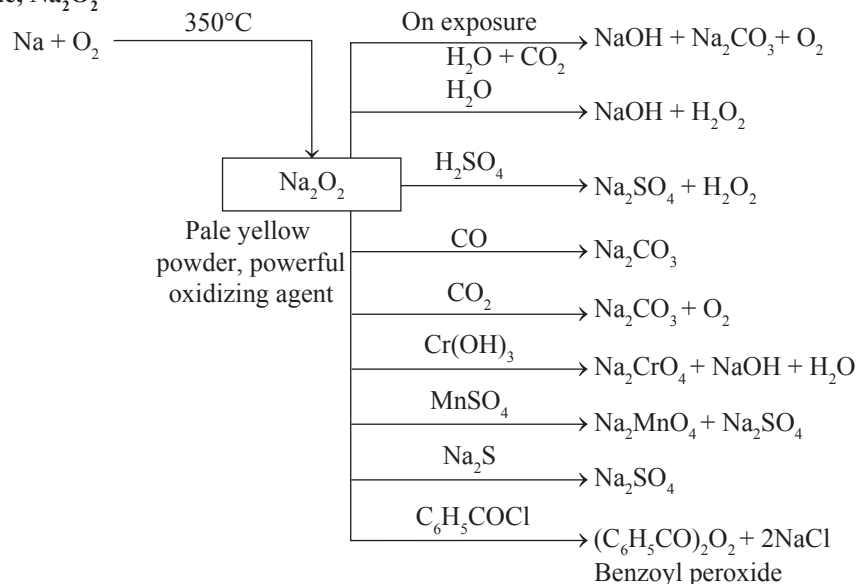
2.18 Alkali Metals (+2)



Compounds of Na

(A) Sodium Oxide, Na_2O –



(B) Sodium peroxide, Na_2O_2 

Uses: It is used:

- (i) as an oxidizing agent in laboratory.
- (ii) for the production of oxygen under the name oxone.
- (iii) for the purification of air.
- (iv) as a bleaching agent for delicate fibres like wool, silk, etc.
- (v) for the preparation of benzoyl peroxide.
- (vi) for the preparation of H_2O_2 .
 - A mixture of Na_2O_2 and dilute HCl is called soda bleach, commercially known as oxone and is used for bleaching of delicate fibres.

Worked-out Examples

17. The oxide that gives hydrogen peroxide on treatment with a dilute acid is: (IITJEE 1985)
- (a) PbO_2 (b) Na_2O_2 (c) MnO_2 (d) TiO_2

Ans. (b)

18. The compound(s) formed upon combustion of sodium metal in excess of air is (are) (IITJEE 2009)
- (a) Na_2O_2 (b) Na_2O (c) NaO_2 (d) NaOH

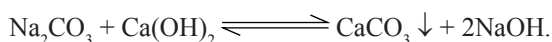
Ans. (a, b, d)

19. The species that do not contain peroxide ions are: (IITJEE 1992)
- (a) PbO_2 (b) H_2O_2 (c) SrO_2 (d) BaO_2

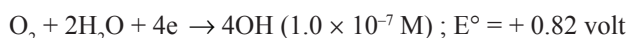
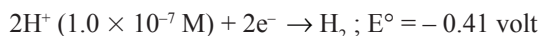
Ans. (a, c)

(C) Sodium Hydroxide (Caustic Soda), NaOH **Preparation**

- (i) **Caustic process** (Grossage process). In this process a 10-20% solution of sodium carbonate is reacted with calcium hydroxide.



- (ii) **Electrolytic process (Sodium Hydroxide).** This is produced commercially by electrolysis of a sodium chloride solution using mercury cathode and carbon anode. Sodium is discharged on mercury and forms an amalgam. This amalgam is treated with water to produce sodium hydroxide. The process is accomplished in three-chambered electrolysis cell with slate partitions reaching almost to the bottom. The cell has a thick layer of mercury on the floor. The side compartments carry the carbon anodes and the mercury serves as the cathode. The central compartment carries water and the other two a strong brine solution. On electrolysis, chlorine is given off at the anode and sodium on the mercury cathode forming sodium amalgam, which is then made to move into the middle compartment by a shaking device, where the amalgam reacts with water to form sodium hydroxide.
- (iii) An inspection of standard electrode potentials convinces us that it is more difficult to reduce Na^+ ion than H^+ ion. On electrolysis of a neutral brine solution we would expect release of H_2 at the cathode instead of deposition of sodium.



But proton transfer from water to a mercury surface needs a high activation energy. *Mercury has a high hydrogen overvoltage while sodium has a very low overvoltage on mercury.* So on electrolysis sodium is discharged on mercury in preference to hydrogen. The resulting amalgam is then reacted with water in the middle compartment of the cell to form sodium hydroxide (and H_2). It also follows from the above standard potentials that OH^- ion is a stronger reducing agent than Cl^- ion so that it will be easier to oxidize OH^- ion to O_2 at the anode than Cl^- to Cl_2 . Once again, our prediction goes wrong. At the anode, chlorine is evolved in preference to oxygen. *Oxygen has a high overvoltage on mercury.*

The reader should note that if we used an open cell then the anode product and the cathode product would have interacted, in the cold, to give NaOCl and, in the hot, NaClO_3 .

- (i) Nelson cell
(ii) Castner-Kellner cell

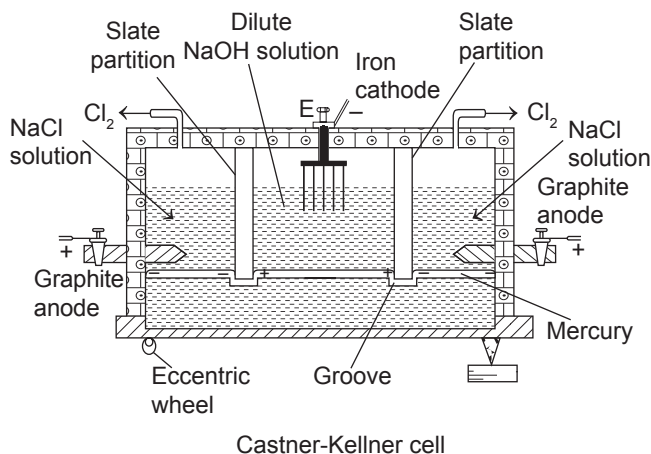
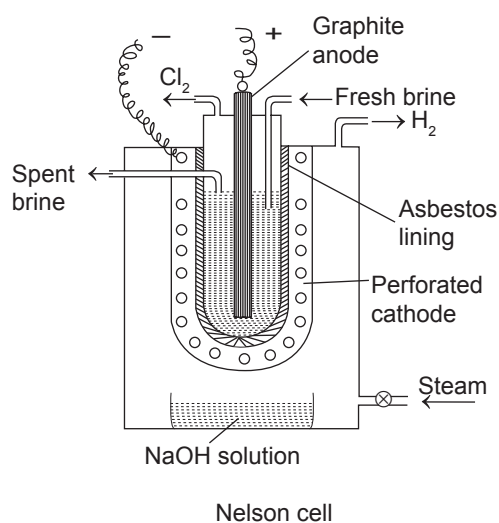


Fig. 2.2

Properties

NaOH Cannot be kept in atmosphere	1	CO_2	$\text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$
	2	HCl or H_2SO_4 or HNO_3 or H_3PO_4 or CH_3COOH	Corresponding salt + H_2O
	3	CO_2 (acidic oxide)	$\text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$
	4	SO_2	$\text{Na}_2\text{SO}_3 + \text{H}_2\text{O}$
	5	NO_2	$\text{NaNO}_3 + \text{NaNO}_2 + \text{H}_2\text{O}$
	6	Al_2O_3	$\text{NaAlO}_2 + \text{H}_2\text{O}$
	7	ZnO	$\text{Na}_2\text{ZnO}_2 + \text{H}_2\text{O}$
	8	SnO	$\text{Na}_2\text{SnO}_2 + \text{H}_2\text{O}$
	9	SnO_2	$\text{Na}_2\text{SnO}_3 + \text{H}_2\text{O}$
	10	PbO	$\text{Na}_2\text{PbO}_2 + \text{H}_2\text{O}$
	11	PbO_2	$\text{Na}_2\text{PbO}_3 + \text{H}_2\text{O}$
	12	Cl_2 or Br_2 or I_2	$\text{NaX} + \text{NaXO} + \text{H}_2\text{O}$
	13	when NaOH is (cold and dil.) + Cl_2 or Br_2 or I_2	$\text{NaX} + \text{NaXO}_3 + \text{H}_2\text{O}$
	14	P_4 (yellow phosphorus) + H_2O	$\text{NaH}_2\text{PO}_2 + \text{PH}_3$
	15	S	$\text{Na}_2\text{S}_2\text{O}_3 + \text{Na}_2\text{S} + \text{H}_2\text{O}$
	16	$\text{Si} + \text{H}_2\text{O}$	$\text{Na}_2\text{SiO}_3 + \text{H}_2$
	17	B	$\text{Na}_3\text{BO}_3 + \text{H}_2$
	18	Zn or Sn or Pb	$(\text{Na}_2\text{ZnO}_2 \text{ or } \text{Na}_2\text{SnO}_2 \text{ or } \text{Na}_2\text{PbO}_2)\text{H}_2$
	19	$\text{Al} + \text{H}_2\text{O}$	$\text{NaAlO}_2 + \text{H}_2$
	20	Sn or $\text{Pb} + \text{H}_2\text{O}$	$\text{Na}_2\text{SnO}_3 \text{ or } \text{Na}_2\text{PbO}_3 + \text{H}_2$
	21	Salt of Ni^{+2} , Fe^{+2} , Cr^{+3} , Mn^{+2} , Cu^{+2}	Corresponding insoluble hydroxide
	22	Salt of Zn^{+2} , Al^{+3} , Sn^{+2}	Corresponding insoluble hydroxide which dissolve in excess of NaOH
	23	Salt of Ag^+ , Hg^{+2}	Unstable hydroxides (e.g.,) $\downarrow \text{AgOH} \rightarrow \text{Ag}_2\text{O} + \text{H}_2\text{O}$ $\text{Hg}(\text{OH})_2 \rightarrow \text{HgO} + \text{H}_2\text{O}$
	24	Ammonium salt	NH_3 + sodium salt + H_2O
	25	CO 150–200°C/5–10 atm	HCOONa (Sodium formate)

■ Soda lime is $\text{NaOH} + \text{CaO}$.

Caustic property

Sodium hydroxide breaks down the proteins of skin to a pasty mass. On account of this property, it is commonly called as **Caustic Soda**.

Uses: Sodium hydroxide is used:

- (i) as a reagent in the laboratory.
- (ii) in petroleum refining.
- (iii) in the manufacture of soap.
- (iv) in the manufacture of sodium metal.
- (v) in the manufacture of paper and rayon.
- (vi) in the manufacture of dyes and drugs.
- (vii) as an absorber of gases.
- (viii) in the manufacture of sodium hypochlorite, sodium chlorate and sodium nitrite.
- (ix) for mercerizing cotton.

Worked-out Example

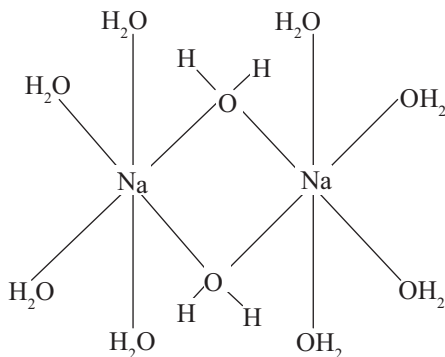
20. Solubility of sodium hydroxide increases with increase in temperature. (True/False)

(IITJEE 1985)

Ans. False Sodium hydroxide dissolves in water with the evolution of heat. Hence, its solubility decreases with increase in temperature.

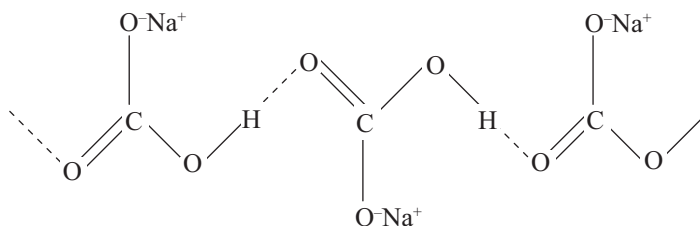
(D) Sodium Carbonate (Washing soda), $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$

The decahydrate $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ consists of the dinuclear $\text{Na}_2(\text{H}_2\text{O})_{10}^{2+}$ ion and CO_3^{2-} in ionic lattice. In the cation, each Na is in octahedral environment with six water molecules. Two water molecules act



The structure of $\text{Na}_2(\text{H}_2\text{O})_{10}^{2+}$ ion present in $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$

As bridging ligands through oxygen atoms. In the bicarbonate, the anion is hydrogen bonded HCO_3^- ion in a chain. The Na ion are held by the electrostatic attraction. Due to the chain anion, NaHCO_3 has a low solubility in water.



Sodium carbonate is generally prepared by a process called the ammonia-soda process or Solvay process as described below:

When carbon dioxide gas is bubbled through a brine solution saturated with ammonia, it results in the formation of sodium hydrogen carbonate.



Sodium hydrogen carbonate so formed precipitates out because of the common ion effect caused due to the presence of excess of sodium chloride. The precipitated NaHCO_3 is filtered off and then ignited to get sodium carbonate.



The filtrate after removal of NaHCO_3 contains ammonium salts such as NH_4HCO_3 and NH_4Cl . In order to recover NH_3 , the filtrate is treated with $\text{Ca}(\text{OH})_2$ and heated with steam.



Calcium chloride is formed as by-product during Solvay process.

It may be mentioned here that Solvay process cannot be employed for the manufacture of potassium carbonate because potassium bicarbonate is fairly soluble in water and hence does not precipitate easily (in carbonation tower).

Sodium carbonate is readily soluble in water and the crystals obtained from the solution are of the decahydrate $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$. This is known as washing soda. Sodium carbonate is an inexpensive alkali. It hydrolyses in water to give an alkaline solution.

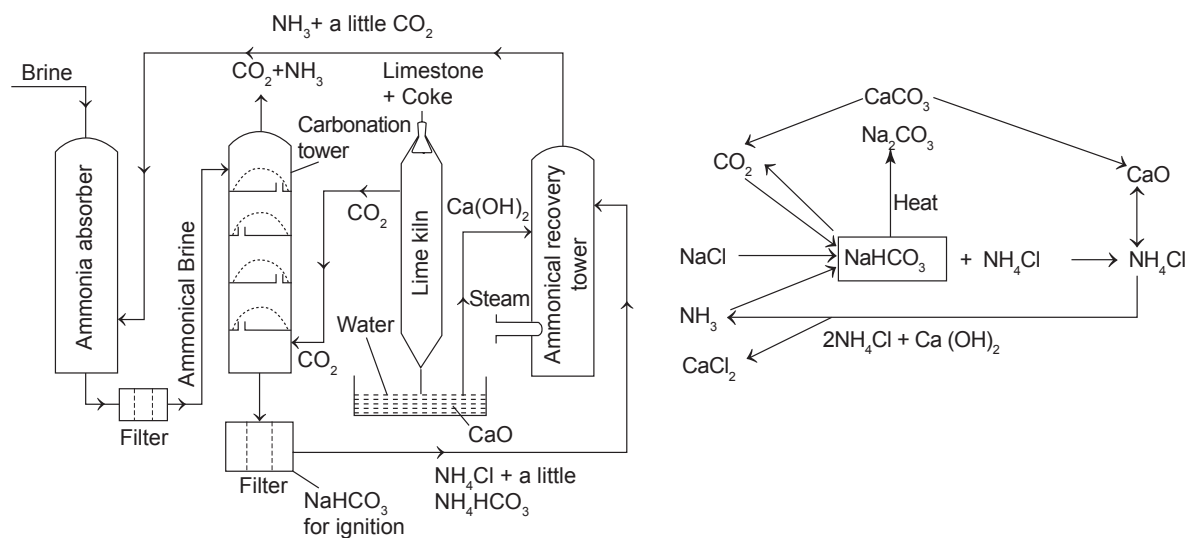
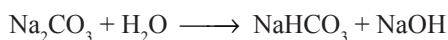
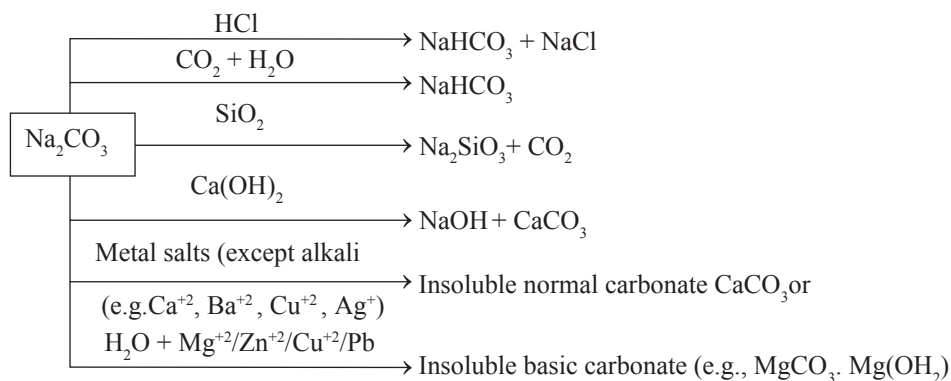


Fig. 2.3

Properties



2.24 Alkali Metals (+2)

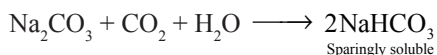
Uses: Sodium carbonate is used:

- (i) in laundries and in softening of water as washing soda.
- (ii) in the manufacture of glass, sodium silicate, paper, borax, soap powders, caustic soda, etc.
- (iii) as a laboratory reagent. The mixture of Na_2CO_3 and K_2CO_3 is used as a fusion mixture. It is used in quantitative analysis to standardise acid solutions. In qualitative analysis, it is used in the detection of acidic radicals especially of insoluble salts.
- (iv) in textile and petroleum refining.
- (v) for the preparation of various carbonates of metals.
- $\text{Na}_2\text{CO}_3 + \text{CaS}$ is called Black ash.

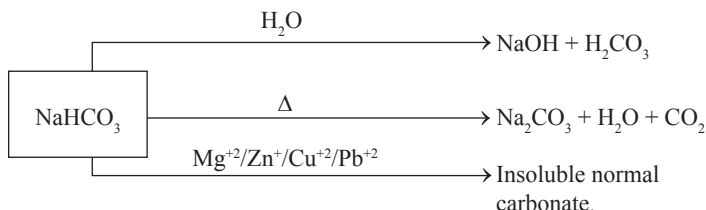
(E) Sodium Bicarbonate (Baking Soda), NaHCO_3

Preparation

It is obtained as the intermediate product in the solvay ammonia soda process. Normal carbonate can be changed to bicarbonate by passing carbon dioxide through its saturated solution.



Properties



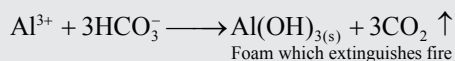
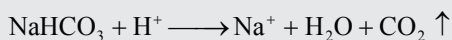
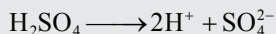
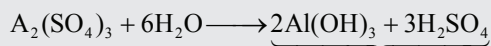
Uses:

- (i) It is used as a medicine (sodabicarb) to neutralize the acidity in the stomach.
- (ii) It is largely used for making baking powder. Baking powder is a mixture of potassium hydrogen tartrate and sodium bicarbonate.
- (iii) It is used in making effervescent drinks.
- (iv) It is used in fire extinguishers.
- (v) It is used for production of carbon dioxide.
- Sodium sesquicarbonate is $\text{Na}_2\text{CO}_3 \cdot \text{NaHCO}_3 \cdot 2\text{H}_2\text{O}$. It is neither deliquescent nor efflorescent and is used for wool washing.
- Fire extinguishers contain conc. H_2SO_4 and NaHCO_3 .
- Seidlitz powder contains NaHCO_3 .

Worked-out Example

21. Write down the chemistry of foam type fire extinguishers.

Ans. Foam type fire extinguishers contain $\text{Al}_2(\text{SO}_4)_3$ and NaHCO_3 .



(F) Sodium chloride (common salt), NaCl (table salt or rock salt or sea salt)

The main source of NaCl is sea water which contains 2.7–2.9% of the salt. It also occurs as salt beds. The evaporation of sea water yields crude sodium chloride which contains calcium sulphate (CaSO_4), sodium sulphate (Na_2SO_4), calcium chloride (CaCl_2), magnesium chloride (MgCl_2), etc., as impurities. CaCl_2 and MgCl_2 are undesirable impurities because they are deliquescent. For purification, a saturated solution of the crude salt is prepared and the insoluble impurities are removed by filtration. The solution is then saturated with HCl gas. Due to common ion effect, crystals of pure NaCl separate out. Chlorides of calcium and magnesium being more soluble remain in the solution.

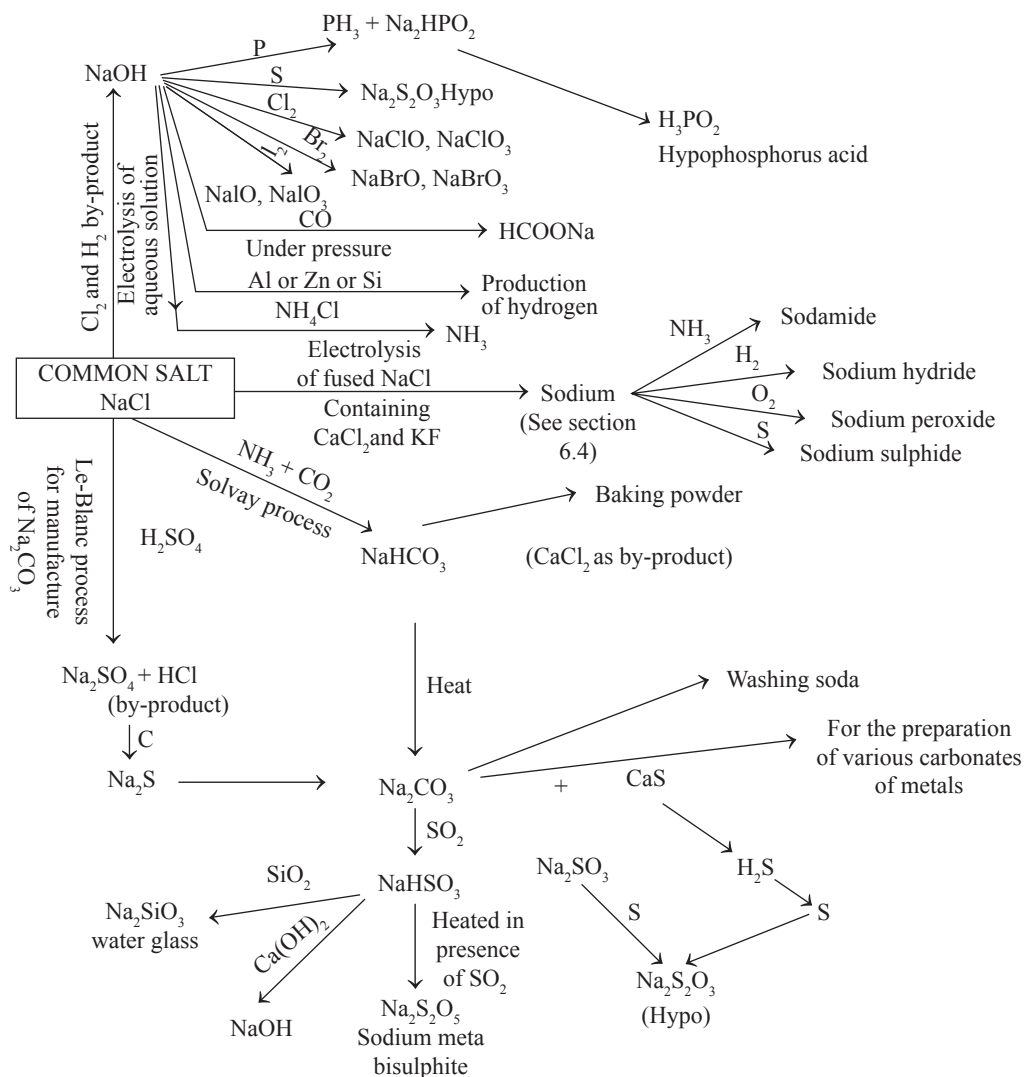


Fig. 2.4 An aqueous solution containing 28% NaCl (by weight) is called Brine

Worked-out Examples

22. A sodium salt on treatment with MgCl_2 gives white precipitate only on heating. The anion of the sodium salt is: (IITJEE 2004)

- (a) HCO_3^- (b) CO_3^{2-} (c) HNO_3^- (d) SO_4^{2-}

Ans. (a)

23. The pair of compounds which cannot exist together in solution is:

(ITJEE 1986)

- (a) NaHCO_3 and NaOH (b) Na_2CO_3 and NaHCO_3
 (c) Na_2CO_3 and NaOH (d) NaHCO_3 and NaCl

Ans. (a)

24. Molecular formula of Glauber's salt is:

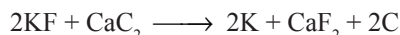
(IITJEE 1985)

- (a) $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ (b) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
 (c) $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ (d) $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$

Ans. (d)

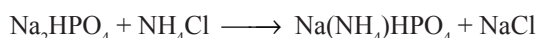
Potassium	Ores
Carnallite	$\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$
Kainite	$\text{KCl} \cdot \text{MgSO}_4 \cdot \text{MgCl}_2 \cdot 3\text{H}_2\text{O}$
Indian salt peter	KNO_3
Feldspar	$\text{K}_2\text{O} \cdot \text{Al}_2\text{O}_3 \cdot 6\text{SiO}_2$ (clay)
Polyhalite	$\text{K}_2\text{SO}_4 \cdot \text{MgSO}_4 \cdot \text{CaSO}_4 \cdot 6\text{H}_2\text{O}$
Sylvine	KCl

Metallic potassium is not of much demand as cheaper sodium can be used for the same purpose. However, it can be prepared by carrying electrolysis of fused potassium hydroxide or by electrolysis of fused mixture of KCl and CaCl_2 . Potassium is manufactured by reducing the fluoride with calcium carbide at 1000°C in steel cylinders.

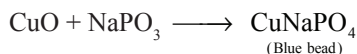
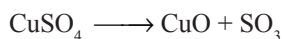
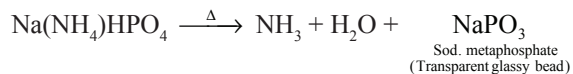


- Microcosmic salt is $\text{Na}(\text{NH}_4)\text{HPO}_4 \cdot 4\text{H}_2\text{O}$

It is prepared by dissolving molecular proportions of Na_2HPO_4 and NH_4Cl in water and crystallizing the contents.



Like Borax, it is used for performing 'bead test' for detecting coloured ions in qualitative inorganic analysis.



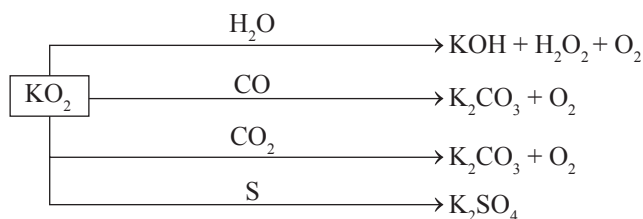
Compounds of Potassium

(A) **Oxides:** Three oxides of potassium are known:

- (a) potassium monoxide, K_2O .
 (b) potassium dioxide (tetraoxide or superoxide), KO_2 or K_2O_4 .
 (c) potassium sesquioxide, K_2O_3 .

Preparation

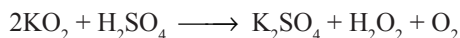
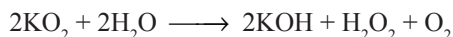
- (i) $\text{K} + \text{O}_2 \longrightarrow \text{KO}_2$
 (ii) $2\text{KOH} + \text{O}_3 \longrightarrow 2\text{KO}_2 + \text{H}_2\text{O}$
 (iii) $4\text{K} \text{ (dissolved in liquid } \text{NH}_3\text{)} \xrightarrow{3\text{O}_2} 2\text{K}_2\text{O}_3$

Properties**Uses:**

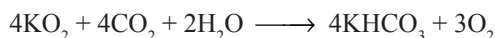
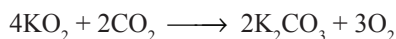
- (i) as an oxidizing agent
- (ii) as air purifier in space capsules, submarines and breathing masks as it both produces oxygen and removes carbon dioxide.

 KO_2 is better than K_2O because of the following reasons.

- (i) It is a better oxidizing agent since it produces oxygen and H_2O_2 both in aqueous solution and the acidic medium.



- (ii) KO_2 finds applications to be used in space capsules, submarines and breathing masks, because it both produces O_2 and removes CO_2 and CO . Both functions are important in life support system.

**(B) KOH**

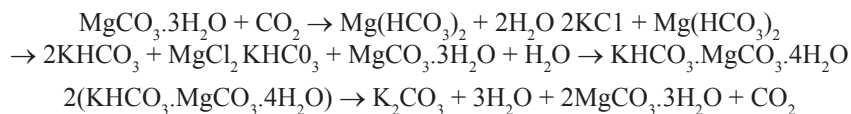
Caustic potash is prepared by methods similar to those used for the preparation of caustic soda.

(C) KCl

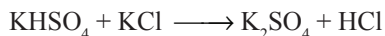
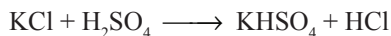
KCl exists in nature as Sylvine. It is prepared by fractional crystallization of carnallite, $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$.

(D) K_2CO_3 (Potassium Carbonate)

It is also called **Potash** or **Pearl ash**: The Solvay process cannot be applied in this case since potassium hydrogencarbonate is too soluble to be precipitated by ammonium hydrogencarbonate. Hydrated magnesium carbonate is added to a cold concentrated solution of KCl and the suspension is treated with CO_2^- . Sparingly soluble potassium hydrogen magnesium carbonate separates. This is filtered and heated with water when magnesium carbonate precipitates out and CO_2 is evolved. The CO_2 and the magnesium carbonate are reused in the cycle. The filtrate is concentrated and potassium carbonate recovered.



Le-Blanc process: KCl is first converted into K_2SO_4 . K_2SO_4 is then heated with CaCO_3 and carbon.



It is a white powder, deliquescent in nature. It is highly soluble in water.

It is used in the manufacture of hard glass. The mixture of K_2CO_3 and Na_2CO_3 is used as a fusion mixture in laboratory.

- A mixture of $\text{K}_2\text{CO}_3 + \text{Na}_2\text{CO}_3$ is known as fusion mixture.
- K_2CO_3 is also known as pearl ash.
- Baking powder has a composition of starch, sodium bicarbonate and potassium hydrogen tartrate.

2.28 Alkali Metals (+2)

- Both NaCN and KCN are poisonous compounds but poisonous nature of KCN is more than that of NaCN.
- Potassium salts of fatty acids are used to make soft soaps because they are more soluble than those of sodium.

Some Noteworthy Points Group 1 Trends

1. All the metals are malleable and they become softer down the column. Lithium is sufficiently hard that it is necessary to cut it with a knife, but rubidium and caesium have the consistency of putty.
2. The reactivities of the metals towards O_2 and water increases down the column. Lithium reacts slowly with water, sodium vigorously, potassium causes the evolved hydrogen to ignite, and rubidium and caesium react explosively. When Li is burned in oxygen (1 atmosphere) it forms Li_2O primarily, sodium forms the peroxide. Na_2O_2 , and K, Rb, and Cs form superoxides, MO_2 .
3. Lithium does not replace the proton in $PhC \equiv CH$ whereas the remaining elements do.
4. They all form a wide range of salts which exhibit typical ionic properties—they have high melting points and are water soluble, giving conducting solutions. The halides are ionic and are not hydrolysed. The oxides and hydroxides are basic and the hydrides are ionic, basic, and strong reducing agents.
5. The thermal stabilities of the carbonates, nitrates, sulphates, peroxides, and superoxides increase down the column.
6. The reverse order of reactivity is observed for the reactions of the alkali metals with N_2 and carbon. Lithium is unique in reacting with N_2 to form the purple nitride, Li_3N , and only Li and sodium react with carbon to form the carbides M_2C_2 ($M = Li$ or Na). Both of these reactions are thermodynamically favourable because of the high lattice enthalpies of the nitride and carbide anions associated with small cations.
7. The high charge/size ratio for lithium leads to other anomalies, E.g., LiH is more thermally stable than the other hydrides of the Group; the carbonate is much less stable. Also, lithium salts are much more soluble in organic solvents. Lithium and sodium form a number of hydrated solid salts, potassium forms a few and Rb and Cs none.
8. The solubilities of the hydroxides in water increase down the column:



9. Alkali metals do not readily form complexes with ligands such as NH_3 and CN^- , but polydentate ligands such as crown ethers and cryptands with oxygen and nitrogen donor atoms do form stable complexes. The bonding in the complexes is primarily electrostatic and the relative sizes of the cation and the cavity is important. For example, for complexes with the 18-crown-6 ether the order of stability is:



Smaller ring sizes lead to a preference for coordination to Na^+ and ultimately Li^+ (12-crown-4 and 15-crown-5) and larger ring sizes for Rb^+ and Cs^+ (24-crown-8).

10. Lithium alkyls are widely used as reagents in organic chemistry because they are readily synthesized, they are soluble in organic solvents and they provide a slightly more reactive source of carbanions than Grignard reagents. These properties arise because the alkyl lithium compounds form oligomers in the form of rings or polyhedra which have the hydrocarbon residues dominating the surface. In contrast, sodium and potassium alkyls are more ionic and generally adopt infinite 3-D structures in the solid state. They are not soluble in hydrocarbons and are extremely air and moisture sensitive.

Anomalous properties associated with lithium

The high enthalpy of atomization of lithium metal, the high hydration enthalpy of Li^+ and its small size make it stand out from the other members of the series in the following ways.

1. Lithium reacts very slowly with water, yet it is the only alkali metal to form a nitride with N_2 .
2. Lithium salts are generally less soluble in water, but more soluble in organic solvents.
3. Its salts are much less thermally stable because, for lithium, the lattice energy of the oxide which is formed is much greater than those of the original salts.



4. The greater electronegativity of lithium leads to the formation of less polar organometallic compounds with oligomeric structures, rather than the 3-D infinite structures for the heavier elements of the Group.
5. The high charge/size ratio of Li^+ leads to more stable complexes with simple ligands containing oxygen and nitrogen donors.

ONE CORRECT OPTION TYPE QUESTIONS

- Mostly metals are hard, but metals of Ist A group are soft, because:
 - their metallic bond is weak
 - they have close packing
 - they have face-centered cubic lattice
 - their b.p. is low
- In the preparation of sodium carbonate which of the following is used:
 - slaked lime
 - lime stone
 - lime
 - quick lime
- Molten sodium is used in nuclear reactors to
 - absorb neutrons in order to control the chain reaction
 - slow down the fast neutrons
 - absorb the heat generated by nuclear fission
 - extract radio-isotopes produced in the reactor
- Which is used in purification of air in the space craft?
 - Slaked lime
 - Quick lime
 - Potassium superoxide
 - CaCl₂
- KO₂ finds use in oxygen cylinders used for space and submarines. The fact(s) related to such use of KO₂ is/are:
 - it produces O₂
 - it produces O₃
 - it absorbs CO₂
 - it absorbs both CO and CO₂
- Li does not resemble other alkali metals in following properties:
 - Li₂CO₃ decomposes into oxides while other alkali carbonates are thermally stable
 - LiCl is predominantly covalent
 - Li₃N is stable
 - All of the above
- Molten sodium chloride conducts electricity due to the presence of:
 - free electrons
 - ions
 - Na atom
 - Cl atom
- Na and Li are placed in dry air. We get:
 - NaOH, Na₂O, Li₂O
 - Na₂O, Li₂O
 - Na₂O, Li₂O, Li₃N, NH₃
 - Na₂O, Li₃N, Li₂O
- The solubility of MClO₄ (M = Li, Na, K, Rb) increases in the order:
 - LiClO₄ < NaClO₄ < KClO₄ < RbClO₄
 - RbClO₄ < KClO₄ < NaClO₄ < LiClO₄
 - NaClO₄ < RbClO₄ < LiClO₄ < KClO₄
 - KClO₄ < LiClO₄ < NaClO₄ < RbClO₄
- The reaction of sodium is highly exothermic with water. The rate of reaction is lowered by:
 - lowering the temperature
 - mixing with alcohol
 - mixing with acetic acid
 - making an amalgam
- What product is obtained after electrolysis in Solvay through cell?
 - NaOH
 - Sodium amalgam
 - Na
 - Na₂CO₃
- Sodium carbonate solution is alkaline due to:
 - hydrolysis of Na⁺
 - hydrolysis of CO₃²⁻
 - hydrolysis of both Na⁺ and CO₃²⁻ ions
 - none of the above
- Consider the following statements:
 - Cs⁺ is highly hydrated.
 - Among Li, Na, K and Rb, Li has highest mp.
 - Among the alkali metals only Li form a nitride.
 Out of these statements:
 - I, II and III are correct
 - I and II are correct
 - II and III are correct
 - I and III are correct
- The most abundant alkali metal in nature is
 - lithium
 - sodium
 - potassium
 - caesium

2.30 Alkali Metals (+2)

15. In the Castner process for the extraction of sodium, the anode is made of
 - (a) nickel
 - (b) iron
 - (c) copper
 - (d) sodium
16. Sodium sulphate is soluble in water whereas barium sulphate is sparingly soluble because:
 - (a) the hydration energy of sodium sulphate is more than its lattice energy
 - (b) the lattice energy of barium sulphate is less than its hydration energy
 - (c) the lattice energy has no role to play in solubility
 - (d) the hydration energy of sodium sulphate is less than its lattice energy
17. Sodium hasas compared to potassium.
 - (a) less electronegativity
 - (b) more ionization potential
 - (c) larger atomic radius
 - (d) lower melting point
18. Which of the following statements is false for alkali metals?
 - (a) Li is strongest reducing agent
 - (d) Na is amphoteric in nature
 - (c) Li^+ is exceptionally small
 - (d) All alkali metals give blue coloured solutions in liquid ammonia
19. The metallic luster exhibited by sodium is explained by:
 - (a) diffusion of sodium ions
 - (b) oscillation of mobile valence electrons
 - (c) existence of free protons
 - (d) existence of body centred cubic lattice
20. Li_2SO_4 is not isomorphous with sodium sulphate:
 - (a) due to small size of lithium
 - (b) due to high hydrogen number of lithium
 - (c) due to high ionization energy of lithium
 - (d) none of the above
21. For the preparation of sodium thiosulphate by "Spring's reaction", the reactants used are:
 - (a) $\text{Na}_2\text{S} + \text{Na}_2\text{SO}_3 + \text{Cl}_2$
 - (b) $\text{Na}_2\text{S} + \text{SO}_2$
 - (c) $\text{Na}_2\text{S} + \text{Na}_2\text{SO}_3 + \text{I}_2$
 - (d) $\text{Na}_2\text{SO}_3 + \text{S}$
22. Which hydrides are formed from Li to Cs with ease?
 - (a) decreases
 - (b) increases
 - (c) remains the same
 - (d) none of these
23. In certain matters, lithium differs from other alkali metals, the main reason for this is
 - (a) small size of lithium atom and Li^+
 - (b) extremely high electropositivity of Li
 - (c) greater hardness of Li
 - (d) hydration of Li^+ ion
24. Identify the correct statement. Elemental sodium
 - (a) can be prepared and isolated by electrolyzing an aqueous solution of sodium chloride
 - (b) is a strong oxidizing agent
 - (c) is insoluble in ammonia
 - (d) is easily oxidized
25. In view of their low ionization energies, the alkali metals are:
 - (a) weak oxidizing agents
 - (b) strong reducing agents
 - (c) strong oxidizing agents
 - (d) weak reducing agents
26. When sodium is treated with sufficient oxygen/air, the product obtained is
 - (a) Na_2O
 - (b) Na_2O_2
 - (c) NaO_2
 - (d) NaO
27. What is the reaction occurring at the anode in Down's process for the extraction of sodium?
 - (a) $2\text{Cl}^- \longrightarrow \text{Cl}_2 + 2\text{e}$
 - (b) $4\text{OH}^- \longrightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}$
 - (c) $\text{Na}^+ + \text{e} \longrightarrow \text{Na}$
 - (d) $\text{NaOH} \longrightarrow \text{Na}^+ + \text{OH}^-$
28. On dissolving moderate amount of sodium metal in liquid NH_3 at low temperature, which one of the following does not occur?
 - (a) Blue coloured, solution is obtained
 - (b) Na^+ ions are formed in the sodium
 - (c) Liquid NH_3 because good conductor of electricity
 - (d) Liquid NH_3 remains diamagnetic

COMPREHENSION BASED QUESTIONS

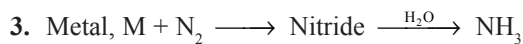
The first element of group in many ways differs from the rest of the elements of the group. These differences may be attributed due to following factors:

- (i) The small size
- (ii) The high electronegativity and
- (iii) The non-availability of d-orbitals.

The first element of group often shows resemblance to the second element of the neighbouring group on the right. This is termed diagonal relationship. Lithium and magnesium show resemblance in a number of properties.

1. Lithium is very similar to that of magnesium. The reason is:
 - (a) both have same size
 - (b) both are found in native state
 - (c) both have same electronic configuration
 - (d) both have same ionization potentials
2. Which one is the wrong statement?
 - (a) Lithium has greater hardness in comparison to other alkali metals.

- (b) LiHCO_3 like $\text{Mg}(\text{HCO}_3)_2$ is not known in solid state.
- (c) Lithium and magnesium form nitrides with nitrogen but other alkali metals do not.
- (d) Fluorides of alkali metals are highly soluble in water.



Metal M can be:

- (a) Na
- (b) K
- (c) Li
- (d) Mg

4. Sodium and lithium are placed in dry air, we get:

- (a) NaOH , Na_2O , Li_2O
- (b) Na_2O , Li_2O
- (c) Na_2O , Li_2O , Li_3N , NH_3
- (d) Na_2O , Li_3N , Li_2O

5. Which will give NO_2 on heating?

- (a) NaNO_3
- (b) LiNO_3
- (c) NH_4NO_3
- (d) NH_4NO_2

ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

1. The pairs of compounds which cannot exist together in aqueous solution are:
 - (a) NaH_2PO_4 and Na_2HPO_4
 - (b) Na_2CO_3 and NaHCO_3
 - (c) NaOH and NaH_2PO_4
 - (d) NaHCO_3 and NaOH
2. Highly pure dilute solution of sodium in liquid ammonia:
 - (a) shows blue colour
 - (b) exhibits electrical conductivity
 - (c) produces sodium amide
 - (d) produces hydrogen gas
3. Alkali metals are characterized by:
 - (a) good conductor of heat and electricity
 - (b) high oxidation potentials
 - (c) high melting points
 - (d) solubility in liquid ammonia
4. Li has following abnormal behaviour in its group:
 - (a) Lithium carbonate decomposes into its oxide on heating unlike other elements
 - (b) LiCl is covalent in nature

- (c) Li_3N is stable compound
- (d) LiCl is poor conductor of electricity in molten state

5. Select wrong statements about alkali metals:

- (a) all form (M NH_2) amide
- (b) all form superoxides (MO_2)
- (c) all form ionic hydrides (M H)
- (d) all form nitrides

6. Which among the following compounds are paramagnetic?

- (a) KO_2
- (b) K_2O_2
- (c) K_2O
- (d) NO_2

7. Which of the following carbonates do not evolve CO_2 on heating?

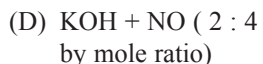
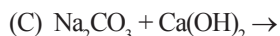
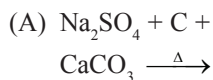
- (a) Li_2CO_3
- (b) MgCO_3
- (c) Na_2CO_3
- (d) K_2CO_3

8. Which of the following is/are correct?

- (a) sodium thiosulphate is called hypo
- (b) sodium peroxide is called oxone
- (c) potassium carbonate is called pearl ash
- (d) sodium nitrate is called Indian nitre

MATCH THE COLUMN TYPE QUESTIONS

1. Match the products of given reaction column (I) with the properties in column (II).

Column-I**Column-II**

(P) One of the products has sp^2 hybridization of central atom.

(Q) One of the products has sp^3 hybridization of central atom.

(R) One of the products is obtained as insoluble precipitate

(S) One of the products is neutral oxide

2. Match the particular given in column (I) with the compounds (s) in column (II)

Column-I

(A) Solvay process

(B) Evolve CO_2 – on heating

Column-II

(P) NaCl

(Q) Na_2O_2

(C) Aqueous solution is neutral towards litmus

(D) Oxone

(S) Na_2CO_3

3. Match the compounds listed in Column-I with the related uses/particulars listed in column-II using the given codes

Column-I

(A) Calcium hypochlorite

(B) Calcium sulphate hemi hydrate

(C) Potassium super oxide

(D) Calcium oxide

Column-II

(P) Used as air purified in submarine

(Q) Used for 'Softening' water

(R) Plaster of Paris

(S) Used in preparation of sodium hydroxide

ASSERTION/REASON TYPE QUESTIONS

The following questions consist of an Assertion (A) and Reason (R). Use the following key to choose the appropriate answer.

- If both assertion and reason are correct, and reason is the correct explanation of the assertion.
- If both assertion and reason are correct, but reason is not the correct explanation of the assertion.
- If assertion is correct but reason is incorrect.
- If assertion is incorrect but reason is correct.

1. **Assertion:** Lithium resembles magnesium, diagonally placed in IIA group.

Reason: The sizes of lithium and magnesium atoms and their ions (Li^+ and Mg^{2+}) are nearly the same.

2. **Assertion:** Alkali metals do not occur in native state.

Reason: Alkali metals are highly reactive metals.

3. **Assertion:** Sodium metal when dissolved in liquid ammonia forms a blue coloured solution.

Reason: The blue solution is a good conductor of electricity.

4. **Assertion:** Among the alkali metals, caesium salts exhibit the maximum electrical conductance in aqueous solutions.

Reason: Bigger the radius of the hydrated cation, higher is the electrical conductance of the aqueous solution.

5. **Assertion:** Sodium thiosulphate is used in photography.

Reason: Sodium thiosulphate is photosensitive.

6. **Assertion:** The alkali metals form ionic hydrides which contain the hydride ion, H^- .

Reason: The alkali metals have low electronegativities.

RECENT YEARS' QUESTIONS OF IIT-JEE AND AIEEE

1. Which of the following statements is correct for CsBr_3 ? [IIT 1996]
 - (a) It is a covalent compound
 - (b) It contains Cs^{2+} and Br^- ions
 - (c) It contains Cs^+ , Br^- and Br_2 lattice molecules
 - (d) It contains Cs^+ and Br_3^- ions
2. Which of the alkali metal chloride is expected to have highest m.p.? [AIEEE 2005]

(a) LiCl	(b) NaCl
(c) KCl	(d) RbCl
3. KO_2 is used in oxygen cylinder in space air craft and submarines because it: [AIEEE 2002]
 - (a) absorbs CO_2 and increase O_2 content
 - (b) eliminate moisture
 - (c) absorbs CO_2
 - (d) produces O_2
4. The compound(s) formed upon combustion of sodium metal in excess air is(are):

(a) Na_2O_2	(b) Na_2O
(c) NaO_2	(d) NaOH

ANSWER KEYS

ONE CORRECT OPTION TYPE QUESTIONS

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (a) | 2. (b) | 3. (c) | 4. (c) | 5. (ac) | 6. (d) | 7. (b) | 8. (d) | 9. (b) | 10. (d) |
| 11. (b) | 12. (b) | 13. (c) | 14. (b) | 15. (a) | 16. (a) | 17. (b) | 18. (b) | 19. (b) | 20. (a) |
| 21. (c) | 22. (a) | 23. (a) | 24. (d) | 25. (b) | 26. (b) | 27. (a) | 28. (d) | | |

COMPREHENSION BASED QUESTIONS

- | | | | | |
|--------|--------|-----------|--------|--------|
| 1. (a) | 2. (d) | 3. (c, d) | 4. (d) | 5. (b) |
|--------|--------|-----------|--------|--------|

ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

- | | | | | | | | |
|-----------|-----------|--------------|--------------|-----------|-----------|-----------|--------------|
| 1. (c, d) | 2. (a, b) | 3. (a, b, d) | 4. (a, b, c) | 5. (b, d) | 6. (a, d) | 7. (c, d) | 8. (a, b, c) |
|-----------|-----------|--------------|--------------|-----------|-----------|-----------|--------------|

MATCH THE COLUMN TYPE QUESTIONS

- | | | | |
|-----------------|----------------|------------|---------------|
| 1. (A–P, Q, R); | (B – P, Q, R); | (C – P, R) | (D – P, Q, S) |
| 2. (A– R, S); | (B – R); | (C – P) | (D – P) |
| 3. (A– S); | (B – R); | (C – P); | (D – P) |

ASSERTION/REASON TYPE QUESTIONS

- | | | | | | |
|--------|--------|--------|--------|--------|--------|
| 1. (a) | 2. (a) | 3. (b) | 4. (c) | 5. (c) | 6. (a) |
|--------|--------|--------|--------|--------|--------|

RECENT YEARS' QUESTIONS OF IIT-JEE AND AIEEE

- | | | | |
|--------|--------|--------|--------|
| 1. (d) | 2. (a) | 3. (a) | 4. (a) |
|--------|--------|--------|--------|

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Group-II Alkaline Earth Metals

INTRODUCTION

Table 3.1 Physical Data of Group 2 Elements

	<i>Atomic number</i>	<i>Electronic configuration</i>	<i>First ionization energy (ev)</i>	<i>Standard electrode potential (volts) $E^\circ [M^{+2}(aq) + 2e^-]$</i>	<i>Atomic radius (Å)</i>	<i>Ionic radius(Å)</i>	<i>M.P. °C</i>	<i>B.P. °C</i>
BE	4	$1s^2 2s^2$	9.32	-1.85	0.90	0.31	1278	2500
MG	12	$-2p^6 3s^2$	7.64	-2.37	1.3	0.72	651	1105
CA	20	$-3p^6 3s^2$	6.11	-2.87	1.74	0.99	851	1494
SR	38	$-3d^{10} 4p^6 5s^2$	5.69	-2.89	1.91	1.13	800	1381
BA	56	$-4d^{10} 5p^6 6s^2$	5.21	-2.90	1.98	1.35	850	1850

Some General Remarks about Group 2

- The elements beryllium, magnesium, calcium, strontium, barium and radium are collectively called alkaline earth metals. The oxides of three principal—members calcium, strontium and barium were known much earlier than the metals themselves. These oxides were alkaline in nature and existed in the earth and were named alkaline earths. The metals when discovered were also called alkaline earths. This term is now applied to all the six elements of group 2.
- The last member radium and its all isotopes are radioactive in nature.
- They are all silvery white metals. They have greyish white luster when freshly cut, but tarnish soon after their exposure in air due to surface oxidation. They are soft in nature but harder than alkali metals because metallic bonding is stronger than group 1 elements due to possession of 2 valency electrons. Moreover, the sizes of atoms are smaller than the alkali metal atoms. As a result of these two factors, alkaline earth metals are:
 - (i) harder (However, hardness decreases with increase in atomic number.)
 - (ii) denser
 - (iii) have higher melting and boiling point
 - (iv) higher heats of atomization than the alkali metals
 - (v) these elements are malleable and ductile but less than alkali metals

3.2 Group-II Alkaline Earth Metals

The metal bond strength (as is evident from their heats of atomization) decreases with increasing atomic number in the group as the size of the atoms increases.

Formation of Dipositive (M^{2+}) Ions

The formation of predominantly Mn^{2+} ions of the alkaline earth metals can be explained in terms of their:

- (i) electronic configuration
- (ii) ionization energies
- (iii) size

M^{+2} is more hydrated in solution and heat released is sufficient to remove 2nd electron.

Further ionization to give M^{3+} ions is ruled out because values for the third stage ionization energies are extremely high and cannot be recovered by the formation of an additional bond in hypothetical MX molecules.

The first and second ionization energies decrease from beryllium to barium because of increasing size of the atoms down the group. The higher values of the ionization energies of radium indicate that it is slightly less reactive than barium. The reduced reactivity of radium is due to the fact that it comes after the lanthanide series of elements in the periodic table which exhibit lanthanide contraction. Therefore, size of the radium atom also decreases resulting in an increase of the ionization energy.

- All are solids and have crystal structure. But unlike group 1 metals, crystal structures are not similar.

Be	HCP
Mg	
Ca	CCP AND BCC
Sr	CCP, HCP AND BCC
Ba	BCC
HCP	HEXAGONAL CLOSE PACKED
CCP	CUBIC CLOSE PACKED
BCC	BODY CENTRED CUBIC

- **Atomic radii and Ionic radii**

The atoms and ions of alkaline earth metals are smaller than those of the preceding alkali metals because of the increased nuclear charge. This contraction in size leads to higher ionization energy of these metals as compared to alkali metals. In moving down the group, the sizes of atoms and ions increases as new shells are added.

- **Density**

- (a) These metals are denser than alkali metals due to their greater nuclear charge and smaller size.
- (b) Density decreases slightly upto calcium and then increases upto radium.
- (c) Irregular trend is due to the difference in the crystal structure of these elements.

- From Be to Ca

- (a) Density decreases
 - (b) Metallic nature increases
 - (c) Reactivity increases
 - (d) Reducing nature increases

- Density of Ca is less than then that of Mg due to the pressure of vacant 3d-subshell leading to much increase in atomic volume.

- **Ionization Energy and Electropositive Character**

- (a) Ionization energy of these metals is higher than alkali metals, yet, it is sufficiently low to make M^{2+} ion.
- (b) These metals are thus strongly electropositive in nature but less than corresponding alkali metal.
- (c) IE_1 and IE_2 of metals decreases from Be to Ba. Due to presence of full filled electronic configuration ionization energy is very high.

- It should be noted that although the first ionization energy of alkaline earth metals are higher than those of the alkali metals, the second ionization energy values are much lower than those of alkali metals. The first ionization energy values of alkaline earth metals are higher because of their smaller size and greater nuclear charge than alkali metals. But when an electron is removed from alkali metals, their ions acquire highly stable noble gas configuration, it is difficult to remove the second electron from their ions. On the other hand, the unipositive ions of alkaline earth metals do not acquire stable noble gas configuration and hence the second electron can be removed comparatively easily.
- Unlike alkali metals, these element do not emit electrons on exposure to light.
- The ionization potential of radium is higher than that of barium.

Worked-out Example

1. The set representing the correct order of first ionization potential is:

(IIT JEE 2001)

- (a) $K > Na > Li$ (b) $Be > Mg > Ca$
 (c) $B > C > N$ (d) $Ge > Si > C$

Ans. (b)

- **Oxidation states**

All metal show a stable oxidation state of +2.

Worked-out Example

2. Why do alkaline earth metals not form monovalent ions?

Ans. The alkaline earth metal ions possess smaller size and thus have high heat of hydration which is responsible for the formation of divalent nature. Also, divalent cations of alkaline metals attain inert gas configuration.

- **Flame Colouration**

In the case of Ca, Sr, Ba and Ra, the electrons can be excited by the supply of energy to higher energy levels. When the excited electrons return to the original level, the energy is released in the form of light. **In beryllium and magnesium, the electrons are tightly held and hence excitation is rather difficult, thus do not show flame colouration.** Ca, Sr, Ba and Ra impart a characteristic colour to the flame.

Ca – brick red, Sr – crimson, Ba – green, Ra – crimson

- Since the divalent ions have no unpaired electron, these are diamagnetic and colourless. Their compounds are colourless provided their anions are also colourless.
- Be and Mg do not give any flame colour due to high ionization energy

- **Reducing Nature**

(a) The alkaline earth metals have the tendency to loose electrons and change into bivalent cation.



Hence they act as strong reducing agents.

(b) The reducing nature increases as the atomic number increases.

(c)

	Be	Mg	Ca	Sr	Ba
OXID. POTENTIAL (VOLT)	1.85	2.37	2.87	2.89	2.90

3.4 Group-II Alkaline Earth Metals

The oxidation potentials are lower than those of the alkali metals, hence, the alkaline earth metals are weaker reducing agents than alkali metals. The reason for the lower values of oxidation potentials is due to high heats of atomization (sublimation) and ionization energies. The high negative E^0 value of reduction potential of these elements mean that they all react vigorously with water also.

Worked-out Examples

3. Alkaline earth metals are weaker reducing agent than alkali metals. Why?

Ans. Due to high sublimation energy and high ionization energy, alkaline earth metals possess lower oxidation potential than alkali metals and thus they are less electropositive and weaker reducing agent than alkali metals.

4. E^0 for $M_{(aq)}^{2+} + 2e \longrightarrow M_{(s)}$ (where $M = Ca, Sr$ or Ba) is nearly constant. Explain.

Ans. Heat of atomization and ionization energy are complimentary to heat of hydration.

■ Melting and boiling points

The melting and boiling points of these elements are higher than corresponding alkali metals. This is due to the presence of two electrons in the valency shell and thus, strongly bonded in the solid state. However, melting and boiling points do not show any regular trend because atoms adopt different crystal structures.

- They are silvery white metals. These are harder than alkali metals
- Mg has exceptionally high melting and boiling point.

■ Conductors of heat and electricity

In the metallic state the two s-electron are delocalized. These electrons can move through the crystal lattice under the influence of an electrical potential. Consequently, these elements are good conductors of heat and electricity. Due to the above reason, these elements reflect light and have a silvery shine.

■ Enthalpies of hydration of M^{2+}

The hydration energies of the Group II ions are four or five times greater than for Group I ions. This is largely due to their smaller size and increased charge, and $\Delta H_{\text{hydration}}$ decreases down the group as the size of the ions increases. In the case of Be, a further factor is the very strong complex $[\text{Be}(\text{H}_2\text{O})_4]^{2+}$ that is formed. The crystalline compounds of Group II contain more water of crystallization than the corresponding Group I compounds. Thus NaCl and KCl are anhydrous but $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$, $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ and $\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$ all have water of crystallization. Note that the number of molecules of water of crystallization decreases as the ions become larger.

Since the divalent ions have a noble gas structure with no unpaired electrons, their compounds are diamagnetic and colourless, unless the acid radical is coloured.

Table 3.2 Hydration Energies

	<i>Ionic radius ($\text{\AA}$)</i>	<i>$\Delta H_{\text{hydration}}$ (kJ mol^{-1})</i>
Be^{2+}	0.31*	-2494
Mg^{2+}	0.72	-1921
Ca^{2+}	1.00	-1577
Sr^{2+}	1.18	-1443
Ba^{2+}	1.35	-1305

Electronegativity

From beryllium to radium, electronegativity values decrease indicating greater tendency for ionic compound formation down the group. However, the electronegativity values of alkaline earth metals are very close to those alkali metals. The high electronegativity of beryllium is suggestive of its tendency to form covalent compounds.

Chemical Properties

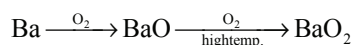
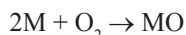
The metals are very reactive, but less than the alkali metals; the reactivity increases with increasing atomic number. The important chemical properties of alkaline earth metals are listed below:

Table 3.3 Some Reactions of Group II Elements

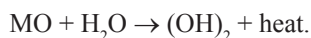
<i>Reaction</i>	<i>Comment</i>
$M + 2H_2O \rightarrow M(OH)_2 + H_2$	Be probably reacts with steam, Mg with water, and Ca, Sr and Ba react rapidly cold water
$M + 2HCl \rightarrow MCl_2 + H_2$	All the metals react with acids, liberating hydrogen
$Be + NaOH \rightarrow Na_2[Be(OH)_4] + H_2$	Be is amphoteric
$2M + O_2 \rightarrow 2MO$ with excess oxygen	Normal oxide formed by all group meml
$Ba + O_2 \rightarrow BaO_2$	Ba also forms the peroxide
$M + H_2 \rightarrow MH_2$	Ionic 'salt-like' hydrides formed at high temperatures by Ca, Sr and Ba
$3M + N_2 \rightarrow M_3N_2$	All form nitrides at high temperatures
$3M + 2P \rightarrow M_3P_2$	All the metals form phosphides at high temperatures
$M + S \rightarrow MS$	All the metals form sulphides
$M + Se \rightarrow MSe$	All the metals form selenides
$M + Te \rightarrow MTe$	All the metals form tellurides
$M + F_2 \rightarrow MF_2$	All the metals form fluorides
$M + Cl_2 \rightarrow MCl_2$	All the metals form chlorides
$M + Br_2 \rightarrow MBr_2$	All the metals form bromides
$M + I_2 \rightarrow MI_2$	All the metals form iodides
$3M + 2NH_3 \rightarrow 2M(NH_2)_2$	All the metals form amides at high temperatures

1. Reaction with Oxygen

(a) Except Ba and Ra, all other metals when burnt in oxygen form oxide of type MO.



(b) BeO is amphoteric in nature while other oxides are basic in nature.

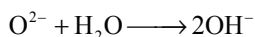


(c) Basic nature of oxide increases from Be to Ba.

Be(OH) ₂	amphoteric
Mg(OH) ₂	mildly basic
Ca(OH) ₂	Moderate bases
Sr(OH) ₂	Moderate bases
Ba(OH) ₂	strong base; as strong as Alkali metal hydroxides

(d) The normal oxide $M^{2+}O^{2-}$ is formed on heating the metals in oxygen; strontium and barium also form the peroxide $M^{2+}O_2^{2-}$ on prolonged heating, particularly if pressure is used. The normal oxides are conveniently prepared by decomposition of the carbonates.

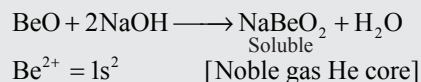
Beryllium oxide, BeO, is not attacked by water, presumably because it is covalent. The other oxides are ionic and the oxide and peroxide ions act as bases, abstracting protons from water molecules:



- Burning Mg wire continues to burn in atmosphere of N_2 and CO_2 due to its high electropositive nature and release sufficient amount of heat to decompose the carbon oxide.

Worked-out Examples

5. Name an element which is invariably bivalent and whose oxide is soluble in excess of NaOH and its dipositive ion has a noble gas core



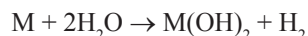
6. BeO and MgO have a high melting point, why?

Ans. The lattice energy of BeO and MgO is appreciably higher due to their small size.

2. Action of Water

Thus beryllium fails to decompose steam, magnesium burns in steam and calcium, strontium and barium decompose water with increasing vigour to give the hydroxide and hydrogen.

- (a) Mg decomposes boiling water.
 (b) Be is not attacked by water at any temperature.
 (c) Ca, Sr, Ba, Ra decomposes cold water with evolution of H_2 gas.



- (d) $\text{Be}(\text{OH})_2$ is amphoteric, but other hydroxides are basic. The basic strength increase from Be to Ba.
 (e) Solubility of hydroxides also increases from $\text{Be}(\text{OH})_2$ to $\text{Ba}(\text{OH})_2$.
 ■ Aqueous suspension of $\text{Mg}(\text{OH})_2$ is called Milk of Magnesia and it is used as antacid.
 ■ Aqueous $\text{Ba}(\text{OH})_2$ is known as baryta water.

Worked-out Examples

7. KOH is a stronger base than $\text{Ba}(\text{OH})_2$. Explain.

Ans. The greater ionic radius and lower ionization energy of potassium as compared to Ba results in stronger basic nature of KOH.

8. Explain why $\text{Be}(\text{OH})_2$ is soluble in NaOH but $\text{Ba}(\text{OH})_2$ is insoluble in NaOH?

Ans. $\text{Be}(\text{OH})_2$ is amphoteric whereas $\text{Ba}(\text{OH})_2$ is alkaline.

9. $\text{Be}(\text{OH})_2$ is amphoteric but $\text{Mg}(\text{OH})_2$ is basic. Explain.

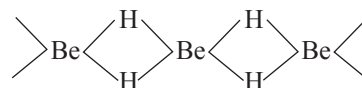
Ans. Be has smaller size, so it is less ionised to release OH^- due to more covalent character.

3. Hydride formation

- (a) Except Be, all other metals directly combines with hydrogen to form hydrides of the type MH_2 . BeH_2 is not obtained by direct combination of beryllium and hydrogen. It is formed by reacting beryllium chloride with lithium aluminium hydride.

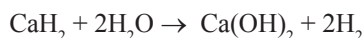


It is polymeric. $(\text{BeH}_2)_n$ possesses hydrogen bridges. Three centre bonds are present in which a banana shaped molecular orbital covers three atom $\text{Be} \dots \text{H} \dots \text{Be}$ and contains two electrons. Hydrogen atom lies in the plane perpendicular to the plane of molecule containing beryllium atoms.

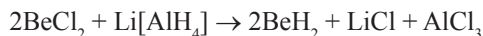


The stability of the hydrides decreases with increasing atomic number because the metallic nature of the elements increases.

- (b) BeH_2 and MgH_2 are covalent hydride while other hydrides are ionic in nature.
 (c) Ionic hydrides are violently decomposed by water evolving hydrogen.



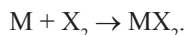
- The hydride of beryllium can be prepared indirectly by reducing beryllium chloride with lithium aluminium hydride.



- CaH_2 is known as Hydrolith.
- Be is not affected by dry air and does not decompose even boiling water.

4. Halides

- (a) Metals directly combines with X_2 to form halides



- (b) BeX_2 is covalent in nature, whereas other metal halides are ionic solids.
 (c) These halides are hygroscopic in nature and readily form hydrates e.g., $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$, $\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$ etc.,
- Anhydrous BeCl_2 acts as a lewis acid.
 - Halides of group 2 elements are hygroscopic and form hydrates like $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$.
 - Compounds of Mg are both covalent and ionic. MgBr_2 and MgI_2 due to their covalent nature are soluble in acetone.
 - Anhydrous CaCl_2 is a good dessicant (drying agent) due to its hygroscopic nature ($\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$). It cannot be used to dry alcohol or ammonia as it forms addition products with them.

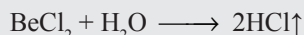
Worked-out Examples

10. BeCl_2 is soluble in ether but BaCl_2 is insoluble in ether. Explain.

Ans. BeCl_2 being non-polar covalent molecule and thus soluble in non-polar or weak polar solvents. BaCl_2 is ionic and soluble in water.

11. Beryllium halides fume in air, why?

Ans. Due to formation of halogen acids on hydrolysis.



12. BeCl_2 can be easily hydrated, why?

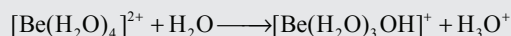
Ans. Be^+ ion has high charge-size ratio. BeCl_2 on hydration forms $\text{BeCl}_2 \cdot 4\text{H}_2\text{O}$. The hydrated ion has the formula $[\text{Be}(\text{H}_2\text{O})_4]^{2+}$ which acts as Lewis acid as well as Bronsted acid.

13. BeCl_2 in aqueous solution exists as $[\text{Be}(\text{H}_2\text{O})_4]^{2+}$ rather than Be^{2+} and forms acidic solution. Explain.

Ans.



Hydrolysis of $[\text{Be}(\text{H}_2\text{O})_4]^{2+}$ gives $[\text{Be}(\text{H}_2\text{O})_3\text{OH}]^+$ and H_3O^+ to produce acidic nature.



14. Alkaline earth metal salts are less soluble than the corresponding alkali metal salts. Explain.

Ans. Due to high charge-size ratio, lattice energy of alkaline earth metal salts are more than corresponding alkali metal salts.

3.8 Group-II Alkaline Earth Metals

5. Carbonates and Bicarbonates

- (a) Carbonates of metals are insoluble in water, they dissolve in presence of CO_2 only.



- (b) Solubility of carbonates decreases on moving down the group, while stability increases.

Worked-out Examples

15. The following compounds have been arranged in order of their increasing thermal stabilities. Identify the correct order



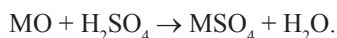
- (a) $\text{I} < \text{II} < \text{III} < \text{IV}$ (b) $\text{IV} < \text{II} < \text{III} < \text{I}$ (c) $\text{IV} < \text{II} < \text{I} < \text{III}$ (d) $\text{II} < \text{IV} < \text{III} < \text{I}$ (IIT JEE 1996)

Ans. (b)

16. NaHCO_3 is known in solid state but $\text{Ca}(\text{HCO}_3)_2$ is not isolated in solid state. Explain

Ans. Bicarbonates of alkaline earth metals are thermally unstable.

6. Sulphates



- (a) Solubility of sulphates decreases on moving down the group.



- (b) As we descend down the group the hydration energy decreases more than the lattice energy (discussed in the 'chemical bonding' section) Thus, BeSO_4 and MgSO_4 are soluble, but CaSO_4 is sparingly soluble and the sulphates of Sr, Ba and Ra are virtually insoluble.

- Be reacts slowly with acids, Mg reacts at a faster rate while Ca, Sr and Ba react explosively with acids.
- Only Mg displaces hydrogen from a very dilute HNO_3 .

Worked-out Examples

17. Solubility of the sulphates, chromates and carbonates of alkaline earth metals decreases from Be to Ba. Explain.

Ans. The solubility of an ionic compound depends: (i) lattice energy, and (ii) hydration energy. These two factors oppose each other. If lattice energy is high, the ions will be tightly packed in the crystal, therefore, solubility will be low. If hydration energy is high, the ions will have greater tendency to be hydrated, therefore, the solubility will be high. In the case of carbonates, sulphates and chromates the anions are larger in size and small changes in cation size do not alter the lattice energies i.e., lattice energies are about the same. However, the hydration energies decrease from Ca^+ to Ba^+ . Hence, the solubility of carbonates, sulphates and chromates decreases from calcium to barium.

18. Solubility of hydroxides, fluorides or oxalate of alkaline earth metals increases from Be to Ba. Explain.

Ans. The solubility of an ionic compound depends: (i) lattice energy and (ii) hydration energy. In the case of hydroxides, fluorides and oxalates the lattice energies are different i.e., lattice energy decreases as the size of cation increases. This tends to increase the solubility as it overcomes the counter effect of decrease in hydration energy. Hence, the solubility of hydroxides, fluorides and oxalates increases from Ca to Ba.

19. The alkaline earth metals are found in nature as carbonates and sulphates whereas mercury and silver are found as sulphides.

Ans. The ions Ca^{2+} , Mg^{2+} , Ba^{2+} etc., are hard acids while SO_4^{2-} and CO_3^{2-} ions are also hard bases. Since compounds of hard acids and hard bases are stable, alkaline earth metals are found in nature as sulphates and carbonates. However, Hg^{2+} and Ag^+ ions are soft acids. They combine with soft bases such as S^{2-} to form stable compounds.

20. The solubilities of carbonates decreases down the magnesium group due to decrease in:

[AIEEE 2003]

- (a) Inter-ionic attraction
- (b) Entropy of solution formation
- (c) Lattice energy of solids
- (d) Hydration energy of cations

Ans. (d) The hydration energy of ion decreases with increasing size and there by solubility decreases.

7. To Complex Formation

Be^{2+} and Mg^{2+} due to their small size show some tendency of complex formation.

Example $[\text{BeF}_3]^-$, $[\text{BeF}_4]^{2-}$ and $[\text{Be}(\text{H}_2\text{O})_4]^{2+}$, chlorophyll (Complex of Mg^{2+})

Worked-out Examples

21. Complex formation in alkaline earth metals is shown predominantly by Be and Mg comment on it.

Ans. Magnesium, having a smaller ion than calcium, forms complexes more readily and the solubility of its bromide and iodide in alcohols, ketones and ethers is probably due to the formation of datively bonded complexes involving bonds between the magnesium ion and the oxygen of the organic liquids. The complex MgCl_4^{2-} ion is present in the compound $\text{NH}_3\text{MgCl}_4^{2-}$ and the chloride ions are probably arranged tetrahedrally around the central magnesium ion. Beryllium having the smallest ion of this group of elements naturally tends to form complexes the most readily, since the small positively charged ion is able to exert considerable attraction on other ions or molecules that have lone pairs of electrons.

22. Calcium forms only one complex i.e., calcium tetraethanolate. Discuss its structure.

Ans. The tendency for the Group 2 metal ions to form complexes decreases with increasing atomic number, thus of the heavier ions, only calcium forms a complex with ethanol. The calcium tetraethanolate ion presumably involves dative bonding from the oxygen atoms of ethanol molecules, with the four molecules of ethanol arranged tetrahedrally.

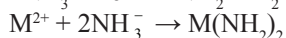
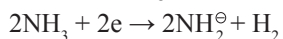
8. Organometallic Compounds

Be and Mg forms a number of compounds with M – C bonds.

e.g., Grignard reagent.

9. Solutions of metals in liquid ammonia

Like alkali metals, alkaline earth metals dissolve also in liquid ammonia to form coloured solutions. Dilute solutions are bright blue in colour due to solvated electrons. These solutions decompose very slowly forming amides and evolving hydrogen.



Worked-out Examples

23. When metals dissolve in NH_3 , there is marked increase in volume. Explain why?

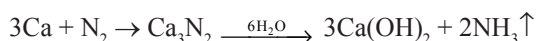
Ans. When metals dissolve in ammonia, the electrons are thought to occupy cavities with surrounding NH_3 molecules oriented such that the hydrogens are directed inward.

24. A metal M readily forms its sulphate MSO_4 which is water soluble. It forms oxide MO which becomes inert on heating. It forms insoluble hydroxide which is soluble in NaOH. The metal M is: **[AIEEE 2002]**

- (a) Mg (b) Ba (c) Ca (d) Be

Ans. (d) BeSO_4 and MgSO_4 are soluble in water. Only Be(OH)_2 being amphoteric reacts with NaOH.

10. All of them react with nitrogen to form nitrides of M_3N_2 type. These nitrides liberate NH_3 on reaction with water as



11. Alkaline earth metals form different types of carbides with C, which are useful in preparing different saturated and unsaturated hydrocarbons on reaction with water.

- (i) $\text{Be} \xrightarrow{\text{forms}} \text{Be}_2\text{C}$ which form CH_4 on reaction with water
 (ii) $\text{Mg} \xrightarrow{\text{forms}} \text{Mg}_2\text{C}_3$ which form $\text{CH}_3\text{—C}\equiv\text{CH}$ on reaction with water
 (iii) $\text{Mg} \xrightarrow{\text{forms}} \text{MgC}_2$ which forms $\text{CH}\equiv\text{CH}$ with water
 (iv) Ca to Ba all metals form carbides of MC_2 type which on reaction with water form $\text{CH}\equiv\text{CH}$.

Similarities (Diagonal Relationship) Between Beryllium and Aluminium

Beryllium shows some similarities in properties with aluminium, the second typical element of group 13 of the next higher period.

	Group 2	Group 13
Second period	Be	B
Third period	Mg	Al

The following are the causes of diagonal relationship:

- (a) The polarising power of Be^{2+} and Al^{3+} ions is high. Consequently, the compounds develop covalent nature.
 (b) The standard oxidation potential values of both the elements are very close to each other.
 (c) The electronegativity values of both the elements are the same.

Worked-out Examples

25. Beryllium turns passive in presence of HNO_3 . Explain.

Ans. HNO_3 acts as an oxidizing agent to oxidize surface of Be to BeO and forms a layer of BeO on Be surface which prevents further action of HNO_3 .

26. Arrange the following in order of increasing

- (a) Be^{2+} , Mg^{2+} , Ca^{2+} , Sr^{2+} , Ba^{2+} (Heat of hydration)
 (b) CaI_2 , CaF_2 , CaCl_2 , CaBr_2 (Melting point)
 (c) CaCl_2 , BeCl_2 , BaCl_2 , MgCl_2 , SrCl_2 (Ionic character)
 (d) MgO , SrO , K_2O , Cs_2O (Basic character)
 (e) BeCO_3 , MgCO_3 , CaCO_3 , BaCO_3 (Thermal stability)
 (f) Sr, Ba, Ca, Mg (Reducing nature)
 (g) CaSO_4 , BaSO_4 , MgSO_4 (Solubility in water)
 (h) CaCl_2 , CaBr_2 , CaF_2 (Solubility in water)

- Ans.** (a) $\text{Ba}^{3+} < \text{Sr}^{2+} < \text{Ca}^{2+} < \text{Mg}^{2+} < \text{Be}^{2+}$ (b) $\text{CaI}_2 < \text{CaBr}_2 < \text{CaCl}_2 < \text{CaF}_2$
 (c) $\text{BeCl}_2 < \text{MgCl}_2 < \text{CaCl}_2 < \text{SrCl}_2 < \text{BaCl}_2$ (d) $\text{MgO} < \text{SrO} < \text{K}_2\text{O} < \text{Cs}_2\text{O}$
 (e) $\text{BeCO}_3 < \text{MgCO}_3 < \text{CaCO}_3 < \text{BaCO}_3$ (f) $\text{Mg} < \text{Ca} < \text{Sr} < \text{Ba}$
 (g) $\text{BaSO}_4 < \text{CaSO}_4 < \text{MgSO}_4$ (h) $\text{CaF}_2 < \text{CaCl}_2 < \text{CaBr}_2$

27. Arrange the following properties as listed in the increasing order:

- (a) Lattice energy : $\text{BeCl}_2, \text{MgCl}_2, \text{CaCl}_2, \text{SrCl}_2$
 (b) Solubility in H_2O : $\text{BeSO}_4, \text{MgSO}_4, \text{CaSO}_4, \text{SrSO}_4, \text{BaSO}_4$
 (c) Stability : $\text{BeCl}_2, \text{BeF}_2, \text{BeBr}_2, \text{BeI}_2$
 (d) Thermal Stability : $\text{BeSO}_4, \text{MgSO}_4, \text{CaSO}_4$
 (e) Polarizing power : $\text{Be}^{2+}, \text{Mg}^{2+}, \text{Ca}^{2+}$
 (f) Solubility in H_2O : $\text{Be}(\text{OH})_2, \text{Mg}(\text{OH})_2, \text{Ca}(\text{OH})_2$
 (g) Covalent nature : $\text{BeCl}_2, \text{MgCl}_2, \text{CaCl}_2$
 (h) Hydrolysis nature : $\text{BeCl}_2, \text{MgCl}_2, \text{CaCl}_2$
 (i) Lattice energy : $\text{CaF}_2, \text{MgF}_2, \text{BaF}_2$
 (j) Hydration energy : $\text{Be}^{2+}, \text{Mg}^{2+}, \text{Ba}^{2+}$
 (k) Solubility in water : $\text{MgF}_2, \text{BaF}_2, \text{BeF}_2$
 (l) Basic nature : $\text{Be}, \text{Mg}, \text{Ca}, \text{Sr}$
 (m) Refractory properties: $\text{BeO}, \text{MgO}, \text{CaO}$

- Ans.** (a) $\text{SrCl}_2 < \text{CaCl}_2 < \text{MgCl}_2 < \text{BeCl}_2$ (b) $\text{BaSO}_4 < \text{SrSO}_4 < \text{CaSO}_4 < \text{MgSO}_4 < \text{BeSO}_4$
 (c) $\text{BeI}_2 < \text{BeBr}_2 < \text{BeCl}_2 < \text{BeF}_2$ (d) $\text{BeSO}_4 < \text{MgSO}_4 < \text{CaSO}_4$
 (e) $\text{Ca}^{2+} < \text{Mg}^{2+} < \text{Be}^{2+}$ (f) $\text{Be}(\text{OH})_2 < \text{Mg}(\text{OH})_2 < \text{Ca}(\text{OH})_2$
 (g) $\text{BaF}_2 < \text{CaF}_2 < \text{MgF}_2$ (h) $\text{Ba}^{2+} < \text{Mg}^{2+} < \text{Be}^{2+}$
 (i) $\text{BaF}_2 < \text{MgF}_2 < \text{BeF}_2$ (j) $\text{Be} < \text{Mg} < \text{Ca} < \text{Sr}$
 (k) $\text{CaO} < \text{MgO} < \text{BeO}$

ALKALINE EARTH METALS

Magnesium

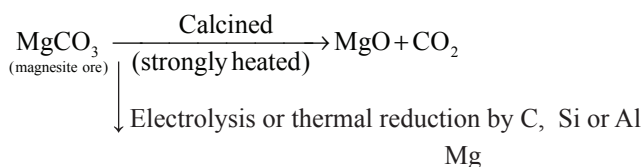
Occurrence

Magnesium does not occur free in nature. In addition to minerals such as magnesite (MgCO_3), dolomite ($\text{MgCO}_3 \cdot \text{CaCO}_3$), kieserite ($\text{MgSO}_4 \cdot \text{H}_2\text{O}$), Epsom salt ($\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$), carnallite ($\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$), asbestos [$\text{CaMg}_3(\text{SiO}_3)_4$], it is found in sea water. It is an essential constituent of chlorophyll, the green colouring matter of plants.

Extraction

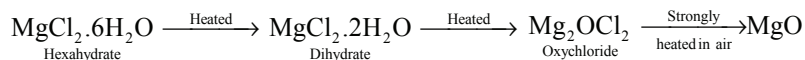
It is usually extracted by the electrolysis of fused oxide or fused anhydrous magnesium chloride.

(1) From Magnesite

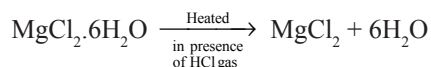


3.12 Group-II Alkaline Earth Metals

- (2) **From Carnallite** Carnallite is a hydrated compound. It cannot be dehydrated by heating because the water of crystallization reacts chemically with MgCl_2 to form magnesium oxychloride and finally magnesium oxide.



To make it anhydrous, it is first heated in air and then in a current of HCl gas.



We can get Mg by electrolysis of anhydrous MgCl_2

Electrolyte	35% MgCl_2 + 50% NaCl + 15% CaCl_2
Cathode	Fe
Anode	Graphite

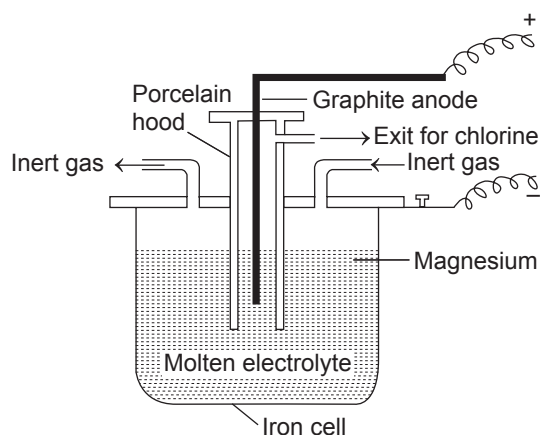


Fig. 3.1

Mg is discharged at cathode. Magnesium being lighter than the fused mass floats on the surface from where it is removed periodically by perforated ladles.

The purpose of addition of NaCl and CaCl_2 to anhydrous MgCl_2 is to lower the fusion temperature and make the fused mass good conductor of electricity.

Worked-out Examples

28. Which of the following processes is used in the extractive metallurgy of magnesium?

(IIT JEE 2002)

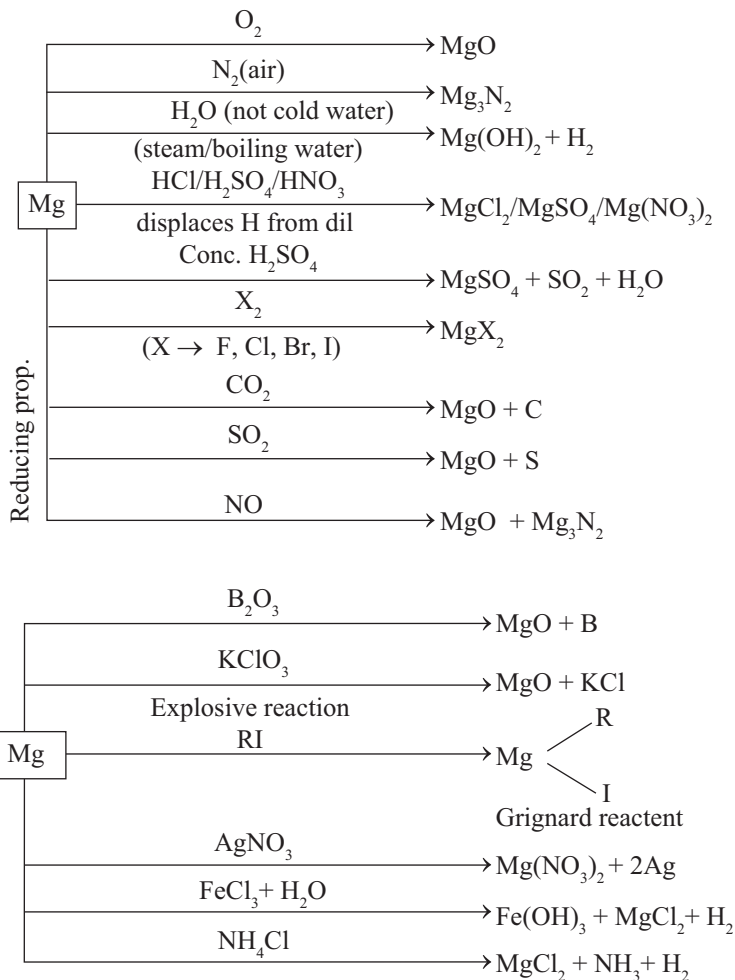
- | | |
|-----------------------------------|------------------------|
| (a) Fused salt electrolysis | (b) Self reduction |
| (c) Aqueous solution electrolysis | (d) Thermite reduction |

Ans. (a)

29. NaCl is added to anhydrous MgCl_2 (fused) before electrolysis. Why?

Ans. The addition of NaCl to anhydrous MgCl_2 during its electrolysis is done to lower the fusion point of MgCl_2 as well as to increase the conducting power of fused mixture.

Chemical Properties

**Worked-out Examples**

30. A piece of burning magnesium ribbon continues to burn in SO₂. Why?

Ans. Mg acts as strong reducing agent and reduces SO₂ to S, using its oxygen for burning.

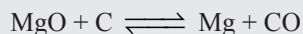


31. Anhydrous MgCl₂ is prepared by heating MgCl₂·6H₂O in a current of HCl gas. Explain with reason.

Ans. Presence of HCl gas checks up the hydrolysis of magnesium chloride by its own water of crystallization.

32. In the manufacture of Mg by carbon reduction of MgO, the product is cooled in the stream of an inert gas. Explain.

Ans. The reduction of MgO by carbon is a reversible reaction.



In order to check backward reaction the products are cooled in a stream of inert gas. Inert gas does not permit reaction of Mg with CO.

3.14 Group-II Alkaline Earth Metals

Uses:

- It is used for the extraction of boron and silicon.
- Magnesium ribbon along with barium peroxide is used as ignition mixture in aluminothermite process.
- Magnesium powder mixed with potassium chlorate is used in flash bulbs for photography.
- It is used for making Grignard reagents.
- It is employed as a deoxidizer in metallurgy and for the removal of last traces of air from radio-valves.
- It finds application in the manufacture of various alloys which are tough and light in nature. The important alloys are:

Mg-alium	95% Al 5% Mg	Used in the construction of airships, balances and pistons of motor engines.
Electron metal	95% Mg 4.5% Zn 0.5% Cu	Used in the construction of aircrafts.
Duralu min	95% Al 0.5% Mg 0.5% Mn 4% Cu	Its strength and toughness is comparable with mild steel. It is used for making airships.

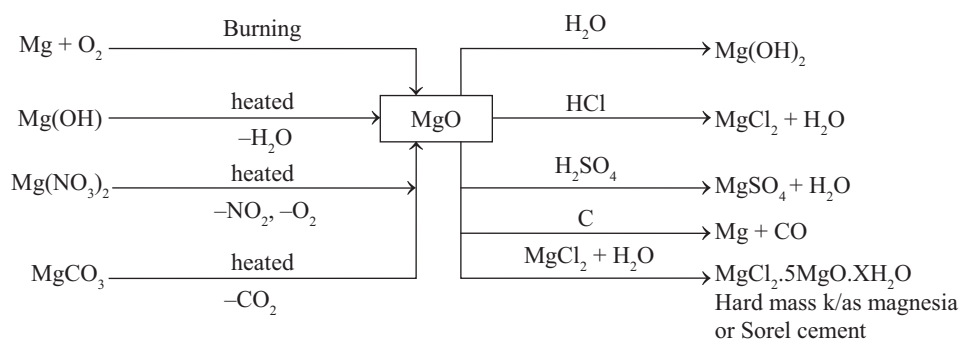
Worked-out Example

33. MgO is used for lining of steel making furnace. Why?

Ans. MgO acts as basic flux and facilitates the removal of acidic impurities of Si, P and S from steel through slag formation.

Compounds of Magnesium

(a) Magnesium Oxide, MgO (Magnesia)



Uses: It is used:

- in medicine as antacid.
- refractory material in furnaces.
- as an insulator when mixed with asbestos for lagging steam pipes and boilers.
- as an absorbent in the manufacture of dynamite and in the vulcanization of rubber.
- for making sorel cement which is employed in dental filling, in cementing glass and porcelain, in plaster casting, etc.,

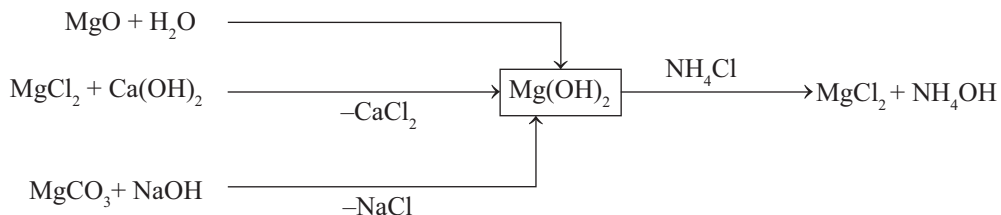
Worked-out Example

34. MgO is used as a refractory material.

Ans. MgO has very high melting point and does not decompose as its lattice energy is very high. It is, thus, used as refractory material.

(b) Magnesium Hydroxide, Mg(OH)₂

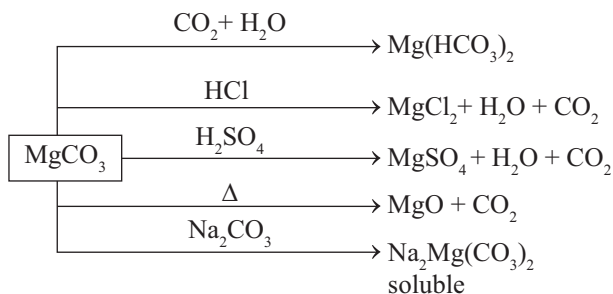
- A suspension of hydroxide of magnesium (Milk of magnesia) is used as a stomach antacid.



(c) $\text{MgCO}_3 \rightarrow$ It is found in nature as magnesite (MgCO_3) and dolomite ($\text{MgCO}_3 \cdot \text{CaCO}_3$)

Preparation

The magnesium carbonate cannot be obtained by the addition of sodium carbonate to the solution of magnesium salt. A white precipitate of a basic carbonate of composition $3\text{MgCO}_3 \cdot \text{Mg(OH)}_2 \cdot 3\text{H}_2\text{O}$ is obtained. It is known as magnesia alva. It is suspended in water and CO_2 is passed when magnesium bicarbonate known as fluid magnesia is formed.

Properties**Uses:**

- as a filler for paper, rubber and pigments
- in the form of basic magnesium carbonate (magnesia alva) for making tooth powders, cosmetics and in medicine as antacid.
- in the form of magnesite as a refractory material.
- in glass manufacture, ceramics and links.

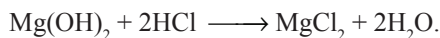
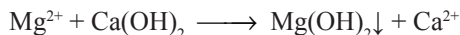
(d) Magnesium Chloride, MgCl₂·6H₂O

It occurs as mineral carnallite, $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ and bischofite, $\text{MgCl}_2 \cdot \text{H}_2\text{O}$. It is found in sea water, mineral springs etc.,

Preparation

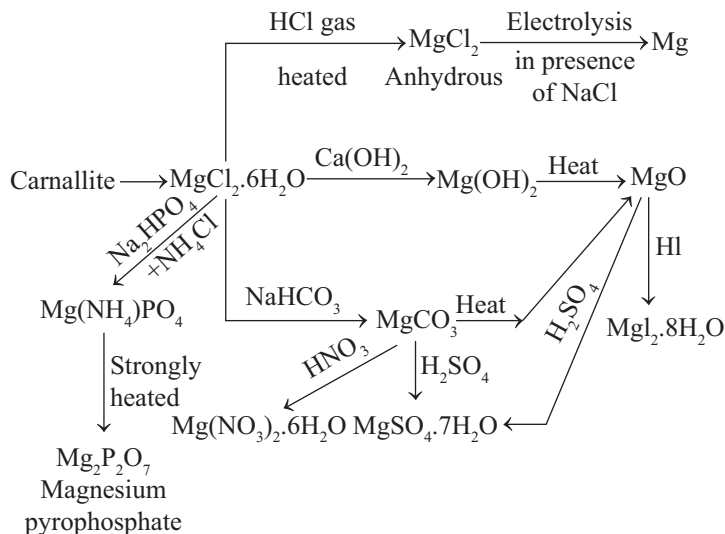
Magnesium chloride can be prepared by fractional crystallization of carnallite, $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$. It can also be obtained from sea water. Magnesium salts present in sea-water are precipitated as Mg(OH)_2 by adding lime. The precipitate is filtered and dissolved in HCl solution. The solution on crystallization yields crystals of $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$.

3.16 Group-II Alkaline Earth Metals



A mixture of magnesium chloride and magnesium oxide is known as **Sorel cement** or **magnesia cement**. This mixture reacts with water and sets into a solid mass having composition $\text{MgCl}_2 \cdot 5\text{MgO} \cdot x\text{H}_2\text{O}$.

Properties



Uses: It is used:

- (i) for making Sorel cement.
- (ii) in the lubrication of cotton threads for spinning.
- (iii) in the preparation of xylotite which is used as covering for floors, laboratory tables, etc.,

Worked-out Examples

Fill-in-the-Blanks Type

35. Anhydrous MgCl_2 is obtained by heating the hydrated salt with _____. (IIT JEE 1980)

Ans. anhydrous HCl

True/False Type

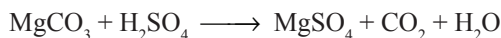
36. $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ on heating gives anhydrous MgCl_2 . (IIT JEE 1982)

Ans. False

(e) Magnesium Sulphate, $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ (Epsom salt)

Preparation

Magnesium sulphate can be prepared from magnesite (MgCO_3) or from kieserite ($\text{MgSO}_4 \cdot \text{H}_2\text{O}$)

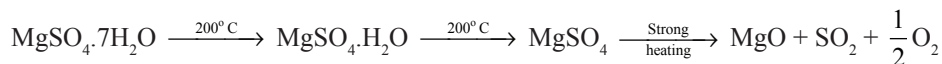


The resulting solution on concentration and cooling gives crystals of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$.

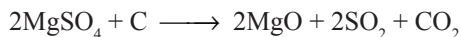
Alternatively mineral kieserite is dissolved in water. The solute on crystallization yields crystals of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$. $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ is **efflorescent**, i.e., it loses water of crystallization on exposure to air.

Properties

1. Bitter in taste and isomorphous with $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$.
2. Heating effect



Magnesium sulphate is reduced by lampblack at 800°C .



Uses: It is used:

- (a) as a purgative in medicine.
- (b) as a filler for paper.
- (c) as a mordant in dyeing and tanning industry.
- (d) in the manufacture of paints and soaps and in fireproofing fabrics.

Worked-out Examples

37. MgSO_4 on reaction with NH_4OH and Na_2HPO_4 forms a white crystalline precipitate. What is the formula?

(IIT JEE 2006)

- (a) $\text{Mg}(\text{NH}_4)\text{PO}_4$ (b) $\text{Mg}_3(\text{PO}_4)_2$ (c) $\text{MgCl}_2 \cdot \text{MgSO}_4$ (d) MgSO_4

Ans. (a)

38. What is an anhydron and for what purposes it is used?

Ans. Anhydron is $\text{Mg}(\text{ClO}_4)_2$ or magnesium perchlorate. It is used as drying agent. It should not be used with organic materials as it is strong oxidizing agent.

39. Be-salts rarely have more than four water of crystallization while Mg-salts generally have six water molecules. Explain.

Ans. Because in Be^{2+} ions, second shell may provide only four orbitals (1s and 3p) for co-ordination with water while Mg can use some 3d orbitals also i.e., why it may co-ordinate with six water molecules.

40. Several blocks of magnesium are fixed into the bottom of ship to:

[AIEEE 2003]

- (a) Prevent action of water and salt (b) Prevent puncturing by under sea rocks
(c) Keep away the sharks (d) Make the ship lighter

Ans. (d) It is a fact

41. One mole of magnesium nitride on reaction with excess of water gives:

[AIEEE 2004]

- (a) Two mole of HNO_3 (b) Two-mole of NH_3
(c) 1 mole of NH_3 (d) 1 mole of HNO_3

Ans. (b) $\text{Mg}_3\text{N}_2 + 6\text{H}_2\text{O} \longrightarrow 3\text{Mg}(\text{OH})_2 + 2\text{NH}_3$

Calcium

Most abundant alkaline earth metal in the earth's crust is calcium (5th most abundant metal) and least abundant is radium

- (i) **Carbonate**—Limestone, marble, chalk, Iceland spar calcite and calcspar as CaCO_3 and dolomite as $\text{MgCO}_3 \cdot \text{CaCO}_3$.
- (ii) **Sulphate**—Gypsum $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ and anhydrite CaSO_4 .
- (iii) **Fluoride**—Fluorspar CaF_2 ,
Fluorapatite $3\text{Ca}_3(\text{PO}_4)_2 \cdot \text{CaF}_2$.
- (iv) **Phosphate**—Phosphorite $\text{Ca}_3(\text{PO}_4)_2$.

3.18 Group-II Alkaline Earth Metals

It is present in natural waters and causes hardness in water. It is an essential constituent of bones and teeth. Egg and seashells contain calcium carbonate.

Electrolytic Extraction of Calcium

Electrolyte : fused mixture of anhydrous CaCl_2 (6 part) and CaF_2 (1 part) at 700°C

Cathode : Steel cathode (water cooled)

Anode : graphite

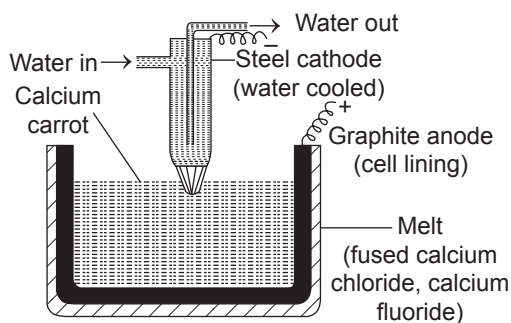
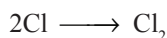
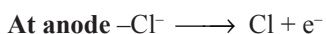
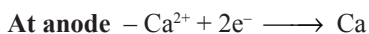


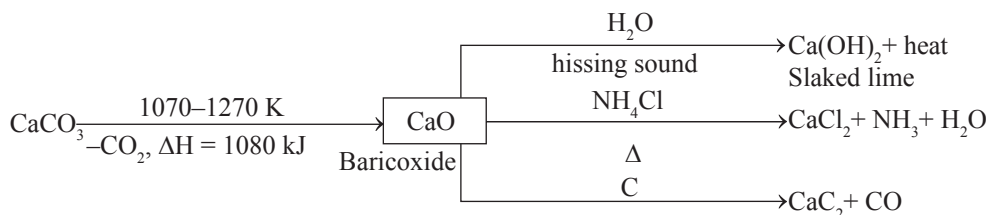
Fig. 3.2



Note:

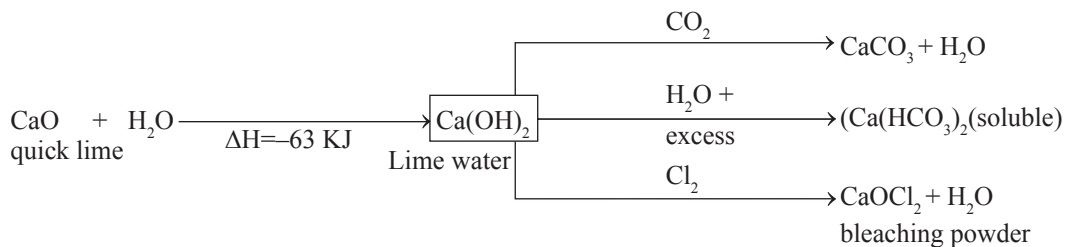
Compounds of Calcium

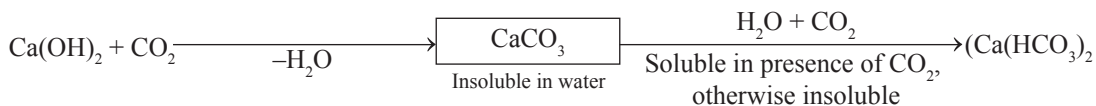
(A) Calcium oxide, Quick lime, Burnt lime, Lime $\text{CaO} \rightarrow$



Uses: It is used as basic flux, for removing hardness of water for preparing mortar ($\text{CaO} + \text{sand} + \text{water}$)

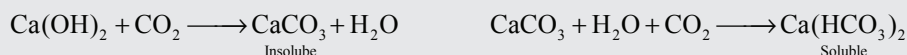
(B) Calcium hydroxide, slaked lime, lime water Ca(OH)_2



(C) Calcium carbonate, limestone, marble, chalk, slate, calcite (CaCO_3)**Worked-out Examples**

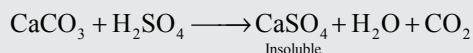
42. Lime water becomes turbid on passing CO_2 through it, but becomes clear when more CO_2 is passed. Why?

Ans. On passing CO_2 in lime water, insoluble CaCO_3 is precipitated out which produces turbidity in water. However, on passage of enough CO_2 , CaCO_3 further reacts to form soluble bicarbonate and solution becomes clear.



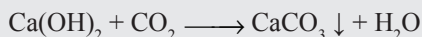
43. The reaction between marble and dilute sulphuric acid stops soon. Why?

Ans. The reaction between marble and dilute sulphuric acid gives insoluble CaSO_4 which covers the surface of marble and thus reaction ceases soon.



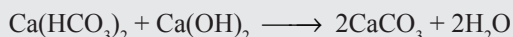
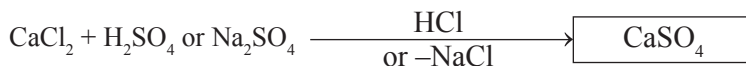
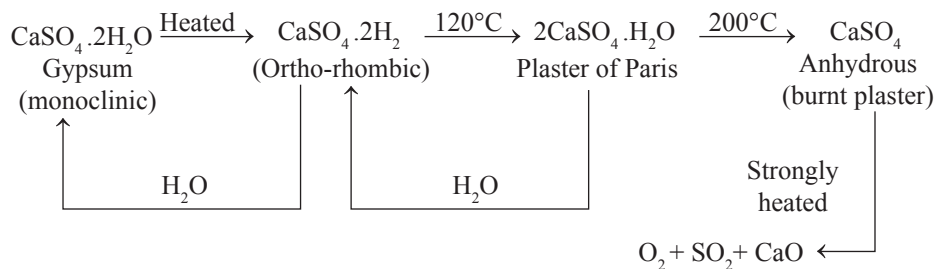
44. What is precipitated chalk?

Ans. The precipitate of CaCO_3 obtained during passage of CO_2 through lime water.



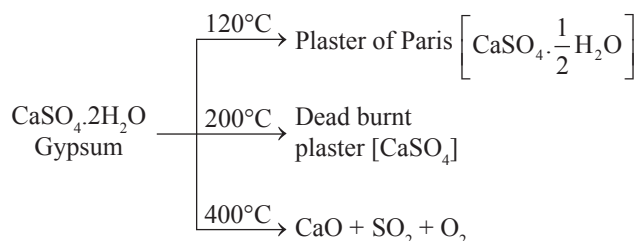
45. $\text{Ca(HCO}_3)_2$ and Ca(OH)_2 solutions can not be stored in one vessel. Why?

Ans. $\text{Ca(HCO}_3)_2$ solution behaves as acid towards Ca(OH)_2 .

**(D) Calcium sulphate, $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ (Gypsum)****Preparation****Properties**

3.20 Group-II Alkaline Earth Metals

Gypsum when heated to different temperatures gives different products.



Worked-out Examples

46. The substance not likely to contain CaCO_3 is:

[AIEEE 2003]

- (a) Sea shells (b) Dolomite
(c) A marble statue (d) Calcined gypsum

Ans. (d) Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$)

47. Gypsum is

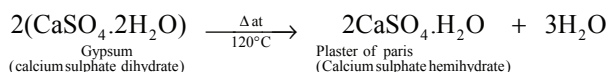
(IIT JEE 1978)

- (a) $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ (b) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
(c) $\text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O}$ (d) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$

Ans. (d)

(E) Plaster of Paris, $2\text{CaSO}_4 \cdot \text{H}_2\text{O}$ (Calcium sulphate hemihydrate)

Preparation



Properties

1. White powder

2. It has the property of setting to a hard mass $2\text{CaSO}_4 \cdot \text{H}_2\text{O} \xrightarrow[\text{H}_2\text{O}]{\text{Setting}} \text{CaSO}_4 \cdot 2\text{H}_2\text{O} \xrightarrow{\text{Hardening}} \text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
Orthorhombic Monoclinic

The setting of Plaster of Paris may be catalysed by sodium chloride while it is retarded by borax or alum. Addition of alum to plaster of Paris makes the setting very hard. The mixture is known as Keene cement.

3. When Plaster of Paris is heated at 200°C , it forms anhydrous calcium sulphate which is known as dead plaster. It has no setting property as it takes up water only very slowly.

Uses: It is used:

- in surgery for setting broken or dislocated bones.
- in making casts for statues, toys, surgical instruments, etc.,
- in making casts in dentistry.
- in making blackboard chalk.

Worked-out Examples

48. The compound insoluble in acetic acid is

- (a) CaO (b) CaCO_3
(c) Calcium oxalate (d) $\text{Ca}(\text{OH})_2$

(IIT JEE 1986)

Ans. (c)

49. Why anhydrite (anhydrous calcium sulphate) cannot be used as Plaster of Paris?

Ans. Anhydrite (anhydrous CaSO_4) cannot be used as Plaster of Paris because it does not set on the water and is known as 'dead burnt plaster'.

50. Anhydrous calcium sulphate cannot be used as Plaster of Paris. Why?

Ans. On moistening with water, anhydrous CaSO_4 does not set like Plaster of Paris.

51. Anhydrous calcium sulphate is used as drying agent. Why?

Ans. Anhydrous CaSO_4 absorbs water and readily forms hydrated sulphate $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$.

52. The number and types of bonds between two carbon atoms in calcium carbide are:

- (a) One sigma, one pi (b) One sigma, two-pi
(c) Two sigma, one pi (d) Two sigma, two pi

[AIEEE 2005]

Ans. (b) CaC_2 is $\text{Ca}^{2+}\text{C}^{2-}(\text{C} \equiv \text{C})$

(F) Portland Cement

Cement is essentially a mixture of complex silicates and aluminates of Ca containing less than 1.0% free lime and some gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$)

Composition: An approximate composition is as follows:

1.	Lime	CaO	60-69%	62%
2.	Silica	SiO_2	17-25%	22%
3.	Alumina	Al_2O_3	3-8%	7.5%
4.	Magnesia	MgO	1-5%	2.5%
5.	Iron oxide	Fe_2O_3	0.5-5%	2.5%
6.	Sulphur trioxide	SO_3	1-3%	1.5%
7.	Sodium oxide	Na_2O	0.3-1.5%	1.0%
8.	Potassium oxide	K_2O	0.3-1.5%	1.0%

Ratio of Silica and alumina

$$\frac{\% \text{ of } \text{SiO}_2}{\% \text{ of } \text{Al}_2\text{O}_3} = 2.5 \text{ to } 4$$

$$\frac{\% \text{ of } \text{CaO}}{\% \text{ of } \text{SiO}_2 + \% \text{ of } \text{Al}_2\text{O}_3 + \% \text{ of } \text{Fe}_2\text{O}_3} = 1.9 \text{ to } 2.1$$

White Cement: It does not contain ferric oxide.

Process: Two processes are employed:

- (i) Wet process
(ii) Dry process

3.22 Group-II Alkaline Earth Metals

Raw material: Lime and Clay

Manufacturing



Gypsum regulates the setting time

Setting of Cement: When mixed with water, the cement forms a gelatinous mass sets to hard mass when three dimensional cross links are formed between.... Si-O-Si— and —Si-O-Al— chains.

The reactions involved in the setting of cement are:

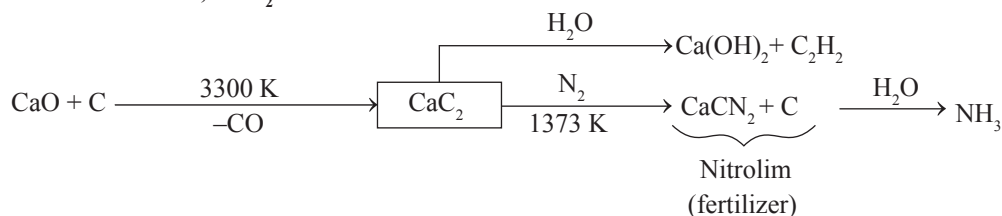
- (i) **Hydration:** of $3\text{CaO} \cdot \text{Al}_2\text{O}_3$ and 2CaOSiO_2 forming colloidal gel.
- (ii) **Hydrolysis:** of $3\text{CaO} \cdot \text{Al}_2\text{O}_3$ and 3CaOSiO_2 forming precipitates of $\text{Ca}(\text{OH})_2$ and $\text{Al}(\text{OH})_3$.

Fly ash: A water product of steel industry possess properties similar to cement. It is added to cement to reduce its cost.

Rice Husk: It has high silica content and employed to make cement.

- Sorrel's cement or Magnesia cement is $\text{MgCl}_2 \cdot 5\text{MgO} \cdot x\text{H}_2\text{O}$

(G) Calcium carbide, CaC_2 ,



Worked-out Examples

53. Why is reduction of CaO is not possible with carbon?

Ans. Reduction of CaO is not possible with carbon because CaO is very reactive and reacts with carbon to produce carbides.

54. How will the solutions of following compounds behave towards litmus?

- (a) NaHSO_4 (b) K_2CO_3 (c) MgSO_4 (d) CaCl_2 (E) BaSO_4

Ans. (a) Acidic (b) Alkaline (c) Acidic (d) Neutral (e) Neutral

55. In curing cement plasters, water is sprinkled from time to time. This helps in:

[AIEEE 2003]

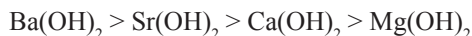
- (a) Hydrating sand and gravel mixed with cement (b) Converting sand into silicate
(c) Developing interlocking needle like crystals of hydrated silicates (d) Keeping it cool

Ans. (c) It is hydration of cement.

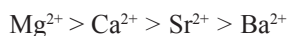
Summary of Group 2 Trends

1. Beryllium is the least reactive and does not react with water even at red heat and does not react with N_2 . Magnesium only reacts at a reasonable rate with steam, calcium and strontium readily tarnish in moist air, and barium tarnishes readily.
2. The ionic character of the compounds increases down the Group. Beryllium forms highly covalent compounds generally with sp^3 tetrahedral geometries, e.g., $[\text{BeF}_4]^{2-}$, $[\text{BeCl}_2]_n$ (infinite linear polymer). Magnesium forms more polar compounds with 6-coordination. Calcium, strontium, and barium form increasingly ionic compounds with higher coordination numbers (8 is particularly common).

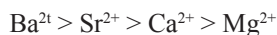
3. The organometallic compounds of beryllium are covalent and rather unreactive. Magnesium forms two important series of organometallic compounds, RMgX and R_2Mg , which provide convenient sources of carbanions for organic syntheses. The organometallic compounds of calcium, strontium, and barium are generally more reactive and are insoluble in organic solvents.
4. The oxides become progressively more basic down the Group. BeO and $\text{Be}(\text{OH})_2$ are amphoteric and react with acids and strong bases such as NaOH . MgO is basic and $\text{Mg}(\text{OH})_2$ weakly basic and do not dissolve in NaOH solution. The oxides of calcium, strontium, and barium are basic and their hydroxides are strongly basic. The solubilities of the hydroxides in water follow the order:



5. BeX_2 ($\text{X} = \text{F}, \text{Cl}, \text{Br}$ or I) are covalent polymers which are readily hydrolysed and are Lewis acids forming adducts BeX_2L_2 ($\text{L} = \text{Lewis base}$). Magnesium, calcium, strontium, and barium halides are essentially ionic and are soluble in water.
6. BeH_2 is a covalent polymer, magnesium hydride is partially ionic and the hydrides of calcium, strontium, and barium are very ionic and hydridic in their properties.
7. Mg^{2+} and Ca^{2+} show the greatest tendency to form complexes especially with ligands which have oxygen donor atoms. For small highly charged anions the order of stability is generally:

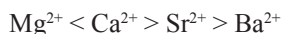


but for the anions, NO_3^- , SO_4^{2-} , and IO_4^- the stability order is:



The most important complexes of these metals are with EDTA^{4-} .

The order of stability for this and related polydentate ligands is;



The calcium complex, $[\text{Ca}(\text{edta})]^{2-}$, is particularly important because it is water soluble and allows edta to solubilize calcium carbonate. Polyphosphates, e.g., $\text{P}_2\text{O}_7^{4-}$ and $\text{P}_3\text{O}_{10}^{5-}$ are able to function similarly to solubilize hard water deposits of CaCO_3 .

The crown polyethers and cryptate ligands also form stable complexes with Ca^{2+} and Mg^{2+} .

8. Both magnesium and calcium have important biological roles. The former occurs in chlorophyll in which it is coordinated to a tetrapyrrole molecule and the latter has an important structural role cross-linking polymer chains, e.g., in collagen.

Because of their important biological roles Ca^{2+} and Mg^{2+} have to be transported through cell membranes and ligands which wrap round the metal ions and retain a hydrophobic surface are able to achieve this, e.g., the *l*-aspartate complex of Mg^{2+} .

9. The thermal stabilities of the nitrates, carbonates, and peroxides increase down the column.
10. The solubilities of the sulfates, nitrates, and chlorides increase down the group.
11. The solubilities of the halides in alcohols increase down the group.

Anomalous Nature of Beryllium

In common with lithium, and for the same underlying causes, beryllium exhibits some anomalous properties.

1. The high enthalpy of atomization of beryllium causes it to be harder, higher melting, more dense, and less reactive than the heavier members of the group.
2. The high size/charge ratio of Be^{2+} leads to compounds which are more covalent and complexes which are more stable than those of the remaining Group 2 cations. Many of its compounds have anomalously low melting points, enthalpies of formation, and are more soluble in organic solvents. The compounds are stronger Lewis acids. The halides are hygroscopic and fume when exposed to moist air.
3. Unlike magnesium and the heavier metals of the group beryllium oxide and hydroxide are amphoteric.
4. Beryllium salts are much less thermally stable because the high lattice energy of the oxide lowers the Gibbs energy change for the decomposition reaction. Similarly, it does not form a peroxide or superoxide. With ethyne, beryllium forms the carbide, Be_2C , rather than the ethynide presumably because the lattice energy of the carbide is very favourable.

ONE CORRECT OPTION TYPE QUESTIONS

- Which of the following configurations stands for an alkaline earth element?
 (a) $[\text{Ar}]4s^2$ (b) $[\text{Ar}]3d^{10}4s^1$
 (c) $[\text{Ne}]3s^23p^2$ (d) $[\text{Ar}]3d^{10}4p^1$
- Which of the following alkaline earth metal shows properties similar to Al?
 (a) Ca (b) Be
 (c) Ba (d) Sr
- Which of the following alkaline earth metals forms covalent compounds?
 (a) magnesium (b) calcium
 (c) barium (d) beryllium
- Which of the following alkaline earth metals has the highest ionic mobility in an aqueous solution?
 (a) Ba^{2+} (b) Mg^{2+}
 (c) Ca^{2+} (d) Be^{2+}
- Which of the following is the correct order of hydration energy of alkaline earth metal ions?
 (a) $\text{Ca}^{2+} > \text{Mg}^{2+} > \text{Sr}^{2+} > \text{Be}^{2+} > \text{Ba}^{2+}$
 (b) $\text{Ba}^{2+} > \text{Be}^{2+} > \text{Ca}^{2+} > \text{Mg}^{2+} > \text{Sr}^{2+}$
 (c) $\text{Be}^{2+} > \text{Mg}^{2+} > \text{Ca}^{2+} > \text{Sr}^{2+} > \text{Ba}^{2+}$
 (d) $\text{Mg}^{2+} > \text{Be}^{2+} > \text{Ba}^{2+} > \text{Ca}^{2+} > \text{Sr}^{2+}$
- Which of the following alkaline earth metal is used in manufacture of aeroplane parts?
 (a) Sr (b) Ca
 (c) Mg (d) Be
- Which of the following is formed when beryllium burns in oxygen?
 (a) Be_2O (b) BeO
 (c) BeO_2 (d) Be_2O_2
- Calcium decomposes water and liberates:
 (a) H_2 (b) O_2
 (c) CaO (d) CaO_2
- Strontium burns in oxygen to form:
 (a) Sr_2O (b) Sr_2O_2
 (c) SrO (d) SrO_2
- The mixture of MgCl_2 and MgO is called:
 (a) Pseudo alum (b) Sorel's cement
 (c) Portland cement (d) Alum
- Which of the following ores contains both calcium and magnesium?
 (a) lime stone (b) dolomite
 (c) Epsom salt (d) carnallite
- Magnesium does not decompose in:
 (a) hot water (b) steam
 (c) semi hot water (d) cold water
- Which of the following is the composition of carnallite?
 (a) $\text{MgCl}_2 \cdot \text{KCl} \cdot 6\text{H}_2\text{O}$ (b) Fe_3O_4
 (c) Na_3AlF_6 (d) $\text{Na}_2\text{Al}_2\text{O}_3$
- $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$, when heated, gives:
 (a) magnesium hydride
 (b) magnesium chloride
 (c) magnesium oxide
 (d) magnesium oxychloride
- A certain metal M is used to prepare an antacid, which is used as a medicine for acidity. This metal accidentally catches fire, which can not be extinguished by using CO_2 based extinguishers. The metal M is:
 (a) Mg (b) Ba
 (c) Be (d) Ca
- Cement does not contain:
 (a) iron (b) aluminium
 (c) calcium (d) sulphur
- Nitrolim is mixture of:
 (a) CaCN_2 and N_2 (b) CaCN_2 and graphite
 (c) CaC_2 and graphite (d) $\text{Ca}(\text{CN})_2$ and graphite
- Calcium decomposes cold water into:
 (a) $\text{Ca}(\text{OH})_2$ (b) CaH_2
 (c) O_2 (d) CaO
- On heating CaO with C, we get:
 (a) Ca and CO (b) CaCO_3 and Ca
 (c) CaC_2 and CO (d) CaC_2 and CO_2
- Calcium cyanamide, on reaction with steam under pressure, gives:
 (a) NH_3 and CaO (b) NH_3 and $\text{Ca}(\text{OH})_2$
 (c) NH_3 and CaCO_3 (d) NH_3 and $\text{Ca}(\text{HCO}_3)_2$
- What is precipitated out when dolomite is dissolved in boiling dilute H_2SO_4 ?
 (a) CaSO_4 (b) MgSO_4
 (c) CO_2 (d) MgH_2

22. The metals X and Y, form oxide and nitride when burnt in air. These are:
 (a) Li and Na (b) Mg and Ca
 (c) Cs and K (d) K and Mg
23. Which of the following metal ions plays an important role in the muscle contraction?
 (a) Na^+ (b) K^+
 (c) Mg^{2+} (d) Ca^{2+}
24. Dead burnt is:
 (a) anhydrous Na_2SO_4
 (b) Na_2CO_3 which is useful for softening
 (c) CaSO_4 which is not useful for setting
 (d) anhydrous CuSO_4
25. Out of the following elements which one do you expect to be most reactive chemically?
 (a) Magnesium (b) Calcium
 (c) Strontium (d) Barium
26. Density is highest for:
 (a) Ca (b) Sr
 (c) Ba (d) Mg
27. Compounds of alkaline earth metals are less soluble in water than the corresponding alkali metal salts due to:
 (a) their increased covalent character
 (b) high lattice energy
 (c) their high ionization energy
 (d) none of the above
28. Beryllium exhibits diagonal relationship with:
 (a) aluminium (b) lithium
 (c) magnesium (d) silicon
29. The strongest reducing agent among the following is:
 (a) Ba (b) Be
 (c) Ca (d) Mg
30. Which of the following metals does not form an ionic hydride?
 (a) Barium (b) Calcium
 (c) magnesium (d) strontium
31. Magnesium with powdered potassium chlorate:
 (a) forms MgCl_2 (b) forms MgO
 (c) liberates O_2 (d) burns with intense light
32. Which of the following metal carbonates decomposes on heating:
 (a) MgCO_3 (b) Na_2CO_3
 (c) K_2CO_3 (d) Rb_2CO_3
33. A piece of magnesium ribbon is heated to redness in the atmosphere of nitrogen and cooled with water. The evolved gas is:
 (a) H_2 (b) N_2
 (c) O_2 (d) NH_3
34. Property of the alkaline earth metals that increases with their atomic number is:
 (a) ionization energy
 (b) solubility of their hydroxides
 (c) solubility of their sulphates
 (d) electronegativity
35. Which of the following dissolves in water with a hissing sound?
 (a) bleaching powder (b) marble
 (c) quick lime (d) slaked lime
36. The formula for plaster of paris is:
 (a) CaSO_4 (b) $2\text{CaSO}_4 \cdot \text{H}_2\text{O}$
 (c) $\text{CaSO}_4 \cdot 3/2\text{H}_2\text{O}$ (d) $\text{CaSO}_4 \cdot \text{H}_2\text{O}$
37. The following is obtained on heating $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$:
 (a) MgO (b) MgCl_2
 (c) MgOCl_2 (d) MgCl
38. Mortar is the mixture of:
 (a) slaked lime, sand and water
 (b) slaked lime, plaster of paris and water
 (c) magnesium chloride, tar and lime
 (d) lime, portland cement and water
39. The substance not likely to contain CaCO_3 is:
 (a) sea shells (b) calcined gypsum
 (c) dolomite (d) a marble statue
40. Which is insoluble in water?
 (a) CaF_2 (b) CaCl
 (c) HgCl_2 (d) $\text{Ca}(\text{NO}_3)_2$
41. Several blocks of magnesium are fixed to the bottom of a ship to:
 (a) keep away the sharks
 (b) prevent action of water and salt
 (c) make the ship lighter
 (d) prevent puncturing by under sea rocks
42. The product obtained on fusion of BaSO_4 and Na_2CO_3 is:
 (a) BaO (b) BaCO_3
 (c) BaHSO_4 (d) $\text{Ba}(\text{OH})_2$
43. Which one of the following is dead burnt plaster?
 (a) Anhydrous CaSO_4
 (b) CaSO_3

3.26 Group-II Alkaline Earth Metals

- (c) $(\text{CaSO}_4)_2 \cdot \text{H}_2\text{O}$
(d) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
44. Be and Al show diagonal relationship hence, both have:
(a) same degree of electronegativity
(b) amphoteric nature of oxides
(c) approximately same charge/radius ratio
(d) all the properties above
45. Be and Al do not resemble in:
(a) both become passive on reaction with HNO_3 due to formation of oxide layer
(b) their chlorides are Lewis bases
(c) chlorides exist in polymeric form
(d) hydroxides are soluble in alkali as well as in acid
46. Going down to II A group, following properties increase except:
(a) solubility of hydroxides in H_2O
(b) hydration energy
(c) thermal stability of carbonates
(d) ionic radius
47. Identify the correct statement:
(a) Gypsum is obtained by heating Plaster of Paris
(b) Plaster of Paris can be obtained by hydration of gypsum
(c) Plaster of Paris contains higher percentage of calcium than does gypsum
(d) Plaster of Paris is obtained from gypsum by oxidation

COMPREHENSION BASED QUESTIONS

COMPREHENSION

Both alkaline earth metals and alkali metals are s-block elements. They resemble with each other in many respects but still there are certain dissimilarities in their properties on account of different number of electrons in the valency shell, smaller atomic radii, higher ionization potential, higher electronegativity, etc.,

Like lithium, beryllium also differs from rest of the alkaline earth metals on account of its small atomic size and high electronegativity. Be^{2+} ion is very small and exerts a high polarizing effect on any anion associated with it.

1. The correct sequence of increasing covalent character is represented by:
(a) $\text{BeCl}_2 < \text{NaCl} < \text{LiCl}$
(b) $\text{NaCl} < \text{LiCl} < \text{BeCl}_2$
(c) $\text{BeCl}_2 < \text{LiCl} < \text{NaCl}$
(d) $\text{LiCl} < \text{NaCl} < \text{BeCl}_2$
2. Which is least thermally stable?
(a) Li_2CO_3 (b) MgCO_3
(c) BaCO_3 (d) BeCO_3
3. Which of the following statements are true for II A group elements?
(a) Lattice energy of oxides, carbonates, fluorides decreases from Be to Ba.
(b) All form nitrides in air.

- (c) The solubility of the hydroxides increases from Be to Ba.
(d) All are correct.

4. The alkaline earth metal which does not directly combine with hydrogen is:
(a) Be (b) Ca
(c) Sr (d) Ba
5. The solubility in water of sulphates down the IIA group is:



This is due to:

- (a) increase in melting point
(b) increasing molecular mass
(c) decreasing lattice energy
(d) high heat of solvation for smaller ions
6. Which of the bicarbonate does not exist in solid state?
(a) NaHCO_3
(b) KHCO_3
(c) $\text{Ca}(\text{HCO}_3)_2$
(d) RbHCO_3
7. The element which does not directly combine with carbon on strong heating:
(a) Li (b) Be
(c) K (d) Ca

ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

- Sodium sulphate is soluble in water but barium sulphate is sparingly soluble because:
 - the hydration energy of Na_2SO_4 is more than its lattice energy
 - the lattice energy of BaSO_4 is more than its hydration energy
 - the lattice energy has no role to play in solubility
 - the lattice energy of Na_2SO_4 is more than its hydration energy
- Select the correct statements about barium:
 - it shows photoelectric effect
 - it is silvery white metal
 - it forms $\text{Ba}(\text{NO}_3)_2$ which is used in preparation of green fire
 - its ionization energy is less than radium
- Be and Al have following resemblance due to diagonal relationship:
 - have nearly equal electronegativity
 - form amphoteric oxides
 - have same charge/radius ratio
 - both form dimeric halides
- The correct statement/s is/are:
 - BeCl_2 is a covalent compound
 - BeCl_2 can form dimer
 - BeCl_2 is an electron deficient molecule
 - The hybrid state of Be in BeCl_2 is sp^2
- The compounds of alkaline earth metals which are amphoteric in nature are:
 - BeO
 - MgO
 - $\text{Be}(\text{OH})_2$
 - $\text{Mg}(\text{OH})_2$
- In which of the following reactions, MgO is formed?
 - $\text{Mg} + \text{CO}_2 \longrightarrow$
 - $\text{Mg} + \text{dil. HNO}_3 \longrightarrow$
 - $\text{Mg} + \text{NO} \longrightarrow$
 - $\text{Mg} + \text{B}_2\text{O}_3 \longrightarrow$
- Gypsum on heating gives:
 - $\text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O}$
 - CaSO_4
 - $\text{CaO} + \text{SO}_3$
 - $\text{CaS} + \text{O}_2$

ASSERTION/REASON TYPE QUESTIONS

The following questions consist of an Assertion (A) and Reason (R). Use the following key to choose the appropriate answer.

- If both assertion and reason are correct, and reason is the correct explanation of the assertion.
 - If both assertion and reason are correct, but reason is not the correct explanation of the assertion.
 - If assertion is correct but reason is incorrect.
 - If assertion is incorrect but reason is correct.
- Assertion:** Beryllium and magnesium do not impart characteristic colour to the Bunsen-burner flame.
Reason: Both Beryllium and magnesium have high ionization energy.
 - Assertion:** $\text{Be}(\text{OH})_2$ dissolves in excess of NaOH.
Reason: $\text{Be}(\text{OH})_2$ is an amphoteric compound.
 - Assertion:** Alkaline earth metals are softer than alkali metals.
Reason: Atomic radii of alkaline earth metals are smaller than corresponding alkali metals in the same periods of periodic table.
 - Assertion:** Beryllium compounds are covalent in nature.
Reason: The size of Be^{2+} ion is larger in comparison to the radii of the other divalent ions of alkaline earth metals.
 - Assertion:** Lithium resembles magnesium.
Reason: Li^+ has same size as Mg^{2+} .
 - Assertion:** The fluorides of alkaline earth metals are almost insoluble in water.
Reason: The lattice energies of the fluorides of alkaline earth metal are very high.
 - Assertion:** Sulphur is estimated as BaSO_4 and not as MgSO_4 .
Reason: The ionic radius of Mg^{2+} is smaller than that of Ba^{2+} ion.
 - Assertion:** Na_2SO_4 is soluble in water but BaSO_4 is insoluble.
Reason: Lattice energy of barium sulphate exceeds its hydration energy.

RECENT YEARS' QUESTIONS OF IIT-JEE AND AIEEE

- Be and Al exhibit many properties which are similar. But the two elements differ in: [AIEEE 2004]
 - exhibiting amphoteric nature of oxides
 - forming polymeric hydrides
 - forming covalent halides
 - exhibiting maximum covalence in compounds
- A metal M readily forms its sulphate MSO_4 which is water soluble. It forms oxide MO which becomes inert on heating. It forms insoluble hydroxide which is soluble in NaOH. The metal M is: [AIEEE 2002]
 - Mg
 - Ba
 - Ca
 - Be
- In curing cement plasters, water is sprinkled from time to time. This helps in: [AIEEE 2003]
 - Hydrating sand and gravel mixed with cement
 - Converting sand into silicate
 - Developing interlocking needle like crystals of hydrated silicates
 - Keeping it cool
- The substance not likely to contain CaCO_3 is: [AIEEE 2003]
 - Sea shells
 - Dolomite
 - A marble statue
 - Calcined gypsum
- The solubilities of carbonates decreases down the magnesium group due to decrease in: [AIEEE 2003]
 - Inter-ionic attraction
 - Entropy of solution formation
 - Lattice energy of solids
 - Hydration energy of cations
- Several blocks of magnesium are fixed into the bottom of ship to: [AIEEE 2003]
 - Prevent action of water and salt
 - Prevent puncturing by under sea rocks
 - Keep away the sharks
 - Make the ship lighter
- One mole of magnesium nitride on reaction with excess of water gives: [AIEEE 2004]
 - Two mole of HNO_3
 - Two mole of NH_3
 - 1 mole of NH_3
 - 1 mole of HNO_3
- The number and types of bonds between two carbon atoms in calcium carbide are: [AIEEE 2005]
 - One sigma, one pi
 - One sigma, two pi
 - Two sigma, one pi
 - Two sigma, two pi
- Which process is used in the extractive metallurgy of Mg? [IIT 2002]
 - Fused salt electrolysis
 - Self reduction
 - Aqueous solution electrolysis
 - Thermite reduction
- A sodium salt on treatment with MgCl_2 gives white precipitate only on heating. The anion of sodium salt is: [IIT 2004]
 - HCO_3^-
 - CO_4^{2-}
 - NO_3^-
 - SO_4^{2-}

ANSWER KEYS

ONE CORRECT OPTION TYPE QUESTIONS

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (a) | 2. (b) | 3. (d) | 4. (a) | 5. (c) | 6. (c) | 7. (b) | 8. (a) | 9. (c) | 10. (b) |
| 11. (b) | 12. (d) | 13. (a) | 14. (b) | 15. (d) | 16. (d) | 17. (b) | 18. (a) | 19. (c) | 20. (c) |
| 21. (a) | 22. (b) | 23. (d) | 24. (c) | 25. (d) | 26. (c) | 27. (a) | 28. (a) | 29. (a) | 30. (c) |
| 31. (d) | 32. (a) | 33. (d) | 34. (b) | 35. (c) | 36. (b) | 37. (b) | 38. (a) | 39. (b) | 40. (a) |
| 41. (c) | 42. (b) | 43. (a) | 44. (d) | 45. (b) | 46. (b) | 47. (c) | | | |

COMPREHENSION TYPE QUESTIONS

- | | | | | | | |
|--------|--------|--------|--------|--------|--------|--------|
| 1. (b) | 2. (d) | 3. (d) | 4. (a) | 5. (d) | 6. (c) | 7. (c) |
|--------|--------|--------|--------|--------|--------|--------|

ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

- | | | | | | | |
|-----------|--------------|-----------------|--------------|-----------|--------------|--------------|
| 1. (a, b) | 2. (b, c, d) | 3. (a, b, c, d) | 4. (a, b, c) | 5. (a, c) | 6. (a, c, d) | 7. (a, b, c) |
|-----------|--------------|-----------------|--------------|-----------|--------------|--------------|

ASSERTION/REASON TYPE QUESTIONS

- | | | | | | | | |
|--------|--------|--------|--------|--------|--------|--------|--------|
| 1. (a) | 2. (b) | 3. (d) | 4. (c) | 5. (d) | 6. (a) | 7. (b) | 8. (a) |
|--------|--------|--------|--------|--------|--------|--------|--------|

RECENT YEARS' QUESTIONS OF IIT-JEE AND AIEEE

- | | | | | | | | | | |
|--------|--------|--------|--------|--------|--------|--------|--------|--------|---------|
| 1. (b) | 2. (d) | 3. (c) | 4. (d) | 5. (d) | 6. (d) | 7. (b) | 8. (b) | 9. (a) | 10. (a) |
|--------|--------|--------|--------|--------|--------|--------|--------|--------|---------|

CHAPTER 4

Group 13- Boron Family

INTRODUCTION

Table 4.1 General Properties

Property	Element				
	Boron B	Aluminium Al	Gallium Ga	Indium In	Thallium Tl
Atomic number	5	13	31	49	81
Atomic mass(g mol ⁻¹)	10.81	26.98	69.72	114.82	204.38
Electronic	[He]2s ² 2p ¹	[Ne]3s ² 3p ¹	[Ar]3d ¹⁰ 4s ² 4p ¹	[Kr]4d ¹⁰ 5s ² 5p ¹	[Xe]4f ¹⁴ 5d ¹⁰ 6s ² 6p ¹
Atomic radius/pm ^a	(85)	143	135	167	170
Ionic radius M ³⁺ /pm ^b	(27)	53.5	62.0	80.0	88.5
Ionic radius M ⁺ /pm	—	—	120	140	150
Ionization $\Delta_f H_1$	801	577	579	558	589
enthalpy $\Delta_f H_2$	2427	1816	1979	1820	1971
(kJ mol ⁻¹) $\Delta_f H_3$	3659	2744	2962	2704	2877
Electronegativity ^c	20.0	1.5	1.6	1.7	1.8
Density/g cm ⁻³ at 298 K	2.35	2.70	5.90	7.31	11.85
Melting point/K	2453	933	303	430	576
Boiling point/K	3923	2740	2676	2353	1730
E ⁰ /V for (M ³⁺ /M)	—	−1.66	−0.56	−0.34	+1.26
E ⁰ /V for (M ⁺ /M)	—	+0.55	−0.79 (acid) −1.39 (alkali)	−0.18	−0.34

^aMetallic radius, ^b6-coordination ^cPauling scale.

Boron is a non-metal and always forms covalent bonds. Normally, it forms three covalent bonds at 120° using sp² hybrid orbitals. There is no tendency to form univalent compounds. If we apply Fajan's rule we can see that the polarizing power of B³⁺ is so high that the bond will always have considerably high covalent character.

All BX₃ compounds are electron deficient and may accept an electron pair from another atom, thus forming a coordinate bond. BF₃ is commercially important as a catalyst. The four elements Al, Ga, In and Tl all form trivalent compounds. The heavier members show the 'inert pair effect' and univalent compounds become increasingly important in the order

4.2 Group13-Boron Family

Ga $\rightarrow$ In $\rightarrow$ Tl. These four elements are more metallic and more ionic, than B, They are moderately reactive metals. Their compounds are on the borderline between ionic and covalent. Many of their compounds are covalent when anhydrous, but they form ions in solution.

Aluminium is the most abundant metal and the third most abundant element (after oxygen and silicon) by weight in the earth's crust. It is well known and is commercially important. Aluminium metal is produced on a vast scale The most important source of aluminium is bauxite. This is a generic name for several minerals with formulae varying between $\text{Al}_2\text{O}_3 \cdot \text{H}_2\text{O}$ and $\text{Al}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$. Aluminium also occurs in large amounts in aluminosilicates rocks such as feldspars and micas.

Table 4.2 Abundance of the Elements in the Earth's Crust by Weight

	<i>ppm</i>	<i>Relative abundance</i>
B	9	38
Al	83000	3
Ga	19	33 =
In	0.24	63
Ti	0.5	60

Gallium is twice as abundant as boron, but indium and thallium are much less common. All three elements, Ga, In and Tl, occur as sulphides. Ga, In and Tl are not very well known. This is partly because they do not occur as concentrated ores and partly because there are no major uses for them. Small amounts of Ga are found in the ores of the elements adjacent to it in the periodic table (Al, Zn and Ge). Traces of In and Tl are found in ZnS and PbS ores.

Atomic and Ionic Radii

Size of atoms and ions

The metallic radii of the atoms do not increase regularly on descending the group (Table 4.1). However, the values are not strictly comparable. Boron is not a metal and the radius given is half the closest approach in the structure. Ga has an unusual structure and the value given is half the closest approach. The others have close-packed metal structures.

The ionic radii for M^{3+} increase down the group, though not in the regular way observed in Groups I and II. There are two reasons for this:

- There is no evidence for the existence of B^{3+} under normal conditions and the value is an estimate.
- The electronic structures of the elements are different. Ga, In and Tl follow immediately after a row of ten transition elements. They therefore have ten d electrons, which are less efficient at shielding the nuclear charge than the 5 s and p electrons. (Shielding is in the order $s > p > d > f$.) Poor shielding of the nuclear charge results in the outer electrons being more firmly held by the nucleus. Thus, atoms with a d^{10} inner shell are smaller and so have higher ionization energies than would otherwise be expected. This contraction in size is sometimes called the d-block contraction. In a similar way, Tl follows immediately after 14 f-block elements. The size and ionization energy of Tl are affected even more by the presence of 14f electrons, which shield the nuclear charge even less effectively.

Table 4.3 Ionic and Covalent Radii and Electronegativity Values

	<i>Metallic radius</i> (Å)	<i>Ionic radius</i> M^{3+} (Å) M^+ (Å)		<i>Pauling's electronegativity</i>
B	(0.885)	(0.27)	—	2.0
Al	1.43	0.535	—	1.5
Ga	(1.225)	0.620	1.20	1.6
In	1.67	0.800	1.40	1.7
Tl	1.70	0.885	1.50	1.8

The large difference in size between B and Al results in many differences in properties. Thus B is a non-metal, has a very high melting point, always forms covalent bonds and forms an acidic oxide. In contrast, Al is a metal has a much lower melting point and its oxide is amphoteric.

In a nutshell

- Down the group, as the number of shell increases, atomic radii increase.
- There is a small decrease in the size from Al to Ga.

The small size of Ga is due the poor screening ability of the ten 'd' electrons which results in the high Z_{eff} value for Ga. [Al $\rightarrow$ (Ne)3s²3p¹] [Ga $\rightarrow$ (Ar)3d¹⁰4s² 4p¹]

Ionization energies The ionization energies increase as expected (first ionization energy < second ionization energy < third ionization energy). The sum of the first three ionization energies for each of the elements is very high. Thus boron has no tendency to form ions and always forms covalent bonds. The other elements normally form covalent compounds except in solution.

Table 4.4 Ionization Energies

	<i>Ionization energy (kJ mol⁻¹)</i>			
	<i>1st</i>	<i>2nd</i>	<i>3rd</i>	<i>Sum of three</i>
B	801	2427	3659	6887
Al	577	1816	2744	5137
Ga	579	1979	2962	5520
In	558	1820	2704	5082
Tl	589	1971	2877	5437

The Group III Elements

The ionization energy values do not decrease smoothly down the group. The decrease from B to Al is the usual trend on descending a group associated with increased size. The poor shielding by d electrons and the resulting d-block contraction affect the values for the later elements.

In a nutshell

- Ionization energy generally decreases on moving down but the small increase in the ionization energy from Al to Ga is due to poor screening ability of the ten 'd' electrons which results in the high Z_{eff} value for Ga. [Al $\rightarrow$ (Ne)3s²3p¹] [Ga $\rightarrow$ (Ar)3d¹⁰4s² 4p¹]
- The considerable increase in the ionization energy from In to Tl is due to very poor screening ability of the fourteen 'f' electrons which results in the high Z_{eff} value for Tl

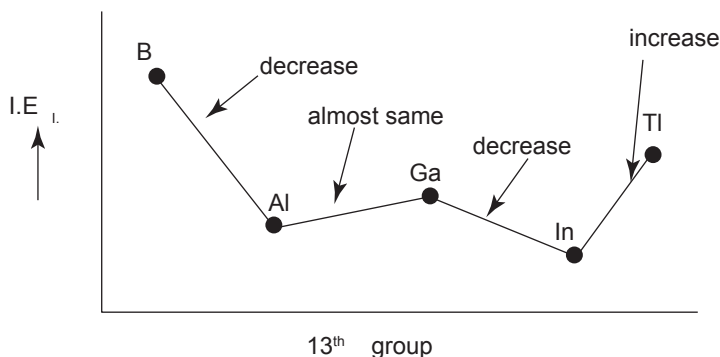


Fig. 4.1

Worked-out Examples

1. Boron does not form B^{3+} ion. Explain.

Ans. The tendency to form cations depends upon ionisation energy. Boron being smaller in size has higher values of ionization energy and thus it is difficult to liberate three electrons and form B^{3+} ions.

2. Arrange the following in increasing order of $(IE_1 + IE_2 + IE_3)$ B, Al, Tl, Ga, In.

Ans. $In < Al < Tl < Ga < B$. The I.E. values do not decrease smoothly down the group. The decrease from B to Al is usual trend due to increased size. The poor shielding by 'd' electrons resulting in d-block contraction affect the values in later elements.

3. Although the ionization energy of boron (8.30 eV) is less than gold (9.22 eV), yet former is a semi-metal while later is a metal.

Ans. Here, metallic and non-metallic characters are explained in terms of the lattice structure of the solid. Boron has 8 or less atoms as nearest neighbours in solid state while gold has 12 atoms. In general, good metals have a large number of neighbours atoms, while non-metals have relatively less atoms.

Density Down the group, density increases as the increase in mass is greater than increase in volume.

(gm/C.C)	B	Al	Ga	In	Tl
	2.34	2.70	5.91	7.31	11.85

Melting Points, Boiling Points and Structures

The melting points of the Group III elements do not show a regular trend as did the metals of Groups I and II. The Group III values are not strictly comparable because B and Ga have unusual crystal structures.

Boron has an unusual crystal structure which results in the melting point being very high. There are at least four different allotropic forms. Boron has insufficient electrons to fill the valence shell even after forming bonds. The variety and complexity of the allotropic forms illustrates the number of ways in which boron attempts to solve this problem. Other elements solve this problem by metallic bonding, but small size and high ionization energy make this impossible for boron. All four allotropic forms contain icosahedral units with boron atoms at all 12 corners. (Note that an icosahedron is a regular shape with 12 corners and 20 faces.) In these units twelve B atoms form a regular shape and each B atom is bonded to five equivalent neighbors (at a distance of 1.77 Å). The difference between the allotropic forms arises in the way the icosahedra are bonded together. The simplest form is β -rhombohedral boron. In this, half the atoms are bonded to one atom in another icosahedron (at a distance of 1.71 Å) and half the atoms are bonded to atoms in two different icosahedra (at a distance of 2.03 Å). Plainly, this is neither a regular structure nor a metallic structure. Only 37% of space is occupied by the atoms, compared with 74% for a close-packed arrangement. This shows that icosahedra fill up space ineffectively. The other allotropes have even more complicated structures.

The elements Al, In and Tl all have close-packed metal structures. Gallium has an unusual structure. Each metal atom has one close neighbour at a distance of 2.43 Å and six more distant neighbours at distances between 2.70 Å and 2.79 Å. This remarkable structure tends towards discrete diatomic molecules rather than a metallic structure. This accounts for the incredibly low melting point of gallium of 30°C. Ga is also unusual because the liquid expands when it forms the solid, i.e., the solid is less dense than the liquid.

Table 4.5 Melting and Boiling Points

	Melting point (°C)	Boiling point (°C)
B	2180	3650
Al	660	2567
Ga	30	2403
In	157	2080
Tl	303	1457

This property is unique to Ga, Ge and Bi.

Though the melting points decrease from Al to In, as expected on descending a group, it increases again for Tl. The boiling point of B is unusually high, but the values for Ga, In and Tl decrease on descending the group as expected. Note that the boiling point for Ga is in line with the others, whereas its melting point is not. The very low melting point is due to the unusual crystal structure, but the structure no longer exists in the liquid.

In a nutshell

Melting points decrease from B to Ga and then increase from Ga to Tl. ($B > Al > Tl > In > Ga$)

Explanation

- (i) Boron has a very high melting point because it exists as a giant covalent polymer in both solid and liquid states.
- (ii) The low melting point of Ga is due to the fact that it consists of only Ga_2 molecules. Gallium remains liquid upto 2273 K and hence, it is used in high temperature thermometry.
The exceptionally low value of M.P. of Ga is due to its structure consists of Ga_2 molecules.

Oxidation State

Oxidation states and type of bonds

The (+III) oxidation state

The elements all have three outer electrons. Apart from Tl they normally use these to form three bonds, giving an oxidation state of (+III). Are the bonds ionic or covalent? Covalency is suggested by the following:

1. Fajans' rules—small size of the ions and their high charge of $3+$ favours the formation of covalent bonds.
2. The sum of the first three ionization energies is very large and this also suggests that bonds will be largely covalent.
3. The electronegativity values are higher than for Groups I and II and when reacting with other elements the difference is not likely to be large.

Boron is considerably smaller than the other elements and thus has higher ionization energy than the others. The ionization energy is so high that B is always covalent.

Many simple compounds of the remaining elements, such as $AlCl_3$ and $GaCl_3$, are covalent when anhydrous. However, Al, Ga, In and Tl all form metal ions in solution. The type of bonds formed depends on which is most favourable in terms of energy. This change from covalent to ionic happens because the ions are hydrated and the amount of hydration energy evolved exceeds the ionization energy. Consider $AlCl_3$: 5137 kJ mol⁻¹ are required to convert Al to Al^{3+} , $\rightarrow H_{\text{hydration}}$ for Al^{3+}

$$\begin{aligned} & -4665 \text{ kJ mol}^{-1} \text{ and } \rightarrow H_{\text{hydra}} \text{ hydration energy is: for Cr is } -381 \text{ kJ mol}^{-1}. \text{ Thus the total} \\ & -4665 + (3 \times -381) = -5808 \text{ kJ mol}^{-1} \end{aligned}$$

This exceeds the ionization energy, so $AlCl_3$ ionizes in solution.

The hydrated metal ions have six molecules of water which are held very strongly in an octahedral structure $[M(H_2O)_6]^{3+}$. The metal-oxygen bonds are very strong. This weakens the O—H bonds and protons are released to neighbouring water molecules, forming H_3O^+ (hydrolysis).



The (+1) oxidation state-the 'inert pair effect'

In the 5-block, Group I elements are univalent and Group II elements are divalent. In Group III we would expect the elements to be trivalent. In most of their compounds this is the case, but some of the elements show lower valency states as well. There is an increasing tendency to form univalent compounds on descending the group. Compounds with Ga(I), In(I) and Tl(I) are known. With Ga and In the (+1) oxidation state is less stable than the (+III) state. However, the stability of the lower oxidation state increases on descending the group. Tl(I) thallos compounds are more stable than Tl(III) thallic compounds.

How and why does monovalency occur? The atoms in this group have an outer electronic configuration of s^2p^1 . Monovalency is explained by the s electrons in the outer shell remaining paired and not participating in bonding. This is called

4.6 Group13-Boron Family

the ‘inert pair effect’. If the energy required to unpair them exceeds the energy evolved when they form bonds, then the *s*-electrons will remain paired. The strength of the bonds in MX_3 compounds decreases down the group. The mean bond energy for chlorides are $\text{GaCl}_3 = 242$, $\text{InCl}_3 = 206$ and $\text{TlCl}_3 = 153 \text{ kJ mol}^{-1}$. Thus, the *s* electrons are most likely to be inert in thallium.

The inert pair effect is not the explanation of why monovalency occurs in Group III. It merely describes what happens, i.e., two electrons do not participate in bonding. The reason that they do not take part in bonding is energy. The univalent ions are much larger than the trivalent ions and (+1) compounds are ionic and are similar in many ways to Group I elements.

The inert pair effect is not restricted to Group III, but also occurs among the heavier elements in other groups in the *p*-block. Examples from Group IV are Sn^{2+} and Pb^{2+} and examples from Group V are Sb^{3+} and Bi^{3+} . The lower oxidation state becomes more stable on descending the group. Thus Sn^{2+} is a reducing agent but Pb^{2+} is stable and Sb^{3+} is a reducing agent but Bi^{3+} is stable. When the *s* electrons remain paired, the oxidation state is always two lower than the usual oxidation state for the group.

Thus in the *s*-block, Groups I and II show only the group valency. Groups in the *p*-block show variable valency, differing in steps of two. Variable valency also occurs with elements in the *d*-block. This arises from using different numbers of *d* electrons for bonding, so in this case the valency can change in steps of one (e.g., Cu^+ and Cu^{2+} , Fe^{2+} and Fe^{3+}).

The (+1) oxidation state

Gallium is apparently divalent in a few compounds, such as GaCl_2 . However, Ga is not really divalent, as the structure of GaCl_2 has been shown to be $\text{Ga}^+[\text{GaCl}_4]^-$ which contains Ga(I) and Ga(III).

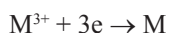
B	–	+3
Al	–	+3
Ga	–	+3, +1
In	–	+3, +1
Tl	–	+3, +1

As we move down the group +1 oxidation state become more stable and stability of +3 oxidation state decreases.

- Note:** → B forms only covalent compounds.
→ Al, Ga → Both covalent and ionic (Al_2O_3 , AlN , AlF_3 – ionic
 AlCl_3 , AlBr_3 – covalent)
→ Tl → resembles with alkali metals due to its stable +1 state.
(TlOH – strong base
 TlCl – (CsCl type lattice
 Tl_2O – Anti fluorite Na_2O)
Reactive – stored in kerosene)
→ Nature of compounds of M^{3+} is decided by Fajan’s rule that why from B^{3+} to Tl^{3+} , the tendency of these ions to form covalent compounds decreases.

Electropositive Character

The electropositive or metallic nature of the elements increases from B to Al, but then decreases from Al to Tl. This is shown by the stand; electrode potentials for the reaction:



The increase in metallic character from B to Al is the usual trend descending a group associated with increasing size. However, Ga, In and Tl do not continue the trend. The elements are less likely to lose electrons (and are thus less electropositive), because of the poor shielding by *d* electrons described previously.

Table 4.6 Standard Electrode Potentials E°

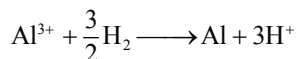
	$M^{3+} M$ (volts)	$M^+ M$ (volts)
B	(-0.87*)	
Al	-1.66	+0.55
Ga	-0.56	-0.79†
In	-0.34	-0.18
Tl	+1.26	-0.34

*For $H_3BO_3 + 3H^+ + 3e^- \rightarrow B + 3H_2O$

†Value in acidic solution.

The standard electrode potentials E° for $M^{3+}|M$ become less negative from Al to Ga to In and the potential becomes positive for Tl. Since $G = -nFE^\circ$ it follows that ΔG , the free energy of formation of the metal, e.g., $Al^{3+} + 3e^- \rightarrow Al$, is positive. Thus it is difficult to make this reaction work. (The reverse reaction $Al \rightarrow Al^{3+} + 3e^-$ occurs spontaneously.) The standard potential becomes less negative on descending the group so it becomes less difficult for the reaction $M^{3+} \rightarrow M$ to occur. Thus, the (+III) oxidation state becomes less stable in aqueous solution on descending the group. In a similar way, the E° values for $M^+|M$ show that the (+1) state increases in stability. With Tl, the (+1) state is more stable than the (+III) state.

It should be remembered that in this type of argument E° and ΔG refer to the reaction with H_2 :



In a nutshell

The electropositive character down the group first increases from boron to aluminium and then decreases from aluminium to thallium.

Explanation. The high metallic character of Al than B is due to its larger size and lower ionization energy. While the decrease in the electropositive character beyond Al is due to the presence of electrons in d and f-orbitals which do not shield the valence electrons from the force of attraction of the nucleus effectively. Consequently, the valence electrons in these elements are firmly held by the nucleus and hence, are not easily removed.

Worked-out Examples

4. From boron to thallium, +1 oxidation state becomes more stable. Explain.

Ans. Due to inert pair effect.

5. Boron forms no compound in unipositive state while thallium is quite stable in unipositive state. Explain.

Ans. M^+ ionic state of group 13 elements exists due to inert pair effect when ns^2 electrons penetrate in $(n-1)$ d-subshell to become inert. The inert pair effect begins from $n \rightarrow 4$ and increases with, increase in value of. The n i.e., outermost shell for boron is 2 and thus it does not form B^+ ion.

6. The first ionization energies of boron and aluminium are lesser than those of Be and Mg respectively. Explain.

Ans. Both B and Al has ns^2np^1 configuration and thus removal of first electrons occurs from 2p and 3p respectively. In Be and Mg (ns^2), the removal of first electron takes place from 2s and 3s respectively. More closer are subshells to the nucleus, more tightly are held electrons and thus difficult is their removal.

Some Noteworthy Points

- Aluminium is bluish white metal with a brilliant luster which is soon destroyed by the formation of an oxide layer on it.
- Gallium is a low melting solid (M.pt. $29.8^\circ C$) and is liquid on a particular warm day.
- Gallium readily supercools, i.e., remains liquid even at temperatures several degrees below its melting (freezing) point.

4.8 Group13-Boron Family

- Maximum covalency of boron is four while that of Al is six.
- Highly toxic element of Group 13 is thallium.
- All the metalloids (e.g., B, Si, As, Sb, Tl etc.,) belong to p-block.

Chemical properties

Table 4.7 Some Reactions of the Other Group III Metals

Reaction	Comment
$4M + 3O_2 \rightarrow M_2O_3$	All react at high temperature Al very strongly exothermic Ga only superficially oxidized Tl forms some Tl_2O as well
$2Al + N_2 \rightarrow 2AlN$	Only Al at high temperature
$2M + 3F_2 \rightarrow 2MF_3$	All the metals } form } Tl^+ also formed trihalides }
$2M + 3Cl_2 \rightarrow 2MCl_3$	
$2M + 3Br_2 \rightarrow 2MBr_3$	
$2M + 3I_2 \rightarrow 2MI_3$	
$TlI + I_2 \rightarrow Tl^+[I_3]^-$	Al, Ga, In only thallium (I) triiodide formed
$2M + 6HCl \rightarrow 2MCl_3 + 3H_2$	All react with dilute mineral acids, Al passivated by HNO_3 particularly when concentrated
$Al + NaOH + H_2O \rightarrow NaAlO_2 + H_2$ $Na_3AlO_3 + H_2$	Al and Ga only
$M + NH_3 \rightarrow MNH_2$	All the metals form amides

Action with air–reaction with water and air

The metals Al, Ga, In and Tl are silvery white. Thermodynamically should react with water and with air, but in fact it is stable in both. The reason is that a very thin oxide film forms on the surface and protects the metal from further attack. This layer is only 10^{-4} to 10^{-6} mm thick. If the protective oxide covering is removed, for example by amalgamating with mercury, then the metal readily decomposes cold water, forming Al_2O_3 and liberating hydrogen.

Table 4.8 Some Reactions of the Other Group III Metals

Reaction	Comment
$4M + 3O_2 \rightarrow M_2O_3$	All react at high temperature Al very strongly exothermic Ga only superficially oxidized Tl forms some Tl_2O as well
$2Al + N_2 \rightarrow 2AlN$	Only Al at high temperature
$2M + 3F_2 \rightarrow 2MF_3$	All the metals } form } Tl^+ also formed trihalides }
$2M + 3Cl_2 \rightarrow 2MCl_3$	
$2M + 3Br_2 \rightarrow 2MBr_3$	
$2M + 3I_2 \rightarrow 2MI_3$	
$TlI + I_2 \rightarrow Tl^+[I_3]^-$	Al, Ga, In only thallium (I) triiodide formed
$2M + 6HCl \rightarrow 2MCl_3 + 3H_2$	All react with dilute mineral acids, Al passivated by HNO_3 particularly when concentrated
$Al + NaOH + H_2O \rightarrow NaAlO_2 + H_2$ $Na_3AlO_3 + H_2$	Al and Ga only
$M + NH_3 \rightarrow MNH_2$	All the metals form amides

Aluminium articles are often 'anodized' to give a decorative finish. This is done by electrolysing dilute H_2SO_4 with the aluminium as the anode. This produces a much thicker layer of oxide on the surface (10^{-2} mm). This layer can take up pigments, thus colouring the aluminium.

Aluminium burns in nitrogen at high temperatures, forming AlN . The other elements do not react.

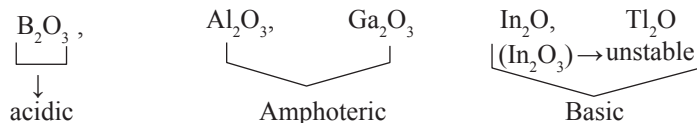
M_2O_3 type oxide (sesquioxides)

B $\rightarrow$ inert, reacts only on heating $\rightarrow \text{B}_2\text{O}_3$

Al $\rightarrow$ at room temperature $\rightarrow \text{Al}_2\text{O}_3$

(Reaction takes place only on surface)

Tl $\rightarrow$ highly reactive, stored in kerosene

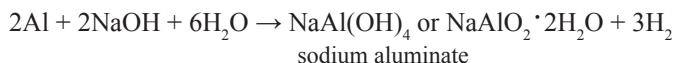


Reaction with acids and alkalis

Aluminium dissolves in dilute mineral acids liberating hydrogen.

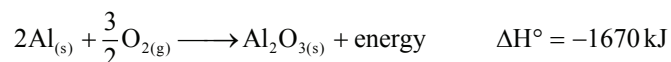


However, concentrated HNO_3 renders the metal passive because it is an oxidizing agent and produces a protective layer of oxide on the surface. Aluminium also dissolves in aqueous NaOH (and is therefore amphoteric), liberating hydrogen and forming aluminates. (The nature of aluminates is discussed later.)

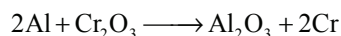


Reaction with oxygen

Aluminium burns readily in air or oxygen and the reaction is strongly exothermic. This is known as the Thermite reaction.



The Thermite reaction evolves so much energy that it can be dangerous. The aluminium becomes white hot and often causes fires. For this precise reason, mixtures of Al and an oxide such as Fe_2O_3 or SiO_2 (to provide the oxygen) were used to make incendiary bombs during World War II. Warships are sometimes made of aluminium alloys to reduce their weight. A thermite reaction can be started if the ship is hit by a missile. Such fire on ships caused considerable casualties in the Falklands Islands conflict. The very strong affinity of Al for oxygen is used in the metallurgies extraction of other metals from their oxides.



Reaction with the halogens and sulphate

Aluminium reacts with the halogens quite readily, even when cold forming trihalides.

Aluminium sulphate is used in large amounts (3.4 million tonnes in 1985). It is made by treating bauxite with H_2SO_4 . It is used as a coagulant and precipitant in treating both drinking water and sewage. It is also used in the paper industry and as a mordant in dyeing.

Reactions of Ga, In and Tl

Gallium and indium are stable in air and are not attacked by water unless free oxygen is present. Thallium is a little more reactive and is superficially oxidized by air. All three metals dissolve in dilute acids, liberating hydrogen. Gallium is

4.10 Group13-Boron Family

amphoteric like aluminium and it dissolves in aqueous NaOH, liberating H₂ and forming gallates. The oxides and hydroxides of Al and Ga are also amphoteric. In contrast, the oxides and hydroxides of In and Tl are purely basic. All three metals react with the halogens on gentle warming.

Some Properties of Thallium (I)

Thallium(I) or thallos compounds are well known. They are typically colourless. They are also extremely poisonous. When ingested, traces turn the hair very black, but larger doses cause loss of hair and death. They are toxic because they upset the enzyme systems in the body.

In aqueous solution the Tl⁺ ion is much more stable than Tl(III). The ionic radius of Tl⁺ (1.50 Å) is between that of K⁺ (1.38 Å) and Rb⁺ (1.52 Å). For this reason Tl⁺ resembles Group I ions in a number of ways. TlOH and Tl₂O are both soluble in water and are strongly basic. They absorb CO₂ from the air, forming Tl₂CO₃. The solubility of most of the salts is slightly lower than for potassium salts. Tl⁺ can replace K⁺ in some enzymes and can thus be used as a biological tracer in the body, there are some differences. TlOH is yellow and on heating to 100°C it turns into black Tl₂O. The coordination number of Tl⁺ is usually 6 or 8, compared with 6 for Group I ions. TlF is soluble in water, but the other halides are almost insoluble. There is also some similarity with Ag⁺, as TlCl is sensitive to light. It darkens when exposed to light rather like AgCl, but TlCl is not soluble in NH₄OH whilst AgCl is soluble.

Boron (5B): 1s²2s²2p¹

- The credit of isolation of boron goes to Davy.
- Boron exhibits allotropy. It exists as amorphous boron (**black boron**) and crystalline boron (**adamanite boron**).
- The relative abundance of boron in earth's crust is $\approx 0.001\%$.
- The important minerals of boron are:

<i>Borax</i>	:	Na ₂ [B ₄ O ₅ (OH) ₄].8H ₂ O or Na ₂ B ₄ O ₇ .10H ₂ O
<i>Orthodoric acid</i>	:	H ₃ BO ₃
<i>Kernite</i>	:	Na ₂ [B ₄ O ₅ (OH) ₄] or Na ₂ B ₄ O ₇ .2H ₂ O
<i>Colemanite</i>	:	Ca ₂ [B ₃ O ₄ (OH) ₃] ₂ .2H ₂ O or Ca ₂ B ₆ O ₁₁ .5H ₂ O

In India, deposits of borates occur in *Puga valley* in Ladakh and *Samber lake* in Rajasthan

Worked-out Example

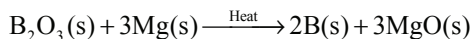
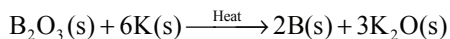
7. What are special features of structure of Boron?

Ans. It is a symmetrical solid with icosahedral shape. There are 20 faces (equilateral triangle). The faces meet at 12 corners. Each icosahedron consists of 12 boron atoms; six of them are bonded to another icosahedron by two centre bonds (1.71 Å); each of the other six is bonded by a three centre bond (2.02 Å) in separated icosahedra.

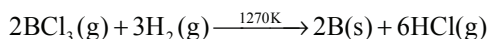
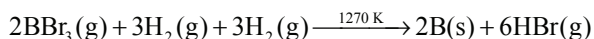
Isolation of boron

Elemental boron is obtained by any of the following methods:

- (a) *By reduction of boron oxides.* Using metals like potassium or magnesium at high temperature.



- (b) *By the reduction of boron halides.* Using dihydrogen at high temperatures.



Use of boron

An important use of boron is to make boron steel or boron carbide control rods for nuclear reactors. Boron has a very high cross-section for capturing neutrons. Control rods made of boron steel or boron carbide may be lowered into a reactor to absorb neutrons and thus slow the reactor down. Boron carbide is also used as an abrasive. Boron is used to make impact resistant steel, as it increases the hardenability (that is the depth to which it will harden) of steel.

Borax $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$, orthoboric acid H_3BO_3 and boron sesquioxide B_2O_3 find many uses. Borax is mixed with NaOH and sold as Tolybor' and 'Timbor' for treating timber and hardboard against attack by wood-boring insects. Other uses are as a food preservative, a mild antiseptic and a flame retardant for fabric and wood. Borax is used as a flux in brazing and in silver soldering. The borax reacts with many metal oxides, forming easily melted borates. The flux removes oxides such as Cu_2O from the surface of hot brass and thus allows the clean metal surfaces to fuse with the solder. Borax is also used in making enamel and in leather tanning.

Orthoboric acid is made by hydrolysing many boron compounds. About 170000 tonnes is made each year by acidifying solutions of borax. Reaction of H_3BO_3 with H_2O_2 gives the mono peroxoboric acid which probably has the structure $[(\text{HO})_3\text{B}(\text{OOH})]$. Sodium peroxoborate $\text{Na}_2[\text{B}_2(\text{O}_2)_2(\text{OH})_4] \cdot 6\text{H}_2\text{O}$ is a constituent of many washing powders. Peroxo-borates act as brighteners, because they absorb UV light and emit visible light. This makes both white and coloured fabrics appear extra bright. If used at temperatures over 80°C , peroxoborates decompose to hydrogen peroxide H_2O_2 , which acts as a bleach.

Boron sesquioxide B_2O_3 is used in making borosilicate (heat resistant) glass (e.g., Pyrex which contains 14% B_2O_3). Borosilicate glass has a lower coefficient of thermal expansion and is easier to work than normal 'soda glass'. H_3BO_3 , B_2O_3 and calcium borate are used to make soda-free glass fibre, which is used for thermal insulation in houses.

Worked-out Example

8. It is difficult to obtain pure crystalline boron, why?

Ans. Its melting point is very high and liquid is corrosive.

Properties of boron

- Boron is extremely hard solid (only next to diamond) due to its three dimensional network structure.
- Its melting point is $\approx 2450\text{ K}$ and its boiling point is $\approx 2825\text{ K}$.
- It has low electrical conductivity.
- It occurs in two *isotopic* forms, i.e., $^{10}_5\text{B}$ and $^{11}_5\text{B}$ in natural abundances of approximately 20% and 80% respectively.

Worked-out Examples

9. Boron has high m.pt. and b.pt. Explain.

Ans. Boron has very high m.pt. and b.pt. because it exists as a giant covalent, polymeric structure both in solid as well as in liquid state.

10. STATEMENT-1: Boron always forms covalent bond.

(IIT 2007)

because

STATEMENT-2: The small size of B^{3+} favours formation of covalent bond.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
- (c) Statement-1 is True, Statement-2 is False
- (d) Statement-1 is False, Statement-2 is True

Ans. (a) Both the statements are correct. Also, the statement 2 is correct explanation of the statement 1. It follows from Fajan's rule that smaller cation with high charge density have a tendency to form covalent bond due to the larger polarization effect.

4.12 Group13-Boron Family

Chemical reactions of boron

It has low reactivity because of its small size and high values of ionization potential. It is almost inert to most of the chemical reagents. Some of its chemical reaction are listed below:

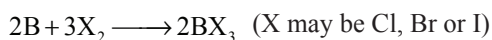
- **Metals** such as **Cr**, **Mg** react at high temperature to form respective **borides**.



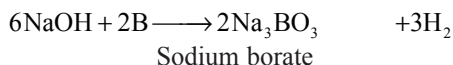
- Non-metals also react at high temperature.



Halogens react to give *boron trihalide*:



- Boron trihalides are **Lewis acids** and the Lewis acid strength decreases as $\text{BBr}_3 > \text{BCl}_3 > \text{BF}_3$.
- *Alkalies in fused state* react to give *dihydrogen* and *borates*



- *Oxidising acids* such as *concentrated H₂SO₄* and *concentrated HNO₃* give **boric acid**

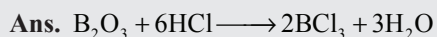


- Boron reduces CO₂ to C and SiO₂ to Si



Worked-out Example

11. Show that B₂O₃ is amphoteric oxide.



Use of boron

Boron is used:

- in steel industry for increasing hardness of steel.
- in making light and composite materials for aircrafts.
- in glass industry, for making *borosilicate* glass.
- as a semiconductor for making electronic devices.
- in nuclear industry in the form of metal borides. These are used as protective shields because of ability of B-10 isotope to absorb neutrons.
- in preparing *boron carbide* (B₄C) also called *norbide* which is the hardest artificial substance.

Hydrides of Boron

None of the Group III elements react directly with hydrogen, but several interesting hydrides are known. The boron hydrides are sometimes called boranes by analogy with the alkanes (hydrocarbons). Almost 20 boranes have been reported and 11 are well characterized. They fall into two series:

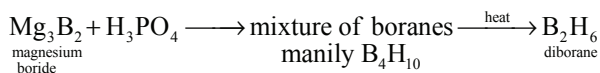
1. $B_n H_{(n+4)}$ (called nido-boranes).
2. A less stable series $B_n H_{(n+6)}$ (called arachno-boranes).

In cases where the nomenclature is ambiguous, as for example for pentaborane, it is usual to include the number of hydrogen atoms in the name.

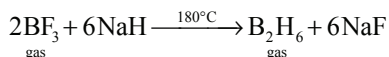
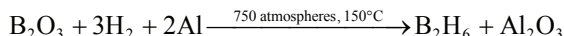
Table 4.9 The Two Series of Boranes

<i>Nido-boranes</i> $B_n H_{(n+4)}$		<i>m.p.</i> (°C)	<i>b.p.</i> (°C)	<i>Arachno-boranes</i> $B_n H_{(n+6)}$		<i>m.p.</i> (°C)	<i>b.p.</i> (°C)
B_2H_6	diborane	-165	-93	B_4H_{10}	tetraborane	-120	18
B_3H_9	pentaborane-9	-47	60	B_5H_{11}	pentaborane-11	-122	65
B_6H_{10}	Hexaborane-10	-62	108	B_6H_{12}	hexaborane-12	-82	
B_8H_{12}	octaborane-12	dec		B_8H_{14}	octaborane-14	dec	
$B_{10}H_{14}$	decaborane	-100	213	B_9H_{15}	(nonaborane) or (enneaborane)	3	

Diborane may be prepared by a variety of methods. Boranes were first prepared by Alfred Stock, who pioneered this branch of chemistry between 1912 and 1936. He heated Mg and B to give magnesium boride Mg_3B_2 and then treated this with orthophosphoric acid. The reaction gives a mixture of products. There were enormous difficulties with this early work because the compounds were highly reactive, flammable and hydrolysed by water. Stock developed vacuum line techniques, which were previously unknown, to handle these reactive compounds. This preparative method has now been superseded except for making B_6H_{10} .

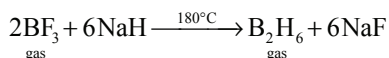


Many other methods have been used:

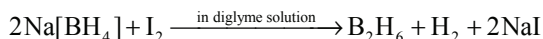


There are several convenient laboratory preparations:

1. Reducing the etherate complexes of the boron halides with



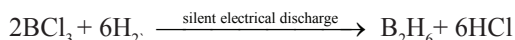
2. Reacting $Na[BH_4]$ and iodine in the solvent diglyme. Diglyme is a polyether $CH_3OCH_2CH_2OCH_2CH_2OCH_3$.



3. Reducing BF_3 with $Na[BH_4]$ in diglyme.



4. By passing electrical discharge through BCl_3 and H_2



Method (3) is particularly useful when diborane is required as a reaction intermediary. It is produced in situ and used without the need to isolate or purify it.

Structures of the boranes

The bonding and structures of the boranes are of great interest. They are different from all other hydrides. There are not enough valency electrons to form conventional two-electron bonds between all of the adjacent pairs of atoms and so these compounds are termed electron deficient.

In diborane, there are 12 valency electrons, three from each B atom and six from the H atoms. Electron diffraction results indicate the structure shown in figure.

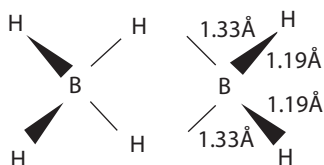


Fig. 4.2 The structure of diborane

The two bridging H atoms are in a plane perpendicular to the rest of the molecule and prevent rotation between the two B atoms. Specific heat measurements confirm that rotation is hindered. Four of the H atoms are in a different environment from the other two. This is confirmed by Raman spectra and by the fact that diborane cannot be methylated beyond $\text{Me}_4\text{B}_2\text{H}_2$ without breaking the molecule into BMe_3 .

The terminal B—H distances are the same as the bond lengths measured in non-electron-deficient compounds. These are assumed to be normal covalent bonds, with two electrons shared between two atoms. We can describe these bonds as two-centre two-electron bonds (2c-2e).

Thus, the electron deficiency must be associated with the bridge groups. The nature of the bonds in the hydrogen bridges is now well established. Obviously, they are abnormal bonds as the two bridges involve only one electron from each boron atom and one from each hydrogen atom, making a total of four electrons. An sp^3 hybrid orbital from each boron atom overlaps with the $1s$ orbital of the hydrogen. This gives a delocalized molecular orbital covering all three nuclei, containing one pair of electrons and making up one of the bridges. This is a three-centre two-electron bond (3c-2e). A second three-centre bond is also formed.

The higher boranes have an open cage structure. Both normal and multi-centre bonds are required to explain these structures:

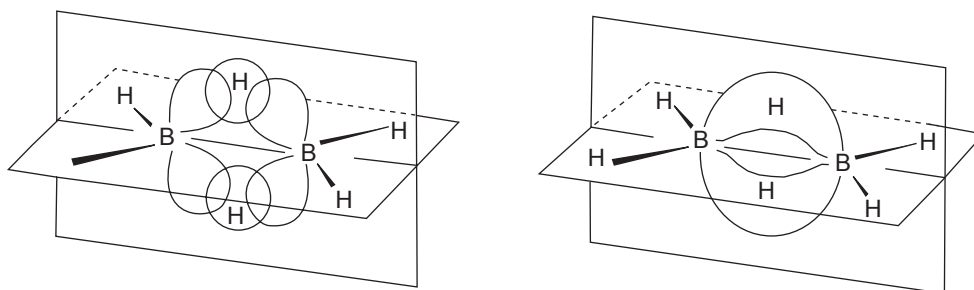
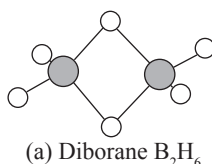


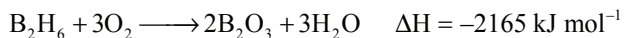
Fig. 4.3 Overlap of approximately sp^2 hybrid orbitals from B with an s -orbital from H to give a 'banana-shaped' three-centre two-electron bond



Properties

Diborane is a colourless gas and must be handled with care as it is highly reactive. It catches fire spontaneously in air and explodes with oxygen. The heat of combustion is very high. In the laboratory it is handled in a vacuum frame. Since it

reacts with the grease used to lubricate taps, special taps must be used. It is instantly hydrolysed by water, or aqueous alkali. At red heat, the boranes decompose to boron and hydrogen.



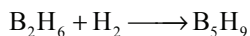
Reactions of the boranes

Most syntheses of higher boranes involve heating B_2H_6 , sometimes with hydrogen. A complex reaction occurs when B_2H_6 is heated in a sealed tube. Various higher boranes are formed (B_4H_{10} , B_5H_{11} , B_6H_{12} and $\text{B}_{10}\text{H}_{14}$). The B_2H_6 molecule probably breaks into the very reactive intermediate $\{\text{BH}_3\}$ which has only a transient existence and reacts with B_2H_6 , giving another intermediate $\{\text{B}_3\text{H}_9\}$. This loses hydrogen, forming $\{\text{B}_3\text{H}_7\}$, which reacts with more $\{\text{BH}_3\}$ to give B_2H_{10} . In a similar way, a variety of higher boranes are formed depending on the exact conditions. For example:

(80°C–90°C, 200 atmospheres)



(rapid at 200–250°C)



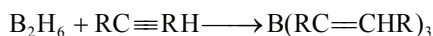
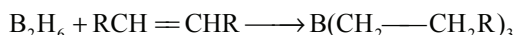
(slow pyrolysis in sealed tube 150°C)



Most of the higher boranes are liquids, but B_6H_{10} and $\text{B}_{10}\text{H}_{14}$ are solids. As the molecular weight increases, they gradually become more stable in air and less sensitive to water. $\text{B}_{10}\text{H}_{14}$ is inert in air and can be recovered from aqueous solutions.

Hydroboration

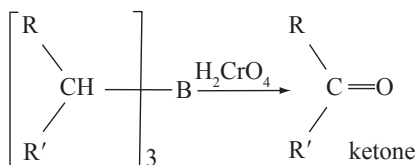
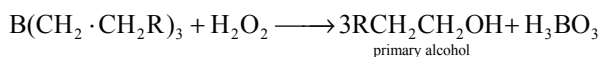
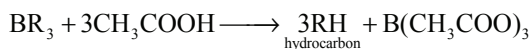
A very important reaction occurs between B_2H_6 (or $\text{BF}_3 + \text{NaBH}_4$) and alkenes and alkynes.



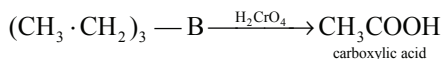
The reactions are carried out in dry ether under an atmosphere of nitrogen because B_2H_6 and the products are very reactive. The alkylborane products BR_3 are not usually isolated. They may be converted as follows:

1. to hydrocarbons by treatment with carboxylic acids,
2. to alcohols by reaction with alkaline H_2O_2 , or
3. to either ketones or carboxylic acids by oxidation with chromic acid.

The complete process is called hydroboration and results in *cis*-hydrogenation, or *cis*-hydration. Where the organic molecule is not symmetrical, the reaction follows the anti-Markovnikov rule, that is B attaches to the least substituted C atom.



4.16 Group13-Boron Family

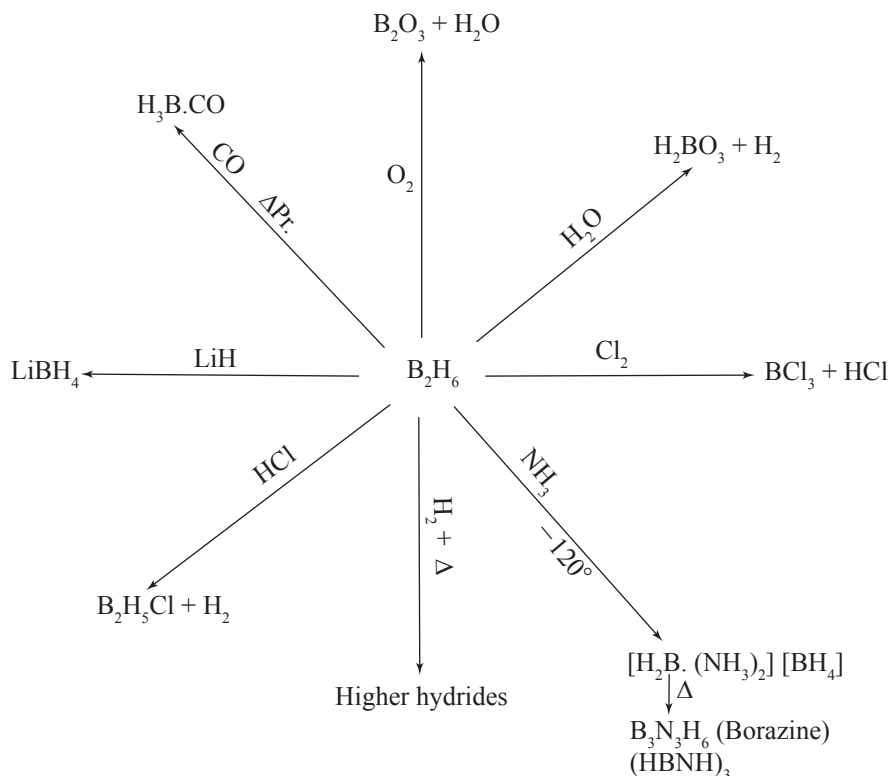
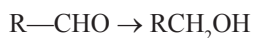
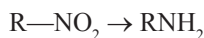
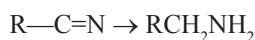


Hydroboration is a simple and useful process for two main reasons:

1. The mild conditions required for the initial hydride addition.
2. The variety of products which can be produced using different reagents to break the B — C bond.

H.C. Brown won the Nobel Prize for Chemistry in 1979 for work on these organoboron compounds.

Diborane is the simplest and most studied of the hydrides. It is used to prepare the higher boranes and is an important reagent in synthetic organic chemistry. For the latter purpose, it is normally generated in situ. It is a versatile reagent for the production of organoboranes, which are useful intermediates in organic synthesis. Alternatively diborane is used as a powerful electrophilic reducing agent for certain functional groups. It attacks sites with a high electron density such as N in cyanides and nitrites and O in carbonyl compounds.



Worked-out Examples

12. The two types of bonds present in B_2H_6 are covalent and _____.

(IIT 1994)

Ans. Three-centre electron pair bond or banana bond

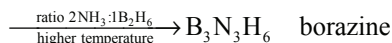
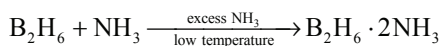
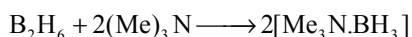
13. The structure of diborane (B_2H_6) contains
- four 2c-2e bonds and four 3c-2e bonds
 - two 2c-2e bonds and two 3c-3e bonds
 - two 2c-2e bonds and four 3c-2e bonds
 - four 2c-2e bonds and two 3c-2e bonds

[AIEEE 2005]

Ans. (d) In diborane structure B_2H_6 there are two 2c-2e bonds and two 3c-2e bonds (see structure of diborane).

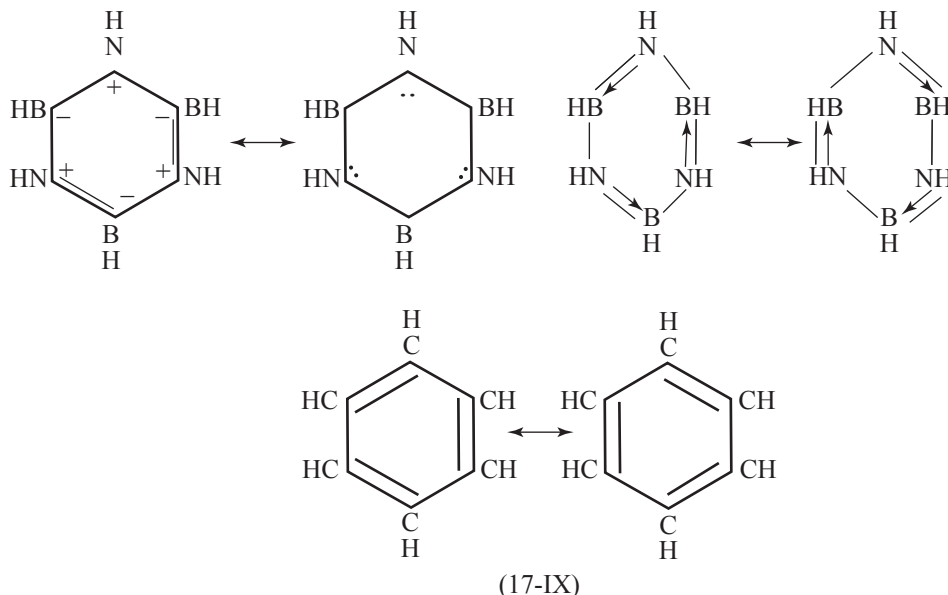
Reaction with ammonia

All the boranes act as Lewis acids and can accept electron pairs. Thus, they react with amines, forming simple adducts. They also react with ammonia, but the products depend on the conditions:

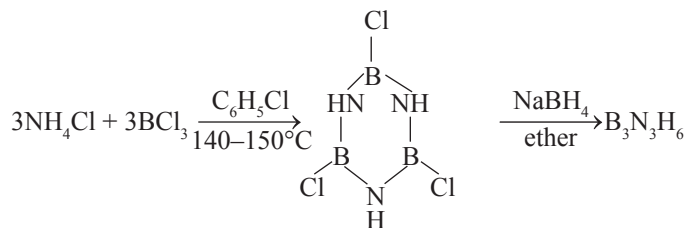


The compound $B_2H_6 \cdot 2NH_3$ is ionic and comprises $[H_3N \rightarrow BH_2 \rightarrow NH_3]^+$ and $[BH_4]^-$ ions. On heating, it forms borazine.

Borazine (Borazole) Inorganic Benzene: The electron deficient character of boron is manifested in a number of compounds containing boron bound to N, P, S, etc.,. One such interesting compound is borazine (17-VIII). $B_3N_3H_6$, which has resemblance with benzene (17-IX) in having two resonating structures containing alternate NH and BH groups.

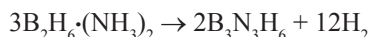


It is conveniently obtained by heating NH_4Cl and BCl_3 , the product being chloroborazine, $B_3N_3H_3Cl_3$, $(3BClNH)_3$ (17-X)



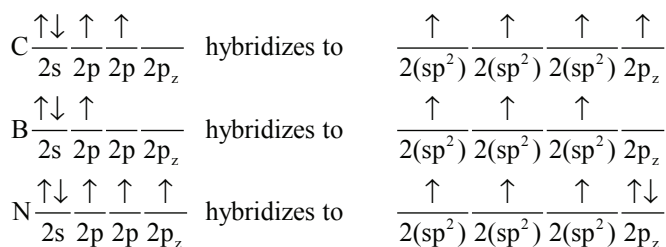
The chloroborazine is next reduced by sodium tetrahydroborate in ether: $2(\text{BClNH})_3 + 6\text{NaBH}_4 \rightarrow 2\text{B}_3\text{N}_3\text{H}_6 + 6\text{NaCl} + 3\text{B}_2\text{H}_6$

Borazine has also been prepared by the pyrolysis of diborane-diammoniate in a sealed tube at $\sim 200^\circ\text{C}$:



The so-called diborane-diammoniate is obtained as a white solid by the reaction of diborane and ammonia. Its structure shows that it is really $[\text{H}_2\text{B}(\text{NH}_3)_2][\text{BH}_4]$.

In borazine, the boron and the nitrogen atoms together are like the six carbon atoms in benzene. In benzene each carbon is sp^2 hybridised and is bonded to two carbons and one hydrogen. The remaining π orbital can overlap with another p_z orbital of a neighbouring carbon to make a double bond. Alternatively, all the p_z orbitals of six carbons may be assumed to overlap to form π -type M.O.'s. In borazine, the three nitrogens and the three borons are sp^2 hybridized.

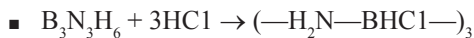


Overlap of the sp^2 orbitals gives three covalent bonds to nitrogen and boron while overlap of the p_z orbitals provides a π -bond from nitrogen to boron (17-VIIIb). Resonance will lead to equivalent B-N distances. In effect, a π -system comparable to that of benzene results. Many physical properties of borazine are similar to those of benzene (Table).

Table 4.10 Comparison of Borazine and Benzene

<i>Properties</i>	<i>Borazine</i>	<i>Benzene</i>
B-N distance	1.42 (142 pm)	—
C-C distance	—	1.41Å (141 pm)
B.P. ($^\circ\text{C}$)	80.5	78
Critical temperature	53	80
Heat of vaporisation	21.4	21.1
kcal/mole (kJ/mole)	7(29)	7.4(31)
Density (g/cm ³)	1.00	1.01

Because of these similarities borazine has been nicknamed *inorganic benzene*. However, borazine is more reactive than benzene. Unlike benzene, borazine undergoes addition reactions, say with HCl:



Boron Nitride: Inorganic Graphite and Diamond

A count of the electrons of boron, carbon and nitrogen shows that a B-N unit is isoelectronic with a C-C unit. Furthermore, electronegativity of carbon (2.5) is just the mean of those of boron (2.0) and nitrogen (3.0). It is small wonder that the binary compound boron nitride will be akin to graphite and diamond.

Boron reacts with nitrogen or ammonia on heating to form a binary nitride BN which is a white, refractory solid. The compound is also obtained by heating ammonium chloride and borax:

$\text{Na}_2\text{B}_4\text{O}_5(\text{OH})_4 + 2\text{NH}_4\text{Cl} \rightarrow 2\text{NaCl} + 2\text{BN} + \text{B}_2\text{O}_3 + 6\text{H}_2\text{O}$ Fusion of urea with boric acid in an atmosphere of ammonia in the temperature range $500^\circ - 950^\circ\text{C}$ gives boron nitride. This form of boron nitride is similar to graphite in physical properties and has a graphite-like layer structure of alternate boron and nitrogen. The boron and nitrogen atoms are $1.45\text{\AA}$ (145pm) from each other. The layers are at a distance of $3.34\text{\AA}$ (334pm). In boron nitride, the atoms of one layer are directly above the atoms of the next layer but that a boron is above a nitrogen and a nitrogen is above a boron and so on. In graphite, the carbon atoms of a particular layer are over the carbon atoms of alternate layers and not over the atoms of the adjacent layers. This form of boron nitride is often called *white graphite*. In this form, each boron and nitrogen has sp^2 hybridization. The overlap of the hybrid orbitals of one boron with three hybrid orbitals of three neighbouring nitrogens give three planar bonds at 120° to each other.

Boron	$\begin{array}{c} \uparrow\downarrow \quad \uparrow \\ \hline 2s \quad 2p \quad 2p \quad 2p \end{array}$	hybridizes to	$\begin{array}{c} \uparrow \quad \uparrow \quad \uparrow \\ \hline 2(\text{sp}^2) \quad 2(\text{sp}^2) \quad 2(\text{sp}^2) \quad 2p \end{array}$
Nitrogen	$\begin{array}{c} \uparrow\downarrow \quad \uparrow \quad \uparrow \quad \uparrow \\ \hline 2s \quad 2p \quad 2p \quad 2p \end{array}$	hybridizes to	$\begin{array}{c} \uparrow \quad \uparrow \quad \uparrow \quad \uparrow\downarrow \\ \hline 2(\text{sp}^2) \quad 2(\text{sp}^2) \quad 2(\text{sp}^2) \quad 2p \end{array}$

The lone pair of nitrogen in the remaining 2p orbital then forms a $\rightarrow$ r-bond with the vacant 2p orbital of boron. Note that $(\text{BN})_3$ is isoelectronic with the C_6 hexagonal structure in graphite. Through resonance all the B-N distance as also the bond order become equivalent. We can also say that the three lone pairs of the three nitrogen atoms of the hexagonal $(\text{BN})_3$ overlap with the three empty pz orbitals of the three borons giving π -type M.O.'s. The graphite-like variety is used as an antisticking material in moulds in glass industry.

B-N distance, $\text{\AA}$ (pm)	1.45(145)	—
C-C distance, $\text{\AA}$ (pm)	—	1.415 (141.5)
Layer spacing, $\text{\AA}$ (pm)	3.34 (334)	3.35 (335)
Density (g/cm^3)	2.29	2.25

Like the conversion of graphite to diamond, the above layer structure of boron nitride can be converted to the tetrahedral non-ending diamond-like structure. High temperature (1800°C) and High pressure (85,000 atm.) and an alkali metal or alkaline earth metal catalyst are required to bring about this conversion. The diamond-like variety of BN is called *borazon*. In borazon each tetrahedral nitrogen is bonded covalently to four tetrahedral boron (and vice versa)—to three borons with usual shared pair covalent bond and to the fourth boron via a coordinate link from nitrogen to boron. Each boron is connected to three nitrogen through shared pair covalent bonds and to the fourth nitrogen by a coordinate link from nitrogen to boron.

Boron	$\begin{array}{c} \uparrow\downarrow \quad \uparrow \\ \hline 2s \quad 2p \quad 2p \quad 2p \end{array}$	hybridizes to	$\begin{array}{c} \uparrow \quad \uparrow \quad \uparrow \\ \hline 2(\text{sp}^3) \quad 2(\text{sp}^3) \quad 2(\text{sp}^3) \quad 2(\text{sp}^3) \end{array}$
Nitrogen	$\begin{array}{c} \uparrow\downarrow \quad \uparrow \quad \uparrow \quad \uparrow \\ \hline 2s \quad 2p \quad 2p \quad 2p \end{array}$	hybridizes to	$\begin{array}{c} \uparrow \quad \uparrow \quad \uparrow \quad \uparrow\downarrow \\ \hline 2(\text{sp}^3) \quad 2(\text{sp}^3) \quad 2(\text{sp}^3) \quad 2(\text{sp}^3) \end{array}$

Worked-out Examples

14. Borazole or borazine is more reactive than benzene. Explain.

Ans. In Borazine B=N bonds are polar and therefore, reactivity of borazole is more than benzene which possesses non polar C=C and C—C bonds. The more negative group attacks boron atom.

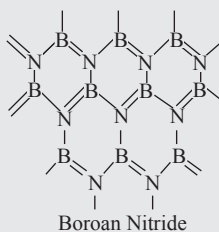
15. How will you distinguish between inorganic and organic benzene?

Ans. Borazine ($B_3N_3H_6$) is known as inorganic benzene. Unlike benzene borazine, adds HX , H_2O , ROH etc., It is hydrolysed to boric acid and NH_3 at elevated temperature.



16. Boron nitride is slippery solid. Explain on the basis of structure.

Ans. $(BN)_x$ has sheet like structure as graphite with sheets made up of hexagonal rings of alternate B and N atoms joined together. The sheets are stacked one on top of the other giving a layer structure. One B-atom and one N-atom together have same number of valency electrons as two C-atoms in graphite.



17. B_2H_6 form adduct with famines, why?

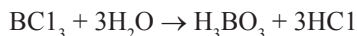
Ans. B atom in BH_3 (monomer of B_2H_6) is electron deficient and so BH_3 acts as Lewis acid.



Boric acid

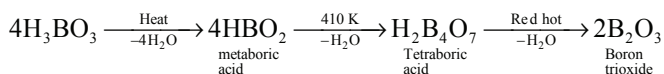
It is a crystalline solid with soapy touch.

Boric acid is obtained in the form of scales by boiling borax with acids and then cystallising. The acid is also produced during hydrolysis of boron halides:

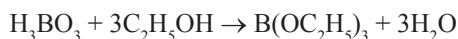


The acid has distinct volatility and can be removed by repeated evaporation with HCl . At around $100^\circ C$, the acid is changed to metaboric acid (HBO_2), at $140^\circ C$ to pyroboric acid ($H_2B_4O_7$) and finally at higher temperature to B_2O_3 .

■ Effect of heat



With alcohols (in presence of sulphuric acid) alkyl borates are formed, this being the basis of detecting borates and boric acid in qualitative analysis.

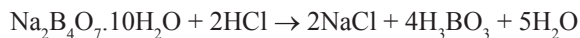
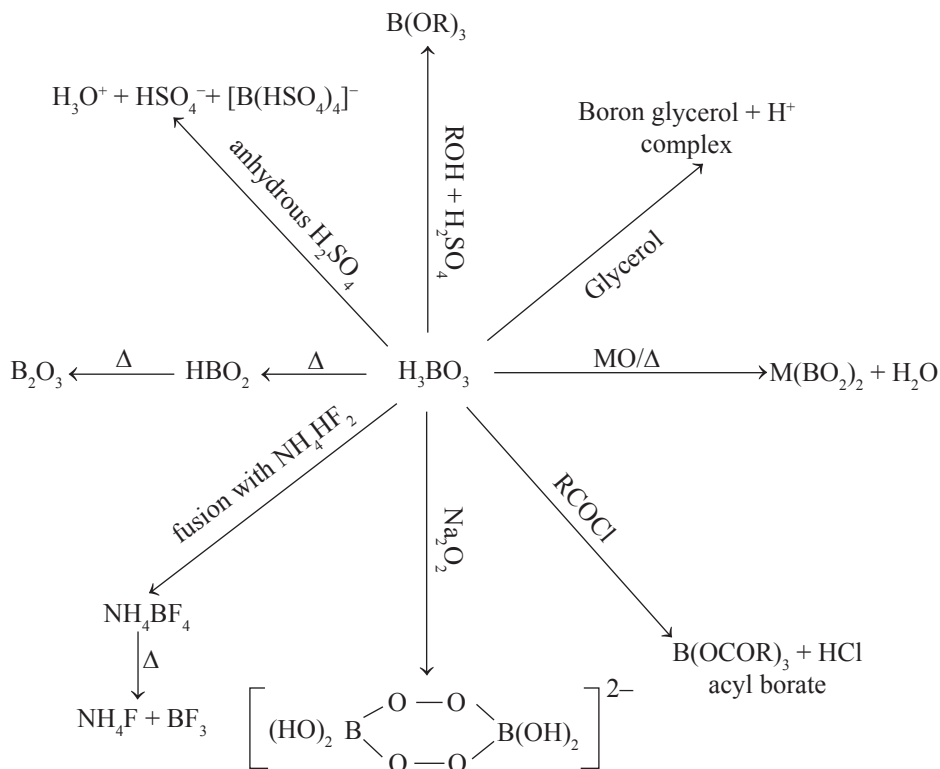


The esters burn in air with a green flame. On fusion with metal oxides coloured borate glasses are formed.

On fusion with NH_4HF_2 ultimately BF_3 is obtained. Boric acid acts as a strong acid in anhydrous H_2SO_4 :

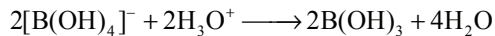
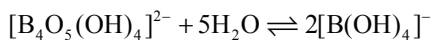


Treatment of boric acid with sodium peroxide provides sodium peroxoborate which is now known to have the structure, of $[B_3(O_2)_2(OH)_4]^{2+}$ in which two tetrahedral (sp^3 hybridized) borons are linked by bridging μ -peroxo groups (17-XIII).



One of the products H_3BO_3 is itself a weak acid. Thus, the indicator used to detect the end point of this reaction must be one that is unaffected by H_3BO_3 . Methyl orange is normally used, which changes in the pH range 3.1-4.4.

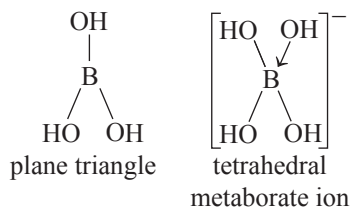
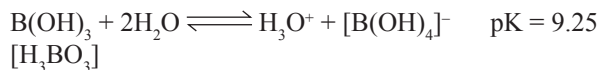
One mole of borax reacts with two moles of acid. This is because when borax is dissolved in water both B(OH)_3 and $[\text{B(OH)}_4]^-$ are formed, but only the $[\text{B(OH)}_4]^-$ reacts with HCl .



The last reaction will titrate at pH 9.2, so the indicator must have $\text{pK}_a < 8$. Borax is also used as a buffer since its aqueous solution contains equal amounts of weak acid and its salt.

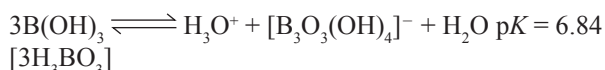
Acidic properties of H_3BO_3 or B(OH)_3

Orthoboric acid H_3BO_3 is soluble in water and behaves as a weak monobasic acid. It does not donate protons like most acids, but rather it accepts OH^- . It is therefore a Lewis acid and is better written as B(OH)_3 .

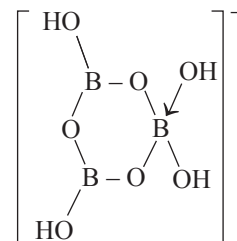


4.22 Group13-Boron Family

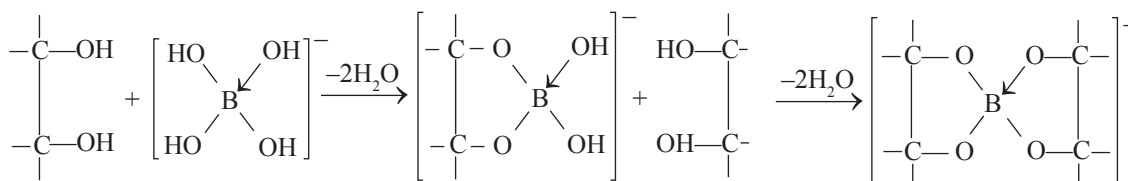
Polymeric metaborate species are formed at higher concentrations, for example:



Since B(OH)_3 only partially reacts with water to form H_3O^+ and $[\text{B(OH)}_4]^-$, it behaves as a weak acid. Thus H_3BO_3 or B(OH)_3 cannot be titrated satisfactorily with NaOH , as a sharp end point is not obtained. If certain organic polyhydroxy compounds such as glycerol, mannitol or sugars are added to the titration mixture, then B(OH)_3 behaves as a strong monobasic acid. It can now be titrated with NaOH and the end point is detected using phenolphthalein as indicator (indicator changes at pH 8.3-10.0).

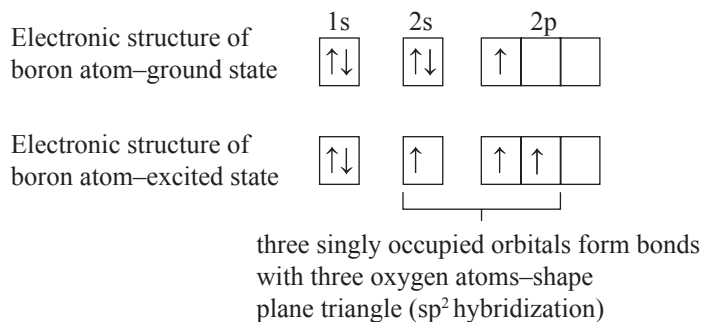


The added compound must be a cis-diol, to enhance the acidic properties in this way. (This means that it has OH groups on adjacent carbon atoms in the cis configuration.) The cis-diol forms very stable complexes with the $[\text{B(OH)}_4]^-$ formed by the forward reaction above, thus effectively removing it from solution. The reaction is reversible. Thus, removal of one of the products at the right hand side of the equation upsets the balance and the reaction proceeds completely to the right. Thus all the B(OH)_3 reacts with NaOH : in effect it acts as a strong acid in the presence of the cis-diol.

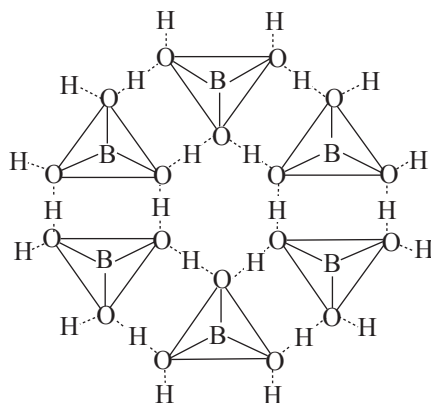


Structures of Borates

In the simple borates, each B atom is bonded to three oxygen atom arranged at the corners of an equilateral triangle. This would be predicted from the orbitals used for bonding.



Thus orthoboric acid contains triangular BO_3^{3-} units. In the solid, B(OH)_3 units are hydrogen bonded together into two-dimensional sheets with almost hexagonal symmetry. (See Figure) The layers are quite large distance apart ($3.18\text{\AA}$) and thus the crystal breaks quite easily in very fine particles. At one time orthoboric acid was used as a mildly anti-septic talcum powder for babies, because it forms a fine powder. It sometimes causes rashes if used longer. The orthoborates contain discrete BO_3^{3-} ions and examples include $\text{Mg}_3(\text{BO}_3)_2$ and the lanthanide orthoborates $\text{Ln}^{\text{III}}\text{BO}_3$.

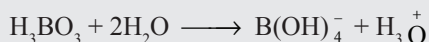


In the metaborates, simple units (BO_3 planar triangular units or BO_4 tetrahedral) join together to form a variety of polymeric chain and ring structures.

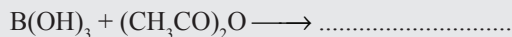
Worked-out Examples

18. The correct equivalence point of boric acid and NaOH titration cannot be determined. Why?

Ans. H_3BO_3 or $\text{B}(\text{OH})_3$ is weak mono-basic acid and partially reacts with H_2O to form H_3O^+ and $\text{B}(\text{OH})_4^-$ ion.



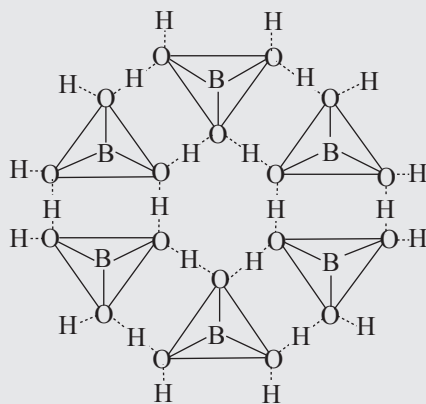
19. Complete the following reaction:



Ans. $2\text{B}(\text{OH})_3 + 5(\text{CH}_3\text{CO})_2\text{O} \longrightarrow (\text{CH}_3\text{COO})_2\text{BOB}(\text{OOCCH}_3)_2 + 6\text{CH}_3\text{COOH}$
Boron acetate

20. Draw the structure of solid orthoboric acid.

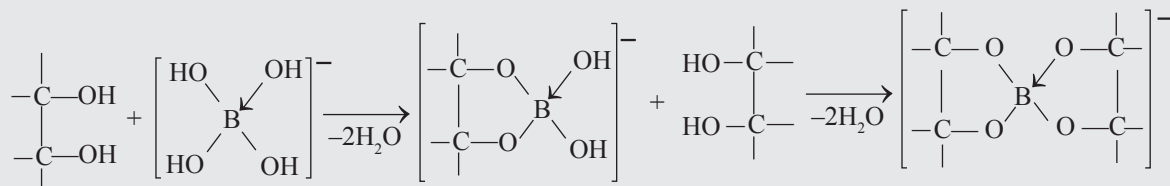
Ans. In solid, $\text{B}(\text{OH})_3$ units are hydrogen bonded together into 2-D sheets with almost hexagonal symmetry.



Hydrogen bonded structure
of orthoboric acid

21. Boric acid can be titrated against sodium hydroxide using methyl orange as indicator only in presence of polyhydroxy compounds like catechol, mannitol, glycerol or sugars. Explain.

Ans. Boric acid, B(OH)_3 is a very weak Lewis acid and in aqueous solution $[\text{B(OH)}_4]^-$ ions are formed which form a stable complex with polyhydroxy compounds. Due to formation of this stable complex, boric acid acts as a strong acid because of removal of $[\text{B(OH)}_4]^-$ in the form of complex and hence can be titrated against NaOH.



22. Boric acid is a

(IIT 2003)

- (a) weak monobasic Lewis acid
- (b) weak monobasic Arrhenius acid only
- (c) weak monobasic Bronsted acid only
- (d) weak monobasic Arrhenius as well as Bronsted acid

Ans. (a) Boric acid acts as a weak monobasic acid by accepting a lone pair of electrons from OH^- ion, thereby, acting as a Lewis acid.

23. STATEMENT-1: In water, orthoboric acid behaves as a weak monobasic acid.

because

STATEMENT-2: In water, orthoboric acid acts as a proton donor.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1
- (c) Statement-1 is True, Statement-2 is False
- (d) Statement-1 is False, Statement-2 is True

(IIT 2007)

Ans. (c)

24. In the presence of which of the following species, boric acid can be titrated against sodium hydroxide solution?

(IIT 2006)

- (a) trans-1, 2-diol
- (b) cis-1, 2-diol
- (c) Borax
- (d) Sodium hydrogen phosphate

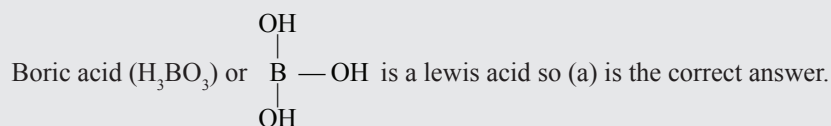
Ans. (b)

25. Which one of the following is the correct statement?

(AIEEE 2008)

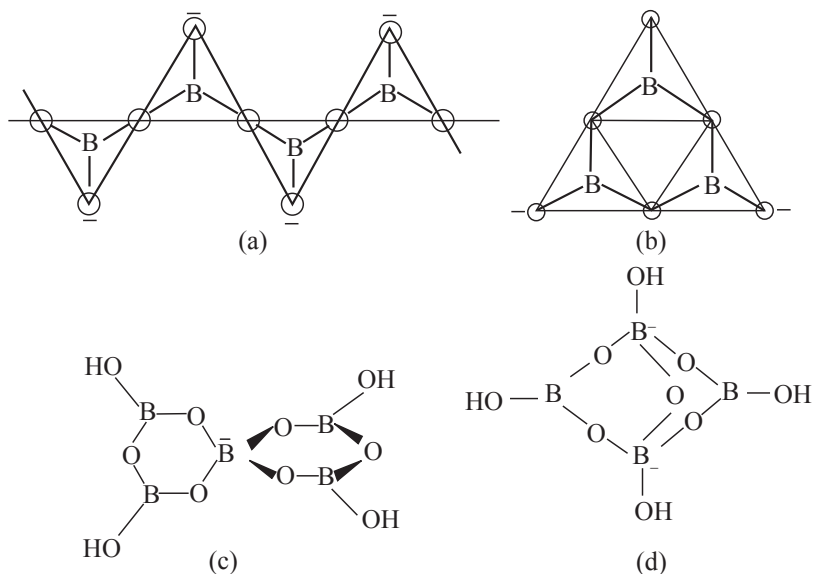
- (a) Boric acid is a protonic acid
- (b) Beryllium exhibits coordination number of six
- (c) Chlorides of both beryllium and aluminium have bridged chloride structures in solid phase
- (d) $\text{B}_2\text{H}_6 \cdot 2\text{NH}_3$ is known as 'inorganic benzene'

Ans. (d) The correct formula of inorganic benzene is $\text{B}_3\text{N}_3\text{H}_6$ so is incorrect statement



Borax

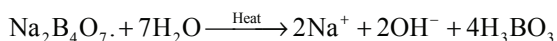
The most common metaborate is borax $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$. It is a useful primary standard for titrating against acids.



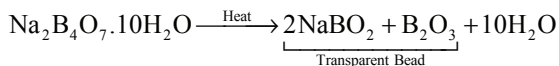
Structures of some borates. (a) Metaborate chain $[\text{Ca}(\text{BO}_2)_2]_n$ made up of triangular BO_3 units. (b) Metaborate ring $\text{K}_3[\text{B}_3\text{O}_6]$ made up of triangular BO_3 units. (c) Complex metaborate $\text{K}[\text{B}_5\text{O}_6(\text{OH})_4]$ called the spiro anion is made up of four triangular BO_3 units and one tetrahedral BO_4 unit. (d) Borax ($\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$) is made up of two triangular and two tetrahedral units. This ion is $[\text{B}_4\text{O}_5(\text{OH})_4]^{2-}$ and the other water molecules are associated with the metal ions.

Properties of Borax

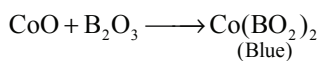
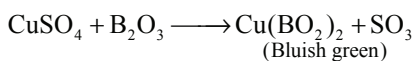
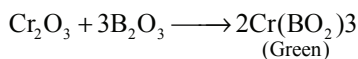
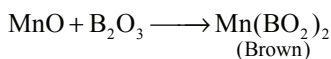
- It is a white crystalline solid.
- It is less soluble in cold water but more soluble in hot water.
- It gets *hydrolysed* with water to form *alkaline solution*.



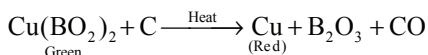
- *Action of heat.* It forms *colourless transparent glassy bead*.



The hot transparent glassy bead reacts with metal ions of colored salts to form metaborates of different colours on heating in oxidizing flame. Some examples are:



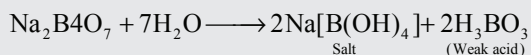
Certain coloured metaborates get reduced to free metal on heating in reducing flame.



Worked-out Examples

26. Why borax can be used as a buffer?

Ans. Borax solution in water produces weak acid H_3BO_3 and its salt $\text{Na}[\text{B}(\text{OH})_4]$.



27. Which indicator can be used to detect the end point of borax-acid titration?

Ans. As the end point, the pH is about 4 and thus methyl orange is used as an indicator.

28. $\text{Na}_2\text{B}_4\text{O}_7 + \text{Conc. H}_2\text{SO}_4 \longrightarrow \text{A} \xrightarrow[\text{(ii) Ignite}]{\text{(i) C}_2\text{H}_5\text{OH}}$

Ans. A: H_3BO_3 ; B: $(\text{C}_2\text{H}_5)_3\text{BO}_3$ (burns with green edge name).

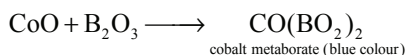
The spiro compound $\text{K}[\text{B}_5\text{O}_6(\text{OH})_4]$ contains one tetrahedral unit and four triangular units. Borax is usually written as $\text{Na}_2\text{B}_4\text{O}_{10} \cdot 10\text{H}_2\text{O}$ but is actually made from two tetrahedral and two triangular units joined and should be written $\text{Na}_2[\text{B}_4\text{O}_5(\text{OH})_4] \cdot 8\text{H}_2\text{O}$.

Boron sesquioxide and the borates

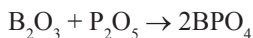
These are the most important compounds of boron. Sesqui means one and a half, so the oxide should have a formula $\text{MO}_{1.5}$ or M_2O_3 . All the elements in the group form sesquioxides when heated in oxygen. B_2O_3 is made more conveniently by dehydrating boric acid.



B_2O_3 is a typical non-metallic oxide and is acidic in its properties. It is the anhydride of orthoboric acid and it reacts with basic (metallic) oxides, forming salts called borates or metaborates. In the borax bead test, B_2O_3 or borax $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$ is heated in a Bunsen burner flame with metals oxides on a loop of platinum wire. The mixture fuses to give a glass-like metaborate bead. Metaborate beads of many transition metals have characteristic colours and so this reaction provides a means of identifying the metal. This simple test provided the first proof that vitamin B_{12} contained cobalt.



However, it is possible to force B_2O_3 to behave as a basic oxide by reacting with very strongly acidic compounds. Thus with P_2O_5 or As_2O_5 , boron phosphate or boron arsenate are formed.

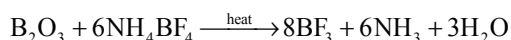
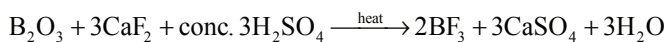
**Trihalides****Halides**

All the elements form trihalides. The boron halides are covalent. BF_3 is gaseous, BCl_3 liquid and BI_3 is solid. BF_3 is by far the most important.

Boron halides are known for all the halogens. These are all covalent and monomeric. In keeping with the normal behaviour, the boiling points increase with increasing molecular weights of the halides.

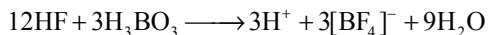
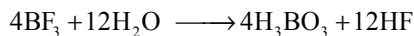
Halide	BF_3	BCl_3	BBr_3
B.P. ($^\circ\text{C}$)	-101	12.5	90

The triiodide, BI_3 , is a white solid (M.P. 43°C). It is a colourless gas, boiling point - 101°C and is made in large quantities.

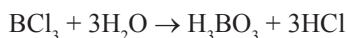


Both BF_3 gas and its complex with diethyl ether $(\text{C}_2\text{H}_5)_2\text{O}$ viscous liquid) are commercially available.

The boron halides are all hydrolysed by water. BF_3 hydrolyses incompletely and forms fluoborates. This is because the HF first formed reacts with the H_3BO_3 .

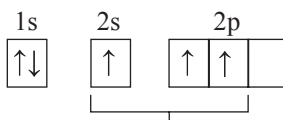


The other halides hydrolyse completely, giving boric acid.



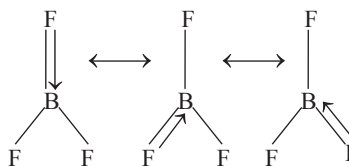
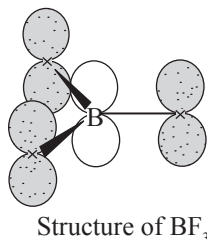
The shape of the BF_3 molecule is a planar triangle with bond angles of 120° . This is predicted by VSEPR theory as the most stable shape for three outer electron pairs round B. The valence bond theory also predicts a planar triangle with hybridization of one s and two p orbitals used for bonding. However, the B atom only has six electrons in its outer shell and this is termed electron deficient.

Electronic structure of
boron atom – excited state

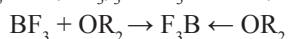
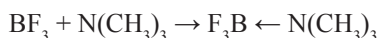


three singly occupied orbitals form bonds
with unpaired electrons from three halogen
atoms – shape plane triangle
(sp^2 hybridization)

The bond lengths in BF_3 are $1.30 \text{ \AA}$ and are significantly shorter than the sum of the covalent radii ($\text{B} = 0.80 \text{ \AA}$, $\text{F} = 0.72 \text{ \AA}$). The bond energy is very high: 646 kJ mol^{-1} , which is higher than for any single bond. The shortness and strength of the bonds is interpreted in terms of a $\pi\text{p}-\text{p}\pi$ interaction, that is the bonds possess some double bond character. The empty 2p_z atomic orbital on B which is not involved in hybridization is perpendicular to the triangle containing the sp^2 hybrid orbitals. This p_z orbital may accept an electron pair from a full p_z orbital on any one of the three fluorine atoms. Thus a dative π bond is formed and the B atom attains an octet of electrons. If one localized double bond existed, then there would be one short bond and two longer ones. However, all measurements show that the three bond lengths are identical. The old valence bond explanation of this was resonance between three structures with the double bond in different positions. The modern explanation is that the double bond is delocalized.

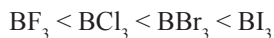


These are all strong Lewis acids and form numerous addition compounds:

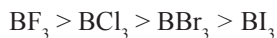


4.28 Group13-Boron Family

For a given Lewis base, L, the stability of the resulting adduct LBX_3 follows the Lewis acidity order:



Consideration of the electronegativity of the halogens alone would indicate the reverse order of Lewis acidity i.e.,



To cite one experimental result, the heats of reaction of BX_3 with Lewis base pyridine in nitrobenzene (i.e., the electron-acceptor strength of BX_3) follows the order:



In order to change over from the π -bonded three-coordinate planar structure to four-coordinate tetrahedral, the π -bond has to be ruptured. By virtue of the smallest size of fluorine BF_3 is the most strongly π -bonded and hence, the adduct of BF_3 is expected to be the least stable, electronegativity effect is evidently less important.

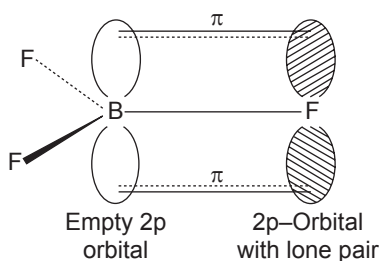


Fig. 4.4 $p\pi = p\pi$ back bonding

Now, the tendency for back donation is maximum in the case of fluorine due to its small size and more interelectronic repulsions, therefore, it is the least acidic. The tendency of back bonding falls as we move from BF_3 to BCl_3 and BCl_3 and BBr_3 due to increase in the size of halogen atoms. Consequently, the acidic character increases accordingly.

For a given BX_3 , the stability of BX_3 depends on several factors of which the chemical nature of the donor atom of L is the most important.

Among the halides of Al, Ga, In and Tl, the fluorides are all expected to be ionic with high melting points. However, it has recently been shown that these have polymeric six co-ordinated covalent structures with all the fluorine atoms-being bridging in nature. The other halides are largely covalent and attain octet through dimerisation. The dimeric structures are retained in non-polar solvents such as benzene but in water they break down, being assisted by a large solvation energy, into the hexaaqua ions $[M(H_2O)_6]^{3+}$.

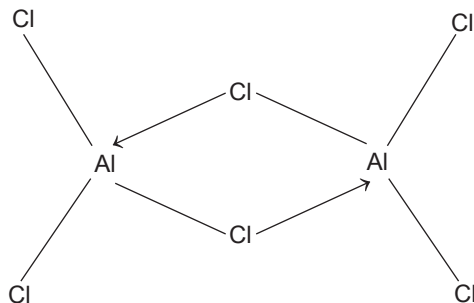
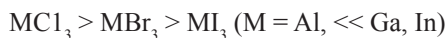


Fig. 4.5

The dimers also break down while reacting with other donor centres (Lewis bases) forming adducts of the type $(CH_3)_3N \rightarrow AlCl_3$. These adducts widely vary in stability and in the mode of bonding, properties and structure. Towards amines, ethers and phosphanes (which are Lewis bases) the acceptor strengths of the metal halides follow the order.

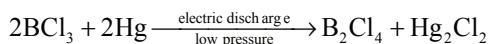


But for mild sulphur donors (soft bases), Me_2S , Et_2S aluminium halides follow the same sequence as given above but for GaX_3 and InX_3 the sequence is reversed.

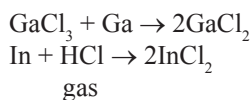


Dihalides

Boron forms halides of formula B_2X_4 . These decompose slowly at room temperature. B_2Cl_4 can be made as follows:



There is free rotation about the B — B bond and in the gaseous and liquid states the molecule adopts a non-eclipsed conformation. In the solid state the molecule is planar, because of crystal forces and ease of packing. Gallium and indium also form ‘dihalides’.

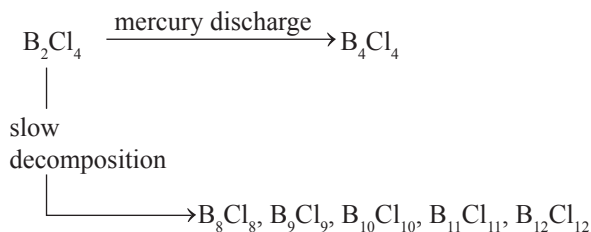


These are more properly written $\text{Ga}^+[\text{GaCl}_4]^-$ and $\text{In}^+[\text{InCl}_4]^-$ and contain M(I) and M(III) rather than Ga(II) and In(II).

TlI_3 is an unusual compound. It is isomorphous with CsI_3 and NH_4I_3 and contains the linear triiodide ion I_3^- . Thus the metal is present as Tl^+ , in the oxidation state (+I), not (+III).

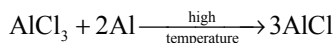
Monohalides

Boron forms several stable polymeric monohalides $(\text{BX})_n$.



The compounds B_4Cl_4 , B_8Cl_8 and B_9Cl_9 are crystalline solids and their structures (Figure) have a closed cage or polyhedron of B atoms, where each B atom is bonded to three other B atoms and to one Cl atom. Since B has only three valency electrons, there are not enough electrons to form normal electron pair bonds. It is probable that multi-centre σ bonds cover all the B atoms in the cage.

Al, Ga and In all form monohalides MX in the gas phase at elevated temperatures, e.g.,



These compounds are not very stable and are covalent.

Thallium forms univalent thallos halides which are more stable than the thallium trihalides. This illustrates the inert pair effect and TlF is ionic.

Worked-out Examples

29. Aluminium forms $[\text{AlF}_6]^{3-}$ ion but boron does not form $[\text{BF}_6]^{3-}$ ion. Why?

Ans. Al forms $[\text{AlF}_6]^{3-}$ ion because Al ($3s^2 3p^1$) can achieve co-ordination number of six by using *d*-orbitals of 3rd shell. On the other hand, boron ($2s^2 2p^1$) cannot extend its co-ordination number by four (it has vacant p-orbitals and no d-orbitals in 2nd shell). Also due to small size B^{3+} cannot show six co-ordination number.

30. Boron tribromide is stronger acid than boron trifluoride.

[CBSE (Mains) 2004]

Ans. Back bonding occurs in boron halides to compensate electron deficiency and hence Lewis acidity is decreased. The tendency of back bonding is maximum in BF_3 and decreases from BF_3 to BI_3 as the size of the halogen increases, *p-orbitals* of high energy shells in bromine overlap less effectively with vacant *p-orbital* of boron in BBr_3 . Thus, BBr_3 is stronger acid than BF_3 as it has still tendency to accept electrons to remove electron deficiency.

31. BX_3 is known but BH_3 is not known. Explain.

Ans. BH_3 undergoes dimerization to form B_2H_6 giving rise to hydrogen bridge bond forced by electron deficiency of the compound. No such bonds exist in BX_3 . Because due to back bonding ($p\pi-p\pi$ bonding) in $\text{B}-\text{X}$ bond, B atom in BX_3 is not as much electron deficient as in BH_3 .

32. Arrange the following in increasing order of heat of formation: BF_3 , BI_3 , BBr_3 , BCl_3 .

Ans. $\text{BI}_3 < \text{BBr}_3 < \text{BCl}_3 < \text{BF}_3$

33. Among Boron trihalides, BF_3 has highest value for H_p , why?

Ans. Because back bonding ($p\pi-p\pi$ bonding) is maximum in $\text{B}-\text{F}$ bond as it is shortest.

34. Pure BBr_3 and BI_3 are colourless, but they become coloured on exposure to light, why?

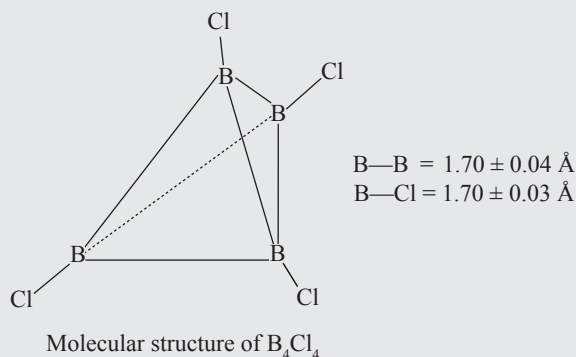
Ans. Due to liberation of free halogen by photolysis.

35. BBr_3 and BCl_3 both form complex with arsine (AsH_3) while BF_3 does not. Why?

Ans. BF_3 is weaker Lewis acid than BCl_3 and BBr_3 .

36. Draw the geometry of tetraboron tetrachloride (B_4Cl_4).

Ans.



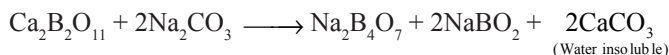
1. The chemistry of boron is quite different from that of the heavier Group 13 elements. It differs from aluminium in the following ways:

- Its oxide and hydroxide are acidic whereas those of aluminium are amphoteric.
- Boron is a semi-conductor which has various polymorphs based on icosahedral boron cages, whereas aluminium is a metal with a close packed structure. Boron is very inert and only attacked by hot concentrated oxidizing acids.
- No simple salts of B^{3+} are known, whereas those of Al^{3+} are numerous and well documented.
- Boron forms a wide range of hydrides which have cage structures. $(\text{AlH}_3)_4$ has a polymeric structure which resembles that of AlF_3 .
- The stereochemistries of many boron compounds are based on trigonal sp^2 and tetrahedral sp^3 geometries. In the latter, the octet rule is obeyed. Aluminium forms many compounds with tetrahedral, trigonal bipyramidal and octahedral geometries.

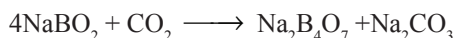
(f) Multiple $\pi\pi$ - $\pi\pi$ bonding in boron-nitrogen, boron-oxygen and boron-fluorine compounds is much more significant than that for the corresponding aluminium compounds. $\{\text{BN}\}_x$ for example, adopts a graphite structure.

- BF_3 is a colourless gas, BCl_3 is a colourless fuming liquid (B.pt 13°C). BBr_3 is also a colourless fuming liquid (B.pt. 90°C) while BI_3 is a white fusible solid (M.pt. 310°C).
- The important mineral of boron is borax ($\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$). The crude form of it is called Trincal.

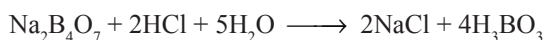
Borax can be prepared from colemanite.



CaCO_3 is removed and the filtrate is concentrated to get crystals of borax. Sodium metaborate is converted to borax by passing CO_2 .

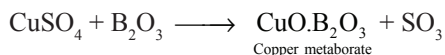
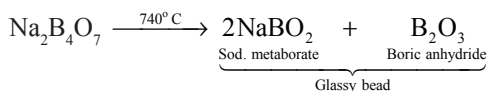
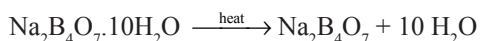


- Borax when treated with conc. HCl or H_2SO_4 gives crystals of boric acid.



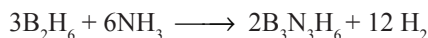
Slippery nature of boric acid is due to hydrogen bonding in it which gives it a layer structure.

- Borax is widely used in qualitative analysis, since it gives coloured beads with coloured salts. The chemistry of the Borax bead test is as follows:



Thus, coloured beads are formed due to formation of metal metaborates.

- $\text{B}_3\text{N}_3\text{H}_6$ is called Inorganic benzene or Borazole or Triborane triamine. It is prepared by heating diborane and NH_3 in the ratio 1: 2.



It was first prepared by Alfred stock. It is a colourless liquid and has melting point of -57°C . Pure sample of borazole are light sensitive and have been known to explode unless stored in dark. Both C and N are sp^2 -hybridised. It is isosteric with benzene.

- Inorganic naphthalene: On heating borazine or by passing silent electric discharge through it, gives a product analogous to naphthalene (known as inorganic naphthalene) and inorganic diphenyl.
- Hybridisation of boron in diborane is sp^3 while in BCl_3 it is sp^2 .
- Due to back bonding and resonance the B-F bond in BF_3 gets a bond order of 1.33.
- Boric acid is used as an antiseptic and eye wash order in the name 'Boric lotion'.
- Aluminium displace H_2 from acid but boron does not.
- Basic nature of oxides and hydroxides show following order:



- B_4C_3 (Boron carbide) is one of the hardest known artificial substance and is called Norbia.

Aluminium (Al)

Occurrence

The important ores of aluminium are:

- (i) Bauxite— $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$
- (ii) Cryolite— Na_3AlF_6
- (iii) Feldspar— $\text{K}_a\text{Si}_3\text{O}_8$
- (iv) Mica— $\text{K}_a\text{SiO}_{10}(\text{OH})_2$
- (v) Alunite— $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 4\text{Al}(\text{OH})_3$

Extraction Aluminium is generally extracted from bauxite ($\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$) by electrolysis. The process involves:

1. Purification of bauxite
2. Electrolytic reduction of alumina
3. Purification of aluminium

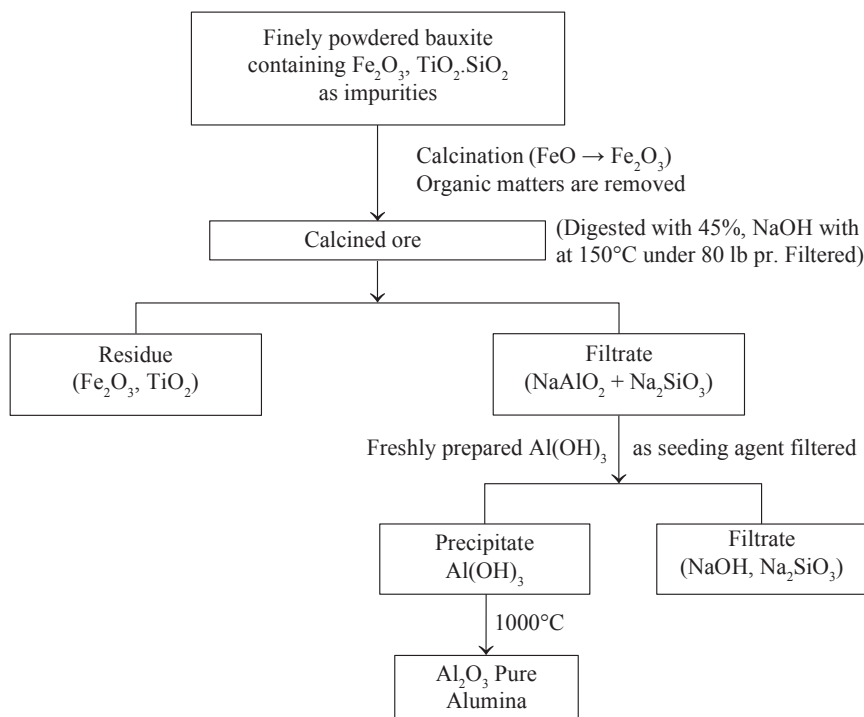
Worked-out Example

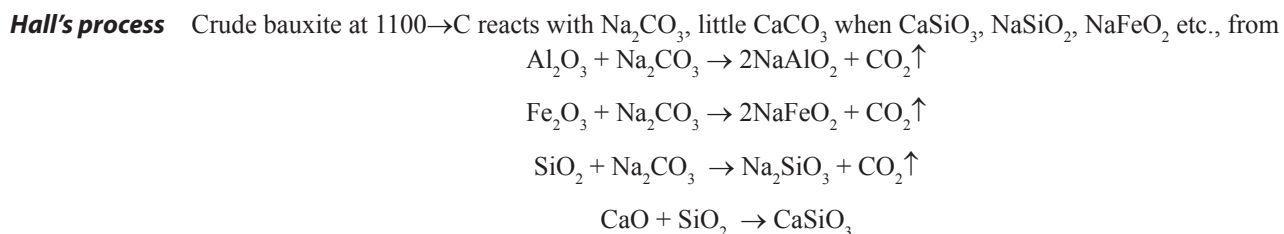
37. Pure alumina cannot be electrolysed. Why?

Ans. Pure alumina has very high melting point (2000°C) as well as it is poor conductor of electricity.

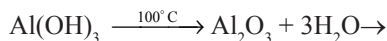
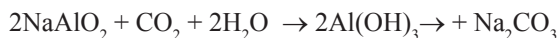
Purification of Bauxite: By Bayer's process, commercially it is being carried out (for red bauxite not for the white bauxite).

Flow sheet of Bayer's process for the preparation of pure Al_2O_3

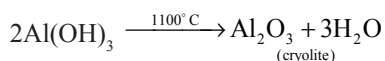
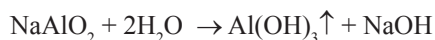
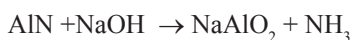
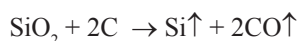
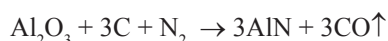




Then at 50°C – 60°C CO_2 is formed through NaAlO_2 solution and produces thereby $\text{Al}(\text{OH})_3$



Serpeck's process



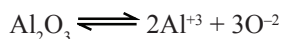
Electrolytic reduction of Al_2O_3 : At 900°C, $\text{AlF}_3 \cdot 3\text{NaF}$, CaF_2 (Fluorspar) to make Al_2O_3 a good conductor of electricity. Metallic Al is liberated at the cathode.

Electrolysis of molten mixture

Cathode	:	Carbon
Anode	:	Graphite rods
Electrolyte	:	60 parts cryolite + 20 parts fluorospar + 20 parts pure Al_2O_3
Temperature	:	900°C

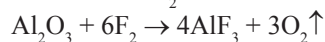
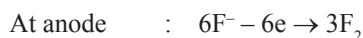
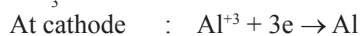
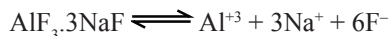
Reactions

According to the 1st theory, the following reaction occurs:

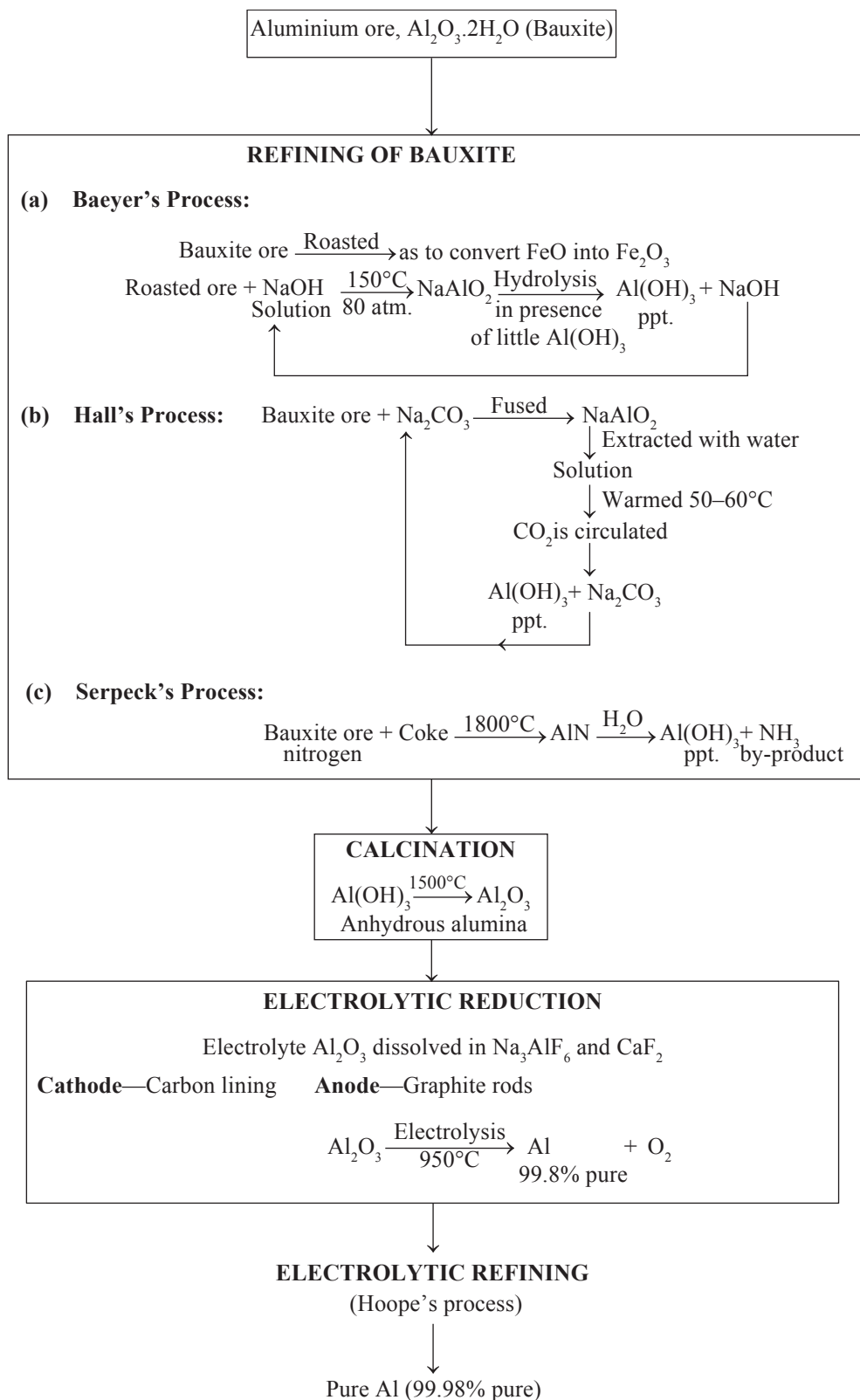


As cryolite has greater electrochemical stability, it does not dissociate. It only increases the dissociation of Al_2O_3 .

But the second theory states that, cryolite undergoes electrolytic dissociation first then Al^{+3} goes to the cathode, produced F_2 at anode then reacts with Al_2O_3 produces AlF_3 .



FLOW SHEET FOR THE EXTRACTION OF ALUMINIUM



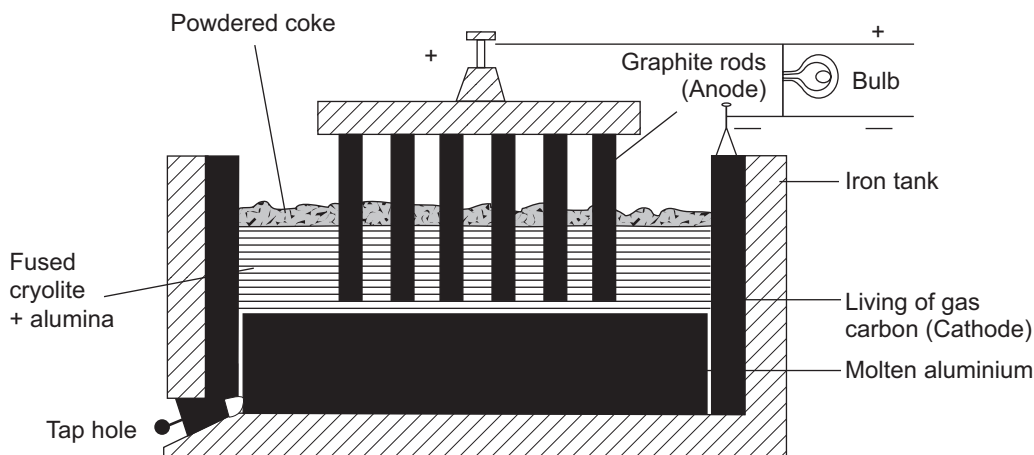


Fig. 4.6

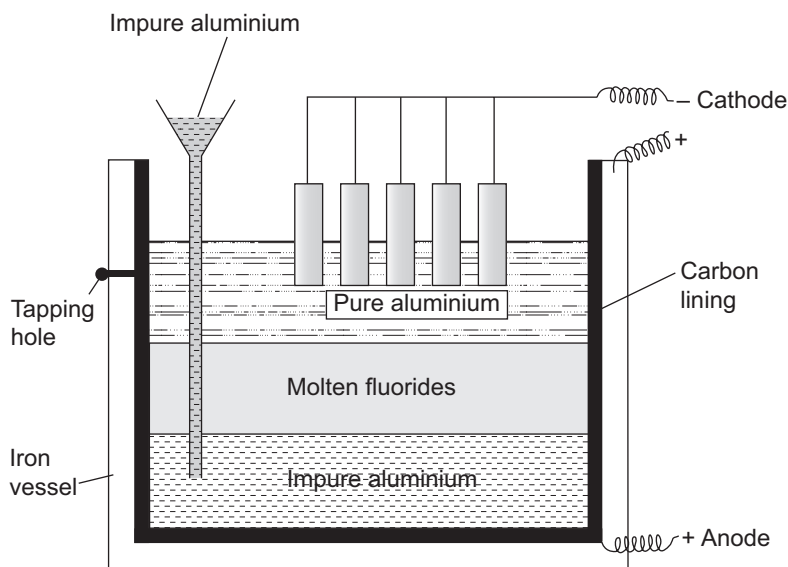


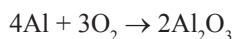
Fig. 4.7

Properties of Al

Physical Properties: Al is silver white, light metal having sp gravity, 2.7. It is very soft, malleable and ductile. Al is a good conductor of heat and electricity.

Chemical Properties

1. With air aluminium oxide aluminium from with bright white flame.

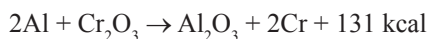


2. By thermit process Al reduces Fe, Cr etc., from their stable oxides

At 1900 °C. Al combines with O_2 of Cr_2O_3 , Fe_2O_3 and reduces consequently to Cr, Al. Exothermicity of the reactions increases the temperature of the molten pool 3000 °C – 3500 °C Exothermicity

4.36 Group13-Boron Family

The reactions are:



Arrangement is like the following diagram; Ignition mixture ($\text{KClO}_3 + \text{Mg}$ powder)

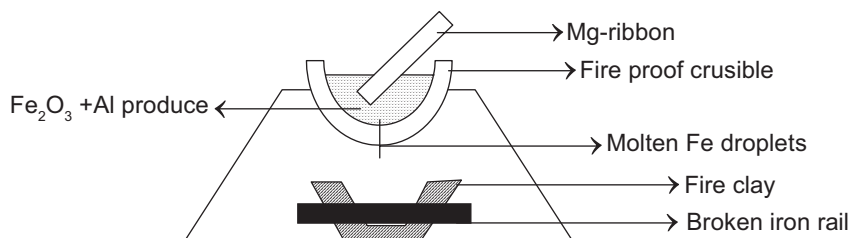
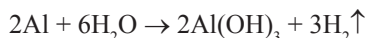


Fig. 4.8

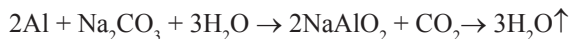
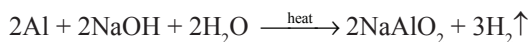
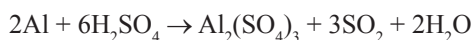
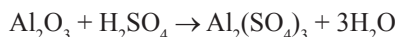
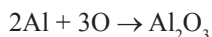
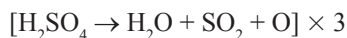
3. Al powder reacts with boiling water (not water at ordinary temp) and produce $\text{Al}(\text{OH})_3$.



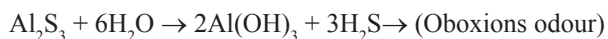
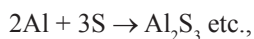
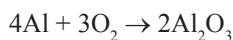
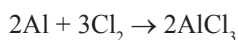
4. Action with acids and alkalis both are as follows:

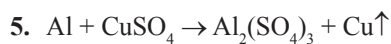


The partial reactions are as follows:



There is no action of dilute or concentrated.





This displacement reaction is because of $E_{\text{Al}/\text{Al}^{+3}}^0 > E_{\text{Cu}/\text{Cu}^{+2}}^0$

Al	O_2 (air)	$\rightarrow \text{Al}_2\text{O}_3 + 772,00 \text{ cal.}$
	$\text{H}_2\text{O}(\text{boiling})$	$\rightarrow \text{Al}(\text{OH})_3 + \text{H}_2$
	HCl	$\rightarrow \text{AlCl}_3 + \text{H}_2$
	$\text{H}_2\text{SO}_4(\text{dil})$	$\rightarrow \text{Al}_2(\text{SO}_4)_3 + \text{H}_2$
	Conc H_2SO_4	$\rightarrow \text{Al}_2(\text{SO}_4)_3 + \text{SO}_2 + \text{H}_2\text{O}$
	HNO_3 (dil)	$\rightarrow \text{Al}(\text{NO}_3)_3 + \text{NH}_4\text{NO}_3 + \text{H}_2\text{O}$
	HNO_2 (Conc.)	$\rightarrow \text{No reaction}$
	$\text{NaOH} + \text{H}_2\text{O}$ solution	$\rightarrow \text{NaAlO}_2 + \text{H}_2$
	NaOH fused	$\rightarrow \text{Na}_3\text{AlO}_3 + \text{H}_2$
	N_2	$\rightarrow \text{AlN} \xrightarrow{\text{H}_2\text{O}} \text{Al}(\text{OH})_3 + \text{NH}_3$
	$\Delta \text{ s}$	$\rightarrow \text{Al}_2\text{S}_2 \xrightarrow{\text{H}_2\text{O}} \text{Al}(\text{OH})_3 + \text{H}_2\text{S}$
	X_2 (X-halogen)	$\rightarrow \text{AlX}_3 \xrightarrow{\text{H}_2\text{O}} \text{Al}(\text{OH})_3 + \text{HX}$
	Cr_2O_3	$\rightarrow \text{Cr} + \text{Al}_2\text{O}_3 + \text{heat}$
	Fe_2O_3	$\rightarrow \text{Fe} + \text{Al}_2\text{O}_3 + \text{heat}$
	CO_2	$\rightarrow \text{C} + \text{Al}_2\text{O}_3$
	SiO_2	$\rightarrow \text{Si} + \text{Al}_2\text{O}_3$
	ZnSO_4	$\rightarrow \text{Zn} + \text{Al}_2(\text{SO}_4)_3$
	CuSO_4	$\rightarrow \text{Cu} + \text{Al}_2(\text{SO}_4)_3$

Worked-out Examples

38. Aluminium containers can be used to store concentrated HNO_3 or Al becomes passive in concentrated HNO_3 . Explain.

Ans. Al on coming in contact with concentrated HNO_3 becomes passive due to the coating of aluminium oxide on its surface and thus Al containers can be used to store conc. HNO_3 .

39. Why is Al stable in air and water?

Ans. The surface of Al reacts readily with water to decompose it even at ordinary temperature to form a protective layer of its oxide (Al_2O_3) which protects it from further action.

40. Explain why aluminium, though an electropositive metal, finds extensive use as a structural material.

Ans. It is cheap and light metal with high tensile strength and possesses tendency to form alloys with several other metals like Cu, Mg, Mn, Zn etc., Al does not corrode easily.

41. Why is aluminium used in making electrical cables despite its low conductivity as compared to copper?

Ans. It is cheaply available. However, since it has lesser conducting nature than copper, thicker aluminium wires are used.

42. In the electrolytic manufacture of aluminium, why are carbon anodes consumed?

Ans. During electrolysis of alumina and cryolite for the manufacture of Al, oxygen evolved at the anode reacts with carbon of anode to form CO and CO_2 and thus anode is slowly consumed.

43. Aluminium cannot be prepared by the electrolysis of aqueous solution of its salt. Why?

Ans. Aluminium has large discharge potential than H^+ ion and thus during electrolysis of its aqueous salt solutions, Al^{3+} ions are not discharged at cathode. On the contrary, H^+ ions are discharged at cathode to liberate H_2 .

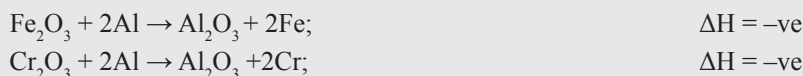
44. Finely powdered aluminium when exposed to air becomes hazardous. Why?

Ans. Al has great affinity for oxygen and forms Al_2O_3 with evolution of excessive amount of heat. Finely divided Al powder has larger surface area and thus reacts vigorously with air and is hazardous.

45. Why is alum used for settling muddy water?

Ans. Alum act as strong coagulating agent for the suspended negatively charged impurities (of dust, dirt) particles in colloidal state.

46. Aluminium metal is frequently used as reducing agent for the extraction of metals such as Cr, Mn , Fe. Why?



Ans. Al has great affinity for oxygen and thus it reduces a large number, of less electropositive metal oxides to metals.

57. Thermite reactions can not be stopped by water like ordinary fire, why?

Ans. Ordinary fire needs burning of material in atmospheric oxygen. Water not only cools the burning material but also keeps oxygen away by covering material. In thermite reaction, oxygen involved in metal oxide is used during combustion as well a high temperature produced during thermite reactions give rise to decomposition of H_2O by Al to H_2 and O_2 .

48. No visible reaction occurs when aluminium is left in contact with concentrated nitric acid.

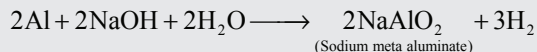
Ans. Al becomes passive due to the coating of Al_2O_3 when it comes in contact with concentrated HNO_3 and becomes inert for further reaction.

59. Aluminium does not react readily with concentrated HNO_3 but reacts immediately with dil. HNO_3 . Explain.

Ans. In presence of concentrated HNO_3 , the Al surface gets oxidised and covered by an oxide film which prevents it for further action.

50. Aluminium vessels should not be cleaned with cleansing agent containing washing soda. Why?

Ans. Washing soda (Na_2CO_3) reacts with water to form NaOH and Al gets dissolved in caustic soda.



51. Aluminium metal is used in nuclear reactors. Why?

Ans. Naturally occurring isotope has thermal neutron cross section of only 0.21 barn (1 barn = 10^{-24} cm²). This low cross section and short half life of the irradiation product (Al^{128}) makes this metal useful for use within nuclear reactors.

52. Beryllium and aluminium exhibit many properties which are similar. But, the two elements differ in

- (a) forming covalent halides (b) forming polymeric hydrides
(c) exhibiting maximum covalency in compounds (d) exhibiting amphoteric nature in their oxides

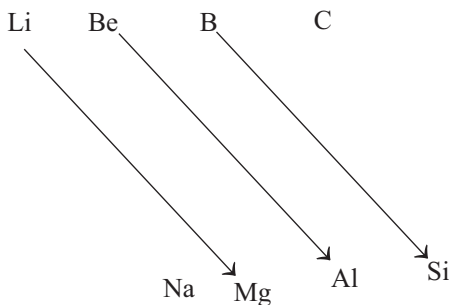
Ans. (c) The valency of beryllium is 2 while that of aluminium is 3.

53. Although aluminium is above hydrogen in the electrochemical series, it is stable in air and water.

Ans. Aluminium is stable in air and water as a thin protective layer of oxide is formed on its surface which prevents the further reaction.

13 Diagonal Relationships in Rows 2 and 3

In descending a column of the Periodic Table, the element's electronegativity generally decreases (electropositive character increases) and the atomic radius increases. These differences are particularly dramatic between rows 2 and 3 of the Periodic Table and lead to the elements Li-F showing distinctly anomalous physical and chemical properties. Diagonal relationships are frequently observed between the elements of rows 2 and 3 as shown below.



Lithium and Magnesium

Table 4.11 Enthalpies of Vaporization, Metallic Radii and Electronegativity Values of Some Elements

	<i>Li</i>	<i>Na</i>	<i>Mg</i>
Enthalpy of vaporization (kJ mol ⁻¹)	135	96	132
Metallic radius/pm	155	190	160
Electronegativity coefficient	1.15	1.0	1.25

- Lithium has an enthalpy of vapourization, metallic radius and electronegativity coefficient which are more similar to those of magnesium than sodium and therefore, there are many similarities in the chemistries of lithium and

4.40 Group13-Boron Family

magnesium. Lithium has an anomalously high enthalpy of vapourization, melting point, boiling point, density and hardness.

- LiOH and Li_2CO_3 are much less soluble in water than the corresponding Na and K compounds. Their fluorides, carbonates and phosphates are insoluble.
- The oxo-anion salts of Li and Mg (CO_3^{2-} , NO_3^- and SO_4^{2-}) decompose quite readily on heating to give the corresponding oxides.
- Lithium and magnesium are unusual in reacting directly with N_2 to form the nitrides Li_3N and Mg_3N_2 .
- Neither lithium nor magnesium form stable peroxides when burned in an atmosphere of dioxygen which remains in excess.
- Lithium and magnesium both form organometallic compounds which have considerable covalent character and are therefore soluble in organic solvents. Organolithium, LiR and Grignard reagents, RMgX , are used in an analogous fashion in organic syntheses.

Beryllium and Aluminium

- The enthalpy of atomization of beryllium is much higher than that of magnesium and therefore it has higher melting and boiling points, enthalpy of fusion and density and it is much harder.

Table 4.12 Physical Properties of Beryllium, Magnesium and Aluminium

	<i>Be</i>	<i>Mg</i>	<i>Al</i>
Melting point (°C)	2970	1107	2467
Enthalpy of vaporization /kJmol ⁻¹	310	132	284
Metallic radius/pm	112	160	143
Electronegativity coefficient	1.5	1.25	1.45

- The bonds in compounds of beryllium and aluminium are highly covalent because of the high electronegativities of the elements. In many of the more covalent compounds Be and Al adopt sp^3 hybridized tetrahedral geometries. The larger size of Al^{3+} compared to Be^{2+} (50 pm versus 31 pm) leads to some octahedral ions for the former, e.g., AlF_6^{3-} and $[\text{Al}(\text{OH}_2)_6]^{3+}$ and AlCl_3 and AlF_3 in the solid state. The organometallic and halide compounds of Be and Al are more soluble in organic solvents than those of the other Group 2 elements.
- Beryllium and aluminium have similar standard reduction potentials:

Be	Mg	Al
$E^\circ/\text{V} -1.85$	-2.37	-1.66

Despite the similarities in their standard reduction potentials, neither of the metals react with water because they both form an impermeable oxide film. Unlike the other alkaline earth metals and the alkali metals beryllium and magnesium do not form coloured solutions in liquid ammonia.

- Beryllium and aluminium both form amphoteric oxides. Both dissolve in alkali to evolve H_2 .
- The Halides of beryllium are hygroscopic and fume in air and in water AlCl_3 hydrolyzes to yield $[\text{Al}(\text{OH})_4]^-$. Positive ions of both beryllium and aluminium have high hydration enthalpies (see Table) and this leads to highly 'soluble salts particularly for beryllium.

$\Delta_{\text{hyd}}H^\circ$	<i>Be</i> ²⁺ -2455	<i>Mg</i> ²⁺ -1900	<i>Al</i> ³⁺ -4630
$[\text{Be}(\text{OH}_2)_4]^{2+}$	$\xrightleftharpoons[\text{H}_3\text{O}^+]{\text{H}_2\text{O}}$	$[(\text{H}_2\text{O})_3\text{Be}-\text{O}-\text{Be}(\text{OH}_2)_3]^{2+} \xrightleftharpoons{\text{OH}^-}$	$\text{Be}(\text{OH})_2$ $\parallel \text{OH}^-$ $[\text{Be}(\text{OH})_4]^{2-}$
$[\text{Al}(\text{OH}_2)_6]^{3+}$	$\xrightleftharpoons[\text{H}_3\text{O}^+]{\text{H}_2\text{O}}$	$[\text{Al}(\text{OH})(\text{OH}_2)_5]^{2+} \xrightleftharpoons{\text{OH}^-}$	$\text{Al}(\text{OH})_3$ $\parallel \text{OH}^-$ $[\text{Al}(\text{OH})_4]^-$

Although Be^{2+} is smaller than Al^{3+} , the larger positive charge of the latter leads to a much more negative value for its hydration enthalpy.

- Both beryllium and aluminium form reasonably stable complexes particularly with oxygen and nitrogen based ligands.
- Unlike magnesium, but similar to aluminium, beryllium forms the ethynide BeC_2 rather than the carbide Be_2C .

Lithium and beryllium are significantly different from the heavier elements of their groups. Specific points follow.

- They exhibit lower coordination numbers because of their small radii. For example, Be in BeCl_2 and BeO is four coordinate, whereas Ca in CaCl_2 and CaO is octahedral.
- The large hydration enthalpies of Li^+ and Be^{2+} lead to many of their salts having hydrated metal cations.
- Their salts with oxoanions, e.g., NO_3^- , CO_3^{2-} , SO_4^{2-} , are much less stable than those of the heavier elements of Groups 1 and 2. Lithium also does not form stable bicarbonates, I_3^- , SH^- and O_2^- salts and BeCO_3 and BeSO_4 decompose more readily on heating than the heavier Group 2 salts.
- The organometallic compounds of lithium and beryllium are more covalent and consequently more soluble in organic solvents and less reactive.
- Their complexes are more stable and numerous.
- They both react more slowly with water. Lithium reacts more slowly with cold water than the heavier alkali metals and beryllium only reacts with steam.

Boron and Silicon

Table summarizes the elemental radii, enthalpies of vapourization and electronegativity coefficients of B, Al. and Si. The diagonal similarities are less pronounced than those discussed above. Boron and silicon do have the following properties in common.

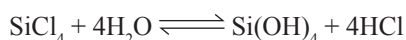
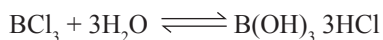
Table 4.13 Physical Properties of Boron, Aluminium and Silicon

	<i>B</i>	<i>Al</i>	<i>Si</i>
Enthalpy of vaporization /kJ mol ⁻¹	536	284	297
Metallic/covalent radius /pm	98	143	132
Electronegativity coefficient	2.0	1.45	1.74

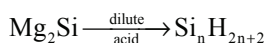
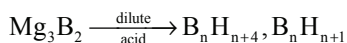
For both elements, the exceptional stabilities of the bonds to oxygen lead to a particularly wide range of borates and silicates. Also the similarities in bond strengths for B-H and Si-H bonds lead to hydrides with analogous properties.

The following similarities may be identified for the elements:

1. Boron is semi-metallic and silicon is a semiconductor. Boron is a black, very hard solid with a high melting point and is chemically inert. Silicon has a metallic appearance and is lustrous, relatively unreactive and has a melting point of 1410°C (cf. boron 2300°C).
2. Boron and silicon do not exist as aquated ions, $[\text{B}(\text{OH}_2)_4]^{3+}$ and $[\text{Si}(\text{OH}_2)_6]^{4+}$. Both form a wide range of oligomers based on boron-oxygen and silicon-oxygen bonds. In borates, BO_3 triangles and BO_4 tetrahedra link through oxygen bridges to form chains and rings. In silicates, SiO_4 tetrahedra build up into chains, multiple chains, rings, sheets and 3-dimensional networks. Both boric oxide, B_2O_3 and silicon dioxide, SiO_2 , form glassy materials.
3. The halides of boron and silicon are molecular covalent solids or liquids which readily hydrolyze and behave as Lewis acids.



4. The hydrides of boron and silicon are volatile, flammable covalent compounds. They are both formed by similar hydrolysis reactions.



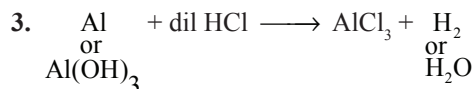
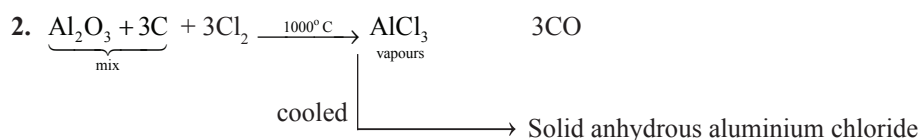
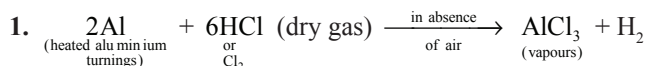
Summary

Diagonal demarcation lines in the Periodic Table separating either chemical or structural properties occur when the atomic characteristic responsible for the changes in the properties occurs with approximately equal weight across and down a column of the Periodic Table. When the changes in atomic properties are much more dramatic down a column than those between elements which occur in a common row, the traditional Mendeleev vertical trends predominate.

Compounds of aluminium

1. Aluminium Chloride ($\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$)

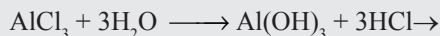
Preparation



Worked-out Example

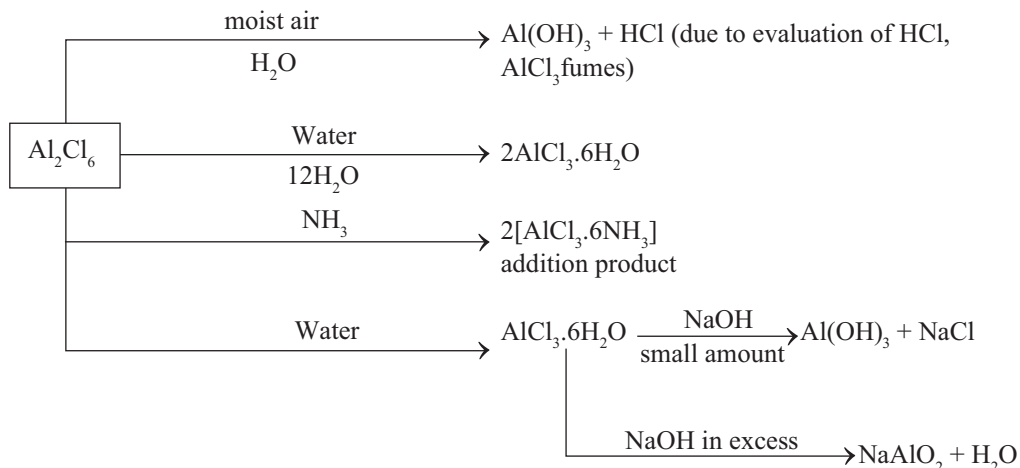
54. Why does anhydrous AlCl_3 fume in air?

Ans. Anhydrous AlCl_3 hydrolyses in moist air and gives fumes of HCl.



Properties

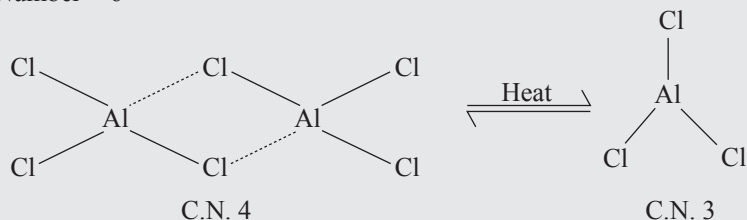
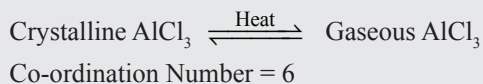
- (i) On heating, it sublimes at 180°C and the vapour density corresponds to the formula Al_2Cl_6 . It is covalent when anhydrous as it does not conduct current in fused state. It is soluble in organic solvents such as alsocho, ether and benzene. The dimeric formula is retained in non-polar solvents but is broken into $[\text{Al}(\text{H}_2\text{O})_6]\text{Cl}_3$ on dissolution in water on account of high heat of hydration. The molecule (dimer) is an autocomplex and is represented as,



Worked-out Examples

55. Discuss the changes in co-ordination number when crystalline AlCl_3 is heated.

Ans. AlCl_3 also crystallizes in an essentially ionic lattice containing 6 co-ordinate metal ions, but dissolves in non-polar solvents.



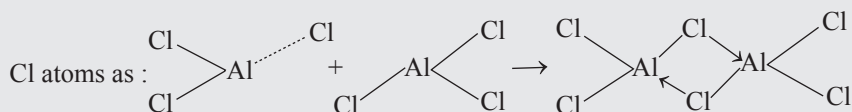
56. Aluminium chloride exists as dimer, Al_2Cl_6 in solid state as well as in solution of non-polar solvents such as benzene. When dissolved in water it gives **[AIEEE 2004]**

- (a) $[\text{Al}(\text{OH})_6]^{3-} + 3\text{HCl}$ (b) $[\text{Al}(\text{H}_2\text{O})_6]^{3+} + 3\text{Cl}^-$
 (c) $\text{Al}^{3+} + 3\text{Cl}^-$ (d) $\text{Al}_2\text{O}_3 + 6\text{HCl}$

Ans. (b) $\text{Al}_2\text{Cl}_6 + 12\text{H}_2\text{O} \rightleftharpoons 2[\text{Al}(\text{H}_2\text{O})_6]^{3+} + 6\text{Cl}^-$

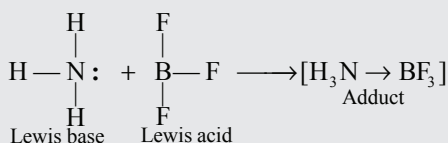
57. AlCl_3 forms a dimer but BCl_3 does not form dimer. Explain.

Ans. AlCl_3 forms dimer by completing octet of Al involving p-orbitals to accept electron pairs from The non-existence of the dimer of boron trifluoride is that the energy released by the formation of an additional bond to another boron atom (as required for the formation of a dimer) is not sufficient to compensate for the loss in energy to the system in overcoming the B—F $p\pi-p\pi$ bonding in the monomer itself.



58. BF_3 , AlCl_3 , SnCl_4 are considered as acid. Why?

Ans. These substances accept electron pair e.g., BF_3 from NH_3 , AlCl_3 from water (Lewis concept).



59. Aluminium fluoride is ionic while AlCl_3 is covalent. Why?

Ans. The size of chloride ion is larger than fluoride ion. Due to higher polarisation in AlCl_3 , it behaves as a covalent compound. The electronegativity difference between Al and Cl is 1.5 while between Al and F it is 2.5. Thus, the bond between Al and F is ionic and between Al and Cl covalent in nature.

60. AlF_3 is a high melting point solid while AlCl_3 is low melting point volatile solid. Explain.

Ans. Smaller anion F^- in AlF_3 is least polarized in comparison to large anion Cl^- in AlCl_3 by Al^{3+} cation. Thus AlF_3 is more ionic in comparison to AlCl_3 and has high melting point in solid state.

61. Molten AlBr_3 is a poor conductor of current. Explain.

Ans. AlBr_3 is covalent in nature and thus during fusion does not produce Al^+ and Br^- ions. Hence, it is poor conductor of current.

62. Which of the following statements about anhydrous aluminium chloride is correct? (IIT 1981)

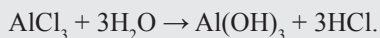
- (a) It exists as AlCl_3 molecules (b) It is not easily hydrolysed
(c) It sublimes at 100°C under vacuum (d) It is a strong Lewis base

Ans. (c)

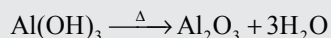
63. Heating an aqueous solution of aluminium chloride to dryness will give [AIEEE 2005]

- (a) $\text{Al}(\text{OH})\text{Cl}_2$ (b) Al_2O_3 (c) Al_2Cl_6 (d) AlCl_3

Ans. (b) The solution of aluminium chloride in water is acidic in due to hydrolysis.

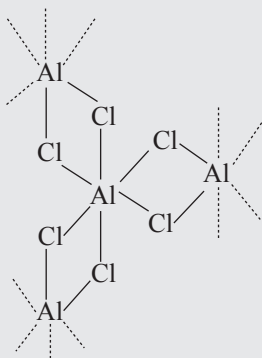


On heating it till dryness, $\text{Al}(\text{OH})_3$ is converted into Al_2O_3 .



64. The coordination number of Al in the crystalline state of AlCl_3 is _____. (IIT 2009)

Ans. (6) The structure of AlCl_3 in solid state is



Each Al is in an octahedral environment with each Cl serving in a bridging capacity. Its coordination number is 6.

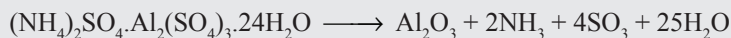
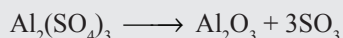
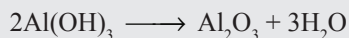
65. All the Al—Cl bonds in Al_2Cl_6 are equivalent. (State whether True or False) (1989)

Ans. False

The terminal bonds are different from the bridge Al—Cl bonds.

66. How is aluminium oxide or alumina (Al_2O_3) prepared?

Ans. It is prepared by igniting aluminium hydroxide, aluminium sulphate or ammonium alum.



Note: $\text{Al}_2\text{O}_3 \rightarrow \alpha$ Alumina $\rightarrow$ (Hard, High M.P., abrasive, precious stone with

blue impurity of Cu $\rightarrow$ Topaz

blue impurity of Co $\rightarrow$ Ruby

β -Alumina $\rightarrow$ Less dense, soft, chromatography separation

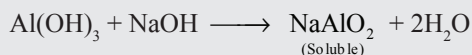
($\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$ — bauxite while red bauxite — Fe_2O_3 , grey — SiO_2)

67. $B(OH)_3$ is an acid, $Al(OH)_3$ is amphoteric and $Tl(OH)_3$ is basic. Explain.

Ans. The electropositive or metallic character of gp. 13 elements increases from B to Tl. B being nonmetal and thus forms acidic hydroxide. Al forms amphoteric hydroxide and Tl, the most metallic form basic hydroxides.

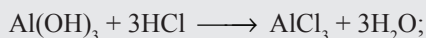
68. $Al(OH)_3$ is insoluble in excess of NH_4OH but dissolves in $NaOH$. Why?

Ans. $NaOH$ is strong alkali. $Al(OH)_3$ being amphoteric dissolves in $NaOH$ and forms soluble sodium meta aluminate.



69. $Al(OH)_3$ is amphoteric. Explain.

Ans. $Al(OH)_3$ reacts with acids and alkalies both:



70. What is emery and for what purpose it is used?

Ans. Impure form of Al_2O_3 containing Fe_2O_3 and silica is called emery. It is coated over paper to produce sand paper or emery paper, used for polishing metals.

71. Hydrogen gas will not reduce

(IIT 1985)

(a) heated cupric oxide

(b) heated ferric oxide

(c) heated stannic oxide

(d) heated aluminium oxide

Ans. (d)

Reasoning Type

72. The question below consist of an 'Assertion' in column 1 and the 'Reason' in column 2. Use the following key to choose the appropriate answer.

(a) If both assertion and reason are CORRECT and reason is the CORRECT explanation of the assertion.

(b) If both assertion and reason are CORRECT, but reason is NOT the CORRECT explanation of the assertion.

(c) If assertion is CORRECT, but reason is INCORRECT.

(d) If assertion is INCORRECT, but reason is CORRECT.

Assertion

$Al(OH)_3$ is amphoteric in nature

Reason

$Al-O$ and $O-H$ bonds can be broken with equal ease in $Al(OH)_3$.

(1998)

2. (c)

Alums

Alums have the general formula $M^I M^{III} (XO_4)_2 \cdot 12H_2O$ where M^I is a monovalent ion (Na, K, Rb, Cs, NH₄ etc.), M^{III} is a trivalent ion (Al, Fe, Cr, Mn, Ir, Ga, In, Tl, V, Co, Rh) and X is sulphur or selenium. The size factors are such that neither too small a monovalent cation, Li^+ , nor too big a trivalent ion like the lanthanides form stable alums. Six of the twelve water molecules are coordinated to the trivalent cation in the first sphere of attraction and the other six water molecules surround the monovalent cation. The arrangement of $[M^I(H_2O)_6]^+$, $[M^{III}(H_2O)_6]^{3+}$ and SO_4^{2-} groups is not always the same.

Alums are double sulphates having general formula,



4.46 Group13-Boron Family

Where, X = monovalent cation such as Na^+ , K^+ , NH_4^+ , etc.,

M = trivalent cation such as Al^{3+} , Cr^{3+} , Fe^{3+} etc.,

Some important alums and their names are given below:

Potash Alum : $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$

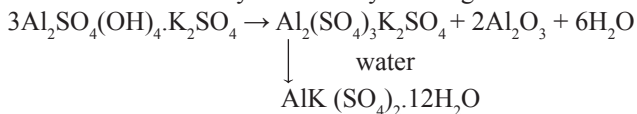
Chromo Alum : $\text{K}_2\text{SO}_4 \cdot \text{Cr}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$

Soda Alum : $\text{Na}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$

Ferric Alum : $(\text{NH}_4)_2\text{SO}_4 \cdot \text{Fe}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$

Alums are isomorphous crystalline solids which are soluble in water. Due to hydrolysis, their aqueous solutions have acidic character.

Potash alum is usually obtained by roasting *alumstone* $3\text{Al}_2\text{S}_4(\text{OH})_4 \cdot \text{K}_2\text{SO}_4$ in air and crystallizing from water:



Aluminium ammonium alum can be made by the addition of ammonium sulphate to aluminium sulphate in aqueous medium and crystallising. Alums are dissociated completely into ions in aqueous medium and therefore, they cannot be regarded as complex entities in aqueous solution. They respond to the usual tests of aquametal ions.

Aluminium alums are chiefly used in dyeing as a mordant. There are numerous dyes whose molecules, although they adhere readily to fibres of animal origin such as silk or wool, do not attach firmly to vegetable fibres such as cotton. However, if cotton is first soaked in a solution of ammonium alum (or any Al^{3+} solution) and then treated to precipitate aluminium hydroxide many of these dyes are adsorbed on the aluminium oxide and hence are strongly held in the cloth. Common mordants include salts of Al, Fe. Aluminium alums also find use in purification of water for human consumption. Water is treated with alum and then made basic with sodium carbonate. Aluminium hydroxide precipitates and this carries down with it suspended foreign matter including many microorganisms. The water is finally treated with chlorine, iodine or in some cases with ozone.

Note: Li cannot form alum due to its small size.

Worked-out Examples

73. Which type of cations are capable of replacing aluminium in alums?

Ans. The trivalent metal cations of about the same size as that of Al^{3+} are capable of replacing aluminium in alums e.g., Ti^{3+} , Cr^{3+} , Mn^{3+} , Fe^{3+} and Co^{3+} .

74. Alum helps in purifying water by

[AIEEE 2002]

- | | |
|--|---|
| (a) forming Si complex with clay particles | (b) sulphate part which combines with the dirt and removes it |
| (c) coagulating the mud particles | (d) making mud water soluble |

Ans. (c) Alum furnishes Al^{3+} ions which bring about coagulation of negatively charged clay particles, bacteria etc.,

Some Noteworthy Points

- Aluminium is the most abundant metal in the earth crust.
- Aluminium is the chief constituent of silvery paints which consists of aluminum powder in linseed oil.
- Concentrated HNO_3 renders aluminium passive.
- Fused bauxite is used as an abrasive under the trade name alundum.
- Ammonal, a mixture of aluminium powder and ammonium nitrate issued is bombs.
- The crystalline Al_2O_3 is a component of many costly stones e.g., rubies, sapphire, topaz etc.,
- Aqueous solutions of alums are acidic due to cationic hydrolysis.

- Aluminium is the most electropositive element in its group. It also have the lowest density among all the elements in the group.
- Aluminium acetate is known as red liquor. It is used as a mordant in dyeing and calico printing.
- Hydrated AlCl_3 on heating to 460 K gives anhydrous dimer $(\text{AlCl}_3)_2$.
- Aluminium, gallium, indium and thallium all form a range of compounds in the +3 oxidation state and compounds in the +1 oxidation state become progressively more stable down Group 13.
- The oxides of aluminium and gallium are amphoteric and indium and thallium oxides are more basic.
- The octahedral aqua-ions $[\text{M}(\text{OH}_2)_6]^{3+}$ are acidic and the pK_a values for the equilibria:



- The MX_3 compounds are Lewis acids and the Lewis acid strengths decrease in the order: $\text{Al} > \text{Ga} > \text{In}$.
- The stability of the hydrides decreases down the Group and there are no stable Tl-H compounds. Extraordinary precautions are required to exclude air and moisture in order to isolate Ga_2H_6 .
- Aluminium is resistant to corrosion because of an impermeable oxide layer, but is soluble in non-oxidizing mineral acids. Gallium, In and Tl dissolve readily in acids, but thallium dissolves only slowly in H_2SO_4 and HCl .
- AlN , GaN and InN have wurtzite structures analogous to cubic BN , but show no analogue of the graphite-type structure of BN .
- AlCl_3 is a low melting solid which sublimes at 183°C .

ONE CORRECT OPTION TYPE QUESTIONS

- Thallium shows different oxidation states because:
 - of its high reactivity
 - of inert pair of electrons
 - of its amphoteric nature
 - it is a transition metal
- Al and Ga have nearly same covalent radii because of:
 - greater shielding power of s-electrons of Ga atoms
 - poor shielding power of s-electrons of Ga atoms
 - poor shielding power of d-electrons of Ga atoms
 - greater shielding power of d-electrons of Ga atoms
- Fluorine is more electronegative than either boron or phosphorus. What conclusion can be drawn from the fact that BF_3 has no dipole moment but PF_3 does?
 - BF_3 is not spherically symmetrical, PF_3 is not
 - BF_3 molecule must be linear
 - The atomic radius of P is larger than the atomic radius of B
 - The BF_3 molecule must be planar triangular
- Correct match is:
 - Ordinary form of borax: $\text{Na}_2\text{B}_4\text{O}_7 \cdot 5\text{H}_2\text{O}$
 - Colemanite: $\text{Ca}_2\text{B}_6\text{O}_{11} \cdot 5\text{H}_2\text{O}$
 - Boronatocalcite: $2\text{Mg}_3\text{B}_8\text{O}_{15} \cdot \text{MgCl}_2$
 - Octahedral form of borax: $\text{Na}_2\text{B}_4\text{O}_7 \cdot \text{OH}_2\text{O}$
- Boron carbide, B_4C , is widely used for:
 - making acetylene
 - making plaster of Paris
 - as a hardest substance after diamond
 - making boric acid
- Boron nitride is isoelectronic with:
 - C_2
 - B_2
 - N_2
 - O_2
- Inorganic benzene is:
 - B_2H_6
 - $\text{B}_3\text{N}_3\text{O}_3$
 - $\text{B}_3\text{N}_3\text{H}_6$
 - B_4H_{10}
- Inorganic graphite is:
 - $\text{B}_3\text{N}_3\text{H}_6$
 - B_2H_6
 - BN
 - BF_3
- Boron compounds behave as Lewis acids because of their:
 - acidic nature
 - covalent nature
 - ionisation property
 - electron deficient nature
- Possible oxidation states of boron family elements are:

(a) +1	(b) +2
(c) +3	(d) +4
- Which of the following is correct?
 - The members of B_nH_{n+6} are less stable than B_nH_{n+4} series
 - Diborane is coloured and unstable at room temperature
 - The reaction of diborane with oxygen is endothermic
 - All of the above
- In diborane:
 - 4 bridged hydrogens and two terminal hydrogens are present
 - 2 bridged hydrogens and four terminal hydrogens are present
 - 3 bridged hydrogens and three terminal hydrogens are present
 - none of these
- Which one is not a borane?

(a) B_5H_9	(b) B_5H_{10}
(c) B_5H_n	(d) B_6H_{10}
- From B_2H_6 , all the following can be prepared except:

(a) B_2O_3	(b) H_3BO_3
(c) $\text{B}_2(\text{CH}_3)_6$	(d) NaBH_4
- Which of the following is a false statement?
 - BH_3 is not a stable compound
 - Boron hydrides are formed when dil. HCl reacts with Mg_3B_2
 - All the B—H bond distances in B_2H_6 are equal
 - The boron hydrides are readily hydrolysed

16. In B_2H_6 :
- there is direct boron-boron bond
 - the boron atoms are linked through hydrogen bridges
 - the structure is similar to C_2H_6
 - all the atoms are in one plane
17. Identify the statement that is not correct as far as structure of diborane is concerned.
- There are two bridging hydrogen atoms in diborane
 - Each boron atom forms four bonds in diborane
 - The hydrogen atoms are not in the same plane
 - All B—H bonds in diborane are similar
18. Select the correct statement about diborane.
- B_2H_6 has three centred bond
 - Each boron atom lies in sp^3 hybrid state
 - $H_b \dots B \dots H_b$ bond angle is 122°
 - All hydrogens in B_2H_6 lie in the same plane
19. The two type of bonds present in B_2H_6 are covalent and.....
- Ionic
 - Co-ordinate
 - Hydrogen bridge bond
 - None of these
20. B—H—B bridge in B_2H_6 is formed by the sharing of:
- 2 electrons
 - 4 electrons
 - 1 electron
 - 3 electrons
21. When strongly heated, orthoboric acid leaves a residue of:
- metaboric acid
 - tetraboric acid
 - boric anhydride
 - boron
22. Boric acid is prepared from borax by the action of:
- hydrochloric acid
 - sodium hydroxide
 - carbon dioxide
 - sodium carbonate
23. Which one of the following statements about H_3BO_3 is not correct?
- It is a strong tribasic acid.
 - It is prepared by acidifying an aqueous solution of borax.
 - It has a layer structure in which planar BO_3 units are joined by hydrogen bonds.
 - It does not act as proton donor as it acts as a Lewis acid by accepting hydroxyl ions.
24. Boric acid is a very weak acid but in presence of certain organic compounds, it acts as a strong acid. Which one of the following organic compounds can affect such change?
- Glycerol
 - Acetic acid
 - Ethyl alcohol
 - Ethylene
25. H_3BO_3 is: [IIT 2003]
- monobasic and weak Lewis acid
 - monobasic and weak Bronsted acid
 - monobasic acid and strong Lewis acid
 - tribasic acid and weak Bronsted acid
26. Orthoboric acid (H_3BO_3) and metaboric acid (HBO_2) differ in respect of:
- basicity
 - structure
 - melting point
 - oxidation
27. (A) Boric acid is a tribasic acid.
(R) Boric acid contains three hydroxyl groups.
- If both assertion and reason are true and reason is the correct explanation of assertion.
 - If both assertion and reason are true but reason is not the correct explanation of assertion.
 - If assertion is true but reason is false.
 - If both assertion and reason are false.
28. Which of the following compound is formed in borax bead test?
- Metaborate
 - Tetraborate
 - Double oxide
 - Orthoborate
29. Which one is borax?
- $Na_2B_4O_7 \cdot 10H_2O$
 - $Na_2B_4O_7 \cdot 4H_2O$
 - $Ca_2B_6O_{11} \cdot 5H_2O$
 - CaB_4O_7
30. Borax is prepared by treating colemanite with:
- $NaNO_3$
 - $NaCl$
 - $NaHCO_3$
 - Na_2CO_3
31. On the addition of mineral acid to an aqueous solution of borax, the following compound is formed:
- boron hydride
 - orthoboric acid
 - metaboric acid
 - pyroboric acid
32. Which of the following compounds is formed when boron trichloride is treated with water?
- $H_3BO_3 + HCl$
 - $B_2H_6 + HCl$
 - $B_2O_3 + HCl$
 - None of these
33. Which one of the following statements regarding BF_3 is not correct?

4.50 Group13-Boron Family

- (a) It is a Lewis acid
(b) It is an ionic compound
(c) It is an electron deficient compound
(d) It forms adducts
34. BF_3 is used as catalyst in several industrial processes due to its:
(a) strong reducing nature
(b) weak reducing action
(c) strong Lewis acid nature
(d) weak Lewis acid character
35. Specify the coordination geometry around and hybridization of N and B atoms in 1:1 complex of BF_3 and NH_3 :
(a) N: tetrahedral, sp^3 ; B: tetrahedral, sp^3
(b) N: pyramidal, sp^3 ; B: pyramidal, sp^3
(c) N: pyramidal, sp^3 ; B: planar, sp^3
(d) N: pyramidal, sp^3 ; B: tetrahedral, sp^3
36. The weakest Lewis acid is:
(a) BF_3 (b) BCl_3
(c) BBr_3 (d) BI_3
37. B—F bond order in BF_3 is:
(a) 1 (b) 2
(c) 3 (d) 4/3
38. Identify the correct statement:
(a) BF_3 , BCl_3 are gases
(b) BF_3 has partially double bond character due to $\text{p}\pi\text{-p}\pi$ bonding and the molecule shows resonance
(c) BF_3 is electron pair donor
(d) BBr_3 is solid
39. In the commercial electrochemical process for aluminium extraction, the electrolyte used is:
(a) $\text{Al}(\text{OH})_3$ in NaOH solution
(b) an aqueous solution of $\text{Al}_2(\text{SO}_4)_3$
(c) a molten mixture of Al_2O_3 and Na_3AlF_6
(d) a molten mixture of $\text{AlO}(\text{OH})$ and $\text{Al}(\text{OH})_3$
40. Aluminium oxide is not reduced by chemical reactions since:
(a) aluminium oxide is highly stable
(b) aluminum oxide is stable
(c) reducing agent contaminates
(d) the process pollutes the environment
41. In the commercial electrochemical process for aluminium extraction the electrolyte used is
(a) $\text{Al}(\text{OH})_3$ in NaOH solution
(b) an aqueous solution of $\text{Al}_2(\text{SO}_4)_3$
(c) a molten mixture of Al_2O_3 and Na_3AlF_6
(d) a molten mixture of $\text{AlO}(\text{OH})$ and $\text{Al}(\text{OH})_3$
42. Al dissolves in molten NaOH with the formation of:
(a) Sodium aluminate (Na_3AlO_3)
(b) Sodium metaluminate (NaAlO_2)
(c) Aluminium hydroxide
(d) Alumina
43. Al_2O_3 formation involves large quantity of heat evolution which makes its use in:
(a) Deoxidizer
(b) Confectionary
(c) Indoor photography
(d) Thermite welding
44. Aluminium vessels should not be washed with materials containing washing soda because:
(a) washing soda is expensive
(b) washing soda is easily decomposed
(c) washing soda reacts with aluminium to form soluble aluminate
(d) washing soda reacts with aluminium to form insoluble aluminate
45. In Al_2Cl_6 , which statement is incorrect?
(a) Four Al—Cl bonds are of same length and four of different length
(b) Six Al—Cl bonds are of same length and two of different length
(c) The angle Cl—Al—Cl is 110° and 93°
(d) The angle Al—Cl—Al is 87°
46. AlCl_3 on hydrolysis gives:
(a) $\text{Al}_2\text{O}_3 \cdot \text{H}_2\text{O}$ (b) $\text{Al}(\text{OH})_3$
(c) Al_2O_3 (d) $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$
47. The bonds present in borazole are:
(a) 12σ , 3σ (b) 9π , 6π
(c) 6π , 6π (d) 9π , 9π
48. Reactivity of borazole is greater than that of benzene because:
(a) Borazole is non-polar compound
(b) Borazole is polar compound
(c) Borazole is electron deficient compound
(d) Of localized electrons in it
49. BCl_3 does not exist as dimer but BH_3 exist as dimer (B_2H_6) because:
(a) Chlorine is more electronegative than hydrogen
(b) There is $\text{p}\pi\text{-p}\pi$ back bonding in BCl_3 but does not contain such multiple bonding

- (c) Large sized chlorine atoms do not fit in between the small boron atoms whereas small sized hydrogen atoms get fitted between boron atoms
(d) None of these
50. The gaseous product formed by the action of BCl_3 and lithium tetrahydro aluminate is:
- (a) Cl_2 (b) HCl
(c) H_2 (d) B_2H_6

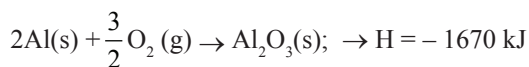
51. Which of the following statement about boron carbide is wrong?
- (a) Its molecular formula is B_4C
(b) It is also called Norbion
(c) It is the hardest substance
(d) It is used for cutting glasses

COMPREHENSION BASED QUESTIONS

COMPREHENSION-I

Aluminium is stable in air and water in spite of the fact that it is reactive metal. The reason is that a thin film of its oxide is formed on its surface which makes it passive for further attack. The layer is so useful that in industry, it is purposely deposited by an electrolytic process called anodizing.

Reaction of aluminium with oxygen is highly exothermic and is called thermite reaction.

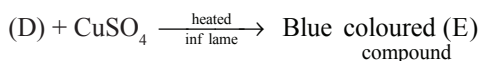
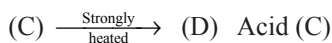
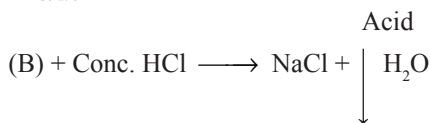
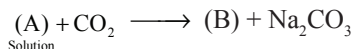
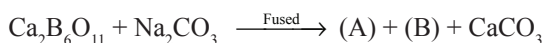


Thermite reaction finds applications in the metallurgical extraction of many metals from their oxides and for welding of metals. The drawback is that to start the reaction, high temperature is required for which an ignition mixture is used.

- Anodizing can be done by electrolyzing dilute H_2SO_4 with aluminium as anode. This results in:
 - the formation of $\text{Al}_2(\text{SO}_4)_3$ on the surface of aluminium anode
 - the formation of oxide film (Al_2O_3) on the surface of aluminium anode
 - the formation of polymeric aluminium hydride film on the surface of aluminium anode
 - None of the above
- The reaction which is not involved in thermite process:
 - $3\text{Mn}_3\text{O}_4 + 8\text{Al} \rightarrow 9\text{Mn} + 4\text{Al}_2\text{O}_3$
 - $\text{Cr}_2\text{O}_3 + 2\text{Al} \rightarrow 2\text{Cr} + \text{Al}_2\text{O}_3$
 - $2\text{Fe} + \text{Al}_2\text{O}_3 \rightarrow \text{Fe}_2\text{O}_3 + 2\text{Al}$
 - $\text{B}_2\text{O}_3 + 2\text{Al} \rightarrow 2\text{B} + \text{Al}_2\text{O}_3$
- Thermite a mixture used for welding is:
 - Fe and Al
 - BaO and Mg powder
 - Cu and Al
 - Fe_2O_3 and Al powder

- Anodized aluminium is:
 - Al obtained at anode
 - Al prepared electrolytically
 - alloy of Al containing 95% Al
 - Al electrolytically coated with aluminium oxide
- Which one of the following metals cannot be extracted by using Al as a reducing agent?
 - Na from Na_2O
 - Cr from Cr_2O_3
 - W from WO_3
 - Mn from Mn_3O_4
- Aluminium becomes passive in:
 - conc. HNO_3
 - H_2CrO_4
 - HClO_4
 - All of these

COMPREHENSION-II



- Compound (A) is:
 - NaBO_2
 - $\text{Na}_2\text{B}_4\text{O}_7$
 - Na_3BO_3
 - NaOH
- Compound (B) is:
 - NaBO_2
 - $\text{Na}_2\text{B}_4\text{O}_7$
 - Na_3BO_3
 - NaOH
- Compound (C) is:
 - $\text{H}_2\text{B}_4\text{O}_7$
 - HBO_2
 - H_3BO_3
 - HB_3O_5

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4. Compound (D) is:

- (a) H_3BO_3 (b) B_2O_3
(c) B (d) none of these

5. Compound (E) is:

- (a) Cu_2O (b) CuS
(c) CuSO_3 (d) $\text{Cu}(\text{BO}_2)_2$

COMPREHENSION-III

- (i) $\text{Boron} + \text{O}_2 \xrightarrow{700^\circ\text{C}} (\text{X})$
(ii) $(\text{X}) + \text{C}(\text{carbon}) + \text{Cl}_2 \longrightarrow (\text{Y}) + \text{CO}$
(iii) $(\text{Y}) + \text{LiAlH}_4 \longrightarrow (\text{Z}) + \text{LiCl} + \text{AlCl}_3$
(iv) $(\text{Z}) + \text{NH}_3 \longrightarrow (\text{A}) \xrightarrow{\text{heat}} (\text{B})$
(v) $(\text{Z}) + \text{NaH} \longrightarrow (\text{D})$

1. Compounds (X) and (Y) are:

- (a) (X) = BO_2 , (Y) = BCl_2
(b) (X) = BO_3 , (Y) = BCl_4
(c) (X) = B_2O_3 , (Y) = BCl_3
(d) (X) = BO_3 , (Y) = B_4C

2. Compound (D) is used as a/an:

- (a) oxidizing agent (b) complexing agent
(c) buffer agent (d) reducing agent

3. Compound (Z) is:

- (a) an ionic compound
(b) an electron deficient compound
(c) $3\text{C}-2\text{e}$ compound
(d) having ethane like structure

4. Compound (B) is:

- (a) borazole (b) inorganic benzene
(c) borazon (d) boron nitride

5. Compound Y:

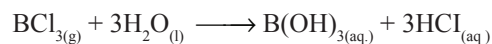
- (a) has boron in sp^2 hybridized state
(b) is a planar molecule
(c) has zero dipole moment
(d) is a Lewis base

COMPREHENSION-IV

All the boron trihalides except BI_3 may be prepared by direct reaction between the elements. Boron trihalides consist of trigonal-planar BX_3 molecules. Unlike the halides of the other elements in the group they are monomeric in the gas, liquid and solid states, BF_3 and BCl_3 are gases, BBr_3 is a volatile liquid and BI_3 is a solid. Boron trihalides are Lewis acids because they form Lewis complexes with suitable bases.



However, boron chlorides, bromides and iodides are susceptible (sensitive) to protolysis by mild proton sources such as water, alcohols and even amines for example BCl_3 undergoes rapid hydrolysis.



It is supposed that the first step in the above reaction is the formation of the complex $\text{Cl}_3\text{B} \rightarrow \text{OH}_2$ which then eliminates HCl and reacts further with water.

1. Which of the following is the best order of Lewis acid strength of BF_3 , BCl_3 and BBr_3 ?

- (a) $\text{BF}_3 > \text{BCl}_3 > \text{BBr}_3$
(b) $\text{BF}_3 = \text{BCl}_3 = \text{BBr}_3$
(c) $\text{BF}_3 < \text{BCl}_3 < \text{BBr}_3$
(d) $\text{BBr}_3 > \text{BF}_3 > \text{BCl}_3$

2. Which of the following is the correct prediction about observed B—X bond length, in BX_3 molecule?

- (a) B—F bond length in BF_3 is found to be less than theoretical value because the electronegativity values of B (2.04) and F (4.0) suggest the bond to be ionic and hence the attraction between oppositely charged ions must decrease the bond length.
(b) BF_3 and $[\text{BF}_4]^-$ have equal B—F bond length.
(c) The decrease in the B—F bond length in BF_3 , is due to delocalised $\text{p}\pi - \text{p}\pi$ bonding between vacant 2p orbital of B and filled 2p orbital of F.
(d) The correct B—X bond length order is $\text{B—F} > \text{B—Cl} > \text{B—Br} > \text{B—I}$.

3. Which is correct about the hydrolysis of BX_3 ?

- (a) All BX_3 undergo hydrolysis to produce $\text{B}(\text{OH})_{3(\text{aq})}$ and $\text{HX}_{(\text{aq})}$.
(b) BF_3 does not undergo complete hydrolysis due to formation of $\text{H}^+[\text{BF}_3\text{OH}]^-$.
(c) BBr_3 does not undergo hydrolysis at all because it can not form H-bonds with water
(d) All of the above are correct

4. Which of the following reactions is incorrect?

- (a) $\text{BF}_{3(\text{g})} + \text{F}_{(\text{aq})}^- \longrightarrow [\text{BF}_4]_{(\text{aq})}^-$
(b) $\text{BCl}_{3(\text{g})} + 3\text{EtOH}_{(\text{l})} \longrightarrow \text{B}(\text{OEt})_{3(\text{l})} + 3\text{HCl}_{(\text{g})}$
(c) $\text{BBr}_{3(\text{l})} + \text{F}_3\text{BN}(\text{CH}_3)_{3(\text{g})} \longrightarrow \text{Br}_{3(\text{g})} + \text{Br}_3\text{BN}(\text{CH}_3)_{3(\text{g})}$
(d) $\text{BCl}_{3(\text{g})} + \underset{\text{Excess}}{2\text{C}_5\text{H}_5\text{N}_{(\text{l})}} \longrightarrow \text{Cl}_3\text{B}(\text{C}_5\text{H}_5\text{N})_2(\text{s})$

5. Which of the following is correct?

- (a) $\text{B}(\text{OCH}_3)_3$ is much weaker Lewis acid than BBr_3
(b) $\text{B}(\text{OH})_{3(\text{aq})}$ behaves as a triprotic acid
(c) $[\text{H}_2\text{BO}_3]_{(\text{aq})}^-$ is a conjugate base of $\text{H}_3\text{BO}_{3(\text{aq})}$
(d) BF_3 does not react with ethers

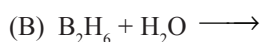
ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

- In the electrolysis of alumina, cryolite is added to:
 - lower the melting point of alumina
 - increase the electrical conductivity
 - minimize the anode effect
 - remove impurities from alumina
- Which of the following oxides are basic?
 - B_2O_3
 - Tl_2O
 - In_2O_3
 - Al_2O_3
- Stability of monovalent and trivalent cations of Ga, In, Tl lie in following sequence:
 - $Ga^{3+} < In^{3+} > Tl^{3+}$
 - $Ga^{3+} > In^{3+} > Tl^{3+}$
 - $Tl^+ > In^+ > Ga^+$
 - $Ga^{3+} > In^+ > Tl^+$
- Which of the following elements do not form carbide?
 - B
 - Al
 - In
 - Ga
- Orthoboric acid (H_3BO_3) and metaboric acid (HBO_2) differ in respect of:
 - basicity
 - structure
 - melting point
 - oxidation
- Select the correct statements about diborane:
 - B_2H_6 has three centred bond
 - Each boron atom lies in sp^3 hybrid state
 - $H_b \dots B \dots H_b$ bond angle is 122°
 - All hydrogens in B_2H_6 lie in the same plane
- Possible oxidation states of boron family elements are:
 - +1
 - +2
 - +3
 - +4
- $Al_2(SO_4)_3 + NH_4OH \longrightarrow X-X$ is:
 - a white gelatinous precipitate
 - insoluble in excess of NH_4OH
 - soluble in excess of NaOH
 - amphoteric in nature
- Which of the following metals are extracted by using Al as a reducing agent?
 - Na from Na_2O
 - Cr from Cr_2O_3
 - W from WO_3
 - Mg from MgO
- Aluminium becomes passive in:
 - conc. HNO_3
 - H_2CrO_4
 - $HClO_4$
 - conc. HCl
- Which of the following statements are true for H_3BO_3 ?
 - It is mainly monobasic acid and a Lewis acid.
 - It does not act as a proton donor but acts as an acid by accepting hydroxyl ions.
 - It has a layer structure in which BO_3 units are joined by hydrogen bonds.
 - It is obtained by treating borax with concentrated H_2SO_4 .
- Boron carbide:
 - is used as an abrasive for polishing
 - is used in brake linings for cars
 - is used in making bullet proof cloths
 - has enormous tensile strength
- Aluminium alkyl R_3Al :
 - is used in the dimerisation of propene to give finally isoprene
 - have three centre bonds and exist as dimer
 - show sp^2 hybridisation
 - show dp^3 hybridisation

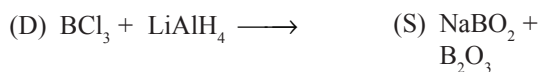
MATCH THE COLUMN TYPE QUESTIONS

1.

Column-I



Column-II



ASSERTION/REASON TYPE QUESTIONS

Directions:

- (a) **Statement-1** is True, **Statement-2** is True; **Statement-2** is a correct explanation for **Statement-1**
 (b) **Statement-1** is True, **Statement-2** is True; **Statement-2** is NOT a correct explanation for **Statement-1**
 (c) **Statement-1** is True, **Statement-2** is False
 (d) **Statement-1** is False, **Statement-2** is True

- Statement-1:** $\text{Al}(\text{OH})_3$ is amphoteric in nature
Statement-2: $\text{Al}-\text{O}$ and $\text{O}-\text{H}$ bonds can be broken with equal ease in $\text{Al}(\text{OH})_3$
- Statement-1:** In water, orthoboric acid behaves as a weak monobasic acid.
Statement-2: In water, orthoboric acid acts as a proton donor.
- Statement-1:** Boron always forms covalent bond.
because
Statement-2: The small size of B^{3+} favours formation of covalent bond.
- Statement-1:** Boron has high melting point and boiling point.
Statement-2: Boron exist as a giant covalent, polymeric structure both in solid as well as in liquid state.

- Statement-1:** Pure born is difficult to be formed
Statement-2: Melting point of pure boron is very high and it is corrosive in liquid state.
- Statement-1:** Boron halides involves $\text{p}\pi\text{-p}\pi$ back bonding whereas Al halides do not.
Statement-2: Al has larger size than boron.
- Statement-1:** Calculated bond length of $\text{B}-\text{F}$ bond in BF_3 is 152 pm whereas observed bond length is 130 pm.
Statement-2: Boron fluoride possesses $\text{B} \text{---} \text{F}$ (or double bond) character due to $\text{p}\pi\text{-p}\pi$ back bonding.
- Statement-1:** Borax solution in water may be used as buffer solution.
Statement-2: Borax on dissolution of water gives H_3BO_3 and its conjugate base $\text{B}(\text{OH})_4^-$.
- Statement-1:** AlF_3 is soluble in KF but addition of BF_3 brings its precipitation.
Statement-2: BF_3 is more acidic than AlF_3 .
- Statement-1:** BF_3 is a weaker Lewis acid than BCl_3 .
Statement-2: BF_3 molecule is stabilized to a greater extent than BCl_3 by $\text{B}-\text{F} \rightarrow$ -bonding.

RECENT YEARS' QUESTION OF IIT-JEE AND AIEEE

- Electrolytic reduction of alumina to aluminium by Hall-Heroult process is carried out [IIT 2000]
 (a) in the presence of NaCl
 (b) in the presence of fluorite
 (c) in the presence of cryolite which forms a melt with lower melting temperature
 (d) in the presence of cryolite which forms a melt with higher melting temperature
- How can the following reaction be made to proceed in forward direction? [IIT 2006]

$$\text{B}(\text{OH})_3 + \text{NaOH} \rightleftharpoons \text{NaBO}_2 + \text{Na}[\text{B}(\text{OH})_4] + \text{H}_2\text{O}$$
 (a) addition of borax
 (b) addition of cis-1, 2-diol
 (c) addition of Na_2HPO_4
 (d) addition of trans-1, 2-diol
- Alum helps in purifying water by (AIEEE 2002)
 (a) forming Si complex with clay partiles
 (b) sulphate part which combines with the dirt and removes it
 (c) coagulating the mud particles
 (d) making mud water soluble
- Aluminium is extracted by the electrolysis of (AIEEE 2002)
 (a) bauxite
 (b) alumina
 (c) alumina mixed with molten cryolite
 (d) molten cryolite
- Aluminium chloride exists as dimer, Al_2Cl_6 in solid state as well as in solution of non-polar solvents such as benzene. When dissolved in water, it gives (AIEEE 2004)

- (a) $[\text{Al}(\text{OH})_6]^{3-} + 3\text{HCl}$
 (b) $[\text{Al}(\text{H}_2\text{O})_6]^{3+} + 3\text{Cl}^-$
 (c) $\text{Al}^{3+} + 3\text{Cl}^-$
 (d) $\text{Al}_2\text{O}_3 + 6\text{HCl}$
6. Specify the coordination geometry around and hybridization of Na and B atoms in 1:1 complex of BF_3 and NH_3 : [IIT (S) 2002]
 (a) N: tetrahedral, sp^3 ; B: tetrahedral, sp^3
 (b) N: pyramidal, sp^3 ; B: pyramidal, sp^3
 (c) N: pyramidal, sp^3 ; B: planar, sp^3
 (d) N: pyramidal, sp^3 ; B: tetrahedral, sp^3
7. H_3BO_3 is [IIT (S) 2003]
 (a) monobasic and weak Lewis acid
 (b) monobasic and weak Bronsted acid
 (c) monobasic acid and strong Lewis acid
 (d) tribasic acid and weak Bronsted acid
8. Be and Al exhibit many properties which are similar, but the two elements differ in: [AIEEE 2004]
 (a) Exhibiting amphoteric nature in their oxides
 (b) Forming polymeric hydrides
 (c) Exhibiting maximum covalency in compounds
 (d) Forming covalent halides
9. Heating an aqueous solution of aluminium chloride to dryness will give: [AIEEE 2005]
 (a) AlCl_3 (b) Al_2Cl_6
 (c) Al_2O_3 (d) $\text{Al}(\text{OH})\text{Cl}_2$
10. The structure of diborane contains: [AIEEE 2005]
 (a) Four 2C-2e bonds and two 3C-2e bonds
 (b) Two 2C-2e bonds and two 3C-2e bonds
 (c) Two 2C-2e bonds and two 3C-3e bonds
 (d) Four 2C-2e bonds and two 3C-2e bonds
11. How the reaction below, can be made to proceed in forward direction?

$$\text{B}(\text{OH})_3 + \text{NaOH} \longrightarrow [\text{NaB}(\text{OH})_4]$$
 or
$$[\text{NaBO}_2 + \text{H}_2\text{O}]$$
 [IIT 2006]
 (a) addition of cis-1, 2-diol
 (b) addition of borax
 (c) addition of trans-1, 2-diol
 (d) addition of Na_2HPO_4
12. Which one of the following is the correct statement? [AIEEE 2008]
 (a) $\text{B}_2\text{H}_6 \cdot 2\text{NH}_3$ is known as 'inorganic benzene'
 (b) Boric acid is a protonic acid
 (c) Beryllium exhibits coordination number of six
 (d) Chlorides of both beryllium and aluminium have bridged chloride structures in solid phase
13. In the reaction

$$2\text{X} + \text{B}_2\text{H}_6 \longrightarrow [\text{BH}_2(\text{X})_2]^+ [\text{BH}_4]^-$$
 the amine(s) X is (are):
 (a) NH_3 (b) CH_3NH_2
 (c) $(\text{CH}_3)_2\text{NH}$ (d) $(\text{CH}_3)_3\text{N}$

ANSWER KEYS

ONE CORRECT OPTION TYPE QUESTIONS

- | | | | | | | | | |
|----------|---------|---------|---------|---------|---------|---------|-----------|-----------|
| 1. (b) | 2. (c) | 3. (d) | 4. (b) | 5. (c) | 6. (a) | 7. (c) | 8. (c) | 9. (d) |
| 10. (ac) | 11. (a) | 12. (b) | 13. (b) | 14. (c) | 15. (c) | 16. (b) | 17. (d) | 18. (abc) |
| 19. (c) | 20. (a) | 21. (c) | 22. (a) | 23. (a) | 24. (a) | 25. (a) | 26. (abc) | 27. (d) |
| 28. (a) | 29. (a) | 30. (d) | 31. (b) | 32. (a) | 33. (b) | 34. (c) | 35. (a) | 36. (a) |
| 37. (d) | 38. (b) | 39. (c) | 40. (a) | 41. (d) | 42. (a) | 43. (d) | 44. (c) | 45. (b) |
| 46. (b) | 47. (a) | 48. (b) | 49. (c) | 50. (d) | 51. (c) | | | |

COMPREHENSION BASED QUESTIONS

Comprehension-I

1. (b) 2. (c) 3. (d) 4. (d) 5. (a) 6. (d)

Comprehension-II

1. (a) 2. (b) 3. (c) 4. (b) 5. (d)

Comprehension-III

1. (c) 2. (d) 3. (bc) 4. (ab) 5. (a, b, c)

Comprehension-IV

1. (c) 2. (c) 3. (b) 4. (d) 5. (a)

ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

1. (a, b) 2. (b, c) 3. (b, c) 4. (c, d) 5. (a, b, c) 6. (a, b, c) 7. (a, c) 8. (a, b, c, d)
9. (b, c) 10. (a, b, c) 11. (a, b, c, d) 12. (a, b, c, d) 13. (a, b, d)

MATCH THE COLUMN TYPE QUESTIONS

1. $A \rightarrow S$; $B \rightarrow R$; $C \rightarrow P$; $D \rightarrow Q$

ASSERTION/REASON TYPE QUESTIONS

1. (a) 2. (c) 3. (a) 4. (c) 5. (c) 6. (c) 7. (c) 8. (c) 9. (c) 10. (a)

RECENT YEARS' QUESTION OF IIT-JEE AND AIEEE

1. (c) 2. (b) 3. (c) 4. (c) 5. (b) 6. (a) 7. (a) 8. (b) 9. (c) 10. (a)
11. (a) 12. (d) 13. (a, b, c)

CHAPTER 5

Group 14- Carbon Family

INTRODUCTION

	<i>Atomic number</i>	<i>Electronic configuration</i>	<i>Atomic radius (Å)</i>	<i>Ionic M^{2+}</i>	<i>Radius(Å) M^{4+}</i>	<i>M.p. °C</i>	<i>B.p. °C</i>	<i>I.P (ev)</i>
C	6	$1s^2 2s^2 2p^2$	0.77					11.2
Si	14	$-2s^2 2p^6 3s^2 3p^2$	1.17		0.41	1410	2680	8.1
Ge	32	$-3s^2 3p^6 3d^{10} 4s^2 4p^2$	1.22	0.93	0.53	937	2830	7.8
Sn	50	$4s^2 4p^6 4d^{10} 5s^2 5p^2$	1.41	1.12	0.71	232	2687	7.3
Pb	82	$-5s^2 5p^6 5d^{10} 6s^2 6p^2$	1.54	1.20	0.84	327	1751	7.4

All these elements exhibit a group valency of four, but because an enormous amount of energy is needed to remove for electrons from their atoms, they form compound which are predominantly covalent. Similarly, the gain of four electrons to give the 4-valent anion is energetically impossible.

Germanium, tin and lead form 2-valent compounds in which the two s electrons are inert (inert pair effect). The stability of this state relative to the 4-valent state increases steadily from germanium to lead, i.e., 2-valent state, while for lead this is the predominant valency state. 4-valent state while for lead this is the predominant valency state. 2-valent compounds of tin and lead are often predominantly ionic.

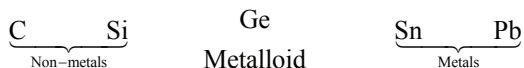
Carbon is non-metallic and so too is silicon. Germanium has properties of both metals and non-metals (it is a metalloid), while the elements tin and lead are definitely metallic. There is a smooth transition from non-metallic to metallic properties on passing down the series silicon, germanium, tin and lead, but the first member carbon differs considerably from silicon. The chemistry of silicon is very similar to that of boron, its diagonal neighbour in the periodic table.

The chemistry of carbon is dominated by its tendency to form chains and rings of carbon atoms in which other atoms, particularly hydrogen, play an important part. The chemistry of silicon is completely different and here, the important feature is the formation of silicon-oxygen bonds which are present in the giant molecule silica, and in the polymeric silicate anions.

5.2 Group 14-Carbon Family

General Properties

- Non-metallic and Metallic character:



The change from non-metal to metal with increasing atomic number is clearly observed in Group 14, where C and Si are non-metals. Ge has some metallic properties, and Sn and Pb are metals. The increase in metallic character shows itself in the structures and appearance of the elements, in physical properties such as malleability and electrical conductivity, and in chemical properties such as the increased tendency to form M^{2+} ions and the acidic or basic properties of the oxides and hydroxides.

- Catenation:** The linking of identical atoms with each other to form long chains is called catenation. All the elements of this group have the property of catenation. However, this property decreases from carbon to lead. Unlike the first element of group 15, 16 or 17, there is no lone pair-lone pair repulsion in the single bond. Thus, carbon has the maximum property of catenation, silicon has much lesser tendency, germanium has still lesser tendency whereas tin and lead hardly show this property. The decrease of this property is associated with $M-M$ bond energy which decreases from carbon to lead.

C—C	83 kcal/mol
Si—Si	54 kcal/mol
Ge—Ge	40 kcal/mol
Sn—Sn	37 kcal/mol

The reason for greater tendency of carbon for catenation than other elements may further be explained by the fact that C—C bond energy is approximately of the same value as the energies of bonds between carbon and other elements. On the other hand, Si—Si bond is weaker than the bonds between silicon and other elements.

C—C	83 kcal/mol	Si—Si	54 kcal/mol
C—O	86 kcal/mol	Si—O	88 kcal/mol
C—Cl	81 kcal/mol	Si—C	186 kcal/mol

Thus, carbon forms a number of compounds in which a large number of carbon atoms are linked together in the form of straight chains, branched chains or closed rings. The property of catenation is responsible for a very large number of compounds of carbon.

- Ionization potential:** The values of ionization energy (IE_1) decreases on moving down the group, though not in a regular order.

Element	C	Si	Ge	Sn	Pb
Ionization Energy (I.E.) (kJ mol ⁻¹)	1086	786	760	707	715



The ionization energy of group 14 elements decreases steadily on moving down the group carbon to tin after that there is a slight increase in value of first ionization energy of lead. The decrease in ionization energy down the group is not regular. From carbon to silicon it is very sharp but the decrease from silicon to germanium is not much.

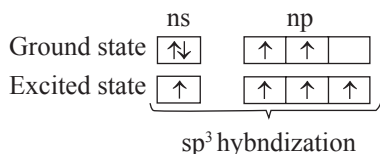
Explanation: In general, the decrease in ionization energy down a group is due to an increased atomic size and screening effect of inner electrons. This results in a decrease of the attractive influence of the nucleus on the outer electrons. As such the release of the electron becomes easier leading to a corresponding decrease in the value of ionization energies. Due to poor shielding effect of d-electrons in Ge and Sn the ionization energies of these elements are somewhat more than the expected values. The ionization energies of Pb are more than those of corresponding values of Sn due to very poor shielding of the valence shell electrons by the electrons present in d and f-subshells.

- Ionization energies (IE_1 , IE_2 , IE_3 and IE_4) of tin is less than Ge and Pb. Thus, the order of ionization energies is,



■ Valency, Oxidation State

Valency: All these elements show a covalency of 4. The tetravalency is explained on the basis of the shifting of one of the paired ns electrons to the vacant np orbital in the same shell.



These four half filled orbitals mix together to give four sp^3 hybrid orbitals resulting in a tetrahedral structure and each having one unpaired electron.

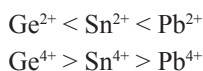
Inert Pair Effect

The inert pair effect shows itself increasingly in the heavier members of the group. There is a decrease in stability of the (+IV) oxidation state and an increase in the stability of the (+II) state on descending the group. Ge(+II) is a strong reducing agent whereas, Ge(+IV) is stable. Sn(+II) exists as simple ions which are strongly reducing but Sn(+IV) is covalent and stable. Pb(+II) is ionic, stable and more common than Pb(+IV), which is oxidizing. The lower valencies are more ionic because the radius of M^{2+} is greater than that of M^{4+} and according to Fajans' rules, the smaller the ion the greater the tendency to covalency.

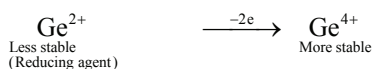
Table 5.1 Standard Reduction Potentials (Volts)

<i>Acid solution</i>			<i>Basic solution</i>		
Oxidation state					
+IV	+II	0	+IV	+II	0
Sn^{4+}	$\xrightarrow{0.15} \text{Sn}^{2+}$	$\xrightarrow{-0.14} \text{Sn}$	$[\text{Sn}(\text{OH})_6]^{2-} \xrightarrow{-0.90} \text{HSnO}_2^- \xrightarrow{-0.91} \text{Sn}$		
PbO_2	$\xrightarrow{+1.46} \text{Pb}^{2+}$	$\xrightarrow{-0.13} \text{Pb}$	PbO_2	$\xrightarrow{+0.28} \text{PbO}$	$\xrightarrow{-0.54} \text{Pb}$

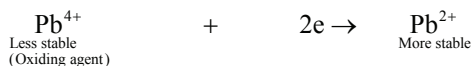
Thus, the stability of these ions follow the following order:



The compounds of Ge^{2+} are unstable while compounds of Ge^{4+} are stable. Thus, the compounds of Ge^{2+} get oxidized into Ge^{4+} easily and act as reducing agents.



The compounds of Sn^{2+} are less stable than Sn^{4+} . Thus, Sn^{2+} compounds also act as reducing agents. In the case of lead, Pb^{2+} compounds are more stable than Pb^{4+} compounds. The Pb^{4+} compounds, thus, act as oxidizing agents.



Since, M^{4+} ions are smaller than M^{2+} ions, in general, M^{4+} compounds behave as covalent due to high polarization (small size and high charge) and M^{2+} compounds are ionic.

Worked-out Example

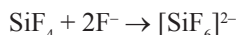
1. Ge^{2+} is strong reducing agent while Pb^{4+} is strong oxidizing agent, why?

Ans. Ge^{2+} is less stable than Ge^{4+} thus, it is readily converted into Ge^{4+} when brought in contact of oxidizing agent. On the other hand, Pb^{4+} is less stable than Pb^{2+} , so Pb^{4+} behaves as oxidizing agent. Stabilities of $\text{Ge}^{2+}/\text{Ge}^{4+}$ and $\text{Pb}^{2+}/\text{Pb}^{4+}$ pairs can be explained on the basis of inert pair effect.

Extention of covalency

Maximum covalency of carbon is four because it has no d-orbitals or vacant orbitals which can be used to accommodate more electrons.

The remaining elements, however, have vacant d-orbitals. These permit the formation of coordinate bonds with other atoms or ions having lone pairs of electrons. For example, SiF_4 can combine with 2F^- ions.



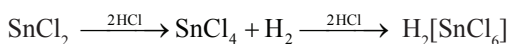
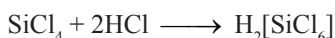
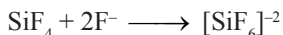
Thus, the maximum covalency of silicon can be 6. Similarly, other members can show maximum covalency of six involving sp^3d^2 hybridization.

Since all these elements except carbon satisfy the following three conditions for the formation of complexes, they have the tendency to form complexes using d-orbitals.

- (i) small size of the atom or ion
- (ii) high charge
- (iii) availability of vacant orbitals of appropriate energy

Due to high effective nuclear charge.

$\text{C} \rightarrow$ no extention possible



All di, tetra halides are Lewis acids and the acidic strength decreases down the gp.

Worked-out Examples

2. Carbon atoms are tetravalent inspite of the fact that there are only two unpaired electron in 2p-orbitals.

Ans. 2s electron pair becomes unpaired by excitation of 2s electron to 2p empty orbital giving rise to four, unpaired electron in excited state.

3. PbBr_4 and PbI_4 does not exists. Explain.

Ans. The non-existence nature of PbBr_4 and PbI_4 is due to the fact that Pb^{4+} ion is a strong oxidant while Br^- and I^- are strong reductants. Thus, Pb^{4+} ion cannot survive in presence of Br^- or I^-

4. Which of the following halides is least stable and has doubtful existence?

(IIT 1996)

- (a) Cl_4 (b) GeI_4 (c) SnI_4 (d) PbI_4

Ans. (d)

5. Compounds that formally contain Pb^{4+} are easily reduced to Pb^{2+} . The stability of the lower oxidation state is due to _____. (IIT 1997)

Ans. inert-pair effect.

6. Statement-1: Pb^{4+} compounds are stronger oxidizing agents than Sn^{4+} compounds.

and

Statement-2: The higher oxidation states for the group 14 elements are more stable for the heavier members of the group due to 'inert pair effect'. (IIT 2008)

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
 (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1
 (c) Statement-1 is True, Statement-2 is False
 (d) Statement-1 is False, Statement-2 is True

Ans. (c)

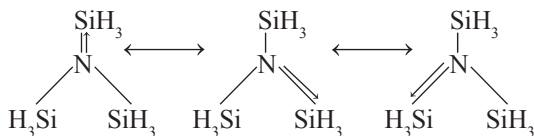
True/False Type

7. The tendency for catenation is much higher for C than for Si. (IIT 1993)

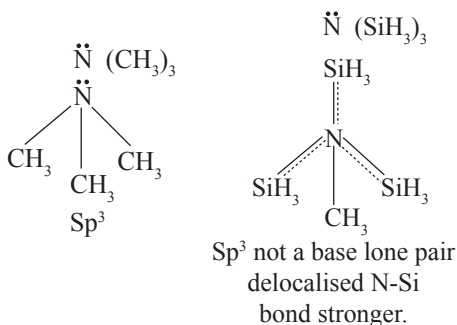
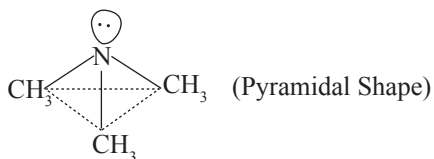
Ans. True

Multiple bond:

Structure of Trimethyl ammonia and trisilyl ammonia: Trimethyl ammonia is pyramidal whereas trisilyl ammonia is a planar molecule. In silicon atom, there is vacant d-orbital hence, electron pair of nitrogen is donated to silicon by $p\pi-d\pi$ overlapping in trisilyl ammonia. Thus, trisilyl ammonia has planar structure.



Trimethyl ammonia has pyramidal shape due to presence of lone pair of electron at nitrogen.



Allotropic Forms:

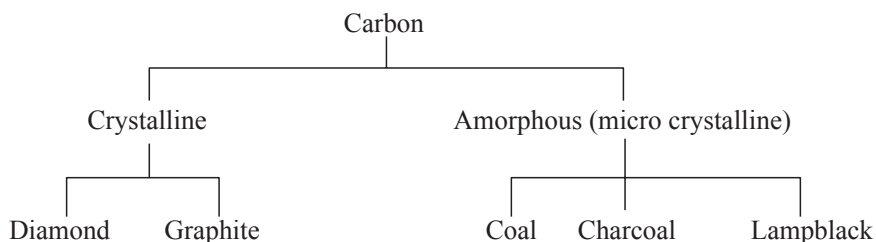
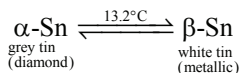
Existence of a substance in various physical forms but same chemical forms is k/as allotropy.

Note: Except Pb all elements of this group show allotropy.

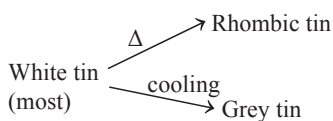
- Carbon exists in 2 crystalline allotropic forms. These are diamond and graphite, the rare hexagonal form of diamond, and two others produced by very high temperature or radiation treatment, which are thought to contain some triple bonds.

5.6 Group 14-Carbon Family

Si, Ge and Sn also exists as a metallic form. Pb exists only in the metallic form. Ge is unusual because the liquid expands when it forms the solid. This property is unique to Ga, Ge and Bi.



- Sn has 3 allotropes



Tin plague or tin disease

On cooling white tin crumbles to powdered form and changes to grey tin. This process is known as tin plague. Tin cannot be used as packaging material in cold countries. On heating, it changes to another allotrope rhombic tin.

- Tin is the only element which has maximum number of isotopes.

Worked-out Example

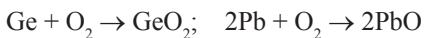
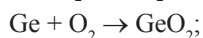
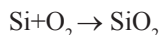
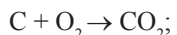
8. Tin vessels are not used for packing in cold countries. Explain.

Ans. Tin at low temperature converts to grey tin which is very brittle and easily crumbles down to powder and thus tin vessels are not used for packing in cold countries.

Chemical Properties

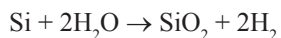
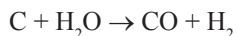
Reactivity: The elements of this group are relatively less reactive but reactivity increases down the group. Lead often appears more unreactive than expected. This is partly due to surface coating of the oxide and partly due to the high over-potential for the reduction of H^+ to H_2 at a lead surface.

- **Action of air:** These elements react with the oxygen of the air on strong heating when their oxides are formed.



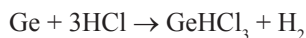
Mono-oxides CO, SiO, GeO are also known.

- **Action of water:** C, Si and Ge are unaffected by water. However, on red heating, these elements except lead decompose steam.

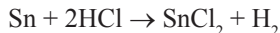


Lead is unaffected by water due to protective oxide film.

- **Action of acids:** Non-oxidizing acids do not attack carbon and silicon. Germanium is not attacked by dilute HCl. However, when metal is heated in a stream of HCl gas, germanium chloroform is formed.

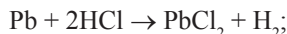


Tin dissolves slowly in dilute HCl but readily in concentrated HCl.



Lead also dissolves in dilute HCl.

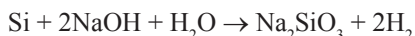
Lead dissolves in concentrated HCl forming chloro-plumbous acid, but the reaction stops after sometime due to deposition of PbCl_2 .



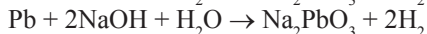
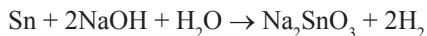
Concentrated aqueous solution of H_2SO_4 and HNO_3 dissolve Ge, Sn and Pb. Sn reacts with inorganic mineral acids only but Pb reacts with organic and inorganic acids.

Sn reacts with mineral not organic acids. This property is required for tinning utensils made up of Cu.

Action of alkalis: Carbon is unaffected by alkalis. Silicon reacts slowly with cold aqueous NaOH and readily with hot solution giving solution of silicate.



Sn and Pb are slowly attacked by cold alkali but readily by hot alkali giving stannates and plumbates. Thus, Sn and Pb are somewhat amphoteric.



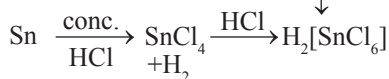
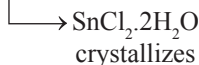
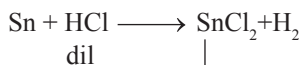
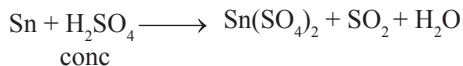
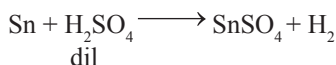
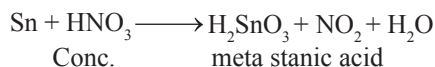
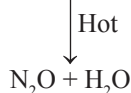
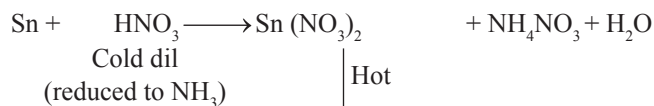
Action with Acid

C, Si – Do not react $\longrightarrow$ Can react with oxidizing acids

Rest of the elements react with acid as well as base to displace H_2 gas.

- Sn reacts with mineral acids to displace H_2 except HNO_3

Sn reacts with mineral acids to displace H_2 except HNO_3



Worked-out Example

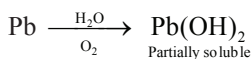
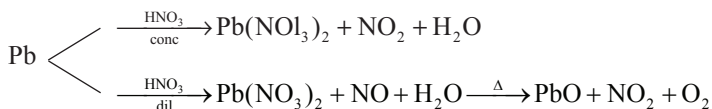
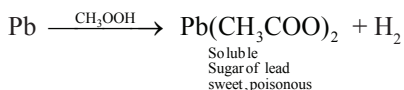
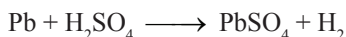
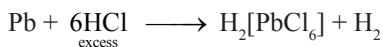
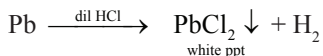
9. A piece of tin foil is added to SnCl_2 solution for preserving it. Explain.

Ans. SnCl_2 in solution state undergoes the hydrolysis.



Presence of tin foils reacts with HCl to reverse the reaction or prevent the hydrolysis of SnCl_2 .

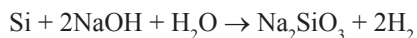
- Pb – reacts with organic, mineral acids and base.



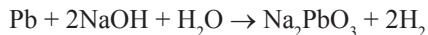
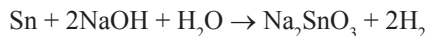
In presence of moisture, Pb reacts with air to form Pb(OH)_2 —toxic

This is called plumbo solvency. It increases in presence of NH_4^+ , NO_3^- acid—(organic).

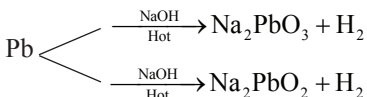
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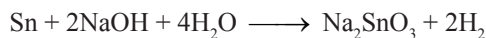
- Pb pipes are not suitable for soft water. Before passing soft water through Pb pipes, hard water should be passed to avoid plumbo solvency.
- Reason: Formation of insoluble SO_4^{2-} , CO_3^{2-}



Note: Pb – soft metal, gives black mark on paper like graphite

- Graphite is called plumbago. graphite pencils—called lead pencil.

- Sn upon reaction with NaOH forms sodium stannite and with conc. HNO_3 forms metastannic acid.



Worked-out Examples

10. Lead pencil contains

- (a) Pb (b) FeS (c) Graphite (d) PbS

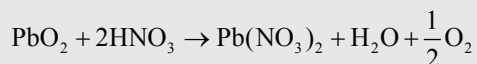
Ans. (c)

11. Which gas is evolved when PbO_2 is treated with concentrated HNO_3 ?

(IIT 2005)

- (a) NO_2 (b) O_2 (c) N_2 (d) N_2O

Ans. (b) The reaction is



12. When PbO_2 reacts with a dilute acid, it gives hydrogen peroxide. (True/False)

(IIT 1982)

Ans. False PbO_2 is a dioxide and not peroxide. (False)

Hydrides: All the members of this group form covalent hydrides of the type MH_4 . Besides MC_4 , carbon forms a large number of hydrides, saturated as well as unsaturated, whereas rest of the elements form only saturated hydrides of formula $\text{M}_n\text{H}_{2n+2}$

Si	$n = 1 - 7$	↓	Silane	
Ge	$n = 1-5$		germane	catenation
Sn	$n = 1, 2$		stannane	decreases
Pb	$n = 1$		Plumbane	

- Thermal stability of hydride decreases down the group and reducing character increases.
- Reactivity towards H_2O (hydrolysis) maximum for silane, after that decreases, least for alkane.



The hydrides of MH_4 type are gaseous and their thermal stability decreases and consequently the reducing nature increases from top to bottom. However, among the hydrides, silicon hydrides are least stable to hydrolysis.

Hydrides	CH_4	SiH_4	GeH_4	SnH_4	PbH_4
Decomposition temp. ($^\circ\text{C}$)	800	450	285	150	0

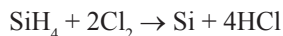
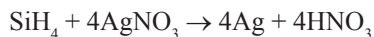
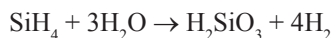
[The hydrides of silicon (silanes) are much more reactive than the alkanes. The difference in behaviour is attributed to various factors:

- (i) Electronegativity values are: C = 2.5, Si = 1.8 and H = 2.1. Thus, the bonding electrons between C and H or Si and H are not equally shared and silicon is vulnerable to attack by nucleophilic reagents.

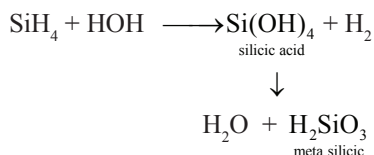


5.10 Group14-Carbon Family

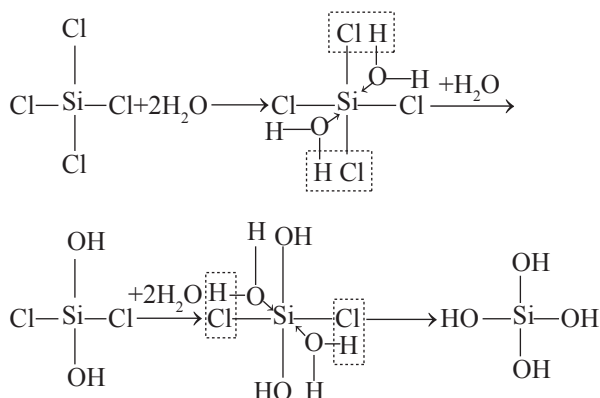
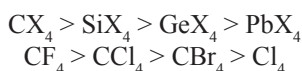
- (ii) The larger size of Si is easier to be attacked.
 (iii) Si has low energy d-orbitals which may be used to form an intermediate compound and thus lower the activation energy of the process].



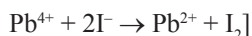
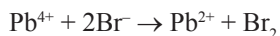
The low stability of GeH_4 , SnH_4 and PbH_4 is due to weak M—H bond. It is due to large difference in the size of M and hydrogen atom leading to poor overlapping and weak covalent bond.



Halides: The members of this group form tetrahalides of the type MX_4 except PbBr_4 and PbI_4 . The halides are covalent and formed by sp^3 hybridization. The thermal stability of halides of different elements with a common halogen decreases with increasing atomic number. The thermal stability of tetrahalides of the same element decreases with increase in molecular mass of the tetrahalide.



[Non-existence of PbBr_4 and PbI_4 can be explained on the basis of strong oxidizing nature of Pb^{4+} . The ions Br^- and I^- are reducing agents, i.e., in presence of these ions, Pb^{4+} ions are reduced to Pb^{2+} ions.



Except carbon halides, other halides are readily hydrolysed by water. The trend towards hydrolysis, however, decreases down the group. The hydrolysis is due to utilization of d-orbitals to which water molecules can get attached.

The tetrahalides of Si, Ge, Sn and Pb can form hexahalo complexes like $[\text{SiF}_6]^{2-}$, $[\text{GeF}_6]^{2-}$, $[\text{GeCl}_6]^{2-}$, $[\text{SnCl}_6]^{2-}$, $[\text{PbCl}_6]^{2-}$ with the corresponding halide ions. Thus, tetrahalides of Si, Ge, Sn and Pb act as strong Lewis acids.

■ Oxides

Carbon forms gaseous oxide but rest of the elements form solid oxides. It is because carbon can satisfy tetravalency by forming multiple bond with oxygen and exist as a discrete molecule with weak Van der Waals forces b/w molecules. Rest of the elements will form maximum single bonds and exist as polymeric solid.

- All the elements of this group form oxides of the type MO_2 .



- In all its oxide carbon is sp hybridized.



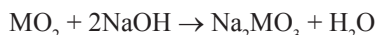
CO_2 is acidic, but CO , C_3O_2 are neutral, SiO , SiO_2 are acidic, GeO , GeO_2 are acidic

SnO , SnO_2 are amphoteric, PbO , PbO_2 , Pb_3O_4 ($2\text{PbO} + \text{PbO}_2$) are amphoteric

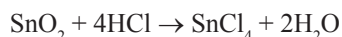
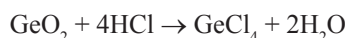
Red lead

more basic

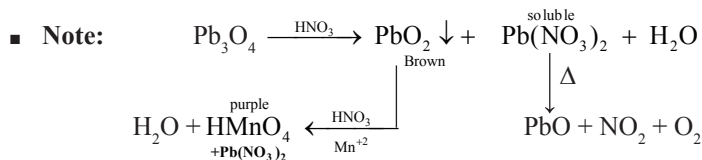
- The acidic nature decreases with increase of atomic number. CO_2 and SiO_2 are acidic while GeO_2 , SnO_2 and PbO_2 are amphoteric. All these dissolve in alkalis giving carbonates, silicates, germanates, stannates and plumbates, respectively.



The dioxides of Ge, Sn and Pb also dissolve in acids.



- They also form oxides of the type MO . SiO is unstable. CO is neutral while SnO and PbO are amphoteric. Basic character of both mono and dioxide increases down the group.



- PbO_2 is used in lead storage battery.
- SiO_2 , GeO_2 , SnO_2 and PbO_2 are all solids.
- CO_2 and SiO_2 is acidic, GeO_2 is weakly acidic while SnO_2 and PbO_2 are amphoteric in nature.
- All the elements of group 14 except silicon form monoxides e.g., CO , GeO , SnO and PbO . Out of these monoxides only CO is neutral, while all other monoxides are basic.

CARBON

Occurrence and Use

Carbon black is produced in large amounts. It is made by the incomplete combustion of hydrocarbons from natural gas or oil. The particle size is very small. Over 90% is used in the rubber industry to make car tyres. Its other main use is in newspaper ink. Coke is produced in very large amounts. It is produced by high temperature carbonization of coal. The coal is heated in large ovens in the absence of air. Coke is extremely important in the metallurgical extraction of iron and many other metals. The distillation of coal also provides a valuable source of organic chemicals.

This is usually found as a mixture with mica, quartz and silicates, which contains 10-60% C. Graphite is separated from most of the impurities by flotation. Finally, it is purified by heating with HCl and HF in a vacuum to remove the last traces of silicon compounds as SiF_4 . Sedimentary deposits of carbon are mined in Mexico. This was once thought to be amorphous carbon, but is now regarded as microcrystalline (very finely divided) graphite. Nearly as much graphite is made synthetically as is mined.



Carbon

Some of the planets are thought to be surrounded by a layer of carbon dioxide, and it is probable that all the deposits of carbon compound in the atmospheric carbon earth originally came from carbon dioxide in the atmosphere carbon dioxide for growth and animals feed on plants. Atmospheric carbon dioxide for plants. The decomposition of dead animals and vegetation, followed by geological upheavals which subjected the decaying matter to great temperatures and pressures for long periods of time in the absence of oxygen, almost certain!; accounts for coal, oil and natural gas. The carbonate minerals were form by the sedimentation of the shells of microscopic sea animals, followed by compression of the layers and the redistribution of the oceans. Pure carbon has two allotropes and both of these—diamond and graphite found naturally.

The percentage of carbon dioxide in the atmosphere remains constant at a figure of about 0-0.3 per cent by volume by the operation of a carbon cycle in nature.

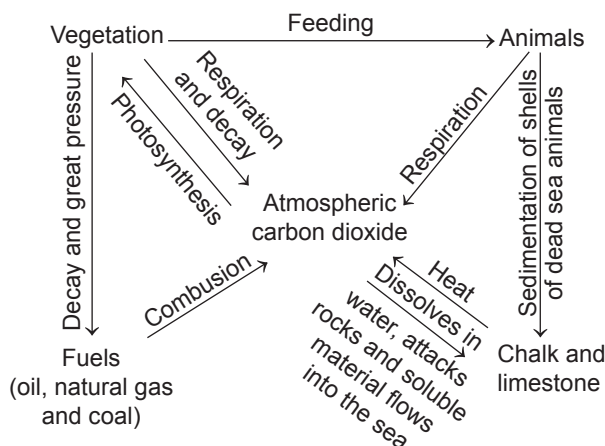


Fig. 5.1 The carbon cycle

Graphite is used for making electrodes, in steel making and metal foundries, for crucibles, as a lubricant, and in pencils, brake linings and brushes for electric motors. It is also used as the moderator in the cores of gas cooled nuclear reactors, where it slows down neutrons.

Activated charcoal is made by heating or chemically oxidizing sawdust or peat. Active carbon has an enormous surface area, and is used to purify and decolourize sugar and other chemicals. It is also used to absorb poisonous gases in gas masks, in filter beds at sewage plants and as a catalyst for some reactions.

About 30% of diamond are used as gemstones, and 70% are used for industrial purposes, mainly for making drills, or as an abrasive powder for cutting and polishing, as diamond is very hard. It is economic to make small industrial quality diamonds synthetically, by high temperature and pressure treatment of graphite.

Carbon exists in two allotropic forms

(a) Crystalline and (b) amorphous.

The crystalline forms are diamond and graphite while the amorphous forms are coal, charcoal, lamp black, etc.

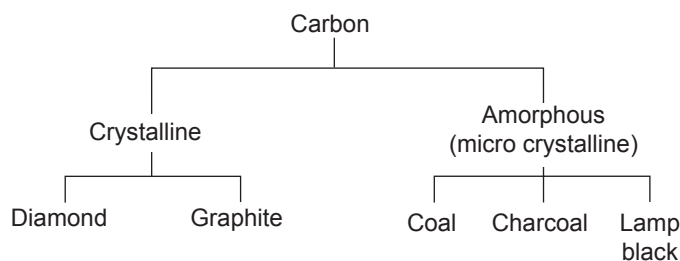
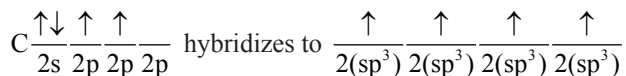


Fig. 5.2

- (a) **Diamond:** Diamond is extremely unreactive, and in contrast graphite is quite reactive. Diamonds are typically colourless, though industrial diamonds are often black. Most naturally occurring diamonds contain a trace of nitrogen, but 'blue diamonds' contain a trace of Al instead. Hybridization of carbon 2s orbital and the three 2p orbitals gives four equivalent sp^3 orbitals each of which contains one unpaired electron. If all the carbons are assumed to have this hybridization then overlap of these hybrid orbitals of neighbouring



carbon atoms will lead to a non-ending tetrahedral structure. This is what obtains in diamond. Diamond thus has covalent bonds but these bonds do not end to produce discrete 'molecules', instead three-dimensional 'giant' 'molecules' are formed. Diamond is therefore an extremely hard and high-melting substance. The C — C bonds (length 1.54 Å) (154 pm) are extraordinarily stable, the non-volatility (sublimes at 3500°C) reflecting the difficulty of breaking these bonds. Thus the melting point is abnormally high (about 3930°C) and the structure is very hard (see Figure). (In a rare modification of diamond, the tetrahedra are arranged differently to give a wurtzite-like structure and a hexagonal unit cell.)

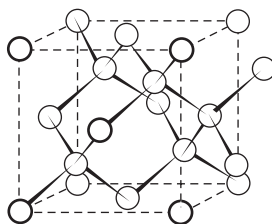
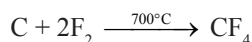
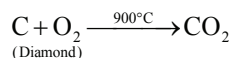


Fig. 5.3

- It is the purest form of carbon.
- It is found naturally as well as obtained artificially.
- It is the hardest natural substance known.
- It is transparent and has a specific gravity 3.52.
- It is a bad conductor of heat and electricity. It is transparent to X-rays and glows in ultraviolet rays with bluish green colour.
- Its refractive index is high (2.45) and when properly cut, it produces maximum total internal reflection which is responsible for its brilliance.
- It is chemically inert. It is not attacked by acids, alkalies and salts. It burns in air on heating at 900°C to form CO_2 . It reacts with fluorine at 700°C to form CF_4 .

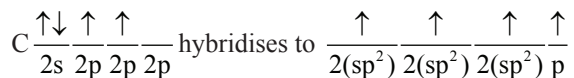


The important properties of diamond are related to its structure. In diamond, each carbon atom is in sp^3 hybridized state and linked to four other carbon atoms tetrahedrally by covalent bonds. This gives a giant three dimensional polymeric structure in which C—C distance is 1.54 Å and bond angle is 109.5° .

As the atoms are held firmly by strong covalent bonds, diamond is the hardest substance, possesses very high melting point (3600°C) and chemically inert. Since there is no mobile electron present, diamond is non-conductor of electricity.

Diamond is used as a gem stone on account of reflection and refraction of light. Impure diamonds (black) are used in knives for cutting glass and rock borers.

- (b) **Graphite:** An alternative hybridization of the 2s and two 2p orbitals gives three equivalent sp^2 hybrid orbitals at 120° to each other on the same plane. The third 2p orbital is perpendicular to the plane of the sp^2 orbitals. If all carbons are assumed to have this



hybridization, and if each carbon overlaps with three neighbouring carbon atoms we have a non-ending two dimensional hexagonal planar structure. As there are four valence electrons in each carbon atom, after forming three C—C bonds, each carbon atom is left with one spare electron in its p-orbital. This electron then overlaps with each other to form a π -bond. Hence, the C—C distance in graphite is shorter ($1.42\text{\AA}$) than that of diamond ($1.54\text{\AA}$). This is equivalent to a resonating alternate double bond on the valence bond model of graphite and their layers of graphite rub off as we write.

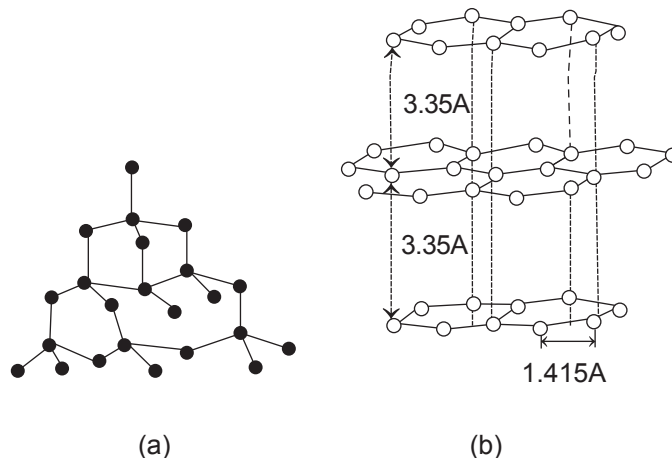


Fig. 5.4 Models of (a) diamond and (b) graphite

The layers are held together by relatively weak van der Waals forces. In α -graphite the layers are arranged in the sequence ABAB. . . with the third layer exactly above the first layer. In β -graphite the order of layers is ABCABC. . . The two forms are interconvertible. Heating turns β into α , and grinding turns α into β . In both forms the C—C bond lengths within a sheet are $1.41\text{ \AA}$ (similar to the C—C distance of $1.40\text{ \AA}$ in benzene). The distance between layers is $3.35\text{ \AA}$. This interlayer distance is large—appreciably more than twice the covalent radius of carbon ($2 \times 1.54\text{ \AA} = 3.08\text{ \AA}$). Thus bonding between layers is weak. Graphite cleaves easily between the layers, which account for the remarkable softness of the crystals (>1 on Mohs' scale). The ability of the planar layer to slip over one another makes graphite a good lubricant. Our 'lead' pencils are really made by graphite. Graphite is used as a lubricant, either on its own or in graphited oil. In contrast, diamond is hard (10 on Mohs' scale) and has abrasive properties. The wide spacing of sheets in graphite also means that the atoms do not pack together to fill space very effectively. Thus the density of graphite (2.22 g cm^{-3}) is lower than that of diamond (3.51 g cm^{-3}).

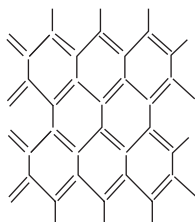


Fig. 5.5 The structure of a graphite sheet

In graphite only three of the valency electrons of each carbon atom are involved in forming σ bonds (using sp^2 hybrid orbitals). The fourth electron forms a π bond. The π electrons are delocalized over the whole sheet, and as they are mobile, graphite conducts electricity. Conduction can occur in a sheet, but not from one sheet to another.

Graphite is thermodynamically more stable than diamond, and its free energy of formation is 1.9 kJ mol^{-1} lower at room temperature and ordinary pressure. Thermodynamically it is favourable for diamonds to turn into graphite. They

do not normally do so because there is a high energy of activation for the process. If this activation energy is available, the change does occur, and diamond tipped drills do burn out and form graphite if they get too hot. The reverse process is not thermodynamically possible, and it requires very forcing high energy conditions to convert graphite to diamond. Graphite can be converted to synthetic diamonds at 1600°C by a pressure of 50000-60000 atmospheres.

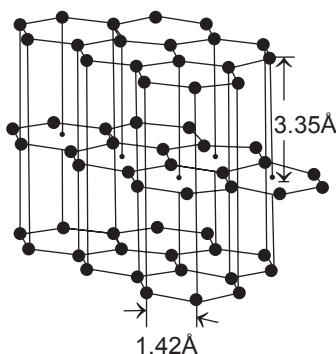
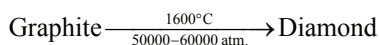


Fig. 5.6 Structure of graphite

When graphite is heated with vapours of K, Rb, Cs at 300°C , it forms C_8M . Graphite is diamagnetic but C_8M is paramagnetic.

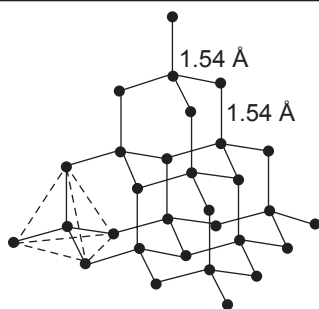
Uses: Graphite is used:

- (i) for lining and making electrodes of electric furnaces.
- (ii) in making refractory crucibles.
- (iii) in making lead pencils.
- (iv) as a moderator in nuclear reactor.
- (v) as a lubricant in machinery.
 - It leaves a black mark on paper and is called black lead or plumbago.
 - It is chemically more active than diamond. It ignites in air or oxygen at 700°C to form CO_2 . It is not attacked by alkalis and dilute acids. However, when treated with concentrated HNO_3 or concentrated H_2SO_4 , it is oxidized to insoluble yellowish green substance known as graphitic acid, $\text{C}_{11}\text{H}_4\text{O}_5$. With alkaline potassium permanganate, it is oxidised to mellitic acid $[\text{C}_6(\text{COOH})_6]$ and oxalic acid. It forms CO_2 with chromic acid. Carbon can occur both in very hard and very soft forms.
 - Diamond is insoluble in any solvent and on heating to $1800\text{--}2000^{\circ}\text{C}$ in the absence of air gives graphite.
 - Artificial graphite is made by Acheson process.
 - Industrial lubricant, oilclay is a suspension of graphite in oil and colloidal solution of graphite is called aquadag.
 - Anthracite is the purest form of coal while lamp black is the softest form.
 - Carbon in group 14 elements has maximum tendency to form multiple bonds.
 - Graphite on oxidation with conc. HNO_3 gives graphitic acid ($\text{C}_{11}\text{H}_4\text{O}_5$).
 - Graphite on oxidation with alkaline KMnO_4 gives mellitic acid or benzene hexacarboxylic acid $[\text{C}_6(\text{COOH})_6]$ and oxalic acid.

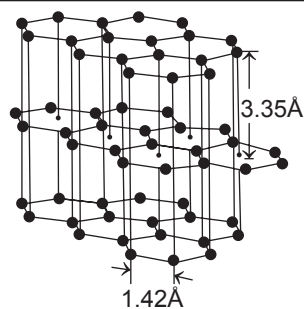
Table 5.2 Difference Between Diamond and Graphite

Diamond	Graphite
1. In diamond, each carbon atom is in sp^3 hybridized state and linked to four other carbon atoms to tetrahedrally by covalent bonds. This gives a giant three dimensional polymeric structure in which C—C distance is $1.54 \text{ \AA}$ and bond angle is 109.5° .	1. In graphite, each carbon atom is in sp^2 hybridized state and is linked to three other carbon atoms in a hexagonal planar structure. The fourth electron present in an unhybridized p-orbital of each carbon atom overlaps with each other to form a π bond. Thus, it has two dimensional sheet structure. In which C—C distance is $1.42 \text{ \AA}$.

(Contd)

Diamond

- Density of diamond is 3.514 g cm^{-3}
- Diamond is hardest natural substance
- Diamond is bad conductor of electricity
- Extremely chemically unreactive
- Its refractive index is high (2.45) and when properly cut, it produces maximum total internal reflection which is responsible for its brilliance.
- It is used as a gem stone on account of reflection and refraction of light. Impure diamonds (black) are used in knives for cutting glass and rock borers.

Graphite

- Density of graphite is 2.226 g cm^{-3}
- Graphite is soft and has lubricating properties
- It is good conductor electricity
- Chemically quite reactive.
- It has loosely bound pi-electrons which can be easily be excited by photons. Hence, it absorbs the light and appears black. It leaves a black mark on paper and is called black lead or plumbago.
- It is used for lining and making electrodes of electric furnaces, in making lead pencils as moderator in nuclear reactor and as a lubricant in machinery.

Fullerenes (Buckyballs): In the two allotropes, diamond and graphite, there are a non-ending (non-finite) number of carbon atoms arranged tetrahedrally and hexagonally respectively. Recently, a strange type of allotrope with finite number of carbon atoms with closed caged structure has been identified. Carbon atom clusters with 28, 32, 50, 60, 70, 84 and 90 atoms have been confirmed, the C_{60} cluster being the most dominant.

The structures of these clusters have been determined to be near-spherical, being close to a football. It was not possible to make a closed-cage structure model of C_{60} cluster with hexagons alone. Ultimately it was possible to have a closed structure with 20 hexagons and 12 pentagons. On top of this model there could be seen 60 vertices representing the 60 carbon atoms. These clusters being a strange new form of carbon have been named after the American architect, Buckminster Fuller, who has invented the geodesic dome. These clusters are fondly called “fullerene” or “buckyball” for short.

The discovery of the fullerenes was made by a group of physicists in 1985 at Rice University in Houston under the leadership of Richard Smalley. This group was studying the formation and structures of atomic clusters and large molecules. They developed a vacuum chamber inside which there was a hollowed out steel block. Different elements were placed in the hollow block and heated with intense laser beam. This vapourized the element which was immediately driven by inert helium jet into another chamber. The vapour was allowed to cool and to form atom clusters. These clusters were transferred to a mass spectrometer for identification of the mass. In the case of carbon the mass spectrometer registered signals for several carbon clusters of which C_{60} was the most abundant.

The original attempt by Smalley and coworkers produced only a microscopic amount of C_{60} cluster. Later studies (1989) by Huffman at Arizona and Kratschmer at Munich gave a better yield, the technique being to pass electric arc between two graphite electrodes in a helium environment.

Back home at the Indian Institute of Science, Bangalore scientists have identified C_{60} and C_{90} clusters by burning benzene and leading the resulting soot by argon jet into a water-cooled copper vessel to collect the fullerenes formed. The soot thus obtained can be dissolved in benzene and the C_{60} and C_{70} clusters separated chromatographically, the ratio of the two clusters being 5 and 1 respectively.

Thermal decomposition of naphthalene in an inert atmosphere of argon around 1000°C also gives these C_{60} and C_{70} clusters.

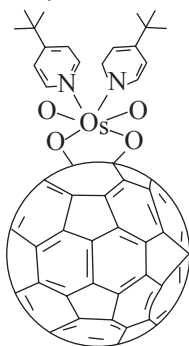
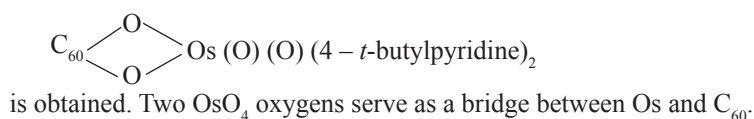
The structure of C_{60} cluster could not be explained by conventional means. A closed football-like structure ultimately emerged from sewing together of 20 hexagons and 12 pentagons.

Crystal structure studies of the C_{60} fullerene by X-ray and electron diffraction techniques in 1990 confirmed the closed-cage structure.

Every 6-membered hexagon ring is surrounded alternately by hexagons and pentagons of carbon. Every carbon pentagon is connected to five carbon hexagons. Such fusion of pentagons and hexagons force the structure to curve. Note that in graphite each hexagon is fused to six hexagons giving a flat structure. Eventually fusion of 60 carbons (C_{60}) gives a dome-like structure curving around itself to a football (soccer ball)-like shape. ^{13}C NMR studies indicate all 60 carbon atoms are equivalent. Although all the atoms are equivalent all the carbon-carbon bonds are not. X-ray studies on C_{60} have revealed that C-C bond length at the fusion of two 6-membered rings is shorter (1.355 Å ; 135 pm) than C-C bond length at the fusion of 5- and 6-membered rings (1.467 Å ; 146.7 pm). A greater degree of pi-bonding is present at the fusion of the 6-membered rings.

C_{70} cluster has the shape of a rugby ball (somewhat prolate structure) while C_{60} has the shape of a dumb-bell, two CGQ units being linked through four carbon-carbon bonds.

Since fullerenes possess extended π -system a number of reaction have been conducted on them. Treatment of C_{60} and C_{70} with Zn a conc. HCl gives $C_{60}\text{H}_{36}$ $C_{70}\text{H}_{36}/C_{70}\text{H}_{38}$. Fluorination of C_{60} by F_2 produces $C_{60}\text{F}_{48}/C_{60}\text{F}_{60}$. On reacting OsO_4 , a good oxidant, can add across the double bond of fullerene. On reacting OsO_4 with C_{60} and 4-*t*-butylpyridine, a complexes of the type:



The complex $C_{60}(\text{OsO}_4)(4-t\text{-butylpyridine})_2$

Smalley has been awarded the Nobel prize for Chemistry in 1996.

Worked-out Examples

13. Graphite is a better lubricant on moon than on the surface of earth. Explain.

Ans. Graphite is a better lubricant on moon because of lack of gravity on moon where friction is already less than the earth.

14. Diamond is the hardest substance but graphite is soft in nature. Explain.

Ans. Diamond possesses very big three dimensional polymeric structure involving each carbon of sp^3 -hybridized nature linked to four other neighbouring carbon atoms which makes it hardest. On the other hand, graphite has layer structure involving each carbon of sp^2 -hybridized nature having wide separation and weak interlayer bonds and thus two adjacent layers can easily slide over each other to produce soft nature. This makes the graphite to show lubricant nature.

15. Diamond is bad conductor of current. Explain.

Ans. All valence electrons of each carbon are used in bond formation with four neighbouring atoms and there is no mobile electron in the system and hence the diamond crystal is bad conductor of current.

16. Diamond is used as precious stone. Explain

Ans. Diamond has a very high refractive index and possesses the ability of reflection and refraction. Moreover, the refraction of different colours of light is not equal and therefore white light passing through it give rise to brilliant play of colours which makes diamond as a valuable gem stone.

17. Diamond (impure) is used as abrasive. Explain.

Ans. Diamond (black) an impure form is the hardest substance known and thus used as abrasive and for cutting glass.

18. C—C bond length in graphite is shorter than C—C bond of diamond. Explain,

Ans. Graphite has sp^2 hybridization and C—C bond involves $sp-sp^2$ hybridized carbon. Diamond has sp^3 hybridization and C—C bond involves sp^3-sp^3 hybridization. Furthermore, more is s character (in sp^2) in hybridized atom, smaller is size of hybridized orbital, more is overlapping, shorter is bond length.

19. Graphite is good conductor of current. Explain.

Ans. Graphite has sp hybridized carbons with layer structure along with n electrons ($p-p$ bond) free to move throughout the entire layers hence graphite is good conductor of current.

20. Diamond is covalent yet its m.pt. is very high. Explain.

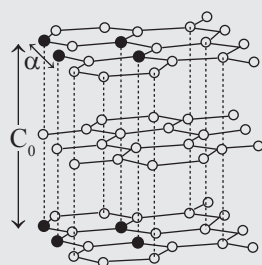
Ans. Diamond has three dimensional polymeric structure involving very strong covalent bonds by which the atoms are held together. Hence diamond possesses high m.pt. inspite of covalent nature.

21. Graphite is thermodynamically more stable than diamond but naturally diamond is not converted into graphite, why?

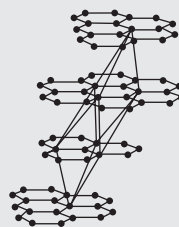
Ans. Because activation energy for the reaction is very high.

22. Compare the structures of α - and β -graphite.

Ans.



α -graphite



β -graphite

- Marks the corners of the unit cell
- α -form is hexagonal form while β form is rhombohedral

23. When graphite is heated at approximately 673 K with vapours of K, Rb or Cs, bronze colour is developed. Explain the chemical changes involved.

Ans. Due to larger difference between layers of graphite. The metal atoms invade the space between sheets forming '**Inter Calation Compounds**' of different composition. In the compounds formed, inter sheet distance increases making electrons more free.



24. One recently discovered allotrope of carbon (e.g., C_{60}) is commonly known as _____. (IIT 1994)

Ans. Fullerene

25. Moderate electrical conductivity is shown by
(a) silica (b) graphite (c) diamond (d) carborundum

Ans. (b)

26. The unit cell of diamond is _____. (IIT 1991)

Ans. Face-centered cube

27. Graphite is a better lubricant on the moon than on the earth. (IIT 1987)

Ans. False, the lubricating properties of graphite depend not only on the slippage between the planes, but also on a film of moisture or gas molecules adsorbed on the surface of the graphite layers. The adsorbed substance decreases the friction as the layers in graphite slide past each other. Dry graphite in vacuum is not slippery.

28. Diamond is harder than graphite. (IIT 1993)

Ans. True, in diamond, all the four valencies of carbon are involved in stronger a bonding whereas in graphite, three are involved in σ bonding whereas the fourth one is involved in weak n bonding.

29. Graphite is used as a solid lubricant. (IIT 1985)

Ans. Graphite has layered-type structure. One layer can easily slide over the other giving graphite a greasy touch.

30. Graphite is a soft solid lubricant extremely difficult to melt. The reason for this anomalous behaviour is that graphite
(a) is an allotropic form of diamond [AIEEE 2003]

(b) has molecules of variable molecular masses like polymers

(c) has carbon atoms arranged in large plates of rings of strongly bound carbon atoms with weak interpolate bonds

(d) is a non-crystalline substance

Ans. (c) In graphite, carbon is sp^2 hybridized. Each carbon is thus linked to three other carbon atoms forming hexagonal rings. Since only three electrons of each carbon are used in making hexagonal ring, fourth electron of each carbon is free to move. This makes graphite a good conductors of heat and electricity. Further, graphite has a two dimensional shell like structure. These various sheets are held together by weak vander Waal's force of attraction, due to these weak forces of attraction, one layer can slip over the other. This makes graphite soft and a good lubricating agent.

Compounds of carbon

1. Oxide

Carbon forms more oxides than the other elements, and these oxides differ from those of the other elements because they contain $p\pi-p\pi$ multiple bonds between C and O. Two of these oxides, CO and CO_2 , are extremely stable and important. Three are less stable: C_3O_2 , C_5O_2 and $C_{12}O_9$. Others which are even less stable include graphite oxide, C_2O and C_2O_3 .

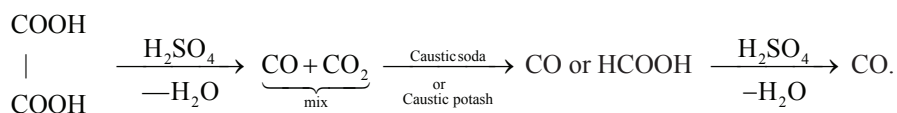
(A) Carbon monoxide

Carbon monoxide CO

CO is a colourless, odourless, poisonous gas. It is formed when C is burned in a limited amount of air.

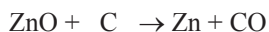
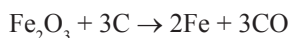
Preparation:

- In the laboratory it is prepared by dehydrating formic acid with concentrated H_2SO_4 or by heating oxalic acid and with conc. H_2SO_4

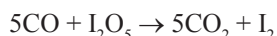
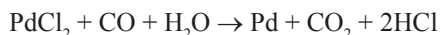


5.20 Group14-Carbon Family

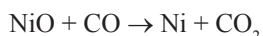
- By the reduction of oxides of heavy metals with carbon.



- When CO_2 is passed over red hot charcoal, the reduction of CO_2 into CO occurs with the formation of a mixture of CO_2 and CO. The gaseous mixture is passed through NaOH or KOH solution to remove CO_2 .
- By heating potassium ferrocyanide with concentrated H_2SO_4 :** In this reaction, dilute H_2SO_4 should not be used because it gives hydrocyanic acid (HCN), an extremely poisonous substance.
 - CO can be detected because it burns with a blue flame. It also reduces an aqueous PdCl_2 solution to metallic Pd, and when passed through a solution of I_2O_5 it liberates I_2 , i.e., it reduces I_2O_5 to I_2 . The latter reaction is used to estimate CO quantitatively. The I_2 is titrated with $\text{Na}_2\text{S}_2\text{O}_3$.



Carbon monoxide is a colourless, odourless gas which burns in air with a blue flame, forming carbon dioxide. It is exceedingly poisonous, combining with the haemoglobin in the blood more readily than oxygen so that normal respiration is impeded very quickly. Ordinary gasmasks are no protection against the gas, since it is not readily absorbed by active charcoal. In the presence of air, a mixture of manganese (IV) oxide and copper (II) oxide catalytically oxidizes it to carbon dioxide, and this mixed catalyst is used in the breathing apparatus worn by rescue teams in mine disasters. Carbon monoxide is a powerful reducing agent, being employed industrially in the extraction of iron and nickel:



It reacts with many transition metals, forming volatile carbonyls. The formation of nickel carbonyl followed by its decomposition is the basis of the Mond process for obtaining very pure nickel:



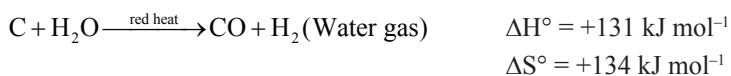
CO is toxic because it forms a complex with haemoglobin in the blood, and this complex is more stable than oxy-haemoglobin. This prevents the haemoglobin in the red blood corpuscles from carrying oxygen round the body. This causes an oxygen deficiency, leading to unconsciousness and then death. CO is sparingly soluble in water and is a neutral oxide. CO is an important fuel, because it evolves a considerable amount of heat when it burns in air.



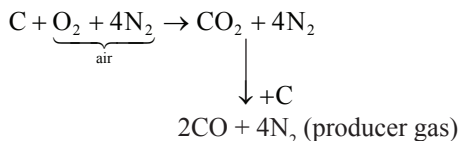
The following are all important industrial fuels:

- Water gas: an equimolecular mixture of CO and H_2 .
- Producer gas: a mixture of CO and N_2 .
- Coal gas: a mixture of CO, H_2 , CH_4 and CO_2 , produced at a gasworks by distilling coal, and stored in large gas holders. This was the 'town gas' supplied to peoples' homes for cooking and heating. In the UK it has now been replaced by natural gas (CH_4), but town gas is still used in some countries.

Water gas is made by blowing steam through red or white hot coke.



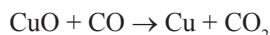
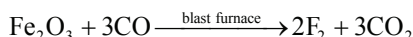
The water gas reaction is strongly endothermic ($\Delta C = \Delta H - T\Delta S$). Thus the coke cools down, and at intervals the steam must be turned off and air blown through to reheat the coke. It is a particularly good fuel, i.e., it has a high calorific value, because both CO and H_2 burn and evolve heat. Producer gas is made by blowing air through red hot coke.



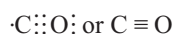
The overall reaction is exothermic, so the coke does not cool down as with water gas.



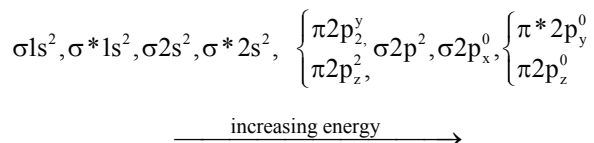
Producer gas is a less efficient fuel than water gas, i.e., it has a lower calorific value, as only part of the gas will burn. The approximate composition of producer gas is 70% N₂, 25% CO, 4% CO₂ with traces of CH₄, H₂ and O₂. CO is a good reducing agent and can reduce many metal oxides to the metal.



The bonding in CO may be represented as three electron pairs shared between the two atoms:



It is better represented using the molecular orbital theory (see Chapter 4).



The carbon-metal bond in carbonyls may be represented as the donation of an electron pair from carbon to the metal $\text{M} \leftarrow \text{C} \equiv \text{O}$. This original σ bond is weak. A stronger second bond is formed by back bonding, sometimes called dative π bonding. This arises from sideways overlap of a full d_{xy} orbital on the metal with the empty anti-bonding $\pi^* 2p_y$ orbital of the carbon, thus forming a $\pi \text{M} \rightarrow \text{C}$ bond. The total bonding is thus $\text{M}=\text{C}=\text{O}$. The filling, or partial filling, of the antibonding orbital on C reduces the bond order of the C—O bond from the triple bond in CO towards a double bond. This is shown by the increase in C—O bond length from 1.128 Å in CO to about 1.15 Å in many carbonyls.

CO is the most studied organometallic ligand. Because of the back bonding it is sometimes called a π acceptor ligand. The drift of 71 electron density from M to C makes the ligand more negative, which in turn enhances its σ donating power. Thus CO forms weak bonds to Lewis acids (electron pair acceptors) such as BF₃ as only σ bonding is involved. In contrast CO forms strong bonds to transition metals where both σ and π bonding can occur. Other π acceptor ligands include CN[−], RNC, and NO⁺.

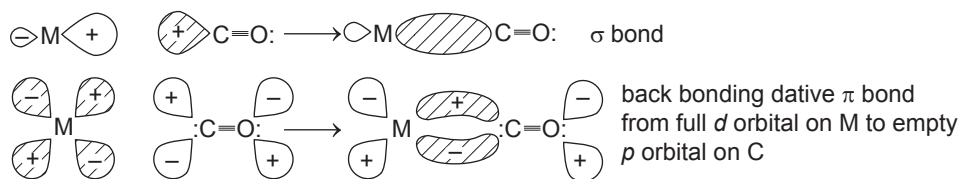
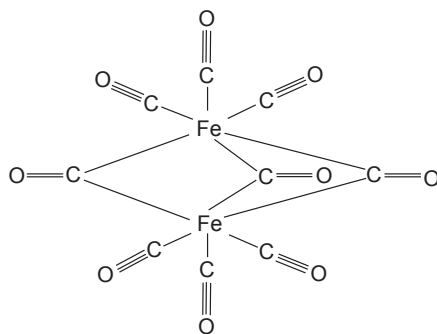


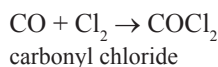
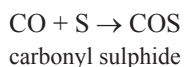
Fig. 5.7 Schematic of orbital overlaps in metal carbonyls. (After N.N. Greenwood and A. Earnshaw, *Chemistry of the Elements*, Pergamon, 1984)

Comparing these ligands, the strengths of the σ bonds are in the order $\text{CN}^- > \text{RNC} > \text{CO} > \text{NO}^+$, whilst their π acceptor properties are in the reverse order.

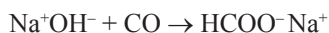
CO is a very versatile ligand. It may act as a bridging group between the two metal atoms, for example in di-iron enneacarbonyl $\text{Fe}_2(\text{CO})_9$. CO may stabilize metal clusters by the C forming a multi-centre bond with three metal atoms, and the π^* orbitals in CO may be involved in bonding to other metal atoms.

Fig. 5.8 Structure of $\text{Fe}_2(\text{CO})_9$.

In addition to reacting with oxygen, carbon monoxide combines with sulphur to give carbonyl sulphide and with chlorine in the presence of light to give carbonyl chloride (phosgene). The latter is an exceedingly poisonous gas.

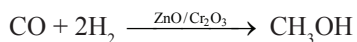


Although carbon monoxide is not a true acid anhydride since it does not react with water to produce an acid, it reacts under pressure with fused sodium hydroxide to give sodium formate:

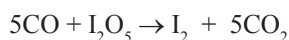


Formic acid is produced from the sodium salt by the addition of dilute hydrochloric acid.

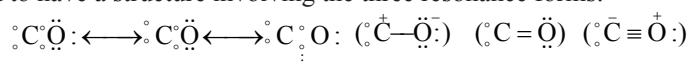
With hydrogen under pressure and in the presence of a zinc oxide/chromium (III) oxide catalyst it reacts to give methanol; this reaction is of industrial importance.



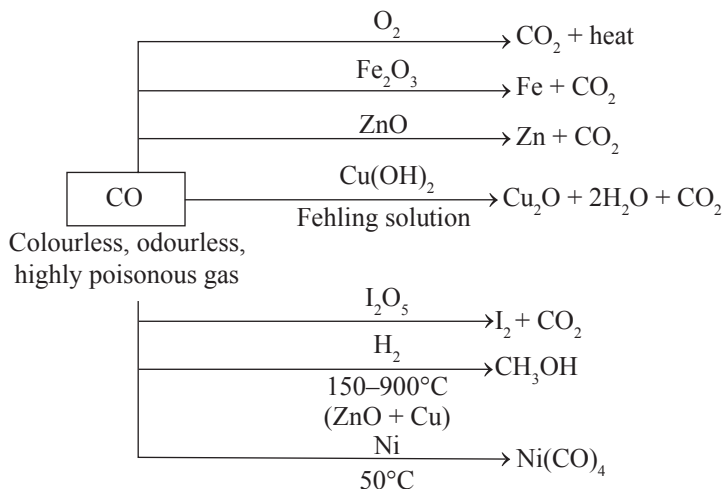
Carbon monoxide is readily absorbed by an ammoniacal solution of copper (I) chloride to give $\text{CuCl} \cdot \text{CO} \cdot 2\text{H}_2\text{O}$. It reduces an ammoniacal solution of silver nitrate to silver (black) and, in the absence of other gaseous reducing agents, this serves as a test for the gas. It can be estimated by reaction with iodine pentoxide, the iodine which is produced quantitatively being titrated with standard sodium thiosulphate solution.



Carbon monoxide is considered to have a structure involving the three resonance forms:



Properties



Structure

- CO converts I_2O_5 to form I_2 which gives violet colour with CCl_4 or CS_2



- CO act as a good ligand due to the presence of lone pair of electrons on carbon $[\text{C} = \ddot{\text{O}}:]$. CO acts as a π -acid ligand.

Worked-out Examples

31. Carbon monoxide is poisonous. Explain.

Ans. The poisonous nature of CO is due to the fact that it combines with haemoglobin (a red colouring matter of blood which is oxygen carrier) to form a stable compound carboxyhaemoglobin. Thus oxygen transportation is disturbed and tissues do not get necessary oxygen and ultimately death occurs.

A victim of CO gas poisoning (people sleeping in closed rooms with charcoal fire inside), if unconscious, should be brought in the open place and give artificial respiration with carbogen (mixture of $\text{O}_2 + 5 - 10\% \text{CO}_2$).

32. Which one of the following oxides is neutral?

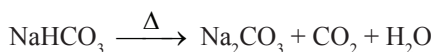
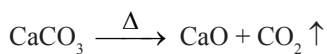
(IIT 1996)

- (a) CO (b) SnO_2 (c) ZnO (d) SiO_2

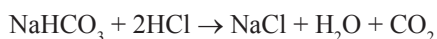
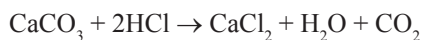
Ans. (a)

(B) Carbon Dioxide (CO_2)**Preparation:**

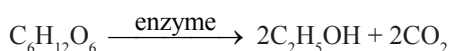
1. By heating carbonates of heavy metals, like Ca, Mg, Zn $\rightarrow$ or bicarbonates of alkalimetals



2. Lab method



3. By Alcoholic fermentation

**Properties**

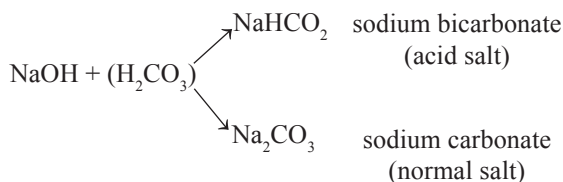
It is a colourless, odourless and heavy gas which dissolves in its own volume of water at ordinary temperature and pressure. Like all gases, it dissolves much more readily in water when the pressure is increased and, this principle is used in the manufacture of soda water and fizzy drinks.

Carbon dioxide is the acid anhydride of carbonic acid, and a solution of it in water will slowly turn blue litmus wine-red. An equilibrium is set up and, when the solution is boiled, all the carbon dioxide is evolved:

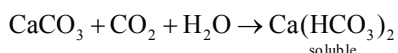
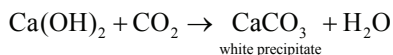


Carbonic acid has never been isolated, but it gives rise to two series of salts, hydrogen-carbonates (otherwise called bicarbonates), and carbonates.

5.24 Group 14-Carbon Family

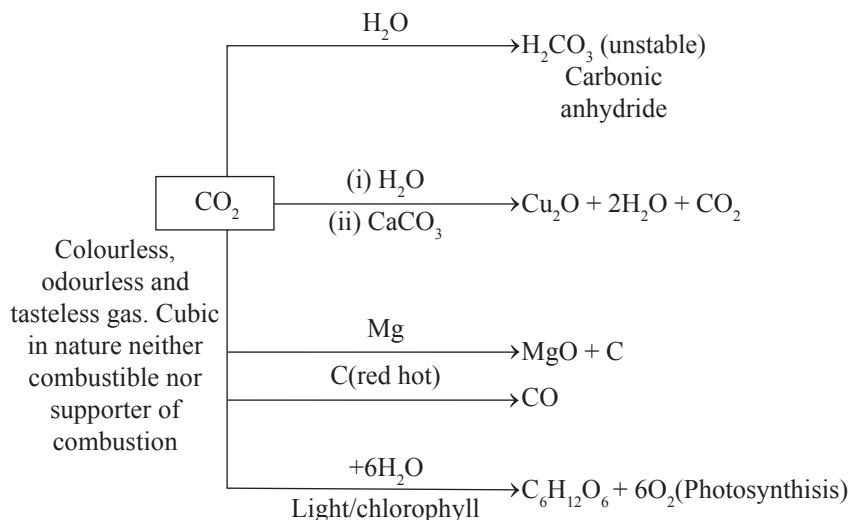


- CO₂ gas is detected by its action on lime water Ca(OH)₂ or baryta water Ba(OH)₂, as a white insoluble precipitate of CaCO₃ or BaCO₃ is formed. If more CO₂ is passed through the mixture, the cloudiness disappears as the soluble bicarbonate is formed.



- Solid carbon oxide is known as **dry ice**. It is obtained when the compressed carbon dioxide gas (CO₂ under pressure 50-60 atm) is cooled. When solid carbon dioxide is allowed to evaporate in air at atmospheric pressure, it changes directly into the gaseous state without liquefying. As a result, unlike ordinary ice it does not wet the surface on which it melts. Therefore, it is called dry ice.

Small scale uses of CO₂ include use in fire extinguishers, blasting in coal mines, as an aerosol propellant, and for inflating life-rafts.

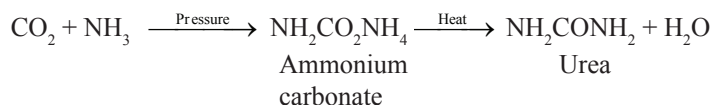


Structure



Uses of CO₂

- CO₂(liquid) in the form of super critical fluid (a substance above its critical pressure) has been used as solvent to extract organic compounds from their natural source (e.g., caffeine from coffee beans; perfumes from flowers, etc.).
- One of the major use of CO₂ is the manufacture of urea by its reaction with ammonia.

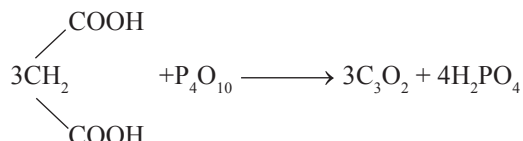


(iii) It is also used as fire extinguisher.

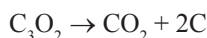
- CO_2 is heavier than air and can be poured from one container to another like a liquid
- Solid CO_2 is called dry ice or drikold because it sublimates without melting. With ether it produces -100°C temperature and forms an excellent freezing mixture
- Dry ice (solid CO_2) is a molecular solid (van der Waal's solid).
- Carbogen (mixture of $\text{O}_2 + 5-10\% \text{CO}_2$) is given to pneumonia patients and victims of CO poisoning for artificial respiration

Carbon suboxide, C_3O_2

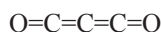
This is an evil-smelling gas and can be made by dehydrating propanedioic (malonic) acid, of which it is the anhydride, with phosphorus pentoxide:



When heated to about 200°C , it decomposes into carbon dioxide and carbon:

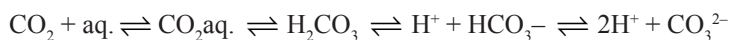


The molecule is thought to have a linear structure:



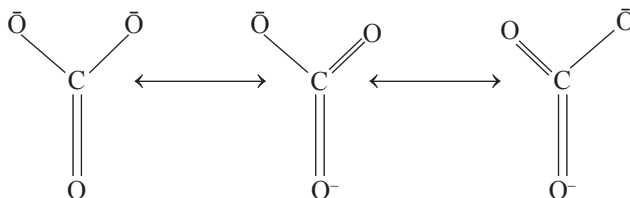
Carbonic acid

When carbon dioxide dissolves in water the greater part of it is only loosely hydrated. This hydrated species is in equilibrium with carbonic acid, hydrogen ions, and hydrogen carbonate and carbonate anions:



Although pure carbonic acid cannot be isolated, solid carbonates are plentiful and the Group 1A metals form solid hydrogen carbonates.

Carbonates contain discrete planar ions in which the three carbon-oxygen bond lengths are identical, the ion is thus a resonance hybrid containing equal contributions from the three forms:



Silicon does not form an analogous discrete silicate anion, SiO_3^{2-} , because of the reluctance of the larger silicon atom to form a multiple bond with oxygen. Silicates contain polymeric silicate anions, some of which contain the anion which approximates to the empirical formula SiO_3^{2-} .

Worked-out Examples

33. Solid CO_2 is known as dry ice. Explain.

Ans. Solid CO_2 technically known as dry ice because it sublimates without leaving any stain on surface.

34. CO_2 cannot be used for extinguishing fires caused by the burning of Mg metal.

Ans. Fire obtained by Mg is highly exothermic reaction and Mg continuously burns in CO_2 also.



35. When CO_2 under high pressure is released from a fire extinguisher, particles of solid CO_2 are formed, despite the low sublimation temperature of CO_2 (1 atm P, -77°C). Explain.

Ans. The gas does work pushing the atmosphere. The energy for doing work is provided by the gas itself and thus kinetic energy of gas molecules is lowered and hence the temperature is lowered.

36. CO_2 is chief gas responsible for global warming (Green house effect), why?

Ans. CO_2 absorbs strongly in IR region and its increased concentration in atmosphere decreases the loss of heat from earth by radiation.

37. Among the following, the molecule that is linear is (IIT 1982)
(a) CO_2 (b) NO_2 (c) SO_2 (d) ClO_2

Ans. (a)

38. CS_2 is stored in dark coloured bottles, why?

Ans. Because sun light changes CS_2 into CS.

39. A mixture of $\text{CS}_{2(\text{v})}$ and $\text{N}_{2(\text{g})}$ gives a vivid blue flame on ignition and a yellow solid is left as residue. Identify it and write reactions.

Ans. $2\text{NO}_{(\text{g})} + \text{CS}_{2(\text{v})} \longrightarrow \underset{\text{(Yellow solid)}}{2\text{S}} + \text{N}_{2(\text{g})} + \text{CO}_{2(\text{g})}$

40. Write down the structure of C_3O_2 .

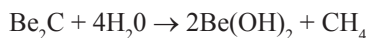
Ans. $\text{O} = \text{C} = \text{C} = \text{C} = \text{O}$, Each carbon is sp -hybridized having delocalization of π -electrons from one end to other.

Carbides

Carbides are the binary compounds of carbon with elements of low or equivalent electronegativity. Some important examples are CaC_2 , Al_4C_3 , Be_2C , SiC etc. These are further classified according to the type of bonding involved in them.

1. **Ionic or salt like carbides:** These are the compounds of carbon with strongly electropositive elements such as Mg, Ca, Al etc. Salt-like carbides are further classified according to the type of hydrocarbon they give on hydrolysis as shown below:

- **Methanides:** The carbides which give methane on hydrolysis are called methanides. For example,

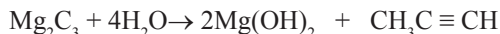


- **Acetylides:** The carbides which on hydrolysis give acetylene are called acetylides. Some examples are:



Some other carbides having similar structures are Cu_2C_2 , Ag_2C_2 etc. However, these carbides on hydrolysis do not give acetylene and hence are not regarded as true carbides.

- **Allylides:** The carbides which on hydrolysis give allylene or propyne are called allylides. For example, magnesium carbide (Mg_2C_3).



Allylene or propyne These carbides contain C_3^{4-} ions.

2. **Covalent carbides:** These are the compounds of carbon with elements having electronegativity slightly less than that of C. These carbides contain covalent bonds between carbon and other elements. Some examples are: SiC , B_4C , B_4C_3 etc.

3. Interstitial carbides: These are the compounds of carbon with transition elements. Some examples are: TiC, WC, ZrC, VC, V_3C , V_4C_3 etc.

Uses of Carbides

- Calcium carbide is the mainly used for preparing acetylene gas. Acetylene is the starting material for a number of industrially important compounds such as vinyl plastics, ethyl alcohol, acetic acid etc. It is also used in oxy-acetylene flame for welding purposes.
- Silicon carbide (SiC) is very hard and is used as an abrasive under the name carborundum.
- Tungsten carbide (WC) is also very hard. It is used for polishing, drilling and cutting tools.
- B_4C (boron carbide) which is even harder than silicon carbide is being used now increasingly as an abrasive. It is also used as a shield for radioactive radiations and for making bullet-proof vests (i.e., clothing).

Worked-out Examples

41. Point out the number of σ and π bonds in (a) C_2^{2-} , (b) C_3^{4-} , (c) CN_2^{2-}

Ans. (a) 1 σ and two π ($\bar{C} \equiv \bar{C}$)

(b) 2 σ and two π ($\bar{C} \equiv C - C^{3-}$)

(c) 2 σ and two π ($\bar{N} = C = N^-$)

42. The number and type of bonds between two carbon atoms in CaC_2 are

(IIT 1996)

Ans. (a) one sigma (σ) and one pi (π) bond

(b) one sigma (σ) and two pi(π) bonds

(c) one sigma (σ) and one and a half pi (π) bonds

(d) one sigma (σ) bond

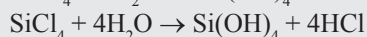
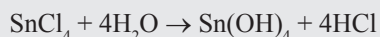
Halides of carbon

- Carbon combines with halogens to form both simple and mixed halides.
- The simple halides of carbon are: CF_4 , CCl_4 , CBr_4 and CI_4 . All these halides are known to exist.
- The stability of these tetrahalides decreases as the size of the halogen increases, i.e.,
 $CF_4 > CCl_4 > CBr_4 > CI_4$.
- This is due to the reason that the bond energies of the carbon-halogen bonds decrease in order:
 $C-F > C-Cl > C-Br > C-I$.
- Some common examples of mixed halides are $CFCI_3$, CF_2Cl_2 , CCl_3Br etc.

Worked-out Example

43. CCl_4 does not act as Lewis acid, while $SiCl_4$ and $SnCl_4$ acts as Lewis acid as well as their aqueous solutions are acidic. Explain.

Ans. $SiCl_4$ and $SnCl_4$ are hydrolysed to form acidic solution as well as they can act as Lewis acid because they can increase their co-ordination number beyond four due to availability of d -orbitals.



On the other hand C cannot increase its co-ordination number beyond four due to non-availability of d -orbitals and thus not hydrolyzed.

5.28 Group 14-Carbon Family

Properties

- All the tetrahalides are covalent compounds having tetrahedral shapes.
- They are chemically inert, non-flammable compounds.
- They are not hydrolyzed by water since carbon cannot expand its coordination number beyond four because of the absence of vacant d-orbitals.

Uses

- (i) Carbon tetrachloride is a common solvent. It is also used as a fire extinguisher under the name pyrene. Carbon tetrachloride is also used in medicines as anti-hookworm in intestines.

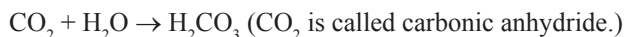
Worked-out Example

44. CCl_4 is used as fire extinguisher but CS_2 not. Why?

Ans. CCl_4 is heavy non combustible liquid but CS_2 is highly combustible liquid.

- (ii) Mixed halides of fluorine and chlorine are especially important as coolants or refrigerants because of being easily liquefiable gases having low b.p.'s and low specific heats. For example, CF_2Cl_2 (dichlorodifluoromethane) is widely used as a refrigerant under the name freon.

Carbon dioxide is an acidic oxide. The solution of CO_2 in water shows feeble acidic nature. This is believed due to the formation of carbonic acid.



Carbonic acid is known only in solution and has never been isolated in free state.

The carbonic acid is a dibasic acid and it forms two series of salts by replacement of one or both the hydrogen atoms.



- (a) **Acid salts**, are also called dicarbonates or hydrogen carbonates. These contain HCO_3^- ions.
(b) **Normal salts**, are also called carbonates. These contain CO_3^{2-} ions.

Fuels

- (i) **Water Gas**

Water gas is mainly a mixture of CO and H_2 . It is made by blowing steam through the layers of incandescent (red hot) coal. The efficient working temperature is 1000–1400°C.

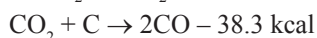


Due to endothermic nature of this reaction, the coal cools down after sometime and the formation of carbon dioxide starts at low temperature. To check this formation, the current of steam is replaced by a blast of air. This raises the temperature of coal and steam is readmitted. In general, water gas consists of 40 volumes of CO, 50 volumes of H_2 , 5 volumes of CO_2 and 4–5 volumes of nitrogen.

- Water gas or synthesis is also known as blue gas (burns with blue flame).

- (ii) **Producer Gas**

Producer gas is mainly a mixture of carbon monoxide and nitrogen. It is prepared by the incomplete combustion of coal, coke or charcoal. The combustion is done in restricted supply of air.



Initially, carbon dioxide is the main product but when it rises through the bed of coal, it combines with carbon, provided the temperature is 1000°C , forming carbon monoxide. Thus, heat is generated during its preparation. This heat can also be utilized if producer gas is generated on the spot and used while still hot.

- (iii) **Coal Gas:** It is a mixture of H_2 , CH_4 , CO and other gases like N_2 , C_2H_4 , O_2 and others.
- (iv) **Oil Gas:** It is a mixture of H_2 , CH_4 , C_2H_4 , CO and other gases like CO_2 .
- (v) **Natural Gas:** It is a mixture of gaseous hydrocarbons, namely, methane, ethane, propane, butane and so on. Liquefied petroleum gas (LPG) is also a mixture of gaseous hydrocarbons mainly butane and isobutane.

Worked-out Examples

45. Gaseous fuels are better fuels than solid and liquid fuels. Explain.

Ans. Gaseous fuels leave no residue after burning, also they have high calorific value because their combustion is complete.

46. Producer gas is less efficient fuel than water gas, why?

Ans. In water gas- $(\text{CO} + \text{H}_2)$, both components burn and evolve heat while in producer gas $(\text{CO} + \text{N}_2)$, only CO burns, i.e., why former has higher calorific value and is good fuel.

Silicon (Si)

Occurrence It occurs as silica (in the form of sand, quartz, rock crystal and so on) and silicates. The common silicate minerals are feldspar ($\text{K}_2\text{O} \cdot \text{Al}_2\text{O}_3 \cdot 6\text{SiO}_2$) and kaolinite ($\text{Al}_2\text{O}_3 \cdot \text{SiO}_2 \cdot 2\text{H}_2\text{O}$).

Preparation

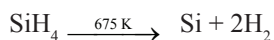
- (i) **From silica (sand)** Elemental silicon is obtained by the reduction of sand (silica) with coke in an electric furnace.



Excess of sand should be used so that any silicon carbide formed will be converted to silicon.

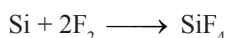
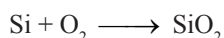


- (ii) **From silicon tetrachloride** Silicon is prepared by the reaction of dihydrogen followed by purification of zone refining.
- (iii) **From silicon hydride** Silicon can be prepared by the decomposition of silicon hydride at 675 K .

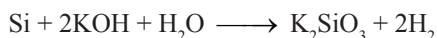
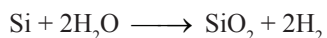


Properties

- (i) Silicon is a very hard solid.
- (ii) It exists in two allotropic forms, namely amorphous silicon and crystalline silicon.
- (iii) Amorphous silicon burns brilliantly in oxygen and ignites spontaneously in fluorine.



- (iv) It decomposes steam at red heat. It dissolves in a mixture of HNO_3 and HCl . However, it dissolves readily in alkalis.



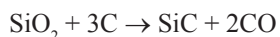
Worked-out Example

47. Why is SiH_4 more reactive than CH_4 ?

Ans. SiH_4 is more reactive than CH_4 due to the polarity of the Si—H bond and the availability of d-orbitals with silicon.

Compounds of silicon**1. Silicon Carbide (Carborundum)****Preparation**

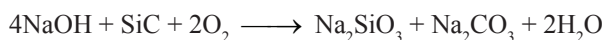
It obtained when a mixture of sand, carbon, common salt and saw dust is strongly heated in an electric furnace.



Salt is added to act as flux and saw dust makes the mass porous. Two carbon rods connected by a thin carbon core act as electrodes in the furnace. Carborundum is formed around the central core of carbon. It is crushed, washed with H_2SO_4 , NaOH , H_2O and then dried.

Properties

- (i) Pure carborundum is colourless but commercial samples possess yellow, green or blue colour.
- (ii) It is very hard mass. However, its hardness is less than diamond.
- (iii) It is chemically inert and resists the attack of almost all the reagents except fused caustic soda. The fused alkali dissolves it in presence of air.



Its structure is similar to diamond.

Uses: It is used as an abrasive instead of emery. It is used to make grind-stones, knife sharpeners, etc.

Worked-out Example

48. The material used in solar cells contains

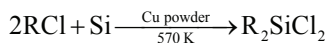
(IIT 1993)

- (a) Cs (b) Si
- (c) Sn (d) Ti

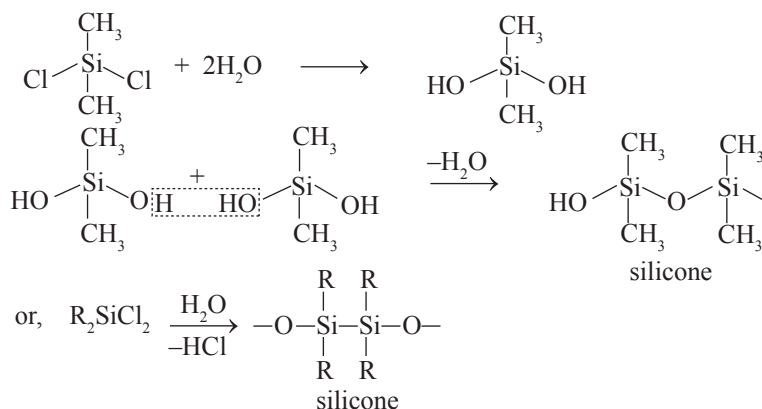
Ans. (b)

2. Silicons

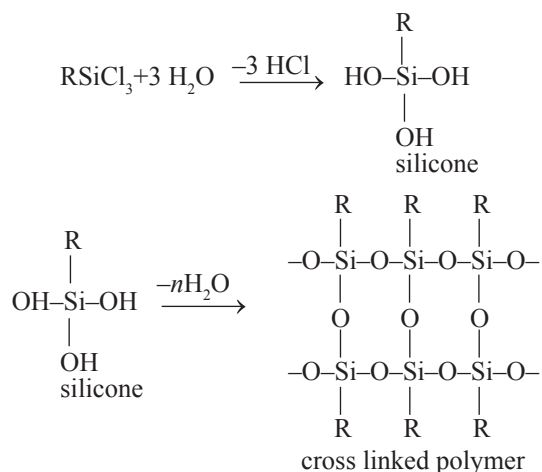
Silicons are polymeric compounds containing repeated R_2SiO units. The name silicon has been coined for these polymeric compounds because their empirical formula is analogous to that of ketones (R_2CO), the starting material for the manufacture of silicones in alkyl substituted chlorosilanes. These alkyl chlorosilanes are obtained by the reaction of alkyl halide with silicon in the presence of metallic copper which acts as a catalyst.



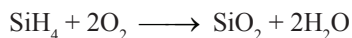
On carrying out the hydrolysis of these alkyl dichlorosilanes the Si—Cl bonds are ruptured and the chlorine atoms are replaced by OH groups. As soon as Si—OH bonds are formed, condensation sets in with the elimination of water, yielding silicone, as shown ahead.



The condensation can be stopped at any stage by regulating the conditions to get the polymeric chains of desired length. It is interesting to note that hydrolysis of alkyl trichlorosilanes, RSiCl_3 gives cross-linked polymers instead of chain polymer.



- When $(\text{SiH}_4 + \text{H}_2)$ mixture is bubbled through water and the bubbles come in contact with air, it burns with a luminous flame. As a result vortex rings of finely divided silica are obtained.



Properties:

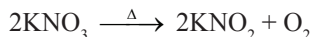
- The lower silicones are oily liquids but higher members containing long chains or ring structures are waxy and rubber like solids.
- (Silicones are stable towards heat.
- Chemical reagents have no action on silicones.
- These are non-toxic.
- Viscosity of silicone oils remains the same at different temperatures.
- Silicones are good electrical insulators.
- These are water repellants.

Uses:

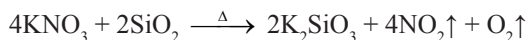
- Silicone oils are used for high temperature oil baths, high vacuum pumps and low temperature lubrication.
- These are used in making water-proof cloth and paper by exposing cloth or paper to the silicone vapour.

5.32 Group14-Carbon Family

- These are used as insulating materials for electric motors.
- KNO_3 never decompose to give N_2O_5 . On heating it gives O_2 and KNO_2



However, when KNO_3 is heated with SiO_2 , SiO_2 replaces N_2O_5



Worked-out Examples

49. Why does element silicon not form a graphite like structure, whereas carbon does?

Ans. $\text{p}\pi\text{-p}\pi$ multiple bonding is not prominent for silicon and other heavier congeners of group 14 and this reluctance for $\text{p}\pi\text{-p}\pi$ bonds in silicon does not allow to form graphite like structure in silicon.

50. SiF_6^{2-} is known but SiCl_6^{2-} is not, why?

Ans. (i) Smaller size of F give rise to less steric repulsion in SiF_6^{2-} .

(ii) Interaction of lone pair of F with Si is stronger than that of chlorine lone pair.

51. The hydrolysis of CCl_4 is not possible but SiCl_4 is easily hydrolysed. Explain.

Ans. Silicon can accommodate OH^- ion in 3d-subshell whereas C-atom does not process d-orbitals in its 2nd shell.

52. Why silanes and other analogous compounds less stable than corresponding hydrocarbons?

Ans. (i) Si—Si bond energy is slightly weaker than C—C bond (≈ 340 and 368 kJ mol^{-1} respectively).

(ii) Si—H bonds are weaker than C—H bonds (393 and 435 kJ mol^{-1}).

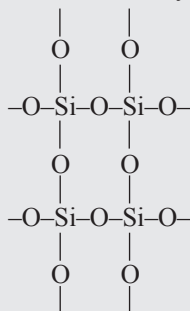
(iii) Si is less electronegative than H (in contrast to carbon which is more electronegative) and is therefore more susceptible to nucleophilic attack.

(iv) Si-atoms are larger and provide greater surface area for attack of nucleophiles.

(v) Si-atoms have low lying J-orbital's that can act as acceptors of electron pairs from

53. CO_2 is gas while silica is solid. Explain.

Ans. Due to large electronegativity difference in Si and O atoms, Si—O bond possess considerable ionic nature and thus silica has three dimensional infinite structure having silicon atom tetrahedrally bonded to four oxygen atoms and each oxygen atom is bound to two silicon atoms by covalent bonds.



The entire crystal of silica shows a giant molecule and thus is a high melting solid. On the other hand, CO_2 shows simple covalent nature having carbon atom sp hybridized. The molecules of CO_2 are held together by weak van der Waals forces and thus it exists as gas.

54. During reduction of SiO_2 into Si, SiO_2 is taken in excess, Why?

Ans. Reduction takes place as:



55. Silicones are chemically inert and have high strength. Why?

Ans. Silicones are inorganic polymers and contain Si—O and Si—C bonds, which are fairly strong bonds.

56. $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ on heating gives Sn(OH)Cl . How can anhydrous SnCl_2 be prepared?

Ans. By Passing dry HCl gas over hot tin.



57. Why $(\text{CH}_3)_3\text{COH}$ is less acidic than $(\text{CH}_3)_3\text{SiOH}$ although carbon is more electronegative than Si.

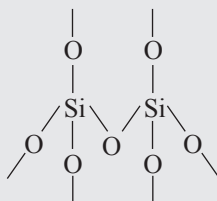
Ans. In $(\text{CH}_3)_3\text{SiOH}$, d-orbitals of Si give rise to $p\pi-d\pi$ back bonding and thus it can easily accommodate lone pair of electron from a base.

58. In silicon dioxide

[AIEEE 2005]

- (a) there are double bonds between silicon and oxygen atoms
- (b) silicon atom is bonded to two oxygen atoms
- (c) each silicon atom is surrounded by two oxygen atoms and each oxygen atoms is bonded to two silicon atoms
- (d) each silicon atom is surrounded by four oxygen atoms and each oxygen atom is bonded to two silicon atoms

Ans. (d) In SiO_2 (quartz), each of O-atom is shared between two SiO_4^{4-} tetrahedra.



59. The hydrolysis of alkyl substituted chlorosilane gives ____.

(IIT 1991)

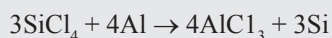
Ans. silicones

60. Write balanced equations for the preparation of crystalline silicon from SiCl_4 .

(IIT 1990)

Ans. Preparation of crystalline Si from SiCl_4 :

The vapours of SiCl_4 are passed over molten Al:

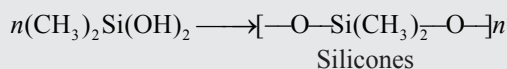
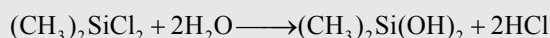


61. $(\text{CH}_3)_2\text{SiCl}_2$ on hydrolysis produces

(IIT 2003)

- (a) $(\text{CH}_3)_2\text{Si(OH)}_2$
- (b) $(\text{CH}_3)_2\text{Si=O}$
- (c) $[-\text{O}-(\text{CH}_3)_2\text{Si}-\text{O}-]_n$
- (d) $(\text{CH}_3)_2\text{SiCl(OH)}$

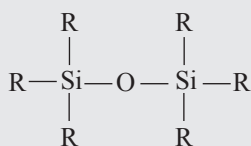
Ans. (c) The involved reactions are



62. The hydrolysis of trialkylchlorosilane, R_3SiCl , yields ____.

(IIT 1994)

Ans.



Reasoning Type

The questions below consist of an 'Assertion' in column 1 and the 'Reason' in column 2. Use the following key to choose the appropriate answer.

- (a) If both assertion and reason are CORRECT, and *reason* is the CORRECT explanation of the *assertion*.
 (b) If both *assertion* and *reason* are CORRECT, but *reason* is NOT the CORRECT explanation of the *assertion*.
 (c) If *assertion* is CORRECT, but *reason* is INCORRECT.
 (d) If *assertion* is INCORRECT, but *reason* is CORRECT.

Assertion (column 1)**Reason (column 2)**

63. Between SiCl_4 and CCl_4 , SiCl_4 is ionic and CCl_4 is covalent.

Only SiCl_4 reacts with water.

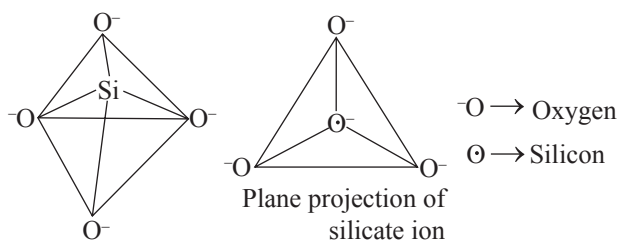
(IIT 2001)

Ans. (c) The statement 1 is correct but the statement 2 is not correct as orthoboric acid acts as a Lewis acid. It does not liberate H^+ ion but accepts a OH^- ion from water.

3. Silicates

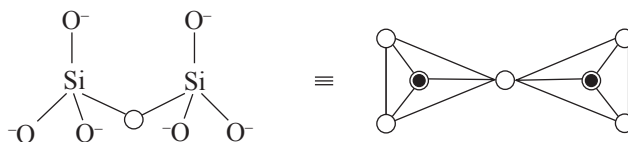
Silicates are metal derivatives of silicic acid, H_4SiO_4 or $\text{Si}(\text{OH})_4$.

e.g., quartz, asbestos, feldspar (KAISi_3O_8) and zeolites ($\text{Na}_2\text{Al}_2\text{Si}_2\text{O}_8 \cdot x\text{H}_2\text{O}$). The silicates are complex network solids having silicate ion (SiO_4) $^{4-}$ as the basic structural unit. The silicate ion has a tetrahedral structure. As a result of sharing of one or more oxygen atoms between such tetrahedrons, a complex structure arises.

**Type of Silicates**

- (i) **Orthosilicates:** These silicates contain single discrete unit of SiO_4 $^{4-}$ tetrahedra, For example, zircon (ZrSiO_4). Fenacite (Be_4SiO_4), forsterite (Mg_2SiO_4) and wellemite (ZnSiO_4).
- (ii) **Pyrosilicates:** These silicates contain two units of SiO_4 $^{4-}$ joined along a corner containing oxygen atom. These are also called as island silicates.

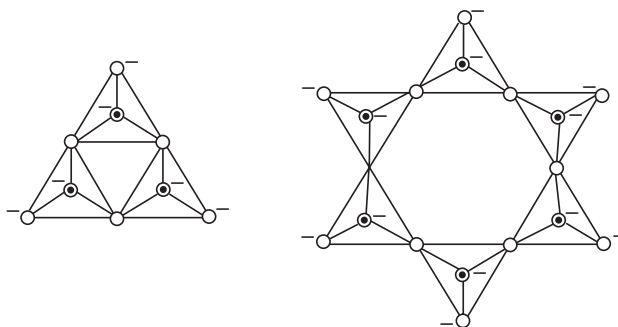
Pyrosilicate ion Si_2O_7 $^{6-}$,



e.g., Thortevitite $\text{Sc}_2\text{Si}_2\text{O}_7$;
 Hemimorphite $\text{Zn}_3(\text{Si}_2\text{O}_7)\text{Zn}(\text{OH})_2 \cdot \text{H}_2\text{O}$

- (iii) **Cyclic structure:** Cyclic or ring silicates have general formula $(\text{SiO}_3)^{2-}_n$ or $(\text{SiO}_3)^{2n-}_n$.

Structure and example of cyclic silicates containing Si_3O_9 $^{6-}$ and Si_6O_{18} $^{12-}$ ions are given ahead:



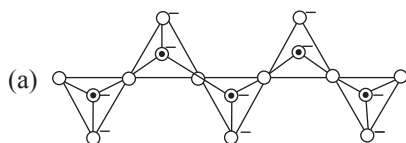
e.g., Beryl $\text{Be}_3\text{Al}_2\text{Si}_6\text{O}_{18}$;

Benitoite $\text{BaTiSi}_3\text{O}_9$

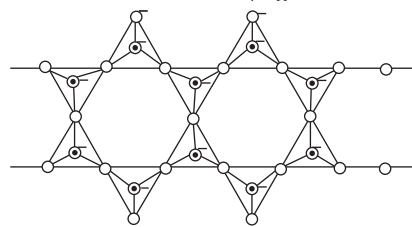
Wollastonite $\text{Ca}_3\text{Si}_3\text{O}_9$;

Catapleite $\text{Na}_2\text{ZrSi}_3\text{O}_9 \cdot 2\text{H}_2\text{O}$

(iv) **Chain silicates:** Chain silicates are formed by sharing two oxygen atoms by each tetrahedral. Anions of chain silicate have two general formulae,



(b)



e.g.,

Synthetic silicates

Spodumene

Jadeite

Enstatite

Diopside

Tremolite

Li_2SiO_3 , Na_2SiO_3 ;

$\text{LiAl}(\text{SiO}_3)_2$;

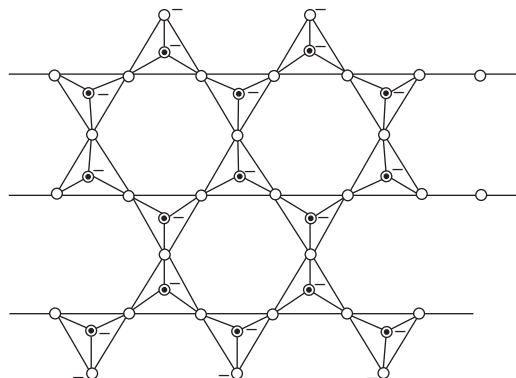
$\text{NaAl}(\text{SiO}_3)_2$;

MgSiO_3 ;

$\text{CaMg}(\text{SiO}_3)_2$;

$\text{Ca}_2\text{Mg}_5(\text{Si}_4\text{O}_{11})_2(\text{OH})_2$

(v) **Two dimensional sheet silicates:** In such silicates, three oxygen atoms of each tetrahedral are shared with adjacent SiO_4^{4-} tetrahedral, such sharing forms two dimensional sheet structure with general formula $(\text{Si}_2\text{O}_5)_n^{2n-}$.



e.g., Talc
Kaolin

$\text{Mg}(\text{Si}_2\text{O}_5)_2\text{Mg}(\text{OH})_2$
 $\text{Al}_2(\text{OH})_4(\text{Si}_2\text{O}_5)_2$

5.36 Group14-Carbon Family

(vi) Three dimensional sheet silicates:

These silicates involve all four oxygen atoms in sharing with adjacent SiO_4^{4-} tetrahedra.

e.g., Quartz, Tridymite, Cristobalite, Feldspars, Zeolites and Ultramarines.

Worked-out Example

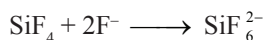
64. Which of the following silicates is formed when three oxygen atoms of SiO_4^{4-} tetrahedral units are shared? (IIT 2005)

- (a) Pyrosilicates (b) Cyclic silicates
(c) Linear chain silicates (d) Sheet silicates

Ans. (d) Pyrosilicates involve sharing of one oxygen atom between two SiO_4^{4-} tetrahedral.

Chain and cyclic silicates involve sharing of two oxygen atoms of a SiO_4^{4-} tetrahedron with two other tetrahedra. Sheet silicates involve sharing of three oxygen atoms by each tetrahedron.

- Silanes: The hydrides of silicon are called silanes. For example, SiH_4
- SiF_4 behaves as a lewis acid and can accept lone pair of electron of ligand due to the presence of vacant d-orbital in silicon.



- High strength silicon rubber withstands extreme surface temperatures. Hence soles of Lunar boots were prepared from this rubber and were used by apollo astronauts.

Glass

Amorphous, hard, brittle transparent, translucent super cooled solution of various silicates and borates of K, Ca and Pb. It has no definite formula but roughly can be represented as $\text{M}'_2\text{O} \cdot \text{M}''\text{O} \cdot x\text{SiO}_2$ where $\text{M}' = \text{Na, K}$ and $\text{M}'' = \text{Ba, Zn, Pb}$

Raw material used in the manufacture of glass.

- (i) SiO_2
- (ii) Na_2CO_3 , K_2CO_3 or NaNO_3 or KNO_3
- (iii) Alkaline earth metals e.g., CaCO_3 , BaCO_3
- (iv) oxides of heavy metals
- (v) Cullets (pieces of glass)
- (vi) Colouring matter

Table 5.3 Colouring Materials

Colour	Substances	Colour	Substances
Green	Cr_2O_3 , FeSO_4	Red	Cu_2O , Selenium oxide
Yellow	Ferric iron, uranate of Na	Lemon yellow	CdS
Purple	MnO_2	Black	NiO
Ruby	AuCl_3	Milky	Na_3AlF_6
Blue	CuO , CoO	Peacock Blue	cupric Salts

Manufacture – Mixture of raw materials $\xrightarrow{1200^\circ\text{C}}$ till CO_2 escapes $\rightarrow$ clear liquid. After some cooling it is used for casting articles

Table 5.4 Types of Glass

Type	Batch	Properties	Uses
(i) Soft or soda glass	$\text{Na}_2\text{O} \cdot \text{CaO} \cdot \text{SiO}_2$	Soft	Window glass and bottles
(ii) Hard or potash glass	Sand+lime stone + K_2CO_3	Hard	Hard glass apparatus
(iii) Flint glass	Sand+Red lead+ K_2CO_3	High refractive index	Optical instrument, prisms
(iv) Pyrex glass	$\text{Na}_2\text{O} \cdot \text{B}_2\text{O}_3 \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2$	Low coefficient	Glass ware of expansion
(v) Jenaglass	Zn and Barium Borosilicate	Low coefficient of expansion	Glass wares
(vi) Crookes glass	$\text{K}_2\text{O} \cdot \text{PbO} \cdot \text{CeO}_2 \cdot \text{SiO}_2$	Cuts off U.V. rays	Eye lenses
(vii) Quartz glass or silica glass	Pure SiO_2	Low coefficient of expansion	Scientific wares
(viii) Safety glass or shatter proof glass	Between two glass layers a transparent plastic layer of vinyl acetate resin is placed		Automobile wind shields, goggles etc.
(ix) Reinforced glass	Has network of wires	Does not shatter	
(x) Ground glass	Soft glass with ground surface		

- Broken glass pieces added during glass making is known as cullet.
- Glass is a supercooled liquid.
- Quartz glass used in the manufacture of optical instruments and chemical apparatus is called Vitreosil.
- Quartz is purest and most common variety of silica.
- HF is used for etching of glass.
- Hydrofluoric acid can easily attack glass, SiO_2 etc., but it cannot attack plastic, wax, copper etc. As such HF is generally kept in bottles made up of wax or plastic.

Worked-out Examples

65. Glass is a [AIEEE 2003]

- (a) super-cooled liquid (b) gel
(c) polymeric mixture (d) micro-crystalline solid

Ans. (a) Glass is a translucent or transparent amorphous supercooled solid solution or we can say super cooled liquid of silicates and borates having a general formula $\text{R}_2\text{O} \cdot \text{MO} \cdot 6\text{SiO}_2$, where R = Na or K and M = Ca, Ba, Zn or Pb.

66. For making good quality mirrors, plates of float glass are used. These are obtained by floating molten glass over a liquid metal which does not solidify before glass. The metal used can be

[AIEEE 2003]

- (a) tin (b) sodium (c) magnesium (d) mercury

Ans. (d) It is mercury because it exists as liquid at room temperature.

67. SiO_4^{4-} ion exists while CO_4^{4-} does not, why?

Ans. Due to small size of C-atom

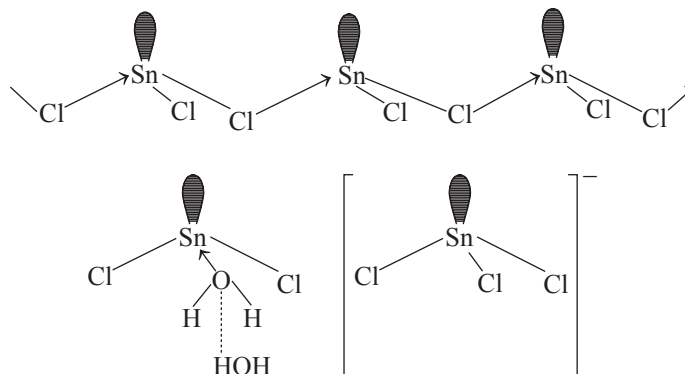
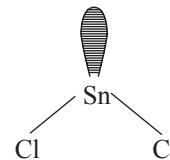
Oxides and chlorides of tin and lead**Oxides and halides of tin and lead**

Tin (II) Chloride: Metallic tin (Sn^{2+}/Sn ; $E^\circ = -0.136$ volt) liberates hydrogen from hydrochloric acid to form SnCl_2 which crystallises as the dihydrate. The anhydrous salt can be prepared by heating the metal in an atmosphere of dry HCl gas.

5.38 Group14-Carbon Family

The hydrated salt may also be dehydrated by acetic anhydride. Solution of SnCl_2 is protected against oxidation and hydrolysis by maintaining the solution acidic (HCl) and adding tin pieces. It is used as a reducing agent—reducing Fe^{3+} to Fe^{2+} , Hg^{2+} to Hg^{2+}_{22} , Cu^{2+} to Cu^+ and organic nitrocompounds to amino compounds. It is used in mordant dyeing.

Anhydrous SnCl_2 in the gas phase has a bent V-shaped structure as predicted by the VSEPR model. Crystalline SnCl_2 has a polymeric chlorobridged structure. Each tin (II) has one terminal chlorine and two bridging chlorines. $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ has two terminal chlorines and one coordinated aqua group. The second water molecule is hydrogen bonded to the coordinated aqua group. In $[\text{SnCl}_3]^-$ all the three chlorines are terminal. Sn-Cl distances in all these compounds are in the range 2.4–2.7 Å (240–270 pm). In $[\text{SnCl}_3]^-$ there are three other chlorines at longer distances 3.2–3.8 Å (320–380 pm). The shaded orbitals carry the lone pairs of electrons.



Tin(IV) Chloride: This is made by oxidizing tin by gaseous chlorine and condensing the gas. It is collected as a colourless liquid being protected from moist air by a CaCl_2 guard tube. Because of the availability of d-orbitals tin(IV) halides are readily hydrolysed, and also readily form halo complexes.

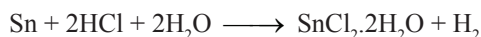


Chlorides of tin

Stannous chloride, SnCl_2

Preparation

Tin dissolved in hot concentrated hydrochloric acid yields $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ on concentrating and crystallization.



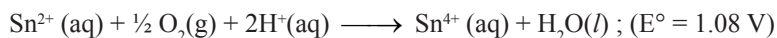
Properties

The hydrated salt on heating forms the oxychloride.

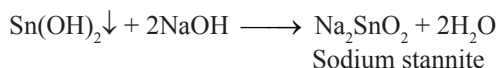
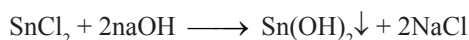


This hydrolysis can be prevented by the presence of excess HCl , with some pieces of tin added.

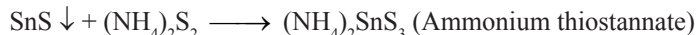
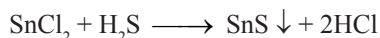
Aqueous and non-aqueous solutions of tin(II) salts are capable of acting as reducing agents, but they must be stored under an inert atmosphere because air oxidation is spontaneous and rapid.



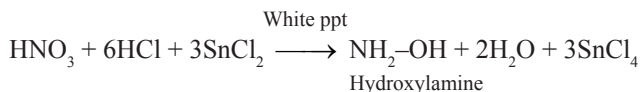
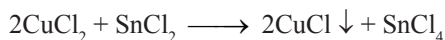
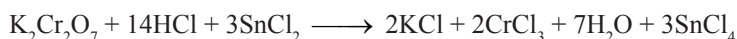
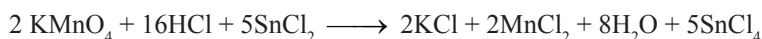
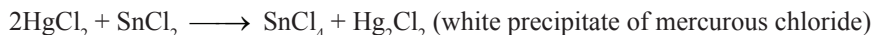
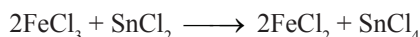
Stannous chloride reacts with NaOH forming a white precipitate of tin (II) hydroxide which dissolves in excess of NaOH forming sodium stannite.



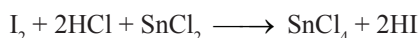
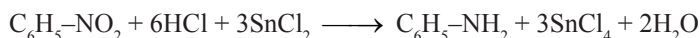
Sodium stannite is oxidized by atmospheric oxygen to form sodium stannate, Na_2SnO_3 . From a solution of stannous chloride, H_2S precipitates brown SnS , soluble in ammonium polysulphides.



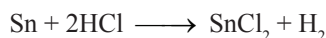
SnCl_2 is a powerful reducing agent, as the following reactions illustrate.



Stannous chloride reduces nitro compounds to amino compounds and iodine to iodides.



Anhydrous stannous chloride, a glassy substance is prepared by heating tin in a stream of HCl or with mercuric chloride.



Excess

Anhydrous SnCl_2 forms a dimer (Sn_2Cl_4) in the vapour, dissolves in organic solvents and forms many addition compounds with NH_3 . e.g., $\text{SnCl}_2 \cdot 2\text{NH}_3$.

In aqueous and non-aqueous solutions Sn(II) forms trihalo complexes, such as $[\text{SnCl}_3]^-$, where the pyramidal structure indicates the presence of a stereochemically active lone pair. The $[\text{SnCl}_3]^-$ ion can serve as a soft donor to d-metal ions. One unusual example of this ability is the red cluster compound $\text{Pt}_3\text{Sn}_8\text{Cl}_{20}$, which is trigonal bipyramidal.

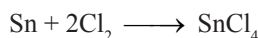
Use

As a reducing agent in the laboratory, as a mordant in dyeing and in the preparation of purple of Cassius.

Stannic chloride, SnCl_4 (Butter of Tin)

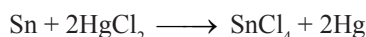
Preparation

Dry Cl_2 gas when passed over heated tin in a retort forms SnCl_4 .

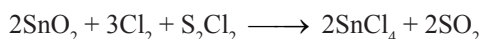


Liquid SnCl_4 is thus collected in a cooled receiver protected from moisture.

It is also made by heating Sn with excess of HgCl_2 .



Another method of preparation is from heated SnO_2 by passing Cl_2 and S_2Cl_2 (sulphur monochloride) vapour over it.

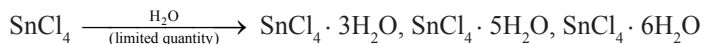


It is also obtained by the removal of tin from (i.e., detinning of) scrap tin plates by chlorine.

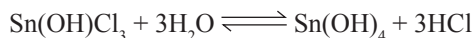
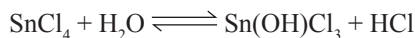
5.40 Group14-Carbon Family

Properties

It is a colourless fuming liquid, soluble in organic solvents and volatile in nature. These properties indicate its covalent nature. It forms hydrates with a limited quantity of water but undergoes hydrolysis with excess of water.

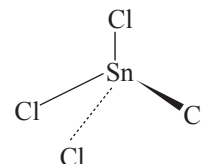


$\text{SnCl}_4 \cdot 5\text{H}_2\text{O}$ is known as '**butter of tin**' or '**oxymuriate of tin**'. It is used as a mordant and also for weighing silk.



This hydrolytic reaction is slow, reversible and can be suppressed by HCl, with which the following reaction occurs: $\text{SnCl}_4 + 2\text{Cl}^- \longrightarrow [\text{SnCl}_6]^{2-}$ Salts with this ion e.g., $(\text{NH}_4)_2\text{SnCl}_6$ are known as chlorostannates. Other addition compounds are obtained with NH_3 , PCl_5 etc., e.g., $\text{SnCl}_4 \cdot 4\text{NH}_3$.

The tetrachloride, bromide and iodide of tin are molecular compounds, but the tetrafluoride has a structure consistent with it being an ionic solid because the small F^- ion permits a six coordinate structure.

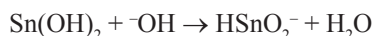


Uses

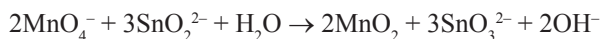
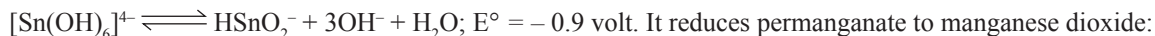
Butter of tin is used as a mordant and for weighing silk.

- $\text{SnCl}_4 \cdot 5\text{H}_2\text{O}$ (Butter of tin or oxymuriate of tin) and pink salt $(\text{NH}_4)_2\text{SnCl}_6$ are used as mordant in dyeing.

Tin (II) and Tin(IV) Oxides: Addition of alkali to tin(II) solution first precipitates hydrated tin(II) oxide. This dissolves in excess alkali to give stannate (II) ion:



The stannate(II) ion is a strong reducing agent as is indicated by its redox potential:



It also reduces a bismuth(III) compound to black metallic bismuth.

The dioxide occurs in nature. Hydrated tin dioxide called metastannic acid (or β -stannic acid) is obtained by the action of concentrated HNO_3 on metallic tin:



This variety of hydrated tin dioxide dissolves in alkali but not in acids. The α -stannic acid is obtained by the acidification of aqueous solutions of alkali metal stannates. This form is readily soluble in acids to form tin (IV) compounds and in alkali to form alkali stannates. Crystalline alkali stannates such as $\text{K}_2\text{SnO}_3 \cdot 3\text{H}_2\text{O}$ really contain the anions $[\text{Sn(OH)}_6]^{2-}$. Six OH^- groups surround tin (IV) octahedrally. Both forms of the hydrated oxide have the same crystal structure. The difference in their reactivity is due to a difference in particle sizes and in the nature of the surfaces of the particles.

Mosaic Gold: Hydrogen sulphide precipitates from a dilute acid solution of tin(IV) compounds a yellow precipitate of tin(IV) sulphide SnS_2 . On a technical scale tin(IV) sulphide is prepared in the form of golden yellow scales by heating metallic tin, mercury (or tin amalgam), flowers of sulphur and ammonium chloride. This product is used as a pigment under the name *mosaic gold* (gold for mosaic work) and under the name *tin bronze* for bronzing.

Worked-out Examples

68. Which of the following oxides is/are amphoteric? (IIT 1993)

- (a) Na_2O (b) CaO (c) Al_2O_3 (d) SnO_2

Ans. (c), (d)

69. SnCl_2 is a nonlinear molecules. (True/False) (IIT 1985)

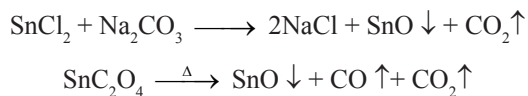
Ans. True

Oxides of tin

Tin(II) oxide, SnO

Preparation

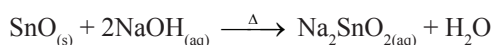
SnO is precipitated by boiling stannous chloride solution with sodium carbonate or by heating the hydroxide or oxalate in absence of air.



When freshly precipitated, the oxide has the composition $2\text{SnO} \cdot 2\text{H}_2\text{O}$.

Properties

It is usually an olive green powder, which gives greyish crystals in contact with water. When heated in air, it forms the dioxide. Both the oxide and hydrated oxide dissolve in acids forming stannous salts and in alkalies, forming stannites.



Uses

SnO acts as strong reducing agent.

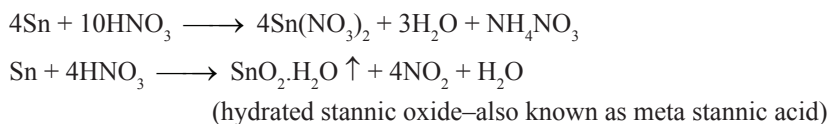
Tin(IV) oxide, SnO₂

SnO₂ occurs in nature as tinstone or cassiterite.

Preparation

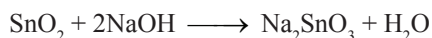
It is easily obtained by the combustion of tin in air, by ignition of metastannic acid produced from the action of nitric acid on tin.

Cold and dilute nitric acid reacts with tin forming stannous nitrate, while concentrated nitric acid attacks tin with the formation of hydrated stannic oxide.



Properties

It is a soft, white solid sparingly soluble in water and acids except concentrated sulphuric acid but readily soluble in fused alkalies to form stannate.



Uses

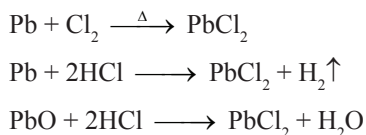
Tin dioxide is used as a polishing powder and the name “putty powder” and for making milky glass and white glazes for tiles and enamels.

Chlorides of lead

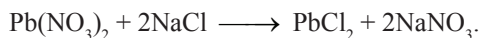
Lead chloride, PbCl₂

Preparation

Prepared by slow direct combination or by the action of boiling concentrated HCl on lead (its oxide or carbonate).



The usual method of preparation is to precipitate PbCl₂ as a white crystalline precipitate by adding a soluble chloride to a lead salt solution.



5.42 Group14-Carbon Family

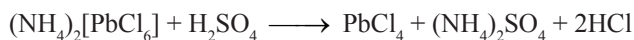
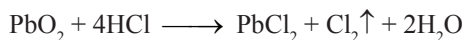
Properties

It is sparingly soluble in cold water but more soluble in hot water. In concentrated solutions of Cl^- ions, it dissolves forming complex ions, $[\text{PbCl}_3]^-$ and $[\text{PbCl}_4]^{2-}$.

Lead tetrachloride, PbCl_4

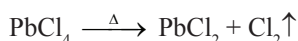
Preparation

This is made by dissolving PbO_2 in ice-cold concentrated HCl . Concentrated H_2SO_4 decomposes ammonium chloroplumbate to yield PbCl_4 .

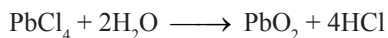


Properties

PbCl_4 is a yellow oily liquid. It is heavy and dissolves in organic solvents. It is a covalent and unstable compound, readily decomposes on heating.



It is easily hydrolysed by water and forms a double salt with NH_4Cl .



Lead tetrabromide and tetraiodide are unknown, so the dihalides dominate the halogen chemistry of lead.

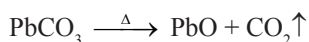
Oxides of lead

Lead monoxide, PbO

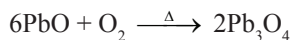
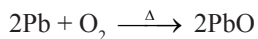
The colour of this oxide is yellow or red depending on the mode of preparation.

Preparation

When lead is gently heated in air, yellow powder is formed as the monoxide, called massicot. When heating is continued it melts and on cooling gives the reddish yellow scales of litharge. These differ only in crystalline structure. The transition temperature being 558°C . Lead monoxide can also be prepared by thermal decomposition of lead nitrate as well as lead carbonate.

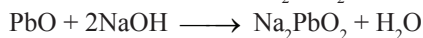
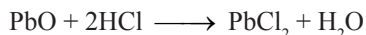


Dry air has no action on lead, but in moist air it tarnishes, forming a film of oxide first and finally basic carbonate, which protects it from further action. On heating in air or oxygen, it forms litharge, PbO . But prolonged heating gives red lead, Pb_3O_4 .



Properties

- (i) At room temperature, it is a yellow amorphous powder that is insoluble in water but dissolves in acids as well as alkalis.



Thus, it behaves as an amphoteric oxide. The acidic properties being rather feeble.

- (ii) It is easily reduced to the metallic state by hydrogen, carbon or carbon monoxide.
(iii) In the red form of PbO , the Pb(II) ions are four coordinate, but the O^{2-} ions around the Pb(II) lie in a square.

Uses

Used in paints, in the vulcanization of rubber and in the preparation of red lead and lead salts.

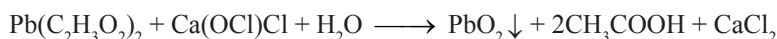
Lead dioxide, PbO₂

Preparation

- (i) Action of cold concentrated nitric acid on red lead gives lead nitrate in solution while lead dioxide is thrown as a chocolate powder.

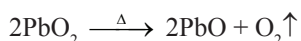


- (ii) Action of powerful oxidizing agents like chlorine, bromine or bleaching powder on alkaline lead salt solution.

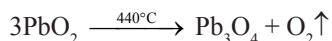


Properties

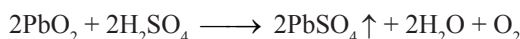
- (i) It is a chocolate coloured powder insoluble in water and dilute acids.
 (ii) It liberates oxygen on gentle heating.



- (iii) At 440°C, it is converted into red lead, Pb₃O₄.



- (iv) PbO₂ is an amphoteric oxide.



- (v) It is a good oxidizing agent. It oxidizes manganous salts to pink permanganic acid when boiled in nitric acid solution.



- (iv) In alkaline medium, chromium hydroxide is oxidized to yellow chromate by PbO₂.



The maroon form of lead(IV) oxide, PbO₂, crystallizes in the rutile structure. This oxide is a component of the cathode of a lead-acid battery.

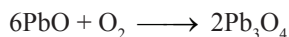
Uses

In the laboratory, lead dioxide finds application as an oxidizing agent. It is also used as the cathode in lead storage battery.

Red lead, Pb₃O₄

Preparation

Roasting of litharge in air 450°C gives a bright red powder.



It is also known as sindur.

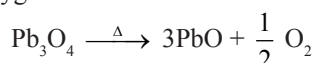
Properties

- (i) Sparingly soluble in water but dissolves in dilute nitric acid.



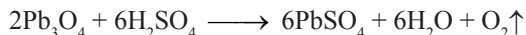
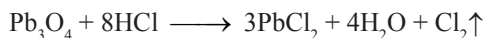
The above reaction indicates that red lead may be considered as plumbous ortho plumbate, 2PbO.PbO₂.

- (ii) It turns dark when heated but restores the original colour on cooling. At about 550°C, it decomposes giving off oxygen.



5.44 Group14-Carbon Family

(iii) It reacts with concentrated HCl and sulphuric acid liberating chlorine and oxygen respectively.

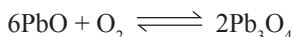


Red lead, Pb_3O_4 contains Pb(IV) in an octahedral arrangement and Pb(II) in an irregular six coordinate environment. The assignment of different oxidation number to the lead in these two sites is based on the shorter Pb–O distance for the atom identified as Pb(IV).

Uses

Red lead, mixed with linseed oil is extensively used as a red point and also for plumbing work.

Red Lead: It is made by heating ($\sim 350^\circ\text{C}$) lead monoxide in a current of air in a muffle furnace:

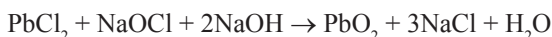


Higher temperature favours the backward reaction. It is a brilliant red powder. With HCl it gives PbCl_2 and Cl_2 :

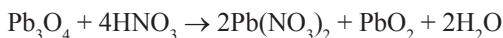


It is used as a red pigment.

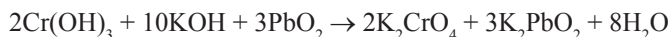
Lead Dioxide: This is made by oxidizing an alkaline solution of a lead salt by hypochlorite:



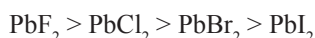
Lead acetate solution also reacts with bleaching powder on steam bath to give PbO_2 . The product is digested with hot dilute HNO_3 to remove any bleaching powder. Pb_3O_4 may also be oxidized by dilute HNO_3 to PbO_2 :



Lead dioxide is a strong oxidant:

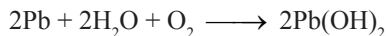


- PbCl_2 is ionic while PbCl_4 is covalent.
- Lead compounds are poisonous.
- Pb reacts with NaOH to form sod. Plumbite and Si acts to give sod. Silicate.
- $\text{Pb} + 2\text{NaOH} \longrightarrow \text{Na}_2\text{PbO}_2 + \text{H}_2\uparrow$
- $2\text{PbCO}_3 \cdot \text{Pb}(\text{OH})_2$ White lead. It can be prepared by Dutch process or Carter process and is used in white paints.
- Lead pipes are not used for carrying water due to the formation of soluble $\text{Pb}(\text{OH})_2$ which gives Pb^{2+} which are highly poisonous. This is called Plumbosolvency.
- Matlockite ($\text{PbCl}_2 \cdot \text{PbO}$) and Lanarkite ($\text{PbO} \cdot \text{PbSO}_4$) are ores of lead.
- Lead marks the paper. It is so soft that it can be cut with a knife and scratched with the finger nail.
- The dihalides of lead are ionic in character and their ionic character decreases with increase in size of halogen in following order:



- Mixture of massicot with glycerine is used for joining broken pieces of stones and glass.
- Sugar of lead is $\text{Pb}(\text{CH}_3\text{COO})_2$.
- Tetraethyl lead (TEL) $[\text{C}_2\text{H}_5]_4\text{Pb}$ is an organometallic compound used as an anti-knock compound in petrol. Petrol containing TEL is known as Leaded petrol or Ethyl petrol.
- Basic lead carbonate (white lead) is $2\text{PbCO}_3 \cdot \text{Pb}(\text{OH})_2$
- Chromoyellow or Lemon chrome is PbCrO_4 .

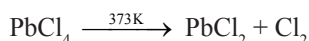
- Lead is readily corroded by water containing dissolved air forming lead hydroxide which is appreciably soluble in water. This is called Plumbosolvency.



- PbO_2 can neither be prepared by direct combination of Pb and O_2 nor by decomposing any salt (carbonate, nitrate etc.). It can only be obtained by treated Pb_3O_4 with dilute nitric acid.



- Lead is quite malleable but not so ductile.
- $\text{Pb} + \text{O}_2 + \text{H}_2\text{O} \longrightarrow \text{Pb}(\text{OH})_2$
- PbO_2 cannot be obtained by heating lead or lead (III) oxide in air.
- PbCl_4 is highly unstable and decompose on mild heating



Worked-out Examples

70. The species that do not contain peroxide ions are

- (a) PbO_2 (b) H_2O_2 (c) SrO_2 (d) BaO_2 (IIT 1992)

Ans. (a), (c)

71. White lead [$2\text{PbCO}_3 \cdot \text{Pb}(\text{OH})_2$] and TiO_2 , both are used as white pigment in paint industry, but use of TiO_2 is preferred. Why?

Ans. (i) Pb is toxic,
(ii) Pb-pigments are blackened by H_2S while Ti-pigments are not.

72. Write the formulae of the following compounds.

- (a) Butter of tin (b) White lead
(c) Sugar of lead (d) Litharge
(e) Red lead (f) Carborundum
(g) Phosgene (h) Chrome yellow
(i) Drykold (j) Water glass

Ans. (a) $\text{SnCl}_4 \cdot 5\text{H}_2\text{O}$ (Stannic chloride) (b) $2\text{PbCO}_3 \cdot \text{Pb}(\text{OH})_2$ (Basic lead carbonate)
(c) $\text{Pb}(\text{CH}_3\text{COO})_2$ (Lead acetate) (d) PbO (Lead monoxide)
(e) Pb_3O_4 (Trilead tetra-oxide) (f) SiC (Silicon carbide)
(g) CoCl_2 (Carbonyl chloride) (h) PbCrO_4 (Lead chromate)
(i) CO_2 (Carbon dioxide in solid state) (j) Na_2SiO_3 (Sodium silicate)

Noteworthy Points of Group 14

1. Stabilization of (+2) oxidation state relative to (+4) of the elements down the group.

The halides, oxides, and sulfides of the M^{2+} ions become more stable on descending the group. For example SiCl_4 , SiBr_4 , and SiI_4 are all stable. PbCl_4 decomposes at 105°C and PbI_4 does not exist. Similarly the ease of oxidation of the M^{2+} halides increases down the column.

PbCl_2 may only be converted to PbCl_4 by heating in a stream of chlorine.

Similarly, PbO_2 is an oxidizing agent, whereas SnO_2 , GeO_2 , SiO_2 are not.

The stabilities of the M^{II} and M^{IV} organometallic derivatives of the elements behave differently. PbEt_4 which can be readily stored is more stable than PbEt_2 , which is not isolable as a solid.

5.46 Group 14-Carbon Family

The compounds in the lower oxidation state are in general more ionic, less likely to form molecular structures, the halides are less readily hydrolysed and the oxides are less acidic.

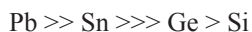
2. Hydrides and alkyls become less stable down the column.

The M–H and M–C mean bond enthalpies decrease down the column and consequently the hydrides and alkyls become thermodynamically less stable and kinetically more reactive. Carbon, of course, forms a very wide range of hydrides, silicon forms primarily SiH_4 and Si_2H_6 which are spontaneously inflammable. The higher silanes decompose readily to Si_2H_6 . The H–H bond polarities are opposite to C–H.

The silanes are strong reducing agents. The germanes GeH_4 , Ge_2H_6 , and Ge_3H_8 are less flammable than SiH_4 and are resistant to hydrolysis. SnH_4 decomposes at 0°C to Sn and PbH_4 is extremely unstable.

The organometallic derivatives of silicon and germanium are very similar. They are more reactive than the carbon analogues because the M–C bonds are more polar and the central atom can expand its coordination number more easily.

The rates of hydrolysis are in the order:



Organotin compounds more readily expand their coordination geometries and more readily form cationic species. Organolead compounds decompose readily at $100\text{--}200^\circ\text{C}$ by free radical processes.

3. Catenation

The element-element mean bond enthalpies decrease in the order



and therefore, the range of ring and polyhedral molecules diminishes down the group. Carbon not only forms an extensive range of chain and ring compounds, but also polyhedral molecules such as prismatic, C_6H_6 , and cubane, C_8H_8 . Analogous compounds are known for Si, Ge, and Sn if the hydrogens are replaced by bulky organic substituents.

However, few examples exist for Pb, which form compound containing the anionic Zintl polyhedral anion Pb_9^{4-} analogous to Sn_5^{2-} .

4. Multiply bonded compounds

The ability of the element to form multiple bonds diminishes in the series.



Because the $p_\pi - p_\pi$ overlap becomes less favourable. This has the following manifestations:

- For the element below C, the allotropes which would structurally resemble graphite which has a delocalised two dimensional π -system are not observed
- There are no simple analogues of ethane (C_2H_6) and ethyne (C_2H_2) and compounds of Si, Ge and Sn with multiple bonds may only be isolated when there are bulky organic substituents on the group 14 atoms. Furthermore, the Ge and Sn compounds do not have planar geometries.
- The analogues of CO_2 and CS_2 have polymeric structures rather than triatomic molecular geometries.
- The heavier elements do not form analogues of carbides with C_2^{2-} and C_3^{2-} multiply bonded ions.

5. The elements become progressively more metallic down the column. Carbon especially in its diamond and polyhedral forms is a typical non-metal, silicon is a semiconductor and tin and lead are typical. Although tin has one modification (grey tin) which is isostructural with Ge, Si, and diamond. Lead only occurs in close packed structure forms.

6. The oxides become more basic down the column.

CO_2 and SiO_2 are acidic oxides, SnO_2 is amphoteric and GeO_2 is mainly acidic with slight amphoteric character. The Si–O mean bond enthalpies are particularly large and this leads to a wide range of silicates. In general for the elements below carbon the M–O bonds are sufficiently strong that the oxides are susceptible to hydrolysis.

7. The typical coordination numbers increase down the group.

For carbon, the tetrahedral geometry predominates unless multiple bonds are formed. For the heavier elements the tetrahedral geometry is also widespread but the larger size of the central atoms leads to the formation of compounds with higher coordination numbers, e.g.,



SnCl_5^-	trigonal bipyramidal
$\text{SnPH}_2(\text{NO}_3)_2(\text{OPPH}_3)$	pentagonal bipyramidal –7–coordinate
$\text{Sn}(\text{NO}_3)_4$	dodecahedral 8-coordinate

The increased facility to achieve the higher co-ordination numbers is also reflected in the transition from molecular to polymeric, e.g., CF_4 , SiF_4 , and GeF_4 are molecular, whereas SnF_4 and PbF_4 have infinite lattices based on octahedral metal centres.

Of course all of these compounds with coordination numbers greater than four have electron configurations which exceed the octet rule. The MX_4 compounds ($\text{M} = \text{Si}, \text{Ge}, \text{Sn}, \text{Pb}$) are therefore Lewis acids, although they do not have empty p valence orbitals.

The higher coordination numbers are also reflected in the geometries of the oxo-anions:

CO_3^{2-}	SiO_4^{4-}	$\text{Ge}(\text{OH})_6^{2-}$	$\text{Sn}(\text{OH})_6^{2-}$	$\text{Pb}(\text{OH})_6^{2-}$
Trigonal planar	Tetrahedral			Octahedral

The chlorides of Ge, Sn, and Pb react with aqueous HCl to form the $[\text{MCl}_6]^{2-}$ anions, whereas SiCl_4 hydrolyses and CCl_4 is unreactive. However, SiF_4 does form $[\text{SiF}_6]^{2-}$ with HF.

8. The heavier elements form a wider range of complexes and more cationic complexes. Si, Ge, Sn, and Pb all form oxalato-complexes $[\text{M}(\text{ox})_3]^{2-}$ and cationic complexes $[\text{M}(\text{acac})_3]^+$.

9. *Ease of reduction of halides.*

Although C–Cl and Ge–Cl bonds in four valent compounds are reduced to the corresponding hydrides by Zn and HCl, Si–Cl and Sn–Cl bonds are not.

10. SiH_4 hydrolyses in the presence of trace amounts of base more readily than CH_4 , GeH_4 , and SnH_4 .

ONE CORRECT OPTION TYPE QUESTIONS

- Silica reacts with magnesium to form a magnesium compound (X). (X) reacts with dilute HCl and forms (Y). (Y) is:
 - MgO
 - MgCl₂
 - MgSiO
 - SiCl₄
- Silica is reacted with sodium carbonate. What is the gas liberated?
 - CO
 - O₂
 - CO₂
 - O₃
- Lead pencil contains
 - Pb
 - FeS
 - Graphite
 - PbS
- Which of the following halides is least stable and has doubtful existence?
 - Cl₄
 - GeI₄
 - SnI₄
 - PbI₄
- Water gas is produced by:
 - Saturating hydrogen with moisture
 - Passing steam through a red hot coke bed.
 - Mixing oxygen and hydrogen in the ratio of 1: 2.
 - Heating a mixture of CO₂ and CH₄ in petroleum refineries.
- Which of the following is similar to graphite?
 - B
 - BN
 - B₂H₆
 - B₄C
- Carbon burns in air and forms two oxides CO and CO₂. This shows that carbon has:
 - Two allotropic forms
 - Two oxidation states
 - Two isotopes
 - 4 electrons in valence shell
- Carbon dioxide is a gas but silica is a solid because:
 - Carbon dioxide is composed of discrete covalent CO₂ molecules whereas silica has continuous tetrahedral structure
 - CO₂ molecules are lighter than SiO₂ molecules
 - CO₂ is more acidic than SiO₂
 - Melting point silica is very high
- Man dies in an atmosphere of carbon monoxide, because it:
 - Combines with the O₂ present in the body to form CO₂
 - Reduces the organic matter of tissues
 - Combines with haemoglobin of blood, making it incapable of absorbing O₂
 - Dries up the blood
- Which statement is correct with respect to the property of the elements with increase in atomic number in the carbon family?
 - Their metallic character decreases
 - The stability of +2 oxidation state increases
 - Their ionization energy increases
 - Their atomic size decreases
- C and Si belong to IV group or gp. 14. The maximum co-ordination number of carbon in commonly occurring compounds is 4, whereas that of silicon is 6. This is due to:
 - Large size of silicon
 - Availability of vacant d-orbitals in silicon
 - More electropositive nature of silicon
 - Silicon being vulnerable to attack by nucleophilic reagents
- Which is not a characteristic property of carbon?
 - Catenation
 - Multiple bond formation
 - Availability of d-orbitals for bonding
 - Highest electronegativity in the group
- Red lead is:
 - PbO
 - Pb₃O₄
 - PbO₂
 - HgS
- Which gas is an essential constituent of almost all fuel gases?
 - CO₂
 - N₂
 - CO
 - H₂O
- Newly shaped glass articles when cooled suddenly become brittle, therefore these are cooled slowly, this process is known as:
 - Tempering
 - Annealing
 - Quenching
 - Galvanising
- CO₂ and N₂ are non-supporters of combustion. However, for putting out fires CO₂ is preferred over N₂ because CO₂:
 - Does not burn
 - Forms non-combustible products with burning substances
 - Is denser than nitrogen
 - Is a more reactive gas

17. The glass having smallest coefficient of thermal expansion is:
 (a) Soda lime glass (b) Soft glass
 (c) Safety glass (d) Pyrex glass
18. The C—X bond energy order and thermal stability for carbon tetra-halides is:
 (a) $\text{CF}_4 > \text{CCl}_4 > \text{CBr}_4 > \text{CI}_4$
 (b) $\text{CCl}_4 > \text{CBr}_4 > \text{CI}_4 > \text{CF}_4$
 (c) $\text{CI}_4 > \text{CBr}_4 > \text{CCl}_4 > \text{CF}_4$
 (d) None of these
19. If a person is injured by the short of a gun and all the pellets could not be removed, it may cause poisoning by:
 (a) Hg (b) Pb
 (c) Fe (d) Au
20. Least stable hydride is:
 (a) Methane (b) Plumbane
 (c) Silane (d) Stibine
21. The correct order of decreasing ionic nature of lead dihalides is:
 (a) $\text{PbF}_2 > \text{PbCl}_2 > \text{PbBr}_2 > \text{PbI}_2$
 (b) $\text{PbF}_2 > \text{PbBr}_2 > \text{PbCl}_2 > \text{PbI}_2$
 (c) $\text{PbF}_2 < \text{PbI}_2 < \text{PbBr}_2 < \text{PbCl}_2$
 (d) $\text{PbI}_2 > \text{PbBr}_2 > \text{PbCl}_2 > \text{PbF}_2$
22. Which gas is responsible for greenhouse effect?
 (a) CO_2 (b) SO_2
 (c) CO (d) SO_3
23. Carbon reacts with conc. H_2SO_4 to give:
 (a) $\text{CO}_2, \text{SO}_2, \text{H}_2\text{O}$
 (b) $\text{SO}_2, \text{H}_2\text{O}, \text{CO}$
 (c) CO, H_2O
 (d) $\text{CO}_2, \text{H}_2\text{O}$
24. In silica, each silicon atom is bonded to:
 (a) Two oxygen atom
 (b) Four oxygen atom
 (c) One Si and four oxygen atom
 (d) Octahedrally to six oxygen atom
25. Which one of the following bonds has the highest bond energy?
 (a) C—C (b) Si—Si
 (c) Ge—Ge (d) Sn—Sn
26. Which one of the following is opaque to ultraviolet rays?
 (a) Crookes glass (b) Flint glass
 (c) Jena glass (d) Pyrex glass
27. The correct order for m.pt. and b.pt. of IV gp. hydrides respectively are:
 1. $\text{CH}_4 < \text{SiH}_4 < \text{GeH}_4 < \text{SnH}_4$
 2. $\text{CH}_4 > \text{SiH}_4 < \text{GeH}_4 < \text{SnH}_4$
 3. $\text{SnH}_4 < \text{GeH}_4 < \text{SiH}_4 < \text{CH}_4$
 4. $\text{CH}_4 < \text{SiH}_4 > \text{GeH}_4 > \text{SnH}_4$
 (a) 2, 1 (b) 1, 2
 (c) 3, 4 (d) 4, 3
28. Lead pipes are readily corroded by
 (a) H_2SO_4 (b) HCl
 (c) CH_3COOH (d) Pure water
29. R_3SiCl on hydrolysis forms
 (a) R_3SiOH (b) $\text{R}_3\text{Si—O—SiR}_3$
 (c) $\text{R}_2\text{Si} = \text{O}$ (d) None
30. H_2SO_4 is not used for preparation of CO_2 from marble chips because
 (a) It does not react
 (b) Huge amount of heat is evolved
 (c) The reaction is vigorous
 (d) Calcium sulphate is sparingly soluble and gets deposited on marble chips and stops the reaction
31. CO_2 is used for extinguishing fire because
 (a) It has a relatively high critical temperature
 (b) In solid state, it is called dry ice
 (c) It is neither combustible nor a supporter of combustion
 (d) It is colorless gas
32. Tin plague is the
 (a) Conversion of stannous to stannic
 (b) Conversion of white tin to grey tin
 (c) Emission of sound while bending at tin rod
 (d) Atmospheric oxidation of tin
33. Tin cry refers to
 (a) Conversion of white to grey tin
 (b) Tin plating
 (c) Conversion of white tetrahedral tin to white rhombohedral tin
 (d) Emission of sound while bending a tin rod.
34. Lead is not affected by dil. HCl in cold because
 (a) Pb is less electronegative than H
 (b) PbO film is formed which resists chemical attack by acid
 (c) A protective coating of PbCl_2 is formed on Pb surface
 (d) PbO_2 film is always present on Pb surface, which resist chemical attack

5.50 Group14-Carbon Family

35. Producer gas, a fuel and also a source of nitrogen is obtained by:
- Passing a mixture of steam and air over incandescent coke
 - Spraying oil into hot retorts.
 - Restricted supply of air through a bed of incandescent coke.
 - Passing steam over incandescent coke.
36. Which of the following is most stable?
- Sn^{2+}
 - Ge^{2+}
 - Si^{2+}
 - Pb^{2+}
37. Ge(II) compounds are powerful reducing agents whereas Pb(IV) compounds are strong oxidants. It can be due to
- Pb is more electropositive than Ge
 - Ionization potential of lead is less than that of Ge
 - Ionic radii of Pb^{2+} and Pb^{4+} are larger than those of Ge^{2+} and Ge^{4+}
 - More pronounced inert pair effect in lead than in Ge
38. Which among CH_4 , SiH_4 , GeH_4 , SnH_4 is most volatile?
- CH_4
 - SiH_4
 - GeH_4
 - SnH_4
39. An oxide of lead which is used in lead storage batterie, in safety matches and is a powerful oxidizing agent is
- PbO
 - PbO_2
 - Pb_3O_4
 - $2\text{PbO} \cdot \text{PbO}_2$
40. Which of the following lead oxide in 'Sindhur'?
- PbO
 - PbO_2
 - Pb_2O_3
 - Pb_3O_4
41. The oxide which cannot act as reducing agent is
- SO_2
 - NO_2
 - CO_2
 - ClO_2
42. Fire extinguishers contain a bottle of H_2SO_4 and:
- CaCO_3
 - MgCO_3
 - NaHCO_3
 - any carbonate
43. A dry ice piece is composed of:
- solid He
 - solid CO_2
 - solid SO_2
 - solid C_6H_6
44. Graphite is a good conductor of electricity because it contains:
- bonded electrons
 - mobile electrons
 - strong C—C bonds
 - strong C=C bonds
45. The material used in solar cells contains:
- Si
 - Sn
 - Ti
 - Cs
46. 'Softening of lead' means:
- conversion of lead to PbO
 - conversion of lead to Pb_3O_4
 - removal of impurities(metallic) from lead
 - washing lead with HNO_3 followed by a dil. alkali solution
47. The carbide which gives propyne on hydrolysis:
- Al_4C_3
 - CaC_2
 - Fe_3C
 - Mg_2C_3
48. Which among the following carbide is methanide?
- Al_4C_3
 - CaC_2
 - Be_2C
 - SiC
49. CCl_4 is used as fire extinguisher because:
- its m.pt. is high
 - it forms covalent bond
 - its b.pt. is low
 - it gives incombustible vapours

COMPREHENSION BASED QUESTIONS

COMPREHENSION-I

When a mixture of sodium carbonate and calcium carbonate is fused with silica at 1500°C , a liquid consisting silicates of sodium and calcium is formed. When this liquid is cooled, it becomes viscous and eventually ceases to flow. It becomes solid and called glass. By varying the proportions of the three basic ingredients and by adding other substances, the properties of glass can be altered. An approximate formula for ordinary glass may be given as,



where $R = \text{Na}$ or K and $M = \text{Ca}$, Ba , Zn and Pb .

SiO_2 may be replaced by Al_2O_3 , B_2O_3 , P_2O_5 . Coloured glasses are obtained by adding certain metallic oxides or salts in the fused mass. Glass is attacked by HF and this property is used to make marking on the glass. This is known as etching.

The glass if cooled rapidly becomes brittle and fragile. The articles of glass are cooled neither very slowly nor very rapidly. The articles are cooled gradually. This process is termed annealing.

- Glass is described as a:
 - solid
 - liquid
 - super cooled liquid
 - colloidal solution
- Which acid cannot be stored in glass?
 - HF
 - HCl
 - HBr
 - HI
- Ordinary glass is:
 - sodium silicate
 - potassium silicate
 - calcium silicate
 - a mixture of sodium and calcium silicates with silica
- Annealing is best described as:
 - slow and gradual cooling
 - sudden and rapid cooling
 - cooling by water
 - very slow cooling
- A blue colour can be imparted to glass by use of:
 - Fe_2O_3
 - CoO
 - NiO
 - Cu_2O
- A special type of glass which contains cerium oxide and does not allow the passage of ultraviolet rays. This glass is used for making lenses. The glass is called:
 - flint glass
 - Crooke's glass
 - hard glass
 - pyrex glass

COMPREHENSION-II

Gaseous fuels have become very popular these days. The following are the advantages of gaseous fuels,

- They have high calorific value,
- They do not produce smoke and leave no ash after combustion.
- Gaseous fuels can flow through pipes and can be ignited at the moments notice at any place. No special devices are required for their combustion.

Coal gas is a good gaseous fuel as it contains 95% combustible gases such as hydrogen, methane, carbon monoxide, etc. It is obtained by destructive distillation of coal at about 1000°C . Water gas which is essentially a mixture of hydrogen and carbon monoxide is prepared by passing steam over incandescent coke. The reaction is however endothermic. Producer gas which possesses lowest calorific value is prepared by passing air over red

hot coke. It contains mainly nitrogen and carbon monoxide. Semiwater gas is a mixture of water gas and producer gas. Oil gas which is used in laboratories is obtained by cracking of kerosene. It is a mixture of hydrocarbons (saturated and unsaturated) mainly lower hydrocarbons. LPG (liquefied petroleum gas), which contains C_3 and C_4 hydrocarbons of the alkane and alkene series supplied in cylinders for domestic uses is very popular these days.

- Which one is the best fuel?
 - Wood
 - Coal
 - Kerosene
 - LPG
- Which gaseous fuel has highest calorific value?
 - Producer gas
 - Semiwater gas
 - Coal gas
 - Water gas
- Which of the following contains highest percentage of carbon monoxide?
 - Coal gas
 - Water gas
 - Producer gas
 - Natural gas
- Which of the following is a cracking process?
 - $\text{C}_3\text{H}_6 + \text{H}_2 \longrightarrow \text{C}_3\text{H}_8$
 - $n\text{C}_2\text{H}_4 \longrightarrow (\text{C}_2\text{H}_4)_n$
 - $\text{C}_{16}\text{H}_{34} \longrightarrow 6\text{CH}_4 + 2\text{C}_2\text{H}_4 + \text{C}_2\text{H}_2 + 4\text{C}$
 - $3\text{C}_2\text{H}_2 \longrightarrow \text{C}_6\text{H}_6$
- Which gas is essential constituent of most of the fuel gases?
 - CO
 - CO_2
 - N_2
 - H_2O

COMPREHENSION-III

The term carbide is generally applied to compounds in which carbon is bonded to the elements of lower or approximately same electronegativity. This definition excludes the compounds in which oxygen, sulphur, phosphorus nitrogen and halogens are united with carbon. Reactive metals (i.e., of gp1 and 2) form ionic carbides. They hydrolyse to liberate hydrocarbons. Most of them resembles with NaCl in crystal structure. Transition metals forms interstitial carbides.

- Consider the following Carbides:



Select the carbide which gives different product on hydrolysis, than other carbides:

- I
- II
- III
- IV

- What is the co-ordination number of Ca^{2+} in solid CaC_2 ?

5.52 Group 14-Carbon Family

- (a) 4 (b) 6
(c) 8 (d) 12
3. Select the methanides from compounds give below:
 I. Al_4C_3 II. Be_2C
 III. MgC_2 IV. Be_2C
 (a) I only (b) I and IV
 (c) I and II (d) I, II, III and IV
4. The conductance of transition metal is not much affected when it forms interstitial carbide because:
 (a) The carbide anion helps in conduction.
 (b) The carbon atoms occupy octahedral holes and so does not affect electrical conductivity.
 (c) The carbon atoms react with metal and liberate electrons.
 (d) The conduction is due to holes.

ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

1. Carbon dioxide is isostructural with:
 (a) HgCl_2 (b) SnCl_2
 (c) C_2H_2 (d) NO_2
2. CO_2 is isostructural with:
 (a) SnCl_2 (b) HgCl_2
 (c) SCl_2 (d) ZnI_2
3. Which of the given is/are amphoteric?
 (a) BeO (b) Ag_2O
 (c) CO_2 (d) SnO_2
4. Decomposition of oxalic acid in presence of conc. H_2SO_4
 (a) CO (b) CO_2
 (c) formic acid (d) H_2O
5. Which of the following is true about silicones?
 (a) They are formed by hydrolysis of R_2SiCl_2
 (b) They are polymer, made up of R_2SiO_2 units
 (c) They are made up of SiO_4^{4-} units
 (d) They are macromolecules
6. Which among the following statements are correct?
 (a) Aqua dag and oil dag are made up of graphite
 (b) Graphite reacts with conc. HNO_3 to form mellitic acid $\text{C}_6(\text{COOH})_6$
 (c) C_3O_2 is also toxic like CO
 (d) Zircon, ZrSiO_4 is a gemstone
7. The non-existence of PbI_4 is due to:
 (a) highly oxidizing nature of Pb^{4+}
 (b) highly reducing nature of Pb^{4+}
 (c) sufficiently large covalent character
 (d) highly reducing nature of Pb^{4+}
8. Which is not correct?
 (a) $\text{Ge}(\text{OH})_2$ is amphoteric
 (b) SnCl_4 is more stable than SnCl_2
 (c) Trisilylamine is pyramidal
 (d) GeCl_4 in HCl forms $\text{H}_2[\text{GeCl}_6]$
9. Carbon differs from the rest of the family members because of:
 (a) number of unpaired electrons in valence shell
 (b) small size
 (c) non-availability of vacant orbitals in valence shell
 (d) non-availability of d-orbitals in valence shell
10. Which of the following carbides on treatment with water give methane?
 (a) CaC_2 (b) Be_2C
 (c) Al_4C_3 (d) Mg_2C_3
11. Which of the metal oxides are reduced by CO ?
 (a) ZnO (b) Fe_2O_3
 (c) CaO (d) Na_2O

MATCH THE COLUMN TYPE QUESTIONS

1. Match the types of glass (in List-I) with their characteristics (in List-II).

List-I

- (a) Alumino-silicate glass

List-II

1. Very high transparency

- (b) calcium-alkali silicate glass
(c) lead glass
(d) soda glass

2. Cheap laboratory glass-ware
3. Optical glass
4. Domestic glass for window
5. Low coefficient of expansion

ASSERTION/REASON TYPE QUESTIONS

In each question below a statement S and an explanation E is given. Choose the correct answers from the codes a, b, c, d given for each question.

- (a) S is correct but E is wrong.
 (b) S is wrong but E is correct.
 (c) Both S and E are correct and E is correct explanation of S.
 (d) Both S and E are correct but E is not correct explanation of S.

1. **Statement:** In between TiCl_4 and SiCl_4 is hydrolysed.
Explanation: SiCl_4 is ionic and CCl_4 is covalent compound.

2. **Statement:** CCl_4 is hydrolysed by super heated steam in presence of iron or Cu.
Explanation: $\text{CCl}_4 + \text{H}_2\text{O} \xrightarrow{\text{(Steam)}} \text{COCl}_2 + 2\text{HCl}$

3. **Statement:** Pb^{2+} is more stable than Pb^{4+} .

Explanation: Inert pair effect in Pb is more predominant.

4. **Statement:** PbI_4 does not exist although PbCl_4 exist.

Explanation: Both Pb^{4+} and I^- are strong oxidant and strong reductant respectively.

5. **Statement:** Diamond is tetrahedral, graphite is planar and C_{60} has bucky ball structure.

Explanation: Carbon in diamond, graphite and C_{60} is sp^3 , sp^2 and sp hybridized respectively.

6. **Statement:** Lead leaves a black mark on paper.

Explanation: Lead is used for making lead pencils.

7. **Statement:** SiF_6^{2-} is known but SiCl_6^{2-} is not.

Explanation: Size of fluorine is small and its lone pair of electrons interacts with d-orbitals of silicon strongly.

RECENT YEARS' QUESTIONS OF IIT-JEE AND AIEEE

1. In $(\text{SiO}_4)^{4-}$ (tetrahedral structure), three oxygen atoms are shared in which of the following? (IIT 2005)

- (a) Sheet silicates (b) Three dimensional silicates
 (c) Linear silicate (d) Pyrosilicate

2. In silicon dioxide: [AIEEE 2005]

- (a) each silicon atom is surrounded by four oxygen atoms and each oxygen atom is bonded to two silicon atoms.
 (b) each silicon atom is surrounded by two oxygen atoms and each oxygen atom is bonded to two silicon atoms.
 (c) silicon atom is bonded to two oxygen atoms.
 (d) there are double bonds between silicon and oxygen atoms.

3. Pb and Sn are extracted from their chief ores by: [IIT 2004]

- (a) carbon reduction and self reduction
 (b) self reduction and carbon reduction
 (c) electrolysis and self reduction
 (d) self reduction and electrolysis

4. The soldiers of Napoleon army while at Alps during freezing winter suffered a serious problem as regards tin buttons of their uniforms. White metallic buttons of tin got converted to grey powder. This transformation is related to: [AIEEE 2004]

- (a) an interaction with nitrogen of the air with very low temperature
 (b) an interaction with water vapour present in humid air
 (c) a change in the partial pressure of oxygen present in the air
 (d) a change in the crystalline structure of tin

5. $(\text{Me})_2\text{SiCl}_2$ on hydrolysis will produce (IIT 2003)

- (a) $(\text{Me})_2\text{Si}(\text{OH})_2$ (b) $(\text{Me})_2\text{Si}=\text{O}$
 (c) $\text{Me}_2\text{SiCl}(\text{OH})$ (d) $-\text{[O-(Me)}_2\text{Si-O]}_n^-$

6. Name of the structure of silicates in which three oxygen atoms of $[\text{SiO}_4]^{4-}$ are shared. (IIT 2005)

- (a) Pyrosilicate (b) Linear chain silicate
 (c) Sheet silicate (d) Three dimensional silicate

7. Graphite is a soft solid lubricant extremely difficult to melt. The reason for this anomalous behaviour is that graphite (AIEEE 2003)

- (a) is an allotropic form of diamond
 (b) has molecules of variable molecular masses like polymers
 (c) has carbon atoms arranged in large plates of rings of strongly bound carbon atoms with weak interplate bonds
 (d) is a non-crystalline substance

5.54 Group14-Carbon Family

8. Glass is a (AIEEE 2003)
 (a) gel (b) super-cooled liquid
 (c) polymeric mixture (d) micro-crystalline solid
9. For making good quality mirrors, plates of float glass are used. These are obtained by floating molten glass over a liquid metal which does not solidify before glass. The metal used can be (AIEEE 2003)
 (a) tin (b) sodium
 (c) magnesium (d) mercury
10. Which of the following oxide is amphoteric? (AIEEE 2005)
 (a) CaO (b) CO₂
 (c) SiO₂ (d) SnO₂
11. An aqueous solution of a substance gives a white precipitate on treatment with dil. HCl, which dissolves on heating. When H₂S is passed through the hot acidic solution, a black precipitate is obtained. The substance is a: (IIT 2004)
 (a) Hg₂²⁺ salt (b) Cu²⁺ salt
 (c) Ag⁺ salt (d) Pb²⁺ salt
12. When PbO₂ reacts with conc. HNO₃, the gas evolved is: (IIT 2005)
 (a) NO₂ (b) O₂
 (c) N₂ (d) N₂O
13. In context with the industrial preparation of hydrogen from water gas (CO + H₂), which of the following is correct statement? (AIEEE 2008)
 (a) CO is oxidized to CO₂ with steam in the presence of a catalyst followed by absorption of CO₂ in alkali
 (b) CO and H₂ are fractionally separated using differences in their densities
 (c) CO is removed by absorption in aqueous Cu₂Cl₂
 (d) H₂ is removed through occlusion with Pd
14. Among the following substituted silanes, the one which will give rise to cross linked silicone polymer on hydrolysis is: (AIEEE 2008)
 (a) R₃SiCl (b) R₄Si
 (c) RSiCl₃ (d) R₂SiCl₂

ANSWER KEYS

ONE CORRECT OPTION TYPE QUESTIONS

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (b) | 2. (c) | 3. (c) | 4. (d) | 5. (b) | 6. (b) | 7. (b) | 8. (a) | 9. (c) | 10. (b) |
| 11. (b) | 12. (c) | 13. (b) | 14. (c) | 15. (b) | 16. (c) | 17. (d) | 18. (a) | 19. (b) | 20. (b) |
| 21. (a) | 22. (a) | 23. (a) | 24. (b) | 25. (a) | 26. (a) | 27. (a) | 28. (c) | 29. (b) | 30. (d) |
| 31. (c) | 32. (b) | 33. (d) | 34. (c) | 35. (c) | 36. (d) | 37. (d) | 38. (a) | 39. (b) | 40. (d) |
| 41. (c) | 42. (c) | 43. (b) | 44. (b) | 45. (a) | 46. (c) | 47. (d) | 48. (a) | 49. (d) | |

COMPREHENSION BASED QUESTIONS

Comprehension-I

1. (c) 2. (a) 3. (d) 4. (a) 5. (b) 6. (b)

Comprehension-II

1. (d) 2. (c) 3. (b) 4. (c) 5. (a)

Comprehension-III

1. (b) 2. (b) 3. (c) 4. (b)

ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

1. (a, c) 2. (b, d) 3. (a, d) 4. (a, b, d) 5. (a, b, d) 6. (a, b, d) 7. (ad) 8. (b, c) 9. (b, d)
 10. (b, c) 11. (a, b)

MATCH THE COLUMN TYPE QUESTIONS

1. (a)–(5); (b)–(4); (c)–(3); (d)–(2)

ASSERTION/REASON TYPE QUESTIONS

1. (a) 2. (d) 3. (c) 4. (c) 5. (a) 6. (c) 7. (a)

RECENT YEARS' QUESTIONS OF IIT-JEE AND AIEEE

- | | | | | | | | | | |
|---------|---------|---------|---------|--------|--------|--------|--------|--------|---------|
| 1. (a) | 2. (a) | 3. (b) | 4. (d) | 5. (d) | 6. (c) | 7. (c) | 8. (b) | 9. (d) | 10. (d) |
| 11. (d) | 12. (b) | 13. (a) | 14. (c) | | | | | | |

CHAPTER 6

Group 15- Nitrogen Family

INTRODUCTION

The Group 5B elements can complete the octet in chemical combination by gaining three electrons to form the 3-valent anion, by forming three covalent bonds, or by losing five electrons; the last possibility is ruled out on energetic grounds. Only nitrogen (and possibly phosphorus to a slight extent) forms the 3-valent ion and reactive metals are required for it to be possible; the N^{3-} ion is present in ionic nitrides, e.g., $(\text{Li}^+)_3\text{N}^{3-}$ and $(\text{Ca}^{2+})_3(\text{N}^{3-})_2$. The majority of compounds formed by this group of elements are covalent.

Antimony and bismuth can form the 3-valent cation X^{3+} (the inert-pair effect), the Sb^{3+} ion being present in $(\text{Sb}^{3+})_2(\text{SO}_4^{2-})_3$ and the Bi^{3+} ion in $\text{Bi}^{3+}(\text{F}^-)_3$ and $\text{Bi}^{3+}(\text{NO}_3^-)_3 \cdot 5\text{H}_2\text{O}$.

Because phosphorus, arsenic, antimony and bismuth have vacant d orbitals they are able to form 5-covalent compounds which are not possible for nitrogen, e.g., in the formation of PCl_5 , one of the 3s electrons of the phosphorus atom is promoted to the 3d level, giving five unpaired electrons for valency purposes. Nitrogen and phosphorus are non-metallic; metallic properties first become apparent with arsenic and become progressively more important for antimony and bismuth. Of these elements only nitrogen is able to multiple bond with itself, the triple bond being present in the nitrogen molecule, $\text{N} \equiv \text{N}$. Phosphorus, arsenic and antimony are allotropic, the less dense allotrope containing X_4 tetrahedra with $\text{X}-\text{X}-\text{X}$ bond angles of 60° which introduce considerable strain.

The structure of one allotropic form of phosphorus, arsenic and antimony.

The denser allotropic forms of phosphorus, arsenic and antimony are more stable and more 'metallic' (closer packing of the atoms). Bismuth also adopts a metallic structure.

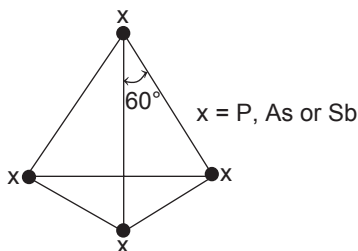


Fig. 6.1

6.2 Group15-Nitrogen Family

There is little resemblance between the chemistry of nitrogen and phosphorus and these two elements are considered separately; arsenic, antimony and bismuth are then treated together.

Red Alert

Some Noteworthy Points

- Out of the various allotropic forms of phosphorus, black phosphorus is a good conductor of electricity (similarity with graphite).
- Proteins, the building blocks of our body contain 16% of nitrogen in them.
- Phosphorescence. White phosphorus glows in dark because of its slow combustion in air. This property is known as phosphorescence (cold fire) or chemiluminescence. The energy of oxidation is not liberated as heat but it is emitted as light. Red phosphorus do not show this property.
- Radioactive phosphorus (P^{32}) is used in the treatment of leukemia (blood cancer).
- The disease caused by the constant touch with white Phosphorus is called Phossy Jaw.

Nitrogen(N), Phosphorus (P), Arsenic(As), Antimony(Sb) and Bismuth(Bi)

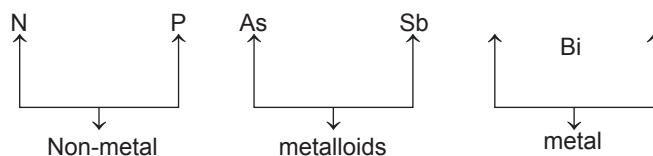


Fig. 6.2

Occurrence

Molecular nitrogen comprises 78% by volume of the atmosphere.

In the earth's crust, it occurs as sodium nitrate, NaNO_3 (called Chile saltpetre) and potassium nitrate (Indian saltpetre).

It is found in the form of proteins, in plants and animals.

Phosphorus occurs in minerals of the apatite family, $\text{Ca}_3(\text{PO}_4)_2$, CaX_2 ($X = \text{F}, \text{Cl}$ or OH) (e.g., fluorapatite $\text{Ca}_5(\text{PO}_4)_3\text{F}$, CaF_2) which are the main components of phosphate rocks. Phosphorus is an essential constituent of animal and plant matter.

It is present in bones as well as in living cells.

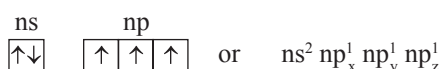
Phosphoproteins are present in milk and eggs.

Physical Properties

- 1. Electronic configuration:** The distribution of electrons in various energy shells of the atoms of these elements is given below:

Element	At. No.		Electronic configuration	Inert gas core
N	7	2, 5	$1s^2, 2s^2, 2p^3$	$[\text{He}] 2s^2, 2p^3$
P	15	2, 8, 5	$1s^2, 2s^2, 2p^6, 3s^2, 3p^3$	$[\text{Ne}] 3s^2, 3p^3$
As	33	2, 8, 18, 5	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^{10}, 4s^2, 4p^3$	$[\text{Ar}] 3d^{10}, 4s^2, 4p^3$
Sb	51	2, 8, 18, 18, 5	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^{10}, 4s^2, 4p^6, 4d^{10}, 5s^2, 5p^3$	$[\text{Kr}] 4d^{10}, 5s^2, 5p^3$
Bi	83	2, 8, 18, 32, 18, 5	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^{10}, 4s^2, 4p^6, 4d^{10}, 4f^{14}, 5s^2, 5p^6, 5d^{10}, 6s^2, 6p^3$	$[\text{Xe}] 4f^{14}, 5d^{10}, 6s^2, 6p^3$

The configurations show that these elements have same number of electrons in the valency shell, i.e., 5 electrons in the valency shell, two of which are in s-orbital and remaining three in three p-orbitals. Thus, they have $ns^2 np^3$ configuration, i.e., 5-orbital is paired and three p-orbitals are unpaired.



The penultimate shell, in nitrogen contains 2 electrons (saturated), in phosphorus contains 8 electrons (saturated), in arsenic contains 18 electrons (saturated) while in antimony and bismuth contain 18 electrons (unsaturated) each. This shows why nitrogen differs from phosphorus in some respects and these two differ from the remaining elements of this group.

In accordance with the Hund's rule, electronic configurations involving fully filled or exactly half filled orbitals are the most stable, the elements of group VA, having exactly half filled orbitals, are also fairly stable and not so reactive. Nitrogen behaves as a noble element under ordinary conditions.

2. Similarities and gradation in physical properties:

- (a) **Metallic and non-metallic character:** Metallic nature increases as the atomic number increases. N and P are purely non-metals while Sb and Bi are metals. Arsenic behaves as a metalloid.

N	P	As	Sb	Bi
Non-metals		Metalloid	Metals	
Metallic nature increases →				

N and P combine with electropositive elements and form nitrides and phosphides, respectively. Nitrides are more stable than phosphides. Bi and Sb form alloys with other metals. Thus, N and P are non-metals while Sb and Bi are metals.

- (b) **Physical state:** There is a gradual change in physical state of these elements under ordinary conditions. Nitrogen, the first element, is a gas while phosphorus, the second member, though a solid, can pass readily into vapour state. It is a soft waxy solid. The remaining elements are solids. These are hard and possess metallic lustre.
- (c) **Atomic radii:** Atomic radii increase with the increase of atomic number.

Element	N	P	As	Sb	Bi
Atomic radii (Å)	0.74	1.10	1.21	1.41	1.52

The difference between N and P is comparatively high, this is due to effective shielding of s- and p-electrons of the penultimate shell in phosphorus.

- (d) **Ionization potential:** The ionization potential decreases regularly on descending the group as atomic radii increases.

Element	N	P	As	Sb	Bi
Ionization potential (Kcal/mol)	336.0	253.9	231.0	199.1	184.9

The ionization potential of nitrogen is very high; this is due to its small size. The difference between N and P is comparatively high but the difference afterwards between two consecutive members is small. This is due to less shielding effect of d-electrons in As and Sb, d and f-electrons in Bi.

Worked-out Example

1. The element with the highest first ionization potential is

(IIT 1982)

- (a) boron (b) carbon (c) nitrogen (d) oxygen

Ans. (c)

- (e) **Electronegativity:** Electronegativity decreases gradually on descending the group from N to Bi.

Element	N	P	As	Sb	Bi
Electronegativity	3.0	2.1	2.0	1.9	1.9

Nitrogen is most electronegative element, i.e., typical non-metal.

- (f) **Density:** It increases gradually on descending group.

Element	N	P	As	Sb	Bi
Density (g/mL)	0.809	1.823	5.73	6.62	9.78

6.4 Group15-Nitrogen Family

- (g) **Allotropy:** All the elements except bismuth show allotropy. Nitrogen exists in two solid allotropic forms α -nitrogen with cubic crystalline structure and β -nitrogen with hexagonal crystalline structure. The transition temperature is -238.5°C . Phosphorus exists in number of allotropic forms such as white, red, scarlet, cc-black, p-black and violet. Arsenic is known in three allotropic forms, i.e., grey, yellow and black Antimony also exists in three forms, i.e., yellow, black and explosive.
- (h) **Catenation:** N, P and As exhibit the property catenation but this property is much less than IVA element No nitrogen compound containing nitrogen chain with more than two atoms is stable although compounds having a 2 chain of eight nitrogen atoms have been prepared. The bond energies of C—C, N—N, P—P and As—As linkages are 85.0, 48.0 39.0 and 35.0 kcal/mole. Thus, the tendency of catenation decreases from N to P and from P to As. P and As can form chain of only two atoms.
- (i) **Atomicity:** Nitrogen is diatomic gaseous molecule at ordinary temperature. The existence of nitrogen as diatomic molecule is due to its ability to form multiple bonds. The molecule has one sigma and two π -bonds.



The triple bond in nitrogen molecule is very stable as the dissociation energy is very high (225 kcal/mole). **It is, therefore, inert under ordinary conditions.**

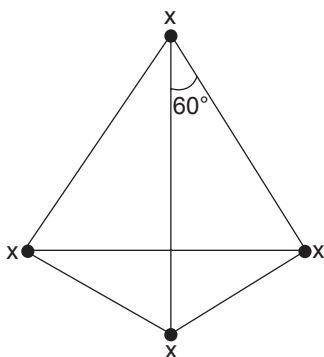


Fig. 6.3 X=P, As or Sb

Phosphorus, arsenic and antimony all exist as discrete tetramictetrahedral molecules, viz. P_4 , As_4 and Sb_4 (Fig.) as these are not capable of forming multiple bonds due to repulsion between non-bonded electrons of the inner core. The angle between X—X—X is 60° . The $p\pi-p\pi$ bonding in phosphorus, arsenic, etc., is therefore not possible. Elements of this group form giant lattice in which bonding changes from covalent to metallic, e.g., violet and red phosphorus has layer structure involving covalent bonding. Bismuth on the other hand involve purely metallic bonding. Metallic layer structure of bismuth is given below:

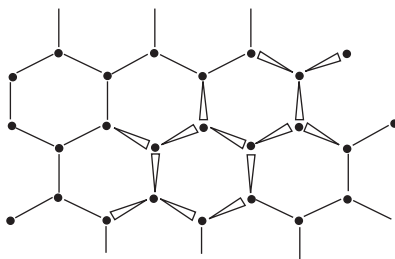


Fig. 6.4

- (j) **Oxidation states:** The elements of this group have five electrons in their outer shell. They exhibit a maximum oxidation state of +5 towards oxygen by using all the five electrons of outer shell. The tendency of the pair of ns electrons to remain inert (the inert pair effect) increases with increase of atomic number. Thus, only the p-electrons are used in bonding, i.e., +3 oxidation state is observed. +3 and +5 oxidation states of these elements are

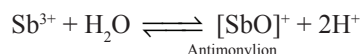
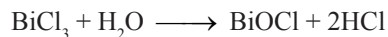
observed with halogens and sulphur. The stability of +3 oxidation state increases and that of +5 oxidation state decreases on moving down from N to Bi.

Nitrogen and phosphorus generally exhibit -3 oxidation state due to high electronegativity and small size. Nitrogen forms nitride ion (N^{3-}) with highly electropositive elements. Phosphorus also forms phosphide ion (P^{3-}) to some extent. The tendency to show -3 oxidation state decreases from nitrogen to bismuth.

Note: Nitrogen besides -3, +3 and +5 oxidation states, exhibits a large number of oxidation states from -3 to +5; -3 in NH_3 and nitrides (Li_3N , Ca_3N_2 , AlN), -2 in NH_2NH_2 , -1 in NH_2OH , 0 in N_2 , +1 in N_2O , +2 in NO , +3 in N_2O_3 and HNO_2 , +4 in NO_2 and +5 in HNO_3 and N_2O_5 .

(k) Valency and nature of bonding: There are five electrons present in the valence shell of these elements. To achieve stable configuration, they require three more electrons which is not easy to accept as to form trinegative ion (M^{3-}) due to energy considerations. Such an ion is formed by nitrogen which is smallest and most electronegative element of the family when it combines with highly electropositive metals. This tendency is much less in phosphorus. These elements easily attain stable configuration by sharing three p-electrons, i.e., by forming three covalent bonds. Except nitrogen, all other elements have d-orbitals in the valence shell. As a result, one of the two ns-electrons may be promoted to nd-orbitals to have five unpaired orbitals in the valence shell. These can be used to form five covalent bonds. This is not possible in nitrogen as it has no d-orbitals (nitrogen never shows pentavalency). However, nitrogen can show a maximum covalency of 4 when it donates the ns electron pair to Lewis acids.

Sb and Bi can lose three electrons forming M^{3+} ions but ionization energy is too high for the other elements. Both SbF_3 and BiF_3 , exist as ionic solids. The M^{3+} ions are not very stable in solution. They can exist in fairly strong acid solution but are rapidly hydrolyzed in water to give the antimony oxide ion or bismuth oxide ion, SbO^+ and BiO^+ . The change is reversed by adding 5M HCl.



Worked-out Example

2. There are ____ π bonds in a nitrogen molecule.

(IIT 1982)

Ans. Two

Thermal and electrical conductivity: Both these properties increase on moving down the group because delocalization of electrons increases from nitrogen to bismuth.

N, P
Poor conductor

As
Semiconductor

Sb, Bi
Good conductor

Occurrence: Nitrogen is found in free as well as in combined state. Nitrogen forms 75% by mass and 78% by volume of the atmosphere. In the combined state, it occurs as nitrates, proteins, amino acids, etc. Phosphorus does not occur free in nature. Phosphorus is the eleventh most abundant element in the earth's crust. Phosphorus is essential for life of both plants and animals. About 60% of bones and teeth are $\text{Ca}_3(\text{PO}_4)_2$ or $[\text{3}(\text{Ca}_3(\text{PO}_4)_2) \cdot \text{CaF}_2]$ and an average person has 3.5 kg of calcium phosphate in his body. Phosphorus is present, in DNA, RNA, ATP, ADP, etc. It is also found as phosphates in nature. As, Sb and Bi are not abundant. They occur as sulphides or oxides.

Chemical properties

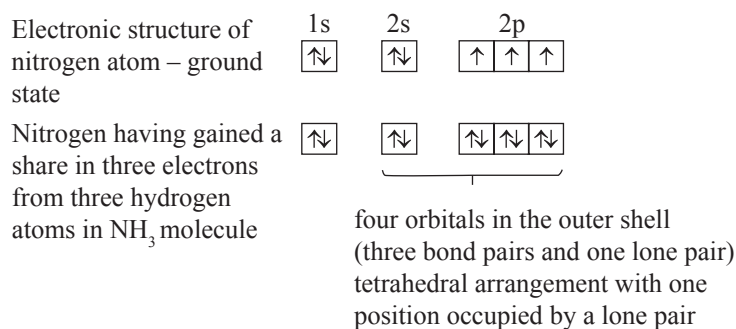
(i) **Hydrides of Group 15** – MH_3 type and have a pyramidal structure.

NH_3	PH_3	AsH_3	SbH_3	BiH_3
Bond Angle	decreases →			
Basic Character	decreases →			
Stability	decreases →			
Reducing Nature	increases →			

Structure of the hydrides

The structure of ammonia may either be described as pyramidal, or tetrahedral with one position occupied by a lone pair. This shape is predicted using the VSEPR theory since there are four electron pairs in the outer shell. These comprise three bonding pairs and one lone pair. The repulsion between a lone pair and a bond pair of electrons always exceeds that between two bond pairs. Thus the bond angles are reduced from $109^\circ 27'$ to $107^\circ 48'$, and the regular tetrahedral shape is slightly distorted.

The hydrides PH_3 , AsH_3 and SbH_3 would be expected to be similar. However, the bond pairs of electrons are much further away from the central atom than they are in NH_3 . Thus the lone pair causes even greater distortion in PH_3 , AsH_3 and SbH_3 . The bond angle decreases to $91^\circ 18'$ (Table). These bond angles suggest that in PH_3 , AsH_3 , SbH_3 and BiH_3 the orbitals used for bonding are close to pure p orbitals.



The melting and boiling points of the hydrides increase from PH_3 through AsH_3 to SbH_3 . The values for NH_3 seem out of line with this trend: one might have expected the boiling point of NH_3 to be -110°C or -120°C . The reason why NH_3 has a higher boiling point and is much less volatile than expected is that it is hydrogen bonded in the liquid state. The other hydrides do not form hydrogen bonds.

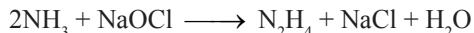
These hydrides are strong reducing agents and react with solutions of metal ions to give phosphides, arsenides and stibnides. They are flammable and extremely poisonous.

Table 6.1 Some Properties of the Hydrides

	<i>m.p.</i> ($^\circ\text{C}$)	<i>b.p.</i> ($^\circ\text{C}$)	<i>Bond energy</i> (kJ mol^{-1})	<i>Bond angle</i>	<i>Bond length</i> ($\text{\AA}$)
NH_3	-77.8	-34.5	N-H = 389	H-N-H = $107^\circ 48'$	1.017
PH_3	-133.5	-87.5	P-H = 318	H-O-H = $93^\circ 36'$	1.419
AsH_3	-116.3	-62.4	As - H = 247	H-As -H = $91^\circ 48'$	1.519
SbH_3	-88	-18.4	Sb - H = 225	H-Sb-H = $91^\circ 18'$	1.707

Some Noteworthy Points

- Disphosphine (P_2H_4) and hydrazine (N_2H_4) are other two important hydrides. Hydrazine a strong reducing agent, is used in organic synthesis and rocket fuels and is prepared as follows:



- Phosphine is poisonous and does not form any complexes.
- Phosphine forms vortex rings of P_2O_5 in the form of white smoke when it comes in contact with air due to combustion. This is due to impurities of diphosphine (P_2H_4).
- Phosphine is used to prepare smoke screens in warfare. Calcium phosphide reacts with water to form phosphine which burns to give clouds of P_2O_5 which acts as smoke screens.
- Liquor ammonia is a concentrated solution of NH_3 in water.
- Ammonia will not burn in air, but burns in pure O_2 with a yellowish flame to produce N_2 and H_2O .

(ii) Oxides of Group 15 – M_2O_3 and M_2O_5 type

(a) Stability of higher oxidation state decreases with increasing atomic number.

(b) Basicity of the oxides, that is, the metallic behaviour increases atomic number.

Except N_2O and NO (which are neutral oxides) a distinct trend from acidic to basic nature is observed. The reason for such a trend is that N^{3+} is much smaller than Bi^{3+} , so it interacts with water more strongly and thus results in acidic properties.

- In the oxides of a particular element, the acidic nature increases as the percentage of oxygen increases or as the oxidation state increases. For example, N_2O_5 is most acidic while N_2O_3 is less acidic, similarly P_2O_5 is more acidic than P_2O_3 and P_2O_4 .

- Stability of oxides of higher oxidation states decreases with increasing atomic number.

(c) The reluctance of P, As, Sb and Bi to enter into $p\pi-p\pi$ multiple bonding leads to cage structure for their oxides and they exist as the dimmers, M_4O_6 and M_4O_{10} .

(iii) Oxyacids of Group 15 –

<i>N</i>	<i>P</i>	<i>As</i>	<i>Sb</i>	<i>Bi</i>
HNO_2 (nitrous acid)	H_3PO_2 (hypophosphorus acid)	H_3AsO_3 (arsenous acid)	H_3SbO_3	$HBiO_3$ (meta-bismuthic acid)
HNO_3 (nitric acid)	H_3PO_3 (phosphorus acid)	H_3AsO_4 (arsenic acid)		
	$H_4P_2O_4$ (hypophosphoric acid)			
	H_3PO_4 (orthophosphoric acid)			
	HPO_3 (metaphosphoric acid)			
	$H_4P_2O_7$ (pyrophosphoric acid)			

(iv) Halides of group 15– except nitrogen all element of this group form MX_3 and MX_5 type halide. Nitrogen forms only MX_3 type.

- Pentahalide are more covalent than trihalides.
- N does not form MX_5 type halide due to non availability of d-orbitals in its valence.
- In case of nitrogen, only NF_3 is known to be stable due to:
 - (i) Low polarity of the N–X bond.
 - (ii) Weakness of the N–X bond due to large size difference between nitrogen and halogen atoms (Cl, Br and I).
- PF_5 is molecular in both the gaseous and solid states and has a trigonal bipyramidal structure but PCl_5 is molecular in the gas and liquid phases but in the solid state exists as $[PCl_4]^+[PCl_6]^-$ containing tetra and hexa-coordinated phosphorus species.

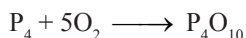
Some Noteworthy Points

- Down the group the basic character of oxides increases due to increase in metallic character.
- Unlike other oxides of nitrogen, nitrous oxide, N_2O is not poisonous. It has a faint sweet smell. It is considerably soluble in fats, a property which has been used in making self-whipping cream. Cream is packaged with N_2O under pressure

6.8 Group15-Nitrogen Family

to increase its solubility. When the pressure is released, the N_2O escapes to form tiny bubbles which produces whipped cream.

- All oxides of nitrogen (except NO and N_2O) and phosphorus are acidic in nature.
- NO and NO_2 are paramagnetic. It is due to the presence of unpaired electron in their molecules.
- P_4S_3 is used in strike anywhere matches.
- Flowers of phosphorus. P_4O_{10} can be prepared by burning phosphorus in excess of air and collected as snowy powder called flowers of phosphorus.



- As_4O_6 is called white arsenic and is a poison.
- As oxidation state of an element increases, the acidic nature of its oxide also increases e.g., As_2O_5 is more acidic than As_2O_3 .
- NCl_3 is an explosive compound.
- NO_2 is called mixed anhydride because it gives a mixture of HNO_2 and HNO_3 upon dissolution in water.
- Hypophosphoric acid, $\text{H}_4\text{P}_2\text{O}_6$ and pyrophosphoric acid, $\text{H}_4\text{P}_2\text{O}_7$ are both tetrabasic acids, yet these two acids form only two series of salts each.
- Cyclic polymetaphosphates can be represented by the general formula $(\text{HPO}_3)_n$.
- SbF_3 is called Swarts reagent. It is used as a fluorinating agent for various compounds of non-metals.
- Nitric acid, HNO_3 is called aqua fortis meaning strong water because it attacks nearly all the metals.
- Fuming nitric acid contains dissolved NO_2 in conc. HNO_3 . It is obtained by distilling conc. HNO_3 with a little starch. Some of the nitric acid is reduced by starch to NO_2 which dissolve in the remaining acid to form fuming nitric acid.
- The yellow colour of commercial nitric acid is due to the presence of dissolved nitrogen dioxide, NO_2 . This colour can be removed by warming the acid to $60\text{--}80^\circ\text{C}$ and bubbling dry air through it.
- Nitric acid forms a constant boiling mixture with water (b.p. 121°C) which contains 68% HNO_3 . As such, nitric acid cannot be concentrated beyond 68% by heating.

Red Alert

Despite the presence of a lone pair of electrons on nitrogen; NF_3 does not act as a Lewis base. There is no known compound in which it donates a pair of electrons to other reagents.

Explanation. The apparent lack of basic properties of NF_3 is because of the high electronegativity of fluorine atoms. The fluorine atoms in NF_3 draw the electron away from the nitrogen atom and thus making NF_3 a poor electron donor.

Red Alert

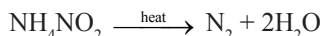
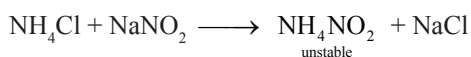
- Solid PCl_5 is an ionic compound consisting of $[\text{PCl}_4]^+ [\text{PCl}_6]^-$. $[\text{PCl}_4]^+$ has a tetrahedral structure, while $[\text{PCl}_6]^-$ has an octahedral structure.
- Since, PCl_5 reacts readily with moisture it is kept in well stoppered bottled.
- PI_5 does not exist due to large size of I atoms and lesser electronegativity difference between phosphorus and iodine.
- Down the group, the tendency to form pentahalides decreases due to inert pair effects. E.g., BiF_5 does not exist.

Nitrogen

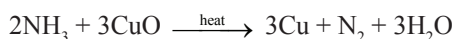
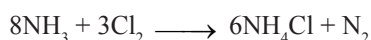
Preparation

Nitrogen can be obtained from the following sources.

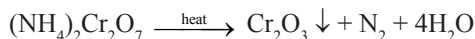
- (a) By heating a mixture of NH_4Cl and NaNO_2 (Laboratory method)



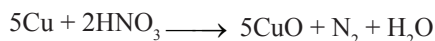
- (b) By oxidation of ammonia



(c) By heating ammonium dichromate



(d) By passing vapours of HNO_3 on strongly heated copper



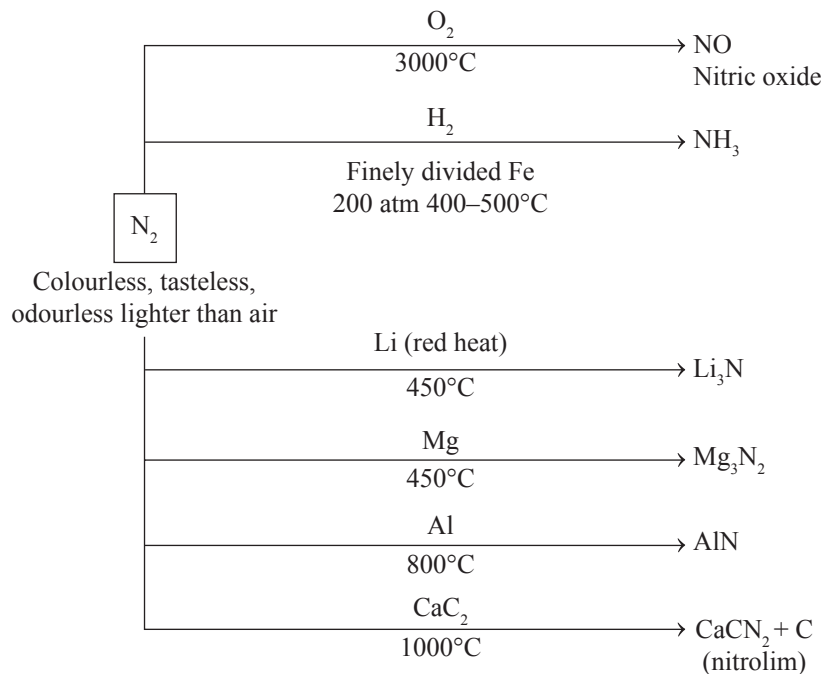
From Air: Commercially, nitrogen is obtained by liquifying air followed by fractional evaporation. Nitrogen having a lower boiling point (77.3 K), boils off more readily than the less volatile oxygen (boiling point 90.04 K).

Worked-out Example

3. What is active nitrogen?

Ans. When electric discharge is passed through molecular nitrogen at very low pressure (0.1–2 mm). The glowing gas is produced which is known as active nitrogen due to its higher chemical reactivity. The high reactivity is due to presence of ground state $\text{N}(4s)$ atoms.

Properties

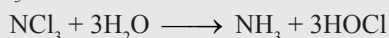


Worked-out Examples

4. NF_3 is not hydrolyzed but NCl_3 is readily hydrolyzed to show explosion. Explain.

Ans. Nitrogen tri-fluoride is not hydrolyzed because neither nitrogen nor fluorine possess vacant d-orbitals to form extra bond with lone pair of electron on oxygen of water molecule. However,

NCl_3 has vacant d-orbitals on Cl atom and hydrolyses violently to show explosion.



5. The electronegativity of both nitrogen and chlorine is almost the same. But nitrogen is inert at room temperature whereas chlorine is highly reactive. Why?

Ans. Nitrogen molecule consists of a strong triple bond whereas a chlorine molecule has only a single bond.

6.10 Group 15-Nitrogen Family

Uses of Dinitrogen

The main use of dinitrogen is in the manufacture of ammonia and other industrial chemicals containing nitrogen, e.g., calcium cyanamide.

It also finds use where an inert atmosphere is required (e.g., in iron and steel industry, inert diluent for reactive chemicals).

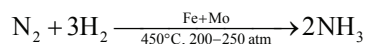
Liquid dinitrogen is used as a refrigerant to preserve biological materials, food items and in cryosurgery.

Ammonia

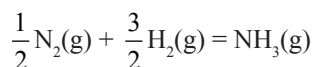
Elemental nitrogen exists as a diatomic molecule whose bond dissociation energy, 225 kcal, is second in magnitude only to that of carbon monoxide, nitrogen and carbon monoxide are isoelectronic and both possess a triple bond. Although carbon monoxide is a moderately reactive molecule, nitrogen is quite inert, particularly at temperatures near 300°K. In some cases, this inert nature is due to thermodynamic considerations. For example, all the nitrogen oxides have positive standard free energies of formation, and thus are intrinsically unstable with respect to nitrogen and oxygen. In other instances, however, the inert character of nitrogen is a result of kinetic factors, for its reactions with many reagents are quite slow. The formation of ammonia from hydrogen and nitrogen is such an example, and it is worth considering this reaction in some detail.

Preparation

By Haber's process (From air)



The standard free energy of formation of ammonia is -3.98 kcal/mole at 25°C , and thus the equilibrium constant of the reaction.



is given at 298°K by

$$K = e^{-\Delta G^\circ/RT} = 8.3 \times 10^2,$$

where the concentration units are atmospheres. This equilibrium constant is quite favorable to the synthesis, and ammonia should be formed in good yield at room temperature if equilibrium could be reached. The reaction is immeasurably slow, however, and no ammonia is formed when nitrogen and hydrogen are mixed at 25°C .

At elevated temperatures and in the presence of iron catalysts, the rate of reaction between nitrogen and hydrogen is large enough to make the ammonia synthesis practical. There is still a difficulty, for the standard enthalpy change for the reaction is -11.04 kcal at 298°K and is -13.3 kcal at 450°C , the temperature at which the reaction rate is conveniently large. Since the reaction is exothermic, the equilibrium constant at the elevated temperature is smaller than at 298°K , and it is important to know whether this seriously affects the maximum yield of ammonia. The value of the equilibrium constant at 723°K might be calculated from the expression

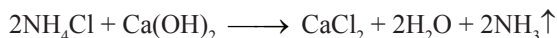
$$\ln \frac{K_2}{K_1} = -\frac{\Delta H^\circ}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right) \quad (1)$$

but a certain amount of care is necessary. Equation (1) is based on the assumption that ΔH° is independent of temperature, and this is not true for the ammonia synthesis. We can calculate a *fair* approximation to the equilibrium constant at 723°K , however, if we use for ΔH° the average of the values at 298°K and 723°K , or -12.2 kcal/mole. Then we have

$$\begin{aligned} 2.3 \log \frac{K_{723}}{8.3 \times 10^2} &= \frac{12,200}{1.98} \left(\frac{1}{723} - \frac{1}{298} \right), \\ \log K_{723} &= -2.35, \\ K_{723} &= 4.5 \times 10^{-3} \end{aligned}$$

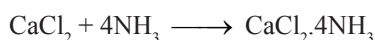
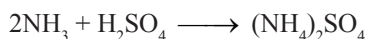
Because of the approximate treatment of the temperature variation of this answer is not particularly accurate, and the experimentally measured value of K_{723} is 6.5×10^{-3} . Nevertheless, the approximate value would be good enough to tell us that the equilibrium yield of ammonia at 723°K is much smaller than at 298°K. This unfavorable equilibrium constant is counteracted somewhat by carrying out the reaction at a high total pressure. While this does not change the equilibrium constant, it does result in a higher percentage conversion of nitrogen and hydrogen to ammonia and makes this direct synthesis an important industrial process.

(i) By heating ammonium salt with a base, for example, *slaked lime* (Laboratory method).

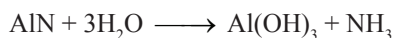
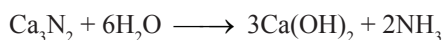


The gas so formed is dried with quick lime CaO

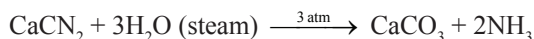
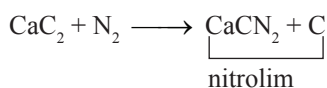
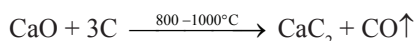
Note: Since NH_3 reacts with other dehydrating agents like H_2SO_4 , CaCl_2 and P_2O_5 , these cannot be used during its preparation.



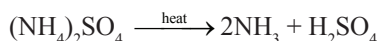
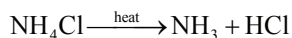
(ii) By the action of water on nitrides of Ca, Mg and Al.



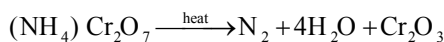
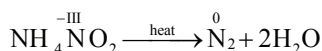
(iii) Cyanamide process (Commercial method)



(iv) Ammonium salts decompose quite readily on heating. If the anion is not particularly oxidizing (e.g., Cl^- , CO_3^{2-} or SO_4^{2-}) then ammonia is evolved:



If the anion is more oxidizing (e.g., NO_2^- , NO_3^- , ClO_4^- , $\text{Cr}_2\text{O}_7^{2-}$) then NH_4^+ is oxidized to N_2 or N_2O .



Worked-out Examples

6. Why is high pressure used in the manufacture of ammonia in Haber process?

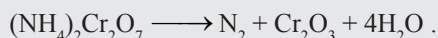
Ans. Formation of NH_3 shows a decrease in volume in forward direction.

7. NF_3 is stable but NCl_3 and NI_3 are explosives. Explain.

Ans. NF_3 is not hydrolyzed because neither N nor F can expand their octet by using d-orbitals. On the other hand NCl_3 and NI_3 hydrolyze violently because Cl and I can expand their octet by using d-orbitals.

8. Which chromium compound on heating gives N_2 and Cr_2O_3 ?

Ans. Ammonium dichromate,



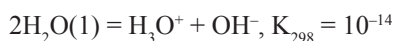
9. Name a compound of N, H, C and O used as fertilizer,

Ans. Urea, NH_2CONH_2

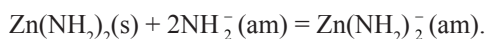
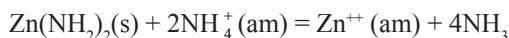
Properties of Ammonia

Ammonia is a colorless gas with an exceedingly pungent odor. It condenses to a liquid which has a normal boiling temperature of -33°C . Both the boiling point and enthalpy of vaporization' (5.64 kcal/ mole) are abnormally high for a substance of this molecular weight, because of hydrogen bonding.

Liquid ammonia is in some ways similar to liquid water. Salts dissolve in ammonia to form conducting solutions, but solubilities are usually smaller in ammonia than in water. Exceptions to this generalization are the salts which contain cations which form stable ammonia complexes. Thus the silver halides, which are very sparingly soluble in water, are quite soluble in ammonia, because of the formation of the very stable $\text{Ag}(\text{NH}_3)_2^+$ complex ion. Liquid ammonia undergoes autoionization, as does water, but to a much smaller degree;



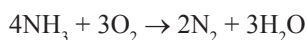
We see that in the liquid ammonia solvent system, NH_4^+ is the acid analogous to H_3O^+ , and NH_2^- is the base analogous to OH^- . The analogy extends to amphoteric behaviour as well. Thus, just as zinc hydroxide dissolves in either strong aqueous acid or strong base, zinc amide, $\text{Zn}(\text{NH}_2)_2$, reacts with excess NH_4^+ or NH_2^- to dissolve in liquid ammonia:



Perhaps the most remarkable characteristic of liquid ammonia is its ability to dissolve the alkali metals to form deep blue solutions of high electrical conductivity. The solubilities of the alkali metals range from 10 to 20 molal, depending on the temperature and the metal. In dilute (~ 0.01 M) solutions, the principal dissolved species are believed to be the alkali metal ions and independent electrons which are loosely trapped in solvent cages by electrostatic interaction with the dipole moment of NH_3 . The blue color of these solutions is attributed to transitions of the quasi-free electrons between energy levels defined by their interaction with the solvent cage. As might be expected, these solutions of alkali metals in liquid ammonia are excellent reducing agents, and are frequently used for this purpose in preparative organic and inorganic chemistry.

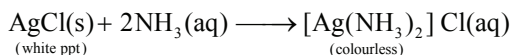
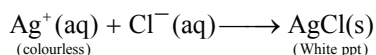
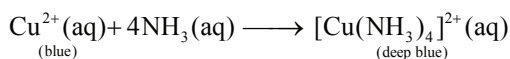
In nutshell:

NH_3 burns in dioxygen with a pale yellow flame:

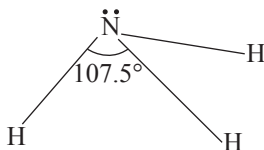


The NH_3 evolved may be detected:

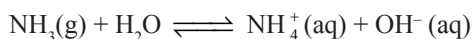
1. By its characteristic smell.
2. By turning moist litmus paper blue.
3. By forming dense white clouds of NH_4Cl with the stopper from a bottle of HCl .
4. By forming a yellow-orange-brown precipitate with Nessler's solution.
 - Ammonia is a colourless gas with a pungent odour.
 - It's freezing and boiling points are 198.4 and 239.7 K respectively.
 - In the solid and liquid states, it is associated through hydrogen bonds as in the case of water due to which it has higher melting and boiling points than expected on the basis of its molecular mass.
 - The presence of a lone pair of electrons on the nitrogen atom of the ammonia molecule makes it a Lewis base. It donates the electron pair and forms linkage with metal ions and the formation of such complex compounds finds applications in detection of metal ions such as Cu^{2+} , Ag^+ .



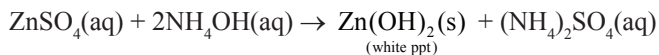
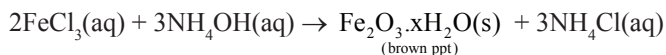
- The ammonia molecule is trigonal pyramidal with the nitrogen atom at the apex. It has three bond pairs and one lone pair of electrons as shown in the structure.



- Ammonia gas is highly soluble in water. It's aqueous solution is weakly basic due to the formation of OH^- ions.



- It forms ammonium salts with acids, e.g., NH_4Cl , $(\text{NH}_4)_2\text{SO}_4$, etc.
As a weak base, it precipitates the hydroxides of many metals from their salt solutions. For example,



Worked-out Examples

10. A bottle of liquor ammonia should be cooled before opening. Explain.

Ans. Liquor ammonia possesses high vapour pressure at room temperature and thus before opening a bottle of liquor ammonia, it should be cooled to lower the pressure of NH_3 inside the bottle, otherwise NH_3 will bump out.

11. Conc. H_2SO_4 , anhydrous CaCl_2 or P_4O_{10} cannot be used as drying agent for ammonia, why.

Ans. Ammonia is basic in nature and cannot be dried by dehydrating agents of acidic nature (i.e., H_2SO_4 or P_2O_5) as they react with NH_3 . Also CaCl_2 forms addition compound with NH_3 .

12. Water has the highest boiling point among NH_3 , PH_3 , HCl and H_2O . Why?

Ans. Maximum hydrogen bonding is present in water as oxygen is the most electronegative among N, P, Cl and O.

13. In trimethyl amine, the nitrogen has a pyramidal geometry whereas in trisilyl amine $[\text{N}(\text{SiH}_3)_3]$, it has a planar geometry. Explain.

Ans. The N atom in $\text{N}(\text{CH}_3)_3$ and $\text{N}(\text{SiH}_3)_3$ both involves sp^3 hybridization having one lone pair on N atom. In $\text{N}(\text{CH}_3)_3$ this produces pyramidal geometry. However, in $\text{N}(\text{SiH}_3)_3$, the lone pair of N atom is transferred to the empty d-orbitals of Si to give $p\pi-d\pi$ overlapping and thereby causing planar structure to it.

14. Nitrogen cannot be stored as liquid in sealed containers but ammonia can be, why?

Ans. Nitrogen cannot be kept as a liquid in a sealed cylinder at ordinary temperature because the critical temperature of nitrogen is very low, whereas the critical temperature of ammonia is higher than the ordinary temperature.

15. NF_3 is an exothermic compound ($\Delta H = -109 \text{ kJ/mol}$) whereas NCl_3 is an endothermic compound ($\Delta H_f = +230 \text{ kJ/mol}$). Explain.

Ans. NF_3 is an exothermic compound whereas NCl_3 is an endothermic compound because in case of NF_3 , N—F bond strength is greater than the F—F bond strength while in case of NCl_3 , N—Cl bond strength is lower than the Cl—Cl bond strength. Thus the formation of NF_3 is spontaneous while energy has to be supplied during the formation of NCl_3 .

16. The compound $(\text{CF}_3)_3\text{N}$ has no basic properties but $(\text{CH}_3)_3\text{N}$ is a strong base. Explain.

Ans. This is due to different electronegativities of H and F. In $(\text{CH}_3)_3\text{N}$ the lone pair on N is concentrated on N and so it can act as a donor (Lewis base). In $(\text{CF}_3)_3\text{N}$ the electron density on N is reduced due to the more electronegative F atom. Hence it cannot act as a Lewis base.

Uses of Ammonia

Ammonia is used to produce various nitrogenous fertilizers (ammonium nitrate, urea, ammonium phosphate and ammonium sulphate) and in the manufacture of some inorganic nitrogen compounds, the most important one being nitric acid.

Liquid ammonia is also used as a refrigerant.

Worked-out Examples

17. Calcium cyanamide is used as fertilizer. Why?

Ans. CaCN_2 is hydrolyzed in the soil to form ammonia which is assimilated by the plants.



18. Amongst the trihalides of nitrogen, the least basic is (IIT 1987)

(a) NF_3 (b) NCl_3 (c) NBr_3 (d) NI_3

Ans. (a)

19. Which one of the following is the strongest base? (IIT 1989)

(a) AsH_3 (b) NH_3 (c) PH_3 (d) SbH_3

Ans. (b)

20. Ammonia can be dried by (IIT 2000)

(a) conc. H_2SO_4 (b) P_4O_{10} (c) CaO (d) anhydrous CaCl_2

Ans. (c) Ammonia cannot be dried over concentrated sulphuric acid, calcium chloride or phosphorus pentoxide as the gas reacts with these substances. However, quicklime can be used.

21. ____ hybrid orbitals of nitrogen atom are involved in the formation of ammonium ion. (IIT 1982)

Ans. sp^3

22. Ammonia can be used as a refrigerant. (IIT 1978)

Ans. True

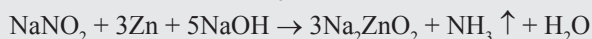
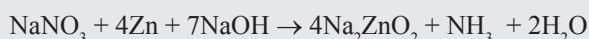
23. The H—N—H bond angle in NH_3 is greater than H—As—H bond angle in AsH_3 . (IIT 1984)

Ans. True

24. A solution of colourless salt H on boiling with excess NaOH produces a non-flammable gas. The gas evolution ceases after sometime. Upon addition of Zn dust to the same solution, the gas evolution restarts. The colourless salt(s) H is (are) (IIT 2008)

- (a) NH_4NO_3 (b) NH_4NO_2 (c) NH_4Cl (d) $(\text{NH}_4)_2\text{SO}_4$

Ans. (AB) All the four choices produce NH_3 gas on boiling with excess of NaOH. However, only the choices NH_4NO_3 and NH_4NO_2 will show re-evolution of the gas due to the reduction of NO_3^- and NO_2^- by Zn in alkaline medium.



25. Regular use of the following fertilizers increases the acidity of soil? [AIEEE 2007]

- (a) Ammonium sulphate (b) Potassium nitrate
(c) Urea (d) Superphosphate of lime

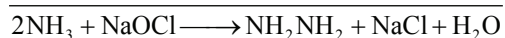
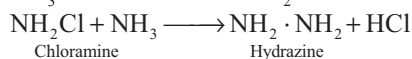
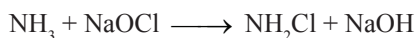
Ans. (a) $(\text{NH}_4)_2\text{SO}_4 + 2\text{H}_2\text{O} \rightarrow 2\text{H}_2\text{SO}_4 + \text{NH}_4\text{OH}$

H_2SO_4 is strong acid and increases the acidity of soil.

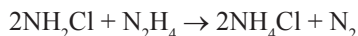
Hydrazine or Diamide, NH_2NH_2 or N_2H_4

This is another hydride of nitrogen. It is prepared by following methods:

Raschig's method: A strong aqueous solution of ammonia is boiled with sodium hypochlorite in presence of a little glue.

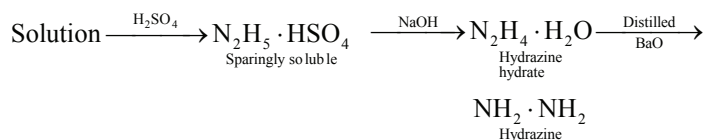


[Hydrazine produced may be decomposed by chloramine to N_2 .

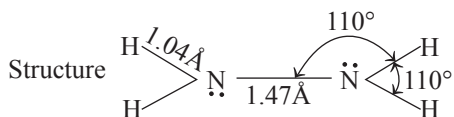


The function of glue is to prevent this reaction. The yield is about 60–70%.]

Recovery of hydrazine from the above solution is made in the following manner:



It reduces Fehling's solution to red cuprous oxide, iodates to iodides and decolourizes acidified KMnO_4 solution. It is used as a fuel for rockets, reducing agent and a reagent in organic chemistry.



Worked-out Example

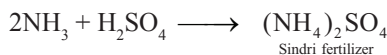
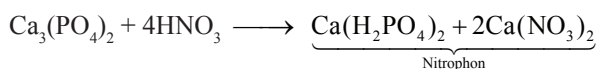
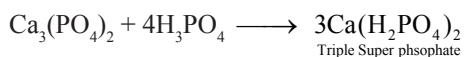
26. Ammonia, on reaction with hypochlorite anion, can form

(IIT 1999)

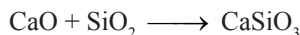
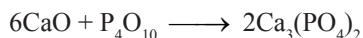
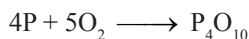
- (a) NO (b)
- NH_4Cl
- (c)
- N_2H_4
- (d)
- HNO_2

Ans. (c) The reaction occurring is $2\text{NH}_3 + \text{OCl}^- \rightarrow \text{H}_2\text{NNH}_2 + \text{H}_2\text{O} + \text{Cl}^-$ **Some Noteworthy Points**

- Fertilizers. $\text{Ca}_3(\text{PO}_4)_2 + 2\text{H}_2\text{SO}_4 \longrightarrow \underbrace{\text{Ca}(\text{H}_2\text{PO}_4)_2 + 2\text{CaSO}_4}_{\text{Superphosphate of lime}}$



- Phosphatic slag or Thomas slag: It is a byproduct of steel industry. When iron is burnt with lime stone in the presence of air, calcium phosphate and calcium silicate appear as slag.

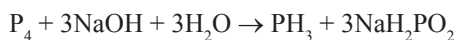
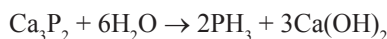


The slag being lighter rises on the top of molten metal and is easily separated. It is cooled. The solid mass so obtained is grinded to a fine powder and used as a fertilizer. It has 14–18% of P_2O_5 .

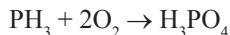
- In tooth paste, $\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$ is added as a mild abrasive and polishing agent.
- Nitric acid attacks proteins giving a yellow nitro compound called Xanthoprotein. Hence nitric acid stains skin. This is used as a test for proteins.
- HNO_3 has smell resembling with bleaching powder.
- Amatol is 80% NH_4NO_3 + 20% TNT and is used as an explosive.
- Certain types of bacteria (Acetobacter) are helpful in the fixation of nitrogen by leguminous plants.
- Traces of Arsenic is detected using Marsh test.
- BiOCl is called pearl white.
- Tartaremetic (pot. Antimonyl tartarate) contains antimony and is used in medicine (as antipoison)
- All trihalides except NI_3 are known. It exists only as a highly explosive complex $\text{NI}_3 \cdot \text{NH}_3$.
- CAN [Calcium ammonium nitrate] is hygroscopic. The pellets are therefore coated with calcium silicate so as to protect from moisture. Ammonium nitrate is explosive but CAN does not explode.
- Out of the three types of cal. Phosphates, $\text{Ca}_3(\text{PO}_4)_2$, CaHPO_4 and $\text{Ca}(\text{H}_2\text{PO}_4)_2$, only $\text{Ca}(\text{H}_2\text{PO}_4)_2$ is soluble in water. It is an important constituent of superphosphate fertilizers. It is acidic in nature. It is used as an acidic component in american baking powder which is a mixture of NaHCO_3 , $\text{Ca}(\text{H}_2\text{PO}_4)_2$ and corn starch.

Phosphine PH_3

Phosphine PH_3 is a colourless and extremely toxic gas, which smells slightly of garlic or bad fish. It is highly reactive. It can be formed either by hydrolysing metal phosphides such as Na_3P or Ca_3P_2 with water, or by hydrolyzing white phosphorus with NaOH solution.



PH_3 , unlike NH_3 , is not very soluble in water: aqueous solutions are neutral. It is more soluble in CS_2 and other organic solvents. Phosphonium salts such as $[\text{PH}_4]^+\text{Cl}^-$ can be formed, but require PH_3 and anhydrous HCl (in contrast to the ready formation of NH_4X in aqueous solution). Pure PH_3 is stable in air, but it catches fire when heated to about 150°C .



PH_3 frequently contains traces of diphosphine P_2H_6 which cause it to catch fire spontaneously. This is the origin of the flickering light called will-o-the-wisp, which is sometimes seen in marshes.

Some Noteworthy Points

- Phosphorus is tenth most abundant element in the earth crust.
- White phosphorus is the most reactive form of phosphorus while red phosphorus is the most stable form.
- White phosphorus is used as rat poison.
- The substance used in the Holmes signals of the ship is a mixture of $\text{CaC}_2 + \text{Ca}_3\text{P}_2$.
- P_4O_{10} is used as a powerful dehydrating agent because of its tendency to react with water.
- Bone and teeth contain 58% P as $\text{Ca}_3(\text{PO}_4)_2$.
- Safety and strike any where matches: P_4S_3 is used in strike any where matches. The head of a safety match box contains KClO_3 , KNO_2 or red lead along with grounded glass pieces and antimony sulphide. Sides of match box contain red phosphorus and sand powder.
- Oxy-acids of phosphorus which do not contain P–H bond do not act as reducing agents.

Some Noteworthy Points

- Only yellow phosphorus reacts with $\text{NaOH}(\text{aq})$ or $\text{KOH}(\text{aq})$.
- Pure PH_3 is not spontaneously inflammable. Ordinary phosphene is spontaneously inflammable due to the presence of P_2H_4 . P_2H_4 can be removed from PH_3 by passing the gas through a long spiral tube cooled by a freezing mixture.
- PH_3 does not react with $\text{HCl}(\text{aq})$, $\text{HBr}(\text{aq})$ or $\text{HI}(\text{aq})$. It reacts only with anhydrous $\text{HCl}(\text{g})$, $\text{HBr}(\text{g})$ or $\text{HI}(\text{g})$. This is because phosphonium halides are decomposed by water to give PH_3 .

Arsine AsH_3 , stibine SbH_3 and bismuthine BiH_3

The bond energy and the stability of the hydrides both decrease on descending the group. Consequently, arsine AsH_3 , stibine SbH_3 and bismuthine BiH_3 are only obtained in small amounts. AsH_3 and SbH_3 are both very poisonous gases. AsH_3 , SbH_3 and BiH_3 can be prepared by hydrolysing binary metal compounds such as Zn_3As_2 , Mg_3Sb_2 or Mg_3Bi_2 with water or dilute acid. AsH_3 and SbH_3 are formed in Marsh's test for As and Sb compounds. Before the use of instruments for analysis, this test was used as a forensic test. Practically all As or Sb compounds can be reduced with Zn and acid, forming AsH_3 or SbH_3 .

Oxides of Nitrogen

Nitrogen forms number of oxides in different oxidation states. The oxides and oxoacids of nitrogen all exhibit $p\pi - p\pi$ multiple bonding between the nitrogen and oxygen atoms. This does not occur with the heavier elements in the group, and consequently nitrogen forms a number of compounds which have no P, As, Sb or Bi analogues. Nitrogen forms a very wide range of oxides, exhibiting all the oxidation states from (+I) to (+VI). The lower oxides are neutral, and the higher ones are acidic.

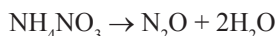
Table 6.2 Oxides of Nitrogen

Formula	Oxidation number	Name
N_2O	+I	Nitrous oxide
NO	+II	Nitric oxide
N_2O_3	+III	Nitrogen sesquioxide
$\text{NO}_2, \text{N}_2\text{O}_4$	+ IV	Nitrogen dioxide, dinitrogen tetroxide
N_2O_5	+ V	Dinitrogen pentoxide
$(\text{NO}_3, \text{N}_2\text{O}_6)$	+VI	Nitrogen trioxide, dinitrogen hexoxide
Very unstable		

6.18 Group 15-Nitrogen Family

Nitrous oxide N_2O

N_2O is a stable, relatively unreactive colourless gas. It is prepared by careful thermal decomposition of molten ammonium nitrate at about 280°C . If heated strongly it explodes. N_2O can also be made by heating a solution of NH_4NO_3 acidified with HCl.



N_2O is a neutral oxide and does not form hyponitrous acid $H_2N_2O_2$ with water nor hyponitrites with alkali. It is important in the preparation of sodium azide, and hence also of the other azides:



The largest use of N_2O is as a propellant for whipped ice cream. Because it has no taste, and is non-toxic, it meets the strict food and health regulations.

N_2O is used as an anaesthetic, particularly by dentists. It is sometimes called 'laughing gas', because small amounts cause euphoria.

The molecule is linear as would be expected for a triatomic molecule with 16 outer shell electrons (see also NO_3^- and CO_2). However, CO_2 is symmetrical ($O=C=O$), whereas in N_2O the orbital energies favour the formation of the asymmetrical molecule $N \equiv N-O$ rather than the symmetrical molecule $N=O=N$. The bond lengths are short, and the bond orders have been calculated as $N \equiv N$ 2.73 and $N-O$ 1.61.

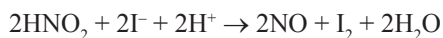


Nitric oxide NO

NO is a colourless gas

NO is a neutral oxide and is not an acid anhydride.

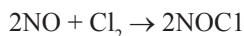
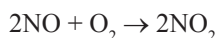
NO is prepared in the laboratory by the reduction of dilute HNO_3 with Cu, or reduction of HNO_2 with I^- :



NO has 11 valency electrons. It is impossible for them all to be paired, and hence this is an odd electron molecule and the gas is paramagnetic. It is diamagnetic in the liquid and solid states, because the molecule dimerizes, forming $O=N-N=O$. The asymmetrical dimer $O=N-O=N$ has been observed to be formed as a red solid in the presence of HCl or other Lewis acids.

The bond length $N-O$ is $1.15 \text{ \AA}$, which is intermediate between a double and a triple bond. Bonding is best described using the molecular orbital theory. The bonding is similar to that in N_2 and CO which both have 10 outer electrons, NO has 11 outer electrons, and the extra unpaired electron occupies in antibonding π^*2p orbital. This reduces the bond order from 3 in N_2 to $2\frac{1}{2}$ in NO. If this electron is removed by oxidizing NO, the nitrosonium ion NO^+ is formed. In NO^+ the bond order is 3, and the $N-O$ bond length contracts from $1.15 \text{ \AA}$ in NO to $1.06 \text{ \AA}$ in NO^+ .

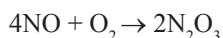
NO readily forms coordination complexes with transition metal ions. These complexes are called nitrosyls. Odd electron molecules are usually highly reactive and tend to dimerize. NO is unusually stable for an odd electron molecule. Nevertheless it reacts instantly with dioxygen to give NO_2 , and with the halogens it gives nitrosyl halides, e.g., NOCl.



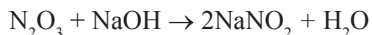
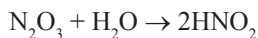
Nitrogen sesquioxide N_2O_3

It can be made by condensing equimolar amounts of NO and NO_2 together, or by reacting NO with the appropriate amount of O_2 . This gives a blue liquid or solid, which is unstable and dissociates into NO and NO_2 at -30°C .

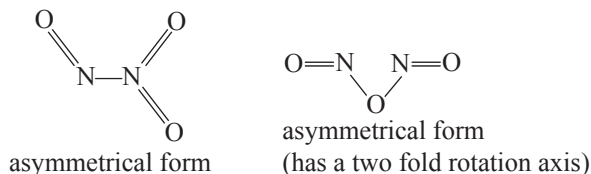
N_2O_3 can only be obtained at low temperatures



It is an acidic oxide and is the anhydride of nitrous acid HNO_2 . With alkali it forms nitrites.



The oxide exists in two different forms. These may be interconverted by irradiation with light of the appropriate wavelength. The N—N bond length from microwave spectra is 1.864 Å in the asymmetrical form. This is exceptionally long and thus the bond is exceptionally weak compared with the N—N bond found in hydrazine (length of 1.45 Å).

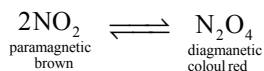


Nitrogen dioxide NO_2 and dinitrogen tetroxide N_2O_4 : Is produced on a large scale by oxidizing NO in the Ostwald process for the manufacture of nitric acid. In the laboratory it is prepared by heating lead nitrate:

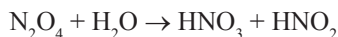


The gaseous products O_2 and NO_2 are passed through a U-tube cooled in ice. The NO_2 (b.p. 21 °C) condenses. The $\text{Pb}(\text{NO}_3)_2$ must be carefully dried, since NO_2 reacts with water NO_2 is a red -brown poisonous gas and

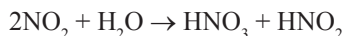
The NO_2 is obtained as a brown liquid which turns paler on cooling, and eventually becomes a colourless solid. This is because NO_2 dimerizes into colourless N_2O_4 . NO_2 is an odd electron molecule, and is paramagnetic and very reactive. It dimerizes to N_2O_4 , pairing the previously unpaired electrons. N_2O_4 has no unpaired electrons and is diamagnetic.



N_2O_4 is a mixed anhydride, because it reacts with water to give a mixture of nitric and nitrous acids:

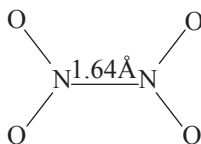


The HNO_2 formed decomposes to give NO.



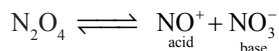
Thus moist NO_2 or N_2O_4 gases are strongly acidic.

The NO_2 molecule is angular with an O—N—O angle of 132°. The bond length O—N of 1.20 Å is intermediate between a single and a double bond X-ray diffraction on solid N_2O_4 shows the structure to be planar.



The N—N bond is very long (1.64 Å), and is therefore weak. It is longer than the single bond N—N distance of 1.47 Å in N_2H_4 , but there is satisfactory explanation of why it is long.

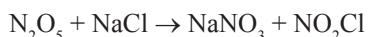
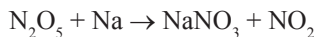
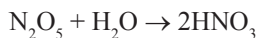
Liquid N_2O_4 is useful as a non-aqueous solvent. It self-ionizes:



It is a colourless deliquescent solid, which is highly reactive, is a strong oxidizing agent, and is light sensitive. It is the anhydride of HNO_3 .

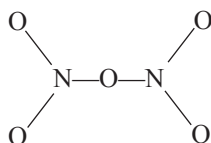
6.20 Group15-Nitrogen Family

N_2O_5 is prepared by carefully dehydrating HNO_3 with P_2O_5 at low temperatures.



In the gas phase N_2O_5 decomposes into NO_2 , NO and O_2 . Nitrogen trioxide NO_3 may be formed by treating N_2O_5 with O_3 .

X-ray diffraction shows that solid N_2O_5 is ionic $\text{NO}_2^+ \text{NO}_3^-$; it should in reality be called nitronium nitrate. It is covalent in solution and in the gas phase, and probably has the structure:



Name	Formula	Oxidation state of nitrogen	Common methods of preparation	Physical appears and chemical nature
Dinitrogen Oxide [nitrogen (I) oxide]	N_2O	+1	$\text{NH}_4\text{NO}_3 \xrightarrow{\text{heat}} \text{N}_2\text{O} + 2\text{H}_2\text{O}$	Colourless gas, neutral
Nitrogenmono oxide	NO	+2	$2\text{NaNO}_2 + 2\text{FeSO}_4 + 3\text{H}_2\text{SO}_4 \rightarrow \text{Fe}_2(\text{SO}_4)_3 + 2\text{NaHSO}_4 + 2\text{H}_2\text{O} + 2\text{NO}$	Colourless gas, neutral
Dinitrogen trioxide	N_2O_3	+3	$2\text{NO} + \text{N}_2\text{O}_4 \xrightarrow{250 \text{ K}} 2\text{N}_2\text{O}_3$	Blue, solid acidic
Nitrogen dioxide	NO_2	+4	$2\text{Pb}(\text{NO}_3)_2 \xrightarrow{673 \text{ K}} 4\text{NO}_2 + 2\text{PbO}$	Brown gas, acidic
Dinitrogen tetraoxide	N_2O_4	+4	$2\text{NO}_2 \xrightleftharpoons[\text{heat}]{\text{cool}} \text{N}_2\text{O}_4$	Colourless solid or liquid acidic
Dinitrogen pentaoxide	N_2O_5	+5	$4\text{HNO}_3 + \text{P}_4\text{O}_{10} \rightarrow 4\text{HPO}_3 + 2\text{N}_2\text{O}_5$	Colourless solid acidic

Table 6.3 Structures of Oxides of Nitrogen

Formula	Resonance structures	Bond parameters
N_2O	$\begin{array}{c} \cdot\cdot \\ \text{N} = \text{N} = \text{O} \\ \cdot\cdot \end{array} \longleftrightarrow \begin{array}{c} \cdot\cdot \\ \text{N} \equiv \text{N} - \text{O} \\ \cdot\cdot \end{array}$	$\text{N} - \text{N} - \text{O}$ 113 pm 119 pm $\text{N} = \text{O}$ 115 pm
NO	$\begin{array}{c} \cdot\cdot \\ \text{N} = \text{O} \\ \cdot\cdot \end{array} \longleftrightarrow \begin{array}{c} \cdot\cdot \\ \text{N} = \text{O} \\ \cdot\cdot \end{array}$	
N_2O_3		

(Contd)

Formula	Resonance structures	Bond parameters
NO ₂		
N ₂ O ₄		
N ₂ O ₅		

Worked-out Examples

(1996)

27. Among the following species, identify the isostructural pairs.

- (a) [NF₃, NO₃⁻] and [BF₃, H₃O⁺] (b) [NF₃, HN₃] and [NO₃⁻, BF₃]
 (c) [NF₃, H₃O⁺] and [NO₃⁻, BF₃] (d) [NF₃, H₃O⁺] and [HN₃, BF₃]

Ans. (c)

28. Nitrous oxide supports combustion more vigorously than air. Explain.

Ans. N₂O is better source of oxygen than air because air has oxygen about 1/5 of its volume, while N₂O on decomposition produces oxygen about 1/3 of the volume.

29. Nitric oxide is paramagnetic. Explain.

Ans. Nitric oxide in all has 15 electrons in its molecule *i.e.*, it possesses at least one odd electron which is responsible for paramagnetic nature of NO.

30. Nitric oxide becomes brown when released into air. Explain.

Ans. Nitric oxide reacts with air to get oxidized to NO₂ which has brown yellow vapours.

31. N₂, CO, CN⁻ and NO⁺ are isoelectronic but the former is chemically inert and later three are very reactive, Why?

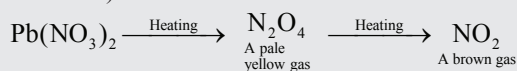
Ans. Although each of the above has same electronic distribution in bonding orbitals but N₂ is inert because of (i) non-polar nature of bond, (ii) high dissociation energy of N → N. Rest all (CO, CN⁻, NO⁺) are polar.

32. Name a compound of N, H and O which is used as freezing salt and decomposes on heating to give a neutral oxide?

Ans. Ammonium nitrate. $\text{NH}_4\text{NO}_3 \xrightarrow{\Delta} \text{N}_2\text{O} + 2\text{H}_2\text{O}$

33. On heating Pb(NO₃)₂ produces a pale yellow gas which on strong heating produces a brown gas. Why?

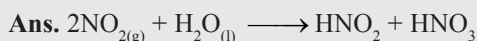
Ans. Lead nitrate first produces N₂O₄ which is pale yellow. On further heating it decomposes into nitrogen dioxide (brown in colour).



34. On cooling paramagnetic NO_2 becomes diamagnetic, why?

Ans. On cooling NO_2 dimerises to N_2O_4 which does not contain unpaired electrons.

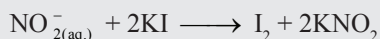
35. Show that NO_2 is mixed anhydride.



$\text{NO}_{2(\text{g})}$ on dissolution in water forms two oxyacids HNO_2 and HNO_3 . Also the reaction shows disproportionation of N in NO_2 where it is oxidised from +4 (NO_2) to +5 (HNO_3) and reduced from +4 (NO_2) to +3 (HNO_2).

36. How NO_3^- and NO_2^- ions present in separate solutions can be identified?

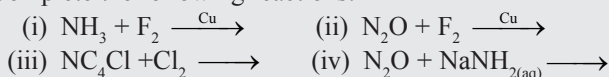
Ans. NO_2^- on reaction with I^- liberates I_2 which turns starch paper blue.



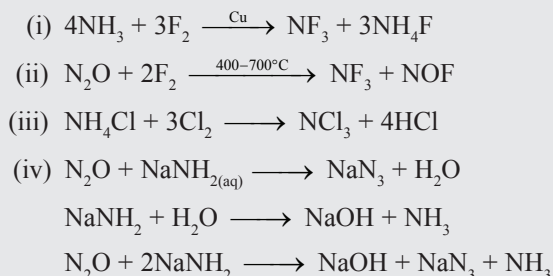
37. NO_2 readily forms a dimer. Explain.

Ans. NO_2 is the odd electron molecule ($17e^-$, N – 2, 5 and O – 2, 6). Hence it is very reactive; when such reactive molecules unite the pairing of the unpaired electron on each NO_2 takes place and N—N bond of the dimer is formed.

38. Complete the following reactions:



Ans.



39. NO is paramagnetic in gaseous state but diamagnetic in liquid or solid state, why?

Ans. The NO molecule has eleven valency electrons and thus paramagnetic in nature due to unpaired electron. In solution or solid state, it forms a loose dimer in such a way that magnetic field of unpaired electrons of two molecules cancel out and it behaves as diamagnetic in nature.

40. Why is magnetic moment of NO is temperature dependent?

Ans. The magnetic moment of NO increases upto 520 K and remains constant above this. Also the variation below 520 K is not in accordance with the Curie's law. The two lowest electronic energy levels of NO molecule are separated by 121 cm^{-1} , which is comparable with KT at normal laboratory temperature. The distribution of molecules between two states of similar energy but different J values is responsible for the anomalous variation of magnetic moment with temp.

41. The reddish-brown coloured gas formed when nitric oxide is oxidized by air is

- (a) N_2O_5 (b) N_2O_4 (c) NO_2 (d) N_2O_2 (IIT 1979)

Ans. (c)

42. Nitrogen dioxide cannot be obtained by heating

- (a) KNO_3 (b) $\text{Pb}(\text{NO}_3)_2$ (c) $\text{Cu}(\text{NO}_3)_2$ (d) AgNO_3 (IIT 1985)

Ans. (a)

43. Which of the following oxides of nitrogen is a coloured gas? (IIT 1987)
 (a) N_2O (b) NO (c) N_2O_4 (d) NO_2

Ans. (d)

44. Nitrogen(I) oxide is produced by
 (a) thermal decomposition of ammonium nitrate
 (b) disproportionation of N_2O_4
 (c) thermal decomposition of ammonium nitrite
 (d) interaction of hydroxylamine and nitrous acid (IIT 1989)

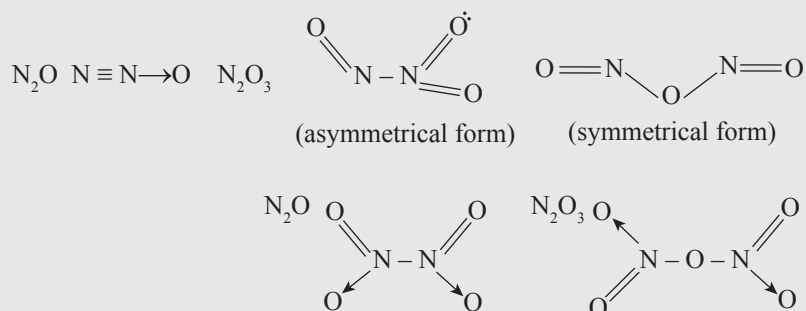
Ans. (a), (d)

45. A pale blue liquid obtained by equimolar mixture of two gases at -30°C is (IIT 2005)
 (a) nitric oxide (b) nitrogen trioxide
 (c) nitrogen tetroxide (d) nitrogen pentoxide

Ans. (b) N_2O_3 is obtained by condensing equimolar mixture of NO and NO_2 .

46. The nitrogen oxide(s) that contain N—N bond(s) is/are (IIT 2009)
 (a) N_2O (b) N_2O_3 (c) N_2O_4 (d) N_2O_5

Ans. (abc) The structures of given species are as follows.



47. Nitrous oxide is heavier than oxygen. (True/False) (IIT 1978)

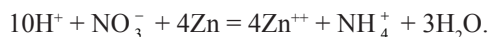
Ans. True

48. Nitric oxide, though an odd electron molecule, is diamagnetic in liquid state. (True/False) (IIT 1991)

Ans. False There occurs only partial dimerization of nitric oxide in the liquid state.

Oxyacids of Nitrogen

The most important of these is nitric acid, HNO_3 . In concentrated solutions, nitric acid is an extremely powerful oxidizing agent and reacts with metals like copper and silver with the production of NO $8\text{H}^+ + 2\text{NO}_2 + 3\text{Cu} = 3\text{Cu}^{+2} + 2\text{NO} + 4\text{H}_2\text{O}$. Reaction of the active metals with concentrated nitric acid produces ammonium ion:



In solutions more dilute than 2 M, the oxidizing power of the nitrate group i.e., greatly diminished, and only the protons of the dissociated acid react with the active metals to evolve hydrogen. This behaviour should not be too surprising, for the half-reaction



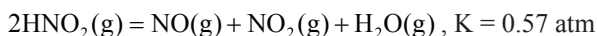
6.24 Group15-Nitrogen Family

shows that the power of NO_3^- as an oxidant should be very sensitive to the concentration of the acid, since four protons appear on the left-hand side of the equation.

We noted above that reaction of equimolar mixtures of NO and NO_2 with aqueous alkali gives solutions of nitrites. Solutions of nitrous acid can be made by acidification of solutions of nitrites. Nitrous acid is a weak acid, as the dissociation constant for



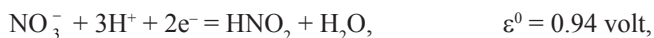
shows. The pure liquid acid is unknown, and in the gas phase it is noticeably dissociated:



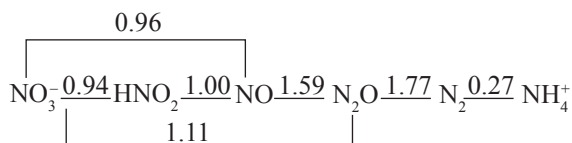
As noted earlier, even aqueous solutions of nitrous acid are unstable and decompose when heated according to



Nitrogen appears in at least eight oxidation states in its water-soluble species, and there is a very large number of half-reactions that involve these oxidation states. In addition to the nitrate-nitric oxide half-reaction given earlier, some of the important half-reactions are the following:



A convenient summary of this series of half-reactions is provided by the following reduction-potential diagram.

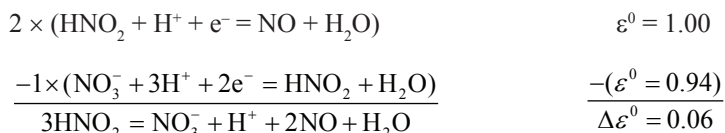


In this diagram only the nitrogen-containing species are indicated. A line represents the properly balanced half-reaction, written as a reduction, that interconverts the two species, and the voltage associated with the half-reaction is written over the line. Thus from this diagram, it is possible to tell at a glance that nitrate ion is reduced to nitric oxide with a standard potential of +0.96 volt, that nitrogen gas going to ammonium ion is a spontaneous process with a standard voltage of +0.27 volt, and so on.

Reduction-potential diagrams make it easy to pick out species that are thermodynamically unstable with respect to disproportionation. For example, consider the following partial diagram.

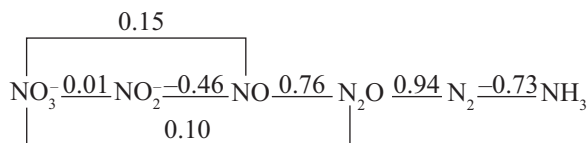


It appears that HNO_2 is reduced to NO with a potential of +1.00 volt, but is oxidized to NO_3^- with a potential of -0.94 volt. Therefore HNO_2 should be converted to NO_3^- and NO with an associated voltage of $1.00 - 0.94 = 0.06$ volt. Writing out the relevant half-reactions in more detailed form confirms the following conclusion.

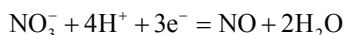


In general, a species having an intermediate oxidation number will be unstable with respect to disproportionation if it appears in the reduction-potential diagram with a voltage to its right that is larger than the voltage to its left.

The reduction-potential diagrams above apply to reactions that occur in acidic solutions. It is instructive to examine the corresponding diagram for basic solutions :



The most remarkable feature of the composition of the standard potentials in acidic and basic solutions is that they show that nitrate and nitrite ions are much weaker oxidants in basic solution than they are in acidic solution. This is not too surprising, for as was noted above, the half-reactions for reduction of nitrate involve several hydrogen ions as *reactants*, as in



A change from 1 M H^+ to 1 M OH^- reduces the hydrogen-ion concentration enormously, and the oxidizing power of the nitrate ion decreases accordingly. The same generalization applies to other oxyanions: in acidic solution MnO_4^- and $\text{Cr}_2\text{O}_7^{2-}$ are stronger oxidizing agents than they are in basic solution. An alternative way of stating this generalization is that it is easier to oxidize an element to an oxyanion of high oxidation number in basic solution than in acidic solution.

Another noteworthy feature of the reduction-potential diagram for basic solution is that it shows that nitrite ion is stable under these conditions:

Writing out the disproportionation reaction in detail we find that



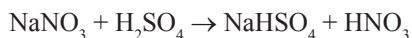
which shows that nitrites can exist without spontaneous decomposition in basic aqueous solution.

Nitric Acid (aqua fortis)

Nitrogen forms oxoacids such as $\text{H}_2\text{N}_2\text{O}_2$ (hyponitrous acid), HNO_2 (nitrous acid) and HNO_3 (nitric acid). Amongst the HNO_3 is the most important.

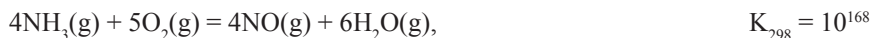
Preparation

In laboratory, it is prepared by heating KNO_3 or NaNO_3 with concentrated H_2SO_4 .

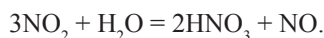


Manufacture of HNO_3

HNO_3 is manufactured by the following important processes. The ammonia synthesis is the first step in the commercial “fixation” of nitrogen. Combustion of ammonia under catalytic conditions produces nitric oxide, NO , which is eventually converted to nitric acid, HNO_3 . The combustion of ammonia may take either of two courses.



Although the equilibrium constant of the first reaction is much larger than that of the second, the latter is selectively catalyzed by platinum metal, and NO is produced in quantity on the platinum surface at a temperature of about 1000°K . To complete the synthesis of nitric acid and the “fixation” of nitrogen, nitric oxide is treated with oxygen and water:



Another reaction by which elemental nitrogen can be converted to a combined form is the nitric oxide synthesis:



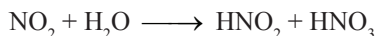
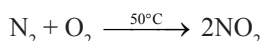
Although ΔG_{298}° for this reaction is 20.72 kcal and the equilibrium constant at 298°K is only 1.6×10^{-15} , the fact that ΔH° is positive means that at elevated temperatures the equilibrium constant will be more favorable for the synthesis. The experimental values of the equilibrium constant are, in fact, 2×10^{-2} at 2000°K , and 6×10^{-2} at 2500°K . One method of synthesizing nitric oxide is to pass nitrogen and oxygen through an electric arc discharge, which creates a high, if ill-defined, temperature.

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In internal combustion engines operating on fuel-air mixtures, the combustion temperature is high enough so that small but significant amounts of nitric oxide are formed. When nitric oxide enters the atmosphere as engine exhaust, it is converted to nitrogen dioxide, NO_2 , by reaction with oxygen. Photo-dissociation of NO_2 to NO and O initiates smog forming reactions when hydrocarbons are present in the atmosphere. Attempts to increase the performance of engines and lower the emission of unburned hydrocarbons by increasing compression ratios and combustion temperature lead to increased formation of NO . Therefore one presently favoured method of decreasing both hydrocarbon and nitric oxide emission from engines is to decrease the cylinder combustion temperature to diminish NO formation, and eliminate unburned hydrocarbons from the exhaust by catalyzed combustion at low temperatures.

Elemental nitrogen does undergo direct reaction with some of the metallic elements. The reaction of nitrogen with lithium to give lithium nitride, Li_3N , proceeds slowly at room temperature, and rapidly at 250°C . The reaction of nitrogen with the alkaline-earth metals to form nitrides like Mg_3N_2 is rapid at temperatures above 500°C , and at an even higher temperature, nitrogen reacts with boron, aluminum, silicon, and many of the transition metals. Thus, while nitrogen is, in general, inert at temperatures near 25°C , it does react directly with a number of elements at somewhat elevated temperatures.

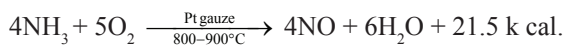
- (a) **Birkland and Eyde process or arc process:** In this method, air is first made free from CO_2 and moisture. It is then passed through a chamber containing an electric arc struck between two copper electrodes.



- (b) **Chile saltpetre method:** From early days, HNO_3 , has been prepared from NaNO_3 extracted from chile saltpetre or nitre (KNO_3), by heating a mixture of about equal weights of NaNO_3 or KNO_3 and 93% H_2SO_4 in a cast iron retort fitted with an outlet pipe near the bottom to withdraw KHSO_4 solution.



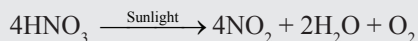
- (c) **Ostwald's process :** This is the modern method of the manufacture of HNO_3 . This process is based on the following fundamental steps.



Worked-out Example

49. Concentrated HNO_3 turns yellow in sun light. Explain.

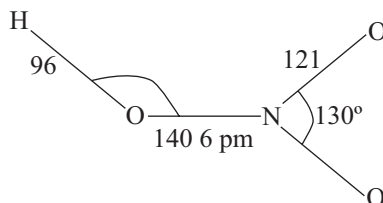
Ans. HNO_3 is partially decomposed in sun light to produce NO_2 . The NO_2 formed dissolves in HNO_3 to produce yellow colour.



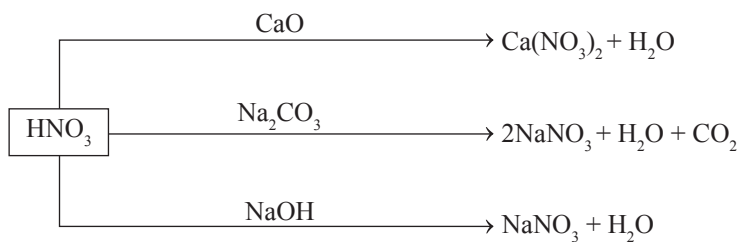
Properties

It is a colourless liquid (f.p. 231.4 K and b.p. 355.6 K). Laboratory grade nitric acid contains $\sim 68\%$ of the HNO_3 by mass and has a specific gravity of 1.504.

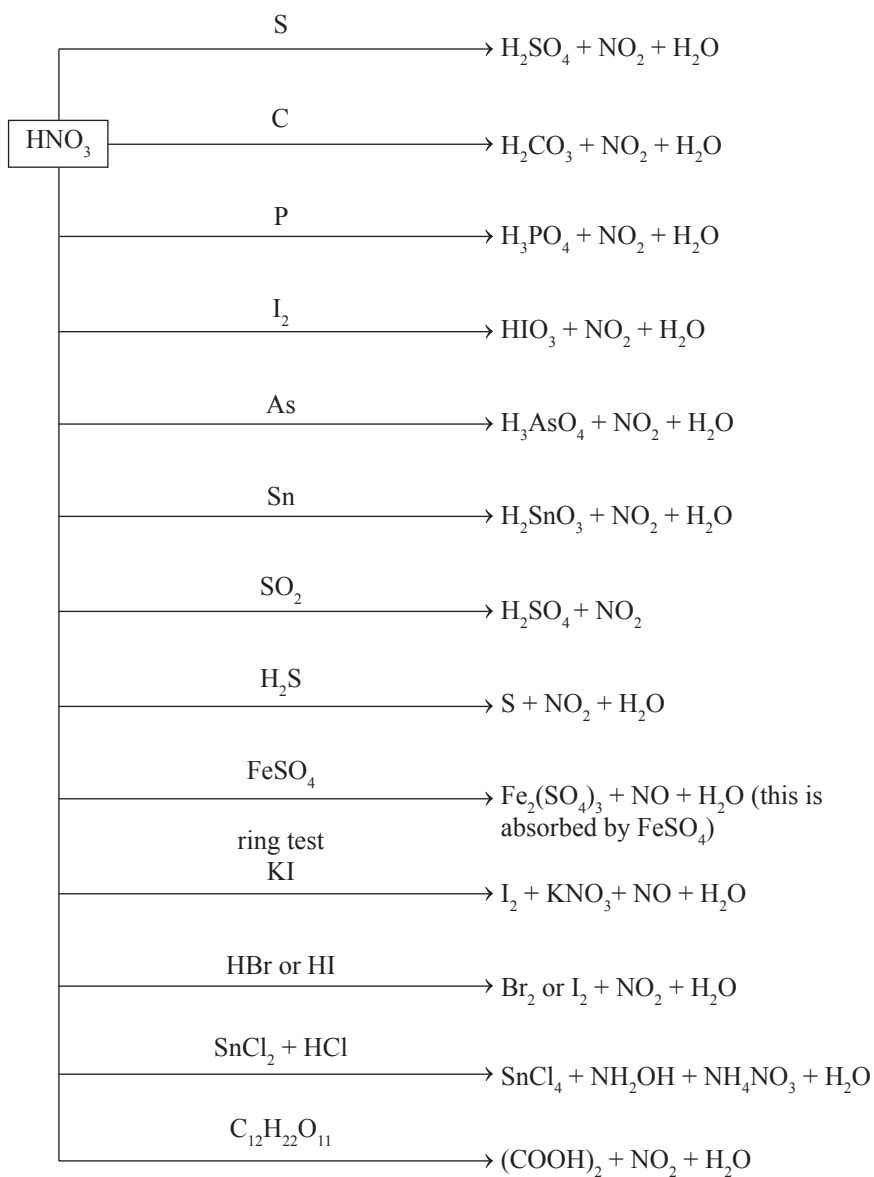
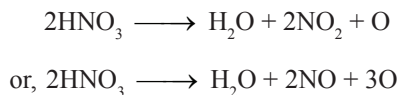
In the gaseous state, HNO_3 exists as a planar molecule with the structure as shown.



- It is very strong acid.

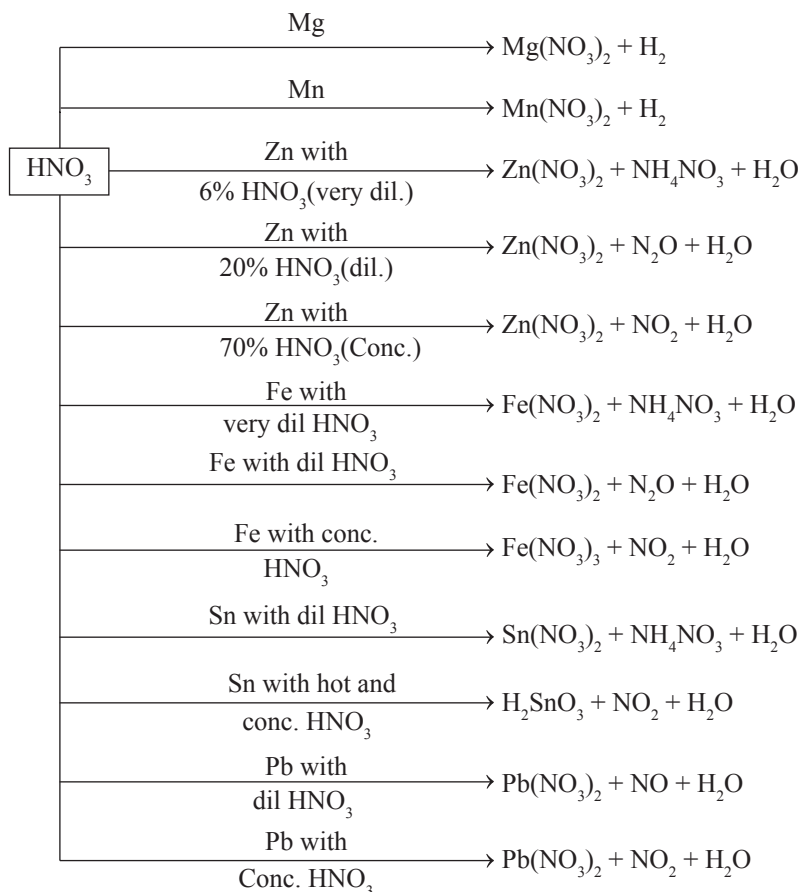


Oxidizing nature

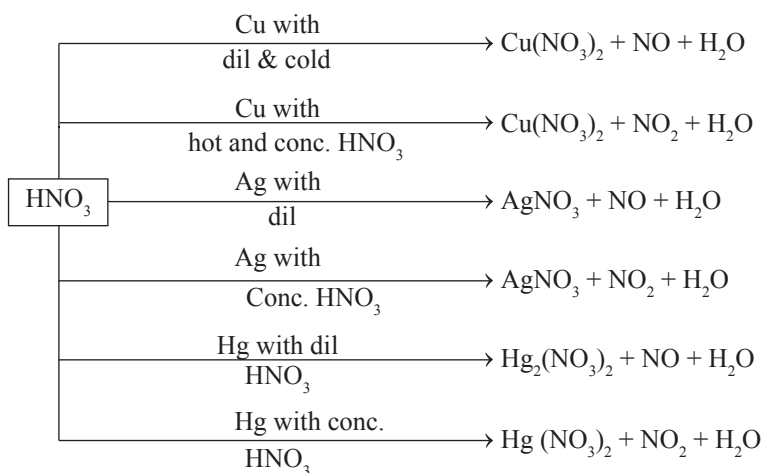


6.28 Group15-Nitrogen Family

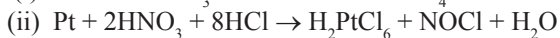
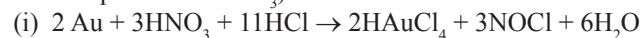
■ Action on Metals which are above H in ECS



■ Metals which are below H in ECS.



■ Metals which do not react → noble metals like gold, platinum, rhodium and so on but these dissolve in aquaregia (3 part HCl + 1 part conc. HNO₃)



Worked-out Examples

50. HNO_3 is strong acid than HNO_2 , Explain.

Ans. Higher is the oxidation state of central non-metal atom, more is acidic character.

51. Aqua-regia dissolves noble metals in it. Explain.

Ans. Aqua-regia is a mixture of 1 part conc. HNO_3 and 3 part conc. HCl . The mixture liberate nascent chlorine which dissolves noble metals.

52. Aluminium containers can be used for storing conc. HNO_3 . Why?

Ans. Al becomes passive on coming in contact with HNO_3 due to a fine coating of its oxide on its surface and thus Al containers can be used to store conc. HNO_3 .

53. Copper dissolves in HNO_3 , but not in HCl . Why?

Ans. Copper is placed below H in electrochemical series and does not liberate H_2 from acid. However, HNO_3 oxidises Cu due to its strong oxidant nature.



54. What is the hybridized state of N in HNO_2 and HNO_3 ?

Ans. sp^2 in both the cases.

55. The oxidizing action of nitric acid can be inhibited by the presence of urea. Explain.

Ans. HNO_3 acts an oxidizing agent due to traces of HNO_2 or NO_2 that is why fuming nitric acid containing excess of NO_2 is much stronger oxidizing agent. Addition of urea consumes NO_2 and thus retards the oxidizing action.

56. HNO_3 is a stronger acid than HNO_2 . In HNO_3 , there are two nitrogen-to-oxygen bonds whereas in HNO_2 there is only one. (IIT 1998)

Ans. (a) Hydrolysis of SiC_4 is due to the presence of d orbitals.

57. Distillation of concentrated HNO_3 with phosphorus pentoxide produces (IIT 2004)

(a) NO (b) N_2O (c) NO_2 (d) N_2O_5

Ans. (d) The reaction occurring is $2\text{HNO}_3 \xrightarrow[-\text{H}_2\text{O}]{\text{P}_2\text{O}_5} \text{N}_2\text{O}_5$

Phosphorus

Allotropic forms of phosphorus

Phosphorus exists in a number of allotropic forms. These forms are

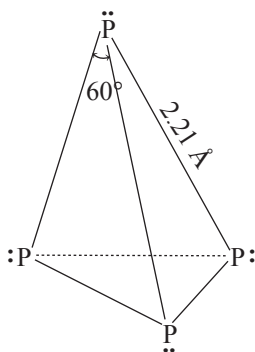
- (i) yellow or white phosphorus
- (ii) red phosphorus
- (iii) scarlet phosphorus
- (iv) α -black phosphorus
- (v) β -black phosphorus
- (vi) violet phosphorus

The main allotropic forms are white red and black phosphorus.

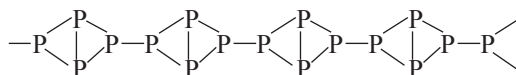
White phosphorus: Molecular weight determination suggests that white phosphorus exists as P_4 molecules. Electron diffraction data shows that P_4 molecule is tetrahedral. These molecules are attracted to each other by Van der Waals forces to form the solid. The P – P – P bond angles are only 60° (theoretically required angle is 90° if pure 3p orbitals are involved). This suggests that the structure is under strain and unstable. The strained nature of P_4 is confirmed by the high chemical

6.30 Group15-Nitrogen Family

reactivity of white phosphorus and its tendency to change under favourable conditions of single bonded structure of unstrained angles. White phosphorus inflames in air. It is soluble in organic solvents such as C_6H_6 and CS_2 . White phosphorus requires relatively high little energy to decompose.



Red phosphorus: It is prepared by heating white P at $400^\circ C$. It exists as long chains of phosphorus atoms which are covalently bonded to give a giant molecular crystal.



Black phosphorous: White phosphorus, when subjected under very high pressure to $200^\circ C$, changes into 'black phosphorus'. It has metallic luster and conducts heat and electricity moderately. The crystal lattice consists of double layers, each layer made of adjacent zig-zag chains with P-P-P angles of 99° . Each phosphorus atoms in a particular layers is $2.18\text{\AA}$ from its nearest neighboring phosphorus atoms, two in the same chain and one in a chain in the layer above or below. Distance between two adjacent layer is $3.68\text{\AA}$. Binding between the layer is thus weaker than within a particular layer.

Comparison of allotropic forms of phosphorus: White phosphorus is the most reactive while black phosphorus is the least reactive. Therefore, white phosphorus is stored under water to protect it from air while red and black are stable in air.

Table 6.4 Comparison Between White and Red Phosphorus

Property	White phosphorus	Red phosphorus
1. Physical state	Soft waxy solid	Brittle powder
2. Colour	White when pure. Attains yellow colour on standing	Red
3. Odour	Garlic	Odourless
4. Specific gravity	1.8	2.1
5. Melting point	$44^\circ C$	Sublimes in absence of air at $290^\circ C$
6. Ignition temperature	Low, $30^\circ C$	High, $260^\circ C$
7. Solubility in water	Insoluble	Insoluble
8. Solubility in CS_2	Soluble	Insoluble
9. Physiological action	Poisonous	Non-poisonous.
10. Chemical activity	Very active	Less active
11. Stability	Unstable	Stable
12. Phosphorescence	Glow in dark.	Does not glow in dark
13. Burning in air	Forms P_4O_{10}	Forms P_4O_{10}
14. Reaction with NaOH	Evolves phosphine	No action
15. Action of Cl_2	Combines spontaneously to form PCl_3 and PCl_5	Reacts on heating to form PCl_3 and PCl_5
16. Reaction with hot HNO_3	Forms H_3PO_4	Forms H_3PO_4
17. Molecular formula	P_4	Complex polymer

Worked-out Examples

58. Phosphorus glows in dark. Explain.

Ans. Phosphorus undergoes spontaneous slow oxidation in air giving a greenish glow visible in dark. This is chemiluminescence.

59. Explain, why white phosphorus is highly reactive?

Ans. The high reactivity of white phosphorus is due to an unusual bonding that produces considerable strain in the P_4 molecule. If each P atom uses 3p-orbitals to form three P—P bonds, all the bond angles should be 90° . The geometry of P_4 , however reveals that all the bonds have 60° angles, which means that p-orbitals can not overlap in a head-on fashion. As a result P—P bonds are 'bent', relatively weak, and highly reactive.

60. Red phosphorus is denser less volatile and chemically less reactive than yellow phosphorus. Explain.

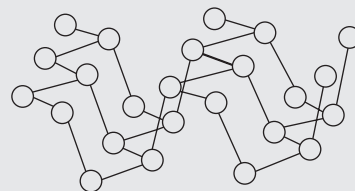
Ans. Red phosphorus exists as chains of P_4 tetrahedron linked together, whereas in yellow phosphorus individual P_4 molecules are held together by weak van der Waals' forces.

61. Red phosphorus is used for making matches, explain.

Ans. Red phosphorus is non poisonous and has high ignition temperature.

62. What is black phosphorus?

Ans. Heating of white P to 200°C under pressure of 12,000 atmosphere results in highly polymerized form of phosphorus known as black phosphorus. It may occur in orthorhombic, rhombohedral and cubic forms. This is thermodynamically most stable allotrope and is inert.



63. Write an equation for disproportionation of P_4 in NaOH.

Ans. $P_4 + 4OH^- + 2H_2O \longrightarrow 2PH_3 + 2HPO_3^{2-}$

64. In case of nitrogen, NCl_3 is possible but not NCl_5 while in case of phosphorous, PCl_3 as well as PCl_5 are possible. It is due to [AIEEE 2002]

- (a) availability of vacant d orbitals in P but not in N
- (b) lower electronegativity of P than N
- (c) lower tendency of H-bond formation in P than N
- (d) occurrence of P in solid while N in gaseous state at room temperature

Ans. (a) ${}_7N = 1s^2 2s^2 2p^3$; ${}_{15}P = 1s^2 2s^2 2p^6 3s^2 3p^3$

In phosphorous, the 3d-orbitals are available. Hence phosphorous can form pentahalides also but nitrogen can not form pentahalide due to absence of d-orbitals.

65. White phosphorus (P_4) has (IIT 1998)

- (a) six P — P single bonds
- (b) four P — P single bonds
- (c) four lone pairs of electrons
- (d) PPP angle of 60°

Ans. (a, c, d) The four atoms in a P_4 molecule are situated at the corners of a tetrahedron. There are six P — P single bonds with PPP bond angle equal to 60° . Each phosphorus has a lone pair of electrons.

66. ____ phosphorus is reactive because of its highly strained tetrahedral structure. (IIT 1987)

Ans. White

67. The angle P—P—P in P_4 molecule is ____ degree. (IIT 1997)

Ans. 60°

68. Red phosphorus is less volatile than white phosphorus because the former has a tetrahedral structure. (IIT 1982)

Ans. False Red phosphorus has polymeric structure and not a tetrahedral, due to which red phosphorus is less volatile than white phosphorus.

69. (a) Give reason(s) why elemental nitrogen exists as a diatomic molecule whereas elemental phosphorous is a tetraatomic molecule.

Ans. Because of the smaller size of nitrogen atom, it can form multiple bonds involving its valence orbitals 2s and 2p's. The size of phosphorus atom is comparatively larger and it cannot form multiple bonds involving its valence orbitals 3s, 3p's and 3d's.

From the data on bond enthalpies, it can be concluded that the species $\text{N}=\text{N}$ is more stable than three $\text{N}-\text{N}$ bonds whereas $\text{P}=\text{P}$ is less stable than three $\text{P}-\text{P}$ bonds.

$$\epsilon(\text{N}=\text{N}) = 946 \text{ kJ mol}^{-1}$$

$$\epsilon(>\text{N}-\text{N}<) = 160 \text{ kJ mol}^{-1}$$

$$\epsilon(\text{P}=\text{P}) = 490 \text{ kJ mol}^{-1}$$

$$\epsilon(>\text{P}-\text{P}<) = 200 \text{ kJ mol}^{-1}$$

The elementary phosphorus is tetraatomic molecule having tetrahedral geometry. Each phosphorus atom is linked to three phosphorus atoms.

Reasoning Type

The questions below (1, 2 and 3) consist of an 'Assertion' in column 1 and the 'Reason' in column 2. Use the following key to choose the appropriate answer.

- (a) If both assertion and reason are CORRECT, and reason is the CORRECT explanation of the assertion.
- (b) If both assertion and reason are CORRECT, but reason is NOT the CORRECT explanation of the assertion.
- (c) If assertion is CORRECT, but reason is INCORRECT.
- (d) If assertion is INCORRECT, but reason is CORRECT.

Assertion (column 1)

Reason (column 2)

70. Although PF_5 , PCl_5 and PBr_5 are known, the pentahalides of nitrogen have not been observed.

Phosphorus has lower electronegativity than nitrogen.

(IIT 1994)

Ans. (b)

Passage

There are some deposits of nitrates and phosphates in earth's crust. Nitrates are more soluble in water. Nitrates are difficult to reduce under the laboratory conditions but microbes do it easily. Ammonia forms larger number of complexes with transition metal ions. Hybridization easily explains the ease of sigma donation capability of NH_3 and PH_3 . Phosphine is a flammable gas and is prepared from white phosphorous.

71. Among the following, the correct statements is

- (a) Phosphates have no biological significance in humans
- (b) Between nitrates and phosphates, phosphates are less abundant in earth's crust
- (c) Between nitrates and phosphates, nitrates are less abundant in earth's crust
- (d) Oxidation of nitrates is possible in soil

Ans. (c)

Between nitrates and phosphates, nitrates are less abundant in earth's crust because nitrates are more soluble in water

72. Among the following, the correct statement is

- (a) Between NH_3 and PH_3 , NH_3 is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional

- (b) Between NH_3 and PH_3 , PH_3 is a better electron donor because the lone pair of electrons occupies sp^3 orbital and is more directional.
- (c) Between NH_3 and PH_3 , NH_3 is a better electron donor because the lone pair of electrons occupies sp^3 orbital and is more directional.
- (d) Between NH_3 and PH_3 , PH_3 is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional.

Ans. (c)

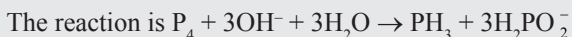
NH_3 is a better electron donor than PH_3 because the lone pair of electrons occupies sp^3 orbital formed from the more compact 2s and 2p orbitals as compared to 3s and 3p orbitals of phosphorous in PH_3 .

73. White phosphorus on reaction with NaOH gives PH_3 as one of the products. This is a

- (a) dimerization reaction (b) disproportionation reaction
(c) condensation reaction (d) precipitation reaction

(IIT 2008)

Ans. (b)

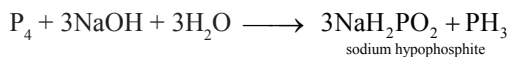


The oxidation state of phosphorus in P_4 , PH_3 and H_2PO_2^- are 0, -3 and +1, respectively. Thus, the reaction is disproportionation reaction.

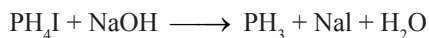
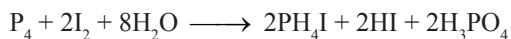
(2) Phosphine (PH_3)

Preparation

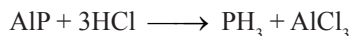
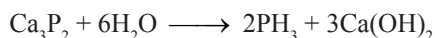
- (i) By boiling white phosphorus with aqueous solution of NaOH or alcoholic solution of KOH .



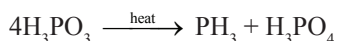
- (ii) By treating phosphorus with iodine solution.



- (iii) By hydrolyzing metallic phosphide with water or dilute mineral acids.



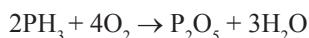
- (iv) By heating phosphorus acid at 210°C .



Phosphine is a covalent molecule. Phosphorus atom has a lone pair of electrons. The molecule has a pyramidal structure, the bond angle is $\approx 93^\circ$.

Properties

- (i) It has an unpleasant odour like that of rotten fish.
(ii) It burns in air or oxygen giving phosphorus pentoxide and water.



- (iii) Phosphine precipitates out metals from their aqueous salt solutions as metallic phosphides.
(iv) It is a powerful reducing agent and precipitates Cu , Ag and Au as phosphides from solutions of their salts at ordinary temperatures. $3\text{CuSO}_4 + 2\text{PH}_3 \rightarrow \text{Cu}_3\text{P}_2 \downarrow + 3\text{H}_2\text{SO}_4$

Uses of Phosphine

The spontaneous combustion of phosphine is technically used in Holme's signals.

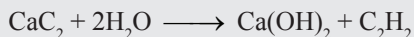
Containers containing calcium carbide and calcium phosphide are pierced and thrown in the sea when the gases evolved burn and serve as a signal.

It is also used in smoke screens.

Worked-out Examples

74. What is Holme's Signal?

Ans. A mixture of CaC_2 and Ca_3P_2 filled in a container having small holes in it is thrown in sea or river. The water coming through holes gives.



P_2H_4 catches fire and C_2H_2 burns with luminous flame which serves as signal to approaching boat or ship.

75. Pure PH_3 Does Not Burn In Air, But Impure Sample Of PH_3 Burns In Air. Explain.

Ans. The impurity of P_2H_4 , the liquid hydride present in phosphine catches fire as soon as it comes in contact with air and forms vortex rings of smoke.

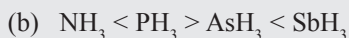
76. Ammonia has a higher boiling point than phosphine. Explain.

Ans. N-atom has higher electronegativity and small size and shows H-bonding and thus ammonia molecule shows association whereas phosphine does not.

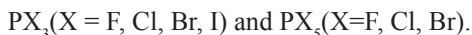
77. Arrange NH_3 , PH_3 , AsH_3 and SbH_3 in the increasing order of

- (a) covalent nature (c) bond angle
(b) reducing nature (d) thermal stability

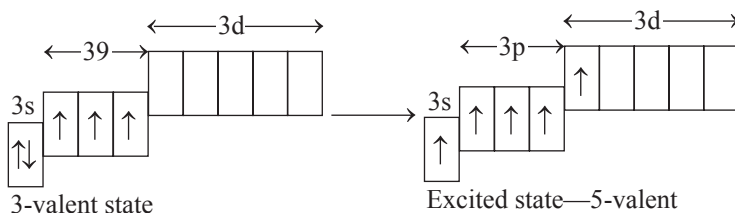
Ans.

**Phosphorus Halides**

Phosphorus forms two types of halides.

**Preparation of Phosphorus Halides****Trichloride or Trihalides of phosphorus**

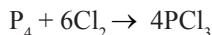
Whereas nitrogen is restricted to a covalency of 3, phosphorus can also form compounds in which it is 5-covalent. This is possible since phosphorus has d orbitals available, and promotion of one 3s electron to the 3d level results in five unpaired electrons:



The trihalides, PX_3

All the four trihalides exist and, except for the trifluoride, can be made by direct synthesis. The trifluoride is a gas, the trichloride and tribromide are liquids, and the tri-iodide is a solid. Their molecules have the expected pyramidal structure.

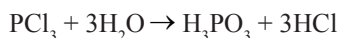
The trichloride, which is the most important of these trihalides, is obtained by passing chlorine over white phosphorus. The phosphorus burns with a pale green flame and phosphorus trichloride distils and is condensed as a colourless liquid. Since it is attacked by air and water, it is necessary to displace the air from the apparatus with a stream of carbon dioxide and to include a soda-lime drying tube.



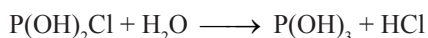
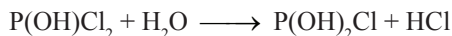
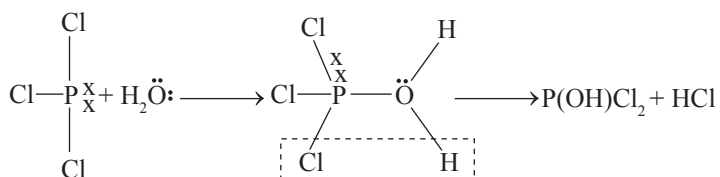
It is also obtained by the action of thionyl chloride with white phosphorus.

**Properties**

Phosphorus trichloride is readily hydrolysed by water to orthophosphorous acid, H_3PO_3 , and hydrogen chloride:

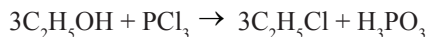
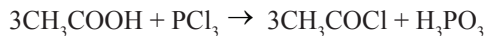


It is thought that the reaction takes place in stages, with the formation of complexes in which the oxygen atom of a water molecule is attached to the phosphorus atom (expansion of the octet can occur since the phosphorus atom has d orbitals available):

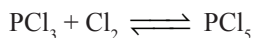
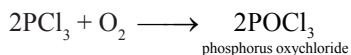


(This should be compared with the hydrolysis of silicon tetrachloride where d orbitals are also available, and contrasted with the hydrolysis of nitrogen trichloride where they are not.)

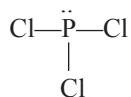
Phosphorus trichloride reacts with many compounds containing the —OH group, and it is used in organic chemistry for the preparation of acid chlorides and alkyl chlorides, e.g.,



It readily combines with oxygen and chlorine (reversibly), the phosphorus atom increasing its covalency from three to five:



It has a pyramidal shape as shown, in which phosphorus is sp^3 hybridized.



Worked-out Examples

78. Why PCl_3 on hydrolysis readily gives phosphorus acid and hydrochloric acid whereas NCl_3 on hydrolysis yields NH_3 and HOCl ?

Ans. This is due to the fact that nitrogen cannot expand the octet. Water can attack phosphorus as it has d-orbitals leading to hydrolysis. NCl_3 cannot accept any other nucleophile till the chlorine first leaves and hydrolysis is slower and different.

79. H_3PO_3 disproportionates whereas H_3PO_4 does not. Explain.

Ans. P-atom in H_3PO_3 has +3 oxidation state (i.e., in between +5 and -3, the maximum and minimum value) and thus it can disproportionate.

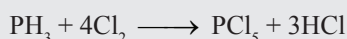
However, H_3PO_4 has P atom in highest oxidation state i.e., +5 which can slow only decrease in oxidation number.

80. What may be expected to happen when phosphine gas is mixed with chlorine gas?

[AIEEE 2003]

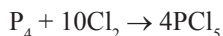
- (a) PCl_3 and HCl are formed and the mixture warms up
- (b) PCl_5 and HCl are formed and the mixture cools down
- (c) $\text{PH}_3\cdot\text{Cl}_2$ is formed with warming up
- (d) The mixture only cools down

Ans. (d) On mixing phosphine with chlorine gas PCl_5 and HCl are formed.

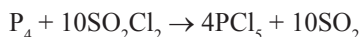
**Pentachloride****Preparation****The pentahalides, PX_5**

Except for the pentafluoride, all the pentahalides have been prepared. Phosphorus pentafluoride is a gas; the pentachloride and pentabromide are solids.

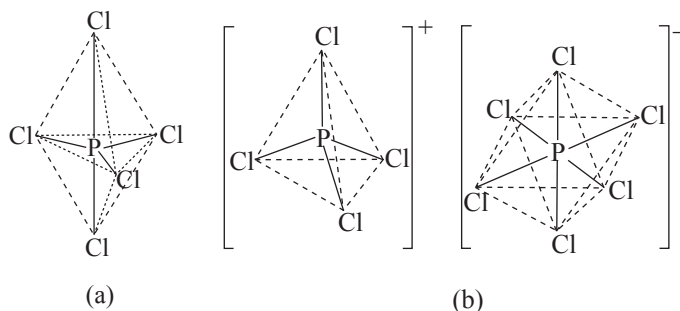
Phosphorus pentachloride is prepared by passing excess dry chlorine through a flask into which phosphorus trichloride is dripping. Since it dissociates into the trichloride and chlorine very readily, the experiment is conducted in an ice-cooled apparatus.



It can also be prepared by the action of SO_2Cl_2 on phosphorus.

**Properties**

In the vapour state the phosphorus pentachloride molecule has a trigonal bipyramidal structure in the solid state it is ionic, having the structure $(\text{PCl}_4^+)(\text{PCl}_6^-)$

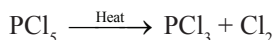


- (a) The structure of phosphorus pentachloride vapour (b) The structure of solid phosphorus penta-chloride. Phosphorus pentabromide is also ionic in the solid state, but its structure is $(\text{PBr}_4^+)\text{Br}^-$. This has been partially attributed to the impossibility of packing six bromines round a central phosphorous atom to give a stable PBr_6^- ion—a steric effect. Steric hindrance is also a possible explanation of why PI_5 does not exist—the iodine atom being larger than the bromine atom.

PCl_5 is a yellowish white powder and in moist air, it hydrolyses to POCl_3 and finally gets converted to phosphoric acid.



When heated, it sublimes but decomposes stronger heating.



It reacts with organic compounds containing $-\text{OH}$ group converting them to chloro derivatives.



Finely divided metals on heating with PCl_5 give corresponding chlorides.



It is used in the synthesis of some organic compounds, e.g., $\text{C}_2\text{H}_5\text{Cl}$, CH_3COCl .

Worked-out Examples

81. In compounds of type ECl_3 , where $\text{E} = \text{B}, \text{P}, \text{As}$ or Bi , the angles $\text{Cl}-\text{E}-\text{Cl}$ for different E are in order
(a) $\text{B} > \text{P} = \text{As} = \text{Bi}$ (b) $\text{B} > \text{P} > \text{As} > \text{Bi}$ (c) $\text{B} < \text{P} = \text{As} = \text{Bi}$ (d) $\text{B} < \text{P} < \text{As} < \text{Bi}$

Ans. (b)

82. PCl_5 is well known while PI_5 is unknown. Why?

Ans. Due to larger size of I , the overlapping in between P and I does not take place to an appreciable extent in order to form covalent bond.

83. Discuss the changes in geometry when PCl_3 is chlorinated under controlled conditions and the product obtained is solidified.

Ans. When PCl_3 is chlorinated, PCl_5 is formed.

- (i) In PCl_3 , the geometry of molecule is pyramidal.
- (ii) In vapour state PCl_5 is trigonal bipyramidal with sp^3d hybridization of phosphorus atom,
- (iii) When PCl_5 is in solid state it has ionic structure corresponding to $[\text{PCl}_4]^+[\text{PCl}_6]^-$. In $(\text{PCl}_4)^+$, P -atom is sp^3 hybridized while in $(\text{PCl}_6)^-$ it is sp^3d^2 hybridized.

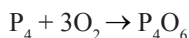
84. What is the geometry of $[\text{PBr}_4]^+$ in solid phase where it exists as $[\text{PBr}_4]^+ \text{Br}^-$ in PBr_5 ?

Ans. Tetrahedral.

The oxides of phosphorus, P_4O_6 and P_4O_{10}

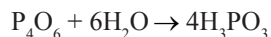
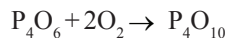
Phosphorus trioxide, P_2O_3

Phosphorus trioxide, once thought to be P_2O_3 but now known to be the dimer, is obtained as a white solid (m.p. 23.8°C) by burning phosphorus in a limited amount of air:



It can be freed from small amounts of phosphorus pentoxide, which are also formed, by making use of its higher volatility.

Phosphorus trioxide reacts readily with oxygen when heated, to give phosphorus pentoxide, and produces orthophosphorous acid with water:



Its molecular weight in solution and in the vapour state corresponds with the formula P_4O_6 . This structural unit is based on the tetrahedral P_4 molecule, each oxygen atom bridging two phosphorus atoms.

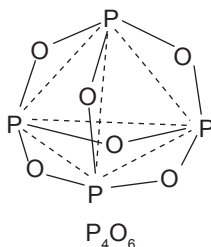
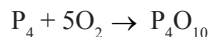


Fig. 6.5 The structure of phosphorus trioxide

Phosphorus pentoxide, P_4O_{10}

Phosphorus pentoxide is formed as a white solid when phosphorus is burnt in an excess of air:



It reacts with water, forming metaphosphoric acid (empirical formula HPO_3) which on boiling gives orthophosphoric acid, H_3PO_4 :

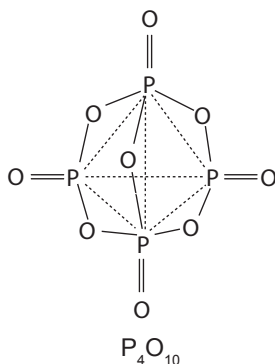
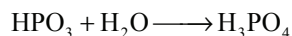
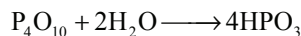
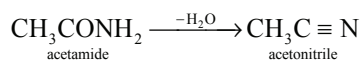
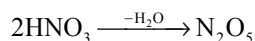


Fig. 6.6 The structure of phosphorus pentoxide

Because of its high affinity for water it is often used as a drying agent. It will also remove the elements of water from compounds and has been used for dehydrating nitric acid to dinitrogen pentoxide, and for converting acid amides to acid nitriles, e.g.,



Phosphorus pentoxide exists in a number of polymorphic forms, at least three of which are crystalline. One of these crystalline forms is composed of P_4O_{10} units which are structurally related to those of phosphorus trioxide, except that each phosphorus atom is bonded to an extra oxygen. Because these bonds are shorter than the other P—O bonds it is thought that double bonding is involved, i.e., phosphorus expands its octet instead of datively bonding to oxygen by means of its lone pair.

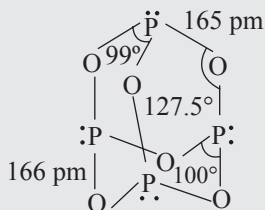
Worked-out Examples

85. Number of sigma bonds in P_4O_{10} is

[AIEEE 2002]

- (a) 6 (b) 7 (c) 17 (d) 16

Ans. (d)



86. P_2O_5 is used in the formation of N_2O_5 , why?

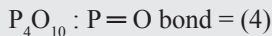
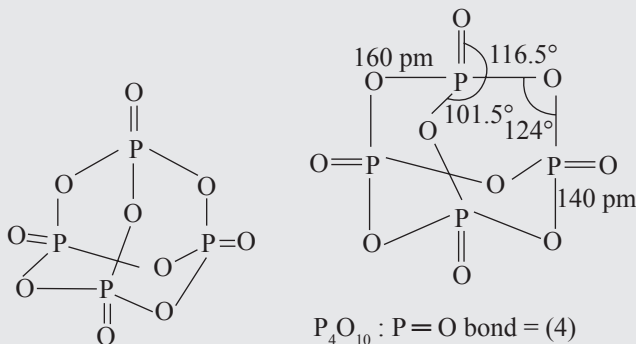
Ans. P_4O_{10} is strong dehydrating agent and dehydrates HNO_3 to N_2O_5 .



87. How many P—O bonds are there in P_2O_3 and P_2O_5 ? Also draw their structures.

Ans. P_2O_3 and P_2O_5 exist as dimers having the following shape:

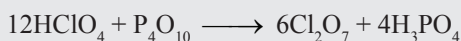
- (i) P_4O_6 : In P_4O_6 , The molecule has tetrahedral symmetry with phosphorus atoms located at the corners of regular tetrahedron and each of six oxygen atoms located between two phosphorus atoms in the plane of symmetry.



- (ii) P_4O_{10} : In vapour state P_4O_{10} molecule has following arrangement. The phosphorus and shared oxygen bond has no π bonding and there is 1.5–2.0 π bond between phosphorus and unshared oxygen in addition to δ -bond.

88. Which among P_4O_{10} , Cl_2O_7 , I_2O_5 has greatest affinity for water.

Ans. P_4O_{10} , one of the most powerful dehydrating agents known. Cl_2O_7 is the product of dehydration of $HClO_4$ by P_4O_{10} .



Also I_2O_5 is obtained by heating HIO_3 and thus it can not have too great affinity for water.

Oxoacids of Phosphorus

Phosphorus forms a number of oxoacids.

Oxoacids or oxyacids are the compounds of an element with oxygen and hydrogen.

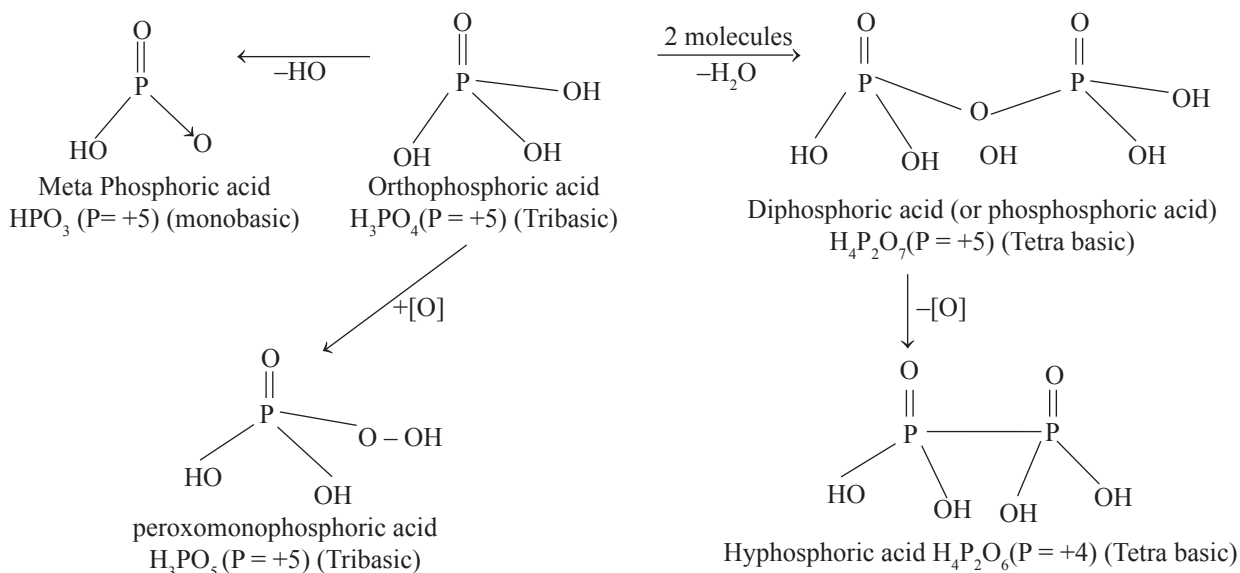
Phosphorus forms numerous oxoacids, all of which are based on tetrahedral four coordinated phosphorus containing at least one P = O unit and one P – OH group.

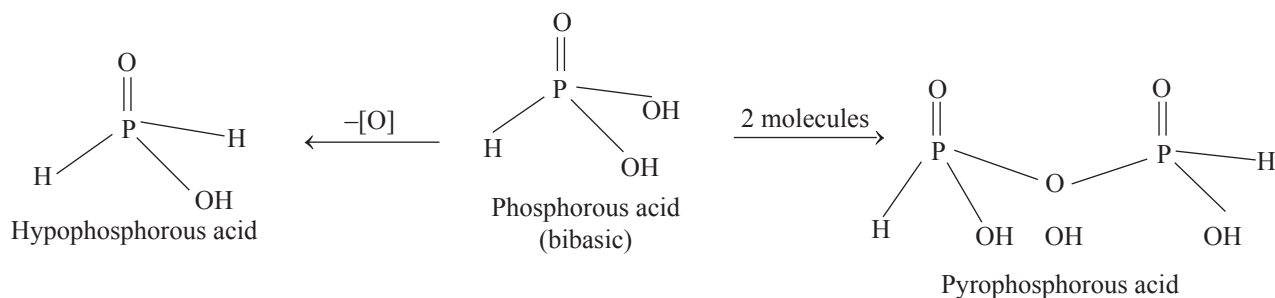
The important oxoacids of phosphorus with their formulas, methods of preparation and the presence of some characteristic bonds in their structures are given in Table.

Note that the oxoacid with P-H bond can function as reducing agents.

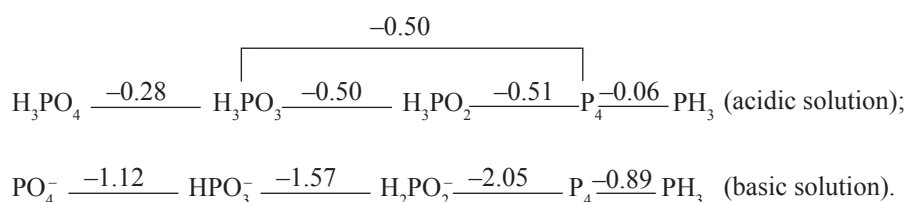
Name	Formula	Oxidation state of Phosphorus	Characteristic bonds and their number	Preparation
Hypophosphorous	H_3PO_2	+1	One P–OH Two P–H One P=O	White P_4 + alkali
Orthophosphorous	H_3PO_3	+3	Two P–OH One P–H One P=O	$\text{P}_2\text{O}_3 + \text{H}_2\text{O}$
Pyrophosphorous	$\text{H}_4\text{P}_2\text{O}_5$	+3	Two P–OH Two P–H Two P=O	$\text{PCl}_3 + \text{H}_3\text{PO}_3$
Hypophosphoric	$\text{H}_4\text{P}_2\text{O}_6$	+4	Four P–OH Two P–H One P=P	Red P_4 + alkali
Orthophosphoric	H_3PO_4	+5	Three P–OH One P=O	
Pyrophosphoric	$\text{H}_4\text{P}_2\text{O}_7$	+5	Four P–OH Two P–H One P–O–P	Heat Phosphoric acid
Metaphosphoric	$[\text{HPO}_3]_n$	+5	Three P–OH Three P=O Three P–O–P	Phosphorous acid + Br_2 heat in a sealed tube

The compositions of the oxoacids are interrelated in terms of loss or gain of H_2O molecule or O-atom.

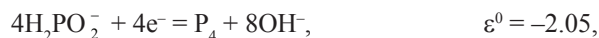
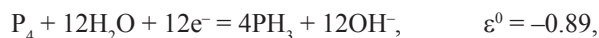




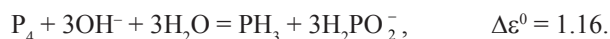
Because phosphorus displays a number of oxidation states, and because its oxyanions have a tendency to polymerize, the solution chemistry of this element is moderately complex. We can get a general idea of the properties of the various oxidation states of phosphorus, however, by consulting the following reduction-potential diagrams:



We can see that the oxyanions of phosphorus are very poor oxidizing agents, particularly in basic solutions. On the contrary, all but the highest oxidation state of phosphorus have strong reducing properties. In addition, elemental phosphorus is unstable with respect to disproportionation, from the potential diagram for basic solutions we can deduce the following:



Combination of these half-reactions gives



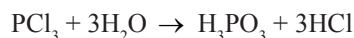
The main oxyacids of phosphorus are the phosphorous acids and the phosphoric acids, of which phosphorus trioxide and phosphorus pentoxide respectively are the acid anhydrides. Other oxyacids of phosphorus exist, but they are not directly based on these oxides and will not be considered here.

Orthophosphorous acid, H_3PO_3

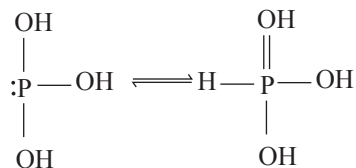
This is the most important acid based on phosphorus trioxide, and can be obtained from the latter by the addition of water:



It can also be obtained by hydrolysis of phosphorus trichloride:



The pure acid is a deliquescent solid and in aqueous solution it is moderately strong. Since it is dibasic its structure is thought to contain only two hydroxyl groups, although its formation by hydrolysis of phosphorus trichloride suggests three. It is thought that an equilibrium exists of the type shown below, which is very largely displaced to the right:



The phosphorus atom is therefore essentially 5-covalent.

The alkali metal salts of orthophosphorous acid, e.g., $\text{Na}^+\text{H}_2\text{PO}_3^-$ and $(\text{Na}^+)_2\text{HPO}_3^{2-}$, are soluble in water; the majority of other salts are insoluble. The acid and its salts are strong reducing agents, the free acid decomposing into phosphine—which is a powerful reducing agent itself—and orthophosphoric acid when heated:

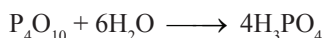
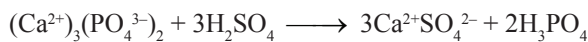


The phosphoric acids and phosphates

A variety of phosphoric acids based on phosphorus pentoxide exists; in many respects the structural chemistry of these acids and their anions is similar to that of the silicates.

Orthophosphoric acid, H_3PO_4 , and orthophosphates

Orthophosphoric acid is manufactured by heating rock phosphate with sulphuric acid; a purer product is obtained by reacting phosphorus pentoxide with water and then boiling.



Orthophosphoric acid is a deliquescent crystalline solid, although it is usually encountered as a concentrated and very viscous aqueous solution; this high viscosity is due to the presence of hydrogen bonding which links individual molecules of the acid into very large aggregates.

The acid is tribasic, and relatively weak, neutralization with alkalis taking place in stages. When an M/10 solution of sodium hydroxide is added to an M/10 solution of the acid, methyl orange changes colour at the stage corresponding to conversion into sodium dihydrogen orthophosphate, $\text{Na}^+\text{H}_2\text{PO}_4^-$:



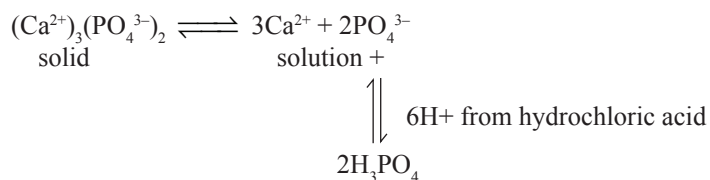
If phenolphthalein is used as indicator, twice as much alkali is needed before the indicator changes colour and this corresponds to the formation of disodium hydrogen orthophosphate, $(\text{Na}^+)_2\text{HPO}_4^{2-}$:



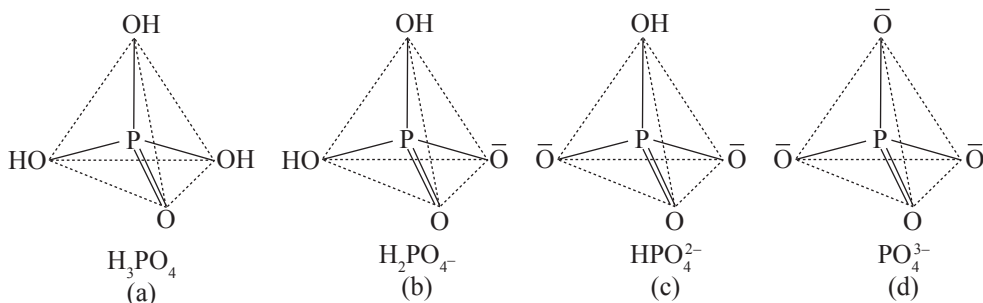
The third stage leading to the formation of trisodium orthophosphate, $(\text{Na}^+)_3\text{PO}_4^{3-}$, cannot be realized in practice since this salt is extensively hydrolyzed in aqueous solution, i.e., the orthophosphate ion, PO_4^{3-} , is a strong base:



Most phosphates are insoluble in water but they usually dissolve in the presence of a strong acid such as hydrochloric acid. This is due to the formation of weakly dissociated phosphoric acid which allows the equilibrium below to move over to the right, e.g.,



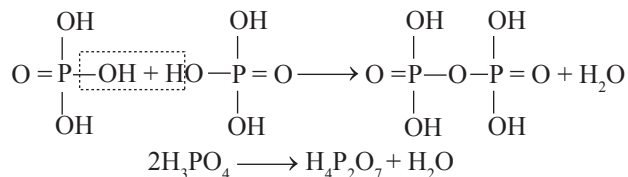
Orthophosphoric acid has an approximately tetrahedrally shaped molecule (a); the orthophosphate anions also adopt a tetrahedral configuration (b), (c) and (d) and in these instances resonance is involved.



- The structure of orthophosphoric acid
- The structure of the dihydrogen orthophosphate anion
- The structure of the hydrogen orthophosphate anion
- The structure of the orthophosphate anion

Pyrophosphoric acid, $\text{H}_4\text{P}_2\text{O}_7$, pyrophosphates and other chain poly-phosphates

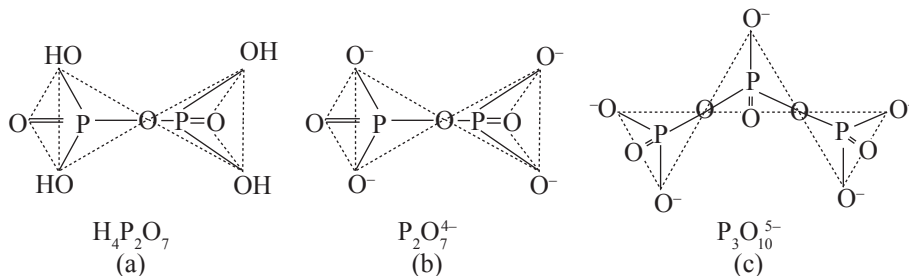
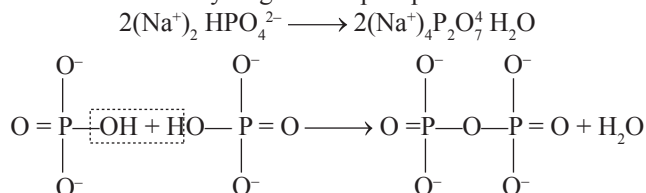
Pyrophosphoric acid can be obtained by heating orthophosphoric acid to 220°C , when two molecules condense, with the elimination of water:



Although pyrophosphoric acid is tetrabasic only two sodium salts are known:

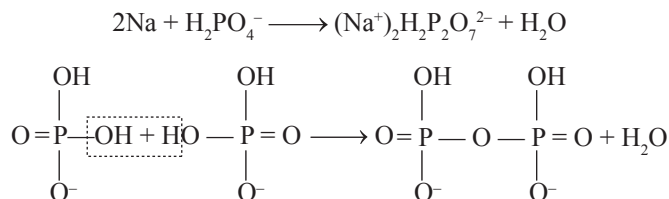


The tetrasodium salt is obtained when disodium hydrogen orthophosphate is heated at 500°C :



- The structure of pyrophosphoric acid
- The structure of the pyrophosphate anion
- The structure of the tripolyphosphate anion

The disodium salt is conveniently obtained by heating sodium dihydrogen orthophosphate at 200°C :



Pyrophosphoric acid and its anions have a structure based on tetrahedral PO_4 units linked together by an oxygen atom common to both.

The linking together of more PO_4 tetrahedral units gives rise to linear (or chain) polyphosphates, e.g., the tripolyphosphate anion $\text{P}_3\text{O}_{10}^{5-}$ contains three PO_4 units.

6.44 Group15-Nitrogen Family

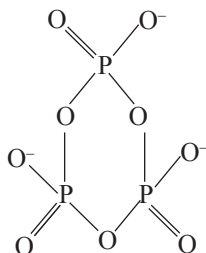
Chain polyphosphates are able to form water soluble complexes with many metals and are used in the softening of hard water. They complex with calcium and magnesium ions and so prevent them forming a scum with soap.

Metaphosphoric acid, HPO_3 , and metaphosphates

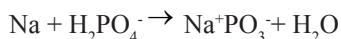
Metaphosphoric acid is obtained by the dehydration of orthophosphoric acid at 316°C . It is a glassy polymeric solid and a mixture of several acids, with the empirical formula HPO_3 .



Although the structures of the various metaphosphoric acids are still uncertain, mixtures of metaphosphates have been separated by chromatographic methods. These metaphosphate ions are built up from PO_4 units and are cyclic as opposed to linear. The trimetaphosphate ion, $\text{P}_3\text{O}_9^{3-}$, is shown below:



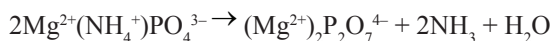
Sodium metaphosphate, empirical formula Na^+PO_3^- , is formed when sodium dihydrogen orthophosphate is heated:



It is a mixture, containing cyclic metaphosphate ions of differing ring size.

Tests for phosphates

Orthophosphates give a yellow precipitate of ammonium phosphomolybdate when warmed to about 60°C with nitric acid and ammonium molybdate solution. This same test is also given by condensed phosphates, since hydrolysis to orthophosphates takes place on warming; precipitation of ammonium phosphomolybdate in these cases, however, is a much slower process phosphates precipitate magnesium ammonium phosphate when added to an ammoniacal solution of magnesium chloride. If this precipitate is washed; filtered and ignited it forms magnesium pyrophosphate. This can be weighed and thus orthophosphates can be estimated quantitatively.



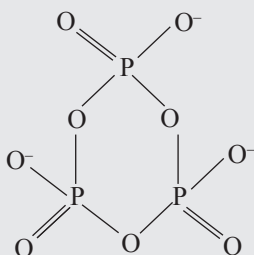
Worked-out Examples

89. The number of P—O—P bonds in cyclic metaphosphoric acid is

(2000)

- (a) zero (b) two (c) three (d) four

Ans. (c) The structure of metaphosphoric acid is



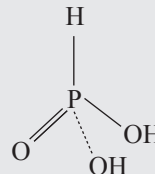
90. Which of the following statements is correct?

(2003)

- (a) H_3PO_3 is dibasic and reducing
- (b) H_3PO_3 is tribasic and reducing
- (c) H_3PO_3 is tribasic and nonreducing
- (d) H_3PO_3 is dibasic and nonreducing

Ans. (a) The structure of H_3PO_3 is

Since it contains two $-\text{OH}$ groups, it is dibasic. Also it contains H atom directly attached to P, hence, it is reducing



91. The basicity of phosphorous acid (H_3PO_3) is ____.

(1990)

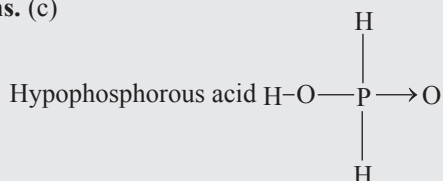
Ans. two

92. The number of hydrogen atom(s) attached to phosphorus atom in hypophosphorous acid is

(AIEEE 2005)

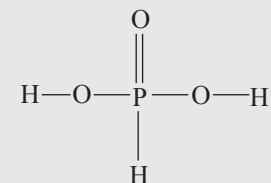
- (a) three
- (b) one
- (c) two
- (d) zero

Ans. (c)

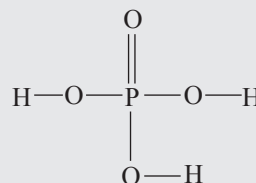


93. Ortho-phosphorus acid is dibasic while ortho-phosphoric acid is tribasic. Explain.

Ans.



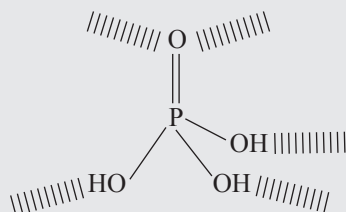
(Ortho-phosphorus acid)



(Ortho-phosphoric acid)

94. Ortho-phosphoric acid is syrupy liquid. Explain.

Ans. Ortho-phosphoric acid is crystalline solid with deliquescent nature and thus absorbs water to form syrupy liquid. Also H-bonding enhances its viscosity.

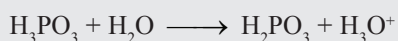


H-bonding with other molecules

95. H_3PO_3 and H_3BO_3 both are acid. Classify then on account of acid-base theories. Also report hybridization in each.

Ans. H_3PO_3 an oxoacid of P $\left[\begin{array}{c} \text{O} \\ \parallel \\ \text{HO}-\text{P}-\text{OH} \\ | \\ \text{H} \end{array} \right]$ and is Bronsted dibasic acid as it acts as protons donor.

P in H_3PO_3 is sp^3 hybridized.



$\text{H}_3\text{BO}_3 \left[\begin{array}{c} \text{OH} \\ | \\ \text{HO} - \text{B} - \text{OH} \end{array} \right]$ and Lewis mono basic acid as it acts electron pair acceptor because Boron is sp^2 hybridized and have only six electrons (an incomplete octet).



96. HNO_3 is an oxidizing agent but H_3PO_4 is not. Explain.

Ans. The inability of nitrogen atom to impair and promote its 2s electron results in the pentapositive state of nitrogen being less stable than tripositive. Thus HNO_3 where nitrogen is in +5 oxidation state is an oxidizing agent. On the other hand P has, d-orbitals to expand its octet and also shows no inert pair effect, is quite stable in V state. Thus H_3PO_4 in which P is in +5 oxidation state is not oxidizing.

97. Although HNO_3 is stronger than HNO_2 , the oxoacids of P i.e., H_3PO_2 , H_3PO_3 and H_3PO_4 are of approximately same strength. Explain.

Ans. In the oxoacids the acidic character of the acid increases with the increase in the oxidation state of central atom. However, this is not so in the oxoacids of phosphorus, viz. H_3PO_2 , H_3PO_3 and H_3PO_4 in which acidic strength does not vary much with the oxidation number of phosphorus atom as is evidenced by their dissociation constants.

The order of K_a : $\text{H}_3\text{PO}_3 > \text{H}_3\text{PO}_2 > \text{H}_3\text{PO}_4$

In the above three acids, although the number of —OH groups (ionisable hydrogen) increases yet the acidity does not increase very much. This is due to the fact that the number of unprotonated oxygen, responsible for the enhancement of acidic character due to inductive effect, remains the same with the result dissociation constant also remains nearly same.

98. Polyphosphates are used as water softening agents because they

(2002)

- form soluble complexes with anionic species
- precipitate anionic species
- form soluble complexes with cationic species
- precipitate cationic species

Ans. (c)

99. Write balanced equations for the following reactions:

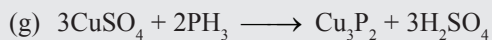
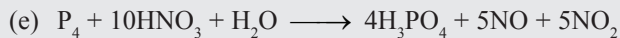
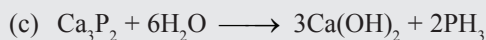
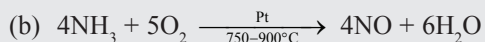
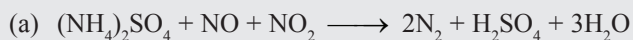
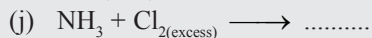
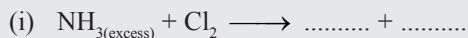
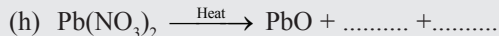
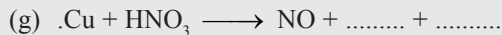
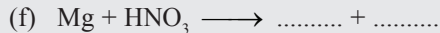
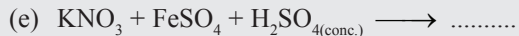
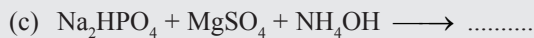
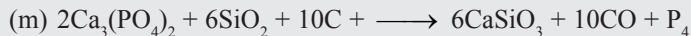
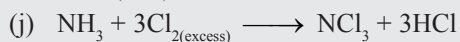
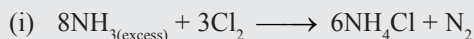
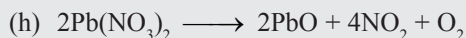
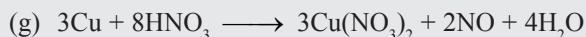
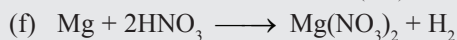
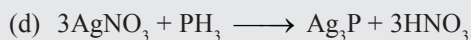
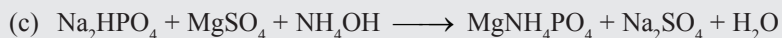
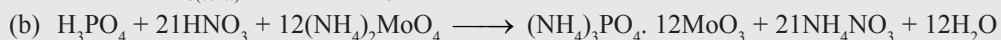
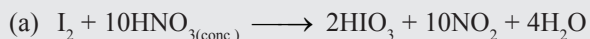
- $\text{Ca}_3\text{P}_2 + \text{H}_2\text{O} \longrightarrow$
- $\text{P}_4\text{O}_{10} + \text{H}_2\text{O} \longrightarrow$
- $\text{As}_4 + \text{Cl}_2 \text{ (excess)} \longrightarrow$
- $\text{P}_4 + \text{KOH} + \text{H}_2\text{O} \longrightarrow$

Ans.

- $\text{Ca}_3\text{P}_2 + 6\text{H}_2\text{O} \longrightarrow 3\text{Ca(OH)}_2 + 2\text{PH}_3$
- $\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \longrightarrow 4\text{H}_3\text{PO}_4$
- $\text{As}_4 + 10\text{Cl}_2 \longrightarrow \text{As}_2\text{Cl}_5$
- $\text{P}_4 + 3\text{KOH} + 3\text{H}_2\text{O} \longrightarrow 3\text{KH}_2\text{PO}_2 + \text{PH}_3$

100. What happens when?

- Ammonium sulphate is heated with a mixture of NO and NO_2 .
- A mixture of air and ammonia is passed over heated platinum gauze.
- Water is added to calcium phosphide.
- Calcium phosphate is heated with a mixture of sand and carbon.
- Phosphorus reacts with nitric acid to give equimolar ratio of nitric oxide and nitrogen dioxide.
- Zinc is treated with very very dilute nitric acid.
- Phosphine is treated with an acidified CuSO_4 solution.

Ans.**101.** Complete and balance the following equations:**Ans.**

ONE CORRECT OPTION TYPE QUESTIONS

- The reddish brown coloured gas formed when nitric oxide is oxidized by air is
 (a) N_2O_5 (b) N_2O_4
 (c) NO_2 (d) N_2O_3
- White P reacts with caustic soda. The products are PH_3 and NaH_2PO_2 . This reaction is an example of
 (a) Oxidation
 (b) Reduction
 (c) Oxidation and reduction
 (d) Neutralization
- Which forms $p\pi-p\pi$ multiple bonds with itself and with C and O?
 (a) P, As (b) N, As
 (c) N, P (d) N
- Nitrogen dioxide cannot be obtained by heating:
 (a) KNO_3 (b) $\text{Pb}(\text{NO}_3)_2$
 (c) $\text{Cu}(\text{NO}_3)_2$ (d) AgNO_3
- The bonds present in N_2O_5 are:
 (a) only ionic (b) covalent and coordinate
 (c) only covalent (d) aluminum oxide
- Which of the following oxides of nitrogen is a coloured gas?
 (a) N_2O (b) NO
 (c) N_2O_5 (d) NO_2
- Amongst the trihalides of nitrogen which one is least basic?
 (a) NF_3 (b) NCl_3
 (c) NBr_3 (d) NI_3
- In P_4O_{10} each P atom is linked with O atoms.
 (a) 2 (b) 3
 (c) 4 (d) 5
- One mole of calcium phosphide on reaction with excess water gives
 (a) one mole of phosphine
 (b) two moles of phosphoric acid
 (c) two moles of phosphine
 (d) one mole of phosphorus pentoxide
- When PbO_2 reacts with conc. HNO_3 the gas evolved is
 (a) NO_2 (b) O_2
 (c) N_2 (d) N_2O
- Ammonia, on reaction with hypochlorite anion, can form
 (a) NO (b) NH_4Cl
 (c) N_2H_4 (d) HNO_2
- White phosphorous on reaction with lime water gives calcium salt of an acid(A) along with a gas (X). Which of the following is correct?
 (a) (A) on heating gives (X) and O_2
 (b) The bond angle in (X) is less than that in case of ammonia
 (c) (A) is a dibasic acid
 (d) (X) is more basic than ammonia
- One mole of H_3PO_3 on reaction with excess of NaOH gives
 (a) one mole of Na_2HPO_3
 (b) two moles of $\text{Na}_2\text{H}_2\text{PO}_3$
 (c) two moles of Na_2HPO_3
 (d) one mole of Na_3PO_3
- A diatomic gas will be obtained in:
 (a) $\text{Cu} + \text{dil. HNO}_3 \longrightarrow$
 (b) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7 \xrightarrow{\text{heat}}$
 (c) both
 (d) none
- For $\text{NH}_2\text{OH} \cdot \text{HCl} + \text{NaNO}_2 \longrightarrow (\text{A}) \xrightarrow{\text{Cu}} (\text{B}) + (\text{X})_{\text{gas}}$, which of the following is correct?
 (a) (B) is an amphoteric oxide
 (b) (X) is a colourless, diamagnetic gas which combines with Al on heating.
 (c) (X) can be produced by action of $(\text{Zn} + \text{NaOH})$ on NaNO_2
 (d) none of the above
- If O_2 is removed from the formula of anhydride of HNO_2 , then the formula of the resulting compound satisfies which of the following properties?
 (a) It produces tear in eyes
 (b) It supports combustion
 (c) It is paramagnetic
 (d) It cannot react with red hot copper
- Which of the following is correct?
 (a) N_2O is a laughing gas and is angular in shape
 (b) NO_2 is a sweet smelling and is angular in shape
 (c) NO is a colourless gas and acidic in nature
 (d) NO_2 on reaction with NaOH gives a mixture of two salts

18. When a mixture of NO and NO₂ is passed through an aqueous solution of ammonium sulphate, we get:
- a dibasic acid which has no dehydrating property
 - a diatomic gas which on reaction with acetylene under electric spark gives a monobasic acid whose anion is pseudohalide
 - a dibasic acid whose dinegative anion has all bonds equal with bond angle 90°
 - a dibasic acid whose anhydride in solid state forms a cyclic trimer
19. A tetra-atomic molecule (A) on reaction with nitrogen (I) oxide, produces two substances (B) and (C). (B) is a dehydrating agent while substance (C) is a diatomic gas which shows almost inert behaviour. The substances (A), (B) and (C) are:
- P₄, P₄O₁₀, N₂
 - P₄, N₂O₅, N₂
 - P₄, P₂O₃, Ar
 - P₄, P₂O₃, O₂
20. The number of P—O—P and P—O—H bonds present respectively in pyrophosphoric acid molecule are:
- 1, 2
 - 2, 2
 - 1, 4
 - 1, 8
21. Which of the following is a cyclic oxoacid?
- H₄P₂O₇
 - H₂P₂O₆
 - H₅P₅O₁₅
 - H₃P₃O₉
22. The equivalent mass of phosphoric acid (H₃PO₄) in the reaction, NaOH + H₃PO₄ → NaH₂PO₄ + H₂O, is:
- 25
 - 49
 - 59
 - 98
23. The oxyacid of phosphorus in which phosphorus has the lowest oxidation state is:
- hypophosphorus acid
 - orthophosphoric acid
 - pyrophosphoric acid
 - metaphosphoric acid
24. Which is/are correct statements about P₄O₆ and P₄O₁₀?
- Both form oxoacids H₃PO₃ and H₃PO₄ respectively
 - In P₄O₆ each P is joined to three O and in P₄O₁₀ each P is linked to four O atoms
 - Both
 - none
25. Blue liquid which is formed at -30°C by mixing of two gases is:
- N₂O
 - N₂O₅
 - N₂O₃
 - N₂O₄
26. For H₃PO₃ and H₃PO₄, the correct choice is:
- H₃PO₃ is dibasic and reducing
 - H₃PO₃ is dibasic and non-reducing
 - H₃PO₄ is tribasic and reducing
 - H₃PO₃ is tribasic and non-reducing
27. In nitrogen family, the H-M-H bond angle in the hydrides gradually becomes closer to 90° on going from N to Sb. This shows that gradually
- The basic strength of the hydrides increases
 - Almost pure p-orbitals are used for M-H bonding
 - The bond energies of M-H bonds increase
 - The bond pairs of electrons become nearer to the central atom
28. Among the trihalides of nitrogen which one is most basic?
- NF₃
 - NCl₃
 - NI₃
 - NBr₃
29. Which of the following compounds does not exist?
- AsCl₅
 - SbCl₃
 - BiCl₃
 - SbCl₅
30. Among the following oxides, the least acidic is
- P₄O₆
 - P₄O₁₀
 - As₄O₆
 - As₄O₁₀
31. What causes nitrogen to be chemically inert?
- Multiple bond formation in the molecule
 - Absence of bond polarity
 - Short internuclear distance
 - High bond energy
32. Ammonia can be dried by
- Conc. H₂SO₄
 - P₄O₁₀
 - CaO
 - Anhydrous CaCl₂
33. Which of the following phosphorus is most reactive?
- Red phosphorus
 - White phosphorus
 - Scarlet phosphorus
 - Violet phosphorus
34. White phosphorus is
- A monoatomic gas
 - P₄, a tetrahedral solid
 - P₈, a crown
 - P₈, a crown
35. One of the oxidants used with liquid propellants is
- Ammonium perchlorate
 - Nitrocellulose
 - Sulphuric acid
 - Nitrogen tetroxide (N₂O₄)

6.50 Group15-Nitrogen Family

36. Which of the following compounds is explosive in nature?
- Phosphorus trichloride
 - Nitrogen trichloride
 - Hyponitrous acid
 - Nitrosyl chloride
37. Among the 15th group elements, as we move from nitrogen to bismuth, the pentavalency becomes less pronounced and trivalency becomes more pronounced due to
- None metallic character
 - Inert pair effect
 - High electronegativity
 - Large ionization energy
38. Of the following compounds, the most acidic is
- As_2O_3
 - P_2O_5
 - Sb_2O_3
 - Bi_2O_3
39. Liquid ammonia bottles be opened after cooling them in ice for sometime. It is because liquid NH_3
- Brings tears to the eyes
 - Has a high vapour pressure
 - Is a corrosive liquid
 - Is a mild explosive
40. The deep blue colour produced on adding excess of ammonia to copper sulphate is due to presence of
- Cu^{2+}
 - $\text{Cu}(\text{NH}_3)_4^{2+}$
 - $\text{Cu}(\text{NH}_3)_6^{2+}$
 - $\text{Cu}(\text{NH}_3)_2^{2+}$
41. Ammonia on catalytic oxidation gives an oxide from which nitric acid is obtained. The oxide is:
- N_2O_3
 - NO
 - NO_2
 - N_2O_5
42. Fixation of nitrogen means:
- Reaction of nitrogen with oxygen
 - Conversion of free atmospheric nitrogen into nitrogen compounds
 - The action of denitrifying bacteria on nitrogen compounds.
 - Decomposition of nitrogenous compounds to yield free nitrogen.
43. In smoke screens calcium phosphide is used, because it:
- Catches fire easily
 - Burns and gives soot
 - Forms phosphine which gives smoke
 - None of these
44. There is very little different in acid strength in the acids H_3PO_4 , H_3PO_3 and H_3PO_2 because:
- Phosphorus in these acids exists in different oxidation states
 - The hydrogens in these acids are not all bound to the oxygen and have same number of unprotonated oxygen
 - Phosphorus is highly electronegative element
 - Phosphorus oxides are less basic
45. Bones glow in the dark because:
- They contain shining material
 - They contain red phosphorus
 - White phosphorus undergoes slow combustion in contact with air
 - White phosphorus changes into red form

COMPREHENSION BASED QUESTIONS

COMPREHENSION-I

Nitrogen, phosphorus and potassium are regarded as essential plant nutrients. After a few years of continuous cultivation, the soil becomes poorer and poorer in materials supplying nitrogen, phosphorus and potassium to plants. If the soil is not compensated, it would become less productive and infertile. In order to maintain soil fertility, it is necessary to add materials containing nitrogen, phosphorus and potassium in the form of fertilizers. Chemical substances which are added to the soil as to make up the deficiency of essential elements are called chemical fertilizers. Every chemical compound of nitrogen, phosphorus and potassium cannot be used as a fertilizer.

Substances to be used as fertilizers must have the following characteristics:

- it must be soluble in water
- it should be easily assimilated by plants
- It should be stable
- it should not disturb the pH of the soil
- it should be cheap

The important nitrogenous fertilizers are:

- ammonium sulphate
- calcium cyanamide
- calcium ammonium nitrate
- urea
- basic calcium nitrate

The important phosphatic fertilizers are:

- (a) calcium superphosphate
- (b) nitrophosphate
- (c) triple phosphate
- (d) phosphatic slag

The commonly used potassium fertilizers are:

- (a) potassium nitrate
- (b) potassium chloride
- (c) potassium sulphate

Fertilizers containing N, P and K is suitable adjusted proportions are known as NPK fertilizers.

1. Which one of the following substances is used as a fertilizer?
 - (a) $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
 - (b) $\text{Ca}(\text{CN})_2$
 - (c) $\text{Ca}(\text{H}_2\text{PO}_4)_2 \cdot \text{H}_2\text{O} \cdot 2(\text{CaSO}_4 \cdot 2\text{H}_2\text{O})$
 - (d) CaSO_4
2. Fertilizer having highest percentage of nitrogen is:
 - (a) urea
 - (b) calcium cyanamide
 - (c) ammonium nitrate
 - (d) ammonium sulphate
3. Sindhri fertilizer is chemically known as:
 - (a) ammonium nitrate
 - (b) ammonium sulphate
 - (c) urea
 - (d) calcium nitrate
4. Ammonium salts are oxidized in the soil to nitrites by:
 - (a) denitrifying bacteria
 - (b) ammonifying bacteria
 - (c) nitrosifying bacteria
 - (d) nitrifying bacteria
5. The actual composition of nitrolim is:
 - (a) $\text{Ca}(\text{CN})_2$
 - (b) $\text{CaC}_2 + \text{Ca}(\text{CN})_2$
 - (c) $\text{CaCN}_2 + \text{C}$
 - (d) CaH_2

6. Expression like 4-8-2 used for a NPK mixed fertilizer indicates that it contains:

- (a) 4% N_2 , 8% P_2O_5 and 2% K_2O
- (b) 4% urea, 8% $\text{Ca}(\text{H}_2\text{PO}_4)_2$ and 2% KCl
- (c) 4% NH_3 , 8% H_3PO_4 and 2% K_2SO_4
- (d) 4% NH_4NO_3 , 8% P_2O_5 and 2% KNO_3

COMPREHENSION-II

The pronounced change from non-metallic to metallic behaviour and also increase in the basicity of oxides from nitrogen to bismuth in group 15 is principally due to increasing size of the atoms. The ionization potential of nitrogen is very high on account of its small size. However, ionization potential decreases regularly on descending the group.

1. Which one of the following is a strongest base?
 - (a) AsH_3
 - (b) SbH_3
 - (c) PH_3
 - (d) NH_3
2. Among the trihalides of nitrogen, which one is least basic?
 - (a) NF_3
 - (b) NCl_3
 - (c) NBr_3
 - (d) NI_3
3. Which one of the following fluorides does not exist?
 - (a) NF_5
 - (b) PF_5
 - (c) AsF_5
 - (d) SbF_5
4. Which of the following oxides is most acidic?
 - (a) Bi_2O_3
 - (b) P_2O_3
 - (c) As_2O_3
 - (d) Sb_2O_3
5. The most unstable hydride is:
 - (a) NH_3
 - (b) PH_3
 - (c) BiH_3
 - (d) SbH_3
6. In all the group 15 elements, the number of unpaired electrons in the valence shell is:
 - (a) 4
 - (b) 3
 - (c) 2
 - (d) 5
7. Which trihalide is most ionic among the following?
 - (a) NCl_3
 - (b) PCl_3
 - (c) BiF_3
 - (d) SbF_3

ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

1. Nitrogen (I) oxide is produced by:
 - (a) thermal decomposition of ammonium nitrate
 - (b) disproportionation of N_2O_4
 - (c) thermal decomposition of ammonium nitrite
 - (d) interaction of hydroxylamine and nitrous acid
2. Ammonia on reaction with hypochlorite anion can form:
 - (a) NO
 - (b) NH_4Cl
 - (c) N_2H_2
 - (d) HNO_2

6.52 Group 15-Nitrogen Family

- Which of the following metals become passive when dropped into conc. HNO_3 ?
 (a) Cu (b) Fe
 (c) Cr (d) Al
- White phosphorus has:
 (a) six P—P single bonds
 (b) four P—P single bonds
 (c) four lone pairs of electrons
 (d) P P P angle of 60°
- Which of the following substances are used as fertilizers?
 (a) nitrolim
 (b) urea
 (c) superphosphate of lime
 (d) phosphorite mineral
- Which elements of group 15 are metalloids?
 (a) P (b) As
 (c) Sb (d) Bi
- The metal which produce hydrogen only with very dilute nitric acid are:
 (a) Zn (b) Sn
 (c) Mg (d) Mn
- Which of the following form oxychlorides as precipitate on hydrolysis?
 (a) BiCl_3 (b) SbCl_3
 (c) CCl_4 (d) PbCl_2
- Which of the elements show allotropy?
 (a) N (b) P
 (c) As (d) Bi
- Which of the following reactions can evolve phosphine?
 (a) $\text{White P} + \text{Ca(OH)}_2 \longrightarrow$
 (b) $\text{AlP} + \text{H}_2\text{O} \longrightarrow$
 (c) $\text{H}_3\text{PO}_4 \xrightarrow{\text{Heat}}$
 (d) $\text{PH}_4\text{I} + \text{NaOH} \longrightarrow$
- Sodium nitrate decomposes above 8000°C to give
 (a) N_2 (b) O_2
 (c) NO_2 (d) Na_2O

MATCH THE COLUMN TYPE QUESTIONS

- Match the reactions of metals with dilute HNO_3 (in List I) with the nitrogen compounds (obtained by oxidation/reduction (in List II).

List I

A. $\text{Mg} + \text{dil. HNO}_3$

List II

1. NO

B. $\text{Zn} + \text{dil. HNO}_3$

C. $\text{Sn} + \text{dil. HNO}_3$

D. $\text{Pb} + \text{dil. HNO}_3$

2. H_2

3. N_2O

4. NH_4NO_3

ASSERTION/REASON TYPE QUESTIONS

The following questions consist of two statements as Assertion(A) and Reason(R). While answering these questions choose correctly any one of the following responses.

- If both assertion and reason are true and reason is the correct explanation of assertion.
 - If both assertion and reason are true and reason is not the correct explanation of assertion
 - If assertion is true but reason is false
 - If assertion is false but reason is true.
 - If both assertion and reason are false.
- Assertion:** Although PF_5 , PCl_5 and PBr_5 are known, the pentahalides of nitrogen have not been observed
Reason: Phosphorus has lower electronegativity than nitrogen.
 - Assertion:** HNO_3 is stronger than HNO_2
Reason: In HNO_3 there are two nitrogen to oxygen bonds whereas in HNO_2 there is only one.
 - Assertion:** P_4 is more reactive than N_2 .
Reason: P—P single bond is much more weaker as compared to $\text{N}\equiv\text{N}$ triple bond.
 - Assertion:** Although PF_5 , PCl_5 and PBr_5 are known, the pentahalides of nitrogen have not been observed.
Reason: Phosphorus has lower electronegativity than nitrogen.
 - Assertion:** Nitrogen is unreactive at room temperature but becomes reactive at elevated temperatures or in presence of a catalyst.

Reason: In nitrogen molecule, there is extensive delocalization of electrons.

6. **Assertion:** PH_3 is more basic than NH_3 .
(R) Electronegativity of N is more than phosphorus.
7. **Assertion:** H_3PO_3 and H_3PO_4 are tribasic acids as they contain three hydrogen atoms each.
Reason: Both H_3PO_3 and H_3PO_4 are reducing in nature.

8. **Assertion:** For drying ammonia gas, the common dehydrating agents like H_2SO_4 , CaCl_2 or P_2O_5 cannot be used.

Reason: H_2SO_4 , CaCl_2 or P_2O_5 react with NH_3

9. **Assertion:** Phosphorite mineral cannot be used as a fertilizer of phosphorus.

Reason: Superphosphate of lime is used as a fertilizer of phosphorus.

RECENT YEARS' QUESTIONS OF IIT-JEE AND AIEEE

- The number of P – O – P bonds in cyclic metaphosphoric acid is (IIT JEE 2000)
 - zero
 - two
 - three
 - four
- Ammonia can be dried by (IIT JEE 2000)
 - conc. H_2SO_4
 - P_4O_{10}
 - CaO
 - anhydrous CaCl_2
- Polyphosphates are used as water softening agents because they (IIT JEE 2002)
 - form soluble complexes with anionic species
 - precipitate anionic species
 - form soluble complexes with cationic species
 - precipitate cationic species
- Which is the most thermodynamically stable allotropic form of phosphorus? (IIT JEE 2005)
 - red
 - white
 - black
 - yellow
- Blue liquid which is obtained on reacting equimolar amounts of two gases at -30°C is
 - N_2O
 - N_2O_3
 - N_2O_4
 - N_2O_5
- In case of nitrogen, NCl_3 is possible but not NCl_5 while in case of phosphorus, PCl_3 as well as PCl_5 are possible. It is due to (AIEEE 2002)
 - availability of vacant d orbitals in P but not in N
 - lower electronegativity of P than N
 - lower tendency of H-bond formation in P than N
 - occurrence of P in solid while N in gaseous state at room temperature.
- Number of sigma bonds in P_4O_{10} is (AIEEE 2002)
 - 6
 - 7
 - 17
 - 16
- What may be expected to happen when phosphine gas is mixed with chlorine gas? (AIEEE 2003)
 - PCl_3 and HCl are formed and the mixture warms up
 - PCl_5 and HCl are formed and the mixture cools down
 - PH_3Cl_2 is formed with warming up
 - The mixture only cools down
- When rain is accompanied by a thunderstorm, the collected rain water will have a pH value (AIEEE 2003)
 - slightly higher than that when the thunderstorm is not there
 - uninfluenced by occurrence of thunderstorm
 - which depends on the amount of dust in air
 - slightly lower than that of rain water without thunderstorm.
- Which one of the following substances has the highest proton affinity? (AIEEE 2003)
 - H_2S
 - NH_3
 - PH_3
 - H_2O
- Regular use of which of the following fertilizers increases the acidity of soil? (AIEEE 2007)
 - KNO_3
 - NH_2CONH_2
 - $(\text{NH}_4)_2\text{SO}_4$
 - $(\text{NH}_4)_2\text{SO}_4$

6.54 Group 15-Nitrogen Family

12. The percentage of p-character in the orbitals forming P-P bonds in P_4 is: (IIT 2007)
 (a) 25 (b) 33
 (c) 50 (d) 75
13. The reaction of P_4 with X leads selectively to P_4O_6 . The X is: (IIT 2009)
 (a) Dry O_2
 (b) a mixture of O_2 and N_2
- (c) Moist O_2
 (d) O_2 in the presence of aqueous NaOH
14. The nitrogen oxide(s) that contain(s) N-N bond(s) is (are): (IIT 2009)
 (a) N_2O (b) N_2O_3
 (c) N_2O_4 (d) N_2O_5

ANSWER KEYS

ONE CORRECT OPTION TYPE QUESTIONS

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (c) | 2. (c) | 3. (d) | 4. (a) | 5. (b) | 6. (d) | 7. (a) | 8. (c) | 9. (c) | 10. (b) |
| 11. (c) | 12. (b) | 13. (a) | 14. (c) | 15. (b) | 16. (b) | 17. (d) | 18. (b) | 19. (a) | 20. (c) |
| 21. (d) | 22. (d) | 23. (a) | 24. (c) | 25. (c) | 26. (a) | 27. (b) | 28. (c) | 29. (c) | 30. (c) |
| 31. (d) | 32. (c) | 33. (b) | 34. (b) | 35. (d) | 36. (b) | 37. (b) | 38. (b) | 39. (b) | 40. (b) |
| 41. (b) | 42. (b) | 43. (c) | 44. (b) | 45. (c) | | | | | |

COMPREHENSION BASED QUESTIONS

Comprehension-I

1. (c) 2. (a) 3. (b) 4. (d) 5. (c) 6. (a)

Comprehension-II

1. (d) 2. (a) 3. (a) 4. (b) 5. (c) 6. (b) 7. (c)

ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

1. (a, d) 2. (b, c) 3. (b, c, d) 4. (a, c, d) 5. (a, b, c) 6. (b, c) 7. (c, d) 8. (a, b)
 9. (b, c) 10. (a, b, d) 11. (a, d)

MATCH THE COLUMN TYPE QUESTIONS

1. (A)-(2); (B)-(3); (C)-(4); (D)-(1)

ASSERTION/REASON TYPE QUESTIONS

1. (b) 2. (c) 3. (a) 4. (b) 5. (c) 6. (d) 7. (e) 8. (a) 9. (b)

RECENT YEARS' QUESTIONS OF IIT-JEE AND AIEEE

1. (c) 2. (c) 3. (c) 4. (c) 5. (b) 6. (a) 7. (d) 8. (a)
 9. (d) 10. (b) 11. (d) 12. (d) 13. (b) 14. (a, b, c)

CHAPTER 7

Group 16- Oxygen Family

INTRODUCTION

Physical Properties

1. The atomic radius, ionic radius, atomic volume and metallic character increases while I.P., electronegativity and electron affinity decreases down the group.

2.

O ₂	S ₈	Se ₈	Te ₈	PO ₄
↑	└──────────┘			
diatomic Gas	Solids having complex structure			
	S exists as 8 atoms ring			

Worked-out Example

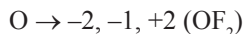
1. Oxygen is a gas while sulphur is solid. Why?

Ans. Oxygen exist as a diatomic molecule while sulphur exist as staggered 8 atom rings or zigzag chains. The former involves π -bonds of p-p orbitals whereas sulphur is incapable of forming π -bonds of p-p orbitals.

3. M.P. and B.P. →

increases down the group. The large difference between the melting and boiling points of O and S may be explained on the basis of their atomicity; (O as O₂, S as S₈)

4. Oxidation State → Stability of -2 oxidation state decreases down the group.



Others also show +2, +4, +6 oxidation state, due to availability of d-orbitals in these elements.

Worked-out Examples

2. Why oxygen does not show positive oxidation state like its other group members? Also Report the compound where oxygen shows +ve oxidation state.

Ans. Oxygen is most electronegative element next to fluorine. It shows only +2 oxidation state in OF_2 .

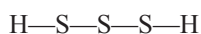
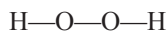
3. Oxygen invariably exhibits an oxidation state of -2 but the other members of family exhibit negative as well as positive oxidation state of $+2$, $+4$ and $+6$. Explain.

Ans. Oxygen is the second most electronegative element after the fluorine and thus invariably show negative oxidation state. Furthermore, it has $2s^2 2p^4$ configuration and thus requires only two electrons to complete its octet to show -2 oxidation state. No doubt, other members of this group also possess $ns^2 np^4$ configuration but due to availability of d-orbitals in their outer most shell -2 , $+2$, $+4$, $+6$ oxidation state are also shown. Oxygen, however, shows only -2 oxidation state due to non-availability of d-orbitals in its outermost shell.

5. Allotropy $\rightarrow$ All the elements of this group show allotropy.

Element	Allotropic forms
O	Ordinary oxygen and Ozone
S	Rhombic, monoclinic, Plastic, Amorphous
Se	Red form (non-metallic), grey form (metallic form)
Te	Crystalline and amorphous
Po	α and β forms (both are metallic)

6. Catenation $\rightarrow$ Only O and S show catenation. The property is more in S.

**Anomalous behaviour of Oxygen—This is due to**

- Small size
- High electronegativity
- non-availability of d-orbitals in the valency shell.

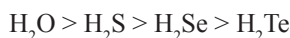
Chemical Properties

- (i) Hydrides of group-16 – H_2M type and have bent shape.

	H_2O	H_2S	H_2Se	H_2Te
Bond angle	————— decreases —————>			
Acidic character	————— increases —————>			
Thermal stability	————— decreases —————>			

Reducing nature	————— $\xrightarrow{\text{increases}}$ —————			
Dipole moment	————— $\xrightarrow{\text{decreases}}$ —————			
B.P. of H_2M	H_2O	H_2S	H_2Se	H_2Te
	373 K	213 K	232 K	269 K

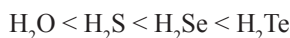
- H_2S is called sulphureted hydrogen. H_2S is poisonous and its large amount proves fatal. Antidote for it is dilute chlorine which destroy the effect of H_2S by oxidizing it to sulphur.
- All hydrides of group 16, except water are volatile, poisonous and foul smelling.
- Melting and boiling points of hydrides decrease in the order



- Volatile nature increases in the order



- Acidic, covalent and reducing character increase in the order



Worked-out Example

4. The volatility of the group 16 halides follows the order:



Ans. The b.pt. increases (or volatility decreases) with increase in molecular weight. The higher b.pt. of H_2O is due to hydrogen bonding. On moving from H_2S to H_2Te the molecular weight increases so volatility decreases.

- The acidic character of hydrides is quite reverse of water is expected on the basis of polarity of the bond between group 16 element and hydrogen. This bond is most polar in H_2O (O has the highest electronegativity in the group), yet H_2O is least acidic. The acidic character increases in the order



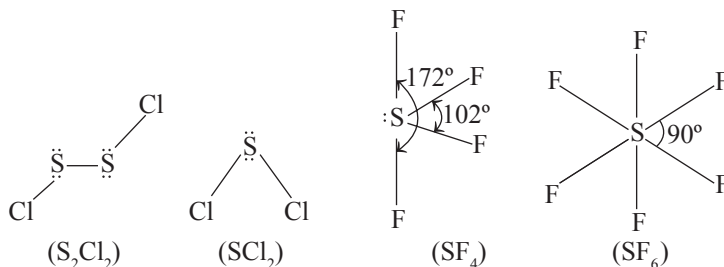
- In fact H_2O is neutral. This, observed trend can best be explained on the basis of bond strength.
- Water has the highest melting point (273K) and boiling point (373 K) in the group. Melting point and boiling point of H_2Te are 222 K and 271 K respectively. This is quite contrary to what is expected on the basis of van der Waal's forces which depend upon the molecular mass. This is due to extensive H-bonding which exist even in liquid state.
- From 273 K to 277 K, density of water increases i.e., when heated it contracts. After 277 K it starts expanding with increase in temperature. Thus, water has maximum density at 277 K (1g mL^{-1}).

Halides

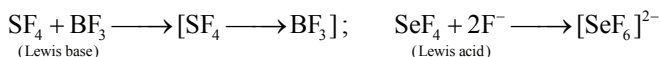
- (i) Oxygen forms only dihalides (OF_2 , Cl_2O , Br_2O , etc.) because it cannot expand its valency due to non-availability of vacant d-orbitals. These dihalides are V-shaped with O assuming sp^3 hybrid state.

7.4 Group 16-Oxygen Family

- (ii) Other elements form tetrahalides and hexahalides in addition to dihalides. This is because of their ability to expand their octets. The shapes associated with S_2Cl_2 , SCl_2 , SF_4 , SF_6 are given as follows:



- (iii) *Tetrahalides* have *see-saw* shape with central atom assuming sp^3d hybrid state: They can act as *Lewis acids* as well as *Lewis bases*.

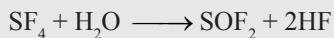


- (iv) Hexahalides of these elements have *octahedral shape* with central atom assuming sp^3d^2 hybrid state. In general, the thermal stability of the halides for the same central atom is:

fluorides > chlorides > bromides > iodides.

Worked-out Example

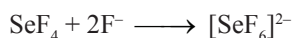
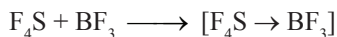
5. Why is SF_6 not hydrolyzed while SF_4 is hydrolyzed?



Ans. SF_6 cannot be hydrolyzed

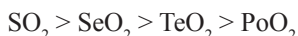
- (i) S—F bond strength is very high
- (ii) S atom is co-ordinately saturated
- (iii) it is a non-polar molecule. But in the case of SF_4 there are still empty d-orbitals present. Hence it is easily hydrolysed to SOF_2 .

- All elements form hexafluorides, i.e., among halogens, only fluorine shows the maximum valency of six which is due to its small size.
- As the size of halogen atom increases, the maximum coordination number of elements of this group decreases. Thus, no hexachloride is known, only three tetrabromides and two tetraiodides are known.
- Although S_2Cl_2 and Se_2Cl_2 are known, the corresponding Te_2Cl_2 and Po_2Cl_2 are not known. This is due to weaker Te—Te and Po—Po bonds as these are very large atoms. Moreover, chlorine being more electronegative will withdraw electrons from Te and Po. Thus making Te—Te and Po—Po bonds weaker.
- All the hexafluorides have a high degree of covalency and have low boiling points.
- SF_4 is a gas, SeF_4 is a liquid and TeF_4 is solid.
- The tetrahalides act as Lewis bases (electron donors) and Lewis acids (electron acceptors)

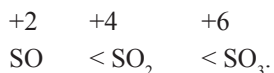


Oxides

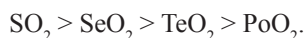
- (i) These elements form variety of oxides in different oxidation states ranging from +2 $\longrightarrow$ +6.
 (ii) Acidic strength of oxides (*in same oxidation state*) decreases as we move down the group.



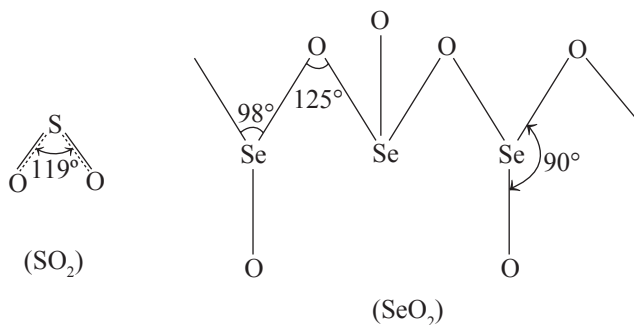
- (iii) for a particular element, the acidic character increases with increase in oxidation state. For example, acidic character among oxides of sulphur increases as:



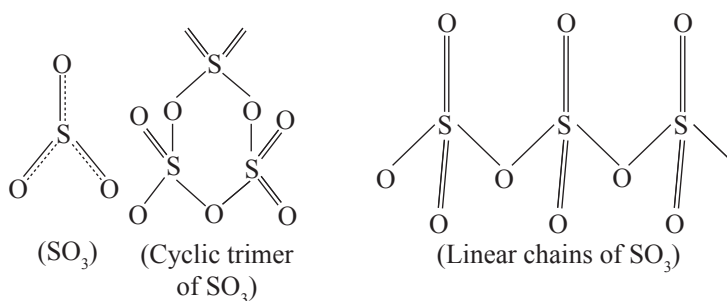
- (iv) Oxides of sulphur are relatively more stable than the oxides of other elements. For example, among dioxides, the stability decreases as:



- (v) Gaseous SO_2 is a bent molecule with S atom assuming sp^2 hybrid state: SeO_2 and TeO_2 are solids having polymeric chains.



- (vi) Gaseous SO_3 is a trigonal planar molecule with O—S—O angle equal to 120° and S atom in sp^2 hybrid state. Solid SO_3 exists in trimeric cyclic form or polymeric chain of SO_3 units.



Worked-out Example

6. Explain the given trend in the bond order of S—O bond in oxyhalides of sulphur:



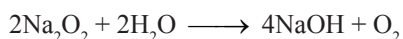
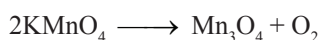
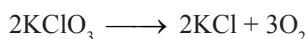
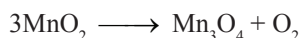
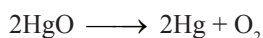
Ans. From fluorine to chlorine and bromine electronegativity decreases. As a consequence the extent of $p\pi-p\pi$ back bonding in S—O bond decreases in the same order giving S—O bond in SOF_2 greatest bond order. In case of SOCl_2 and SOBr_2 the Cl and Br atoms are less electronegative than oxygen. Thus the O—S $p\pi-p\pi$ bonding is not encouraged and bond order is less.

7.6 Group 16-Oxygen Family

Oxygen (O₂)

Oxygen occurs both in the free as well as in combined states. In free state it occurs in air to the extent of 21% by volume and 23% by weight. In combined state it is present in water, nitrates, sulphate and so on. It is the most abundant element (46.5%) in the earth's crust. In combined state, it is also present in plants and animal tissues.

Preparation: It can be prepared by thermal decomposition of metallic oxides such as PbO, HgO, MnO₂, Ag₂O, and BaO₂ or by thermal decomposition of peroxides and hypochlorites or by heating salts such as KNO₃, KClO₃, KClO₄, KMnO₄ and K₂Cr₂O₇, which are rich in O₂. In laboratory, it is prepared by heating potassium chlorate above its melting point in the presence of MnO₂ (catalyst) or by the action of water on sodium peroxide.



Properties of Oxygen

- Dioxygen is colourless, tasteless and odourless gas.
- It is slightly soluble in water. Its solubility being approximately 30 cm³ per litre of water at 298 K.
- It is appreciably soluble in **alkaline pyrogallol** solution.
- Liquid oxygen exhibits, paramagnetism, i.e., it is slightly attracted by magnet.
- It is neutral to litmus.

Worked-out Examples

7. Why is oxygen paramagnetic?

Ans. Oxygen molecule, as its structure suggests, contains two unpaired electron and thus shows paramagnetism.

8. Gaseous oxygen is colourless, liquid oxygen is pale blue and solid O₂ is blue, explain why?

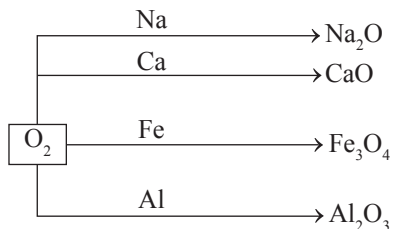
Ans. The colour arises from electronic transitions which excite the ground state (a triplet state) to singlet state (excited state). This triplet-singlet transition is forbidden in gaseous oxygen. In liquid or solid oxygen, a single photon may collide with two molecules simultaneously and promote both to excited states, absorbing red-yellow-green light, so O₂ appears blue. The origin of excited singlet states in O₂ lies in the rearrangement of electrons in π^*2p_y and π^*2p_z M.Os.

Second excited state	<table><tr><td>↑</td><td>↓</td></tr></table>	↑	↓	singlet	157 kJ
↑	↓				
First excited state	<table><tr><td>↑↓</td><td></td></tr></table>	↑↓		singlet	92 kJ
↑↓					
Ground state	<table><tr><td>↑</td><td>↑</td></tr></table>	↑	↑	Triplet	0
↑	↑				
	π^*2p_y	π^*2p_z			

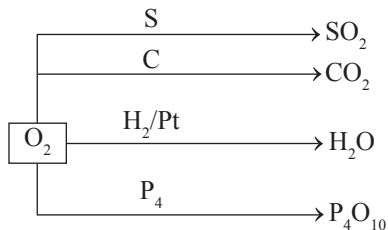
9. The O—O bond energy in O₂ is much lesser than the S—S bond energy in S₈ or sulphur possesses most catenating nature among group 16 elements.

Ans. The bond energy of catenation in sulphur is appreciably higher and maximum in group 16 elements. Oxygen form *p-p* π -bonds to develop diatomic nature where S forms puckered ring structure and no *p-p* π -bonding due to larger size.

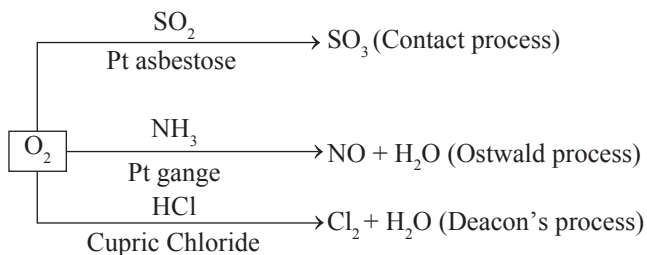
(i) **Reaction with metals:** Metals oxides are basic in nature.



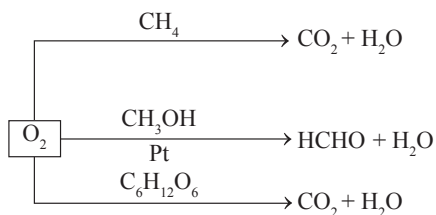
(ii) **Reaction with non-metals:** Non-metallic oxides are acidic in nature.



(iii) **Reaction with compounds:**



(iv) **Oxidation of organic compounds:**



Worked-out Examples

10. The reaction of P_4 with X leads selectively to P_4O_6 . The X is

(IIT 2009)

- (a) Dry O_2
- (b) A mixture of O_2 and N_2
- (c) Moist O_2
- (d) O_2 in the presence of aqueous NaOH

Ans. (b) P_4O_6 is obtained when P_4 is burned in a deficit of oxygen.

11. In P_4O_{10} , the number of oxygen atoms bonded to each phosphorus atom is ____.

(IIT 1992)

Ans. Four

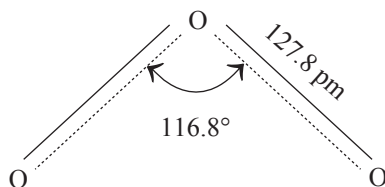
Uses of Dioxygen

Some of the important uses of dioxygen are:

- In the oxy-hydrogen or oxy-acetylene torches, which are used for cutting and welding of metals.
- In the metallurgical processes for the removal of metallic impurities by oxidation.
- In the manufacture of phenol, ethylene oxide, etc.
- A mixture of carbon dust and liquid oxygen is used as explosive for coal mining.
- Liquid oxygen is used as a fuel for rocket and missiles.
- Commercially, oxygen is obtained by fractional distillation of liquid air.
- Priestley and Scheele discovered oxygen in 1774.
- The name oxygen was given by A. L. Lavoisier in 1777.
- M. Faraday, in 1848, noted that oxygen was paramagnetic.
- The other elements of this group have less tendency to have -2 oxidation state due to low electronegativity and low ionization energy. The last element, Po does not show oxidation state of -2 at all.
- OF_2 is known as oxygen difluoride and not fluorine oxide because fluorine is more electronegative than oxygen. Since oxygen is more electronegative than chlorine, bromine or iodine hence the compounds of oxygen with these halogens are known as oxides such as chlorine dioxide ClO_2 .

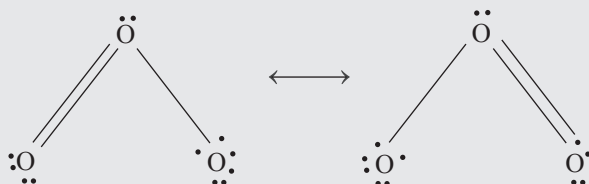
OZONE OR TRIOXYGEN

- Ozone was discovered by *Van Marum* and its name was proposed by *Schenbein*.
- It is an allotrope of oxygen.
- It is present in the upper atmosphere where it is believed to be formed by action of U.V. radiation on O_2 .
- It protects us from harmful U.V. radiations.
- The ozone layer, these days is getting depleted due to NO or freons, i.e., chlorofluoro carbons.
- Ozone has a bent structure with O—O—O angle 116.8° and O—O bond lengths as 127.8 pm .

**Worked-out Examples**

12. Why is O—O bond length in ozone ($1.28 \text{ \AA}$) more than in O_2 ($1.21 \text{ \AA}$)?

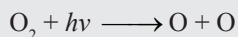
Ans. Ozone has two resonance forms:



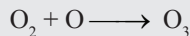
The molecular structure of O_3 is bent with an O—O—O bond angle of approximately 120° . The π -bond in ozone is delocalised over the entire molecule; neither individual O—O bond is a full double bond, so the observed O—O distance of $1.28 \text{ \AA}$ is greater than the $1.21 \text{ \AA}$ distance in O_2 which has a full O—O double bond.

13. Ozone is not present in lower atmosphere. Explain.

Ans. In the upper atmosphere (about 90 km from the surface of the earth) oxygen molecules absorb UV radiation and produce oxygen atom.



The oxygen atoms collide with O_2 molecules and produce ozone.



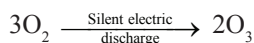
High energy UV rays do not reach the lower atmosphere in high concentration and hence do not cause the dissociation of oxygen molecule into oxygen atoms. As a result, ozone is not produced in the lower layer of atmosphere.

14. Ozone layer acts as an umbrella for the earth against the harmful radiations from sun. Why?

Ans. Ozone possesses the tendency to absorb U.V. rays of sunlight and thus ozone layer acts as a screen and prevents the harmful radiations from reaching the surface of earth.

Preparation of Ozone

Ozone is prepared by the action of silent (i.e., sparkless) electric discharge through pure and dry oxygen.



- The apparatus used for preparing ozone is called **ozonizer**.
- Two types of ozonisers are generally used. **Siemen's ozonizer** and **Brodie's ozonizer**.
- The conditions for better yield of ozone are:
 - (a) *low temperature* ($\approx 273 \text{ K}$); (b) *sparkless discharge*; (c) *pure and dry oxygen*.

Properties of Ozone

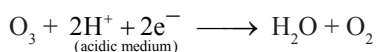
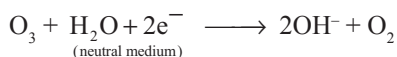
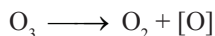
- Ozone is light blue gas with pungent odour.
- It condenses to deep blue liquid (b.p. 161.2 K) and to violet black solid (m.p. 80.6 K).
- It is slightly soluble in water but appreciably soluble in turpentine oil.
- It is poisonous in nature, and is used as disinfectant. It produces headaches when inhaled in small amount but larger doses can be fatal.
- It is neutral to litmus.

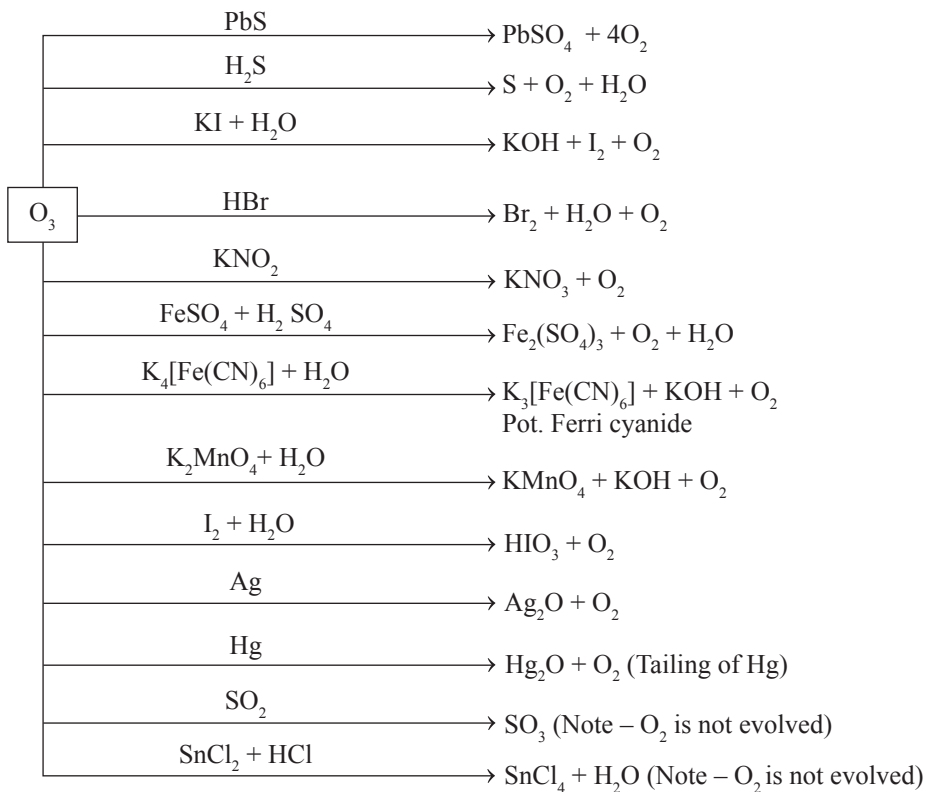
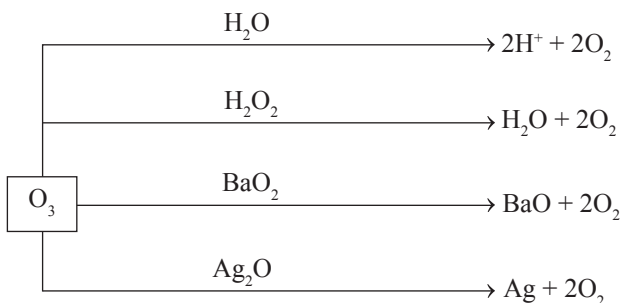
Chemical properties

- (i) **Decomposition:** Ozonized oxygen decomposes slowly, while pure O_3 decomposes with an explosive violence.

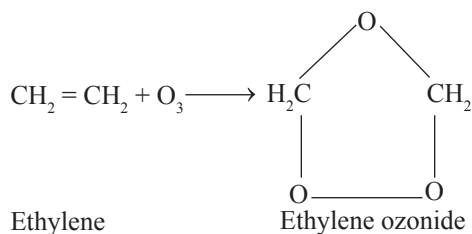


- (ii) **Oxidation reactions:** Since ozone can easily lose oxygen atom (nascent oxygen), it acts as a more powerful oxidizing agent than the molecular oxygen.



Properties of Ozone**Reducing Reaction**

- Alkenes are oxidized to ozonides

**Bleaching action of O_3**

Colouring matter + $O_3 \longrightarrow$ Oxidized matter (Colourless) + O_2

Since ozone bleaches even in the absence of moisture. It is also referred to as **dry bleach**.

Worked-out Examples

15. Ozone destroys mercury meniscus. Why?

Ans. Ozone oxidizes Hg to mercury suboxide which starts sticking to glass and loses its mobility. Hence, Hg loses its meniscus in contact with ozone. However, it can be regained by the action of H_2O_2 .



16. Ozone is more reactive than oxygen. Why?

Ans. Formation of O_3 takes place with absorption of energy and thus it possesses higher energy i.e., an endothermic compound which is less stable.

17. Discuss the product formed in the following reaction: $\text{KOH} + \text{O}_3(\text{g}) \longrightarrow$

Ans. The reaction occurs as:



The product formed is potassium ozonide and is orange red solid and is paramagnetic in nature with one unpaired electron.

Uses of Ozone

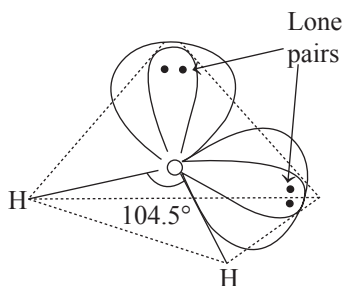
Ozone is used:

- as a disinfectant and germicide in the purification of drinking water.
- for purification of air in hospitals, railway tunnels, cinema houses and for destroying disagreeable smell in slaughter houses.
- for bleaching of delicate articles, fabrics, oils, starch, etc.
- as an oxidizing agent. Industrially, it is used for the oxidation of fatty acids and K_2MnO_4 .

Worked-out Example

18. Ozone is used in improving the atmosphere of crowded places such as cinema hall, tube Railways, etc. why?

Ans. Ozone easily decomposes to give O_2 and thus improves the percentage of oxygen at crowded places.

Compound of Oxygen**Water**

Physical Properties of Water

- Water is universal polar solvent because a large number of substances (most of the inorganic substances) are soluble in it.
- Water has maximum density of 1.00 g cm^{-3} at 277 K i.e., 4°C
- Water has strong hydrogen bonding in its molecules and exists as associated molecules $(\text{H}_2\text{O})_n$.
- Water has high specific heat and thermal conductivity. The high heat of vapourization and the high specific heat of water are responsible for moderating the influence of water on the climate and body temperature of living organisms.

Worked-out Examples

19. What is co-ordination number of O in $\text{H}_2\text{O}_{(l)}$?

Ans. Four.

20. Among H_2O , HCl and NH_3 , H_2O has the highest boiling point and HCl the lowest. Why?

Ans. H_2O is capable of showing more stronger H-bonding than NH_3 , while HCl has no H—bonding. The b.pt. order is $\text{H}_2\text{O} > \text{NH}_3 > \text{HCl}$.

21. Water is liquid while H_2S gas. Explain.

Ans. Oxygen has small size and more electronegativity in comparison to sulphur and thus H_2O molecules exists as associated molecule to form liquid state because of H-bonding. H_2S does not show H-bonding and thus exists as unassociated molecules to have gaseous state.

22. The maximum possible number of hydrogen bonds a water molecule can form is (IIT 1992)

- (a) 2 (b) 4 (c) 3 (d) 1

Ans. (b)

23. Amongst H_2O , H_2S , H_2Se and H_2Te , the one with the highest boiling point is (IIT 2000)

- (a) H_2O because of hydrogen bonding. (b) H_2Te because of higher molecular weight
(c) H_2S because of hydrogen bonding. (d) H_2Se because of lower molecular weight.

Ans. (a)

24. Heavy water is (IIT 1983)

- (a) H_2^{18}O (b) Water obtained by repeated distillation
(c) D_2O (d) Water at 4°C

Ans. (c)

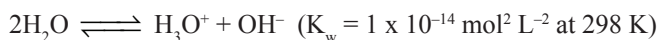
25. The formula of heavy water is H_2O . (True or False) (IIT 1978)

Ans. False.

Chemical Properties of Water

(i) **Action towards litmus** Pure water is neutral to litmus.

(ii) **Decomposition**



(iii) **Acid-Base reactions**

Water is amphoteric in nature i.e., it behaves both as an acid as well as a base.

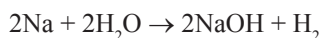


However, water is neutral towards litmus and its pH is 7.

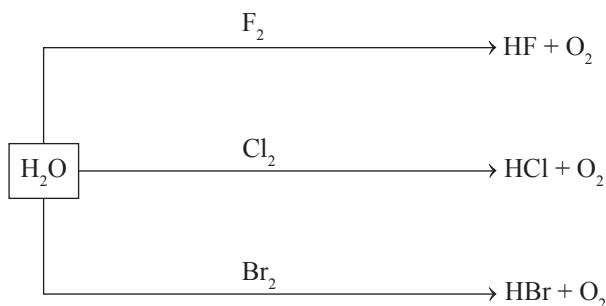
(iv) Oxidation-Reduction Reactions

Water can act both as an oxidizing as well as a reducing agent.

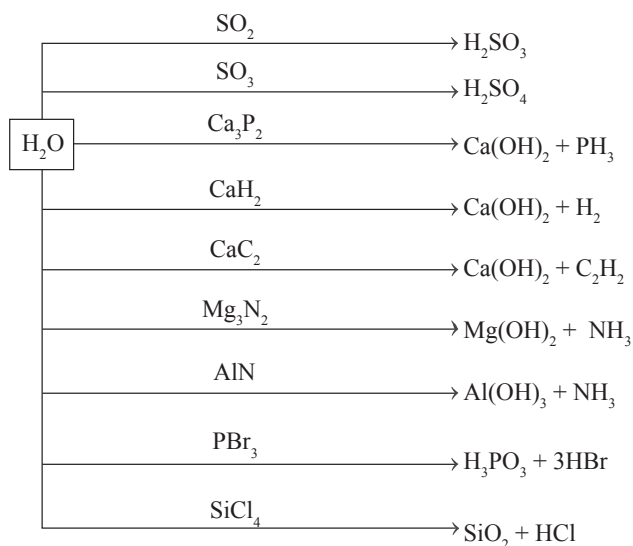
Water behaves as weak oxidizing agent with active metals.



Water behaves as a reducing agent with highly electronegative elements.

**(v) Hydrolytic Reactions**

Water hydrolyses a variety of compounds like oxides and halides of non-metals.

**(vi) Hydration Reactions**

Water has a strong ability to form compounds with same metal salts known as hydrates.

Water form three categories of hydrates:

(a) by combining metal ions through co-ordinate bonds in a complex.



Here, all the water molecules are bonded to the metal ion.

(b) Water molecules may be hydrogen bonded to certain oxygen containing anions. E.g., $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, in which four water molecules are co-ordinated to a central Cu^{2+} ion while the fifth water molecule is hydrogen bonded to the sulphate group.

(c) Water may also occupy interstitial sites. E.g., in $\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$

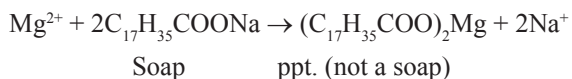
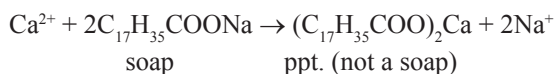
Hard and Soft Water

Soft Water: Water which produces lather with soap solution readily is called soft water.

Hard Water: Water which doesn't produce lather with soap solution readily is called hard water.

Cause of hardness of water

- The hardness of water is due to the presence of Mg and Ca salts like bicarbonates, chlorides and sulphates.
- Hard water doesn't produce lather with soap because the Mg^{+2} and Ca^{+2} cations react with soap to form Ca and Mg salts of fatty acids (which are not soaps)



Types of Hardness

- Temporary hardness (Carbonate hardness)**
It is due to the presence of soluble bicarbonates of calcium and magnesium.
- Permanent hardness (Non-carbonate hardness)**
It is due to the presence of chlorides and sulphates of Ca and Mg.

Softening of hard water

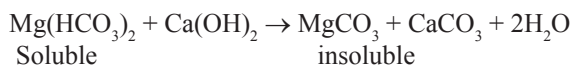
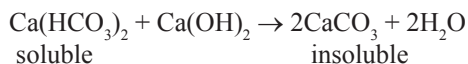
The process of removal of metallic ions (Ca^{2+} and Mg^{2+}) responsible for hardness of water is known as softening of water.

Removal of temporary hardness

- Boiling:** Temporary hard water is taken in large boilers and boiled for about fifteen minutes. During this process, the soluble bicarbonates are converted into insoluble carbonates. These insoluble carbonates are removed from the vessel and in the process the calcium and magnesium ions are also removed from water, thereby removing hardness.



- Calcium hydroxide (or CLARK'S) method:** Here a calculate quantity of calcium hydroxide is added when the soluble bicarbonates are converted into insoluble carbonates.



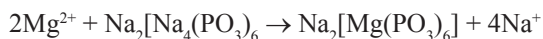
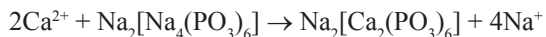
Removal of Permanent Hardness

- By the addition of washing soda**
Here, the Mg^{+2} or Ca^{+2} ions combine with the CO_3^{2-} ions of the washing soda ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$.) These insoluble carbonates are then removed, thereby removing the ions responsible for hardness.



(ii) **By the addition of sodium polymetaphosphate (CALGON Process)**

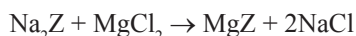
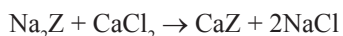
Here, sodium polymetaphosphate $\text{Na}_2[\text{Na}_4(\text{PO}_3)_6]$, is added and the calcium and magnesium from complexes with sodium polymetaphosphate.



The complexes of Calcium and Magnesium so formed remain dissolved in water. But still they cannot cause hardness. This is because Ca and Mg ions are not free to react with soap as these have been tied up in a stable complex i.e., $[\text{Ca}_2(\text{PO}_3)_6]$. This is known as **Sequestration** of Ca^{2+} and Mg^{2+} ions.

(iii) **Permutit Method**

Here, hard water is passed through zeolite, whereby the Calcium and Magnesium ions of hard water are replaced with Na^+ ions of the permutit.



Where Na_2Z is the zeolite that is hydrated sodium aluminium silicate ($\text{Na}_2\text{Al}_2\text{Si}_2\text{O}_8 \cdot x\text{H}_2\text{O}$).

After some time the Na^+ ions of the zeolite are used up.

After this, NaCl solution is passed through the vessel during which the CaZ and MgZ are reconverted into Na_2Z .

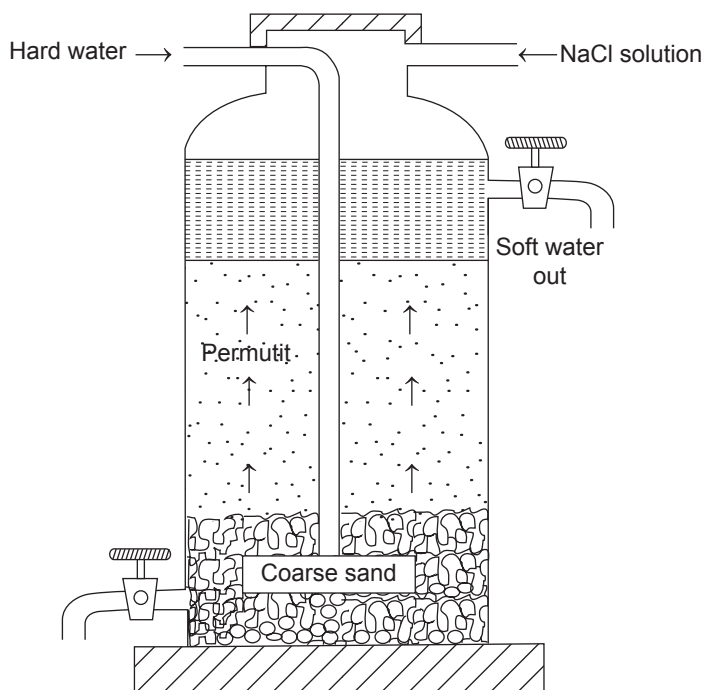
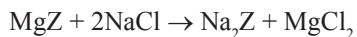
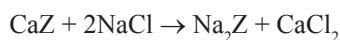


Fig. 7.1

The reaction taking place during place during regeneration of zeolite is



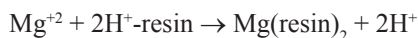
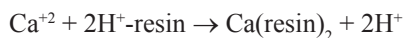
Permutit process for softening of hard water

(iv) **Ion Exchange Methods**

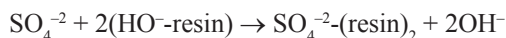
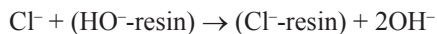
Hard water is first passed through a bed of cation

exchange resin (i.e., H^+ -resin) in the first tank where the cations are exchanged with the H^+ of the resin.

7.16 Group16-Oxygen Family



The water which comes out of the first tank is therefore free from Ca^{+2} and Mg^{+2} ions and this water is then passed through the second tank which contains a bed of anion exchange resins (i.e., $\text{HO}^-\text{-resin}$). Here, the anions of hard water are exchanged with the OH^- ions of the resin.



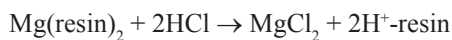
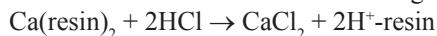
The H^+ ions formed in the first tank combine with the OH^- ions formed in the second tank to produce water.



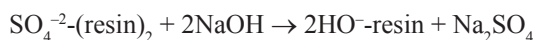
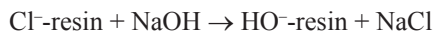
The water obtained after this is known as Deionized Water as this does not contain any of the ions.

Regeneration of the Resins

The exhausted resin of the first tank is regenerated by treatment with moderately concentrated HCl or H_2SO_4 .



The exhausted resin of the second tank is regenerated by treatment with moderately concentrated solution of sodium hydroxide.

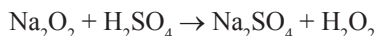


Hydrogen peroxide (H_2O_2)

Hydrogen peroxide was discovered by Thenard (1918).

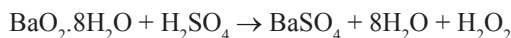
Preparation

- (i) From sodium peroxide (Merck's process)

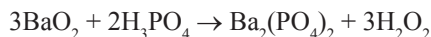


- (ii) From barium peroxide

- (a) By reaction with sulphuric acid



- (b) By reaction with phosphoric acid



Remarks about the methods of preparation

From sodium peroxide (Merck's process)

- In the Merck's Process, calculated amounts of sodium peroxide (Na_2O_2) are added to ice cold dilute (20%) solution of sulphuric acid slowly with constant stirring. Cooling the solution gives crystals of $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ which are decanted leaving behind a solution of hydrogen peroxide.
- Merck's process give 30% hydrogen peroxide solution

From barium peroxide

- While preparing hydrogen peroxide from Barium Peroxide (BaO_2), a paste of Barium peroxide is prepared in ice cold water and is treated with about 20% ice cold solution of sulphuric acid. The white precipitate of barium sulphate is removed by filtration leaving behind about 5% solution of hydrogen peroxide.

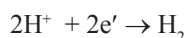
- Here, anhydrous Barium peroxide cannot be used because in this case, the barium sulphate produced forms a protective layer around the unreacted peroxide which prevents the reaction of the remaining barium peroxide.
- The limitation of this method is that the hydrogen peroxide produced by this method contains some heavy Ba^{+2} ions. These Ba^{+2} catalyst the decomposition of hydrogen peroxide and hence the hydrogen peroxide produced by this method cannot be stored for a long time.
- In order to prevent the presence of Ba^{+2} ions in the hydrogen peroxide solution, Phosphoric acid is used in place of sulphuric acid. When phosphoric acid is used, the barium phosphate produced gets completely precipitated and the solution does not contain any free Ba^{+2} ions and hence the decomposition of hydrogen peroxide is not catalysed.

Manufacture of hydrogen peroxide

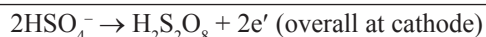
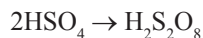
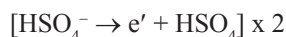
1. By electrolysis of 50% H_2SO_4



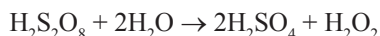
At cathode:



At anode:



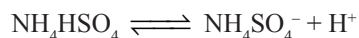
The peroxydisulphuric acid ($\text{H}_2\text{S}_2\text{O}_8$; Marshall's acid) is drawn off from the cell and is hydrolysed with water to give hydrogen peroxide.



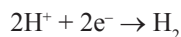
The resulting solution is distilled under reduced pressure when hydrogen peroxide gets distilled while the sulphuric acid with high boiling point remains undistilled.

2. By electrolysis of 50% NH_4HSO_4

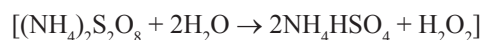
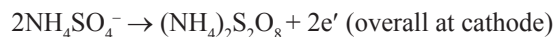
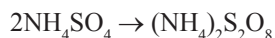
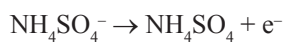
The above method has been modified by using equimolar mixture of sulphuric acid and ammonium sulphate for electrolysis.



At cathode:

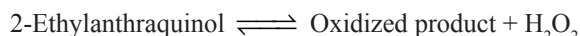


At anode:

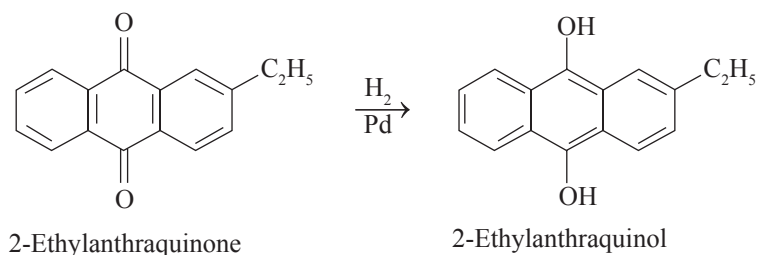


3. From 2-Ethyl anthraquinol

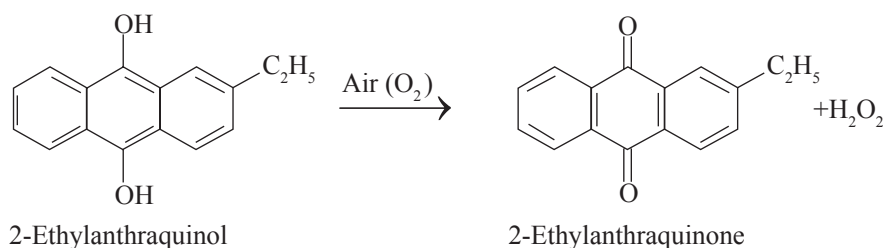
auto oxidation



Here, 2-Ethylanthraquinone is dissolved in benzene and hydrogen gas is passed through the solution in the presence of palladium(Pd) catalyst. Here, 2-Ethylanthraquinone gets reduced to 2-Ethylanthraquinol.



The reduced product is dissolved in a mixture of benzene and cyclohexanol and when air is passed through this, it is oxidized back to 2-Ethylanthraquinone and hydrogen peroxide is also produced alongwith.



This process is very cheap because the 2-Ethylanthraquinone can be used again and again.

Physical Properties of Hydrogen Peroxide

Pure H₂O₂ is thick syrupy pale blue liquid. It is completely miscible with water, alcohol and ether. It is more viscous and dense than water.

The density of H₂O₂ is 1.44 g cm⁻³, melting point is 272.4 K and boiling point is 358 K at 68 mm of Hg.

Chemical properties of hydrogen peroxide

(i) Decomposition

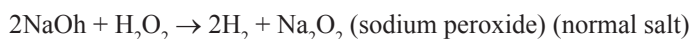
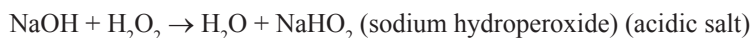


The decomposition is further accelerated with the presence of metals like platinum, gold, metal oxides (MnO₂), and certain metal ions like Fe⁺² ions.

(ii) Acidic Behaviour

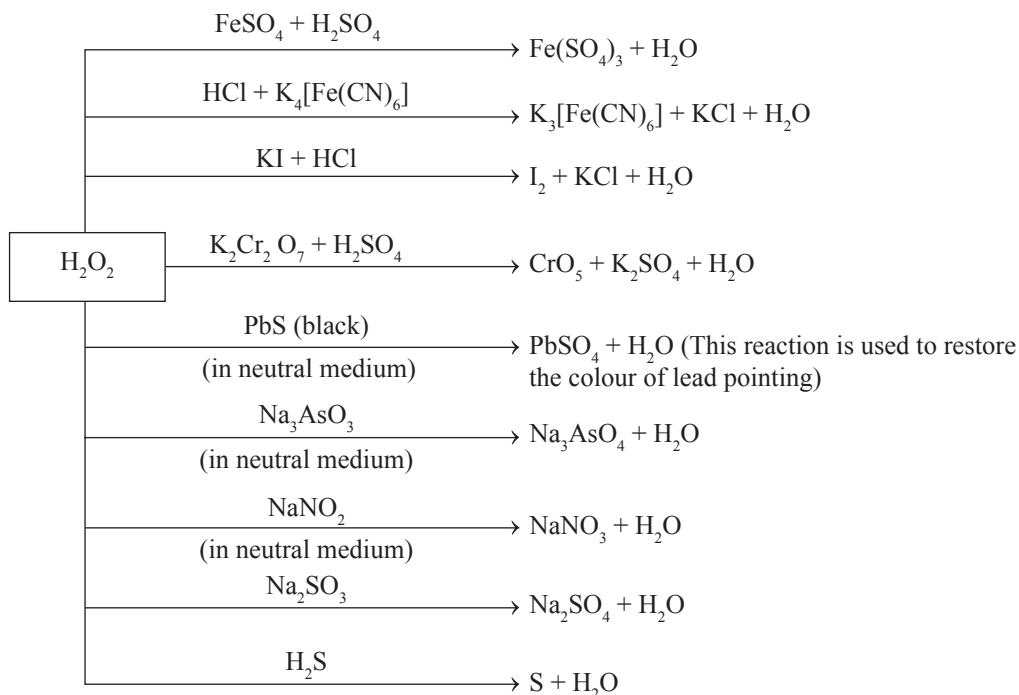
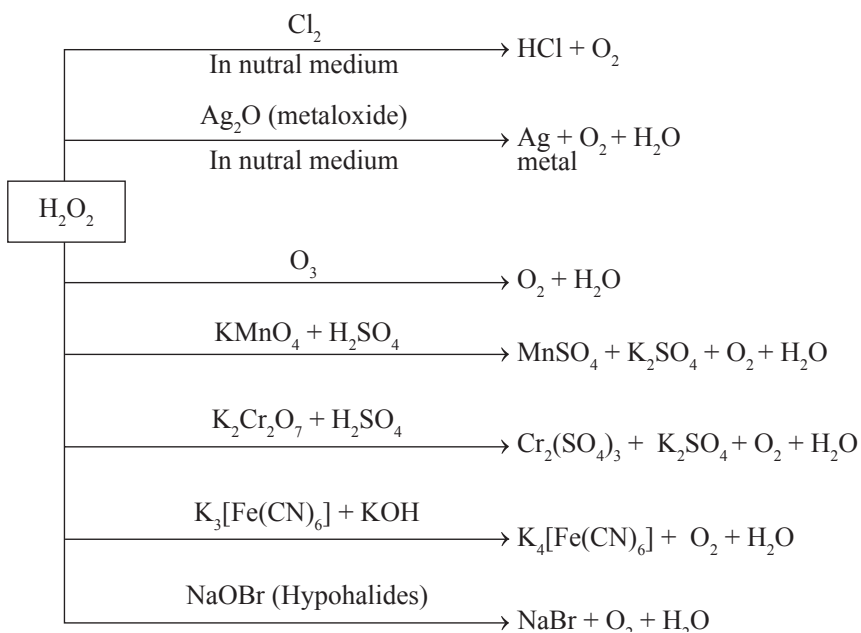
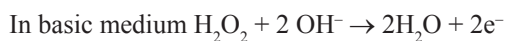
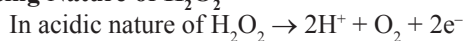
Hydrogen peroxide is weak dibasic/diprotic acid with a K_a value of 1.55 × 10⁻¹² at 298 K.

Hydrogen peroxide forms two types of salts i.e.,



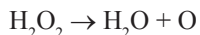
(iii) Oxidizing and Reducing Character of Hydrogen Peroxide

- Hydrogen peroxide behaves both as a reducing agent and oxidizing agent in both acidic and alkaline medium.
- The oxidation state of oxygen in hydrogen peroxide is -1. It can be oxidized to zero oxidation state (e.g., in O₂) or it can be reduced to -2 oxidation state (e.g., in H₂O).
- Hydrogen peroxide is a strong oxidizing agent.
- Hydrogen peroxide is a weak reducing agent.
- Hydrogen peroxide acts as an oxidizing agent in neutral, acidic or in alkaline medium.
- Hydrogen peroxide acts as a reducing agent in basic, or even neutral medium.

Oxidizing Nature of H_2O_2 **(iv) Reducing Nature of H_2O_2** 

Bleaching Action of Hydrogen Peroxide

Hydrogen peroxide acts as a Bleaching Agent due to the release of nascent oxygen.



The nascent oxygen produced oxidizes the coloured matter and a colourless matter is obtained.

Uses of Hydrogen Peroxide

- as a bleaching agent.
- as an antiseptic perhydrol.
- as an antichlor in bleaching.
- to restore colour of lead paintings.
- to manufacture many organic compounds and inorganic compounds like sodium perborates, epoxides etc.
- to preserve milk, wine and other liquors.
- to impart golden colour to hair.

Strength of H_2O_2

The method used to express the strength of H_2O_2 solution are

- (i) **By-percentage:** X% of H_2O_2 means X gm of H_2O_2 are present in 100 gm of solution
- (ii) **By number volume:** "X Volume" H_2O_2 is a solution of H_2O_2 whose 1 litre will give X litre of oxygen at STP.

Worked-out Example

26. The volume strength of 1.5 N H_2O_2 solution is

(IIT 1991)

- (a) 4.8 (b) 8.4 (c) 3.0 (d) 8.0

Ans. (b)

Structure of H_2O_2

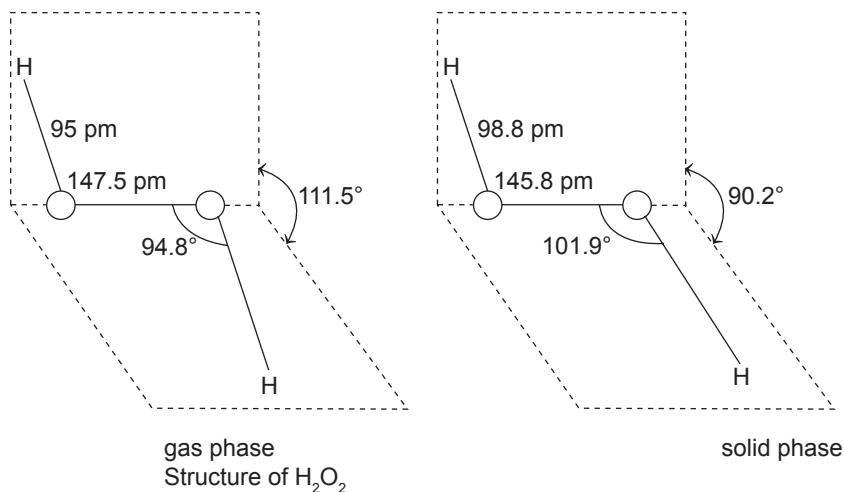


Fig. 7.2

Allotropic forms of sulphur

It exists in

- (i) **Yellow orthorhombic α -form:** Which is stable at room temperature but transforms to monoclinic sulphur on heating above 369 K.
- (ii) **β -monoclinic form:** Both orthorhombic and monoclinic sulphur are molecular solids containing S_8 puckered rings and have a crown shape. In S_8 molecule, each sulphur atom undergoes sp^3 -hybridization involving both bonding and non-bonding pairs of electrons. There are two single covalent bonds and two lone pairs of electrons associated with each sulphur atom and there are eight total number of single covalent bonds and sixteen lone pairs of electrons in one S_8 molecule. The S—S—S bond angle is 107° and S—S bond length is 204 pm.
- (iii) **Cyclo S_{6-20} form:** In cyclo- S_6 , the ring adopts the chair form.
- (iv) **Chain polymers catena- S_n :** Unstable small molecules, S_n ($n = 2, 5$) exist in liquid sulphur at elevated temperature and in sulphur vapour. At 1000 K, the dominant species is S_2 which is paramagnetic.

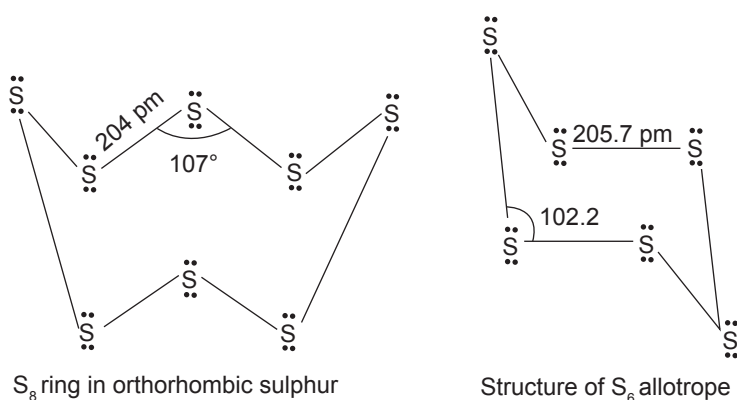


Fig. 7.3

The tendency to exist in 8 atom rings is greatest with sulphur and decreases as we go down the group. The special stability of the eight membered ring in sulphur is due to sp^3 -hybridization of sulphur atoms involving both bonding and non-bonding pairs of electrons.

Worked-out Example

27. Why does sulphur begin to melt below its actual melting points.

Ans. Monoclinic sulphur has a true melting point of 392 K. However, it often melts a few degree lower due to the break down of some of the S_8 molecules.

- Vulcanization of rubber by heating with sulphur was discovered by Charles Goodyear in 1839.
- The name sulphur has been derived from Sanskrit word 'sulveri' meaning killer of copper.
- Another name for sulphur is brimstone. In its group, sulphur has maximum tendency of catenation.
- Mining of sulphur is carried out by Sicilian and Frasch (Louisiana) processes.
- Gun powder is a mixture of S, charcoal and KNO_3 .
- The group 16 elements which show catenation are S, Se and Te. They all exist as eight membered rings (S_8 , Se_8 and Te_8). Oxygen is diatomic whereas Po is monoatomic.
- Ordinary sulphur (rhombic sulphur) melts at 387.5 K to a thin pale yellow liquid. However, at 453 K it suddenly becomes so viscous that it cannot be poured out of the vessel. On further, heating the viscosity decreases ultimately boiling at 717.6K

Worked-out Examples

28. SF_6 known but SCl_6 is not known. Explain.

Ans. As the size of halogen atoms increases, the maximum co-ordination number of the elements of group 16 decreases. Thus hexahalides other than hexafluorides are not known.

29. Which of the following should have the greatest enthalpy of combustion at 25°C and why?

Rhombic S, Monoclinic S, Plastic sulphur.

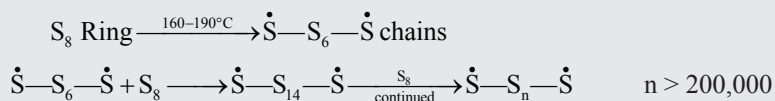
Ans. Plastic sulphur possess maximum energy and thus releases more energy during the combustion to form SO_3 .

30. The S_8 ring is non-planar why?

Ans. Each S is sp^3 hybridized with two lone pair of electrons. The bond angles are therefore $109^\circ 28'$. A planar ring would require bond angle of 135° .

31. Monoclinic sulphur melts to clear mobile liquid at 119°C , but on further heating to 160°C 190°C , it becomes viscous. Also at 200°C , the viscosity again decreases. Explain.

Ans. Molten sulphur at 119°C exists as S_8 puckered ring structure which can roll over one another to show mobile nature. Above 160°C , cleavage of ring structure takes place leading to open S_8 chains tangled together to form long polymeric chain with more than 200,000 S atoms in chain and thus sulphur becomes more viscous (about 10000 times more viscous).



Above 200°C , the polymer chain begin to fragment in smaller pieces and thus viscosity decreases.

32. Sulphur acts as ____ agent in vulcanization of rubber.

(IIT 1989)

Ans. polymerizing

Oxides of sulphur

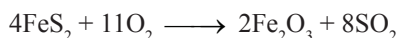
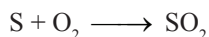
(i) **Sulphur dioxide (SO_2)**

Preparation

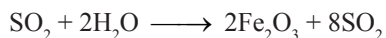
(i) $\text{Cu} + 2\text{H}_2\text{SO}_4$ (hot and conc.) $\longrightarrow$ $\text{CuSO}_4 + \text{SO}_2 + 2\text{H}_2\text{O}$ (lab method)

Copper turnings

(ii) On large scale, SO_2 is prepared by burning sulphur or iron pyrites.

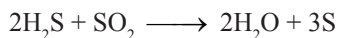
**Properties**

- (a) It turns blue litmus red. It acts as an oxidizing as well as reducing agent.
- (b) It acts as a strong reducing agent because in the presence of moisture it can liberate nascent hydrogen or it can take up an oxygen atom in the presence of an oxidizing agent.

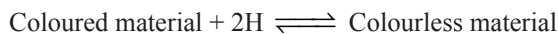
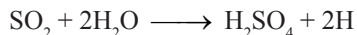


Thus, it decolourizes acidified KMnO_4 , turns acidified potassium dichromate green, reduces halogenes to halogen acids and ferric salts to ferrous salts.

- (c) Sulphur dioxide also acts as an oxidizing agent and oxidizes moist H_2S to S.



- (d) SO_2 also acts as a bleaching agent. In the presence of water, it is oxidized with the liberation of nascent hydrogen which reduces the coloured matter to colourless matter.



It should be noted that bleaching by SO_2 is not permanent because reduced colourless material is reoxidized by air to coloured compound. The bleaching action of chlorine is due to oxidation.

Uses of SO_2

Sulphur dioxide is used

- (i) in refining petroleum and sugar
- (ii) in bleaching wool and silk and
- (iii) as an anti-chlor disinfectant and preservative.

Sulphuric acid, sodium hydrogen sulphite and calcium hydrogen sulphite (industrial chemicals) are manufactured from sulphur dioxide.

Liquid SO_2 is used as a solvent to dissolve a number of organic and inorganic chemicals.

Worked-out Examples

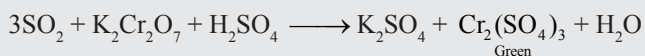
33. Dry SO_2 does not bleach dry flowers. Explain.

Ans. Only in presence of moisture, the nascent hydrogen responsible for bleaching (reduction) is produced.



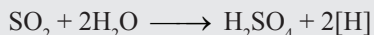
34. An acidified $\text{K}_2\text{Cr}_2\text{O}_7$ paper turns green when exposed to SO_2 . Explain.

Ans. SO_2 reduces $\text{K}_2\text{Cr}_2\text{O}_7$ to green chromium sulphate.



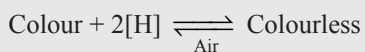
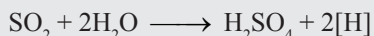
35. SO_2 acts as bleaching agent. Explain.

Ans. SO_2 in presence of water is oxidized to H_2SO_4 liberating nascent hydrogen which reduces the colouring matter.



36. Bleaching by SO_2 is temporary but bleaching by Cl_2 is permanent. Explain,

Ans. Bleaching by SO_2 takes place due to reduction, since SO_2 liberates nascent hydrogen. The colours are restored by the oxidation of colourless substance by air.



7.24 Group16-Oxygen Family

On the other hand Cl_2 acts as bleaching agent due to oxidation and permanent.



37. A gas that cannot be collected over water is (IIT 1985)

- (a) N_2 (b) O_2 (c) SO_2 (d) PH_3

Ans. (c)

38. Which one of the following substances has the highest proton affinity? [AIEEE 2003]

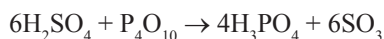
- (a) H_2S (b) NH_3 (c) PH_3 (d) H_2O

Ans. (b) Among the given compounds, the NH_3 is most basic. Hence has highest proton affinity

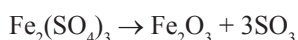
Sulphur Trioxide $\rightarrow \text{SO}_3$

Preparation

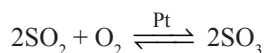
- (i) by dehydrating sulphuric acid with excess of phosphorus pentoxide.



- (ii) by heating ferric sulphate strongly.



- (iii) by oxidation of SO_2 with oxygen in presence of a catalyst (commercial method).



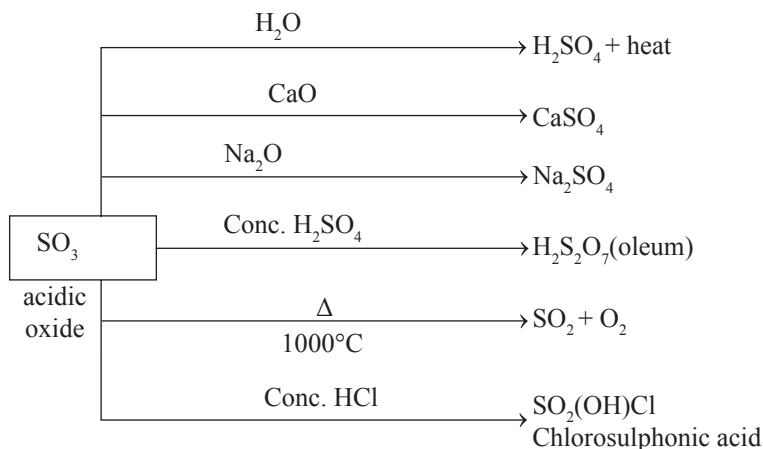
Both SO_2 and O_2 should be completely dry. The mixture is passed over platinized asbestos at 400°C , when SO_3 is formed.

Properties

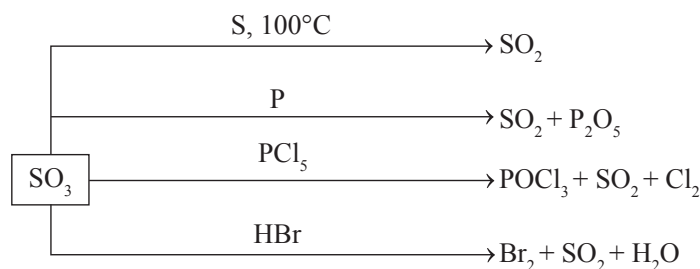
- (i) SO_3 exists in three allotropic forms:

- (a) $\alpha\text{-SO}_3$: It is chemically active form. It forms long transparent ice like crystals. The melting point of this form is 17°C .
 (b) $\beta\text{-SO}_3$: It is dimeric (S_2O_6). It forms needle like silky white crystals. It melts at 32.5°C . Above 50°C , it changes to α -form.
 (c) $\gamma\text{-SO}_3$: It is like β -form and obtained by completely drying $\beta\text{-SO}_3$. It melts at 62.2°C under 2 atmospheric pressure.

- (ii)



(iii) Oxidizing properties

**Uses:**

- (i) In the manufacture of sulphuric acid and oleum.
- (ii) It is used as a drying agent for gases.

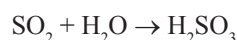
Worked-out Example

39. Write most favourable conditions for the conversion of SO₂ into SO₃.

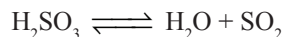
Ans. The reaction is $2\text{SO}_2 + \text{O}_2 \longrightarrow 2\text{SO}_3$; $\Delta H = -\text{ve}$. According to Le Chatelier principle, high pressure, low temperature (450°C) will favour the conversion of SO₂ to O₂. Also the presence of catalyst will enhance the rate of reaction.

Sulphurous Acid, (H₂SO₃)**Preparation**

This acid is known only in solution. The solution is obtained by dissolving sulphur dioxide in water.

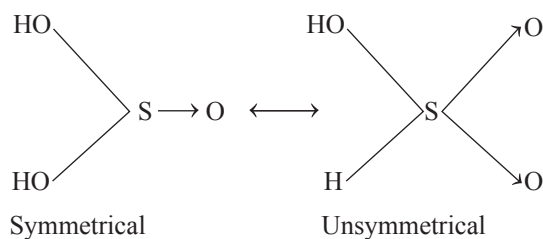


The solution gives a smell of sulphur dioxide which is evolved completely on heating. It is, thus, believed that the acid is present in equilibrium with the free gas.

**Properties**

- (i) Dibasic acid
- (ii) Like SO₂, it acts as a reducing agent and shows bleaching properties.

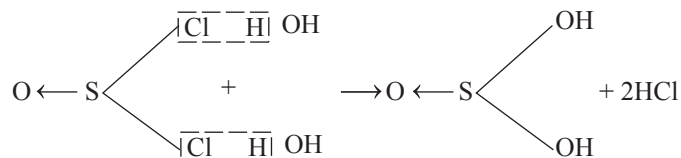
Structure: Sulphurous acid is believed to exist in two forms which are always in equilibrium with each other.



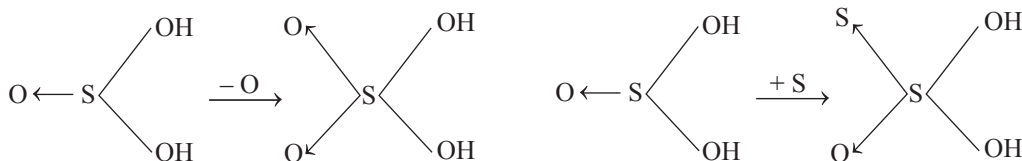
7.26 Group16-Oxygen Family

There is more evidence in favour of symmetrical formula.

- (i) The preparation of sulphurous acid by the action of H_2O on thionyl chloride supports symmetrical formula.

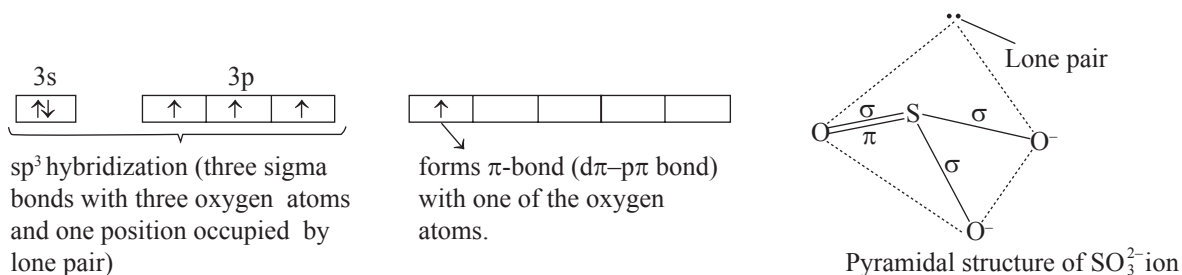


- (ii) The molecule of sulphurous acid readily takes up oxygen or sulphur atom. This can be explained if symmetrical formula is accepted.



The unsymmetrical formula explains the reducing character due to presence of $\text{H}-\text{S}$ bond and it also accounts for the formation of bisulphate HSO_3^- ion.

X-ray analysis of the crystal of Na_2SO_3 has revealed that SO_3^{2-} ion has a pyramidal structure with three oxygen atoms in a triangle. This structure can be possible from sp^3 hybridization of excited sulphur atom giving a tetrahedral configuration in which one position is occupied by a lone pair. This distorted tetrahedral structure reduces to pyramidal structure.



Worked-out Example

40. Using VSEPR theory, predict the probable structures of SO_3^{2-} , IF_6^- , ClO_4^- , ICl_4^- and IBr_2^- .

Ans. SO_3^{2-} : Pyramidal

IF_6^- : Pentagonal pyramidal

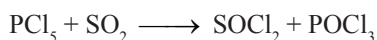
ClO_4^- : Tetrahedral

ICl_4^- : Square planar

IBr_2^- : Linear

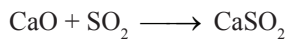
Some Noteworthy Points

- SO_2 gas is collected by upward displacement of air (heavier than air) or over mercury. This is because it is quite soluble in water at room temperature.
- PCl_5 on reaction with SO_2 give thionyl chloride which fumes in moist air and is used in organic chemistry.



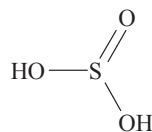
- SO_2 is an anhydride of H_2SO_3 and is called sulphurous anhydride whereas SO_3 is an anhydride of H_2SO_4 and is called sulphuric anhydride.
- SO_2 also turns lime water milky due to the formation of calcium sulphite. Milkiness disappears on passing excess SO_2 due to the formation of calcium bisulphate.

- SO_2 gas is dried by bubbling the mixture through conc. H_2SO_4 . It is not dried over quick lime as it reacts with it to form calcium sulphite.

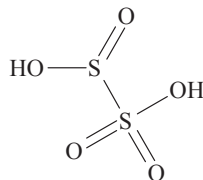


- Out of S^{2-} , SO_3^{2-} , SO_4^{2-} and $\text{S}_2\text{O}_3^{2-}$, S^{2-} does not give a white ppt, with $\text{BaCl}_2(\text{aq})$. Out of SO_3^{2-} , SO_4^{2-} and $\text{S}_2\text{O}_3^{2-}$, which ppt. of only BaSO_4 is insoluble in conc. HCl .

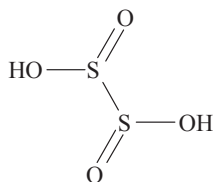
H_2SO_3 Sulphurous acid



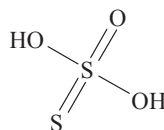
$\text{H}_2\text{S}_2\text{O}_5$ di-or pyrosulphurous acid



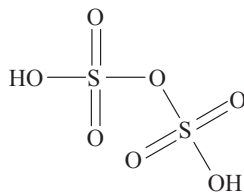
$\text{H}_2\text{S}_2\text{O}_4$ Dithionous acid



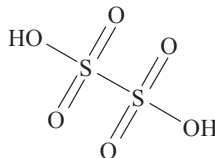
$\text{H}_2\text{S}_2\text{O}_3$ Thiosulphuric acid



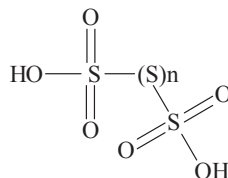
$\text{H}_2\text{S}_2\text{O}_7$ di or pyrosulphuric acid



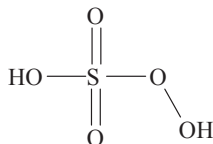
$\text{H}_2\text{S}_2\text{O}_6$ dithionic acid



$\text{H}_2\text{S}_n\text{O}_6$: Polythionic acid ($n = 1$ to 12)



H_2SO_5 Peroxymonosulphuric acid

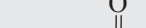


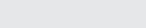
(a) $\text{S}_2\text{O}_4^{2-}$
(b) $\text{S}_2\text{O}_5^{2-}$
(c) $\text{S}_2\text{O}_3^{2-}$
(d) $\text{S}_2\text{O}_7^{2-}$

(a) three (b) two
(c) one (d) zero

O=S(=O)(O)O=S(=O)(O)O

(a) $\text{H}_2\text{S}_2\text{O}_3$ (b) $\text{H}_2\text{S}_2\text{O}_5$ (c) $\text{H}_2\text{S}_2\text{O}_6$ (d) $\text{H}_2\text{S}_2\text{O}_8$

$\text{H}_2\text{S}_2\text{O}_3$

 (Thiosulphuric acid)

$\text{H}_2\text{S}_2\text{O}_5$

 (Pyrosulphurous acid)

$$\text{H}_2\text{S}_2\text{O}_6$$

$$\text{HO}-\text{S}(=\text{O})_2-\text{S}(=\text{O})_2-\text{OH}$$

(Dithionic acid)

$$\text{H}_2\text{S}_2\text{O}_8$$

$$\text{HO}-\text{S}(=\text{O})_2-\text{O}-\text{O}-\text{S}(=\text{O})_2-\text{OH}$$

(Peroxodisulphuric acid)

Thus, the compound $\text{H}_2\text{S}_2\text{O}_8$ contains $-\text{O}-\text{O}-$ linkage.

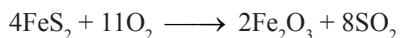
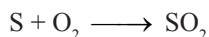
Sulphuric Acid

Preparation

Sulphuric acid is manufactured by two important process.

Lead chamber process

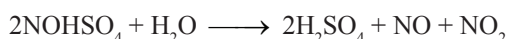
In this process, SO_2 is first obtained by burning sulphur or roasting iron pyrites.



This SO_2 is then oxidized catalytically by means of oxides of nitrogen to form nitroso sulphuric acid.



The nitroso sulphuric acid on hydrolysis with water yields sulphuric acid.



The oxides of nitrogen simply act as carriers of oxygen from air to SO_2 . The above reactions take place at $400 - 450^\circ\text{C}$ and $1.5 - 2.0$ atmospheric pressure. This method involves homogeneous catalyst and lead chambers are used because the metal is passive to the acid.

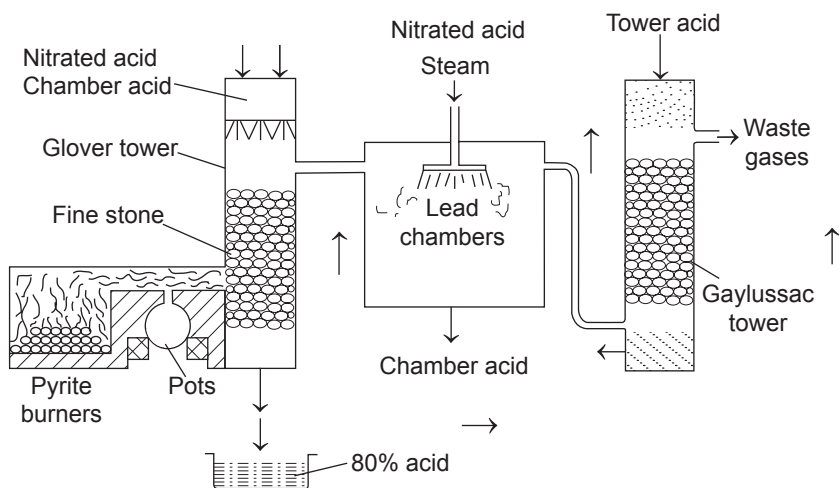
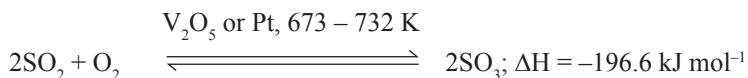
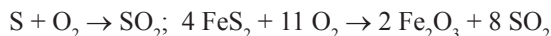


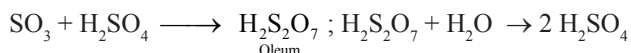
Fig. 7.4

Contact Process

In the contact-process, SO_2 obtained by burning of S or iron pyrites is catalytically oxidized to SO_3 in presence of finely divided Pt or V_2O_5 as catalyst.



V_2O_5 is, however, preferred since it is much cheaper than platinum and is also not poisoned by arsenic impurities. The favourable conditions for maximum yield of SO_3 are: (i) high concentration of SO_2 and O_2 , (ii) low temperature of 673 to 723 K and (iii) high pressure of about 2 atmospheres. Sulphur trioxide thus obtained is absorbed in 98% H_2SO_4 to form oleum which on dilution with water gives H_2SO_4 of desired concentration.



Contact process is preferred over lead chamber process since it gives H_2SO_4 of greater purity.

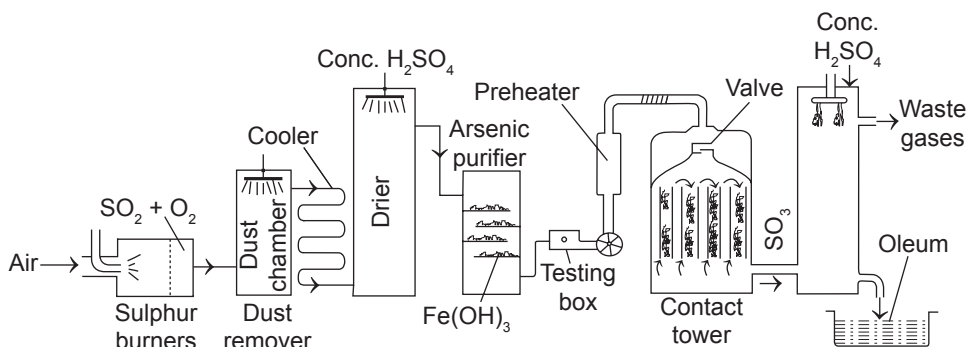
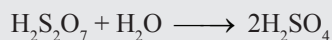
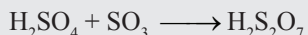


Fig. 7.5

Worked-out Examples

46. In the manufacture of sulphuric acid by the contact process, sulphur trioxide is not directly dissolved in water. Why?

Ans. SO_3 reacts with water to form H_2SO_4 which is not easily condensed and escapes in the atmosphere. It is, therefore, absorbed in Conc. H_2SO_4 to form oleum which may be diluted to form H_2SO_4 .

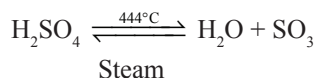


47. What catalysts are used in contact process. Write their names and optimum temperature for their efficient working

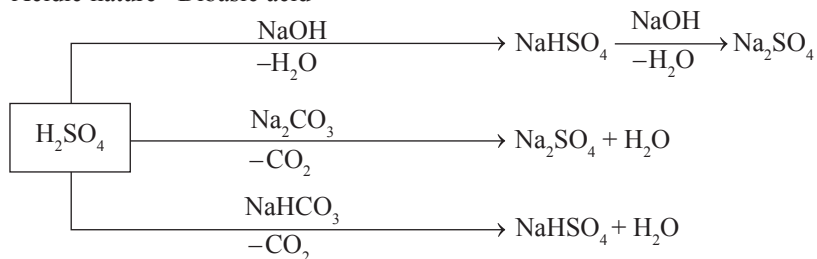
Ans. The commonly used catalysts in contact process are Pt, Fe_2O_3 or V_2O_5 . Out of these, the best yield is obtained by the use of Pt. But it is very expensive and easily poisoned by impurities. Now a days V_2O_5 is preferred as it is cheaper and not poisoned by impurities. The optimum temperature is 450°C .

Properties

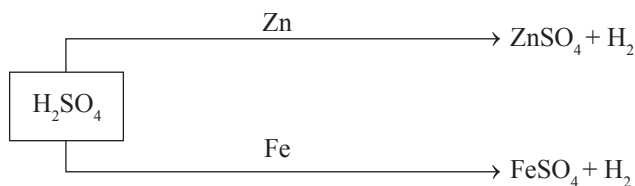
- (i) It is highly soluble in water. Various hydrates such as $\text{H}_2\text{SO}_4 \cdot \text{H}_2\text{O}$, $\text{H}_2\text{SO}_4 \cdot 2\text{H}_2\text{O}$, $\text{H}_2\text{SO}_4 \cdot 3\text{H}_2\text{O}$, $\text{H}_2\text{SO}_4 \cdot 4\text{H}_2\text{O}$ are known. Water should not be added to concentrated H_2SO_4 for dilution but concentrated H_2SO_4 should be added slowly to cold water with constant stirring.
- (ii) Dissociation



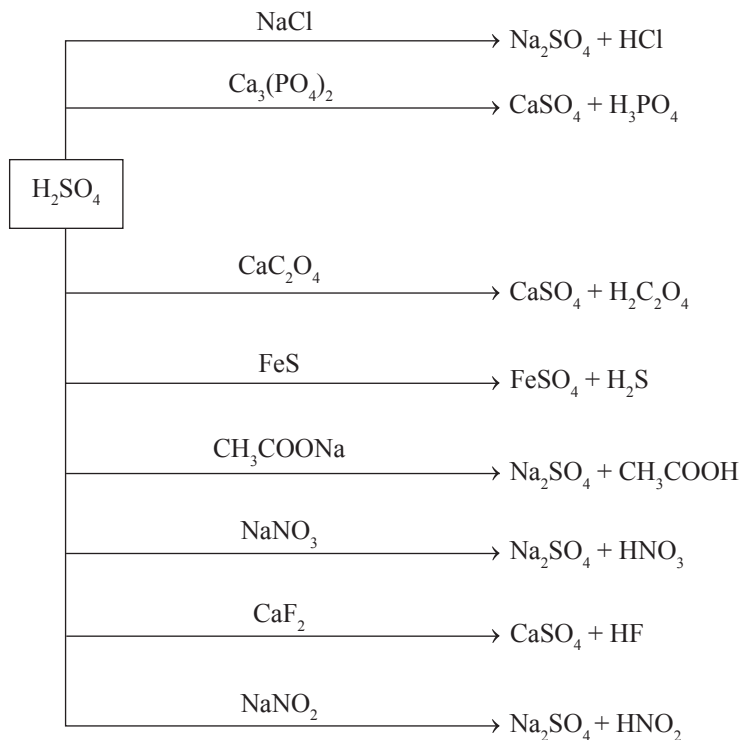
- (iii) Acidic nature— Dibasic acid



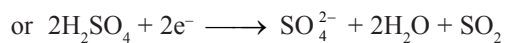
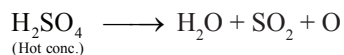
(iv) It reacts with electropositive metals evolving hydrogen.



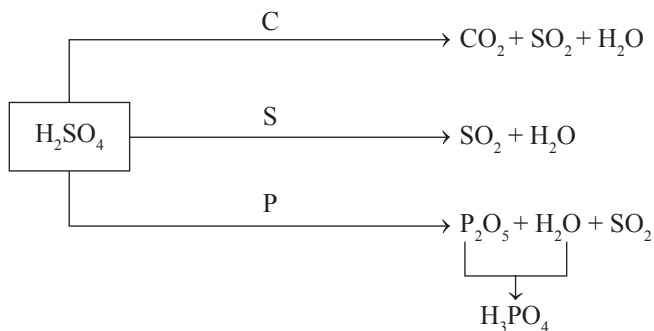
(v) It displaces more volatile acids from their metal salts.



(vi) Oxidizing nature– H_2SO_4 is strong oxidizing agent.

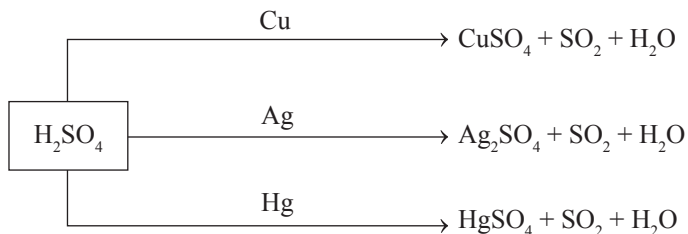


(a) Non-metals such as carbon and sulphur are oxidized to their oxides.



7.32 Group16-Oxygen Family

- (b) Metals like copper, silver, mercury, etc., are first oxidized by conc. H_2SO_4 and then the oxides combine with acid to form corresponding sulphates.



- (c) Iodine is liberated from KI.



- (d) In the presence of Hg as a catalyst, naphthalene is oxidized to phthalic acid.



- (v) **Dehydrating nature** As H_2SO_4 has great affinity for water, it acts as a powerful dehydrating agent. Its corrosive action on the skin is also due to dehydration of skin. It absorbs water from organic compounds.



Some Noteworthy Points

- Lead chamber process for the manufacture of H_2SO_4 was introduced by John Roebuck in 1746.
- Brown oil of Vitriol and Rectified oil of Vitriol. The sulphuric acid obtained from Glover tower contains about 80% H_2SO_4 and is known as Brown oil of Vitriol (B.O.V) due to its colour. It can be further concentrated and the concentrated acid is called Rectified oil of Vitriol (R.O.V).
- Corrosive action of conc. H_2SO_4 on skin is due to dehydration of skin which then burns and produces itching sensation
- H_2SO_4 due to its wide applications is called king of chemicals. It is also called oil of vitriol.
- Conc. H_2SO_4 containing SO_3 is called fuming sulphuric acid (Oleum).
- Conc. HNO_3 + conc. H_2SO_4 is called nitrating mixture or mixed acid.
- For diluting conc. H_2SO_4 , always add conc. H_2SO_4 to water taken in excess. Never add water to conc. H_2SO_4 .

Worked-out Examples

48. Conc. sulphuric acid cannot be used for drying H_2 . Why?

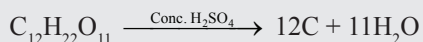
Ans. Conc. H_2SO_4 on absorbing water liberates so much heat that the hydrogen ignites.

49. For the dilution of H_2SO_4 , water should not be added to conc. H_2SO_4 . Explain.

Ans. Conc. H_2SO_4 has a strong affinity for water and a large amount of heat is produced on mixing water with acid. The temperature rises suddenly and bumping occurs usually. Thus acid should be added into water slowly with constant stirring to avoid fast temperature rise and bumping.

50. Sugar turns black on addition of Conc. H_2SO_4 . Why?

Ans. Conc. H_2SO_4 has great affinity for H_2O as it is strong dehydrating agent. It dehydrates sugar leaving carbon behind.



51. Sulphuric acid has high b.pt. and viscosity. Explain.

Ans. High b.pt. and viscous nature of H_2SO_4 is due to hydrogen bonding which results in binding of simple molecules together to form clusters.

52. Colic. H_2SO_4 is used as dehydrating agent. Explain.

Ans. Concentrated H_2SO_4 has a strong affinity for water and it prevents the reversibility of the reaction where water is formed. Thus it is used as dehydrating agent.

53. The wooden shelf under the reagent bottle containing concentrated H_2SO_4 blackens after some time. Explain.

Ans. Conc. H_2SO_4 being strong dehydrating agent, on coming in contact with wood (cellulose material) of shelf, takes out water from it leaving behind black carbon.

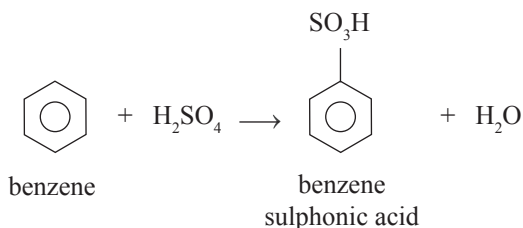
54. The conjugate base of HSO_4^- in aqueous solution is ____.

(IIT 1989)

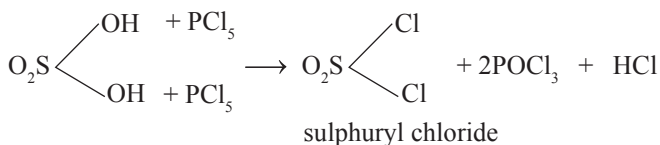
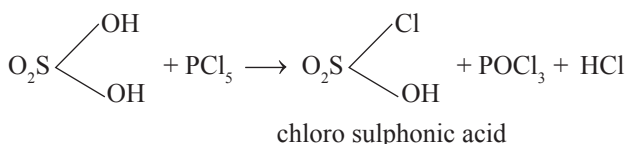
Ans. SO_4^{2-}

(vi) **Miscellaneous reaction**

(a) **Sulphonation:** Aromatic compounds react with H_2SO_4 forming sulphonic acids.



(b) **Action of PCl_5**



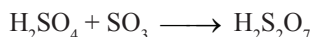
(c) On heating potassium ferrocyanide with concentrated H_2SO_4 , carbon monoxide is evolved.



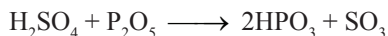
(d) On heating potassium chlorate with concentrated H_2SO_4 , chlorine dioxide is evolved with explosion.



H_2SO_4 absorbs SO_3 forming oleum or fuming sulphuric acid.



(e) H_2SO_4 when treated with P_2O_5 loses water and forms SO_3 .



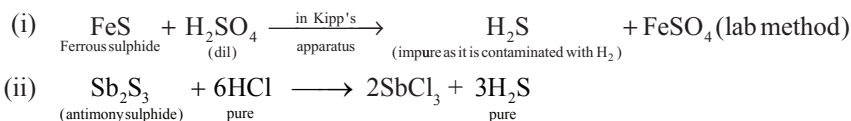
7.34 Group16-Oxygen Family

Uses: It is used:

- for the manufacture of fertilizers such as ammonium sulphate and superphosphate of lime.
- in the manufacture of other acids such as hydrochloric acid, orthophosphoric acid, nitric acid, etc.
- in the manufacture of explosives such as, nitroglycerine, gun cotton, TNT (trinitro-toluene), picric acid, etc.
- in the manufacture of important chemicals such as sodium carbonate, alums, metal sulphates, ethers, phosphorus, etc.
- in the refining of petroleum.
- in the manufacture of coal-tar dyes and a number of drugs and disinfectants.
- in the textile, paper and dyeing industries.
- in the leather industry for tanning.
- for cleansing metals before electroplating, enameling, galvanizing, etc., (Pickling).
- in storage batteries.
- in manufacturing paints and pigments.
- as an important laboratory reagent. In organic chemistry it is used for synthesis of various organic compounds and sulphonation reactions. It is also used as an oxidizing and a dehydrating agent.

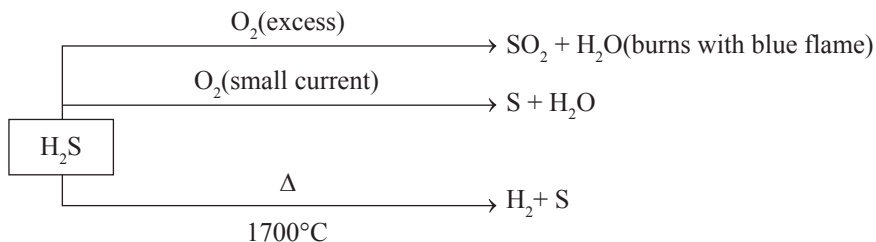
H_2S

Preparation

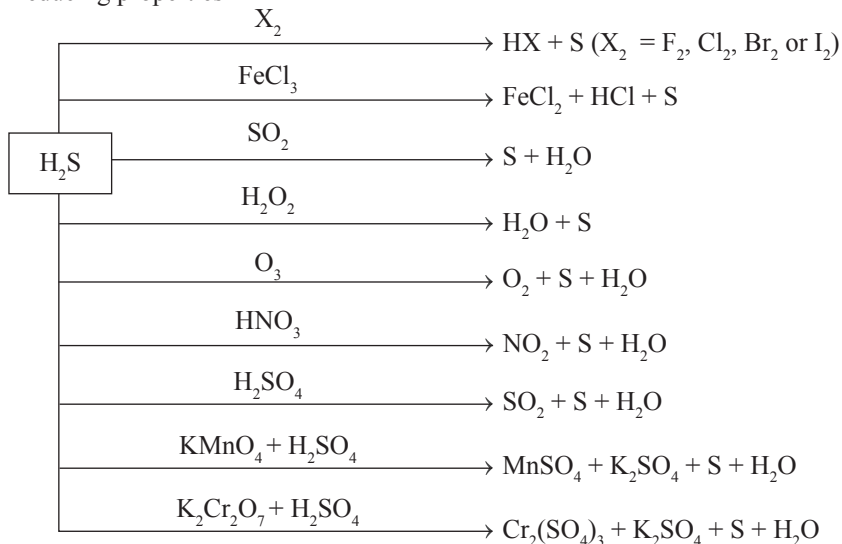


Properties

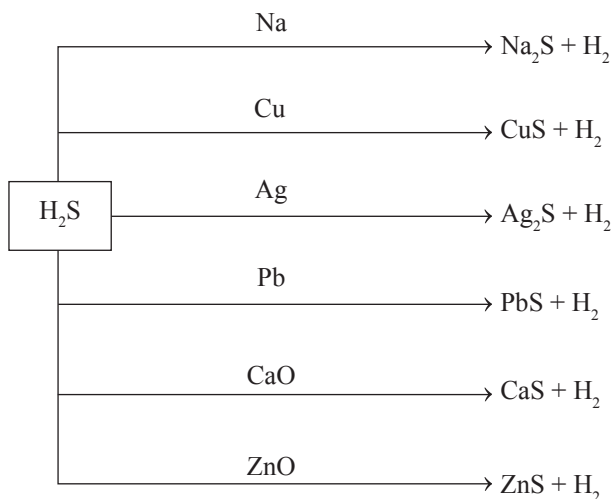
- Colourless gas with unpleasant odour of rotten eggs.
- Slightly heavier than air and moderately soluble in cold water.
- Poisonous in nature. Produces headache when inhaled in small quantities but may cause death when inhaled for a long time.



- Reducing properties



- Acidic nature H_2S is week dibasic acid and forms two series of salts hydrosulphide and sulphides
- Reaction with metal and metal oxides



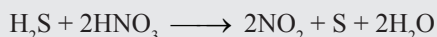
Uses:

- It is an important laboratory reagent for the detection of basic radicals in qualitative analysis.
- It is used for the preparation metallic sulphides many of which find use in paint industry.
- It is used as a reducing agent.

Worked-out Examples

55. Nitric acid cannot be used to prepare H_2S . Explain.

Ans. Nitric acid is an oxidizing agent. It will oxidize H_2S .

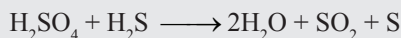


56. H_2S acts only as reducing agent while SO_2 can act both as reducing agent and oxidizing agent. Explain.

Ans. S in SO_2 has oxidation state +4. It lies between the minimum oxidation state (–2) and maximum oxidation state (+6) of S. Thus S in SO_2 can show an increase in its ox.no. (i.e., act as reductant) or can show a decrease in its ox. no. (i.e., acts as oxidant). On the other hand in H_2S , S is in –2 oxidation state and can only increase its oxidation state to act as reductant.

57. H_2SO_4 cannot be used for drying up H_2S . Explain.

Ans. Conc. H_2SO_4 oxidizes H_2S to S and thus cannot be used as for drying up H_2S .

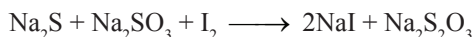
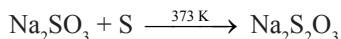


58. A white turbidity is obtained by passing H_2S in aqueous solution of SO_2 . Why?

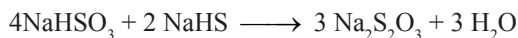
Ans. Aqueous solution of SO_2 and H_2S undergoes redox changes to produce colloidal sulphur as white turbidity.



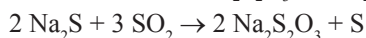
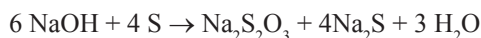
Sodium thiosulphate or Hypo ($\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$) is prepared (i) by heating a solution of Na_2SO_3 with finely powdered sulphur in an alkaline medium until the yellow colour due to sulphur disappears (ii) by adding iodine to a mixture of sodium sulphide and sodium sulphite. (Spring's reaction)



- (i) Pure sodium thiosulphate may be prepared by treating sodium bisulphate with sodium bisulphide followed by crystallization.



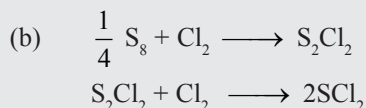
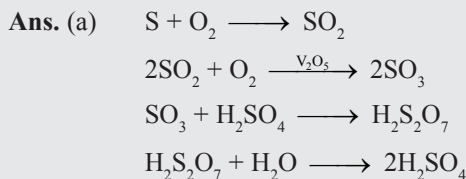
- (ii) It is also formed when sodium hydroxide solution is boiled with sulphur powder or (v) by passing SO_2 gas into Na_2S solution.



Worked-out Examples

59. Starting from elemental sulphur, how would you prepare:

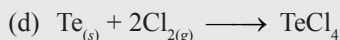
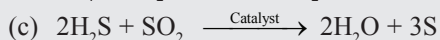
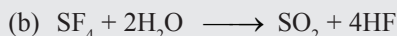
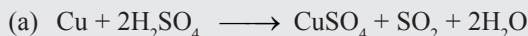
- (a) H_2SO_4 , (b) SCl_2 , (c) SF_6 .



60. Write balanced equations for the following reactions and write down the products:

- (a) $\text{Cu} + \text{Conc. H}_2\text{SO}_4 \longrightarrow$ (b) $\text{SF}_4 + \text{H}_2\text{O} \longrightarrow$
(c) $\text{H}_2\text{S} + \text{SO}_2 \xrightarrow{\text{Catalyst}}$ (d) $\text{Te}_{(s)} + \text{Cl}_{2(g)} \longrightarrow$

Ans.



Structure of thiosulphate ion: Like SO_4^{2-} ion, thiosulphate ion ($\text{S}_2\text{O}_3^{2-}$) is also tetrahedral. The only difference being that in $\text{S}_2\text{O}_3^{2-}$ ion one of the positions of the tetrahedron is occupied by a sulphur atom. Thus, the two sulphur atoms in $\text{Na}_2\text{S}_2\text{O}_3$ are non-equivalent; that is why when it is treated with dilute acids, one sulphur atom forms SO_2 while the other gets precipitated.



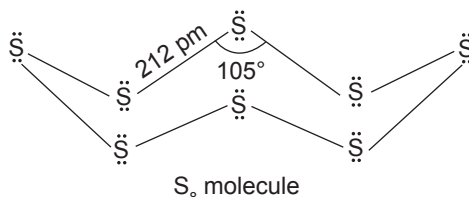


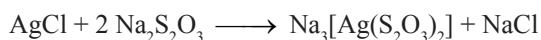
Fig. 7.6

Properties

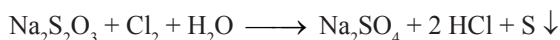
- (i) A dilute solution of hypo when treated with AgNO₃ solution first gives a white ppt. of silver thiosulphate which readily changes to yellow, orange, brown and finally black due to the formation of silver sulphide.



- (ii) Hypo solution dissolves silver halides (AgCl, AgBr, AgI) to form colourless complex of sodium argentothio-sulphate.



- (iii) Cl₂ and Br₂ oxidize sodium thiosulphate to sodium sulphate. But hypo decolourizes solution of I₂ due to formation of sodium tetrathionate.



This reaction is used for the estimation of iodine in iodimetry.

Na₂S₂O₃·5 H₂O crystals are efflorescent and lose water of crystallization at 490 K.

Uses: Sodium thiosulphate is used (i) in photography as a fixer since it dissolves unexposed silver bromide (ii) as an antichlor for removing excess of Cl₂ used in bleaching fabrics (iii) in metallurgy for extraction of Au and Ag and (iv) in the estimation of iodine.

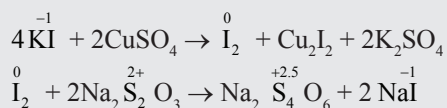
Chlorine oxidizes sodium thiosulphate to sodium sulphate. Hence hypo is used as antichlor in textile industry to remove excess of chlorine from clothes.

Worked-out Examples

61. Excess of KI reacts with CuSO₄ solution and then Na₂S₂O₃ solution is added to it. Which of the statements is incorrect for this reaction? [AIEEE 2004]

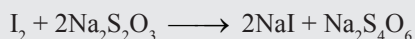
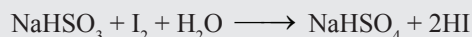
- (a) Na₂S₂O₃ is oxidized (b) CuI₂ is formed
(c) Cu₂I₂ is formed (d) Evolved I₂ is reduced

Ans. (c)



62. How can NaHSO₃ can be estimated volumetrically?

Ans. NaHSO₃ solution is treated with I₂ solution of known concentration in excess and the left I₂ is titrated using hypo solution with starch indicator.



$$\begin{aligned} \text{Meq. of NaHSO}_3 &= \text{Meq. of I}_2 \text{ taken} - \text{Meq. of I}_2 \text{ left (estimated by Na}_2\text{S}_2\text{O}_3) \\ &= \text{Meq. of I}_2 \text{ taken} - \text{Meq. of Na}_2\text{S}_2\text{O}_3 \text{ used} \end{aligned}$$

Some Noteworthy Points

- The elements become progressively more metallic down the column. Polonium has chemical and physical properties characteristic of a metal and tellurium lies on the borderline.
- Chemically, the metallic character of the heavier elements is reflected in their increased tendency to form cationic species, the ionic character and basicities of their oxides, and their increased tendency to form complexes.
- The non-metal character of the earlier members of the Group is the molecular nature of the stable elemental allotrope, the ability to form anions, e.g., O^{2-} and S^{2-} , which result from completing the octet. Compounds resulting from these anions, e.g., Na_2E , CaE ($E = O, S, Se \dots$) become progressively more covalent down the Group as a result of the decreasing electronegativity of the chalcogens.
- The thermal stabilities of the hydrides EH_2 decreasing down the column primarily because of the decreasing EH mean bond enthalpies.
- The ability of these hydrides to form hydrogen bonds decreases rapidly down the Group after oxygen and this has a dramatic effect on the boiling and melting points of the hydrides
- The formation of molecular compounds with strong multiple bonds is particularly important for oxygen which forms strong $p_\pi - p_\pi$ bonds, e.g., CO , $d_\pi - d_\pi$ bonds with transition metals, e.g., OsO_4 and main group atoms, e.g., R_3PO and even $p_\pi - f_\pi$ bonds in UO_2^{2+} . Multiple bonding is less significant for the heavier elements, although sulphur and selenium provide some examples of multiple bonded compound with $p_\pi - p_\pi$ and $d_\pi - d_\pi$ bonding, e.g., selenoketones.
- The increasing size of the atoms leads to compounds with progressively larger maximum coordination numbers. Oxygen usually has a coordination number of 2 or 3 with a few examples of 4 coordination sulfur exhibits a maximum coordination number of 6 and higher coordination number of 8 have been observed for Te.

These maximum coordination numbers have an impact on the case of hydrolysis of the halides, e.g., the rate of hydrolysis is:



Also, the octahedral anionic complexes $[MX_6]^{2-}$ ($X = \text{halide}$) are more commonly observed for Se, Te, and Po.

- Oxygen has strong preference for the -2 formal oxidation state, whereas the heavier elements exhibit oxidation states of 2, 4 and 6. The alternation effect, which is discussed earlier in this chapter, is observed for this Group.
- The tendency towards catenation reaches a maximum at sulphur which forms a wide range of ring and chain compounds. e.g., S_n , XS_nX , $O_3S(S_n)SO_3^{2-}$, and S_8^{2+}

ONE CORRECT OPTION TYPE QUESTIONS

- Sodium thiosulphate is prepared by
 - reducing Na_2SO_4 solution with H_2S
 - boiling Na_2SO_3 solution with S in alkaline medium
 - neutralizing $\text{H}_2\text{S}_2\text{O}_3$ solution with NaOH
 - boiling Na_2SO_3 solution with S in acidic medium
- The number of S — S bonds in sulphur trioxide trimer (S_3O_9) is
 - three
 - two
 - one
 - zero
- The acid having O — O bond is
 - $\text{H}_2\text{S}_2\text{O}_3$
 - $\text{H}_2\text{S}_2\text{O}_6$
 - $\text{H}_2\text{S}_2\text{O}_8$
 - $\text{H}_2\text{S}_4\text{O}_6$
- Which of the following is not oxidized by O_3 ?
 - KI
 - FeSO_4
 - KMnO_4
 - K_2MnO_4
- KO_2 (potassium super oxide) is used in oxygen cylinders in space and submarines because it
 - absorbs CO_2 and increases O_2 content
 - eliminates moisture
 - absorbs CO_2
 - produces ozone.
- When sulphur is boiled with Na_2SO_3 , a compound (A) is produced. (A) with excess of AgNO_3 solution gives a compound (B) which is water soluble and produces a black coloured sulphide (C). Compounds (A), (B) and (C) will be respectively:
 - $\text{Na}_2\text{S}_2\text{O}_3$, $\text{Ag}_2\text{S}_2\text{O}_3$, Ag_2S
 - Na_2SO_4 , Ag_2SO_4 , Ag_2S
 - $\text{Na}_2\text{S}_2\text{O}_7$, Ag_2SO_4 , Ag_2S
 - Na_2SO_5 , Ag_2SO_5 , Ag_2SO_4
- A yellow coloured crystalline substance gave a colourless gas X on reaction with fluorine, which is thermally stable and has octahedral geometry. X can be:
 - SF_4
 - S_2F_2
 - SF_6
 - S_2F_6
- The number of lone pairs and the number of S—S bonds in S_8 molecules are respectively:
 - 8, 8
 - 16, 8
 - 8, 16
 - 8, 4
- Tailing of mercury test can be used for which of the following gas?
 - Dioxygen
 - Dihydrogen
 - Dinitrogen
 - Ozone
- Which of the following solutions does not change its colour on passing ozone through it?
 - Starch iodide solution
 - Alcoholic solution of benzidine
 - Acidic solution of $\text{K}_2\text{Cr}_2\text{O}_7$
 - Acidified solution of FeSO_4
- A green coloured solution of a salt changes its colour to light pink on passing ozone through it. Which of the following species represent pink and green colour respectively?
 - Mn^{2+} and MnO_2
 - MnO_4^- and MnO_4^{2-}
 - Co^{2+} and Co^{3+}
 - MnO_4^{2-} and MnO_4^-
- Oxygen is more electronegative than sulphur, yet H_2S is acidic in nature while H_2O is neutral because:
 - water molecules are associated due to intermolecular hydrogen bonding.
 - water has higher boiling point than H_2S
 - H—S is weaker than O—H bond
 - H_2S is a gas at ordinary temperature while H_2O is a liquid
- O_3 cannot oxidize:
 - KI
 - FeSO_4
 - KMnO_4
 - K_2MnO_4
- Which of the following is/are correct?
 - $\text{O}_3 + \text{moist iodine} \longrightarrow \text{HIO}_3$
 - $\text{FeCl}_3 + \text{H}_2\text{S} \longrightarrow \text{colloidal sulphur}$
 - $\text{O}_3 + \text{Ag} \longrightarrow \text{black silver}$
 - all
- Which of the following is correct structure of S_2Cl_2 ?
 - $$\begin{array}{c} \text{Cl} \\ \diagup \\ \text{S} = \text{S} \\ \diagdown \\ \text{Cl} \end{array}$$
 - $\text{Cl}-\text{S}-\text{S}-\text{Cl}$
 - $$\begin{array}{c} \text{Cl} \\ \diagup \\ \text{S}-\text{S} \\ \diagdown \\ \text{Cl} \end{array}$$
 - $\text{Cl}=\text{S}=\text{S}=\text{Cl}$
- What is the hybrid state and oxidation state of sulphur in Caro's acid?
 - sp^2 , +10
 - sp^3 , +10
 - sp^2 , +6
 - sp^3 , +6

17. Identify the correct sequence of increasing number of π -bonds in structures of the following molecules.
 (i) $\text{H}_2\text{S}_2\text{O}_6$ (ii) H_2SO_3
 (iii) $\text{H}_2\text{S}_2\text{O}_5$
 (a) I, II, III (b) II, III, I
 (c) II, I, III (d) I, III, II
18. Sulphur reacts with chlorine in 1 : 2 ratio and forms (X). (X) on hydrolysis gives a sulphur compound (Y). What is the hybridized state of central atom in the anion of (Y)?
 (a) sp (b) sp^3
 (c) sp^2 (d) sp^3d
19. Ozone is:
 (a) An isobar of oxygen
 (b) An isotope of oxygen
 (c) A polymer of oxygen
 (d) An allotrope of oxygen
20. Ozone reacts with dry iodine to give:
 (a) IO_2 (b) I_2O_3
 (c) I_2O_4 (d) I_4O_9
21. Starch paper moistened with KI solution turns blue in ozone because of:
 (a) Iodine liberation
 (b) Oxygen liberation
 (c) Alkali formation
 (d) Ozone reacts with litmus paper
22. A considerable part of the harmful ultraviolet radiation of the sun does not reach the surface of the earth. This is because high above the earth's atmosphere there is a layer of:
 (a) Carbon dioxide (b) Hydrogen
 (c) Ozone (d) Ammonia
23. Oxygen is gas but sulphur is solid because:
 (a) Oxygen is composed of discrete molecules while sulphur is polymeric
 (b) Molecular weight of sulphur is much higher than that of oxygen
 (c) Oxygen is a stronger oxidizing agent than sulphur
 (d) Boiling point of sulphur is much higher than that of oxygen
24. Elements O, S, Se and Te are usually known as:
 (a) Metals
 (b) Rare earth metals
 (c) Coinage metals
 (d) Chalcogens
25. The bond angle in H_2S is:
 (a) $109^\circ 28'$ (b) $104^\circ 51'$
 (c) 120° (d) 92.5°
26. Low volatile nature of H_2SO_4 is due to:
 (a) Hydrogen bonding
 (b) van der Waal's forces
 (c) Strong bonds
 (d) None of these
27. Dry bleaching is done by:
 (a) Cl_2 (b) SO_2
 (c) O_3 (d) H_2O_2
28. There is no S—S bond in:
 (a) $\text{S}_2\text{O}_4^{2-}$ (b) $\text{S}_4\text{O}_6^{2-}$
 (c) $\text{S}_2\text{O}_3^{2-}$ (d) $\text{S}_2\text{O}_7^{2-}$
29. Liquid oxygen:
 (a) is an important constituent of rocket fuels
 (b) is used for artificial respiration with CO_2
 (c) mixed with finely divided carbons is explosive
 (d) all of the above
30. The reaction in the Kipp's apparatus stops on closing the outlet, because:
 (a) The acid becomes weak
 (b) Gas starts coming out from top
 (c) A protective film is formed on iron sulphide
 (d) The contact between sulphide and the acid is broken by the presence of gas collected in the free surface of the middle chamber
31. Oxygen is more electronegative than sulphur, yet H_2S is acidic while H_2O is neutral. This is because.
 (a) Water is highly associated compound
 (b) H—S bond is weaker than H—O bond
 (c) H_2S is a gas while H_2O is a liquid
 (d) The molecular weight of H_2S is more than that of H_2O
32. H_2S cannot be dried by passing over conc. H_2SO_4 because:
 (a) The acid oxidizes it
 (b) The acid combines with H_2S to form a salt
 (c) Both form complex
 (d) It dissolves in the acid
33. The bond angles in the following compounds are:
 (a) $\text{TeH}_2 > \text{SeH}_2 > \text{SH}_2 > \text{OH}_2$
 (b) $\text{TeH}_2 > \text{SH}_2 > \text{SeH}_2 > \text{OH}_2$
 (c) $\text{OH}_2 > \text{SH}_2 > \text{SeH}_2 > \text{TeH}_2$
 (d) $\text{SH}_2 > \text{SeH}_2 > \text{TeH}_2 > \text{OH}_2$

34. Cyclic trimer structure of SO_3 contains:
- 6 S = O bonds and three S—O—S bonds
 - 3 S = O bonds and six S—O—S bonds
 - 6 S = O bonds and six S—O—S bonds
 - None of these
35. The abundance of sulphur in volcanic region is due to:
- $\text{SO}_2 + \text{H}_2\text{S} \longrightarrow 3\text{S} + 2\text{H}_2\text{O}$
 - $\text{H}_2\text{S} + \text{FeCl}_3 \longrightarrow 2\text{FeCl}_2 + 2\text{HCl} + \text{S}$
 - $\text{H}_2\text{S} + \text{H}_2\text{SO}_4 \longrightarrow \text{SO}_2 + \text{S} + 2\text{H}_2\text{O}$
 - None of these
36. Ozone is more reactive than oxygen because:
- Ozone is endothermic compound
 - Ozone possess higher energy
 - The formation of ozone takes place with absorption of energy
 - All of these
37. SO_2 is more powerful reducing agent in:
- Acid medium
 - Alkaline medium
 - Neutral medium
 - None of these

COMPREHENSION BASED QUESTIONS

COMPREHENSION-I

Sulphuric acid is considered as the king of chemicals. The prosperity of any country is measured by the amount of sulphuric acid it consumes. Sulphuric acid is, thus, a substance of very great commercial importance as it is used practically in every important industry. This is due to the following properties of sulphuric acid:

- acidic nature
- oxidizing nature
- dehydrating nature
- sulphonation

1. Sulphuric acid has very corrosive action on skin because:

- it reacts with proteins
- it acts as an oxidizing agent
- it acts as a dehydrating agent
- it acts as a dehydrating agent and absorption of water is highly exothermic.

2. Which of the following reactions depict the oxidising behaviour of H_2SO_4 ?

- $2\text{HI} + \text{H}_2\text{SO}_4 \longrightarrow \text{I}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$
- $\text{NaCl} + \text{H}_2\text{SO}_4 \longrightarrow \text{NaHSO}_4 + \text{HCl}$
- $2\text{NaOH} + \text{H}_2\text{SO}_4 \longrightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$
- $2\text{PCl}_5 + \text{H}_2\text{SO}_4 \longrightarrow 2\text{POCl}_3 + 2\text{HCl} + \text{SO}_2\text{Cl}_2$

3. Sulphuric acid is used:

- in lead storage batteries
- in making fertilizers
- in making explosives
- in all of these

4. The formation of nitroglycerine is done by the use of concentrated nitric acid and concentrated sulphuric acid. The process of conversion of glycerine into nitroglycerine is termed as:

- sulphonation
- oxidation
- nitration
- dehydration

5. Only carbon is obtained when concentrated H_2SO_4 is added to:

- formic acid
- cane sugar
- oxalic acid
- ethyl alcohol

6. Concentrated H_2SO_4 cannot be used to prepare HBr or HI from KBr or KI because it:

- reacts too slowly with KBr or KI
- reduces HBr or HI
- oxidizes HBr or HI
- oxidizes KBr to KBrO_3 or KI to KIO_3

COMPREHENSION-II

The binary compounds of oxygen with other elements are called oxides. They are classified either depending upon their acid-base characteristics or on the basis of oxygen content.

(a) **Normal oxides:** These oxides which contain oxygen atoms as permitted by the normal oxidation number, i.e., -2 . Normal oxide may be acidic, basic, amphoteric or neutral.

(b) **Polyoxides:** The oxides which contain oxygen atoms different than those permitted by the normal oxidation number of -2 .

(i) **Peroxides:** Two oxygen atoms are linked to each other and each oxygen has -1 oxidation number. They contain $(\text{O—O})^{2-}$ unit.

(ii) **Superoxides:** These oxides contain $(\text{O—O})^-$ unit, i.e., each O-atom has oxidation number $-1/2$.

(c) **Suboxides:** These contain low content of oxygen than expected.

(d) **Mixed oxides:** These oxides are made of two simpler oxides.

7.42 Group16-Oxygen Family

- Which pair of species is referred to as suboxides?
(a) CO, NO (b) SO_2 , CaO
(c) N_2O , CO (d) N_2O , C_3O_2
- Which of the following pairs contains neutral oxides?
(a) SO_2 , SO_3 (b) N_2O_3 , N_2O_5
(c) CO, NO (d) Na_2O , CaO
- Which of the following pairs contains mixed oxides?
(a) Pb_3O_4 , Fe_3O_4 (b) MnO_2 , BaO_2
(c) KO_2 , Na_2O_2 (d) Mn_3O_4 , N_2O_5
- Which of the following pairs contains amphoteric oxides?
(a) BeO, BaO
(b) BeO, Al_2O_3
(c) Al_2O_3 , P_2O_5
(d) FeO, CuO
- Which of the following oxides is paramagnetic in nature?
(a) KC_2 (b) BaO_2
(c) H_2O (d) CO_2

ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

- Which among the following are peroxo acid of sulphur?
(a) H_2SO_3 (b) H_2SO_5
(c) $\text{H}_2\text{S}_2\text{O}_8$ (d) H_2SO_4
- Select the correct statements about oxygen molecule:
(a) It is paramagnetic
(b) Its bond order is two
(c) In liquid state it is blue coloured
(d) It has two unpaired electrons
- SO_2 acts as:
(a) bleaching agent
(b) oxidizing agent
(c) reducing agent
(d) disinfectant
- Sulphuric acid can be used as:
(a) hygroscopic agent
(b) oxidizing agent
(c) sulphonating agent
(d) efflorescent
- H_2S can be used as:
(a) acid (b) base
(c) oxidizing agent (d) reducing agent
- Select the correct statements about $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$;
(a) It is also called as hypo
(b) It is used in photography to form complex with AgBr
(c) It can be used as antichlor
(d) It is used to remove stains of I_2
- Which of the following are amphoteric?
(a) BeO (b) Al_2O_3
(c) ZnO (d) SO_2
- Which reagent does not give oxygen as one of the product during oxidation with ozone?
(a) SO_2 (b) SnCl_2/HCl
(c) H_2S (d) PbS
- In which of the following S—S link is present?
(a) Caro's acid
(b) Dithionic acid
(c) Thiosulphuric acid
(d) Chlorosulphonic acid
- Peroxo acids of sulphur are:
(a) H_2SO_4 (b) H_2SO_3
(c) H_2SO_5 (d) $\text{H}_2\text{S}_2\text{O}_8$

MATCH THE COLUMN TYPE QUESTIONS

1.

Acid	No. of π bond
(i) H_2SO_5	(a) 1
(ii) $\text{H}_2\text{S}_2\text{O}_6$	(b) 4

Acid	No. of π bond
(iii) H_2SO_3	(c) 2
(iv) $\text{H}_2\text{S}_2\text{O}_5$	(d) 3

ASSERTION/REASON TYPE QUESTIONS

Each question contains Assertion and Reason. Examine the statements carefully and mark the correct answer according to the instructions given below:

- (a) If both the statement are TRUE and Reason is the correct explanation of Assertion
- (b) If both the statements are TRUE but Reason is NOT the correct explanation of Assertion
- (c) If Assertion is TRUE and Reason is FALSE
- (d) If Assertion is FALSE and Reason is TRUE

1. **Assertion:** Tailing of mercury occurs on passing O_3 through it.

Reason: This is due to oxidation of mercury.

2. **Assertion:** H_2O is the only hydride of the chalcogens which is liquid

Reason: In ice, each O atom is surrounded by 4H atoms

3. **Assertion:** Ozone is a powerful oxidizing agent in comparison to O_2 .

Reason: Ozone is diamagnetic while O_2 is paramagnetic.

4. **Assertion:** $SeCl_4$ does not have a tetrahedral structure.

Reason: Se in $SeCl_4$ has two lone pairs.

5. **Assertion:** H_2SO_4 forms only one series of salts.

Reason: The molecule of H_2SO_4 consists of only one OH group.

6. **Assertion:** SO_3 acts as an oxidizing as well as reducing agent.

Reason: SO_2 acts as an oxidizing as well as a reducing agent.

RECENT YEARS' QUESTIONS OF IIT-JEE AND AIEEE

1. The correct order for acid strength is:

(AIEEE 2004)

- (a) $Al_2O_3 < SiO_2 < P_2O_3 < SO_2$
- (b) $SiO_2 < SO_2 < Al_2O_3 < P_2O_3$
- (c) $Al_2O_3 < SiO_2 < Al_2O_3 < P_2O_3$
- (d) $SO_2 < P_2O_3 < SiO_2 < Al_2O_3$

2. The number of S—S bonds in sulphur trioxide trimer (S_3O_9) is:

(IIT 2001)

- (a) 3
- (b) 2
- (c) 1
- (d) 0

3. $[X] + H_2SO_4 \longrightarrow [Y]$ a colourless gas with irritating smell, $[Y] + K_2Cr_2O_7 + H_2SO_4 \longrightarrow$ Green solution, $[X]$ and $[Y]$ is:

(IIT 2003)

- (a) SO_3^{2-}, SO_2
- (b) Cl^-, HCl
- (c) S^{2-}, H_2S
- (d) CO_3^{2-}, CO_2

4. The acid having O—O bond is: (IIT 2004)

- (a) $H_2S_2O_3$
- (b) $H_2S_2O_6$
- (c) $H_2S_2O_8$
- (d) $H_2S_4O_6$

5. Which of the following is not oxidized by O_3 :

(IIT 2005)

- (a) KI
- (b) $FeSO_4$
- (c) $KMnO_4$
- (d) K_2MnO_4

6. Aqueous solution of $Na_2S_2O_3$ on reaction with Cl_2 gives:

(IIT 2008)

- (a) $Na_2S_4O_6$
- (b) $NaHSO_4$
- (c) NaCl
- (d) NaOH

ANSWER KEYS

ONE CORRECT OPTION TYPE QUESTIONS

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (b) | 2. (d) | 3. (c) | 4. (c) | 5. (a) | 6. (a) | 7. (c) | 8. (b) | 9. (d) | 10. (c) |
| 11. (d) | 12. (c) | 13. (c) | 14. (d) | 15. (c) | 16. (d) | 17. (b) | 18. (b) | 19. (d) | 20. (a) |
| 21. (d) | 22. (c) | 23. (b) | 24. (c) | 25. (d) | 26. (c) | 27. (a) | 28. (b) | 29. (a) | 30. (d) |
| 31. (d) | 32. (a) | 33. (c) | 34. (a) | 35. (d) | 36. (d) | 37. (a) | | | |

COMPREHENSION BASED QUESTIONS

Comprehension-I

1. (c) 2. (d) 3. (b) 4. (d) 5. (b) 6. (a)

Comprehension-II

1. (c) 2. (a) 3. (a) 4. (d) 5. (b)

ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

1. (b, c) 2. (a, b, c, d) 3. (a, b, c, d) 4. (a, b, c) 5. (a, d)
6. (a, b, c, d) 7. (a, b, c) 8. (a, b) 9. (b, c) 10. (c, d)

MATCH THE COLUMN TYPE QUESTIONS

1. (i) – c (ii) – b (iii) – a (iv) – d

ASSERTION/REASON TYPE QUESTIONS

1. (a) 2. (b) 3. (b) 4. (c) 5. (e) 6. (d)

RECENT YEARS' QUESTIONS OF IIT-JEE AND AIEEE

1. (a) 2. (d) 3. (a) 4. (c) 5. (c) 6. (b)

CHAPTER 8

Group17-
Halogens

INTRODUCTION

	<i>F</i>	<i>Cl</i>	<i>Br</i>	<i>I</i>
Atomic Number	9	17	35	53
Electronic Configuration	[He] 2s ² 2p ⁵	[Ne] 3s ² 3p ⁵	[Ar] 4s ² 4p ⁵	[Kr] 5s ² 5p ⁵
First Ionization Energy (ev)	17.42	13.01	11.84	10.44
Standard reduction electrode potential (volts)	+2.85	+1.36	+1.06	+0.53
$\begin{array}{c} \text{X}_2(\text{g}) + 2\text{e}^- \\ \rightleftharpoons \\ 2\text{X}^-(\text{aq}) \end{array}$ reaction				
Atomic Radius (Å)	0.72	0.99	1.14	1.33
Ionic Radius (Å)	1.36 (−1)	1.81 (−1), 0.26 (+7)	1.95 (−1), 0.39 (+7)	2.16(−1), 0.50 (+7)
M.p. °C	−218.6	−101.0	−7.2	113.5
B.p. °C	−188.1	−34.6	59.5	185.2
Heat of dissociation of molecules (kcal/mole)	37.7	56.7	45.2	35.4
Electron affinity (ev)	3.62	3.79	3.56	3.28
Electronegativity	4.0	3.0	2.8	2.5

ns²np⁵
₉F (He) 2s²2p⁵ → Pale yellow gas F₂
₁₇Cl (Ne) 3s²3p⁵ → Greenish yellow gas Cl₂
₃₅Br(Ar) 3d¹⁰4s²4p⁵ → Red liquid Br₂
₅₃I (Kr) 4d¹⁰5s²5p⁵ → Purple solid
₈₅At (Xe) 4f¹⁴5d¹⁰6s²6p⁵ → radio active (half life 8.3 hour)

8.2 Group 17-Halogens

- The word Halogen is derived from Greek words Hals = Sea salt, gennao = producer i.e., sea salt producer.
- Halogens have the electronic configuration ns^2np^5 i.e., they are just one electron short of noble gas configuration. They have a very strong tendency to take up one electron to acquire a stable configuration. That's why they all are typical nonmetals.
- They are reactive nonmetals. Down the group reactivity decreases. Fluorine is most reactive and known as superhalogen.
- I_2 has some metallic properties—metallic luster—can form I^+ ion.
- All are diatomic molecules.
- Down the group sizes of halogens increase—Van der Waal force increase—m.p and b.p increase down the group. (note that physical state changes from gas to solid)
- Ionization energies of all the halogens are very high.
I.P. decreases down the group and iodine forms I^+ ion. IP : $F > Cl > Br > I$

Some Noteworthy Points

- Electron affinity in general decreases down the group. However, electron affinity of Cl is more than F.
Explanation: The unexpectedly low electron affinity of fluorine as compared to chlorine is due to very small size of the fluorine atom (72 pm). As such, there are strong inter-electronic repulsions in the relatively compact 2p-subshell of fluorine. Thus, the incoming electron does not feel much attraction and as such the electron affinity of fluorine decreases and becomes less than that of chlorine.
- Bond energy, in general, decreases as the bond length increases. However, bond energy of F_2 is less than Cl_2 , this is despite the fact that F–F bond is shorter than Cl–Cl bond.
- Bond energies $Cl-Cl > Br-Br > F-F > I-I$
The small bond dissociation energy of F_2 is due to $e^- - e^-$ repulsion between non-bonded electrons $\begin{array}{c} \cdot\cdot \\ \cdot\cdot \\ :\ddot{F}-\ddot{F}: \\ \cdot\cdot \\ \cdot\cdot \end{array}$ (small size $\Rightarrow$ greater repulsion)
- **It is because** the lesser bond energy of F_2 is due to greater inter electronic repulsion between the lone pairs of electrons on the two bonded fluorine atoms. Such repulsions are almost not there in other halogens due to greater X-X bond length.
- At room temperature, liquid halogen is bromine while solid halogen is iodine.
- Chlorine gas is collected by upward displacement of air (Cl_2 gas is heavier than air and quite soluble in water).
- Bromine was discovered by Ballard. Iodine was discovered by Courtois.
- Iodine is purified by sublimation.
- On careful evaporation for Cl_2 water or Br_2 water, crystals of the composition $Cl_2 \cdot 8H_2O$ and $Br_2 \cdot 8H_2O$ are formed.
- Aqueous solution of I_2 is brown in colour while organic solution of I_2 are violet in colour.
- Iodine is used in medicine in the form of tincture of iodine [2-3% alcoholic solution of I_2] and iodox ointment (contains I_2 and methyl salicylate in a veseline base).
- The mother liquor containing 0.25% bromine as $MgBr_2$ is known as Bittern.
- NaBr and KBr are used as sedatives.
- The order of electron affinity: $F < Cl > Br > I$.
- Increasing order of electronegativity : $I < Br < Cl < F$.

Worked-out Examples

1. Iodine forms a cation. Explain.

Ans. The atomic radii increase down the gp. i.e., F to I and thus ionization energy decreases down the group. Hence, I possesses appreciable low values of i.e., to show cation formation.

2. The increasing order of the first ionization enthalpies of the elements B, P, S and F (Lowest first) is [AIEEE 2006]
(a) $B < P < S < F$ (b) $B < S < P < F$ (c) $F < S < P < B$ (d) $P < S < B < F$

Ans. (b) The correct order of ionization enthalpies is



On moving along a period, ionization enthalpy increases from left to right and decreases from top to bottom in a group. But this trend breaks up in case of atom having fully or half filled stable half orbitals. In this case P has a stable orbital half filled electronic configuration hence its ionization enthalpy is greater in comparison to S. Hence the correct order is $B < S < P < F$.

- Colour of halogen – Halogens are coloured. The colour of the halogen is due to absorption of certain wavelengths of visible light by their molecules, resulting in the excitation of outer electron to higher energy orbitals. Fluorine being smaller in size absorbs shorter wavelengths corresponding to violet colour for excitation and appears pale yellow. Iodine on the other hand absorbs longer wavelengths corresponding to yellow colour for excitation and therefore appears violet. In between fluorine and iodine, the colour of chlorine is greenish yellow and of bromine is reddish brown. Thus, the colour deepens down the group.

Worked-out Examples

3. Why are halogens coloured?

Ans. Halogens are coloured because of their low ionization energy. As a result, the electrons get excited in the visible region of light. The remaining light falls in the visible region and halogens appear to be coloured. For example, fluorine absorbs wave lengths corresponding to violet light and the remaining light appears to be yellow. Thus, fluorine has a yellow colour. Similarly, chlorine is greenish yellow, bromine reddish and iodine is a violet solid.

4. On moving from F_2 to I_2 , colour of halogens becomes dark, why?

Ans. Colour of halogens arises due to absorption of light, promoting an electron from ground state to higher state. On moving from F_2 to I_2 , the size of atom increases and so energy levels become closer and so promotional energy becomes less so colour deepens.

5. The enthalpy of sublimation of I_2 is 14.92 k cal/mol. What will be heat of formation of $I_2(g)$?

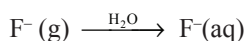
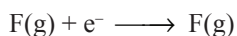
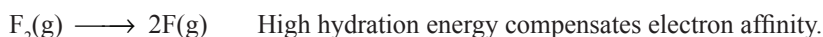
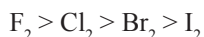
Ans. The enthalpy of sublimation of nearly equal to heat of formation of $I_2(g)$.

6. Which among the following factors is the most important in making fluorine the strongest oxidizing halogen? **[AIEEE 2004]**

- | | |
|------------------------|------------------------------|
| (a) Hydration enthalpy | (b) Ionization enthalpy |
| (c) Electron affinity | (d) Bond dissociation energy |

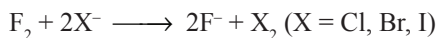
Ans. (d) The fluorine has low dissociation energy of F-F bond and reaction of atomic fluorine is exothermic in nature.

- Electron affinities of halogens are highest in their respective periods. I.A. of halogens $Cl > F > Br > I$. The lower value for F is due to $e^- - e^-$ repulsion which arises due to compactness of 2p orbital.
- These elements are strong oxidizing agents. Oxidizing power decreases down the group. F_2 is the strongest oxidizing agent.

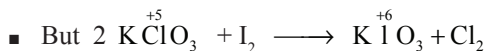


8.4 Group 17-Halogens

Hence,



But $\text{I}_2 + 2\text{KBr} \longrightarrow$ no reaction [less electronegative halogen cannot displace more electronegative halogen in normal salt]



[Note that the halogens in those oxysalts are in +ve oxidation state]

- F is the most electronegative element and it does not 'have vacant 'd' orbital. Hence, its oxidation number is -ve and only one oxidation number is observed i.e., '-1' state.
- Rest of the halogens show variable oxidation state due to vacant 'd' orbital.

Worked-out Examples

7. Fluorine exists only in oxidation state of -1, Explain.

Ans. Fluorine is the most electronegative element and thus shows only negative oxidation state. Furthermore it has $2s^2 2p^5$ configuration and thus shows only -1 oxidation state in order to complete its octet. Also like other halogens, it does not have vacant d-orbitals in its valence shell

Halogens (Cl, Br, I)	ns^2	np^5	nd	
Ground state	$\uparrow\downarrow$	$\uparrow\downarrow \uparrow\downarrow \uparrow$		Covalence = 1
I excited state	$\uparrow\downarrow$	$\uparrow\downarrow \uparrow \uparrow$	$\uparrow \quad \quad \quad \quad$	Covalence = 3
II excited state	$\uparrow\downarrow$	$\uparrow \uparrow \uparrow \uparrow$	$\uparrow \uparrow \quad \quad \quad$	Covalence = 5
III excited state	$\uparrow$	$\uparrow \uparrow \uparrow \uparrow$	$\uparrow \uparrow \quad \quad \quad$	Covalence = 7

8. Fluorine brings about the highest co-ordination number of elements. Explain.

Ans. Fluorine possesses the smallest size and maximum electronegativity and thus, elements combining with fluorine during.

9. Fluorine usually does not form oxoacids. Explain.

Ans. Fluorine is most electronegative element. Its electronegativity is higher than oxygen and thus, OF_2 is not an oxide of fluorine but it is oxygen difluoride.

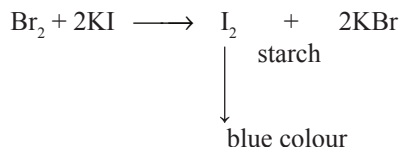
10. Fluorine does not form F_3^- ion while iodine form I_3^- ions. Explain.

Ans. It is due to unavailability of vacant d-orbital in second shell of F-atom.

- Highly soluble in non polar solvent like CS_2 , CCl_4 .
 $\text{Br}_2 \rightarrow$ Orange layer when dissolved.
 $\text{I}_2 \rightarrow$ purple layer when dissolved

Halogens (except F_2) are soluble in some polar solvent due to charge transfer compounds.

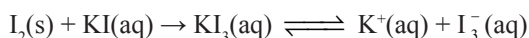
- I_2 gives blue colour with starch solution whereas, Cl_2 and Br_2 give yellow colour with starch solution. With starch iodide solution, Br_2 and Cl_2 give blue colour because I^- is oxidized to I_2 which gives blue colour.



Note:

Solubility of I_2 is more in $KI(aq)$ than in pure water.

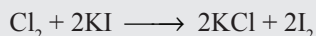
Explanation. I_2 when dissolved in $KI(aq)$ results in the formation of polyhalide, I_3^- ion which increases the solubility of I_2 in water.



Worked-out Example

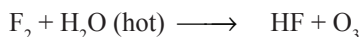
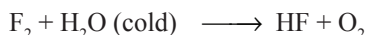
11. Colour of KI solution containing starch turns blue when chlorine water is added to it. Explain.

Ans. Cl_2 replaces I_2 from its aqueous solution getting itself reduced due to higher standard reduction potential than iodine and thus, colour of KI solution containing starch develops blue colour due to absorption of I_2 by starch.



■ **Action of water**

F_2 reacts vigorously with H_2O . With cold H_2O , O_2 is produced, with hot water O_3 is produced: Hence, F_2 is soluble in water. Reaction takes place even in the dark.



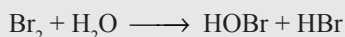
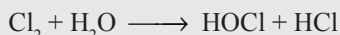
Worked-out Examples

12. Fluorine gives fumes with moist air. Explain.

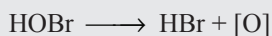
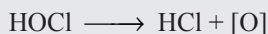
Ans. Fluorine reacts with H_2O to give HF which being liquid absorbs liquid droplets to form fumes.



Cl_2 , Br_2 react in presence of sunlight. They disproportionate in H_2O giving halogen acid and halogen oxyacid.

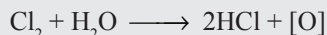


Their aqueous solution shows bleaching property due to the decomposition of halogen oxyacid to nascent oxygen.

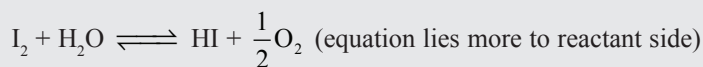


13. Dry chlorine does not act as bleaching agent.

Ans. The bleaching action of Cl_2 is due to its reaction with H_2O to liberate nascent oxygen.



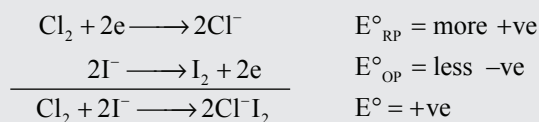
I_2 does not react with H_2O (insoluble in water)



I_2 dissolves in water containing I^- . This is due to the formation of I_3^- .

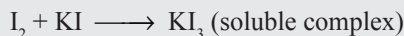
14. Chlorine displaces iodine from aqueous solution of KI. Explain.

Ans. The standard reduction potential of Cl is higher than iodine and thus Cl_2 get reduced to oxidize I^- ion to I_2 .



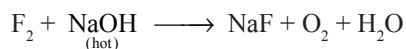
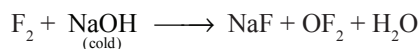
15. Iodine dissolves more in KI solution, than in pure water. Explain.

Ans. I_2 is non-polar in nature and thus its solubility in water (a polar solvent) is less which however becomes more if KI is present due to complex formation.

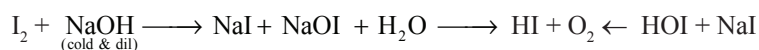
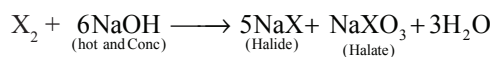
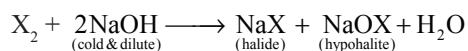


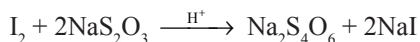
Reaction with base

Fluorine reacts with cold alkali and produces OF_2 , with hot alkali O_2 is produced

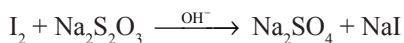


Rest of the halogens behave in the same manner. They produce halide (X^-) and hypohalite (XO^-) with cold alkali and a mixture of halide (X^-) and halite (XO_3^-) in concentrated. alkali. Hence, disproportion takes place.



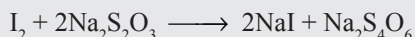
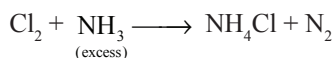
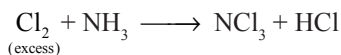
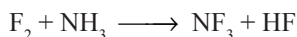
Reaction with thiosulphate

In alkaline medium,

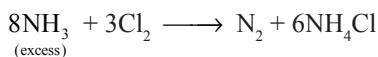
**Worked-out Example**

16. Iodine stains on clothes can be removed by hypo.

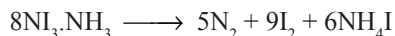
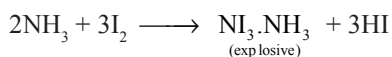
Ans. Hypo reacts with iodine to form water soluble sodium tetra thionate and sodium iodide.

**Reaction with NH_3** 

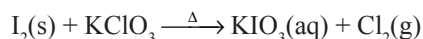
- Halogens react with NH_3 and give different products.



- Br_2 also reacts with NH_3 in the same way as Cl_2 . But the reaction of iodine is different.



- Fluorine due to its high reactivity is called superhalogen.
- I_2 gives a blue colour with starch solution. This colour disappears on heating. However, on cooling, the colour reappears.
- Reaction of F_2 with NaOH or KOH (hot or cold) is different from the reaction of other halogens (Cl_2 , Br_2 and I_2) with NaOH or KOH . This is because fluorine does not form oxoanions.
- UF_6 is more stable than UCl_6 .
- Br_2 turns moist starch paper brown.
- When I_2 is heated with $\text{KClO}_3(\text{s})$ or I_2 is added to hot concentrated solution of KClO_3 in the presence of a little strong nitric acid, Cl_2 is given out. The reaction can be written as,



However, the exact reaction is quite complicated.

- NF_3 is nonexplosive while NCl_3 is highly explosive.

8.8 Group 17-Halogens

- With sulphur, fluorine give SF_6 , chlorine forms SCl_4 , bromine give SBr_2 while iodine does not react with sulphur.
- AgF is soluble in water while AgCl , AgBr and AgI are insoluble.
- CaF_2 is insoluble while all other calcium halides (CaCl_2 , CaBr_2 , CaI_2) are soluble in water.
- Fluorochlorobromocarbons are used for fire lighting in homes, offices industry, transport etc. Common brand name 'Cease fire'. Examples are: Halon-1211 (CF_2ClBr), Halon-1301 (CF_3Br), Halon-2402 ($\text{C}_2\text{F}_4\text{Br}_2$). Halones destroy O_3 layer like chlorofluorocarbons (CFCs).
- On careful evaporation of chlorine, water (or bromine water) crystals of composition $\text{Cl}_2 \cdot 8\text{H}_2\text{O}$ (or $\text{Br}_2 \cdot 8\text{H}_2\text{O}$) are obtained.
- The ionic character of the halides of the same element (M) decreases in the order $\text{MF} > \text{MCl} > \text{MBr} > \text{MI}$.

Worked-out Examples

17. How would you prepare:

- | | |
|---|---------------------------------------|
| (a) Cl_2 gas in the laboratory | (b) HF from CaF_2 |
| (c) Br_2 from sea water | (d) I_2 from NaIO_3 |
| (e) HBr from NaBr ? | |

Ans.

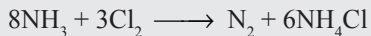
- (a) $\text{MnO}_2 + 4\text{HCl} \rightarrow \text{MnCl}_2 + 2\text{H}_2\text{O} + \text{Cl}_2$
 (b) $\text{CaF}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + 2\text{HF}$
 (c) $2\text{Br}^- + \text{Cl}_2 \rightarrow \text{Br}_2 + 2\text{Cl}^-$
 (d) $\text{NaIO}_3 + 3\text{HNaHSO}_3 \rightarrow \text{Hal} + 3\text{NaHSO}_4$
 $\text{NaIO}_3 + 5\text{NaI} + 3\text{H}_2\text{SO}_4 \rightarrow 3\text{I}_2 + 3\text{H}_2\text{O} + 3\text{Na}_2\text{SO}_4$
 (e) $2\text{NaBr} + \text{Cl}_2 \rightarrow 2\text{NaCl} + \text{Br}_2$
 $\text{Br}_2 + \text{H}_2 \xrightarrow{\text{Pt}} 2\text{HBr}$

18. Identify the incorrect statement among the following.

[AIEEE 2007]

- (a) Br_2 reacts with hot and strong NaOH solution to give NaBr and H_2O
 (b) Ozone reacts with SO_2 to give SO_3 .
 (c) Silicon reacts with SO_2 to give SO_3 .
 (c) Silicon reacts with $\text{NaOH}_{(\text{aq})}$ in the presence of air to give Na_2SiO_3 and H_2O .
 (d) Cl_2 reacts with excess of NH_3 to give N_2 and HCl .

Ans. (d) Chlorine reacts with excess of ammonia to produce ammonium chloride and nitrogen.



19. Aqueous solution of $\text{Na}_2\text{S}_2\text{O}_3$ on reaction with Cl_2 gives

(IIT 2008)

- (a) $\text{Na}_2\text{S}_4\text{O}_6$ (b) NaHSO_4 (c) NaCl (d) NaOH

Ans. (b)

20. In an aqueous solution, chlorine is a stronger oxidizing agent than fluorine. (True/False)

(IIT 1984)

Ans. False because the hydration energy of F^- is more than that of Cl^- .

21. The decreasing order of electron affinity of F, Cl, Br is $\text{F} > \text{Cl} > \text{Br}$. (True/False)

(IIT 1993)

Ans. False the correct order is $\text{F} < \text{Cl} > \text{Br}$. The larger electronic repulsion in F due to smaller size makes its electron affinity less than that of Cl.

22. Chlorine acts as a bleaching agent only in presence of

- (a) dry air (b) moisture (c) sunlight (d) pure oxygen

Ans. (b)

23. Iodine reacts with hot NaOH solution. The products are NaI and _____. (IIT 1980)
Ans. NaIO_3
24. The increase in the solubility of iodine in an aqueous solution of potassium iodide is due to the formation of _____. (IIT 1982)
Ans. KI_3 or I_3^-
25. Solubility of iodine in water is greatly increased by the addition of iodide ions because of the formation of _____. (IIT 1994)
Ans. I_3^-
26. The compounds used as refrigerant are
 (a) NH_3 (b) CCl_4 (c) CF_4 (d) CF_2Cl_2 (IIT 1990)
Ans. (a), (d)
27. Of the halide ions, _____ is the most powerful reducing agent. (IIT 1978)
Ans. iodide
28. Give an example of oxidation of one halide by another halogen. Explain the feasibility of the reaction. (IIT 2000)
Ans. The example is

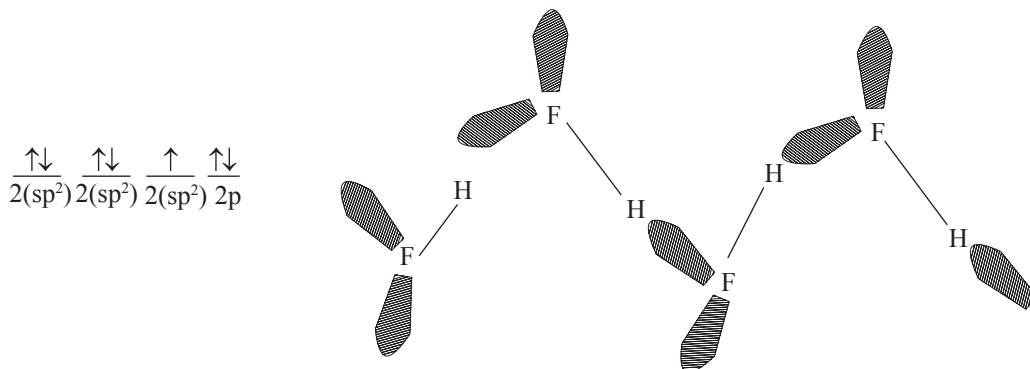
$$2\text{I}^-(\text{aq}) + \text{Cl}_2(\text{g}) \rightarrow \text{I}_2 + 2\text{Cl}^-(\text{aq})$$
 The standard reduction potential of Cl_2 to Cl^- is larger than that of I_2 to I^- .
29. Which of the following statements is correct for CsBr_3 ? (IIT 1996)
 (a) It is a covalent compound
 (b) It contains Cs^{3+} and Br^- ions
 (c) It contains Cs^+ and Br_3^- ions molecule
 (d) It contains Cs^+ , Br^- and lattice of Br_2
Ans. (c)

Hydrohalic acid

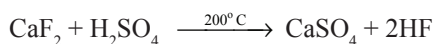
All four halogens give hydrides HX (X = F, Cl, Br, I). HF is a colourless volatile liquid and is considerably associated $(\text{HF})_n$. The other hydrides are colourless molecular species. Some properties are recorded in the table below.

Hydride	Bond energy kcal/mole (kJ/mole)	M.P. (°C)	B.P(°C)	Density (g/cm ³)	Dipole moment (D)
HF	135 (565)	-83.4	-19.5	1.002	1.74a
HCl	103(431)	-114.7	-84.2	1.187	1.07
HBr	88(368)	-88.6	-67.1	2.603	0.788
HI	71(297)	-51.0	-35.1	2.85	0.382

A strong hydrogen bonding is indicated in HF. Decreasing dipole moment values from HF to HI indicate decreasing ionic character of the H-X bonds. Crystal structure of HF reveals a zig-zig planar chain polymer with F...F distance of 2.49 Å (249 pm) F...F...F angle of ~ 130°. The planarity of associated $(\text{HF})_n$ along with the bond angle may tempt one to speculate an sp^2 hybridization of fluorine. The p_z orbital with paired spin remains outside the purview of the hybridization and is not shown in (21-IV). The nature of bonding is not yet fully explained.

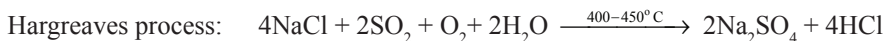
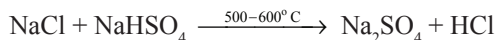
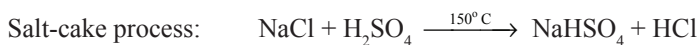


Anhydrous HF is commercially made by the interaction of concentrated H_2SO_4 on fluorspar:



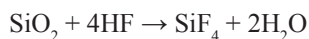
HF is used in the syntheses of chlorofluorocarbons (for refrigeration), synthetic cryolite Na_3AlF_6 , uranium hexafluoride (in nuclear plants) etc. The sodium salt, NaF is used as wood preservative, insecticide and fungicide.

HCl is commercially obtained by a number of methods:



Some Noteworthy Points

1. HCl is also called Muriatic acid.
2. HI is least stable at all hydrogenhalides and decomposes to yield H_2 and I_2 that is why a bottle containing HI acquires brown colour after some time.
3. Etching of glass: HF reacts with glass to form hydrofluosilicic acid, H_2SiF_6 so it is used for etching of glass.



4. Concentrated HCl + con. HNO_3 is called aqua regia and Kingly Water or Royal Water.
5. HF is not stored in glass vessels since it reacts with SiO_2 of glass. It is stored in copper, wax, polythene or gutta percha.
6. HF is more corrosive and reactive than fluorine.
7. Fremy's salt is potassium hydrogen fluoride, KHF_2 .

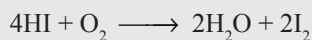
Worked-out Examples

30. Anhydrous HCl is a bad conductor of electricity but aqueous HCl is a good conductor. Explain.

Ans. HCl is a covalent molecule and thus it is a bad conductor of current, however in aqueous medium, HCl is ionized (Fajan's rule) to give H^+ and Cl^- ions to show good conductor nature.

31. Pure HI kept in a bottle acquires a brown colour after some time. Explain.

Ans. HI is strong reducing agent and it reduces oxygen of the air getting itself oxidized to I_2 . The liberated iodine is dissolved there and imparts brown colour to solution.



32. HF is the least volatile and HCl is most volatile amongst hydrogen halide.
or

HF is liquid whereas, HCl, HBr and HI are gases. Explain.

Ans. The boiling point of covalent compounds (HF, HCl, HBr, HI) increases with increase in their molecular weight due to increase in van der Waal's forces of attraction among molecules. Furthermore, HF showing H-bonding possess higher boiling point or less volatile nature. The volatile nature of halogen acids may be represented as

Volatile nature	$\text{HCl} > \text{HBr} > \text{HI} > \text{HF}$
or Boiling point order	$\text{HCl} < \text{HBr} < \text{HI} < \text{HF}$

33. HF is stored in wax coated glass bottles. Explain.

Ans. HF reacts with glass (sodium and calcium silicate present in glass) containers and thus it is stored in wax coated glass bottle.

34. KHF_2 is well known, whereas KCl_2 or KBr_2 does not exist. Explain.

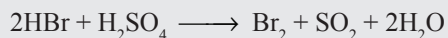
Ans. H_2F_2 exists as dimeric molecule due to H-bonding and thus shows dibasic nature. Hence it gives two series of salts KHF_2 as $[\text{K}^+ \text{ and } \text{F}^- \dots \text{H}-\text{F}^-]$ and $\text{KF}[\text{K}^+ \text{ and } \text{F}^-]$ whereas HCl and HBr do not show hydrogen bonding and thus formation of KCl_2 or KBr_2 is not possible.

35. Why has HF a lower boiling point than water even though it has stronger hydrogen bonds than water?

Ans. HF cannot form three dimensional array of hydrogen bonds as in water and so has less boiling point than water.

36. HBr can not be prepared by the action of concentrated H_2SO_4 on NaBr.

Ans. HBr acts as reducing agent and H_2SO_4 as oxidizing agent,



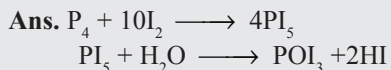
37. HF is a weaker acid in comparison to HCl, HBr and HI. Explain.

Ans. The acidic character of halogen acids has been explained as larger is the H—X bond, lesser is the bond energy, more is the tendency to give H^+ ion and thus HF is the weakest acid among all because of the shortest H—F bond length.

38. HF exists as dimeric molecule i.e., H_2F_2 or HF has higher viscosity or high b.pt. Explain.

Ans. H atom attached on N, O or F possesses the tendency show H-bonding and thus HF molecules undergo cluster formation due to H-bonding and acquires dimeric nature as well as higher viscosity and high b.pt.

39. Calculate the weight of HI obtained by the reaction of 62 g red phosphorus with iodine in the presence of water.



1 mole = $(31 \times 4) = 124$ g Phosphorus produce 4 mole PI_5

which produces 8 mole of HI = $8(1 + 127) = 1024$ g

$\therefore$ 62 g will produce 512 g HI.

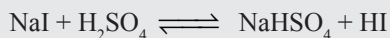
40. Arrange the halogen acids in the increasing order of:

- (i) dipole moment (ii) reducing power
- (iii) thermal stability (iv) bond length
- (v) ionic character

- Ans.** (i) $\text{HI} < \text{HBr} < \text{HCl} < \text{HF}$ (Increasing dipole moment)
 (ii) $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$ (Increasing reducing nature)
 (iii) $\text{HI} < \text{HBr} < \text{HCl} < \text{HF}$ (Increasing thermal stability)
 (iv) $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$ (Increasing bond length)
 (v) $\text{HI} < \text{HBr} < \text{HCl} < \text{HF}$ (Increasing ionic character)

41. In preparation of HI from alkali iodide, why is H_3PO_4 preferred in place of H_2SO_4 ? Explain.

Ans. HI formed during the action of alkali iodide with H_2SO_4 acts as reducing agent and reduces H_2SO_4 to SO_2 , getting itself oxidized to I_2 .



42. HF is corrosive and causes painful skin wounds which are slow to heal, why?

Ans. HF dehydrates the tissue and is acidic in nature i.e., why it causes skin wounds. Healing of tissue requires Ca^{2+} ions which are removed by F^- forming insoluble CaF_2 so healing is slow.

43. HBr is a stronger acid than HI because of hydrogen bonding. (True/False) (IIT 1993)

Ans. False Stronger the hydrogen bonding, weaker the acid.

44. ____ is a weak acid (HF, HCl, HBr). (IIT 1981)

Ans. HF

45. KF combines with HF to form KHF_2 . The compound contains the species (IIT 1996)

- (a) K^+ , F^- and H^+ (b) K^+ , F^- and HF (c) K^+ and $[\text{HF}_2]^-$ (d) $[\text{KHF}]^+$ and F^-

Ans. (c)

46. Which of the following statements is true? [AIEEE 2002]

- (a) HF is less polar than HBr
 (b) Absolutely pure water does not contain any ions
 (c) Chemical bond formation take place when forces of attraction overcome the forces of repulsion
 (d) In covalency, transference of electron takes place

Ans. (c)

47. The correct order of the thermal stability of hydrogen halides (H-X) is [AIEEE 2005]

- (a) $\text{HI} > \text{HCl} < \text{HF} > \text{HBr}$ (b) $\text{HCl} < \text{HF} > \text{HBr} < \text{HI}$
 (c) $\text{HF} > \text{HCl} < \text{HBr} > \text{HI}$ (d) $\text{HI} < \text{HBr} > \text{HCl} < \text{HF}$

Ans. (c) The H-X bond strength decreases from HF to HI, i.e., $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$. Thus HF is most stable While HI is least stable. This is evident from their decomposition reaction: HF and HCl are stable upto 1473K, HBr decreases slightly and HI dissociates considerably at 713K. The decreasing stability of the hydrogen halide is also reflected in the values of dissociation energy of the H-X bond

H – F	H – Cl	H – Br	H – I
135kcal mol ⁻¹	103kcal mol ⁻¹	87kcal mol ⁻¹	71 kcal mol ⁻¹

48. Concentrated hydrochloric acid when kept in open air sometimes produces a cloud of white fumes. The explanation for it is that [AIEEE 2003]

- (a) oxygen in air reacts with the emitted HCl gas to form a cloud of chlorine gas.
 (b) strong affinity of HCl gas for moisture in air results in forming of droplets of liquid solution which appears like a cloudy smoke.

- (c) due to strong affinity for water, concentrated hydrochloride acid pulls moisture of air towards itself. This moisture forms droplets of water and hence the cloud.
- (d) concentrated hydrochloric acid emits strongly smelling HCl gas all the time.

Ans. (b)

Oxides of Chlorine

There are four well-characterized oxides of chlorine. They are generally prepared by indirect methods. These are: (i) Chlorine monoxide (Cl_2O), (ii) Chlorine dioxide (ClO_2), (iii) Chlorine hexoxide (Cl_2O_6) and (iv) Chlorine heptoxide (Cl_2O_7). Their best methods of preparation, state of aggregation and accepted structure are given in the below table.

Name and formula	Method preparation	Important properties	Accepted structure
Chlorine monoxide (Cl_2O)	Obtained by passing Cl_2 over precipitated HgO , previously heated to 300°C : $2\text{Cl}_2 + 2\text{HgO} = \text{HgO} \cdot \text{HgCl}_2 + \text{Cl}_2\text{O} \uparrow$.	Brownish yellow gas; condenses to an orange liquid (b. p. 2°C) on cooling. Dissolves in water to give a soln. of hypochlorous acid: $\text{Cl}_2\text{O} + \text{H}_2\text{O} = 2\text{HOCl}$.	
Chlorine dioxide (ClO_2)	Prepared by passing dry Cl_2 over silver chlorate at 90°C : $2\text{AgClO}_3 + \text{Cl}_2 = 2\text{AgCl} + 2\text{ClO}_2 + \text{O}_2$.	Yellow gas; condenses to a dark-red liquid (b.p. 11°C) and freezes to orange crystals at -59°C . Aqueous soln. is neutral; but reacts with bases to give chlorites and chlorates: $2\text{ClO}_2 + 2\text{KOH} = \text{KClO}_3 + \text{KClO}_2 + \text{H}_2\text{O}$	
Chlorine hexoxide (Cl_2O_6)	Obtained by the action of ozonized oxygen on ClO_2 at 0°C and by fractionating the products: $2\text{ClO}_2 + 2\text{O}_3 = \text{Cl}_2\text{O}_6 + 2\text{O}_2$.	Dark-red unstable liquid, freezing at 3.5°C . Reacts with alkalis to give chlorates and perchlorates: $\text{Cl}_2\text{O}_6 + 2\text{KOH} = \text{KClO}_3 + \text{KClO}_4 + \text{H}_2\text{O}$	
Chlorine heptoxide (Cl_2O_7)	HClO_4 is added to P_2O_5 at -10°C . The mixture is kept for 24 hours and then distilled at 85°C : $2\text{HClO}_4 + \text{P}_2\text{O}_5 = \text{Cl}_2\text{O}_7 \uparrow + 2\text{HPO}_3$.	An oily, colourless liquid (b.p. 82°C). Reacts with water to give perchloric acid: $\text{Cl}_2\text{O}_7 + \text{H}_2\text{O} = 2\text{HClO}_4$.	

8.14 Group 17-Halogens

Some Noteworthy Points

- Solid Cl_2O_6 is ionic and exists as ClO_2^+ and ClO_4^- .
- KClO_3 is called as Berthelot's salt.
- Warming KClO_3 with concentrated HCl gives a mixture of Cl_2 and ClO_2 known as Euchlorine (bleaching agent).
The species Cl_2O_7 , SO_2 and P_4O_{11} are the anhydrides of HClO_4 , H_2SO_3 and H_3PO_3 , respectively. The acid strength of these acids follows the order $\text{HClO}_4 > \text{H}_2\text{SO}_3 > \text{H}_3\text{PO}_3$. The corresponding anhydrides also follow the same order.

Worked-out Examples

49. ____ acid gives hypo ____ ion. (hydrobromic, hypobromous, perbromic, bromide, bromite, perbromate).
(IIT 1988)

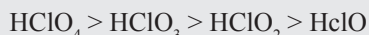
Ans. hypobromous bromite.

50. Explain the acidic nature of HOCl , HOBr and HOI .

Ans. The decreasing trend of thermal stability and acidic nature of oxoacid of halogens having same oxidation number of halogen i.e., as $\text{HOCl} > \text{HOBr} > \text{HOI}$ has been explained in terms of decreasing electronegativity of halogen due to increasing size. As the electronegativity difference in $\text{O}-\text{X}$ bond decreases, the bond strength of $\text{H}-\text{O}$ increases which results in decrease in acidic character of oxoacid of halogens.

51. Explain the acidic nature of HClO_4 , HClO_3 , HClO_2 and HClO .

Ans. The acidic character of oxoacid of same halogen but in different oxidation state is $\text{HClO} < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}_4$ increases with increase in oxidation number of halogen atom. The conjugate base of HCl , HClO_2 , HClO_3 and HClO_4 are ClO^- , ClO_2^- , ClO_3^- and ClO_4^- respectively. As the number of oxygen atom increases, the negative charge dispersal becomes more and more from Cl atom due to more electronegativity of oxygen atom and thus lesser is the charge on Cl atom, more will be its stability. Consequently, stability order for anions is $\text{ClO}_4^- > \text{ClO}_3^- > \text{ClO}_2^- > \text{ClO}^-$ or basic character for anions is $\text{ClO}^- > \text{ClO}_2^- > \text{ClO}_3^- > \text{ClO}_4^-$. Thus acidic character of conjugate acid may be given as:



52. Write balanced equation for the following:

- NaCl is heated with sulphuric acid in the presence of MnO_2 .
- Chlorine gas is passed into a solution of NaI in water.
- SiO_2 is treated with HF .
- NaClO_3 is treated with SO_2 .
- Iodine is treated with concentrated HNO_3 .

Ans. (a) $2\text{NaCl} + 2\text{H}_2\text{SO}_4 + \text{MnO}_2 \longrightarrow \text{MnSO}_4 + \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O} + \text{Cl}_2$

(b) $\text{Cl}_2 + 2\text{NaI} \longrightarrow 2\text{NaCl} + \text{I}_2$

(c) $\text{SiO}_2 + 6\text{HF} \longrightarrow \text{H}_2\text{SiF}_6 + 2\text{H}_2\text{O}$

(d) $2\text{NaClO}_3 + \text{SO}_2 \longrightarrow 2\text{ClO}_2 + \text{Na}_2\text{SO}_4$

(e) $\text{I}_2 + 10\text{HNO}_3 \longrightarrow 2\text{HIO}_3 + 10\text{NO}_2 + 4\text{H}_2\text{O}$

53. How will you prepare DCl ?

Ans. $\text{PCl}_5 + \text{D}_2\text{O} \xrightarrow{\text{Hydrolysis}} \text{POCl}_3 + 2\text{DCl}$

54. Arrange the following:

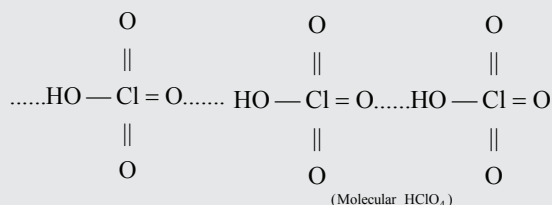
- Increasing order of thermal stability HOCl , HClO_2 , HClO_3 , HClO_4
- Increasing acid strength HClO , HClO_2 , HClO_3 , HClO_4 .
- Increasing oxidizing power HClO , HClO_2 , HClO_3 , HClO_4 .
- Increasing reducing nature F^- , Cl^- , Br^- , I^- .

- (v) Increasing oxidation number of iodine I_2 , HI, HIO_4 , ICl.
- (vi) Increasing acid strength HOF, HOCl, HOBr, HOI.
- (vii) Increasing oxidizing power F_2 , Cl_2 , Br_2 , I_2 .
- (viii) Increasing acid strength HF, HCl, HBr, HI.
- (ix) Increasing electronegativity or reactivity F, Cl, Br, I.
- (x) Increasing electron affinity F, Cl, Br, I.

- Ans.**
- (i) $HClO < HClO_2 < HClO_3 < HClO_4$
 - (ii) $HClO < HClO_2 < HClO_3 < HClO_4$
 - (iii) $HClO_4 < HClO_3 < HClO_2 < HClO$
 - (iv) $F^- < Cl^- < Br^- < I^-$
 - (v) $HI < I_2 < ICl < HIO_4$
 - (vi) $HOI < HOBr < HOCl < HOF$
 - (vii) $I_2 < Br_2 < Cl_2 < F_2$
 - (viii) $HF < HCl < HBr < HI$
 - (ix) $I < Br < Cl < F$
 - (x) $I < Br < F < Cl$

55. Pure $HClO_4$ is a liquid which does not conduct electricity. When melted, the solid hydrate $HClO_4 \cdot H_2O$ does conduct electricity. Explain.

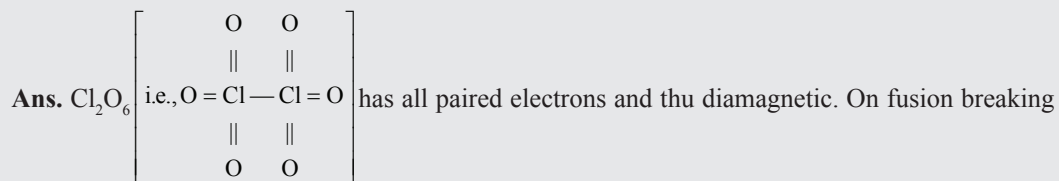
Ans. In molecular $HClO_4$, H-bonding is extensive. In its hydrate, the ions are localized because of the lattice forces, the hydrogen bonding might not be as extensive and gives $[H_3O^+][ClO_4^-]$ ions.



56. In OF_2 , F—O—F angle is 103° while in Cl_2O , Cl—O—Cl bond angle is 111° , Explain. Or Explain the bond angle in OF_2 and give reason why it is different in Cl_2O ?

Ans. In both OF_2 and Cl_2O central O-atom is sp^3 hybridized and the due to lone pair-bond pair repulsion in OF_2 bond angle is reduced from normal $109^\circ 28'$ to 103° . In Cl_2O angle is more than $109^\circ 28'$, due to steric crowding produced by larger Cl-atoms.

57. Cl_2O_6 is diamagnetic but in liquid state it exhibits paramagnetism.



leads to formation of ClO_3 having unpaired electron and thus paramagnetic.

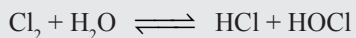
of Cl—Cl bond

58. A gas 'X' is passed through water to form a saturated solution. The aqueous solution on treatment with silver nitrate gives a white precipitate. The saturated aqueous solution also dissolves magnesium ribbon with evolution of a colourless gas 'Y'. Identify 'X' and 'Y'

- (a) $X = CO_2$, $Y = Cl_2$
- (b) $X = Cl_2$, $Y = CO_2$
- (c) $X = Cl_2$, $Y = H_2$
- (d) $X = H_2$, $Y = Cl_2$

(IIT 2002)

Ans. (c) Cl_2 in water undergoes disproportionation reaction.



A white precipitate with AgNO_3 is due to Cl^- ions. The dissolution of magnesium ribbon liberates H_2 gas.



59. The type of hybrid orbitals used by the chlorine atom in ClO_2^- is **(IIT 1992)**

- (a) sp^3 (b) sp^2 (c) sp (d) None of these

Ans. (a)

60. Concentrated HNO_3 reacts with iodine to give **(IIT 1989)**

- (a) HI (b) HOI (c) HOIO_2 (d) HOIO_3

Ans. (c)

61. The set with correct order of acidity is **(IIT 2001)**

- (a) $\text{HClO} < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}_4$ (b) $\text{HClO}_4 < \text{HClO}_3 < \text{HClO}_2 < \text{HClO}$
 (c) $\text{HClO} < \text{HClO}_4 < \text{HClO}_3 < \text{HClO}_2$ (d) $\text{HClO}_4 < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}$

Ans. (a) The larger the number of oxygen atoms attached to the chlorine, the larger the acidity.

62. Which of the following statements is true? **[AIEEE 2006]**

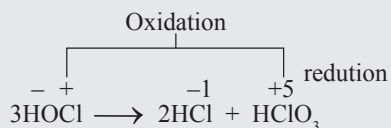
- (a) HClO_4 is a weaker acid than HClO_3
 (b) HNO_3 is a stronger acid than HNO_2
 (c) H_3PO_3 is a stronger acid than H_2SO_3
 (d) In aqueous medium HF is a stronger acid than HCl

Ans. (b) The HNO_3 is stronger than HNO_2 . The more the oxidation state of N, the more is the acid character.

63. What products are expected from the disproportionation reaction of hypochlorous acid? **[AIEEE 2006]**

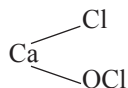
- (a) HCl and Cl_2O (b) HCl and HClO_3
 (c) HClO_3 and Cl_2O (d) HClO_2 and HClO_4

Ans. (b) During disproportionation same compound undergo simultaneous oxidation reduction.

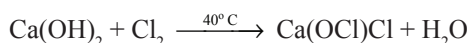


Bleaching Powder

Bleaching powder is also called calcium chlorohypochlorite because it is considered as a mixed salt of hydrochloric acid and hypochlorous acid.



Preparation



Properties

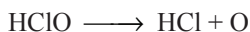
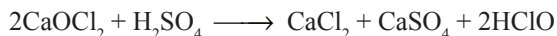
- (a) It is a pale yellow powder. It has a strong smell of chlorine. It is soluble in water but a clear solution is never formed due to the presence of impurities.

(b) On long standing, it undergoes auto-oxidation into calcium chlorate and calcium chloride.



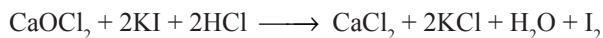
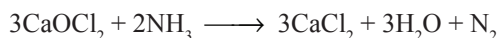
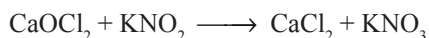
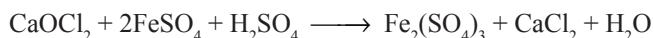
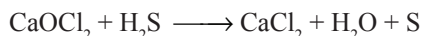
(c) $2\text{CaOCl}_2 \xrightarrow{\text{CoCl}_2} 2\text{CaCl}_2 + \text{O}_2$

(d) In presence of a slight amount of a dilute acid, it loses oxygen.



On account of the formation of nascent oxygen, it shows oxidizing and bleaching properties.

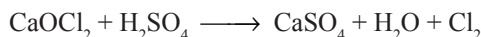
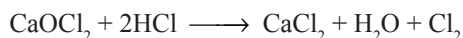
(i) **Oxidizing properties**



(ii) **Bleaching action**

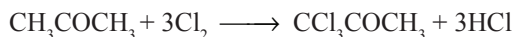
Coloured matter + [O] $\longrightarrow$ colourless product.

(e) It loses its chlorine by the action of dilute acids (in excess or CO_2 .)



The amount of chlorine obtained from a sample of bleaching powder by treatment with excess of dilute acids or CO_2 is called available chlorine. A good sample of bleaching powder contains 35-38% of available chlorine.

(f) Bleaching powder converts acetone or ethyl alcohol into CHCl_3



- **Available chlorine:** The amount of chlorine obtained from a sample of bleaching powder by treatment with excess of dilute acids or carbon dioxide is called available chlorine. A very good sample of bleaching powder contains 35-38% of available chlorine.

Worked-out Examples

64. Bleaching powder loses its bleaching property when it is kept in open bottle for long time. Explain.

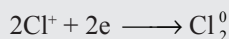
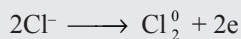
Ans. Bleaching powder, if left open for long time in contact with air, gets converted to $\text{Ca}(\text{ClO}_3)_2$ and CaCl_2 due to auto oxidation and loses the available Cl_2 in it and thereby the bleaching action is also decreased.



65. Name the type of reaction as given below:



Ans. Cl in CaOCl_2 has -1 (Cl) and $+1$ (OCl) i.e., $\text{Ca} \begin{matrix} \nearrow \text{Cl} \\ \searrow \text{OCl} \end{matrix}$ oxidation state.



Thus, the above reaction shows disproportionation of CaOCl_2 .

66. Oxidation number of Cl in CaOCl_2 (bleaching power) is:

[AIEEE 2002]

- (a) zero, since it contains Cl_2 (b) -1 , since it contains Cl^-
 (c) $+1$, since it contains ClO^- (d) $+1$ and -1 since it contains ClO^- and Cl^-

Ans. (d)

INTERHALOGEN COMPOUNDS

We know that halogen atoms have different electronegativity. Due to this difference in electronegativity, the halogen atoms combine with each other and give rise to the formation of binary covalent compounds, which are called interhalogen compounds. These compounds are of four types: (i) AB type ($+1$), e.g., ClF , BrF , IF (detected spectroscopically), BrCl , ICl , IBr (ii) AB_3 type ($+3$), e.g., ClF_3 , BrF_3 , IF_3 , ICl_3 (iii) AB_5 type ($+5$), e.g., ClF_5 , BrF_5 , IF_5 (iv) AB_7 type ($+7$), e.g., IF_7 . The number given in the bracket indicates the oxidation number of A atom. A halogen atom is larger than B halogen atom, i.e., A atom is less electronegative than B atom. In interhalogen compounds, the halogen atoms having less electronegativity show positive oxidation state. It is for this reason that interhalogen compounds are named as the halides of less electronegative halogen atom. Thus, ClF is called chlorine monofluoride and not fluorine monochloride, since the electronegativity of Cl atom is less than that of F atom.

Halogen atom A acts as the central atom with which odd number of B atoms are covalently bonded. The central atom A may be Cl, Br or I but never F, since F atom seldom achieves a positive oxidation state and also it cannot expand its coordination number beyond 4.

With the exception of BrCl , ICl , IBr and ICl_3 all the remaining interhalogen compounds are halogen fluorides. Greater is the electronegativity difference, $(x_F - x_A)$ (A may be I or Br), greater is the number of fluorides viz. IF , IF_3 , IF_5 and IF_7 , while Br atom $(x_F - x_{\text{Br}} = 4.0 - 2.8 = 1.2)$ gives only three fluorides namely BrF , BrF_3 and BrF_5 . The formation of three fluorides by Cl-atom $(x_F - x_{\text{Cl}} = 4.0 - 3.0 = 1.0)$ is taken to be an exception. Actually, Cl-atom should give less than three fluorides.

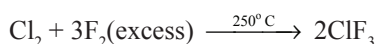
It has not been possible to prepare the interhalogen compounds containing more than two halogen atoms (ternary interhalogen compounds). The non-formation of ternary interhalogen compounds is due to the fact that these compounds so formed get decomposed into a mixture of stable binary interhalogen compounds and/or elemental halogens. Polyhalides such as MIBrF , MIClBr , MIFCl_3 etc., which contain three different halogen atoms which have, however, been prepared.

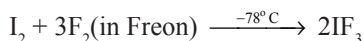
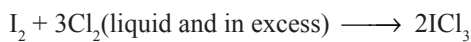
General Methods of Preparation

Following are the important general methods by which interhalogen compounds can be prepared.

(i) By the direct combination of halogens

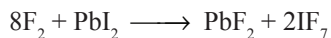
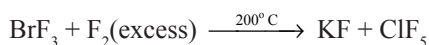
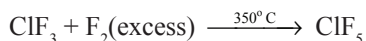
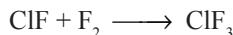
All the interhalogens, except IF_7 , can be prepared by the direct combination of the halogens under appropriate conditions. The following examples are noteworthy. For the preparation of AB_3 and AB_5 , interhalogen compounds excess of halogen atom must be used in the reaction mixture.





(ii) **From lower interhalogens**

Higher interhalogens can be prepared by the interaction of lower interhalogens with halogens. This method is particularly used for the preparation of halogen fluorides. The following examples illustrate the method.



Worked-out Example

67. BF_3 and BrF_3 have different shapes, why?

Ans. BF_3 has coplanar geometry due to sp^2 hybridization on boron atom whereas BrF_3 has sp^3 d hybridization and possesses two lone pairs of electron on Br atom which leads to T shaped geometry.

General Properties

(i) **Physical state**

The interhalogen compounds may be covalent gases (e.g., ClF , BrF , ClF_3 , IF_7) liquids (e.g., BrF_3 , BrF_5) or solids (e.g., ICl , IBr , IF_3 , ICl_3).

(ii) **Colour**

Although many of the interhalogen compounds containing fluorine are colourless, yet those made up of heavier halogens are coloured. The colour becomes deeper with the increase of the molecular weight of the compound. In this behaviour, these compounds resemble the halogens themselves.

(iii) **Diamagnetic nature**

Since all the valence electrons in interhalogen compounds are present as bonding or non-bonding (i.e., lone pairs) electron pairs, these compounds are diamagnetic in nature.

Worked-out Example

68. Interhalogen compounds are always diamagnetic. Explain.

Ans. Due to absence of unpaired electrons.

(iv) **Heats of formation**

All these compounds have low heats of formation. For example, these values for IBr and BrCl are -2.5 and -0.34 kcal/mole respectively. These are extremely volatile compounds. In general, compounds containing fluorine are more volatile than those containing chlorine, bromine or iodine.

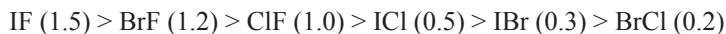
8.20 Group 17-Halogens

(v) Boiling points

For each type of interhalogen compounds, the boiling points increase with the increase in the electronegativity difference between A and B atoms.

(vi) Thermal stability

Thermal stability of AB type interhalogen compounds decreases with the decrease in electronegativity difference between A and B atoms. Thus, the order of stability of some AB compounds is as,



In parentheses are given the electronegativity difference between A and B atoms. The above order is also explained by saying that greater is the difference between the electronegativity values of A and B, the more polar is the A–B bond and hence greater is the thermal stability of AB compounds.

(vii) Reactivity

AB type compounds are more reactive than A_2 and B_2 molecules, since A–B bond in AB compounds is weaker than A–A and B–B bonds in A_2 and B_2 molecules respectively. Thus, AB type compounds convert the metals into a mixture of two halides. For example,



The order of reactivity of some interhalogen compounds has been found as $\text{ClF}_3 > \text{BrF}_3 > \text{IF}_7 > \text{BrF}_5 > \text{BrF}$.

Worked-out Examples

69. Inter halogen compounds are more reactive than halogens. Explain.

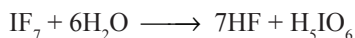
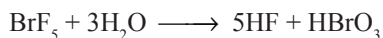
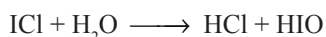
Ans. X—X bond is diatomic halogen i.e., Cl—Cl is stronger than X—Y bond in inter halogen compounds i.e., I—Cl. The difference in strength is due to different size of overlapping orbitals in inter halogen compounds which results in less effective overlapping.

70. IF_3 exists but I_4 does not, though both have four halogen atoms.

Ans. Iodine atom is larger in size. The +3 oxidation state by iodine is not shown with bigger iodine atoms. Steric crowding is also responsible for nonexistence of I_4 .

(viii) Hydrolysis

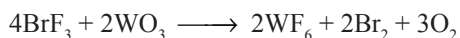
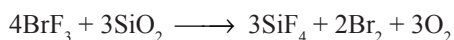
Hydrolysis gives halogen acid and oxy-halogen acid. The oxy-halogen acid is of larger (i.e., central) halogen atom. Examples are,



It may be noted that the oxidation state of A atom does not change during hydrolysis.

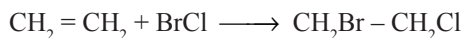
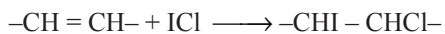
(ix) Reaction with non-metallic and metallic oxides

Non-metallic and metallic oxides are fluorinated by halogen fluorides to give the corresponding fluorides.

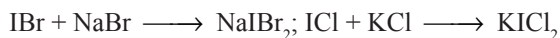


(x) **Addition reactions**

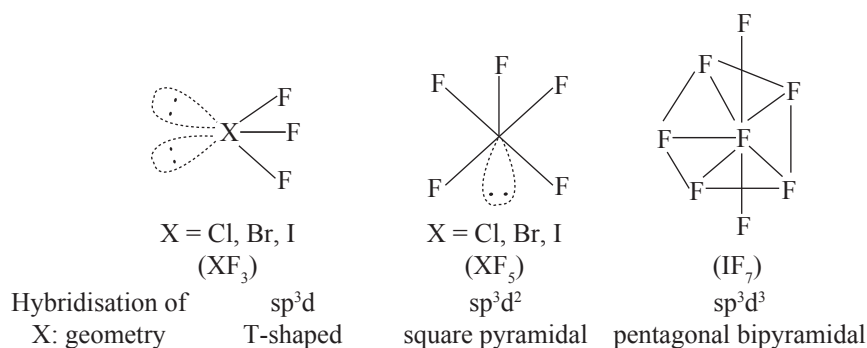
The diatomic interhalogens (AB type compounds) add at olefinic double bond sites.

(xi) **Reaction with alkali metal halides**

Reaction with alkali metal halides gives polyhalides like NaIBr_2 , CsIBrCl , KICl_4 etc.



The molecular structures of interhalogen compounds are very interesting and can be explained on the basis of VSEPR theory. The structure of some of these compounds are shown below.



Structures of some interhalogen compounds

Worked-out Examples

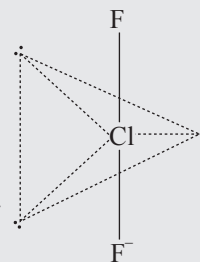
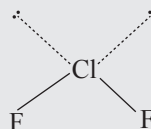
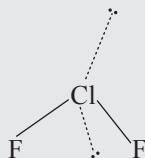
71. Why is IBr known but CBr is not known?

Ans. As we move from chlorine to bromine to iodine, the electropositive character goes on increasing due to decreasing ionization energy. It is more difficult for chlorine to show positive oxidation state than for iodine.

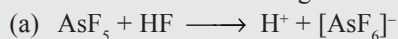
72. ClF_2^- is linear but ClF_2^+ is a V-shape molecule. Explain.

Ans. In ClF_2^- , Cl atom possesses sp^3d hybridization with three lone pairs on the corners of triangular plane.

In ClF_2^+ , Cl atom possesses sp^3 hybridization with two lone pairs at two corners of regular tetra-hedron.



73. Select the acid-base in the given changes.



Ans. AsF_5 and BF_3 are electron pair acceptors (F^-) and thus behaves as Lewis acid whereas, HF behaves as base.

74. Complete and balance the following reactions:

- (a) $\text{CuSO}_4 + \text{KI} \longrightarrow$ (b) $\text{NH}_3 + \text{NaOCl} \longrightarrow$
 (c) $\text{Cl}_2 + \text{H}_2\text{O} + \text{HgO} \longrightarrow$ (d) $\text{P}_4 + \text{I}_2 + \text{H}_2\text{O} \longrightarrow$
 (e) $\text{NaBr} + \text{MnO}_2 + \text{H}_2\text{SO}_4 \longrightarrow$ (f) $\text{Cl}_2 + \text{OH}^- \longrightarrow$
 (g) $\text{MnO}_2 + \text{HCl} \longrightarrow$ (h) $\text{CH}_3\text{CH}=\text{CH}_2 + \text{ICl} \longrightarrow$
 (i) $\text{CsI}_3 \xrightarrow{\Delta}$ (j) $\text{CsF} + \text{IF}_7 \xrightarrow{\Delta}$
 (k) $\text{Rb}[\text{ICl}_2] \xrightarrow{\Delta}$ (l) $\text{I}_2 + \text{ICl} \xrightarrow[\Delta]{\text{AlCl}_3}$

Ans. (a) $2\text{CuSO}_4 + 4\text{KI} \longrightarrow 2\text{K}_2\text{SO}_4 + \text{Cu}_2\text{I}_2 + \text{I}_2$ (b) $2\text{NH}_3 + 3\text{NaOCl} \longrightarrow 3\text{NaCl} + \text{N}_2 + 3\text{H}_2\text{O}$ (c) $2\text{Cl}_2 + \text{H}_2\text{O} + 2\text{HgO} \longrightarrow 2\text{HClO} + \text{HgO} \cdot \text{HgCl}_2$ (d) $\text{P}_4 + 6\text{I}_2 + 12\text{H}_2\text{O} \longrightarrow 4\text{H}_3\text{PO}_3 + 12\text{HI}$ (e) $2\text{NaBr} + \text{MnO}_2 + 3\text{H}_2\text{SO}_4 \longrightarrow 2\text{NaHSO}_4 + \text{MnSO}_4 + 2\text{H}_2\text{O} + \text{Br}_2$ (f) $3\text{Cl}_2 + 6\text{OH}^- \longrightarrow 5\text{Cl}^- + 3\text{H}_2\text{O} + \text{ClO}_3^-$ (g) $\text{MnO}_2 + 4\text{HCl} \longrightarrow \text{MnCl}_2 + \text{Cl}_2 + 2\text{H}_2\text{O}$ (h) $\text{CH}_3\text{CH}=\text{CH}_2 \xrightarrow{\text{I}^{+8}\text{Cl}^{-6}} \text{CH}_3\underset{\text{Cl}}{\underset{|}{\text{CH}}}\cdot\underset{\text{I}}{\underset{|}{\text{CH}_2}}$ (i) $\text{CsI}_3 \rightleftharpoons \text{CsI} + \text{I}_2$ (j) $\text{CsF} + \text{IF}_7 \xrightarrow{\Delta} \text{CsI} + \text{I}_2$ (k) $\text{Rb}[\text{ICl}_2] \xrightarrow{\Delta} \text{RbCl} + \text{ICl}$ (l) $\text{I}_2 + \text{ICl} + \text{AlCl}_3 \xrightarrow[\Delta]{\text{AlCl}_3} [\text{I}_3]^+[\text{AlCl}_4]^-$ 75. On heating, $\text{Rb}(\text{ICl}_2)$ will decompose giving _____ and _____. (IIT 1997)**Ans.** RbCl , ICl

PSEUDOHALOGENS AND PSEUDOHALIDES

A few ions are known, consisting of two or more atoms of which at least one is N, that have properties similar to those of the halide ions. They are therefore called pseudohalide ions. Pseudohalide ions are univalent and these form salts resembling the halide salts. For example, the sodium salts are soluble in water, but the silver salts are insoluble. The hydrogen compounds are acids like the halogen acids HX . Some of the pseudohalide ions combine to form dimers comparable with the halogen molecules X_2 . These include cyanogens $(\text{CN})_2$, thiocyanogen $(\text{SCN})_2$ and selenocyanogen $(\text{SeCN})_2$ and selenocyanogen $(\text{SeCN})_2$.

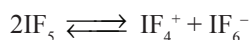
Anion	Acid	Dimer
CN^- cyanide ion	HCN hydrogen cyanide	$(\text{CN})_2$ Cyanogens
SCN^- thiocyanate ion	HSCN thiocyanic acid	$(\text{SCN})_2$ Thiocyanogen
SeCN^- selenocyanate ion		$(\text{SeCN})_2$ Selenocyanogen
OCN^- cyanate ion	HOCN cyanic acid	
NCN^{2-} cyanamide ion	H_2NCN cyanamide	
N_3^- azide ion	HN_3 hydrogen azide	

The best known pseudohalide is CN^- . This resembles Cl^- , Br^- and I^- in the following respects:

1. It forms an acid HCN.
2. It can be oxidized to form a molecule cyanogens $(\text{CN})_2$.
3. It forms insoluble salts with Ag^+ , Pb^{2+} and Hg^+ .
4. Interpseudohalogen compounds ClCN , BrCN and ICN can be formed.
5. AgCN is insoluble in water but soluble in ammonia, as is AgCl .
6. It forms a large number of complexes similar to halide complexes, e.g., $[\text{Cu}(\text{CN})_4]^{2-}$ and $[\text{CuCl}_4]^{2-}$ and $[\text{Co}(\text{CN})_6]^{3-}$ and $[\text{CoCl}_6]^{3-}$

Polyhalides (Interhalogen Ions)

Halogens or interhalogens combine with halide ions to form polyhalide ions (interhalogen ions). The polyhalides are known mostly as salts of strongly electropositive large cations, like the alkalis, heavy alkaline earths or large organic cations e.g., $[\text{N}(\text{CN})_4]^+$. As yet no anhydrous acid of the polyhalide ions has been prepared. As in the interhalogens, here too the heavier halogen plays the role of the central atom. Some examples are included in table. The AB_2^+ and AB_4^+ type polyhalogen cations arise out of the unusual dissociation of the interhalogens, AB_3 and AB_5 :

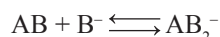


In interhalogens, we have all odd numbers of B atoms for a central A atom but in polyhalides for one A atom, we have all even numbers of B atoms. The even number of the B atoms in AB_n^+ or AB_n^- allows the polyhalide ions to attain all paired spin system.

AB_2^-	AB_4^-	AB_6^-	AB_8^-	AB_2^+	AB_4^+	ABC^-
I_3^-	ICl_4^-	IF_6^-	I_9^-	BrF_2^+	IF_4^+	IBrCl^-
ICl_2^-	I_5^-	I_7^-		ICl_2^+		
IBr_2^-	BrF_4^-					
BrCl_2^-						
Br_3^-						
I_2Cl^-						
I_2Br^-						
Br_2Cl^-						
Cl_3^-						

Ready formation of polyhalide is shown by the enhanced solubility of iodine in aqueous medium in the presence of iodide ions: $\text{I}_2 + \text{I}^- \rightleftharpoons \text{I}_3^-$. That the triiodide ion is made up of one I^- and one I_2 is shown by its reaction with two moles of $\text{Na}_2\text{S}_2\text{O}_3$: $2\text{Na}_2\text{S}_2\text{O}_3 + \text{KI}_3 \rightarrow \text{Na}_2\text{S}_4\text{O}_6 + 2\text{NaI} + \text{KI}$. The pentaiodide, heptaoidide and the enneaiodide will react with 4, 6 and 8 moles of sodium thiosulphate respectively.

The stability against dissociation increases with the size of the central atom and the attached groups. The stability constants, K, of the following equilibrium



have been determined for a number of polyhalides.

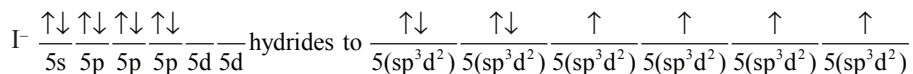
K=	Cl_3^-	Br_3^-	ICl_2^-	IBr_2^-	I_3^-
	0.01	17.8	167.0	370.0	725.0

Polyhalides are formed by the action of a halogen on a halide, or by the action of an interhalogen such as ICl , ICl_3 or IBr on a halide. The salts are simply made by crystallizing the halide salt in the presence of the appropriate halogen. $\text{KI}_3 \cdot \text{H}_2\text{O}$ forms black crystals.

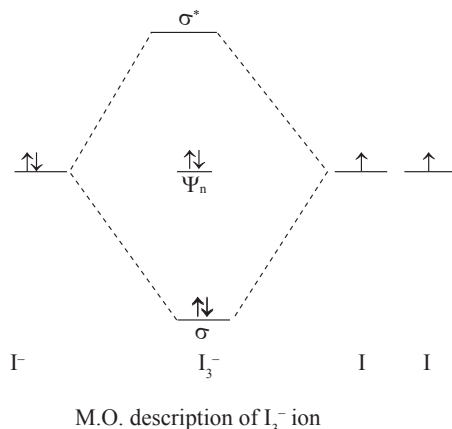
The two I-I distances in linear triiodide (e.g., In tetraphenylarsonium triiodide) is the same 2.9 Å (290 pm).

8.24 Group17-Halogens

The boiling in polyhalides follows the pattern of interhalogens. For example, in ICl_4^- the central iodine has a total of $7 + 1 + 4 = 12$ electrons around itself. Iodine is sp^3d^2 hybridized.



The four hybrid orbitals with unpaired spins now overlap with chlorine orbitals (with one electron on each of the four chlorine atoms) The fifth and the sixth hybrid orbitals carry the lone pairs. The bond angles Cl-I-Cl are 90° so that ICl_4^- ion may be said to have a square planar arrangement of the four chlorine atoms around the iodine. The two lone pairs will point above and below the square plane.

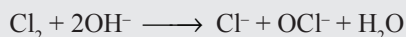
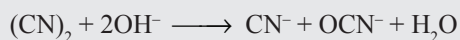


For the linear triiodide ion we may develop a simple M.O. picture. The central iodine is taken as I^- , with all the p orbitals having paired spins. One such p orbital will combine with one p orbital of a terminal iodine (with one unpaired spin) to generate one $\sigma_{\text{M.O.}}$ and one $\sigma^*_{\text{M.O.}}$. The other p-orbital of the second terminal iodine will remain non-bonding. The four electrons (two of I^- and one each of the two terminal iodines) will be housed a paired spins in the $\sigma_{\text{M.O.}}$ and the non-bonding (ψ_n) p-orbital. (Fig. 8.1). We may also assume linear sp hybridization for the central I^- . The two sp hybrid orbitals will have two paired spins (2 from the s orbital and $(1 + 1)$ for the p-orbital). One sp of the central iodine and one p of one terminal iodine may interact to generate one $\sigma_{\text{M.O.}}$ and one $\sigma^*_{\text{M.O.}}$. One p-orbital of a terminal iodine will remain non-bonding as indicated earlier.

Worked-out Example

76. $(\text{CN})_2$ is called pseudo halogens.

Ans. The pseudohalogens show a general similarity to the halogens in their reactions e.g.,



- ICl is a good conductor of a electricity in liquid state.
- Interhalogen compounds: Compounds made up of two or more different halogens are called inter halogen compounds. Example: ClF , ICl_3 , IF_5 , BrF_5 etc.
- All neutral interhalogen molecules and uninegative interhalogen ions are diamagnetic because of the absence of unpaired electron.
- Polyhalide ions: These ions contain more than two halogen atoms. These can be prepared by reacting interhalogens or halogen compounds with halide ions. Example, I_3^- , Cl_3^- , ICl_2^- , IF_5^- , I_7^- , IBr_2^- etc.
- $(\text{CN})_2$ is used as poisonous gas in war.

- Cationic iodine (Electropositive nature of Iodine). In the halogen family, the last element iodine has low ionization energy and exhibit some metallic character and has a tendency to lose electron to form cations I^+ and I^{3+} .
- Example: ICl , IBr , ICN contain I^+ , ICl_3 , $I(CH_3COO)_3$ contain I^{3+}

Occurrence and Preparation of Fluorine

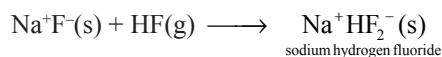
The most important naturally occurring compound of fluorine is fluorspar (calcium fluoride), $Ca^{2+}(F^-)_2$; cryolite, $(Na^+)_3AlF_6^{3-}$, is of importance in the extraction of aluminium.

The only possible method of preparing fluorine is by electrolysis and even then the chemicals used must be free from traces of water which react rapidly with the free halogen. The electrolyte used is hydrogen fluoride to which some potassium fluoride has been added to increase the electrical conductivity. One type of cell which operates at about $250^\circ C$ employs an equimolar mixture $HF-KF$; carbon is used as the anode and a steel vessel serves as the cathode. The overall process involves discharge of fluorine at the anode and hydrogen at the cathode; hydrogen fluoride is added to the cell periodically to maintain the correct composition of the mixture.

Fluorine discharged at the anode: $2F^- \longrightarrow F_2 + 2e^-$

Hydrogen discharged at the cathode: $2H^+ + 2e^- \longrightarrow H_2$

The fluorine is contaminated with hydrogen fluoride which is removed by passing it over sodium fluoride pellets:



Worked-out Examples

77. During preparation of fluorine by electrolysis anode and cathodes are separated by a diaphragm, why?

Ans. Because H_2 liberated at cathode will explosively react with F_2 liberated at anode, if the two electrodes are not separated by diaphragm.

78. During preparation of F_2 by electrolysis, graphite anode is not used why?

or

In the electrolysis of $HF + KHF_2$ mixture to produce F_2 , cathode is made by ungraphitized carbon, why?

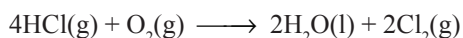
Ans. It is due to two reasons:

- F_2 liberated attacks on graphite to form graphite compounds CF_x .
- In graphite compounds fluorine atoms progressively invade the spaces between sheets of graphite forcing them apart. This gradually stops the graphite from conducting. Due to increased resistance. The more power is required and more heat is produced which may lead to explosion.

Chlorine, Bromine and Iodine

The main source of chlorine is sodium chloride, which occurs in sea water and in salt deposits, e.g., the Dead Sea and Stassfurt deposits and the rock salt deposits in Cheshire. Electrolysis of an aqueous solution of sodium chloride produces sodium hydroxide, chlorine and hydrogen for a description of the Castner-Kellner process and the Gibbs diaphragm cell process.

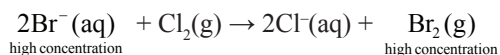
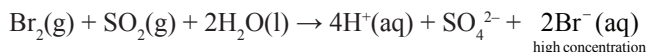
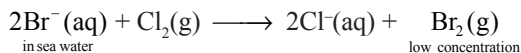
Large quantities of chlorine are used in the production of organic chlorides and often appreciable quantities of hydrogen chloride are formed in these process. This 'by-product' hydrogen chloride is catalytically oxidized back again to chlorine and this process is now of major industrial importance; the catalyst used is copper mixed with rare earth chlorides and supported on silica.



Bromides, e.g., those of potassium, sodium and magnesium, occur in sea water and in salt deposits, but they are not nearly so abundant as chlorides.

8.26 Group 17-Halogens

Most of the world's supply of bromine is obtained from sea water (bromide content about 1 part in 15000). The sea water is first acidified (to prevent appreciable hydrolysis of chlorine and bromine) and then treated with chlorine. The bromide ions are oxidized to bromine which is then expelled as a vapour by passing air through the liquid mixture. The air, containing a very small concentration of bromine, is mixed with sulphur dioxide and then water is added; the bromine is reduced back again to bromide ions and the air recycled to pick up more bromine, which is reduced to bromide and added to the aqueous solution. By this means, a solution containing a high concentration of bromide ions is obtained. Treatment of this solution with chlorine results in the displacement of bromine in high concentration, which can be distilled off on heating the solution. Dilute sulphuric acid is produced in the process and is used to treat the sea water prior to addition of chlorine.

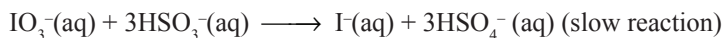


Iodides occur only in minute concentrations in sea water, but certain species of seaweed can use these minute amounts and concentrate the iodine in themselves as organic compounds. This source, together with the brine wells in California, supply some of the world's iodine, but the main source of this element is sodium iodate (V), $\text{Na}^{+}\text{IO}_3^{-}$, which occurs in the sodium nitrate deposits in Chile.

The mother liquor containing sodium iodate (V), i.e., the liquid remaining after crystallization of the sodium nitrate, is treated with sodium hydrogen sulphite, when the iodate (V) is reduced to iodine:



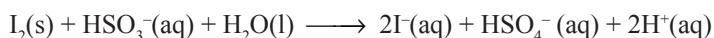
The reaction involves three stages; first of all the iodate (V) is slowly reduced to iodide:



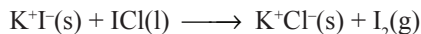
The iodide then rapidly reacts with unchanged iodate (V) in the presence of hydrogen ions:



A reaction occurs between the iodine and unchanged hydrogen sulphite and iodine is only precipitated once all the hydrogen sulphite has been consumed:

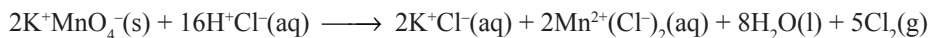
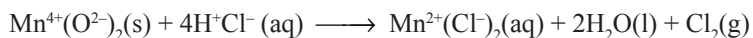


Crude iodine may be contaminated with iodine monochloride, ICl and iodine monobromide, IBr . It is purified by sublimation in the presence of potassium iodide which removes these impurities:



Laboratory preparation of chlorine, bromine and iodine

Chlorine can be obtained in the laboratory by oxidizing concentrated hydrochloric acid with either manganese (IV) oxide (heat required) or potassium manganate (VII) (no heat required):

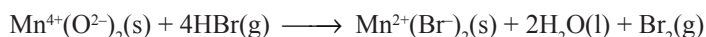
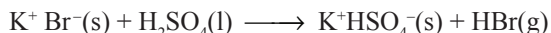


Bleaching powder also evolves chlorine when treated with dilute acids.

Chlorine from any of these sources can be dried by passing it through concentrated sulphuric acid and collected by downward delivery. When it is obtained by oxidation of concentrated hydrochloric acid it is passed through water to dissolve any hydrogen chloride fumes before being dried.

Hydrobromic and hydriodic acids, the bromine and iodine equivalents of hydrochloric acid, are unstable in the presence of air and are not often used as laboratory reagents; therefore the laboratory preparations of bromine and iodine, although similar, are not quite analogous to the methods used for obtaining chlorine.

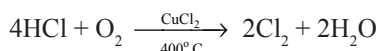
Bromine is obtained by heating a mixture of potassium bromide, manganese (IV) oxide and concentrated sulphuric acid. The reaction is carried out in a 'Quickfit' distillation apparatus and the bromine vapour is condensed in a receiver cooled under running water. The reaction takes place in two stages:



Iodine is obtained by a similar reaction using a mixture of potassium iodide, manganese (IV) oxide and concentrated sulphuric acid.

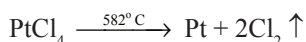
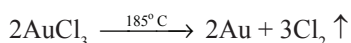
Note:

1. Fluorine was first isolated by Moissan in 1886. It can be prepared by electrolysis of a mixture of KHF_2 and anhydrous H_2SO_4 using an apparatus made up of Monel metal (Cu , Ni , Fe). This is because reaction of F_2 on monel metal stops after the initial formation of a layer of fluoride. This layer of fluoride protects the remaining metal from the action of F_2 . Two methods i.e.,
Dennis and Whytlaw—Gray methods are used.
2. Cl_2 is also called oxymuriatic acid and was discovered by Scheele in 1774.
3. Cl_2 is prepared by Deacon's process,



Pure chlorine is obtained by heating dry PtCl_4 or AuCl_3 in a hard glass tube.

4. Deep sea weeds (Laminaria species) are main source of iodine. Their ashes (known as kelp) contain 0.5% of iodine in the form of iodides. Caliche or crude Chile salt petre also contains 0.2% of NaIO_3 (sodium iodate).
5. Pure Cl_2 can be obtained by heating AuCl_3 or PtCl_4



Worked-out Examples

79. (i) Bromine is manufactured by the action of _____ on the bromide present in sea water.
(ii) Bromine reacts with moist sulphur dioxide to give _____.
(iii) Iodine reacts with potassium chlorate in the presence of nitric acid to produce _____ and chlorine.

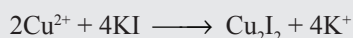
Ans. (i) Chlorine (ii) Sulphuric acid (iii) iodate

80. Bromine can be liberated from potassium bromide solution by action of
(a) iodine solution (b) chlorine solution (c) sodium chloride (d) potassium iodide

Ans. (b)

81. Iodine is liberated in the reaction between KI and Cu^{2+} but chlorine is not liberated when KCl is added to Cu^{2+} . Explain.

Ans. The I^- ion acts as strong reducing agent where, Cl^- acts as reductant only in presence of strong oxidant and therefore, KI reduces Cu^{2+} to Cu^+ .

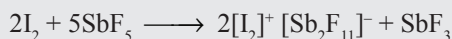


82. The brown colour of an acidified dilute solution of iodine is aqueous, KI is intensified by the addition of nitrite but it is discharged by the addition of a sulphite. Explain.

Ans. Nitrite oxidizes iodide ion to iodine and thus liberated iodine gets dissolved in KI solution to intensify its colour. On the other hand, sulphites are themselves oxidized by I_2 of solution and thus, reducing I_2 to discharge the colour of solution.

83. When I_2 is dissolved in strong oxidizing agents like oleum or SbF_5 etc., a blue coloured compound is formed which is paramagnetic in nature, why?

Ans. In presence of strong oxidizing agent, I_2 shows basic behaviour and is oxidized to I_2^+ which is paramagnetic.

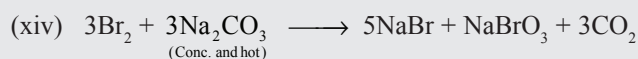
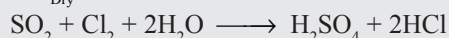
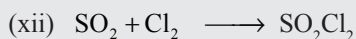
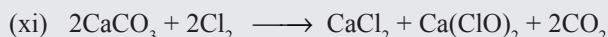
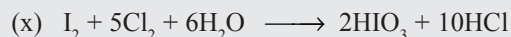
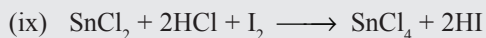
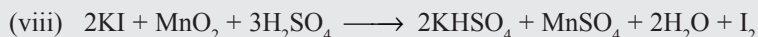
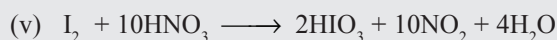
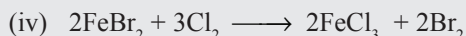
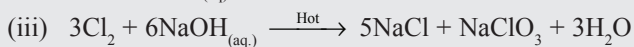
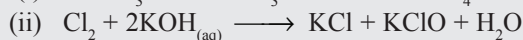


84. $FeCl_3$ is stable but FeI_3 is unstable.

Ans. I^- is strong reductant and Fe^{3+} is oxidant and thus I^- reduces Fe^{3+} to Fe^{2+} ; on the other hand Cl^- is weak reductant.

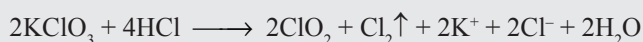
85. What happens when?

- (i) Sodium iodate is treated with sodium bisulphate solution.
- (ii) Chlorine is passed into aqueous potassium hydroxide.
- (iii) Chlorine is passed through hot NaOH solution.
- (iv) Chlorine gas is bubbled through a solution of ferrous bromide.
- (v) Iodine reacts with concentrated HNO_3 .
- (vi) Chlorine is passed over slaked lime.
- (vii) Chlorine reacts with Na_2SO_3 solution.
- (viii) Potassium iodide is heated with MnO_2 and concentrated H_2SO_4 .
- (ix) Iodine is added to stannous chloride solution.
- (x) Chlorine is passed through a suspension of $CaCO_3$.
- (xi) Cl_2 is passed through a suspension of $CaCO_3$.
- (xii) Chlorine gas is passed through dry and aqueous sulphur dioxide.
- (xiii) Potassium iodide is added to bleaching powder containing dilute acetic acid.
- (xiv) Bromine reacts with Na_2CO_3 solution.



86. Explain, what is 'Euchlorine'?

Ans. All the chlorates are decomposed by concentrated HCl, ClO_2 , thus formed impart yellow colour to acid solution. The mixture of various gases thus evolved is known as **Euchlorine**.



Some Noteworthy Points

1. All the elements are diatomic and molecular and the boiling and melting points increase as a result of the increasing van der Waals interactions between the diatomic molecules for the heavier elements.
2. The elements are typical non-metals in their physical and chemical properties. They form anionic compounds based on X^- (X = Halogen) which is associated with the completed octet.

The ionic compounds MX become progressively less ionic as the relative atomic mass of X increases, because of the decreasing electronegativity of the halogens. Iodine has the greatest tendency to form cationic species, e.g., I_2^+ , I_5^+ , because it has the lowest ionization energy. The cation Br_2^+ is known in $Br_2^+Sb_3F_{16}^-$ and Br_5^+ has been reported.

3. The atoms also form strong covalent bonds with other non-metals. The mean bond enthalpies for $E-X$ bonds are particularly large for fluorine and therefore a wide range of molecular fluorides is known and fluorine is particularly effective at bringing out the highest valencies of the non-metals and highest oxidation states of the metals.
4. The oxidizing ability of the halogens decreases markedly down the Group; $F_2 > Cl_2 > Br_2 > I_2$ and only iodine is oxidized by nitric acid.
5. The stabilities of the hydrogen halides decreases down the Group, but their acid strengths increase.
6. Only $H-F$ forms strong hydrogen bonds and this is reflected in the boiling and melting points of the hydrogen halides.
7. The halogens form many interhalogen compounds with the less electronegative halogen surrounded by the more electronegative halogens. Neutral, anionic and cationic interhalogen compounds are known. ICl and IBr are widely used in organic synthesis and are commercially available.

The most extensive series of compounds exists for iodine, e.g., IF_7 , IF_5 , ICl_4^- , ICl_2^- . Fluorine does not form any interhalogen compounds where it occupies the central position within the molecule.

8. Oxygen fluorides are extremely strong and reactive oxidants and have been explored as potential rocket fuels, the oxides become less reactive down the column and more numerous. Iodine forms a particularly wide range of oxides.
9. The perchlorates, EO_4^- , are only known for Cl , Br and I . They exhibit an alternation in their oxidizing abilities and the perbromates are particularly strong oxidizing agents.
10. In the highest oxidation state (+7) the relative oxidizing ability is:



and results in the formation of the corresponding +5 oxoanions.

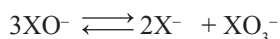
	E°/V
$ClO_4^- \rightleftharpoons ClO_3^-$	1.20
$BrO_4^- \rightleftharpoons BrO_3^-$	1.85
$IO_4^- \rightleftharpoons IO_3^-$	1.63

In common with the examples discussed above the oxoanions are less strongly oxidizing in basic solution, at $pH = 1$, for ClO_4^-/ClO_3^- $E^\circ = 0.37$ V and even then readily disproportionate to $HOCl$ and $HOClO$.

The +3 oxidation state is only significant for chlorine and the +1 oxidation state is well established for Cl , Br and I . Solutions of alkali metal hypochlorites and hypobromites are prepared by passing Cl_2 or Br_2 respectively into cold solutions of the Group 1 hydroxide. They are strong oxidizing agents:

	$E^\circ/V, a_{OH^-} = 1$
$ClO^- \rightleftharpoons Cl^-$	0.89
$BrO^- \rightleftharpoons Br^-$	0.76
$IO^- \rightleftharpoons I^-$	0.48

The hypohalite ions disproportionate according to the equation:



The equilibrium constants are 10^{27} for ClO^-/Cl^- (the reaction is slow at room temperature), 10^{15} for BrO^-/Br^- and 10^{20} for IO^-/I^- . HOI has been prepared from ice + F_2 but is very reactive, decomposing to $HF + O_2$.

ONE CORRECT OPTION TYPE QUESTIONS

- The most powerful oxidizing agent is:
 - fluorine
 - chlorine
 - bromine
 - iodine
- Which of the following elements does not show positive oxidation state?
 - fluorine
 - chlorine
 - oxygen
 - sulphur
- Fluorine reacts with water to give:
 - hydrogen fluoride and oxygen
 - hydrogen fluoride and ozone
 - hydrogen fluoride and oxygen fluoride
 - hydrogen fluoride, oxygen and ozone
- Super halogen is:
 - F_2
 - Cl_2
 - Br_2
 - I_2
- The great reactivity of fluorine is due to:
 - Low energy of the F—F bond
 - Small size
 - High heat of hydration
 - All of these
- Chlorine acts as a bleaching agent only in presence of
 - dry air
 - moisture
 - sunlight
 - pure oxygen
- When chlorine is passed through concentrated hot solution of KOH, the compound formed is:
 - KClO
 - KClO₂
 - KClO₃
 - KClO₄
- Chlorine is prepared in the laboratory by the action of:
 - Zn and HCl
 - CaCO₃ and HCl
 - Pb(NO₃)₂ and HCl
 - MnO₂ and HCl
- Chlorine cannot be prepared by the action of HCl on:
 - MnO₂
 - KMnO₄
 - NaCl
 - K₂Cr₂O₇
- Pure chlorine is obtained:
 - by heating PtCl₄
 - by heating MnO₂ with HCl
 - by treating bleaching powder with HCl
 - by heating mixture of NaCl and MnO₂ with concentrated H₂SO₄
- Chlorine gas is dried over:
 - CaO
 - NaOH
 - H₂SO₄
 - HBr
- The catalyst used in the Deacon's process for the manufacture of chlorine is:
 - Cu
 - An alloy of copper
 - CuCl₂
 - CuS
- Which one of the following reacts with chlorine to form phosgene?
 - SO₂
 - CO₂
 - NO
 - CO
- Chlorine is mixed with drinking water so that:
 - bacteria are killed
 - dirt is removed
 - water is cleaned
 - suspension is removed
- Which of the following possesses the highest bond energy?
 - F₂
 - Cl₂
 - Br₂
 - I₂
- Which of the following halogen disproportionates in water?
 - F₂
 - Cl₂
 - I₂
 - All three
- Predict the correct product when Cl₂ passed through $H-\overset{18}{O}-\overset{18}{O}-H$ solution.
 - $H^+ + Cl^- + O_2$ (both oxygen having 18)
 - HOCl and HClO₂ (all oxygen having 18)
 - HClO₄ and HCl (all oxygen having 18)
 - Cl₂O and H₂O (all oxygen having 18)
- $$Cl_2(g) + Ba(OH)_2 \longrightarrow X(aq.) + BaCl_2 + H_2O$$

$$X + H_2SO_4 \longrightarrow Y + BaSO_4$$

$$Y \xrightarrow[\Delta > 365\text{ K}]{\Delta} Z + H_2O + O_2$$

Y and Z are respectively:

 - HClO₄, ClO₂
 - HClO₃, ClO₂
 - HClO₃, ClO₆
 - HClO₄, Cl₂O₇
- A solution of KBr is treated with each of the following. Which one would liberate bromine.
 - Cl₂
 - HI
 - I₂
 - SO₂

20. Bromine occurs as a bromide in a mineral called:
 (a) nitre (b) tincal
 (c) common salt (d) carnallite
21. Bromine can be liberated from potassium bromide solution by the action of:
 (a) iodine solution
 (b) chlorine water
 (c) sodium chloride
 (d) potassium iodide
22. The solubility of iodine in H_2O may be increased by the addition of:
 (a) $\text{Na}_2\text{S}_2\text{O}_3$ (b) CHCl_3
 (c) KI (d) CS_2
23. The colour of iodine solution is discharged by shaking with:
 (a) aqueous sulphur dioxide
 (b) sodium sulphide
 (c) sodium sulphate
 (d) sodium chloride
24. Iodine vapours are violet in colour because:
 (a) the molecules of iodine in vapour phase absorb green and yellow radiations.
 (b) the molecules of iodine absorb red and violet radiations and then emit them.
 (c) it is a solid.
 (d) it is easily volatile.
25. Iodine may be liberated from sodium iodate by reacting with:
 (a) H_2SO_4 (b) NaHSO_3
 (c) KMnO_4 (d) HCl
26. Concentrated HNO_3 reacts with iodine to give:
 (a) HI (b) HOI
 (c) HOIO_2 (d) HOIO_3
27. When iodine is dissolved in CCl_4 , the colour that results is:
 (a) brown (b) bluish green
 (c) violet (d) colourless
28. When thiosulphate ion is oxidized by iodine, the new product formed is:
 (a) SO_3^- (b) SO_4^-
 (c) $\text{S}_4\text{O}_6^{2-}$ (d) $\text{S}_2\text{O}_6^{2-}$
29. The X—X bond dissociation energy is minimum in:
 (a) F_2 (b) Cl_2
 (c) Br_2 (d) I_2
30. Which halogen can be purified by sublimation?
 (a) F_2 (b) Cl_2
 (c) Br_2 (d) I_2
31. HBr and HI reduce sulphuric acid, HCl can reduce KMnO_4 and HF can reduce
 (a) H_2SO_4
 (b) KMnO_4
 (c) $\text{K}_2\text{Cr}_2\text{O}_7$
 (d) none of the above
32. H_2SO_4 cannot be used to prepare HBr from NaBr as it:
 (a) reacts slowly with NaBr
 (b) oxidizes HBr
 (c) reduces HBr
 (d) disproportionates HBr
33. Which one of the hydrides does not form any precipitate with AgNO_3 ?
 (a) HF (b) HCl
 (c) HBr (d) HI
34. Which one of the following is strongest acid?
 (a) HF (b) HI
 (c) HBr (d) HCl
35. Which one of the following is most volatile acid?
 (a) HF (b) HCl
 (c) HBr (d) HI
36. The acid which cannot be kept in glass bottles is:
 (a) HI (b) HBr
 (c) HF (d) HCl
37. Which one of the following has hydrogen bonding?
 (a) HI (b) HBr
 (c) HF (d) HCl
38. Hydrogen fluoride is a liquid unlike other hydrogen halides, because:
 (a) H—F bond is strong
 (b) F-atom is small in size
 (c) Hydrogen bonding is present
 (d) HF is a weak acid
39. Which amongst the following reactions cannot be used for the respective preparation?
 (a) $2\text{KBr} + \text{H}_2\text{SO}_4(\text{concentrated}) \longrightarrow \text{K}_2\text{SO}_4 + 2\text{HBr}$
 (b) $\text{NaCl} + \text{H}_2\text{SO}_4(\text{concentrated}) \longrightarrow \text{NaHSO}_4 + \text{HCl}$

8.32 Group17-Halogens

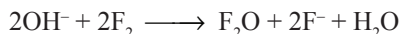
- (c) $\text{NaHSO}_4 + \text{NaCl} \longrightarrow \text{Na}_2\text{SO}_4 + \text{HCl}$
 (d) $\text{CaF}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{CaSO}_4 + 2\text{HF}$
40. Order of boiling point is:
 (a) $\text{HF} > \text{HI} > \text{HBr} > \text{HCl}$
 (b) $\text{HF} > \text{HBr} > \text{HI} > \text{HCl}$
 (c) $\text{HCl} > \text{HBr} > \text{HI} > \text{HF}$
 (d) $\text{HCl} > \text{HI} > \text{HBr} > \text{HF}$
41. The correct order of thermal stability of hydrogen halides ($\text{H}-\text{X}$) is:
 (a) $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$
 (b) $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$
 (c) $\text{HCl} > \text{HF} > \text{HBr} > \text{HI}$
 (d) $\text{HI} > \text{HCl} > \text{HF} > \text{HBr}$
42. Which of the following has highest bond strength?
 (a) HI (b) HCl
 (c) HF (d) HBr
43. Which one of the following is the strongest acid?
 (a) HClO (b) HClO_2
 (c) HClO_3 (d) HClO_4
44. Which one of the following is the strongest oxidizing agent?
 (a) HClO (b) HClO_2
 (c) HClO_3 (d) HClO_4
45. The following acids have been arranged in order of decreasing acid strength. Identify the correct order:
 I. ClOH
 II. BrOH
 III. IOH
 (a) $\text{I} > \text{II} > \text{III}$ (b) $\text{II} > \text{I} > \text{III}$
 (c) $\text{III} > \text{II} > \text{I}$ (d) $\text{I} > \text{III} > \text{II}$
46. The shape of O_2F_2 is similar to:
 (a) H_2O_2 (b) C_2F_2
 (c) H_2F_2 (d) C_2H_2
47. Which of the following statements is correct?
 (a) All halogens form oxyacids
 (b) Only chlorine and bromine form oxyacids
 (c) All halogens except fluorine form oxyacids
 (d) Only iodine forms oxyacid
48. The set with correct order of acidity is;
 (a) $\text{HClO} < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}_4$
 (b) $\text{HClO}_4 < \text{HClO}_3 < \text{HClO}_2 < \text{HClO}$
 (c) $\text{HClO} < \text{HClO}_4 < \text{HClO}_3 < \text{HClO}_2$
 (d) $\text{HClO}_4 < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}$
49. Slaked lime reacts with chlorine to form:
 (a) $\text{Ca}(\text{OCl})_2$ (b) $\text{Ca}(\text{OCl})\text{Cl}$
 (c) $\text{Ca}(\text{ClO}_3)_2$ (d) CaCl_2
50. Bleaching properties of bleaching powder are due to its:
 (a) oxidizing properties
 (b) reducing properties
 (c) basic properties
 (d) disinfecting properties
51. When CO_2 is passed through an aqueous solution of bleaching powder, the gas evolved is:
 (a) chlorine
 (b) oxygen
 (c) ozone
 (d) chlorine dioxide
52. Aqueous solution of bleaching powder gives a test for:
 (a) only Ca^{2+} ions
 (b) only OCl^- ions
 (c) only Cl^- ions
 (d) all Cl^- , OCl^- and Ca^{2+} ions
53. In a given sample of bleaching powder the percentage of available chlorine is 49. The volume of chlorine obtained if 10 g of the sample is treated with HCl at NTP is:
 (a) 1.5 litre (b) 3.0 litre
 (c) 15.0 litre (d) 150 litre
54. The interhalogen which does not exist is:
 (a) IF_5 (b) ClF_3
 (c) BrCl (d) ICl_4
55. Only iodine forms hepta-fluoride IF_7 , but chlorine and bromine give penta-fluorides. The reason for this is:
 (a) low electron affinity of iodine.
 (b) unusual pentagonal bipyramidal structure of IF_7 .
 (c) that the larger iodine atom can accommodate more number of smaller fluorine atom around it.
 (d) low chemical reactivity of IF_7 .
56. Which of the following inter halogen compounds does not exist?
 (a) ClF_3 (b) BrCl_7
 (c) ICl (d) IF_5
57. The T-shaped interhalogen compound is:
 (a) ClF_5 (b) IF_5
 (c) ICl (d) ClF_3

58. Charge distribution in iodine monochloride is best represented as:
 (a) $\text{I}^+ \text{Cl}^-$ (b) $\text{I}^+ \text{Cl}^-$
 (c) $\text{I}^{\delta-} \text{Cl}^{\delta+}$ (d) $\text{I}^{\delta+} + \text{Cl}^{\delta-}$
59. The shape of IF_7 molecule is:
 (a) pentagonal bipyramidal
 (b) triangular bipyramidal
 (c) tetrahedral
 (d) octahedral
60. Halogens show usually:
 (a) electrovalency
 (b) covalency
 (c) both electrovalency and covalency
 (d) coordinate valency
61. The strongest reducing agent is:
 (a) F^- (b) Cl^-
 (c) Br^- (d) I^-
62. Which one of the following acts as an anti-chlor?
 (a) MnO_2 (b) $\text{Na}_2\text{S}_2\text{O}_3$
 (c) $\text{K}_2\text{Cr}_2\text{O}_7$ (d) Na_2SO_4
63. Which statement is correct about halogen?
 (a) They are all diatomic and form univalent ions
 (b) They are all capable of exhibiting several oxidation states
 (c) They are all diatomic and form diatomic ions
 (d) They are all reducing agents
64. Glass is soluble in:
 (a) Aqua-regia (b) HClO_4
 (c) H_2SO_4 (d) HF
65. Which of the following arrangements for the three halogens Cl, Br, I when placed in order of their increasing electron affinity is correct?
 (a) Cl, Br, I (b) I, Br, Cl
 (c) Br, Cl, I (d) I, Cl, Br
66. Oxidizing action increases in the following order:
 (a) $\text{Cl} < \text{Br} < \text{I} < \text{F}$
 (b) $\text{Cl} < \text{I} < \text{Br} < \text{F}$
 (c) $\text{I} < \text{F} < \text{Cl} < \text{Br}$
 (d) $\text{I} < \text{Br} < \text{Cl} < \text{F}$
67. Which of the following is not oxidized by MnO_2 ?
 (a) F^- (b) Cl^-
 (c) Br^- (d) I^-
 [Hint: The fluoride ion; F^- is extremely stable. Hence, it is not oxidized by chemical reagents.]
68. Which of the following pairs is not correctly matched?
 (a) A halogen which is liquid at room temperature—*Bromine*
 (b) The most electronegative element—*Fluorine*
 (c) The most reactive halogen—*Fluorine*
 (d) The strongest oxidising agent—*Iodine*
69. Astatine is the element below iodine in the group VIIA of the periodic table. Which of the following statements is not true for astatine?
 (a) It is less electronegative than iodine
 (b) It will exhibit only -1 oxidation state
 (c) Intermolecular forces between the astatine molecules will be larger than between iodine molecules
 (d) It is composed of diatomic molecules
70. Which of the following is a pseudohalogen?
 (a) IF_7 (b) $(\text{CN})_2$
 (c) ICl_2 (d) I_3^-
71. Which of the following is correct statement?
 (a) F_2 has higher dissociation energy than Cl_2
 (b) F has higher electron affinity than Cl
 (c) HF is stronger acid than HCl
 (d) Boiling point increases down the group in halogens.
72. In the oxoacids of chlorine Cl—O bond contains:
 (a) $\text{d}\pi\text{-d}\pi$ bonding
 (b) $\text{p}\pi\text{-d}\pi$ bonding
 (c) $\text{p}\pi\text{-p}\pi$ bonding
 (d) None of these
73. Maximum covalency of iodine in its interhalogen compounds is shown with:
 (a) Fluorine (b) Chlorine
 (c) Bromine (d) Iodine
74. The cyanide ion is referred as pseudo halide ion because:
 (a) It can be produced by reduction of an uncharged dimer C_2N_2 like halogen molecule
 (b) Cyanogen disproportionates to give CN^- and CNO^- like halogens disproportionation
 (c) CN^- is reasonably reducing agent like Cl^-
 (d) The solubility behaviour is very similar to halide ions
75. Euchlorine is a mixture of:
 (a) Cl_2 and SO_2 (b) Cl_2 and ClO_2
 (c) Cl_2 and CO (d) None of these

COMPREHENSION BASED QUESTIONS

COMPREHENSION-I

F_2O gas can be prepared by action of on cold dilute NaOH.



It is covalent and analogous to H_2S in structure. It is strong oxidizing agent. It gives fluorides and oxides with metals S, P and halogens.

- $F-O-F$ bond angle in F_2O is:
 - $109^\circ 28'$
 - 180°
 - $< 109^\circ 28'$
 - $> 109^\circ 28'$
- In the preparation of F_2O by action of NaOH on F_2 gas, the atom which is oxidized is:
 - H
 - F
 - O
 - Both F and O
- How many electrons are transferred from oxygen atoms to fluorine atoms in the preparation of 1 mole F_2O ?
 - 2
 - 2 mole
 - 4
 - 4 mole
- The only halogen atom which show -1 oxidation state in its oxides is:
 - F
 - Cl
 - Br
 - I
- The correct order of stability of halogen oxides is:
 - $I > Br > Cl$
 - $I < Br < Cl$
 - $I < Cl < Br$
 - $I > Br < Cl$

COMPREHENSION-II

Pseudo halides are anions having resemblance with halide ions. Group I metals can form salts with pseudo halides.

Pseudo halogens can act as ligands and form co-ordinate complexes. Their hydrides are weakly acidic and can be prepared in analogous way as halogen hydrides are prepared. Azides, cyanides, selenocyanides are examples of pseudo halogens.

- CN^- is pseudo halide, when cyanogens are heated with alkali, the products are:
 - HCN, H_2O
 - NH_3 , $(NH_4)_2CO_3$
 - NaCN, NaOCN
 - $HCOONa$, NH_3
- When sodium cyanide reacts with H_2SO_4 . The products are:
 - HCN and Na_2SO_4
 - HCN and $NaHSO_4$
 - $(CN)_2$, Na_2SO_4 , H_2O
 - None of the above
- When sodium pseudohalide salts are dissolved in water it resembles with:
 - NaCl
 - NaBr
 - NaF
 - NaI
- $(CN)_2$ when react with Na metal, the product is:
 - NaCN
 - Na_2C , N_2
 - NaN_3 , carbon black
 - $[Na(CN)_2]^-$
- When sodium azide reacts with concentrated sulphuric acid, the product is:
 - $(NH_4)_2SO_4$, Na_2SO_4
 - NH_3 , Na_2SO_4
 - N_3H , $NaHSO_4$
 - None of these

ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

- Identify the correct statements:
 - Fluorine is a super halogen
 - Iodine shows basic nature
 - AgF is insoluble in water
 - SCN^- is a pseudohalide
- Which of the following statement(s) is/are correct?
 - Chlorine dioxide (ClO_2) is powerful oxidizing agent but bleaching action is lower than Cl_2
 - ClO_2 in alkaline solution undergoes disproportionation
 - ClO_2 is diamagnetic in nature
 - ClO_2 is a yellow gas but deep red liquid
- $Cl_2O_6 + NaOH \longrightarrow ?$
 - NaClO₄
 - NaOCl
 - NaClO₂
 - NaClO₃
- Predict product(s) in the following reaction, $Cl_2 + OH^- \xrightarrow{\text{hot}} ?$
 - Cl^-
 - ClO_2
 - OCl^-
 - ClO_3^-

5. In the isolation of fluorine a number of difficulties were encountered. Which statements are correct?
- Fluorine reacts with moist glass vessels
 - Fluorine has great affinity for hydrogen
 - Electrolysis of aqueous HF gives ozonized oxygen
 - The potential required for the discharge of the fluoride ions lowest
6. Available chlorine is liberated from bleaching powder when it:
- is heated
 - reacts with acid
 - reacts with H_2O
 - reacts with CO_2

ASSERTION/REASON TYPE QUESTIONS

The following questions consist of two statements as Assertion(A) and Reason(R). While answering these questions, choose correctly, any of the following responses.

- If both assertion and reason are correct and reason is the correct explanation of the assertion.
- If both assertion and reason are correct, but reason is not the correct explanation of the assertion.
- If assertion is correct but reason is incorrect.
- If assertion is incorrect but reason is correct.

1. **Assertion:** F atom has less electron affinity than Cl atom.

Reason: Additional electrons are repelled more effectively by 3p electrons in Cl atom than by 2p electrons in F atom.

2. **Assertion:** Concentrated H_2SO_4 cannot be used to prepare pure HBr from NaBr.

Reason: It reacts slowly with NaBr.

3. **Assertion:** Liquid IF_5 conducts electricity.

Reason: Liquid IF_5 conducts as, $2\text{IF}_5 \rightleftharpoons \text{IF}_4^+ + \text{IF}_6^-$

4. **Assertion:** Chlorine gas disproportionates in hot and concentrated NaOH solution.

Reason: NaCl and NaOCl are formed in the above reaction.

5. **Assertion:** HF forms two series of salts but HCl does not.

Reason: F atom is more electronegative than Cl atom.

6. **Assertion:** FeCl_3 is stable but FeI_3 is unstable.

Reason: Fe^{3+} is strong oxidant and I^- is strong reductant.

7. **Assertion:** HF is corrosive, etches glass and also causes painful skin wounds.

Reason: HF reacts with silica and dehydrates tissue as well as acidic action of HF on tissue give rise to painful wounds.

8. **Assertion:** Iodine chloride acts as chlorinating as well as iodinating agent.

Reason: The nature of ICl to act as chlorinating ($\text{ICl}_{\text{vapour}}$) or iodinating (ICl in nitrobenzene) agent depends upon the conditions.

MATCH THE COLUMN TYPE QUESTIONS

1.

Column-I

- Maximum solubility in water
- Corrosive liquid

Column-II

- F_2
- Cl_2

Column-I

- Maximum intermolecular distance
- Enthalpy of dissociation (maximum)

Column-II

- Br_2
- I_2

RECENT YEARS' QUESTIONS OF IIT-JEE AND AIEEE

- Concentrated hydrochloric acid when kept in open air sometimes produces a cloud of white fumes. This is due to: [AIEEE 2003]
 - strong affinity of HCl gas for moisture in air results in forming of droplets of liquid solution which appears like a cloudy smoke
 - strong affinity for water, concentrated HCl pulls moisture of air towards self. The moisture forms droplets of water and hence the cloud.
 - concentrated HCl emits strongly smelling HCl gas all the time
 - oxygen in air reacts with emitted HCl gas to form a cloud of Cl_2 gas.
- The correct order of the thermal stability of hydrogen halide ($\text{H}-\text{X}$) is: [AIEEE 2005]
 - $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$
 - $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$
 - $\text{HCl} < \text{HF} < \text{HBr} < \text{HI}$
 - $\text{HI} > \text{HCl} < \text{HF} < \text{HBr}$
- Which products are expected from the disproportionation of hypochlorous acid: [AIEEE 2006]
 - HClO_3 and Cl_2O
 - HClO_2 and HClO_4
 - HCl and Cl_2O
 - HCl and HClO_3
- The correct order of acidic nature is: [IIT 2000]
 - $\text{CO}_2 > \text{N}_2\text{O}_5 > \text{SO}_3$
 - $\text{Cl}_2\text{O}_7 > \text{SO}_2 > \text{P}_4\text{O}_{10}$
 - $\text{K}_2\text{O} > \text{CaO} > \text{MgO}$
 - $\text{Na}_2\text{O} > \text{MgO} > \text{Al}_2\text{O}_3$
- A gas X is passed through water to form a saturated solution. The aqueous solution on treatment with silver nitrate gives a white precipitate. The saturated aqueous solution also dissolves magnesium ribbon with evolution of colour less gas Y. 'X' and 'Y' are respectively: [IIT 2002]
 - CO_2, Cl_2
 - Cl_2, CO_2
 - Cl_2, H_2
 - H_2, Cl_2
- The product of oxidation of I^- with MnO_4^- in alkaline medium is: [IIT 2004]
 - IO_3^-
 - I_2
 - IO^-
 - IO_4^-

ANSWER KEYS

ONE CORRECT OPTION TYPE QUESTIONS

- | | | | | | | | | | |
|---------|---------|---------|------------------|---------|---------|---------|---------|---------|---------|
| 1. (a) | 2. (a) | 3. (d) | 4. (a) | 5. (d) | 6. (b) | 7. (c) | 8. (d) | 9. (c) | 10. (a) |
| 11. (c) | 12. (c) | 13. (d) | 14. (a) | 15. (b) | 16. (b) | 17. (a) | 18. (b) | 19. (a) | 20. (d) |
| 21. (b) | 22. (c) | 23. (a) | 24. (a) | 25. (b) | 26. (c) | 27. (c) | 28. (c) | 29. (d) | 30. (d) |
| 31. (d) | 32. (b) | 33. (a) | 34. (b) | 35. (b) | 36. (c) | 37. (c) | 38. (c) | 39. (a) | 40. (a) |
| 41. (b) | 42. (c) | 43. (d) | 44. (a) | 45. (a) | 46. (a) | 47. (c) | 48. (a) | 49. (b) | 50. (a) |
| 51. (a) | 52. (d) | 53. (b) | 54. (d) | 55. (c) | 56. (b) | 57. (d) | 58. (d) | 59. (a) | 60. (c) |
| 61. (d) | 62. (b) | 63. (a) | 64. (d) | 65. (b) | 66. (d) | 67. (a) | 68. (d) | 69. (b) | 70. (b) |
| 71. (d) | 72. (b) | 73. (a) | 74. (a, b, c, d) | | 75. (b) | | | | |

COMPREHENSION BASED QUESTIONS

Comprehension-I

1. (c) 2. (c) 3. (d) 4. (a) 5. (d)

Comprehension-II

1. (c) 2. (a) 3. (c) 4. (a) 5. (c)

ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

1. (a, b, d) 2. (b, d) 3. (a, d) 4. (a, d) 5. (a, b, c) 6. (b, c, d)

ASSERTION/REASON TYPE QUESTIONS

1. (c) 2. (c) 3. (a) 4. (c) 5. (b) 6. (c) 7. (c) 8. (c)

MATCH THE COLUMN TYPE QUESTIONS

1. $\text{A} \rightarrow \text{P}$; $\text{B} \rightarrow \text{R}$; $\text{C} \rightarrow \text{P}$; $\text{D} \rightarrow \text{O}$

RECENT YEAR'S QUESTIONS OF IIT-JEE AND AIEEE

1. (c) 2. (b) 3. (d) 4. (b) 5. (c) 6. (a)

CHAPTER 9

Group 18- Noble Gases

INTRODUCTION

- Group –18 (Also called Zero group) includes 6 elements.
 - (i) ${}_2\text{He}$ (Helium) $1s^2$
 - (ii) ${}_{10}\text{Ne}$ (Neon) $2s^22p^6$
 - (iii) ${}_{18}\text{Ar}$ (Argon) $3s^23p^6$
 - (iv) ${}_{36}\text{Kr}$ (Krypton) $3d^{10}4s^24p^6$
 - (v) ${}_{54}\text{Xe}$ (Xenon) $4d^{10}5s^25p^6$
 - (vi) ${}_{86}\text{Rn}$ (Radon) $4f^{14}5d^{10}6s^26p^6$
- All are colourless, tasteless, odourless gases.
- These elements are neither inflammable nor supporter of combustion.
- These elements are monoatomic gases (cp/cv = 1.667) at room (ordinary) temperature. (Bond order zero by M.O theory)
- Except He, all these gases have eight valence electrons.
- All the elements have completely filled valence shell (no unpaired electron). Hence these are chemically inert and known as inert gases.
- Recent researches proved that under special condition, some of them can form chemical compounds (specially Xe with oxygen and fluorine). Hence these are called noble gases.
- The noble gases occupy only ~ 1% of the atmosphere. Hence they are called rare gases.
- They are known as aerogens as they are found in free state in air.
- (except Rn which is a decay product of Ra) ${}_{88}\text{Ra}^{226} \longrightarrow {}_{86}\text{Rn}^{222} + 2\text{He}^4$
- (Argon) Ar is most abundant of all the inert gases in atmosphere, but He is second most abundant gas in universe (after H_2).

Worked-out Example

1. Why is helium molecule, (He_2) not formed?

Ans. When two helium atoms come close to each other, repulsive forces overcome attractive forces (needs increases in energy).

Properties of Noble Gases

- Noble gases join the two extremely reactive groups of elements. Halogens have maximum capacity to gain electrons while Alkali metals are in a mood to get rid of valence electron but their link i.e., noble gases neither loose nor accept electrons.’
- They have maximum I.P. and minimum electron affinity.
- Their atoms are held together by weak van der Waal (London force) force of attraction. Down the group, atomic mass and surface area (volume) increase. Hence the extent of force of attraction increases. As a result of that, down the group
 - (i) Polarizability increases
 - (ii) Solubility increases
 - (iii) M.p and b.p increase (He has the lowest b.p 4.2 K of any known compounds)
 - (iv) Ease of adsorption increases
 - (v) Ease of liquefaction increases
 - (vi) Enthalpy of fusion and enthalpy of vaporization increase.
- Down the group
 - (i) I.P. decreases
 - (ii) Electrical conductivity decreases
- Separation of inert gases is possible based upon adsorption–desorption method on charcoal (Dewar’s method).

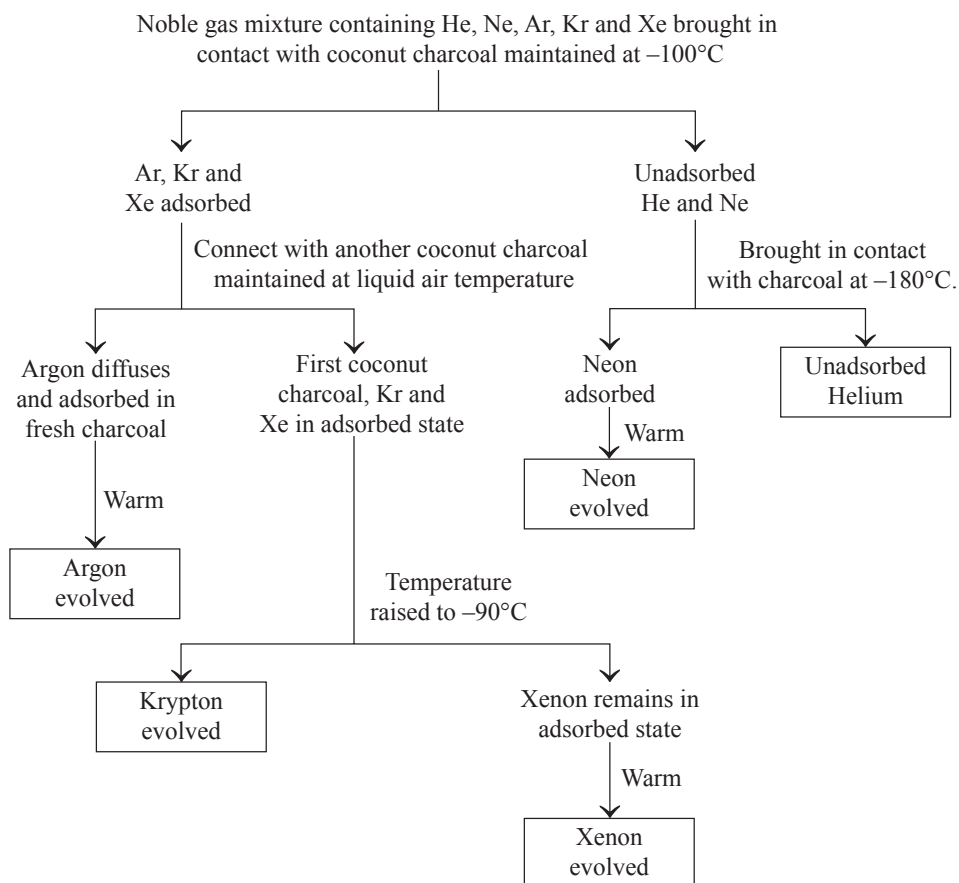
Principle involved: The greater the molar mass and surface area (down the group), the more easily it is absorbed.

Xe → more strongly adsorbed.

He → cannot be absorbed

Ne → adsorbed at -180°

Ar, Kr, Xe → adsorbed at -100°C



Worked-out Examples

2. Helium and neon do not form compounds with fluorine. Explain.

Ans. He and Ne being smaller in size possess very high ionization energy and thus their electrons cannot be excited to higher energy level like xenon and thus they do not form compounds with fluorine.

3. Xenon has closed shell configuration but forms compounds with fluorine. Explain.

Ans. Xe has relatively lower ionization energy among inert gases and thus the outermost shell electrons of Xe are excited to d-subshell and thereby showing unpaired electronic structure. Therefore Xe shows covalent bonding with fluorine showing sharing of electron pair.

4. Explain why solubility of noble gases in water is more than of hydrogen?

Ans. In presence of polar H_2O molecule, noble gas atom is polarized and weak dipole-induced dipole attraction forces are responsible for solubility. Larger atoms are more polarized i.e., why solubility increases with increase in size of atom. H_2 being less polarized is less soluble than noble gases.

5. Which one of the following statement regarding helium is incorrect?

[AIEEE 2004]

- It is used to produce and sustain powerful superconducting magnets.
- It is used as a cryogenic agent for carrying out experiments at low temperatures.
- It is used to fill gas balloons instead of hydrogen because it is lighter and non-inflammable.
- It is used in gas-cooled nuclear reactors.

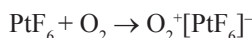
Ans. (c) Helium is heavier than hydrogen although it is non-inflammable.

Some Noteworthy Points

- Percentage of argon in air is nearly thirty times more than that of carbon dioxide. Percentage of argon in air is nearly 1% where as percentage of CO_2 in air is nearly 0.03% (by volume).
- Argon is the most abundant noble gas in atmosphere.
- The noble gases neither act as oxidizing agent nor as reducing agent.
- The different noble gases are absorbed by coconut charcoal at different temperatures. It provides a means to separate these gases from one another.
- The only compound of krypton studied in detail is krypton difluoride KrF_2 .
- Percentage of argon in air ($\approx 1\%$) is nearly thirty times more than carbon dioxide (0.03%).
- No true compound of He, Ne and Ar is as yet reported.

Chemical reactions

The first real compound of the noble gases was made in 1962. Bartlett and Lohman had previously used the highly oxidizing compound platinum hexafluoride to oxidize dioxygen.

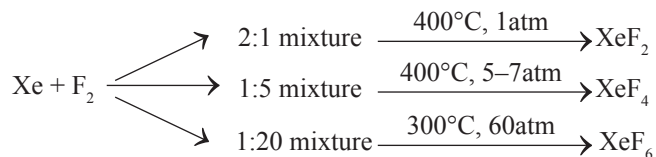
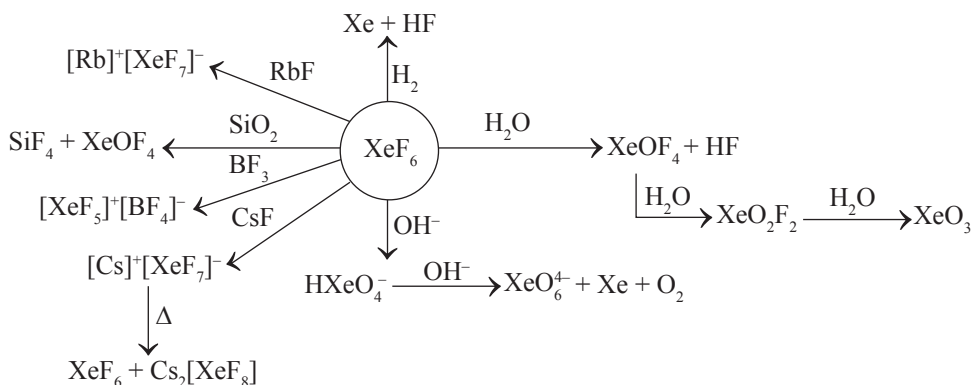
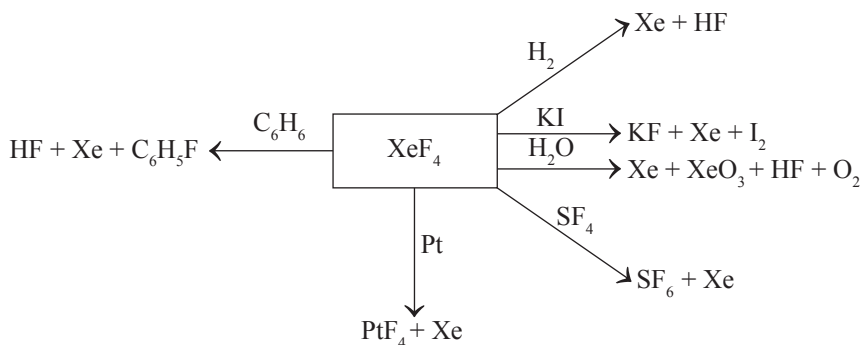
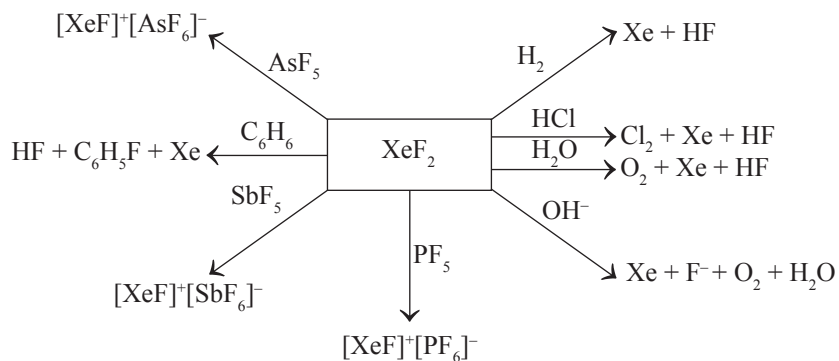


The first ionization energy for $\text{O}_2 \rightarrow \text{O}_2^+$ is 1165 kJ mol^{-1} (Note that IP of $\text{O}_2 < \text{IP of N}_2$ because the e^- in oxygen is removed from anti-bonding π_{2p}^* orbital), which is almost the same as the value of 1170 kJ mol^{-1} for $\text{Xe} \rightarrow \text{Xe}^+$. It was predicted that xenon should react with PtF_6 . Experiments showed that when deep red PtF_6 vapour was mixed with an equal volume of Xe, the gases combined immediately at room temperature to produce a yellow solid. They (incorrectly) thought the product obtained was xenon hexafluoroplatinate (V), $\text{Xe}^+[\text{PtF}_6]^-$. The reaction has since been shown to be more complicated, and the product is really $[\text{XeF}]^+[\text{Pt}_2\text{F}_{11}]^-$.



Xenon fluorides**Preparation**

- Xenon reacts directly with fluorine when the gases are heated at 300–600°C in a sealed nickel vessel, and the products formed depend on the Xe/F₂ ratio.

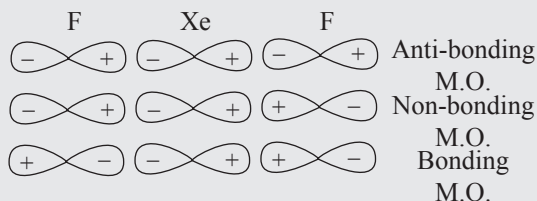
**Properties**

Worked-out Examples

6. Discuss the structure of XeF_2 using molecular orbital theory. Why do we need M.O. explanation of structure?

Ans. There are two main objections on VB structure of XeF_2 .

- Size of 5d orbital is large enough for effective overlapping.
- In hybridization, only orbitals of almost same energy take part and 5p and 5d of Xe differ by $\approx 960 \text{ kJ mol}^{-1}$, then in sp^3d hybridization, contribution of more energetic 5d orbital is objectionable.



One way to remove these objections is M.O. explanation of structure. M.O. structure involves three centre four electron 6 bonding. The two $2p_z$ orbitals and one sp^2 orbital of Xe combine linearly give rising to 3 M.O.

Bonding M.O. < non-bonding M.O. < Anti-bonding M.O. Linear combination of atomic orbitals and filling of four electrons in these M.Os. is represented in the diagram given below. The four electron occupy bonding and non-bonding M.O.

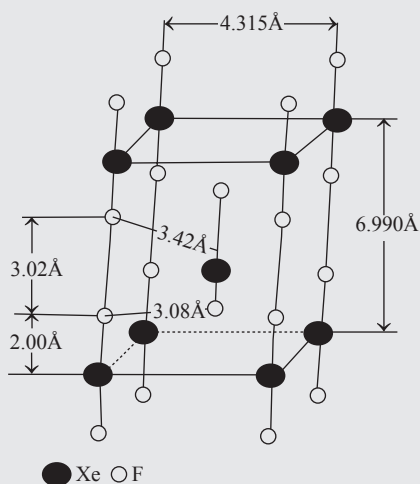
7. Arrange the XeF_2 , XeF_4 , XeF_6 in decreasing order of Xe—F bond length give reason also.

Ans. $\text{XeF}_2 > \text{XeF}_4 > \text{XeF}_6$
 $2.00 \text{ \AA} \quad 1.95 \text{ \AA} \quad 1.89 \text{ \AA}$

bond shortening may be accounted in terms of increase in charge on xenon atom.

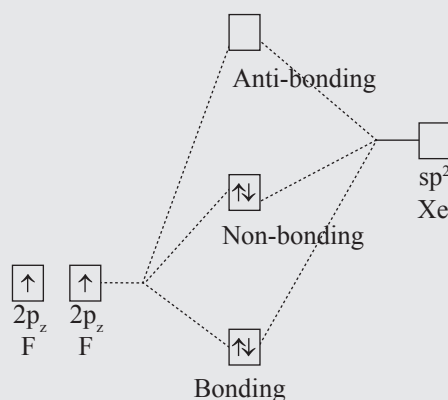
8. Discuss the shape of crystalline XeF_2 .

Ans. XeF_2 (linear) molecules are aligned parallel in body centred tetragonal array. The structure is shown below.

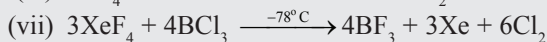
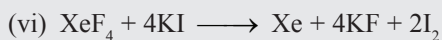
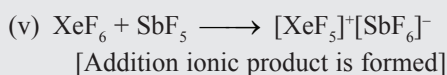
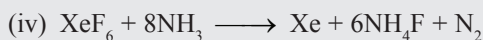
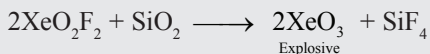
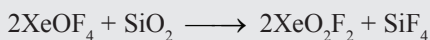
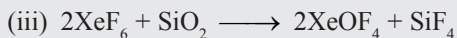
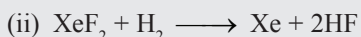
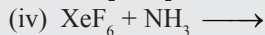
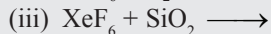
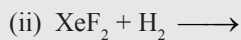


9. What is stability order of Xenon halides?

Ans. $\text{XeF}_2 > \text{XeCl}_2 > \text{XeBr}_2$



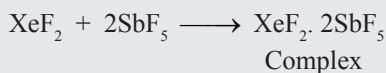
10. Complete the following reactions:



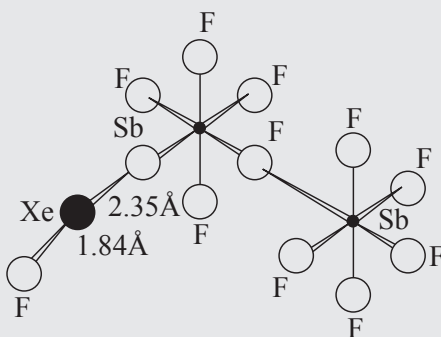
11. In XeF_2 , the two Xe—F bonds are of same bond length ($\approx 2.0\text{\AA}$) but when SbF_5 reacts with XeF_2 the Xe—F bond length is changed with one Xe—F bond equal to $1.84\text{\AA}$ and another $2.35\text{\AA}$. Comment upon structure of complex formed on the basis of results obtained.

Ans.

The following reaction occurs when XeF_2 reacts with SbF_5 .



The shortening of Xe—F bond ($1.84\text{\AA}$) shows that there is formation of XeF^+ and formula of complex must be $[\text{XeF}]^+[\text{Sb}_2\text{F}_{11}]^-$. The second bond is of $2.35\text{\AA}$, which is much shorter than van der Waals' distance $3.50\text{\AA}$. This suggests bridging F-atom between Xe and Sb. The structure is intermediate between ionic and covalent bridge structure.



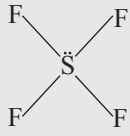
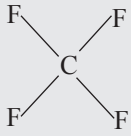
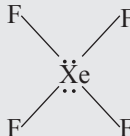
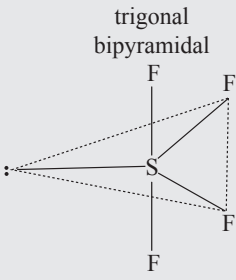
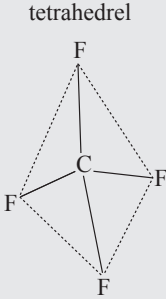
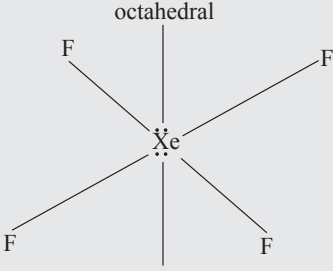
12. Molecular shapes of SF_4 , CF_4 and XeF_4 are

(IIT 2000)

- the same with 2, 0 and 1 lone pairs of electron respectively
- the same with 1, 1 and 1 lone pairs of electron respectively

- (c) different with 0, 1 and 2 lone pairs of electrons respectively
 (d) different with 1, 0 and 2 lone pairs of electrons respectively

Ans. (d) For the given molecules, we have

	SF_4	CF_4	XeF_4
Number of valence electrons	34	32	36
Lewis structure			
Number of lone pairs on the central atom	1	0	2
Number of pairs of electrons (bonded + lone) around the central atom	5	4	6
Probable structure			

13. In XeF_2 , XeF_6 the number of lone pairs on Xe are respectively

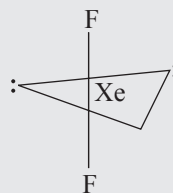
[AIEEE 2002]

- (a) 2, 3, 1 (b) 1, 2, 3
 (c) 4, 1, 2 (d) 3, 2, 1

Ans. (d) In the formation of XeF_2 , sp^3d hybridization occurs which gives the molecule a trigonal bipyramidal structure.

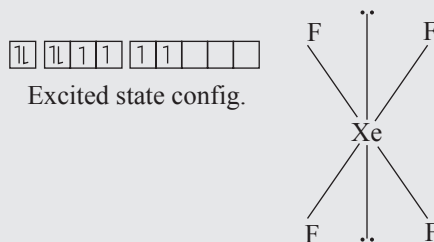


Ground state
configuration

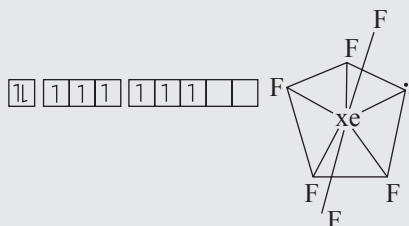


Excited state
configuration

In the formation of XeF_4 , sp^3d^2 hybridization occurs which gives the molecule an octahedral structure.



In the formation of XeF_6 , sp^3d^3 hybridization occurs which gives the molecule a pentagonal bipyramidal structure.



Clathrates: Noble gases form a number of combinations in which gases are trapped into the cavities of crystal lattices of certain organic and inorganic compounds. These combinations are called **enclosure or cage or clathrate compounds**. (Latin : clathrate = enclosed or protected by cross bars). The clathrates are normally non-stoichiometric compounds.

When quinol (o-dihydroxy benzene) is allowed to crystallise its solution in water in the presence of heavier noble gases such as argon, krypton or xenon under a pressure of 10–40 atmosphere, the noble gases get trapped within the lattice of quinol crystals. These crystals are quite stable. However, when heated or dissolved in solvent, the gas escapes.

The crystal structure with cavities is called the 'host' and the atom or molecule entrapped in it is called the 'guest'. The formation of a clathrate is only possible when the size of the cavity is such that it can fit in tightly the guest atom or molecule.

Thus, helium and neon do not form clathrate compounds with quinol as the size of the cavity is more than the size of helium or neon atom.

Some Noteworthy Points

- Van der Waal's radii of noble gases are lesser than the van der Waal's radii of preceding halogens. It is only when we compare covalent radii of halogens with van der Waal's radii of noble gases it appears that the atomic radii of noble gases are larger than the preceding halogens.
- Percentage of argon in air is thirty times more than that of carbon dioxide.
- He, Ne and Ar do not form any clathrate.
- No true compound of He, Ne and Ar is as yet reported.
- All noble gases except He have 8 electrons in its valence shell. He has only 2 electrons in its valence shell.
- Only He forms interstitial compounds with metals.
- XeF_6 acts as a fluoride ion acceptor in its reaction with alkali metal fluorides. XeF_2 and XeF_4 act as fluoride ion donor with super-acid (SbF_5).

TABLE 9.1 Structures of Some Xenon Compounds

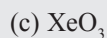
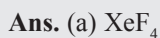
Formula	Oxid. State	Number of		Hybridization of Xe	Structure
		Bonding pairs	lone pairs		
XeF_2	+2	2	3	sp^3d	Linear with three lone pairs in equatorial position.
XeF_4	+4	4	2	sp^3d^2	Square planar with two positions occupied by lone pairs.

(Contd)

Formula	Oxid. State	Number of		Hybridization of Xe	Structure
		Bonding pairs	lone pairs		
XeF ₆	+6	6	1	sp ³ d ³	Distorted octahedron.
XeO ₃	+6	6	1	sp ³	Trigonal pyramidal with three π bonds and one lone pair.
XeO ₂ F ₂	+6	6	1	sp ³ d ²	Trigonal pyramidal with two π bonds and one lone pair
XeOF ₄	+6	6	1	sp ³ d ²	Square pyramidal with one π bond and one lone pair.

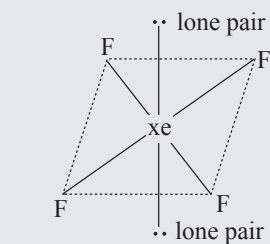
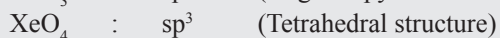
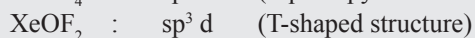
Worked-out Examples

14. Write down the formula of the noble gas species which are isostructural with:

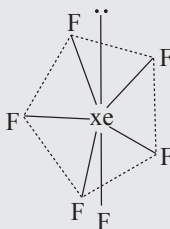


15. Point out hybridization and draw the structures of XeF₄, XeF₆, XeOF₄, XeOF₂, XeO₃ and XeO₄.

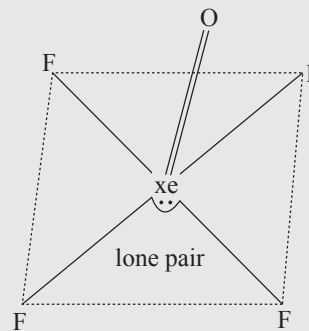
Ans.



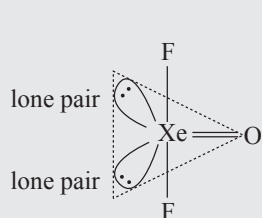
XeF₄(sp³d²)
(Square planar structure)



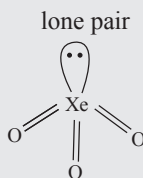
XeF₆(sp³d³)
(Pentagonal pyramidal structure)



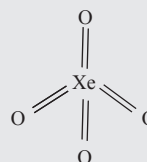
XeOF₄(sp³d²)
(Square pyramidal structure)



XeOF₂(sp³d)
(T-shaped structure)



XeO₃(sp³)
(Trigonal pyramidal structure)



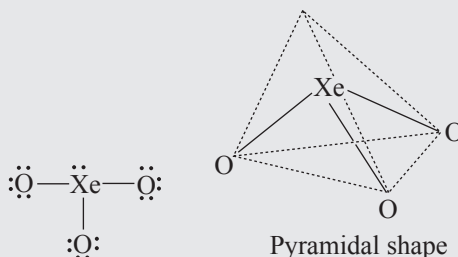
XeO₄(sp³)
(Tetrahedral structure)

16. Argon is used in arc welding because of its
- low reactivity with metal
 - ability to lower the melting point of metal
 - flammability
 - high calorific value

Ans. (a) Argon is used to provide an inert atmosphere in arc welding.

17. The structure of XeO_3 is
- linear
 - planar
 - pyramidal
 - T-shaped

Solution: (c) The total number of valence electrons in XeO_3 is $8 + 3 \times 6 = 26$. There are four electrons around Xe. Hence, these electrons acquire tetrahedral orientations. The shape of the molecule is pyramidal.



18. XeF_4 and XeF_6 are expected to be
- oxidizing
 - reducing
 - unreactive
 - strongly basic

(IIT 2007)

Solution: (a) XeF_4 and XeF_6 are oxidizing agents. These can oxidize H_2 to H^+ , Cl^- to Cl_2 , I^- to I_2 and Ce(III) to Ce(IV) .

The noble gases have closed-shell electronic configuration and are monoatomic gases under normal conditions. The low boiling points of the lighter noble gases are due to weak dispersion forces between the atoms and the absence of other interatomic interactions.

The direct reaction of xenon with fluorine leads to a series of compounds with oxidation numbers +2, +4 and +6. XeF_4 reacts violently with water to give XeO_3 . The compounds of xenon exhibit rich stereochemistry and their geometries can be deduced considering the total number of electron pairs in the valence shell.

Uses of Noble Gases

Noble gas	Uses	Reasons
Helium	- Filling of observation balloons and other lighter-than-air craft He/O_2 mixture for deep-sea breathing, instead of N_2/O_2 mixture - Diluent for gaseous anaesthetics He/O_2 mixtures for respiratory patients - Heat transfer medium for nuclear reactions - Liquid He to maintain very low temperature in research (cryogenics)	Non-flammable; 93% lifting power as compared to flammable H_2 Low solubility in blood; prevents nitrogen narcosis and "bends" Nonflammable, non-reactive Low density flows easily through restricted passages Transfers heat readily; does not become radioactive; chemically inert Extremely low boiling point

(Contd)

Noble gas	Uses	Reasons
Neon	1. Neon signs	Even at low Ne pressure, moderate electric current causes bright orange-red glow; can be modified by coloured glass or mixing with Ar or Hg vapour.
Argon	1. Inert atmosphere for welding 2. Filling incandescent light bulbs	Chemically inert Inert; prevents vaporization of tungsten and blackening of bulbs
Krypton	Airport runway and approach lights	Gives longer life to incandescent lights than Ar, but more expensive
Xenon	Xe and Kr mixture in high-intensity, short-exposure photographic flash tubes	Both have fast response to electric current.

Allotropes of Helium: Since the van der Waals forces are weaker in the helium than in any other substance, helium is the most difficult gas to liquefy, and helium approaches most closely to as 'ideal gas'. Liquid helium occurs in two allotropic forms, He I and He II. He I at 4.12 K is a normal liquid, but on cooling to 2.178 K another allotrope of helium, He II appears. The transition from He I to He II is accompanied by a sharp transition in physical properties. For example, He II expands on cooling has a thermal conductivity many hundreds of times greater than that of copper at ordinary temperature, and has no viscosity under certain conditions. Liquids with such low viscosity are known as super-fluids. There is effectively no resistance to their flow. Therefore, when kept in a container He II even goes uphill (defies gravity) and flow out of vessel. This state is sometimes referred to as the fourth state of matter. Helium can only be solidified under pressure. For example, its freezing point at a pressure of 20 atmospheres is 0.95 K, only just above absolute zero.

Worked-out Examples

19. Neon is generally used in warning signal illuminations. Explain.

Ans. Neon lights are visible from long distances and even visible through fog and mist.

20. Why is He preferred for filling meteorological balloons?

Ans. He is non inflammable and has 93% lifting power as compared to flammable H_2 .

21. What is quantum mechanical liquid i.e., helium (II)? Give its important characteristics.

Ans. Liquid helium is unique in that it exists in two forms He I and He II (liquid helium is obtained by Joule-Thomson expansion of the gas previously cooled to 15 K which is below the inversion temperature of He i.e., 35 K). The liquid helium I boils at 4.2 K and has normal liquid properties. On cooling He (I) to 2.19 K and 38 mm pressure, it changes to He (II) with abrupt changes in many physical properties such as density, dielectric constant and specific heat. He (II) is super fluid having so low energy that thermal motion of atoms do not take place, however, interatomic forces are also so weak that it does not occupy solid state. Thus He (II) is liquid with properties of gas. It has following characteristics :

- It has very high thermal conductance i.e., 800 times of copper.
- It has very low viscosity $\approx 1/100$ of H_2 gas.
- It has a very flat meniscus and very low surface tension and creeps over the surface of glass container.
- Its electrical resistance is zero i.e., it is super conducting liquid.
- He (II) has much lower entropy.

22. What is K—Ar dating?

Ans. By electron Capture, K^{40} gets converted into Ar^{40} and ratio of Ar and K in rock containing K can be used to determine the age of rock. This technique is called potassium-argon dating.

ONE CORRECT OPTION TYPE QUESTIONS

- The credit for the discovery of noble gases is given to:
 - Cavendish
 - Ramsay
 - Rayleigh
 - Frankland and Lockyer
- Which of the following is most abundant in air?
 - Ar
 - He
 - Ne
 - CO₂
- The source of most of the noble gases is:
 - decay of radioactive minerals
 - the atmospheric air
 - the natural gases coming out of the earth
 - the decay of rocks
- Which one of the following noble gas is not found in atmosphere?
 - Rn
 - Kr
 - Ne
 - Ar
- Noble gases are also known as aerogens because
 - they are rarely found in atmosphere
 - argon, a noble gas is the most abundant gas of the atmosphere
 - they occur in air
 - None of the above
- For the isolation of inert gases from air, the nitrogen and oxygen present are removed by passing air over heated
 - Calcium carbide
 - Copper metal
 - Activated charcoal
 - Finely divided mixture of Pt and Pd.
- Noble gases are chemically inactive because of:
 - mono-atomic nature
 - low boiling points
 - closed valency shell
 - rare occurrence
- The lightest gas which is non-inflammable:
 - H₂
 - N₂
 - Ar
 - He
- Which of the following does not react with fluorine?
 - Kr
 - Xe
 - Ar
 - all of these
- In solid argon, the atoms are held together by:
 - ionic bonds
 - hydrogen bonds
 - van der Waals' forces
 - none of these
- The density of neon will be highest at: [IIT 1990]
 - S.T.P.
 - 0°C, 2 atm
 - 273°C, 1 atm
 - 273°C, 2 atm
- Which noble gas is the least polarizable?
 - He
 - Ne
 - Kr
 - Rn
- The correct order of enthalpy of vaporization of noble gases is:
 - Xe > Kr > Ar > Ne > He
 - Xe > Ar > He > Ne > Kr
 - He > Ne > Kr > Ar > Xe
 - Ne > Xe > Kr > He > Ar
- The solubility of noble gases in water shows the order:
 - He > Ar > Kr > Ne > Xe
 - He > Ne > Ar > Kr > Xe
 - Xe > Kr > Ar > Ne > He
 - None of these
- The ease of liquefaction of noble gases decreases in the order:
 - He > Ne > Ar > Kr > Xe
 - Xe > Kr > Ar > Ne > He
 - Kr > Xe > He > Ar > Ne
 - Ar > Kr > Xe > He > Ne
- Which of the following does not form clathrates?
 - He
 - Ar
 - Kr
 - Xe
- Noble gases are sparingly soluble in water due to
 - dipole-dipole interactions
 - dipole-induced dipole interaction
 - Induced dipole-induced dipole interaction
 - Hydrogen bonding.
- In the Dewar's method of separation of noble gases, the mixture of noble gases is kept in contact with

- coconut charcoal at 173 K. Which one of the following gaseous mixture is not adsorbed on the charcoal?
- (a) Ar, Kr (b) He, Ne
(c) Xe, Ar (d) Xe, Kr
19. In the Dewar's method of separation of noble gases, the mixture of noble gases is kept in contact with coconut charcoal at 173 K. Which one of the following gaseous mixture is not adsorbed on the charcoal?
- (a) Ar, Kr (b) He, Ne
(c) Xe, Ar (d) Xe, Kr
20. Which of the following is not the correct use of clathrates?
- (a) In the separation of noble gases
(b) In transporting of isotopes of noble gases
(c) Kr-85 clathrate provide a useful source of β -radiations
(d) Clathrates compounds are used for producing compounds of noble gases.
21. What is formed when xenon combines with fluorine in the presence of mercury vapour?
- (a) XeF (b) Xe₂F
(c) XeF₂ (d) XeF₄
22. The idea which prompted Bartlett to prepare first ever compound of noble gas was:
- (a) High bond energy of Xe—F
(b) Low bond energy of F—F in
(c) Ionisation energies of O₂ and xenon were almost similar
(d) None of these
23. Among the noble gases Xenon reacts with fluorine to give stable Xenon fluorides because Xenon
- (a) has the largest size
(b) has lowest ionization energy
(c) has highest heat of vapourization
(d) is the most readily available noble gas.
24. Of the following species, one which is non-existent:
- (a) XeF₆ (b) XeF₅
(c) XeF₄ (d) XeF₂
25. The formation of XeF₄ is due to hybridization of excited xenon.
- (a) sp² (b) sp³
(c) sp³d (d) sp³d²
26. XeF₆ on complete hydrolysis gives:
- (a) Xe (b) XeO₂
(c) XeO₃ (d) XeO₄
27. Which of the following is planar?
- (a) XeF₄ (b) XeO₄
(c) XeO₂F₂ (d) XeOF₄
28. Xenon best reacts with:
- (a) most electropositive elements
(b) most electronegative elements
(c) hydrogen halides
(d) alkali metals
29. What are the products formed in the reaction of xenon hexafluoride with silicon dioxide?
- (a) XeSiO₄ + HF (b) XeF₂ + SiF₄
(c) XeOF₄ + SiF₄ (d) XeO₃ + SiF₂
30. XeF₆ on complete hydrolysis gives:
- (a) XeO₄ (b) XeOF₂
(c) XeOF₄ (d) XeO₃
31. XeF₆ on reaction with CsF gives:
- (a) [XeF₅]⁺ [CsF₂]⁻ (b) XeF₈
(c) Cs⁺[XeF₇]⁻ (d) [XeF₄]²⁺ [CsF₃]²⁻
32. XeF₄ on reaction with H₂ gives:
- (a) Xenon (b) Hydrogen fluoride
(c) Both (a) and (b) (d) None of these
33. The mixture which is used by sea diverse for respiration is of:
- (a) N₂ and O₂ (b) helium and O₂
(c) argon and O₂ (d) neon and O₂
34. The gas which is filled in tungsten lamp is:
- (a) H₂ (b) He
(c) Ar (d) O₂
35. The coloured discharge tube for advertisement mainly contains:
- (a) xenon (b) helium
(c) neon (d) argon
36. In Kroll and I.C.I, process of the production of titanium, the inert gas used is:
- (a) Ne (b) Ar
(c) Kr (d) Xe
37. Helium is added to oxygen used by deep sea divers because:
- (a) it is less soluble in blood than nitrogen under high pressure
(b) it is lighter than nitrogen
(c) it is readily miscible with oxygen
(d) it is less poisonous than nitrogen

9.14 Group 18-Noble Gases

38. Which of the following statements is not correct?
 (a) Argon is used in electric bulbs
 (b) Krypton is obtained during radioactive disintegration
 (c) Half life of radon is only 3.8 days
 (d) Helium is used in producing very low temperatures
39. The gaseous mixture used by divers for respiration is:
 (a) $N_2 + O_2$ mixture (b) $He + O_2$ mixture
 (c) $Ar + O_2$ mixture (d) $Neon + O_2$ mixture
40. The gas used in gas thermometer is:
 (a) He (b) O_2
 (c) Xe (d) Ne
41. Welding of magnesium can be done in an atmosphere of
 (a) O_2 (b) N_2
 (c) He (d) All
42. The noble gas used for the treatment of cancer is
 (a) Helium (b) Argon
 (c) Krypton (d) Radon
43. A noble gas which is not adsorbed by coconut charcoal is:
 (a) He (b) Ne
 (c) Ar (d) Kr
44. In XeO_3 and XeF_6 the oxidation state of Xe is:
 (a) +4 (b) +6
 (c) +1 (d) +3
45. Which of the following is not correct?
 (a) XeO_3 has four σ and four π bonds
 (b) The hybridization of Xe in XeF_4 is sp^3d^2
 (c) Among noble gases, the occurrence of argon is highest in air
 (d) Liquid helium is used as cryogenic liquid.
46. The xenon compound(s) that are iso-structural with IBr_2^- and BrO_3^- respectively are:
 (a) linear XeF_2 and pyramidal XeO_3
 (b) bent XeF_2 and pyramidal XeO_3
 (c) bent XeF_2 and planar XeO_3
 (d) linear XeF_2 and tetrahedral XeO_4 .
47. Total number of lone pair of electrons in $XeOF_4$ is
 (a) 0 (b) 1
 (c) 2 (d) 3
48. The Poisson's ratio for inert gases is:
 (a) 1.40 (b) 1.66
 (c) 1.34 (d) None of these

COMPREHENSION BASED QUESTIONS

The noble gases have closed-shell electronic configuration and are monoatomic gases under normal conditions. The low boiling points of the lighter noble gases are due to weak dispersion forces between the atoms and the absence of other interatomic interactions.

The direct reaction of xenon with fluorine leads to a series of compounds with oxidation numbers +2, +4 and +6. XeF_4 reacts violently with water to give XeO_3 . The compounds of xenon exhibit rich stereochemistry and their geometries can be deduced considering the total number of electron pairs in the valence shell.

- Argon is used in arc welding because of its
 (a) low reactivity with metal
 (b) ability to lower the melting point of metal
 (c) flammability
 (d) high calorific value
- The structure of XeO_3 is
 (a) linear (b) planar
 (c) pyramidal (d) T-shaped
- XeF_4 and XeF_6 are expected to be
 (a) oxidizing (b) reducing
 (c) unreactive (d) strongly basic

ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

- XeF_6 on reaction with H_2 gives:
 (a) Xe (b) HF
 (c) XeF_2 (d) XeF_4
- XeF_6 on hydrolysis gives:
 (a) $XeOF_4$ (b) XeO_2F_2
 (c) XeO_3 (d) XeO_4

3. Which among the following statements is/are correct?
 - (a) XeF_4 and SbF_5 combine to form salt
 - (b) He and Ne do not form clathrates
 - (c) He has lowest b.pt. in its group
 - (d) He diffuses through rubber and polyvinyl chloride
4. The noble gases which do not form clathrates are
 - (a) He
 - (b) Ne
 - (c) Kr
 - (d) Xe
5. The only known true chemical compounds of noble gases are with
 - (a) F
 - (b) O
 - (c) N
 - (d) S
6. Xe shows oxidation states of
 - (a) 2
 - (b) 4
 - (c) 6
 - (d) 8
7. Which of the following reacts with F_2 ?
 - (a) Ne
 - (b) Xe
 - (c) Ar
 - (d) Kr
8. Clathrates are obtained from noble gases and
 - (a) Water
 - (b) p-Quinol
 - (c) Ammonia
 - (d) Phenol
9. The noble gases heavier than air are
 - (a) Ar
 - (b) He
 - (c) Ne
 - (d) Kr
10. Noble gases used in flash tubes for high speed photography are
 - (a) Ne
 - (b) Kr
 - (c) Ar
 - (d) Xe

ASSERTION/REASON TYPE QUESTIONS

The following questions consist of two statements as Assertion (A) and Reason (R). While answering these questions, you are required to choose any one of the following responses.

- (a) If both assertion and reason are true and reason is the correct explanation of assertion.
- (b) If both assertion and reason are true and reason is not the correct explanation of assertion.
- (c) If assertion is true but reason is false.
- (d) If both assertion and reason are false.

1. **Assertion:** Helium and neon do not form clathrates.
Reason: Both have smallest atomic size among all the elements of group 18.
2. **Assertion:** The elements of group 18 are called inert gases.

Reason: The elements of group 18 are mono-atomic in nature.

3. **Assertion:** Noble gases have highest ionization energies in their respective periods.
Reason: The outermost shell of the noble gases is completely filled.
4. **Assertion:** Noble gases can combine with fluorine to form respective fluorides under specific conditions.
Reason: Fluorides of He, Ar and Ne are unstable in nature.
5. **Assertion:** Noble gases are diamagnetic atoms.
Reason: The atomic numbers of noble gases are even and all the orbitals are doubly occupied by the electrons.

MATCH THE COLUMN TYPE QUESTIONS

1. Select the correct matching:

List I

A : XeF_4

B : XeF_6

C : XeO_3

D : XeO_2F_2

List II

1. Pyramidal

2. T-shape

3. Distorted octahedral

4. Square planar

A	B	C	D
(a) 4	3	1	2
(b) 1	2	3	4
(c) 2	1	3	4
(d) 4	1	3	2

RECENT YEARS' QUESTIONS OF IIT-JEE AND AIEEE

- Number of lone pairs of electrons on Xe atoms in XeF_2 , XeF_4 and XeF_6 molecules are respectively: [AIEEE 2002]
 (a) 3, 2 and 1 (b) 4, 3 and 2
 (c) 2, 3 and 1 (d) 3, 2 and 0
- Among the following molecules, [AIEEE 2003]
 (i) XeO_3 (ii) XeOF_4
 (iii) XeF_6
 Those having same number of lone pairs on Xe are:
 (a) (i) and (ii) only (b) (i) and (iii) only
 (c) (ii) and (iii) only (d) (i), (ii) and (iii)
- Which one of the following statements regarding helium is incorrect? [AIEEE 2004]
 (a) It is used to fill gas balloons instead of hydrogen because it is lighter and non-inflammable
 (b) It is used in gas-cooled nuclear reactions
 (c) It is used to produce and sustain powerful superconducting magnets
 (d) It is used as a cryogenic agent for carrying out experiments at low temperatures.
- Which one is incorrect statement?
 (a) He is used in gas cooled nuclear reactors
 (b) He is used as a cryogenic agent for carrying out experiments at low temperature
 (c) He is used to produce and sustain powerful superconducting magnets
 (d) He is used to fill gas balloons instead of H_2 because it is lighter and non-inflammable
- How many lone pair of electrons are present on Xe in XeOF_4 ? [IIT 2004]
 (a) 1 (b) 2
 (c) 3 (d) 4

ANSWER KEYS

ONE CORRECT OPTION TYPE QUESTIONS

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (c) | 2. (a) | 3. (b) | 4. (a) | 5. (c) | 6. (a) | 7. (c) | 8. (d) | 9. (c) | 10. (c) |
| 11. (b) | 12. (a) | 13. (a) | 14. (c) | 15. (b) | 16. (a) | 17. (b) | 18. (a) | 19. (b) | 20. (d) |
| 21. (c) | 22. (c) | 23. (b) | 24. (b) | 25. (d) | 26. (c) | 27. (a) | 28. (b) | 29. (c) | 30. (d) |
| 31. (c) | 32. (c) | 33. (b) | 34. (c) | 35. (c) | 36. (b) | 37. (a) | 38. (b) | 39. (b) | 40. (a) |
| 41. (c) | 42. (d) | 43. (a) | 44. (b) | 45. (a) | 46. (a) | 47. (b) | 48. (b) | | |

ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

- | | | | | | | | | | |
|-----------|--------------|-----------------|--------|--------|--------|--------|--------|--------|---------|
| 1. (a, b) | 2. (a, b, c) | 3. (a, b, c, d) | 4. (a) | 5. (c) | 6. (c) | 7. (c) | 8. (b) | 9. (b) | 10. (b) |
|-----------|--------------|-----------------|--------|--------|--------|--------|--------|--------|---------|

COMPREHENSION BASED QUESTIONS

- | | | |
|--------|--------|--------|
| 1. (a) | 2. (c) | 3. (a) |
|--------|--------|--------|

ASSERTION/REASON TYPE QUESTIONS

- | | | | | |
|--------|--------|--------|--------|--------|
| 1. (a) | 2. (b) | 3. (c) | 4. (d) | 5. (a) |
|--------|--------|--------|--------|--------|

MATCH THE COLUMN TYPE QUESTIONS

- | |
|--------|
| 1. (b) |
|--------|

RECENT YEARS' QUESTIONS OF IIT-JEE AND AIEEE

- | | | | | | |
|--------|--------|--------|--------|--------|--------|
| 1. (a) | 2. (d) | 3. (a) | 4. (d) | 5. (d) | 6. (a) |
|--------|--------|--------|--------|--------|--------|

Hydrogen

INTRODUCTION

Hydrogen

- Hydrogen is the first element in the periodic table.
- Hydrogen is the lightest element known.
- Hydrogen was discovered by Henry Cavendish in 1766. He prepared the gas by treating iron with dilute H_2SO_4 .
- The name hydrogen was proposed by Lavoisier because it produces water on burning with oxygen (Greek; hydro= water; gene = producing).
- The various isotopes of hydrogen are:
 - (a) Protium (P or $^1\text{H}_1$)
 - (b) Deuterium (D or $^2\text{H}_1$)
 - (c) Tritium (T of $^3\text{H}_1$).
- The position of hydrogen in the periodic table is doubtful because hydrogen has properties similar to both alkali metals and halogens.

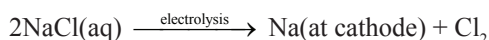
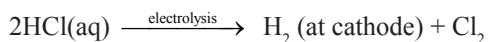
Resemblances of hydrogen with alkali metals

- *Electronic configuration*
 $\text{H} \rightarrow 1s^1$ and Alkali metals $\rightarrow ns^1$
- *Electropositive character*
 $\text{H} \rightarrow \text{H}^+ + e^-$ $\text{M} \rightarrow \text{M}^+ + e^-$
- *Valency and oxidation state*
 Both show valency and oxidation state of +1
 H^+Cl^- Na^+Cl^-
- *Reducing nature*
 Both hydrogen and alkali metals behave as reducing agents.
 - $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$
 - $\text{Fe}_3\text{O}_4 + 4\text{H}_2 \rightarrow 3\text{Fe} + 4\text{H}_2\text{O}$
 - $\text{B}_2\text{O}_3 + 6\text{K} \rightarrow 3\text{K}_2\text{O} + 2\text{B}$

10.2 Hydrogen

■ Liberation at cathode

Both H^+ and M^+ are liberated at cathode when salts of hydrogen and alkali metals (HCl and MCl etc.,) are electrolysed.



■ Both form similar halides.

E.g., HCl, NaCl, KCl

■ Both form similar oxides

E.g., H_2O , Na_2O , K_2O

■ Both form similar sulphides

E.g., H_2S , Na_2S , K_2S

Differences between hydrogen and alkali metals

■ Physical state

$H_2 \rightarrow$ gas whereas, alkali metals $\rightarrow$ solids at room temperatures.

■ Non-metallic character

$H_2 \rightarrow$ non-metal whereas, alkali metals $\rightarrow$ metallic nature

■ Atomicity

$H_2 \rightarrow$ diatomic and alkali metals $\rightarrow$ monoatomic.

■ Ionization energy

Ionization energy of H_2 is much higher (1312 kJ/mol) as compared to that of alkali metals (–500 kJ/mol).

■ Nature of compounds

- Hydrogen compound $\rightarrow$ covalent in nature whereas, compounds of alkali $\rightarrow$ ionic in nature.
- The oxides of alkali metals are basic whereas, the oxide of hydrogen are neutral.

Na_2O (basic) and H_2O (neutral)

■ Existence in solution

H^+ exists as H_3O^+ (represented as H_3O^+ for simplicity) whereas alkali metals exist as hexahydrated ions $[M(H_2O)_6]^+$.

Resemblances of hydrogen with halogens (X)

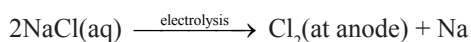
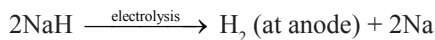
Hydrogen resembles with halogens also as shown in the following properties:

■ Electronic configuration

$H \rightarrow 1s^1$ and Halogens $\rightarrow ns^2np^5$

■ Electrochemical nature/Liberation at Anode

During electrolysis of compounds like LiH, CaH_2 , $CaCl_2$, LiCl etc., both H and X are liberated at the anode (i.e., liberation at anode).



■ Electronegative character



■ Oxidation state

Both hydrogen and halogens exhibit oxidation state of –1. e.g.,
e.g., Na^+H^- and Na^+Cl^-

- Atomicity
Both show atomicity of 2 as hydrogen and halogens are diatomic in nature.
- Both form similar compounds on combination with metals.



- Both form similar compounds on combination with other non-metals.



- Physical nature
Both H_2 and X_2 are gases at room temperature.
- Non-metallic nature
Both H and X are non-metallic.
- Ionization energy
The ionization energy of hydrogen is similar to that of halogens.
Ionization energy of hydrogen is 1312 kJ/mol
Ionization energy of fluorine is 1680 kJ/mol
Ionization energy of chlorine is 1255 kJ/mol
Ionization energy of bromine is 1143 kJ/mol

Differences between hydrogen and halogens

1. Lower tendency of hydride formation

The tendency of hydrogen to form hydrides (MH) is much less as compared to tendency of halogens to form halides (MX).

2. Absence of unshared electrons/lone pairs

Hydrogen molecule does not contain any unshared electrons whereas halogens have six lone pairs of electrons on each molecules (X_2).

3. Nature of Oxides

The oxides of hydrogen (H_2O) is neutral whereas the oxides of halogens (Cl_2O_7) are acidic.

H_2 Molecule (dihydrogen)

Occurrence of Hydrogen

- In the 'free state', it exists in small traces in volcanic and natural gases and in the outer atmosphere of sun and other stars of the universe. Nearly half of the mass of sun and the stars is due to hydrogen. Jupiter and Saturn consist mainly of hydrogen.
- Hydrogen constitutes about 09% by mass of earth's crust.
- Hydrogen is the ninth element in order of abundance in earth's crust.
- Ordinary, dihydrogen contains 75% ortho and 25% para hydrogen.

Isotopes of hydrogen

The various isotopes of hydrogen are:

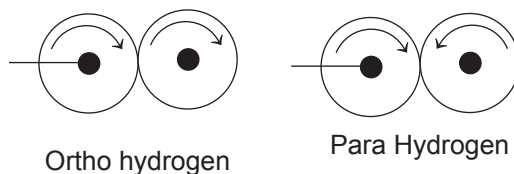
- Protium or Ordinary hydrogen (P or $^1\text{H}_1$)
- Deuterium or Heavy hydrogen (D or $^2\text{H}_1$)
- Tritium (T or $^3\text{H}_1$)

The relative abundance of Protium, Deuterium and Tritium is $1 : 1.5 \times 10^{-2} : 1 \times 10^{-7}$ (or 99.985% : 0.015% : $10^{-15}\%$) respectively.

Ortho and Para Hydrogen

The dihydrogen molecule is known as ortho hydrogen if the spins of the nuclei of the two atoms of the molecule are in the same direction.

The dihydrogen molecule is known as para hydrogen if the spins of the nuclei of the two atoms of the molecule are in the opposite direction.

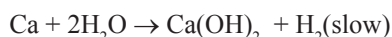
**Fig. 10.1**

- Ordinary hydrogen is an equilibrium mixture of ortho and para hydrogen.
- At 0°K, hydrogen contains mainly para hydrogen because the para hydrogen is more stable.
- At the temperature of liquefaction of air, the ratio of ortho and para hydrogen is 1 : 1.
- At room temperature, the ratio of ortho and para hydrogen is 3 : 1.
- Even at higher temperatures, the ratio of ortho and para hydrogen can never be more than 3 : 1.
- It is possible to obtain pure para hydrogen by cooling ordinary hydrogen gas to very low temperatures of 20 K, containing more than 75% of the ortho hydrogen.

Preparation of dihydrogen

(A) By the action of water on metals

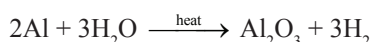
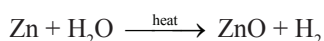
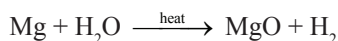
(i) With highly reactive metals (at room temperature):



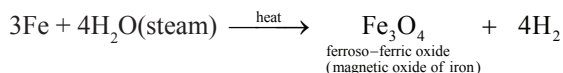
In the case of Na and K, the reaction is very vigorous/fast and exothermic. Due to the heat released, the hydrogen produced catches fire and hence, we say that sodium catches fire on being dropped in water.

In order to slow down these reactions, the amalgams of these metals with mercury are used.

(ii) With less reactive metals like Mg, Al and Zn (on heating):

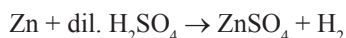
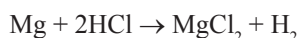
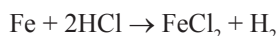


(iii) With very less reactive metals like nickel and iron, hydrogen can be produced only by passing steam since they are very less reactive.

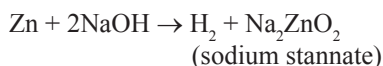
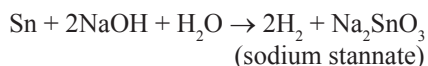
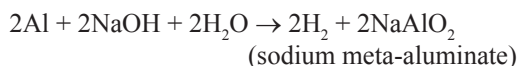


This reaction is used for the large scale production of dihydrogen by lane's process.

(B) By action of acids on metals



(C) By action of strong alkalis on metals

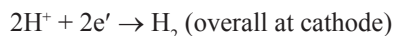
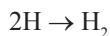
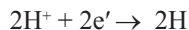


(D) Electrolysis of water

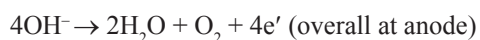
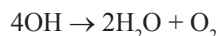
The hydrogen produced by electrolysis of water is very pure but is also very expensive, because of which it cannot be used as a commercial method of preparation.



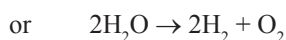
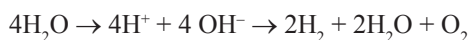
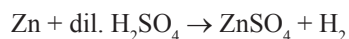
At cathode:



At anode:



Overall Reaction

**Laboratory Preparation of Dihydrogen**

Note: Concentrated H_2SO_4 is not used because it acts as an acid as well as an oxidizing agent and the following reaction takes place:

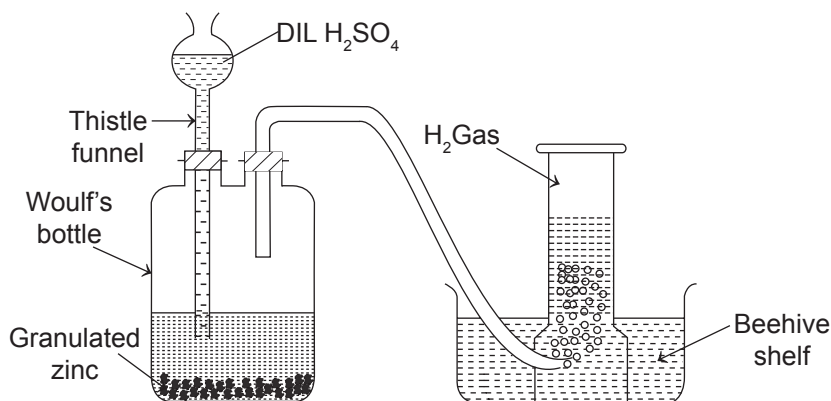
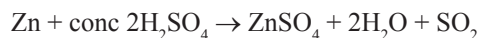


Fig. 10.2

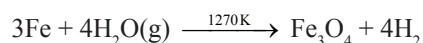
- Pure zinc is not used in the laboratory preparation because its reaction with sulphuric acid is slow. The presence of impurities increases the rate of reaction due to the formation of electrochemical couples.

Manufacture of Dihydrogen

(Industrial/Commercial production/preparation of hydrogen)

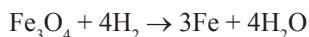
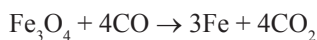
(A) By Lane's Process

(a) Oxidation state



10.6 Hydrogen

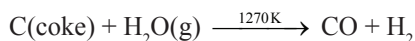
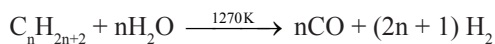
(b) Reduction State



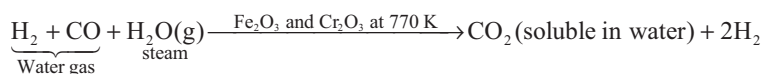
Here oxidation and reduction are carried out alternatively using two or more furnaces depending upon the requirement of hydrogen.

(B) Bosch's Process

(a) Preparation of water gas



(b) Separation of hydrogen



(C) As a byproduct in various industrial process like NaOH manufacture by electrolysis of sodium chloride solution.

(D) By electrolysis of water

This is the best method when electricity is cheap.

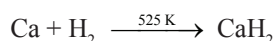
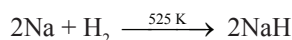
Physical properties

1. It is colourless, tasteless and odourless gas
2. It is slightly soluble in water
3. It is highly combustible
4. It is the lightest substance known (1 L of H_2 weighs 0.098 g)

Chemical properties

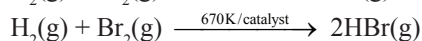
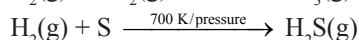
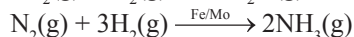
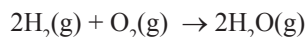
- Dihydrogen is relatively less reactive due to the high bond dissociation energy (4.36 kJ/mol).

(a) Reaction with metals

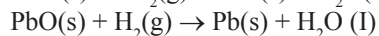
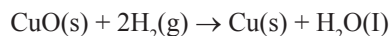


Hydrogen forms interstitial hydrides with metals like Pt, Pd, Ni, etc. In these hydrides, hydrogen occupies interstitial holes in the metallic crystals. This is called OCCLUSION.

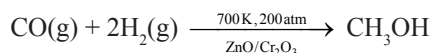
(b) Reaction with non-metals

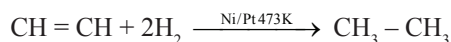
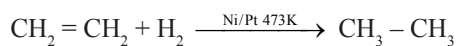
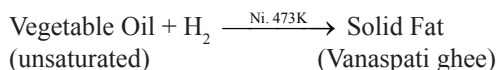


(c) Reaction with metal oxides



(d) Reaction with carbon monoxide



(e) Reaction with unsaturated hydrocarbons**(f) Action with vegetable oil****(Hydrogenation/Hardening of Oils)****Uses of Dihydrogen**

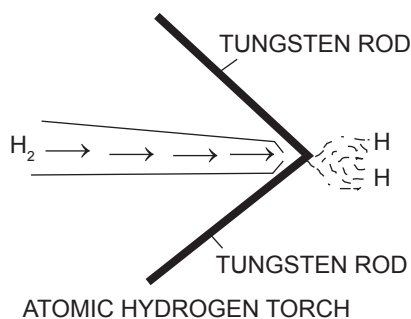
- It is used in the manufacture of synthetic petrol.
- It is used in the hydrogenation of vegetable oils.
- It is used in oxy-hydrogen torch for welding for the attainment of high temperatures of around 2500°C.
- It is used as a rocket fuel when mixed in liquid form with liquid oxygen.
- It is used in atomic-hydrogen torch for welding, when even higher temperatures of around 4000°C are required.
- It is used in the manufacture of ammonia. Its significance lies in the fact that ammonia is a starting material for the manufacture of various fertilizers like urea, $(\text{NH}_4)_2\text{SO}_4$ etc.,.

Nascent Hydrogen (Newly born hydrogen)

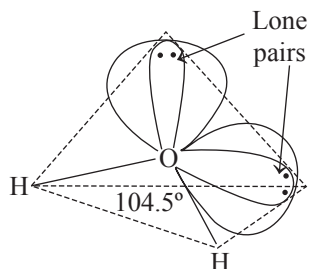
It is the form of hydrogen at the moment of its generation from chemical reactions in aqueous solutions.

Atomic Hydrogen

It is formed by passing dihydrogen gas through an electric arc struck between tungsten filaments when the dihydrogen molecules dissociate into hydrogen atoms.

**Fig. 10.3****Order of reactivity**

$\text{H}_2 < \text{Nascent hydrogen} < \text{Atomic hydrogen}$

Water

10.8 Hydrogen

Physical Properties of Water

- Universal polar solvent because a large number of substances (most of the inorganic substances) are soluble in it.
- Water has maximum density of 1.00 g cm^{-3} at 277 K i.e., 4°C
- Has strong hydrogen bonding in its molecules and exists as associated molecules $(\text{H}_2\text{O})_n$.
- Water has high specific heat and thermal conductivity. The high heat of vapourization and the high specific heat of water are responsible for moderating the influence of water on the climate and body temperature of living organisms.

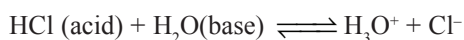
Chemical Properties of Water

(i) **Action towards litmus** – Pure water is neutral to litmus

(ii) **Decomposition**– $2\text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{O}^+ + \text{OH}^-$
($K_w = 1 \times 10^{-14} \text{ mol}^2 \text{ L}^{-2}$ at 298 K)

(iii) **Acid-Base reactions**

Water is amphoteric in nature i.e., it behaves both as an acid as well as a base.

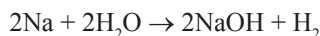


However, water is neutral towards litmus and its pH is 7.

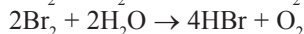
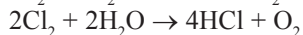
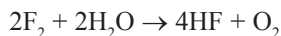
(iv) **Oxidation-Reduction reactions**

Water can act both as an oxidizing as well as a reducing agent.

Water behaves as weak oxidizing agent with active metals.

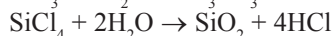
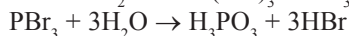
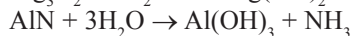
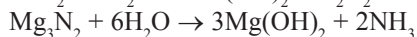
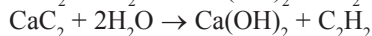
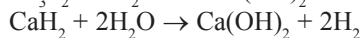
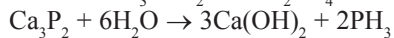
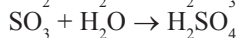
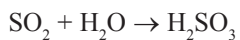


Water behaves as a reducing agent with highly electronegative elements.



(v) **Hydrolytic reactions**

Water hydrolyses a variety of compounds like oxides and halides of non-metals.



(vi) **Hydration reactions**

Water has a strong ability to form compounds with same metal salts known as hydrates.

Water forms three categories of hydrates:

(a) By combining metal ions through co-ordinate bonds in a complex.



Here, all the water molecules are bonded to the metal ion.

(b) Water molecules may be hydrogen bonded to certain oxygen containing anions. e.g., $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, in which four water molecules are co-ordinated to a central Cu^{2+} ion while the fifth water molecule is hydrogen bonded to the sulphate group.

(c) Water may also occupy interstitial sites. e.g., in $\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$

Hard and soft water**Soft water**

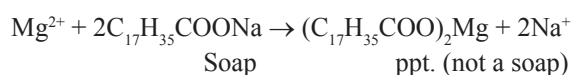
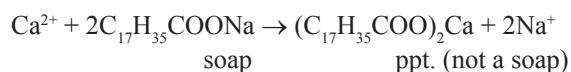
Water which produces lather with soap solution readily is called soft water.

Hard water

Water which does not produce lather with soap solution readily is called hard water.

Cause of Hardness of Water

- The hardness of water is due to the presence of Mg and Ca salts like bicarbonates, chlorides and sulphates.
- Hard water does not produce lather with soap because the Mg^{+2} and Ca^{+2} cations react with soap to form Ca and Mg salts of fatty acids (which are not soaps).

**Types of Hardness**

- Temporary hardness (Carbonate hardness)
It is due to the presence of soluble bicarbonates of calcium and magnesium.
- Permanent hardness (Non-carbonate hardness)
It is due to the presence of chlorides and sulphates of Ca and Mg.

Softening of Hard Water

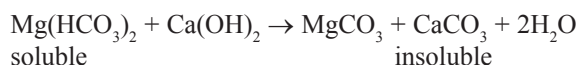
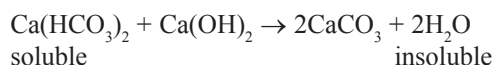
The process of removal of metallic ions (Ca^{2+} and Mg^{2+}) responsible for hardness of water is known as softening of water.

Removal of Temporary Hardness

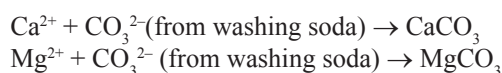
- Boiling
Temporary hard water is taken in large boilers and boiled for about fifteen minutes. During this process, the soluble bicarbonates are converted into insoluble carbonates. These insoluble carbonates are removed from the vessel and in the process the calcium and magnesium ions are also removed from water, thereby removing hardness.



- Calcium hydroxide (or CLARK'S) method
Here, a calculated quantity of calcium hydroxide is added when the soluble bicarbonates are converted into insoluble carbonates.

**1. Removal of Permanent Hardness**

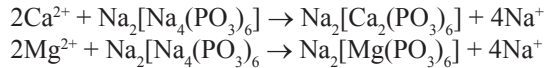
- By the addition of washing soda
Here, the Mg^{+2} or Ca^{+2} ions combine with the CO_3^{2-} ions of the washing soda ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$). These insoluble carbonates are then removed, thereby removing the ions responsible for hardness.



10.10 Hydrogen

(ii) By the addition of sodium polymetaphosphate (CALGON Process)

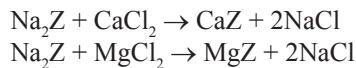
Here, sodium polymetaphosphate $\text{Na}_2[\text{Na}_4(\text{PO}_3)_6]$, is added and the calcium and magnesium form complexes with sodium polymetaphosphate.



The complexes of Calcium and Magnesium so formed remain dissolved in water. But still they cannot cause hardness. This is because Ca and Mg ions are not free to react with soap as these have been tied up in a stable complex i.e., $[\text{Ca}_2(\text{PO}_3)_6]$. This is known as Sequestration of Ca^{2+} and Mg^{2+} ions.

2. Permutit Method

Here, hard water is passed through zeolite, whereby, the Calcium and Magnesium ions of hard water are replaced with Na^+ ions of the permutit.



Where, Na_2Z is the zeolite that is hydrated sodium aluminium silicate ($\text{Na}_2\text{Al}_2\text{Si}_2\text{O}_8 \cdot x\text{H}_2\text{O}$).

After some time the Na^+ ions of the zeolite are used up.

After this NaCl solution is passed through the vessel during which the CaZ and MgZ are reconverted into Na_2Z .

The reaction taking place during place during regeneration of zeolite is,

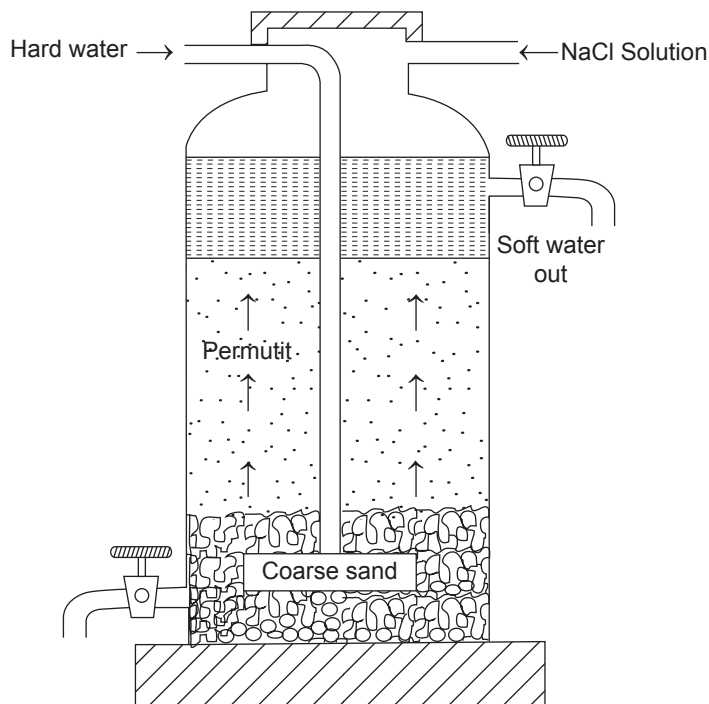
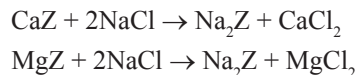
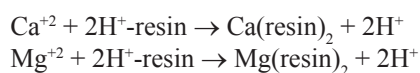


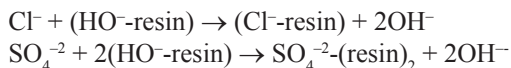
Fig. 10.4 Permutit process for softening of hard water

3. Ion Exchange Resins Method

Hard water is first passed through a bed of cation exchange resin (i.e., H^+ -resin) in the first tank where the cations are exchanged with the H^+ of the resin.



The water which comes out of the first tank is therefore free from Ca^{+2} and Mg^{+2} ions and this water is then passed through the second tank which contains a bed of anion exchange resins (i.e., HO^- -resin). Here, the anions of hard water are exchanged with the OH^- ions of the resin.



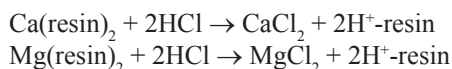
The H^+ ions formed in the first tank combine with the OH^- ions formed in the second tank to produce water.



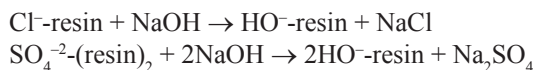
The water obtained after this is known as Deionised Water as this does not contain any of the ions.

Regeneration of the Resins

The exhausted resin of the first tank is regenerated by treatment with moderately concentrated HCl or H_2SO_4 .



The exhausted resin of the second tank is regenerated by treatment with moderately concentrated solution of sodium hydroxide.

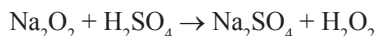


Hydrogen Peroxide (H_2O_2)

Hydrogen peroxide was discovered by Thenard (1918)

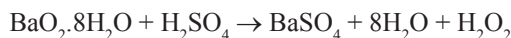
Preparation

- (i) From sodium peroxide (Merck's process)

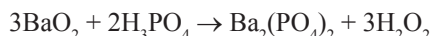


- (ii) From barium peroxide

- (a) By reaction with sulphuric acid



- (b) By reaction with phosphoric acid



Remarks About the Methods of Preparation

From sodium peroxide (Merck's process)

- In the Merck's Process, calculated amounts of sodium peroxide (Na_2O_2) are added to ice cold dilute (20%) solution of sulphuric acid slowly with constant stirring. Cooling the solution gives crystals of $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ which are decanted leaving behind a solution of hydrogen peroxide.
- Merck's process give 30% hydrogen peroxide solution.

From barium peroxide

- While preparing hydrogen peroxide from Barium Peroxide (BaO_2), a paste of Barium peroxide is prepared in ice cold water and is treated with about 20% ice cold solution of sulphuric acid. The white precipitate of barium sulphate is removed by filtration leaving behind about 5% solution of hydrogen peroxide.
- Here, anhydrous Barium peroxide cannot be used because in this case, the barium sulphate produced forms a protective layer around the unreacted peroxide which prevents the reaction of the remaining barium peroxide.
- The limitation of this method is that the hydrogen peroxide produced by this method contains some heavy Ba^{+2} ions. These Ba^{+2} catalyst the decomposition of hydrogen peroxide and hence the hydrogen peroxide produced by this method cannot be stored for a long time.

10.12 Hydrogen

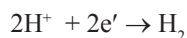
- In order to prevent the presence of Ba^{+2} ions in the hydrogen peroxide solution, Phosphoric acid is used in place of sulphuric acid. When phosphoric acid is used, the barium phosphate produced gets completely precipitated and the solution does not contain any free Ba^{+2} ions and hence the decomposition of hydrogen peroxide is not catalysed.

Manufacture of Hydrogen Peroxide

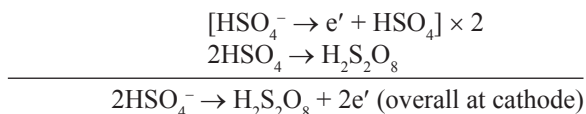
- By electrolysis of 50% H_2SO_4



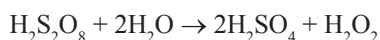
At cathode:



At anode:



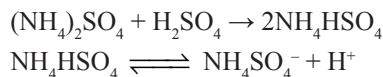
The peroxydisulphuric acid ($\text{H}_2\text{S}_2\text{O}_8$; Marshall's acid) is drawn off from the cell and is hydrolysed with water to give hydrogen peroxide.



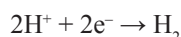
The resulting solution is distilled under reduced pressure when hydrogen peroxide gets distilled while the sulphuric acid with high boiling point remains undistilled.

- By electrolysis of 50% NH_4HSO_4

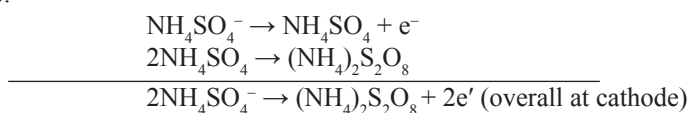
The above method has been modified by using equimolar mixture of sulphuric acid and ammonium sulphate for electrolysis.



At cathode:



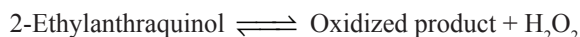
At anode:



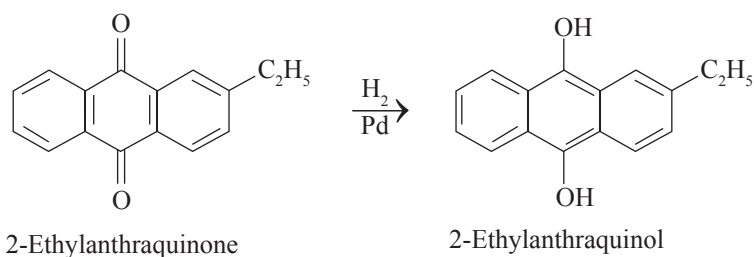
The ammonium $[(\text{NH}_4)_2\text{S}_2\text{O}_8 + 2\text{H}_2\text{O} \rightarrow 2\text{NH}_4\text{HSO}_4 + \text{H}_2\text{O}_2]$

- Manufacture of Hydrogen Peroxide from 2-Ethyl Anthraquinol**

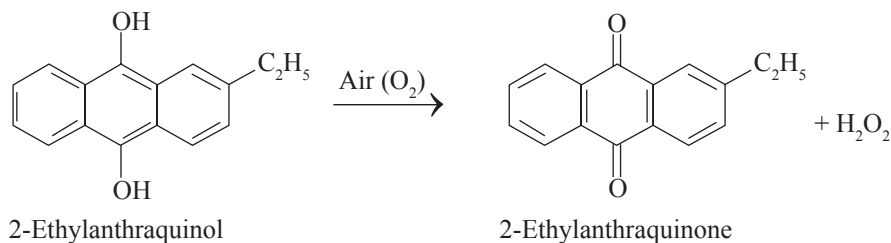
On an industrial scale, the hydrogen peroxide can be obtained by the auto-oxidation of 2-Ethyl anthraquinol.



Here, 2-Ethylantraquinone is dissolved in benzene and hydrogen gas is passed through the solution in the presence of palladium(Pd) catalyst. Here, 2-Ethylantraquinone gets reduced to 2-Ethylantraquinol.



The reduced product is dissolved in a mixture of benzene and cyclohexanol and when air is passed through this, it is oxidized back to 2-Ethylanthraquinone and hydrogen peroxide is also produced alongwith.



This process is very cheap because the 2-Ethylanthraquinone can be used again and again.

Storage of Hydrogen Peroxide

The following precautions have to be taken during the storage of hydrogen peroxide:

- Hydrogen peroxide has to be stored in wax-lined bottles because the rough glass surface causes its decomposition.
- A small amount of phosphoric acid, glycerol or acetanilide is generally added to the solution. These substances retard the decomposition of H₂O₂. Such substances which retard the reactions are known as Negative Catalysts.

Physical Properties of Hydrogen Peroxide

Pure H₂O₂ is thick syrupy pale blue liquid. It is completely miscible with water, alcohol and ether. It is more viscous and dense than water.

The density of H₂O₂ is 1.44 g cm⁻³, melting point is 272.4 K and boiling point is 358 K at 68 mm of Hg.

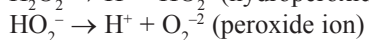
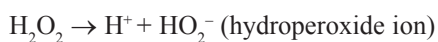
Chemical Properties of Hydrogen Peroxide

(i) Decomposition



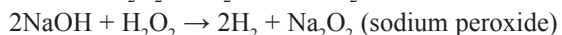
The decomposition is further accelerated with the presence of metals like platinum, gold, metal oxides (MnO₂), and certain metal ions like Fe⁺² ions.

(ii) Acidic behaviour



Hydrogen is a weak dibasic/diprotic acid with a K_a value of 1.55 × 10⁻¹² at 298 K.

Hydrogen peroxide forms two types of salts i.e.,



Here, sodium hydroperoxide is an acidic salt and sodium peroxide is normal salt.

(iii) Oxidizing and reducing character of hydrogen peroxide

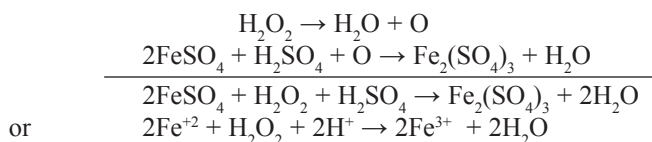
- Hydrogen peroxide behaves both as a reducing agent and oxidizing agent in both acidic and alkaline medium.
- The oxidation state of oxygen in hydrogen peroxide is -1. It can be oxidized to zero oxidation state (e.g., in O₂) or it can be reduced to -2 oxidation state (e.g., in H₂O).
- Hydrogen peroxide is a strong oxidizing agent.
- Hydrogen peroxide is a weak reducing agent.
- Hydrogen peroxide acts as an oxidizing agent in neutral, acidic or in alkaline medium.
- Hydrogen peroxide acts as a reducing agent in basic, or even neutral medium.

(iv) Oxidizing nature of H₂O₂

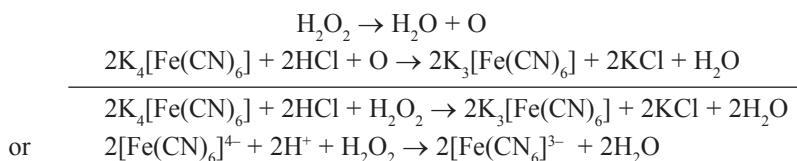


10.14 Hydrogen

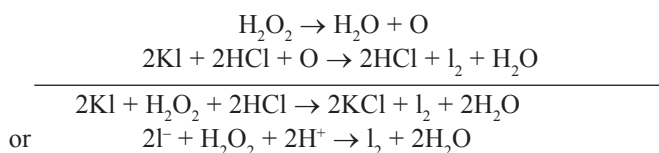
1. Oxidation of Ferrous sulphate into Ferric sulphate in acidic medium



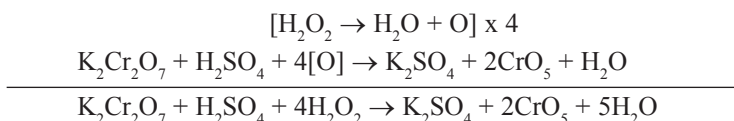
2. Oxidation of Potassium ferrocyanide into Potassium ferricyanide in acidic medium



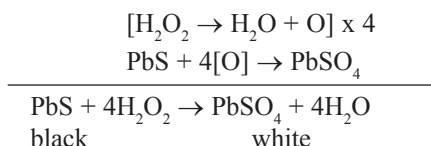
3. Oxidation of Potassium Iodide to I_2 in acidic solution (I^- to I_2) or (Iodides to Iodine) in acidic medium



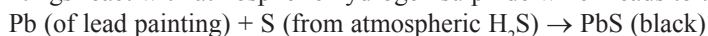
4. Oxidation of Potassium dichromate to chromium pentaoxide in acidic medium



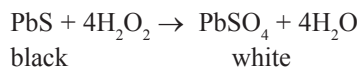
5. Oxidation of Lead sulphide into Lead sulphate in neutral medium



Lead paintings react with atmospheric hydrogen sulphide which leads to the formation of black PbS.

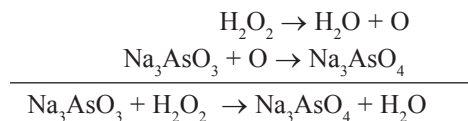


Due to this the original white colour of lead paintings is blackened. These paintings are then treated by,

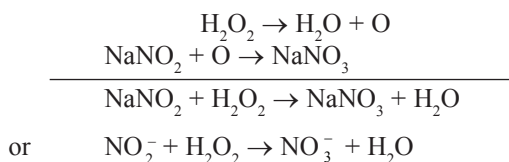


The above reaction is used to restore the colour of lead paintings as the black lead sulphide is oxidized again into white lead sulphate.

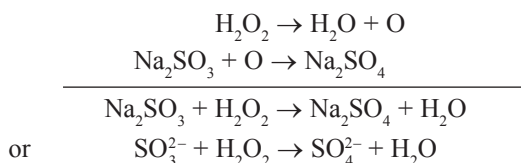
6. Oxidation of Arsenites into Arsenate in neutral medium



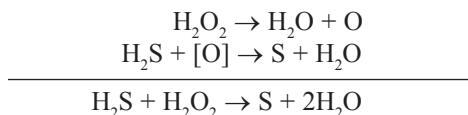
7. Oxidation of Nitrites to nitrates in neutral medium



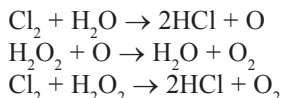
8. Oxidation of sulphites to sulphates in neutral medium



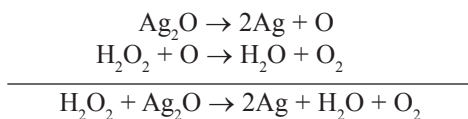
9. Oxidation of hydrogen sulphide into Sulphur in neutral medium

(v) Reducing Nature of H_2O_2 In acidic nature of $\text{H}_2\text{O}_2 \rightarrow 2\text{H}^+ + \text{O}_2 + 2\text{e}^-$ In basic medium $\text{H}_2\text{O}_2 + 2\text{OH}^- \rightarrow 2\text{H}_2\text{O} + 2\text{e}^-$

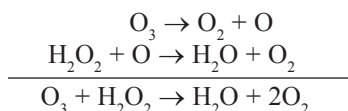
1. Halogens are reduced into halogen acids in neutral medium



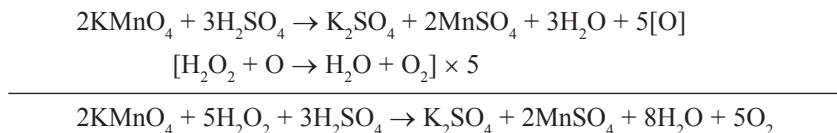
2. The metal oxides are reduced into metals in the neutral medium



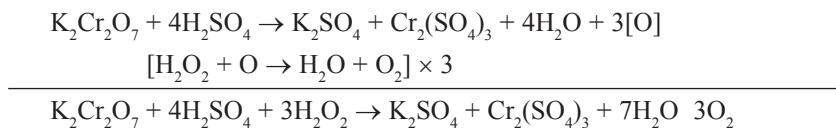
3. Ozone is reduced into dioxygen



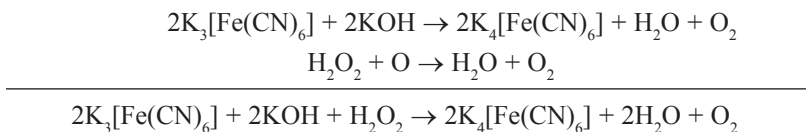
4. Acidified Potassium permanganate solution is reduced to colourless manganese sulphate



5. Acidified Potassium dichromate solution is reduced into green chromium salts

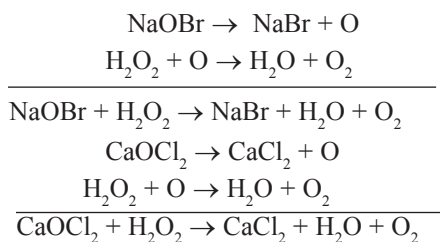


6. Alkaline Potassium ferrocyanide is reduced to potassium ferricyanide

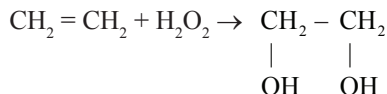


10.16 Hydrogen

7. Hypohalides are reduced in halides

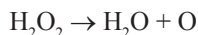


(vi) Formation of Addition compounds



(vii) Bleaching Action of Hydrogen Peroxide

Hydrogen peroxide acts as a bleaching agent due to the release of nascent oxygen.



The nascent oxygen produced oxidizes the coloured matter and a colourless matter is obtained.

Uses of Hydrogen Peroxide

- as a bleaching agent
- as an antiseptic perhydrol
- as an antichlor in bleaching
- to restore colour of lead paintings
- to manufacture many organic compounds and inorganic compounds like sodium perborates, epoxides etc.,
- to preserve milk, wine and other liquors
- to impart golden colour to hair

Strength of H_2O_2

The methods used to express the strength of H_2O_2 solution are

(i) By-percentage

X% of H_2O_2 means X gm of H_2O_2 are present in 100 g of solution

(ii) By number volume

“X Volume” H_2O_2 is a solution of H_2O_2 whose 1 litre will give X litre of oxygen at STP.

Structure of H_2O_2

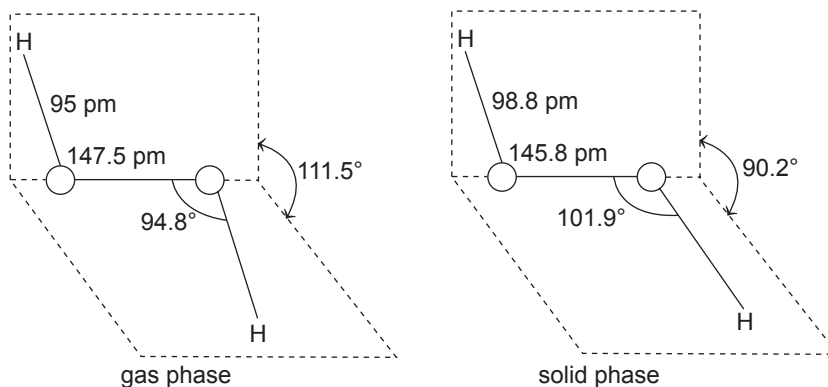


Fig. 10.5 Structure of H_2O_2

Hydrides

The binary compounds of hydrogen with other elements are called hydrides. On the basis of their physical and chemical properties and the types of bonding, they are classified into three main types:

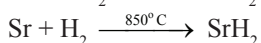
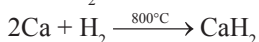
1. Ionic or salt like or saline hydrides
2. Covalent or molecular hydrides
3. Metallic or interstitial hydrides

(i) Ionic or Saline Hydrides

- These are formed by the combination of hydrogen with metals which have low electronegativity values and are electropositive with respect to hydrogen.
- Ionic hydrides are prepared by the direct combination of the metals with hydrogen at high temperatures usually close to 750°C.



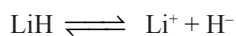
molten



Characteristics of Ionic Hydrides

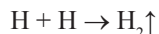
1. They conduct electricity in the fused state liberating hydrogen at the anode. The negative charge on the hydrogen in these hydrides is proved by electrolysis of molten lithium hydrides at 700°C. During electrolysis, lithium metal deposits at the cathode and hydrogen is liberated at the anode.

700°C

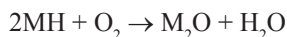


At cathode: $\text{Li}^+ + \text{e}^- \rightarrow \text{Li}$ (deposited)

At anode: $\text{H}^- \rightarrow \text{H} + \text{e}^-$

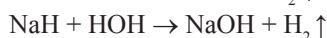
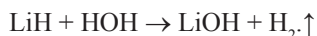


2. Their densities are higher than those of metals.
3. LiH, CaH₂ and SrH₂ are the most stable hydrides of this class, others undergo thermal decomposition above 500°C.
 - Larger the heats of formation, greater are their stabilities.
 - The thermal stability of the alkali metal hydrides decreases from LiH to CsH.
4. The ionic hydrides are oxidized by air to form metal oxide and water.



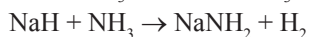
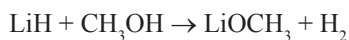
- RbH and CsH burn spontaneously at room temperature.

5. The hydrides are hydrolysed by water.



Therefore, their aqueous solutions are basic.

6. They react with protonic solvents liberating hydrogen

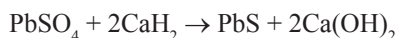


7. They act as powerful reducing agents, especially at high temperatures. For example, NaH reduces CO and CO₂ to formates on heating.

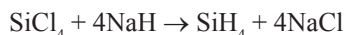
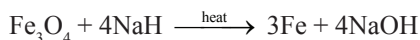


10.18 Hydrogen

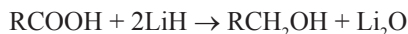
8. They reduce sulphates to sulphides



9. They also reduce some halides and oxides as:



10. They are capable of reducing organic acids to alcohols.



11. Some hydrides are used to prepare complex hydrides of other metals.

- The complex hydrides also act as reducing agents. For example, LiH reacts with aluminium chloride in ether solution to form Lithium aluminium hydride, LiAlH_4 (complex hydride).



- NaH reacts with diborane (B_2H_6) to form sodium borohydride as:



- The hydrides of lanthanides are also regarded as ionic hydrides

Uses of Ionic Hydrides

The ionic hydrides and their complexes are used as reducing agents. Since they evolve hydrogen when heated and, therefore, ignite spontaneously, they are used as solid fuels.

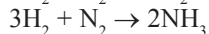
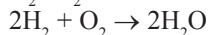
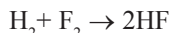
(ii) Molecular or Covalent Hydrides

- These hydrides are formed by the combination of elements of comparatively higher electronegativity as p-block elements.
- The bonds formed are mostly covalent in character. But sometimes e.g., in HF, the bond may be partly ionic in character as well.
- These hydrides have molecular lattices where they are held together by weak Van der Waal forces.
- The general formula of a covalent hydride is XH_{8-n} where X stands for the symbol of a metal and n is the group number.

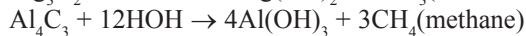
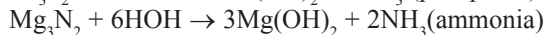
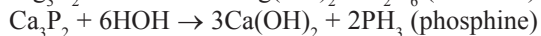
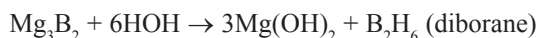
Preparation of Covalent Hydrides

Covalent hydrides are prepared by the following methods:

(i) By direct combination of elements



(ii) By hydrolysis of metal borides, carbides, nitrides, phosphides etc.



(iii) By reduction of certain compounds by nascent hydrogen or by using reducing agents like LiAlH_4



General Characteristics of Covalent Hydrides

- The molecular hydrides have low electrical conductivity
- The molecular hydrides are soft
- The molecular hydrides have low melting points/boiling points
- The molecular hydrides are generally volatile in nature
- Along any given row of periodic table, covalent hydrides become acidic in nature in going from left to right



Interstitial Hydrides

- They are binary compounds of hydrogen and transition elements.
- They are generally formed by transition metals of group 3, 4, 5 of d-block, Cr, and some *f*-block elements.
- They are generally powders or brittle solids.
- They have good thermal as well as electrical conductivities.
- They are harder than the parent metals.
- They liberates hydrogen reversibly on strong heating.
- In these hydrides, the hydrogen atoms occupy the interstitial holes in the crystal lattice of metal atoms.

ONE CORRECT OPTION TYPE QUESTIONS

- Hydrogen is evolved by the action of cold dilute HNO_3 on
(a) Fe (b) Mn
(c) Cu (d) Al
- A hydride of nitrogen which is acidic is
(a) NH_3 (b) N_2H_4
(c) N_2H_2 (d) N_3H
- When the same amount of zinc is treated separately with excess of sulphuric acid and excess of sodium hydroxide, the ratio of volumes of hydrogen evolved is
(a) 1 : 1 (b) 1 : 2
(c) 2 : 1 (d) 9 : 4
- HCl is added to the following oxides. Which one would give H_2O_2 ?
(a) MnO_2
(b) PbO_2
(c) BaO
(d) None of the above
- Heavy water is
(a) H_2^{18}O
(b) Water obtained by repeated distillation
(c) D_2O
(d) Water at 4°C
- Hydrogen will not reduce
(a) Heated cupric oxide
(b) Heated ferric oxide
(c) Heated stannic oxide
(d) Heated aluminium oxide
- The sum of the number of neutrons and protons in the isotope of hydrogen is
(a) 6 (b) 5
(c) 4 (d) 3
- The oxidation states exhibited by hydrogen in its various compounds are
(a) -1 only (b) Zero only
(c) +1, -1 and zero (d) +1 only
- The strongest base is
(a) NH_3 (b) PH_3
(c) AsH_3 (d) SbH_3
- When zeolite, which is hydrated sodium aluminium silicate, is treated with hard water the sodium ions are exchanged with
(a) H^+ ions (b) Ca^{2+} ions
(c) OH^- ions (d) SO_3^{2-} ions
- Which of the following statements is not correct?
(a) H_2O_2 oxidizes Fe(II) to Fe(III)
(b) H_2O_2 can be obtained by electrolysis of dil. H_2SO_4
(c) H_2O_2 reduces Mn(VII) to Mn(II)
(d) H_2O_2 is a weak base
- The volume strength of 1.5 N H_2O_2 solution is
(a) 4.8 (b) 8.4
(c) 3.0 (d) 8.0
- The maximum possible number of hydrogen bonds a water molecule can form in ice is
(a) 1 (b) 2
(c) 3 (d) 4
- 30 volumes of H_2O_2 means
(a) 30% H_2O_2 solution
(b) 30 cm^3 of the solution contains 1 g of H_2O_2
(c) 1 cm^3 of the solution liberates 30 cm^3 of O_2 at STP
(d) 30 cm^3 of the solution contains 1 mole of H_2O_2
- The oxide that gives H_2O_2 on treatment with dilute acid is
(a) PbO_2 (b) Na_2O_2
(c) MnO_2 (d) TiO_2
- 11.2 volumes of H_2O_2 is about
(a) 3.4% (b) 0.34%
(c) 34% (d) 0.34%
- 2 g of aluminium is treated separately with excess of dilute H_2SO_4 and excess of NaOH. The ratio of the volumes of hydrogen evolved is
(a) 2 : 3 (b) 1 : 1
(c) 2 : 1 (d) 1 : 2
- A 6 volume sample of H_2O_2
(a) will contain 6% v/v of H_2O_2
(b) will contain 6 % w/v of H_2O_2
(c) would give 6 volumes of oxygen per unit volume of H_2O_2 sample at STP
(d) would give 6 volumes of oxygen per unit weight of H_2O_2 sample at STP

19. Amongst H_2O , H_2S , H_2Se and H_2Te , the one with the highest boiling point is
 (a) H_2O because of hydrogen bonding
 (b) H_2Te because of higher molecular weight
 (c) H_2S because of hydrogen bonding
 (d) H_2Se because of lower molecular weight
20. Match list I with list II and select the correct answer using the codes given below the lists:
- | List I | List II |
|-------------------------|--|
| 1. Heavy water | A. Bicarbonates of Mg and Ca in water |
| 2. Temporary hard water | B. No foreign ions in water |
| 3. Soft water | C. D_2O |
| 4. Permanent hard water | D. Sulphates and chlorides of Mg and Ca in water |
- Codes:**
 (a) 1-C, 2-D, 3-B, 4-A
 (b) 1-B, 2-A, 3-C, 4-D
 (c) 1-B, 2-D, 3-C, 4-A
 (d) 1-C, 2-A, 3-B, 4-D
21. The correct order of bond angles in the molecules H_2O , NH_3 , CH_4 and CO_2 is
 (a) $\text{H}_2\text{O} > \text{NH}_3 > \text{CH}_4 > \text{CO}_2$
 (b) $\text{H}_2\text{O} < \text{NH}_3 < \text{CO}_2 < \text{CH}_4$
 (c) $\text{H}_2\text{O} < \text{NH}_3 > \text{CO}_2 > \text{CH}_4$
 (d) $\text{H}_2\text{O} > \text{NH}_3 < \text{CH}_4 > \text{CO}_2$
 (e) $\text{H}_2\text{O} < \text{NH}_3 < \text{CH}_4 < \text{CO}_2$
22. Polyphosphates are used as water softening agents because they
 (a) form soluble complexes with anionic species
 (b) precipitate anionic species
 (c) form soluble complexes with cationic species
 (d) precipitate cationic species
23. The volume strength of perhydrol is
 (a) 20 (b) 30
 (c) 100 (d) 10
24. Semi water gas is a mixture of
 (a) $\text{SO} + \text{H}_2$ (b) $\text{CO} + \text{N}_2$
 (c) $\text{CO} + \text{H}_2 + \text{N}_2$ (d) None
25. Y is
 (a) V_2O_5 (b) Pt
 (c) $\text{Fe}_2\text{O}_3/\text{Cr}_2\text{O}_3$ (d) Ni
26. X is
 (a) producer gas
 (b) carbon dioxide
 (c) mixture of CO and H_2
 (d) $\text{H}_2\text{O}_2 + \text{CO}_2$
27. CO_2 and H_2 in the sequence are respectively represented by
 (a) Z, W (b) W, X
 (c) W, Z (d) X, Y
28. The whole sequence refers to the
 (a) Lane's process
 (b) Reforming of hydrocarbons
 (c) Bosch's process
 (d) Linde's method
29. The process refers to the manufacture of
 (a) Nitrogen (b) Hydrogen
 (c) Oxygen (d) Carbon dioxide
30. Which hydride is also called alanate?
 (a) $\text{Li}^+ [\text{AlH}_4]^-$ (b) $\text{Na}^+ [\text{BH}_4]^-$
 (c) CaH_2 (d) AlH_3
31. The maximum possible number of hydrogen bonds a water molecule can form in ice is
 (a) 1 (b) 2
 (c) 3 (d) 4
32. Calculate the normality of 10 volume H_2O_2 ?
 (a) 1.7 N (b) 12 N
 (c) 30.3 N (d) 0.0303 N
33. The volume strength of 1.5 N H_2O_2 solution is
 (a) 4.8 (b) 5.2
 (c) 8.8 (d) 8.4
34. The hydride gap refers to inability of
 (a) noble gases to form hydrides
 (b) elements of group 7, 8, 9 of d-block to form hydrides
 (c) s-electrons of p-block element to participate in bonding
 (d) hydrogen to react with saturated hydrocarbons
35. Among O^{2-} , O_2^{2-} and O_2^- ions which will react with water to liberate OH^- ions along with O_2 gas?
 (a) O^{2-} and O_2^{2-} (b) O_2^{2-} and O_2^-
 (c) O^{2-} and O_2^- (d) All the three
36. Which pair of nitrogen compounds react with water to liberate azane gas?
 (a) CaCN_2 and $\text{Ca}(\text{CN})_2$
 (b) Mg_3N_2 and $\text{Ca}(\text{CN})_2$
 (c) Mg_3N_2 and $(\text{NH}_4)_2\text{SO}_4$
 (d) Ca_3N_2 and CaCN_2

10.22 Hydrogen

37. Hydride of certain non-metallic element X is amphoteric in nature. It also reacts with sodium hydride as well as with metallic sodium to liberate dihydrogen gas. The element X can be
 (a) Nitrogen (b) carbon
 (c) Oxygen (d) sulphur
38. How much volume of 40 volume- H_2O_2 will oxidize 0.2 mol of black lead sulphate to white lead sulphate?
 (a) 112 mL (b) 22.4 mL
 (c) 448 mL (d) 224 mL
39. Which of the following pair of species will not produce dihydrogen gas?
 (a) Cu + dil. HCl (b) Fe + H_2SO_4
 (c) Mg + steam (d) Na + alcohol
40. The hydrolysis of one mole of peroxodisulphuric acid produces
 (a) two moles of sulphuric acid
 (b) two moles of peroxomonosulphuric acid
 (c) one mole of sulphuric acid and one mole of peroxomonosulphuric acid
 (d) one mole of sulphuric acid, one mole of peroxomonosulphuric acid and one mole of hydrogen peroxide
41. The critical temperature of water is higher than that of O_2 because the H_2O molecule has
 (a) fewer electrons than oxygen
 (b) two covalent bonds
 (c) V-shape
 (d) high dipole moment
42. One mole of calcium phosphide on reaction with excess of water gives
 (a) one mole of phosphine
 (b) two moles of phosphoric acid
 (c) two moles of phosphine
 (d) one mole of phosphorus pentoxide
43. Polyphosphates are used as water softening agents because they
 (a) form soluble complexes with anionic species
 (b) precipitate anionic species
 (c) form soluble complexes with cationic species
 (d) precipitate cationic species.
44. 10 ml of a given solution H_2O_2 contains 0.91 g of H_2O_2 . The volume strength of the solution is
 (a) 30 volume (b) 20 volume
 (c) 10 volume (d) 9.1 volume
45. Steam $\xrightarrow[\text{Iron}]{\text{Red Hot}}$ Solid (A) + Gas (B)
 $\text{X} \xleftarrow[\text{Cr}_2\text{O}_3]{\text{CO}}$ 700 K
- In the above sequence of reaction, X is
 (a) COH_2 (b) Mixture of CO_2 and H_2
 (c) CH_3OH (d) Water gas
46. In the sequence given in Q. 45. A and B are respectively
 (a) Fe, H_2 (b) Fe_2O_3 , H_2O
 (c) Fe_3O_4 , H_2 (d) Fe_3H_4 + O_2
47. The reactivity of halogens towards reaction with dihydrogen to form halogen acid is
 (a) $\text{Cl}_2 > \text{Br}_2 > \text{F}_2 > \text{I}_2$ (b) $\text{F}_2 > \text{Cl}_2 > \text{I}_2 > \text{Br}_2$
 (c) $\text{F}_2 > \text{I}_2 > \text{Br}_2 > \text{Cl}_2$ (d) $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$
48. In the above sequence of reaction. Z is
 (a) protium oxide (b) diprotium
 (c) deuterium oxide (d) deuterium protium oxide
49. Y and B are respectively
 (a) C_2D_2 , CaCl_2 (b) C_2D_2 , CaH_2
 (c) CaCl_2 , C_2D_2 (d) C_2H_2 , D_2
50. A and X are respectively
 (a) Ca(OD)_2 , NH_3 (b) Ca(OH)_2 , ND_3
 (c) Ca(OD)_2 , ND_3 (d) CaCl_2 , H_2O_2

HINTS AND EXPLANATIONS

- (b) $\text{Mn} + 2 \text{HNO}_3 (\text{dil.}) \rightarrow \text{Mn(NO}_3)_2 + \text{H}_2$
- (d) N_3H (hydrazoic acid) is the acidic hydride of nitrogen
- (a) $\text{Zn} + \text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + \text{H}_2$;
 $\text{Zn} + 2 \text{NaOH} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2$;
 The ratio of volumes of H_2 evolved in both the cases is 1 : 1.
- (d) None of the oxides is peroxide
- (c) D_2O is called heavy water.
- (d) Al is more electropositive than hydrogen, therefore, its oxide will not be reduced by hydrogen.
- (d) In tritium nucleus there is one proton and 2 neutrons.

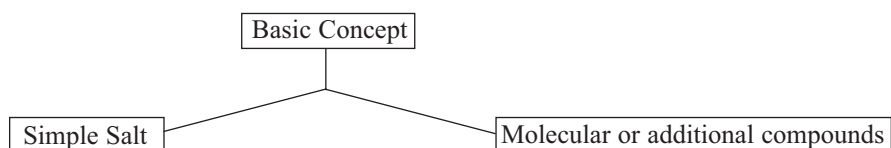
13. (d) In ice, each H_2O molecule is surrounded by four other water molecules, bonded through H-bonds.
 14. (c) It is a factual statement.
 15. (c) $\text{Na}_2\text{O}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}_2$
All other oxides are dioxides.
 16. (a) $11.2 = \text{Normality} \times 5.6$;
Normality = $\frac{11.2}{5.6}$
Strength (g/l) = $\frac{11.2}{5.6} \times 17$;
Percentage strength = $\frac{11.2 \times 17}{5.6 \times 10} = 3.4$
 17. (b) $2\text{Al} + 3\text{H}_2\text{SO}_4 \rightarrow \text{Al}_2(\text{SO}_4)_3 + 3\text{H}_2$
 $2\text{Al} + 2\text{NaOH} + 2\text{H}_2\text{O} \rightarrow 2\text{NaAlO}_2 + 3\text{H}_2$
Hence, the ratio of volumes of hydrogen evolved is 1 : 1
 18. (c) A 6 vol. sample of H_2O_2 means 6 vol. of oxygen are given out per unit vol. of H_2O_2
 19. (a) H_2O has the highest b.p. due to H-bonding in it.
 20. (d) Heavy water is D_2O , temporary hardness is due to bicarbonates of Mg and Ca in water, soft water does not contain any foreign ions and permanent hardness is due to sulphates and chlorides of Mg and Ca in water.
 21. (e) Bond angle in CO_2 is 180° , in CH_4 is 109.5° , in NH_3 is 107° and in H_2O it is 105° .
 22. (c)
 23. (c) Perhydrol is 30% H_2O_2 .
10 vol. $\text{H}_2\text{O}_2 = 30\%$ then 30% H_2O_2 has 100 volume strength.
 24. (c) Mixture of $\text{CO} + \text{H}_2 + \text{N}_2$ is called semi water gas.
 25. (c) Y is $\text{Fe}_2\text{O}_3/\text{Cr}_2\text{O}_3$, a catalyst used during the process
 26. (c) X is water gas
 27. (a) Z, W represent CO_2 , H_2 respectively
 28. (c) Factual question
 29. (b) Dihydrogen gas by Bosch's process
 30. (a) Alanate is lithium aluminium hydride.
 31. (d) H_2O molecule can form 4 H-bonds with surrounding molecules
 32. (a) Normality = $10/5.6 = 1.7\text{ N}$
 33. (d) Volume strength = Normality $\times 5.6 = 1.5 \times 5.6 = 8.4$ volume
 34. (b) It is a conceptual fact
 35. (b) $\text{O}^{2-} \xrightarrow{\text{H}_2\text{O}} 2\text{OH}^-$ $2\text{O}_2^{2-} \xrightarrow{2\text{H}_2\text{O}} \text{O}_2 + 4\text{OH}^-$
 $4\text{O}_2^{2-} \xrightarrow{2\text{H}_2\text{O}} 3\text{O}_2 + 4\text{OH}^-$
 36. (d) $\text{Ca}_3\text{N}_2 + 6\text{H}_2\text{O} \rightarrow 3\text{Ca}(\text{OH})_2 + 2\text{NH}_3$
 $\text{CaCN}_2 + 3\text{H}_2\text{O} \rightarrow \text{CaCO}_3 + 2\text{NH}_3$
 37. (c) X is oxygen. Hydride of O is H_2O which is amphoteric
 $2\text{H}_2\text{O} + 2\text{NaH} \rightarrow 2\text{NaOH} + 2\text{H}_2$ $2\text{H}_2\text{O} + 2\text{Na} \rightarrow 2\text{NaOH} + \text{H}_2$
 38. (d) $4\text{H}_2\text{O}_2 + \text{PbS} \rightarrow \text{PbSO}_4 + 4\text{H}_2\text{O}$.
4 mol 1 mol
 H_2O_2 required to oxidize 0.2 mol PbS = 0.8 Mol
Now, molarity of 40 volume $\text{H}_2\text{O}_2 = \frac{40}{11.2}$
 39. (a) Cu cannot displace H from HCl
 40. (c) $\text{H}_2\text{S}_2\text{O}_8 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4 + \text{H}_2\text{SO}_5$
Peroxodi- Peroxomono-
Sulphuric acid sulphuric acid.
 41. (d) Conceptual question.
 42. (c)
 43. (c)
 44. (a) $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$
10 ml of sol. Produces $\text{O}_2 = \frac{22400 \times 0.91}{68}$ mL
= 299.76 mL at STP
1 mL of sol. produce $\text{O}_2 = 29.96 \approx 30$ mL.
 45. (c) $\text{H}_2\text{O}(\text{g}) + \text{Fe} \rightarrow \text{Fe}_3\text{O}_4 + \text{H}_2$
(A) (B)
 $2\text{H}_2 + 2\text{CO} \xrightarrow[\text{ZnO, 700 K}]{\text{Cr}_2\text{O}_3} 2\text{CH}_3\text{OH}$
(X) (Y) (Z)
 46. (c)
 47. (d) Factual question
Sequence of reactions for Q. 30-32
 $\text{D}_2\text{O} + \text{CaC}_2 \rightarrow \text{Ca}(\text{OD})_2 + \text{C}_2\text{D}_2$
(A) (B)
 $\text{Ca}(\text{OD})_2 + \text{NH}_4\text{Cl} \rightarrow \text{NH}_3 + \text{CaCl}_2(\text{aq}) + 2\text{HDO}$
(X) (Y) (Z)
 48. (b) Refer above sequence
 49. (c)
 50. (a) Factual statement.

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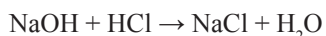
CHAPTER 11

Co-ordination Chemistry

INTRODUCTION

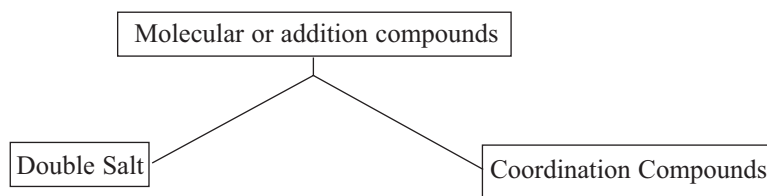


When an acid reacts with an alkali, neutralization takes place and a simple salt is produced,



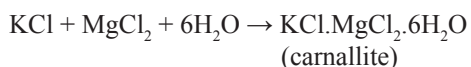
When dissolved in water, these salts ionize and produce ions in solution. Depending upon the extent of neutralization of the acid or base, these may be normal, acid or basic salts. Mixed salts contain more than one acidic or basic radicals, e.g., NaKSO_4 .

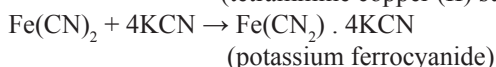
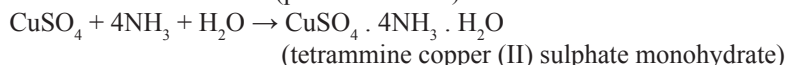
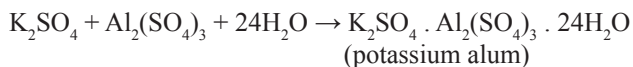
When solutions containing two or more salts in stoichiometric (i.e., simple molecular) proportions are allowed to evaporate, we get crystals of compounds known as molecular or addition compounds. These are of two types depending on their behaviour in aqueous solution.



Examples of Addition Compounds

Addition compounds are formed when stoichiometric amounts of two or more stable compounds join together. For example:





<i>Double salts or lattice compounds</i>	<i>Coordination compounds</i>
Those which lose their identity in solution (double salts)	Those which retain their identity in solution (complexes)
When crystals of carnallite are dissolved in water, the solution shows the properties of K^+ , Mg^{2+} and Cl^- ions. In a similar way, a solution of potassium alum shows the properties of K^+ , Al^{3+} and SO_4^{2-} ions. These are both examples of double salts which exist only in the crystalline state.	When the other two examples of coordination compounds dissolve, they do not form simple ions – Cu^{2+} , or Fe^{2+} and CN^- – but instead their complex ions remain intact. Thus the cuproammonium ion $[\text{Cu}(\text{H}_2\text{O})_2(\text{NH}_3)_4]^{2+}$ and the ferrocyanide ion $[\text{Fe}(\text{CN})_6]^{4-}$ exist as distinct entities both in the solid and in solution. Complex ions are shown by the use of square brackets.

- Double salts and mixed salts. Mohr salt, $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ is a double salt. Bleaching powder, CaOCl_2 is a mixed salt. It is a salt of Ca^{2+} in which the anions present are Cl^- and OCl^- . Similarly, NaKSO_4 is a mixed salt in which there are two cations Na^+ and K^+ and one anion SO_4^{2-} .

Definitions: Complex, Ligand and Coordination Number

Basic definition

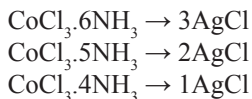
Werner's Theory

In his coordination theory, Werner introduced two great ideas:

- In a coordination complex, a metal ion is engaged in strong binding with a certain number of neutral and/or anionic groups in the first sphere of attraction, now called the coordination zone or sphere.
- The metal ion is surrounded by the neutral and/or anionic groups in the coordination zone in a definite geometrical arrangement.

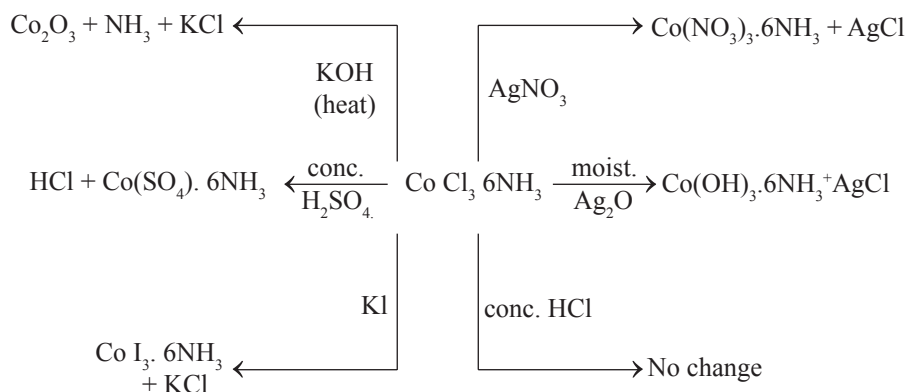
Werner was able to explain the nature of bonding in complexes, and he concluded that in complexes the metal shows two different sorts of valency:

- Primary valencies:** These are non-directional. The modern explanation would be as follows. The complex commonly exists as a positive ion. The primary valency is the number of charges on the complex ion. In compounds, this charge is matched by the same number of charges from negative ions. Primary valency applies equally well to simple salts and to complexes. Thus, in CoCl_2 ($\text{Co}^{2+} + 2\text{Cl}^-$) there are two primary valencies, i.e., two ionic bonds. The complex $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ actually exists as $[\text{Co}(\text{NH}_3)_6]^{3+}$ and 3Cl^- . Thus, the primary valency is 3, as there are three ionic bonds.
- Secondary valencies:** These are directional. In modern terms, the number of secondary valencies equals the number of ligand atoms coordinated to the metal. This is now called the coordination number. Ligands are commonly negative ions such as Cl^- , or neutral molecules such as NH_3 . Less commonly, ligands may be positive ions such as NO^+ . Each metal has a characteristic number of secondary valencies. Thus in $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$, the three Cl^- are held by primary valencies. The six NH_3 groups are held by secondary valencies.



Formulae of some cobalt complexes

<i>Old</i>	<i>New</i>
$\text{CoCl}_3 \cdot 6\text{NH}_3$	$[\text{Co}(\text{NH}_3)_6]^{3+} 3\text{Cl}^-$
$\text{CoCl}_3 \cdot 5\text{NH}_3$	$[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+} 2\text{Cl}^-$
$\text{CoCl}_3 \cdot 4\text{NH}_3$	$[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+ \text{Cl}^-$



Ligands

The neutral molecules or ions (usually anions) which are linked directly with the central metal atom/ion are called ligands. In most of the complexes, ligands act as donor of one or more lone pairs to the central metal atom/ion. It should be noted that in metallic carbonyls, the ligand, CO, acts as both donor and acceptor ($\text{M} \rightleftharpoons \text{CO}$).

Coordination number (or liganacy)

Number of sigma coordinate bonds attached to the central metal ion or atom. For example, in $[\text{Fe}(\text{CN})_6]^{4-}$, the coordination number of Fe^{2+} ion is 6.

Coordination sphere

The central metal ion and the ligands that are directly attached to it, are enclosed in a square bracket, called coordination sphere or first sphere of attraction.

Effective atomic number (Ean)

Sidgwick extended the Lewis theory to account for the bonding in the coordination compounds. He introduced the term coordinate bond for a shared electron pair, if it initially belonged to one atom (donor atom) only. In this case, the donor atom acts as a Lewis base and the metal ion acts as a Lewis acid. The metal ion accepts the electron pairs till it achieves the next inert gas configuration. This is called the effective number rule.

The total number of electrons, which the central metal atom appears to possess in the complex, including those gained by it in bonding, is called effective atomic number of central metal ion. When the EAN was 36 (Kr), 54 (Xe) or 86(Rn), the EAN rule was said to be followed.

For example, in $[\text{Co}(\text{NH}_3)_6]^{3+}$, cobalt has an atomic number 27. In Co^{3+} number of electrons is 24. Each ammonia molecule donates a pair of electrons. So, EAN becomes $24 + (2 \times 6) = 36$.

In many cases, it was found EAN in a complex should be equal to number of electrons present in next noble gas.

There are exceptions as well. For example,

EAN of $[\text{Ni}(\text{NH}_3)_6]^{+2}$ is 38 and $[\text{Cr}(\text{NH}_3)_6]^{3+}$ is 33.

Table 11.1 The EAN of Metals in Some Metal Complexes

Metal complex	Atomic number of metal	Electrons on metal ion	Electrons donated by the ligands	EAN
$[\text{Co}(\text{NO}_2)_6]^{2+}$	Co (27)	24	$6 \times 2 = 12$	$24 + 12 = 36$
$[\text{Cd}(\text{NH}_3)_4]^{2+}$	Cd(48)	46	$4 \times 2 = 8$	$46 + 8 = 54$
$[\text{PtCl}_6]^{2-}$	Pt(78)	74	$6 \times 2 = 12$	$74 + 12 = 86$
$[\text{Cr}(\text{CO})_6]$	Cr(24)	24	$6 \times 2 = 12$	$24 + 12 = 36$
$[\text{Ni}(\text{CO})_4]$	Ni(28)	28	$4 \times 2 = 8$	$28 + 8 = 36$

(Contd)

<i>Metal complex</i>	<i>Atomic number of metal</i>	<i>Electrons on metal ion</i>	<i>Electrons donated by the ligands</i>	<i>EAN</i>
$[\text{Ag}(\text{NH}_3)_2]\text{Cl}$	Ag (47)	46	$2 \times 2 = 4$	$46 + 4 = 50$
$\text{K}_4[\text{Fe}(\text{CN})_6]$	Fe (26)	24	$6 \times 2 = 12$	$24 + 12 = 36$
$[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$	Cu (29)	27	$4 \times 2 = 8$	$27 + 8 = 35$
$\text{K}_2[\text{Ni}(\text{CN})_4]$	Ni (28)	26	$4 \times 2 = 8$	$26 + 8 = 34$
$\text{K}_2[\text{PtCl}_6]$	Pt (78)	74	$6 \times 2 = 12$	$74 + 12 = 86$
$\text{K}_3[\text{Cr}(\text{C}_2\text{O}_4)_3]$	Cr (24)	21	$6 \times 2 = 12$	$21 + 12 = 33$
$\text{K}_2[\text{HgI}_4]$	Hg (80)	78	$4 \times 2 = 8$	$78 + 8 = 86$

- Use of Sidgwick theory or Effective number rule to calculate number of ligands. This rule can be used, to some extent, to calculate the number of ligands (especially CO) attached to a metal atom or ion. This is because carbonyls, in general, obey E.A.N. rule. This application is illustrated by the following examples.
- Calculate the number of CO ligands that can be attached to Ni, [Atomic Number of Ni = 28]. Atomic Number of next noble gas, Kr = 36

∴ Number of electrons to be provided by ligands = $36 - 28 = 8$

As each CO provides two electrons, number of CO groups that will be attached to Ni = $8/2 = 4$

∴ Formula of complex compound formed – $[\text{Ni}(\text{CO})_4]$

(ii) Calculate the value of (x) in the carbonyls $\text{H}_x\text{Cr}(\text{CO})_5$ [At. no. of Cr = 24]

Electrons from 5 CO groups = $5 \times 2 = 10$

At. number of Kr (next noble gas) = 36

∴ Number of (1 electrons to be provided by H-atoms, $x = 36 - (24 + 10) = 2$

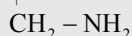
∴ Formula of the complex compound formed is $\text{H}_2\text{Cr}(\text{CO})_5$.

Worked-out Examples

1. The coordination number and the oxidation state of the element 'E' in the complex $[\text{E}(\text{en})_2(\text{C}_2\text{O}_4)]\text{NO}_2$ (where (en) is ethylene diamine) are, respectively (AIEEE 2008)

(a) 6 and 2 (b) 4 and 2 (c) 4 and 3 (d) 6 and 3

Ans. en, $\text{CH}_2 - \text{NH}_2$ is bidentate ligand



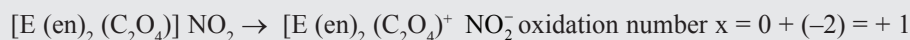
$\text{C}_2\text{O}_4^{2-}$ is also bidentate ligand

Hence, coordinating head are six

(en: $2 \times 2 = 4 + (\text{ox}) 2$)

Coordination number 6

Complex can be ionized as,

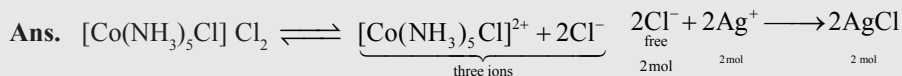
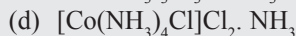
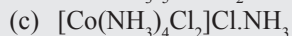


2. The coordination number of central metal atom in a complex is determined by

(a) the number of ligands around a metal ion bonded by sigma bonds
 (b) the number of ligands around a metal ion bonded by pi bonds
 (c) the number of ligands around a metal ion bonded by sigma and pi bonds both
 (d) the number of only anionic ligands bonded to the metal ion

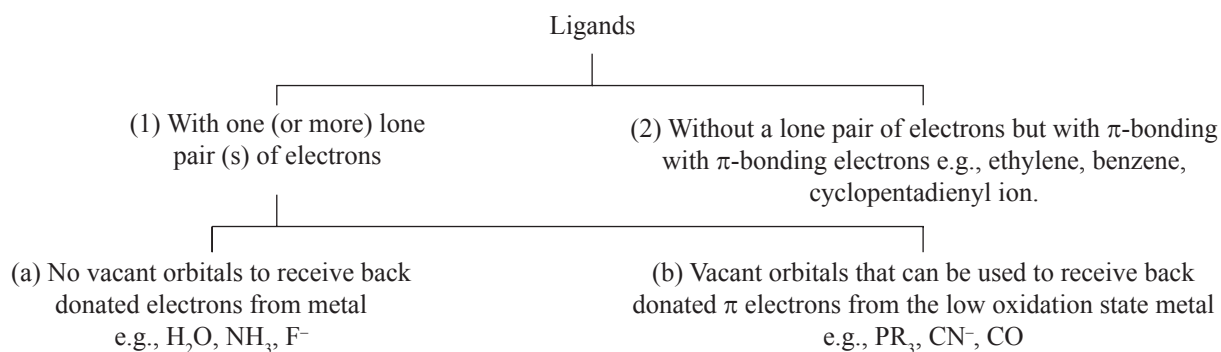
Ans. Coordinate number is the maximum covalency shown by a metal or metal ion. It is the maximum number of ligands attached to metal by sigma or coordinate bonds.

3. One mole of the complex compound $\text{Co}(\text{NH}_3)_4\text{Cl}_2$, gives 3 moles of ions on dissolution in water. One mole of the same complex reacts with two mole of AgNO_3 solution to yield two moles of $\text{AgCl}(\text{s})$. The structure of the complex is (AIEEE 2003)

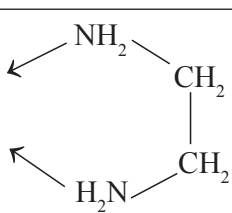
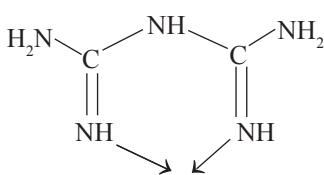
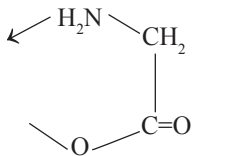


Classifications of ligands

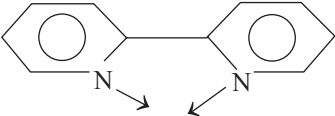
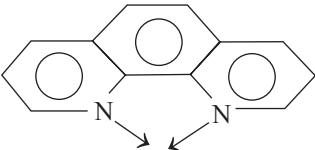
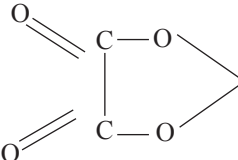
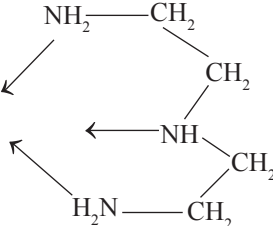
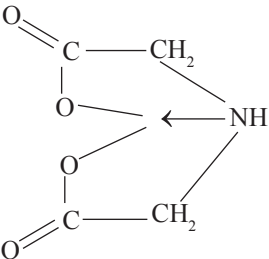
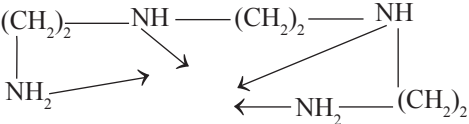
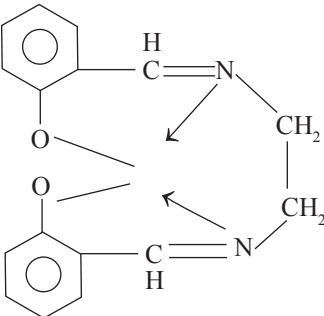
Based on donor and acceptor properties, ligands may be of the following types:



Ligands may also be classified according to the number of coordination positions they occupy around the metal ion. A ligand having only one lone pair of electrons can only be bonded to a metal ion at just one stereochemical point, that is, it occupies one coordination position only. This is called a unidentate or a monodentate ligand. When the ligand can occupy two positions around the metal ion it is di(bi) dentate; with three, four, five or six such links, the ligands are called tri (ter) denate, (tetra) quadridentate, penta (qunique) dentate or (hexa) sexa dentate respectively.

Ligand type	Name	Formula
Didentate	Ethylenediamine (en)	
	Biguanide (Hbig)	
	Glycinate ion (gly)	

(Contd)

Ligand type	Name	Formula
	Bipyridine (bpy)	
	o-Phenanthroline (o-phen) (phen)	
Tridentate (Terdentate)	Oxalate ion (ox)	
	Diethylenetriamine (dien)	
	Iminodiacetate ion (ida)	
Tetradentate (Quadri- dentate)	Triethylenetetramine (trine)	
	Ethylenebis (salicyladimine) anion (salen)	

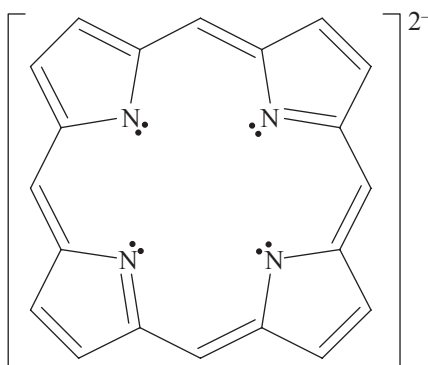
Some authors prefer to classify ligands according to the number of donor atoms they can offer to metal ions. Thus, when a ligand has two atoms which can simultaneously serve as donors it is called didentate and so on. It follows that a didentate ligand has to have two lone pairs, a tridentate three lone pairs and so on. The polydentate ligands may be further subdivided according to the nature of their donor centres. Thus, ethylenediamine is a didentate ligand with two neutral donors (N) whereas, oxalate ion is didentate with two acidic (anionic) donors (O^-). Glycinate ion is again a didentate with one neutral donor (N) and an acidic donor (O^-). Such discussions may be extended to other polydentates as well.

A chelating ligand is one that contains two or more donor centres so disposed that they can simultaneously occupy more than one position around the same metal ion in the first sphere of coordination (A chelating ligand offers more than one sigma-electron pair donor groups to the same metal ion.) The resulting complexes are called metal chelates. Thus all polydentate ligands, when bonded to the same metal ion, are also chelating ligands. A chelating ligand must possess two or more lone pairs such distance apart that it may form suitable strain-free rings at the metal ion. Too small-sized and too large-sized rings are unsuitable for stable binding. Usual strain-free rings are 5— or 6— membered ones. 4— and 7— membered chelate rings are also known but are less stable than 5— or 6— membered ones.

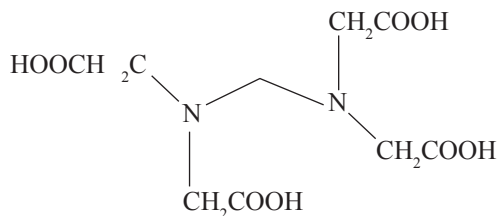
A bridging ligand is one that can simultaneously bond itself to more than one metal ion. OH , Cl^- , NH_2^- etc., can function as monodentate and also as bridging ligands. A bridging ligand must have at least two lone pairs so that it can get linked to at least two metal ions. Hydrazine, NH_2-NH_2 , can serve as a monodentate and as a bridging ligand but not as a chelating ligand because a 3-membered ring will be too strained to be stable.

A flexidentate ligand is one that can chelate a metal ion in more than one way. For example cysteine, $HS-CH_2-CH(NH_2)-COOH$ can function as a tridentate (SNO) and as a didentate in three ways again (SN; NO; SO).

The most important tetradentate ligands are the many 'biological' macrocyclic ligands, such as porphyrin and its derivatives and a host of similar molecules that can be readily synthesised.



Hexadentate ligands: The most common example for a hexadentate ligand is the EDTA (ethylene diamine tetra acetic acid).

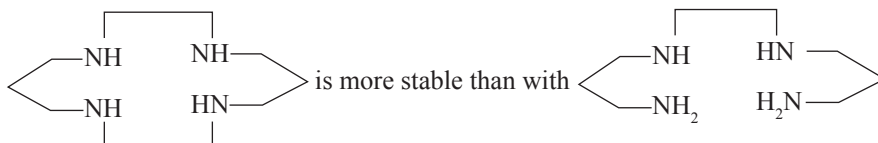


Edta

- Flexidentate character: Polydentate ligands are said to have flexidentate character if they do not use all its donor atoms to get coordinated to the metal ion e.g., EDTA generally act as a hexadentate ligand but it can also act as a pentadentate and tetradentate ligand.
- Certain polydentat ligands have flexidentate character e.g., EDTA generally act as a hexadentate ligand but it can also act as a pentadentate or a tetradentate ligand.

11.8 Co-ordination Chemistry

- In complex compounds, the central metal atom or ion can act as Lewis acid and ligands as Lewis bases.
- Transition metal can form complexes in the zero, positive and low negative oxidation states i.e., the centre of co-ordination can be a neutral atom, cation or an anion in low negative oxidation state.
- Chelating ligands is a neutral molecule, free radical or ion (cation or anion) with two or more lonepairs of electrons on different atoms.
- Chelates with five membered rings are very stable if these do not contain a double bond e.g., in ethylenediamine. However, chelates with six membered rings are very stable if the ligands contain double bond e.g., acac.
- Macrocyclic effect: This term refers to the greater thermodynamic stability of a complex with a cyclic polydenate ligand when compared to the complex formed with a non-cyclic ligand.
e.g., Zn(II) complex with ligand;

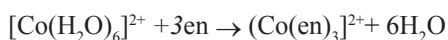


- π -acid ligand: Ligands which are capable of accepting an appreciable amount of π -electron density from the metal atom into empty π or π^* orbital of their own are called π -acceptor or π -acid ligands e.g., CO.
- Certain ligands like NO can act as a neutral as well as unipositive (NO^+) ligand. In such case, it is very difficult to find the oxidation state and the charge on the ligand from the given formula of the complex. For example, in the complex $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]^{2+}$ Fe is in +1 oxidation state with five H_2O neutral ligands and one Unipositive NO^+ ligand. During the formation of this complex, NO gives its unpaired electron to Fe^{2+} and change it into Fe^+ . Another example of this type is the complex nitroprusside ion, $[\text{Fe}(\text{CN})_5\text{NO}]^{2-}$, in which Fe is in +2 oxidation state with five CN^- ligands and one NO ligand.

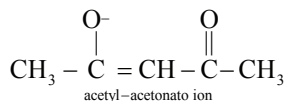
Similarly, NO_2 can be a uninegative ligand (nitrito $\text{O}^- - \text{N} = \text{O}$ or $\text{N}^- = \text{O} - \text{O}$) or a positive ligand (NO_2^+).

In such cases, the experimentally observed magnetic moment of the complex in solid state can decide the oxidation state of the central metal ion. From it, nature of the ligand can be decided.

- Chelating rings form more stable complexes than monodentate ligands. This is because replacing monodentate ligands by chelating ring is a process in which entropy increases e.g.,



- Ligands with more than six donor atoms or larger groups form unstable ring. This is due to the steric hinderance.
- Chelates with five membered rings including the metal atom are more stable if these do not contain a double bond. On the other hand chelates with six membered rings are more stable if the ligands contain conjugated double bonds e.g., acetyl acetonato complexes.



- The central atom in a complex compound or ion can be even a non-metal e.g., in $\text{Ba}[\text{BrF}_4]_2$ central atom in complex ion is Br.

NOMENCLATURE OF COORDINATION COMPOUNDS

The following rules are used when naming coordination compounds.

- If a coordination compound is ionic the name of cation is given first followed by the name of anion. The names of cation and anion are separated by a space.
- Within a complex ion, ligands are named first followed by the metal ion. And no space are left between parts of the name of same coordination entity.

- Ligand names are listed in alphabetical order (numerical prefixes indicating the number of ligands are not considered in determining that order).

Example: 1. $[\text{CoCl}(\text{NH}_3)_5]\text{Cl}_2$ (pentaamminechloridocobalt(2+) chloride)

2. $[\text{AuXe}_4]^{2+}$ (tetraxenonidogold(2+))

- The oxidation number of the central atom is shown by Roman numeral in bracket immediately following its name but only if the oxidation state can be defined without ambiguity.
- Alternatively, the charge on a coordination entity may be indicated in Arabic numbers with the number preceding the charge sign and enclosed in bracket.

Example: $\text{K}_4[\text{Fe}(\text{CN})_6]$

potassium hexacyanidoferrate(II), or

potassium hexacyanidoferrate(4-), or tetrapotassium hexacyanidoferrate

- Complex positive ions and neutral coordination compound have no special ending but complex negative ions always end in the suffix *-ate*. In most of the cases, the suffix *-ate* is fixed to English names of the metals but in some cases *-ate* is fixed with Latin names of the metal.

Example: $\text{K}_3[\text{Fe}(\text{CN})_5\text{NO}]$ Potassium pentacyanonitrosylferrate(II)

Continuum

<i>Element</i>	<i>Metal as named in anionic complex</i>
Cobalt	Cobaltate
Nickel	Nickelate
Chromium	Chromate
Iron	Ferrate
Copper	Cuprate
Silver	Argenate
Lead	Plumbate

Note: In the case of mercury, we may use both mercurate and hydrazinate.

- The name of the neutral coordination compound is given in one word only as the name of $\text{Ni}(\text{CO})_4$ is tetracarbonylnickel (O)
- Name of the Ligands
 - Anionic ligands ending with 'ide' 'ite' or 'ate' are named by replacing the final 'e' by 'O' giving 'ido', 'ito' and 'ato' respectively.

Example:

Formula	Ligand name
1. Cl^-	chlorido
2. CN^-	cyanido
3. H^-	hydrido
4. D^- or $^2\text{H}^-$	deuterido or $[\text{D}]\text{hydrido}$
5. $\text{PhCH}_2\text{CH}_2\text{Se}^-$	2-phenylethaneselenolato
6. MeCOO^-	acetato or ethanoato
7. Me_2N	dimethylazanido or dimethylamido
8. MeCONH_2	acetamide (not acetamido)
9. MeCONH^-	acetylazanido or acetylamido (not acetamido)
10. MeNH_2	methanamine
11. MeNH^-	methylazanido or methanaminido
12. MePH_2	methylphosphane
13. MePH^-	methylphosphanido
14. $\text{MeOS}(\text{O})\text{OH}$	methyl hydrogen sulfite
15. $\text{MeOS}(\text{O})\text{O}^-$	methyl sulfito, or methoxidodioxidosulfato(1-)

Some Noteworthy Points

- (i) In its complexes, hydrogen is always treated as anionic and named as Hydrido.
- (ii) Names of neutral and cationic ligand names are used without modification (even if they carry 'ide', 'ite' or 'ate')
- (iii) Prefixes di, tri, etc., are generally used for simple ligands. Enclosing marks are not required.
- (iv) Prefixes bis, tris, tetrakis, etc., are used with complex expressions and when required to avoid ambiguity. Enclosing marks be placed around the multiplicand.

Example: For example, one would use diammine, for $(\text{NH}_3)_2$, but bis(methylamine), for $(\text{NH}_2\text{Me})_2$.

Formulae of coordination compounds

The following rules are used when formulating coordination compounds.

- (i) The central atom is listed first.
- (ii) The ligands are then listed in alphabetical order. If an abbreviation is used in a formula to represent a ligand, the first letter of the abbreviation is used to determine the position of the ligand in the alphabetical list.
- (iii) The formula for the entire coordination entity, whether charged or not, is enclosed in square brackets. When ligands are polyatomic, their formulae are enclosed in parentheses. Ligand abbreviations are also usually enclosed in parentheses.
- (iv) There should be no space between representations of ionic species within a coordination formulae.
- (v) If the formula of a charged coordination entity is to be written without that of the counter-ion, the charge is indicated outside the square bracket as a right superscript, with the number before the sign.
- (vi) The oxidation number of a central atom may be represented by a Roman numeral, which should be placed as a right superscript on the element symbol.

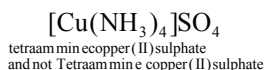
Some more examples

1. $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
hexaamminecobalt(III) chloride
2. $[\text{CoCl}(\text{NH}_3)_5]\text{Cl}_2$
pentaamminechloridocobalt(2⁺) chloride
3. $[\text{CoCl}(\text{NH}_3)_4(\text{NO}_2)]\text{Cl}$
tetraamminechloridonitrito- κ N-cobalt(III) chloride
4. $[\text{PtCl}(\text{NH}_2\text{Me})(\text{NH}_3)_2]\text{Cl}$
diamminechlorido(methanamine)platinum(II) chloride
5. $[\text{CuCl}_2\{\text{O}=\text{C}(\text{NH}_2)_2\}_2]$
dichloridobis(urea)copper(II)
6. $\text{K}_2[\text{PdCl}_4]$
potassium tetrachloridopalladate(II)
7. $\text{K}_2[\text{OsCl}_5\text{N}]$
potassium pentachloridonitridoosmate(2-)
8. $\text{Na}[\text{PtBrCl}(\text{NH}_3)(\text{NO}_2)]$
sodium amminebromidochloridonitrito- κ N-platinate(1₋)
9. $[\text{Fe}(\text{CNMe})_6]\text{Br}_2$
hexakis(methyl isocyanide)iron(II) bromide
10. $[\text{Co}(\text{en})_3]\text{Cl}_3$
tris(ethane-1,2-diamine)cobalt(III) trichloride

- Transition metals can form complexes both in zero, positive and also in low negative oxidation states. For example,



- The name of a complex compound should not start with a capital letter e.g.,



- The full name of the complex should be written as one word without any gap.
 - In optical isomers, the designation (+) and (-) or (*d*) and (?) are put before the complete formula of the complex compound and not before the complex ion showing optical isomerism
- e.g., (+) – $K_3 [Cr(C_2O_4)_3]$ and not $K_3 (+) - [Cr(C_2O_4)_3]$

Worked-out Examples

4. The IUPAC name for the complex $[Co(NO_2)(NH_3)_5]Cl_2$ is (AIEEE 2006)
 (a) nitrito-N-pentamminecobalt (III) chloride (b) nitrito-N-pentamminecobalt (II) chloride
 (c) pentammine nitrito-N-cobalt (II) chloride (d) pentammine nitrito-N-cobalt (III) chloride

Ans. (d)

5. The IUPAC name of the coordination compound $K_3 [Fe(CN)_6]$ is (AIEEE 2005)
 (a) tripotassium hexacyanorin (II) (b) potassium hexacyanoferrate (III)
 (c) potassium hexacyanoferrate (III) (d) potassium hexacyanoferrate (II)

Ans. $\underset{\text{cation}}{K_3} \underset{\text{anion}}{[Fe(CN)_6]}$

Oxidation state of Fe in anion = + 3

6. Write the IUPAC name for $[Cr(NH_3)_5CO_3]Cl$. [IIT 1996]

Ans. Pentaamminecarbonatochromium(III) chloride.

7. Write the formulae of the following complexes. [IIT, July 1997]
 (i) Pentamminechlorocobalt (III) ion (ii) Lithium tetrahydridoaluminate (III)

Ans. (i) $[Co(NH_3)_5Cl]^{2+}$; (ii) $Li[AlH_4]$

8. Write the IUPAC name of this compound $[Cr(NH_3)_5(NCS)][ZnCl_4]$. Is this compound coloured? [IIT, May, 1997]

Ans. Pentammineisothiocyanatochromium (III) tetrachlorozincate (II). Compound will be coloured because Cr (III) has d^3 configuration and so d–d transition is possible.

9. Write IUPAC names of the following compounds: [IIT Quali. 1995]
 (i) $[Co(NH_3)_5ONO]Cl_2$; (ii) $K_3[Cr(CN)_6]$

Ans. (i) Pentaamminenitrito-O-cobalt (III) chloride.
 (ii) Potassium hexacyanochromate (III)**The Valence Bond Theory****Introduction**

The valence bond theory deals with the electronic structure of the central metal ion in its ground state, kind of bonding, geometry and magnetic properties of the complexes. This theory takes into account the hybridization of vacant orbitals of central metal ion and was proposed by Linus Pauling, using hybridized orbitals. The main points of the valence bond theory are as follows:

- The central metal loses requisite number of electrons and forms the cation. The number of electrons lost corresponds to the valency of the resulting cation.
- The central metal ion makes available a number of empty s, p and d atomic orbitals equal to its coordination number.
- These vacant orbitals hybridise together to form hybrid orbitals which are the same in number as the atomic orbitals hybridizing together. These hybrid orbitals are vacant, equivalent in energy and have a definite geometry.

- The non-bonding metal electrons occupies the inner orbitals and they do not take part in the hybridization. The electrons are grouped in accordance with the Hund's rule of maximum multiplicity. However, under the influence of a strong ligand, they may be forced to pair up against the Hund's rule.
 - The d-orbitals involved in the hybridization may be either inner $(n - 1)$ d-orbitals or outer nd-orbitals.
 - Each ligand (donor group) must contain a lone pair of electrons.
 - Vacant hybrid orbitals of the metal atom or ion overlap with the filled (for example, containing lone-pair of electrons) σ -orbitals of the ligands to form a covalent σ -bond represented as $M \xleftarrow{\sigma} L$. The bond is also referred to as coordinate bond.
 - In addition to the σ -bond, a π -bond may be formed by overlap of a filled metal d-orbital with a vacant ligand orbital (M L). This usually happens in complexes of metal ions of low oxidation states.
 - If the complex contains unpaired electrons, the complex is paramagnetic in nature, whereas, if it does not contain any unpaired electron, the complex is diamagnetic in nature.
- Let us consider a few examples to explain the valance bond theory over the complexes.

VBT-octahedral complexes

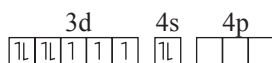
On the basis of VBT, octahedral complexes are of two types:

1. **Inner orbital octahedral complexes**, which result from d^2sp^3 hybridization of the central metal atom/ion.
2. **Outer orbital octahedral complexes**, which result from sp^3d^2 hybridization.

(a) Inner orbital octahedral complexes

The formation of these complexes can be explained on the basis of VBT by considering the complex ion, viz., $[\text{Co}(\text{NH}_3)_6]^{3+}$.

(a) Co atom
($3d^7 4s^2 4p^0$)



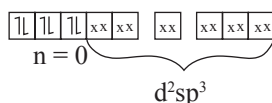
(b) Free Co^{3+} ion
($3d^6 4s^0 4p^0$)



(c) Cr^{3+} ion in $[\text{Cr}(\text{NH}_3)_6]^{3+}$



(d) $[\text{Co}(\text{NH}_3)_6]^{3+}$

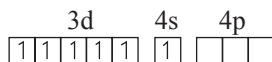


Here n represents the number of unpaired electrons and 'XX' represents an electron pair donated by each of free six NH_3 ligands. The two electrons of the electron pair have opposite spin. The above complex ion is diamagnetic as all the electrons are paired.

In order to make 3d electrons paired, the two unpaired electrons residing in $3d_{z^2}$ and $3d_{x^2-y^2}$ orbitals are forced by the six NH_3 ligands to occupy $3d_{yz}$ and $3d_{zx}$ orbitals. By doing so, all the 3d electrons become paired and also at the same time, two 3d orbitals namely $3d_{z^2}$, $3d_{x^2-y^2}$, hybridise together with 4s, $4p_x$, $4p_y$ and $4p_z$ orbitals to give d^2sp^3 hybrid orbitals which being empty, accepts the six electron pairs donated by six NH_3 ligand molecules.

$[\text{Cr}(\text{NH}_3)_6]^{4+}$.

(a) Cr atom
($3d^5 4s^1$)



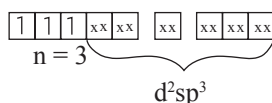
(b) Free Cr^{3+} ion ($3d^3 4s^0 4p^0$)
in ground state



(c) Cr^{3+} ion in $[\text{Cr}(\text{NH}_3)_6]^{3+}$



(d) $[\text{Cr}(\text{NH}_3)_6]^{3+}$



The above complex ion is paramagnetic as there are three unpaired electrons.

$$\mu = \sqrt{n(n+2)} = \sqrt{3(3+2)} = \sqrt{15} \text{ B.M.}$$

[Cr(NH)₆]⁴⁺.

(a) Cr atom

(3d⁵ 4s¹)



(b) Cr²⁺ ion

(3d⁴) in ground state

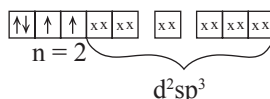


(c) Cr²⁺ ion in

[Cr(CN)₆]⁴⁺



(d) [Cr(CN)₆]⁴⁺



The above complex ion is paramagnetic since two unpaired electrons are present.

$$\mu = \sqrt{2(2+2)} = \sqrt{8} \text{ B.M.}$$

Other examples of inner orbital octahedral paramagnetic complexes are [Ti(H₂O)₆]³⁺ (n=1), [Mn(CN)₆]⁴⁻ (n = 1), [Mn(CN)₆]³⁻ (n = 2), [Fe(CN)₆]³⁻ (n = 1), [Co(CN)₆]⁴⁻ (n = 1 in 5s-orbital). While the examples of inner orbital octahedral diamagnetic complexes are: [Fe(CN)₆]⁴⁻, [Co(CN)₆]³⁻, [Co(H₂O)₆]³⁺, [Co(NH₃)₆]³⁺, [Co(NO₂)₆]³⁻, [Pt(NH₃)₆]⁴⁺ etc.,

All these complexes result from d²sp³ hybridization, using outer d and outer s and p-orbitals are called outer orbital octahedral complexes.

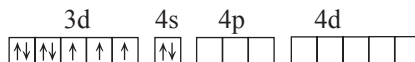
(b) Outer orbital octahedral complexes

Octahedral complexes resulted from sp³d² hybridization, using outer d and outer s and p-orbitals are called outer orbital octahedral complexes.

[CoF₆]³⁻.

(a) Co atom

(3d⁷ 4s²)



(b) Co³⁺ ion

in ground state

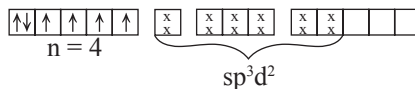


(c) Co³⁺ ion in

[CoF₆]³⁻ ion in excited state



(d) [CoF₆]³⁻ ion



In this complex ion, it is 4d_{x²-y²} and 4d_{z²} orbitals that mix with one 4s and three 4p orbitals to give six sp³d² hybrid orbitals, which being empty, accept the six electron pairs denoted by each of the six F⁻ ligands. It is paramagnetic as there are four unpaired electrons.

$$\mu = \sqrt{n(n+2)} = \sqrt{4(4+2)} = \sqrt{24} \text{ B.M.}$$

Some other examples of outer orbital octahedral paramagnetic complex are:

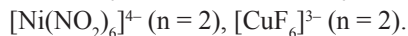
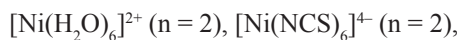
[Cr(H₂O)₆]²⁺ (n = 4), [Mn(H₂O)₆]²⁺ (n = 5), [Fe(H₂O)₆]²⁺ (n = 4), [Fe(NH₃)₆]²⁺ (n = 4),

[Fe(H₂O)₆]³⁺ (n = 5), [Fe(F)₆]³⁻ (n = 5), [CoF₆]³⁻ (n = 4),

[Co(NH₃)₆]²⁺ (n = 3),

[Co(H₂O)₆]²⁺ (n = 3), [Ni(NH₃)₆]²⁺ (n = 2),

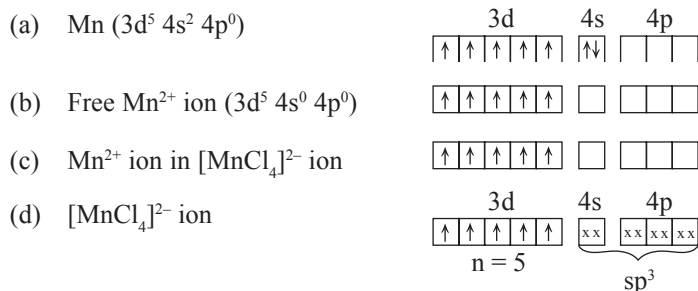
11.14 Co-ordination Chemistry



$[\text{Zn}(\text{NH}_3)_6]^{2+} (n = 0)$ is an example of outer orbital octahedral diamagnetic complex

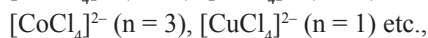
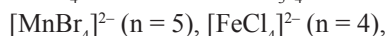
VBT-tetrahedral complexes (coordination no. = 4, sp^3 hybridization)

Tetrahedral complexes result from sp^3 hybridization. In sp^3 hybridization, the s and three p-orbitals should belong to the same shell. The formation of tetrahedral complexes by VBT can be explained by considering the complex ion like $[\text{MnCl}_4]^{2-}$. This complex ion is paramagnetic corresponding to the presence of five unpaired electrons and hence the configuration of Mn^{2+} ion in the free state and in the complex ion remains the same.

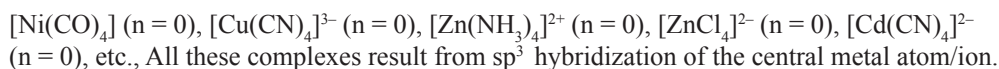


$$\mu = \sqrt{n(n+2)} = \sqrt{5(5+2)} = \sqrt{35} \text{ B.M.}$$

Examples of some paramagnetic tetrahedral complexes are:

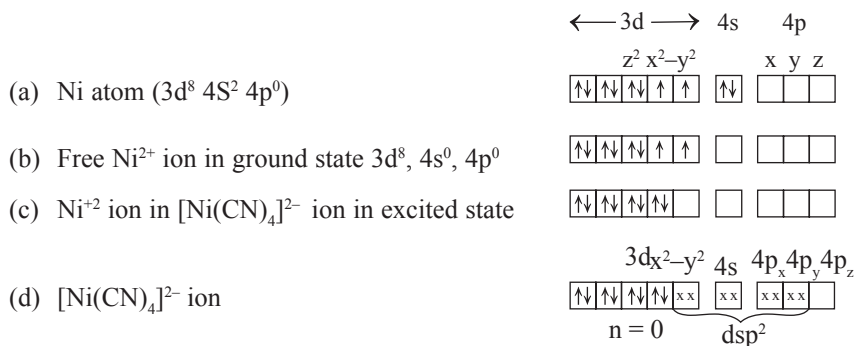


While the examples of some diamagnetic tetrahedral complexes are:



VBT-square planar complexes (coordination no. = 4, dsp^2 hybridization)

Square planar complexes result from dsp^2 hybridization. In dsp^2 hybridization, d-orbital should be $d_{x^2-y^2}$ orbital (belonging to the lower shell) while s and p orbitals should be from the higher shell. The two p-orbitals should be p_x and p_y orbitals. The selection of $d_{x^2-y^2}$, p_x and p_y orbitals is based on the fact that all these orbitals lie in the same plane. The formation of square planar complexes by VBT can be explained by considering the complex ion like $[\text{Ni}(\text{CN})_4]^{2-}$. The measurement of magnetic moment value for $[\text{Ni}(\text{CN})_4]^{2-}$ ion has shown that $\mu = 0$ i.e., the complex ion has no unpaired electron and hence, it is diamagnetic.



In order to make all the 3d-electrons paired, one unpaired electron residing in $3d_{x^2-y^2}$ orbital is forced by the four CN^- ligand to occupy $3d_{z^2}$ orbital. Now, $3d_{x^2-y^2}$, $4s$, $4p_x$, $4p_y$ orbitals mix together to form four dsp^2 hybrid orbitals which being empty, accept the four electron pairs donated by the four CN^- ligand ions.

Examples of paramagnetic square planar complex are:

$[\text{Cu}(\text{CN})_4]^{2-}$ ($n = 1$), $[\text{Cu}(\text{NH}_3)_4]^{2+}$ ($n = 1$), $[\text{CuCl}_4]^{2-}$ derived from $(\text{NH}_4)_2[\text{CuCl}_4]$ ($n = 1$) etc., While the examples of diamagnetic square planar complexes are: $[\text{Ni}(\text{CN})_4]^{2-}$ ($n = 0$), $[\text{PtCl}_4]^{2-}$ ($n = 0$), $[\text{Pt}(\text{NH}_3)_4]^{2+}$ ($n = 0$) etc.,

- We can predict the magnetic moments of complexes on the basis of the number of unpaired electrons, n by using the formula,

$$\text{Magnetic moment, } \mu_{\text{spin only}} = \sqrt{n(n+2)} \text{ B.M.}$$

where B.M. stands for Bohr magnetons—unit for magnetic moment. For example,

Complex	No. of unpaired electrons	Predicted magnetic moment
$[\text{Cr}(\text{NH}_3)_6]^{3+}$	3	$\mu_{\text{s.a.}} = \sqrt{3(3+2)} \text{ B.M.} = \sqrt{15} \text{ B.M.} = 3.873 \text{ B.M.}$
$[\text{Fe}(\text{CN})_6]^{3-}$	1	$\mu_{\text{s.a.}} = \sqrt{1(1+2)} \text{ B.M.} = \sqrt{3} \text{ B.M.} = 1.732 \text{ B.M.}$
$[\text{Fe}(\text{CN})_6]^{4-}$	0	$\mu_{\text{s.a.}} = \sqrt{0(0+2)} = 0 \text{ B.M.}$
$[\text{CoP}_6]^{3-}$	4	$\mu_{\text{s.a.}} = \sqrt{4(4+2)} \text{ B.M.} = \sqrt{24} \text{ B.M.} = 4.9 \text{ B.M.}$
$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$	5	$\mu_{\text{s.a.}} = \sqrt{5(5+2)} \text{ B.M.} = \sqrt{35} \text{ B.M.} = 5.9 \text{ B.M.}$

However, in actual practice, this method is generally used to cross-check the state of hybridization predicted on the basis of the geometry of the complex obtained from X-ray diffraction studies.

Certain complexes involve special types of electron transfer before hybridization, as explained below.

- **$[\text{Cu}(\text{NH}_3)_4]^{2+}$:** Electronic configuration of Cu^{2+} : $[\text{Ar}]^{18}3d^9$. In this complex, one electron from 3d-subshell is promoted to 4p-subshell. The remaining eight 3d-electrons pair up in four 3d-orbitals followed by dsp^2 hybridization. The resulting complex is square planar with one unpaired electron in 4p-orbital. It is thus paramagnetic.
- **$[\text{Co}(\text{NH}_3)_6]^{2+}$:** Electronic configuration of Co^{2+} : $[\text{Ar}]^{18} 3d^7$. In this complex, one electron from 3d-subshell is promoted to 4p-subshell and the remaining six 3d-electrons pair up in three 3d-orbitals. This is followed by d^2sp^3 -hybridization. The resulting complex is octahedral with one unpaired electron in 4p-orbital. It is thus paramagnetic.
- **$[\text{Ni}(\text{CO})_4]$:** Electronic configuration of Ni: $[\text{Ar}]^{18} 4s^2 3d^8$. In this complex, the two 4s-electrons jump to 3d-subshell. The ten 3d-electrons pair up in five 3d-orbitals. This is followed by sp^3 -hybridization giving a tetrahedral complex with no unpaired electrons. The complex is as such diamagnetic.
- **All octahedral complexes of Ni^+ , Cu^+ , Cu^{2+} , Zn^{2+} and Co^{2+} are outer orbital complexes. Explanation.** Let us explain it by taking the example of Ni^+ complexes. Outer electronic configuration of Ni^{2+} ion is



As such, even if pairing takes place, inner octahedral complex is not possible. Similar explanation is possible in other cases.

- **If there are three electrons or fewer in $(n-1)$ d-subshell of central metal atom or ion**, the number of unpaired electrons is not effected by inner orbital bonding in octahedral complexes. **Explanation.** If there are three unpaired electrons in $(n-1)$ d-subshell, it leaves two $(n-1)$ d-orbitals empty. As such inner orbital bonding (involving d^2sp^3 -hybridization) is possible even without pairing.
- **Square planar complexes are possible** only in case of central metal atoms or ions with $(n-1)d^8$ or $(n-1)d^9$ configuration. In case of $(n-1)d^9$ configuration one of the electron is promoted to np-subshell before hybridization (e.g., in $[\text{Cu}(\text{NH}_3)_4]^{2+}$ complex). However, this generalization is not always true e.g., $[\text{NiCl}_4]^{2-}$ complex with d^8 configuration

is tetrahedral and not square planar. This may be due to the fact that Cl^- ion with a Weak ligand field is unable to pair up the unpaired d-electrons. On the other hand, $[\text{Ni}(\text{CN})_4]^-$ is square planar. In this case, pairing of electrons in $(n-1)$ d-subshell is possible due to CN^- ion with a strong ligand field.

Square planar complexes are always inner. A sp^2d -hybridization is not possible.

Limitations of VBT (Valence Bond Theory)

- Although, valence bond theory provides a satisfactory representation of the complex compound based upon the concept of orbital hybridization, it cannot account for the relative stabilities for different shapes and coordination numbers in metal complexes.
- VBT cannot explain as to why Cu^{+2} forms only distorted octahedral complexes even when all the six ligands are identical.
- The valence bond theory does not provide any satisfactory explanation for the existence of inner orbital and outer orbital complexes.
- Sometimes the theory requires the transfer of electron from lower energy to the higher energy level, which is very much unrealistic in absence of any energy supplier (for example, this happens in the case of $[\text{CuX}_4]^{-2}$).
- The changes in the properties of the metal ion along with the ligands and the simple metal ions cannot be explained. For example, the colour changes associated with electronic transition within d-orbitals are affected on formation of complex, but the valence bond theory does not offer any explanation.
- Sometimes the same metal acquires different geometry when formation of complex takes place with different ligands. The theory does not explain as to why at one time the electrons must be rearranged against Hund's rule while, at other times the electronic configuration is not disturbed.
- The energy change of the metal orbitals on formation of complex is difficult to be calculated mathematically.
- VBT fails to explain the finer details of magnetic properties including the magnitude of the orbital contribution to the magnetic moments.
- The VBT does not explain why certain complexes are more labile than the others.
- It does not give quantitative interpretation of thermodynamic or kinetic stabilities of coordination compounds.
- It does not make exact predictions regarding the tetrahedral and square planar structure of 4 coordinate complexes.
- It does not tell about the spectral properties of coordination compounds.
- The above points may be made clear with the help of the following examples.

Worked-out Examples

10. Which one of the following has a square planar geometry?

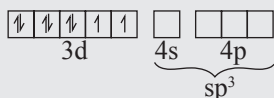
(AIEEE 2007)

(Atomic number Co = 27, Ni = Fe = 26, Pt = 78)

Ans. Cl^- is a weak ligand but Cl^- cause the pairing of electrons with large Pt^{2+} and consequently give dsp^2 hybridization and square planar geometry.

11. Nickel ($Z = 28$) combines with a uninegative monodentate ligand X^- to form a paramagnetic complex $[NiX_4]^{2-}$. The number of unpaired electron (s) in the nickel and geometry of this complex ion are, respectively **(AIEEE 2006)**

Ans. ${}_{28}\text{Ni} = [\text{Ar}] 4s^2, 3d^8$

$$\text{Ni}^{2+} = [\text{Ar}] 3d^8$$


Nickel has two unpaired electrons and geometry is tetrahedral due to sp^3 hybridization.

12. Which one of the following cyano complexes would exhibit the lowest value of paramagnetic behaviour?

(AIEEE 2005)

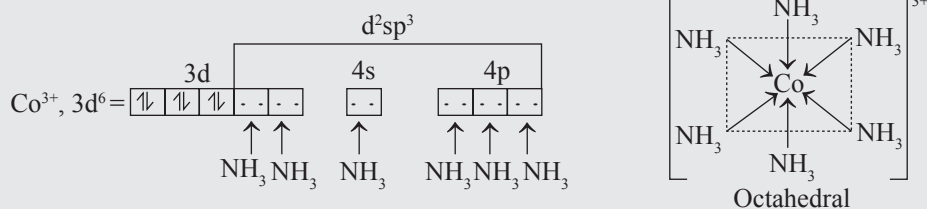
(Atomic number Cr = 24, Mn = 25, Fe = 26, Co = 27)

- (a) $[\text{Co}(\text{CN})_6]^{3-}$ (b) $[\text{Fe}(\text{CN})_6]^{3-}$ (c) $[\text{Mn}(\text{CN})_6]^{3-}$ (d) $[\text{Cr}(\text{CN})_6]^{3-}$

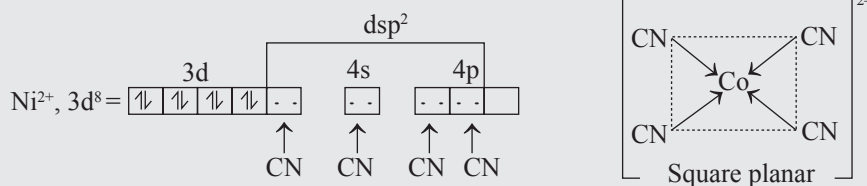
Ans. Hybridi	Unpaired electrons	Magnetic moment
(a) $[\text{Co}(\text{CN})_6]^{3-}$	d^2sp^3	0
(b) $[\text{Fe}(\text{CN})_6]^{3-}$	d^2sp^3	$\sqrt{3}$ BM
(c) $[\text{Mn}(\text{CN})_6]^{3-}$	d^2sp^3	$\sqrt{8}$ BM
(d) $[\text{Cr}(\text{CN})_6]^{3-}$	d^2sp^3	$\sqrt{15}$ BM

13. Draw the structures of $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{Ni}(\text{CN})_4]^{2-}$ and $\text{Ni}(\text{CO})_4$ and write the hybridization of each. [IIT 2000]

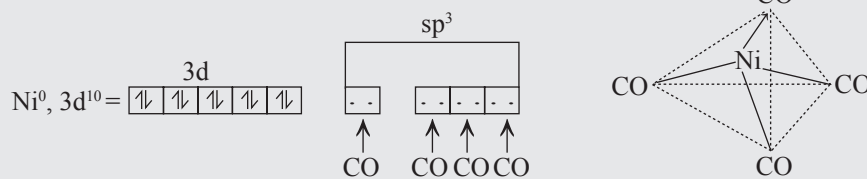
Ans. (i) $[\text{Co}(\text{NH}_3)_6]^{3+}$ = Octahedral; d^2sp^3



(ii) $[\text{Ni}(\text{CN})_4]^{2-}$ = Square planar; dsp^2

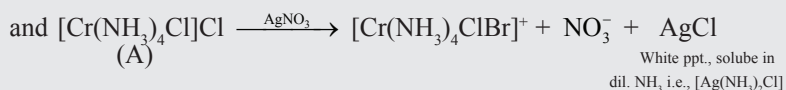
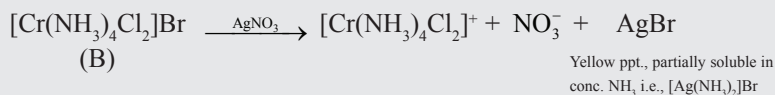


(iii) $[\text{Ni}(\text{CO})_4]$ = Tetrahedral; sp^3



14. A metal complex having composition $\text{Cr}(\text{NH}_3)_4\text{Cl}_2\text{Br}$ has been isolated in two forms (A) and (B). The form (A) reacts with AgNO_3 to give a white precipitate readily soluble in dilute aqueous ammonia, whereas, (B) gives a pale yellow precipitate soluble in concentrated ammonia. Write the formula of (A) and (B) and state the hybridization of chromium in each. Calculate their magnetic moments (spin-only value). [IIT 2001]

Ans. Complex $\text{Cr}(\text{NH}_3)_4\text{Cl}_2\text{Br}$ has two isomers. Since co-ordination number of Cr is six and thus two forms may be:



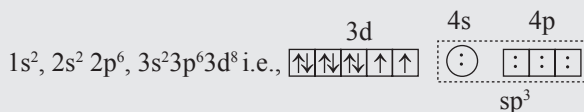
Hybridization of Cr in (A) and (B) is d^2sp^3 having 3 unpaired electrons ($3d^3$)

$$\text{Magnetic moment} = \sqrt{n(n+2)} \text{ B.M.} = \sqrt{3 \times (3+2)} = 3.87 \text{ B.M.}$$

15. Deduce the structures of $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$ considering the hybridization of the metal ion. Calculate the magnetic moment (spin only) of the species. [IIT 2002]

Ans. $[\text{NiCl}_4]^{2-}$: Ni in $[\text{NiCl}_4]^{2-}$ is sp^3 hybridized; Cl^- being weak ligand is not capable to pair up the electrons of Ni^{2+} .

Electronic configuration of Ni^{2+} :

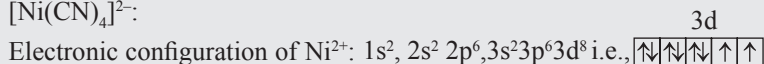


The electron pair from ligand Cl^- is provided to sp^3 hybridized orbitals of 4th shell of Ni^{2+} . Thus, $[\text{NiCl}_4]^{2-}$ is tetrahedral.

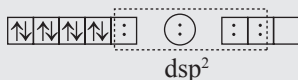
$$\text{Magnetic moment of } [\text{NiCl}_4]^{2-} = \sqrt{n(n+2)} = \sqrt{2 \times (2+2)} = \sqrt{8} \text{ B.M.} = 2.82 \text{ B.M.}$$

n is number of unpaired electrons = 2

$[\text{Ni}(\text{CN})_4]^{2-}$:



In presence of CN^- , a strong ligand, electronic configuration of Ni^{2+} changes to $\begin{array}{|c|c|c|c|} \hline \uparrow\downarrow & \uparrow\downarrow & \uparrow\downarrow & \uparrow\downarrow \\ \hline \end{array}$ i.e., pairing of 3d electrons takes place. In $[\text{Ni}(\text{CN})_4]^{2-}$, Ni^{2+} is dsp^2 hybridized to which 4 electron pairs from CN^- are provided.



Thus, $[\text{Ni}(\text{CN})_4]^{2-}$ is square planar having no unpaired electron i.e., magnetic moment is zero.

16. Write the IUPAC name of given complex along with its hybridization and structure.



[IIT 2003]

Ans. Potassium amminetetracyanonitrosylchromate (I), Cr is in +1 oxidation state and possesses d^2sp^3 hybridization with one unpaired electron.

$$\mu = \sqrt{n(n+2)} = \sqrt{1 \times (1+2)} = \sqrt{3} = 1.73 \text{ BM}$$

17. Identify (A) and (B) in the given Sequence of reaction. Also write their IUPAC names and calculate spin only magnetic moment of (B). [IIT 2005]



Ans. The given reaction sequence is:



(A) : Trithiocyanoferrum (III)

(B) : Hexafluoroferrate (III) ion

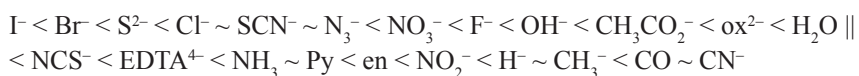
In $[\text{FeF}_6]^{3-}$, Fe^{3+} has configuration $3d^5$, F^- being weak field ligand and thus it can not pair up the electrons and $[\text{FeF}_6]^{3-}$ shows sp^3d^2 hybridization to possess octahedral shape with five unpaired electrons. Thus spin magnetic moment.

$$= \sqrt{n(n+2)} = \sqrt{5 \times (5+2)} = \sqrt{35} = 5.92 \text{ BM}$$

Crystal Field Theory (CFT)

In crystal field theory, we assume the ligands to be the point charges and there is interaction between the electrons of the ligands and the electrons of the central metal atom or ion. The five d-orbitals in an isolated gaseous metal atom or ion are degenerate. The degeneracy is maintained if an spherically symmetrical negative field surrounds the metal atom/ion. However, when ligands approach the central metal atom/ion, the field created is not exactly spherically symmetrical and the degeneracy of the d-orbitals is lifted. It results in the splitting of d-orbitals and the pattern of splitting depends upon the nature of the crystal field. This splitting of d-orbitals energies and its effects, form the basis of the crystal field treatment of the coordination compounds. It is important to record that the crystal field theory considers the ligand atoms as point charges or point dipoles, and does not consider any overlap between ligand orbitals and metal ion orbitals.

Ligands that cause large degree of crystal field splitting are termed as **strong field ligands**. Ligands that cause only a small degree of crystal field splitting are termed as **weak field ligands**. The common ligands can be arranged in ascending order of crystal field splitting energy. The order remains practically constant for different metals and this series is called the **spectrochemical series**.



The spectrochemical series is an experimentally determined series. It is difficult to explain the order as it incorporates both the effect of σ and π -bonding. The halides are in the order expected from electrostatic effects. In other cases, we must consider covalent bonding to explain the order. A pattern of increasing σ -donation is as follows:

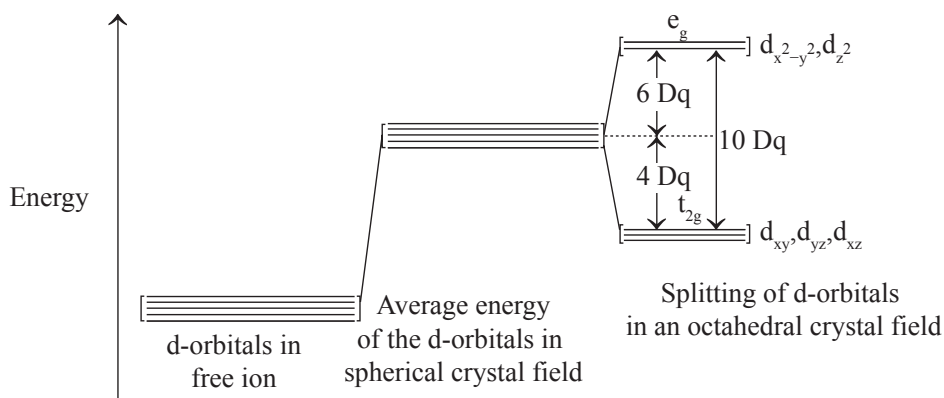
Halides donors < O donors < N donors < C donors

The crystal field stabilization produced by the strong CN^- is almost double than that of halide ions. This is attributing π -bonding in which the metal donates electrons from a filled t_{2g} orbitals into a vacant orbital on the ligand. In a similar way, many unsaturated N donors and C donors may also act as π -acceptors.

Crystal field effects in octahedral coordination entities

The five d-orbitals are degenerate in a field-free ion. The five d-orbitals are not all alike. The three orbitals d_{xy} , d_{xz} , d_{yz} have their four lobes concentrated in between the coordinate axes: for example, d_{xy} orbital has its four lobes between the x and y axes. The $d_{x^2-y^2}$ orbital has its lobes along the x and the y axes. The d_{z^2} orbital has two lobes along the z-axis and a concentric lobe all around the nucleus in the xy plane. When such a set of five d-orbitals is subjected to an octahedral crystal field the d-orbitals can no longer remain degenerate. Their overall energy is raised from that in the absence of any field, and they undergo splitting. Since, the $d_{x^2-y^2}$ and the d_{z^2} orbitals are face to face, with the six ligands they are now raised in energy whereas, the d_{xy} , d_{xz} , d_{yz} orbitals are lowered in energy. The splitting is so adjusted that the centre of gravity (baricentre) of the five degenerate d-orbitals in a uniformly smeared out field of six negative charges is maintained. The difference in energy of the orbitals d_{xy} , d_{xz} , d_{yz} (t_{2g} set) and the $d_{x^2-y^2}$, d_{z^2} orbitals (e_g set) is given as symbol Δ or $10 Dq$, this difference being a measure of the strength of the crystal field operating on the metal ion. Δ or $10 Dq$, is the separation between the t_{2g} , (d_{xy}) and the e_g ($d_{x^2-y^2}$) and is known as the *crystal field splitting parameter* or as the *crystal field strength*.

The energy separation is denoted by Δ_0 or $10 Dq$. (where Δ_0 stands for octahedral field), as a shown below,



Significance of Δ_o

A strong field ligand approaches the central metal atom/ion strongly and thus the magnitude of Δ_o is high. Hence, in the case of strong field ligand, the magnitude of Δ_o is greater than the pairing energy (the energy required to pair up two negatively charged electrons having opposite spin in an orbital). However, under the influence of weak field ligand, $\Delta_o < P$ (where P represents the pairing energy).

Now, let us consider the d^4 configuration of the central metal atom/ion. The first three electrons will go into t_{2g} orbitals using Hund's rule of maximum multiplicity. The fourth electron will go in the e_g orbital when the ligands are weak as, $\Delta_o < P$ giving the configuration $t_{2g}^3 e_g^1$. But if the ligands are strong, then the fourth electron will pair up with any of the singly occupied t_{2g} orbitals (as $\Delta_o > P$) to give the configuration $t_{2g}^4 e_g^0$.

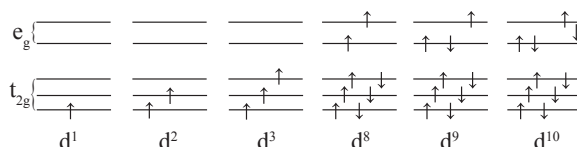
Since the pre-splitting centre of gravity is to be maintained, the e_g set will go up by say x and the t_{2g} set will go down by say y . If E is the energy of each electron prior to splitting it follows that:

$$10E = 4(E + x) + 6(E - y) \text{ so that } 2x = 3y$$

Also, since $x + y = \Delta$ we have, $x = \frac{3}{5} \Delta = 6Dq$ and $y = \frac{2}{5} \Delta = 4Dq$. Thus, each t_{2g} electron is stabilized by $-4Dq$ and each e_g electron is destabilized by $+6Dq$ with respect to the unsplit baricentre energy.

The symbol t_{2g} indicates a triplet orbital degeneracy i.e., a three fold degeneracy comprising in this case the orbitals d_{xy}, d_{yz}, d_{xz} . The symbol e_g speaks of a two fold degeneracy comprising in the present case the orbitals $d_{x^2-y^2}$ and d_{z^2} . The subscript g stands for 'gerade' or even. In an octahedral complex, the metal ion is at the centre of inversion, and with respect to this inversion centre, the d-orbitals maintain the same sign of their wave function (i.e., same sign on the lobes) on inversion. But a tetrahedron has no centre of inversion and hence the g subscripts are dropped in a tetrahedral crystal field. The subscript 2 has its origin in group theory and refers to some particular symmetry operations. It may be noted that p orbitals change their sign on inversion in octahedral field and hence they are 'ungerade' or uneven and carry the symbol 'u'.

The splitting of the d-orbitals has significant consequences on the colour and magnetic properties of transition metal complexes. The d-orbital electrons in a crystal field will now have a choice of either going to the t_{2g} or to the e_g . There will be two opposing forces in this regard. The crystal field splitting Δ ($10 Dq$) will tend to force as many electrons into the more stable t_{2g} set whereas, the electron pairing energy (energy required to cause pairing of two electrons in the same orbital) P , will counter any move to induce spin pairing in the t_{2g} set.



The d^1, d^2, d^3, d^8, d^9 and d^{10} configurations in octahedral field

In practice the relative magnitude of Δ ($10 Dq$), which for a given metal ion varies from ligand to ligand, and P^* (which is fixed for a metal ion but varies from metal ion to metal ion) will decide whether or not spin pairing will take place, that is, will decide whether or not the t_{2g} set will be occupied in preference to the e_g set. The d^1, d^2, d^3 systems will occupy the t_{2g} set with their electron spins parallel and this will be true irrespective of the strength of the crystal field. Again d^8, d^9 and d^{10} systems can be arranged in one way only with two, one and no unpaired electrons in the e_g set.

HIGH SPIN STATE	LOW SPIN STATE	HIGH SPIN STATE	LOW SPIN STATE
e_g { } t_{2g} { } $d^4 = t_{2g}^3 e_g^1$ $E = -\frac{3}{5} \Delta$	 $d^4 = t_{2g}^4$ $E = -\frac{8}{5} \Delta + P$	 $d^5 = t_{2g}^3 e_g^2$ $E =$	 $d^5 = t_{2g}^5$ $E = -2\Delta + 2P$
e_g { } t_{2g} { } $d^6 = t_{2g}^4 e_g^2$ $E = -\frac{2}{5} \Delta + P$	 $d^6 = t_{2g}^6$ $E = -\frac{12}{5} \Delta + 3P$	 $d^7 = t_{2g}^5 e_g^2$ $E = -\frac{4}{5} \Delta + 2P$	 $d^7 = t_{2g}^6 e_g^1$ $E = -\frac{9}{5} \Delta + 3P$

High-spin and low-spin configuration and their energies for d^4, d^5, d^6 and d^7 ion in octahedral field

- Pairing energy P is dictated by the principal quantum number n of the electrons. The 4d and the 5d orbitals extend more in space i.e., they are larger in size than the 3d orbitals. Therefore, spin pairing i.e., double occupation of an orbital involves much less interelectronic repulsion in 4d and 5d orbitals than in 3d orbitals. Thus, P falls in the order: $5d < 4d < 3d$.

But in the cases of d^4 , d^5 , d^6 and d^1 , high-spin (maximum possible number of unpaired electrons) and low-spin (minimum possible number of unpaired electrons) may arise. According to Hund's rule of maximum spin multiplicity, n unpaired electrons will tend to occupy n orbitals to give rise to n unpaired spins. Thus, pairing of spin is not normally a favourable process and the pairing energy P will have to be used to counter the electrostatic repulsion between two electrons in the same orbital. A d^4 high-spin system in a crystal field will have three electrons in the t_{2g} set (energy = $3 \times (-2/5 \Delta) = -6/5 \Delta = -12 \text{ Dq}$) and one in the e set (energy = $3/5 \Delta = 6 \text{ Dq}$), the total energy due to the crystal field being $-6/5 \Delta + 3/5 \Delta = -3/5 \Delta = -6 \text{ Dq}$. On the contrary, if the crystal field is strong enough to induce spin pairing, all four electrons will remain in the t_{2g} set and taking the pairing energy into consideration, the energy is $4 \times (-2/5 \Delta) + P = -8/5 \Delta + P = -16 \text{ Dq} + P$. The energies of the d^4 , d^5 , d^6 and d^7 systems in the two states are enumerated in. These energies are with respect to the unsplit barycentre whose energy is conveniently taken to be zero.

A perusal of the above energy values show that on putting $\Delta = P$, each of the d^4 , d^5 , d^6 and d^7 systems provides the same energy for both the spin-states. With $\Delta > P$ the energy of the low spin state becomes smaller than that of the high spin state i.e., the low spin state becomes the preferred state. With a $\Delta < P$ the reverse holds. This condition for high spin d^6 is demonstrated below:

$$\begin{array}{ll}
 \text{For the high spin state,} & \Delta < P \text{ i.e., } P > \Delta \text{ so } P = \Delta + x \Delta \\
 \text{Energy of the high spin state} & = -2/5 \Delta + P = -2/5 \Delta + \Delta + x \Delta \\
 & (t_{2g}^4 e_g^2) = 3/5 \Delta + x \Delta \\
 \text{Energy of the low spin state} & = -12/5 \Delta + 3P = -12/5 \Delta + 3\Delta + 3x \Delta \\
 & (t_{2g}^4 e_g^0) = 3/5 \Delta + 3x \Delta
 \end{array}$$

Thus the energy of the low spin state is greater than the energy of the high spin state. Therefore, the high spin state will result when $\Delta < P$ and conversely, the low spin state will be the ground state when $\Delta > P$. These two conditions are apparent on mere inspection of the energies of the two spin states of d^5 . When $\Delta > P$, the energy of the low spin state is negative while the energy of the high spin state is zero. Again when $\Delta < P$, the energy of the low spin state is positive while the energy of the high spin state is still zero.

Thus, the spin-state of an ion in a complex depends on whether the crystal field splitting is bigger or smaller than the pairing energy. When Δ is greater than P , electrons will tend to pair their spins and when Δ is smaller than P , electrons will tend to remain unpaired and will spread out in the d-orbitals. A weak-field ($\Delta < P$) produces high-spin complexes while a strong field ($\Delta > P$) gives low-spin complexes. Note also that the spin arrangement in weak field complexes is the same as in free ion, although, the orbitals are no longer degenerate.

For each of d^4 , d^5 , d^6 , d^1 ions there is a critical crystal field strength ($\Delta = P$) below which all ligands will produce high spin state and above which all ligands will produce low spin state. The critical 10Dq (Δ) is sometimes referred to as the cross-over region. A ligand, whose crystal field strength is close to the critical 10 Dq of a particular d^n ion ($n = 4, 5, 6, 7$), can give rise to spin-state equilibrium.

Crystal field effects in tetrahedral coordination entities

Tetrahedral Complexes: A tetrahedron is best conceived by taking the alternate corners of a cube. The centre of the cube

is $\left(\begin{array}{c} \text{S} \\ \text{R}_2\text{N}-\text{C} \\ \text{S}^- \end{array} \right)_3 \text{Fe}$, occupied by the metal ion. The dispositions of the d_{xy} , d_{xz} , d_{yz} , $d_{x^2-y^2}$ and d_{z^2} orbitals relative to

the tetrahedral points (where we place four ligands) show that d_{xy} , d_{xz} and d_{yz} orbitals lie just half an edge of a face of the cube away from the nearest ligands whereas, the $d_{x^2-y^2}$ and d_{z^2} orbitals lie half a diagonal of a face of the cube away. For example, each of the two lobes of the d_{z^2} orbital along the z -axis will be placed in between the two ligands which occupy diagonally opposite corners of a face of the cube. Thus, in contrast to the octahedral geometry, in a tetrahedral crystal field

d_{xy} , d_{xz} , d_{yz} orbitals. (t_2 set) are repelled more and hence, destabilized compared to the $d_{x^2-y^2}$ and d_{z^2} orbitals (e set).

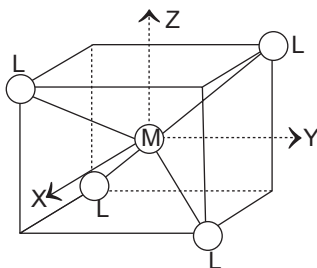


Fig. 11.1 A tetrahedral ML_4 complex with the four ligands at the alternate corners of a cube, the centre of which is occupied by the metal ion

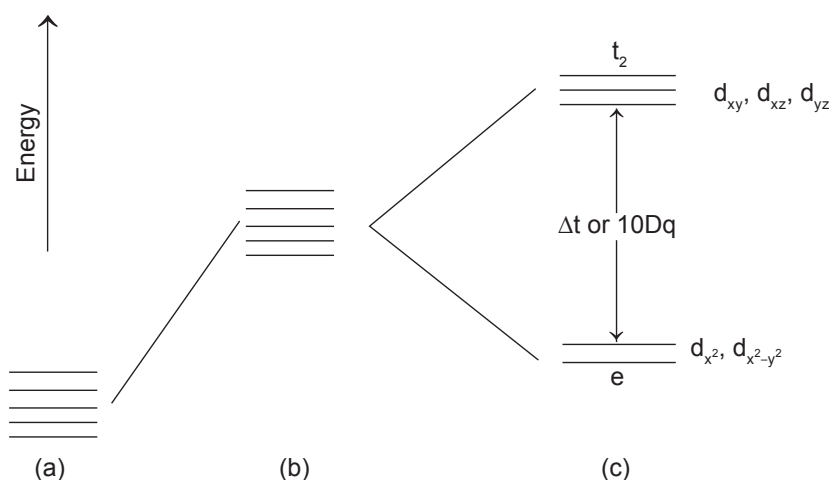


Fig. 11.2 Splitting of the d orbitals in a tetrahedral field. (a) free ion
(b) smeared out field (c) splitting

Further, calculations show that for a given metal ion and a given set of ligands and the same internuclear distance the tetrahedral splitting Δ^t is about 4/9th of the octahedral splitting between the same metal ion and the same six ligands. In principle only d^3 , d^4 , d^5 and d^6 may exist in both spin-states. The low spin-state will be favoured only if Δ^t is greater than P . A tetrahedral field produces only a weak splitting. No tetrahedral field is strong enough to initiate spin pairing (Δ^t is always less than P).

Splitting of d-orbitals in other Crystal Fields: We give the d-orbital ordering in several other fields.

In a square planar geometry the $d_{x^2-y^2}$ orbital alone directly faces the four ligands along the x and y axes. Hence $d_{x^2-y^2}$ orbital is the most destabilised, d_{xy} orbital is also repelled by the 4 ligands but expectedly to a lower extent. The d_{xz} and d_{yz} orbitals are not in the immediate vicinity of the ligands. In an idealised square planar stereochemistry there are no ligands along the z direction and thus the d_{z^2} orbital is the most stabilised. Splitting in a square planar geometry may be likened to the splitting in a tetragonal elongation case.

In square pyramidal geometry the $d_{x^2-y^2}$ orbital is face to face with the four ligands in the equatorial xy plane and hence this is the most destabilised orbital. Next in order will be the d_{z^2} , d_{xy} and the doublet set d_{xz} , d_{yz} .

In the trigonal bipyramidal geometry the d_{z^2} orbital is the most destabilised as it faces two ligands directly. However, the equatorial orbitals d_{xy} , $d_{x^2-y^2}$ do not face the three ligands (in the xy plane) directly but are equidispersed with respect to them. The d_{xz} , d_{yz} orbitals, as expected, are the most stabilised. In the trigonal bipyramidal stereochemistry, the energy difference (δ_1) between the lowest doublet set (d_{xz} , d_{yz}) and the upper doublet set (d_{xy} , $d_{x^2-y^2}$) is usually much smaller than

the energy difference (δ_2) between the set ($d_{xy}, d_{x^2-y^2}$) and the highest energy d_z^2 orbital. It has been established that δ_1 can never exceed pairing energy P of a metal ion. But δ_2 may be less than or greater than P . Therefore, in this geometry d^1 , d^2 , d^3 and d^4 systems will always give the high-spin form while d^5 , d^6 , d^7 and d^8 systems may be either high-spin or low-spin. Hence, with favourable ligands i.e., $\delta_2 \approx P$, we may observe anomalous moments with the d^5 , d^6 , d^7 and d^8 systems. In reality, however, for d^8 nickel (II) alone high spin and low-spin trigonal bipyramidal complexes are known. For all the other, i.e., d^5 , d^6 and d^7 systems all the known trigonal bipyramidal complexes are high-spin. Academically, however, the spin arrangements will be say for d^5 as:

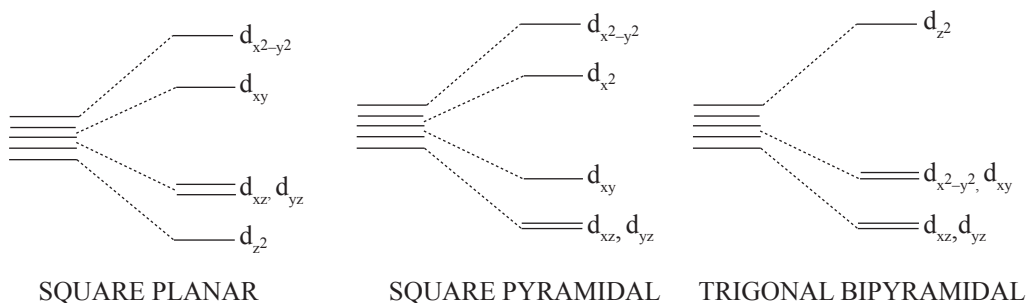


Fig. 11.3 Splitting of the d -orbitals in square planar, square pyramidal and trigonal bipyramidal crystal fields

$$\begin{aligned} \text{high-spin } d^5 &: (d_{xy}, d_{yz})^2, (d_{xy}, d_{x^2-y^2})^2, d_{z^2}^1 \\ \text{low-spin } d^5 &: (d_{xz}, d_{yz})^3, (d_{xy}, d_{x^2-y^2})^2, d_{z^2}^1 \end{aligned}$$

In trigonal planar geometry, the two ligands along the z -axis are missing and hence the d_{z^2} orbital is now the most stabilised orbital. The energy order will be: $d_{x^2-y^2}, d_{xy} > d_{xz}, d_{yz} > d_{z^2}$. In a linear geometry, the z -axis is taken as the unique axis and the two ligands are placed on either side of the metal ion. It can be guessed that the energy order is: $d_{z^2} > d_{xz}, d_{yz} > d_{xy}, d_{x^2-y^2}$.

Under the influence of tetrahedral field, the d -orbital splitting is smaller as compared to the octahedral field splitting. For the same metal, the same ligands and metal-ligand distances, it can be shown that $\Delta_t = \frac{4}{9} \Delta_o$. Consequently, the orbital splitting energies are not sufficiently large for forcing pair and therefore, low spin paired configurations are rarely observed.

Variation of 10 Dq

- Given the same metal ion and the same geometry, it varies from ligand to ligand. This variation gives us the following spectrochemical series of ligands in order of increasing crystal field splitting:



- Given the same metal ion and the same ligand it varies from geometry to geometry:

square planar	octahedral	tetrahedral
13 Dq (oct)	10 Dq (oct)	4.5 Dq (oct)

- Given the same ligand, the same geometry and the same oxidation state 10 Dq varies from metal ion to metal ion:

M^{2+}	V^{2+}	Cr^{2+}	Mn^{2+}	Fe^{2+}	Co^{2+}	Ni^{2+}	Cu^{2+}
10 Dq (kK)* of $[M(H_2O)_6]^{2+}$	12.4	13.9	8.5	10.4	9.3	8.5	12.6

- Given the same ligand, the same geometry, the same metal but varying oxidation state 10 Dq increases with increasing oxidation state. 10 Dq increases due to substantial increase in Δ_o bonding due to an obvious decrease in the radius

11.24 Co-ordination Chemistry

of the metal ion. The increased charge will drag the ligands more strongly, thus, perturbing the d-orbitals to a greater extent.

$[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$	$[\text{Mn}(\text{H}_2\text{O})_6]^{3+}$	$[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$	$[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$
10 Dq (kK) 8.5	21.00	13.9	17.4

- Given the same ligand, the same geometry, the same oxidation state 10 Dq increases in a group from the first transition series to the second and then to the third. It is likely that the 4d and 5d orbitals, due to their larger size interact better with the ligand orbitals than the 3d orbitals. Increased nuclear charge of the metal ions has also an effect.

	$[\text{Co}(\text{en})_3]^{3+}$	$[\text{Rh}(\text{en})_3]^{3+}$	$[\text{Ir}(\text{en})_3]^{3+}$
10Dq (kK)	24.0	35.0	41.0

Worked-out Examples

18. In which of the following octahedral complexes of Co (at no 27), will the magnitude of Δ_0 be the highest? (AIEEE 2006)

- (a) $[\text{Co}(\text{CN})_6]^{3-}$ (b) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$
 (c) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ (d) $[\text{Co}(\text{NH}_3)_6]^{3+}$

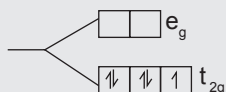
Ans. CFSE (crystal field splitting energy) for octahedral complex, Δ_0 depends on the strength of negative ligand. Spectrochemically, it has been found that the strength of splitting is as follows:



19. The value of the 'spin only' magnetic moment for one of the following configuration is
- (a) d^5 (in strong ligands field) (b) d^3 (in weak as well as in strong fields)
 (c) d^4 (in weak ligand field) (d) d^4 (in strong ligands field)

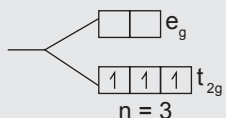
Ans.

- (a) d^5 in strong field
 $n = \text{unpaired electron} = 1$



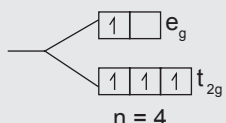
$$\begin{aligned} \text{Magnetic moment} &= \sqrt{n(n+2)} \text{ BM} \\ &= \sqrt{1(1+2)} \text{ BM} = 1.73 \text{ BM} \end{aligned}$$

- (b) d^3 in strong/weak field

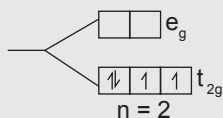


$$\text{Magnetic moment} = \sqrt{15} = 3.87 \text{ BM}$$

- (c) d^4 in weak field



$$\text{Magnetic moment} = \sqrt{24} = 4.90 \text{ BM}$$

(d) d^4 in strong field

$$\text{Magnetic moment} = \sqrt{8} = 2.83 \text{ BM}$$

Colour of Coordination Complexes: The colour of complexes originates from two sources: (1) d-d transition and (2) charge transfer transition.

A complex absorbs the required energy for a particular transition from the surrounding white light and obviously transmits the remaining complementary colour. This transmitted complementary colour is the colour that we observe.

d-d transition: In its ground state the d^1 electron of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ is the t_{2g} set. On irradiation with blue-green light of energy equal to the Δ ($493 \text{ m}\mu = 20,300 \text{ cm}^{-1}$) of the hexaaquatitanium (III) ion, the complex will absorb such energy to allow transition of the t_{2g} electron to the excited e_g set. When we see the complex through transmitted light it appears purple violet. For multielectron system interactions between d-electrons gives rise to many electronic states so that more than one transition is possible. Quantitative expressions have been derived for spectral transitions on the crystal field model connecting energy of the transition, crystal field strength and interelectronic repulsion. Furthermore, quantitative expressions are also available connecting magnetic moments with $10 Dq$, which connections were not forthcoming from VB theory.

It can be guessed that if H_2O of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ is substituted by a stronger ligand, say by bipyridine giving $[\text{Ti}(\text{bpy})_3]^{3+}$, then light of much higher energy (presumably of the violet region ; $\sim \lambda = 400 \text{ nm}$; $\sim 25 \text{ kK}$) will be required to effect the $t_{2g} \rightarrow e_g$ electronic transition since Δ will now be much higher. The transmitted light will be correspondingly of lower energy and thus, the observed colour will be yellow-green or so. Thus, we have a simple crystal field explanation of changing colour of a complex with change in the coordinating ligand, assuming however, the geometry remains the same. This is, in fact, how the spectrochemical series is obtained.

It is also of interest to note that high-spin cobalt (III) sd^6 fluoro complex $\text{K}_3[\text{CoF}_6]$ looks blue while all low-spin octahedral complexes of cobalt (III) eg: $(\text{Co}(\text{NH}_3)_6)\text{Cl}_3$ are yellow to orange. Evidently $\text{K}_3[\text{CoF}_6]$ absorbs low energy light and transmits high energy light while the diamagnetic complex does the reverse i.e., absorbs high energy light and transmits low energy light. A simple undergraduate level explanation is that for the high-spin complex Δ is less than the pairing energy P of cobalt (III) while for the low-spin complex Δ is much larger than P i.e., $\Delta(\text{l.s.}) \gg \Delta(\text{h.s.})$. Hence, different amounts of energy are required for the electronic transitions in the two spin states, thereby showing varying colours. However this explanation is too simple.

Charge transfer transition: There are many compounds where either the metal ion is highly oxidising and the ligands are reducing or that the metal ion is highly reducing and the ligands are oxidizing. In such cases there occurs transfer of charge (i.e., electron) from the reducing partner to the oxidizing partner. The metal ions may or may not possess d-electrons. Common examples are intense purple permanganate ($\text{Mn}(\text{VII})$, $3d^0$), brick-red mercury(II) iodide ($\text{Hg}(\text{II})$, $5d^{10}$), yellow chromate ($\text{Cr}(\text{VI})$, $3d^0$), dark red $[\text{Fe}(\text{bpy})_3]^{2+}$ ($\text{Fe}(\text{II})$, $3d^6$) blood red iron(III) thiocyanate ($\text{Fe}(\text{III})$, $3d^5$), etc.,

The selection rules of electronic transitions are:

1. No electronic transition is permitted where the ground state and the excited state have different spin arrangements i.e., have different spin multiplicities. Only those transitions are allowed which $\Delta S = 0$ i.e., transitions are permitted between states of the same spin multiplicity.
2. Only those electronic transitions are allowed for which $\Delta l = \pm 1$. This selection rule due to Laporte points out that no electronic transitions are allowed for which $\Delta l = 0$ or ± 2 . Thus, transitions such as $2s \rightarrow 2p$, $2s \rightarrow 3p$, $3p \rightarrow 3d$ are allowed but $3d \rightarrow 3d$, $1s \rightarrow 2s$ or $3s \rightarrow 3d$ are not. No electronic transition can occur merely because of redistribution of electrons in similar orbitals i.e., d-d transitions (such as $t_{2g} \rightarrow e_g$ transition in $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$) are forbidden.

Charge transfer transitions may be either from ligand to metal ($L \rightarrow M$) or from metal to ligand ($M \rightarrow L$). In such complexes, the ligand p-orbitals and metal d-orbitals interact to form M.O.'s. When these M.O.'s are arranged in order of their energy it is found that some M.O.'s have predominantly ligand (i.e., p) character while some M.O.'s retain predominantly metal (i.e., d) character. Laporte selection rule allows $p \rightarrow d$ transition ($\Delta l = +1$) so that charge transfer transitions are allowed transitions with high molar extinction coefficient (10^3 to $10^5 \text{ l mol}^{-1} \text{ cm}^{-1}$). Such transitions are usually high energy transitions and occur in the ultraviolet region. Charge transfer complexes appear coloured to our eyes only when the transitions occur in the near visible region or when the transition in the u.v. has a long tail in the visible. Common examples of $L \rightarrow M$ charge transfer are mercury(II) iodide, MnO_4^- , CrO_4^{2-} etc., Examples of $M \rightarrow L$ transition include $[\text{Fe}(\text{o-phen})_3]^{2+}$, $[\text{Fe}(\text{bpy})_3]^{2+}$ etc., While d-d transitions can occur only in complexes of transition metal ions with incomplete d-level, charge transfer can occur in both transition metals with incomplete d-level as also in d^0 , d^{10} and non-transition metal compounds. Charge transfer transitions are also known as redox transitions.

The more reducing M is and the more oxidizing L is, the lower is the energy of the $M \rightarrow L$ charge transfer transition. The absorption maximum of $[\text{Fe}(\text{bpy})_3]^{2+}$ occurs at $515 \text{ m}\mu$ while for $[\text{Ni}(\text{bpy})_3]^{2+}$, $[\text{Co}(\text{bpy})_3]^{2+}$ the maximum is at still lower wavelength range. Iron can be estimated down to 10^{-6} g/ml via formation of $[\text{Fe}(\text{bpy}/\text{o-phen})_3]^{2+}$. The more oxidizing M is and the more reducing L is, the lower is the energy of the $L \rightarrow M$ charge transfer transition. Thus iron(III) complexes give more $L \rightarrow M$ charge transfer transition than does chromium(III). Most colorimetric analyses of trace quantities of metal ions utilize charge transfer transitions. d-d transitions, being of low intensity, are of little value for such purposes.

Charge transfer spectra need not be restricted to $M \rightarrow L$ or $L \rightarrow M$ type. A third less common type is exhibited by the so-called Prussian blue or Turnbull's blue precipitate $\text{KFeFe}(\text{CN})_6$. One of the two irons is Fe(III) while the other is Fe(II). Electron transfer is possible from Fe(II) to Fe(III). Note that $\text{K}_2\text{FeFe}(\text{CN})_6$ is colourless because both the irons are Fe(II) and thus there exists no scope for any (redox) charge transfer transition.

Given below are the salient distinctions between d-d transitions and charge transfer transitions.

<i>d-d Transitions</i>	<i>Charge transfer transitions</i>
1. Presence of d electrons essential	1. Presence of d electron not essential. It may occur in d^0 , d^n , d^{10} systems
2. Laporte forbidden	2. Laporte allowed
3. Low molar extinction coefficient	3. Extremely high molar extinction coefficient ($5,000 \sim 35,000 \text{ l mol}^{-1} \text{ cm}^{-1}$)
4. No redox system involved	4. Redox system essential
5. Not useful for spectrophotometric	5. Very useful for spectrophotometric estimation of metal ions

Worked-out Examples

20. Identify the complexes which are expected to be coloured and explain.

[IIT 1994]

- (i) $\text{Ti}(\text{NO}_3)_4$ (ii) $[\text{Cu}(\text{NCCH}_3)_4]^+\text{BF}_4^-$
 (iii) $[\text{Cr}(\text{NH}_3)_6]^{3+}3\text{Cl}^-$ (iv) $\text{K}_3[\text{VF}_6]$

Ans. (iii) and (iv) are coloured because Cr^{3+} in $[\text{Cr}(\text{NH}_3)_6]^{3+}$ and V^{3+} in $[\text{VF}_6]^{3-}$ have $3d^3$ and $3d^2$ configuration respectively and thus show d-d transition.

21. In d-block metal ions, the colour of complex changes with ligand used while in f-block metal ion complexes, the colour remains almost constant and does not depend upon nature of ligand, why?

Ans. Colour of a meta ion depends upon energy difference of two energy levels involved in transition splitting of degenerate d-orbitals into t_{2g} and e_g orbitals is greatly affected by nature of ligand. i.e., why $[\text{Ni}(\text{NH}_3)_6]^{2+}$ is blue, $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ is green and $[\text{Ni}(\text{NO}_2)_6]^{4-}$ is brown.

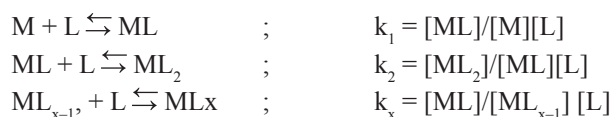
Colour of f-block metal ions is due to f-transition. 4f-orbitals in lanthanoids are deeply embedded in the atom and are deeply shielded by 5s- and 5p-orbitals. Thus, they remain almost unaffected by complex formation i.e., this is why colour does not depend upon nature of ligand.

Merits and demerits of Crystal Field Theory and Valence Bond Theory: The two theories are based on very widely different premises. Yet for the benefit of the young inquisitive readers some of the salient points are given below:

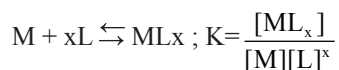
- In order to create a required geometry for the metal complex the VB theory hybridises suitable metal orbitals, sometimes belonging to different principal quantum numbers. On the contrary, the CF theory invokes no mixing of the metal orbitals. Instead, a stereochemistry is assumed, the ligands are placed at these stereochemical positions and then the energetics and hence the distribution of the metal ion d-electrons in the five d-orbitals are examined.
- VB theory, of necessity, has to promote some d-electrons to higher quantum shells particularly in the cases of d^7 , d^8 and d^9 (f ions in inner orbital octahedral cases). CF theory does not have to do this but merely reorganises the available d-electrons in the t_{2g} and e_g levels (of the same quantum shell), although they are not equienergetic.
- The VB theory assumes overlap of filled ligand orbitals with empty hybrid orbitals of the metal ion i.e., covalence is assumed. But CF does not accept any overlap between metal and ligand orbitals i.e., complexes are regarded as ionic.
- The 'outer orbital' and 'inner orbital' octahedral complexes in VB terminology become 'high-spin' and 'low-spin' complexes in CF terminology.
- In the "inner orbital" case, VB theory uses the $d_{x^2-y^2}$ and d_{z^2} in the hybridization and hence does not permit metal d-electrons to be placed in these orbitals. In CF theory, these two d-orbitals form the doublet e_g set and because of their higher energy (compared to the triplet t_{2g} set) disfavours putting d-electrons in this set.
- The explanation of colour of coordination complexes in the low energy visible range is understandable on the CF theory as being due to electron transition from the lower energy t_{2g} set to the higher energy e_g set (in the simplest octahedral cases). However, this has to be carried in our minds that such d-d transitions are Laporte forbidden, the VB theory does not have any such simple explanation.
- CF theory readily admits a gradual transition from high-spin octahedral to low-spin octahedral or even square planar stereochemistry with increasing donor strengths of the ligands. But VB theory is not suitable for such changes.
- As a consequence of recognition of the aforesaid gradual change, the CF theory offers a satisfactory explanation of the cross over region and spin state equilibrium (leading to anomalous magnetic moments). The VB theory is unable to tackle such situation.
- The CF theory offers interesting explanation of orbital contribution to spin magnetic moments while VB theory fails to provide desired answers.

Stability Constants

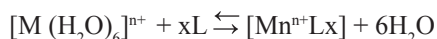
Quantitative study of the formation of a complex offers very useful informations. A complex ML_x may be formed in a number of consecutive steps:



where, $k_1, k_2, \dots, k_x$ are the stepwise formation constants. The overall formation (stability) constant may be represented as:



Reciprocal of the stability constant is called instability constant. The symbols β_x are often used to indicate the xth overall formation constant. Thus, $\beta_1 = k_1$; $\beta_2 = k_1.k_2$; $\beta_3 = k_1.k_2.k_3$. In aqueous medium, the formation reactions involve replacement of the coordinated aqua molecules in the first sphere of attraction by the ligand molecules:



Since the determination is made in the solvent water ($[H_2O] = 55M$) which is present in a constant and very high concentration, the release of coordinated water molecules has significant impact on the value of AT, and hence the reaction is considered without the solvent molecules.

Starting with known concentration of the metal ion and the ligand it is possible to determine the concentrations of all the different species involved in the equilibrium. Largely due to the efforts of Bjerrutn, Sillen, Calvin, Irving and Schwarzenbach several methods, namely potentiometry, spectrophotometry, polarography, 'solvent extraction, etc., are

now available for the evaluation of stability constants. A comparative study of the stability constants; (Table) of metal complexes with different ligands provides interesting conclusions:

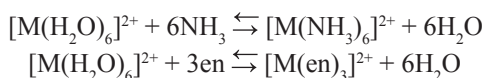
Table 11.2 Stability Constant (log K) for Metal Complexes in Aqueous Medium

Ligand	K_n	Mn^{2+}	Fe^{2+}	$\log K_n$ Co^{2+}	Ni^{2+}	Cu^{2+}	Zn^{2+}
NH_3	K_1	...	...	2.11	2.79	4.15	2.37
	K_2	...	...	1.63	2.24	3.50	2.44
	K_3	...	...	1.05	1.73	2.89	2.50
	K_4	...	...	0.76	1.19	2.13	2.15
	K_5	...	...	0.18	0.75	-0.52	...
	K_6	...	...	-0.62	0.03	...	...
Ethylene diamine	K_1	2.73	4.28	5.89	7.52	10.55	5.71
	K_2	2.06	3.25	4.83	6.28	9.05	4.66
	K_3	0.88	1.99	3.10	4.26	-1.0	1.72
Salicylaldehyde	K_1	3.73	4.22	4.67	5.22	7.40	4.50
	K_2	3.06	3.40	3.63	3.97	5.91	3.60
Ethylenediamine tetraacetic acid	K	14.0	14.3	16.3	18.6	18.8	16.5

1. The successive formation constants of a complex between a given metal ion and a given ligand steadily fall off (from K_1 to K_6 for metal ammines). This is due to the fact that as more ligands move into the coordination zone, less and less aqua molecules are available to fresh ligands for replacement. With progressive intake of ligands the metal ion becomes less electron greedy.
2. For a given metal ion the formation, constants are far larger for a chelating ligand than for a similar non-chelating ligand. This enhanced stability is called *chelate effect*.

Formation constants—a measure of the stability—are far greater for chelated complexes than for similar non-chelated complexes. Thus, the sum of the log K 's for the addition of four ammonia molecules to copper(II) is 12.67 while log K 's for the addition of two ethylenediamine is 19.60 (Table 11.2). Corresponding values for nickel(II) complexes are 7.95 and 13.80 (Table).

When a metal chelate is formed by the displacement of monodentate ligands there is an increase in the disorder of the system because of release of more molecules.



The free energy change of a system, its enthalpy change and entropy change in the standard state are related as,

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

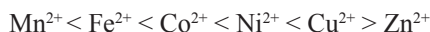
Again equilibrium constant of a reaction indicates the standard free energy change in the reaction i.e.,

$$\Delta G^\circ = -RT \ln K$$

Thus, a lower value of ΔG° indicates a higher value for K . Enthalpy change between chelated and non-chelated structures is small due to similar metal donor atom linkages. It follows from the above relations that more positive value of ΔS° will lead to more negative ΔG° and hence a more stable complex will be formed. In the case of substitution of the hexaaquametal(II) complex by six monodentate ligands there is no net change in the number of reactant species and product species. But in the case of substitution by three didentate chelating ligands the number of product species goes up by three ($7 - 4 = 3$) compared to the reactant species. If a hexadentate ligand were used instead of three didentate ligands then the above difference would have been $7 - 2 = 5$. Thus the product species have much more translational randomness than the reactant species, thus giving higher entropy.

The chelate effect is thus mainly an entropy effect resulting from an increase of molecules in the system. Following Sidgwick the explanation of the chelate effect may be presented more pictorially. If one of the two metal-ligand bonds of a didentate chelate system is broken, the remaining, bond will still hold the molecule in place, so that the broken link can be reestablished. But a ligand bound by a single link will drift away if the bond is broken. In essence a chelate complex dissociates less than a non-chelated complex and hence registers a higher stability constant.

3. An empirical order of the stability constants of metal complexes emerges as:



This order is commonly known as Irving-Williams series of stability constants. Recent determinations of formation constants of oxovanadium (IV), VO^{2+} , have shown that complexes of this ion are even stabler than those of copper(II).

Irving-Williams series reflects continuing increase in the formation constants from manganese (II) to copper (II) but thereafter, there is a sharp drop in zinc (II). From manganese (II) to nickel (II) there is a gradual increase in CFSE, then it drops in copper(II) and becomes zero in zinc(II). The anomalous behaviour of copper(II) is the result of stabilisation due to Jahn Teller distortion. Thus, the

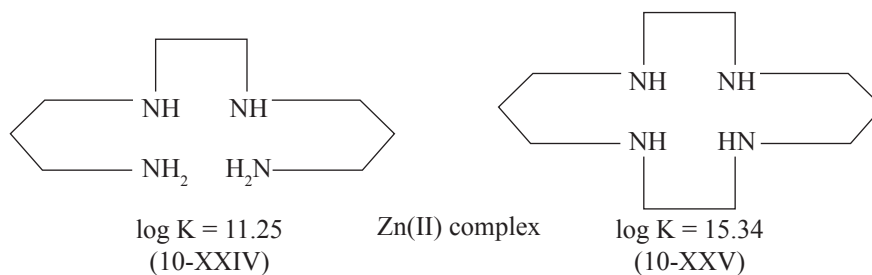
Ion	Mn(II)	Fe(II)	Co(II)	Ni(II)	Cu(II)	Zn(II)
Configuration in	$t_{2g}^3 e_g^2$	$t_{2g}^4 e_g^2$	$t_{2g}^5 e_g^2$	$t_{2g}^6 e_g^2$	$t_{2g}^6 e_g^3$	$t_{2g}^6 e_g^4$
CFSE(Dq)	0	-4	-8	-12	-6,	0

order of the Irving-Williams series is mainly the order of crystal field stabilisation energies.

Several workers have tried to correlate the formation constant data with fundamental properties of metal ions, namely ionisation potential, electronegativity, atomic number, ionic size as also with the basic properties of the ligands. It may be generally stated that stability constants of complexes increase with increasing ionisation potential of the metals (first plus second potentials for M^{2+} ions). Stability constants of a given metal ion for a series of ligands increase with decreasing $\text{p}K_a$ values of the ligands. Although data on complexes of the second and third transition series are scanty, there is little doubt that the stability constant increase within any group (of transition metals) in passing from the first to the second and to the third series. For example, complexes of nickel(II), palladium(II) and platinum(II) show an increase in thermodynamic stability in that order ($\text{Ni}^{2+} < \text{Pd}^{2+} < \text{Pt}^{2+}$)

Steric and inductive effects on formation constants

Substituents close enough to the donor centres of a chelating ligand often influence the magnitude of the formation constant. In some cases, the substituents may obstruct formation of chelate rings at the metal centre. For example, bipyridine readily forms tris (chelate) complexes with metal ions but when the two α -hydrogens are substituted by $-\text{CH}_3$ groups the resulting dimethyl bipyridine gives only mono(chelate) or bis(chelate) complexes. Formation constants of metal complexes of 8-hydroxy-quinoline and quinoline and their variously substituted derivatives have been determined. For a given metal ion (say Cu^{2+}) and a series of substituted derivatives it has been observed that formation constants of the complexes vary linearly with the $\text{p}K_b$ i.e., the basicity of the ligands. It is also observed that the effect is most severe for substituents at the 2-position i.e., α -to the heterocyclic nitrogen i.e., formation constant drops sharply for metal complexes of such α -substituted quinoline and 8-hydroxy quinoline.



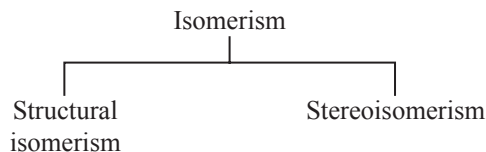
Macrocyclic Effect: As an extension of chelate effect, it has been observed that formation constant of metal complexes of a closed ring macrocyclic ligand is several orders more than that of a comparable open-ended polydentate ligand containing an equal number of equivalent donor atoms. Thus log K for the zinc (II) complexes of the two ligands (10-XXIV) and (10-XXV) are 11.25 and 15.34 respectively. Similar enhancement in the order of log K has also been observed for nickel(II) complexes. This enhanced stability due to a macrocyclic ligand compared to a similar open-ended chelating ligand has been termed *macrocyclic effect*.

Isomerism

The compounds having same chemical formula but different structural arrangement of their atoms and hence different physical and chemical properties are called isomers and the phenomenon is called isomerism. Isomerism in complexes are of two types:

(i) Structural Isomerism

(ii) Stereoisomerism



- (a) Ionization Isomerism
- (b) Hydrate isomerism
- (c) Ligand isomerism
- (d) Coordination isomerism
- (e) Linkage isomerism
- (f) Coordination position isomerism
- (g) Polymerisation isomerism

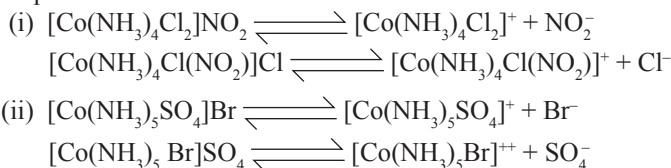
- (a) Geometrical or cis-trans isomerism
- (b) Optical isomerism

Structural isomerism

This isomerism arises due to the difference in structures of coordination compounds and are of the following types.

(a) Ionization isomerism

Complexes that have the same empirical formula and are produced by the interchange of the position of the ligands inside the complex zone and outside the complex zone are called ionization isomers. They give different ions for examples.



The number of ions in a solution can be determined by conductivity measurement. More the number of ions in a solution more is the conductivity. Greater the charge on ions, greater is the conductivity of solution.

T-17

How can $[\text{Cr}(\text{NH}_3)_6]^{3+}$, $[\text{Co}(\text{NO}_2)_6]^{3-}$ and $[\text{Cr}(\text{NH}_3)_4(\text{NO}_2)_2][\text{Co}(\text{NH}_3)_2(\text{NO}_2)_4]$ be distinguished from conductivity measurement in their aqueous solution?

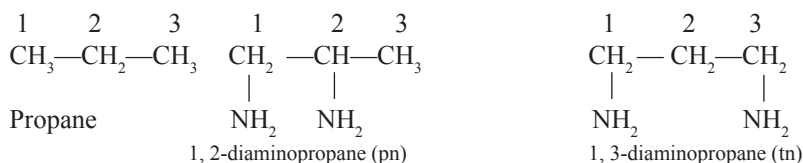
(b) Hydrate isomerism

This type of isomerism arises due to the different position of water molecules inside and outside the coordination sphere. For example,

- (i) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ (violet), does not lose water over H_2SO_4 and all Cl^- ions are immediately precipitated by Ag^+ ions.
- (ii) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$ (green), loses H_2O over H_2SO_4 and two Cl^- ions are precipitated by Ag^+ ions.
- (iii) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$ (green), loses two water molecules over H_2SO_4 and only one Cl^- ion is precipitated by Ag^+ ions.

(c) Ligand isomerism

Some ligands themselves are capable of existing as isomers, for example, diamino propane can exist both as 1, 2-diaminopropane (pn) and 1, 3-diaminopropane, also called trimethylenediamine (tn).

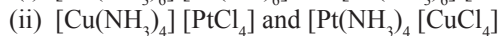
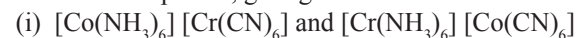


When these ligands (for example, pn an tn) are associated to form complexes, the complexes are isomers of each other.



(d) **Coordination isomerism**

If both cation and anion of a complex compound are complex, there may be an exchange of ligands between the two coordination spheres, giving rise to isomers known as coordination isomers. e.g.,



(e) **Linkage isomerism**

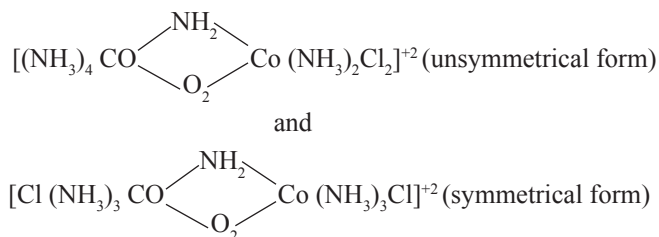
Those complexes in which the ligands can coordinate with the central metal ion through either of the two atoms, give rise to the linkage isomerism.

The best known ligands of this type are NO_2^- , SCN^- and $\text{S}_2\text{O}_3^{2-}$ ions. In complexes containing NO_2^- ion as ligand, NO_2^- ion may attach with the central ion either through O-atom or through N-atom



(f) **Coordination position isomerism**

In some polynuclear complexes, interchange of the ligands between the metal atoms which are present as a part of the complex is possible. This type of interchange of ligands between the metal atoms gives rise to coordination position isomerism for example,



are coordination position isomers.

(g) **Polymerisation isomerism**

This is not the true isomerism because it occurs between compound having the same empirical formula, but different molecular weights. For example, $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$, $[\text{Pt}(\text{NH}_3)_4] [\text{Pt}(\text{Cl}_4)]$, $[\text{Pt}(\text{NH}_3)_4] [\text{Pt}(\text{NH}_3)\text{Cl}_3]_2$ and $[\text{Pt}(\text{NH}_3)_3\text{Cl}]_2 [\text{PtCl}_4]$ all have the same empirical formula.

Stereoisomerism or Space Isomerism

When two compounds contain the same ligands coordinated to the same central ion, but the arrangement of ligands in the space is different, the two compounds are said to be stereoisomers and the type of isomerism is called stereoisomerism.

Stereoisomerism is of two types:

- (a) Geometrical or cis-trans isomerism
- (b) Optical or mirror image isomerism

Geometrical isomerism

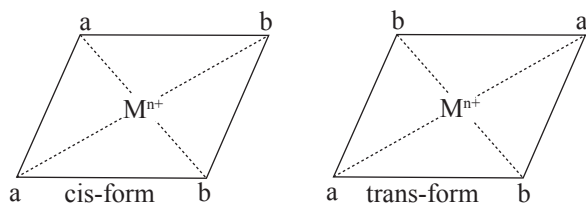
Geometrical isomerism is due to ligands occupying different positions around the central metal ion. The ligands occupy positions either adjacent to one another or opposite to one another. These are referred to as cis-form and trans-form respectively. This type of isomerism is, therefore, also referred to as cis-trans isomerism.

(a) **Geometrical isomerism in 4-coordinate complex**

Complexes having central metal atom with coordination number equal to 4, may be either tetrahedral or square planar. Geometrical isomerism cannot arise in tetrahedral complexes because this geometry contains all the ligands in the cis (i.e., adjacent) position with respect to each other i.e., each ligand is equidistant from the other three ligands and all the bond angles are the same (109.5°). Hence, geometrical isomerism cannot be expected in tetrahedral complexes.

Square planar complexes of $[Ma_4]$, $[Ma_3b]$ and $[Mab_3]$ type (a and b are monodentate ligands) do not show geometrical isomerism, since every conceivable spatial arrangement of the ligands around the metal ion is exactly equivalent.

1. $[Ma_2b_2]$ type complexes

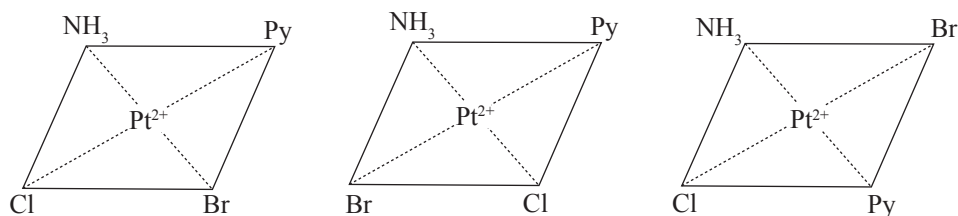


e.g., $[Pt^{+2}(NH_3)_2Cl_2]$, $[Pt^{2+}(NH_3)_2Br_2]$ and

$[Pd^{2+}(NH_3)_2(NO_2)_2]$ are square planar complexes which exhibit cis-trans isomerism.

2. $[Mabcd]$ type complexes

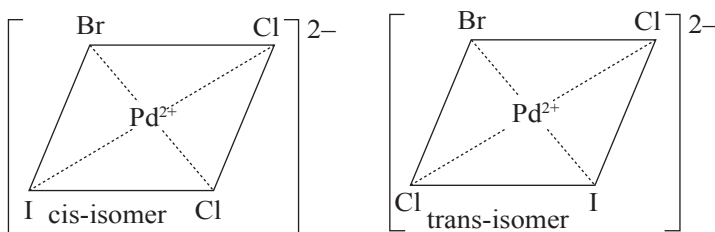
Square planar complexes on this type exist in three isomeric forms for example, $[Pt^{2+}(NH_3)(Py)(Cl)(Br)]$ exist in the following structures.



$[Pt^{2+}(NO_2)(Py)(NH_3)(NH_2OH)]^+$ and $[Pt^{2+}(C_2H_4)(NH_3)(Cl)(Br)]$ are other examples of square planar complexes which exist in three isomeric forms.

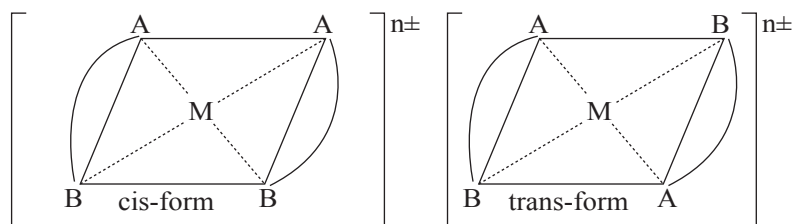
3. $[Ma_2bc]$ type complexes

Square planar complexes of this type also show cis-trans isomerism. For example, $[Pd^{2+}Cl_2BrI]^{2-}$ exists in the following cis-trans.

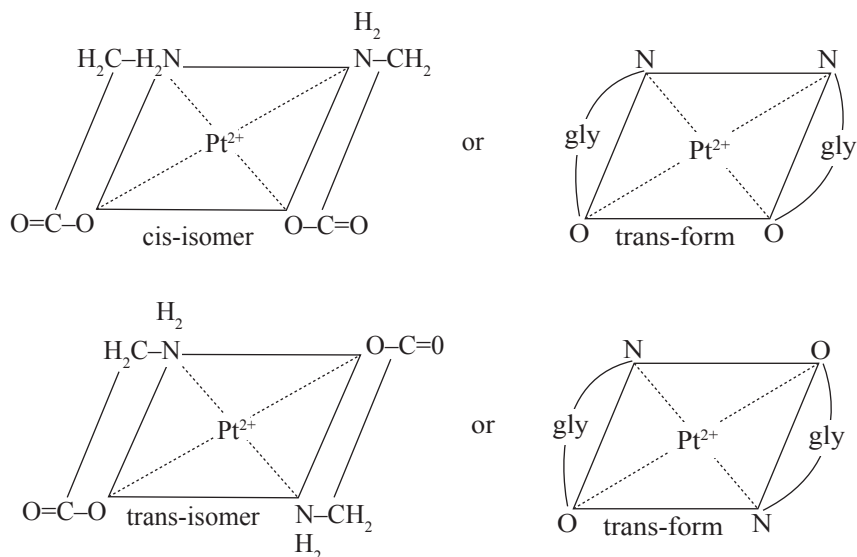


4. $[M(AB)_n]^{n\pm}$ type complexes

Here, M is the central metal ion and (AB) represents an unsymmetrical bidentate ligand. (A) and (B) are the two ends (i.e., coordinating atoms) of the bidentate ligand. Such type of complexes also show cis and trans isomerism.

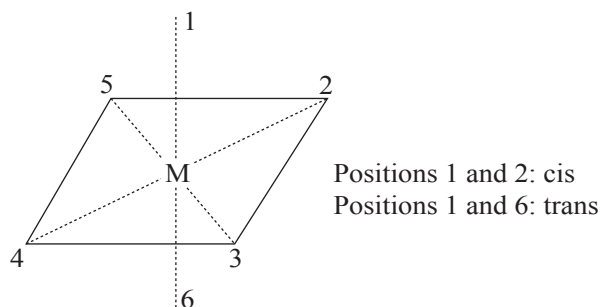


For example, $[\text{Pt}^{2+}(\text{gly})_2]$; Here, gly represents the glycinate ligand, $\text{NH}_2\text{CH}_2\text{COO}^-$ which has N and O atoms as its donor atoms.



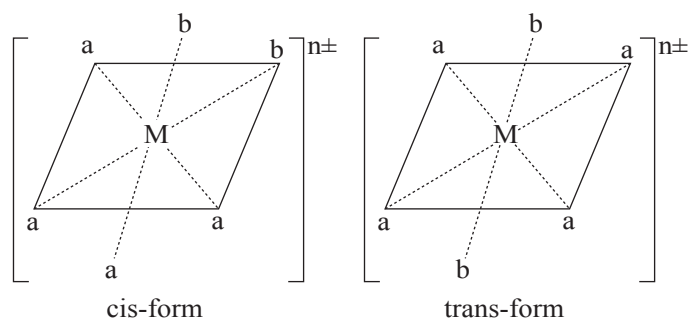
(b) Geometrical isomerism in 6-coordinate complexes

A complex compound having the central metal ion with coordination number equal to 6 has octahedral shape. The system used for numbering different positions of the ligands in an octahedral geometry has been shown below.

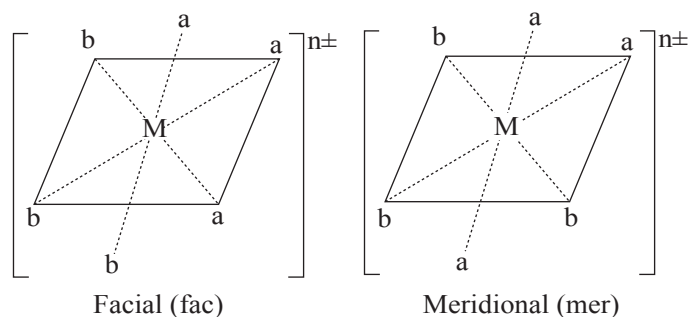


The octahedral complexes of the types $[\text{Ma}_6]$, $[\text{M}(\text{AA})_3]$ and $[\text{Ma}_3\text{b}_3]$ do not show geometrical isomerism. The following octahedral complexes give two or more geometrical isomers.

1. $[\text{Ma}_4\text{b}_2]^{n\pm}$ type complexes



Examples of such complexes are $[\text{Co}^{3+}(\text{NH}_3)_4\text{Cl}_2]^+$, $[\text{Co}^{3+}(\text{NH}_3)_4(\text{NO}_2)_2]^+$ etc.,

2. $[\text{Ma}_3\text{b}_3]^{n\pm}$ type complexes

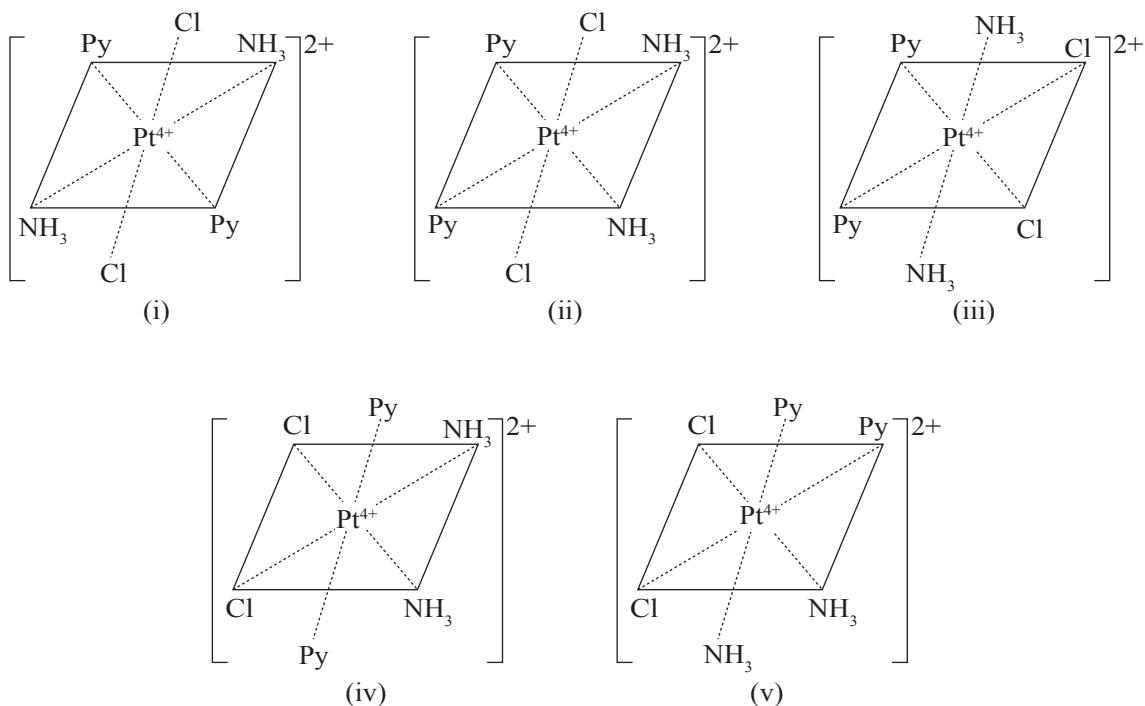
(When each trio of donor atoms (viz. the ligands a, a and a) occupy adjacent positions at the corners of an octahedral face, we have facial (fac) isomer. When the positions are around the meridian of the octahedron, we get Meridional (mer) isomer.

Complexes like $[\text{Co}^{3+}(\text{NH}_3)_3\text{Cl}_3]$, $[\text{Cr}^{3+}(\text{H}_2\text{O})_3\text{F}_3]$, $[\text{Pt}^{3+}(\text{Py})_3\text{Cl}_3]$, $[\text{Cr}^{3+}(\text{NH}_3)_3\text{Cl}_3]$, $[\text{Ru}^{3+}(\text{H}_2\text{O})_3\text{Cl}_3]$, $[\text{Pt}^{4+}(\text{NH}_3)_3\text{Br}_3]^+$ etc., Exhibit such type of isomerism.

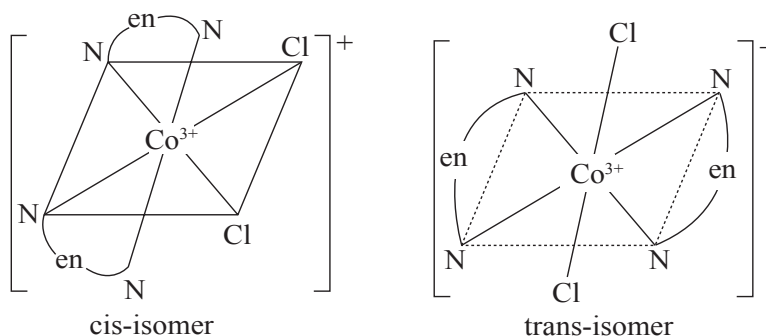
3. $[\text{Ma}_2\text{b}_2\text{c}_2]$ type complexes

Octahedral complexes of this type can exist theoretically in five geometrical isomers. Out of these only three have been prepared.

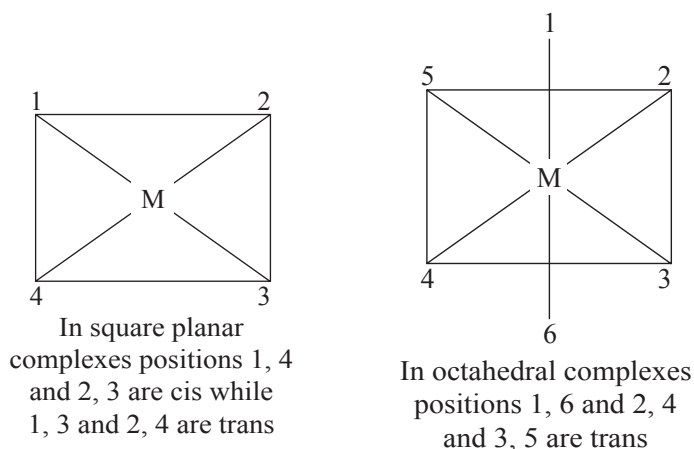
For example, consider $[\text{Pt}^{4+}(\text{NH}_3)_2(\text{Py})_2\text{Cl}_2]^{2+}$ ion. It can theoretically exist in the following structures.

4. $[\text{M}(\text{AA})_2\text{a}_2]$ type complexes: Here, (AA) represents a symmetrical bidentate ligand in which A and A are two identical coordinating atoms.

Examples of such complexes are $[\text{Co}^{3+}(\text{en})_2\text{Cl}_2]^+$, $[\text{Co}^{3+}(\text{en})_2(\text{NO}_3)_2]^+$, $[\text{Cr}^{3+}(\text{en})_2\text{Cl}_2]^+$, $[\text{Cr}^{3+}[\text{C}_2\text{O}_4]_2(\text{H}_2\text{O})_2]^-$ (where en = ethylenediamine).



- *Polymerization isomerism*: This is not a true isomerism. It is used to denote compounds having same empirical formula but different molecular formulae, e.g., the compounds $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ and $[\text{Pt}(\text{NH}_3)_4][\text{PtCl}_4]$.
- The geometrical isomerism is not possible in tetrahedral complexes like MA_4 , MA_2B_2 and MABCD and square planar complexes like MA_4 , MAB_3 , and MA_3B , where A, B, C and D are monodentate ligands.

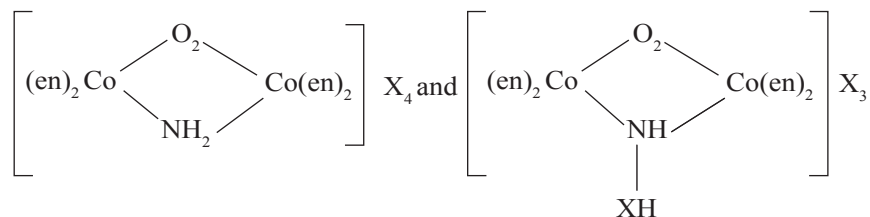


- Octahedral complex of the type $[\text{M}(\text{ABCDEF})]$ have 15 different geometrical isomers with a pair of enantiomers. Although, a few geometrical isomers have been prepared but none has been resolved.

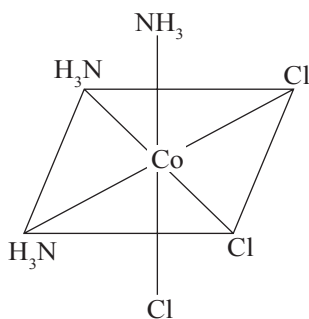
e.g., $[\text{Pt}(\text{py})(\text{NH}_3)(\text{NO}_2)(\text{Cl})(\text{Br})(\text{I})]$

- Square planar complexes of type MA_2B_2 , MA_2BC and MABCD show geometrical isomerism. Geometrical isomerism is not possible in square planar complexes of type MA_4 and MA_3B .
- Square planar complex of type MABCD having four different ligands give three geometrical isomers.
- Complexes of type MA_2B_4 , MA_4BC , MA_3B_3 , MA_2C_2 , exhibit geometric isomerism.
- Complexes of type $\text{M}(\widehat{\text{AA}})_2\text{B}_2$ where $\widehat{\text{AA}}$ is a symmetrical bidentate ligand and B is unidentate ligand show geometric isomerism.
- Geometrical isomerism is not possible in tetrahedral complexes of type MA_4 , MA_2B_2 and MABCD .
- Compound of type $\text{MA}_2\text{X}_2\text{Y}_2 > \text{MA}_2\text{X}_2\text{YZ}$, MA_2XYZL , MABXYZL , $\text{M}(\widehat{\text{AA}})_3$, $\text{M}(\widehat{\text{AA}})_2\text{X}_2$ show optical isomerism.
- $\text{MA}_2\text{X}_2\text{Y}_2$ show both optical and geometrical isomerism.
- Octahedral complexes of type MA_6 and MA_5B do not show geometric isomerism.
- Octahedral complex of type MA_3B_3 exist only in two isomeric forms,
- Octahedral complexes of type MABCDXY exist in 15 different geometrical isomeric forms.
- $[\text{M} \widehat{\text{AA}} \text{B} \widehat{\text{AA}} \text{C}_2]$ can show geometrical isomerism.

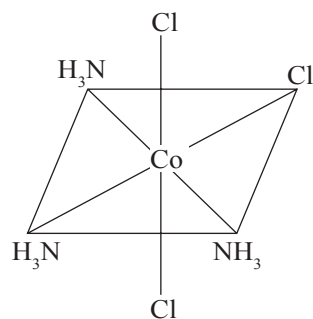
- *Valency isomerism*. This type of isomerism arises when the ligand is held by different types of valency bonds e.g.,



- For the complex of type MA_3B_3 e.g., $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$, $[\text{Rh}(\text{py})_3\text{Cl}_3]$ the isomer is called facial (fac) if all the similar ligands occupy the same face of an octahedron. The other geometric isomer is called meridional (mer).



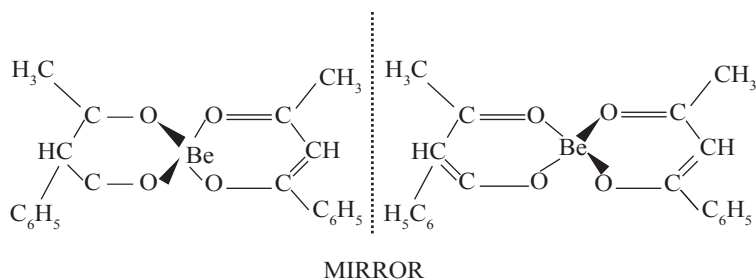
Facial or fac isomer



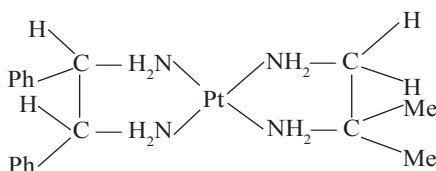
Meridional or mer isomer

Stereoisomerism/Optical Isomerism

- Metal complexes with co-ordination number of four also show optical isomerism but to a much smaller extent.
- Tetrahedral complexes of Be(II), B(III), Ni(II), Cu(II) and Zn(II) containing unsymmetrical bidentate ligands have been resolved into optically active forms. Some examples of this type of complexes are bis(benzoylacetonato)beryllium(II), bis(glycinato)nickle(II) etc.,



- It must be noted that tetrahedral complexes having four different ligands around the central metal atom or ion do not show optical activity because these complexes are labile and cannot be isolated (labile complexes are those complexes which can easily undergo exchange of ligands). (ii) Square planar complexes are seldom optically active. One such complex which has been resolved into two enantiomeric forms is given below:



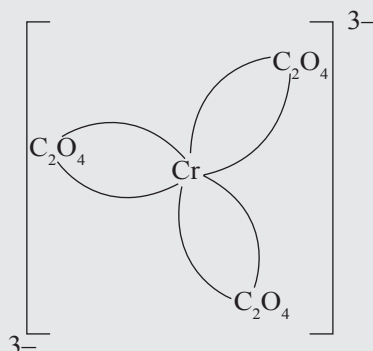
Worked-out Examples

22. Which of the following compounds shows optical isomerism?

(AIEEE 2005)

- (a) $[\text{Co}(\text{CN})_6]^{3-}$ (b) $[\text{Cr}(\text{C}_2\text{O}_4)_6]^{3-}$ (c) $[\text{ZnCl}_4]^{2-}$ (d) $[\text{Cu}(\text{NH}_3)_4]^{2+}$

Ans.



Mirror image is not superimposable, hence, optical isomerism is possible.

23. Which one of the following has larger number of isomers?

(AIEEE 2004)

(R = alkyl group, en = ethylenediamine)

- (a) $[\text{Ru}(\text{NH}_3)_4\text{Cl}_2]^+$ (b) $(\text{Co}(\text{NH}_3)_5\text{Cl})^{2+}$ (c) $[\text{Ir}(\text{PR}_3)_2\text{H}(\text{CO})]^{2+}$ (d) $[\text{Co}(\text{en})_2\text{Cl}_2]^+$

Ans. $[\text{Co}(\text{en})_2\text{Cl}_2]$ forms optical and geometrical isomers.

24. Type of isomerism shown by $[\text{Cr}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$ is

(AIEEE 2002)

- (a) optical (b) ionization (c) geometrical (d) linkage

Ans. $[\text{Cr}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$

Pentamminenitrito-N-chromm (III) chloride exhibits linkage isomerism as $-\text{NO}_2$ is ambidentate ligand. Isomer is $[\text{Cr}(\text{NH}_3)_5\text{ONO}]\text{Cl}_2$.

Metal Carbonyl

CO

The carbon atom has $2 + 4 = 6$ electrons, and the O atom has $2 + 6 = 8$ electrons, so the CO molecule contains 14. In this case, we are rather less certain of the energies of the MOs. since they are different for C and O. Assume the order is the as for atoms like,

$$\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2 \begin{cases} \pi 2p_1^2 \\ \pi p_z^2 \end{cases} \sigma^2 p_1^2$$

The inner shell is non-bonding, and the bonding and antibonding 2s orbitals cancel, leaving one σ and two π bonds and thus an order of 3. Alternatively, the bond order may be calculated using the formula (bonding-antibonding)/2, that is $(10-4)/2 = 3$. This simple picture is not adequate, since if CO is ionized to give CO^+ by removal of one electron from the $\sigma 2p$, orbital then the bond order should be reduced to $2\frac{1}{2}$ and the bond length increased.

In fact, the bond length in CO is $1.128\text{\AA}$ and in CO^+ it is $1.115\text{\AA}$. Thus, the bond length decreases when we expected it to increase, and it indicates that the electron must have been removed from an antibonding orbital. The problem remains if we assume the order of energy for the MOs is the same as for atoms heavier than C, since this only reverses the position of the $\sigma 2p$, and the $(2\pi p_y \text{ and } \pi 2p_z)$ MOs. The likely explanation of the bond shortening when CO is changed to CO^+ is that

the σ_{2s} and σ^*_{2s} molecular orbital differ in energy more than is shown in the figure. This means that they are wider apart, and the σ^*_{2s} MO is higher in energy than the σ_{2p_x} , π_{2p_y} and π_{2p_z} MOs. This illustrates very plainly that the order of MO energy levels for simple homonuclear diatomic molecules used above is not automatically applicable when two different types of atoms are bonded together, and it is certainly incorrect in this particular heteronuclear case.

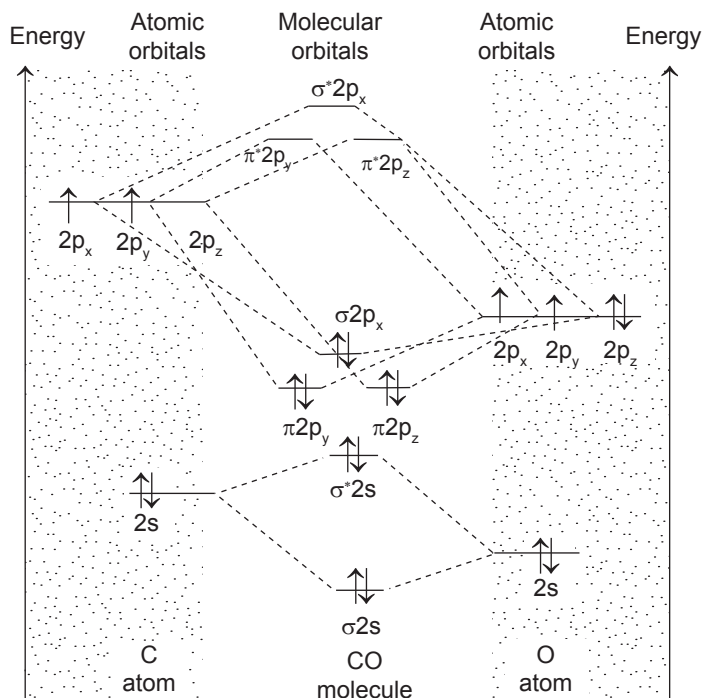


Fig. 11.4 Electronic configuration, atomic orbitals and molecular orbitals for carbon monoxide (The σ_{1s} and σ^*_{1s} MOs are omitted for simplicity)

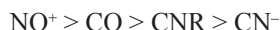
Carbon monoxide is normally a poor donor as revealed by its poor basic strength. For instance, CO does not form an adduct even with a very strong Lewis acid (electron acceptor) BF_3 . But it forms a long series of metal carbonyls. This improved behaviour in presence of low-valent metals is due to the fact that on back donation from the metal to the ligand, the otherwise poor donor CO now becomes a substantially better donor. This effect is called a synergic effect which means a mutually sustaining interaction.

Complexes with CO, NO, CNR, PR_3 , etc., are also called π -acid complexes since the metal to ligand bond is of the π -type and the ligand behaves as a π -acid as it can receive back donated electrons from the low oxidation state metal into its orbitals. Note that in such complexes both the metal and the ligand function both as donors and acceptors.

Studies on the complexing abilities of CN^- , CNR, NO^+ and CO have established that the negative charge on CN^- enhances its σ -donor ability but weakens π -acceptor effectiveness. σ -donor ability decreases in the order:



While the π -acceptor effectiveness increases in the order:

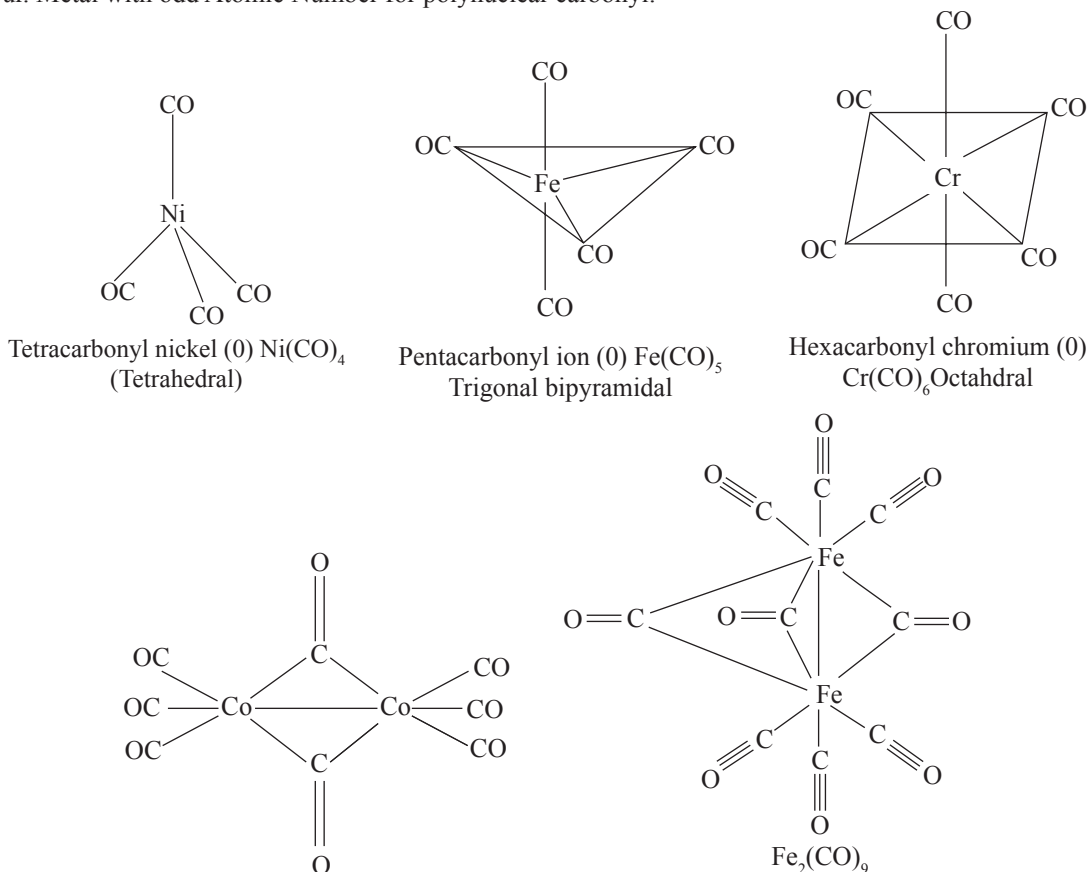


The ligands at the ends, CN^- and NO^+ differ a lot from CO and are not good analogues. Electron-rich CN^- is a good σ -donor but a weak π -acceptor. It forms a large number of stable cyano complexes with metal atoms in high oxidation states.

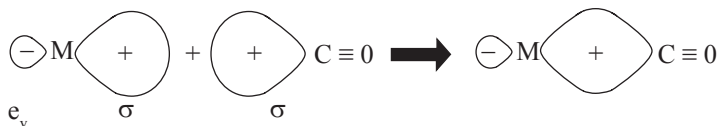
Metal carbonyl

Metal carbonyl compounds formed between metal and carbon monoxide belong to this class. The Homoleptic carbonyls (carbonyl) having only carbonyl ligands are formed by many Transition Metals. They have simple well defined structure. These compounds possess both σ - and π -bonding. The oxidation state of metal atoms in these compounds is zero.

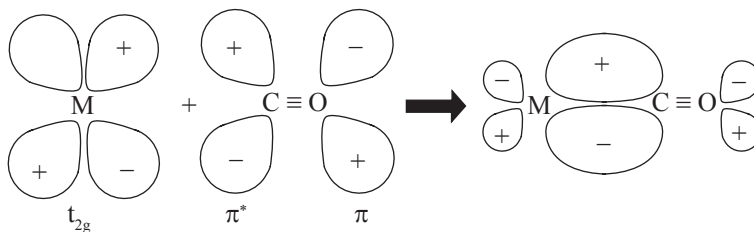
Carbonyls may be monomeric, bridged or polynuclear. Carbonyl are mainly formed by the transition metals of VIth, VIIth and VIIIth groups. Mononuclear carbonyl are volatile and toxic, light colour of colourless. Polynuclear carbonyl have a deep colour. Metal with odd Atomic Number for polynuclear carbonyl.



In a metal carbonyl, the metal-carbon bond possesses both the σ - and π -character. A σ -bond between metal and carbon atom is formed when a vacant hybrid orbital of the metal atom overlaps with a fully filled orbital on C atom of carbon monoxide containing a lone pair of electrons.



Formation of π -bond is caused when a filled orbital of the metal atom overlaps with a vacant antibonding π^* orbital of C atom of carbon monoxide. This overlap is also called back donation of electrons by metal atom to carbon. It has been shown below.



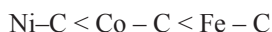
Carbon monoxide, CO, as a ligand binds itself to metal atoms through the carbon atom. It is weak donor (a weak base). It forms a weak σ bond to the central atom. CO is also an acceptor ligand and capable of accepting electron in vacant π or π^* orbital of their own. These are also called π -acid ligand. The $d\pi - \pi$ overlap, apart from providing an additional bond (π -bond) between M and C, stabilizes or strengthens the M–C bond by removing the excess negative charge which gets accumulated on the metal as a result of the coordination of CO with it. The stronger M–C bond brings the atoms closer to each other thus bringing about a better π -overlap. In other words, one type of bonding strengthens the other type. Such an effect in bonding is called as synergic effect. Due to this effect, CO is a good donor for transition metals not for alkali or alkaline earth metals. As transition metals have fully filled orbitals for back donation.

During the formation of the above type of π -bonds, electronic charge is transferred from the filled non-bonding orbitals of the metal to the π^* orbitals of the ligand CO. This reduces the bond order of CO. Thus, while the π back bonding strengthens the M–C bond, it weakens the CO bond. This means that M–C and CO bond orders in carbonyls are roughly complementary to each other and any factor that increases the metal π -electrons transfer to CO would increase the M–C bond order and decrease the CO bond order simultaneously. Since the bond order between M and C is generally higher than that due to π -back bonding, the single line M – C bond simply represents a metal carbon bond without any reference to the actual bond order in this bond.

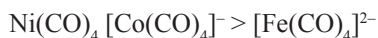
In the following isoelectronic and isostructural species:



The overall negative charge in the case of the last two species is assumed to reside on the metal. The greater the negative charge on the metal, the greater would be ease of electron transfer from the metal to the π^* orbitals of CO. Accordingly, the M–C bond order increases in the direction,



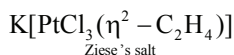
Simultaneously, the C–O bond order decreases in the direction,



The actual bond order as determined from spectroscopic data are given in table.

<i>Metal Carbonyl</i>	<i>M–C Bond order</i>	<i>C–O Bond order</i>	<i>M–C + C–O Bond order</i>
Ni(CO) ₄	1.33	2.64	3.97
[Co(CO) ₄] [–]	1.89	2.14	4.03
[Fe(CO) ₄] ^{2–}	2.16	1.85	4.01

- Ligands are generally capable of donating one or more lone pairs of electrons. Positively charged ligands also donate lone pair of electrons.
- There are certain ligands which do not have lone pair of electrons but still can act as ligand e.g., ethylene or C₂H₄ in Zeise's salt is one such ligand.



- Oxidation state of Fe in ferrocene is Fe²⁺. The two cyclopentadienyl anion (C₅H₅[–]) carries one unit negative charge each.
- Metal carbonyls are not regarded as organometallic compounds in the true sense. This is because CO is not an organic compound. However, metal carbonyls are studied along with other organometallic compounds.
- KCN contains a metal-carbon bond but it is not considered as an organometallic compound.
- In sodium nitroprusside (a complex compound), Na₂[Fe(CN)₅NO], the oxidation state of Fe is +2 and the ligand NO is nitrosonium i.e., NO⁺ and not Fe³⁺ and NO (nitrosyl)—a neutral ligand. This data is based on the magnetic moment measurement of the complex in the solid state.

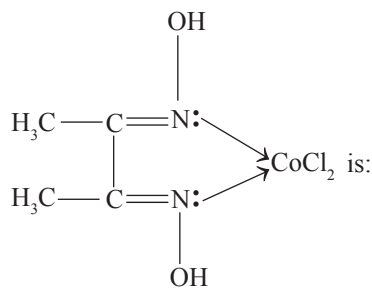
ONE CORRECT OPTION TYPE QUESTIONS

- The number of ions formed in aqueous solution by the compound $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ is:
 - 2
 - 3
 - 4
 - 7
- The groups satisfying the secondary valencies of a cation in a complex are called:
 - Ligands
 - Radicals
 - Primary valencies
 - None of these
- Which one is the most likely structure of $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$, if 1/3 of total chlorine of the compound is precipitated by adding AgNO_3 to its aqueous solution?
 - $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$
 - $[\text{Cr}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot (\text{H}_2\text{O})_3$
 - $[\text{CrCl}_2(\text{H}_2\text{O})_4] \cdot \text{Cl} \cdot 2\text{H}_2\text{O}$
 - $[\text{CrCl}(\text{H}_2\text{O})_5] \cdot \text{Cl}_2 \cdot \text{H}_2\text{O}$
- $\text{K}_4[\text{Fe}(\text{CN})_6]$ is used to detect the presence of:
 - Metallic ion
 - Ferrous ion
 - Ferric ion
 - None of these
- The oxidation state of Fe in $\text{Na}_3[\text{Fe}(\text{NH}_3)(\text{CN})_5] \cdot 2\text{H}_2\text{O}$ is:
 - 1
 - +1
 - +2
 - +3
- The primary valency of Fe in $\text{K}_3[\text{Fe}(\text{CN})_6]$ is:
 - 3
 - 2
 - 1
 - zero
- Co-ordination number of Co in $[\text{Co}(\text{NH}_3)_6]^{2+}$ is:
 - 4
 - 5
 - 6
 - 8
- In $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ the co-ordination number of cobalt is:
 - 4
 - 6
 - 2
 - 7
- The co-ordination number of cobalt in $[\text{Co}(\text{en})_2\text{Br}_2]\text{Cl}_2$
 - 2
 - 4
 - 6
 - 8
- Which statement about co-ordination number of a cation is true?
 - Most metal ions exhibit only a single characteristic co-ordination number.
 - The co-ordination number is equal to the number of ligands bonded to the metal atom.
 - The co-ordination number is determined solely by the tendency to surround the metal atom with the same number of electrons as one of the rare gases.
 - For most cations, the co-ordination number depends on the size, structure and charge of the ligands.
- Which ion shows usually the co-ordination number 6?
 - Cr^{3+}
 - Fe^{3+}
 - Fe^{2+}
 - All
- The co-ordination number of Ni in $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ complex is:
 - 2
 - 4
 - 6
 - 5
- The EAN of nickel in $\text{Ni}(\text{CO})_4$ is:
 - 36
 - 38
 - 28
 - 54
- The EAN of platinum in potassium hexachlorido-platinate (IV) is:
 - 46
 - 86
 - 36
 - 84
- According to effective atomic number rule the central metal acquires:
 - Inert gas configuration
 - Duplet
 - Octet
 - Quartet
- In which of the following compounds, the central metal atom obeys EAN rule?
 - $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$
 - $\text{K}_3[\text{Fe}(\text{CN})_6]$
 - $\text{K}_4[\text{Fe}(\text{CN})_6]$
 - All
- EAN of Fe in $\text{K}_3[\text{Fe}(\text{CN})_6]$ is:
 - 36
 - 37
 - 38
 - 35
- EAN of Cr in $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$ is:
 - 32
 - 33
 - 34
 - 35
- The EAN of Ni in $[\text{Ni}(\text{CN})_4]^{2-}$ is:
 - 34
 - 35
 - 36
 - 28

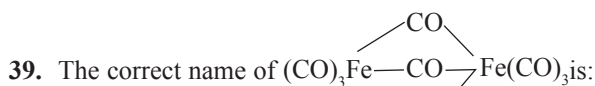
11.42 Co-ordination Chemistry

20. Which does not obey EAN rule?
 (a) $[\text{Cu}(\text{NH}_3)_4]^{2+}$ (b) $[\text{Zn}(\text{OH})_4]^{2-}$
 (c) $[\text{HgI}_4]^{2-}$ (d) $\text{Fe}(\text{CO})_5$
21. Which follows EAN rule?
 (a) $\text{Fe}(\text{CO})_5$ (b) $\text{Ni}(\text{CO})_4$
 (c) $\text{K}_4[\text{Fe}(\text{CN})_6]$ (d) All are correct
22. The EAN of Cr in $[\text{Cr}(\text{SCN})_6]^{3-}$ is:
 (a) 35 (b) 33
 (c) 34 (d) 37
23. In which of the following ions has the metal atom EAN as 36?
 (a) $[\text{Fe}(\text{CN})_6]^{4-}$ (b) $[\text{Fe}(\text{CN})_6]^{3-}$
 (c) $[\text{PdCl}_4]^{2-}$ (d) $[\text{Pd}(\text{CN})_6]^{2-}$
24. All ligands are:
 (a) Lewis acid (b) Lewis base
 (c) Neutral (d) None of these
25. A bridging ligand possesses;
 (a) Polydentate or monodentate nature
 (b) Two or more donor centres
 (c) The tendency to get itself attached to two metal ions
 (d) All of the above
26. Diethylene triamine is:
 (a) Chelating agent
 (b) Polydentate ligand
 (c) Tridentate ligand
 (d) All of these
27. In SCN ligand if N is attached to central atom, the name of ligand is:
 (a) Thiocyanato-N (b) Cyanato-N
 (c) Thiocyanato-S (d) Cyanato-S
28. EDTA is a ligand.
 (a) Monodentate (b) Hexadentate
 (c) Bidentate (d) Tridentate
29. Complexes with bidentate ligands are called:
 (a) Ligands (b) Chelates
 (c) Complexes (d) None of these
30. Which ligand is expected to be bidentate?
 (a) C_2O (b) $\text{CH}_3\text{C}\equiv\text{N}$
 (c) Br^- (d) CH_3NH_2
31. A group of atoms can function as a ligand only when:
 (a) It is a small molecule
 (b) It has an unshared electron pair

- (c) It is negatively charged ion
 (d) It is positively charged ion
32. Bidentate ligand is:
 (a) CN^- (b) Ethylene diamine
 (c) EDTA (d) SCN^-
33. The neutral ligand is:
 (a) Chloro (b) Hydroxo
 (c) Ammine (d) Oxalato
34. Which is not true ligands metal complex?
 (a) Larger the ligand, the more stable is the metal-ligand complex
 (b) Highly charged ligand forms stronger bonds
 (c) Larger the permanent dipole moment of ligand, the more stable is the bond
 (d) Greater the ionization potential of central metal, the stronger the bond
35. Which ligand is capable of forming low spin as well as high spin complexes?
 (a) CO (b) NO_2^-
 (c) CN^- (d) NH_3
36. IUPAC name of $\text{Na}_3[\text{Co}(\text{ONO})_6]$ is:
 (a) Sodium cobaltinitrite
 (b) Sodium hexanitritocobaltate (III)
 (c) Sodium hexanitrocobalt (III)
 (d) Sodium hexanitritocobaltate (II)
37. The IUPAC name of $\text{Ni}(\text{CO})_4$ is:
 (a) Tetracarbonyl nickelate (0)
 (b) Tetracarbonyl nickelate (II)
 (c) Tetracarbonyl nickel (0)
 (d) Tetracarbonyl nickel (II)
38. The correct IUPAC name of the complex



- (a) Dichloridodimethylglyoximate cobalt(II)
 (b) Bis(dimethylglyoxime) dichlorido cobalt(II)
 (c) Dimethylglyoxime cobalt(II) chloride
 (d) Dichloridodimethylglyoxime-N,N-cobalt(II)

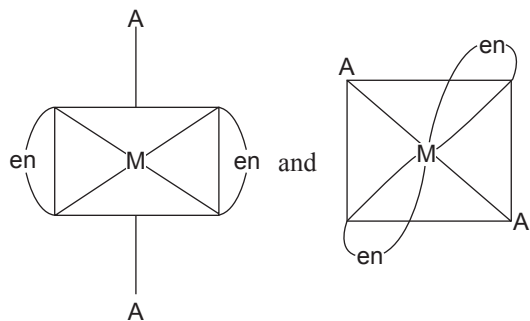


- (a) Tri- μ -carbonyl bis (tricarbonyl) iron(0)
 - (b) Hexacarbonyl iron(III) μ -tricarbonyl ferrate(0)
 - (c) Tricarbonyl iron(0) μ -tricarbonyl iron(0) tricarbonyl
 - (d) Nonacarbonyl iron
40. Lithium tetrahydridoaluminate is correctly represented as:
- (a) $\text{Al}[\text{LiH}_4]$
 - (b) $\text{Al}_2[\text{LiH}_4]_3$
 - (c) $\text{Li}[\text{AlH}_4]$
 - (d) $\text{Li}[\text{AlH}_4]_2$
41. IUPAC name of $[\text{Fe}(\text{O}_2)(\text{CN})_4\text{Cl}]^{4-}$ is:
- (a) Chloridotetracynido dioxoferrate (II) ion
 - (b) Chloridotetracyanido peroxoferrate (II) ion
 - (c) Chloridotetracyanido superoxoferrate (II) ion
 - (d) None of these
42. The IUPAC name of the Wilkinson's catalyst $[\text{RhCl}(\text{PPh}_3)_3]$ is:
- (a) Chloridotris (triphenylphosphine) rhodium (I)
 - (b) Chloridotris (triphenylphosphine) rhodium (IV)
 - (c) Chloridotris (triphenylphosphine) rhodium (0)
 - (d) Chloridotris (triphenylphosphine) rhodium (VI)
43. The IUPAC name for $\text{K}_2[\text{Cr}(\text{CN})_2\text{O}_2(\text{O}_2)\text{NH}_3]$ is:
- (a) Potassium amminedicyanidotetraoxo chromium (III)
 - (b) Potassium amminedicyanidodioxxygenidoxo chromate (IV)
 - (c) Potassium amminedicyanidosuperoxoperoxo chromate (III)
 - (d) Potassium amminedicyanidodioxoperoxo chromate (V)
44. IUPAC name of $\text{H}_2[\text{PtCl}_6]$ is:
- (a) hydrogen hexachloridoplatinate (IV)
 - (b) dihydrogen hexachloridoplatinate (IV)
 - (c) hydrogen hexachloridoplatinic (IV) acid
 - (d) hexachloridoplatinic (IV) acid.
45. The correct name for the complex ion $[\text{CoCl}(\text{ONO})(\text{en})_2]^+$ is:
- (a) chloridobis(ethylenediamine)nitrito-O-cobaltate (III)
 - (b) chloridodiethyldiaminenitrito-O-cobalt (III)
 - (c) chloridonitrito-O-diethyldiamine cobaltate (III)
 - (d) chloridobis(ethylenediamine)nitrito-O-cobalt (III)
46. The IUPAC name for $[\text{PtCl}(\text{NH}_2\text{CH}_3)(\text{NH}_3)_2]\text{Cl}$ is:
- (a) diamminechlorido(methylamine)platinum(II) chloride
 - (b) (dimethylamine) chloridodiamminoplatinum (II) chloride
 - (c) bis (ammine) chlorido (methylamine) platinate (II) chloride
 - (d) diamminechlorido(methylamine)platinum(II)chloride
47. The IUPAC nomenclature for the complex $\text{Na}[\text{PtBrCl}(\text{NO}_2)(\text{NH}_3)]$ is:
- (a) Sodium amminechloridobromidonitro-N-platinum(II)
 - (b) Sodium nitrochloridobromidoammine-N-platinate(II)
 - (c) Sodium amminebromidochloridonitro-N-platinate(II)
 - (d) Sodium amminebromidochloridonitro-N-platinum(II)
48. The IUPAC name of $\text{Xe}[\text{PtF}_6]$ is:
- (a) Hexafluoridoplatinate(VI)xenon
 - (b) Xenonhexafluoridoplatinate(V)
 - (c) Xenonhexafluoridoplatinate(VI)
 - (d) Xenoniumhexafluoridoplatinum(V)
49. Which of the following is correct IUPAC name of any complex compound?
- (a) Tris(acetylacetonato)iron(III)chloride
 - (b) Hexachloroplatinum(IV)tetraammine dicyanoplatinate(IV)
 - (c) Ammine bromochloro methylamine platinum (II)
 - (d) Cis-dichloro (ethylenediamine) platinum (II)
50. $[\text{Cr}(\text{NH}_3)_6]^{3+}$ ion is:
- (a) Paramagnetic
 - (b) Diamagnetic
 - (c) Square planar
 - (d) None of these
51. Among $[\text{Ni}(\text{CN})_4]^{2-}$, $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CO})_4]$:
- (a) $[\text{Ni}(\text{CN})_4]^{2-}$ is square planar and $[\text{NiCl}_4]^{2-}$, $\text{Ni}(\text{CO})_4$ are tetrahedral
 - (b) $[\text{NiCl}_4]^{2-}$ is square planar and $[\text{Ni}(\text{CN})_4]^{2-}$, $\text{Ni}(\text{CO})_4$ are tetrahedral
 - (c) $\text{Ni}(\text{CO})_4$ is square planar and $[\text{Ni}(\text{CN})_4]^{2-}$, $[\text{NiCl}_4]^{2-}$ are tetrahedral
 - (d) None of the above
52. Among $\text{Ni}(\text{CO})_4$, $[\text{Ni}(\text{CN})_4]^{2-}$ and $[\text{Ni}(\text{Cl})_4]^{2-}$:
- (a) $\text{Ni}(\text{CO})_4$, $[\text{NiCl}_4]^{2-}$ are diamagnetic and $[\text{Ni}(\text{CN})_4]^{2-}$ is paramagnetic
 - (b) $[\text{NiCl}_4]^{2-}$, $[\text{Ni}(\text{CN})_4]^{2-}$ are diamagnetic and $\text{Ni}(\text{CO})_4$ is paramagnetic

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- (c) $\text{Ni}(\text{CO})_4$, $[\text{Ni}(\text{CN})_4]^{2-}$ are diamagnetic and $[\text{NiCl}_4]^{2-}$ is paramagnetic
 (d) $\text{Ni}(\text{CO})_4$ is diamagnetic and $[\text{NiCl}_4]^{2-}$, $[\text{Ni}(\text{CN})_4]^{2-}$ are paramagnetic
53. The geometry of $\text{Ni}(\text{CO})_4$ and $\text{Ni}(\text{PPH}_3)_2\text{Cl}_2$ are:
 (a) Both square planar
 (b) Tetrahedral and square planar, respectively
 (c) Both tetrahedral
 (d) Square planar and tetrahedral, respectively
54. The structure of $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ ion is:
 (a) Square planar (b) Tetrahedral
 (c) Octahedral (d) Distorted rectangle
55. The shape of $[\text{Cu}(\text{NH}_3)_4]\text{Cl}_2$ is:
 (a) Tetrahedral (b) Octahedral
 (c) Square planar (d) Pyramidal
56. $\text{Fe}_2(\text{CO})_9$ is diamagnetic. Which of the following reasons is correct?
 (a) Presence of one CO as bridge group
 (b) Presence of monodentate ligand
 (c) Metal-metal (Fe-Fe) bond in molecule
 (d) Resonance hybridization of CO
57. The hybridization involved in $[\text{CoF}_6]^{3-}$ is:
 (a) d^2sp^3 (b) d^3sp^2
 (c) dsp^3 (d) sp^3d^2
58. Which have octahedral shape (d^2sp^3) hybridization of central atom?
 (a) $[\text{Cr}(\text{NH}_3)_6]^{2+}$ (b) $[\text{Fe}(\text{CN})_6]^{3-}$
 (c) $[\text{Cu}(\text{NH}_3)_6]^{+}$ (d) All are correct
59. The number of d-electrons in $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ is:
 (a) 2 (b) 3
 (c) 4 (d) 5
60. Which is the strongest field ligand?
 (a) CN^- (b) NO_2
 (c) NH_3 (d) En
61. Which ion produces a small crystal field splitting (a weak ligand field)?
 (a) I^- (b) Cl^-
 (c) F^- (d) All
62. Which ligand produces a high crystal field splitting (a strong ligand field)?
 (a) CO (b) NO
 (c) CN^- (d) All are correct
63. Of the following complexes, the one with the largest value of the crystal field splitting is:
 (a) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ (b) $[\text{Ru}(\text{CN})_6]^{3-}$
 (c) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ (d) $[\text{Fe}(\text{NH}_3)_6]^{3+}$
64. The tetrahedral crystal field splitting is only of the octahedral splitting,
 (a) 1/9 (b) 2/9
 (c) 4/9 (d) 5/9
65. The colour of $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ is:
 (a) Red (b) White
 (c) Green (d) Blue
66. Addition of KI to HgI_2 forms complex $\text{K}_2[\text{HgI}_4]$ having:
 (a) Red colour (b) Blue colour
 (c) Violet colour (d) Colourless nature
67. Which is colourless complex?
 (a) $\text{Cu}_2(\text{CH}_3\text{COO})_4 \cdot \text{H}_2\text{O}$
 (b) Cu_2Cl_2
 (c) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
 (d) $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4 \cdot \text{SO}_4 \cdot 4\text{H}_2\text{O}$
68. Blue colour/precipitate will be obtained when $\text{K}_4[\text{Fe}(\text{CN})_6]$ reacts with:
 (a) Fe(II) ions (b) Cu(II) ions
 (c) Fe(III) ions (d) Cu(I) ions
69. If a compound absorbs violet colour from the sunlight, then the observed colour is:
 (a) Yellow (b) Orange
 (c) Blue (d) Green
70. The probable formula for prussian blue is:
 (a) $\text{Fe}_3[\text{Fe}(\text{CN})_6]_2$ (b) $\text{Fe}_2[\text{Fe}(\text{CN})_6]_3$
 (c) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$ (d) $\text{Fe}_3[\text{Fe}(\text{CN})_6]_4$
71. Colour of transition metal complexes can be explained by:
 (a) Completely filled d-orbitals
 (b) Vacant d-orbitals
 (c) d-d transition
 (d) None of the above
72. Which of the following compounds is not coloured?
 (a) $\text{Na}_2[\text{CuCl}_4]$ (b) $\text{Na}_2[\text{CdCl}_4]$
 (c) $\text{K}_4[\text{Fe}(\text{CN})_6]$ (d) $\text{K}_3[\text{Fe}(\text{CN})_6]$
73. The number of isomers possible for square planar complex $\text{K}_2[\text{PdClBr}_2\text{SCN}]$ is:
 (a) 2 (b) 3
 (c) 4 (d) 6
74. Which complex is likely to show optical activity?
 (a) $\text{Trans}[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$
 (b) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$
 (c) $\text{Cis}[\text{Co}(\text{NH}_3)_2(\text{en})_2]^{3+}$
 (d) $\text{Trans}[\text{Co}(\text{NH}_3)_2(\text{en})_2]^{3+}$

75. The number of isomeric forms in which $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ ion can occur is:
 (a) 2 (b) 3
 (c) 4 (d) 1
76. $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$ and $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ are the examples of:
 (a) Linkage isomerism
 (b) Geometrical isomerism
 (c) Ionization isomerism
 (d) Optical isomerism
77. The possible number of isomers for the complex $[\text{MCl}_2\text{Br}_2]\text{SO}_4$ is:
 (a) 1 (b) 2
 (c) 4 (d) 5
78. The number of geometrical isomers of $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$ are:
 (a) Zero (b) 2
 (c) 3 (d) 4
79. Geometrical isomerism is found in co-ordination compounds having co-ordination number:
 (a) 2 (b) 3
 (c) 4 (tetrahedral) (d) 6
80. Cis-trans-isomerism is found in square planar complexes of the molecular formula: (a and b are monodentate ligands)
 (a) Ma_4 (b) Ma_3b
 (c) Ma_2b_2 (d) Mab_3
81. The two complexes given below are:



- (a) Geometrical isomers
 (b) Position isomers
 (c) Optical isomers
 (d) Identical
82. The compound $[\text{Co}(\text{NO}_2)(\text{NH}_3)_5]\text{Cl}_2$ and $[\text{Co}(\text{ONO})(\text{NH}_3)_5]\text{Cl}_2$ are examples of:
 (a) Geometrical isomers
 (b) Linkage isomers
 (c) Ligand isomers
 (d) Ionization isomers
83. Which of the following compounds would exhibit co-ordination isomerism?
 (a) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_2$ (b) $[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$
 (c) $[\text{Cr}(\text{en})_2]\text{NO}_2$ (d) $[\text{Ni}(\text{NH}_3)_6][\text{BF}_4]_2$
84. Which one of the following will be able to show cis-trans-isomerism
 (a) MA_3B (b) $\text{M}(\text{AA}')_2$
 (c) MA_2BCD (d) MAA
 (AA' : is unsymmetrical bidentate ligand ABCD are unidentate ligands.)
85. $[(\text{C}_6\text{H}_5)_2\text{Pd}(\text{SCN})_2]$ and $[(\text{C}_6\text{H}_5)_2\text{Pd}(\text{NCS})_2]$ are:
 (a) Linkage isomers
 (b) Co-ordination isomers
 (c) Ionization isomers
 (d) Geometrical isomers
86. The magnetic moments of complexes given below are in the order:
 (I) $[\text{Ni}(\text{CO})_4]$ (II) $[\text{Mn}(\text{CN})_6]^{4-}$
 (III) $[\text{Cr}(\text{NH}_3)_6]^{3+}$ (IV) $[\text{CoF}_6]^{3+}$
 (a) $\text{I} > \text{II} > \text{III} > \text{IV}$ (b) $\text{IV} > \text{II} < \text{II} > \text{III}$
 (c) $\text{IV} > \text{II} > \text{I} > \text{III}$ (d) $\text{IV} < \text{II} < \text{I} < \text{III}$
87. Which is low spin complex:
 (a) $[\text{Fe}(\text{CN})_6]^{3-}$ (b) $[\text{Co}(\text{NO}_2)_6]^{3-}$
 (c) $[\text{Mn}(\text{CN})_6]^{3-}$ (d) All of these
88. The spin magnetic moment of cobalt in $\text{Hg}[\text{Co}(\text{SCN})_4]$ is:
 (a) $\sqrt{3}$ (b) $\sqrt{8}$
 (c) $\sqrt{15}$ (d) $\sqrt{24}$
89. Which of the following complex is an outer orbital complex?
 (a) $[\text{Ni}(\text{NH}_3)_6]^{2+}$ (b) $[\text{Mn}(\text{CN})_6]^{4-}$
 (c) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (d) $[\text{Fe}(\text{CN})_6]^{4-}$
90. The most stable ion is:
 (a) $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ (b) $[\text{Fe}(\text{CN})_6]^{3-}$
 (c) $[\text{Fe}(\text{CN})_6]^{3-}$ (d) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$
91. The magnetic moment of $[\text{MnX}_4]^{2-}$ is 5.9 BM. The geometry of the complex ion is:
 (X - monodentate halide ion)
 (a) tetrahedral (b) square planar
 (c) both are possible (d) None of these
92. A complex has a composition corresponding to the formula $\text{CoBr}_2\text{Cl} \cdot 4\text{NH}_3$. What is the structural formula if conductance measurements show two ions per

formula unit? Silver nitrate solution given an immediate precipitate of AgCl but no AgBr.

- (a) $[\text{CoBrCl}(\text{NH}_3)_4]\text{Br}$
 (b) $[\text{CoCl}(\text{NH}_3)_4]\text{Br}_2$
 (c) $[\text{CoBr}_2\text{Cl}(\text{NH}_3)_4]$
 (d) $[\text{CoBr}_2(\text{NH}_3)_4]\text{Cl}$

93. The correct formula of diammine dichloridodicyanido chromate (III) is:

- (a) $[\text{CrCl}_2(\text{CN})_2(\text{NH}_3)_2]^{3+}$
 (b) $[\text{CrCl}_2(\text{CN})_2(\text{NH}_3)_2]^{3-}$
 (c) $[\text{CrCl}_2(\text{CN})_2(\text{NH}_3)_2]^-$
 (d) $[\text{CrCl}_2(\text{CN})_2(\text{NH}_3)_2]^-$

94. Which one of the following pairs of isomers and types of isomerism are correctly matched?

- (i) $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$ and $[\text{Co}(\text{NH}_3)_5(\text{ONO})]\text{Cl}_2 \dots$ (Linkage)
 (ii) $[\text{Cu}(\text{NH}_3)_4][\text{PtCl}_4]$ and $[\text{Pt}(\text{NH}_3)_4][\text{CuCl}_4] \dots$ (Coordination)

Select the correct answer using the codes given below:

- (a) (ii) and (iii) (b) (i), (ii) and (iii)
 (c) (i) and (iii) (d) (i) and (ii)

95. Which is not true about metal carbonyls?

- (a) Here CO acts as a Lewis base as well as Lewis acid
 (b) Here metal acts as Lewis base as well as Lewis acid
 (c) Here $d\pi$ - $p\pi$ back bonding takes place
 (d) Here $p\pi$ - $p\pi$ back bonding takes place

96. Consider the following complex: $[\text{Co}(\text{NH}_3)_5\text{CO}_3]\text{ClO}_4$

The coordination number, oxidation number, no. of d-electrons and number of unpaired d-electrons on the metal are respectively:

- (a) 6, 2, 7, 3 (b) 7, 2, 7, 1
 (c) 5, 3, 6, 4 (d) 6, 3, 6, 0

97. Which of the following pairs of complexes are isomeric with each other but their aqueous solutions exhibit different molar conductivities?

- (a) $[\text{PtCl}_2(\text{NH}_3)_4]\text{Br}_2$ and $[\text{PtBr}_2(\text{NH}_3)_4]\text{Cl}_2$
 (b) $[\text{CoCl}_2(\text{NH}_3)_4]\text{NO}_2$ and $[\text{CoCl}(\text{NO}_2)(\text{NH}_3)_4]\text{Cl}$
 (c) $[\text{Co}(\text{NO}_2)(\text{NH}_3)_5]\text{Cl}_2$ and $[\text{Co}(\text{ONO})(\text{NH}_3)_5]\text{Cl}_2$
 (d) $[\text{CoBr}(\text{NH}_3)_5\text{SO}_4]$ and $[\text{Co}(\text{SO}_4)(\text{NH}_3)_5]\text{Br}$

98. The degeneracy of d-orbitals is lost under:

- (I) Strong field ligand
 (II) Weak field ligand

- (III) Mixed field ligand
 (IV) Chelated ligand field

Choose the correct code:

- (a) I, II and IV (b) I and II
 (c) I, II, III and IV (d) I, II and III

99. Select the correct code about complex $[\text{Cr}(\text{NO}_2)(\text{NH}_3)_5][\text{ZnCl}_4]$:

- (I) IUPAC name of the compound is pentaamminenitrito-N-chromium (III) tetra chlorozincate(II)
 (II) It shows geometrical isomerism
 (III) It shows linkage isomerism
 (IV) It shows coordination isomerism
 (a) III, IV (b) I, III and IV
 (c) II, III and IV (d) I, II, III and IV

100. Which of the following is true about the complex $[\text{PtCl}_2(\text{H}_2\text{O})(\text{NH}_3)]$?

- (a) It exhibits geometrical isomerism
 (b) It is paramagnetic complex
 (c) Its geometry is tetrahedron
 (d) Platinum is sp^3 hybridised

101. Which of the following is not optically active?

- (a) $[\text{Co}(\text{en})_3]^{3+}$
 (b) $[\text{Cr}(\text{Ox})_3]^{3-}$
 (c) $\text{cis-}[\text{CoCl}_2(\text{en})_2]^+$
 (d) $\text{trans-}[\text{CoCl}_2(\text{en})]^+$

102. Which of the following isomerism, exhibited by $[\text{CrCl}_2(\text{OH})_2(\text{NH}_3)_2]^+$?

- (a) Ionization (b) Geometrical
 (c) Hydrate (d) Linkage

103. An aqueous solution of titanium chloride, when subjected to magnetic measurement, measured zero magnetic moment. Assuming the octahedral complex in aqueous solution, the formulae of the complex is:

- (a) $[\text{Ti}(\text{H}_2\text{O})_6]\text{Cl}_2$ (b) $[\text{Ti}(\text{H}_2\text{O})_6]\text{Cl}_4$
 (c) $[\text{TiCl}_3(\text{H}_2\text{O})_3]$ (d) $[\text{TiCl}_2(\text{H}_2\text{O})_4]$

104. Consider the following complex: $[\text{CO}_3(\text{NH}_3)_5]\text{ClO}_4$
 Mark the correct option:

	Coordination no.	Oxidation no.	No. of d-electrons	Unpaired d-electrons
(a)	6	3	6	0
(b)	7	2	7	1
(c)	7	1	6	4
(d)	6	2	7	3

- 105.** The crystal field stabilisation energy of $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ is:
 (a) $7.2\Delta_0$ (b) $0.4\Delta_0$
 (c) $2.4\Delta_0$ (d) $3.6\Delta_0$
- 106.** Which of the complexes is most stable?
 (a) $[\text{CoFe}]^{3-}$ (b) $[\text{Co}(\text{NH}_3)_6]^{3+}$
 (c) $[\text{Co}(\text{CN})_6]^{3-}$ (d) $[\text{Co}(\text{en})_3]^{3+}$
- 107.** The magnitude of crystal field stabilization energy in octahedral field depends on:
 (I) the nature of the ligand
 (II) the charge on the metal ion
 (III) whether the metal is in the first, second or third row of the transition elements
 (a) I, II, III only correct
 (b) I, II only correct
 (c) II, III only correct
 (d) III only correct
- 108.** Which of the following does not involve back bonding?
 (a) MnO_4^- (b) $\text{Ni}(\text{CO})_4$
 (c) $[\text{Fe}(\text{CN})_6]^{4-}$ (d) $[\text{CoCl}_4]^-$
- 109.** An effective atomic number of $\text{Co}(\text{CO})_4$ is 35 and hence is less stable. It attains stability by:
 (a) Oxidation of Co
 (b) Reduction of Co
 (c) Dimerization
 (d) Both (b) and (c)
- 110.** The value of 'spin only' magnetic moment for one of the following configuration is 2.84 BM.
 The correct one is:
 (a) d^4 (in strong field ligand)
 (b) d^2 (in weak field ligand)
 (c) d^3 (in weak as well as in strong field ligand)
 (d) d^5 (in strong field ligand)
- 111.** Which of the following has largest number of isomers?
 (a) $[\text{Co}(\text{en})_2\text{Cl}_2]^+$ (b) $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$
 (c) $\text{Ir}(\text{PR}_3)_2\text{H}(\text{CO})^{2+}$ (d) $[\text{Ru}(\text{NH}_3)_4\text{Cl}_2]^+$
- 112.** Which of the following statements is incorrect?
 (a) The stability constant of $[\text{Co}(\text{NH}_3)_6]^{3+}$ is greater than that of $[\text{Co}(\text{NH}_3)_6]^{2+}$.
 (b) The cyano complexes are far more stable than those formed by halide ions.
 (c) The stability of halide complexes follows the order $\text{I}^- < \text{Br}^- < \text{Cl}^-$.
 (d) The stability constant of $[\text{Cu}(\text{NH}_3)_4]^{2+}$ is greater than that of $[\text{Cu}(\text{en})_2]^{2+}$.
- 113.** Which of the following statements is true?
 (a) In $[\text{PtCl}_2(\text{NH}_3)_2]^{2+}$ the cis form is optically inactive while trans form is optically active.
 (b) In $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$, geometrical isomerism does not exist while optical isomerism exists.
 (c) In Mabcd , square planar complexes show both optical as well as geometrical isomerism.
 (d) In Mabcd tetrahedral complex, optical isomerism cannot be observed.
- 114.** Which of the following statement is not true for the reaction given below?

$$[\text{Cu}(\text{H}_2\text{O})_4]^{2+} + 4\text{NH}_3 \rightarrow [\text{Cu}(\text{NH}_3)_4]^{2+} + 4\text{H}_2\text{O}$$
 (a) It is a ligand substitution reaction.
 (b) NH_3 is a relatively strong field ligand while H_2O is a weak field ligand.
 (c) During the reaction, there is a change in colour from light blue to dark blue.
 (d) $[\text{Cu}(\text{NH}_3)_4]^{2+}$, has a tetrahedral structure and is paramagnetic.
- 115.** $[\text{M}(\text{H}_2\text{O})_6]^{2+}$ complex typically absorbs at around 600 nm. It is allowed to react with ammonia to form a new complex $[\text{M}(\text{NH}_3)_6]^{2+}$ that should have absorption at:
 (a) 800 nm (b) 580 nm
 (c) 620 nm (d) 320 nm
- 116.** An ion M^{2+} , forms the complexes $[\text{M}(\text{H}_2\text{O})_6]^{2+}$, $[\text{M}(\text{en})_3]^{2+}$ and $[\text{MBr}_6]^{4-}$, match the complex with the appropriate colour:
 (a) Green, blue and red
 (b) Blue, red and green
 (c) Green, red and blue
 (d) Red, blue and green
- 117.** If $\Delta_0 < P$, then what is electronic arrangement of metal atom/ion in complex with d^4 configuration:
 (a) t_{2g}^4, e_g^0 (b) e_g^4, t_{2g}^0
 (c) t_{2g}^3, e_g^1 (d) e_g^2, t_{2g}^2
- 118.** Set of d-orbitals which is used by central metal during formation of
 (a) $d_{x^2-y^2}, d_{z^2}, d_{xy}$
 (b) d_{xy}, d_{yz}, d_{xz}
 (c) $d_{x^2-y^2}, d_{xy}, d_{xz}$
 (d) $d_{x^2-y^2}, d_{z^2}, d_{xz}$
- 119.** The CFSE for $[(\text{CoCl})_6]^{4-}$ complex is 18000 cm^{-1} . The Δ for $[\text{CoCl}_4]^{2-}$ will be:

- (a) 18000 cm^{-1} (b) 16000 cm^{-1}
 (c) 8000 cm^{-1} (d) 2000 cm^{-1}

120. The type of isomerism shown by $[\text{Co}(\text{en})_2(\text{NCS})_2]\text{Cl}$ and $[\text{Co}(\text{en})_2(\text{NCS})\text{Cl}]\text{NCS}$ is:

- (a) Co-ordination (b) Ionization
 (c) Linkage (d) All of these

121. Which of the following complexes will show geometrical as well as optical isomerism? (*en* = ethylene diamine)

- (a) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ (b) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$
 (c) $[\text{Pt}(\text{en})_3]^{4+}$ (d) $[\text{Pt}(\text{en})_2\text{Cl}_2]$

COMPREHENSION BASED QUESTIONS

COMPREHENSION-I

Complex compounds are molecular compounds which retain their identities even when dissolved in water. They do not give all the simple ions in solution but instead furnish complex ions. The complex compounds are often called coordination compounds because certain groups called ligands are attached to the central metal ion by coordinate or dative bonds. Coordination compounds exhibit isomerism, both structural and stereoisomerism. the structure, magnetic property, colour and electrical properties of complexes are explained by various theories.

1. Arrange the following compounds in order of their molar conductance:

- (i) $\text{K}[\text{Co}(\text{NO}_2)_4(\text{NH}_3)_2]$
 (ii) $[\text{Cr}(\text{ONO})_3(\text{NH}_3)_3]$
 (iii) $[\text{Cr}(\text{NO}_2)(\text{NH}_3)_5]_3[\text{Co}(\text{NO}_2)_6]_2$
 (iv) $\text{Mg}[\text{Cr}(\text{NO}_2)_5(\text{NH}_3)]$
 (a) (ii) < (i) < (iv) < (iii)
 (b) (i) < (ii) < (iii) < (iv)
 (c) (ii) < (i) < (iii) < (iv)
 (d) (iv) < (iii) < (ii) < (i)

2. The oxidation number, coordination number and magnetic moment in the following complex is:

- $[\text{Cr}(\text{C}_2\text{O}_4)_2(\text{NH}_3)_2]^-$
 (a) O.N. = +3, C.N. = 6, M.M. = $\sqrt{15}$ BM
 (b) O.N. = -1, C.N. = 6, M.M. = $\sqrt{15}$ BM
 (c) O.N. = +3, C.N. = 6, M.M. = $\sqrt{3}$ BM
 (d) O.N. = +3, C.N. = 6, M.M. = $\sqrt{12}$ BM

3. In which of the following pairs, both the complexes have the same geometry?

- (a) $[\text{NiCl}_4]^{2-}$, $[\text{Ni}(\text{CN})_4]^{2-}$
 (b) $[\text{CoF}_6]^{3-}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$
 (c) $[\text{Ni}(\text{CO})_4]$, $[\text{Ni}(\text{CN})_4]^{2-}$
 (d) $[\text{Cu}(\text{NH}_3)_4]^{2+}$, $[\text{Ni}(\text{NH}_3)_4]^{2+}$

COMPREHENSION-II

Recent X-ray work, IR and other spectroscopic methods have proved that Turnbull's blue is identical to Prussian blue.

1. What is the common formula of Turnbull's blue and Prussian blue?

- (a) $\text{Fe}_3[\text{Fe}(\text{CN})_6]_2$
 (b) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
 (c) $\text{KFe}[\text{Fe}(\text{CN})_6]$
 (d) $\text{KFe}_2[\text{Fe}(\text{CN})_6]$

2. Intense blue colour arises as a result of:

- (a) electron transfer between Fe(II) and Fe(I)
 (b) electron transfer between Fe(II) and Fe(III)
 (c) d-d transition
 (d) spin magnetic moment

COMPREHENSION-III

On the basis of stability of complex in the solution, complexes may be of two types, perfect and imperfect complexes. The stability depends upon the extent of dissociation which in turn depends upon the strength of metal-ligand bond. The stability of complex also depends upon charge on central metal atom, basic nature of ligand, chelation, and nature of metal ion and ligand according to HSAB principle.

1. Which one of the following does not follow EAN rule?

- (a) $\text{Fe}(\text{CO})_5$ (b) $\text{V}(\text{CO})_6$
 (c) $\text{K}_4[\text{Fe}(\text{CN})_6]$ (d) $\text{Mn}_2(\text{CO})_{10}$

2. Which complex is most stable?

- (a) $[\text{Cu}(\text{CN})_2]^-$ $K_d = 1 \times 10^{-16}$
 (b) $[\text{Fe}(\text{CN})_6]^{4-}$ $K_d = 1 \times 10^{-37}$
 (c) $[\text{Fe}(\text{CN})_6]^{3-}$ $K_d = 1 \times 10^{-44}$
 (d) $[\text{Ag}(\text{CN})_2]^-$ $K_d = 1 \times 10^{-20}$

(c) Three are 3 geometrical and 15 stereoisomers possible for $[\text{Pt}(\text{NO}_2)(\text{NH}_3)(\text{NH}_2\text{OH})]$

(py)]⁺ and [PtBrClI(NO₂)(NH₃)(NH₃)(py)] respectively

- (d) cis and trans forms are not diastereomers to each other

3. Which of the following statements is/are true?

- (a) In metal carbonyl complexes d_{C-O} increases compared to that in CO molecule
 (b) The pair of compounds [Cr(H₂O)₆]Cl₃ and [CrCl₃(H₂O)₃].3H₂O show hydrate isomerism
 (c) d_{z^2} orbital of central metal atom/ion is used in dsp^2 hybridization
 (d) Facid and Meridional isomers associated with [Ma₃b₃]^{n±} type complex compound, both are optically inactive

4. Which is not correctly matched?

Column-I

Column-II

- | | |
|---|--|
| (a) K[CrF ₄ O] | Potassium tetrafluorooxochromate (v) |
| (b) Na[BH(OCH ₃) ₃] | Sodium hydrido trimethoxy borate (III) |
| (c) [Be(CN ₃ —CO—CH—CO—C ₆ H ₅) ₂] ⁰ | Bis(Benzoylacetonato)beryllium (III) |
| (d) A[AuCl ₄] | Hydrogen tetrachloroaurate(III) |

5. Which of the following statement is true about the complex [CrCl₃(OH)₂(NH₃)]²⁻ ion?

- (a) It has three geometrical isomers.
 (b) Only one space isomers is optically active and remaining are inactive.
 (c) There are total four space isomers.
 (d) The magnetic moment of complex ion is 3.89 BM.

6. A d-block element forms octahedral complex but its magnetic moment remains same either in strong field or in weak field ligand. Which of the following is/are correct?

- (a) Element always forms colourless compound
 (b) Number of electrons in t_{2g} orbitals are higher than in e_g orbitals
 (c) It can have either d^3 or d^8 configuration
 (d) It can have either d^7 or d^8 configuration

7. Which of the following is correct about

Tetraamminedithiocyanato-s cobalt (III) tris(oxalato)cobaltate(III)?

- (a) Formula of the complex is [Co(SCN)₂(NH₃)₄][Co(ox)₃]
 (b) It is a chelating complex and show linkage isomerism

- (c) It shows optical isomerism
 (d) It shows geometrical isomerism

8. Which is correct statement(s)?

- (a) [Ag(NH₃)₂]⁺ is linear with sp hybridised Ag⁺ ions.
 (b) NiCl₄²⁻, CrO₄²⁻ and MnO₄⁻ have tetrahedral geometry.
 (c) [Cu(NH₃)₄]²⁺, [Pt(NH₃)₄]²⁺ and [Ni(CN)₄]²⁻ have dsp^2 hybridization of the metal ion.
 (d) Fe(CO)₅ has trigonal bipyramidal structure with d_{z^2} sp^3 hybridised ion.

9. Which of the following statements is/are false?

- (a) In [PtCl₂(NH₃)₄]²⁺ complex ion, the cis-form is optically active, while trans form is optically inactive.
 (b) In [Fe(C₂O₄)₃]³⁻, geometrical isomerism does not exist, while optical isomerism exists.
 (c) [Mabcd]^{n±} tetrahedral complexes, optical isomerism cannot be observed.
 (d) In [Mabcd]^{n±} tetrahedral complexes, optical isomerism cannot be observed.

10. For which of the following d^n configuration of octahedral complex(es), cannot exist in both high spin and low spin forms?

- (a) d^3 (b) d^5
 (c) d^6 (d) d^8

11. Select the correct statement:

- (a) Chelation effect is maximum for five and six membered rings.
 (b) Greater the charge on the central metal cation, greater the value of $\Delta(CFSE)$.
 (c) In complex ion [CoF₆]³⁻, F⁻ is a weak field ligand, so that $\Delta_{oct} < P$ (Pairing energy) and it is low spin complex.
 (d) [CoCl₂(NH₃)₂(en)]⁺ complex ion will have four different isomers.

12. Which of the following statements is/are true?

- (a) In ferrocyanide ion, the effective atomic number is 36.
 (b) Chelating ligands are atleast bidentate ligand.
 (c) [CrCl₂(CN)₂(NH₃)₂]⁺ and [CrCl₃(NH₃)₃] both have d^2sp^3 hybridization.
 (d) As the number of rings in complex increases, stability of complex (chelate) also increases.

13. Which of the following statement(s) is/are correct?

- (a) The oxidation state of iron in sodium nitroprusside Na₂[Fe(CN)₅(NO)] is +II
 (b) [Ag(NH₃)₂]⁺ is linear in shape

- (c) In $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, Fe is d^2sp^3 hybridized
 (d) In $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ complex ion, Co is d^2sp^3 hybridized
14. Which one of the following statements(s) is/are false?
 (a) Weak ligands like F^- , Cl^- and OH^- usually form low spin complexes
 (b) Strong ligand like CN^- and NO_2^- , generally form high spin complexes
 (c) $[\text{FeF}_6]^{3-}$ is high spin complex
 (d) $[\text{Ni}(\text{CO})_4]$ is high spin complex
15. In test of NO_3^- ion, the dark brown ring complex is formed, which is true of this complex?
 (a) The colour is due to charge transfer spectra
 (b) Iron and NO both have +1 charge
 (c) The complex species can be represented as $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]^{2+}$
 (d) Iron has +2 oxidation state and NO is neutral
16. Which of the following pairs show coordination isomerism?
 (a) $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$ and $[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$
 (b) $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})_2\text{Cl}]\text{Br}_2$ and $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})\text{Cl}]\text{Br}[\text{Br}.\text{H}_2\text{O}]$
 (c) $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Br}_2$ and $[\text{Pt}(\text{NH}_3)_3\text{B}_2]\text{Cl}_2$
 (d) $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{C}_2\text{O}_4)_3]$ and $[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{C}_2\text{O}_4)_3]$
17. What is/are the coordination number(s) of Au in the complexes formed by Au?
 (a) 6 (b) 4
 (c) 5 (d) 2
18. The d-orbitals involved in sp^3d^2 or d^2sp^3 hybridization of the central metal ion are:
 (a) $d_{x^2-y^2}$ (b) d_{xy}
 (c) d_{yz} (d) d_{z^2}
19. The complex(es) which is/are blue in colour:
 (a) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$ (b) $\text{Zn}_2[\text{Fe}(\text{CN})_6]$
 (c) $\text{Cu}_2[\text{Fe}(\text{CN})_6]$ (d) $\text{Fe}_3[\text{Fe}(\text{CN})_6]_2$
20. Which of the following are coordination isomers of $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$?
 (a) $[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$
 (b) $[\text{Cr}(\text{NH}_3)_2(\text{CN})_4][\text{Co}(\text{CN})_4(\text{NH}_3)_2]$
 (c) $[\text{Cr}(\text{NH}_3)_3(\text{CN})_3][\text{Co}(\text{NH}_3)_3(\text{CN})_3]$
 (d) None of these
21. Which of the following statement(s) is(are) correct?
 (a) The complexes $[\text{NiCl}_4]^{2-}$ and $[\text{NiCl}_4]^{2-}$ differ in the magnetic properties
 (b) The complexes $[\text{NiCl}_4]^{2-}$ and $[\text{NiCN}_4]^{2-}$ differ in the geometry
 (c) The complexes $[\text{NiCl}_4]^{2-}$ and $[\text{NiCN}_4]^{2-}$ differ in primary valencies of nickel
 (d) The complexes $[\text{NiCl}_4]^{2-}$ and $[\text{NiCN}_4]^{2-}$ differ in the state of hybridization of nickel

MATCH THE COLUMN TYPE QUESTIONS

Column-I and Column-II contains four entries each. Entries of Column-I are to be matched with one or more than one entries of Column-II. Each entry of Column-I may have the matching with one or more than one entries of Column-II.

- | 1. Column-I
(Pair of complex compounds) | Column-II
(Property which is different in given pair) |
|--|--|
| (a) $[\text{Ni}(\text{CO})_4]$ and $\text{K}_2[\text{Ni}(\text{CN})_4]$ | (P) Magnetic moment |
| (b) $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$ and $\text{K}_3[\text{Cu}(\text{CN})_4]$ | (Q) Oxidation no. of central metal |
| (c) $\text{K}_2[\text{NiCl}_4]$ and $\text{K}_4[\text{Ni}(\text{CN})_4]$ | (R) Geometry |

- | | |
|---|--------------------------|
| (d) $\text{K}_2[\text{NiCl}_4]$ and $\text{K}_2[\text{PtCl}_4]$ | (S) EAN of central metal |
|---|--------------------------|

- | 2. Column-I
(Pair of complexes) | Column-II
(Property which is similar in given pair) |
|--|--|
| (a) $[\text{Fe}(\text{CN})_6]^{3-}$ and $[\text{Co}(\text{NH}_3)_6]^{2+}$ | (P) Magnetic moment |
| (b) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Fe}(\text{CN})_6]^{4-}$ | (Q) Geometry |
| (c) $[\text{Ni}(\text{CN})_4]^{4-}$ and $[\text{Ni}(\text{CO})_4]$ | (R) Hybridization |
| (d) $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{NiCl}_4]^{2-}$ | (S) Number of d-electrons |

3. Column-I (Molar conductance at infinite dilution)	Column-II (Complex compound)	(b) 0	(Q) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$
(a) 229	(P) $[\text{Pt}(\text{NH}_3)_5\text{Cl}]\text{Cl}_3$	(c) 404	(R) $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}_2$
		(d) 523	(S) $[\text{Pt}(\text{NH}_3)_6]\text{Cl}_4$

ASSERTION/REASON TYPE QUESTIONS

These questions consist of two statements each, printed as assertion and reason, while answering these questions you are required to choose any one of the following responses:

- (a) If assertion is true but the reason is false
 (b) If assertion is false but reason is true
 (c) If both assertion and reason are true and the reason is the correct explanation of assertion
 (d) If both assertion and reason are true but reason is not the correct explanation of assertion

1. Assertion: $[\text{Ti}(\text{H}_2\text{O})_6]^{4+}$ is coloured while $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$ is colourless.

Reason: d-d transition is not possible in $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$.

2. Assertion: $[\text{Cu}(\text{NH}_3)_4]^{2+}$ and $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ both react with $\text{K}_4[\text{Fe}(\text{CN})_6]$ to give brown ppt.

Reason: Both complexes are blue in colour of little different shade.

3. Assertion: $[\text{Fe}(\text{EDTA})]^-$ complex is octahedral in shape.

Reason: EDTA is a hexadentate ligand and undergoes sp^3d^2 hybridization.

4. Assertion: Tetrahedral complexes with chiral structure exhibit optical isomerism.

Reason: They lack plane of symmetry.

5. Assertion: Oxidation state of Fe in $\text{Fe}(\text{CO})_5$ is zero.

Reason: Synergic bonding takes place in this metal carbonyl complex.

6. Assertion: Zeise's salt is a π -bonded organometallic compound.

Reason: Zeise's salt contains C_2H_4 molecules as one of the ligands.

7. Assertion: $[\text{CoCl}_3(\text{NH}_3)_3]$ does not give white precipitate with AgNO_3 solution.

Reason: $[\text{CoCl}_3(\text{NH}_3)_3]$ complex is optically inactive.

8. Assertion: Transition metal ion forming octahedral complexes undergo sp^3d^2 or d^2sp^3 hybridization.

Reason: Strong field ligands force the unpaired electrons of central metal ion to pair up causing d^2sp^3 hybridization whereas weak field ligands do not affect electronic configuration of the metal ion undergoes in sp^3d^2 hybridization.

9. Assertion: Complex ion $[\text{Co}(\text{NH}_3)_6]^{2+}$ is readily oxidized to $[\text{Co}(\text{NH}_3)_6]^{3+}$.

Reason: Unpaired electron in complex ion $[\text{Co}(\text{NH}_3)_6]^{2+}$ is present in 4p orbital.

10. Assertion: Hydrazine is a neutral ligand.

Reason: It has two N as donor atoms and behaves as a chelating ligand.

RECENT YEARS' QUESTIONS OF IIT-JEE AND AIEEE

1. A square planar complex is formed by hybridization of which atomic orbitals: [AIEEE 2002]

- (a) s, p_x, p_y, d_{yz} (b) $s, p_x, p_y, d_{x^2-z^2}$
 (c) s, p_x, p_y, d_{z^2} (d) s, p_y, p_z, d_{xy}

2. Among $[\text{Ni}(\text{CO})_4]$, $[\text{Ni}(\text{CN})_4]^{2-}$ and $[\text{NiCl}_4]^{2-}$ species, the hybridization state of Ni atoms are respectively: [AIEEE 2002]

- (a) $\text{sp}^3, \text{ds}^2\text{p}, \text{dsp}^2$ (b) $\text{sp}^3, \text{dsp}^2, \text{sp}^3$
 (c) $\text{sp}^3, \text{sp}^3, \text{dsp}^2$ (d) $\text{dsp}^2, \text{sp}^3, \text{sp}^3$

3. The type of isomerism present in nitro pentaammine-chromium (III) chloride is: [AIEEE 2002]

- (a) Optical
 (b) Linkage
 (c) Ionisation
 (d) Polymerization

4. The most stable ion is: [AIEEE 2002]

- (a) $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ (b) $[\text{Fe}(\text{Cl})_6]^{3-}$
 (c) $[\text{Fe}(\text{CN})_6]^{3-}$ (d) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$

5. Ammonia forms the complex $[\text{Cu}(\text{NH}_3)_4]^{2+}$ with copper ions alkaline solutions but not in acidic solution. The reason for it is: **[AIEEE 2003]**
- In alkaline solution $\text{Cu}(\text{OH})_2$ is precipitated which is soluble in excess of alkali
 - Copper hydroxide is amphoteric substance
 - In acidic solution hydration protects Cu^{2+} ions
 - In acidic solution protons co-ordinates with ammonia molecule forming NH_4 ions and NH_3 molecules are not available.
6. One mole of complex compound $\text{Co}(\text{NH}_3)_5\text{Cl}_3$ gives 3 moles of ions on dissolution in water. One mole of same complex reacts with two moles of AgNO_3 to yield two moles of $\text{AgCl}_{(s)}$. The complex is: **[AIEEE 2003]**
- $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}.\text{NH}_3$
 - $[\text{Co}(\text{NH}_3)_4\text{Cl}]\text{Cl}_2.\text{NH}_3$
 - $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
 - $[\text{Co}(\text{NH}_3)_3\text{Cl}_3].2\text{NH}_3$
7. Which one of the following statement is correct: **[AIEEE 2003]**
- Ferric ions give a deep green precipitate on adding potassium ferrocyanide solution
 - On boiling a solution having K^+ , Ca^{2+} and HCO_3^- ions, we get a precipitate of $\text{K}_2\text{Ca}(\text{CO}_3)_2$
 - Manganese salt give a violet borax test in reducing flame
 - From a mixed precipitate of AgCl and AgI , ammonia solution dissolves only AgCl
8. Among the properties (a) reducing (b) oxidizing (c) complexing, the set of properties shown by CN^- ion towards metal species is: **[AIEEE 2004]**
- a, b, c
 - b, c
 - c, a
 - a, b
9. Which of the following complex is an outer orbital complex? **[AIEEE 2004]**
- $[\text{Ni}(\text{NH}_3)_6]^{2+}$
 - $[\text{Mn}(\text{CN})_6]^{4-}$
 - $[\text{Co}(\text{NH}_3)_6]^{3+}$
 - $[\text{Fe}(\text{CN})_6]^{4-}$
10. Which one of the following has largest number of isomers? **[AIEEE 2004]**
- $[\text{Co}(\text{en})_2\text{Cl}_2]^+$
 - $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$
 - $[\text{Ir}(\text{PR}_3)_2\text{H}(\text{CO})]^{2+}$
 - $[\text{Ru}(\text{NH}_3)_4\text{Cl}_2]^+$
11. The correct order of magnetic moments (spin values in B.M.) among is: **[AIEEE 2004]**
- $[\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-}$
 - $[\text{MnCl}_4]^{2-} > [\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-}$
 - $[\text{Fe}(\text{CN})_6]^{4-} > [\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-}$
 - $[\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{Fe}(\text{CN})_6]^{4-}$
12. Co-ordination compounds have great importance in biological systems. In this context which of the following statements is incorrect: **[AIEEE 2004]**
- Carboxypeptidase — A is an enzyme and contains zinc
 - Haemoglobin is the red pigment of blood and contains iron
 - Cyanocobalamin is B_{12} and contains cobalt
 - Chlorophylls are green pigments in plants and contain calcium
13. The co-ordination number of a central metal atom in a complex is determined by: **[AIEEE 2004]**
- The number of only anionic ligands bonded to the metal ion
 - The number of ligands around a metal ion bonded by π -bonds
 - The number of ligands around a metal ion bonded by σ - and π -bonds both
 - The number of ligands around a metal ion bonded by σ -bonds
14. Which one of the following has the regular tetrahedral structure: **[AIEEE 2004]**
- $[\text{Ni}(\text{CN})_4]^{2-}$
 - SF_4
 - BF_4^-
 - XeF_4
15. The IUPAC name of the co-ordination compound $\text{K}_3[\text{Fe}(\text{CN})_6]$ is: **[AIEEE 2005]**
- Potassium hexacyanoferrate (II)
 - Potassium hexacyanoferrate (III)
 - Potassium hexacyanoiron (II)
 - Tripotassium hexacyanoiron (II)
16. Which of the following will show optical isomerism: **[AIEEE 2005]**
- $[\text{Cu}(\text{NH}_3)_4]^{2+}$
 - $[\text{ZnCl}_4]^{2-}$
 - $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$
 - $[\text{Co}(\text{CN})_6]^{3-}$
17. Which one of the following has lowest value of paramagnetic behaviour: **[AIEEE 2005]**
- $[\text{Cr}(\text{CN})_6]^{3-}$
 - $[\text{Mn}(\text{CN})_6]^{3-}$
 - $[\text{Fe}(\text{CN})_6]^{3-}$
 - $[\text{Co}(\text{CN})_6]^{3-}$
18. The value of 'spin only' magnetic moment for one of the following configuration is 2.84 BM. The correct one is: **[AIEEE 2005]**
- d^4 (in strong field ligand)
 - d^2 (in weak field ligand)
 - d^3 (in weak as well as in strong field ligand)
 - d^5 (in strong field ligand)

19. The IUPAC name for the complex $[\text{Co}(\text{NO}_2)(\text{NH}_3)_5]\text{Cl}_2$ is: [AIEEE 2006]
 (a) Nitrito-N-pentaamminecobalt(III) chloride
 (b) Nitrito-N-pentaamminecobalt(II) chloride
 (c) Pentaammine nitrito-N-cobalt(II) chloride
 (d) Pentaammine nitrito-N-cobalt(III) chloride
20. Nickel ($Z = 28$) combines with a uninegative monodentate ligand X^- to form a paramagnetic complex $[\text{NiX}_4]^{2-}$. The number of unpaired electron(s) in the nickel and geometry of this complex ion are, respectively: [AIEEE 2006]
 (a) One, tetrahedral
 (b) Two, tetrahedral
 (c) One, square planar
 (d) Two, square planar
21. In $\text{Fe}(\text{CO})_5$, the $\text{Fe}-\text{C}$ bond possesses: [AIEEE 2006]
 (a) π -character only
 (b) Both σ - and π -characters
 (c) Ionic character
 (d) σ -character only
22. Which one of the following has a square planar geometry? [AIEEE 2007]
 (Co = 27, Ni = 28, Fe = 26, Pt = 78)
 (a) $[\text{CoCl}_4]^{2-}$ (b) $[\text{FeCl}_4]^{2-}$
 (c) $[\text{NiCl}_4]^{2-}$ (d) $[\text{PtCl}_4]^{2-}$
23. The complex ion which has no 'd' electron in the central metal atom is: [IIT 2001]
 (a) $[\text{MnO}_4]^-$ (b) $[\text{Co}(\text{NH}_3)_6]^{3+}$
 (c) $[\text{Fe}(\text{CN})_6]^{3-}$ (d) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$
24. Mixture X of 0.02 mole of $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ and 0.02 mole of $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$ was prepared in 2 litre of solution:
 1 litre of mixture X + Excess of $\text{AgNO}_3 \rightarrow \text{Y}$
 1 litre of mixture X + Excess of $\text{BaCl}_2 \rightarrow \text{Z}$
 Number of moles of Y and Z respectively are: [IIT 2003]
 (a) 0.01, 0.01 (b) 0.02, 0.01
 (c) 0.01, 0.02 (d) 0.02, 0.02
25. The species having tetrahedral shape is: [IIT 2004]
 (a) $[\text{PdCl}_4]^{2-}$ (b) $[\text{Ni}(\text{CN})_4]^{2-}$
 (c) $[\text{Pb}(\text{CN})_4]^{2-}$ (d) $[\text{NiCl}_4]^{2-}$
26. The spin magnetic moment of cobalt in $\text{Hg}[\text{Co}(\text{SCN})_4]$ is: [IIT 2004]
 (a) $\sqrt{3}$ (b) $\sqrt{8}$
 (c) $\sqrt{15}$ (d) $\sqrt{24}$
27. Which kind of isomerism is exhibited by octahedral $[\text{Co}(\text{NH}_3)_4\text{Br}_2]\text{Cl}$: [IIT 2005]
 (a) Geometrical and ionization
 (b) Geometrical and optical
 (c) Optical and ionization
 (d) Geometrical only
28. The bond length of $\text{C}-\text{O}$ bond in carbon monoxide is 1.128 Å. The $\text{C}-\text{O}$ bond in $\text{Fe}(\text{CO})_5$ is: [IIT 2006]
 (a) 1.158 Å (b) 1.128 Å
 (c) 1.178 Å (d) 1.118 Å
29. Among the following metal carbonyls, the $\text{C}-\text{O}$ bond order is lowest in: [IIT 2007]
 (a) $[\text{Mn}(\text{CO})_6]^+$ (b) $[\text{Fe}(\text{CO})_5]$
 (c) $[\text{Cr}(\text{CO})_6]$ (d) $[\text{V}(\text{CO})_6]^-$
30. In which of the following octahedral complexes of Co (at. no. 27), will the magnitude of Δ_0 be the highest? [AIEEE 2008]
 (a) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (b) $[\text{Co}(\text{CN})_6]^{3-}$
 (c) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ (d) $[\text{Co}(\text{H}_2\text{O})_6]^{3-}$
31. The co-ordination number and the oxidation state of the element 'E' in the complex $[\text{E}(\text{en})_2(\text{C}_2\text{O}_4)]\text{NO}_2$ (where (en) is ethylene diamine) are, respectively. [AIEEE 2008]
 (a) 6 and 3 (b) 6 and 2
 (c) 4 and 2 (d) 4 and 3
32. Among the following, the coloured compound is: [IIT 2008]
 (a) CuCl (b) $\text{K}_3\text{C}_4(\text{CN})_4$
 (c) CuF_2 (d) $[\text{Cu}(\text{CH}_3\text{CN})_4]\text{BF}_3$
33. The IUPAC name of $[\text{Ni}(\text{NH}_3)_4][\text{NiCl}_4]$ is: [IIT 2008]
 (a) Tetrachloronickel (II)-tetraamminenickel (II)
 (b) Tetraamminenickel (II)-tetrachloronickel (II)
 (c) Tetraamminenickel (II)-tetrachloronickelate (II)
 (d) Tetrachloronickel (II)-tetraamminenickelate (O)
34. Both $[\text{Ni}(\text{CO})_4]$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are diamagnetic. The hybridization of nickel in the compounds respectively are: [IIT 2008]
 (a) sp^3 , sp^3 (b) sp^3 , dsp^2
 (c) dsp^2 , sp^3 (d) dsp^2 , dsp^2
35. The compounds that exhibit geometrical isomerism is(are):
 (a) $[\text{Pt}(\text{en})\text{Cl}_2]$ (b) $[\text{Pt}(\text{en})_2]\text{Cl}_2$
 (c) $[\text{Pt}(\text{en})_2\text{Cl}_2]\text{Cl}_2$ (d) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$

ANSWER KEYS**ONE CORRECT OPTION TYPE QUESTIONS**

- | | | | | | | | | | |
|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| 1. (a) | 2. (a) | 3. (c) | 4. (c) | 5. (c) | 6. (a) | 7. (c) | 8. (b) | 9. (c) | 10. (d) |
| 11. (d) | 12. (c) | 13. (a) | 14. (b) | 15. (a) | 16. (c) | 17. (d) | 18. (b) | 19. (a) | 20. (a) |
| 21. (d) | 22. (b) | 23. (a) | 24. (b) | 25. (d) | 26. (d) | 27. (a) | 28. (b) | 29. (b) | 30. (a) |
| 31. (b) | 32. (b) | 33. (c) | 34. (a) | 35. (d) | 36. (b) | 37. (c) | 38. (a) | 39. (a) | 40. (c) |
| 41. (c) | 42. (a) | 43. (d) | 44. (d) | 45. (d) | 46. (a) | 47. (c) | 48. (b) | 49. (c) | 50. (a) |
| 51. (a) | 52. (c) | 53. (b) | 54. (a) | 55. (c) | 56. (c) | 57. (d) | 58. (d) | 59. (b) | 60. (a) |
| 61. (d) | 62. (d) | 63. (b) | 64. (c) | 65. (c) | 66. (d) | 67. (b) | 68. (c) | 69. (a) | 70. (c) |
| 71. (c) | 72. (b) | 73. (c) | 74. (c) | 75. (a) | 76. (c) | 77. (d) | 78. (b) | 79. (d) | 80. (c) |
| 81. (d) | 82. (b) | 83. (b) | 84. (c) | 85. (a) | 86. (b) | 87. (d) | 88. (c) | 89. (a) | 90. (a) |
| 91. (a) | 92. (d) | 93. (d) | 94. (b) | 95. (d) | 96. (d) | 97. (d) | 98. (c) | 99. (a) | 100. (a) |
| 101. (d) | 102. (b) | 103. (b) | 104. (a) | 105. (b) | 106. (d) | 107. (a) | 108. (d) | 109. (d) | 110. (b) |
| 111. (a) | 112. (d) | 113. (b) | 114. (d) | 115. (b) | 116. (b) | 117. (c) | 118. (b) | 119. (c) | 120. (b) |
| 121. (d) | | | | | | | | | |

COMPREHENSION BASED QUESTIONS**Comprehension-I**

1. (a) 2. (a) 3. (b)

Comprehension-II

1. (c) 2. (b)

Comprehension-III

1. (b) 2. (c)

Comprehension-IV

1. (a) 2. (b) 3. (c) 4. (c) 5. (c) 6. (a) 7. (a)

ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

- | | | | | |
|---------------|------------------|-----------------|---------------|---------------|
| 1. (b, d) | 2. (a, b, c, d) | 3. (a, d) | 4. (b, c, d) | 5. (a, d) |
| 6. (b, c) | 7. (b, c, d) | 8. (a, b, c, d) | 9. (a, c, d) | 10. (a, d) |
| 11. (a, b, d) | 12. (a, b, c, d) | 13. (a, b, d) | 14. (a, b, d) | 15. (a, b, c) |
| 16. (a, d) | 17. (b, d) | 18. (a, d) | 19. (a, d) | 20. (a, b) |
| 21. (a, b, d) | | | | |

MATCH THE COLUMN TYPE QUESTIONS

- | | | | |
|-------------------|-------------------|-------------------|---------------|
| 1. (a) → Q, R, S; | (b) → P, Q, R, S; | (c) → P, Q, S; | (d) → P, R, S |
| 2. (a) → P, Q, R; | (b) → Q, S; | (c) → P, Q, R, S; | (d) → P, S |
| 3. (a) → R; | (b) → Q; | (c) → P; | (d) → S; |

ASSERTION/REASON TYPE QUESTIONS

1. (b) 2. (d) 3. (a) 4. (c) 5. (d) 6. (c) 7. (d) 8. (c) 9. (a) 10. (a)

RECENT YEARS' QUESTIONS OF IIT-JEE AND AIEEE

- | | | | | | | | | | |
|---------|---------|---------|---------|------------|---------|---------|---------|---------|---------|
| 1. (b) | 2. (b) | 3. (b) | 4. (a) | 5. (d) | 6. (c) | 7. (d) | 8. (c) | 9. (a) | 10. (a) |
| 11. (d) | 12. (d) | 13. (d) | 14. (c) | 15. (b) | 16. (c) | 17. (d) | 18. (b) | 19. (d) | 20. (b) |
| 21. (b) | 22. (d) | 23. (a) | 24. (a) | 25. (d) | 26. (c) | 27. (a) | 28. (d) | 29. (b) | 30. (b) |
| 31. (a) | 32. (c) | 33. (c) | 34. (b) | 35. (c, d) | | | | | |

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Transition Elements (d-block) and Their Compounds

INTRODUCTION

The **transition elements** may be defined as elements whose atoms or simple ions in their most common oxidation state contain partially filled d-orbitals. This definition does not cover zinc, cadmium and mercury. However, these are studied with transition elements.

The general electronic configuration of these elements is $(n-1)d^{1-10} ns^{0-2}$.

- Zn, Cd and Hg (elements of group 12), i.e., the last members of each d-block series have a completely filled $(n-1)$ d-subshell in their elementary as well as in the combined state. Thus according to the definition of transition elements, these cannot be included in the transition elements. This is also further evident by the fact that they do not show much resemblance with other transition elements except for their ability to form complexes with ligands like ammonia, amines and halide ions. However, they are generally studied with d-block elements in order to maintain a rational classification of elements.
- Elements of group 3 (Sc, Y, La and Ac) differ markedly in their properties from that of other transition elements, viz., their compounds are uniformly trivalent, diamagnetic and colourless while the typical transition elements show variable oxidation state, form coloured ions and are paramagnetic. But since they have partly filled d-subshell, these are considered as transition elements.
- Thus elements of group 3 (Sc, Y, La and Ac) and group 12 (Zn, Cd and Hg) are referred to as non-typical transition elements while the other transition elements are called typical transition elements.

Classification of d-Block Elements

These are divided into four transition series:

- (i) 3d series (1st series: 4th period) → It has 10 elements from Sc($Z = 21$) to Zn₃₀
 - (ii) 4d series (2nd series: 5th period) – It has 10 elements Y₃₉ to Cd₄₈
 - (iii) 5d series (3rd series: 6th period) – It has 10 elements from La₅₇ (One element) and Hf₇₂ to Hg₈₀ (nine elements)
 - (iv) 6d series (4th series: 7th period) – (incomplete period) from Ac₈₉ and Ku₁₀₄ and extend up to element with atomic number 106.
- Transition elements lie between electropositive s-block elements and electronegative p-block elements in the periodic table.
 - In representative elements, (s and p-block elements) the valence electrons are present only in the outermost shell while in transition elements, the valence electrons are present in the outermost shell as well as d-orbitals of penultimate shell.

12.2 Transition Elements (d-block) and Their Compounds

- All transition elements are d-block elements, but the d-block elements are not transition elements.
- Most abundant and most widely used transition metal is iron.
- The only synthetic transition metal is Tc.
- Fe, Co and Ni are called ferrous metals. They are all strongly attracted by a magnet and are ferromagnetic.
- Ru, Rh, Pd, Os, Ir and Pt are called platinum metals.
- Cu, Ag and Au are called coinage metals.

General characteristics of transition elements

The members of given transition series do not differ so much from one another as those of non-transition elements (representative elements) of the same period. The reason is that the electronic configuration of transition elements differ only in the number of electrons in $(n-1)$ d subshell i.e., the number of electrons in the outermost shell (n) remains the same. Some important Properties of transition elements are as follows.

ATOMIC AND PHYSICAL PROPERTIES

The important properties of the transition metals are given below:

1. Metallic Character

All the transition elements are metal.

Reason – due to low values of their ionization energies, ns electrons can easily be lost by them to form M^+ or M^{+2} cation. i.e., metallic character increases on going down the group. Thus, metallic properties are shown by transition elements. Example,

- (i) They are good conductors of heat and electricity because of the existence of metallic bonding (i.e., presence of free mobile electrons)

Note: Among the atoms of these elements, Cu, Ag, Au show exceptionally high thermal and electrical conductivity.

- (ii) Transition elements exhibit good mechanical properties, i.e, they are hard, malleable and ductile. They have high enthalpies of atomization, high melting and boiling points.
- (iii) Their mechanical properties and high melting as well as boiling points indicate the presence of strong metallic bond. They exhibit all the three types of structures: face centred cubic (fcc), hexagonal closed packed (hcp) and body centred cubic (bcc). These properties reveal that both metallic and covalent bonding are present in the atoms of transition elements.

Worked-out Example

1. Why Cr, Mn and W are very hard metals but Zn, Cd and Hg are not hard?

Ans. (As the no. of unpaired $e^- \rightarrow$ no. of metallic bonds $\rightarrow$, strength of bond $\rightarrow$ and hence hardness $\rightarrow$) because Cr, Mn, W has maximum unpaired e^- and Zn, Cd, Hg do not have any unpaired e^-

- The transition elements show both covalent as well as metallic bonding amongst their atoms.
- The d-block elements do not form ionic compounds so readily as the alkali and alkaline earth metals.
- Unlike the s-block elements, the d-block elements have tendency to form covalent compounds under certain conditions.
- Generally, lower valent compounds are ionic while the higher valent compounds are covalent.

Worked-out Example

2. The enthalpies of atomization of the transition metals are high.

Ans. Due to strong interatomic forces.

2. Atomic radii

	IIIB (3)		IVB (4)	VB (5)	VIB (6)	VII B (7)	← (8)	VIII (9)	→ (10)	IB (11)	IIB (12)
1st series	Sc ₂₁ 162 A°		Ti 147 A°	V 134	Cr 127	Mn 126	Fe 126	Co 125	Ni 124	Cu 128	Zn ₃₀ 138
2nd series	Y39 180		Zr 160	Nb 146	Mo 139	Tc 136	Ru 134	Rh 134	Pd 137	Ag 144	Cd ₄₈ 154
3rd series	La57 187	Ce ₅₈ —Lu ₇₁ 165 156	Hf ₇₂ 158	Ta 146	W 139	Re 137	Os 135	Ir 136	Pt 138	Au 144	Hg ₈₀ 157

(i) Variation of atomic radii in a given series (period)

(a) It decreases gradually from 3rd to 6th gp (Ex Sc to Cr.)

Reason → the increased nuclear charge makes the atom to shrink in size and hence, the size of the atom decreases.

(b) Then atomic radii values remain almost constant up to the element of gp 11th (Ex from Cr to Cu).

Reason → due to the screening effect caused by (n-1) d electrons, the magnitude of nuclear charge decreases and as a result, the size of the atom does not change much from Cr to Cu.

(c) The last element of each series shows an increase in its atomic radius.

(Ex. Zn, Cd, Hg)

Reason → Towards the end of each series, there is an increase in $e^- - e^-$ repulsion between the electrons being added to 3d orbitals because all d-electrons are paired). Thus, increase in repulsion becomes greater than that of the attraction between the nucleus and the 4s e^- , hence size increases.

Note: Upto Mn ($3d^5 4s^2$) electron-electron repulsions do not come into picture because there is no pairing of electrons in 3d orbitals. However, in case of the elements from Fe to Zn, the pairing of electrons in 3d orbitals takes place and hence electron-electron repulsions are produced. Now the increase in electron-electron repulsion makes the atom to expand in size while the nuclear charge makes the atom to shrink in size. In the elements from Fe to Cu, these opposing effects almost counterbalance each other and hence there is no major change in size on going from Fe to Cu (similarly from Ru to Ag in 2nd series and from Os to Au in 3rd series).

Variation of atomic radii in a given group

It increases from top to bottom but it is found that,

(a) radii of 2nd transition series > radii of 1st transition series

Reason – As the at. no. increases down the gp, no. of shells increases

(b) radii of 2nd transition series $\cong$ radii of 3rd transition series

Reason – Lanthanide contraction

Worked-out Example

3. Atomic radius in a transition series does not vary much while that of s and p-block does.

Ans. Due to more pronounced shielding effect of penultimate d-subshell in transition elements.

3. Atomic volumes and densities

- The atomic volumes of transition elements are much lower than those of their neighbouring s and p-block elements. This is due to the fact that in d-block elements, the inner orbitals, i.e., (n-1) d-orbitals are filled. This leads to more nuclear pull resulting in decrease in volume.
- Due to decrease in atomic volume, the density (mass/volume) of these elements increases. Therefore, the transition elements have quite high densities. Following trends are observed among the densities of the transition elements:

12.4 Transition Elements (d-block) and Their Compounds

- (a) In a given transition series, the density increases across the series and reaches a maximum value at the middle of the series. (i.e., for the elements of group 9, 10). Then they start increasing as we proceed further to group 11 and 12.
- (b) The density increases down the group. Because the atomic size of the elements of 2nd and 3rd transition series is nearly the same, their atomic masses increase nearly two fold and the densities of the elements of the 3rd series are generally twice those of the corresponding elements of the 2nd transition series.
- Lightest transition metal is Sc(density 3.1 g cm^{-3}) and heaviest and densest transition element is osmium and Iriridium (both are comparable values).

Worked-out Example

4. Name the lightest and heaviest elements in terms of density among the transition elements.

Ans. Lightest element — Scandium;

Heaviest element — Iriridium and Osmium.

4. Ionization Potential

- (i) The I.P. of most of the d block elements lie in between those of s and p-block.
- (ii) The values of I.E. increases as we move from left to right in each series although increase is quite gradual or slow.

Reason – When we move from left to right in a given series, the nuclear charge increases. The increased nuclear charge would attract the ns^2 electron cloud with greater force and hence the ionization energies are expected to increase at each step. However, as, the electron is added to $(n-1)d$ sub-shell at each next element, the ns^2 electrons are shielded more and more. The effect of increasing nuclear charge is opposed by the additional screening effect of the nucleus and consequently, the ionization energies increase but quite slowly along a given series.

- (iii) When we proceed from top to bottom in gp 3rd, I.E. value decreases as $\text{Sc}_{21} > \text{Y}_{39} > \text{La}_{57}$. However, in most of the remaining groups the I.E increases in the same direction.

Reason – In the atoms of 14 elements following La_{57} (Ce_{58} to Lu_{71}), 4f orbitals are completely filled and hence cannot shield the nucleus effectively. Due to ineffective (i.e., poor) shielding of the nucleus by the 4f electrons, the elements lying after La_{57} in 3rd series have greater effective nuclear charge acting on the ns^2 electrons. Therefore, the ionization energies of the elements following La_{57} in 3rd series are higher than those of their corresponding elements lying in 1st and 2nd series.

- In each series, the elements of group 12th (Zn, Cd and Hg) have the highest values for their ionization energy. This is due to the extra stability associated with their completely filled $(n-1)d$ and ns orbitals ($\text{Zn}_{30} = 3d^{10} 4s^2$, $\text{Cd}_{48} = 4d^{10} 5s^2$ and $\text{Hg}_{80} = 5d^{10} 6s^2$). (a) The second ionization energies of the first transition series increase almost regularly with increase in atomic number. However, the value for Cr and Cu are sufficiently higher than those of their neighbours. This is due to stable configurations of Cr^+ ($3d^5$ -exactly half filled) and Cu^+ ($3d^{10}$ -completely filled).
- The 5d-transition elements(Group 4 to Group 12) possess higher ionization energies than 3d and 4d-transition elements due to the greater effective nuclear charge in 5d-transition elements. This is due to poor shielding effect of 4f-electrons.

Worked-out Examples

5. The ionization energies of elements of 3d-series do not vary much with increasing atomic number. Explain.

Ans. The increase in effective nuclear charge responsible for steady increase in ionization energy is counterbalanced by shielding effect.

6. The ionization energies of first five members of 3d-series increase with increase in atomic number and then become constant or irregular for next five members. Explain.

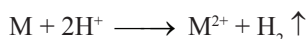
Ans. For first five elements, the effective nuclear charge predominates over shielding effect and thus ionization energy increases. In next five elements, shielding effect increases more rapidly due to dense packing of electrons in d-orbitals (pairing takes place) and the effective nuclear charge and shielding effect counterbalance each other to show a net result of constant or irregular but closer trend in ionization energy values.

7. First ionization energy of 5d-series transition metals are higher than those of 3d and 4d-series. Explain.

Ans. The valence shell electrons of 5d-elements experience greater effective nuclear charge than 3d and 4d-elements due to poor shielding of 4f-subshell electrons.

Electrode potential

In general, transition elements have low negative values of standard reduction electrode potential due to high ionisation energies, high heat of sublimation which are more than offset with large heats of hydration. Consequently, transition elements are **weak reducing agents** and are **less reactive**. The standard reduction potential of all the transition elements (except Cu and Hg in 3d-series etc.) is lower (negative) than that of hydrogen (taken as zero). Thus all the transition elements, with negative reduction potential liberate hydrogen from dilute acids.



However, some metals evolve hydrogen very slowly because they are protected from the attack of acids by the formation of an impervious layer of an inert oxide. For example, chromium is so unreactive that it can be used as a protective non-oxidising metal.

Worked-out Examples

8. Most of the transition elements do not displace hydrogen from dilute acids. Explain.

Ans. They are placed below H in electrochemical series are less reactive than hydrogen.

9. Transition metals are weak reducing agents. Explain.

Ans. Due to high heat of sublimation, high ionization energy and low E_{RP}° .

Melting and boiling point: The melting and boiling points of the transition elements are generally very high due to the fact that these elements have strong metallic bonding between their atoms.

(As the number of unpaired e^- increases, strength of metallic bond increases and hence melting point increases).

As we move along a particular series, since the number of unpaired electrons increases upto Cr (in 1st series), Mo (in 2nd series) and W (in 3rd series), the melting points of the elements also increase upto these elements and then with decrease in the number of unpaired electrons, the melting points also decrease. Since Zn, Cd and Hg do not have any unpaired electrons, their melting points are the lowest in their respective periods. Their melting points are so low that these elements can be volatilized off easily. In fact, Hg is a liquid at room temperature.

Exception: Although Mn, Tc and Re also have the maximum number of unpaired electrons, these elements have lower values of their melting points than Cr, Mo and W respectively. Their low values are due to the fact that the metallic bonding in these elements are relatively weak.

- Since Zn, Cd and Hg have completely filled $(n-1)$ d-orbitals, their atoms are not expected to form covalent bonding amongst themselves, hence they have relatively lower melting points than the other d-block elements.
- Zinc metal is brittle and hard at room temperature.
- Mercury is the only liquid metal at room temperature.
- Transition metal with highest melting point is W and transition metal with lowest melting point is Hg.

Worked-out Examples

10. Transition metals have high melting point and boiling point why?

Ans. The unpaired electrons in d-orbitals get involved in covalent bonding amongst the atoms of transition elements to give rise to high melting point and boiling point.

11. Mercury metal is liquid at room temperature, why?

Ans. The electronic configuration of Hg is $[\text{Xe}]4f^{14}, 5d^{10}, 6s^2$. Assuming 5d and 6s bands overlap, the composite band can accommodate 12 valence electrons per metal atom. Both bonding and anti bonding molecular orbitals are occupied in Hg and thus weak bonding and low melting point.

12. Explain the large difference in melting point of Cr (1920°C) and Zn (420°C).

Ans. Electronic configuration of Cr is $3d^5 4s^1$. Energy of 3d and 4s are very close to each other and the bands of 3d and 4s overlap. The composite band which can accommodate 12 valence electron per metal atom will be half filled for Cr and completely filled for Zn. The melting point depends on the occupancy of the bonding and anti-bonding molecular orbitals.

Variable oxidation state

All transition elements, except the first and the last members of each series exhibit variable oxidation states. The cause of showing different oxidation states is due to the fact that there is only a small difference between the energies of the electrons in the ns orbital and (n – 1) d-orbitals with the result that both ns as well as (n – 1) d-electron may be used for compound formation. Therefore, the variable oxidation states of transition elements are related to their electronic configurations. Transition metals exhibit a wide range of oxidation states. When 'ns' electrons are involved, then compounds with lower oxidation states are formed. In compounds with higher oxidation states, '(n–1) d' electrons are also involved.

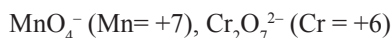
This is clear from the following table of oxidation states of elements of first transition series.

<i>Element</i>	<i>Outer electronic configuration</i>	<i>Oxidation state</i>
Sc	$3d^1 4s^2$	+2, +3
Ti	$3d^2 4s^2$	+2, +3, +4
V	$3d^3 4s^2$	+2, +3, +4, +5
Cr	$3d^5 4s^2$	+1, +2, +3, +4, +5, +6
Mn	$3d^5 4s^2$	+2, +3, +4, +5, +6, +7
Fe	$3d^6 4s^2$	+2, +3, +4, +5, +6
Co	$3d^7 4s^2$	+2, +3, +4
Ni	$3d^8 4s^2$	+2, +3, +4
Cu	$3d^{10} 4s^2$	+1, +2
Zn	$3d^{10} 4s^2$	+2

It must be noted that in some transition elements all of the (n–1) d-electrons are not involved during bond formation, e.g., iron ($3d^6 4s^2$) should have +8 as its highest oxidation states, but actually it is only +6 which is also very rare (+2 and +3 are the common oxidation states of iron). The +6 (and not +8) as the highest oxidation state is explained on the basis that during bond formation only the unpaired electrons of the 3d-subshell take part in bond formation. In iron there are 4 unpaired and 2 paired 3d-electrons and hence the effective electrons for bond formation in 3d-orbitals are only four which may give +6 as the highest oxidation state.

- The highest oxidation state exhibited by any transition metal is +8, i.e., Ruthenium tetroxide (RuO_4) and Osmium tetroxide (OsO_4).
- The highest oxidation state are shown by transition metal when they combine with most electronegative elements such as fluorine or oxygen, i.e., CrO_3 , Mn_2O_7 and VF_5 .

- Maximum number of variable oxidation states are shown by Mn (+2, +3, +4, +5, +6, +7).
- The most common oxidation state for first transition series is +2 which arises from the loss of 4s electrons (except Sc).
- In lower oxidation states as +2 and +3, the bonds are mostly ionic while in higher oxidation states such as +6 or +7, the bonds are essentially covalent as in,



- Many transition elements exhibit oxidation states which differ from each other only by one unit (e.g., Fe(II) and Fe(III)). This is in marked contrast to the s-block elements which exhibit only the group valency and the p-block elements which exhibit either the maximum group valency or valency less by two units due to inert pair effect.
- In general, the minimum oxidation state shown by any transition element is equal to the number of ns-electrons. For the first five numbers of each transition series, the highest oxidation state shown, in general, is equal to the sum of ns and (n-1) d-electrons. For the remaining five elements, the maximum oxidation state is not related to their electronic configuration. The highest oxidation state shown by any transition element is +8.
- The +2 oxidation state becomes more stable while the +3 state becomes less stable from left to right in the periodic table.
- Some transition elements also show an oxidation state of zero in their compounds e.g., $[\text{Ni}(\text{CO})_4]$ and $[\text{Fe}(\text{CO})_5]$
- In general, for the first four elements of a transition series, higher oxidation states are more stable than lower oxidation states. For the next five elements lower oxidation states are more stable than higher oxidation state e.g., Cr(III) is more stable than Cr(II) but Mn(II) is more stable than Mn(III).
- Highest oxidation state shown by any transition elements is +8 (Ru in 4d-series and Os in 5d-series).
- Mn shows maximum state of +7 in KMnO_4 .
- Magnetic property of transition, metals depends upon the member of unpaired electrons in d-orbitals. Metals with five unpaired electrons e.g., Mn have maximum paramagnetic character.
- Transition elements like Sc, Y, La and Ac do not show variable valency.

Complex formation: The cations of transition metals have a great tendency to form complexes with several molecules or ions called ligands. The tendency of formations of complexes is due to following two factors:

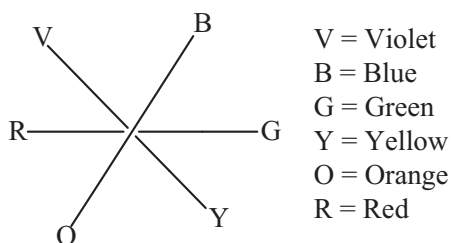
- (a) Due to small size of the transition metal atoms or their cations and high effective nuclear charge, they have a high positive charge density on them. This high positive charge density makes the atoms or cations to attract (accept) the lone pairs of electrons from the ligands.
- (b) The transition metal cations or atoms have vacant d-orbitals [(n – 1) d orbitals] in which they can accommodate the lone pairs of electrons donated by the ligands and thus can form $L \rightarrow M$ coordinate bonds.

Formation of Coloured Compounds: The transition metal ions have unpaired d-electrons, which on absorbing visible light can jump from one d-orbital to another i.e., intra d-d transitions take place. Thus, when light falls certain visible wavelengths are absorbed. The transmitted (unabsorbed) or reflected light (or colour) appears coloured and gives the colour of compound. The ions having no d-d transitions are colourless.

<i>Colour of the absorbed radiation</i>	<i>Range of the wavelength (Å) of the absorbed light</i>	<i>Colour of the transmitted light (This colour is called complimentary colour of that of the absorbed light)</i>
Violet	4000 – 4500	Yellow green
Blue	4500 – 4800	Yellow
Green blue	4800 – 4900	Orange
Blue green	4900 – 5000	Red
Green	5000 – 5600	Violet (almost purple)
Yellow Green	5600 – 5750	Violet
Yellow	5750 – 5900	Blue
Orange	5900 – 6250	Green blue
Red	6250 – 7500	Blue green

12.8 Transition Elements (d-block) and Their Compounds

Complementary colours can be identified using Munshel colour wheel.



Opposite colours are complementary.

Note: the transition metal ions containing empty d-orbitals [e.g., Sc^{3+} and $\text{Ti}^{4+}(\text{TiO}_2)$ ions] or completely-filled d-orbitals [e.g., Cu^+ , Ag^+ , Zn^{2+} (ZnSO_4), Cd^{2+} , Hg^{2+} etc.] are colourless. Because d-d electron transition is not possible in these cases.

To explain the purple colour of octahedral $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ ion by d-d electron transition

Electronic configuration of Ti^{3+} : $[\text{Ar}]3d^1$

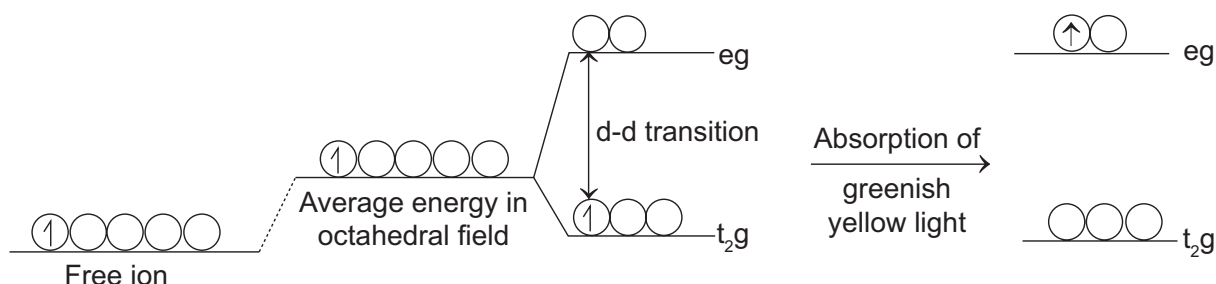


Fig. 12.1

During the formation of complex, as the six water molecules approach Ti^{3+} ion from different sides these develop a negative field around it, as they donate lone pair present on oxygen to metal ion. There occurs repulsions between electrons of metal ion and ligand and energy of degenerate orbitals of metal ions increases and ultimately they split into two set of orbitals. The electron present in Ti^{3+} , prefers lower set of d-orbital. The energy gap between two set of d-orbitals is very less and the energy available in the visible light is sufficient to cause excitation of electron from lower set of d-orbitals to higher set of d-orbitals, called as d-d transitions. Ti^{3+} absorbs greenish yellow component of white light during excitation of electrons, hence its aqueous solution appears as purple. This is because purple is the complimentary colour of greenish yellow in white light.

- Zn, Cd, Hg are colourless and are diamagnetic.
- The d-block metal cations like Sc^{3+} , Ti^{4+} , Cu^+ , Hg^{2+} , Cd^{2+} and Zn^{2+} have either completely empty or fully filled 3d-orbitals i.e., they do not have any unpaired d-electron and hence appear colourless.

Worked-out Examples

13. Unlike transition metals, salt of s- and p-block elements are not coloured, why?

Ans. The energy required to promote s- and p-orbitals into higher energy levels lie in UV-region.

14. Transition metal ions like Cu^+ , Ag^+ , Zn^{2+} , Hg^{2+} and Cu^{2+} are colourless. Explain.

Ans. All the ions reported above have no unpaired electron in them and d-orbitals $[(n-1)d^{10}]$ are also completely filled. Thus, d-d transition responsible for colour is not possible.

Magnetic properties: Most of the transition elements show paramagnetism. Paramagnetism arises from the presence of unpaired electrons in atoms, ions or molecules.

In general, greater the number of unpaired electrons, greater is the magnetic character. The maximum paramagnetism is noticed in d^5 case which has maximum number of unpaired electrons.

- In transition elements, magnetic moment is given by spin only formula because orbital motion is quenched i.e., $\mu = \sqrt{4s(s+1) + l(l+1)}$ where s = Spin Quantum number, l = angular quantum number or Azimuthal quantum number.
- When the metals like Fe, Co, Ni are placed in magnetic field, their magnetic domains are turned so that all points in the same direction giving rise to a large magnetic effect. Such substances are known as ferromagnetic.

Worked-out Examples

15. Transition metals and many of their compounds show paramagnetic behaviour. Why?

Ans. It is due to the presence of unpaired electrons.

16. The paramagnetic character in 3d transition series increases upto Cr and then decreases. Explain.

Ans. The number of unpaired electrons increases from $_{21}\text{Sc}$: $[\text{Ar}]3d^1, 4s^2$ to $_{24}\text{Cr}$: $[\text{Ar}]3d^5, 4s^1$ goes on decreasing continuously to $_{30}\text{Zn}$: $[\text{Ar}]3d^{10}, 4s^2$

Catalytic Properties

Most of the transition metals and their compounds are found to act as catalysts.

Transition Metals as Catalyst: Catalytic power of transition metals is believed to operate either by the formation of interstitial compounds to adsorb and activate the reacting substances e.g., hydrogenation of alkenes in presence of palladium or platinum is thought to take place through this mechanism.

Many of the d-block elements and their compounds act as catalysts in various reactions. Some common examples are:

- Pt-used as a catalyst in the manufacture of H_2SO_4 .
- Fe-used as a catalyst in the manufacture of NH_3 by Haber process. A small amount of molybdenum is added as a promoter.
- Ni-used as a catalyst in the hydrogenation of oils.
- V_2O_5 -used as a catalyst for the oxidation of SO_2 into SO_3 for the manufacture of H_2SO_4 in the contact process.
- MnO_2 -used as a catalyst in the decomposition of KClO_3 for preparation of oxygen.
- Cobalt salts catalyse the decomposition of bleaching powder.

Catalytic property is probably due to the utilization of $(n-1)d$ -orbitals or formation of interstitial compounds.

Some important catalysts having transition metals

- Zeigler Natta catalyst – $\text{TiCl}_4 + (\text{C}_2\text{H}_5)_3\text{Al}$
- Lindlar's catalyst – H_2Pd , BaSO_4 in the presence of quinoline
- Wilkinson's catalyst – H_2Pd , BaSO_4 in the presence of quinoline
- Adam's catalyst – Pt/PtO
- Brown's catalyst – Nickel boride (P-2 catalyst)

Worked-out Example

17. Transition metals in their many compounds act as good catalyst.

Ans. (c) Due to variable oxidation states.

12.10 Transition Elements (d-block) and Their Compounds

Alloy Formation: Formation of Alloys is due to their almost equal atomic sizes, they can mutually substitute one another in the crystal lattice to form alloys.

Example: When Mn is dissolved in molten iron and the solution is cooled, manganese – iron alloy is obtained. Such alloys are hard, have high M.P. and are more resistant to corrosion than the parent metals.

- (a) Molten transition metals are miscible with one another. Therefore, on cooling a mixture of the transition metals results in the formation of alloys.
- (b) Such alloys are usually harder, have higher melting points and are more resistant to corrosion than the parent metals.

Formation of non-stoichiometric and interstitial compound: We know that transition metals have defects in their structures. These defects create vacant interstitial space (also called interstitial holes or sites) between the atoms of those metals. The vacant space can easily accommodate small atoms like those of H, B, C, N etc., (non-metals) and give rise to the formation of interstitial compounds. It is due to the filling of the vacant spaces by the non-metallic atoms that the transition metals become rigid and hard, for example steel and cast iron are hard because of the formation of interstitial compounds with carbon.

The stable interstitial compounds are obtained when the non-metallic to metallic atom radius ratio is less than 0.59. These compounds cannot be represented by a definite composition and hence are non-stoichiometric in nature, i.e., these compounds have variable composition and hence cannot be represented by a definite composition. Non-stoichiometry is produced due to variable valency of transition metals and due to the defects in solid structures.

Chemical Properties

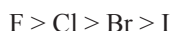
1. Oxides:

In general, the transition metal oxides are:

- (a) Basic, if the metal atom is in low oxidation state.
- (b) Amphoteric if the metal atom is in intermediate oxidation state.
- (c) Acidic if it is in high oxidation state.

2. Halides:

- (a) Transition metals react with halogens at high temperature to form halides.
- (b) The order of reactivity of different halogens is as:



- (c) Fluorides are ionic while chlorides, bromides and iodides are predominantly covalent.
- (d) Halides of metals in higher oxidation states are relatively unstable and get hydrolysed easily.

General Group Trends in the Chemistry of Transition Metals

Group 4: Titanium, zirconium and hafnium.

- (a) Most stable oxidation state for this group is +4.
- (b) Zirconium and hafnium occur together and exhibit similar properties. Their atomic radii are almost same due to lanthanide contraction.

Group 5: Vanadium, niobium and tantalum.

- (a) These elements exhibit all the oxidation state from +1 to +5.
- (b) Vanadium is used as an additive to steel and also used in the form of V_2O_5 as catalyst.

Group 6: Chromium, molybdenum and tungsten.

- (a) Most important oxidation state for Cr is +3, +6.
- (b) Most important oxidation state of Mo and W are +5, +6.
- (c) Very high melting and boiling points.
- (d) Chromium is unreactive at low temperature.
- (e) Chromium is most abundant element of this group.

Group 7: Manganese, technetium and rhenium.

- (a) Elements exhibit all the oxidation states from 0 to +7.
- (b) Elements have quite high melting and boiling points.
- (c) Manganese metal is used in the manufacture of alloys such as *ferromanganese* and *bronze*.

Group 8, 9, 10:

- (a) These metals are known as **iron group** metals.
- (b) Fe, Co, Ni is known as *ferrous metals*. They are ferromagnetic.
- (c) The element of second and third triad are inert and serve as good catalyst.

Group 11: Copper, silver and gold

- (a) These metals are known as **coinage metals**.
- (b) They form alloy with many metals.
- (c) These metals are the best conductor of heat and electricity.
- (d) Metals occur in combined state as well as free state.

Group 12: Zinc, cadmium and mercury.

- (a) Elements do not show characteristic properties of transition metals.
- (b) Elements are moderately electropositive and exhibit an oxidation state of +2.
- (c) Mercury, due to metal-metal bond also shows a formal oxidation state of +1.
- (d) These elements are diamagnetic.

SOME IMPORTANT INFORMATIONS REGARDING 'F-BLOCK ELEMENT'

Actinides

The series of elements involving the filling of 5f-orbitals is called actinide series. They follow Ac (89) and include elements from Th (90) and Lw (103).

Lanthanide contraction

There is a regular decrease in the size of atoms/tripositive ions with increase in atomic number as moving from La to Lu. The steady decrease in size of lanthanide atom/ions with increase in atomic number is called Lanthanide contraction.

Cause of lanthanide contraction

The configurations of lanthanides show that as the additional electron, enters the 4f-subshell. The shielding of one 4f electron by another is very little, being very smaller than that encountered in case of d-electrons (d-transition series). The imperfect shielding of f-electrons is due to the shape of f-orbitals. Which is very much diffused. As the atomic number increases, the nuclear charge increases by unity at each step, while no comparable increase in the mutual shielding effect of 4f electrons occurs. This causes a contraction in the size of the 4f subshell. Consequently, the atomic and ionic size goes on decreasing systematically from La to Lu.

Decrease in atomic radii is continuous but not regular. The decrease is much more in case of first six elements than in subsequent elements. Hence the properties of lanthanide compounds show divergence from regularity. However, decrease in ionic radii is more regular.

Use of Lanthanides

1. A pyrophoric alloy which contains cerium (40.5%), lanthanum and neodymium (44%) ; iron (4.5%); calcium, carbon and silicon (10.5%) is used in cigarette lighters, toys, flame throwing tanks and tracer bullets as well as shells.
2. Cerium cuts off heat and ultraviolet light and so is used in glare reducing spectacles.
3. Cerium magnesium alloys are used in flash light powders
4. Cerium salts are used in analysis, dyeing cotton, lead accumulators, medicines as catalysts.
5. Ceria and thoria are used in gas lamp mantles.

Actinides

The elements with atomic numbers 90 to 103 i.e., thorium to lawrencium which come immediately after actinium, (Z = 89) are called actinides or actinones. These elements involve the filling of 5f orbitals. Their general electronic configuration is $[Rn] 5f^{1-14} 6d^{0-1} 7s^2$.

Table 12.1 Electronic Configuration of Actinides

<i>Element</i>	<i>Symbol</i>	<i>Atomic number</i>	<i>Configuration</i>
Actinium	Ac	89	6d ¹ 7s ²
Thorium	Th	90	6d ² 7s ²
Protactinium	Pa	91	5f ² 6d ¹ 7s ²
Uranium	U	92	5f ³ 6d ¹ 7s ²
Neptunium	Np	93	5f ⁴ 6d ¹ 7s ²
Plutonium	Pu	94	5f ⁶ 7s ²
Americium	Am	95	5f ⁷ 7s ²
Curium	Cm	96	5f ⁷ 6d ¹ 7s ²
Berkelium	Bk	97	5f ⁸ 6d ¹ 7s ²
Californium	Cf	98	5f ¹⁰ 7s ²
Einsteinium	Es	99	5f ¹¹ 7s ²
Fermium	Fm	100	5f ¹² 7s ²
Mendelevium	Md	101	5f ¹³ 7s ²
Nobelium	No	102	5f ¹⁴ 7s ²
Lawrencium	Lr	103	5f ¹⁴ 6d ¹ 7s ²

They include three naturally occurring elements thorium, protoactinium and uranium and eleven transuranium elements or transuranics which are produced artificially by nuclear reactions. They are synthetic or man made.

All actinides are radioactive.

Uses of Actinides

- Thorium** is used in atomic reactors as fuel rods and in the treatment of cancer. A mixture of thorium and cerium nitrate in the ratio of 99: 1 is used for making incandescent gas menthes.
- Uranium** is used as nuclear fuel. Its salts are used in glass industry (for imparting green colour), textile industry and also in medicines.
- Plutonium** is used as fuel for atomic reactors as well as in atomic bombs.

Characteristics of actinides

- **Oxidation States:** These elements usually exhibit oxidation state of +2. However, they also show oxidation states of +4, +5 and +6.
- **Physical State:** These are silvery white metals and get tarnished with alkalies.
- **Density:** All the actinides except thorium and americium have high densities.
- **Colour:** The actinide ions in general, are coloured. The colour depends upon the number of 5f electrons. The ions with 5f⁰ and 5f⁷ configurations are colourless while ions with 5f³ to 5f⁶ configurations are coloured.

U³⁺ (5f³): Red, Np³⁺ (5f⁴): Bluish, Pu³⁺ (5f⁵): Blue, Am³⁺ (5f⁶): Pink

- **Ionization Energies:** These elements have low values of ionization energies.
- **Electropositive Charger:** All the actinides are highly electropositive and are strong reducing agents.
- **Complex Formation:** Actinides have a strong tendency towards complex formation and form cations like UO₂²⁺, PuO₂²⁺, UO⁺ etc.
- **Magnetic properties.** The actinide elements are paramagnetic due to the presence of unpaired electrons.
- **Actinide Contraction:** Actinides show actinide contraction. The size of atom/cation decreases regularly along the actinide series. The steady decrease in ionic radii with increase in atomic number is referred to as actinide contraction. This is due to poor shielding of 5f electrons.

Lanthanum (Z=57) does not have any electron in 4f-subshell. Similarly, actinium does not have any electron in 5f-subshell. Thus, these two elements do not belong to f-block of the periodic table. However, these two elements are studied along

with other elements of 4f and 5f-series as lanthanum is a prototype of 4f-series (i.e., its properties resemble with elements of 4f-series) and actinium is a prototype of 5f-series.

- (i) All the elements of f-block are in group 3 of the periodic table.
- (ii) Samarium and thulium also show +2 state although, Mn^{2+} ions so formed have f^6 and f^7 configurations respectively. These metals do not show +1 oxidation state, although in that case, their M^+ ions would have acquired stable f^7 and f^{14} configuration respectively.

Some Noteworthy Points

- In case of lanthanide ions, colour of ion having nf -electrons is nearly the same as that with $(14-n)$ f -electrons i.e., ions having equal number of unpaired electrons have similar colours.
- Calcium is used as a scavenger of oxygen and sulphur in many metals.
- Super alloys: Super alloys are iron based, nickel based or cobalt based alloy. These possess extremely low cheap rates and are highly resistant to oxidation at high temperature. They are mainly used for blades and rotors of jet engines which operate above 900°C continuously for long times e.g., Nimonic alloys and Haste alloys.

Magnetic moments of Ln^{3+} ions: Magnetic moments of transition metal ion as calculated from “spin only” formula are in close agreement with the corresponding observed magnetic moments. This is because in transition metal ions, the orbital angular momentum of the d -electrons is more or less ‘quenched’ by the interaction of these electrons and the atoms with which bonding takes place. As lanthanoid $3+$ ions have incomplete $4f$ subshell of electrons, the orbital angular momentum of these electrons contribute to the overall magnetic moment. Therefore, magnetic moments of Ln^{3+} ions calculated on the basis of “spin only” formula are not in agreement with the observed values. However, in case of Gd^{3+} the magnetic moment calculated on the basis of “spin only” formula is in agreement with the observed value. This is due to the fact that Gd^{3+} has a half-filled f -subshell. As such the total orbital angular momentum is zero and the “spin only” formula works.

Worked-out Examples

18. What is lanthanoid contraction? What are the consequences of lanthanoid contraction?

Ans. A steady decrease in the size of lanthanoids with increase in atomic number. It leads to similar physical and chemical properties among lanthanoids. Zr and Hf exist together.

19. In d -block metal ions, the colour of complex changes with ligand used while in f -block metal ion complexes, the colour remain almost constant and does not depend upon nature of ligand, why?

Ans. Colour of a metal ion depends upon energy difference of two energy levels involved in transition splitting of degenerate d -orbitals into t_{2g} and e_g orbitals is greatly affected by nature of ligand. i.e., why $[\text{Ni}(\text{NH}_3)_6]^{2+}$ is blue, $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ is green and $[\text{Ni}(\text{NO}_2)_6]^{4-}$ is brown.

Colour of f -block metal ions is due to f -transition. $4f$ -orbitals in lanthanoids are deeply embedded in the atom and are deeply shielded by $5s$ - and $5p$ -orbitals. Thus they remain almost unaffected by complex formation i.e., this is why colour does not depend upon nature of ligand.

20. The lanthanoid contraction is responsible for the fact that:

[AIEEE 2005]

- (a) Zr and Y have about the same radius
- (b) Zr and Nb have similar oxidation state
- (c) Zr and Hf have about the same radius
- (d) Zr and Zn have the same oxidation state

Ans. (c)

21. Which of the following factors may be regarded as the main cause for lanthanoid contraction?

[AIEEE 2005]

- (a) Poor shielding of one of $4f$ electrons by another in the sub shell
- (b) Effective shielding of one of $4f$ electrons by another in the sub shell
- (c) Poor shielding of $5d$ electrons by $4f$ electrons
- (d) Poor shielding of $4f$ electrons by $5d$ electrons

Ans. (a)

22. Cerium ($Z = 58$) is an important member of the lanthanoids. Which of the following statements about cerium is incorrect? [AIEEE 2006]

- (a) Cerium (IV) acts as an oxidizing agent
- (b) The +3 oxidation state of cerium is more stable than the +4 oxidation state
- (c) The +4 oxidation state of cerium is not known in solutions
- (d) The common oxidation states of cerium are +3 and +4

Ans. (c)

23. Identify the incorrect statement among the following: [AIEEE 2007]

- (a) 4f and 5 f-orbitals are equally shielded
- (b) d-block elements show irregular trend and erratic chemical properties among them selves
- (c) La and Lu have partially filled d-orbitals and no other partially filled orbitals
- (d) The chemistry of various lanthanoids is very similar

Ans. (a)

IMPORTANT COMPOUNDS OF SOME TRANSITION METALS

Compounds of Chromium

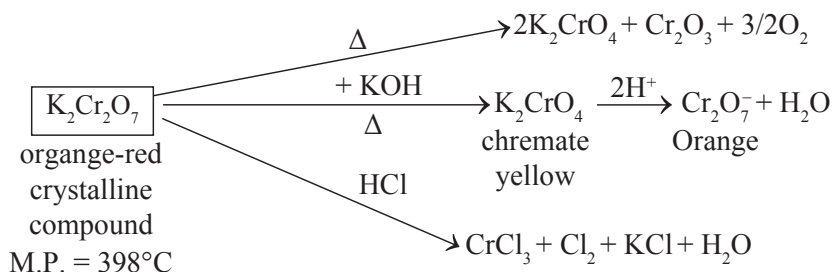
(i) **Potassium dichromate ($K_2Cr_2O_7$)**

Preparation $K_2Cr_2O_7$ is obtained from chromite ore ($FeCr_2O_4$) in following steps:

- Step I Chromite is fused with molten Na_2CO_3 in the presence of air when Na_2CrO_4 and Fe_2O_3 are formed. Fe_2O_3 is water insoluble while Na_2CrO_4 is water soluble hence separation is done.
- $$4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \xrightarrow{\Delta} 8Na_2CrO_4 + Fe_2O_3 + 8CO_2 \uparrow$$
- water so luble water insoluble
- Step II Aqueous solution of Na_2CrO_4 is acidified when $Na_2Cr_2O_7$ is formed.
- $$2Na_2CrO_4 + \text{dil. } H_2SO_4 \longrightarrow Na_2Cr_2O_7 + Na_2SO_4 + H_2O$$
- Step III Sodium dichromate ($Na_2Cr_2O_7$) is more soluble and less stable than $K_2Cr_2O_7$ hence $Na_2Cr_2O_7$ changes to $K_2Cr_2O_7$ on reaction with KCl . On cooling, concentrated solution of $K_2Cr_2O_7$, orange crystals are obtained.
- $$Na_2Cr_2O_7 + 2 KCl \longrightarrow K_2Cr_2O_7 + 2 NaCl$$

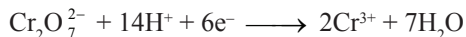
Properties

(i) **Heating effect, reaction with alkali and acid**

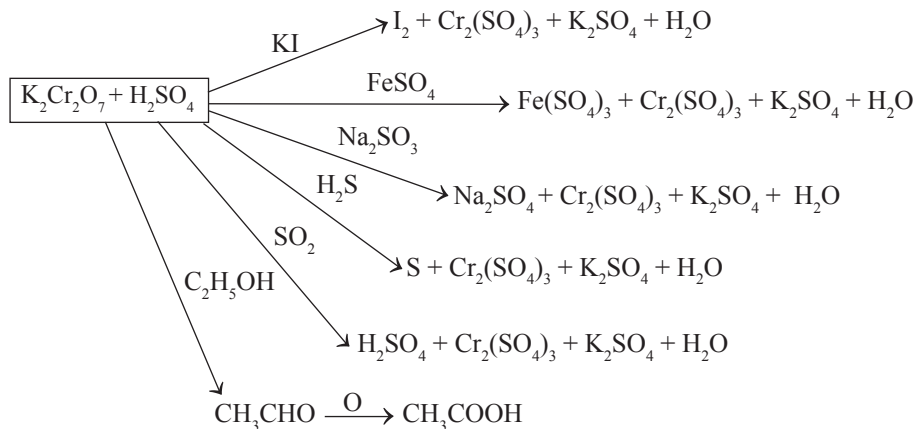


(ii) **Oxidation Reaction**

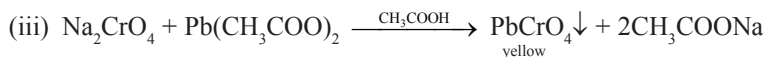
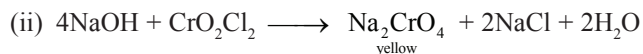
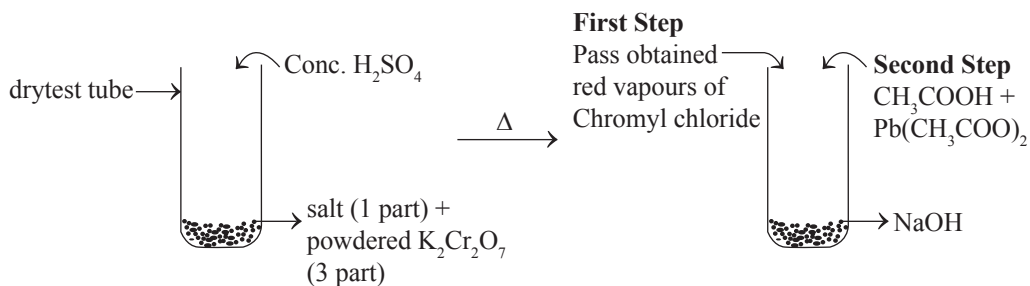
It acts as a powerful oxidizing agent in acidic medium (dilute H_2SO_4).



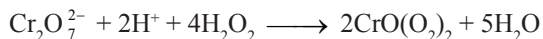
The oxidation state of Cr changes from +6 to 3. Some typical oxidation reactions are given below.



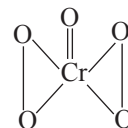
It also oxidizes nitrites to nitrates, arsenites to arsenates, thiosulphate to sulphate and sulphur

(iii) **Chromyl chloride test**

(iv) **Reaction of $\text{K}_2\text{Cr}_2\text{O}_7$ with H_2O_2 solution** When H_2O_2 is added to an acidified solution of a dichromate (or any order Cr(VI) species), a complicated reaction occurs. The product depend on the pH and the concentration of Cr



A deep blue-violet coloured peroxo compound $\text{CrO}(\text{O}_2)_2$ called chromic peroxide is formed. This decomposes rapidly in aqueous solution into Cr^{3+} and oxygen. The peroxo compound can be extracted into ether, where it is stable; it reacts with pyridine forming the adduct: $\text{Py} \cdot \text{CrO}(\text{O}_2)_2$. In less acidic solution, $\text{K}_2\text{Cr}_2\text{O}_7$ and H_2O_2 give salts which are violet coloured and diamagnetic due to formation



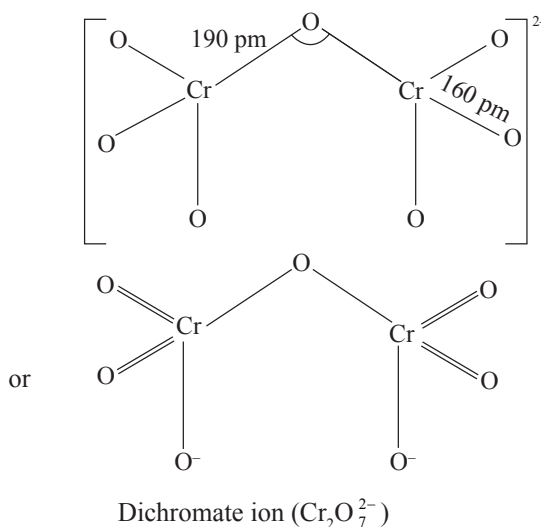
12.16 Transition Elements (d-block) and Their Compounds

of $[\text{CrO}(\text{O}_2)(\text{OH})]^-$. In alkaline solution, with 30 per cent H_2O_2 , a red-brown compound K_3CrO_8 is formed. It is tetraperoxo species $[\text{Cr}(\text{O}_2)_4]^{3-}$ and thus contains Cr(V) ion. In ammoniacal solution $(\text{NH}_3)_3\text{CrO}_4$, a dark red-brown compound with Cr(IV) ion is formed.

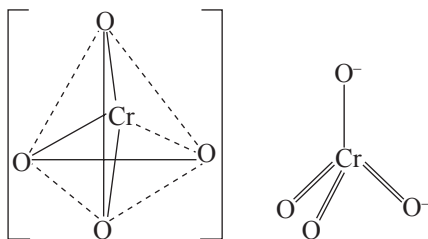
(v) Some important compounds of Cr

PbCrO_4	—	Chrome yellow
$\text{PbCrO}_4 \cdot \text{PbO}$	—	Chrome red
$\text{PbCrO}_4 + [\text{FeFe}(\text{CN})_6]^-$	—	Chrome green
$\text{Zn}(\text{OH})_2 \cdot \text{ZnCrO}_4 \cdot \text{H}_2\text{O}$	—	Yellow
$\text{Cr}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$	—	Guigret's green
CrO_3	—	Orange (called chromic acid)
$\text{K}_3[\text{CrO}_8]$	—	Red-brown

(vi) Structure of dichromate and chromate ions



The chromate ion has tetrahedral structure as shown below.



Discussion

- In CrO_5 , oxidation state of Cr is not 10 because oxidation state of Cr more than 8 is not possible. CrO_5 is in fact a peroxide.

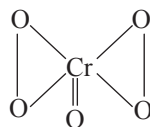
For peroxide oxygen, O.S. of $\text{O} = -1$

There are 4 peroxide oxygens and one oxygen a simple oxide.

$$\therefore 4(-1) + x + (-2) = 0$$

$$x = 6$$

Thus, O.S. of Cr in CrO_5 is +6.



Worked-out Examples

24. How would you account for the following?

- (a) Of the d^4 species, Cr^{2+} is strongly reducing while manganese (III) is strongly oxidizing.
- (b) Cobalt (II) is stable in aqueous solution but in the presence of complexing reagents it is easily oxidized.
- (c) The d^1 configuration is very unstable in ions.

Ans. (a) E_{op}° of $\text{Cr}^{2+}/\text{Cr}^{3+}$ is +0.4 V and E_{op}° of $\text{Mn}^{2+}/\text{Mn}^{3+}$ is -1.5 V and thus Cr^{2+} will be oxidized easily in comparison to Mn^{2+} which possess more tendency to get reduced. Thus Cr^{2+} is reducing agent and Mn^{2+} is oxidised agent.

(b) Co^{2+} converts to Co^{3+} in presence of complexing agents since the later exhibits greater tendency to form co-ordination complexes.

(c) The ions with d^1 configuration either oxidized themselves or disproportionate spontaneously.

25. Why $\text{K}_2\text{Cr}_2\text{O}_7$ is preferred over $\text{Na}_2\text{Cr}_2\text{O}_7$ in volumetric analysis, although both are oxidants?

Ans. $\text{Na}_2\text{Cr}_2\text{O}_7$ is hygroscopic and cannot be used as primary standard in volumetric estimations.

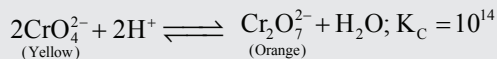
26. Potassium chromate solution does not give a precipitate of $\text{Cr}(\text{OH})_3$ on addition of ammonia. Explain.

Ans. Chromate ion has poor oxidizing nature in basic medium due to negative E_{RP}° .



27. A solution of K_2CrO_4 changes colour on being acidified.

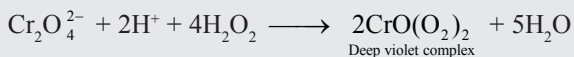
Ans. On being acidified, the chromate ions inter converted by the rapid equilibrium reaction to form dichromate ions which are orange coloured. Thus,



28. What happens when $\text{K}_2\text{Cr}_2\text{O}_7$ reacts with H_2O_2 ?

- (i) In strong acidic medium, (ii) In less acidic medium, (iii) In alkaline medium

Ans. (i) Peroxo compounds are formed.



(ii) In less acidic medium, violet coloured complex $[\text{CrO}(\text{O}_2)(\text{OH})]^-$ is formed.

(iii) In alkaline medium, tetraperoxo species $[\text{Cr}(\text{O}_2)_4]^{3-}$ is formed.

29. In chromic acid, (CrO_3) Cr has d^0 configuration but it is bright orange coloured solid, why?

Ans. Colour is not due to d-d transition but due to charge transfer.

30. CrCl_3 does not dissolve in pure water at significant rate but in presence of $\text{Cr}(\text{II})$ or reducing agent such as SnCl_2 the rate of dissolution increases. Why?

Ans. This is because the process of solubilization can then take place by electron transfer from $\text{Cr}(\text{II})$ in solution via a Cl bridge to the $\text{Cr}(\text{III})$ in the crystal. This $\text{Cr}(\text{II})$ can then leave the crystal and act upon a $\text{Cr}(\text{III})$ ion else where on the crystal surface. It is known as solubilizing effect.

31. Fe^{2+} and Cr^{3+} ions can be distinguished by

[AIEEE 2002]

- (a) By taking excess of NH_4OH solution
- (b) BY increasing concentration of NH_4^+ ion
- (c) By decreasing concentration of OH^- ions
- (d) Both (b) and (c)

Ans. (d)

Compound of Mn

(I) Potassium permanganate (KMnO_4)

Preparation KMnO_4 is prepared from MnO_2 (pyrolusite) as:

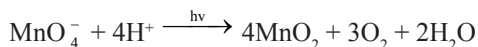
- Step I MnO_2 is fused with KOH and KClO_3 or KNO_3 when potassium manganate (K_2MnO_4) is formed as indicated by its green colour. $\text{KClO}_3/\text{KNO}_3$ acts as oxidizing agent.
- $$3\text{MnO}_2 + 6\text{KOH} + \text{KClO}_3 \xrightarrow{\Delta} 3\text{K}_2\text{MnO}_4 + \text{KCl} + 3\text{H}_2\text{O}$$
- Step II The fused mass is extracted with water and solution is green (MnO_4^{2-})
- $$2\text{K}_2\text{MnO}_4 + \text{Cl}_2 \longrightarrow 2\text{KCl} + 2\text{KMnO}_4$$
- (a) When Cl_2 gas is passed, K_2MnO_4 is oxidized to KMnO_4 (Purple)
- (b) MnO_4^{2-} can be oxidized to MnO_4^- electro-chemically at anode.
- At anode: $\text{MnO}_4^{2-} \longrightarrow \text{MnO}_4^- + \text{e}^-$
 At anode: $2\text{H}^+ + 2\text{e}^- \longrightarrow \text{H}_2$

Note: Strong oxidizing agents such as PbO_2 or sodium bismuthate (NaBiO_3) can also oxidize Mn^{2+} to MnO_4^- .

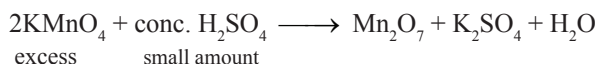
- (i) MnO_4^{2-} is unstable in acidic medium and dissociates disproportionately.



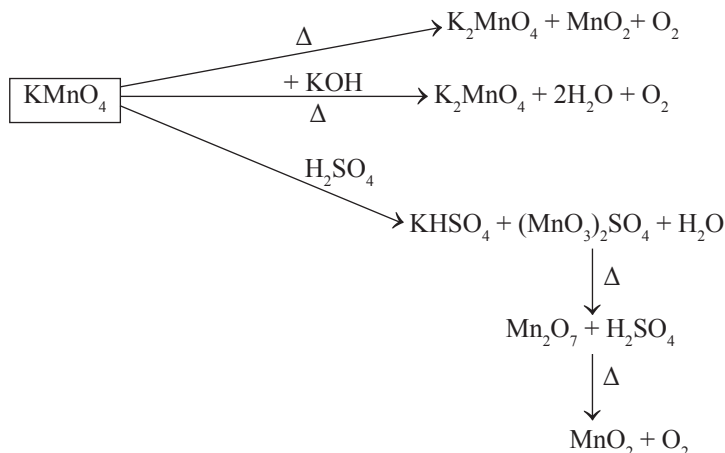
- (ii) Permanganate solutions are intrinsically unstable in acidic solution and decompose slowly. Decomposition is catalysed by sunlight, so KMnO_4 solutions should be stored in dark bottles and they must be standardized at the moment of its use in volumetric analysis.



- (iii) $\text{KMnO}_4 + 3\text{H}_2\text{SO}_4 \longrightarrow \text{K}^+ + \text{MnO}_3^+ + 3\text{HSO}_4^- + \text{H}_2\text{O}^+$
 Small amount excess



- (iv) Heating effect, reaction with alkali and acid.



Oxidation Reaction

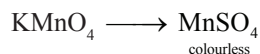
- Potassium permanganate is a powerful oxidizing agent.
- A mixture of sulphur, charcoal and KMnO_4 forms an explosive powder.

- A mixture of oxalic acid and KMnO_4 catches fire spontaneously after a few seconds.
- The same thing happens when glycerine is poured over powdered KMnO_4 .
- Potassium permanganate acts as an oxidizing agent in alkaline, neutral or acidic solutions.

(a) In alkaline solution

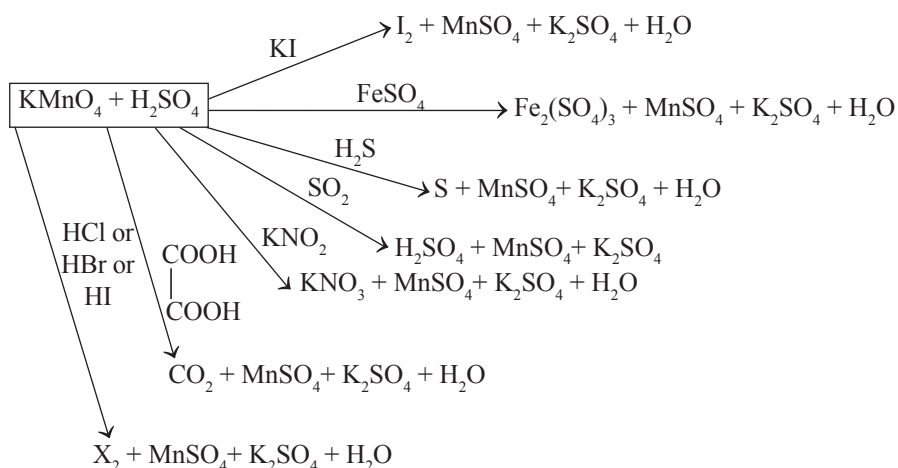


(b) In neutral solution

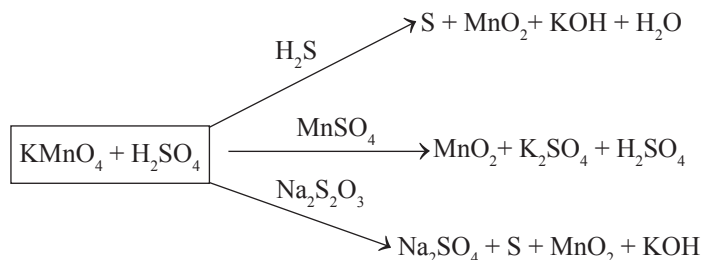
(c) In acidic solution (in the presence of dilute H_2SO_4)

The oxidation reactions of acidified KMnO_4 are catalysed by Mn(II) ion. The important oxidation reactions are:

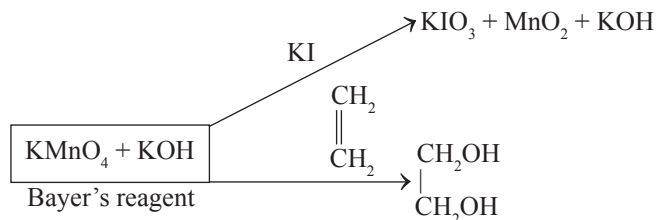
In acidic medium



In neutral medium

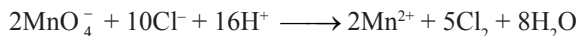


In alkaline medium



Discussion

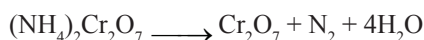
1. Of all the three common mineral acids, only sulphuric acid is found to be suitable for making the solution acidic. Under the titration conditions, it does not react with potassium permanganate or the reducing agent used.
2. Hydrochloric acid cannot be used as there is likelihood of the some potassium permanganate to be consumed in reaction with hydrochloric acid and its volume used is slightly more which introduces an error into the result.



3. Nitric acid itself is an oxidizing agent which reacts with reducing agents and therefore, cannot be used for this purpose.

Volcano experiment

When orange red monoclinic crystals of ammonium dichromate are heated, it decomposes violently evolving nitrogen and large volumes of green coloured chromium(III) oxide which resembles volcanic ash.

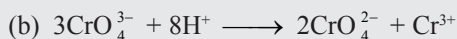


This reaction forms the basis of volcano experiment.

Compounds of transition metals are coloured due to $d-d$ transfer and charge transfer. Colour due to $d-d$ transition is shown by those transition metal ions which have one or more unpaired electrons in d-subshell of valence shell of the ion. Thus, compounds in which transition metal ions have d^0 or d^{10} configuration, are not generally coloured. However, in certain cases compounds in which transition metal has d^0 or d^{10} configuration and they are intensely coloured. This can be explained on the basis of charge transfer. The colour produced due to charge transfer is always intense since the restrictions of selection rules do not apply to transition between atoms. For example, permanganate ion, MnO_4^- has an intense purple colour in solution due to charge transfer. In MnO_4^- , an electron is momentarily transferred from O to the metal, thus momentarily changing O^{2-} to O^- and reducing the oxidation state of Mn from Mn(VII) to Mn(VI). Charge transfer requires that the energy levels on the two different atoms involved are fairly close.

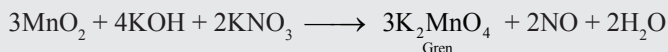
Worked-out Examples

32. What is meant by 'disproportionation'? Give two examples of disproportionation reaction.



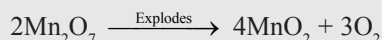
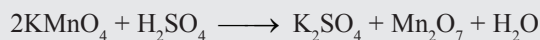
33. How KMnO_4 can be obtained from MnO_2 , KOH and KNO_3 ?

Ans. Equimolar Quantities of MnO_2 , KOH and KNO_3 are melted and the reaction mixture is allowed to solidify to give green mass of K_2MnO_4 .



34. An accident would occur if a student dissolves KMnO_4 in conc. H_2SO_4 instead of dilute H_2SO_4 . Explain.

Ans. KMnO_4 reacts with concentrated H_2SO_4 to form Mn_2O_7 which is a highly explosive substance which exploded.



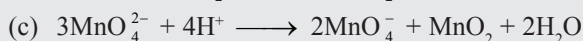
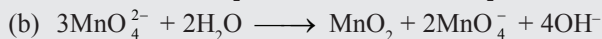
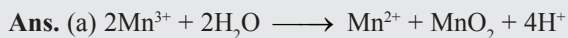
35. In steel manufacture, Mn metal is used as a scavenger to reduce traces of iron oxide and iron sulphide. Explain why Mn is effective for the process?

Ans. (i) Mn is active enough to react with S and O of iron compounds.

- (ii) Small quantities of the products of these reaction dissolves in the metal with out disruption of lattice. (Large quantities of MnO and MnS would form a slag which can be skimmed of the metal). Excess of Mn acts as cathodic protection for iron.

36. Write down the following reactions:

- Disproportionation of Mn^{2+} in acidic medium.
- Disproportionation of MnO_4^{2-} in alkaline medium.
- Disproportionation of MnO_4^{2-} in acidic medium.
- Oxidation of Mn^{2+} to MnO_2 by MnO_4^- in acidic medium.

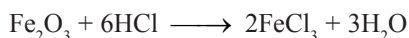
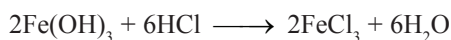


IRON (Fe)

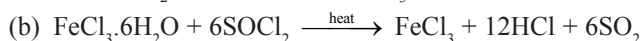
(I) Ferric chloride (FeCl_3)

Preparation

- (i) Hydrated ferric chloride ($\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$) can be prepared by dissolving iron hydroxide, $\text{Fe}(\text{OH})_3$ or ferric oxide in dilute HCl.

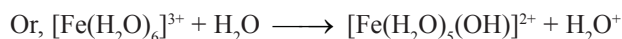
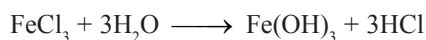


- (ii) Anhydrous ferric chloride may be prepared as:

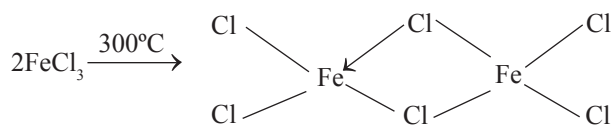


Properties

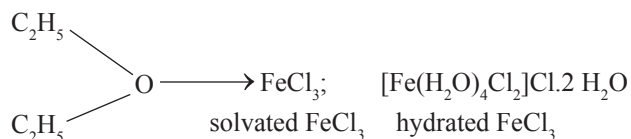
- Ferric chloride forms a number of hydrated salts having 12, 7, 6, 5 and 4 molecules of water of crystallization.
- Anhydrous salt is yellow, deliquescent compound, highly soluble in water. Its aqueous solution is acidic due to hydrolysis.



- (iii) It sublimes at 300°C giving a dimeric gas.

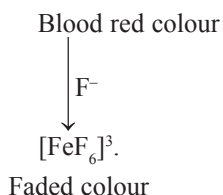
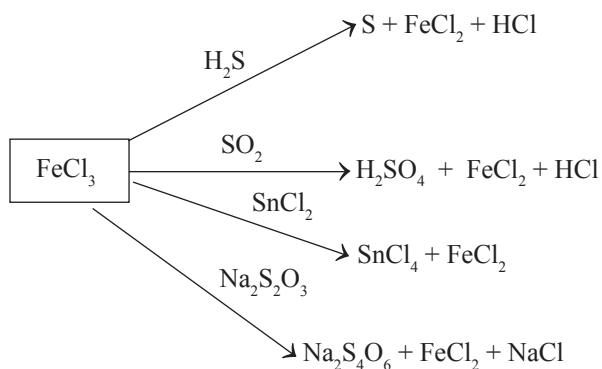


- (iv) FeCl_3 dissolves in both ether and water giving solvated monomeric species.

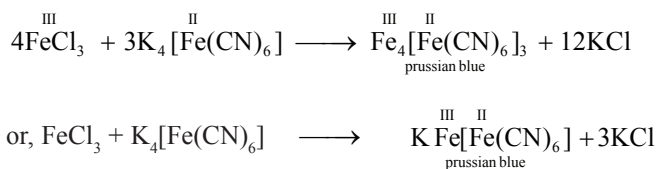


12.22 Transition Elements (d-block) and Their Compounds

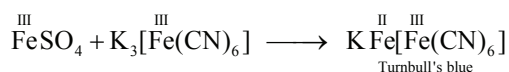
(v) Oxidising property



(vii) Fe^{3+} solution also gives deep blue precipitate of Prussian blue with $\text{K}_4[\text{Fe}(\text{CN})_6]$, potassium ferrocyanide.



(viii) A deep colour is also produced when Fe^{2+} reacts with $\text{K}_3[\text{Fe}(\text{CN})_6]$, potassium ferricyanide.



Recent X-ray work, IR and other spectroscopic methods have proved that turnbull's blue is identical to Prussian blue. The intense colour arises from electron transfer between $\text{Fe}(+II)$ and $\text{Fe}(+III)$.

Fe^{3+} partly oxidizes $[\overset{\text{II}}{\text{Fe}(\text{CN})_6}]^{4-}$ to $[\overset{\text{III}}{\text{Fe}(\text{CN})_6}]^{3-}$ and itself is reduced to Fe^{2+} , and also Fe^{2+} partly reduces $[\overset{\text{III}}{\text{Fe}(\text{CN})_6}]^{3-}$ to $[\overset{\text{II}}{\text{Fe}(\text{CN})_6}]^{4-}$ and itself is oxidized to Fe^{3+} . Thus, $\overset{\text{II}}{\text{Fe}}[\overset{\text{III}}{\text{Fe}(\text{CN})_6}]^-$ and $\overset{\text{III}}{\text{Fe}}[\overset{\text{II}}{\text{Fe}(\text{CN})_6}]^-$ are identical.

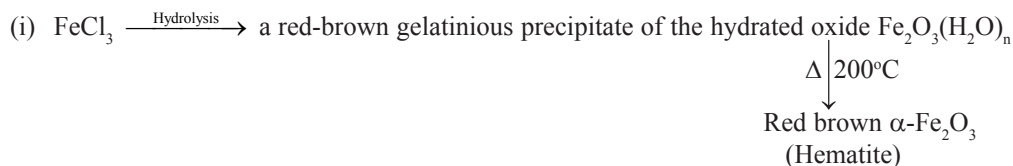
Worked-out Examples

37. Anhydrous ferric chloride is prepared by:

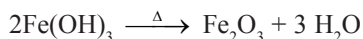
[IIT 2002]

- Heating hydrated ferric chloride at a high temperature in a stream of air
- Heating metallic iron in a steam of dry chlorine gas
- Reactions of ferric oxide with $\text{HCl}_{(\text{aq.})}$
- Reaction of metallic iron with $\text{HCl}_{(\text{aq.})}$

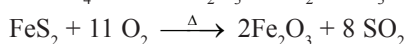
Ans. (b)

(II) Ferric oxide (Fe_2O_3)**Preparation**

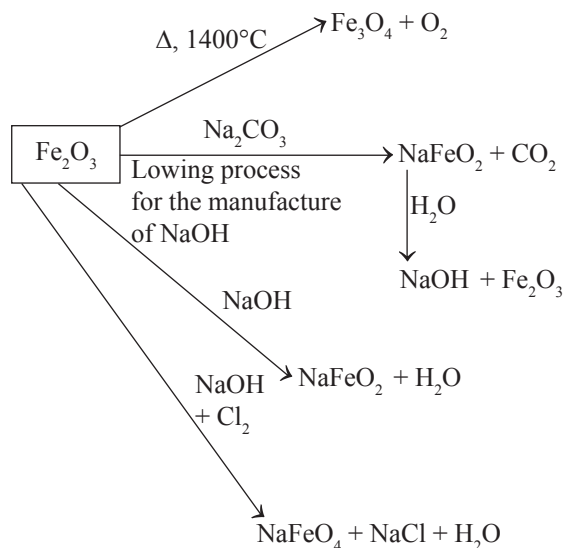
- (ii) On oxidation of Fe_3O_4 , $\gamma\text{-Fe}_2\text{O}_3$ is formed.
 (iii) By heating ferric nitrate, hydroxide, carbonate and oxalate.



- (iv) By heating ferrous sulphate and sulphides

**Properties**

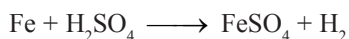
- (i) $\alpha\text{-Fe}_2\text{O}_3$ has hexagonally close-packed lattice of O^{2-} ions with Fe^{3+} ion in two third of the octahedral holes while $\gamma\text{-Fe}_2\text{O}_3$ has cubic close-packed arrangement of O^{2-} ions with Fe^{3+} ions randomly distributed in both the octahedral and tetrahedral sites.
 (ii)



- (iii) Fe_2O_3 is predominantly basic. Freshly precipitated $\text{Fe}_2\text{O}_3 \cdot (\text{H}_2\text{O})_n$ dissolves in acidic solution giving pale violet $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ ions. This $[\text{Fe}_2\text{O}_3 \cdot (\text{H}_2\text{O})_n]$ also dissolves in concentrated NaOH forming $[\text{Fe}(\text{OH})_6]^{3-}$. This shows that Fe_2O_3 is slightly amphoteric.

(III) Ferrous sulphate or green vitriol ($\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$)**Preparation**

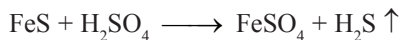
- (i) It is obtained by dissolving scrap iron in dilute sulphuric acid.



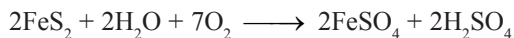
The solution is crystallized by the addition of alcohol as ferrous sulphate is sparingly soluble in it.

12.24 Transition Elements (d-block) and Their Compounds

- (ii) It can also be prepared in the laboratory from Kipp's waste. The excess of sulphuric acid is neutralized by heating with a small quantity of iron sulphide filings. The solution is then crystallized.



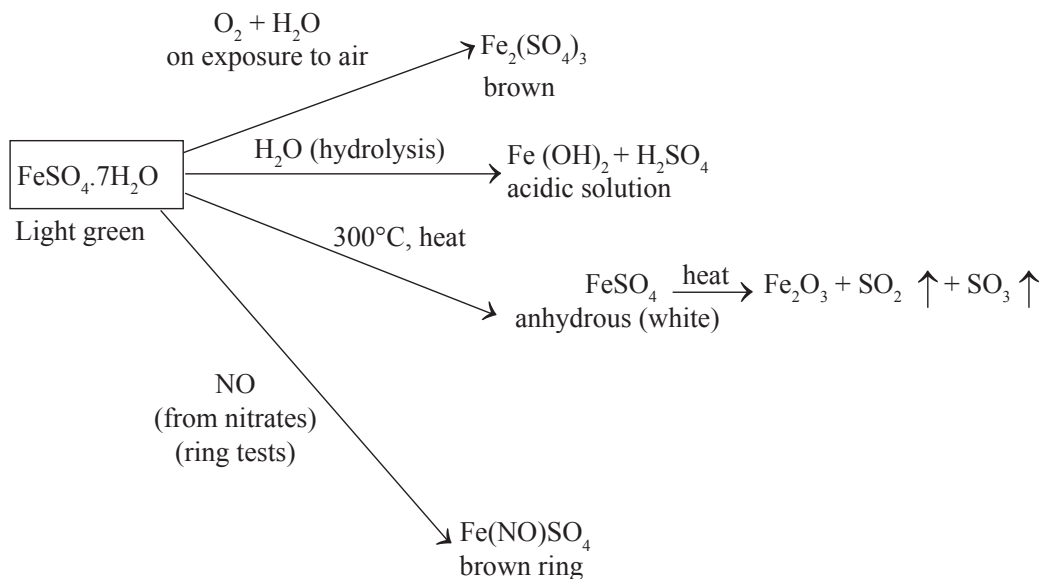
- (iii) Commercially, ferrous sulphate is obtained by the slow oxidation of iron pyrites in the presence of air and moisture. The pyrites are exposed to air in big heaps.



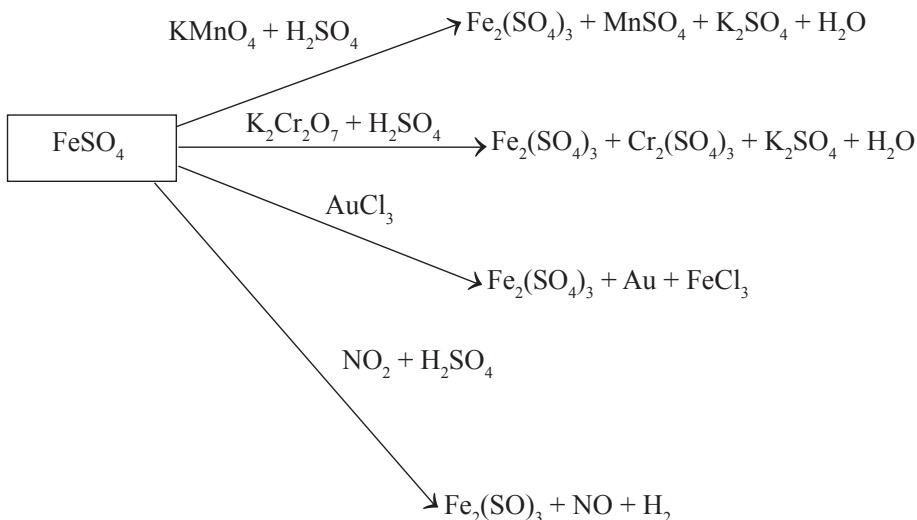
The free sulphuric acid is removed by the addition of scrap iron. Green crystals are obtained on crystallization.

Properties

(i)

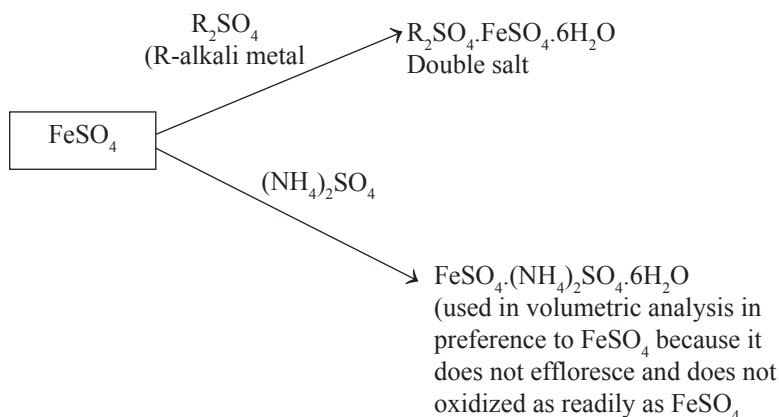


(ii) Reducing properties



(iii) Formation of double salt

(iii) formation of double salt

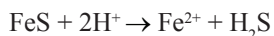
**(IV) Iron (II) sulphide, FeS**

This solid is obtained by heating a mixture of iron filings and sulphur:



Samples of iron (II) sulphide having slightly variable compositions, but all deficient in iron, have been reported, i.e., the sulphide is non-stoichiometric. The formula of one specimen has been reported as $\text{Fe}_{0.858}\text{S}$.

The action of dilute acids on iron (II) sulphide produces hydrogen sulphide, and this is a convenient way of making the gas in the laboratory:



- Fe^{2+} is green and Fe^{3+} is yellow.
- Fe^{3+} is powerful oxidizing agent than Fe^{2+} ion.
- Iron is rendered passive by conc. HNO_3 and other oxidizing agents like $\text{K}_2\text{Cr}_2\text{O}_7$, KMnO_4 , chloric acid, chromic acid, silver nitrate etc. A specimen of passive iron can be rendered active by scratching the film mechanically or chemically (e.g., immersing in conc. alkali solution, heating in reducing atmosphere of H_2 or CO).
- FeCl_3 is used a styptic to stop bleeding from a cut.
- Iron carbide or cementite is Fe_3C .
- Kipp's base is $\text{FeS} + \text{H}_2\text{SO}_4$.
- FeSO_4 is used in manufacture of blue black ink.
- Slag is formed only by ferrous oxide (FeO), ferric oxide (Fe_2O_3) does not form a slag.
- Copper dissolves in an aqueous solution of FeCl_3 .
- Mohr's salt is resistant to atmospheric oxidation.

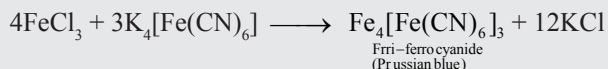
Worked-out Examples

38. Why are the compounds of iron coloured and paramagnetic?

Ans. Iron compounds either has Fe^{2+} or Fe^{3+} oxidation state and each contains unpaired electrons which show strong paramagnetic nature. Furthermore, d-d transition of these unpaired electrons impart complementary colours to iron compounds.

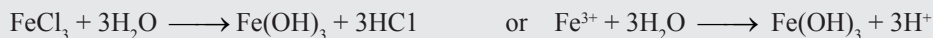
39. Ferric chloride solution becomes blue when $K_4[Fe(CN)_6]$ solution is added to it. Why?

Ans. Ferric chloride reacts with potassium ferrocyanide to give Prussian blue.



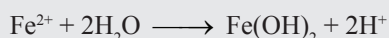
40. The aqueous solution of $FeCl_3$ is acidic. Why?

Ans. $FeCl_3$ hydrolyses in water to form acidic solution.



41. A little acid is always added in the preparation of ferrous ammonium sulphate solution.

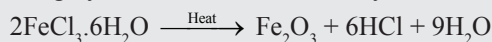
Ans. On dissolution of ferrous ammonium sulphate in water, green turbidity of $Fe(OH)_2$ appears due to hydrolysis of Fe^{2+} ions.



Addition of little acid reverse the direction of reaction i.e., prevents hydrolysis of Fe^{2+} ion to give a clear solution.

42. Anhydrous $FeCl_3$ cannot be obtained by heating hydrated ferric chloride. Why?

Ans. On heating hydrated ferric chloride anhydrous $FeCl_3$ is not formed but it is changed to Fe_2O_3 .

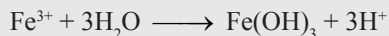


43. Why Mohr's salt is used as primary standard and not simple $FeSO_4$?

Ans. $FeSO_4$ is easily oxidized to $Fe_2(SO_4)_3$ whereas Mohr's salt $[FeSO_4(NH_4)_2SO_4 \cdot 6H_2O]$ not.

44. A solution of $FeCl_3$ in water gives a brown precipitate on standing.

Ans. $FeCl_3$ hydrolysis to give brown precipitate of $Fe(OH)_3$.



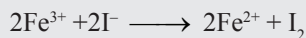
45. Sn^{2+} and Fe^{3+} cannot coexist in appreciable concentration in the same solution.

Ans. Sn^{2+} and Fe^{3+} shows redox change.



46. Ferric iodide is very unstable but ferric chloride is not.

Ans. I^- ion is a stronger reducing agent in comparison to Cl^- ion. Fe^{3+} is easily reduced by iodide ion.



Copper

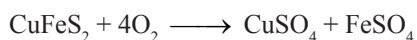
(I) Copper sulphate or blue vitriol ($CuSO_4 \cdot 5H_2O$)

Preparation

(i) It is prepared by treating copper scrap or turning, cuprous oxide, cupric oxide, or malachite with sulphuric acid.



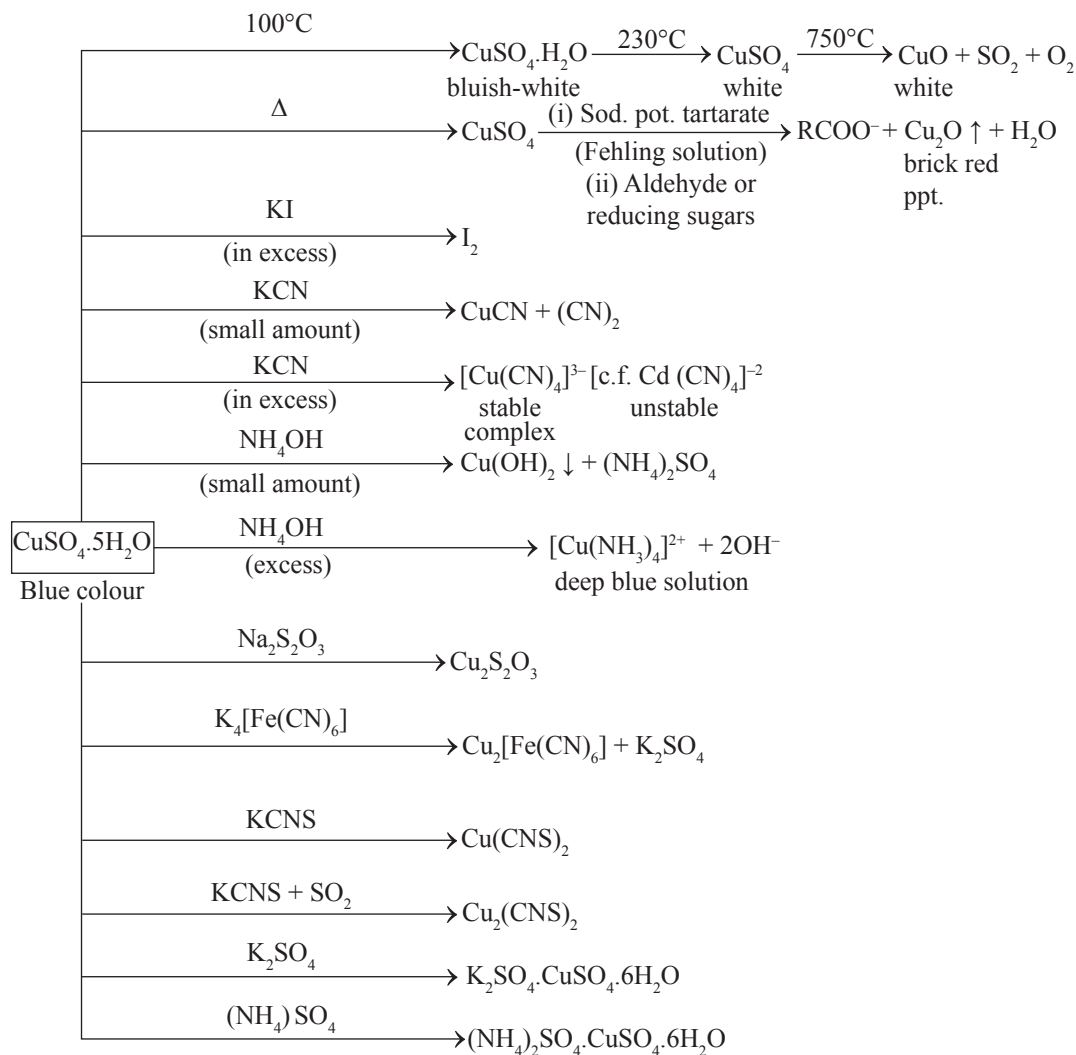
(ii) By roasting copper pyrites, $CuFeS_2$ in air (large scale method).



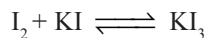
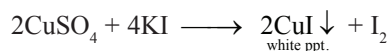
Ferrous sulphate is removed as insoluble ferric oxide obtained at high temperature.

Properties

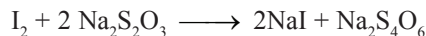
- (i) The Cu(II) ion or cupric ion is written more precisely as $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ when in aqueous solution. It has a bright blue colour. In $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, four of the water molecules are associated with Cu^{2+} , and the fifth is hydrogen-bonded to the sulphate ion as well as to the water molecules on the copper ion, hence written as $[\text{Cu}(\text{H}_2\text{O})_4]\text{SO}_4 \cdot \text{H}_2\text{O}$.
- (ii)



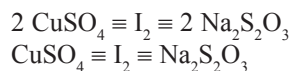
- (iii) If excess of KI is added to an acidified solution of CuSO_4 (Cu^{2+} ions), I_2 is produced by oxidation of I^- which reduces Cu^{2+} to CuI (or Cu_2I_2).



I_2 produced can be determined by titration with hypo solution.



Thus, CuSO_4 (Cu^{2+} salts) can be determined iodometrically.

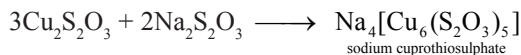
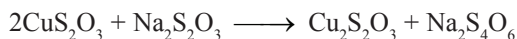
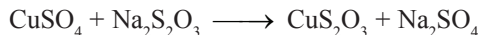


Equivalent weight of CuSO_4 = Molecular weight

12.28 Transition Elements (d-block) and Their Compounds

(iv) Action of sodium thiosulphate

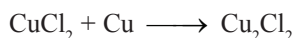
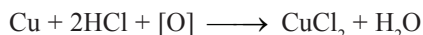
Sodium thiosulphate reacts with copper sulphate to form cupric thiosulphate which is reduced by sodium thiosulphate. The cuprous compound thus formed dissolves in excess of sodium thiosulphate to form a complex.



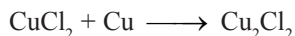
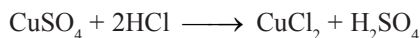
(II) Cuprous chloride (Cu_2Cl_2)

Preparation

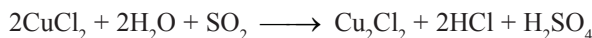
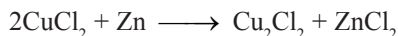
- (i) By heating excess of copper with concentrated hydrochloric acid in the presence of a little potassium chlorate.



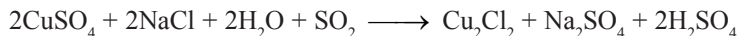
- (ii) By boiling copper sulphate solution with excess of copper turnings in the presence of hydrochloric acid.



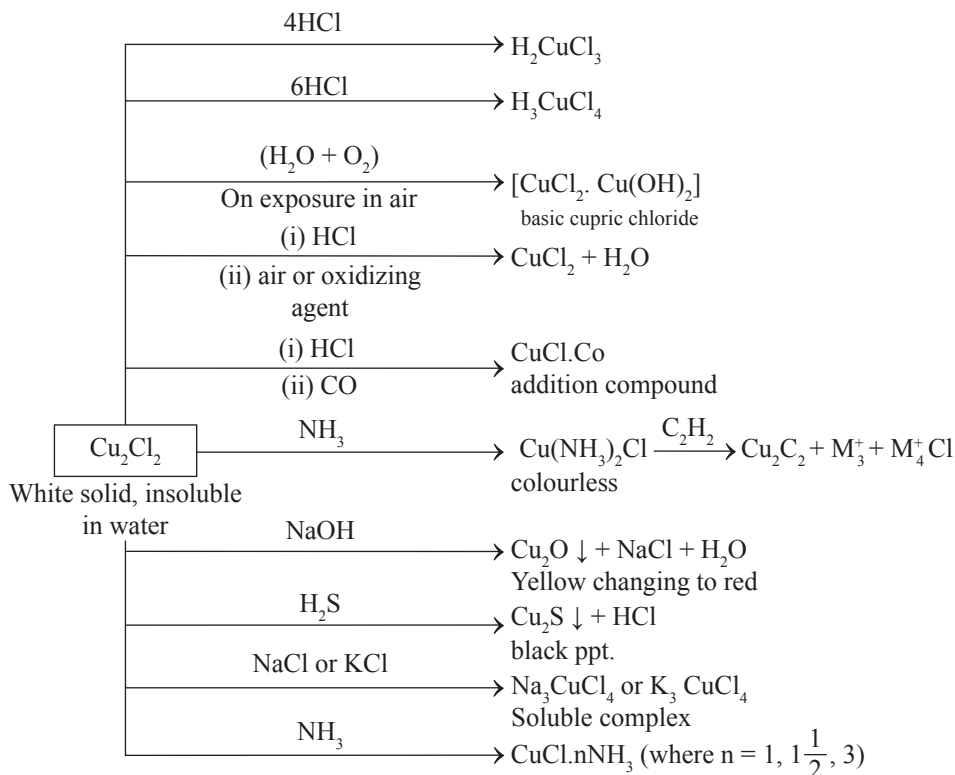
- (iii) By heating cupric chloride with zinc or sulphur dioxide.



- (iv) By passing SO_2 through the solution containing copper sulphate and sodium chloride.



Properties



(III) Cupric Oxide (CuO)

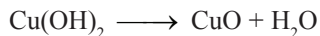
It is called black oxide of copper and is found in nature as tenorite.

Preparation

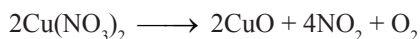
- (i) By heating Cu_2O in air or by heating copper for a long time in air (the temperature should not exceed 1100°C).



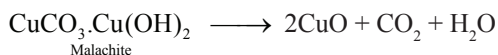
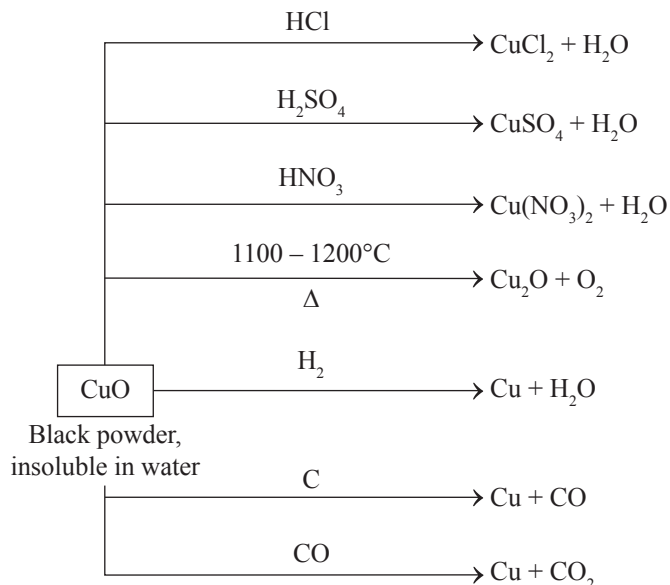
- (ii) By heating cupric hydroxide.



- (iii) By heating copper nitrate.



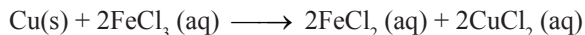
- (iv) On a commercial scale, it is obtained by heating malachite which is found in nature.

**Properties****Some Noteworthy Points**

1. Verdigris is $\text{Cu}(\text{CH}_3\text{COO})_2 \cdot \text{Cu}(\text{OH})_2$ i.e., basic copper acetate and is used as a green pigment in paints.
2. Copper is second most useful metal (the first being Iron).
3. Schweitzer's reagent contains copper and is used for dissolving cellulose in the manufacture of artificial silk or Rayon.
4. Bordeaux mixture (CuSO_4 solution + lime) is used as fungicide.
5. Matte contain $\text{Cu}_2\text{S} + \text{FeS}$.
6. Blister copper contains 96–98% copper with small amounts of Ag and Au as impurity.
7. $\text{Cu}(\text{NO}_3)_2$ is monomeric and $\text{Cu}_2(\text{CH}_3\text{COO})_4 \cdot 2\text{H}_2\text{O}$ is dimeric in vapour state.
8. If an arc is struck between the two copper rods dipped in water, a colloidal solution of copper is obtained.
9. $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is called Blue vitriol or 'Nila thotha'.
10. CuFeS_2 is called Fool's gold.
11. CuCl_2 and CuBr_2 are covalent and exist as polymers.
12. CuF_2 is an ionic solid.

12.30 Transition Elements (d-block) and Their Compounds

13. Cu^+ with d^{10} configuration should be stable but actually Cu^{2+} ion is more stable than Cu^+ ion because of increased hydration.
14. A piece of copper readily dissolves in a solution of FeCl_3 . This is because $E^\circ_{\text{Fe}^{3+}/\text{Fe}^{2+}}$ is more than $E^\circ_{\text{Cu}^{2+}/\text{Cu}}$.



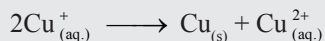
In $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, four molecules of water are ligands and are attached to Cu^{2+} by coordinate bonds while fifth water molecule is hydrogen bonded with SO_4^{2-} ion.

15. The green coloured copper carbonate obtained by the action of $\text{Na}_2\text{CO}_3(\text{aq})$ on $\text{CuSO}_4(\text{aq})$ is in fact basic copper carbonate, $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2 \cdot \frac{1}{2}\text{H}_2\text{O}$. It is not readily decomposed by dilute acids to give CO_2 .
16. A strip of copper dissolves in FeCl_3 , however, Cu cannot displace Fe.
- $$\text{Cu} + 2\text{FeCl}_3 \longrightarrow \text{CuCl}_2 + 2\text{FeCl}_2$$
17. CuSO_4 reacts with KI to give white ppt. of Cu_2I_2 . Such a reaction is not possible with KCl, KBr or KI.
- $$4\text{KI} + 2\text{CuSO}_4 \longrightarrow 2\text{K}_2\text{SO}_4 + \underset{\text{White ppt.}}{\text{Cu}_2\text{I}_2} + \text{I}_2$$
18. Anhydrous CuF_2 is an ionic solid, whereas, CuCl_2 and CuBr_2 are polymeric covalent.

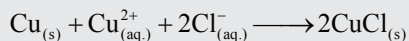
Worked-out Examples

47. Although Cu^+ does not exist in solution state, but $\text{CuCl}_{(\text{s})}$ is formed in presence of Cl^- ions in aqueous solution of $\text{Cu}_{(\text{s})}$ and Cu^{2+} . Explain.

Ans. In presence of Cl^- ion the disproportionation of Cu^+ i.e.,

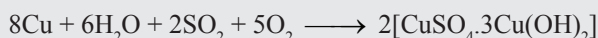


is reversed because precipitation of copper (I) chloride drives the disproportionation reaction to the left.



48. When copper is exposed to atmosphere, a loose layer of green material is formed over its surface. What is this green mass?

Ans. Copper although not affected by dry air at ordinary temperature, but when exposed to air and moisture, it is covered by a green thin film of bronchamite, $\text{CuSO}_4 \cdot 3\text{Cu}(\text{OH})_2$ or atacamite, $\text{CuCl}_2 \cdot 3\text{Cu}(\text{OH})_2$ due to the presence of SO_2 and HCl in the atmosphere.



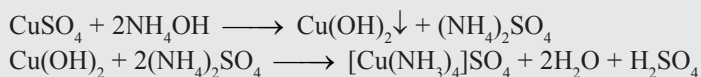
49. Blue copper sulphate turns white on heating. Why?

Ans. Hydrated copper sulphate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$) is blue which on heating loses water of crystallization to form anhydrous CuSO_4 a white compound.



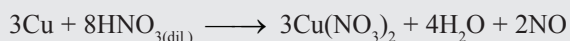
50. Copper sulphate is soluble in excess of NH_4OH .

Ans. Copper sulphate reacts with NH_4OH to form precipitate of $\text{Cu}(\text{OH})_2$ which redissolves in excess of NH_4OH forming a complex, tetramine copper II sulphate.



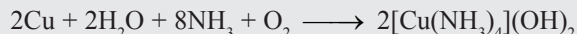
51. Copper dissolves in dil. Nitric acid but not in dil. HCl . Why?

Ans. Copper has less standard oxidation potential (E°_{op}) than H and thus cannot liberate H_2 from acids. However, it dissolves in nitric acid because HNO_3 is strong oxidant.



52. In order to remove oxygen, commercial nitrogen is passed through copper chips placed in ammonia solution. Why?

Ans. Cu reacts with NH_3 in presence of O_2 forming soluble complex, cupraammonium hydroxide



53. Cu^+ is colourless whereas Cu^{2+} is blue in aqueous solution. Explain.

Ans. Cu^+ has $[\text{Ar}] 3d^{10}$ configuration i.e., d-orbitals are completely filled.

54. First ionization energy of $_{29}\text{Cu}$ is higher than those of $_{19}\text{K}$ while second and third energies are lower. Explain.

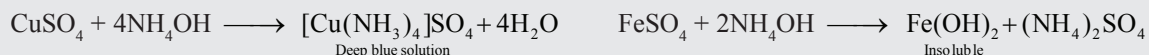
Ans. Cu has $[\text{Ar}]3d^{10}, 4s^1$ configuration 3d-subshell is much less effective in shielding the outer s-electron than a p-subshell in $\text{K}([\text{Ar}]4s^1)$, so s-electron of copper is more tightly bound than in K. Hence, its first ionisation energy is higher than potassium. However, 2nd and 3rd ionization energy values of copper are lower as compared to potassium due to the loss of electrons from diffused d-orbitals.

55. Copper implements were known to mankind much earlier than iron implement.

Ans. Copper occurs in nature also a free metal but iron is always found in nature in combined state. The technology to manufacture iron was developed much later. Hence iron implements were used later than copper implements as copper was known in the early times.

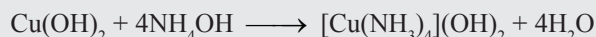
56. Copper sulphate dissolves in NH_4OH solution but FeSO_4 does not. Why?

Ans. Copper sulphate dissolves in the ammonium hydroxide due to formation of a copper complex. Ferrous sulphate reacts with NH_4OH to form insoluble $\text{Fe}(\text{OH})_2$. It does not form any complex with NH_4OH .



57. Copper hydroxide is soluble in ammonium hydroxide but not in sodium hydroxide.

Ans. $\text{Cu}(\text{OH})_2$ dissolves in NH_4OH by forming a complex.



$\text{Cu}(\text{OH})_2$ is insoluble in NaOH as no such complex is formed.

58. Cu_2S is black and Cu_2O is red whereas Cu^+ has no unpaired electron and is colourless. Explain

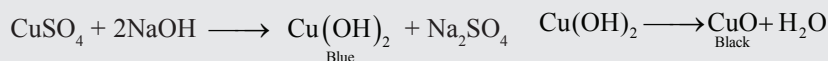
Ans. Cu^+ having $3d^{10}$ configuration and can not show d-d transition. Cu_2O and Cu_2S has O^{2-} and S^{2-} ions respectively. Due to charge transfer from O^{2-} and S^{2-} to the vacant orbitals of Cu^+ they have colour.

59. Why is copper is largely used in electrical wiring.

Ans. Copper is a very good conductor of electricity. Moreover, it is not easily affected by the atmosphere due to E_{RP}° being positive.

60. A dark blue precipitate is formed when sodium hydroxide solution is added to copper sulphate solution. The precipitate darkens on heating. Why?

Ans. NaOH reacts with CuSO_4 when dark blue precipitate of $\text{Cu}(\text{OH})_2$ is formed. This precipitate on heating form CuO which is black in colour. Hence, the colour darkens on heating.



61. The species $[\text{CuCl}_4]^{2-}$ exist but $[\text{CuI}_4]^{2-}$ does not.

Ans. I^- ion is a stronger reducing agent than Cl^- ion. It reduces $^{2+}$ ion into Cu^+ ion. Hence, cupric iodide is converted into cuprous iodide. Thus, the species $[\text{CuI}_4]^{2-}$ does not exist.

62. Which ore contains both iron and copper?

[IIT 2005]

- (a) Cuprite (b) Chalcocite (c) Chalcopyrite (d) Malachite

Ans. (c)

63. Excess of KI reacts with CuSO_4 solution and then $\text{Na}_2\text{S}_2\text{O}_3$ solution is added to it. Which of the statements is incorrect for this reaction: [AIEEE 2004]

- (a) Evolved I_2 is reduced (b) CuI_2 is formed
(c) $\text{Na}_2\text{S}_2\text{O}_3$ is oxidized (d) Cu_2I_2 is formed

Ans. (b)

64. Which pair of compound is expected to show similar colour in aqueous solution: [IIT 2005]

- (a) FeCl_2 , CuCl_2 (b) VOCl_2 , CuCl_2 (c) VOCl_2 , FeCl_2 (d) FeCl_2 , MnCl_2

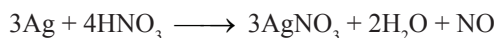
Ans. (a)

Silver

(I) Silver nitrate or lunar caustic (AgNO_3)

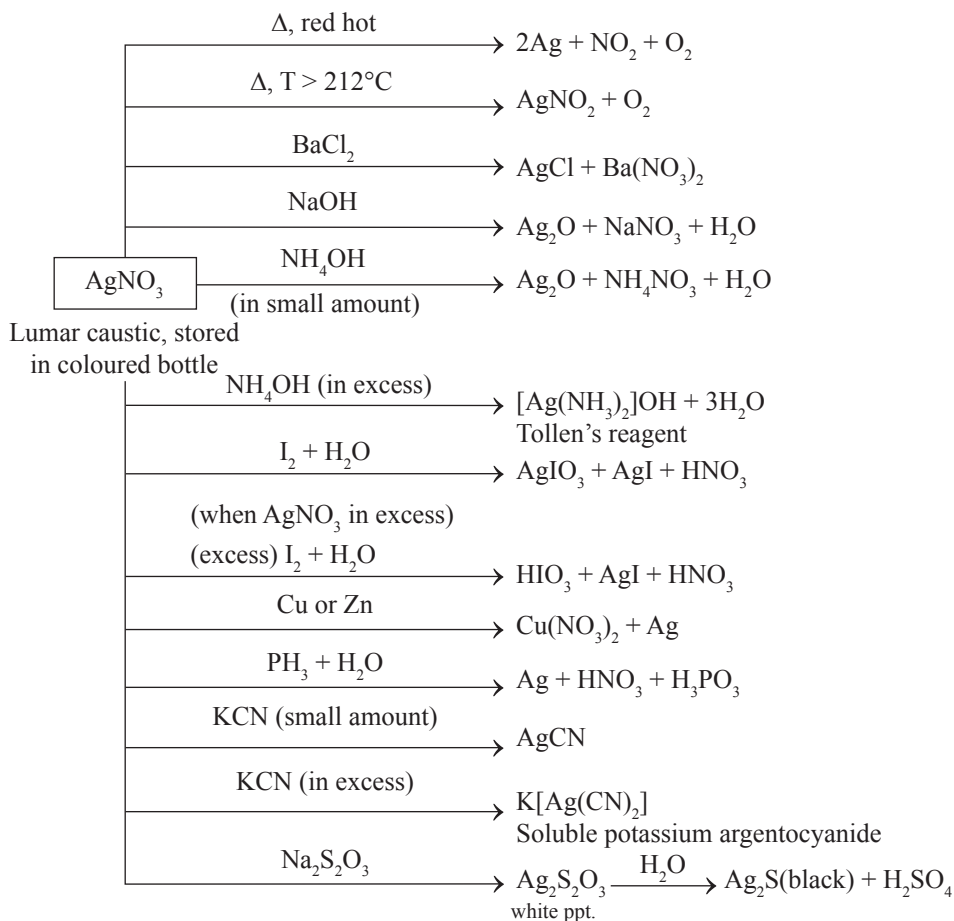
Preparation

Silver nitrate is prepared by dissolving silver in dilute nitric acid.



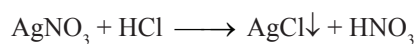
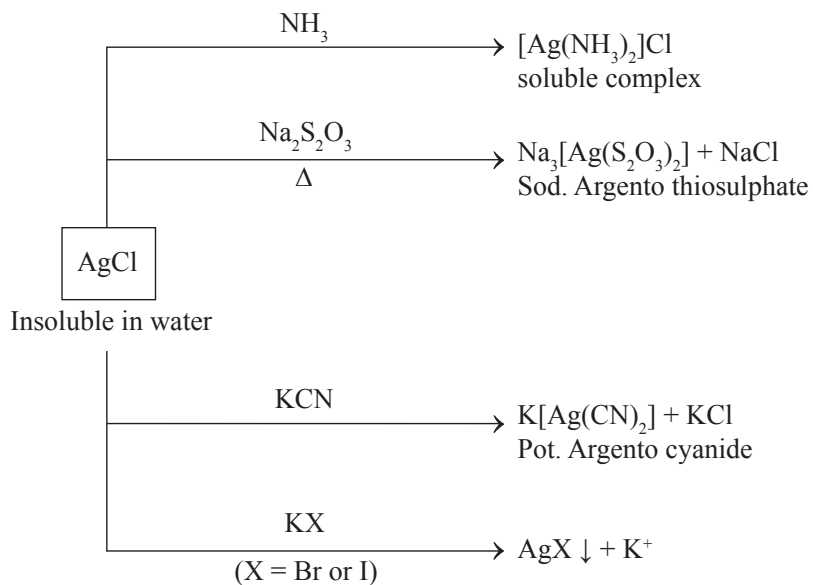
On crystallisation, colourless, rhombic crystals of silver nitrate are formed.

Properties

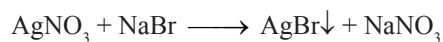
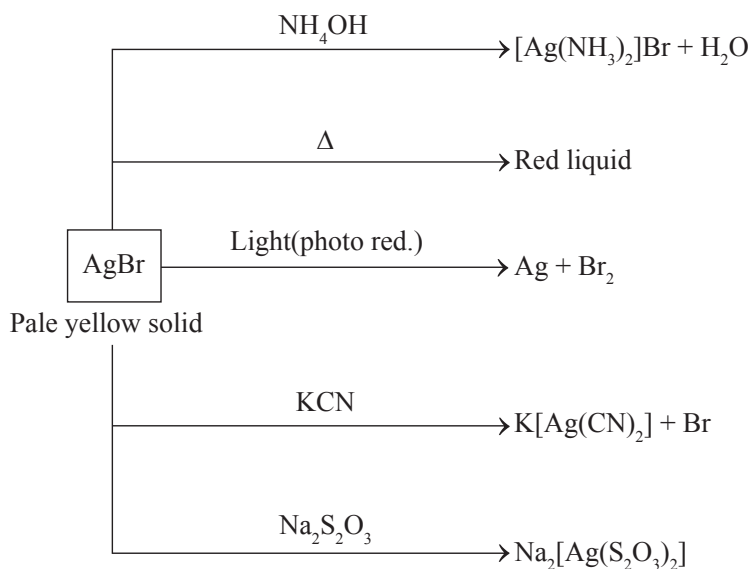


(II) Silver chloride (AgCl)(AIEEE)**Preparation**

Silver chloride, AgCl is obtained as a white precipitate when a solution containing chloride ions is mixed with a solution containing Ag^+ ions.

**Properties****(III) Silver bromide (AgBr)****Preparation**

By adding AgNO_3 solution to soluble bromide solution.

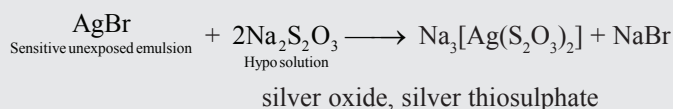
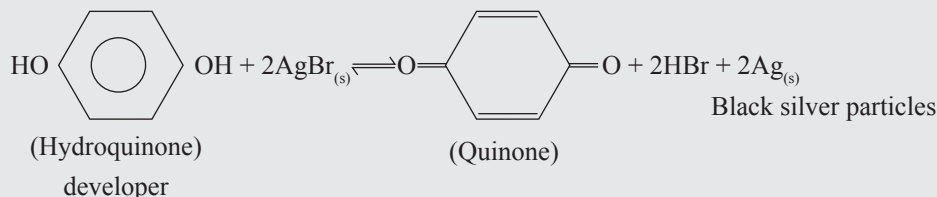
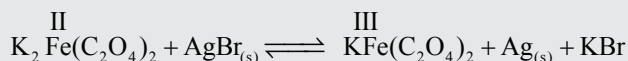
**Properties**

Worked-out Example

65. Write the balanced chemical equation for developing black and white photographic film.

[IIT 2001, 2005]

Ans. AgBr is reduced to Ag by,

**(IV) Silver (I) Compounds or Argentous Compounds:**

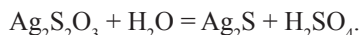
Silver (I) Oxide, Ag₂O, is formed as a brown-black precipitate, on adding alkali solutions to the aqueous solutions of silver salts. It may be assumed that silver hydroxide (AgOH) is first formed, and this immediately passes into silver oxide by loss of water:

$\text{Ag}^+ + (\text{OH})^- = \text{Ag}(\text{OH})$; $2\text{AgOH} = \text{Ag}_2\text{O} + \text{H}_2\text{O}$. The precipitate obstinately occludes some of the precipitant (alkalis).

Properties

1. Argentous oxide is sparingly soluble in water.
2. Its suspension imparts a perceptibly alkaline reaction to water, indicating that the second one of the above reactions can be reversed, and the AgOH formed is almost completely ionized to the Ag⁺ and OH⁻ ions.
3. It begins to decompose into Ag and O₂ on heating even at 190°C and the decomposition is practically complete at 300°C. Decomposition also takes place even at ordinary temperatures on exposure to light.
4. It is reduced by hydrogen even at 100°C.
5. Ag₂O reacts vigorously with hydrogen peroxide at ordinary temperatures and is thereby reduced to metallic silver: $\text{Ag}_2\text{O} + \text{H}_2\text{O}_2 = 2\text{Ag} + \text{H}_2\text{O} + \text{O}_2$. (This reaction shows the reducing property of H₂O₂.)
6. It is a highly basic oxide and absorbs CO₂ from air to form silver carbonate. It dissolves in acids to form Ag(I) salts. An aqueous solution of ammonia readily dissolves Ag₂O, forming ammine complex: $\text{Ag}_2\text{O} + 4\text{NH}_4\text{OH} = 2[\text{Ag}(\text{NH}_3)_2]\text{OH} + 3\text{H}_2\text{O}$. Such a solution, on long standing in air, precipitates a black substance known as fulminating silver, which is extraordinarily explosive even in the moist state, as it contains silver nitride, Ag₃N. (This substance should not be confused with AgONC, and silver azide, AgN₃)

Silver Thiosulphate, Ag₂S₂O₃, is precipitated by adding a solution of sodium thiosulphate to a solution of silver nitrate. It is a white substance, insoluble in water, but soluble in excess of the precipitate, forming complex thiosulphato-argentates of varying compositions: $2\text{AgNO}_3 + \text{Na}_2\text{S}_2\text{O}_3 = \text{Ag}_2\text{S}_2\text{O}_3 \downarrow + 2\text{NaNO}_3$; $\text{Ag}_2\text{S}_2\text{O}_3 + 3\text{Na}_2\text{S}_2\text{O}_3 = 2\text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)_2]$. It is decomposed by dilute mineral acids (and also slowly by water), forming black Ag₂S and H₂SO₄:

**Some Noteworthy Points**

- Silver metal is best conductor of heat and electricity.
- Molten silver dissolves 20 times its volume of oxygen.

- Silver amalgam issued in silvering of mirrors and in filling of teeth.
- Silver leaves are used in Ayurvedic and Yunani medicine.
- Sterling silver (80% Ag, 20% Cu) is used in making Jewellery.
- Silver is not attacked by aqua regia but it reacts with concentrated H_2SO_4 and concentrated HNO_3 .
- Spitting of silver: Molten silver absorbs oxygen which is given off on cooling causing violet spurting. This is called spitting of silver.
- Silver react with sulphur or H_2S forming black Ag_2S hence silver cutlery is tarnished black by food rich in sulphur (mustard and egg yolk).
- Silvering of mirror is based on reduction of ammonical silver nitrate solution by neutral solution of tartarate or glucose in cold.
- AgNO_3 is called Lunar caustic.
- All silver halides except AgF are insoluble in water.
- AgI is insoluble in NH_4OH but AgCl is soluble in NH_4OH due to formation of $[\text{Ag}(\text{NH}_3)_2]\text{Cl}$.
- AgBr is most sensitive to photo-reduction. AgI is not so sensitive to light. Both AgBr and AgI are used in photography.
- Silver nitrate is used for preparing marking inks and hair dyes.
- AgO is a brown coloured solid insoluble in water.

Worked-out Examples

66. Silver fluoride is fairly soluble in water while other silver halides are insoluble.

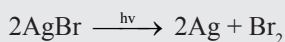
Ans. The hydration energy of AgF is higher than its lattice energy. Hence, it is soluble in water. The hydration energy values of other halides are smaller than their lattice energy values. Hence, these halides are insoluble in water.

67. Strong oxidation of Ag_2O in alkaline solution gives black oxide of formula AgO . Ag (II) has d^9 configuration, so AgO must be paramagnetic, but AgO is diamagnetic, why?

Ans. AgO does not contain Ag(II) but it is mixed oxide having composition Ag(I) Ag(II) O_2 .

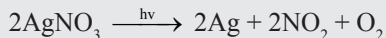
68. Why is AgBr used in photography?

Ans. AgBr is the most sensitive silver halide to photo reduction, hence it is used as the light sensitive material in photographic films. The unchanged AgBr is dissolved in hypo solution to cast an image on photographic plate.



69. Why is AgNO_3 kept in brown coloured bottles?

Ans. AgNO_3 decomposes on exposure to light and is thus stored in brown bottles to prevent the action of light.



70. Why does AgNO_3 produce black stain on the skin?

Ans. In contact with organic substances (skin), it blackens due to decomposition into metallic silver and thus leaves black stain on skin.

71. Why is AgNO_3 also called lunar caustic?

Ans. It produces burning sensation like caustic and leaves black stain like the moon luna on skin and thus called lunar caustic.

72. Which of the following metal will leave behind a metal on strong heating?

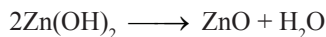
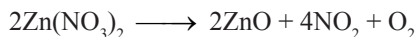
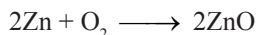
[AIEEE 2003]

- (a) $\text{Mn}(\text{NO}_3)_2$ (b) AgNO_3
(c) $\text{Fe}(\text{NO}_3)_3$ (d) $\text{Cu}(\text{NO}_3)_2$

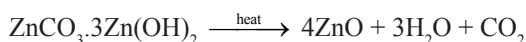
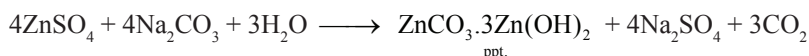
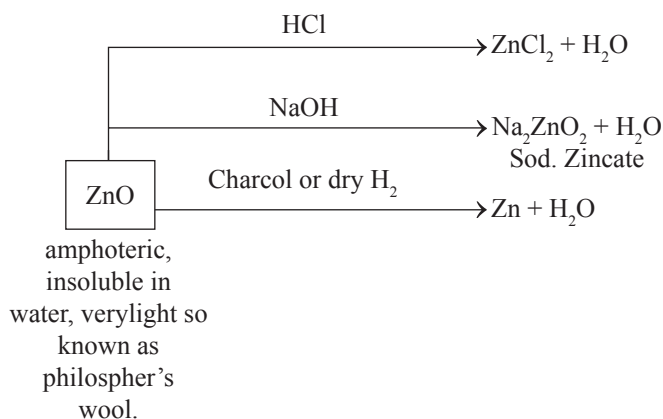
Ans. (b)

ZINC (Zn)**(I) Zinc oxide (ZnO)****Preparation**

It is obtained by the combustion of zinc or by the calcinations of zinc carbonate, zinc nitrate or zinc hydroxide.



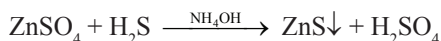
Very pure zinc oxide is prepared by mixing a solution of zinc sulphate with sodium carbonate. The basic zinc carbonate, thus precipitated, on heating gives pure zinc oxide.

**Properties****(II) Zinc Sulphide (ZnS)**

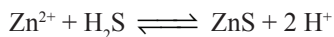
It occurs in nature as zinc blende.

Preparation

- (i) By passing H₂S through ammoniacal (alkaline) solution of a zinc salt.

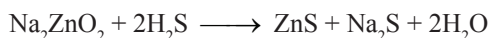


Note: Zinc sulphide, being soluble in dilute acids, cannot be precipitated from an acidic solution. On the other hand, in the neutral aqueous solution of the zinc salt, partial precipitation occurs because the following equilibrium lies towards left.

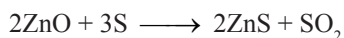


Alkali present in the solution removes H⁺ as water and hence, favours reaction towards right.

- (ii) By passing H₂S gas through a zincate solution.

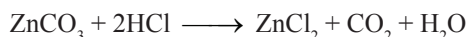


- (iii) By heating zinc oxide with sulphur.

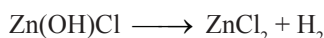
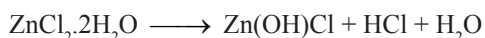


(III) Zinc Chloride ($\text{ZnCl}_2 \cdot 2\text{H}_2\text{O}$)**Preparation**

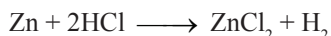
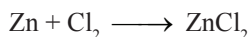
It is obtained by treating zinc oxide or zinc carbonate or zinc hydroxide with dilute hydrochloric acid. The solution on concentration and cooling gives hydrated zinc chloride crystals, $\text{ZnCl}_2 \cdot 2\text{H}_2\text{O}$.



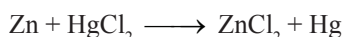
Anhydrous zinc chloride cannot be obtained by heating crystals of hydrated zinc chloride as hydrolysis occurs and basic chloride (zinc hydroxyl chloride) is formed which on further heating gives zinc oxide.



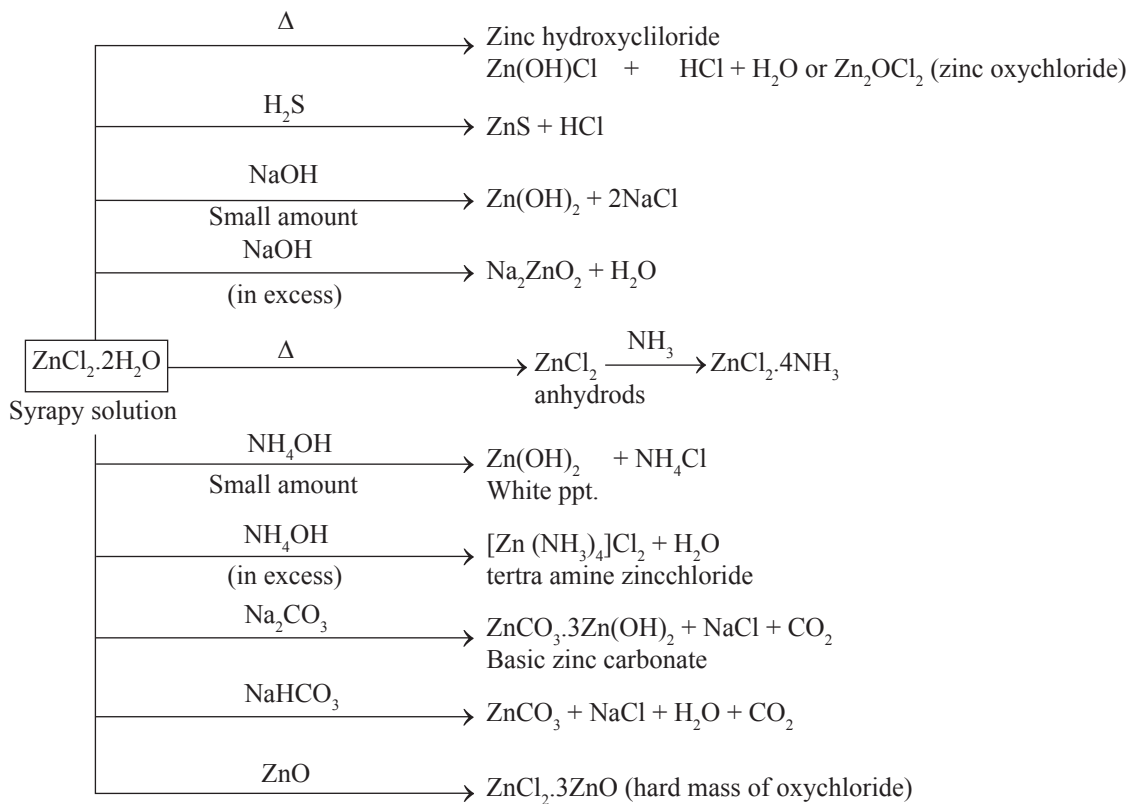
The anhydrous zinc chloride is obtained by heating zinc in an atmosphere of dry chlorine or dry HCl gas.



This can also be formed by distilling zinc powder with mercuric chloride.

**Properties**

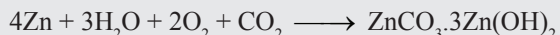
(i) Anhydrous zinc chloride is a white solid



- It is prepared by treating metallic zinc, zinc oxide or zinc carbonate with dilute H_2SO_4 .

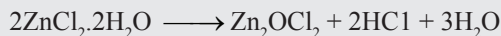
75. Why does zinc become dull in moist air?

Ans. Zinc in the presence of moist air is coated by the thin film of its basic carbonate on its surface and becomes dull.



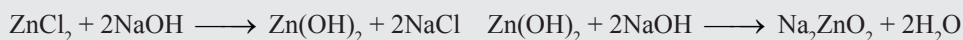
76. Hydrated zinc chloride cannot be dehydrated on heating.

Ans. On heating, it forms zinc oxychloride.



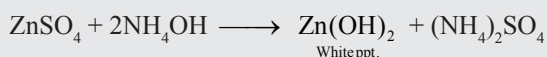
77. The addition of NaOH solution to a solution of zinc chloride produces a white precipitate which dissolves on further addition of NaOH.

Ans. On addition of NaOH, a white precipitate of Zn(OH)_2 is formed which dissolves in excess of NaOH forming sodium zincate.



78. The addition of NH_4OH to ZnSO_4 solution produces white precipitate but no precipitate is formed if it contains NH_4Cl .

Ans. NH_4OH is a weak hydroxide. It ionizes slightly furnishing OH^- ions. However, the OH^- ions are sufficient to cause the precipitation of Zn(OH)_2 as its ionic product is exceeded from its K_{sp} .



In presence of NH_4Cl , the ionization of NH_4OH is further suppressed and sufficient OH^- ions are not available to cause precipitation as the ionic product is not exceeded from its K_{sp} .

79. ZnO is white and is used as white pigment but it changes to pale yellow on heating. Explain.

Ans. On heating the lattice structure of ZnO changes as some defects are developed in lattice.

80. Which of the following is amphoteric oxide?

[AIEEE 2003]

- (a) SO_2 (b) B_2O_3 (c) ZnO (d) Na_2O

Ans. (c)

81. A solution when diluted with H_2O and boiled gives a white precipitate. On addition of excess of $\text{NH}_4\text{Cl}/\text{NH}_4\text{OH}$, the volume of precipitate decreases leaving behind a white gelatinous precipitate. The precipitate which dissolved in $\text{NH}_4\text{OH}/\text{NH}_4\text{Cl}$ is:

[IIT 2006]

- (a) Zn(OH)_2 (b) Al(OH)_3 (c) Mg(OH)_2 (d) Ca(OH)_2

Ans. (a)

82. Match each of the reactions given in **Column I** with the corresponding product(s) given in **Column II**.

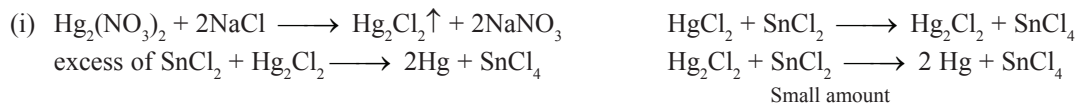
[IIT 2009]

Column I

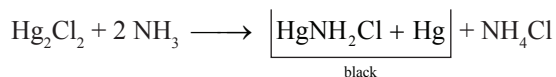
Column II

- | | |
|--------------------------------------|--------------------------|
| (A) $\text{Cu} + \text{dil. HNO}_3$ | (p) NO |
| (B) $\text{Cu} + \text{conc. HNO}_3$ | (q) NO_2 |
| (C) $\text{Zn} + \text{dil. HNO}_3$ | (r) N_2O |
| (D) $\text{Zn} + \text{conc. HNO}_3$ | (s) $\text{Cu(NO}_3)_2$ |
| | (t) $\text{Zn(NO}_3)_2$ |

Ans. (A–p, r, s) (B – q, s) (C – p, r, t) (D – q, t)

Mercury (Hg)**(I) Mercurous chloride or calomel (Hg_2Cl_2)****Preparation****Properties**

(i) If treated with aqueous NH_3 , it turns black due to formation of $\text{Hg}(\text{NH}_2)\text{Cl} \cdot \text{Hg}$.



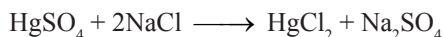
(ii) Mercurous ion is diamagnetic. This indicates that there is no unpaired electron, hence mercurous ion is not Hg^+ but Hg_2^{2+} .

**(II) Mercuric chloride or corrosive sublimate (HgCl_2)****Preparation**

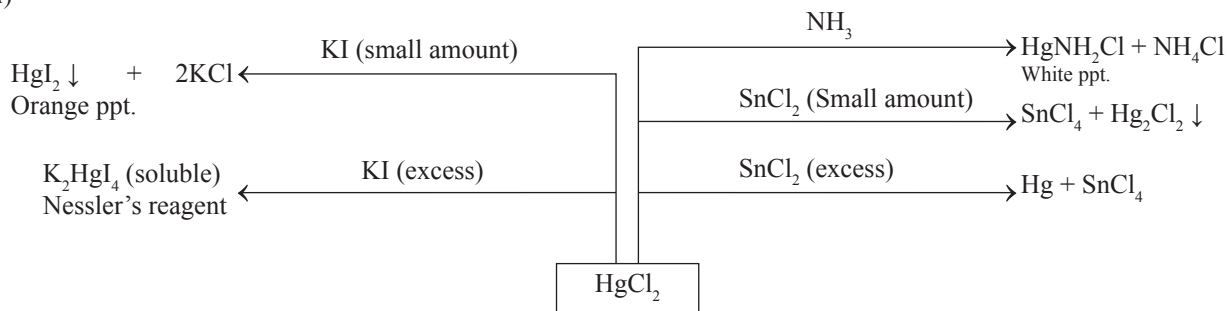
It is formed when chlorine is passed over heated mercury or mercury (I) chloride.



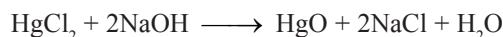
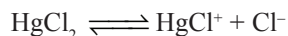
It is manufactured by heating a mixture of mercury (II) sulphate and sodium chloride in the presence of a little MnO_2 which oxidizes any Hg_2Cl_2 salt formed during the reaction.

**Properties**

(i)

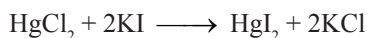


(ii) It dissolves AgCl

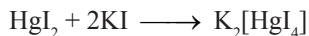


(III) Mercuric iodide (HgI₂)**Preparation**

It is obtained as red precipitate when a requisite amount of potassium iodide solution is added to a solution of mercury (II) chloride.



Mercury (II) iodide readily dissolves in an excess of potassium iodide solution and forms the $[\text{HgI}_4]^{2-}$ complex ion.



Mercurous ion exist as dimer Hg_2^{2+} and not as Hg^+ and this explains its diamagnetic nature.

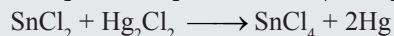
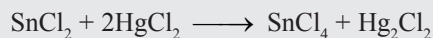
Some Noteworthy Points

- Mercury is the only liquid metal at room temperature.
- Mercury is the only liquid metal at room temperature.
- Mercuric salts are more stable than mercurous salts.
- Mercury forms alloys (called Amalgam) with metals except iron and platinum. Hence, mercury is transported in iron containers.
- When a little of mercury is poured into a AgNO_3 solution, a tree like growth of silver amalgam is produced. This is called Mercury tree.
- Mercuric chloride or Corrosive sublimate is HgCl_2 . It is poisonous and best antidote for it is white of an egg.
- Alkaline solution of K_2HgI_4 is called Nessler's reagent and gives brown precipitate $[\text{HgI}.\text{HgO}.\text{NH}_2]$ or iodide of million's base] with ammonia.
- HgS is used as a cosmetic under the name Vermillion and in Ayurvedic medicine as Makardhwaja. HgS produced by either direct combination or by precipitation by H_2S is black in colour. This black HgS on sublimation gives the bright red coloured product used as vermilion or as Makardhwaja.
- Mercury vapour are invisible and has no shining appearance as the liquid metal has.
- Calomel, Hg_2Cl_2 is used for making standard calomel electrode, used as a reference electrode.
- Mercuric thiocyanate is used for making Pharaoh's serpents. For this purpose, it is mixed with a little gum and made into pellets, which when dried and ignited, yield a serpentlike voluminous ash.

Worked-out Examples

83. HgCl_2 and SnCl_2 cannot exist together in an aqueous solution. Why?

Ans. SnCl_2 is strong reducing agent and reduces HgCl_2 first to Hg_2Cl_2 (white) and then to Hg (black).

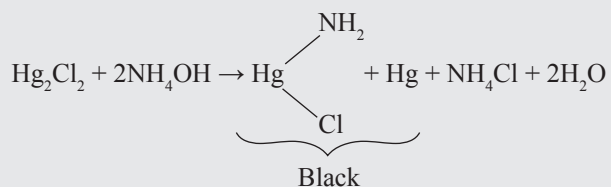


84. Why is mercuric chloride called corrosive sublimate?

Ans. HgCl_2 has corrosive nature as well as it sublimes on heating.

85. The colour of mercurous chloride changes from white to black when treated with ammonia solution. Why?

Ans. Hg_2Cl_2 absorbs NH_3 to form a mixture of mercury and mercuric amino chloride, a black substance.



86. What happens when aqueous ammonia reacts with:

- (a) silver chloride, (b) mercury (I) chloride, (c) mercury (II) chloride?

Ans. (a) $\text{AgCl} + 2\text{NH}_3 \longrightarrow [\text{Ag}(\text{NH}_3)_2]\text{Cl}$

(b) $\text{Hg}_2\text{Cl}_2 + 2\text{NH}_3 \longrightarrow \text{Hg} + \text{Hg}(\text{NH}_2)\text{Cl} + \text{NH}_4\text{Cl}$

(c) $\text{HgCl}_2 + 2\text{NH}_3 \longrightarrow [\text{Hg}(\text{NH}_3)_2]\text{Cl}_2$

87. Zn, Cd, Hg compounds are usually white. Explain.

Ans. All show +2 oxidation state involving $(n-1)d^{10}$ configuration having no unpaired electron in d-subshell. Thus, d-d transition is not possible.

88. Arrange ZnO, CdO and HgO in increasing order thermal stability.

Ans. $\text{HgO} < \text{CdO} < \text{ZnO}$.

89. What happens when HgCl_2 reacts with concentrated HCl?

Ans. $\text{HgCl}_2 + 2\text{HCl} \longrightarrow \text{H}_2\text{HgCl}_4$ (crystallises on cooling)

90. Mercurous ion is written as Hg_2^+ whereas cuprous ion is written as Cu^+ . Explain.

Ans. Hg^+ has $[\text{Xe}] 4f^{14}, 5d^{10}, 6s^1$ configuration and thus Hg^+ ion should be paramagnetic due to unpaired electron but Hg_2^+ ions show diamagnetic nature and thus it is taken as Hg_2^{2+} . Cu^+ ions are diamagnetic with $[\text{Ar}] 3d^{10}$ configuration.

91. Hg is liquid at room temperature, why?

Ans. Very high ionization energy makes it difficult for electrons to participate in metallic bonding.

92. A red solid is insoluble in water. However becomes soluble if some KI is added to water. Heating the red solid in a test tube results in liberation of some violet coloured fumes and droplets of metal appear on the cooler part of the test tube. The red solid is: **[AIEEE 2003]**

- (a) HgO (b) Pb_3O_4 (c) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ (d) HgI_2

Ans. (d)

93. For making good quality mirrors, plates of float glasses are used. These are obtained by floating molten glass over a liquid metal which does not solidify before glass. The metal used can be:

[AIEEE 2003]

- (a) Na (b) Mg (c) Hg (d) Sn

Ans. (c)

94. Calomel on reaction with NH_4OH gives:

[AIEEE 2005]

- (a) HgNH_2Cl (b) $\text{NH}_2\text{—Hg—Hg—Cl}$
(c) Hg_2O (d) HgO

Ans. (a)

Some important reagents having transition metals

- Baeyer's reagent: Dilute alkaline KMnO_4
- Schweitzer reagent: $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$
- Tollen's reagent: Ammonical solution of AgNO_3 $[\text{Ag}(\text{NH}_3)_2]\text{OH}$
- Nessler's reagent: Alkaline solution of K_2HgI_4
- Fehling solution A: CuSO_4 solution.
- Benedict's solution: CuSO_4 solution + sodium citrate + Na_2CO_3 .

- Lucas reagent: $\text{HCl}(\text{conc.}) + \text{anhydrous ZnCl}_2$.
- Fenton's reagent: $\text{FeSO}_4 + \text{H}_2\text{O}_2$.
- Etard's reagent: CrO_2Cl_2 .
- Barfoed's reagent: $\text{Cu}(\text{CH}_3\text{COO})_2 + \text{CH}_3\text{COOH}$.
- Milon's reagent: Solution of mercuric and mercurous nitrate.

Titanox. Another white pigment is

lithophone—a mixture of ZnS and BaSO_4 .

Table 12.2 Some important compounds of transition Metals, their Common Names, Formulae and Uses

S.No.	Common name	Formula	Uses
1.	Green vitriol	$\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$	Lab. Reagents, For making blue black ink.
2.	Blue vitriol	$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$	Bordacux mixture
3.	White vitriol	$\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$	For making ZnO
4.	Philosopher's wool	ZnO	Antiseptic catalyst
5.	Mohr shalt	$\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$	In volumetric analysis
6.	Vermillion	HgS	Red pigment for paints
7.	Calmol	Hg_2Cl_2	In calmol electrode
8.	Lunar caustic	AgNO_3	Photography, lab. Reagent.
9.	Pyrolusite	MnO_2	For making KMnO_4 , in dry cell.
10.	Corrosive sublimate	HgCl_2	For preparing Nessler's reagent.
11.	Fulminating gold	$\text{Au} = \text{NH}$ NH_2	Explosive
12.	Bordeaux mixture	$\text{CuSO}_4 + \text{Ca}(\text{OH})_2$	Fungicide
13.	Pharaoh serpent	$\text{Hg}(\text{CNS})_2$	Fire works
14.	Rouge	Fe_2O_3	Assbrasive
15.	Nessler's Reagent	K_2HgI_4	For testing NH_3 and NH_4^+ ions.
16.	Scheel's green	$\text{Cu}_3(\text{AsO}_3)_2$	Green pigment for paints
17.	Lithopone	$\text{BaSO}_4 + \text{ZnS}$	White pigment for paints
18.	Malachite (green)	$\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$	Green pigment for paints
19.	Azurtite (blue green)	$2\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$	Blue green pigment for paints.

ONE CORRECT OPTION TYPE QUESTIONS

- In $\text{Cr}_2\text{O}_4^{2-}$ every Cr atom is linked to
 - two O atoms
 - three O atoms
 - four O atoms
 - five O atoms
- FeCr_2O_4 (chromite) is converted to Cr by following steps
 Chromite $\xrightarrow{\text{I}}$ NaCrO_4 $\xrightarrow{\text{II}}$ Cr_2O_3 $\xrightarrow{\text{III}}$ Cr
 (I), (II) and (III) are

(I)	(II)	(III)
(a) $\text{Na}_2\text{CO}_3/\text{air}, \Delta$	C	C
(b) $\text{NaOH}/\text{air}, \Delta$	C, Δ	Al, Δ
(c) $\text{NaOH}, \text{air}, \Delta$	C, Δ	C, Δ
(d) conc. $\text{H}_2\text{SO}_4, \Delta$	$\text{NH}_4\text{Cl}, \Delta$	C, Δ
- Chromite ore has the formula
 - FeCr_2O_4
 - $\text{FeO} \cdot \text{Cr}_2\text{O}_3$
 - FeCr_2O_7
 - Both (a) and (b)
- $\text{K}_2\text{Cr}_2\text{O}_7$ is preferred to $\text{Na}_2\text{Cr}_2\text{O}_7$ for use in volumetric analysis as a primary standard because
 - $\text{Na}_2\text{Cr}_2\text{O}_7$ is hygroscopic while $\text{K}_2\text{Cr}_2\text{O}_7$ is not
 - $\text{K}_2\text{Cr}_2\text{O}_7$ is hygroscopic while $\text{Na}_2\text{Cr}_2\text{O}_7$ is not
 - $\text{K}_2\text{Cr}_2\text{O}_7$ is pure while $\text{Na}_2\text{Cr}_2\text{O}_7$ is impure
 - None of the above
- The deep blue colour produced on adding H_2O_2 to acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution is due to the formation of
 - Cr_2O_3
 - CrO_3
 - CrO_5
 - CrO_4^{2-}
- Ammonium dichromate on heating decomposes with spark. This reaction is called
 - A phosphorescence
 - Incandescence
 - Chemical volcano
 - None of these
- Mixture of $\text{K}_2\text{Cr}_2\text{O}_7$ and conc. H_2SO_4 is called
 - Perchromic acid
 - Chromic acid
 - Chromium sulphate
 - None of these
- When acidified solution of $\text{K}_2\text{Cr}_2\text{O}_7$ is shaken with aqueous solution of FeSO_4 , then
 - $\text{Cr}_2\text{O}_7^{2-}$ ion is reduced to Cr^{3+} ions
 - $\text{Cr}_2\text{O}_7^{2-}$ ion is converted to CrO_4^{2-} ions
 - $\text{Cr}_2\text{O}_7^{2-}$ ion is oxidized to Cr
 - $\text{Cr}_2\text{O}_7^{2-}$ ion is oxidized to CrO_3
- One of the products formed when $\text{K}_2\text{Cr}_2\text{O}_7$ reacts with concentrated H_2SO_4 in cold is
 - $\text{Cr}_2(\text{SO}_4)_3$
 - Cr_2O_3
 - CrO_3
 - CrO_4Cl_2
- In the redox reaction involving $\text{Cr}_2\text{O}_7^{2-}$ and Fe^{2+} ions the number of electrons absorbed per chromium atom is
 - 1
 - 3
 - 4
 - 6
- Number of moles of $\text{K}_2\text{Cr}_2\text{O}_7$ reduced by one mole of Sn^{2+} ion is
 - $\frac{1}{3}$
 - 3
 - $\frac{1}{6}$
 - 6
- Ammonium dichromate is used in some fire works. The green coloured powder blown in the air is
 - CrO_3
 - Cr_2O_3
 - Cr
 - $\text{CrO}(\text{O}_2)$
- The mineral from which potassium permanganate is manufactured is
 - Pyrolusite, MnO_2
 - Braunite, Mn_2O_3
 - Hausmannite, Mn_3O_4
 - Manganite, $\text{Mn}_2\text{O}_3 \cdot \text{H}_2\text{O}$
- Which one of the following oxidizing agent is not used for oxidation of potassium manganate to potassium permanganate?
 - CO_2
 - Cl_2
 - O_3
 - H_2O_2
- The electrolysis of potassium manganate solution gives
 - $\text{MnO}_2 + \text{KOH}$
 - $\text{KOH} + \text{H}_2$
 - $\text{KOH} + \text{O}_2$
 - $\text{KMnO}_4 + \text{KOH}$
- Baeyer's reagent is
 - $\text{KMnO}_4 + \text{H}_2\text{SO}_4$
 - $\text{KMnO}_4 + \text{KOH}$
 - $\text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4$
 - $\text{K}_2\text{Cr}_2\text{O}_7 + \text{KOH}$
- Which one of the following is not oxidized by acidified KMnO_4 ?
 - Sodium oxalate
 - Potassium Iodide
 - Ferrous sulphate
 - Sodium sulphate

18. In alkaline medium, the equivalent mass of KMnO_4 is
 (a) $\frac{M}{3}$ (52.66) (b) $\frac{M}{5}$ (31.60)
 (c) $\frac{M}{6}$ (26.33) (d) M (158.00)
 (M = molecular mass)
19. Which of the following cannot reduce acidified solution of MnO_4^- ion?
 (a) $\text{H}_2\text{C}_2\text{O}_4$ (b) H_2
 (c) H(nascent) (d) Fe^{2+} ions
20. The reaction $2 \text{MnO}_4^{2-} + \text{Cl}_2 \rightarrow 2 \text{MnO}_4^- + 2 \text{Cl}^-$ takes place in
 (a) Basic medium (b) Acidic medium
 (c) Neutral medium (d) Both (a) and (b)
21. Oxidation of oxalic acid by acidified KMnO_4 is an example of autocatalysis, it is due to the presence of
 (a) SO_4^{2-} (b) MnO_4^-
 (c) Mn^{2+} (d) K^+
22. The starting material for the manufacture of KMnO_4 is
 (a) Pyrolusite (b) Manganite
 (c) Magnetite (d) Haematite
23. The number of moles of KMnO_4 that will be needed to react with one mole of sulphite ion in acidic medium is
 (a) $\frac{2}{5}$ (b) $\frac{3}{5}$
 (c) $\frac{4}{5}$ (d) 1
24. The number of moles of KMnO_4 that will be needed to react completely with one mole of ferrous oxalate in acidic solution is
 (a) $\frac{3}{5}$ (b) $\frac{2}{5}$
 (c) $\frac{4}{5}$ (d) 1
25. Anhydrous ferric chloride is prepared by
 (a) heating hydrated ferric chloride at a high temperature in a stream of air.
 (b) heating metallic iron in a stream of dry chlorine gas.
 (c) reactions of ferric oxide with hydrochloric acid.
 (d) reaction of metallic iron with hydrochloric acid.
26. $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ is actually
 (a) $[\text{Fe}(\text{H}_2\text{O})_6]\text{Cl}_3$
 (b) $[\text{Fe}(\text{H}_2\text{O}_5\text{Cl})\text{Cl}_2] \cdot \text{H}_2\text{O}$
 (c) $[\text{Fe}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl}_2 \cdot 2\text{H}_2\text{O}$
 (d) $[\text{Fe}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot 3 \text{H}_2\text{O}$
27. Green vitriol is
 (a) $\text{CuSO}_4 \cdot 5 \text{H}_2\text{O}$ (b) $\text{ZnSO}_4 \cdot 7 \text{H}_2\text{O}$
 (c) $\text{Na}_2\text{SO}_4 \cdot 10 \text{H}_2\text{O}$ (d) $\text{FeSO}_4 \cdot 7 \text{H}_2\text{O}$
28. Which of the following is a primary standard in volumetric analysis?
 (a) FeSO_4
 (b) $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$
 (c) NaOH
 (d) HCl
29. Ferrous and ferric ions in solution may be distinguished by
 (a) Silver nitrate solution
 (b) Lead acetate solution
 (c) Acidified solution of KMnO_4
 (d) Sodium chloride solution
30. Which of the following iron salts exists as a dimer?
 (a) Ferric chloride (b) Ferrous chloride
 (c) Ferrous sulphate (d) Mohr's salt.
31. Ferric chloride is used for etching copper and silver metals. This is due to
 (a) oxidising action of Fe^{3+} ions
 (b) reducing action of Fe^{3+} ions
 (c) both oxidizing and reducing actions of Fe^{3+} ions
 (d) reducing action of Fe^{2+} ions
32. Ferric ions can be estimated by
 (a) titrating against acidified KMnO_4 solution
 (b) titrating against Mohr's salt solution
 (c) titrating against oxalic acid solution
 (d) first reducing the ferric ions by stannous chloride and then titrating against a solution of acidified KMnO_4 solution.
33. Mohr salt on oxidation with concentrated HNO_3 gives
 (a) Potash alum (b) Chrome alum
 (c) Ferric alum (d) None of these
34. In the volumetric analysis, Mohr salt is preferred to ferrous sulphate because
 (a) It is more readily oxidizable than FeSO_4
 (b) It is less easily oxidizable than FeSO_4
 (c) it can undergo both oxidation and reduction
 (d) It can be oxidized even in the absence of dilute H_2SO_4
35. Mohr salt is prepared from equimolar solution of
 (a) Green vitriol and potassium sulphate
 (b) Green vitriol and ammonium sulphate
 (c) Blue vitriol and potassium sulphate
 (d) Blue vitriol and ammonium sulphate

12.46 Transition Elements (d-block) and Their Compounds

36. Ferric sulphate on heating gives
(a) SO_2 and SO_3 (b) SO_2 only
(c) SO_3 Only (d) S
37. Argentite is an ore of
(a) Cu (b) Pt
(c) Au (d) Ag
38. CuSO_4 can be estimated volumetrically
(a) by reaction with KI followed by reaction with $\text{Na}_2\text{S}_2\text{O}_3$
(b) by reaction with BaCl_2
(c) by reaction with $\text{K}_4\text{Fe}(\text{CN})_6$
(d) None is correct
39. On adding KI to a solution of CuSO_4
(a) cupric oxide is precipitated
(b) metallic copper is precipitated
(c) cuprous iodide is precipitated with the liberation of iodine
(d) no change takes place
40. Boiling CuCl_2 with excess of copper turnings and conc. HCl gives
(a) $\text{CuCl}(\text{s})$ (b) $\text{Cu}_2\text{O}(\text{s})$
(c) $[\text{CuCl}_3]^-$ (d) $[\text{CuCl}_2]^-$
41. Which of the following on treatment with water turns blue?
(a) $\text{Na}_2\text{SO}_4(\text{anhyd.})$ (b) $\text{ZnSO}_4(\text{anhyd.})$
(c) $\text{CuSO}_4(\text{anhyd.})$ (d) $\text{MgSO}_4(\text{anhyd.})$
42. Copper sulphate pentahydrate on heating at 423 K gives
(a) $\text{CuSO}_4 \cdot 2 \text{H}_2\text{O}$ (b) CuSO_4
(c) $\text{CuSO}_4 \cdot \text{H}_2\text{O}$ (d) $\text{CuO} + \text{SO}_3$
43. Blue vitriol is
(a) $\text{CuSO}_4 \cdot 5 \text{H}_2\text{O}$ (b) $\text{FeSO}_4 \cdot 7 \text{H}_2\text{O}$
(c) $\text{ZnSO}_4 \cdot 7 \text{H}_2\text{O}$ (d) $\text{CaSO}_4 \cdot 2 \text{H}_2\text{O}$
44. $\text{CuSO}_4 \cdot 5 \text{H}_2\text{O}$ is blue in colour because
(a) It contains water of crystallization
(b) SO_4^{2-} ions absorb red light
(c) Cu^{2+} ions absorb red light
(d) Cu^{2+} ions absorb all colours except red from the white light
45. The formula of malachite is
(a) $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$
(b) $2 \text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$
(c) $\text{CuCO}_3 \cdot 2 \text{Cu}(\text{OH})_2$
(d) $\text{CuCl}_2 \cdot 2 \text{Cu}(\text{OH})_2$
46. The formula of azurite is
(a) $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$
(b) $2 \text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$
(c) $\text{CuCO}_3 \cdot 2 \text{Cu}(\text{OH})_2$
(d) $\text{CuSO}_4 \cdot 2 \text{Cu}(\text{OH})_2$
47. Malachite and azurite are used respectively as
(a) Blue and green pigments
(b) Green and blue pigments
(c) Red and green pigments
(d) Green and red pigments
48. Cu^{2+} is more stable than Cu^{2+} . This is due to
(a) extensive hydration of Cu^{2+} ion
(b) higher charge and smaller size of Cu^{2+} ion as compared to Cu^+ ion
(c) Both (a) and (b)
(d) None of these
49. CuSO_4 solution + lime is called
(a) Luca's reagent
(b) Barfoed's reagent
(c) Fehling solution A
(d) Bordeaux mixture
50. Paris green is
(a) double salt of CuCO_3 and $\text{Cu}(\text{NO}_3)_2$
(b) double salt of $\text{Cu}(\text{CH}_3\text{COO})_2$ and $\text{Cu}_3(\text{AsO}_3)_2$, copper arsenite
(c) double salt of $\text{Cu}(\text{CH}_3\text{COO})_2$ and CuSO_4
(d) double salt of Cu and Ag nitrates
51. On heating copper nitrate strongly is finally obtained
(a) copper (b) copper oxide
(c) copper nitrate (d) copper nitride
52. Copper sulphate is commercially made from copper scraps by
(a) dissolving in hot concentrated H_2SO_4
(b) the action of dil. H_2SO_4 and air
(c) heating with sodium sulphate
(d) heating with sulphur
53. On adding KI to a solution of CuSO_4
(a) Cupric oxide is precipitated
(b) Metallic copper is precipitated
(c) Cuprous iodide is precipitated with the liberation of iodine
(d) No change takes place

54. What is the effect of shaking dil. H_2SO_4 with a small quantity of anhydrous CuSO_4 ?
- The white solid dissolves to form a colourless solution
 - The white solid dissolves to form a blue solution
 - The white solid dissolves to form a green solution
 - The white solid turns blue but does not dissolve
55. In solid $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, copper is coordinated to
- Five water molecules
 - Four water molecules
 - One sulphate ion
 - One water molecule
56. AgCl dissolves in a solution of NH_3 but not in water because
- NH_3 is a better solvent than H_2O
 - Ag^+ forms a complex ion with NH_3
 - NH_3 is a stronger base than H_2O
 - dipole moment of water is higher than NH_3
57. Silver ornaments turn black by atmospheric
- O_2
 - N_2
 - Cl_2
 - H_2S
58. Which of the following oxides is least stable at room temperature?
- CuO
 - Ag_2O
 - ZnO
 - Hg_2O
59. Choose the least soluble of the following?
- AgNO_3
 - AgF
 - AgCl
 - AgBr
60. Which of the following does not dissolve in NH_3 solution?
- AgF
 - AgCl
 - AgBr
 - AgI
61. The photographic film is coated with an emulsion of
- AgNO_3
 - AgF
 - AgBr
 - $\text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)_2]$
62. The developer used in photography is an alkaline solution of
- Hydroquinone
 - Glycerol
 - Phenol
 - Picric acid
63. The black image on an exposed and developed photograph film is composed of
- Ag
 - AgBr
 - $\text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)]$
 - Ag_2O
64. The fixer used in photography is
- $\text{Na}_2\text{S}_2\text{O}_3$
 - Na_2S
 - AgI
 - AgF
65. The silver salt most commonly used in photography is
- AgCl
 - AgBr
 - AgI
 - AgF
66. Which of the following metals have the highest thermal and electrical conductivity?
- Copper
 - Silver
 - Zinc
 - Aluminium
67. The image on an expose and developed photographic film is due to
- AgBr
 - $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$
 - Ag
 - Ag_2O
68. Lunar caustic is
- NaOH
 - KOH
 - $\text{Ba}(\text{OH})_2$
 - AgNO_3
69. When AgNO_3 comes in contact with skin, it leaves a black stain. This is because of
- HNO_3 produced by hydrolysis of AgNO_3
 - AgOH produced by hydrolysis of AgNO_3
 - Its reduction to silver
 - Its oxidation to silver oxide
70. AgNO_3 solution is kept in coloured bottles in the laboratory. This is done
- Because colours in glass act as negative catalysts
 - To cut off the entry of light
 - Simply to give it a distinction from other reagent bottles as it is costly
 - As ordinary white glass reacts with AgNO_3
71. The soluble complex formed on adding NH_4OH to AgNO_3 solution is
- $[\text{Ag}(\text{NH}_3)_4]\text{OH}$
 - $[\text{Ag}(\text{NH}_3)_2]\text{OH}$
 - $[\text{Ag}(\text{NH}_3)]\text{OH}$
 - $[\text{Ag}(\text{NH}_3)_2](\text{OH})_2$
72. Which of the following chemicals is used as an eye drop in trancoma?
- AgNO_3
 - KNO_3
 - H_2O_2
 - CuSO_4
73. Which of the following sulphides is soluble in dil. HCl ?
- HgS
 - ZnS
 - CuS
 - PbS

12.48 Transition Elements (d-block) and Their Compounds

74. Which of the following is a constituent of lithopone
 (a) ZnSO_4 (b) ZnS
 (c) ZnCl_2 (d) ZnO
75. When mercury (I) chloride is heated and the vapours so evolved are cooled, the substance on sublimation thus collected consists of
 (a) Mercury and mercury (II) chloride
 (b) Mercury (II) chloride
 (c) Mercury (I) and mercury (II) chloride
 (d) Mercury
76. Yellow mercury (II) oxide is obtained when
 (a) Hg is heated in excess of air at 623 K
 (b) HgCl_2 is treated with NaOH solution
 (c) HgS is roasted in air
 (d) $\text{Hg}(\text{NO}_3)_2$ is heated in presence of Hg
77. Read the yellow mercury (II) oxides differ from each other in
 (a) chemical composition
 (b) particle size
 (c) only in colour
 (d) both in colour and particle size
78. Which one of the following ions exists as a dimer?
 (a) Hg^{2+} (b) Hg^+
 (c) Cu^{2+} (d) Cu^+
79. Calomel contains
 (a) Mercury (II) ion
 (b) Mercury (I) ion
 (c) Mercury (I) and mercury (II) ions
 (d) None of these
80. The salt of mercury which is used to prepare standard electrode is
 (a) Mercuric nitrate
 (b) Mercuric chloride
 (c) Mercuric nitrate
 (d) Mercurous chloride
81. Which of the following reactions is reversible?
 (a) $\text{Hg}_2\text{Cl}_2 \xrightarrow{\Delta} \text{HgCl}_2 + \text{Hg}$
 (b) $\text{CuCl}_2 + \text{Cu} \xrightarrow{\text{conc. HCl}} 2 \text{CuCl}$
 (c) $\text{Hg} + 2 \text{H}_2\text{SO}_4 \rightarrow \text{HgSO}_4 + \text{SO}_2 + 2 \text{H}_2\text{O}$
 (d) $2 \text{HgCl}_2 + \text{SnCl}_2 \rightarrow \text{Hg}_2\text{Cl}_2 + \text{SnCl}_4$
82. Which of the following is not an amphoteric oxide?
 (a) HgO (b) PbO_2
 (c) ZnO (d) SnO_2
83. Mercury (II) chloride is also called
 (a) Calomel (b) Corrosive sublimate
 (c) Nessler's reagent (d) Millon's base
84. When NH_4OH is added to Hg_2Cl_2 , the solid turns black due to the formation of
 (a) HgCl_2 (b) $\text{Hg}(\text{NH}_3)\text{Cl}$
 (c) Finely divided Hg (d) $[\text{Hg}(\text{NH}_3)_4]\text{Cl}$
85. When stannous chloride solution is added to HgCl_2 , the white precipitate produced is due to the formation of
 (a) Hg_2Cl_2 (b) $\text{Hg}(\text{NH}_3)_2\text{Cl}$
 (c) $\text{Hg}(\text{NH}_2)\text{Cl}$ (d) $\text{NH}_2 - \text{Hg} - \text{Hg} - \text{Cl}$
86. The brown precipitate formed when Nessler's reagent is treated with any ammonium salt is due to the formation of
 (a) $\text{NH}_2 - \text{Hg} - \text{I}$
 (b) $\text{NH}_2 - \text{Hg} - \text{O} - \text{Hg} - \text{I}$
 (c) HgI_4^{2-}
 (d) $[\text{Hg}(\text{NH}_3)_2]\text{I}_2$
87. Which of the following statements is incorrect?
 (a) Mercurous ion exists as Hg^+
 (b) Mercurous ion is diamagnetic and exists as dimer Hg_2^{2+}
 (c) Mercurous ion is colourless
 (d) There is a covalent bond between two Hg^+ ions
88. Which of the following statement about corrosive sublimate is incorrect?
 (a) It is prepared by heating mercury in chlorine
 (b) It reduces stannic chloride
 (c) It oxidises stannous chloride
 (d) It sublimates readily
89. The reaction $\text{MnO}_4^- + \text{e}^- \rightleftharpoons \text{MnO}_4^{2-}$ Takes place in
 (a) A basic medium
 (b) An acid medium
 (c) A neutral medium
 (d) Both acidic and basic media
90. Pick out the wrong reaction from
 (a) $2 \text{Na}_2\text{CrO}_4 + \text{H}^+ \rightarrow \text{Na}_2\text{Cr}_2\text{O}_7 + 2 \text{Na}^+ + \text{H}_2\text{O}$
 (b) $2 \text{MnO}_2 + 4 \text{KOH} + \text{O}_2 \rightarrow 4 \text{KmnO}_4 + 2 \text{H}_2\text{O}$
 (c) $\text{MnO}_4^- + 8 \text{H}^+ + 5 \text{Fe}^{2+} \rightarrow 5 \text{Fe}^{3+} + \text{Mn}^{2+} + 4 \text{H}_2\text{O}$
 (d) $2 \text{MnO}_4^- + 5 \text{C}_2\text{O}_4^{2-} + 16 \text{H}^+ \rightarrow 2 \text{Mn}^{2+} + 10 \text{CO}_2 + 8 \text{H}_2\text{O}$

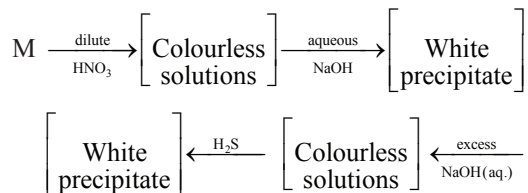
91. CrO_4^{2-} (yellow) changes to $\text{Cr}_2\text{O}_7^{2-}$ (orange) in $\text{pH} = x$ and vice-versa in $\text{pH} = y$. Hence, x and y are:
 (a) MnO_2 (b) Mn_2O_3
 (c) Mn_2O_7 (d) MnO
92. Manganese ions (Mn^{2+}) can be oxidized by Persulphate ions $\text{S}_2\text{O}_8^{2-}$ according to the following half-equations,

$$\text{S}_2\text{O}_8^{2-} + 2e^- \longrightarrow 2\text{SO}_4^{2-}$$

$$\text{Mn}^{2+} + 4\text{H}_2\text{O} \longrightarrow \text{MnO}_4^- + 8\text{H}^+ + 5e^-$$
 How many moles of $\text{S}_2\text{O}_8^{2-}$ are required to oxidize 1 mole of Mn^{2+} ?
 (a) 2.5 (b) 2.0
 (c) 11.0 (d) 0.4
93. AgCl on fusion with Na_2CO_3 forms:
 (a) Ag_2CO_3 (b) Ag_2O
 (c) Ag (d) Ag_2C_2
94. Boiling CuCl_2 with Cu in conc. HCl give:
 (a) CuCl (b) CuCl_2
 (c) $\text{H}[\text{CuCl}_2]$ (d) Cu_2Cl
95. Mercury is transported in the containers made of:
 (a) Ag (b) Pb
 (c) Al (d) Fe
96. Silver nitrate is usually kept in coloured bottles because it is:
 (a) oxidized in air
 (b) decomposes in sunlight
 (c) explodes in sunlight
 (d) reacts with air in sunlight
97. The iron salt used in blue prints is:
 (a) FeC_2O_4 (b) $\text{Fe}_2(\text{C}_2\text{O}_4)_2$
 (c) $\text{K}_4\text{Fe}(\text{CN})_6$ (d) FeSO_4
98. The maximum and minimum melting point of first and second transition series elements respectively are obtained with:
 (a) Cr and Zn (b) Cr and Hg
 (c) Cr and Cd (d) Mo and Cd
99. ZnO shows yellow colour on Heating due to:
 (a) d-d transition
 (b) C-T spectra
 (c) Higher polarization caused by Zn^{2+} ion
 (d) F-centres
100. The product of I^- with MnO_4^- in alkaline medium is:
 (a) I_2 (b) IO_3^-
 (c) IO^- (d) IO_4^-
101. Which of the following statements is incorrect?
 (a) Mercurous ion exists as Hg^+
 (b) Mercurous ion is diamagnetic and exists as dimer Hg_2^{2+}
 (c) Mercurous ion is colourless
 (d) There is a covalent bond between two Hg^+ ions
102. $\text{Na}_2\text{CO}_3 + \text{Fe}_2\text{O}_3 \longrightarrow \text{A} + \text{CO}_2$ what is A in the reaction?
 (a) NaFeO_2 (b) Na_3FeO_3
 (c) Fe_3O_4 (d) Na_2FeO_2
103. Ferrous sulphate on heating gives:
 (a) SO_2 and SO_3 (b) SO_2 only
 (c) SO_3 only (d) H_2S only
104. CuSO_4 solution + lime is called:
 (a) Luca's reagent (b) Bafoed's reagent
 (c) Fehling solution A (d) Bordeaux mixture
105. When ammonia is added to a cupric salt solution, the deep blue colour is observed it is due to the formation of:
 (a) $[\text{Cu}(\text{OH})_4]^{2-}$
 (b) $[\text{Cu}(\text{NH}_3)_4]^{2+}$
 (c) $[\text{Cu}(\text{H}_2\text{O})_2(\text{NH}_3)_2]^{2+}$
 (d) $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$
106. Philosopher's wool when heated with BaO at 1100°C gives the compound:
 (a) BaCdO_2 (b) $\text{Ba} + \text{ZnO}_2$
 (c) $\text{BaO}_2 + \text{Zn}$ (d) BaZnO_2
107. The electrons which take part in order to exhibit variable oxidation states by transition metals are:
 (a) ns only
 (b) $(n-1)$ d only
 (c) ns and $(n-1)$ d only but not np
 (d) $(n-1)$ d and np only but not ns
108. During estimation of oxalic acid Vs KMnO_4 , self indicator is:
 (a) KMnO_4 (b) oxalic acid
 (c) K_2SO_4 (d) MnSO_4
109. When KMnO_4 solution is added to hot oxalic acid solution, the decolourisation is slow in the beginning but becomes instantaneous after some time. This is because:
 (a) Mn^{2+} acts as auto catalyst
 (b) CO_2 is formed
 (c) Reaction is exothermic
 (d) MnO_4^- catalysis the reaction

12.50 Transition Elements (d-block) and Their Compounds

110. A metal M and its compound can give the following observable changes in a consequence of reactions



- (a) Mg (b) Pb
(c) Zn (d) Sn
111. Sodium thiosulphate is used to remove the unexposed AgBr from photographic films by forming a complex. In this complex. In this complex of silver, the coordination number of silver is:
(a) 2 (b) 4
(c) 6 (d) 8
112. Mercury (II) chlorides on reaction with gaseous ammonia forms:
(a) $\text{Hg}(\text{NH}_2)\text{Cl}$ (b) $\text{Hg}(\text{NH}_3)_2\text{Cl}_2$
(c) $[\text{Hg}(\text{NH}_3)_4]\text{Cl}_2$ (d) $[\text{Hg}(\text{NH}_3)_2]\text{Cl}$
113. Copper sulphate is prepared by blowing a current of air through copper scrap and dilute H_2SO_4 . Dilute HNO_3 is also added:
(a) to oxidize copper to Cu^{2+} which then form CuSO_4 with dilute H_2SO_4
(b) to oxidize Fe^{2+} to iron (III) sulphate, which remains in solution after crystallisation of CuSO_4
(c) to speed up the ionization of H_2SO_4 to give SO_4^{2-} ions
(d) which combines with H_2SO_4 to give a very strong oxidizing mixture and oxidizes Cu to Cu^{2+}
114. Which two sets of reactants best represent the amphoteric character of $\text{Zn}(\text{OH})_2$?
Set 1: $\text{Zn}(\text{OH})_2(\text{s})$ and $\text{OH}^-(\text{aq})$
Set 2: $\text{Zn}(\text{OH})_2(\text{s})$ and $\text{H}_2\text{O}(\text{l})$
Set 3: $\text{Zn}(\text{OH})_2(\text{s})$ and $\text{H}^+(\text{l})$
Set 4: $\text{Zn}(\text{OH})_2(\text{s})$ and $\text{NH}_3(\text{aq})$
(a) 1 and 2 (b) 1 and 3
(c) 2 and 4 (d) 3 and 4
115. $\text{AgNO}_3 \xrightarrow{\Delta} (\text{W}) + (\text{X}) + \text{O}_2$
 $(\text{X}) + \text{H}_2\text{O} \longrightarrow \text{HNO}_2 + \text{HNO}_3$
 $(\text{W}) + \text{HNO}_3 \longrightarrow \text{Y} + \text{NO} + \text{H}_2\text{O}$
 $(\text{Y}) + \text{Na}_2\text{S}_2\text{O}_3 (\text{excess}) \longrightarrow (\text{Z}) + \text{NaNO}_3$
Identify (W) to (Z).
(a) $\text{W} = \text{Ag}$ $\text{X} = \text{N}_2\text{O}$ $\text{Y} = \text{AgNO}_3$
 $\text{Z} = \text{Na}_2[\text{Ag}(\text{S}_2\text{O}_3)_2]$
(b) $\text{W} = \text{Ag}_2\text{O}$ $\text{X} = \text{NO}$ $\text{Y} = \text{AgNO}_3$
 $\text{Z} = \text{Na}_2[\text{Ag}(\text{S}_2\text{O}_3)_2]$

- (c) $\text{W} = \text{Ag}$ $\text{X} = \text{NO}_2$ $\text{Y} = \text{AgNO}_3$
 $\text{Z} = \text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)_2]$
(d) $\text{W} = \text{Ag}_2\text{O}$ $\text{X} = \text{N}_2$ $\text{Y} = \text{AgNO}_3$
 $\text{Z} = \text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)_2]$

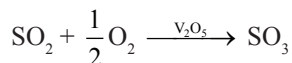
116. The oxidation state of copper changes when aqueous copper (II) ions react with:
(I) $\text{NaOH}(\text{aq})$ (II) $\text{Fe}(\text{s})$
(III) $\text{KI}(\text{aq})$
(a) I, II, III (b) II only
(c) II, III (d) I only
117. The aqueous solution of transition metal salt changes colour from pink to blue, when concentrated hydrochloric acid is added to it. The change in colour is due to:
(a) evolution of hydrogen that changes the oxidation state of the metal ion
(b) change in the coordination number of the metal ion from 6 to 4 and formation of new species in solution
(c) formation of a coordination complex of the metal ion with hydrochloric acid
(d) protonation of the metal ion
118. Which of the comparison regarding Zn, Cd, Hg is/are incorrect?
(I) ZnCl_2 is ionic whereas CdCl_2 and HgCl_2 are covalent
(II) Zn and Cd dissolves in dilute acid (HCl) liberating H_2 but Hg cannot
(III) Zn and Cd forming white ppt. of $\text{Zn}(\text{OH})_2$ and $\text{Cd}(\text{OH})_2$ but Hg forms coloured ppt. of $\text{Hg}(\text{OH})_2$
(IV) all form A_2^{2+} type ion
(a) only III (b) I, III, IV
(c) I and III (d) all of these
119. Which of the following compound can produce Riemann's green with $\text{Co}(\text{NO}_3)_2$ solution?
(a) ZnO
(b) $3\text{Zn}(\text{OH})_2 \cdot \text{ZnCO}_3$
(c) ZnSO_4
(d) All of these
120. Metal-metal bonding is more frequent in 4d or 5d series than in 3d series due to:
(a) their greater enthalpies of atomization
(b) the large size of the orbitals which participates in the metal-metal bond formation
(c) their ability to involve both ns and $(n-1)$ d electrons in the bond formation
(d) the comparable size of 4d and 5d series elements

121. Which of the oxide of manganese is amphoteric?
 (a) MnO_2 (b) Mn_2O_3
 (c) Mn_2O_7 (d) MnO
122. The compound in which nickel has the lowest oxidation state is:
 (a) $\text{Ni}(\text{CO})_4$ (b) $(\text{CH}_3\text{COO})_2\text{Ni}$
 (c) NiO (d) $\text{NiCl}_2(\text{PPh}_3)_2$
123. A metal M which is not affected by strong acids like conc. HNO_3 , concentrated H_2SO_4 and concentrated solution of alkalies like NaOH , KOH forms MCl_3 , which finds use for tanning in photography?
 (a) Ag (b) Hg
 (c) Au (d) Cu
124. Copper (II) ions gives reddish brown precipitate with potassium ferrocyanide. The formula of the precipitate is:
 (a) $\text{Cu}_4[\text{Fe}(\text{CN})_6]$ (b) $\text{Cu}_2[\text{Fe}(\text{CN})_6]$
 (c) $\text{Cu}_3[\text{Fe}(\text{CN})_6]$ (d) $\text{Cu}_3[\text{Fe}(\text{CN})_6]_2$
125. The correct statement about iron includes
 (I) the highest oxidation state of iron is +6 in K_3FeO_4
 (II) that the iron shows +2 oxidation state with six electrons in the unfilled 3d orbitals
 (III) the common oxidation state of iron is +3 with five unpaired electrons in the 3d orbital
 (a) I, II, III (b) I, II
 (c) II, III (d) I only
126. Identify the wrong statement regarding copper sulphate:
 (a) It reacts with KI to give I_2
 (b) It reacts with KCl to give Cl_2
 (c) Its tartarate complex reacts with NaOH and glucose to give Cu_2O
 (d) It gives CuO on strong heating in air
127. Anhydrous mercurous chloride can be prepared by:
 (a) the reduction of HgCl_2 with SnCl_2
 (b) the reaction of HgCl_2 with Hg
 (c) the reaction of Hg with limited amount of Cl_2
 (d) the reaction of Hg with concentrated HCl
128. When aqueous sodium hydroxide is added to an aqueous solution of chromium (III) ions, a green blue precipitate is first formed which re-dissolves to give a green solution. This green colour is due to:
 (a) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ (b) CrO_4^{2-}
 (c) $[\text{Cr}(\text{OH})_4]^-$ (d) $[\text{Cr}(\text{OH})_3(\text{H}_2\text{O})_3]$
129. Give the correct order of initials T or F for following statements. Use T if statement is true and F if it is false.
- (I) Sulphide ions reacts with $\text{Na}_2[\text{Fe}(\text{CN})_5(\text{NO})]$ to form a purple coloured compound $\text{Na}_4[\text{Fe}(\text{CN})_5(\text{NOS})]$. In the reaction, the oxidation state of iron changes.
 (II) $\text{Pt}(\text{IV})$ compounds are relatively more stable than $\text{Ni}(\text{IV})$ compounds.
 (III) The welding of magnesium can be done in the atmosphere of Helium.
 (IV) LiAlH_4 on hydrolysis will give H_2 .
 (a) FFTT (b) FTTT
 (c) TFTF (d) TFFT
130. $(\text{x}) + \text{K}_2\text{CO}_3 + \text{Air} \xrightarrow{\text{heat}} (\text{Y})$
 $(\text{Y}) + \text{Cl}_2 \longrightarrow (\text{Z})$ Pink
 Which of the following is correct?
 (a) X = black, MnO_2 , Y = Blue, K_2CrO_4 , Z = KMnO_4
 (b) X = green, Cr_2O_3 , Y = Yellow, K_2CrO_4 , Z = $\text{K}_2\text{Cr}_2\text{O}_7$
 (c) X = black, MnO_2 , Y = green, K_2MnO_4 , Z = KMnO_4
 (d) X = black; Bi_2O_3 , Y = colourless KBiO_2 , Z = KBiO_3
131. A white precipitate of AgCl dissolves in excess of:
 (I) $\text{NH}_3(\text{aq})$ (II) $\text{Na}_2\text{S}_2\text{O}_3$
 (III) NaCN
 (a) III only (b) I, II, III
 (c) I, II (d) I only
132. Which of the following statements regarding copper salts is not true?
 (a) Copper (I) disproportionates to copper and copper (II) ion in aqueous solution.
 (b) Copper (I) can be stabilized by the formation of insoluble complex compounds such as CuCl_2 and $\text{Cu}(\text{CN})_2$.
 (c) Copper (II) oxide is red powder.
 (d) The water of crystallization of copper sulphate is five.
133. Zinc (II) ion on reaction with NaOH first give a white precipitate which dissolves in excess of NaOH due to the formation of:
 (a) ZnO (b) $\text{Zn}(\text{OH})_2$
 (c) $[\text{Zn}(\text{OH})_4]^{2-}$ (d) $[\text{Zn}(\text{H}_2\text{O})_4]^{2+}$
134. Which of the following double salt does not exist?
 (a) $(\text{NH}_4)_2\text{SO}_4 \cdot \text{CuSO}_4 \cdot 6\text{H}_2\text{O}$
 (b) $(\text{NH}_4)_2\text{SO}_4 \cdot \text{FeSO}_4 \cdot 6\text{H}_2\text{O}$
 (c) $(\text{NH}_4)_2\text{SO}_4 \cdot \text{ZnSO}_4 \cdot 6\text{H}_2\text{O}$
 (d) $(\text{NH}_4)_2\text{SO}_4 \cdot \text{NiSO}_4 \cdot 6\text{H}_2\text{O}$

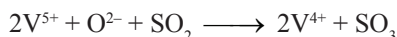
COMPREHENSION BASED QUESTIONS

COMPREHENSION-I

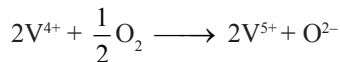
Transition metal and their compounds are used as catalysts in industry and in biological system. For example, in the Contact Process, vanadium compounds in the +5 state (V_2O_5 or VO_3^-) are used to oxidise SO_2 to SO_3 :



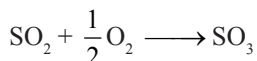
It is thought that the actual oxidation process takes place in two stages. In the first step, V^{5+} in the presence of oxide ions converts SO_2 to SO_3 . At the same time, V^{5+} is reduced to V^{4+} .



In the second step, V^{5+} is regenerated from V^{4+} by oxygen:



The overall process is, of course, the sum of these two steps:



- Transition metals and their compounds catalyse reactions because:
 - they have completely filled s-subshell.
 - they have a comparable size due to poor shielding of d-subshell.
 - they introduce an entirely new reaction mechanism with a lower activation energy.
 - they have variable oxidation states differ by two units.
- During the course of the reaction:
 - catalyst undergoes changes in oxidation state
 - catalyst increases the rate constant
 - catalyst is regenerated in its original form when the reactants form the products
 - All are correct
- Catalytic activity of transition metals depends on:
 - their ability to exist in different oxidation states.
 - the size of the metal atoms.
 - the number of empty atomic orbitals available.
 - None of these
- Which of the following ion involved in the above process will show paramagnetism?
 - V^{5+}
 - V^{4+}
 - O^{2-}
 - VO_3^-

COMPREHENSION-II

MnO_2 is the most important oxide of manganese. MnO_2 occurs naturally as the black coloured mineral pyrolusite. It is an oxidizing agent, and decomposes to Mn_3O_4 on heating to $530^\circ C$. It is used in the preparation of potassium permanganate and in the production of Cl_2 gas. Over half a million tones per year of MnO_2 is used in dry batteries.

- In the laboratory, MnO_2 is made by:
 - heating Mn in O_2
 - oxidizing Mn^{2+} in air
 - electrolytic oxidation of $MnSO_4$
 - precipitating MnO_2 from solution when performing titration of $KMnO_4$ in alkaline medium
- When MnO_2 is fused with KOH in the presence of air, the product formed is:
 - purple colour $KMnO_4$
 - green colour K_2MnO_4
 - colourless MnO_4^-
 - purple colour K_2MnO_4
- MnO_2 dissolves in concentrated HCl to form:
 - Mn^{4+} ion and Cl_2
 - Mn^{2+} ion and Cl_2
 - $[MnCl_4]^{2-}$ and Cl_2
 - only $[MnCl_4]^{2-}$
- In which of the following species, the colour is due to charge transfer.

(I) $[Mn(OH)_4]^{2-}$	(II) MnO_4^{2-}
(III) MnO_2	
(a) I, II, III correct	(b) II, IV correct
(c) I, III correct	(d) only IV correct

COMPREHENSION-III

Iron (+II) is one of the most important oxidation states, and salts are called ferrous salts. Most of the Fe (+II) salts are pale green and contain $[Fe(H_2O)_6]^{2+}$ ion. Fe(+II) compounds are easily oxidized by air and so are difficult to obtain, pure Fe^{2+} form many complexes like $K_4[Fe(CN)_6]$.

- Anhydrous $FeCl_2$ is made by:
 - heating Fe with dilute HCl
 - heating Fe with gaseous HCl
 - reacting Fe with concentrated HCl
 - heating Fe with Cl_2 gas
- $K_3[Fe(CN)_6]$ is used in the detection of Fe^{2+} ion with which it gives a deep blue colour. This colour is due to the formation of:

- (a) $\text{K}_2\text{Fe}[\text{Fe}(\text{CN})_6]$ (b) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
 (c) $\text{Fe}[\text{Fe}(\text{CN})_6]$ (d) $\text{Fe}_3[\text{Fe}(\text{CN})_6]_2$

3. FeSO_4 is used in brown ring test for nitrates and nitrites. In this test, a freshly prepared FeSO_4 solution is mixed with solution containing NO_2^- or NO_3^- and the conc. H_2SO_4 is run down the side of the test tube. If the mixture gets hot or is shaken,
 (I) the brown colour disappears
 (II) NO is evolved
 (III) a yellow solution of $\text{Fe}_2(\text{SO}_4)_3$ is formed
 (a) I, II, III correct (b) I, III correct
 (c) II, III correct (d) only I correct

COMPREHENSION-IV

Iron forms iron halide salts by reacting the metal directly with halogen. FeI_3 does not exist. FeF_3 is white solid in spite of five unpaired electrons with d^5 configuration. FeCl_3 is soluble in water and is used as a mordant in dyeing industry.

1. FeI_3 does not exist because:
 (a) of large size
 (b) Fe^{3+} oxidizes I^- to I_2
 (c) of low lattice energy
 (d) iodine is not highly electronegative enough to oxidise Fe to Fe^{3+}
2. Anhydrous FeCl_3 can be prepared by reaction of:
 (a) Fe with dry chlorine
 (b) Fe with dil. HCl in the presence of O_2
 (c) $\text{Fe}(\text{OH})_3$ with concentrated HCl
 (b) Fe_2O_3 with concentrated HCl
3. FeCl_3 solution added to $\text{K}_4[\text{Fe}(\text{CN})_6]$ gives A while with KSCN gives B. A and B respectively are:
 (a) $\text{Fe}_3[\text{Fe}(\text{CN})_6]_2$, $\text{Fe}(\text{CNS})_3$
 (b) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$, $\text{KFe}(\text{CNS})_3$
 (c) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$, $\text{K}_3[\text{Fe}(\text{CNS})_6]$
 (d) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$, $\text{K}_3[\text{Fe}(\text{SCN})_6]$

ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

1. Mercury is a liquid at 0°C because of:
 (a) very high ionization energy
 (b) weak metallic bonds
 (c) high heat of hydration
 (d) high heat of sublimation
2. An element of 3d-transition series shows two oxidation states x and y, differ by two units then:
 (a) compounds in oxidation state x are ionic if $x > y$
 (b) compounds in oxidation states x are ionic if $x > y$
 (c) compounds in oxidation state y are covalent if $y < x$
3. The metal oxide which decomposes on heating is/are:
 (a) ZnO (b) Al_2O_3
 (c) Ag_2O (d) HgO
4. Which of the following acids attack(s) on copper and silver?
 (a) dilute HNO_3 (b) dilute HCl
 (c) concentrated H_2SO_4 (d) aqua regia
5. Which of the following statements are true for Mohr's salt?
 (a) It decolorizes KMnO_4 solution
 (b) It is a double salt
 (c) It is colourless salt
 (d) It is a primary standard substance
6. Which of the following statement(s) is/are correct?
 (a) The chief ore of zinc is cinnabar
 (b) Mac-Arthur's process is used to extract silver
 (c) $\text{Na}_2\text{S}_2\text{O}_3$ is used to remove the unexposed AgBr from the photographic films
 (d) Nessler's reagent is a complex of zinc in +2 oxidation state
7. Roasting of copper pyrites is done:
 (a) to remove moisture and volatile impurities
 (b) to oxidize free sulphur
 (c) to decompose pyrites into Cu_2S and FeS
 (d) to decompose Cu_2S into blister copper
8. Identify the correct statements:
 (a) Iron belongs to first transition series of the periodic table.
 (b) The purest form of commercial iron is wrought iron.
 (c) Anhydrous ferrous sulphate is called as yellow vitriol.
 (d) Iron is the most abundant transition metal.
9. Which statements about mercury are correct?
 (a) Hg is a liquid metal
 (b) Hg forms a series of salts

12.54 Transition Elements (d-block) and Their Compounds

- (c) Hg forms no amalgam with iron and platinum
(d) Anhydrous Hg does not show variable valency
10. Which statements about corrosive sublimate (HgCl_2) are correct?
(a) It sublimes on heating
(b) It oxidizes stannous chloride
(c) It is highly poisonous
(d) It is prepared by heating mercury in chlorine
11. Which statements are correct regarding copper sulphate?
(a) It reacts with NaOH and glucose to give Cu_2O
(b) It reacts with KCl to give Cu_2O
(c) It gives CuO on strong heating in air
(d) It reacts with KI to give brown colouration
12. To an acidified dichromate solution, a pinch of Na_2O_2 is added and shaken. What is observed.
(a) Blue colour
(b) Orange colour changing to green
(c) Copious evolution of oxygen
(d) Bluish-green precipitate
13. Pick out the correct statement(s):
(a) MnO_2 dissolves in concentrated HCl, but does not form Mn^{4+} ions
(b) Decomposition of acidic KMnO_4 is not catalysed by sunlight
(c) MnO_4^{2-} is strongly oxidizing and stable only in very strong alkali. In dilute alkali, water or acidic solutions it disproportionates
(d) KMnO_4 does not act as oxidising agent in alkaline medium
14. The species that undergoes disproportionation in an alkaline medium are:
(a) Cl_2
(b) MnO_4^{2-}
(c) NO_2
(d) ClO_4^-
15. Which of the following statements regarding d-block elements are true?
(a) The colour of anhydrous CuSO_4 is blue.
(b) "splitting of silver" can be prevented by covering the surface of molten silver with charcoal.
(c) Iodine liberated in a reaction can be estimated by titration against a standard thiosulphate.
(d) Lanthanum is first element of third transition series
16. Which is/are insoluble in NH_3 solution?
(a) AgCl
(b) AgBr
(c) AgI
(d) Ag_2S

MATCH THE COLUMN TYPE QUESTIONS

Column-I and **Column-II** contains four entries each. Entries of Column-I are to be matched with one or more than one entries of Column-II. Each entry of Column-I may have the matching with one or more than one entries of Column-II.

Entries of Column-I are to be matched with entries of Column-II. Each entry of Column-I may have the matching with one or more than one entries of Column-II.

1. Column-I contains four statements following reason and Column-II consists of four options P, Q, R, S

Answer the following:

P → If both statement and reason are true and reason is correct explanation of statement.

Q → If both statement and reason are true and reason is not correct explanation of statement.

R → If statement is correct and reason is incorrect.

S → If both statement and reason are incorrect.

Column-I

- (A) **Statement:** The reaction of oxalic acid with acidified KMnO_4 is first slow and then speeds up by itself
Reason: KMnO_4 decomposes into MnO_2 in sunlight.
- (B) **Statement:** Anhydrous ZnCl_2 cannot be made by heating $\text{ZnCl}_2 \cdot 2\text{H}_2\text{O}$.
Reason: It undergoes hydrolysis to product $\text{Zn}(\text{OH})_2$ and HCl.
- (C) **Statement:** KMnO_4 is not used as a primary standard substance.
Reason: It is deliquescent in nature.
- (D) **Statement:** $\text{K}_2\text{Cr}_2\text{O}_7$ has orange colour due to polarization.
Reason: In dichromate ion all Cr—O bonds are identical.

Column-II

(P)

(Q)

(R)

(S)

2.

Column-I

- (A) Kipp's apparatus waste
 (B) Green coloured compound
 (c) leave(s) brown residue on heating
 (d) leave(s) black residue on heating

Column-II

- (P) $(\text{NH}_4)_2\text{SO}_4 \cdot \text{FeSO}_4 \cdot 6\text{H}_2\text{O}$
 (Q) $\text{Cu}(\text{OH})_2 \cdot \text{CuCO}_3$
 (R) FeSO_4
 (S) $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$

3. Match each of the reactions given in Column I with the corresponding product(s) given in Column II.

[IIT 2009]

Column I

- (A) $\text{Cu} + \text{dil HNO}_3$
 (B) $\text{Cu} + \text{conc. HNO}_3$
 (C) $\text{Zn} + \text{dil HNO}_3$
 (D) $\text{Zn} + \text{conc HNO}_3$

Column II

- (P) NO
 (Q) NO_2
 (R) N_2O
 (S) $\text{Cu}(\text{NO}_3)_2$
 (T) $\text{Zn}(\text{NO}_3)_2$

ASSERTION/REASON TYPE QUESTIONS

These questions consists of two statements each, printed as assertion and reason, while answering these questions you are required to choose any one of the following responses.

- (a) If both assertion and reason are true and the reason is a correct explanation of assertion
 (b) If both assertion and reason are true but reason is not a correct explanation of assertion
 (c) If assertion is true but the reason is false
 (d) If assertion is false but the reason is true

1. **Assertion:** Melting point of Mn is more than that of Fe.

Reason: Mn has higher number of unpaired e^- than Fe in atomic state.

2. **Assertion:** Cu^+ has less stable than Cu^{2+} but Fe^{3+} is more stable than Fe^{2+} .

Reason: Half filled and completely filled subshells are more stable.

3. **Assertion:** Zn gives H_2 gas with dil. HCl and also with dil. H_2SO_4 .

Reason: NO_3^- ion is reduced in preference to hydronium ion.

4. **Assertion:** KMnO_4 has different equivalent weights in acid, neutral or alkaline medium.

Reason: In different medium, change in oxidation number shown by manganese is altogether different.

5. **Assertion:** Cu^{2+} is more stable than Cu^+ .

Reason: Electrode potential is more important in determining stable oxidation state than electronic configuration.

6. **Assertion:** Concentrated aqueous solution of CuCl_2 is green in colour.

Reason: The solution contains two complex ions i.e., $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ and $[\text{CuCl}_4]^{2-}$ in equilibrium.

7. **Assertion:** KMnO_4 is purple in colour due to charge transfer.

Reason: There is no electron present in d-orbitals of manganese in MnO_4^- .

8. **Assertion:** CrO_3 reacts with HCl to form chromyl chloride gas.

Reason: Chromyl chloride (CrO_2Cl_2) has tetrahedral shape.

9. **Assertion:** Hg is the only metal which is liquid at 0°C

Reason: It has very high I.P. and weak metallic bond.

RECENT YEARS' QUESTIONS OF IIT-JEE AND AIEEE

1. When H_2S is passed through HgCl_2 ?

[AIEEE 2002]

- (a) MnO_2 (b) Mn_2O_3
 (c) Mn_2O_7 (d) MnO

2. Fe^{2+} and Cr^{3+} ions can be distinguished by

[AIEEE 2002]

- (a) taking excess of NH_4OH solution
 (b) increasing concentration of NH_4^+ ion

- (c) decreasing concentration of OH^- ions

- (d) Both (b) and (c)

3. Which of the following is amphoteric oxide?

[AIEEE 2003]

- (a) SO_2
 (b) B_2O_3
 (c) ZnO
 (d) Na_2O

12.56 Transition Elements (d-block) and Their Compounds

4. A red solid is insoluble in water. However, becomes soluble if some KI is added to water. Heating the red solid in a test tube results in liberation of some violet coloured fumes and droplets of metal appear on the cooler part of the test tube. The red solid is: **[AIEEE 2003]**
 - (a) HgO
 - (b) Pb_3O_4
 - (c) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$
 - (d) HgI_2
5. Which of the following metal will leave behind a metal on strong heating: **[AIEEE 2003]**
 - (a) $\text{Mn}(\text{NO}_3)_2$
 - (b) AgNO_3
 - (c) $\text{Fe}(\text{NO}_3)_3$
 - (d) $\text{Cu}(\text{NO}_3)_2$
6. For making good quality mirrors, plates of float glasses are used. These are obtained by floating molten glass over a liquid metal which does not solidify before glass. The metal used can be: **[AIEEE 2003]**
 - (a) Na
 - (b) Mg
 - (c) Hg
 - (d) Sn
7. Cerium ($Z = 58$) is an important member of the lanthanoids. Which of the following statements about cerium is incorrect? **[AIEEE 2006]**
 - (a) Cerium (IV) acts as an oxidizing agent.
 - (b) The +3 oxidation state of cerium is more stable than the +4 oxidation state.
 - (c) The +4 oxidation state of cerium is not known in solutions.
 - (d) The common oxidation states of cerium are +3 and +4.
8. Excess of KI reacts with CuSO_4 solution and then $\text{Na}_2\text{S}_2\text{O}_3$ solution is added to it. Which of the statements is incorrect for this reaction: **[AIEEE 2004]**
 - (a) Evolved I_2 is reduced
 - (b) CuI_2 is formed
 - (c) $\text{Na}_2\text{S}_2\text{O}_3$ is oxidized
 - (d) Cu_2I_2 is formed
9. Calomel on reaction with NH_4OH gives: **[AIEEE 2005]**
 - (a) HgNH_2Cl
 - (b) $\text{NH}_2\text{—Hg—Hg—Cl}$
 - (c) Hg_2O
 - (d) HgO
10. The lanthanoid contraction is responsible for the fact that: **[AIEEE 2005]**
 - (a) Zr and Y have about the same radius
 - (b) Zr and Nb have similar oxidation state
 - (c) Zr and Hf have about the same radius
 - (d) Zr and Zn have the same oxidation state
11. Which of the following factors may be regarded as the main cause for lanthanoid contraction: **[AIEEE 2005]**
 - (a) Poor shielding of one of 4f electrons by another in the sub shell
 - (b) Effective shielding of one of 4f electrons by another in the sub shell
 - (c) Poor shielding of 5d electrons by 4f electrons
 - (d) Poor shielding of 4f electrons by 5d electrons
12. Anhydrous ferric chloride is prepared by: **[IIT 2002]**
 - (a) Heating hydrated ferric chloride at a high temperature in a stream of air
 - (b) Heating metallic iron in a steam of dry chlorine gas
 - (c) Reactions of ferric oxide with $\text{HCl}_{(\text{aq.})}$
 - (d) Reaction of metallic iron with $\text{HCl}_{(\text{aq.})}$
13. Which ore contains both iron and copper: **[IIT 2005]**
 - (a) Cuprite
 - (b) Chalcocite
 - (c) Chalcopyrite
 - (d) Malachite
14. Which pair of compound is expected to show similar colour in aqueous solution: **[IIT 2005]**
 - (a) FeCl_2 , CuCl_2
 - (b) VOCl_2 , CuCl_2
 - (c) VOCl_2 , FeCl_2
 - (d) FeCl_2 , MnCl_2
15. A solution when diluted with H_2O and boiled gives a white ppt. On addition of excess of $\text{NH}_4\text{Cl}/\text{NH}_4\text{OH}$, the volume of precipitate decreases leaving behind a white gelatinous ppt. The precipitate which dissolved in $\text{NH}_4\text{OH}/\text{NH}_4\text{Cl}$ is: **[IIT 2006]**
 - (a) $\text{Zn}(\text{OH})_2$
 - (b) $\text{Al}(\text{OH})_3$
 - (c) $\text{Mg}(\text{OH})_2$
 - (d) $\text{Ca}(\text{OH})_2$
16. Identify the incorrect statement among the following: **[AIEEE 2007]**
 - (a) 4f and 5 f-orbitals are equally shielded
 - (b) d-block elements show irregular trend and erratic chemical properties among them selves
 - (c) La and Lu have partially filled d-orbitals and no other partially filled orbitals
 - (d) The chemistry of various lanthanoids is very similar

ANSWER KEYS**ONE CORRECT OPTION TYPE QUESTIONS**

- | | | | | | | | | | |
|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| 1. (c) | 2. (b) | 3. (d) | 4. (a) | 5. (c) | 6. (c) | 7. (b) | 8. (a) | 9. (c) | 10. (b) |
| 11. (a) | 12. (b) | 13. (a) | 14. (d) | 15. (d) | 16. (b) | 17. (d) | 18. (a) | 19. (b) | 20. (a) |
| 21. (c) | 22. (a) | 23. (a) | 24. (a) | 25. (b) | 26. (c) | 27. (d) | 28. (b) | 29. (c) | 30. (a) |
| 31. (a) | 32. (d) | 33. (c) | 34. (b) | 35. (b) | 36. (c) | 37. (d) | 38. (a) | 39. (c) | 40. (a) |
| 41. (c) | 42. (b) | 43. (a) | 44. (c) | 45. (a) | 46. (b) | 47. (b) | 48. (c) | 49. (d) | 50. (b) |
| 51. (b) | 52. (b) | 53. (c) | 54. (b) | 55. (b) | 56. (b) | 57. (d) | 58. (b) | 59. (d) | 60. (d) |
| 61. (c) | 62. (a) | 63. (a) | 64. (a) | 65. (b) | 66. (b) | 67. (c) | 68. (d) | 69. (c) | 70. (b) |
| 71. (b) | 72. (a) | 73. (b) | 74. (b) | 75. (a) | 76. (b) | 77. (d) | 78. (b) | 79. (b) | 80. (d) |
| 81. (a) | 82. (a) | 83. (b) | 84. (c) | 85. (a) | 86. (b) | 87. (a) | 88. (b) | 89. (a) | 90. (b) |
| 91. (a) | 92. (a) | 93. (c) | 94. (a) | 95. (d) | 96. (b) | 97. (b) | 98. (c) | 99. (d) | 100. (b) |
| 101. (a) | 102. (a) | 103. (a) | 104. (d) | 105. (b) | 106. (d) | 107. (c) | 108. (d) | 109. (a) | 110. (c) |
| 111. (b) | 112. (b) | 113. (b) | 114. (b) | 115. (c) | 116. (c) | 117. (b) | 118. (b) | 119. (d) | 120. (a) |
| 121. (a) | 122. (a) | 123. (c) | 124. (b) | 125. (a) | 126. (b) | 127. (b) | 128. (c) | 129. (b) | 130. (c) |
| 131. (b) | 132. (c) | 133. (c) | 134. (c) | | | | | | |

COMPREHENSION BASED QUESTIONS**Comprehension-I**

1. (c) 2. (d) 3. (a) 4. (b)

Comprehension-II

1. (d) 2. (b) 3. (b) 4. (d)

Comprehension-III

1. (b) 2. (d) 3. (a)

Comprehension-IV

1. (b) 2. (a) 3. (d)

ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

- | | | | | |
|---------------|--------------|--------------|-----------------|------------------|
| 1. (a, b) | 2. (b, c) | 3. (c, d) | 4. (a, c, d) | 5. (a, b, d) |
| 6. (b, c) | 7. (a, b, c) | 8. (a, b, d) | 9. (a, b, c, d) | 10. (a, b, c, d) |
| 11. (a, c, d) | 12. (a, c) | 13. (a, c) | 14. (a, b, c) | 15. (b, c, d) |
| 16. (c, d) | | | | |

MATCH THE COLUMN TYPE QUESTIONS

1. A-Q; B-P; C-R; D-S
 2. A-R; B-P, Q, R, S; C-P, R; D-Q
 3. A - P, R, S; B - Q, S; C - P, R, T; D - Q, T

ASSERTION/REASON TYPE QUESTIONS

1. d 2. (b) 3. (b) 4. (a) 5. (a) 6. (a) 7. (b) 8. (b) 9. (a)

RECENT YEARS' QUESTIONS OF IIT-JEE AND AIEEE

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|--------|--------|--------|---------|
| 1. (a) | 2. (d) | 3. (c) | 4. (d) | 5. (b) | 6. (c) | 7. (c) | 8. (b) | 9. (a) | 10. (c) |
| 11. (a) | 12. (b) | 13. (c) | 14. (a) | 15. (a) | 16. (a) | | | | |

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CHAPTER 13

Principles of Metallurgy

INTRODUCTION

Metallurgical Processes

Extractive metallurgy is a science technique employed for the extraction of metals economically on a large scale from their respective ores.

Li	Be	B													
Na	Mg	Al	Si												
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po
Fr	Ra	Ac	Th	Pa	U										
1	2	3				4									

Type 1 metals

These are very electropositive metals found as soluble salts, e.g., chlorides, carbonates, sulphates, etc., in the Stassfurt deposits in North Germany, and as constituents of alumino-silicate rocks. The metals are extracted by electrolytic techniques.

Type 2 metals

These are the electropositive alkaline-earth metals, found principally as insoluble carbonates and sulphates (magnesium sulphate is soluble). They are extracted by electrolysis.

- (a) **Gravity separation:** The ore is powdered. It is then washed in running stream of water. The lighter earthy materials and gangue are washed away while the heavy ore particles remain behind. The oxide ore of iron Fe_2O_3 and Fe_3O_4 are concentrated by this method. This method is also called **lavigation**.
- (b) **Froth floatation process:** *Mostly employed for sulphide ores.* Water is mixed with 3.5 % by weight eucalyptus oil (or some other cheap oil) and the mixture is stirred by compressed air as shown in Fig.13.1. Froth is generated at surface. Sulphide particles in ores are preferentially wetted by this froth and rise to surface. They are skimmed off by a skimmer. Gangue is preferentially wetted by water and it sinks to bottom. Reagents employed in froth floatation are of four types,
- Frothers** Which create froth e.g., palm oil.
 - Collectors** Which help in attachment of ore particle to an air bubble in forth. e.g., Sodium xanthates (containing -C(=S)-S- group similar to -C(=O)-O- group).
 - Activators** Simple inorganic compounds which increase the effect of collectors e.g., Na_2S .
 - Depressants** Which suppress the flotation of a particular selectively e.g., when an ore containing PbS and ZnS is floated in presence of CN^- , flotation of ZnS is suppressed and only PbS is removed. CN^- is then destroyed by any oxidizing agent and ZnS is floated again.

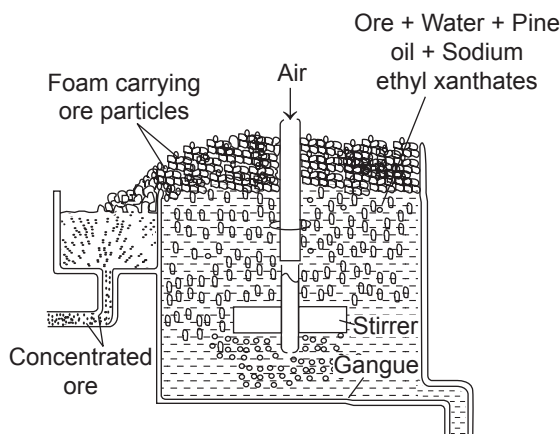


Fig.13.1 Froth Floatation Process

Worked-out Examples

2. Which are of the following ores is best concentrated by froth floatation process?

[AIEEE 2004]

- | | |
|---------------|-----------------|
| (a) Malachite | (b) Cassiterite |
| (c) Galena | (d) Magnetite |

Ans. (c) Froth floatation process is generally used to concentrate sulphide ore (Galena is PbS).

3. Name three ores which are concentrated by froth-floatation process. What is meant by a depressant?

Ans. Zinc blende (ZnS), Galena (PbS) and Argentite (Ag_2S).

Depressant: It prevents certain type of particles to form froths with bubbles.

4. During concentration of sulphide ores by froth-floatation process potassium ethyl xanthate is added, why?

Ans. These compounds behaves as collector and attach themselves by polar groups to ore particles making that hydrophobic and i.e., why ore particles are passed into froth.

Magnetic Separation

Principle: This process is based upon magnetic properties of the ore and gangue. If the ore or the impurity present is magnetic, then this process is used to separate the magnetic substance from the non-magnetic substance.

Method: Ore is concentrated by a magnetic separator. It consists of a leather belt moving over two rollers. One of the roller is magnetic. The powdered ore is put on the belt. The magnetic portion of the ore falls nearer to the magnetic roller. The non-magnetic portion falls further off the magnetic roller.

This method is used to concentrate tin stone (SnO_2). It contains magnetic oxide ion (Fe_3O_4) as impurity. The magnetic impurity of Fe_3O_4 is collected near the magnetic roller while the ore particles of SnO_2 fall further off the magnetic roller.

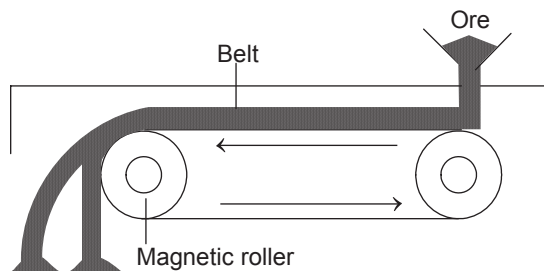
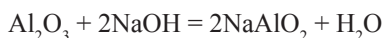


Fig. 13.2

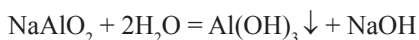
Chemical method:

In this process, the desired metallic component in the ore is dissolved selectively by digesting the powdered ore with a suitable reagent. The impurities are left behind as undissolved residue. From the aqueous solution thus obtained, the derived metallic component is regenerated.

In **Bayer's process**, bauxite ($\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$), an ore of aluminium, is digested with 40% NaOH solution which dissolves all the Al_2O_3 component in the ore to form soluble sodium aluminate and impurities like TiO_2 and Fe_2O_3 are left as undissolved residue.



From the aqueous solution of sodium aluminate, $\text{Al}(\text{OH})_3$ is precipitated by adding a little freshly prepared $\text{Al}(\text{OH})_3$.



The precipitate of $\text{Al}(\text{OH})_3$ is collected and is ignited to obtain pure alumina.



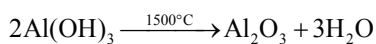
Worked-out Example

5. Name the chief ores of tin, iron and aluminium. What methods are employed for the concentration/purification of their ores?

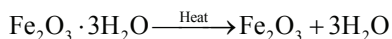
Ans. Sn—Cassiterite—Gravity separation. Fe—Haematite—Magnetic separation.
Al—Bauxite—Leaching.

2. **Calcination:** The concentrated ore is heated in absence of air below its melting point. This process is called calcinations. On calcinations, metal carbonates and metal hydroxides are decomposed to give metal oxide. Moisture, water of hydration and other volatile impurities present in the ore are removed. On calcinations, the ore becomes porous. Thus, ore is heated uniformly. Following are the examples of calcination:

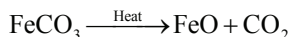
- (a) Aluminium hydroxide on calcination gives alumina.



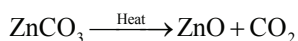
- (b) Limonite is the ore of iron. This on calcination gives anhydrous ferric oxide.



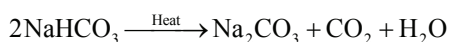
- (c) Siderite is the ore of iron. This on calcination gives ferrous oxide.



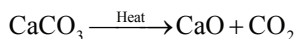
- (d) Calamine is the ore of zinc. This on calcination gives zinc oxide.



- (e) Sodium bicarbonate on calcination gives sodium carbonate.

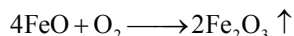
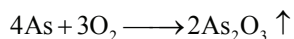
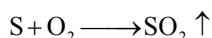


- (f) Calcination of lime stone gives CaO and CO₂.



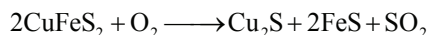
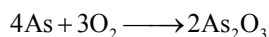
3. Roasting: The concentrated ore is heated alone or mixed with some other substance below its melting point in the presence of air. The ore and impurities get oxidized. Impurities of sulphur and arsenic are removed as volatile oxides. Following are the examples of roasting:

- (a) In the extraction of iron, the concentrated and calcined ore is roasted. The impurities of S and As are removed as volatile SO₂ and As₂O₃. Any ferrous oxide present in the ore is oxidized into ferric oxide.

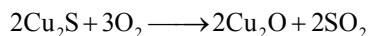
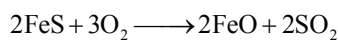


- (b) In the extraction of zinc, its ore zinc blende (ZnS) is roasted to get ZnO.

- (c) On roasting the copper ore chalcopryrite (CuFeS₂), the impurities of S and As present in it are removed as volatile SO₂ and As₂O₃. CuFeS₂ is converted into cuprous sulphide and ferrous sulphide. $\text{S} + \text{O}_2 \longrightarrow \text{SO}_2$



Some part of cuprous sulphide and ferrous sulphide is converted into Cu₂O and FeO.



4. Smelting:

Gangue, Flux and slag.

The ores contain some refractory impurities (gangue) such as silica, FeO, MgO etc. These impurities found in ores are known as gangue. It is difficult to remove these from the metals as they do not melt even at high temperature. Consequently, their presence creates a problem in the isolation and purification of metals. These impurities react chemically with some added substances at high temperature yielding fusible compounds. These compounds melt easily and form a separate layer without mixing with the molten metals and thus can be removed easily.

- The flux is a substance which reacts with the gangue during smelting to form an easily fusible compound which can be removed from the molten metal easily.

13.6 Principles of Metallurgy

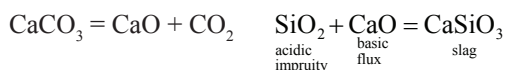
- The slag is the fusible compound which is obtained by the reaction of the flux with the gangue present in the ore.

Thus, **Flux + Gangue = slag**.

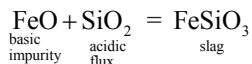
The gangue may be of two types — (i) acidic in nature such as silica, (ii) basic in nature such as CaO, MgO, FeO etc.

The selection of flux depends on the chemical nature of the impurities present in the ore.

- If the impurities present in the ore are acidic in nature e.g., SiO_2 then a suitable basic flux like lime is used to remove the impurities. The basic flux reacts with the acidic impurities to form slag. For example, in the extraction of iron, to remove the acidic impurities such as silica, basic flux such as limestone is added. Limestone at first decomposes to form calcium oxide which reacts with silica to form calcium silicate which is slag.



- When the impurities present in the ore are basic in nature e.g., CaO, FeO etc., then an acidic flux like silica is used to remove the basic impurities. For example, basic FeO is found to be present in the copper ore as impurity. An acidic flux namely silica is added to remove this basic impurity during smelting. A slag ferrous silicate is formed.

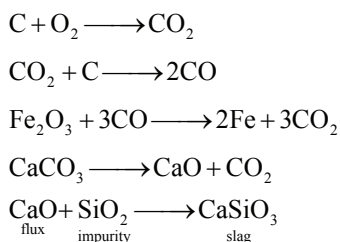


In this process, the roasted ore is mixed with coke and a suitable flux and is then heated in the presence of air at high temperature. Following reactions take place during smelting:

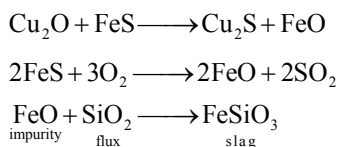
- Roasted ore is reduced into metal or a compound containing that metal.
- Impurity present in the roasted ore is removed as slag.
- Melting point of slag is lower than that of the metal and it is also lighter than the metal. Slag is insoluble in the molten metal and floats over the molten metal. Slag is removed from the upper exit (tap hole).

Following are the examples of smelting:

- In the extraction of iron, the roasted ore is mixed with coke and lime stone (flux) and is then heated in the presence of air at high temperature in a furnace. Impurity of SiO_2 is removed as calcium silicate (CaSiO_3 , slag). Lime stone acts as flux. Iron oxide is reduced into iron.



- In the extraction of copper, the roasted ore is mixed with silica (SiO_2 , flux) and is then heated in the presence of air at high temperature in a furnace. FeS is present as an impurity. It is converted into FeO and then removed as ferrous silicate (FeSiO_3 , slag). SiO_2 acts as flux.



Thus in smelting a mixture of Cu_2S and FeS is obtained. This molten mixture of FeS and Cu_2S is called matte. Copper is then extracted from matte.

- Heavy metals like Cu, Zn, Fe, Pb, Sn etc., are obtained by roasting and smelting process.

- 5. Purification of metals:** Several methods are used to purify metals. The method used depends upon the nature of the metal and the impurity present in it.

Thermal refining:

- (a) **Oxidation by air:** In this process, the crude metal is melted and air is blown through the metal. As the more basic impurities are easily oxidisable, they are readily oxidized by the O_2 of air. The volatile oxides thus formed escape and the non volatile oxides float on the molten metal as scum, which are removed. A small portion of the metal may be oxidized in this process.
- (b) **Poling process:** This process is used for the purification of copper and tin.

Examples.

(a) *Purification of impure copper.* Impure copper contains a number of impurities like S, Sb, As, Fe, CuO etc. In this process, impure copper is remelted in a reverberatory furnace lined with SiO_2 and a blast of O_2 is blown into the furnace. O_2 oxidised S, Sb and As to their respective oxides which, being, volatile, get volatilised and are thus removed. Fe is oxidized to FeO which forms a slag of $FeSiO_3$ with SiO_2 lining of the furnace. Molten copper left behind contains CuO as impurity. This molten copper is treated with powdered anthracite and then stirred with a pole of green wood. Green wood, at high temperature, liberates hydrocarbon gases, which are converted into methane (CH_4). Methane thus obtained reduces CuO to free Cu-metal, which is about 99.5% pure and is called tough pitch copper.



(b) *Purification of impure tin.* Impure tin metal contains the impurities of Cu, Fe, W and SnO_2 . The impurity of SnO_2 is due to the incomplete reduction of tin stone ore (SnO_2) during smelting. In order to remove these impurities, the impure molten tin metal is taken in a big pot and stirred with a pole of green wood. Green wood, at high temperature liberates hydrocarbon gases, which are converted into methane CH_4 . Methane, thus produced, reduces SnO_2 to pure Sn metal while the impurities of Cu, Fe, W etc., come up to the surface, where they come in contact with air and are oxidized to their respective oxides. The oxides form a scum on the surface of pure tin metal. This scum is removed from the surface. Tin metal obtained by this method is 99% pure.



- (c) **Fractional distillation:** In this process, the crude metal is subjected to fractional distillation, when the impurities having lower boiling points than that of the metal, distill off first leaving behind the pure metal as residue. If the boiling points of the impurities are found to be greater than that of the metal, the metal distills off first which is collected in a condenser. The impurities are left behind as residue.

As for example, zinc contains impurities like cadmium, iron and lead. When the crude zinc is heated, cadmium having the least boiling point ($767^\circ C$) distills off first and thus removed. Now the temperature is raised to the boiling point of zinc ($920^\circ C$) when zinc distills off and is collected. The boiling points of iron and lead being higher, are left behind as residue.

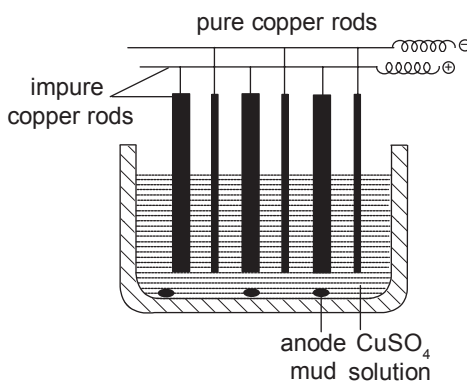


Fig. 13.3 Purification of copper by electrolytic process

- (d) **Liquation method:** Zn, Pb, Sn, Bi etc., metals are purified by this method. This method is used to purify those metals which have low melting point and the impurity present has high melting point. Impure metal is heated on an inclined hearth. Metal melts and flows down. Impurities remain behind.
- (e) **Zone refining:** This process is based on the fact that the presence of impurities lowers the melting point of a metal. Consequently, the impure molten metal on resolidification, the pure metal first crystallizes out leaving behind the impurities in the molten metal.

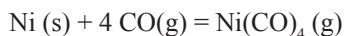
A circular heater is fitted around a rod of impure metal and the heater is slowly moved down the rod. The metal rod melts at the heated zone. As the heater moves down the rod, pure metal recrystallises out of the melt and the impurities in the metal pass into the adjacent molten zone. Thus by moving the heater slowly along the metal rod, the impurities are swept to the other end of the rod which is subsequently discarded. Metals like germanium, and gallium of high purity which are used as semiconductors, are purified by this process.

Thermal decomposition method:

In this method, the impure metal is converted into a suitable volatile compound which, on being heated, gets decomposed, to give pure metal. Some examples of thermal decomposition method are given below:

(i) **Mond's process (Also called carbonyl process):**

In this process the impure metal is converted into its volatile compound when the impurities are left behind. The volatile compounds thus formed are decomposed to pure metal. Thus impure nickel is reacted with carbon monoxide at 60°C when volatile nickel carbonyl is formed leaving behind the solid impurities.



Nickel carbonyl is decomposed at 150°C – 180°C to get back pure nickel. CO gas is recirculated.



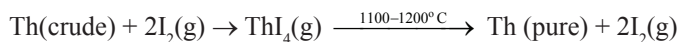
(ii) **Van-Arkel-de-Boer's process:**

Also called iodine refining process (Ti, Zr, etc.). In this process, the impure metal is heated with iodine, in a sealed vessel, so that the metal forms volatile metallic iodide and impurities do not react with iodine. The metallic iodide, so formed, is decomposed to get the pure metal and iodine. The decomposition of the metallic iodide to pure metal is done by passing it over a thin tungsten or tantalum wire (filament) placed in the vessel and electrically heated to a high temperature, when pure metal gets deposited on the surface of the hot wire and iodine vapour gets condensed. The pure metal can be removed from the surface of the tungsten wire. This method is very expensive and hence is used only when very pure metal is required for special uses. The only which the metallic iodide decomposes, i.e., the decomposition temperature of the metallic iodide should be less than the melting point of the metal (e.g., decomposition temperature of $\text{ZrI}_4 = 1800^\circ\text{C}$, while the melting point of Zr-metal = 1857°C).

Van-Arkel process has been used to purify Ti, Zr, V and The metals. Equation showing the purification of Ti-metal by Van-Arkel process can be written as:



Similar equation for the purification of Th, Zr and V can also be written,



Cupellation process:

Zn-Ag alloy is used to separate by this method. In which, the alloy is heated strongly in presence of little carbon in a fire-clay retort. Zn, being more volatile, distills off while Ag remains in the retort, carbon used in the process reduces the oxide of Zn, if formed. Ag obtained from Zn-Ag alloy is contaminated with a little of Pb as impurity. This impurity of Pb is removed by cupellation process. In this process, Ag containing the traces of Pb as impurity, is placed in a cupel (cupel is a boat-shaped dish made of bone ash which is porous in nature) in a reverberatory furnace and heated in the presence of air. By doing so, lead (impurity) is oxidized to PbO (litharge) which volatilizes and pure Ag is left behind in the cupel. Last traces of PbO are absorbed by the porous mass of the cupel.

- (a) **Electrolytic Process:** Copper, silver, gold, lead etc., are purified by this method. It is an important method for the purification of metals. Pure metal is made cathode and impure metal anode. Aqueous solution of a salt of the metal acts as electrolyte.

Anode	:	Impure metal
Cathode	:	Pure metal
Electrolyte	:	Aqueous salt solution of the metal

On passing electric current, pure metal deposits on cathode. Equivalent mass of anode dissolves in the form of a salt and comes in the solution. The impurities either remain dissolved in solution or settle down as anode mud near anode.

In the purification of copper, pure copper made cathode, impure copper is anode, and aqueous solution of copper sulphate acts as electrolyte. On passing electric current, pure copper deposits on cathode. Equivalent mass of anode dissolves in the form of CuSO_4 and comes in solution. Impurities of less reactive metals such as Ag, Au, Pt etc., are removed as anode mud.

Worked-out Examples

6. When the sample Cu with Zn impurity is to be purified by electrolysis, the appropriate electrodes are:

[AIEEE 2002]

Cathode	Anode
(a) Pure Zn	Pure Cu
(b) Impure sample	Pure Cu
(c) Impure Zn	Impure sample
(d) Pure copper	Impure sample

Ans. (d) Anode $\text{Cu} \longrightarrow \text{Cu}^{2+} + 2\text{e}^-$
(Impure sample)

Cathode $\text{Cu}^{2+} + 2\text{e}^- \longrightarrow \text{Cu}$
(Pure copper)

7. During the process of electrorefining of copper some metals present as impurity settle as anode mud. These are:

[AIEEE 2005]

(a) Sn and Ag	(b) Pb and Zn
(c) Ag and Au	(d) Fe and Ni

Ans. (c) More electropositive metals associated with Cu (Fe, Zn, Co, Ni) pass in solution state and less electropositive metals (Au, Ag) collect below the anode as anode mud.

8. You are provided with samples of some impure metals such as zinc, copper and germanium. Which methods would you recommend for the purification of each of these metals?

Ans. Zn—Distillation, Cu—Electrolytic refining, Ge—Zone phase refining.

9. Why Graphite is used as anode but not diamond?

Ans. Graphite is good conductor of current due to sp^2 hybridization having mobile π electrons. Diamond has sp^3 hybridization and is insulator.

10. Name one metal that is refined by each of the following process:

(a) Vacuum distillation	(b) Mond's process
(c) Electrolysis	(d) Van Arkel process
(e) Zone refining	

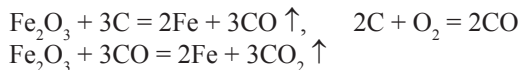
Ans. (a) Mg (b) Ni
(c) Cu (d) Zr (e) Ge

1. Carbon reduction process

The metals which occupy the central positions in the electrochemical series, such as Zn, Fe, Sn, Pb etc., are moderately reactive. Carbon reduction process is employed in the extraction of such metals. The oxide ore or the other ores of such metals which can be converted to oxide by calcinations or roasting, are reduced by coke at high temperature. Carbon monoxide which is produced by the incomplete combustion of coke, can also act as reducing agent.

At present magnesium is extracted by the carbon reduction of its oxide at high temperature (2000°C). As for examples.

- (a) The oxide ore such as haematite is reduced to metallic iron in blast furnace at 1000°C–1200°C by coke and CO.



- (b) The sulphide ore such as zinc blende (ZnS) or galena (PbS) is roasted in air to convert it into ZnO or PbO. The oxides thus produced are reduced by carbon to produce the metals.

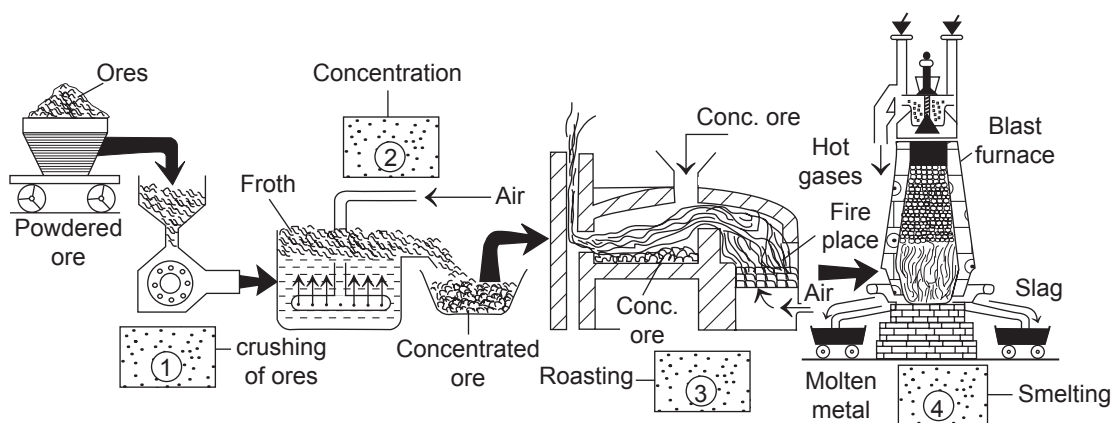
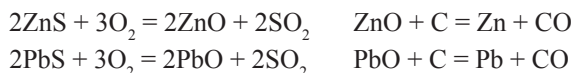
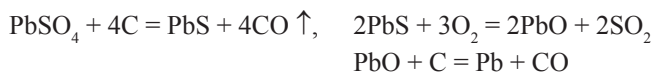


Fig. 13.4

- (c) The carbonate ores such as calamine (ZnCO_3), magnesite (MgCO_3) etc., are first calcined to produce the respective oxides which are subsequently reduced by carbon to produce the respective metals.

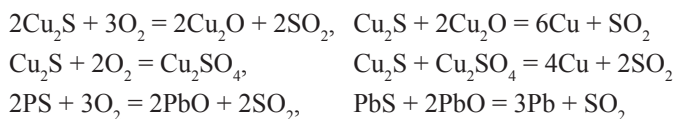


- (d) The sulphate ore is heated with coke in a furnace at high temperature when the sulphate is reduced to sulphide. The sulphide thus produced is roasted to produce oxide, which is reduced by coke at high temperature to produce metal.



2. Self reduction

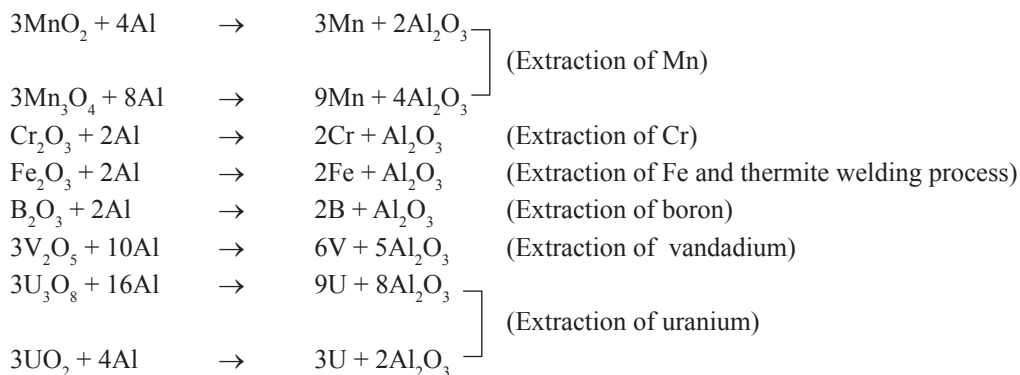
This process of extraction of metals is employed in case of the sulphide ores of lead and copper. The sulphide ore is first roasted in a controlled supply of air when a part of the sulphide is oxidized by air to form oxide and sulphate of the metal. The oxide and sulphate thus formed react with the unchanged sulphide at high temperature in absence of air to produce metal. As no external reducing agent is used in this process, the process is called self reduction process.



Reduction by aluminium—Thermite process:

If the temperature needed for carbon to reduce an oxide is too high for economic or practical purpose, the reduction may be effected by another highly electropositive metal such as aluminium, which liberates a large amount of energy (1675 kJ mol⁻¹) on oxidation to Al₂O₃. This is the basis of the Thermite process.

Metallic oxides can be reduced to free metals by Al-powder.



The reaction between Al-powder and metallic oxides is exothermic and heat evolved produces a high temperature due to which the metals obtained by this process are in the molten state.

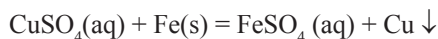
Reduction of Cr₂O₃ to Cr-metal by Al-powder. Cr₂O₃ is mixed with requisite amount of Al-powder (this mixture is called thermite mixture) and is placed in a large fire-clay crucible. An intimate mixture of Na₂O₂ or BaO₂ and Mg powder (called ignition mixture or igniter) is placed in a small depression made in the thermite mixture. The crucible is surrounded by sand which prevents the loss of heat by radiation. A piece of Mg ribbon is struck into the ignition mixture and the charge is covered by a layer of fluorspar (CaF₂) which acts as a heat insulator. Now, Mg-ribbon is ignited so that ignition mixture catches fire and flame is produced, leading to a violent reaction between Mg and BaO₂ with the evolution of large amount of heat.



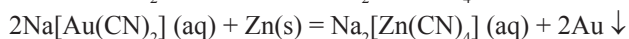
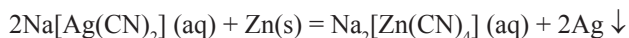
Heat produced in the above reaction makes Cr₂O₃ and Al-powder react together.

**Metal replacement method (Hydrometallurgy):**

This process involves the dissolution of concentrated ore by a suitable reagent. The desired metal is then precipitated from the solution by the addition of a more electro-positive metal. As for example, copper is precipitated from CuSO₄ solution by adding more electro-positive metal iron.



Silver and gold are precipitated by this process. Zinc dust is added to argento cyanide or auro cyanide solution when Ag or Au is precipitated.

**Worked-out Example**

11. The metal extracted by leaching with cyanide is:

[AIEEE 2002]

- (a) Mg (b) Ag
(c) Cu (d) Na

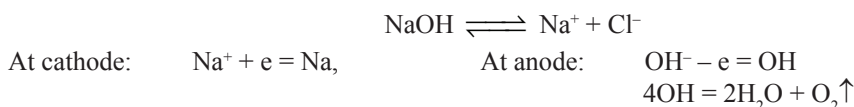
Ans. (b)

Electrolytic reduction:

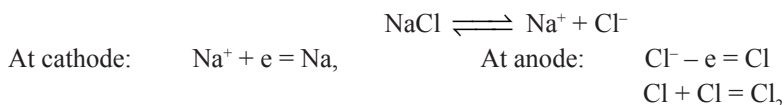
The strong electro-positive metals such as K, Ca, Na, Al, Mg are very active. The oxides of these metals (with the exception of Mg) are very stable and so it is difficult to convert them into metallic state by carbon reduction process. Moreover, even if the metal is formed, carbon reacts with the metal produced to form carbide at high temperature.

Such metals are extracted by passing electricity through their fused chlorides or oxides or hydroxides.

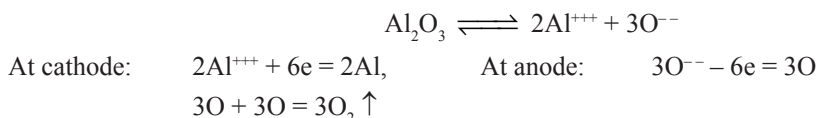
- (a) In the extraction of sodium or potassium, fused hydroxides of the metals are used as electrolytes. When electric current is passed through the fused hydroxides, the metal ions migrate to the cathode and accept necessary electrons and thus undergo reduction to produce free metals,



As the melting point of sodium chloride is high, some amount of calcium chloride is added to it to reduce its melting point. When electric current is passed through this fused NaCl, the metal ions migrate to the cathode and by accepting electrons undergo reduction to produce free metals.



- (b) In the extraction of aluminium, pure alumina is used as electrolyte. The melting point of alumina is very high and molten alumina is a poor conductor of electricity. To overcome this difficulty, alumina is dissolved in molten cryolite (AlF_3 , 3NaF) containing a little fluorspar (CaF_2). The presence of cryolite makes the molten alumina a good conductor of electricity and CaF_2 lowers the melting point of alumina. The fused mass on being electrolysed, aluminum is liberated at the cathode.



- (c) The metals which occur below aluminium in the electro-chemical series, can be extracted by the electrolysis of the aqueous solutions of their salts. This process (wet process) is employed for the extraction of copper and zinc. The sulphide ores of these metals are oxidized to produce sulphates by calcination. The sulphates thus produced are dissolved in dilute sulphuric acid. On passing electric current through the acidified solutions of the salts, metals are deposited at the cathode.

**Worked-out Examples**

12. Which of the following processes is used in the extractive metallurgy of magnesium?

(IITJEE 2002)

- (a) fused salt electrolysis (b) self reduction
 (c) aqueous solution electrolysis (d) thermite reduction

Ans. (a)

13. Giving appropriate examples (at least three), explain how the reactivity of a metal is related to its mode of occurrence in nature.

Ans. (a) Highly electropositive metals are found as oxides, halides, carbonates e.g., Na, K, Al.

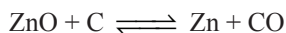
(b) Relatively less electropositive metals are found as sulphides e.g., Zn, Pb, Hg.

(c) Less reactive metals in native form e.g., Au, Pt.

- Noble metals like Au, Ag etc., are obtained by cyanide or amalgamates process,
- Active metals like Li, Na, K (alkali metals) Ca, Mg, Sr, Ba (alkali metals) etc., are obtained by the electrolysis of their chloride, oxides or hydroxides.

Reduction process and thermodynamical aspects:

Carbon is the most versatile of all reducing agents used in the metallurgy. At high temperature, it reduces many metallic oxides e.g., ZnO, Fe₂O₃ etc., to metallic state. The second law of thermodynamics explains why carbon can reduce a metallic oxide at high temperature while at low temperature the reverse is the case. Let us take the example of the reduction of zinc oxide by coke.



The above equilibrium involves (a) formation of ZnO from Zn and oxygen and (b) formation of CO from C and O₂.



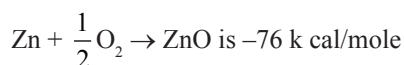
The reduction of ZnO will be spontaneous if the difference between the standard free energy change in the formation of CO and ZnO becomes negative i.e., $\Delta G_{\text{CO}}^\circ - \Delta G_{\text{ZnO}}^\circ$ is negative.

From the second law of thermodynamics, we get,

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

For the reaction, $\text{C} + \frac{1}{2} \text{O}_2 \rightarrow \text{CO}$, ΔH° is negative and ΔS° is positive. Hence ΔG° becomes more negative at high temperature.

For the reaction $\text{Zn} + \frac{1}{2} \text{O}_2 \rightarrow \text{ZnO}$ both H° and ΔS° are negatives. Hence, ΔG° becomes less negative as the temperature rises. Normally, the free energy change ΔG° decreases with the rise of temperature. At a specific temperature ΔG° for the formation of CO will be more negative than that for the formation of ZnO. At this temperature the reduction of ZnO will occur i.e., ΔG° for the reduction $\text{ZnO} + \text{C} \rightleftharpoons \text{Zn} + \text{CO}$ will be negative. Now ΔG° for the reactions,



$$\therefore \Delta G_{\text{CO}}^\circ - \Delta G_{\text{ZnO}}^\circ = -33 - (-76) = +43 \text{ k cal/mole}$$

This is the ΔG° value for the reaction $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$. Since ΔG° value is positive, the reduction of ZnO by carbon is not possible at 25°C.

Now at 1200°C, ΔG° value for the reaction $\text{Zn} + \frac{1}{2} \text{O}_2 \rightarrow \text{ZnO}$ is -42 k cal/mole and ΔG° value for the reaction $\text{C} + \frac{1}{2} \text{O}_2 \rightarrow \text{CO}$ is -57 k cal/mole. So at 1200°C,

$$\Delta G_{\text{CO}}^\circ - \Delta G_{\text{ZnO}}^\circ = -57 - (-42) = -15 \text{ k cal/mole.}$$

Since ΔG° value becomes negative, the reduction of ZnO by coke is spontaneous at 1200°C. This explains why in all the carbon reduction processes the temperature must be high. This is illustrated in the graph which is known as Ellingham

diagram indicating the change of ΔG° with temperatures. The graph shows the temperatures at which the carbon reduction becomes theoretically possible.

A close scrutiny of the graph exhibits the following features:

- (i) Since ΔG° decreases with the rise of temperature the graph will have upward slopes.
- (ii) The free-energy changes follow a straight line. However, when the substance melts or vaporizes, the slope of the line changes. This is due to large change in ΔS value associated with the change of state. Thus, Hg–HgO line changes slope at 356°C (b.p. of Hg), Ca–CaO at 1440°C and Mg–MgO changes at about 1120°C.
- (iii) When the graph crosses $\Delta G^\circ = 0$ line with increase of temperature, the free energy formation of oxide changes from negative to positive value. It means that the oxide is unstable and undergoes thermal decomposition into metal and O_2 . Although from this diagram, it follows that theoretically all metal oxides are decomposed into metal and O_2 , but in practice, only Hg, Au, Ag are generally extracted by thermal decomposition of their oxides.
- (iv) The eligibility of metal as reducing agent in the reduction of metal oxide is also obtained from this diagram. Any metal will reduce the oxides of other metals which lie above in the diagram. This is because, the free energy will become more negative by an amount equal to the difference between the two graphs at that temperature. Thus, Al reduces Cr_2O_3 , Fe_2O_3 , TiO_2 etc.

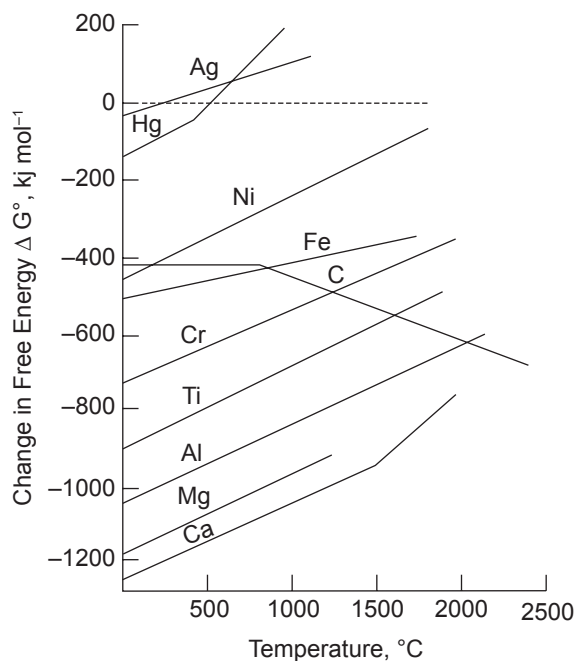


Fig. 13.5

- (v) Carbon line slopes downwards at around 710°C. Hence, carbon can be used as a reducing agent at high temperature. The graph shows the temperature when the oxide line cuts the carbon line. This is the temperature of reduction of the oxide by carbon. Thus, MgO is reduced at 1650°C, ZnO at 950°C, FeO at 725°C and so on.

Thermodynamic arguments regarding the reduction of metal oxides by carbon suffers from three limitations:

- (a) Formation of carbides at high temperature by the metal.
- (b) Incorrect assumption of the equilibrium between the reactants and product.
- (c) Maintaining silence about the rate of reaction (slow or fast) and possibility of side reaction which is a common feature in the metallurgy.

Worked-out Examples

14. Which of the following factors is of no significance for roasting sulphide ores to the oxides and not subjecting the sulphide ores to carbon reduction directly? [AIEEE 2008]

- (a) Metal sulphides are thermodynamically more stable than CS_2
- (b) CO_2 is thermodynamically more stable than CS_2
- (c) Metal sulphides are less stable than the corresponding oxides
- (d) CO_2 is more volatile than CS_2

Ans. (c) The reduction of metal sulphides by carbon reduction process is not spontaneous because ΔG for such a process is positive. The reduction of metal oxide by carbon reduction process is spontaneous as ΔG for such a process is negative. From this we find that on thermodynamic considerations CO_2 is more stable than CS_2 and the metal sulphides are more stable than corresponding oxides.

In view of above, the factor listed in choice (c) is incorrect and so is of no significance.

15. The choice of a reducing agent in a particular case depends on thermodynamic factor. How far do you agree with this statement? Support your opinion with examples.

Ans. Thermodynamic considerations are very important in deciding the temperature and the suitable reducing agent during extraction of metals, e.g., Extraction of Fe from Fe_2O_3 by reducing it at 823 K by CO; SnO_2 is reduced by C at 1473 K.

16. Which is the cheapest and most abundant reducing agent employed in the extraction of metals?

Ans. Carbon in the form of coke.

17. Indicate the temperature at which carbon can be used as a reducing agent for FeO.

Ans. Above 1123 K.

18. Why does carbon reduce copper oxide but not calcium oxide?

Ans. Carbon has more affinity for oxygen than copper and thus, it reduces copper oxide to copper. On the other hand, calcium being more reactive for oxygen than carbon and is not reduced by carbon.

19. Carbon is a cheaper reducing agent than aluminium. Why is then aluminium used for reduction in some cases?

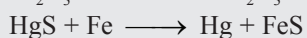
Ans. Metals which are not reduced by carbon are successfully reduced to metals by aluminium.

20. Carbon is good reducing agent for reduction of oxides but not for sulphides, why?

Solution: Ellingham Curve ($\Delta G^\circ/T$) for oxidation of C to CO has negative slope and thus carbon reduces a number of oxides. On the other hand, no such compound (CS) and thus slope for sulphur i.e., why sulphides are not easily reduced and they are roasted to oxides before reduction.

21. Name two metals which are used for reduction in metallurgical processes. Give one chemical equation for each.

Ans. (a) Al (b) Fe.



22. Aluminium metal is frequently used as a reducing agent for the extraction of metals such as chromium or manganese from their respective oxides. Why?

Ans. Aluminium metal is frequently used as a reducing agent for the extraction of metals such as Cr and Mn from their respective oxides because aluminium is more electropositive than Cr or Mn. The process of reduction is known as **Aluminothermy**.



23. Why is aluminium a good reducing agent?

Ans. Aluminium has strong affinity for oxygen and is thus readily oxidized. It thus reduces a large number of metals oxides to metals.



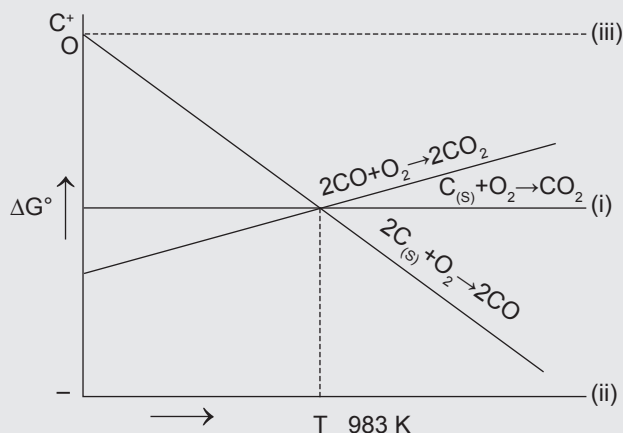
24. Is it true that under certain conditions Mg can reduce SiO_2 and Si can reduce MgO ?

Ans. Yes.



25. Carbon monoxide is more effective reducing agent than carbon below 983 K but above this temperature the reverse is true. How would you explain this?

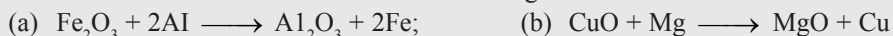
Solution: The two reactions are as:



Below 983 K reaction $2\text{CO} + \text{O}_2 \longrightarrow 2\text{CO}_2$ is more favoured due to more negative ΔG value thus CO is better reducing agent than carbon.

Above 983 K, reaction $2\text{C}_{(\text{s})} + \text{O}_2 \longrightarrow 2\text{CO}$ has more negative AG than oxidation of CO to CO_2 , so carbon will be better reducing agent.

26. Which of the two metals in each of the following is more reactive?



Ans. (a) Al (b) Mg

27. Why reduction of metal oxides by aluminium becomes faster just after ignition of mixture?

Ans. The reaction possesses exothermic nature and huge amount of heat released during the course of reaction makes it fast.



28. Hydrogen is good reducing agent, but is not widely used as reducing agent in metallurgical processes.

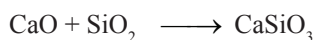
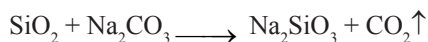
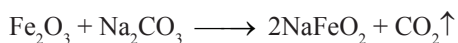
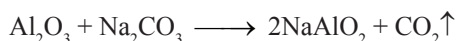
Ans. Yet hydrogen is good reducing agent but is not generally used for the purpose because at elevated temperatures hydrides are formed and mixture of H_2 and O_2 may explode at elevated temperatures. Also $\Delta G^\circ/T$ (Ellingham curve) has positive slope and runs parallel to curve for several metal oxides.

Extraction of Aluminium

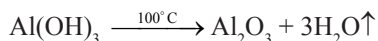
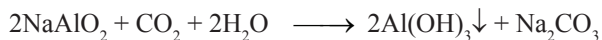
Ores of aluminium

- (a) Bauxite; $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$ (oxide)
- (b) Corundum; Al_2O_3 (oxide)
- (c) Feldspar; KAlSi_3O_6 (oxide)
- (d) Clay silicate; $\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2 \cdot 2\text{H}_2\text{O}$ (oxide)
- (e) Cryolite; $2\text{NaF} \cdot \text{AlF}_3 (\text{Na}_3\text{AlF}_6)$ (halide ore)
- (f) Alum; $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
- (g) Diaspore; $\text{Al}_2\text{O}_3 \cdot \text{H}_2\text{O}$
- (h) Mica; $\text{K}_2\text{O} \cdot 3\text{Al}_2\text{O}_3 \cdot 6\text{SiO}_2 \cdot 2\text{H}_2\text{O}$
- (i) Lapis Lazuli (Sodium aluminium silicate with sulphur)

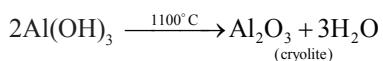
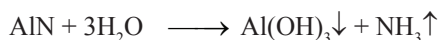
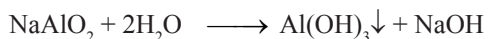
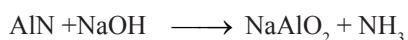
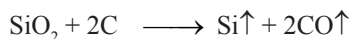
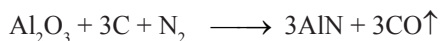
Hall's process: Crude bauxite at 1100°C reacts with Na_2CO_3 , little CaCO_3 when CaSiO_3 , NaSiO_2 , NaFeO_2 etc., from



Then at $50^\circ - 60^\circ\text{C}$ CO_2 is formed through NaAlO_2 solution and produces thereby $\text{Al}(\text{OH})_3$



Serpeck's Process



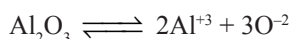
Electrolytic reduction of Al_2O_3 : At 900°C , $\text{AlF}_3 \cdot 3\text{NaF}$, CaF_2 (Fluorspar) to make Al_2O_3 a good conductor of electricity. Metallic Al is liberated at the cathode.

Electrolysis of molten mixture

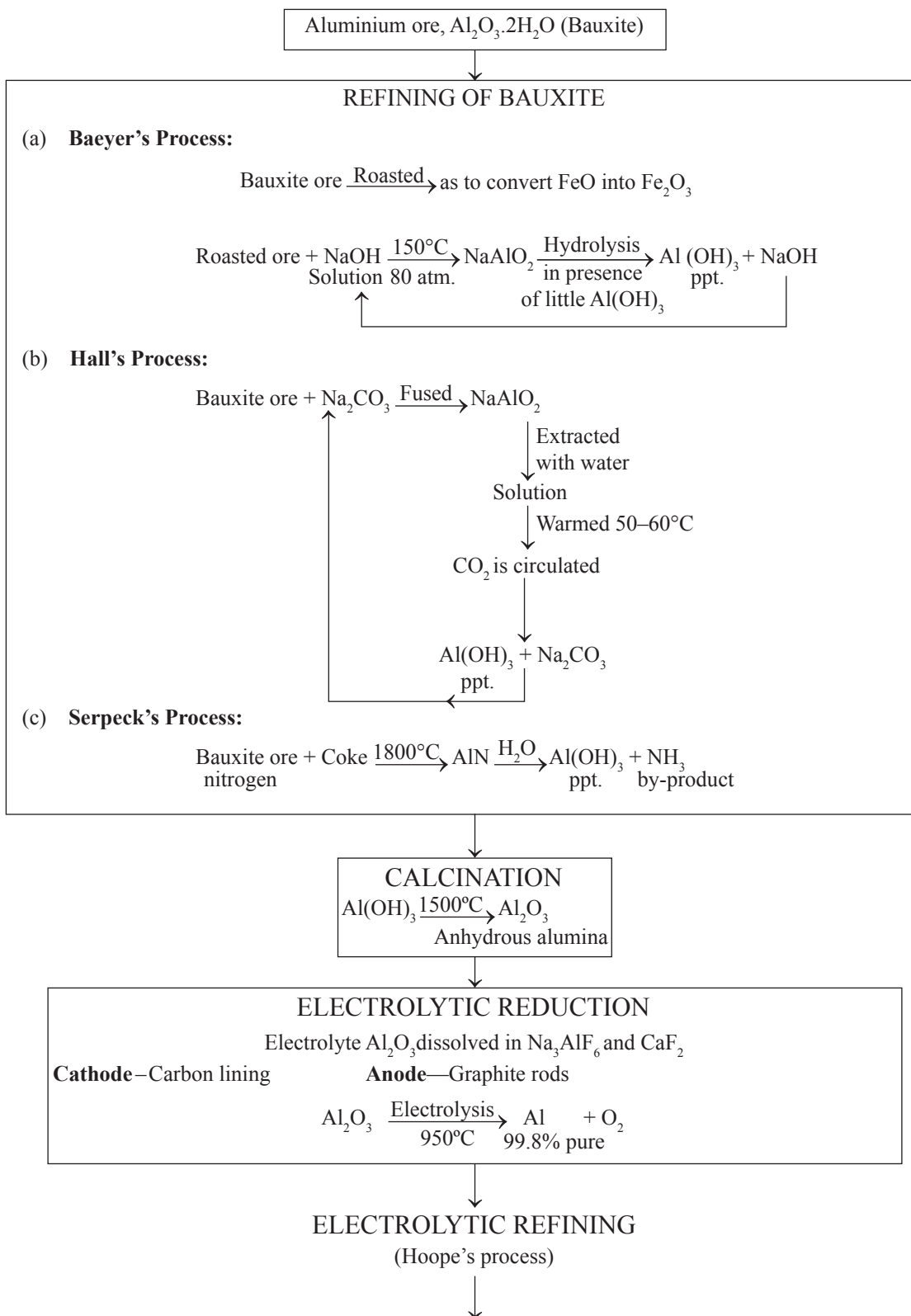
Cathode	:	Carbon
Anode	:	Graphite rods
Electrolyte	:	60 parts cryolite + 20 parts fluorspar + 20 parts pure Al_2O_3
Temperature	:	900°C

Reactions

According to the 1st theory, the following reaction occurs:

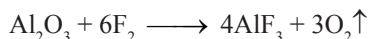
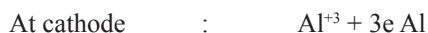
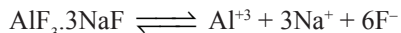


FLOW SHEET FOR THE EXTRACTION OF ALUMINIUM



As cryolyte has greater electrochemical stability, it does not dissociate. It only increases the dissociation of Al_2O_3 .

But the second theory states that, cryolite undergoes electrolytic dissociation first then Al^{+3} goes to the cathode, produced F_2 at anode then reacts with Al_2O_3 produces AlF_3 .



Worked-out Examples

29. Aluminium is extracted by the electrolysis of

[AIEEE 2002]

- (a) Bauxite
- (b) Alumina
- (c) Alumina mixed with molten cryolite
- (d) Molten cryolite

Ans. (c)

30. In the commercial electrochemical process for aluminium extraction, the electrolyte used is

[IITJEE 1999]

- (a) $\text{Al}(\text{OH})_3$ in NaOH solution
- (b) an aqueous solution of $\text{Al}_2(\text{SO}_4)_3$
- (c) a molten mixture of Al_2O_3 and Na_3AlF_6
- (d) a molten mixture of $\text{AlO}(\text{OH})$ and $\text{Al}(\text{OH})_3$

Ans. (c)

31. Electrolytic reduction of alumina to aluminium by Hall-Heroult process is carried out

(IITJEE 2000)

- (a) in the presence of NaCl
- (b) in the presence of fluorite
- (c) in the presence of cryolite which forms a melt with lower melting temperature
- (d) in the presence of cryolite which forms a melt with higher melting point

Ans. (c)

32. The major role of fluorspar (CaF_2) which is added in small quantities in the electrolytic reduction of alumina dissolved in fused cryolite (Na_3AlF_6) is

- (a) that of a catalyst
- (b) to make the fused mixture very conducting
- (c) to lower the temperature of the melt
- (d) to decrease the rate of oxidation of carbon at the anode

(IITJEE 1993)

Ans. (c)

33. Why is cryolite used in the electrolytic manufacture of Al from pure alumina?

Ans. Cryolite lowers the melting point of Al_2O_3 and also makes it more conducting.

34. Why is CaF_2 used in the extraction of aluminium?

Ans. CaF_2 is used to bring down the melting point of alumina as well as it acts as a flux.

Extraction of Mg

Ores

- (a) Carnalite; KCl , $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ (chloride)
- (b) Dolomite; MgCO_3 ; CaCO_3 (carbonate)
- (c) Magnesite; MgCO_3 (carbonate)
- (d) Epsomite; $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ (sulphate)
- (e) Kiesserite; $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ (sulphate)
- (f) Asbestos; $\text{CaMg}(\text{SiO}_3)_4$

FLOW SHEET FOR THE EXTRACTION OF Mg

Electrolytic reduction method

Electrolyte
 Anhydrous MgCl_2 (35%) + 50% NaCl and 15% CaCl_2 ;
 difficult to get added to lower m.p and to increase electrical conductivity.

At cathode : $\text{Mg}^{2+} + 2\text{e}^- \rightarrow \text{Mg}$
 At anode : $\text{Cl}^- + \text{Cl} + \text{e}^- ; \text{Cl} + \text{Cl} \rightarrow \text{Cl}_2 \uparrow$

$E^\circ_{\text{Mg}^{2+}/\text{Mg}} > E^\circ_{\text{Na}^+/\text{Na}}$

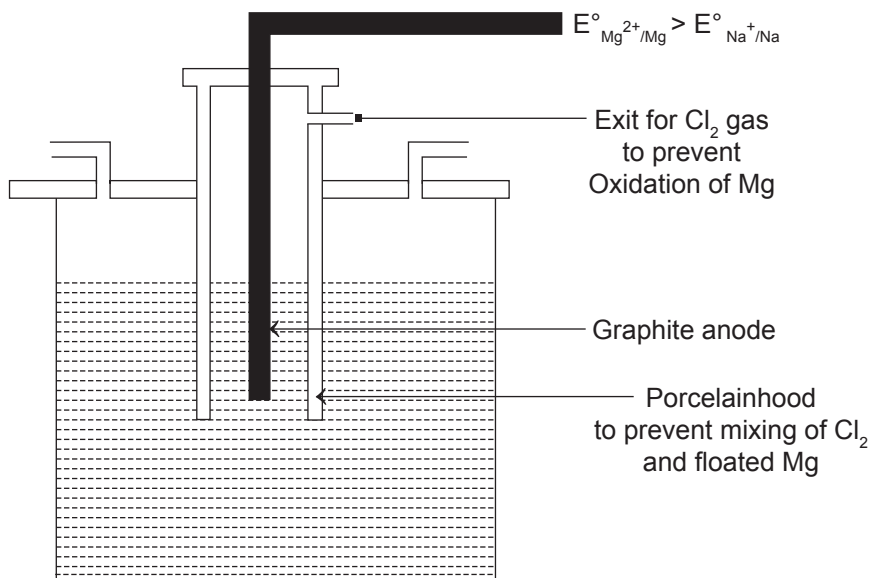


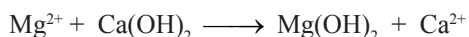
Fig. 13.6

MgO dissolved in a mixture of fluoride of Mg , Ba and Na (magnesia) can also be electrolysed at 650°C .

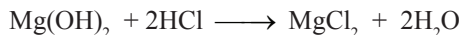
Extraction of Magnesium

Magnesium from sea-water is obtained by the *Dow's process*. It consists of electrolysis of molten magnesium chloride using an iron cathode and a graphite anode. Following are the steps that involved during the process.

- (a) Mg^{+2} is precipitated as magnesium hydroxide by the addition of slaked lime, $\text{Ca}(\text{OH})_2$ to the sea water.



- (b) Magnesium hydroxide on treatment with hydrochloric acid is converted to chloride which is crystallized as $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$



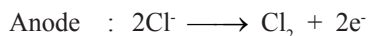
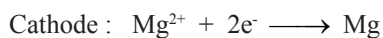
- (c) Now for electrolysis, magnesium chloride is fused as follows:

On passing a current of dry HCl gas, it get partially dehydrated and the chloride thus obtained is added to a molten mixture of sodium chloride and calcium chloride (temperature range 973 – 1023 K). Under this condition, MgCl_2 melts with the loss of water.

If magnesium chloride hydrate is heated strongly, it hydrolyses to yield magnesia (magnesium oxide) which is a refractory.



- (d) Molten mixture of MgCl_2 , NaCl and CaCl_2 is electrolysed. Magnesium is formed at the cathode and chlorine is evolved at the anode. The chlorine is used to make HCl acid which in turn is required for making magnesium chloride.



Worked-out Examples

35. Why is CaF_2 used in the extraction of aluminium?

Ans. CaF_2 is used to bring down the melting point of alumina as well as it acts as a flux.

36. In the manufacture of Mg by the carbon-reduction of MgO , why is it necessary to cool the products in a stream of air?

Solution: $\text{MgO} + \text{C} \rightleftharpoons \text{Mg} + \text{CO}$. A stream of air is passed to prevent the reverse reaction.

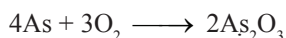
IRON

Extraction of Iron Ores

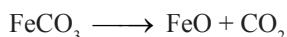
- (a) Iron pyrite; FeS_2
- (b) Siderite; FeCO_3
- (c) Red haematite; Fe_2O_3
- (d) Magnetite; Fe_3O_4
- (e) Limonite or brown haematite; $2\text{Fe}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$

Concentration of the ore

1. **Dressing of the ores:** The iron ores are first broken into small pieces 3-5 cm in size.
2. **Roasting:** During roasting S, As, P are oxidized to the respective oxides.

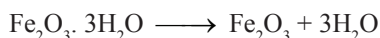


FeCO_3 decomposes as,

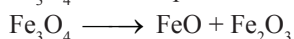


13.22 Principles of Metallurgy

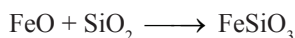
$\text{Fe}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$ loses water



Fe_3O_4 is decomposed to ferrous oxide and ferric oxides.



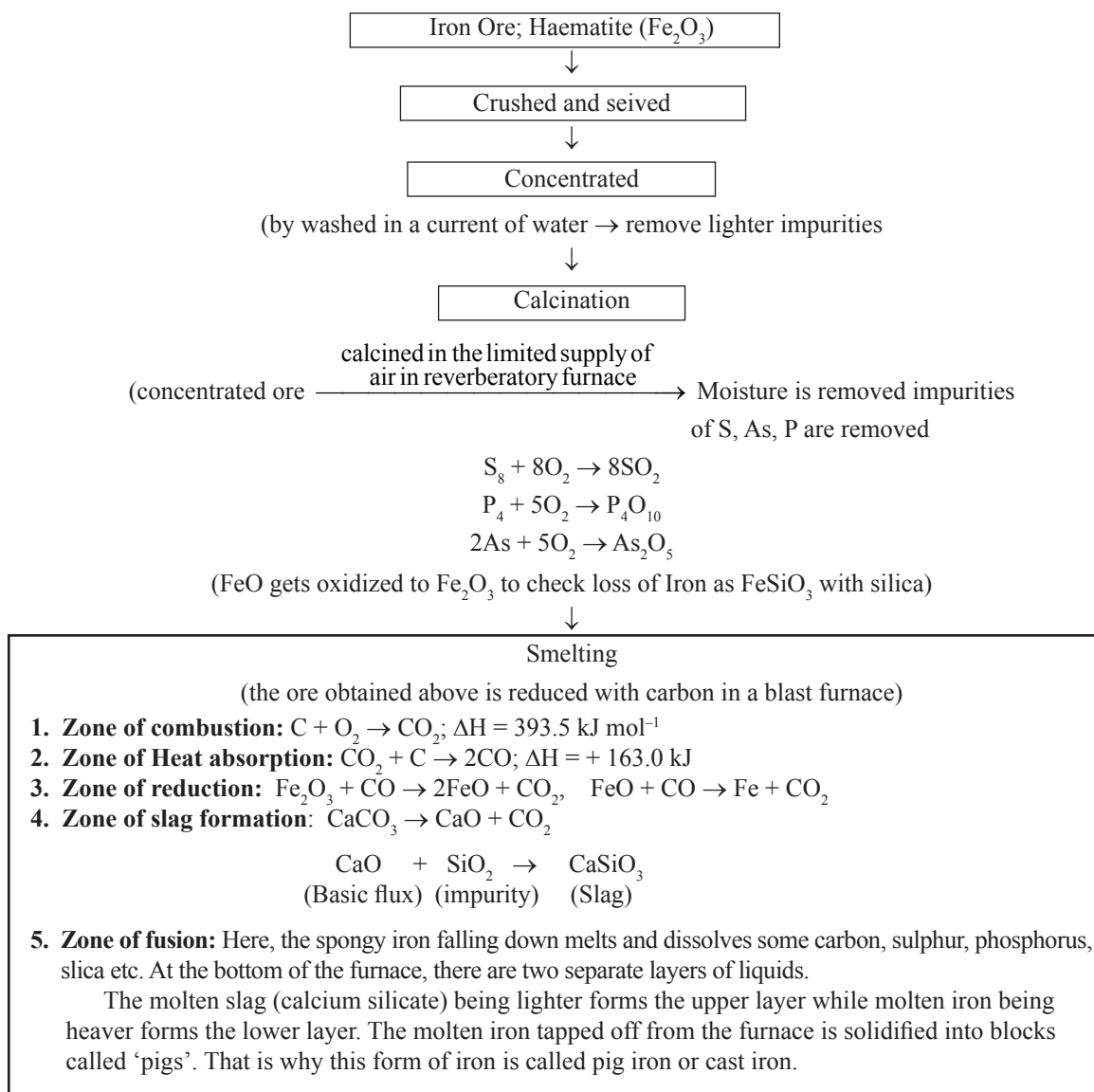
Ferrous oxide reacts with silica to forms ferrous silicate at high temperature.



But the conversion of FeO into Fe_2O_3 will prevent the formation of FeSiO_3 . Thus the mass of the ore becomes porous casting the increase in the effective surface area.

- 3. Smelting in the Blast furnace:** Blast furnace is a shaft furnace made of steel plate of 20–30 inch with 4–6 diameter.

FLOW CHART FOR THE EXTRACTION OF IRON



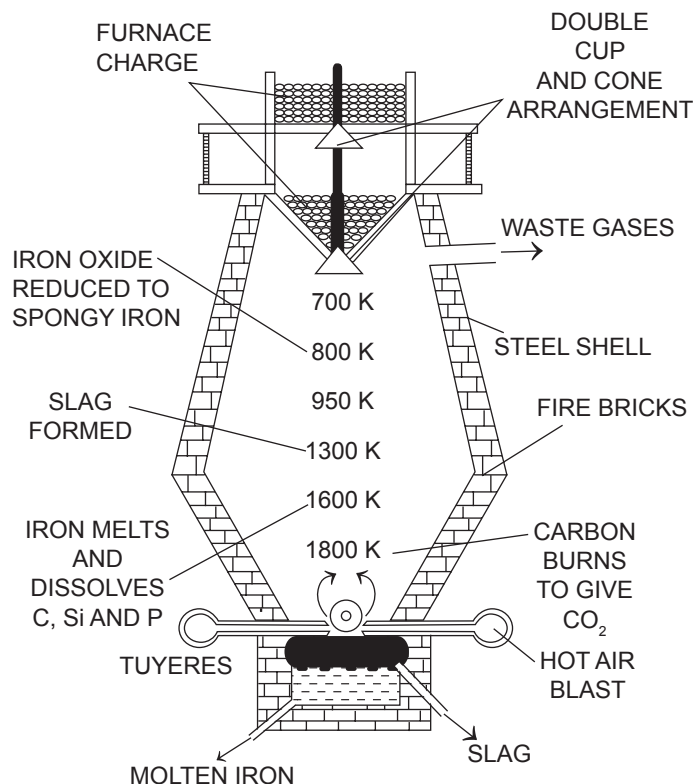
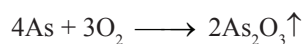
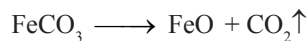


Fig. 13.7 Extraction of Cast Iron

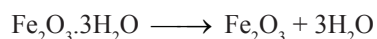
- 1. Dressing of the ores:** The iron ores are first broken into small pieces 3–5 cm in size.
- 2. Roasting or Calcination:** During roasting S, As, P are oxidized to the respective oxides.



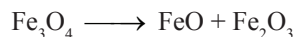
FeCO_3 decomposes as,



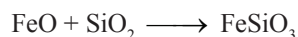
$\text{Fe}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$ loses water



Fe_3O_4 is decomposed to ferrous oxide and ferric oxides.



Ferrous oxide reacts with silica to form ferrous silicate at high temperature.



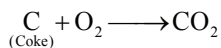
But the conversion of FeO into Fe_2O_3 will prevent the formation of FeSiO_3 . Thus the mass of the ore becomes porous causing the increase in the effective surface area.

Extraction of Cast Iron

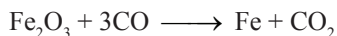
Iron is usually extracted from the haematite. Concentrated ore after calcination is reduced with carbon i.e., smelted in the blast furnace:

Reactions taking place in the blast furnace:

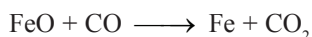
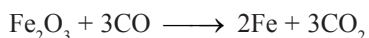
(i) **Zone of combustion**



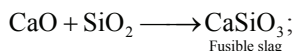
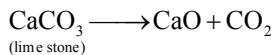
(ii) **Zone of reduction**



(iii) **Zone of reduction**



(iv) **Zone of slag formation**



(v) **Zone of fusion (lower part of furnace)**

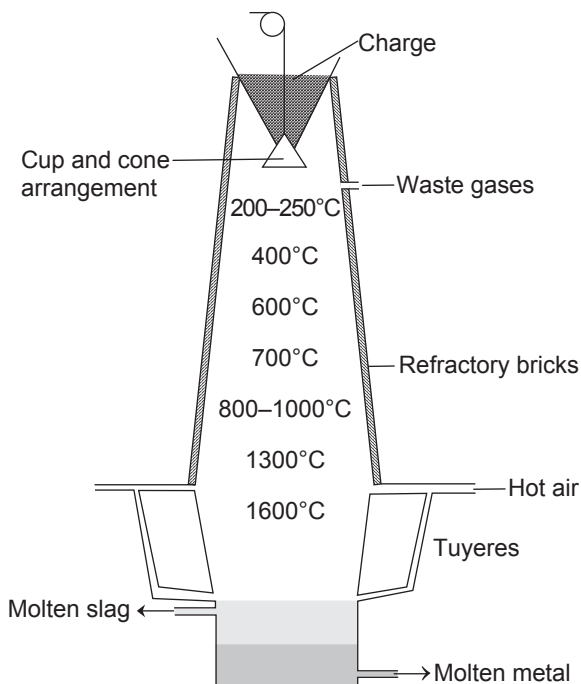
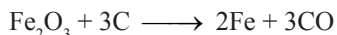


Fig. 13.8 Blast furnace

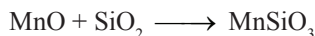
Molten iron is heavier than molten slag. The two liquids are periodically tapped off. The molten iron tapped off from the furnace is solidified into blocks called 'pigs'.

Preparation of Wrought Iron

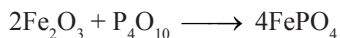
This is done by heating cast iron with haematite (Fe_2O_3) which oxidizes C to CO, S to SO_2 , Si to SiO_2 , P to P_4O_{10} and Mn to MnO



Where CO and SO_2 escape, manganous oxide (MnO) and Silica (SiO_2) combine to form slag.



Similarly, phosphorus pentoxide combines with haematite to form ferric phosphate slag.



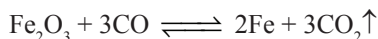
Bosh: The diameter of the furnace gradually increases from the top down wards. Widest part of the furnace is called Bosh. At above 2 m. tuyeres are there through which hot air blast is blown into the furnace.

Hearth: Below the bosh this region exists. (1) slag notch is at higher height and (2) tap hole for metal passage at lower position from the bottom. At the top of the furnace the hopper is there which is cup and cone arrangement. Through this charge is introduced till the charge bed in the furnace is 4/5th of the furnace. Hot air at 700°C is forced into the furnace through the tuyeres. The thermal gradient inside exists from 1800°C (hearth) to $400^\circ\text{C} - 900^\circ\text{C}$ in the upper region. Near the bosh the temperature varies from $1200^\circ\text{C} - 1300^\circ\text{C}$ chemical reactions which take place are.

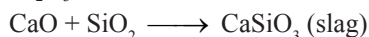
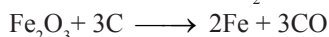
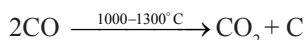
1. At 1200°C near the tuyeres,



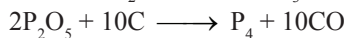
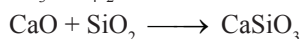
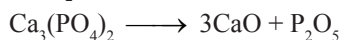
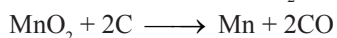
2. Above bosh $600^\circ - 900^\circ\text{C}$, ferric oxide is partially reduced by CO as



3. $\text{CaCO}_3 \xrightarrow{900^\circ\text{C}} \text{CaO} + \text{CO}_2$



4. The reaction at 1500°C , MnO_2 is reduced to Mn.



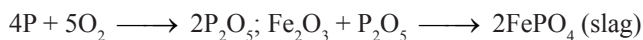
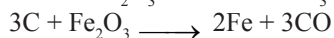
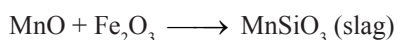
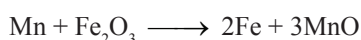
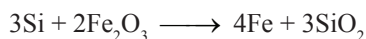
5. **Collection of Cast Iron:** Metal is cast into ingots or in the ladle for further refining like steel making.

Wrought iron

Minimum percentage of carbon, is 0.1 – 0.15% and other impurities like S, P, Mn, Si less than 0.3%.

Manufacturing process

Cast iron takes in puddling furnace and melted by hot blast of air. The chemical reactions which occur are

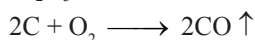
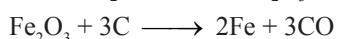
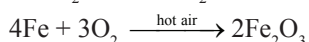
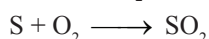
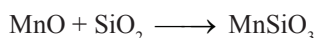
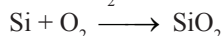
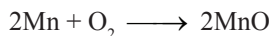


The impurities are removed from iron, the melting point of the metal rises and it becomes a semi solid mass. Metal taken out from the furnace in the form of balls with the help of the rabbles. The balls are then beaten under hammer to separate out the slag. The product thus formed is thus called wrought iron.

STEEL

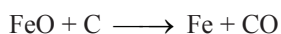
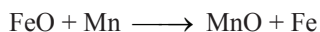
Manufacture from CI (Cast Iron)

Acid Besimer: This process is carried out when percentage of P is very low. The lining of the furnace is made of SiO_2 . The following reactions take place to lessen the impurities.



When the blue flame of CO drops out it, complete oxidation of carbon.

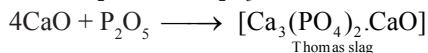
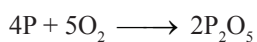
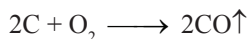
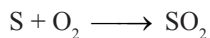
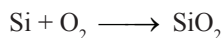
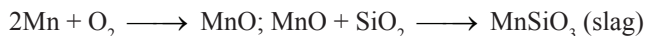
All the reactions are exothermic. After the complete of the reaction calculated amount of speigleisen (an alloy of Fe, Mn, C) is added to the molten mass. After that again air blast is continued for two minutes to engage the thorough mixing of the speigel.



It acts as the deoxidizer to remove oxygen inclusion which in the metals are the excess of elements dissolve in the molten Fe and thus supplies are requisite amount of carbon to iron to produce steel of desired quality. Mn serves to make the steel harder and to increase its tensile strength. The converter is then tilted and molten steel cast into moulds, or taken in ladle for further processing.

Basic besemer process

The lining of the converter is of basic materials like CaO, MgO etc. Mn, Si of CI is first oxidized. S from SO_2 , C, P are then oxidized to form CO and P_2O_5 .



Spiegeleisen is added to the molten Fe when it is converted and steel.

Open hearth process

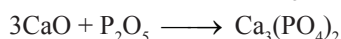
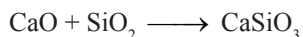
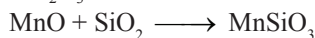
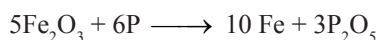
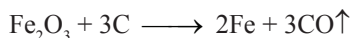
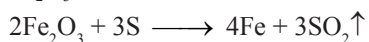
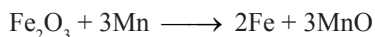
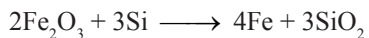
Open hearth process is the modern process for the manufacture of the high grade steel.

- (i) The impurities present in the C.I. are mainly oxidized by the addition of haematite ores and the rest are oxidized by O_2 of air.
- (ii) The percentage of C, Si is lowered down by the addition of scrap Fe.
- (iii) Heat generated by burning of mixture of produce gas air, is used to preheat the fuel gas and air.

This regenerative system causes an economic fuel consumption and maintains high temperature (1800°C) in the furnace.

Reactions taking place in the hearth

Mn, Si, P, S and carbon are oxidized by haematite (Fe_2O_3) to form their respective oxides. The reactions occurring are as follows:



C, P are also oxidized by air to form carbon monoxide and phosphorus pentoxide.



After the removal of impurities slag is separated, requisite amount of spiegeleisen is added to the molten mass to obtain desired quality of steel by deoxidation and recarburisation. Fe, Si and Al are also sometimes added for removal of O_2 to make killed steel.

Duplex process

This is actually a combination of acid bessemer process and open hearth process. Si, Mn, S, C (partially) are oxidized and removed as slag or volatile oxides.

Worked-out Examples

37. The chemical processes in the production of steel from haematite ore involve (IITJEE 2000)

- (a) reduction (b) oxidation
(c) reduction followed by oxidation (d) oxidation followed by reduction

Ans. (d)

Firstly, carbon which is added along with crushed haematite ore is oxidized to CO (and CO_2). The produced CO acts as chief reducing agent for the reduction of haematite to steel.

38. In the metallurgy of iron, when limestone is added to the blast furnace, the calcium ions end up in (IITJEE 1982)

- (a) slag (b) gangue
(c) metallic Ca (d) calcium carbonate

Ans. (a)

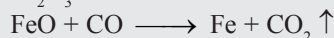
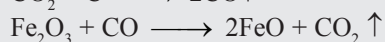
39. In the manufacture of iron from haematite, the function of lime stone is as (IITJEE 1978)

- (a) a reducing agent (b) slag
(c) flux (d) gangue

Ans. (c)

40. What is the actual reducing agent of haematite in blast furnace?

Solution: Carbon monoxide is actual reducing agent of haematite in blast furnace.

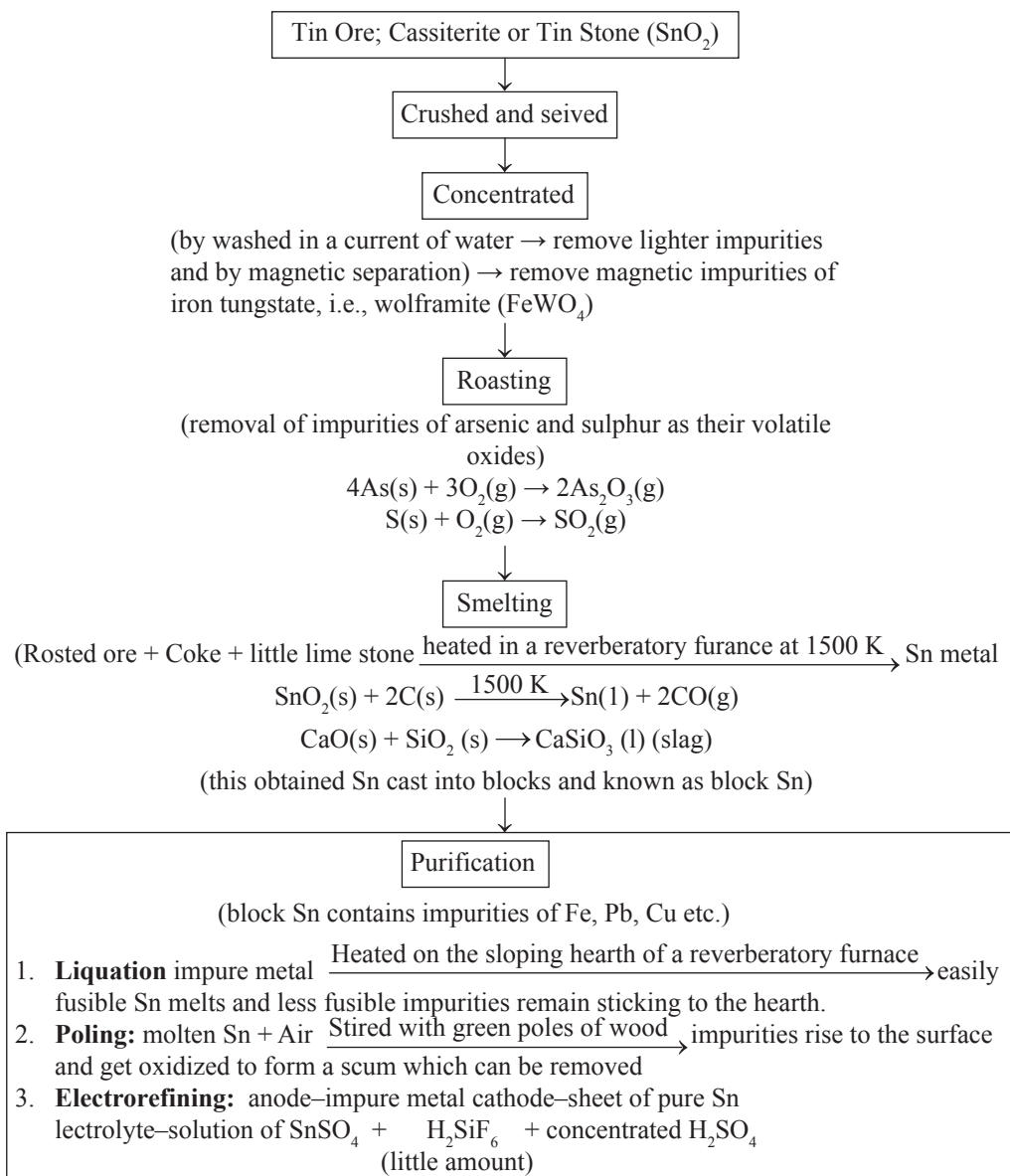


Extraction of Tin

Ores

- (a) Cassiterite or Tin stone; SnO_2
 (b) Stannite; $\text{Cu}_2\text{S} \cdot \text{FeS} \cdot \text{SnS}_2$

FLOW CHART FOR THE EXTRACTION OF TIN

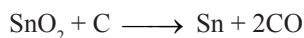


On passing current, pure tin from the anode goes into the solution and an equivalent amount of the metal from the solution gets deposited on the cathode.

Extraction

- Concentration:** Crushed ore is water washed by a stream of water.
- Roasting:** S, As present in one ore are volatilized away in the form of their oxides in a revolving furnace.

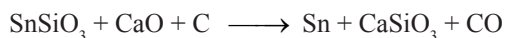
3. **Washing:** It is then washed with hot water and mainly FeSO_4 is present at this stage.
4. **Electromagnetic separation:** From the hopper the dry ore is supplied and from the belt the FeSO_4 is separated by magnetic field. It contains 75% SnO_2 (black tin)
5. **Carbon reduction:**



Temperature $1200 - 1300^\circ\text{C}$

Furnace: Reverberating furnace.

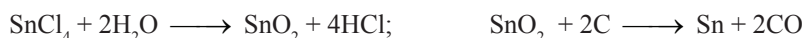
The molten tin is collected as 99% pure Sn. The upper layer of the slag contains 10-20% Sn. This is melted with coke.



6. **Refining:** This is carried out by sweating process. Sn metals leaving behind Fe, CuO, N. Tin thus obtained is 99.5% pure.

Recovery of Sn from scrap:

The scrap tin plates, free from dirt are reacted by Cl_2 to SnCl_4 . SnCl_4 being liquid is volatilized away and is condensed.



Worked-out Example

41. Extraction of zinc from zinc blende is achieved by

(IITJEE 2007)

- (a) electrolytic reduction
- (b) roasting followed by reduction with carbon
- (c) roasting followed by reduction with another metal
- (d) roasting followed by self-reduction

Ans. (b)

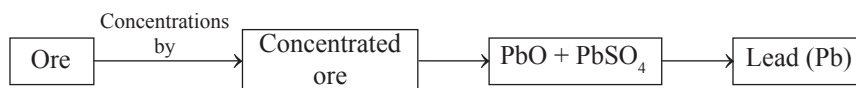
Zinc from zinc blende is achieved by roasting followed by reduction with carbon.

Extraction of Lead

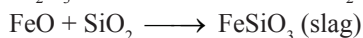
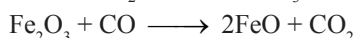
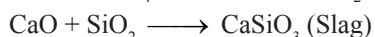
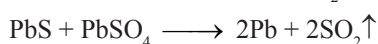
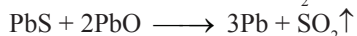
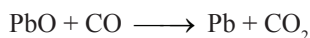
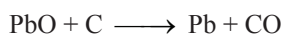
- (a) Galena; PbS
- (b) Cerussite; PbCO_3
- (c) Anglesite; PbSO_4
- (d) White lead; $2\text{Pb(OH)}_2 \cdot \text{PbCO}_3$

Extraction

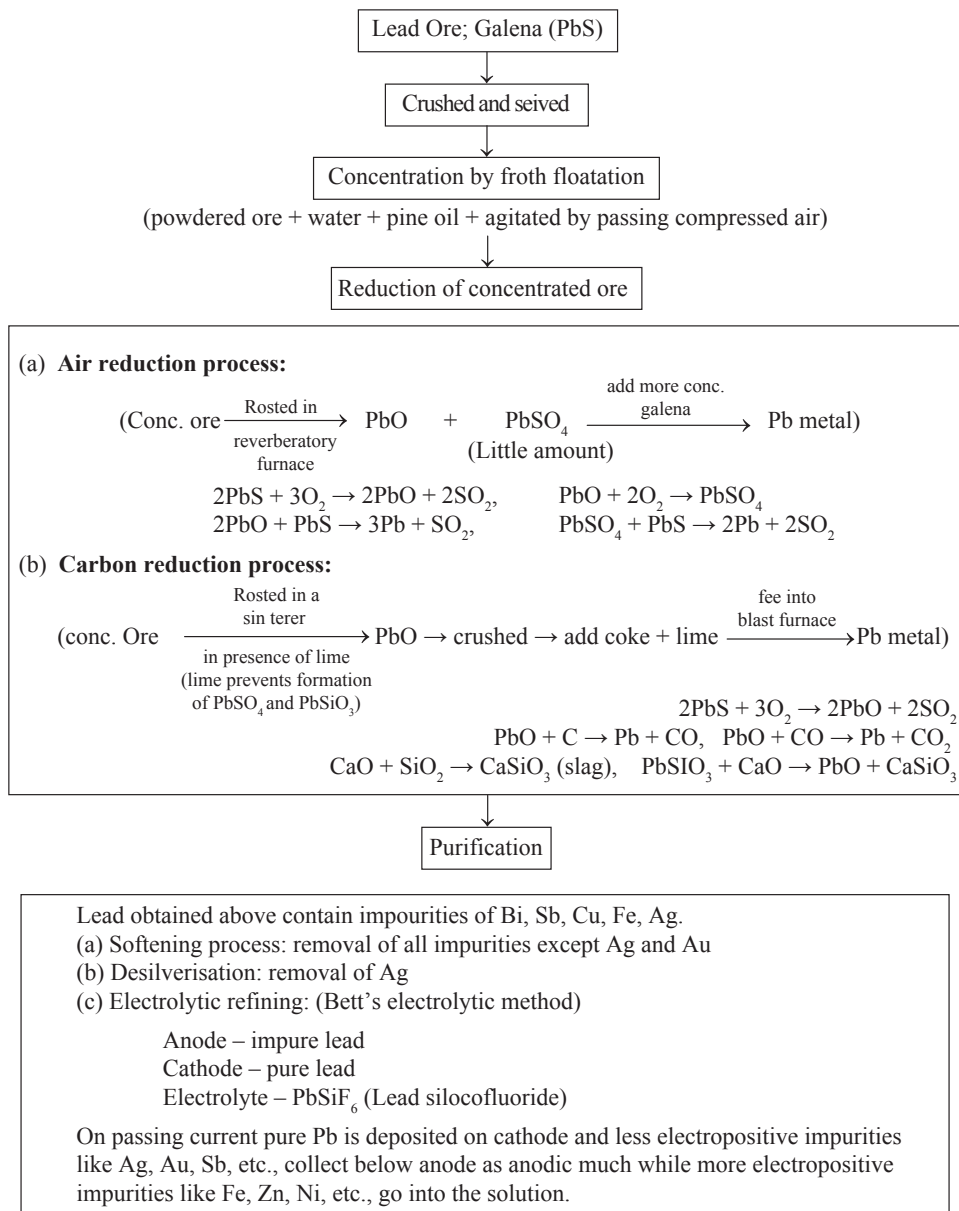
Galena is the commercially and economically viable ore. Steps can be shown by the following flow chart diagram.



The reaction at the that furnace are as follows:



FLOW SHEET FOR THE EXTRACTION OF LEAD OR PLUMBUM



Purification

Lead thus obtained contains Cu, Sb, Sn, As, Ag, Au etc., which make lead hard. In reverberatory furnance, in presence of air impurities are oxidized and form a layer on the surface of molten lead. The larger is removed until lead becomes soft.

Worked-out Example

42. Lead and tin are, respectively, extracted from their chief ore by

(IITJEE 2004)

- (a) carbon reduction and self reduction (b) carbon reduction and electrolytic reduction
- (c) self reduction and carbon reduction (d) electrolytic reduction and carbon reduction

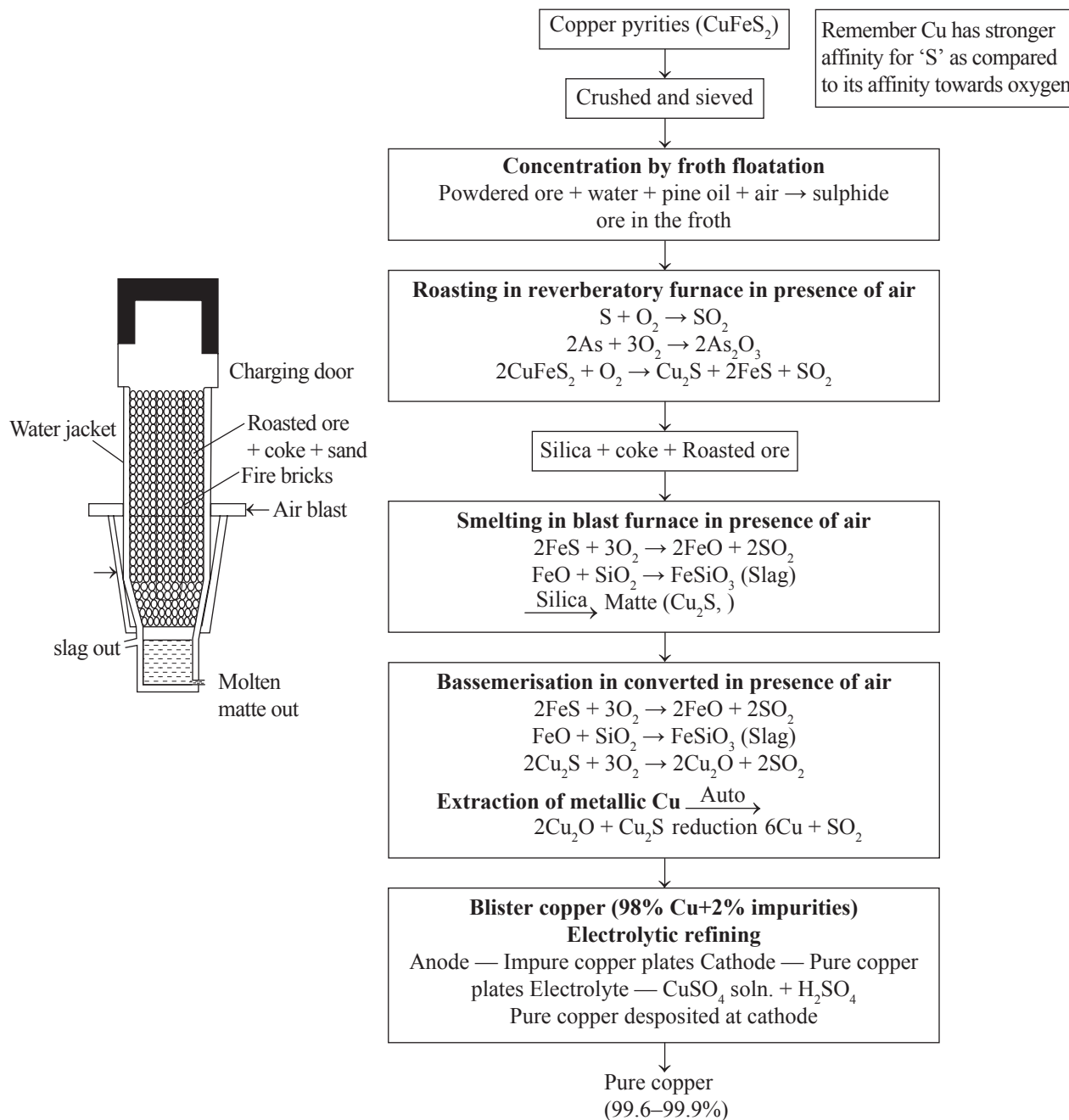
Ans. (c)

Lead is extracted by self-reduction method and tin by carbon-reduction method.

Extraction of Copper

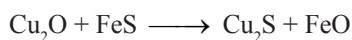
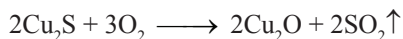
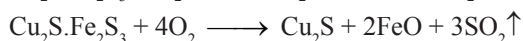
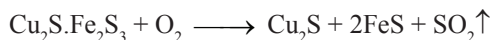
- (a) Cuprite; Cu_2O
- (b) Copper pyrites; CuFeS_2 (as sulphides)
- (c) Copper glance; Cu_2S (as sulphides)
- (d) Malachite; CuCO_3 ; $\text{Cu}(\text{OH})_2$
- (d) Azurite $2\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$

FLOW SHEET FOR THE EXTRACTION OF COPPER



1. Concentration of the ore by oil floatation process: Copper pyrites contains only (2–3)% of copper. The rest of the ore contains iron or sulphide, silica, silicious materials, sulphur, arsenic etc., as impurities. Froth floatation by Xanthate and pine oil. The froth is collected and dried when concentrated ore is obtained which contains 25–30% of Cu.

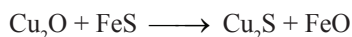
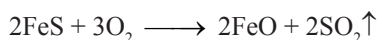
2. Roasting



3. Smelting of the roasted ore in blast furnace: material required

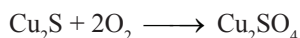
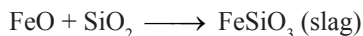
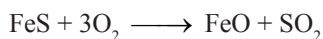
- (i) Roasted ore
- (ii) Lime stone
- (iii) Coke (used as fuel)
- (iv) Silica (used as flux)
- (v) Lime stone (used to remove excess silica)

Reactions occurring ore as follows:

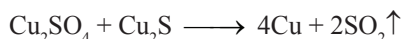
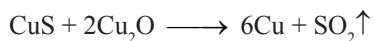


4. Self-reduction in Bessemer Converter

Reactions involved are,



1/3 rd of Cu_2S goes in Cu_2SO_4 . Formation of 2/3 rd of Cu_2S is oxidized while the blast of hot air is stopped CuSO_4 are reduced by the rest of cuprous sulphide to produce metallic copper with the evolution of SO_2



Thus metallic copper is produced at the bottom of the furnace and collected as ingots. This is blister Cu names so, due to the evolution of SO_2 from the surface of metal.

5. Refining

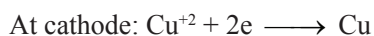
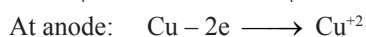
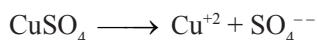
Electro refining

Anode: Impure copper obtained above

Cathode: Pure copper

Electrolyte: 15% CuSO_4 solution + 5% H_2SO_4

Reactions coming are as follows:



Worked-out Examples

43. On heating a mixture of Cu_2O and Cu_2S , we get: [AIEEE 2005]
 (a) $\text{Cu} + \text{SO}_2$ (b) $\text{Cu} + \text{SO}_3$ (c) $\text{CuO} + \text{CuS}$ (d) Cu_2SO_3

Ans. (a) Auto reduction occurs



44. Copper can be extracted from (IITJEE 1978)
 (a) Kupfernickel (b) Dolomite (c) Cinnabar (d) Galena (e) Malachite

Ans. (e)

45. Which of the following ores contains both iron and copper?
 (a) Malachite (b) Azurite (c) Dolomite (d) Chalcopyrite

Ans. (d)

Chalcopyrites is copper pyrites (CuFeS_2).

Malachite is $\text{Cu}(\text{OH})_2 \cdot \text{CuCO}_3$.

Azurite is $\text{Cu}(\text{OH})_2 \cdot 2\text{CuCO}_3$.

Dolomite is $\text{MgCO}_3 \cdot \text{CaCO}_3$

46. The chemical composition of 'slag' formed during the smelting process in the extraction of copper is (IITJEE 2001)
 (a) $\text{Cu}_2\text{O} + \text{FeS}$ (b) FeSiO_3 (c) CuFeS_2 (d) $\text{Cu}_2\text{S} + \text{FeO}$

Ans. (b)

47. Why are sulphide ores generally roasted to oxide for the extraction of metals instead of being directly reduced?

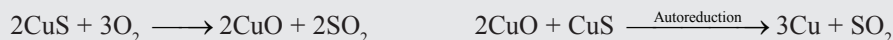
Ans. Carbon is usually used for the reduction of ores and if sulphide ores are used, it will form carbon disulphide which is difficult to get rid off. Therefore, sulphide ores are first roasted to get metal oxides which are then reduced by carbon to give metal and CO_2 escapes easily into the air.

48. Why is chalcocite roasted and not calcinated during recovery of copper?

Ans. Chalcocite being a sulphide ore of copper needs roasting (heating in excess of air) and not calcination.

49. Partial roasting of sulphide ores is done in the metallurgical operation of copper.

Ans. Partial reduction of sulphide ores gives copper oxide which further reacts with copper sulphide to show auto-reductions.

**Extraction of Silver**

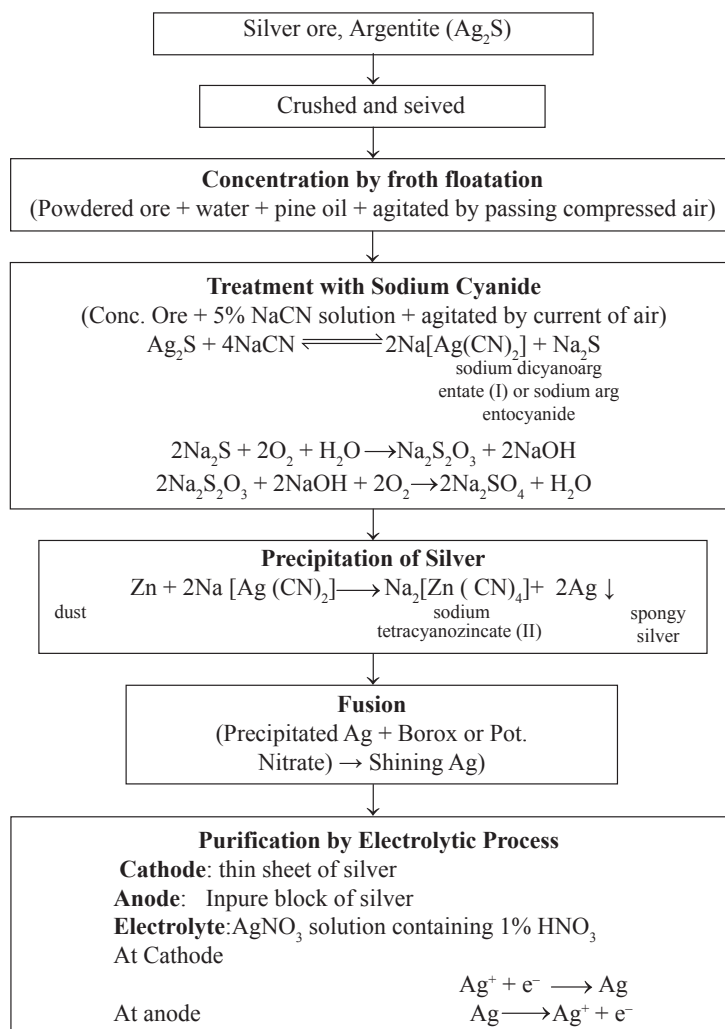
- (a) Horn silver (chloroapryrite); AgCl (as halide)
 (b) Lunar Caustic; AgNO_3 (as nitrate)
 (c) Silver glance or argentite; Ag_2S (as sulphide)
 (d) Ruby silver or pyrogryrite; $3\text{Ag}_2\text{S} \cdot \text{Sb}_2\text{S}_3$

Worked-out Example

50. Copper and silver are below hydrogen in electrochemical series and yet they are found in the combined state as sulphides in nature. Comment.

Ans. Both Cu and Ag combines directly with sulphur at very high temperature to form sulphides under earth crust.

FLOW CHART FOR THE EXTRACTION OF SILVER



Worked-out Examples

51. Zinc but not copper is used for the recovery of metallic silver from the complex $[\text{Ag}(\text{CN})_2]^-$.

Ans. E_{OP}° (standard oxidation potential) of Zn is more positive than E_{OP}° of Cu and thus Zn is more powerful reductant than Cu and thus Zn can easily replace Ag from $[\text{Ag}(\text{CN})_2]^-$. Furthermore, Zn is cheaper than Cu.

52. Give balanced equation for the extraction of silver from sulphide ore.

Ans. $\text{Ag}_2\text{S} + 4\text{NaCN} \longrightarrow 2\text{Na}[\text{Ag}(\text{CN})_2] + \text{Na}_2\text{S}$

$2\text{Na}[\text{Ag}(\text{CN})_2] + \text{Zn} \longrightarrow \text{Na}_2[\text{Zn}(\text{CN})_4] + 2\text{Ag}$

53. Native silver metal forms a water soluble complex with a dilute aqueous solution of NaCN in the presence of

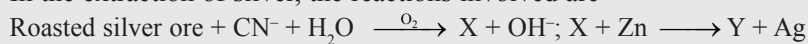
- (a) nitrogen (b) oxygen (c) carbon dioxide (d) argon

Ans. (b)

(IITJEE 2008)

54. In the extraction of silver, the reactions involved are

(IITJEE 2003)

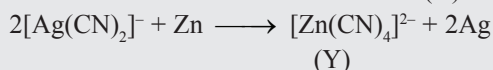
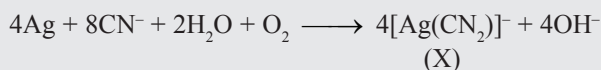


The species X and Y, respectively, are

- (a) $[\text{Ag}(\text{CN})_2]^-$, $[\text{Zn}(\text{CN})_4]^{2-}$ (b) $[\text{Ag}(\text{CN})_2]^-$, $[\text{Zn}(\text{CN})_6]^{2-}$
 (c) $[\text{Ag}(\text{CN})_4]^{3-}$, $[\text{Zn}(\text{CN})_4]^{2-}$ (d) $[\text{Ag}(\text{CN})_4]^-$, $[\text{Zn}(\text{CN})_4]^{2-}$

Ans. (a)

The reactions involved are

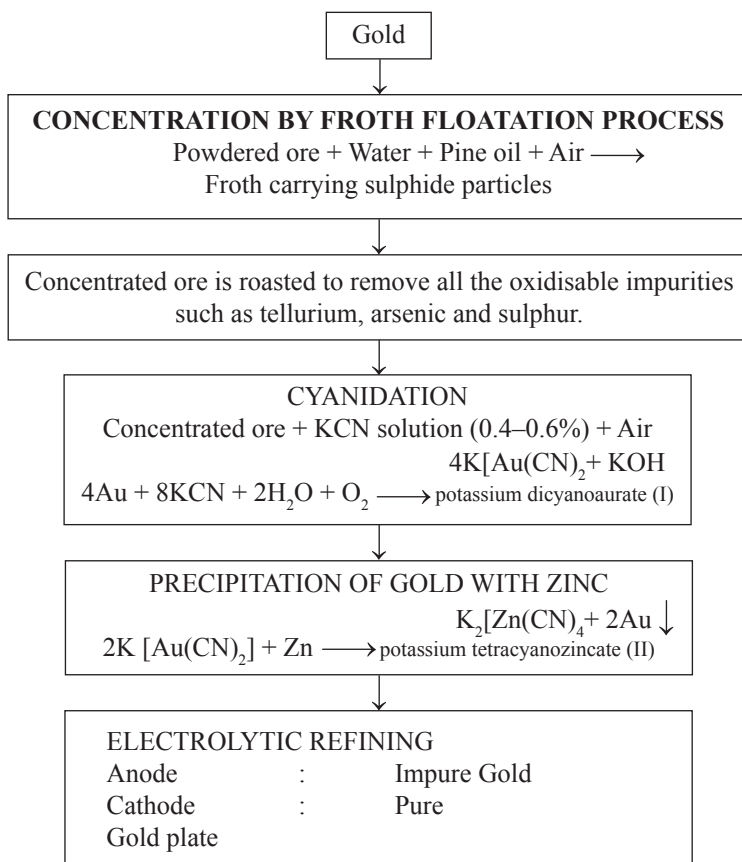


55. In the reduction of gold (III) chloride from AuCl_4^- ion solution, the solution must be stirred readily during electrolysis. Explain why stirring is so important in this process?

Solution: The AuCl_4^- ion, which is negatively charged should be reduced at an electrode with excess electrons i.e., cathode to bring in the change $\text{AuCl}_4^- + 3\text{e}^- \longrightarrow \text{Au} + 4\text{Cl}^-$. The AuCl_4^- ion is negatively charged and it will be repelled by cathode (negatively charged). Thus stirring is essential to get the AuCl_4^- anions at desired electrode i.e., cathode.

Extraction of Gold

FLOW CHART FOR THE EXTRACTION OF GOLD



STRAIGHT OBJECTIVE TYPE QUESTIONS

- In the manufacture of iron from haematite, the function of lime stone is as (1978)
 - a reducing agent
 - slag
 - flux
 - gangue
 - None of these
- Calcium is obtained by (1980)
 - Electrolysis of molten CaCl_2
 - Electrolysis of aqueous solution of CaCl_2
 - Reduction of CaCl_2 with carbon
 - Roasting of lime stone
- In the metallurgy of iron, when limestone is added to the blast furnace, the calcium ions ends up in (1982)
 - slag
 - gangue
 - metallic Ca
 - calcium carbonate
- The major role of fluorspar (CaF_2) which is added in small quantities in the electrolytic reduction of alumina dissolved in fused cryolite (Na_3AlF_6) is (1993)
 - that of a catalyst
 - to make the fused mixture very conducting
 - to lower the temperature of the melt
 - to decrease the rate of oxidation of carbon at the anode
- In the commercial electrochemical process for aluminium extraction, the electrolyte used is (1999)
 - $\text{Al}(\text{OH})_3$ in NaOH solution
 - an aqueous solution of $\text{Al}_2(\text{SO}_4)_3$
 - a molten mixture of $\text{AlO}(\text{OH})$ and $\text{Al}(\text{OH})_3$
 - a molten mixture of Al_2O_3 and Na_3AlF_6
- The chemical processes in the production of steel from haematite ore involve (2000)
 - reduction
 - oxidation
 - reduction followed by oxidation
 - oxidation followed by reduction
- Electrolytic reduction of alumina to aluminium by Hall-Heroult process is carried out (2000)
 - in the presence of NaCl
 - in the presence of fluorite
 - in the presence of cryolite which forms a melt with lower melting temperature
 - in the presence of cryolite which forms a melt with higher melting point
- In the aluminothermite process, aluminium acts as (1983)
 - an oxidizing agent
 - a flux
 - a reducing agent
 - a solder
- Among the following statements, the incorrect one is (1997)
 - calamine and siderite are carbonates;
 - argentite and cuprite are oxides;
 - zinc blende and pyrites are sulphides;
 - malachite and azurite are ores of copper.
- In the commercial electrochemical process for aluminium extraction the electrolyte used is (1999)
 - $\text{Al}(\text{OH})_3$ in NaOH solution
 - an aqueous solution of $\text{Al}_2(\text{SO}_4)_3$
 - a molten mixture of Al_2O_3 and Na_3AlF_6
 - a molten mixture of $\text{AlO}(\text{OH})$ and $\text{Al}(\text{OH})_3$
- The chemical composition of 'slag' formed during the smelting process in the extraction of copper is (2001)
 - $\text{Cu}_2\text{O} + \text{FeS}$
 - FeSiO_3
 - CuFeS_2
 - $\text{Cu}_2\text{S} + \text{FeO}$
- Which of the following processes is used in the extractive metallurgy of magnesium? (2002)
 - fused salt electrolysis
 - self reduction
 - aqueous solution electrolysis
 - thermite reduction
- In the extraction of silver, the reactions involved are roasted silver ore + $\text{CN}^- + \text{H}_2\text{O} \xrightarrow{\text{O}_2} \text{X} + \text{OH}^-$; (2003)

$$\text{X} + \text{Zn} \longrightarrow \text{Y} + \text{Ag}$$
 The species X and Y, respectively, are
 - $[\text{Ag}(\text{CN})_2]^-$, $[\text{Zn}(\text{CN})_4]^{2-}$
 - $[\text{Ag}(\text{CN})_2]^-$, $[\text{Zn}(\text{CN})_6]^{2-}$
 - $[\text{Ag}(\text{CN})_4]^{3-}$, $[\text{Zn}(\text{CN})_4]^{2-}$
 - $[\text{Ag}(\text{CN})_4]^-$, $[\text{Zn}(\text{CN})_4]^{2-}$
- Lead and tin are, respectively, extracted from their chief ore by (2004)
 - carbon reduction and self reduction
 - carbon reduction and electrolytic reduction
 - self reduction and carbon reduction
 - electrolytic reduction and carbon reduction
- Which of the following ores contains both iron and copper? (2005)
 - Malachite
 - Azurite
 - Dolomite
 - Chalcopyrite

16. Extraction of zinc from zinc blende is achieved by (2007)
- electrolytic reduction
 - roasting followed by reduction with carbon
 - roasting followed by reduction with another metal
 - roasting followed by self-reduction
17. Native silver metal forms a water soluble complex with a dilute aqueous solution of NaCN in the presence of (2008)
- nitrogen
 - oxygen
 - carbon dioxide
 - argon

MULTIPLE CORRECT ANSWER TYPE QUESTIONS

1. In the electrolysis of alumina, cryolite is added to (1986)
- lower the melting point of alumina
 - increase the electrical conductivity
 - minimise the anode effect
 - remove impurities from alumina
2. Of the following, the metals that cannot be obtained by electrolysis of the aqueous solution of their salts are (1990)
- Ag
 - Mg
 - Cu
 - Al

FILL-IN-THE-BLANKS TYPE QUESTIONS

1. In the basic Bessemer process for the manufacture of steel, the lining of the converter is made of _____. The slag formed consists of _____. (1980)
2. In the thermit process _____ is used as a reducing agent. (1980)
3. Cassiterite is an ore of _____. (1980)
4. AgCl dissolves in excess KCN solution to give the complex compound. (1980)
5. In extractive metallurgy of zinc partial fusion of ZnO with coke is called _____ and reduction of the ore to the molten metal is called _____. (1988)

ANSWER KEYS

STRAIGHT OBJECTIVE TYPE QUESTIONS

1. (c) 2. (a) 3. (a) 4. (c) 5. (d) 6. (d) 7. (c) 8. (c) 9. (b) 10. (c)
11. (b) 12. (a) 13. (a) 14. (c) 15. (d) 16. (b) 17. (b)

MULTIPLE CORRECT-ANSWER TYPE QUESTIONS

1. (a), (b) 2. (b), (d)

FILL-IN-THE-BLANKS TYPE QUESTIONS

1. magnesia, lime 2. aluminium 3. Tin 4. $KAg(CN)_2$ 5. sintering, smelting

ONE CORRECT OPTION TYPE QUESTIONS

- Which of the following set of elements mostly occur as sulphide ores:
 (a) Zn, Cu, Na (b) Zn, Cu, Pb
 (c) Fe, Al, Ti (d) Cu, Ag, Au
- Which one of the following is not an ore of aluminium?
 (a) Bauxite (b) Corundum
 (c) Epsomite (d) Cryolite
- Cinnabar is the ore of:
 (a) Zn (b) Cd
 (c) Hg (d) Ag
- Match List I with List II and select the correct answer using the codes given below the lists:

List I A. Carnallite B. Calamine C. Ilmanite D. Chalcopryite	List II 1. Zinc 2. Titanium 3. Magnesium 4. Copper
---	---

A B C D	A B C D
(a) 1 3 2 4	(b) 1 3 4 2
(c) 3 1 4 2	(d) 3 1 2 4
- Match the method of concentration of the ore in column I with the ore in column II and select the correct alternate:

I	II
X. magnetic separation	A. Ag_2S
Y. froth flotation	B. FeCr_2O_4
Z. gravity separation	C. $\text{Al}_2(\text{SiO}_3)_3$

X Y Z
(a) A B C
(b) B A C
(c) C A B
(d) B C A
- The maximum temperature 1550°C is obtained in the region of the blast furnace used in the extraction of iron.
 (a) reduction (b) fusion
 (c) combustion (d) slag formation
- The iron obtained from the blast furnace is called:
 (a) pig iron (b) cast iron
 (c) wrought iron (d) steel

- Stainless steel contains iron and:
 (a) Zn (b) Cu
 (c) Al (d) Cr
- Nitriding is a process of heating steel in atmosphere of:
 (a) ammonia (b) oxygen
 (c) carbon dioxide (d) air
- Bessemer converter is used in the manufacture of:
 (a) pig iron (b) steel
 (c) wrought iron (d) cast iron

- Match the reactions

Reactions	Temperature range
(A): $\text{Fe}_2\text{O}_3 + \text{CO} \rightarrow \text{CO}_2 + \text{Fe}$	(I) 1200°C
(B): $\text{C} + \text{CO}_2 \rightarrow \text{CO}$	(II) 900°C
(C): $\text{C} + \text{H}_2\text{O} \rightarrow \text{CO} + \text{H}_2$	(III) 600°C
(D): $\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$	(IV) 1100°C

Choose the correct alternate

- | |
|-----------------|
| A B C D |
| (a) IV III II I |
| (b) II IV III I |
| (c) I II IV III |
| (d) I II III IV |
- In the metallurgy of iron, when CaCO_3 is added to blast furnace, calcium ion appears as
 (a) CaO (b) metallic Ca
 (c) gangue (d) slag
 - Iron obtained from blast furnace is
 (a) wrought iron (b) cast iron
 (c) pig iron (d) steel
 - Calcination is the process of heating the ore
 (a) in inert gas
 (b) in the presence of air
 (c) in the absence of air
 (d) in the presence of CaO and MgO
 - Annealing of steel is the process of heating steel
 (a) to a bright red hot and then cooling it slowly
 (b) is a bright red hot and then cooling it suddenly
 (c) to a temperature much below redness and cooling it slowly
 (d) None of these

16. The method of zone refining of metals is based on the principle of:
- Greater mobility of the pure metal than that of impurity
 - Higher melting point of the impurity than that of the pure metal
 - Greater noble character of the solid metal than that of the impurity
 - Greater solubility of the impurity in the molten state than in the solid
17. Cassiterite is an ore of:
- Mn
 - Ni
 - Sb
 - Sn
18. In the extraction of copper from its sulphide ore, the metal is formed by reduction of Cu_2O with:
- FeS
 - CO
 - Cu_2S
 - SO_2
19. Copper is extracted from sulphide ore using the method:
- carbon reduction
 - carbon monoxide reduction
 - auto reduction
 - none of these
20. In the extraction of copper from copper pyrites, iron is removed as:
- FeSO_4
 - FeSiO_3
 - Fe_3O_4
 - Fe_2O_3
21. In the electrolytic refining of copper ore during extraction in order to remove:
- cuprous sulphide
 - ferrous oxide
 - ferrous sulphide
 - cuprous oxide
22. In the electrolytic refining of copper, Ag and Au are found:
- on anode
 - in electrolyte solution
 - in anode mud
 - in cathode mud
23. Blister copper is:
- pure copper
 - ore of copper
 - alloy of copper
 - impure copper
24. Poling process is used for:
- The removal of Cu_2O from Cu
 - The removal of Al_2O_3 from Al
 - The removal of Fe_2O_3 from Fe
 - All of these
25. Consider the following steps:
- $$\text{CuS} \xrightarrow{\text{roast in air}} (\text{a}) \xrightarrow{\text{roast without air}} (\text{B})$$
- Which is not the correct statement
- it is self-reduction
 - it involves disproportionation $\text{Cu}_2\text{S} \rightarrow \text{Cu} + \text{CuS}$
 - (a) is a mixture of CuO and CuS and (b) is a mixture of Cu and SO_2
 - all are incorrect statements
26. Blister Cu is about
- 60% of Cu
 - 90% Cu
 - 98%
 - 100% Cu
27. Froth floatation process for the concentration of sulphide ores is an illustration of the practical application of:
- adsorption
 - absorption
 - sedimentation
 - coagulation
28. 'Softening of lead' means:
- conversion of lead to PbO
 - conversion of lead to Pb_3O_4
 - removal of impurities (metallic) from lead
 - washing lead with HNO_3 followed by alkali solution
29. The element which is recovered from electrolytic process is:
- iron
 - lead
 - aluminium
 - zinc
30. In the Pidgeon process, Mg is produced by
- electrolysis of fused MgCl_2
 - reducing calcined dolomite with ferrosilicon at high temperature under pressure
 - both correct
 - None is correct
31. During the electrolysis of carnallite, MgCl_2 is decomposed and not KCl. This is because of:
- lower decomposition voltage of MgCl_2 than that of KCl
 - reverse reaction $\text{MgCl}_2 + 2\text{K} \longrightarrow \text{Mg} + 2\text{KCl}$ if KCl is decomposed under other experimental condition
 - both (a) and (b)
 - None of the above
32. Extraction of aluminium from bauxite ore, reduction is carried out by:
- carbon
 - magnesium
 - electrolysis
 - hydrogen
33. Aluminum metal is purified by:
- Hoppe's process
 - Hall's process
 - Serpeck's process
 - Baeyer's process
34. In the alumino-thermite process, al acts as:
- an oxidizing agent
 - a flux
 - solder
 - a reducing agent

13.40 Principles of Metallurgy

35. Which of the following reaction forms the basis of Goldschmidt aluminothermite process?
- $2\text{Al} + \text{N}_2 \rightarrow 2\text{AlN}$
 - $2\text{Al} + 3\text{Cl}_2 \rightarrow 2\text{AlCl}_3$
 - $2\text{Al} + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2$
 - $2\text{Al} + \text{Fe}_2\text{O}_3 \rightarrow \text{Al}_2\text{O}_3 + 2\text{Fe}$
36. The function fluorspar in the electrolytic reduction of alumina dissolved in fused cryolite (Na_3AlF_6) is:
- as a catalyst
 - to lower the temperature of the melt and to make the fused mixture very conducting
 - to decrease the rate of oxidation of carbon at the anode
 - None of the above
37. Following reaction is not involved in Thermite process
- $3\text{Mn}_2\text{O}_3 + 8\text{Al} \rightarrow 9\text{Mn} + 4\text{Al}_2\text{O}_3$
 - $\text{Cr}_2\text{O}_3 + 2\text{Al} \rightarrow \text{Al}_2\text{O}_3 + 2\text{Cr}$
 - $2\text{Fe} + \text{Al}_2\text{O}_3 \rightarrow 2\text{Al} + \text{Fe}_2\text{O}_3$
 - $\text{B}_2\text{O}_3 + 2\text{Al} \rightarrow 2\text{B} + \text{Al}_2\text{O}_3$
38. Following method is not used for extraction of Al:
- van Arkel
 - Serpeck
 - Bayer
 - Hall-Heroult
39. Extraction of aluminium from aluminium oxide (Al_2O_3) is best done by
- electrolytic reduction of Al_2O_3
 - reduction of Al_2O_3 with carbon
 - reduction of Al_2O_3 with sodium
 - reduction of Al_2O_3 with CO
40. The element which could be extracted by electrolytic reduction of its oxide dissolved in a high temperature melt is
- sodium
 - magnesium
 - fluorine
 - aluminium
41. Chemical leaching is useful in the concentration of
- copper pyrites
 - bauxite
 - galena
 - cassiterite
42. Carbon cannot be used in the reduction of Al_2O_3 because:
- it is non-metal
 - the heat of formation of CO_2 is more than that of Al_2O_3
 - pure carbon is not easily available
 - the heat of formation of Al_2O_3 is too high
43. There are following extraction process of silver but not:
- as a side product in electrolytic refining of copper
 - Parke's process in which Zn is used to extract silver by solvent extraction from molten lead
 - by reaction of silver sulphide with KCN and then reaction of soluble complex with Zn
 - by heating $\text{Na}[\text{Ag}(\text{CN})_2]$
44. In the leaching of Ag_2S with NaCN, a stream of air is also passed. It is because of:
- reversible nature of reaction between Ag_2S and NaCN
 - to oxidise Na_2S formed into Na_2SO_4 and sulphur
 - Both (a) and (b)
 - None of the above
45. Leaching of Ag_2S is carried out by heating it with a dilute solution of:
- NaCN only
 - HCl
 - NaOH
 - NaCN in presence of O_2
46. Silver can be separated from lead by:
- fractional crystallization
 - amalgamation
 - cupellation
 - addition of zinc (Parker's method)
47. AgCl on fusion with sodium carbonate, gives:
- Ag_2CO_3
 - Ag_2O
 - Ag
 - Ag_2C_2
48. In the cyanide process for extraction of gold and silver from ores, the cyanide solution acts as a:
- reducing agent to reduce the gold and silver compounds present in the ores into the metallic states
 - Leaching agent to bring the gold and silver into solution as cyanide complexes and thus separate these metals from the ores
 - leaching agent to dissolve all the other constituents of the ores leaving the gold and silver as metals
 - leaching agent to bring the ores into solution.
49. Most abundant metal in earth crust is:
- Al
 - O
 - Fe
 - Si
50. If a metal has low oxygen affinity then the purification of metal may be carried out by:
- liquation
 - distillation
 - zone refining
 - cupellation
51. The metal X is prepared by the electrolysis of fused chloride. It reacts with hydrogen to form a colourless solid from which hydrogen gas is released on treatment with water. The metal is:
- Al
 - Ca
 - Cu
 - Zn

52. In the extraction of nickel by Mond's process, the metal is obtained by
- electrochemical reduction
 - thermal decomposition
 - chemical reduction by aluminium
 - reduction by carbon
53. Four metals and their methods of refinement are given
- Ni, Cu, Zn, Ga
 - Electrolysis, van Arkel process, Zone refining, Mond's process
- Choose the right method for each
- Ni: Electrolysis, Cu: van Arkel process, Zn: Zone refining, Ga: Mond's process
 - Ni: Mond's process, Cu: Electrolysis, Zr: van Arkel process, Ga: Zone refining
 - Ni: Mond's process, Cu: van Arkel process, Zr: Zone refining, Ga: Electrolysis
 - Ni: Electrolysis, Cu: Zone refining, Zr: van Arkel process, Ga: Mond's process
54. When FeCr_2O_4 (chromite) is reduced with C in an electric arc furnace:
- Cr and Fe_2O_3 are formed
 - Fe and Cr_2O_3 are formed
 - Fe and Cr (ferrochrome) are formed
 - FeCrO_4 is formed
- FeCr_2O_4 (chromite) is a good source of Cr and its compounds like Na_2CrO_4 , $\text{Na}_2\text{Cr}_2\text{O}_7$
55. On heating quicklime with coke in an electric furnace, we get:
- Ca and CO_2
 - CaCO_3
 - CaO
 - CaC_2
56. Formation of metallic copper from the sulphide ore in the commercial metallurgical process involves:
- $\text{CuS} + \frac{3}{2}\text{O}_2 \longrightarrow \text{CuO} + \text{SO}_2$; $\text{CuO} + \text{C} \longrightarrow \text{Cu} + \text{CO}$
 - $\text{CuS} + \frac{3}{2}\text{O}_2 \longrightarrow \text{CuO} + \text{SO}_2$; $2\text{CuO} + \text{CuS} \longrightarrow 3\text{Cu} + \text{SO}_2$
 - $\text{CuS} + 2\text{O}_2 \longrightarrow \text{CuSO}_4$; $\text{CuSO}_4 + \text{CuS} \longrightarrow 2\text{Cu} + 2\text{SO}_2$
 - $\text{CuS} + \frac{3}{2}\text{O}_2 \longrightarrow \text{CuO} + \text{SO}_2$; $\text{CuO} + \text{CO} \longrightarrow \text{Cu} + \text{CO}_2$
57. Froth floatation process used for the concentration of sulphide ore:
- is based on the difference in wettability of different minerals
 - uses sodium ethyl xanthate, $\text{C}_2\text{H}_5\text{OCS}_2\text{Na}$ as collector
 - uses NaCN as depressant in the mixture of ZnS and PbS when ZnS forms soluble complex and PbS forms froth
 - All are correct statements
58. In the extraction of aluminium
- Process X: applied for red bauxite to remove iron oxide (chief impurity)
- Process Y: (Serpeck's process): applied for white bauxite to remove SiO_2 (chief impurity) then, process X and impurity Z are:
- X = Hall and Heroult's process and Y = SiO_2
 - X = Baeyer's process and Y = SiO_2
 - X = Serpeck's process and Y = iron oxide
 - X = Baeyer's process and Y = iron oxide
59. Give the correct order of initials T or F for following statements. Use T if statement is true and F if it is false.
- In Gold Schmidt thermite process aluminium acts as a reducing agent,
 - Mg is extracted by electrolysis of aq. solution of MgCl_2
 - Extraction of Pb is possible by smelting
 - Red Bauxite is purified by Serpeck's process
- TTTF
 - TTTF
 - FTTT
 - TFTF
60. Give the correct order of initials T or F for following statements. Use T if statement is true and F if it is false.
- Every mineral is an ore but every ore is not a mineral
 - Slag is product formed during extraction of metal by combination of flux and impurities,
 - Highly pure metals can be obtained by zone refining,
 - Carnallite is an ore of magnesium and sodium,
- TTTF
 - FTTF
 - FTTT
 - TFTF
61. When alumina is heated with carbon in nitrogen atmosphere, the products are:
- Al + CO
 - Al + CO_2
 - Al + CO + CO_2
 - AlN + CO

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62. The electrolysis of pure alumina is not feasible because:
 (a) it is bad conductor of electricity and its fusion temperature is high
 (b) it is volatile in nature
 (c) it is decomposed when fused
 (d) it is amphoteric
63. The anode mud in the electrolytic refining of silver contains:
 (a) Zn, Cu, Ag, Au (b) Zn, Ag, Au
 (c) Cu, Ag, Au (d) Au only
64. In which of the following pair of metals, both are commercially extracted from their respective ores by carbon reduction method?
 (a) Zn, Cu (b) Fe, Cu
 (c) Sn, Zn (d) Al, Ag
65. A sulphide ore is first converted into its oxide before reduction. This is done because:
 (a) a sulphide ore cannot be reduced to metal at all
 (b) no reducing agent is found suitable for reducing a sulphide ore
 (c) the enthalpy of formation of CO_2 is more than that of CS_2
 (d) a metal oxide is generally less stable than the metal sulphide
66. The method of electrolytic refining is not suitable in the extraction of:
 (a) Aluminium (b) Copper
 (c) Mercury (d) Silver
67. Choose the correct code regarding Roasting process.
 (I) It is the process of heating ore in air to obtain the oxide
 (II) It is an exothermic process
 (III) It is used for hydrated oxide and oxysalt ore
 (IV) It is used after the concentration of ore
 (a) I, II and III (b) I, II and IV
 (c) I, III and IV (d) I, II, III and IV
68. Which of the following reaction does not occur in Bessemer's converter?
 (a) $2\text{Cu}_2\text{S} + 5\text{O}_2 \longrightarrow 12\text{CuSO}_4 + 2\text{CuO}$
 (b) $2\text{Cu}_2\text{S} + 3\text{O}_2 \longrightarrow 2\text{Cu}_2\text{O} + 2\text{SO}_2 \uparrow$
 (c) $2\text{CuFeS}_2 + \text{O}_2 \longrightarrow \text{Cu}_2\text{S} + 2\text{FeS} + \text{SO}_2$
 (d) $\text{FeO} + \text{SiO}_2 \longrightarrow \text{FeSiO}_3$
69. Blister copper is refined by stirring molten impure metal with green logs of wood because such a wood liberates hydrocarbon gases (like CH_4). This process X is called and the metal contains impurities of Y is
 (a) X = cupellation, Y = Cu_2O
 (b) X = polling, Y = Cu_2O
 (c) X = polling, Y = CuO
 (d) X = cupellation, Y = CuO

COMPREHENSION BASED QUESTIONS

COMPREHENSION-I

Magnesium is a valuable, light weight metal used as a structural material as well as in alloys, batteries, and in chemical synthesis. Although magnesium is plentiful in Earth's crust, it is mainly found in the sea water (after sodium). There is about 1.3 g of magnesium in every kilogram of sea water. The process for obtaining magnesium from sea water employs all three types of reactions i.e., precipitation, acid-base, and redox-reactions.

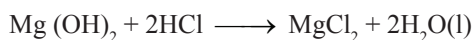
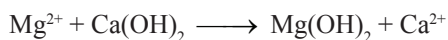
- Precipitation reaction involves formation of:
 (a) insoluble MgCO_3 by adding Na_2CO_3
 (b) insoluble $\text{Mg}(\text{OH})_2$ by adding $\text{Ca}(\text{OH})_2$
 (c) insoluble in MgSO_4 by adding Na_2SO_4
 (b) insoluble MgCl_2 by adding NaCl
- Acid-base reaction involves reaction between:
 (a) MgCO_3 and HCl
 (b) $\text{Mg}(\text{OH})_2$ and H_2SO_4

- $\text{Mg}(\text{OH})_2$ and HCl
- MgCO_3 and H_2SO_4

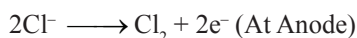
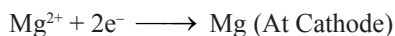
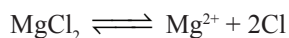
- Redox reaction takes place (in the extraction of Mg):
 (a) in the electrolytic cell when fused MgCl_2 is subjected to electrolysis
 (b) when fused MgCO_3 is heated
 (c) when fused MgCO_3 is strongly heated
 (d) None of the above

COMPREHENSION-II

Dow's process of extraction of Mg involves extraction from sea water. Sea water is concentrated in sun light and is then treated with slaked lime. Magnesium hydroxide is heated in a stream of HCl to give MgCl_2 which is electrolysed to discharge Mg. The mixture is in the ratio 35% MgCl_2 + 50% NaCl + 15% CaCl_2 . NaCl and CaCl_2 are added to lower the fusion temperature and to increase the conductance.



Electrolysis of fused



Mg electrolysed is protected from atmospheric oxidation by a blanket of inert gases.

1. In the hydrated chloride of Mg the value of x is:

- (a) 6 (b) 4
(c) 8 (d) 10

2. Molten mixture contains Mg^{2+} , Na^+ and Ca^{2+} but at cathode only Mg^{2+} is discharged because:

- (a) Standard reduction potential of Mg^{2+} is least among the three
(b) Standard oxidation potential of Mg is the least among the three
(c) Discharge potential of Mg^{2+} is highest
(d) None of these

3. Molten mixture of NaCl and CaCl_2 is added to the heated MgCl_2 because

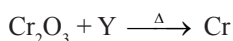
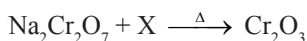
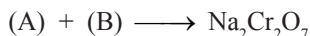
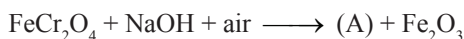
- (a) $\text{MgCl}_2 \cdot x\text{H}_2\text{O} + \text{dry HCl} \xrightarrow{973-1223 \text{ K}} \text{Partially dehydrated MgCl}_2$ and molten mixture ($\text{NaCl} + \text{CaCl}_2$) makes it fully dehydrated

(b) CaCl_2 is dehydrating agent

(c) ($\text{CaCl}_2 + \text{NaCl}$) lowers the melting point of MgCl_2

(d) None of these

COMPREHENSION-III



1. Compound (A) and (B) are:

- (a) Na_2CrO_4 , H_2SO_4
(b) $\text{Na}_2\text{Cr}_2\text{O}_7$, HCl
(c) Na_2CrO_5 , H_2SO_4
(d) $\text{Na}_4[\text{Fe}(\text{OH})_6]$, H_2SO_4

2. CO and (Y) are:

- (a) C and Al (b) Al and C
(c) C in both (d) Al in both

3. Na_2CrO_4 and Fe_2O_3 are separated by:

- (a) dissolving in cone. H_2SO_4
(b) dissolving in NH_3
(c) dissolving in H_2O
(d) dissolving in dil. HCl

ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

1. Pick up the correct statement(s):

- (a) All minerals are ores
(b) All minerals cannot be an ore
(c) All ores are minerals
(d) The minerals from which metals can be extracted profitably are called ores

2. The process (es) by which lighter earthy particles are freed from the heavier particles using water is/are:

- (a) Gravity separation (b) Levigation
(c) Hydraulic washing (d) Leaching

3. Which of the following process makes the ore porous?

- (a) Roasting (b) Calcination
(c) Reduction (d) Distillation

4. The extraction of metals from oxide ores involve:

- (a) Reduction with carbon
(b) Reduction with aluminium

(c) Electrolyte reduction

(d) Reduction with CO

5. Metals which can be extracted by smelting process are:

- (a) Pb (b) Fe
(c) Zn (d) Al

6. Roasting is carried out to:

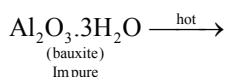
- (a) convert sulphide to oxide and sulphate
(b) remove water of hydration
(c) melt the ore
(d) remove arsenic and sulphur impurities

7. Of the following reduction processes, correct processes are:

- (a) $\text{Fe}_2\text{O}_3 + \text{C} \longrightarrow \text{Fe}$
(b) $\text{ZnO} + \text{C} \longrightarrow \text{Zn}$
(c) $\text{Ca}_3(\text{PO}_4)_2 + \text{C} \longrightarrow \text{P}$
(d) $\text{PbO} + \text{C} \longrightarrow \text{Pb}$

8. Select correct statement regarding silver extraction process.
- When the lead-silver alloy is rich in silver, lead is removed by the cupellation process
 - When the lead-silver alloy is rich in lead, lead is removed by parke's or pattinson's process
 - Zinc forms an alloy with lead, from which lead is separated by distillation
 - Zinc forms an alloy with silver, from which zinc is separated by distillation

9. In the extraction of aluminium metal, one of the process is summarised as follows:
Which of the following entries correctly summarises reagents, electrodes and products of the process?



	I	II	III	IV	V
(a)	NaOH	Al ³⁺	HF	Na ₃ AlF ₆	Cathode
(b)	NaOH	NaAlO ₂	CO ₂	NaF	Anode
(c)	H ₂ SO ₄	Al ₂ (SO ₄) ₃	NH ₃	Na ₃ AlF ₆	Cathode
(d)	NaOH	NaAlO ₂	CO ₂	Na ₃ AlF ₆	Cathode

10. During the production of iron and steel.
- The oxide ore is primarily reduced to iron by solid coke according to the reaction

$$2\text{Fe}_2\text{O}_3 + 3\text{C} \longrightarrow 4\text{Fe} + 3\text{CO}_2$$
 - The oxide ore is reduced by the carbon monoxide according to the reaction

$$\text{Fe}_2\text{O}_3 + 3\text{CO} \longrightarrow 2\text{Fe} + 3\text{CO}_2$$
 - Major silica impurities are removed as calcium silicate slag by addition of a fluxing agent limestone
 - The silicate slag is used in manufacturing cement
11. Which of the following ores is/are oxide ore(s)?
- Tinstone
 - Bauxite
 - Cryolite
 - Carnallite
12. Aluminothermy used for the spot welding of large iron structures is based upon the fact that:
- As compared to iron, aluminium has greater affinity for oxygen
 - As compared to aluminium, iron has greater affinity for oxygen
 - Reaction between aluminium and oxygen is endothermic
 - Reaction between iron oxide and aluminium is exothermic
13. Roasting of copper pyrites is done:
- to remove moisture

- to oxidize free sulphur
- to decompose pyrite into Cu₂S and FeS
- to remove volatile organic impurities

14. The smelting of iron in a blast furnace involves the following processes:
- combustion
 - reduction
 - slag formation
 - sublimation
15. Highly electropositive metals cannot be extracted by carbon reduction process because these:
- Metals combine with carbon to form carbides
 - Metals do not react with carbon
 - Metal oxides are not reduced by carbon
 - Loss of metal is more by vapourisation
16. Which of the following is a correct statement?
- Calamine is the ore of zinc
 - Pyrolusite is the ore of manganese
 - Cassiterite is the ore of tin
 - Calcite is the ore of calcium
17. Which one of the following metals can be extracted by aluminothermic process?
- Manganese
 - Iron
 - Chromium
 - Magnesium
18. For which of the following metal, the carbon reduction cannot be used?
- Lead
 - Manganese
 - Tungsten
 - Iron
19. Which of the following reaction in the blast furnace is/are endothermic?
- $\text{C(s)} + \text{O}_2\text{(g)} \rightleftharpoons \text{CO}_2\text{(g)}$
 - $\text{CO}_2\text{(g)} + \text{C(s)} \rightleftharpoons 2\text{CO(g)}$
 - $\text{CaCO}_3\text{(s)} \rightleftharpoons \text{CaO(s)} + \text{CO}_2\text{(g)}$
 - $\text{Fe}_2\text{O}_3\text{(s)} + 3\text{CO(g)} \rightleftharpoons 2\text{Fe(l)} + 3\text{CO}_2\text{(g)}$
20. The advantage (s) of using carbon to reduce a number of oxides and other compounds are:
- easy availability of coke
 - low cost of carbon
 - tendency of carbon to show catenation
 - presence of carbon lowers the melting point of the oxides
21. The disadvantage of carbon reduction method are:
- high temperature needed which is expensive and requires the use of a blast furnace
 - many metals combine with carbon forming carbides
 - carbon combines with oxygen to form poisonous CO
 - carbon cannot be used with highly electroposi-

tive metals

22. The chemical treatment of the ore for concentration is done in the case of:
 - (a) aluminium (b) silver
 - (c) copper (d) gold
23. In which of the following pairs, both the minerals are oxides?
 - (a) Sylvine, saltpetre (b) Cassiterite, litharge
 - (c) Siderite, corundum (d) Cuprite, tinstone
24. Which of the following metals are extracted from its ore by using self-reduction method?
 - (a) Copper (b) Mercury
 - (c) Lead (d) Silver
25. Froth floatation:
 - (a) is a physical method of separating mineral from the gangue
 - (b) is a method to concentrate the ore depending on the difference in wetability of gangue and the ore
 - (c) is used for the sulphide ores
 - (d) is a method in which impurities sink to the bottom
26. The function of adding cryolite in the electrolytic reduction of alumina by Hall-Heroult process is to:
 - (a) dissolve alumina
 - (b) lower the melting point of alumina
 - (c) lower the fuel bill
 - (d) increase the electrical conductivity of alumina
27. Which of the following pair consists of ore of the same metal?
 - (a) Bauxite, Limonite (b) Haematite, Siderite
 - (c) Cinnabar, Cassiterite
 - (d) Galena, Cerrusite
28. The furnace lining in steel manufacture consists of:
 - (a) CaO (b) SiO₂
 - (c) MgO (d) CaCO₃
29. Which of the following reduction reactions are actually employed in commercial extraction of metals?
 - (a) $\text{Fe}_2\text{O}_3 + 2\text{Al} \longrightarrow \text{Al}_2\text{O}_3 + 2\text{Fe}$
 - (b) $\text{Cr}_2\text{O}_3 + 2\text{Al} \longrightarrow \text{Al}_2\text{O}_3 + 2\text{Cr}$
 - (c) $2\text{Na} [\text{Au}(\text{CN})_2] + \text{Zn} \longrightarrow \text{Na}_2[\text{Zn}(\text{CN})_4] + 2\text{Au}$
 - (d) $\text{Cu}_2\text{S} + \text{Pb} \longrightarrow \text{Cu} + \text{PbS} \downarrow$
30. Which of the following reaction(s) occur during calcination?
 - (a) $\text{CaCO}_3 \longrightarrow \text{CaO} + \text{CO}_2$
 - (b) $4\text{FeS}_2 + 11\text{O}_2 \longrightarrow 2\text{Fe}_2\text{O}_3 + 8\text{SO}_2$
 - (c) $2\text{Al}(\text{OH})_3 \longrightarrow \text{Al}_2\text{O}_3 + 3\text{H}_2\text{O}$
 - (d) $\text{CuS} + \text{CuSO}_4 \longrightarrow 2\text{Cu} + 2\text{SO}_2$
31. Amphoteric nature of aluminium is employed in

which of the following process for extraction of aluminium?

- (a) Baeyer's process (b) Hall's process
 - (c) Serpek's process (d) Bow's process
32. The chief reaction (s) occurring in blast furnace during extraction of iron from haematite is/are:
 - (a) $\text{Fe}_2\text{O}_3 + 3\text{CO} \longrightarrow 2\text{Fe} + 3\text{CO}_2$
 - (b) $\text{FeO} + \text{SiO}_2 \longrightarrow \text{FeSiO}_3$
 - (c) $\text{Fe}_2\text{O}_3 + \text{C} \longrightarrow 2\text{Fe} + 3\text{CO}$
 - (d) $\text{CaO} + \text{SiO}_2 \longrightarrow \text{CaSiO}_3$
 33. Which of the following are true for electrolytic extraction of aluminium?
 - (a) Cathode material contains graphite
 - (b) Anode material contains graphite
 - (c) Cathode reacts away forming CO₂
 - (d) Anode reacts away forming CO₂
 34. Which of the following is true for calcination of a metal ore?
 - (a) It makes the ore more porous.
 - (b) The ore is heated to a temperature when fusion just begins.
 - (c) Hydrated salts lose their water of crystallization.
 - (d) Impurities of S, As and Sb are removed in the form of their volatile oxides.
 35. The difference(s) between roasting and calcination is/are:
 - (a) roasting is highly endothermic while calcination is not
 - (b) partial fusion occurs in calcination but not in roasting
 - (c) calcination is performed in limited amount of air but roasting employs excess air
 - (d) combustion reaction occur in roasting but not in calcinations

The questions given below consist of "Assertion" and their "Reason". Use the following key to choose the appropriate answer.

- (a) If both assertion and reason are CORRECT, and reason is the CORRECT explanation of the assertion
- (b) If both assertion and reason are CORRECT, but reason is NOT the CORRECT explanation of the assertion
- (c) If assertion is CORRECT but reason is INCORRECT
- (d) If assertion is INCORRECT but reason is CORRECT

1. **Assertion:** All the ores are mineral.
Reason: Ores contains metals in combined state.

ASSERTION/REASON TYPE QUESTIONS

2. **Assertion:** Nitriding is the process of heating steel in presence of N_2 to form iron nitrides.
Reason: The surface of steel becomes hard after nitriding process.
3. **Assertion:** Ores are generally converted into oxides, prior to reduction.
Reason: Metal oxides can be easily reduced.
4. **Assertion:** In the extraction of Ag, complex $Na[Ag(CN)_2]$ is reacted with Zn.
Reason: Zn is d-block transition metal.
5. **Assertion:** Thermite mixture $Fe_2O_3 + Al$ (powder) is used in the welding.
Reason: Al is a good reductant.
6. **Assertion:** In froth floatation process sodium ethyl xanthate is used as collector.
Reason: Sulphide ores are water soluble.
7. **Assertion:** Cryolite is used in electrolytic extraction of Al from alumina.
Reason: It dissolves alumina.
8. **Assertion:** $CuFeS_2$ is concentrated by froth floatation method.
Reason: $CuFeS_2$ is main ore of copper.
9. **Assertion:** In the electrolytic reduction of Al_2O_3 , cryolite is used.
Reason: Cryolite is an ore of aluminium.
10. **Assertion:** Wrought iron is more malleable and ductile than steel.
Reason: It contains slightly less percentage of carbon.
11. **Assertion:** Lead, tin and bismuth are purified by liquation method.
Reason: Lead, tin and bismuth have low m.p. as compared to impurities.
12. **Assertion:** Al_2O_3 is converted into Al by reduction with carbon at high temp.
Reason: Carbon has greater affinity for oxygen than aluminium.
13. **Assertion:** Reduction of ZnO with carbon is done at $1100^\circ C$.
Reason: ΔG° is negative at this temperature thus, process is spontaneous.
14. **Assertion:** Desilverisation of lead is done by Parke's method.
Reason: When lead-silver alloy is poor in silver, zinc is added to molten ore.

MATCH THE COLUMN TYPE QUESTIONS

Column-I and Column-II contains four entries each. Entries of column-I are to be matched with some entries of column-II. Each entry of column-I may have the matching with one or more than one entries of column-II.

1. **Column-I**
(A) Ca
(B) Zn
(C) Cr
(D) Ag
- Column-II**
(P) Found as its native state
(Q) Found as its sulphide
(R) Found as its carbonate
(S) Found as its oxide
2. **Column-I**
(A) Cuprite
(B) Cerussite
(C) Kainite
(D) Calamine
- Column-II**
(P) Sulphate ore
(Q) Carbonate ore
(R) Oxide ore
(S) Chloride ore

3. **Column-I**
(A) Poling
(B) Cupellation
(C) Electro-refining
(D) van Arkel method
- Column-II**
(P) Titanium
(Q) Copper
(R) Silver
(S) In

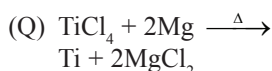
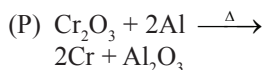
4. **Column-I**
(A) Metal which occur in the native state in nature is
(B) The oxides of metal that can be commercially reduced by Aluminothermic reduction process is
(C) van Arkel method is used for preparing ultrapure metal of
(D) Auto reduction process is employed for the sulphide ore of
- Column-II**
(P) Hg
(Q) Ti
(R) Cr
(S) Ag

5. Column-I

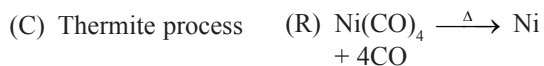
(A) Mond's process

(B) van Arkel process

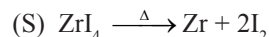
Column-II



(C) Thermite process



(D) Kroll's process



RECENT YEARS' QUESTION OF IIT-JEE AND AIEEE

1. The metal extracted by leaching with cyanide is:

[AIEEE 2002]

- (a) Mg (b) Ag
(c) Cu (d) Na

2. Aluminium is extracted by the electrolysis of:

[AIEEE 2002]

- (a) Bauxite
(b) Alumina
(c) Alumina mixed with molten cryolite
(d) Molten cryolite

3. When the sample Cu with Zn impurity is to be purified by electrolysis, the appropriate electrodes are:

[AIEEE 2002]

- | Cathode | Anode |
|-------------------|---------------|
| (a) Pure Zn | Pure Cu |
| (b) Impure sample | Pure Cu |
| (c) Impure Zn | Impure sample |
| (d) Pure copper | Impure sample |

4. The substance not likely to contain CaCO_3 is:

[AIEEE 2003]

- (a) Sea shells (b) Dolomite
(c) Marble statue (d) Calcined gypsum

5. Which of the following ores is best concentrated by froth floatation process?

[AIEEE 2004]

- (a) Malachite (b) Cassiterite
(c) Galena (d) Magnetite

6. During the process of electrorefining of copper some metals present as impurity settle as anode mud. These are:

[AIEEE 2005]

- (a) Sn and Ag (b) Pb and Zn
(c) Ag and Au (d) Fe and Ni

7. On heating a mixture of Cu_2O and Cu_2S , we get:

[AIEEE 2005]

- (a) $\text{Cu} + \text{SO}_2$ (b) $\text{Cu} + \text{SO}_3$
(c) $\text{CuO} + \text{CuS}$ (d) Cu_2SO_3

8. The chemical process in the production of steel from haematite ore involve:

[IIT 2000]

- (a) Reduction
(b) Oxidation
(c) Reduction followed by oxidation
(d) Oxidation followed by reduction

9. The chemical composition of slag formed during smelting process in the extraction of Cu is:

[IIT 2001]

- (a) $\text{Cu}_2\text{O} + \text{FeS}$ (b) FeSiO_3
(c) CuFeS_2 (d) $\text{Cu}_2\text{S} + \text{FeO}$

10. Electrolyte reduction of alumina to aluminium by Hall-Heroult process is carried out:

[IIT 2000]

- (a) In the presence of NaCl
(b) In the presence of fluorite
(c) In the presence of cryolite which forms a melt with lower melting temperature
(d) In the presence of cryolite which forms a melt with higher melting temperature

11. Which of the process is used in the extractive metallurgy of Mg?

[IIT 2002]

- (a) Fused salt electrolysis
(b) Self reduction
(c) Aqueous solution electrolysis
(d) Thermite reduction

12. Roasted gold ore + $\text{CN}^- + \text{H}_2\text{O} \xrightarrow{\text{O}_2} [\text{X}] + \text{OH}^-$ [X] + Zn $\longrightarrow$ [Y] + Au.

[IIT 2003]

- (a) $\text{X} = [\text{Au}(\text{CN})_2]^-$; $\text{Y} = [\text{Zn}(\text{CN})_4]^{2-}$
(b) $\text{X} = [\text{Au}(\text{CN})_4]^{3-}$; $\text{Y} = [\text{Zn}(\text{CN})_4]^{2-}$
(c) $\text{X} = [\text{Au}(\text{CN})_2]^-$; $\text{Y} = [\text{Zn}(\text{CN})_6]^{4-}$
(d) $\text{X} = [\text{Au}(\text{CN})_4]^{3-}$; $\text{Y} = [\text{Zn}(\text{CN})_6]^{2-}$

13. By which process Pb and Sn are extracted respectively by:

[IIT 2004]

- (a) Carbon reduction: self reduction
(b) Self reduction: carbon reduction

- (c) Electrolytic reduction: cyanide process
(d) Cyanide process: electrolytic reduction

14. Extraction of zinc from zinc blende is achieved by:
[IIT 2007]

- (a) electrolytic reduction
(b) roasting followed by reduction with carbon
(c) roasting followed by reduction with another metal
(d) roasting followed by self-reduction

15. Which of the following factors is of no significance for roasting sulphide ores to the oxides and not subjecting the sulphide ores to carbon reduction directly?
[AIEEE 2008]

- (a) CO_2 is more volatile than CS_2
(b) Metal sulphides are thermodynamically more stable than CS_2
(c) CO_2 is thermodynamically more stable than CS_2
(d) Metal sulphides are less stable than the corresponding oxides

ANSWER KEYS

ONE CORRECT OPTION TYPE QUESTIONS

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (b) | 2. (c) | 3. (c) | 4. (d) | 5. (b) | 6. (c) | 7. (a) | 8. (d) | 9. (a) | 10. (b) |
| 11. (b) | 12. (d) | 13. (c) | 14. (c) | 15. (a) | 16. (d) | 17. (d) | 18. (c) | 19. (c) | 20. (b) |
| 21. (b) | 22. (c) | 23. (d) | 24. (a) | 25. (b) | 26. (c) | 27. (a) | 28. (c) | 29. (c) | 30. (b) |
| 31. (c) | 32. (c) | 33. (a) | 34. (d) | 35. (d) | 36. (b) | 37. (c) | 38. (a) | 39. (a) | 40. (d) |
| 41. (b) | 42. (d) | 43. (d) | 44. (c) | 45. (d) | 46. (d) | 47. (c) | 48. (b) | 49. (a) | 50. (d) |
| 51. (b) | 52. (b) | 53. (b) | 54. (c) | 55. (d) | 56. (b) | 57. (d) | 58. (b) | 59. (d) | 60. (b) |
| 61. (d) | 62. (a) | 63. (d) | 64. (c) | 65. (d) | 66. (c) | 67. (b) | 68. (c) | 69. (b) | |

COMPREHENSION BASED QUESTIONS

Comprehension-I

1. (b) 2. (c) 3. (a)

Comprehension-II

1. (a) 2. (b) 3. (c)

Comprehension-III

1. (a) 2. (a) 3. (c)

ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

- | | | | | |
|------------------|-----------------|------------------|-----------------|------------------|
| 1. (b, c, d) | 2. (a, b, c) | 3. (a, b) | 4. (a, b, c, d) | 5. (a, b, c) |
| 6. (a, b, d) | 7. (a, b, c, d) | 8. (a, b, d) | 9. (d) | 10. (b, c, d) |
| 11. (a, b) | 12. (a, d) | 13. (a, b, c, d) | 14. (a, b, c) | 15. (a, d) |
| 16. (a, b, c, d) | 17. (a, b, c) | 18. (b, c) | 19. (b, c) | 20. (a, b) |
| 21. (a, b) | 22. (a, b, d) | 23. (a, d) | 24. (a, b, c) | 25. (a, b, c, d) |
| 26. (a, b, c, d) | 27. (b, d) | 28. (a, c) | 29. (b, c) | 30. (a, c) |
| 31. (a, b) | 32. (a, d) | 33. (a, b, d) | 34. (a, c) | 35. (c, d) |

ASSERTION/REASON TYPE QUESTIONS

- | | | | | | | | |
|--------|---------|---------|---------|---------|---------|--------|--------|
| 1. (c) | 2. (d) | 3. (c) | 4. (c) | 5. (b) | 6. (c) | 7. (a) | 8. (b) |
| 9. (b) | 10. (a) | 11. (a) | 12. (c) | 13. (a) | 14. (a) | | |

MATCH THE COLUMN TYPE QUESTIONS

- | | | | | | | | |
|--|--|--|------------------------------------|-------------------------------------|----------------------------------|-------------------------------------|---------------------------------|
| 1. $\text{A} \rightarrow \text{R};$ | $\text{B} \rightarrow \text{Q, R, S};$ | $\text{C} \rightarrow \text{S};$ | $\text{D} \rightarrow \text{P, Q}$ | 2. $\text{A} \rightarrow \text{R};$ | $\text{B} \rightarrow \text{Q};$ | $\text{C} \rightarrow \text{P, S};$ | $\text{D} \rightarrow \text{Q}$ |
| 3. $\text{A} \rightarrow \text{Q, S};$ | $\text{B} \rightarrow \text{R},$ | $\text{C} \rightarrow \text{Q, R, S};$ | $\text{D} \rightarrow \text{P}$ | 4. $\text{A} \rightarrow \text{R};$ | $\text{B} \rightarrow \text{S};$ | $\text{C} \rightarrow \text{P};$ | $\text{D} \rightarrow \text{Q}$ |
| 5. $\text{A} \rightarrow \text{R};$ | $\text{B} \rightarrow \text{S};$ | $\text{C} \rightarrow \text{P};$ | $\text{D} \rightarrow \text{Q}$ | | | | |

RECENT YEARS' QUESTIONS OF IIT-JEE AND AIEEE

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|--------|--------|--------|--------|---------|
| 1. (b) | 2. (c) | 3. (d) | 4. (d) | 5. (c) | 6. (c) | 7. (a) | 8. (d) | 9. (b) | 10. (c) |
| 11. (a) | 12. (a) | 13. (b) | 14. (b) | 15. (b) | | | | | |

Salt Analysis

INTRODUCTION

Qualitative inorganic analysis deals with the identification of radicals (cations and anions) in an inorganic salt or in mixture of salts. A salt consists of two parts known as radicals. The positively charged part of salt (cation) which has been derived from a base is termed basic radical and the negatively charged part of the salt (anion) which has been derived from an acid is termed acidic radical.

Preparation of the Soda Extract

Intimately mix the solid sample with 3-4 times (in volume) of anhydrous sodium carbonate, add just sufficient water to dissolve the sodium carbonate. In the case of dissolved samples, add 5 g anhydrous sodium carbonate to 10ml solution. Heat the mixture under reflux for 15 minutes and filter. Wash and discard the residue. Acidify the combined filtrate and washings gently with 8M nitric acid, boil gently to expel carbon dioxide and allow to cool. Add 2M ammonia solution portionwise and with stirring, until the solution just turns alkaline. Boil gently for 1-2 minutes to remove excess ammonia. Cool, and filter again if necessary. The filtrate will be termed here the 'neutralized soda extract'.

The identification of the acidic radicals, the following scheme is followed.

Group I: This group consists of radicals which are decompose by dilute H_2SO_4 or dilute HCl . These radicals are (i) carbonate, (ii) sulphite, (iii) sulphide, (iv) nitrite and (v) acetate.

Group II: This group consists of radicals which are detected by concentrated H_2SO_4 . These radicals are (i) chloride, (ii) bromide, (iii) iodide, (iv) nitrate and (v) oxalate.

Group III: The radicals, which do not give any characteristic gas with dilute and concentrated H_2SO_4 . These radicals are (i) sulphate (ii) phosphate, (iii) borate and (iv) fluoride, (v) thiosulphate.

Group I

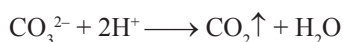
For analysis of anions of I group, sample is treated with dilute H_2SO_4 or HCl and the observations are given below in the table.

Table 14.1 Test with dil. H_2SO_4 or HCl

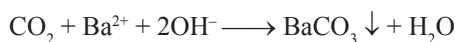
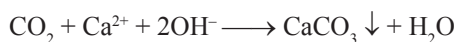
Observations	Inference
1. Colourless gas is evolved with effervescence, gas is odourless and produces turbidity when passed through lime water.	CO_2 from carbonates and bicarbonates
2. Colourless gas evolved with suffocating odour, turns acidified $\text{K}_2\text{Cr}_2\text{O}_7$ paper green.	SO_2 from sulphites
3. Nitrous fumes, reddish-brown colour and pungent odour.	NO_2 from nitrites
4. Colourless gas, odour of rotten eggs, lead acetate paper turns black and cadmium acetate paper yellow.	H_2S from sulphides
5. Odour of vinegar (CH_3COOH), turns neutral FeCl_3 red.	CH_3COOH from acetates

1. Carbonate (CO_3^{2-})

Dilute HCl: Gives effervescence, due to the evolution of carbon dioxide.



The gas gives turbidity with lime water and baryta water.

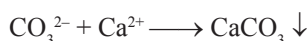
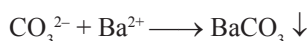


On prolonged passage of carbon dioxide in lime water, the turbidity slowly disappears due to the formation of soluble hydrogen carbonate.

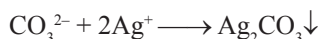


The following tests are performed with the aqueous salt solution.

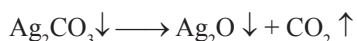
Barium chloride or Calcium chloride solution: White ppt of barium or calcium carbonate is obtained, which is soluble in mineral acid.



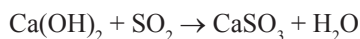
Silver nitrate solution: White ppt of silver carbonate is obtained.



The ppt so obtained is soluble in nitric acid and in ammonia. The ppt becomes yellow or brown on addition of excess reagent and same may also happen if the mix is boiled, due to the formation of silver oxide

**Limitation**

- (a) Carbonates of bismuth and barium are not easily decomposed by dilute H_2SO_4 . Dilute HCl should be used.
- (b) Sulphur dioxide evolved from sulphites also turns lime water milky. In the test of carbonates, the evolved gas must be colourless and odourless (SO_2 is a pungent gas).



White ppt.

- **Carbonate in the presence of sulphite:** Sulphites, on treatment with dilute sulphuric acid, liberate sulphur dioxide which, like carbon dioxide, produces, a turbidity with lime or baryta water. The dichromate test for sulphites is, however, not influenced by the presence of carbonates. To detect carbonates in the presence of sulphites, treat the solid mixture with M sulphuric acid and pass the evolved gases through a small wash bottle or boiling tube containing potassium

dichromate solution and M sulphuric acid. The solution will turn green and the sulphur dioxide will, at the same time, be completely removed; the residual gas is then tested with lime or baryta water in the usual manner.

An alternative procedure is to add a little sodium potassium dichromate or a small volume of 3% hydrogen peroxide solution to the mixture and then to warm with M sulphuric acid, the evolved gas is then passed through lime or baryta water.

2. Sulphites (SO_3^{2-})

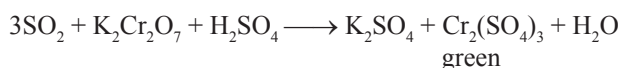
Dilute HCl or Dilute H_2SO_4 : decomposes with the evolution of sulphur dioxide



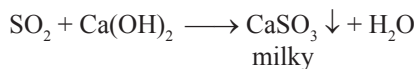
The gas has a suffocating odour of burning sulphur.

The following tests are performed with the aqueous salt solution.

Acidified potassium dichromate solution: Turns filter paper moistened with acidified potassium dichromate solution, green due to the formation of Cr^{3+} ions.



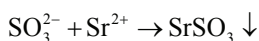
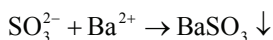
Lime water: On passing the gas through lime water, a milky ppt is formed.



Precipitate dissolves on prolonged passage of the gas, due to the formation of hydrogen sulphite ions.

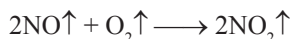
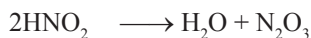
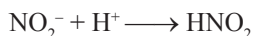


Barium chloride or Strontium chloride solution: Gives white precipitate of barium or strontium sulphite.



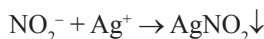
3. Nitrites (NO_2^-)

- (i) **Dilute HCl and Dilute H_2SO_4 :** Adding to solid nitrite in cold HCl yields pale blue liquid (due to the presence of free nitrous acid HNO_2 or its anhydride N_2O_3) and the evolution of brown fumes of nitrogen dioxide, the latter being largely produced by combination of nitric oxide with the oxygen of the air.

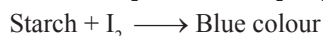
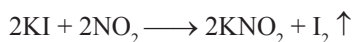


The following tests are performed with the aqueous salt solution.

- (ii) **Silver nitrate solution:** White crystalline precipitate is obtained.

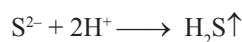


- (iii) **Turns acidified KI-starch paper blue**



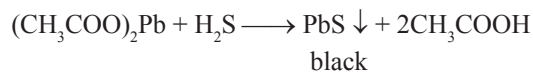
4. Sulphide (S^{2-})

- (i) **Dilute HCl or Dilute H_2SO_4 :** A colourless gas smelling of rotten eggs (H_2S) is evolved.



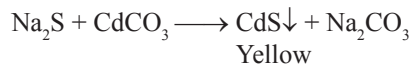
14.4 Salt Analysis

(ii) *The gas turns lead acetate paper black*

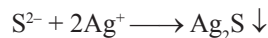


The following tests are performed with the aqueous salt solution.

(iii) *Gives yellow precipitate with CdCO_3*



(iv) *Silver nitrate solution:* black precipitate of silver sulphide insoluble in cold but soluble in hot dil nitric acid.

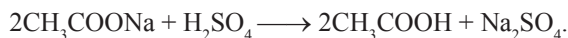


(v) *Sodium nitroprusside solution:* Turns sodium nitroprusside solution purple

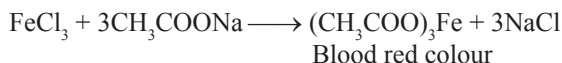


5. Acetate: (All acetates are soluble in water)

Acetates when heated with dilute H_2SO_4 decompose to give acetic acid vapours, which possess characteristic smell of vinegar.



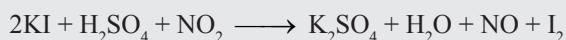
On addition of neutral FeCl_3 solution to the solution of acetate, blood red colouration develops due to the formation of ferric acetate.



Worked-out Example

1. NO_2 turns acidic KI-starch paper blue, why?

Ans. NO_2 turns I^- into I_2 , which gives dark blue complex with starch.



Group II

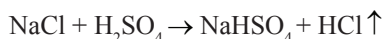
For analysis of anions of II group, sample is treated with conc. H_2SO_4 . The observations are given below in the table.

Table 14.2 Test with Conc. H_2SO_4

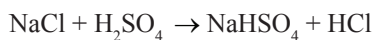
Observations	Inference
1. Colourless gas evolved with pungent odour and white fumes in air, white fumes of NH_4Cl in contact with NH_3 , Cl_2 evolved on addition of MnO_2	HCl from chlorides
2. Gas evolved with pungent odour, reddish colour and fumes in moist air, addition of MnO_2 increased red fumes	HBr and Br_2 from bromides
3. Violet vapours evolved	HI and I_2 from iodides
4. Pungent brown fumes. Colour deepens on addition of copper turnings (if nitrite is absent)	HNO_3 and NO_2 from nitrate and nitrites
5. Colourless gas evolved, turns lime water milky and also burns with a blue flame	CO and CO_2 from oxalates

1. Chloride (Cl⁻)

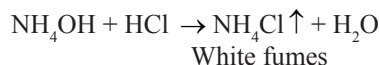
Colourless pungent fumes of hydrogen chloride are evolved on heating the solid chloride with concentrated H₂SO₄.



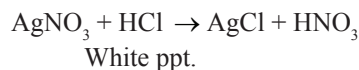
- (a) Yellowish-green gas of chlorine with suffocating odour is evolved when the solid chloride mixed with manganese dioxide is heated with concentrated H₂SO₄.



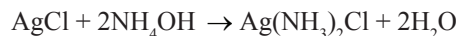
- (b) The gas evolved by heating chloride with sulphuric acid forms white fumes of ammonium chloride with NH₄OH.



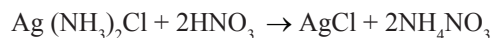
- (c) The gas evolved by heating chloride with H₂SO₄ forms a curdy precipitate of silver chloride with silver nitrate solution.



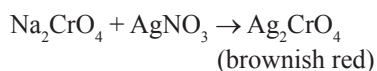
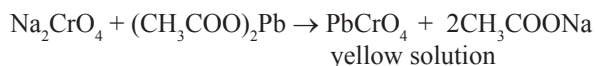
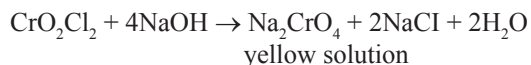
Note: The curdy precipitate dissolves in ammonium hydroxide by forming a complex salt.



When the solution having the silver complex is acidified with dilute nitric acid, a white precipitate of silver chloride is again formed.

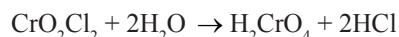
**Chromyl chloride test for chloride**

The chloride salt is mixed with K₂Cr₂O₇ and heated with concentrated H₂SO₄. Under this condition orange – red vapours of chromyl chloride are evolved. These are passed through dilute NaOH solution to give yellow solution of Na₂CrO₄. The yellow solution is acidified with CH₃COOH and lead acetate solution is added. Formation of yellow precipitate of lead chromate shows the presence of chloride.



Note:

- (i) This test is not given by the chlorides of mercury, tin, silver, lead and antimony. (The chlorides of mercury, owing to their slight ionization, do not respond to this test, only partial conversion to CrO₂Cl₂ occurs with the chloride of lead silver, antimony and tin.
- (ii) The chromyl chloride test is always to be performed in a dry test tube; otherwise, the chromyl chloride vapours will be hydrolysed in the test tube.



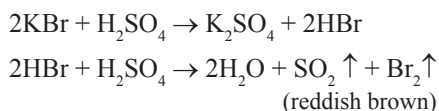
14.6 Salt Analysis

(iii) Bromides and iodides do not give this test.

Bromides and iodides give rise to free halogens, which yield colorless solution with sodium hydroxide; Fluorides give rise to the volatile chromyl fluoride, CrO_2F_2 which is decomposed by water and hence should be absent or removed.

2. Bromide (Br^-)

Conc. H_2SO_4 : Gives reddish brown vapours of bromine accompanying the hydrogen bromide.

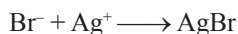


Manganese dioxide and concentrated sulphuric acid: When a mix of solid bromide, MnO_2 and conc. H_2SO_4 is heated reddish brown vapours of bromine are evolved.



The following tests are performed with the aqueous salt solution.

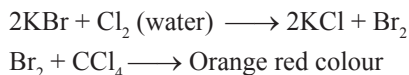
Silver nitrate solution: Pale yellow precipitate of silver bromide is obtained. This precipitate is sparingly soluble in dil but readily soluble in concentrated ammonia solution and insoluble in dilute HNO_3 .



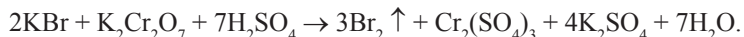
Lead acetate solution: White crystalline precipitate of lead bromide which is soluble in boiling water.



Chlorine water: When chlorine water is added to a mixture of CCl_4 and a solution of bromide liberates free bromine is liberated, which colours the organic layer orange – red.

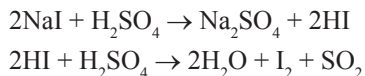


Potassium dichromate and concentrated H_2SO_4 : When a mix of solid bromide, $\text{K}_2\text{Cr}_2\text{O}_7$, and concentrated H_2SO_4 is heated and on passing the evolved vapours into water, a yellowish brown solution is obtained.

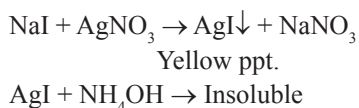


3. Iodide (I^-)

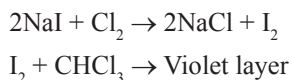
(i) Iodide salt is decomposed by concentrated H_2SO_4 to evolve violet vapours of I_2 .



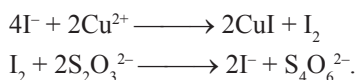
(ii) AgNO_3 reacts with iodide to form a yellow precipitate of AgI which is insoluble in NH_4OH .



(iii) Cl_2 water liberates I_2 from NaI which imparts violet colour to CHCl_3 layer.



(iv) With Copper sulphate solution it gives brown precipitate consisting of a mixture of copper (I) iodide and iodine and on addition of hypo solution brown precipitate changes to white precipitate.



Some Noteworthy Points

- AgCl is not soluble in H_2O and dilute HNO_3 but soluble in NH_4OH , NaCN, KCN, hypo etc., due to formation of complexes
- AgBr is not soluble in HNO_3 but completely soluble in KCN, NaCN and hypo
- AgI is insoluble in HNO_3 and NH_4OH but soluble in hypo
- AgCl, AgBr, AgI all are soluble in hypo
- AgF is soluble in water.

Chloride in the presence of bromide and/or iodide this procedure involves the removal of the bromide and iodide (but not chloride) by oxidation with potassium or ammonium peroxodisulphate in the presence of dilute sulphuric acid. The free halogens are thus liberated, and may be removed either by simple evaporation (addition of water may be necessary to maintain the original volume) or by evaporation at about 80°C in a stream of air.

It must be emphasized that these reactions take place only in acid media. Add solid potassium peroxodisulphate and 2M sulphuric acid to the neutralized soda extract of the mixed halides contained in a conical flask; heat the flask to about 80°C , and aspirate a current of air through the solution with the aid of a filter pump until the solution is colourless (Fig. T is a drawn-out capillary tube). An all-glass device can be assembled from Quick-fit conical flask FE 50/2 and trap head MF 28/52. Add more potassium per-oxodisulphate or water as may be found necessary. Test the residual colourless liquid for chloride with silver nitrate solution and 2M nitric acid.

Do not mistake precipitated silver sulphate for silver chloride. In a modification of this method, lead dioxide is substituted for potassium peroxodisulphate. Acidify the solution with 2M acetic acid, add lead dioxide, and boil the mixture until bromine and iodine are no longer evolved. Filter, and test the filtrate, which should be colourless, with silver nitrate solution and dilute nitric acid.

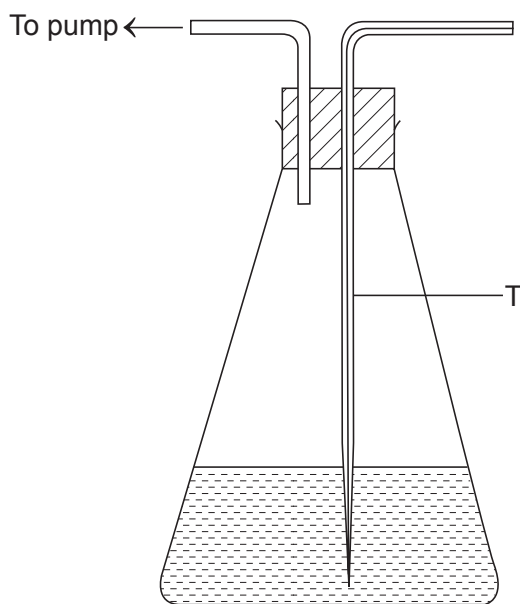


Fig. 14.1

Chloride in the presence of bromide: $\text{Cl}^- + \text{Br}^-$ warm the solid mixture with a little solid potassium dichromate and concentrated sulphuric acid in a small distilling flask (Fig.), and pass the vapours, which contain chromyl chloride, into sodium hydroxide solution. Test for chromate, which proves the presence of a chloride, with hydrogen peroxide and amyl alcohol or with the diphenylcarbazide reagent.

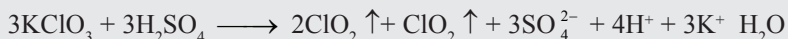
Chloride in the presence of iodide: $\text{Cl}^- + \text{I}^-$ add excess 0.1M silver nitrate solution to the neutralized soda extract and filter; reject the filtrate. Wash the precipitate with 2M ammonia solution and filter again. Add 2M nitric acid to the washings; a white precipitate of silver chloride indicates the presence of chloride.

Bromide and iodide in the presence of each other and of chloride: $\text{Br}^- + \text{I}^- + \text{Cl}^-$ the presence of a chloride does not interfere with the reactions described below. To the soda extract, strongly acidified with 6M hydrochloric acid, add 1–2 drops of chlorine water (the solution obtained by carefully acidifying a M solution of sodium hypochlorite with 2M hydrochloric acid may also be used) and 2–3 ml chloroform; shake; a violet colour indicates iodide. Continue the addition of chlorine water or of acidified sodium hypochlorite solution drop by drop to oxidize the iodine to iodate and shake after each addition. The violet colour will disappear, and a reddish-brown colouration of the chloroform, due to dissolved bromine (and/or bromine monochloride, BrCl), will be obtained if a bromide is present. If iodide alone is present, the solution will be colourless after the violet colour has disappeared.

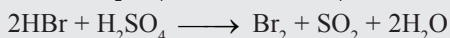
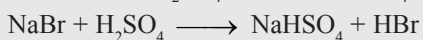
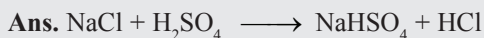
Worked-out Examples

2. Chromyl chloride test for chlorides is not performed if mixture contains chlorates, why?

Ans. Because ClO_3^- ions react with H_2SO_4 to form chlorine dioxide (yellow greenish gas) which dissolves in H_2SO_4 forming orange-yellow solution. This solution is highly explosive.



3. NaCl on heating with conc. H_2SO_4 gives HCl whereas NaBr and NaI give Br_2 and I_2 respectively. Why?



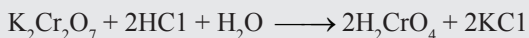
HBr and HI are reducing agent whereas H_2SO_4 is oxidizing agent and thus Br_2 and I_2 are formed.

4. Why bromides and iodides do not respond to chromyl chloride test?

Ans. Because chromyl bromide (CrO_2Br_2) and chromyl iodide (CrO_2I_2) compounds are unknown and are not formed. In such cases bromine and iodine are evolved.

5. Hydrochloric acid contains Cl^- ions but it does not give positive chromyl chloride test, why?

Ans. Because in presence of water chromic acid is obtained in place of chromyl chloride.



6. During testing of Cl and Br by chloroform test, CHCl_3 layer first turns violet and then brown, confirming the presence of I⁻ and Br⁻ respectively. When layer turns brown then violet colour disappears or not and why?

Ans. Violet and brown colours are due to oxidation of I^- to I_2 and Br^- to Br_2 respectively. The violet colour of I_2 disappears when layer is brown due to formation of IO⁻ (iodate ion) which is colourless.

7. Hg, Sn and Sb chlorides do not give chromyl chloride test, why?

Ans. Due to covalent nature, they do not give CrO_2Cl_2 with $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}_2\text{SO}_4$.

8. Chromyl chloride test is performed in dry test tube, why?

Ans. CrO_2Cl_2 is water soluble and gives chromic acid, so gas will not come out.



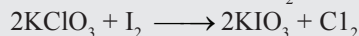
9. I_2 gives brown coloured solution in KI, violet in chloroform and benzene and brown in alcohol, ether and acetone, why?



In C_6H_6 and $CHCl_3$, it remains in molecular form so gives violet colour. In alcohol, ether and acetone it polymerise to form large molecule.

10. Cl_2 is more reactive than I_2 but when $KClO_3$ reacts with I_2 , Cl_2 is liberated, why?

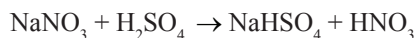
Ans. It is redox reaction and I_2 is oxidized.



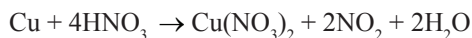
4. Nitrate

Solubility all nitrates are soluble in water. The nitrates of mercury and bismuth yield basic salts on treatment with water; these are soluble in dilute nitric acid.

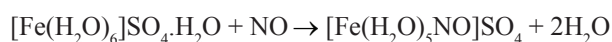
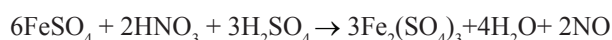
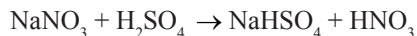
- Light brown fumes of nitrogen dioxide are evolved on heating the nitrate with concentrated H_2SO_4 .



These fumes intensify when copper turnings are added.



Ring Test: When the aqueous solution of the substance is treated with freshly prepared solution of ferrous sulphate and conc. H_2SO_4 a brown ring is formed on account of the formation of a complex at the junction of two liquids.



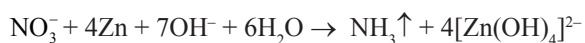
Ferrous sulphate

Brown ring

Note:

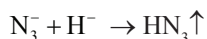
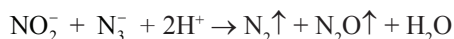
- Bromides and iodides interfere because of the liberated halogen; the test is not trustworthy in the presence of chromates, sulphites, thiosulphates, iodates, cyanides, thiocyanates, hexacyanoferrate (II) and (III) ions. All of these anions may be removed by adding an excess of nitrate-free Ag_2SO_4 to an aqueous solution (or sodium carbonate extract), shaking vigorously for 3–4 minutes, and filtering the insoluble silver salts, etc.
- In the complex $[Fe(H_2O)_5NO]SO_4$, iron remains in the +1 state of oxidation (actually, 'Fe' has been reduced by NO from +2 state to +1 state) and NO remains as NO^+ (actually, 'NO' has been oxidized by Fe from zero state to +1 state)
- Reduction of nitrates in alkaline medium* ammonia is evolved (detected by its odour; by its action upon red litmus paper and upon mercury (I) nitrate paper; or by the tannic acid-silver nitrate test), when a solution of a nitrate is boiled with zinc dust or gently warmed with aluminium powder and sodium hydroxide solution. Excellent results are obtained by the use of Devarda's alloy (45% Al, 5% Zn, 50% Cu).

Ammonium ions must, of course, be removed by boiling the solution with sodium hydroxide solution (and, preferably, evaporating almost to dryness) before the addition of the metal.



14.10 Salt Analysis

Nitrites give a similar reaction and may be removed most simply with the aid of sulphamic acid. Another, but more expensive, procedure involves the addition of sodium azide to the acid solution; the solution is allowed to stand for a short time and then boiled in order to complete the reaction and to expel the readily volatile hydrogen azide:



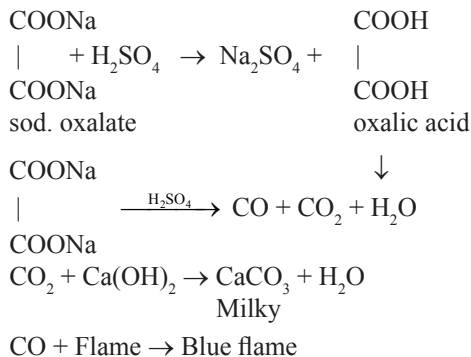
- **Nitrate in the presence of nitrite** the nitrite is readily identified in the presence of a nitrate by treatment with dilute mineral acid, potassium iodide, and starch paste (or potassium iodide-starch paper). The nitrate cannot, however, be detected in the presence of a nitrite since the latter gives the brown ring test with iron (II) sulphate solution and dilute sulphuric acid. The nitrite is therefore completely decomposed first by adding some sulphamic acid to the solution.

The nitrate can then be tested for by the brown ring test.

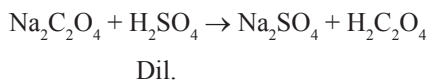
- **Nitrate in the presence of bromide and iodide** the brown ring test for nitrates cannot be applied in the presence of bromides and iodides since the liberation of the free halogen with concentrated sulphuric acid will obscure the brown ring due to the nitrate. The solution is therefore boiled with 2M sodium hydroxide solution until ammonium salts, if present, are completely decomposed; powdered Devarda's alloy or aluminium powder (or wire) is then added and the solution gently warmed. The evolution of ammonia, detected by its smell and its action upon red litmus paper, indicates the presence of a nitrate.

Oxalate ($\text{C}_2\text{O}_4^{2-}$)

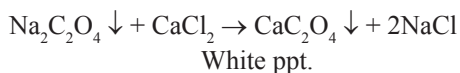
- Oxalates are decomposed by conc. H_2SO_4 to give CO_2 and CO. The latter burns with blue flame, while the former turns lime water milky.



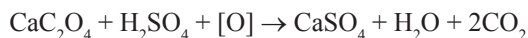
- Oxalate salt on reaction with dil. H_2SO_4 and MnO_2 gives more intensive effervescence of CO_2 .



- Aq. solution of oxalate gives white precipitate with CaCl_2 solution, which is insoluble in CH_3COOH



- In presence of dil. H_2SO_4 , the oxalate reduces permanganate, Hence its pink colour disappears.



Worked-out Examples

11. NO_3^- ions in a mixture can be tested from aqueous extract whereas other acid radicals not?

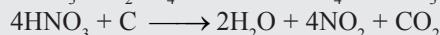
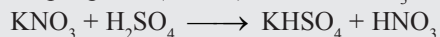
Ans. Nitrates, acetates and chlorates of each metal are water soluble and they come out in aqueous extract.

12. Both NO_2 and Br_2 are brown gases. How can they be identified if placed separately in two containers?

Ans. Water is added in both the containers. The yellowish solution confirms Br_2 in container where colourless solution confirms NO_2 in container.

13. All nitrates on heating with concentrated H_2SO_4 in presence of paper pallet evolve NO_2 gas. What is the function of paper pallet?

Ans. Paper pallet (carbon) reduces HNO_3 to NO_2 .



14. Sodium carbonate extract is treated with CaCl_2 in order to test oxalate or fluoride. Can we use BaCl_2 in place of CaCl_2 ?

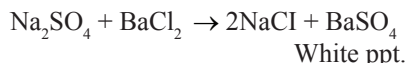
Ans. No, because SO_4^{2-} may interfere and barium oxalate is soluble in acetic acid.

Group III

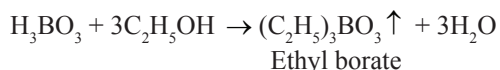
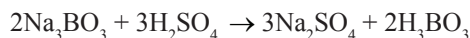
Some anions are identified by their characteristic chemical reactions. These radicals are sulphate, borate, phosphate and fluoride.

Sulphate

Dissolve a little amount of the substance (salt or mixture) and add barium chloride solution. A white precipitate insoluble in concentrated HNO_3 is formed.

**Borate**

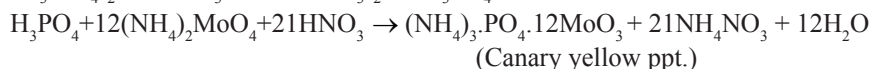
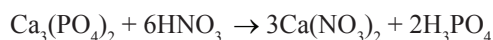
When borate is heated with ethyl alcohol and H_2SO_4 , ethyl borate vapours come out which burn with green edged flame.



In place of ethyl alcohol, methyl alcohol can also be used. This test should be performed in a test tube and not in a porcelain basin because copper or barium salts, if present, will come in contact with the flame which also gives green flame.

Phosphate

The canary yellow precipitate is due to the formation of ammonium phosphomolybdate.

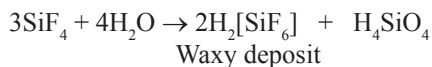
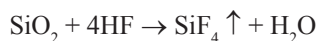
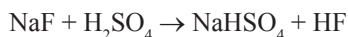


Arsenic under similar conditions also yields a yellow precipitate of $(\text{NH}_4)_3\text{AsO}_4 \cdot 12\text{MoO}_3$ (Ammonium arsenomolybdate). So, in presence of As, phosphate is tested in the filtrate of second group.

The precipitate of ammonium phosphomolybdate dissolves in excess of phosphate. Thus, the reagent (ammonium molybdate) should always be added in excess.

Fluoride (F⁻)

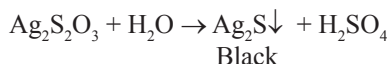
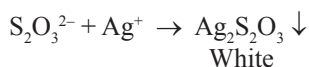
A fluoride reacts with concentrated H₂SO₄ to evolve HF. It reacts with silica to form fumes of silicon tetrafluoride which is hydrolysed by water to give a waxy deposit.



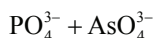
The test should be performed in perfectly dry test tube; otherwise, waxy white deposit will not be formed on the rod.

Thiosulphate (S₂O₃²⁻)

With dilute H₂SO₄, it gives S and SO₂ with AgNO₃ solution, white precipitates turns to brown and ultimately black precipitates are produced.



Phosphate in the presence of arsenate both arsenate and phosphate give a yellow precipitate on warming with ammonium molybdate solution and nitric acid, the latter on gently warming and the former on boiling.



Acidify the soda extract with 6M hydrochloric acid, bubble through sulphur dioxide to reduce the arsenate to arsenite, boil off the excess sulphate dioxide (test with potassium dichromate paper), and pass hydrogen sulphide into the solution to precipitate the arsenic as arsenic(III) sulphide; continue the passage of hydrogen sulphide until no more precipitate forms. Filter, boil off the hydrogen sulphide, and test the filtrate for phosphate by the ammonium molybdate test or with the magnesium nitrate reagent.

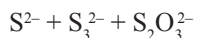
Sulphide, sulphite, thiosulphate, and sulphate in the presence of each other upon the addition of dilute acid to the mixture, the hydrogen sulphide, liberated from the sulphide, and the sulphur dioxide, liberated from the sulphite and thiosulphate, react and sulphur is precipitated; this complication necessitates the use of a special procedure for their separation.

Separation of sulphide, sulphite, thiosulphate, and sulphate shake the solution with excess of freshly precipitated CdCO₃, and filter.

Residue	Filtrate	
CdS and excess of CdCO ₃ . Wash and reject washings. Digest residue with 2M acetic acid to remove excess carbonate. A yellow residue indicates sulphide. Confirm by warming with 2M HCl and test the evolved H ₂ S with lead acetate paper.	Add Sr(NO ₃) ₂ solution in slight excess, shake, allow to stand overnight and filter.	
	Residue	Filtrate
	SrSO ₃ and SrSO ₄ . Wash, treat with 2M HCl and filter.	Contains SrS ₂ O ₃ . Acidify with 2M HCl and boil; SO ₂ is evolved and S is slowly precipitated. Thiosulphate present.
	Residue	Filtrate
	SrSO ₄ . White.	Contains sulphurous acid. Add a few drops of a 0.005M solution of Iodine; the latter is decolourized. Sulphite present.
	Sulphate present. Confirm by fusion with Na ₂ CO ₃ and apply sodium, nitroprusside test.	

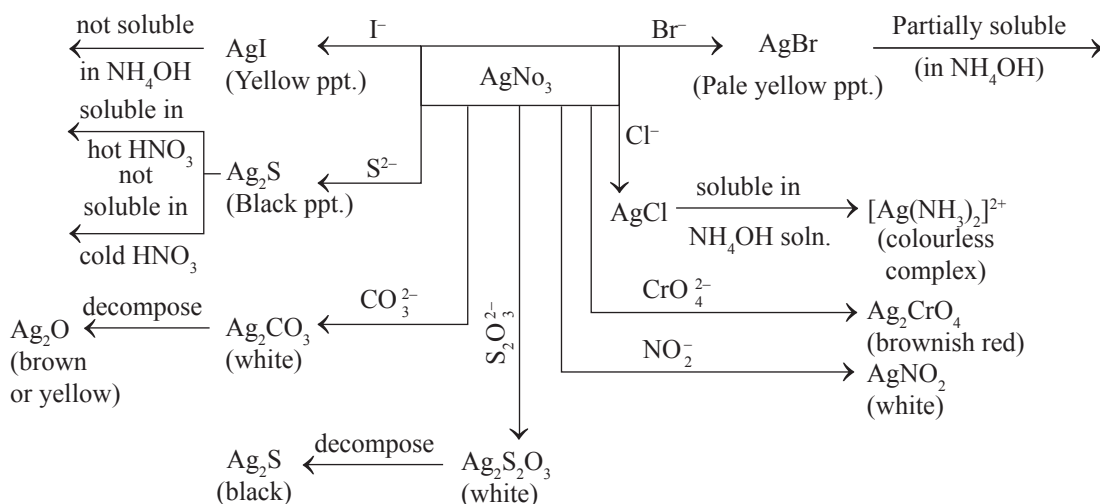
A mixture of the sodium salts or soda extract is employed for this separation.

Sulphide, sulphite, and thiosulphate in the presence of each other it is assumed that the solution is slightly alkaline and contains the sodium salts of the anions, e.g., the soda extract is used.

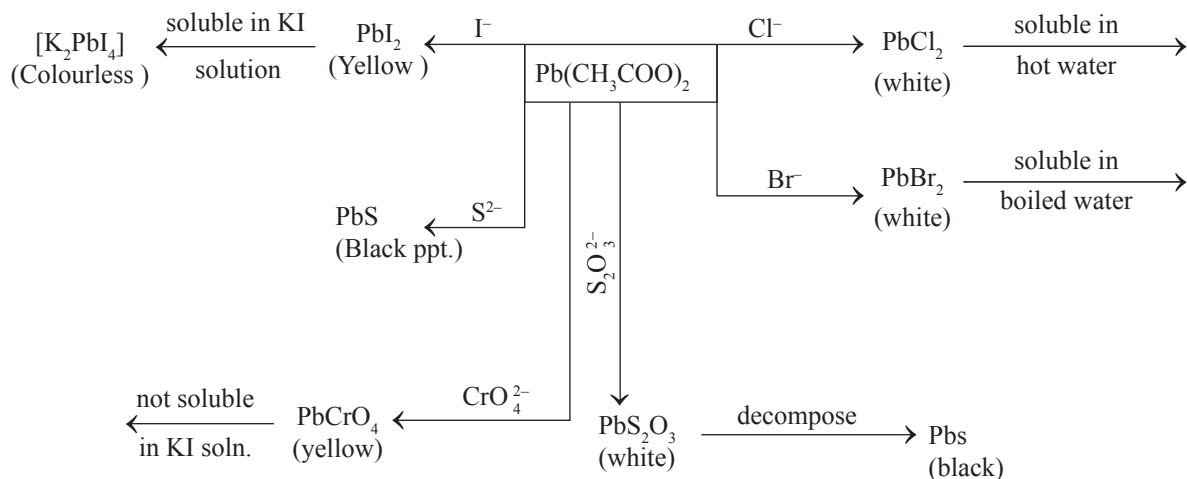


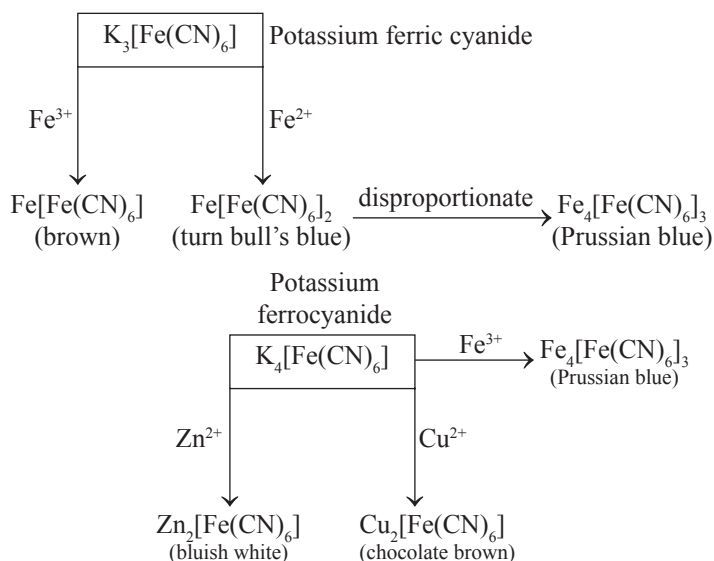
- Test a portion of the solution for sulphide with sodium nitroprusside solution. The formation of a purple colouration indicates the presence of a sulphide.
- If sulphide is present, remove it by shaking with freshly precipitated cadmium carbonate and filter. Test a portion of the filtrate with sodium nitroprusside solution to see that all the sulphide has been removed. When no sulphide is present, treat the remainder of the filtrate with a drop of phenolphthalein solution and bubble through carbon dioxide until the solution is decolourized by it. Test 2–3 ml of the colourless solution with 0.5–1 ml of the 0.1% fuchsin reagent. If the reagent is decolourized, the presence of sulphite is indicated.
- Treat the remainder of the colourless solution with 2M hydrochloric acid and boil for a few minutes. If sulphur separates, a thiosulphate is indicated.

Important Reactions of Group III acid Radicals (with AgNO_3)

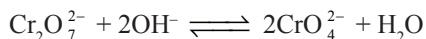
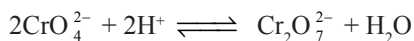


Important Reactions of Group III Acid Radicals (with $\text{Pb}(\text{CH}_3\text{COO})_2$)



Important reactions of Qualitative Analysis**Anions Containing 'd'block Metals****Chromate CrO_4^{2-} and dichromate ($\text{Cr}_2\text{O}_7^{2-}$)**

Metallic chromates give yellow solution when dissolved in water. In the presence of H^+ , chromates are converted into dichromates (orange-red solution).



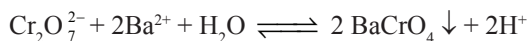
It may also be expressed as:



- (i) **Barium chloride solution:** Pale-yellow precipitate of barium chromate soluble in dilute mineral acids but insoluble in water and acetic acid.

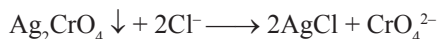
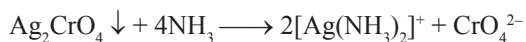
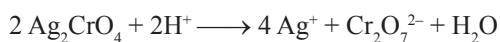
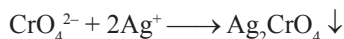


Dichromate ion also gives the same precipitate but due to the formation of strong acid precipitation is partial.

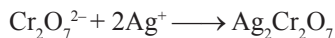


If sodium hydroxide or sodium acetate is added, precipitation becomes quantitative.

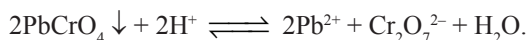
- (ii) **Silver Nitrate Solution:** Brownish-red precipitate of silver chromate Ag_2CrO_4 which is soluble in dilute nitric acid and in ammonia solution, but insoluble in acetic acid.



A reddish brown precipitate of silver dichromate $\text{Ag}_2\text{Cr}_2\text{O}_7$ is formed with a concentrated solution of a dichromate.



- (iii) **Lead acetate solution:** Yellow precipitate of lead chromate PbCrO_4 insoluble in acetic acid, but soluble in dilute nitric acid.



- (iv) **H_2O_2 :** If an acidic solution of a chromate is treated with H_2O_2 , a deep blue solution of chromium penta oxide is obtained.



CrO_5 is unstable and it decomposes yielding oxygen and a green solution of a Cr^{+3} salt.

Permanganate MnO_4^-

- (i) **Hydrogen peroxide:** It decolourises acidified potassium permanganate solution



- (ii) Iron (II) sulphate, in the presence of sulphuric acid, reduces permanganate to manganese (II). The solution becomes yellow because of the formation of iron (III) ions

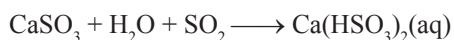
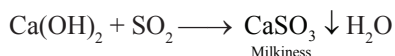


- (iii) **Action of heat:** On heating a black residue of potassium manganate K_2MnO_4 and manganese dioxide remains behind. Upon extracting with water and filtering, a green solution of potassium manganate is obtained



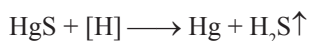
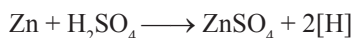
Some Noteworthy Points

- Carbonates of bismuth and barium are not easily decomposed by dilute H_2SO_4 . Use dilute HCl . This is because BaSO_4 and $\text{Bi}_2(\text{SO}_4)_3$ are insoluble in water.
- SO_2 also forms milkiness with lime water. Further, this milkiness dissolves, if SO_2 is passed in excess (c.f. SO_2)



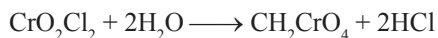
However, CO_2 is odourless while SO_2 has a pungent suffocating smell of burning sulphur.

- PbCO_3 reacts slowly both with dilute HCl or dilute H_2SO_4 as PbCl_2 and PbSO_4 are both insoluble in cold
- Basic copper carbonate reacts very slowly with all dilute acids. The effervescence (if any) is hard to observe
- Sulphides of Pb, Ca, Ni, CO, Sb, Sn are not decomposed with dilute H_2SO_4 . Conc. HCl is to be used to detect the presence of S^{2-} . Brisk evolution of H_2S is possible with dil. H_2SO_4 if a pinch of Zn dust is added to the reaction mixture



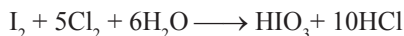
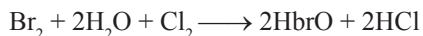
14.16 Salt Analysis

- Use only a dry test tube for chromyl chloride test, water present (if any) hydrolyse the chromyl chloride



Br^- and I^- do not give this test.

- Excess of chlorine water is to be avoided in chlorine water test. Excess Cl_2 reacts with Br_2 or I_2 to give HBrO or HIO_3 respectively, which are colourless



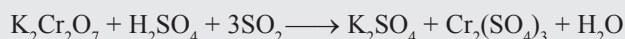
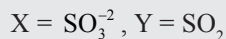
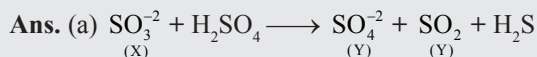
- (a) Arsenic interferes in the ammonium molybdate test. As it also gives a yellow precipitate of $(\text{NH}_4)_3\text{AsO}_4 \cdot 12\text{MoO}_3$. In such a case, test the presence of PO_4^{3-} with the filtrate of II group.
- (b) Reducing agents like SO_3^{2-} , S^{2-} etc., interfere in ammonium molybdate test as they reduce the reagent to molybdenum blue ($\text{M}_3\text{O}_8 \cdot x\text{H}_2$). In such a case, boil the salt with HNO_3 to oxidize these ions (SO_3^{2-} , S^{2-} etc.,) before testing for phosphate.

Worked-out Examples

15. $[\text{X}] + \text{H}_2\text{SO}_4 \longrightarrow [\text{Y}]$ a colourless gas with irritating smell
 $[\text{Y}] + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow$ green solution. $[\text{X}]$ and $[\text{Y}]$ are:

(IIT 2003)

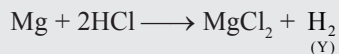
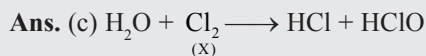
- (a) SO_3^{2-} , SO_2 (b) Cl^- , HCl
 (c) S^{2-} , H_2S (d) CO_3^{2-} , CO_2



16. A gas 'X' is passed through water to form a saturated solution. The aqueous solution on treatment with silver nitrate gives a white precipitate. The saturated aqueous solution also dissolves magnesium ribbon with evolution of a colourless gas 'Y'. Identify 'X' and 'Y'.

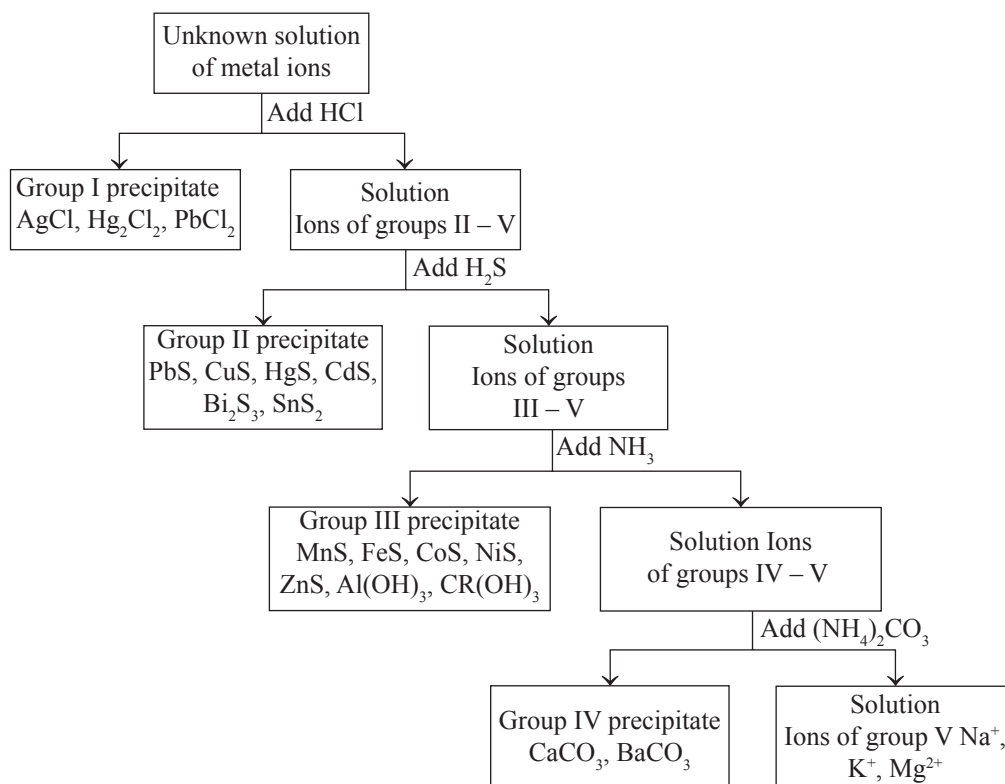
(IIT 2002)

- (a) $\text{X} = \text{CO}_2$, $\text{Y} = \text{Cl}_2$ (b) $\text{X} = \text{Cl}_2$, $\text{Y} = \text{CO}_2$
 (c) $\text{X} = \text{Cl}_2$, $\text{Y} = \text{H}_2$ (d) $\text{X} = \text{H}_2$, $\text{Y} = \text{Cl}_2$



IDENTIFICATION OF BASIC RADICALS

The ions are identified by specific chemical tests, but because one ion can interfere with the test for another, the ions must first be separated. In the traditional scheme of analysis for metal cations, some 20 cations are separated initially into five groups by selective precipitation



Group I: Ag^+ , Hg_2^{2+} , and Pb^{2+} When aqueous HCl is added to the unknown solution, the cations of group I precipitate as insoluble chlorides— AgCl , Hg_2Cl_2 , and PbCl_2 . The cations of groups II–V, which form soluble chlorides, remain in solution. A small amount of the Pb^{2+} also remains in solution because PbCl_2 is slightly soluble.

Group II: Pb^{2+} , Cu^{2+} , Hg^{2+} , Cd^{2+} , Bi^{3+} , and Sn^{4+} After the insoluble chlorides have been removed, the solution is treated with H_2S to precipitate the cations of group II as insoluble sulfides— PbS , CuS , HgS , CdS , Bi_2S_3 , and SnS_2 . Because the solution is strongly acidic at this point ($[\text{H}_3\text{O}^+] = 0.3 \text{ M}$), only the most insoluble sulphides precipitate. The acid-insoluble sulphides are then removed from the solution.

Group III: Mn^{2+} , Fe^{2+} , Co^{2+} , Ni^{2+} , Zn^{2+} , Al^{3+} , and Cr^{3+} At this point aqueous NH_3 is added, neutralizing the acidic solution and giving an $\text{NH}_4^+/\text{NH}_3$ buffer that is slightly basic ($\text{pH} \approx 8$). The decrease in $[\text{H}_3\text{O}^+]$ shifts the metal sulphide solubility equilibrium to the left, thus precipitating the $2+$ cations of group III as insoluble sulphides— MnS , FeS , CoS , NiS , and ZnS . The $3+$ cations precipitate from the basic solution, not as sulphides, but as insoluble hydroxides— $\text{Al}(\text{OH})_3$ and $\text{Cr}(\text{OH})_3$.

Group IV: Ca^{2+} and Ba^{2+} After the base-insoluble sulphides and the insoluble hydroxides have been removed, the solution is treated with $(\text{NH}_4)_2\text{CO}_3$ to precipitate the cations of group IV as insoluble carbonates— CaCO_3 and BaCO_3 . Magnesium carbonate does not precipitate at this point because $[\text{CO}_3^{2-}]$ in the $\text{NH}_4^+/\text{NH}_3$ buffer is maintained at a low value.

Group V: Na^+ , K^+ , and Mg^{2+} The only ions remaining in solution at this point are those whose chlorides, sulphides, and carbonates are soluble under conditions of the previous reactions. Magnesium ion is separated and identified by addition of a solution of $(\text{NH}_4)_2\text{HPO}_4$; if Mg^{2+} is present, a white precipitate of $\text{Mg}(\text{NH}_4)\text{PO}_4$ forms. The alkali metal ions are usually identified by the characteristic colours that they impart to a Bunsen flame.

Once the cations have been separated into groups, further separations and specific tests are carried out to determine the presence or absence of the ions in each group. In group I, for example, lead can be separated from silver and mercury by treating the precipitate with hot water. The more soluble PbCl_2 dissolves, but the less soluble AgCl and Hg_2Cl_2 do not. To test for Pb^{2+} , the solid chlorides are removed, and the solution is treated with a solution of K_2CrO_4 . If Pb^{2+} is present, a yellow precipitate of PbCrO_4 forms.

14.18 Salt Analysis

Detailed procedures for separating and identifying all the ions can be found in general chemistry laboratory manuals. Although, modern methods of metal-ion analysis employ sophisticated analytical instruments, qualitative analysis is still included in many general chemistry laboratory courses because it is an excellent vehicle for developing laboratory skills and for learning about acid-base, solubility, and complex-ion equilibria.

Wet test analysis for basic radicals includes the following steps:

- (a) Preparation of the original solution of the salt or mixture.
- (b) Separation of basic radicals into different groups.
- (c) Analysis of the precipitates obtained in different groups and confirmation of the basic radicals by specific tests.

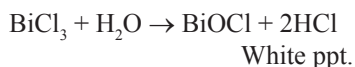
(a) Preparation of the original solution

The following solvents should be tried strictly in the order given below:

- (i) Cold water
- (ii) Hot water
- (iii) Dilute HCl
- (iv) Concentrated HCl

Note:

- When the salt or mixture is soluble in cold dilute HCl, it shows that basic radicals Pb^{2+} , Hg_2^{2+} and Ag^+ are absent.
- Lead chloride, PbCl_2 , is soluble in hot solution but insoluble, in cold solution. Thus, when the hot solution is cooled, if a white substance appears, it shows the presence of Pb^{2+} .
- Sometimes, a white precipitate or milkiness is formed when the solution prepared in concentrated HCl is diluted, even when no member of first group is present. This may be due to formation of the oxy chlorides of antimony, bismuth, and tin.



The precipitate disappears if little concentrated HCl is added. In such a case, H_2S can be passed through milky solution.



- A white crystalline precipitate of NaCl and BaCl_2 may be formed from the concentrated solution by the addition of concentrated HCl due to common ion effect. Such precipitates dissolve on dilution.
- The original solution should not be prepared in concentrated nitric acid and concentrated H_2SO_4 . These act as oxidizing agents and convert H_2S into S in second group. Nitric acid converts sulphides of Ba, Sr, and Pb into insoluble sulphates. Similarly, sulphuric acid converts the salts of Ba, Sr and Pb into insoluble sulphates.

(b) Separation of basic radicals into groups

For systematic analysis, the basic radicals have been classified into groups. Each group has a reagent, which shows the presence of the basic radical belonging to that particular group. It is necessary that the radical is completely removed in the respective group by using excess of group reagent, otherwise its presence creates difficulty in higher group.

Worked-out Examples

17. Why transition state cannot be isolated as independent chemical species?

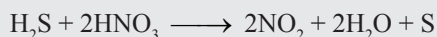
Ans. Transition state have maximum energy. No matter how their bonds vibrate, the resulting molecule possess lower energy and thus more stable.

18. HNO_3 or H_2SO_4 are not used to prepare solution for analysis of basic radicals.

Ans. Both are oxidizing agent as well as sulphates of many of the metals are water insoluble.

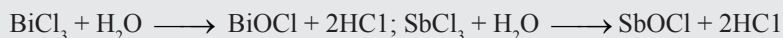
19. Why original solution is not prepared in concentrated HNO_3 ?

Ans. HNO_3 agent which on decomposition gives oxygen. A yellow precipitate of sulphur is obtained in presence of HNO_3 , when H_2S is passed for detecting the radicals of group II and group IV.



20. An original solution is prepared in concentrated HCl. When diluted a white precipitate is formed. What does it indicate?

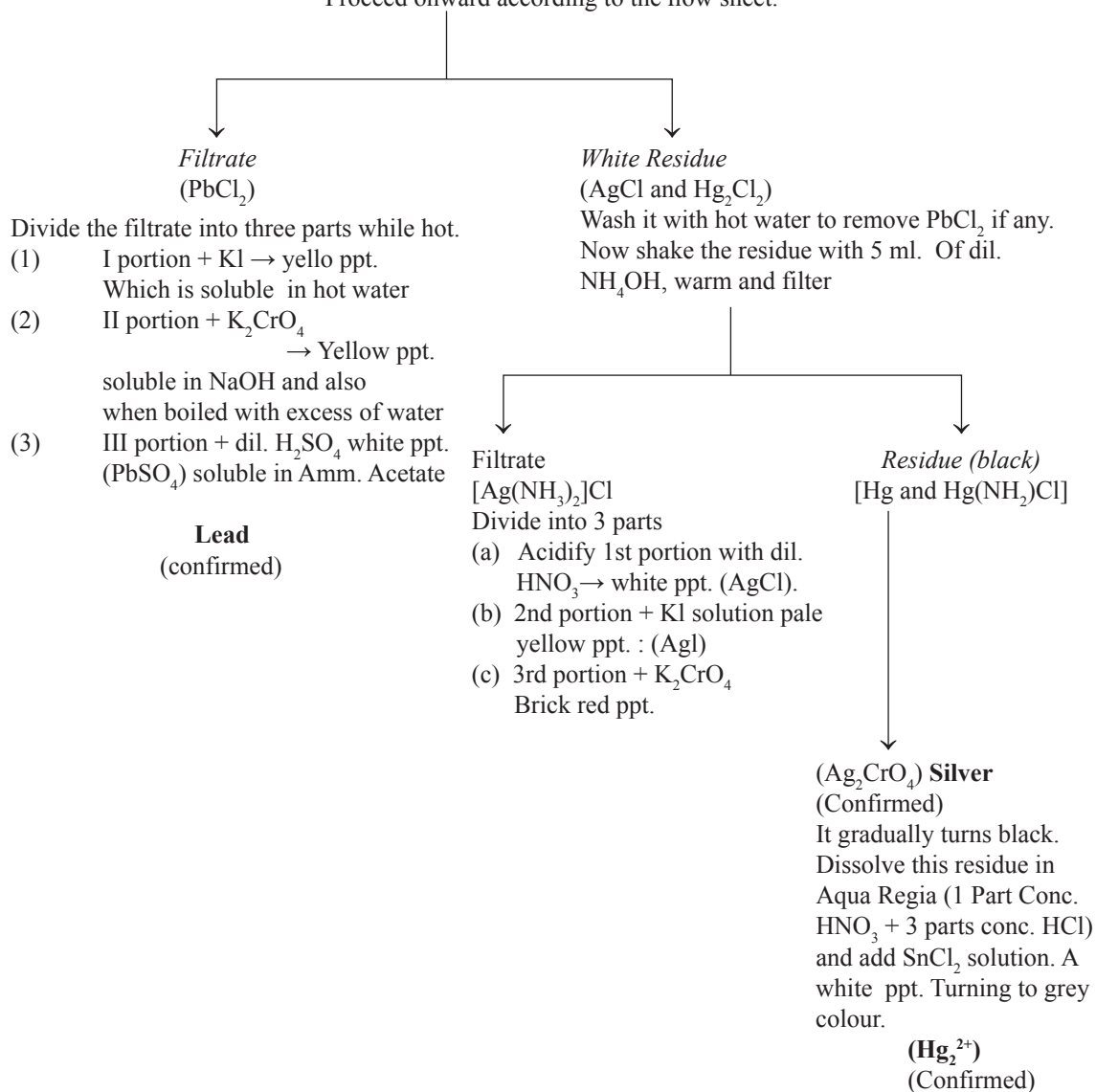
Ans. Presence of Sb^+ or Bi. Their chlorides hydrolyse to oxychlorides in presence of excess of water.



Salt Analysis of Cations Group I

To the original solution dil. HCl is added and a white ppt. Of 1st group chlorides (PbCl_2 , AgCl and Hg_2Cl_2) is obtained. It is filtered and the filtrate is left for second group.

This precipitate is boiled with water and filtered while the solution is hot.
Proceed onward according to the flow sheet.



Note:

- It is **mercuric salt**, it will not be precipitated in the first group because HgCl_2 is soluble. It is precipitated in the second group.
- **Lead** (i) this gives red or reddish violet colouration with p-dimethyl amino benzilidine rhodamide. (ii) to the original solution, add 2 drops of dithiozone and shake—**red colour**.
- **Silver**: Ammonical solution of it, when treated with aqueous solution of resorcinol gives **blue colour** slowly.
- **Mercurous**: Solution when treated with alcoholic solution of gallic acid gives an **orange or yellow colour**.

Enquiry: What are the various chemical reactions involved in I group analysis?

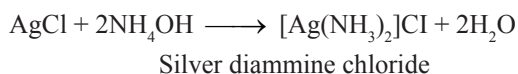
The Ag^+ , Pb_2^+ and Hg_2^{2+} ions present in the solution react with Cl^- ions from HCl , to give white precipitate

1. $\text{Pb}(\text{NO}_3)_2 + 2\text{HCl} \rightarrow \text{PbCl}_2 \downarrow + 2\text{HNO}_3$
2. $\text{AgNO}_3 + \text{HCl} \rightarrow \text{AgCl} \downarrow + 2\text{HNO}_3$
3. $\text{Hg}_2(\text{NO}_3)_2 + 2\text{HCl} \rightarrow \text{Hg}_2\text{Cl}_2 + 2\text{HNO}_3$

Lead: On boiling the precipitate with water, PbCl_2 becomes soluble thus separated by filtration. This when treated with KI , K_2CrO_4 and H_2SO_4 separately by, following reactions take place.

1. Portion – $\text{PbCl}_2 + 2\text{KI} \rightarrow \text{PbI}_2 \downarrow + 2\text{KCl}$.
2. Portion – $\text{PbCl}_2 + \text{K}_2\text{CrO}_4 \rightarrow \text{PbCrO}_4 \downarrow + 2\text{KCl}$.
3. Portion – $\text{PbCl}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{PbSO}_4 \downarrow + 2\text{HCl}$.

Silver: (i) Silver chloride dissolves in NH_4OH forming a soluble complex.



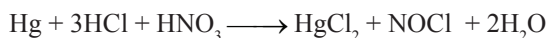
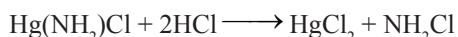
The filtrate containing soluble silver complex is treated in three ways separately.

- (a) I portion – $[\text{Ag}(\text{NH}_3)_2]\text{Cl} + 2\text{HNO}_3 \longrightarrow \text{AgCl} \downarrow + 2\text{NH}_4\text{NO}_3$
- (b) II portion – $[\text{Ag}(\text{NH}_3)_2]\text{Cl} + \text{KI} \longrightarrow \text{AgI} \downarrow + \text{KCl} + 2\text{NH}_3$
- (c) III portion – $[\text{Ag}(\text{NH}_3)_2]\text{Cl} + \text{K}_2\text{CrO}_4 \longrightarrow \text{Ag}_2\text{CrO}_4 \downarrow + 2\text{KCl} + 4\text{NH}_3$

Mercurous: NH_4OH converts Hg_2Cl_2 to a black residue which consists of white amino–mercuric chloride and black finely divided Hg . In this case, internal redox reaction takes place whereby Hg_2^{2+} ions of Hg_2Cl_2 are reduced to metallic Hg and others are oxidised to mercuric state.

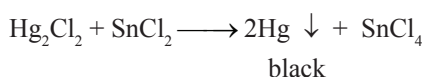
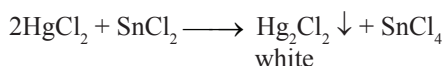


Aqua regia dissolves the black residue forming soluble HgCl_2



Addition of SnCl_2 reduces HgCl_2 to white Hg_2Cl_2 .

Further addition of SnCl_2 reduces white Hg_2Cl_2 to black finely divided Hg . Usually, grey mixture of Hg_2Cl_2 and Hg is obtained.



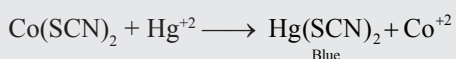
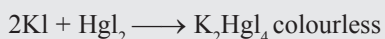
Worked-out Examples

21. Why a salt containing lead turn black in colour, when placed for a long time in laboratory?

Ans. Due to the formation of black lead-sulphide by the action of H_2S in atmosphere.

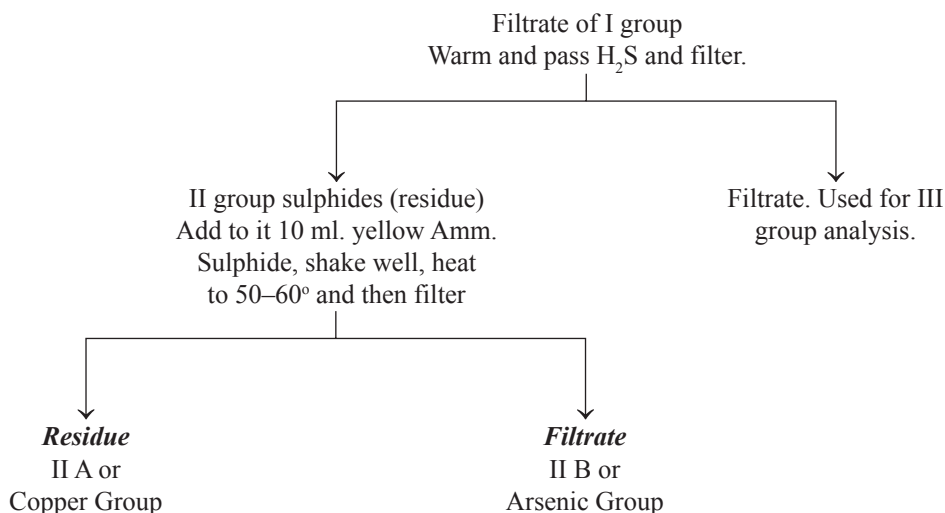
22. A solution of metal ion when treated with KI gives a red precipitate which dissolves in excess KI to give a colourless solution. Moreover, the solution of metal ion on treatment with a solution of cobalt (II) thiocyanate gives rise to deep blue crystalline precipitate. The metal ion is (IIT 2007)

- (a) Pb^{2+} (b) Hg^{2+}
(c) Cu^{2+} (d) Co^{2+}

**Salt Analysis of Cations Group II**

The second group contains largest number of radicals. These are of course further divided into two sub groups:

1. **Cu group or II A group** (2) **Arsenic group or II B group**, with respect to solubility of their sulphides in yellow ammonium sulphide. The procedure of analysis is as follows:



Now, further analysis of both the groups may be done separately.

Boil the filtrate obtained after first group precipitation and reduce it to two-thirds of its volume then pass H_2S . The second group radicals are precipitated as sulphides with their characteristic colours. The following table includes all the second group radicals, their precipitated forms and their colours.

Radicals: Pb^{2+} , Bi^{3+} , Hg^{2+} , Cu^{2+} , Cd^{2+} (Group IIA)

As^{3+} , Sb^{3+} , Sn^{2+} , Sn^{4+} , (Group IIB)

Group reagent: H_2S gas in presence of dil. HCl .

14.22 Salt Analysis

The filtrate of first group or original solution containing dilute HCl (if first group is absent) and pass H₂S gas till the precipitation is complete. Keep the filtrate for subsequent groups.

If the precipitate does not dissolve in **yellow ammonium sulphide**, Group IIA radical may be present.

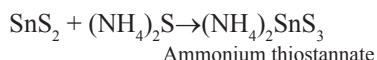
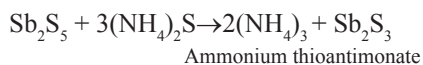
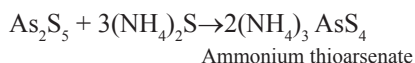
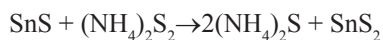
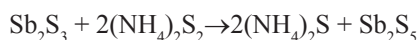
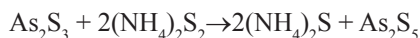
If the precipitate dissolves in **yellow ammonium sulphide**, Group IIB radical may be present.

<i>Insoluble in yellow ammonium sulphide (IIA)</i>		<i>Soluble in yellow ammonium sulphide</i>		
HgS	Black	IIB Group	As ₂ S ₃	Yellow
Pbs	Black		Sb ₂ S ₃	Orange
Bi ₂ S ₃	Black		SnS	Brown
CuS	Black		SnS ₂	Yellow
Cds	Yellow			

Yellow ammonium sulphide is used only for one radical, i.e., Sn²⁺ which is not soluble in ordinary ammonium sulphide. Yellow ammonium sulphide consists excess of sulphur and is formulated as (NH₄)₂S_x, commonly as (NH₄)₂S₂. This converts SnS first into SnS₂ which then dissolves forming thio complex. In case Sn²⁺ is absent, ordinary ammonium sulphide can be used in place of yellow ammonium sulphide.

Group II B

The precipitate is treated with yellow ammonium sulphide. The sulphides of IIB are first oxidized to higher sulphides, which then dissolve to form thio-compounds.

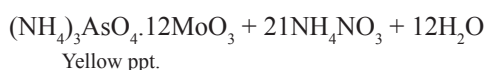
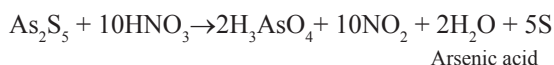


In case the precipitate does not dissolve in yellow ammonium sulphide, it may be either HgS or PbS or Bi₂S₃ or CuS or CdS.

In case the precipitate dissolves in yellow ammonium sulphide, the tests of the radicals arsenic, antimony and tin are performed. The sulphide is treated with concentrated hydrochloric acid. Antimony and tin sulphides dissolve while arsenic sulphide remains insoluble.

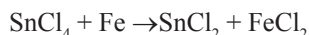
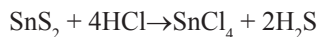
As³⁺ (arsenic)

The insoluble sulphide is treated with concentrated nitric acid which is then heated with ammonium molybdate. Yellow precipitate of ammonium arsenomolybdate is formed.

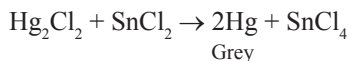
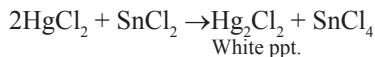


Sn²⁺ or Sn⁴⁺ (tin)

Solution of sulphide in concentrated HCl is reduced with iron fillings to granulated zinc.

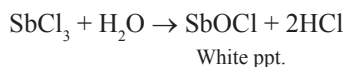


HgCl₂ solution is added to above solution which gives first a white precipitate that turns to grey.

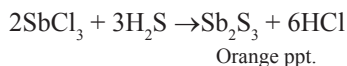
**Sb³⁺ (antimony)**

Filtrate of sulphide in concentrated HCl is divided into two parts.

Part I: On dilution with excess of water, a white precipitate of antimony oxychloride is obtained.



Part II: H₂S is circulated. Orange precipitate is formed.



The copper group i.e., II A group residue (insoluble in yellow amm. Sulphide) is washed with hot water and then boiled with 5–10 ml. HNO₃. Filter and proceed according to flow chart.

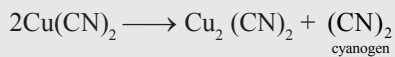
Worked-out Examples

23. CuSO₄ decolourizes on addition of KCN, the product is:

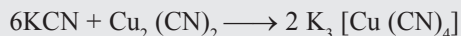
(IIT 2006)

- (a) [Cu(CN)₄]²⁻
- (b) Cu⁺² gets reduced to form [Cu(CN)₄]³⁻
- (c) [Cu(CN)₂]
- (d) CuCn

Ans. (b) $\text{CuSO}_4 + 2 \text{KCN} \longrightarrow \text{Cu(CN)}_2 + \text{K}_2\text{SO}_4$



Cu⁺² is reduced to Cu⁺¹.

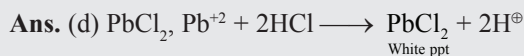


Cuprous compounds are colourless.

24. An aqueous solution of a substance gives a white precipitate on treatment with dilute hydrochloric acid, which dissolves on heating. When hydrogen sulphide is passed through the hot acidic solution, a black precipitate is obtained. The substance is a:

(IIT 2002)

- (a) Hg₂²⁺ salt
- (b) Cu²⁺ salt
- (c) Ag⁺ salt
- (d) Pb²⁺ salt



PbCl₂ is soluble in hot water.



Group (IIA) Analysis

Residue (black)

HgS + sulphur

Dissolve the ppt. In a little aqua regia.
divide in two portions.

- (a) I portion + excess of $\text{SnCl}_2 \rightarrow$
white ppt. (Hg_2Cl_2) turning grey.
- (b) II portion + Cu turning $\rightarrow$ white
or grey deposit on Cu

Filtrate

$[\text{Pb}(\text{NO}_3)_2, \text{Bi}(\text{NO}_3)_3, \text{Cu}(\text{NO}_3)_2$
and $\text{Cd}(\text{NO}_3)_2]$ Add excess of
concentrated NH_4OH until
precipitation is complete.

Filter

Residue
 $[\text{Bi}(\text{OH})_3 \text{ and } \text{Pb}(\text{OH})_2]$
Warm with 5 ml. NaOH
Soln. and filter

Dissolve it in least amount of dil. HCl
and divide soln. In two parts

- (i) I part + excess of water $\rightarrow$ white
turbidity.
- (ii) II part + sodium stannite soln.
(excess) $\rightarrow$ black ppt.

*Bismuth**Filtrate* $\text{Na}_2(\text{PbO}_2)$

Acidify with dil. acetic
acid and add K_2CrO_4 soln. $\rightarrow$
yellow ppt. soluble in
NaOH.

Lead*Filtrate* $[\text{Cu}(\text{NH}_3)_4]^{2+}$ and $[\text{Cd}(\text{NH}_3)_4]^{2+}$

- (1) If filtrate is colourless
Cu is absent. Pass H_2S into
the filtrate for 10 to
20 seconds $\rightarrow$ yellow ppt.
of CdS .

Cadmium

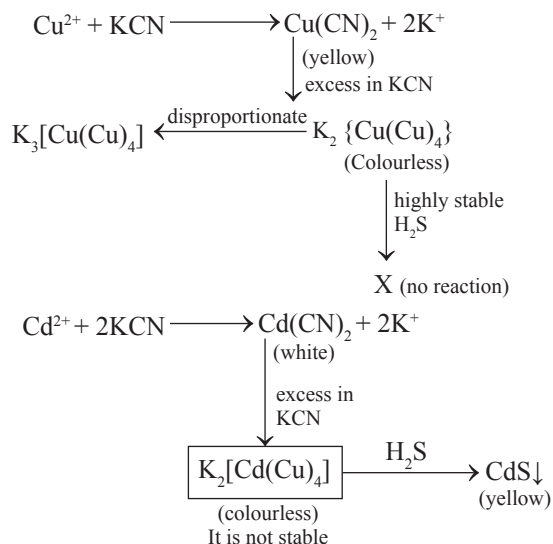
- (2) If filtrate is blue
(a) only Cu is blue
(b) or Cu and Cd both
are present

Note: Sodium stannite is prepared by adding NaOH solution to SnCl_2 solution drop by drop until precipitate of $\text{Sn}(\text{OH})_2$ first formed redissolves.

Enquiry: How do we separate copper from cadmium?

If we get blue colour after adding NH_4OH this shows presence of copper but cadmium may or may not be present. In such a case, divide blue solution into two parts and test these as follows:

- (a) **Test for copper:** To one part add CH_3COOH to acidify, then $\text{K}_4[\text{Fe}(\text{CN})_6]$ solution. Chocolate brown precipitate copper is confirmed.
- (b) **Test for cadmium:** Divide second part into two parts.
- (i) To one part add KCN solution drop by drop until blue colour disappears. Pass H_2S yellow ppt. Shows cadmium.
- (ii) To second part, add excess of dilute H_2SO_4 to make solution, strongly acidic. Now add a little zinc dust, boil and filter. To the filtrate add NH_4OH to neutralize excess of acid, then add 1 ml. Dil. HCl and pass H_2S yellow ppt. of Cd, confirmed.

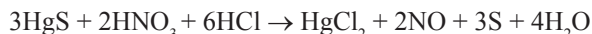


Enquiry: What are various reactions involved in II A (copper group) analysis?

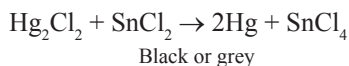
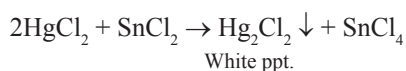
The sulphides are boiled with HNO_3 , whereby, these are converted into their corresponding soluble nitrates.

- $4\text{CuS} + 8\text{HNO}_3 \rightarrow 3\text{Cu}(\text{NO}_3)_2 + 2\text{NO} + 3\text{S} + 4\text{H}_2\text{O}$
- $3\text{CdS} + 8\text{HNO}_3 \rightarrow 3\text{Cd}(\text{NO}_3)_2 + 2\text{NO} + 3\text{S} + 4\text{H}_2\text{O}$
- $3\text{PbS} + 8\text{HNO}_3 \rightarrow 3\text{Pb}(\text{NO}_3)_2 + 2\text{NO} + 3\text{S} + 4\text{H}_2\text{O}$
- $\text{Bi}_2\text{S}_3 + 8\text{HNO}_3 \rightarrow 3\text{Bi}(\text{NO}_3)_3 + 2\text{NO} + 3\text{S} + 4\text{H}_2\text{O}$

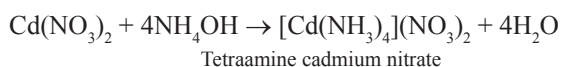
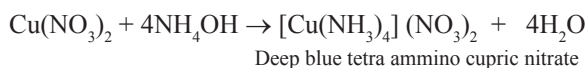
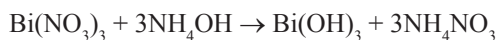
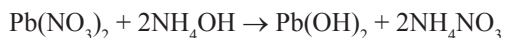
Reactions of mercuric sulphide: The mercuric sulphide being insoluble is treated with aqua regia (3 parts concentrated HCl + 1 part concentrated HNO_3) and it goes into solution:



(a) This mercuric chloride is treated with SnCl_2 .



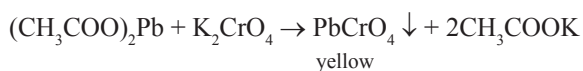
Treatment of filtrate with NH_4OH : The filtrate containing $\text{Pb}(\text{NO}_3)_2$, $\text{Bi}(\text{NO}_3)_3$, $\text{Cu}(\text{NO}_3)_2$ is treated with concentrated NH_4OH whereby, Pb and Bi are precipitated as their hydroxides and Cu and Cd pass into solution in the form of soluble complex cations, $[\text{Cu}(\text{NH}_3)_4]^{2+}$ and $[\text{Cd}(\text{NH}_3)_4]^{2+}$.



Reactions of Pb and Bi: The residue containing $\text{Pb}(\text{OH})_2$, $\text{Bi}(\text{OH})_3$ is warmed with NaOH whereby, Pb is changed into soluble sodium plumbite and $\text{Bi}(\text{OH})_3$ remains undissolved.

14.26 Salt Analysis

Test of Pb: The filtrate containing $\text{Na}_2[\text{PbO}_2]$ when acidified with acetic acid and K_2CrO_4 is added, a yellow precipitate of PbCrO_4 is obtained

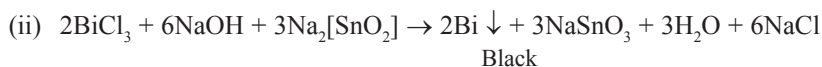
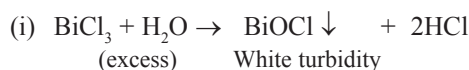


The yellow precipitate is insoluble in acetic acid and NH_4OH , but soluble in alkali and HNO_3

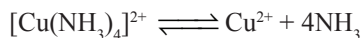


Test for Bi: $\text{Bi}(\text{OH})_3 + 3\text{HCl} \rightarrow \text{BiCl}_3 + 2\text{H}_2\text{O}$
ppt. Soluble

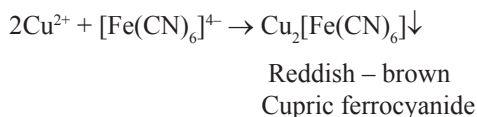
In this solution Bi be confirmed by any of the following reactions:



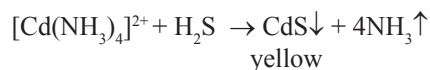
Reactions with Cu: The ionization of tetra-ammine cupric cation



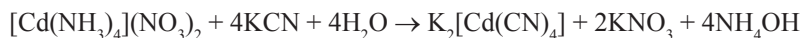
is small in presence of excess of ammonia, even then the Cu^{2+} ions thus produced are sufficient of their identification. Hence, with $\text{K}_4[\text{Fe}(\text{CN})_6]$ a reddish brown precipitate insoluble in CH_3COOH but soluble in NH_4OH is produced.



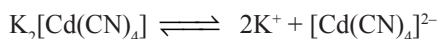
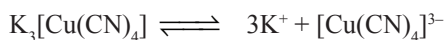
Reactions with Cd:



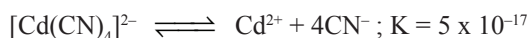
Reactions in Cu, Cd separation: When filtrate is blue: Addition of excess of KCN forms soluble colourless complexes with Cu and Cd respectively.



Cyanide complex of copper is more stable than that of cadmium. The primary ionisation of both complexes are as under:



The secondary ionization and ionization constant of these complexes are:

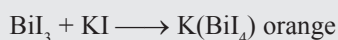


It is thus clear that secondary ionisation of cuprocyanide complex is so low that no appreciable quantity of Cu^+ ions is available in the solution. When H_2S is passed, no black precipitate of Cu_2S is produced; whereas secondary ionisation for cadmium cyanide complex is comparatively high, therefore, enough Cd^{2+} ions are available in the solution and when H_2S is passed, a yellow precipitate of CdS is produced. In this way, cadmium is identified in the presence of copper.

Worked-out Examples

25. A metal nitrate reacts with KI to give a black precipitate which on addition of excess of KI is converted into orange colour solution. The cation of the metal nitrate is: (IIT 2005)

- (a) Hg^{2+} (b) Bi^{3+}
(c) Pb^{2+} (d) Cu^+



26. Dissolution of CuS in hot HNO_3 gives:
(a) A blue solution, blue solid and colourless gas
(b) A yellow solution, blue solid and brown gas
(c) A green solution, yellow solid and brown gas
(d) None

Ans. (a) $\text{Cu}(\text{NO}_3)_2(\text{aq.})$ is blue, $\text{S}_{(\text{s})}$ is yellow and NO is colourless.

27. A solution of Cu^{2+} , Zn^{2+} , Bi^{3+} , Mn^{2+} and Co^{2+} at $\text{pH} = 1$ is treated with H_2S . Which of these ions will give precipitate?

Ans. Cu_2S and Bi_2S_3 (both black precipitate). Rest all ions give soluble sulphides.

28. Sometimes during precipitation of gp.II cations, a red ppt. is obtained which turns black upon dilution, why?

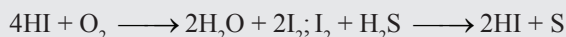
Ans. In presence of concentrated HCl , red precipitate obtained is of lead sulphochloride (Pb_2SCl_2); which is converted to PbS (black precipitate) on dilution.

29. Can we use other reagents in place of yellow ammonium sulphide, for separation of II A and IIB gp.?

Ans. Yes; If Sn^{2+} is absent then NaOH , KOH or colourless ammonium sulphide may be used except SnS , all the sulphides of gp II B are soluble in these solutions also.

30. Before testing II gp. iodide ions are separated, why?

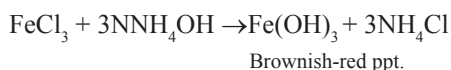
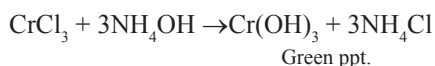
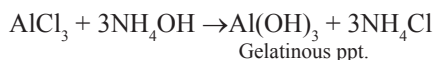
Ans. Because I^- ions react with air to form I_2 , which reacts with H_2S and give white or light yellow precipitate of sulphur i.e., why F are removed by boiling original solution with NaNO_2 .



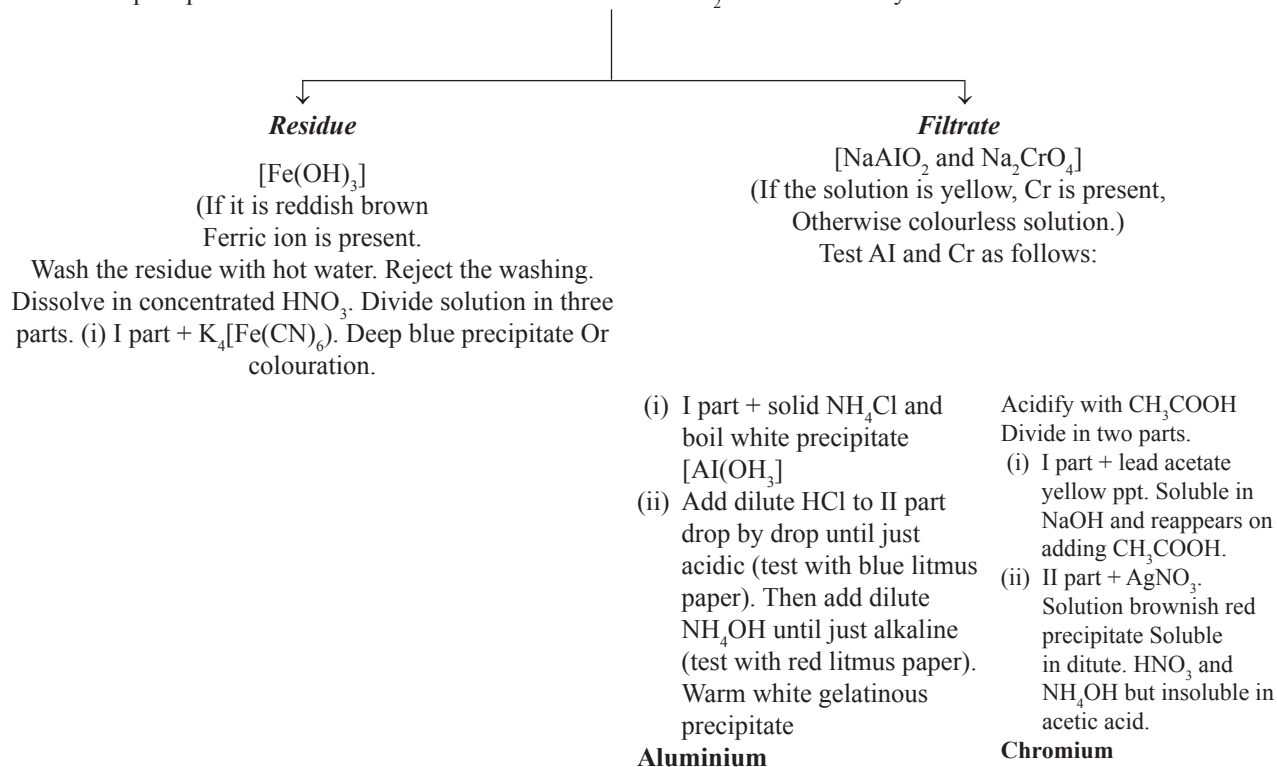
Group III A (Al^{3+} , Fe^{3+} , Cr^{3+})

Group reagent: $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$

Hydroxides are precipitated on addition of excess of ammonium hydroxide in presence of ammonium chloride.

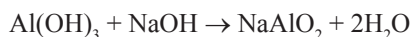


The precipitate of III Group consists of reddish brown $\text{Fe}(\text{OH})_3$; dirty green $\text{Cr}(\text{OH})_3$; and white gelatinous $\text{Al}(\text{OH})_3$. The precipitate is washed 2–3 times with a solution of 10% NH_4Cl and then transferred to a dish. The excess of NaOH is added to the precipitate and the solution is boiled with excess of Br_2 water and finally filtered.

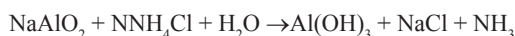


Al^{3+} (aluminium)

- White precipitate of $\text{Al}(\text{OH})_3$ is soluble in NaOH

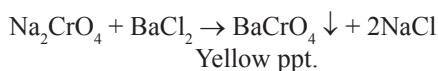
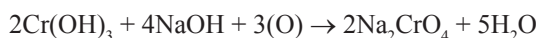
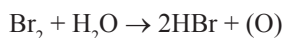


- The solution is boiled with ammonium chloride when $\text{Al}(\text{OH})_3$ is again formed.



Cr^{3+} (chromium)

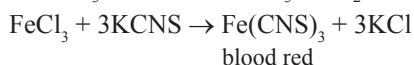
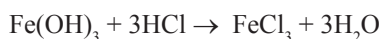
- Precipitate of $\text{Cr}(\text{OH})_3$ is soluble in $\text{NaOH} + \text{Br}_2$ water and addition of BaCl_2 to this solution gives yellow precipitate.



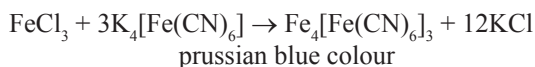
Fe^{3+} (iron)

- $\text{Fe}(\text{OH})_3$ is insoluble in NaOH

Brown precipitate of $\text{Fe}(\text{OH})_3$ is dissolved in HCl and addition of KCNS to this solution gives blood red colour.



Also on addition of $K_4[Fe(CN)_6]$ to this solution, a prussian blue colour is obtained.



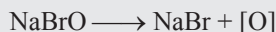
Worked-out Examples

31. If Fe^{3+} and Cr^{3+} both are present in group III of qualitative analysis, then distinction can be made by (AIEEE 2002)

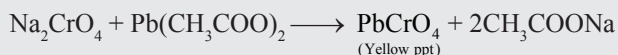
- addition of NH_4OH in presence of NH_4Cl when only $Fe(OH)_3$ is precipitated
- addition of NH_4OH in presence of NH_4Cl when only $Cr(OH)_3$ and $Fe(OH)_3$ both are precipitated and on adding Br_2 water and $NaOH$, $Cr(OH)_3$ dissolves
- precipitate of $Cr(OH)_3$ and $Fe(OH)_3$ as obtained in (b) are treated with conc HCl when only $Fe(OH)_3$ dissolves
- both (b) and (c)

Ans. If Fe^{3+} and Cr^{3+} both are present, then very first solid ammonium chloride and ammonium hydroxide is added slowly till the solution smells of ammonia. Fe^{3+} and Cr^{3+} precipitates in the hydroxide form.

For identification precipitate is treated with $NaOH$ and Br_2 water, yellow colouration confirms Cr^{3+} ion.

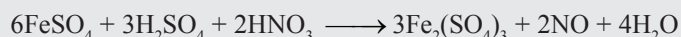


Solution is acidified and treated with lead acetate solution.



32. Why is the original solution boiled with conc. HNO_3 in III group?

Ans. Fe^{2+} salts are oxidized to Fe^{3+} salts by boiling with concentrated HNO_3 , before adding NH_4Cl and NH_4OH ; otherwise Fe^{2+} would not be precipitated in III group.



33. Concentrated HNO_3 is added in original solution before testing III gp. cations, why?

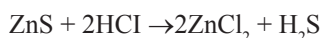
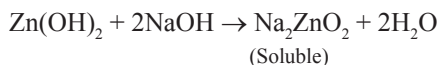
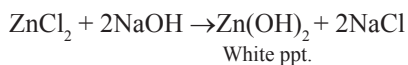
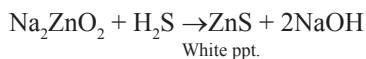
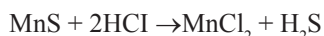
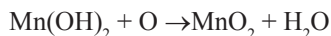
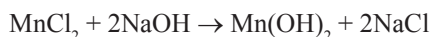
Ans. HNO_3 oxidizes Fe^{2+} to Fe^{3+} . If NH_4OH is added in solution containing Fe^{2+} , then $Fe(OH)_2$ is not completely precipitated but $Fe(OH)_3$ is precipitated completely.

Some Noteworthy Points

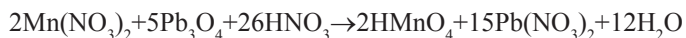
- If Fe^{2+} ions are expected (indicated by light green colour of the mixture) the solution after group II is boiled with a few drops of conc. HNO_3 to oxidise Fe^{2+} to Fe^{3+} . This is because the solubility product of $Fe(OH)_2$ is high as compared to $Fe(OH)_3$ and in presence of NH_4Cl (group reagent), the ionisation of NH_4OH is suppressed and thus only the solubility product of $Fe(OH)_3$ is reached. It is important to note that iron in the II group filtrate will be present in the ferrous state no matter if it was originally in the ferric state. It is because on passing H_2S in II group, Fe^{3+} ions are reduced to Fe^{2+} ion.
- Function of NH_4Cl is to suppress the ionisation of NH_4OH so that only the III group radicals are precipitated because the solubility product of III group hydroxides is less than that of IV and V group hydroxides
- Excess of NH_4Cl should be added otherwise, manganese will be precipitated as barium (if present) as $BaSO_4$
- NH_4NO_3 cannot be used in place of NH_4Cl because NO_3^- ions will oxidise Mn^{2+} to Mn^{3+} and thus $Mn(OH)_3$ will be precipitated in third group.

Group IV**Radicals:** Ni^{2+} , Co^{2+} , Mn^{2+} , Zn^{2+} Precipitate**Group reagent:** H_2S gas in presence of $\text{NH}_4\text{Cl} + \text{NH}_4\text{OH}$

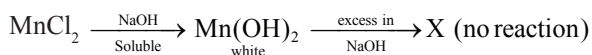
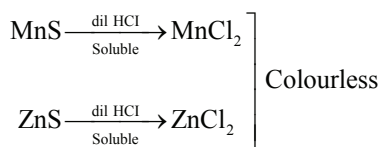
On passing H_2S through the filtrate of the third group, sulphides of fourth group are precipitated. NiS and CoS are black and insoluble in concentrated HCl while MnS (buff coloured), ZnS (colourless) are soluble in concentrated HCl .

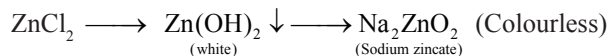
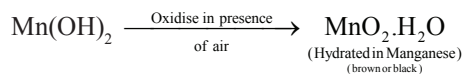
 Zn^{2+} (Zinc)The sulphide dissolves in HCl .When the solution is treated with NaOH , first a white precipitate appears which dissolves in excess of NaOH .On passing H_2S , white precipitate of zinc sulphide is formed. **Mn^{2+} (Manganese)**Manganese sulphide dissolves in HCl .On heating the solution with NaOH and Br_2 -water, manganese dioxide gets precipitated.

The precipitate is treated with excess of nitric acid and PbO_2 or Pb_3O_4 (red lead). The contents are heated. The formation of permanganic acid imparts pink colour to the supernatant liquid.

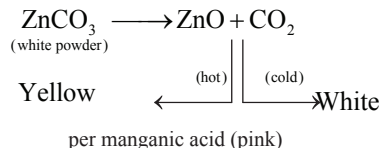
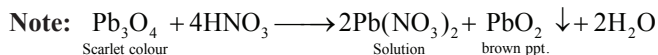
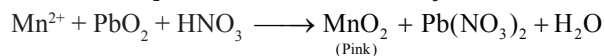


Mind map



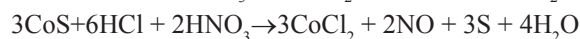


Test of Mn with PbO₂ and concentrated HNO₃.



Ni²⁺ (nickel) and Co²⁺ (cobalt)

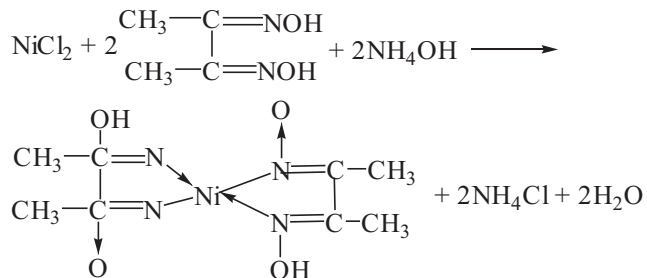
The black precipitate is dissolved in aqua-regia.



The solution is evaporated to dryness and residue extracted with dilute HCl.

It is divided into three parts:

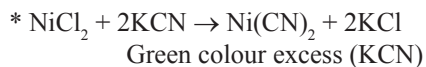
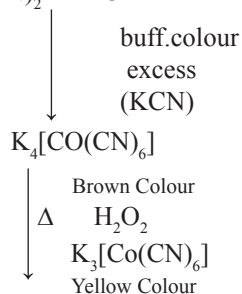
Part I: Add NH₄OH (excess) and dimethyl glyoxime. A rosy red precipitate appears, if nickel is present:



Test used to differentiate NiCl₂ and CoCl₂.

(i) KCN Test

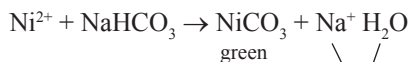
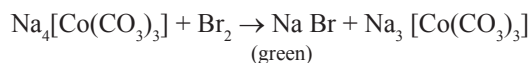
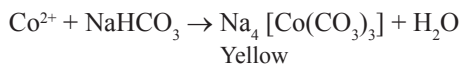
(ii) Palit Test



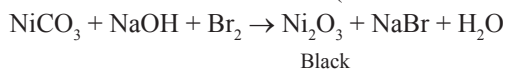
14.32 Salt Analysis

(ii) Palit Test:

Group reagent of palit test $\rightarrow \text{Br}_2$ and NaHCO_3

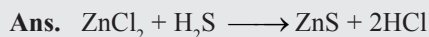


(alkaline solution)



Worked-out Example

34. When H_2S gas is passed through ZnCl_2 solution, ZnS is not precipitated, why?



The HCl formed dissolves ZnS .

Group V

Radicals: Ba^{2+} , Sr^{2+} , Ca^{2+}

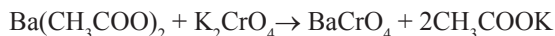
Group reagent: $(\text{NH}_4)_2\text{CO}_3$ in presence of NH_4OH

Ammonium carbonate precipitates Vth group radicals in the form of carbonates. These carbonates are soluble in acetic acid.



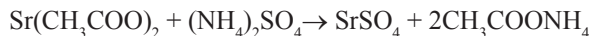
Ba^{2+} (barium)

Barium chromate is insoluble and precipitated by Ole addition of potassium chromate solution.



Sr^{2+} (strontium)

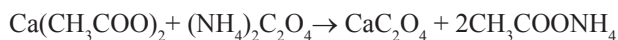
Strontium sulphate is insoluble and precipitated by the addition of ammonium sulphate solution.



White ppt.

Ca^{2+} (calcium)

Calcium oxalate is insoluble and precipitated by the addition of ammonium oxalate.



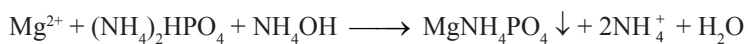
White ppt.

Analysis of Group VI

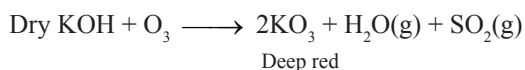
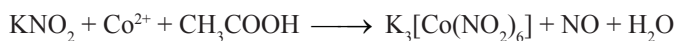


Important reactions of Mg^{2+}

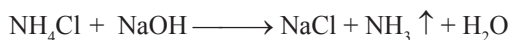
Mg^{2+} ions give white precipitate with NH_4OH and $(\text{NH}_4)_2\text{HPO}_4$.



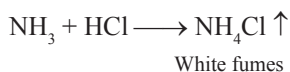
Important reactions of K^+



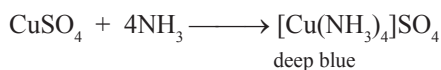
Analysis of NH_4^+ (i) All ammonium salts on heating with alkali give a colourless gas with a pungent smell (NH_3)



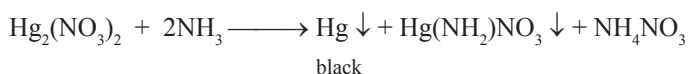
(a) Gas evolved gives white fumes with HCl



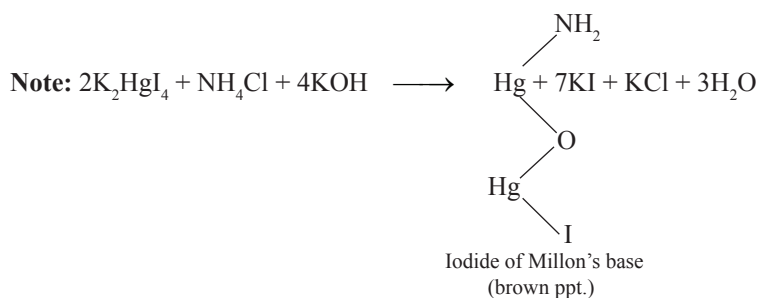
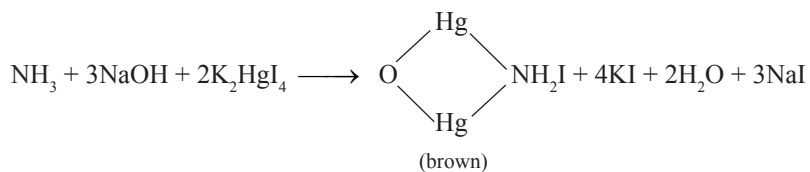
(b) Paper soaked in CuSO_4 solution, is turned deep blue by NH_3 due to complex formation



(c) With $\text{Hg}_2(\text{NO}_3)_2$, a black colour is obtained



(d) With Nessler's reagent, a brown precipitate is obtained.

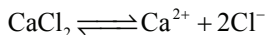
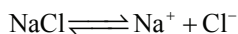


Dry Tests

These test can be performed with the dry sample of the salt.

(i) Flame test

Some volatile salts impart characteristic colour to the non-luminous flame. The chlorides of the metals are more volatile in comparison to other salts. The metal chloride volatilizes and its thermal ionization takes place.



The cations impart a characteristic colour to the flame as these absorb energy from the flame and transmit the same as light of characteristic colour.

<i>Colour on flame</i>	<i>Inference</i>
1. Golden yellow	Sodium
2. Violet	Potassium
3. Brick red	Calcium
4. Crimson red	Strontium
5. Apple green	Barium
6. Green with a blue centre	Copper

Note: Flame test should not be performed in the presence of As, Sb, Bi, Sn and Pb as these radicals form alloy with platinum and hence, the wire is spoiled.

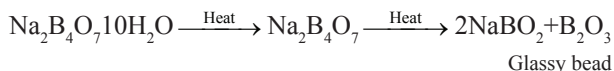
Worked-out Example

35. Cu^{2+} and Ba^{2+} interfere in the flame test for borate, why?

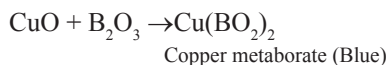
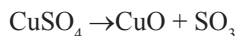
Ans. Because they also impart green colour to flame.

(ii) Borax bead test

When borax is heated, a colourless glassy bead is formed that consists of sodium metaborate and boric anhydride.

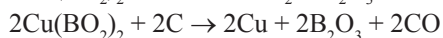
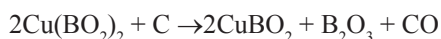


On heating with a coloured salt, the glassy bead forms a coloured metaborate in oxidizing flame.



The metaborates possess different characteristic colours. The shade of the colour gives a clue regarding the presence of the radical. However, in reducing flame the colours may be different due to different reactions.

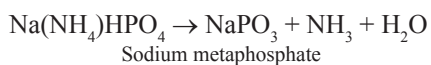
For example, copper metaborate may be reduced to colourless cuprous metaborate or to metallic copper, which appears red and opaque.



<i>Metal</i>	<i>Colour of the bead in</i>			
	<i>Oxidizing-flame</i>		<i>Reducing-flame</i>	
	<i>Hot</i>	<i>Cold</i>	<i>Hot</i>	<i>Cold</i>
Copper	Green	Blue	Colourless	Brown-red
Iron	Brown-yellow	Pale-yellow	Bottle green	Bottle green
Chromium	Green	Green	Green	Green
Cobalt	Blue	Blue	Blue	Blue
Manganese	Violet	Amethyst red	Grey	Grey
Nickel	Violet	Brown	Grey	Grey

(iii) Microcosmic salt bead test

When microcosmic salt is heated, a colourless transparent bead of sodium meta phosphate is formed.

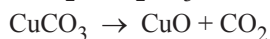
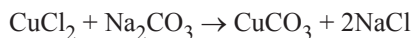


Sodium metaphosphate combines with metallic oxides to form orthophosphates which are usually coloured. The shade of the colour gives a clue regarding the presence of metal. Like borax bead test, colours are noted both in oxidizing and reducing flames in hot and cold conditions.

<i>Metal</i>	<i>Colour of the bead in</i>			
	<i>Oxidizing-flame</i>		<i>Reducing-flame</i>	
	<i>Hot</i>	<i>Cold</i>	<i>Hot</i>	<i>Cold</i>
Copper	Green	Blue	Colourless	Red
Iron	Yellow or reddish-brown	Yellow	Yellow	Colourless
Chromium	Green	Green	Green	Green
Manganese	Violet	Violet	Colourless	Colourless
Cobalt	Blue	Blue	Blue	Blue
Nickel	Brown	Brown	—	Grey

Charcoal Cavity Test

The mixture is mixed with double of its amount of anhydrous sodium carbonate and placed in a charcoal block having a small cavity. The mass is moistened with a drop of water and heated in a reducing flame with a blow pipe. Metal salt is converted into metal via carbonate and oxide. e.g.,



The colour in the cavity bead or incrustation is observed.

Observation**Incrustation****1. Formation of metallic bead**

- | | |
|----------------------------------|----|
| (i) Lustrous white, malleable | Ag |
| (ii) Greyish white, marks paper | Pb |
| (iii) White, does not mark paper | Sn |
| (iv) Red | Cu |

2. Incrustation with metal

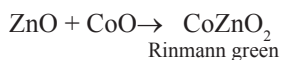
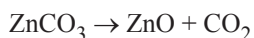
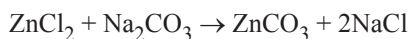
- | | |
|--|----|
| (i) White incrustation, brittle metal | Sb |
| (ii) Yellow incrustation, brittle metal | Bi |
| (iii) Yellow incrustation, malleable metal | Pb |

3. Incrustation without metal

- | | |
|-------------------------------------|--------------------------------|
| (i) White and yellow when hot | ZnO, SnO |
| (ii) Yellow and orange when hot | BiO |
| (iii) Brown | CdO |
| (iv) White (volatile, garlic odour) | As ₂ O ₃ |

Cobalt Nitrate Charcoal Test

The mixture is mixed with double of its amount with sodium carbonate, placed in the charcoal cavity, and moistened with a drop of water and heated in an oxidizing flame with the help of a blow pipe. After cooling, one or two drops of cobalt nitrate solution are added and the mass is again heated in the oxidising flame. Metal oxide combines with cobalt oxide (from cobalt nitrate) forming mixed oxide of characteristic colour.

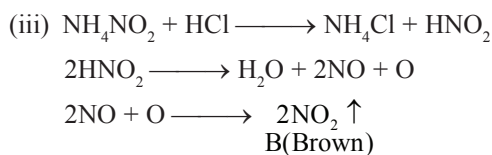
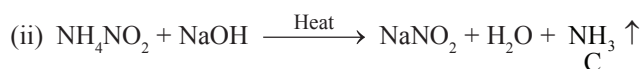
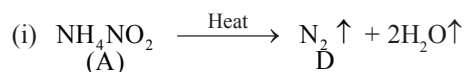


Colour	Incrustation
Blue mass	Al
Green residue	Zn
Pink residue	Mg
Bluish green	Sn

Solved Example

1. A colourless solid A liberates a brown gas B on acidification, a colourless alkaline gas C on treatment with NaOH, and a colourless non-reactive gas D on heating. If heating of the solid A is continued, it completely disappears. Identify A to D.

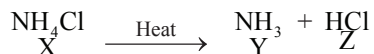
Solution: Treatment of compound A with NaOH to give alkaline gas C(NH₃) indicates that A is an ammonium salt. Further, heating of A to give non-reactive gas D (most probably N₂) indicates that A is nitrite. Thus, the compound A is ammonium nitrite and its different reactions are represented as below.



2. A colourless water soluble solid X on heating gives equimolar quantities of Y and Z. Y gives dense white fumes with HCl and Z does so with ammonia. Y gives brown precipitate with Nessler's reagent and Z gives white precipitate with nitrates of Ag⁺, Pb²⁺ and Hg⁺. What is X?

Solution: The reactions reveal following facts

(i) Y is NH_3 and (ii) Z is HCl . Hence, X must be NH_4Cl .



3. An aqueous solution of gas (X) shows the following reactions

- (i) It turns red litmus blue.
 - (ii) When added in excess to copper sulphate solution, a deep blue colour is obtained.
 - (iii) On addition of FeCl_3 solution a brown ppt. soluble in dilute nitric acid is obtained.
- Identify (X) and give equations for the reactions at step (ii) and (iii)

Solution: X - NH_3

Reactions: (i) $\text{CuSO}_4 + 4\text{NH}_4\text{OH} \longrightarrow [\text{Cu}(\text{NH}_3)_4]\text{SO}_4 + \text{H}_2\text{O}$
deep blue

(ii) $\text{FeCl}_3 + 3\text{NH}_4\text{OH} \longrightarrow \text{Fe}(\text{OH})_3 \downarrow + 3\text{NH}_4\text{Cl}$
brown ppt.

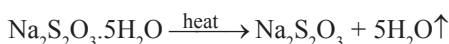
$\text{Fe}(\text{OH})_3 + 3\text{HNO}_3 \longrightarrow \text{Fe}(\text{NO}_3)_3 + 3\text{H}_2\text{O}$
Soluble

4. An unknown inorganic compound (X) loses its water of crystallization on heating and its aqueous solution gives the following reactions.

- (i) It gives a white turbidity with dilute hydrochloric acid solution.
- (ii) It decolourises a solution of iodine in potassium iodide.
- (iii) It gives a white precipitate with silver nitrate solution which turns black on standing.
- (iv) It imparts a golden yellow colour to the flame in the flame test.

Identify the compound (X) and give chemical equations for the reactions at steps (i), (ii) and (iii).

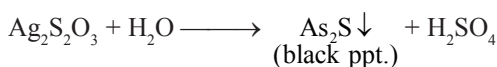
Solution: Since the compound X decolourises a solution of iodine in potassium iodide, it should contain thiosulphate ion, $\text{S}_2\text{O}_3^{2-}$ which also coincides with the two other given facts, i.e., (i) and (iii). Hence, the compound X is sodium thiosulphate, $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$ which explains the given reactions as below.



(i) $\text{Na}_2\text{S}_2\text{O}_3 + 2\text{HCl} \longrightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{SO}_2 + \overset{\text{S}}{\text{(white turbidity)}}$

(ii) $2\text{Na}_2\text{S}_2\text{O}_3 + \text{I}_2 \longrightarrow \text{Na}_2\text{S}_4\text{O}_6 + 2\text{NaI}$
Sod. thiosulphate Sod. tetrathionate

(iii) $\text{Na}_2\text{S}_2\text{O}_3 + 2\text{AgNO}_3 \longrightarrow \text{Ag}_2\text{S}_2\text{O}_3 \downarrow + 2\text{NaNO}_3$
Silver thiosulphate
(white ppt.)

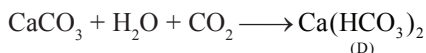
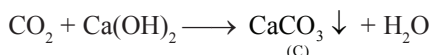
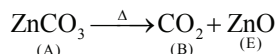


5. When a white powder A is strongly heated, it gives off a colourless, odourless gas B which turns lime water milky due to the formation of C and if the passage of this gas is continued, the milkiness disappears and gives a solution D. The solid residue E is yellow when hot, but turns white on cooling. Identify A to E with the help of the equations.

Solution: (a) ZnCO_3 (b) CO_2 (c) CaCO_3
(d) $\text{Ca}(\text{HCO}_3)_2$ (e) ZnO

14.38 Salt Analysis

The salt must be a carbonate as CO_2 gas is evolved on heating. The residue is yellow when hot and white when cold. This indicates ZnO .

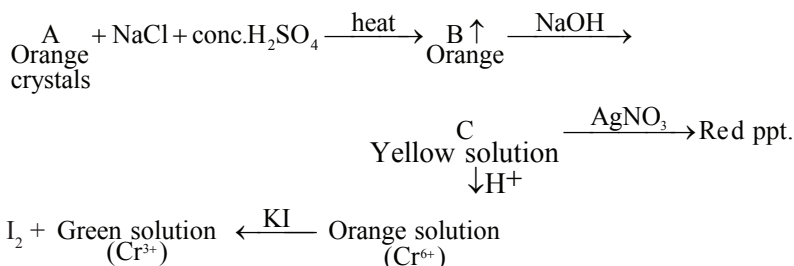


6. When an orange coloured crystalline compound (A) was heated with common salt and concentrated sulphuric acid an orange-yellow coloured gas (B) was evolved. The gas (B) when passed through caustic soda solution gave a yellow solution (C) which in turn gave the following reactions.

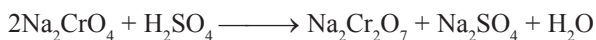
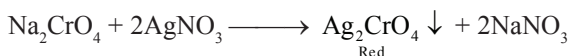
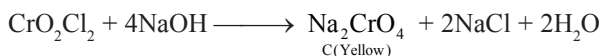
- Addition of silver nitrate solution to (C) gave first a white precipitate which then turns red. Quantitatively, 0.155 g of the gas (B) required 2.0 mmole of AgNO_3 to produce the first trace of red colour.
- Acidification of the solution (C) with dilute H_2SO_4 gave an orange solution which contained chromium in +6 oxidation state. The solution liberated iodine from aqueous potassium iodide, leaving a green solution containing chromium in +3 oxidation state. Quantitatively 0.155 g of the gas (B) liberated 1.5 mmole of iodine.

Deduce the formula of A, B and C and explain the reactions.

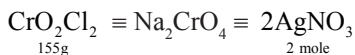
Solution: Let us summarise the given reactions.



The above set of reactions indicates that compound A is $\text{K}_2\text{Cr}_2\text{O}_7$, B is chromyl chloride gas and C is sodium chromate which explains all the given reactions as below.



The quantitative data is explained in the following manner. From the above reactions we observe that,



155 g of CrO_2Cl_2 require 2 mole of AgNO_3

$$0.155 \text{ g of } \text{CrO}_2\text{Cl}_2 \text{ will require } = \frac{2}{155} \times 0.155 = 0.002 \text{ mole} = 2 \text{ mmole}$$

This coincides with the given data.



Thus, 2×155 g of CrO_2Cl_2 liberate 3 moles of I_2

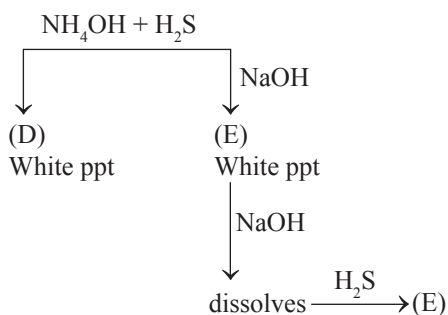
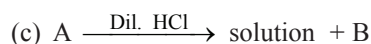
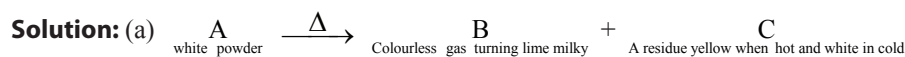
$$0.155 \text{ g } \text{CrO}_2\text{Cl}_2 \text{ will liberate } = \frac{3 \times 0.155}{2 \times 155}$$

$$= 0.0015 \text{ moles}$$

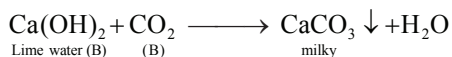
$$= 1.5 \text{ mmole}$$

This also coincides with the given data.

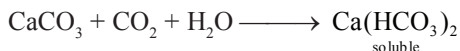
7. A white amorphous powder (A) when heated, gives a colourless gas (B), which turns lime water milky (which dissolves on passing excess of gas (B) and the residue (C) which is yellow while hot but white when cold. The residue (C) dissolves in dilute HCl and the resulting solution gives a white precipitate on addition of potassium ferrocyanide solution. (A) dissolves in dilute HCl with the evolution of a gas which is identical in all respects to gas (B). The solution of (A) in dilute HCl gives a white precipitate (D) on addition of NH_4Cl excess of NH_4OH and on passing H_2S gas. Another portion of this solution gives initially a white ppt. (E) on addition of NaOH solution which dissolves in excess of NaOH. The solution on passing again H_2S gives back the white precipitate of (E), the white precipitate on heating with dilute H_2SO_4 give a gas used in II and IV group analysis. What are (A) to (E)? Give balanced chemical equations of the reactions.



From observations for section (a), one may conclude that the colourless gas is CO_2 because it turns lime water milky, due to formation of insoluble CaCO_3 .



CaCO_3 is soluble in excess of CO_2 due to formation of soluble calcium bicarbonate

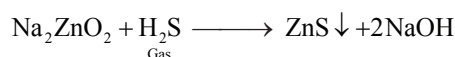
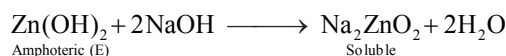
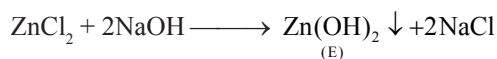
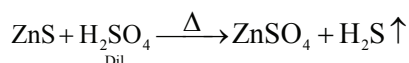
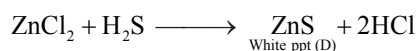
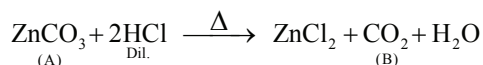
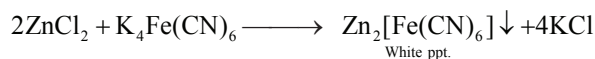
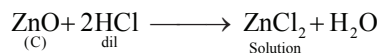
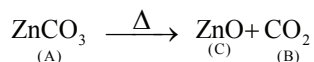


The compound (C) is zinc oxide (ZnO) because it is yellow when hot and white when cold, hence, the initial compound (A) is zinc carbonate (ZnCO_3).

From section (b), it is inferred that (C) is a salt of Zn(II) which dissolves in dilute HCl and white precipitate obtained after addition of $\text{K}_4\text{Fe(CN)}_6$ is due to zinc ferrocyanide.

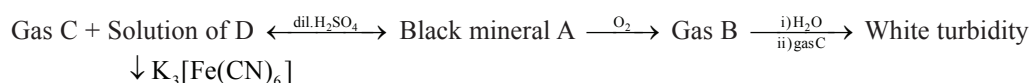
ZnCO_3 because on treatment with dilute HCl it gives gas (B), i.e., CO_2 , while Zn (II) goes in solution, i.e., ZnCl_2 . On passing H_2S gas in presence of NH_4OH , it gives a white precipitate of ZnS (D). ZnS on heating with dilute H_2SO_4 evolves H_2S , which is used for the precipitation of sulphides of group II in acidic medium and of IV group in alka-

line medium. ZnCl_2 reacts with NaOH to give a precipitate of Zn(OH)_2 which dissolves in NaOH , as Zn(OH)_2 is amphoteric in nature. The solution Na_2ZnO_2 again gives ZnS on passing H_2S gas into it. Different chemical equations concerned are given below:

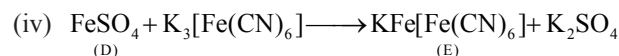
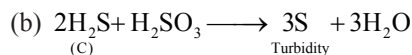
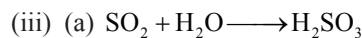
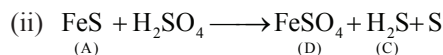


8. (i) A black mineral (A) on heating in presence of air gives a gas (B).
 (ii) The mineral (A) on reaction with dilute H_2SO_4 gives a gas (C) and solution of a compound (D).
 (iii) On passing gas (C) into an aqueous solution of (B) a white turbidity is obtained.
 (iv) The aqueous solution of compound (D) on reaction with potassium ferricyanide gives a blue compound (E).
 Identify (A) to (E) and give chemical equations for reactions at steps (i) to (iv).

Solution: Let us summarize the given statements.



Formation of blue coloured compound E by treating solution of compound D with potassium ferricyanide indicates that compound D contains Fe^{2+} ion which is also the constituent ion of the compound A. Further A, is a black mineral of Fe^{2+} ion, it should be ferrous sulphide (FeS) which is confirmed by the various given reactions.



9. (i) An ore (A) on roasting with sodium carbonate and lime in the presence of air gives two compounds, (B) and (C).
 (ii) The solution of (B) in concentrated HCl on treatment with potassium ferrocyanide gives a blue colour or precipitate of compound (D).

- (iii) The aqueous solution of (C) on treatment with concentrated H_2SO_4 gives a yellow coloured compound (E).
 (iv) Compound (E) when treated with KCl gives an orange-red compound (F) which is used as an oxidizing reagent.
 (v) The solution of (F) on treatment with oxalic acid and then with an excess of potassium oxalate gives blue crystals of compound (G).

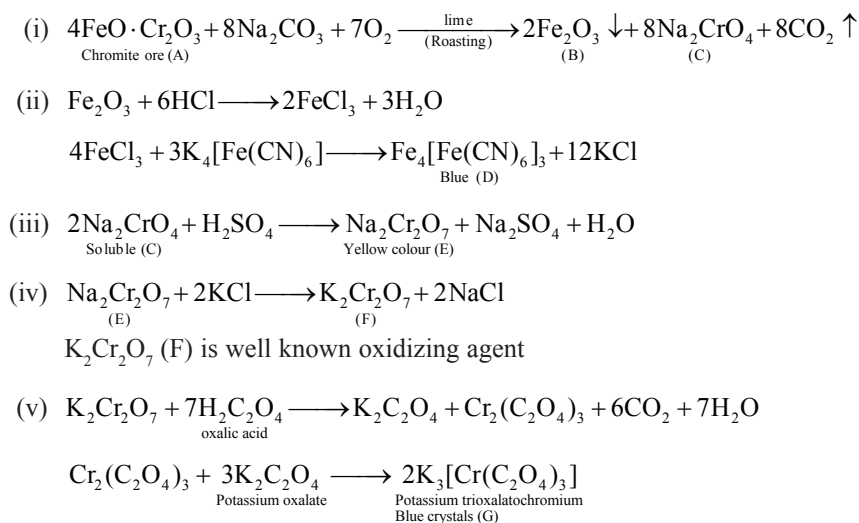
Identify (A) to (G) and give balanced chemical equations for reactions at steps (i) to (v).

Solution: (a) Reaction (i) suggests that the ore contains Fe.

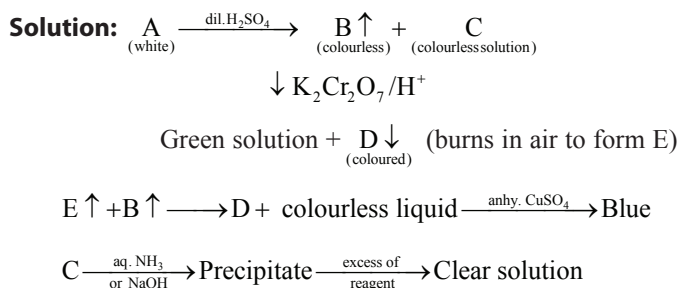
(b) Reaction (ii) and (iv) suggest that the ore also contains chromium.

In fact, if we review our memory, we will remember that the reaction sequence (i), (iii) and (iv) are involved in the preparation of $\text{K}_2\text{Cr}_2\text{O}_7$ from chromite ore, $\text{FeO} \cdot \text{Cr}_2\text{O}_3$.

Hence, the given reactions can be written as below,



10. A white substance A reacts with dilute H_2SO_4 to produce a colourless gas B and a colourless solution C. The reaction between B and acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution produces a green solution and a slightly coloured precipitate D. The substance D burns in air to produce a gas E which reacts with B to yield D and a colourless liquid. Anhydrous copper sulphate is turned blue on addition of this colourless liquid. Addition of aqueous NH_3 or NaOH to C produces first a precipitate which dissolves in the excess of the respective reagent to produce a clear solution in each case. Identify A, B, C, D and E. Write the equation of the reactions involved.

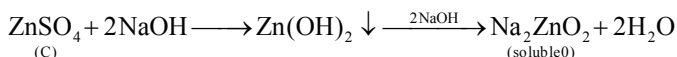
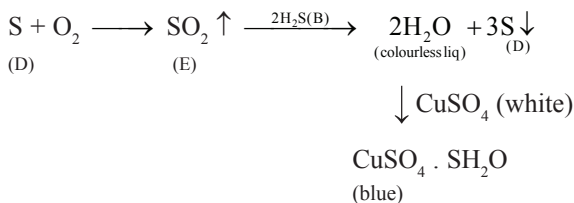
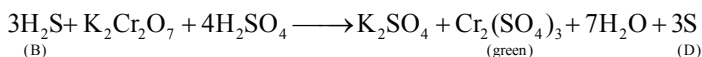
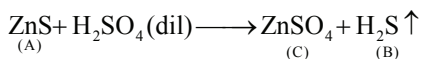


The above set leads to following conclusions.

- (i) Since the gas (B) is colourless and turns acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution green, it should be H_2S .
 (ii) Since H_2S gas is obtained by the reaction of dilute H_2SO_4 on A, the latter must be sulphide.
 (iii) The white colour of the sulphide (A) points out towards ZnS .

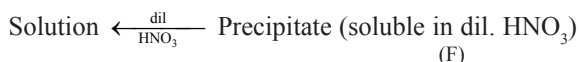
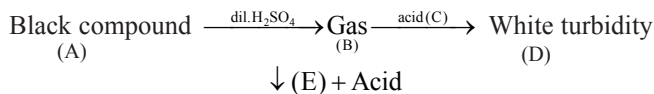
14.42 Salt Analysis

Thus, the various reactions can be written as below.



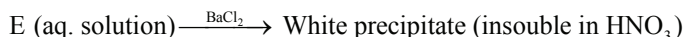
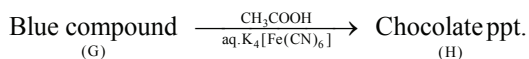
11. A black coloured compound (A) on reaction with dilute sulphuric acid gives a gas (B) which on passing in a solution of an acid (C) gives a white turbidity (D). Gas (B) when passed in acidified solution of a compound (E) gives a precipitate (F) soluble in dilute nitric acid. After boiling this solution, when an excess of ammonium hydroxide is added, a blue coloured compound (G) is formed. To this solution on addition of acetic acid and aqueous potassium ferrocyanide a chocolate precipitate (H) is obtained. On addition of an aqueous solution of barium chloride to an aqueous solution (E), a white precipitate insoluble in HNO_3 is obtained. Identify from (A) to (H).

Solution:



(i) boil

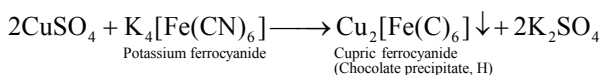
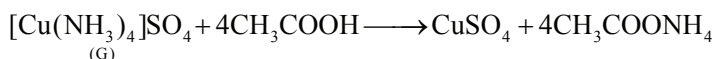
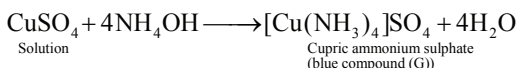
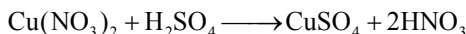
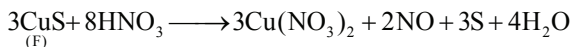
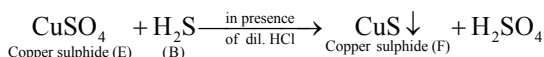
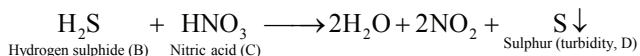
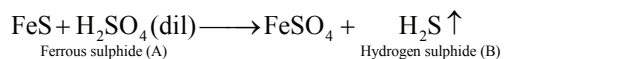
(ii) NH_4OH



Let us draw important conclusion from the given summary:

- Reaction in aqueous solution of the compound E with BaCl_2 to give white precipitate insoluble in nitric acid indicates that the salt (E) contains SO_4^{2-} ion.
- Reaction of compound (G) with potassium ferrocyanide in presence of acetic acid to give chocolate precipitate (H) indicates that (G) must contain Cu^{2+} and hence, (H) must be cupric ferrocyanide, $\text{Cu}_2[\text{Fe}(\text{CN})_6]$.
- Since compound (G) is derived from (F), compound (F) also contains Cu^{2+} . Further, since (F) is derived from the reaction of the gas (B) and compound (E), (E) must contain Cu^{2+} ion. Precipitate of Cu^{2+} ion soluble in dil. nitric acid should be CuS ; hence (F) must be CuS and thus (B) is H_2S .
- According to point (i), compound (E) contains SO_4^{2-} hence (E) must be CuSO_4 .
- Gas (B) which has now been identified as H_2S is obtained by the decomposition of black coloured compound (A) with dil. H_2SO_4 . Hence, (A) must be sulphide of Cu, Pb, Hg, Fe, Co, Ni, etc.,

Thus, the various compounds from (A) to (H) and their reactions can be written as below.



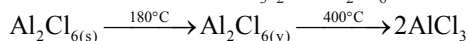
Thus the compounds (A) to (H) are,

A = Ferrous sulphide, FeS, B = Hydrogen sulphide, H_2S , C = Nitric acid, HNO_3 , D = Sulphur S, E = Copper sulphate, CuSO_4 , F = Copper sulphide, CuS, G = Cupric ammonium sulphate, $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$, H = Cupric ferrocyanide, $\text{Cu}_2[\text{Fe}(\text{CN})_6]$.

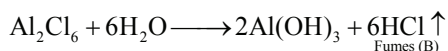
12. An inorganic compound (A) shows the following reactions.

- (i) It is white solid and exists as dimer; gives fumes of (B) with wet air.
- (ii) It sublimes on 180°C and forms monomer if heated to 400°C .
- (iii) Its aqueous solution turns blue litmus to red.
- (iv) Addition of NH_4OH and NaOH separately to a solution of (A) gives white precipitate which is however soluble in excess of NaOH.

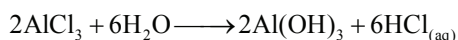
Solution: (i) (a) is a characteristic dimerized compound which sublimes on 180°C and forms monomer if heated to 400°C and thus, (A) is $(\text{AlCl}_3)_2$ or Al_2Cl_6 .



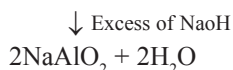
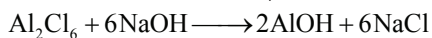
(ii) It fumes with wet air



(iii) Its solution in water is acidic due to hydrolysis



(iv) (a) gives white ppt. with NH_4OH and NaOH, soluble in excess of NaOH.

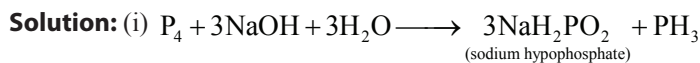


13. (i) Solution salt of an acid (A) is formed on boiling white phosphorus with NaOH solution.
- (ii) On passing chlorine through phosphorus kept fused under water, another acid (B) is formed which on strong heating gives metaphosphorus acid.

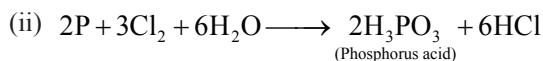
14.44 Salt Analysis

- (iii) Phosphorus on treatment with concentrated HNO_3 gives an acid (C) which is also formed by the action of dilute H_2SO_4 on powdered phosphorite rock.
- (iv) (A) on treatment with a solution of HgCl_2 first gives a white precipitate of compound (D) and then a grey precipitate (E).

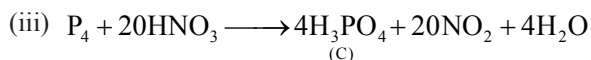
Identify (A) to (E) and write balanced chemical equations for the reactions at step (i) to (iv)



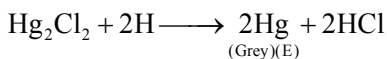
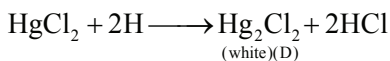
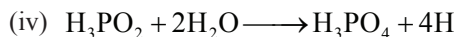
Thus, acid (A) is H_3PO_2 , i.e., hypophosphorus acid



Thus, acid (B) is H_3PO_3



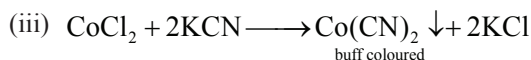
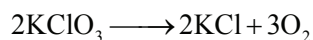
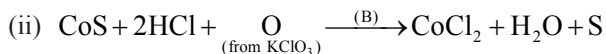
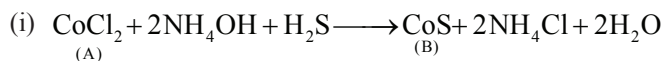
Thus, acid (C) is H_3PO_4

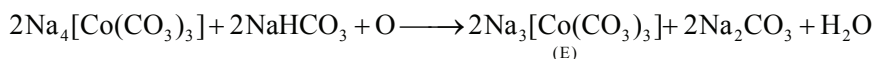
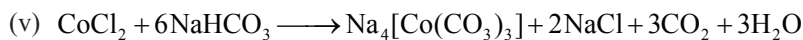
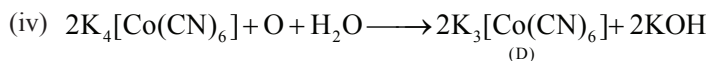


14. (i) A black coloured compound (B) is formed on passing hydrogen sulphide through the solution of a compound (A) in NH_4OH .
- (ii) (B) on treatment with hydrochloric acid and potassium chlorate gives (A).
- (iii) (A) on treatment with potassium cyanide gives a buff coloured precipitate which dissolves in excess of this reagent forming a compound (C).
- (iv) The compound (C) is changed into a compound (D) when its aqueous solution is boiled.
- (v) The solution of (A) was treated with excess of sodium bicarbonate and then with bromine water. On cooling and shaking for some time, a green colour of compound (E) is formed. No change is observed on heating.

Identify (A) to (E) and give chemical equations for the reactions at steps (i) to (v).

Solution: The formation of black coloured compound (B) by passing H_2S through the alkaline solution of the compound (A) indicate that (A) is a salt of the IV group radicals (Co^{2+} , Ni^{2+} or Zn^{2+}). However, the given reactions especially reaction (iii) indicates what compound (A) is a cobalt salt (CoCl_2) which explains all the given reactions.





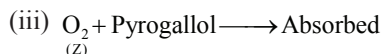
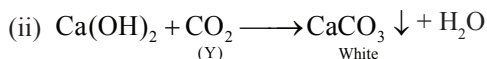
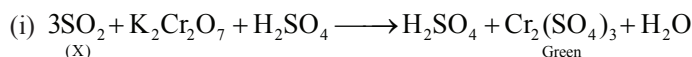
15. A gaseous mixture containing (X), (Y) and (Z) gases, when passed into acidified $K_2Cr_2O_7$ solution, gas (X) was absorbed and the solution was turned green. The remainder gas mixture was then pass through lime water, which turns milky by absorbing gas (Y). The residual gas when passed through alkaline pyrogallol solution, it turns black. Identify gas (X), (Y) and (Z) and explain the reaction involved.

Solution: (i) Gas (X) is absorbed in acidified $K_2Cr_2O_7$ and the solution turns green. So (X) is SO_2 .

(ii) Gas (Y) is absorbed in lime water turning it white so (Y) is CO_2 .

(iii) Gas (Z) is absorbed in pyrogallol so (Z) is O_2 .

Reactions

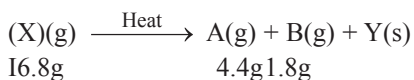


Subjective Problems

1. When 16.8 g of white solid X were heated, 4.4 g of acid gas A that turned lime water milky was driven off together with 1.8 g of a gas B which condensed to a colourless liquid. The solid that remained, Y, dissolved in water to give an alkaline solution, which with excess barium chloride solution gave a white precipitate Z. The precipitate effervesced with acid giving off carbon dioxide. Identify A, B and Y and write down the equation for the thermal decomposition of X. (1984–4 Marks)

Ans. CO_2 , H_2O , Na_2CO_3

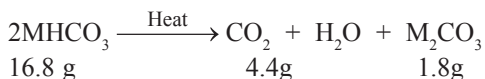
Solution: Representing the given facts in the form of equation.



The above equation leads to the following facts:

- (i) Since the gas A turned lime water milky, it must be CO_2 .
- (ii) The compound Y gives alkaline solution in water which when treated with $BaCl_2$ forms a white precipitate of Z. Since the compound Z when treated with acid gives effervescence of CO_2 , Z and hence Y must be carbonate, CO_3^{2-} . Hence, Y may be written as metal carbonate MCO_3 or M_2CO_3 .
- (iii) When X is heated, it yields a carbonate (Y) along with the evolution of CO_2 (A) and another gas (B), it must be a bicarbonate.
- (iv) The above facts point out that B may be water.

Thus, the above reaction can be written as below.



Calculation of molecular weight of $MHCO_3$

44.g of CO_2 is given by 16.8 g of $MHCO_3$

44g of CO_2 is given by $= \frac{16.8}{4.4} \times 44 = 168\text{ g}$

14.46 Salt Analysis

Since two molecules of MHCO_3 are taking part in the reaction, the molecular weight of

$$\text{MHCO}_3(X) = \frac{168}{2} = 84$$

Calculation of atomic weight of metal M

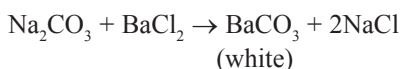
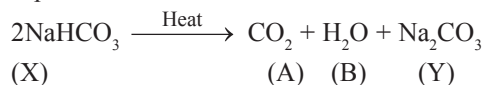
$$\text{MHCO}_3 = 84$$

$$M + 1 + 12 + 48 = 84$$

$$M + 61 = 84$$

$$M = 84 - 61 = 23$$

Thus, the metal must be Na and hence the given salt X is NaHCO_3 . The above facts coincide with the given thermal decomposition.



Thus A is CO_2 , B is H_2O , Y is Na_2CO_3

2. A mixture of two salts was treated as follows:

(1987–5 Marks)

- (i) The mixture was heated with manganese dioxide and concentrated sulphuric acid when yellowish green gas was liberated.
- (ii) The mixture on heating with sodium hydroxide solution gave a gas which turned red litmus blue.
- (iii) Its solution in water gave blue precipitate with potassium ferricyanide and red colouration with ammonium thiocyanate.
- (iv) The mixture was boiled with potassium hydroxide and the liberated gas was bubbled through an alkaline solution of K_2HgI_4 to give brown precipitate.

Identify the two salts. Give ionic equations for reactions involved the tests (i), (ii) and (iii)

Ans. $\text{FeCl}_2, \text{NH}_4\text{Cl}$

Solution: (a) Test (i) of the problem indicates that the mixture contains Cl^- ion which is liberated as Cl_2 (yellowish green gas) when heated with MnO_2 and cone. H_2SO_4 .

(b) Test (ii) indicates the presence of NH_4^+ ion in the mixture which gives ammonia when heated With NaOH solution. Since ammonia is basic in nature, it turns red litmus blue. Presence of NH_4^+ in the mixture is further confirmed by the given test (iv) according to which the gas (NH_3) gives brown precipitate with Nessler's reagent (alkaline solution of K_2HgI_4).

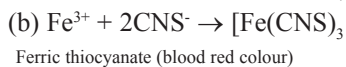
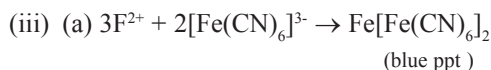
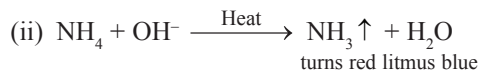
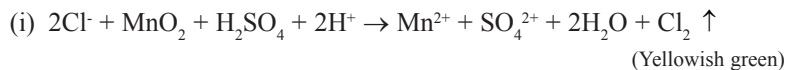
(c) Test (iii) indicates Fe^{2+} ion in the mixture which gives blue precipitate with potassium ferricyanide (note that potassium ferricyanide gives brown precipitate with Fe^{3+} ions).

(d) Red colouration with ammonium thiocyanate indicates that the mixture also contains Fe^{3+} ions which are believed to be formed by the oxidation of Fe^{2+} ions by air.



Thus, the mixture contains **FeCl_2 and NH_4Cl** .

Ionic reactions:

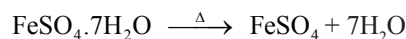


3. A hydrated metallic salt A, light green in colour, on careful heating gives a white anhydrous residue B. B is soluble in water and its aqueous solution reacts with NO to give a dark brown compound C. B on strong heating gives a brown residue D and a mixture of two gases E and F. The gaseous mixture when passed through acidified permanganate, discharges the pink colour and when passed through acidified BaCl_2 solution gave a white precipitate. Identify A, B, C, D, E and F. (1988–3 Marks)

Ans. $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$, FeSO_4 , $\text{FeSO}_4 \cdot \text{NO}$, Fe_2O_3 , SO_2 , SO_3

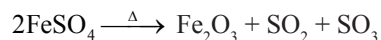
(i) (A) on heating loses water of crystallization and thus it is a hydrated salt.

(ii) Anhydrous salt (B) on heating gives two gases and brown residue and so (B) is FeSO_4 . Thus, (A) is $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$



(A)

(B)



(B)

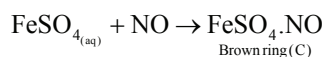
(D)

(E)

(F)

Brown

(iii) (B) is soluble in water and reacts with NO to give brown compound.

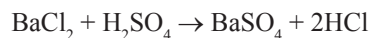
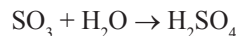


Brown ring (C)

(iv) Gaseous mixture decolourizes acidified KMnO_4 .



(v) Gaseous mixture on passing through BaCl_2 , gives white precipitate of BaSO_4 .

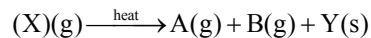


White ppt.

4. When 20.02 g of a white solid X is heated 4.4 g of an acid gas A and 1.8 g of a neutral gas B are evolved, leaving behind a solid residue Y of weight 13.8 g. A turns lime water milky and B condenses into a liquid which changes anhydrous copper sulphate blue. The aqueous solution of Y is alkaline to litmus and gives 19.7 g of white precipitate Z with barium chloride solution. Z gives carbon dioxide with an acid. Identify A, B, X, Y and Z. (1989–5 Marks)

Ans. CO_2 , H_2O , KHCO_3 , K_2CO_3 , BaCO_3

Solution: Representing the given facts in the form of equation.



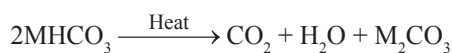
20.02 g

4.4 g

1.8 g

The above equation leads to the following facts:

- Since gas A turned lime water milky, it must be CO_2 .
- The compound Y is alkaline to litmus and when treated with BaCl_2 forms a white precipitate of Z. Since the compound Z when treated with acid gives effervescence of CO_2 , Z and hence Y must be carbonate, CO_3^{2-} . Hence, Y may be written as metal carbonate MCO_3 or M_2CO_3 .
- When X is heated, it yields a carbonate (Y) along with the evolution of CO_2 (A) and a gas (B), it must be a bicarbonate.
- B changes anhydrous CuSO_4 blue, which point out that B is water. Thus, the above reaction can be written as below:



20.02 g

4.4 g

1.08 g

Calculation of molecular weight of MHCO_3

4.4 g of C_2 is given by 20.02 g of MHCO_3

$$44 \text{ g of } \text{C}_2 \text{ is given by } = \frac{22.02}{4.4} \times 44 = 200.2 \text{ g}$$

Since two molecules of MHCO_3 are taking part in the reaction, the molecular weight of

$$\text{MHCO}_3 (\text{X}) = \frac{200.2}{2} = 100$$

Calculation of atomic weight of Metal M

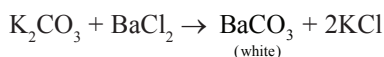
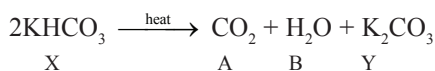
$$\text{MHCO}_3 = 100$$

$$\text{M} + 1 + 12 + 48 = 100$$

$$\text{M} + 61 = 100$$

$$\text{M} = 100 - 61 = 39$$

Thus, the metal must be K and hence the given salt X is KHCO_3 . The above facts coincide with given thermal decomposition.



Hence, we have $\text{X} = \text{KHCO}_3$, $\text{Y} = \text{K}_2\text{CO}_3$, $\text{Z} = \text{BaCO}_3$, $\text{A} = \text{CO}_2$, $\text{B} = \text{H}_2\text{O}$

5. The gas liberated on heating a mixture of two salts with NaOH, gives a reddish brown precipitate with an alkaline solution of K_2HgI_4 . The aqueous solution of the mixture on treatment with BaCl_2 gives a white precipitate which is sparingly soluble in concentrated HCl. On heating the mixture with $\text{K}_2\text{Cr}_2\text{O}_7$ and concentrated H_2SO_4 , red vapours A are produced. The aqueous solution of the mixture gives a deep blue colouration B with potassium ferricyanide solution. Identify the radicals in the given mixture and write the balanced equations for the formation of A and B. **(1991-4 Marks)**

Ans. NH_4^+ , Fe^{2+} , SO_4^{2-} , Cl^-

Solution: Let us summarise the given facts of the question.

Red vapours of (A) $\xleftarrow[\text{conc. H}_2\text{SO}_4, \text{heat}]{\text{K}_2\text{Cr}_2\text{O}_7}$ **Mixture of two salts** $\xrightarrow[\text{NaOH}]{\text{Heat with}}$ Gas $\xrightarrow{\text{Alk. K}_2\text{HgI}_4}$ Reddish brown ppt.

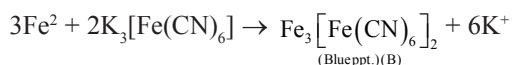
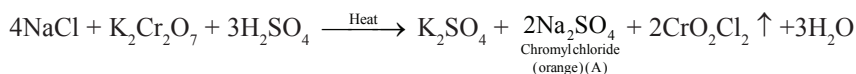
Deep blue colour, (B) $\xleftarrow{\text{K}_3[\text{Fe}(\text{CN})_6]}$ **Aq. solution of the mixture** $\xrightarrow{\text{BaCl}_2}$ White ppt.

The given reactions lead to the following conclusions.

- Heating of mixture with NaOH to give NH_3 gas (indicated by reddish brown precipitate with alkaline solution of K_2HgI_4) indicates the presence of NH_4^+ ion in the mixture.
- Heating of mixture with $\text{K}_2\text{Cr}_2\text{O}_7$ and conc. H_2SO_4 to give red vapours (of chromyl chloride) indicates the presence of Cl^- ion in the mixture.
- Reaction of aqueous solution of the mixture with barium chloride solution to give white precipitate (of BaSO_4) sparingly soluble in concentrated HCl Indicates the presence of SO_4^{2-} ions in the mixture.
- Reaction of aqueous solution of the mixture with potassium ferricyanide solution to give deep blue colour indicates the presence of Fe^{2+} ions in the mixture.

Hence, the mixture contains following four ions: NH_4^+ , Fe^{2+} , SO_4^{2-} and Cl^- .

Equations for the formation of A and B



6. A light bluish green crystalline compound responds to the following tests:

- Its aqueous solution gives a brown precipitate or colour with alkaline $K_2[HgI_4]$ solution.
- Its aqueous solution gives a blue colour with $K_3[Fe(CN)_6]$ solution.
- Its solution in hydrochloric acid gives a white precipitate with $BaCl_2$ solution.

Identify the ions present and suggest the formula of the compound.

(1992-4 Marks)

Ans. NH_4^+ , Fe^{2+} , SO_4^{2-} , $FeSO_4(NH_4)_2SO_4 \cdot 6H_2O$

Solution: 'Compound gives brown precipitate with alkaline K_2HgI_4 and so contain NH_4^+ ions.'

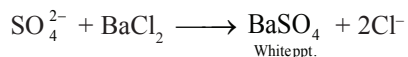
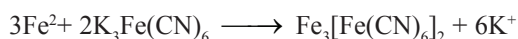
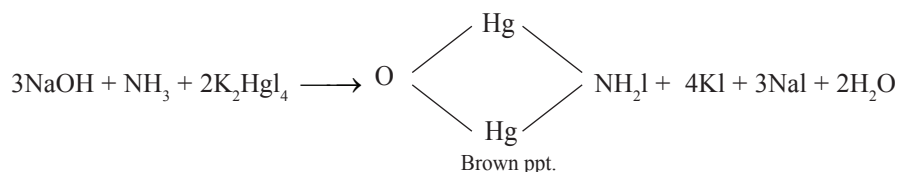
'Compound gives blue colour with $K_3Fe(CN)_6$ and so contains Fe^{2+} ions.'

'Solution of compound in HCl gives white precipitate with $BaCl_2$ and so it contains SO_4^{2-} ions.'

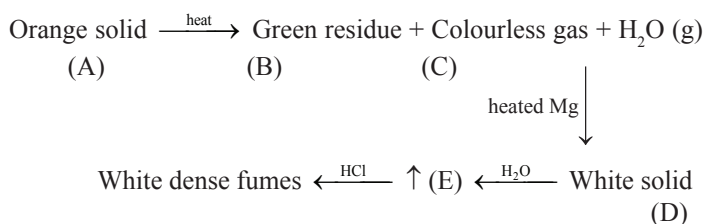
'Bluish green compound with NH_4^+ , Fe^{2+} and SO_4^{2-} suggests that it is Mohr salt i.e.,'



Reactions:



Let us summarise the given facts



7. An orange solid (A) on heating gave a green residue (B), a colourless gas (C) and water vapour. The dry gas (C) on passing over heated Mg gave a white solid (D). (D) on reaction with water gave a gas (E) which formed dense white fumes with HCl. Identify (A) to (E) and give reactions involved.

(1993-3 Marks)

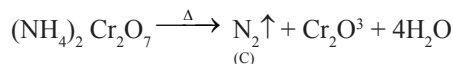
Ans. $(NH_4)_2Cr_2O_7$, Cr_2O_3 , N_2 , Mg_3N_2 , NH_3

Solution:

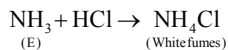
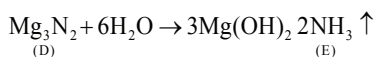
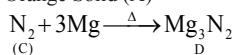
- Formation of white dense fumes by gas (E) with HCl indicates that the gas (E) is ammonia (NH_3).
- Formation of ammonia (E) by the hydrolysis of white solid (D) indicates that (D) should be magnesium nitride, Mg_3N_2 .
- Since compound (D) is formed by reaction of gas (C) with magnesium, the colourless gas (C) must be nitrogen.
- Orange colour of the original compound (A) and green colour of the residue (B) indicates that compound (A) is ammonium dichromate, $(NH_4)_2Cr_2O_7$.

14.50 Salt Analysis

Reactions:



Ammoniumdichromate green (B)
Orange Solid (A) Chromium Oxide



8. A is a binary compound of a univalent metal. 1.422 g of A reacts completely with 0.321 g of sulphur in an evacuated and sealed tube to give 1.743 g of a white crystalline solid B, that forms a hydrated double salt, C with $\text{Al}_2(\text{SO}_4)_3$. Identify A, B and C **(1994–5 Marks)**

Ans. KO_2 , K_2SO_4 , $\text{Al}_2(\text{SO}_4)_3$

Solution: As the solid B forms a hydrated salt C with $\text{Al}_2(\text{SO}_4)_3$; B should be sulphate of a monovalent I cation, i.e., M_2SO_4 . Now since sulphate of a monovalent cation contains one sulphur atom per mol, weight of metal sulphate obtained by 32.1 g (at. wt. of S) should be the molecular weight of the metal sulphate. Thus, 0.321 g of sulphur is present in 1.743 g of B 32.1 g of sulphur is present in = Thus, molecular weight of B

$$32.1 \text{ g of sulphur is present in } = \frac{1.743}{0.321} \times 32.1 = 174.3 \text{ g}$$

$$\text{Thus molecular weight of B } (\text{M}_2\text{SO}_4) = 174.3 \text{ g mol}^{-1}$$

$$2x + 32.1 + 64 = 174.3$$

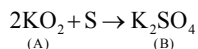
$$2x = 78.2$$

$$x = 39.1$$

Atomic weight 39.1 corresponds to metal potassium, K.

Thus, B is K_2SO_4 , and C is $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$

Nature of compound A: Since A is a binary compound of potassium and it reacts with sulphur to form K_2SO_4 , it must be oxide of potassium, probably potassium superoxide (KO_2) which is supported by the given data.



$$2(39.1 + 32) = 142.2$$

$$32.1 \text{ g of S reacts with } 142.2 \text{ g of KO}_2$$

$$0.321 \text{ g of S reacts with } = \frac{142.2}{32.1} \times 0.321 = 1.422 \text{ g}$$

Similarly,

$$32.1 \text{ g of S gives } 174.3 \text{ g of K}_2\text{SO}_4$$

$$0.321 \text{ g of S gives } = \frac{174.3}{32.1} \times 0.321 = 1.743 \text{ g}$$

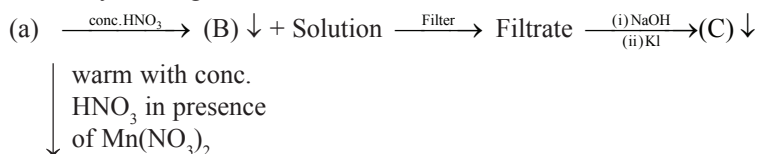
Both these datas are also given in the problem. Thus, A is KO_2 .

9. A scarlet compound A is treated with concentrated HNO_3 to give a chocolate brown precipitate B. The precipitate is filtered and the filtrate is neutralised with NaOH. Addition of KI to the resulting solution gives a yellow precipitate C. The precipitate B on warming with cone. HNO_3 in the presence of $\text{Mn}(\text{NO}_3)_2$ produces a pink-coloured solution due to the formation of D. Identify A, B, C and D. Write the reaction sequence. **(1995–4 Marks)**

Ans. Pb_3O_4 , PbO_2 , PbI_2

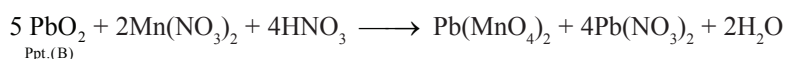
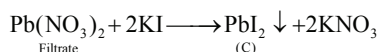
Solution:

Summary of the given facts.



Pink coloured solution (D)

From the colour of the known compound and reaction involved, it seems that (A) is red lead (Pb₃O₄) and its various reactions can be represented as below.

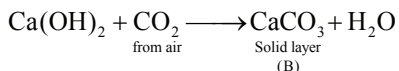
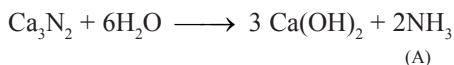
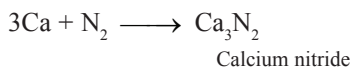


10. Calcium burns in nitrogen to produce a white powder which dissolves in sufficient water to produce a gas (A) and an alkaline solution. The solution on exposure to air produces a thin solid layer of (B) on the surface. Identify the compounds A and B. **(1996–2 Marks)**

Ans. NH₃, CaCO₃

Solution: NH₃, CaCO₃

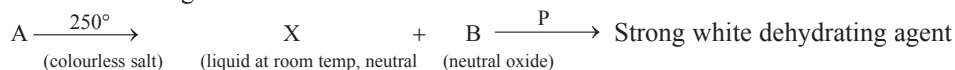
The reactions are given as follows:



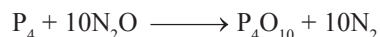
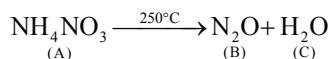
11. A colourless inorganic salt (A) decomposes completely at about 250°C to give only two products, (B) and (C), leaving no residue. The oxide (C) is a liquid at room temperature and neutral to moist litmus paper while the gas (B) is a neutral oxide. White phosphorus burns in excess of (B) to produce a strong white dehydrating agent. Write balanced equations for the reactions involved in the above process. **(1996–3 Marks)**

Solution:

Let us summaries the given facts.



- (i) Since the resulting dehydrating agent is derived from P, it is likely to be P₄O₁₀.
 (ii) P₄O₁₀ is produced by burning phosphorus in excess of neutral oxide (B) which is likely to be NO₂.
 (iii) Thus the salt A should be NH₄NO₃ which explains all given reactions.



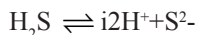
12. During the qualitative analysis of a mixture containing Cu²⁺ and Zn²⁺ ions, H₂S gas is passed through an acidified solution containing these ions in order to test Cu²⁺ alone. Explain briefly. **(1998–2 Marks)**

Solution: The solubility products of CuS and ZnS are

$$K_{\text{sp}}(\text{CuS}) = 10^{-38} \text{ and } K_{\text{sp}}(\text{ZnS}) = 10^{-22}$$

14.52 Salt Analysis

Since $K_{sp}(\text{CuS}) \ll K_{sp}(\text{ZnS})$, very small of concentration of S^{2-} is sufficient to cause the precipitation of Cu^{2+} ions. In order to have very small concentration of S^{2-} ions, acidic medium is used. Due to the common ion H^+ , the ionisation of H_2S is suppressed:



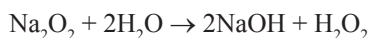
The available concentration of S^{2-} ions in acidic medium causes only the precipitation of CuS and not that of ZnS .

13. A white solid is either Na_2O or Na_2O_2 . A piece of red litmus paper turns white when it is dipped into a freshly made aqueous solution of the white solid. **(1999–4 Marks)**

- (i) Identify the substance and explain with balanced equation.
(ii) Explain what would happen to the red litmus if the white solid were the other compound.

Ans. (i) Na_2O_2 , (ii) turns blue

Solution: (i) The substance is Na_2O_2 . When dissolved in water, the solution becomes alkaline with the liberation of H_2O_2



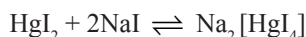
Due to the alkaline solution, the red litmus paper will turn into blue, which subsequently changes into white due to oxidation caused by H_2O_2 .

- (ii) The substance Na_2O merely produces alkaline solution and thus the red litmus paper will turn into blue.



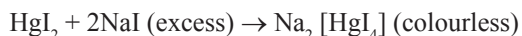
14. An aqueous solution containing one mole of HgI_2 and two moles of NaI is orange in colour. On addition of excess NaI , the solution becomes colourless. The orange colour reappears on subsequent addition of NaOCl . Explain with equations. **(1999–3 Marks)**

Solution: Sodium iodide on reaction with HgI_2 gives colourless complex salt, $\text{Na}_2[\text{HgI}_4]$

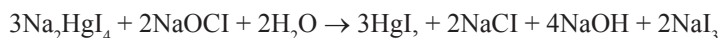


Colour is due to presence of residual HgI_2

But on addition of excess NaI , it becomes colourless due to change of residual HgI_2 into $\text{Na}_2[\text{HgI}_4]$



The orange colour of HgI_2 reappears due to conversion of $\text{Na}_2[\text{HgI}_4]$ into HgI_2 by means of NaOCl

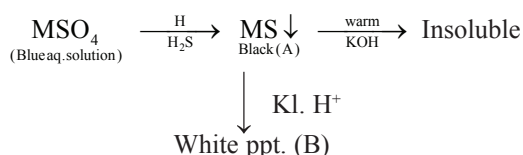


Orange colour

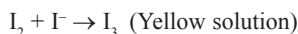
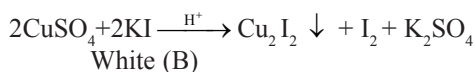
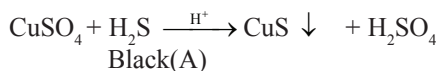
15. An aqueous blue coloured solution of a transition metal sulphate reacts with H_2S in acidic medium to give a black precipitate **A**, which is insoluble in warm aqueous solution of KOH . The blue solution on treatment with KI in weakly acidic medium, turns yellow and produces a white precipitate **B**. Identify the transition metal ion. Write the chemical reactions involved in the formation of **A** and **B**. **(2000–4 Marks)**

Ans. Cu^{2+}

Solution: Summary of the given facts

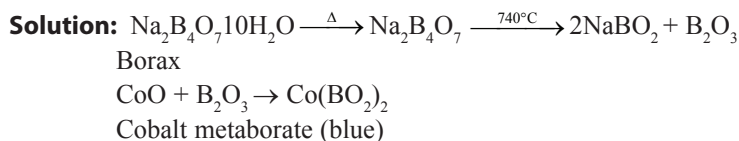


The reaction corresponds to copper sulphate.



16. Write the chemical reactions associated with the 'borax bead test' of cobalt (II) oxide.

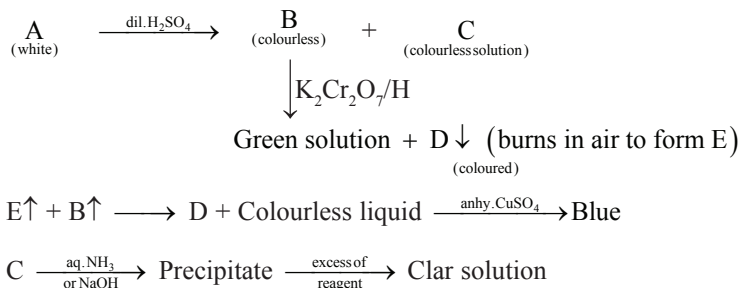
(2000–3 Marks)



17. A white substance (A) reacts with dilute H_2SO_4 to produce a colourless gas (B) and a colourless solution (C). The reaction between (B) and acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution produces a green solution and a slightly coloured precipitate (D). The substance (D) burns in air to produce a gas (E) which reacts with (B) to yield (D) and a colourless liquid. Anhydrous copper sulphate is turned blue on addition of this colourless liquid. Addition of aqueous NH_3 or NaOH to (C) produces first a precipitate, which dissolves in the excess of the respective reagent to produce a clear solution in each case. Identify (A), (B), (C), (D) and (E). Write the equations of the reactions involved. (2001–10 Marks)

Ans. ZnS , H_2S , ZnSO_4 , S , SO_2

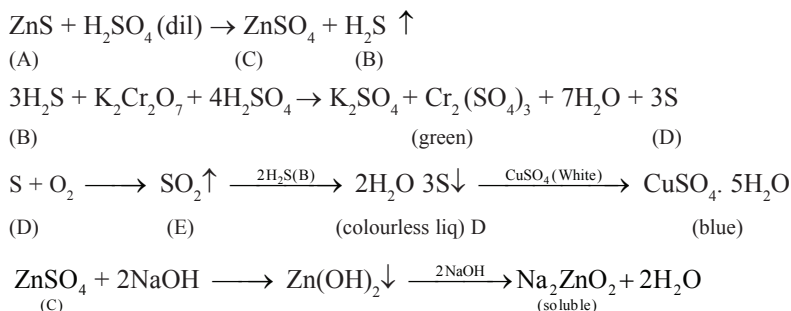
Solution:



The above set leads to following conclusions.

- Since the gas (B) is colourless and turns acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution given, it should be H_2S .
- Since H_2S gas is obtained by the reaction of dilute H_2SO_4 on A, the latter must be sulphide.
- The white colour of the sulphide (A) points out towards ZnS .

Thus, the various reactions can be written as below.



18. When a white crystalline compound X is heated with $\text{K}_2\text{Cr}_2\text{O}_7$ and concentrated H_2SO_4 , a reddish brown gas A is evolved. On passing A into caustic soda solution, a yellow coloured solution of B is obtained. Neutralizing the solution B with acetic acid and on subsequent addition of lead acetate, a yellow precipitate C is obtained. When X is

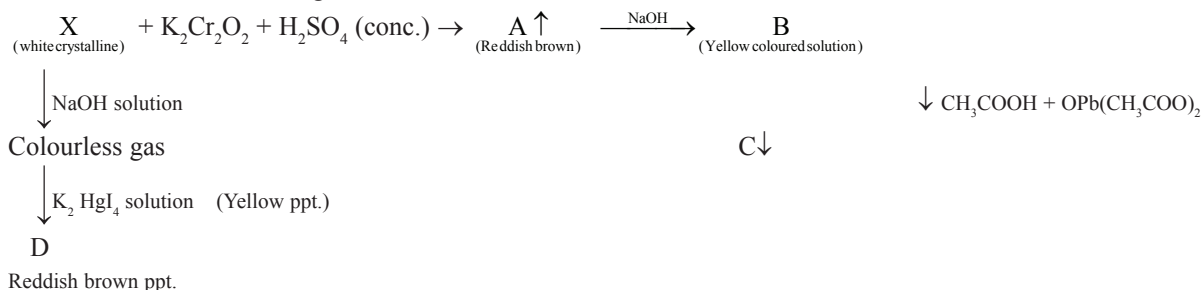
14.54 Salt Analysis

heated with NaOH solution, a colourless gas is evolved and on passing this gas into K_2HgI_4 solution, a reddish brown precipitate **D** is formed. Identify **A**, **B**, **C**, **D** and **X**. Write the equations of reactions involved.

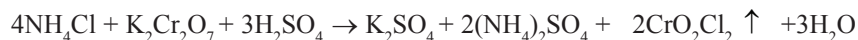
(2002–5 Marks)

Ans. CrO_2Cl_2 , Na_2CrO_4 , $PbCrO_4$, iodide of Millon's base, NH_4Cl

Solution: Let us summarize the given facts.



Reaction of compound X with NaOH solution and subsequent treatments indicate that X has NH_4^+ radical. On the other hand, reaction of X with $K_2Cr_2O_7$ solution, cone. H_2SO_4 and subsequent treatments indicate that A has Cl^- radical. Thus, compound X is NH_4Cl which explains all the above reactions.

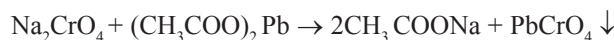


X

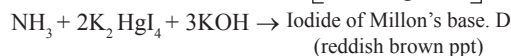
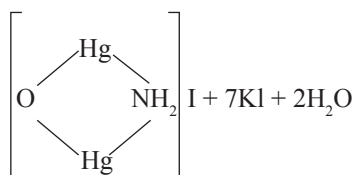
Chromyl chloride, A
(reddish brown)



Sod. chromate, B
(yellow coloured)



Leadchromate, C
(yellow ppt)



19. A mixture consists of A (yellow solid) and B (colourless solid) which gives lilac colour in flame

(a) Mixture gives black precipitate C on passing $H_2S_{(g)}$ through its aqueous solution.

(b) C is soluble in aqua-regia and on evaporation of aqua-regia and adding $SnCl_2$ gives greyish black precipitate D.

The salt solution with NH_4OH gives a brown precipitate.

(i) The sodium extract of the salt with $CCl_4/FeCl_3$ gives a violet layer.

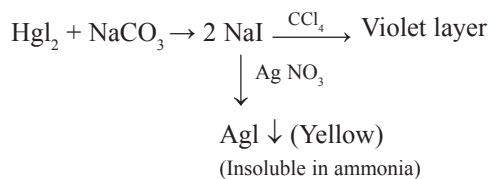
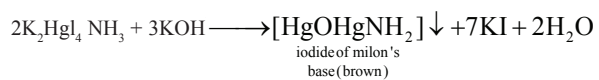
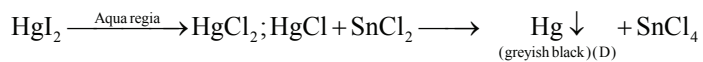
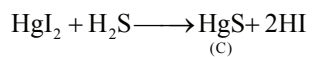
(ii) The sodium extract gives yellow precipitate with $AgNO_3$ solution which is insoluble in NH_3 .

Identify A and B, and the precipitates C and D.

(2003–4 Mark)

Ans. HgI_2 , KI , HgS , Hg

Solution: [A = HgI_2 (yellow), B = KI (colourless)]



ONE CORRECT OPTION TYPE QUESTIONS

- A gas is obtained by addition of dil. H_2SO_4 to a mixture which turns lead acetate paper black. It is
 - SO_2
 - CO_2
 - H_2S
 - NO_2
- Aqueous solution of an acetate salt when treated with ferric chloride solution gives a blood-red solution due to formation of
 - $\text{Fe}(\text{CNS})_3$
 - $(\text{CH}_3\text{COO})_3\text{Fe}$
 - $(\text{CH}_3\text{COO})_2\text{Fe}$
 - $\text{Fe}(\text{OH})_3$
- Which of the following gives a suffocating gas when treated with dilute HCl ?
 - carbonate
 - sulphite
 - sulphate
 - borate
- The acidic solution of a salt produces blue colour with KI starch solution. The salt may be:
 - Sulphite
 - Bromide
 - Nitrite
 - Chloride
- In an alkaline solution, sodium nitroprusside gives a violet colour with:
 - S^{2-}
 - SO_3^{2-}
 - SO_4^{2-}
 - NO_2^-
- In the ring test for nitrates, the ring formed is due to:
 - $\text{FeSO}_4 \cdot \text{NO}$
 - $\text{FeSO}_4 \cdot \text{NO}_2$
 - $\text{Fe}(\text{NO}_3)_3$
 - $\text{Fe}_2(\text{SO}_4)_3 \cdot \text{NO}$
- Violet vapours are given when is treated with conc. H_2SO_4 .
 - bromide
 - iodide
 - chloride
 - nitrate
- Oxalate + MnO_2 + dil. $\text{H}_2\text{SO}_4 \rightarrow$ Gas. The gas evolved is:
 - CO_2
 - CO
 - SO_2
 - O_2
- Brown ring in the test of NO_3^- is formed due to the formation of:
 - $\text{FeSO}_4 \cdot \text{NO}$
 - $[\text{Fe}(\text{SO}_4)_2 \cdot \text{NO}] \cdot \text{H}_2\text{O}$
 - $\text{Fe}(\text{SO}_4)_3 \cdot \text{NO}$
 - None of these
- Which is not easily precipitated from aqueous solution?
 - Cl^-
 - SO_4^{2-}
 - NO_3^-
 - CO_3^{2-}
- Soda extracted is useful when given mixture has any insoluble salt, it is prepared by:
 - fusing soda and mixture and then extracting with water
 - dissolving NaHCO_3 and mixture in dilute HCl
 - boiling Na_2CO_3 and mixture in dilute HCl
 - boiling Na_2CO_3 and mixture in distilled water
- Which of the following reagents can be used to identify bromide and iodide ions in the presence of each other?
 - Chlorine water
 - Silver nitrate solution
 - Starch solution
 - Concentrated sulphuric acid
- On heating a mixture of NaBr and conc. H_2SO_4 we obtained:
 - HOBr
 - HBr
 - Br_2
 - HBrO_3
- A bromide ion does not interfere with the chromyl chloride test because when a bromide is present,
 - Br_2 is liberated, which leaves the NaOH solution colourless
 - CrO_2Br_2 formed does not volatilize as CrO_2Cl_2 does
 - CrO_2Br_2 does not react with NaOH
 - no gaseous substance containing bromine is produced
- Which of the following reagents are used for the detection of acetate and oxalate ions respectively?
 - BaCl_2 and CaCl_2
 - NaOH and BaCl_2
 - FeCl_3 and CaCl_2
 - FeCl_3 and NaOH
- When a mixture containing phosphate is heated with conc. HNO_3 and ammonium molybdate solution, a canary yellow precipitate is formed. The formula of the yellow precipitate is:
 - $(\text{NH}_4)_3\text{PO}_4$
 - $(\text{NH}_4)_3\text{PO}_4 \cdot 12\text{MoO}_4$
 - $(\text{NH}_4)_3\text{PO}_4 \cdot 12\text{MoO}_3$
 - $(\text{NH}_4)_3\text{PO}_4 \cdot (\text{NH}_4)_2\text{MoO}_4$
- In qualitative inorganic analysis, phosphate if present, is to be eliminated in the appropriate group in order to detect the radical.,
 - Pb^{2+}
 - As^{3+}
 - Ca^{2+}
 - Cd^{2+}

18. Mercurous chloride turns black on treatment with ammonium hydroxide. This is due to the formation of:
- $\text{Hg}(\text{NH}_2)\text{Cl}$
 - $\text{Hg}_2\text{Cl}_2 \cdot \text{NH}_4\text{OH}$
 - Hg and HgNH_2Cl
 - $\text{HgCl}_2 \cdot \text{NH}_4\text{OH}$
19. When H_2S is passed through Hg_2^{2+} , we get:
- HgS
 - $\text{HgS} + \text{Hg}_2\text{S}$
 - $\text{HgS} + \text{Hg}$
 - Hg_2S
20. The reagent that can distinguish between silver and lead salt is
- H_2S gas
 - Hot dilute HCl solution
 - NH_4Cl (solid) + NH_4OH (solution)
 - NH_4Cl (solid) + $(\text{NH}_4)_2\text{CO}_3$ solution.
21. On adding water to BiCl_3 solution in HCl , the compound formed is:
- Bi_2O_3
 - $\text{Bi}(\text{OH})_3$
 - BiOCl
 - BiOCl_2
22. Which compound does not dissolve in hot dilute HNO_3 ?
- HgS
 - PbS
 - CuS
 - CdS
23. To a metal nitrate, when KI solution is added, a black precipitate is produced at first, on adding excess of KI , orange solution is produced. Identify the metal ion.
- Hg^{2+}
 - Bi^{3+}
 - Cu^{2+}
 - Pb^{2+}
24. When calomel reacts with NH_4OH solution, the compound formed is:
- $\text{NH}_2(\text{Hg})\text{Cl}$
 - $\text{Hg}_2\text{Cl}_2 \cdot \text{NH}_3$
 - $\text{Hg}(\text{NH}_3)_2\text{Cl}_2$
 - $\text{HgCl}_2 \cdot \text{NH}_3$
25. Which one among the following pair of ions cannot be separated by H_2S in dilute hydrochloric acid?
- Bi^{3+} , Sn^{4+}
 - Al^{3+} , Hg^{2+}
 - Zn^{2+} , Cu^{2+}
 - Ni^{2+} , Cu^{2+}
26. The sulphides of which one of the following groups of elements are soluble in yellow ammonium sulphide:
- As, Sb and Sn
 - As, Cd and Sn
 - Cd, Cu and Bi
 - Hg, Cu and Cd
27. A precipitate of which of the following would be obtained when HCl is added to a solution of stannous sulphide (SnS) in yellow ammonium sulphide?
- SnS
 - SnS_2
 - $(\text{NH}_4)_2\text{SnS}_2$
 - Sn_2S_3
28. Ferric ion forms a Prussian blue coloured precipitate with $\text{K}_4[\text{Fe}(\text{CN})_6]$ due to the formation of:
- $\text{K}_3\text{Fe}(\text{CN})_6$
 - $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
 - $\text{Fe}(\text{OH})_3$
 - $\text{KFe}[\text{Fe}(\text{CN})_6]$
29. Which one among the following is soluble in excess of NaOH ?
- $\text{Fe}(\text{OH})_3$
 - $\text{Al}(\text{OH})_3$
 - $\text{Cr}(\text{OH})_3$
 - $\text{Mn}(\text{OH})_2$
30. The only cations present in a slightly acidic solution are Fe^{3+} , Zn^{2+} and Cu^{2+} . The reagent which when added in excess to this solution would identify and separate Fe^{3+} in one step is:
- 2 M HCl
 - 6 M NH_3
 - 6 M NaOH
 - H_2S gas
31. Conc. Nitric acid is added before proceeding for the detection of radicals of group III. This helps to
- oxidize any remaining H_2S
 - convert ferrous ions into ferric ions
 - form nitrate, which gives granular precipitate
 - increase ionization of ammonium hydroxide
32. Iron is precipitated as $\text{Fe}(\text{OH})_3$ in the third group and not as $\text{Fe}(\text{OH})_2$ because
- $\text{Fe}(\text{OH})_2$ is light green
 - $\text{Fe}(\text{OH})_3$ is reddish brown
 - $\text{Fe}(\text{OH})_2$ is partially soluble and is not completely precipitated
 - $\text{Fe}(\text{OH})_2$ is oxidant
33. To avoid the precipitation of hydroxides of Ni^{2+} , Co^{2+} , Zn^{2+} and Mn^{2+} along with those of Fe^{3+} , Al^{3+} and Cr^{3+} the third group solution should be
- heated with a few drops of concentrated HNO_3
 - boiled by adding excess of NH_4Cl
 - concentrated by evaporation
 - made alkaline by NaOH
34. NiS is separated from ZnS by treating with:
- NaOH
 - concentrated HCl
 - yellow ammonium sulphide
 - aqua-regia
35. Manganese salt + PbO_2 + conc. $\text{HNO}_3 \rightarrow$ The solution acquires purple colour. The colour is due to:
- HMnO_4
 - a lead salt
 - $\text{Mn}(\text{NO}_3)_2$
 - H_2MnO_4

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36. Cobalt salt + $\text{KNO}_2 + \text{CH}_3\text{COOH} \rightarrow$ Yellow ppt. The yellow precipitate is:
 (a) potassium cobaltonitrite
 (b) potassium cobaltinitrite
 (c) cobalt nitrite
 (d) cobalt nitrate
37. Dimethyl glyoxime is used for the test of _____. In alkaline medium:
 (a) cobalt (b) zinc
 (c) manganese (d) nickel
38. In IV group analysis NH_4OH is added before passing H_2S gas because:
 (a) the sulphides of IV group are insoluble in NH_4OH
 (b) the sulphides of other metals are soluble in NH_4OH
 (c) the concentration of S^{2-} ions is high enough to precipitate the sulphides of IV group
 (d) the sulphides of second group are soluble in NH_4OH
39. Reaction of $\text{Zn}(\text{OH})_2$ with NaOH produces:
 (a) Na_2ZnO_2 (b) ZnO
 (c) Na_2O (d) None of these
40. Cobalt salt + $\text{KNO}_2 + \text{CH}_3\text{COOH} \longrightarrow$ yellow ppt. The yellow precipitate is:
 (a) Potassium cobaltonitrate
 (b) Potassium cobaltinitrite
 (c) Cobalt nitrite
 (d) Cobalt nitrate
41. The group reagent for the V group radicals is:
 (a) $(\text{NH}_4)_2\text{CO}_3$
 (b) $(\text{NH}_4)_2\text{SO}_4$
 (c) NH_4Cl
 (d) $(\text{NH}_4)_2\text{C}_2\text{O}_4$
42. Disodium hydrogen phosphate is used to test which ion?
 (a) Mg^{2+} (b) Na^+
 (c) Ca^{2+} (d) All
43. Which of the following compounds does magnesium precipitate when you test for it?
 (a) $\text{MgCO}_3 \cdot \text{MgO}$ (b) MgCO_3
 (c) $\text{Mg}(\text{OH})_2$ (d) $\text{MgNH}_4\text{PO}_4 \cdot 6\text{H}_2\text{O}$
44. MgCO_3 is not precipitated with the carbonates of Vth group radicals in presence of NH_4Cl and NH_4OH because:
 (a) MgCO_3 is soluble in NH_4OH
 (b) MgCO_3 is not precipitated in pressure of NH_4Cl
 (c) MgCO_3 is soluble in water
 (d) MgCO_3 is soluble in $(\text{NH}_4)_2\text{CO}_3$
45. Which of the following gives green colour to the flame?
 (a) CaCO_3 (b) NaCl
 (c) $\text{Sr}(\text{NO}_3)_2$ (d) BaCl_2
46. Which of the following do not respond to borax bead test?
 (a) Nickel salts (b) Copper salts
 (c) Cobalt salts (d) Aluminium salts
47. The salt used for performing 'bead test' in qualitative inorganic analysis is:
 (a) $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
 (b) $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$
 (c) $\text{Na}(\text{NH}_4)\text{HPO}_4 \cdot 4\text{H}_2\text{O}$
 (d) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
48. In borax bead test which compound is formed?
 (a) Orthoborate
 (b) Metaborate
 (c) Double oxide
 (d) Tetraborate
49. Which of the following salt gives green colour mass in cobalt nitrate charcoal cavity test?
 (a) Zn salts (b) Al salts
 (c) Alums (d) Copper salts
50. To remove sulphate ions from a solution containing chloride and sulphate ions, the reagent used is
 (a) HCl (b) H_2SO_4
 (c) $\text{Ba}(\text{OH})_2$ (d) BaSO_4
51. Which one is not the interfering radical?
 (a) PO_4^{3-} (b) BO_3^{3-}
 (c) F^- (d) SO_4^{2-}
52. SO_2 and CO_2 both turn lime water (A) milky, SO_2 also turns $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}^+$ (B) green while O_2 is soluble in pyrogallol (C), turning it black. These gases are to be detected in order by using these reagents. The order is
 (a) (A), (B), (C) (b) (B), (C), (A)
 (c) (B), (A), (C) (d) (A), (C), (B)
53. Which compound will not give positive chromyl chloride test?
 (a) Copper chloride CuCl_2
 (b) Mercuric chloride, HgCl_2
 (c) Zinc chloride, ZnCl_2
 (d) Anilinium chloride, $\text{C}_6\text{H}_5\text{NH}_3^+\text{Cl}^-$

54. For the confirmatory tests of acid radicals, sodium carbonate extract is prepared because
 (a) all anions react with Na
 (b) Na is more reactive
 (c) Na_2CO_3 is water soluble
 (d) sodium salts of almost all anions are water soluble.
55. For the test of halides, the soda extract of the salt is first acidified with
 (a) Dil. H_2SO_4 (b) Dil. HNO_3
 (c) Dil. HCl (d) Any of three
56. Sulphuric acid is not used for the preparation of original solution in the analysis of basic radicals because
 (a) it forms insoluble sulphates with some of the basic radicals
 (b) it is a strong oxidizing agent
 (c) it is a strong reducing agent
 (d) it decomposes many of the anions
57. Nitric acid is generally not used for preparation of original solution in analysis of basic radicals, because
 (a) it is oxidizing agent
 (b) it is reducing agent
 (c) it forms insoluble nitrates
 (d) nitrate ions cause interference with the scheme of analysis.
58. Solution of a chemical compound 'X' reacts with AgNO_3 solution to form a white precipitate 'y' which dissolves in NH_4OH to give a complex 'Z'. When 'Z' is treated with dilute HNO_3 , 'y' reappears. The chemical compound 'X' can be
 (a) NaCl (b) CH_3Cl
 (c) NaBr (d) NaI
59. Which of the sulphide is yellow?
 (a) ZnS (b) CdS
 (d) NiS (d) PbS
60. Which of the following are soluble in excess of NaOH (X) : As_2S_3 ; (Y) : CuS ; (Z) : AlCl_3 ?
 (a) X, Y, Z (b) Y, Z
 (c) X, Z (d) X, Y
61. Which compound does not dissolve in hot dil. HNO_3 ?
 (a) HgS (b) PbS
 (c) CuS (d) CdS
62. A mixture on heating gave a gas used as an anaesthetic, 1.1 g of gas occupies 0.56 l at NTP. The mixture contains
 (a) $\text{NaNO}_3 + \text{NH}_4\text{Cl}$ (b) $\text{NaNO}_2 + \text{NH}_4\text{Cl}$
 (c) $\text{CaCO}_3 + \text{MgCO}_3$ (d) $\text{NH}_4\text{Cl} + \text{Na}_2\text{SO}_4$
63. Colourless salt (X) $\xrightarrow{\Delta}$ (Y) $\xrightarrow{\text{Cu}^{2+}, \Delta}$ coloured bead (Z). (X) can be:
 (a) borax (b) micro cosmic salt
 (c) both (d) none of these
64. Salts of which of the following metal are white?
 (a) Zinc (b) Cobalt
 (c) Chromium (d) Fe
65. A glassy bead formed by heating borax on a platinum wire loop is
 (a) Sodium tetraborate
 (b) Sodium metaborate
 (c) Sodium metaborate and boric anhydride
 (d) Boric anhydride and sodium tetraborate
66. A minute quantity of cupric salt is heated on borax bead in reducing flame of Bunsen burner, the colour of bead after cooling will be
 (a) blue (b) Brown red
 (c) Colourless (d) Green
67. Flame test is not given by
 (a) Ba^{2+} (b) Ca^{2+}
 (c) Na^+ (d) Be^{2+}
68. The salts of which of the following elements are generally dark green coloured?
 (a) Chromium (b) Copper (I)
 (c) Barium (d) Cobalt
69. Which of the following sulphate is insoluble in water?
 (a) CuSO_4 (b) CdSO_4
 (c) PbSO_4 (d) $\text{Bi}_2(\text{SO}_4)_3$
70. The carbonate of which of the following cation is insoluble in water?
 (a) Na^+ (b) Mg^{2+}
 (c) NH_4^+ (d) K^+
71. In the precipitation of the radicals of iron group in qualitative analysis, NH_4Cl is added before adding NH_4OH . This causes
 (a) decrease in the concentration of OH^- ions
 (b) removal of PO_4^{3-} ions
 (c) removal of PO_4^{3-} ions
 (d) increase in the concentration of Cl^- ions
72. A solution of KI is added to CS_2 followed by addition of Cl_2 water. A violet colour appears in CS_2 layer. If excess of Cl_2 water is added the violet colour disappears.

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pears. The disappearance of violet colour is due to the formation of

- (a) I_3^- ion (b) HIO_3 ion
(c) ICl ion (d) I^- ion.

73. Which of the following pairs of ions would be expected to form precipitate when their dilute solutions are mixed?

- (a) Na^+ , SO_3^{2-} (b) NH_4^+ , CO_3^{2-}
(c) Na^+ , S^{2-} (d) Fe^{3+} , PO_4^{3-}

74. The sulphide which is insoluble in 30% HNO_3 is:

- (a) HgS (b) CuS
(c) PbS (d) CdS

75. A precipitate of calcium oxalate will not dissolve in:

- (a) HNO_3 (b) HCl
(c) H_2SO_4 (d) aqua-regia

76. An aqueous solution of $\text{FeSO}_4 \cdot \text{Al}_2(\text{SO}_4)_3$ and chrome alum is heated with excess of Na_2O_2 and filtered. The materials obtained are:

- (a) a colourless filtrate and a green residue
(b) a yellow filtrate and a green residue
(c) a yellow filtrate and a brown residue
(d) a green filtrate and a brown residue

77. In Nessler's reagent for detection of ammonia, the active species is:

- (a) Hg_2Cl_2 (b) Hg^{2+}
(c) Hg_2I_2 (d) HgI_4^{2-}

78. Which of the following statements is correct?

- (a) manganese salts give a violet borax bead test in reducing flame
(b) from a mixed precipitate of AgCl and AgI , ammonia solution dissolves only AgCl
(c) ferric ions give a deep green precipitate, on adding potassium ferrocyanide solution
(d) on boiling the solution having K^+ , Ca^{2+} and HCO_3^- ions we get a precipitate of $\text{K}_2\text{Ca}(\text{CO}_3)_2$

79. Nessler's reagent is:

- (a) NaHgCl_4
(b) K_2HgI_4
(c) $\text{Hg}(\text{NH}_3)_2\text{Cl}$
(d) $\text{K}_2\text{HgI}_4 + \text{KOH}$

80. (A) $\xrightarrow{\text{Fused with Na}_2\text{CO}_3}$ (B) $\xrightarrow{\text{(H}_2\text{SO}_4 + \text{H}_2\text{O})}$ (C)
Green solid Yellow solution Evaporation Orange
 $\downarrow$ $(\text{CH}_3(\text{COO})_2\text{Pb})$
(D) Yellow ppt.

Hence, A, B, C and D are respectively:

A	B	C	D
(a) FeSO_4	FeCO_3	$\text{Fe}(\text{OH})_3$	PbCO_3
(b) Cr_2O_3	Na_2CrO_4	$\text{Na}_2\text{Cr}_2\text{O}_7$	PbCrO_4
(c) FeCl_2	FeSO_4	PbSO_4	$\text{Fe}(\text{OH})_3$
(d) FeSO_4	FeCl_3	$\text{Fe}(\text{OH})_3$	PbCl_2

81. $[\text{X}] + \text{H}_2\text{SO}_4 \longrightarrow [\text{Y}]$ a colourless gas with irritating smell,

$[\text{Y}] + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow$ green solution. $[\text{X}]$ and $[\text{Y}]$ are:

- (a) SO_3^{2-} and SO_2
(b) Cl^- and HCl
(c) S^{2-} and H_2S
(d) CO_3^{2-} and CO_2

82. An aqueous solution containing Hg^{2+} , Hg_2^{2+} , Pb^{2+} and Cd^{2+} ions will give precipitates of with HCl .

- (a) Hg_2Cl_2 only
(b) PbCl_2 only
(c) Hg_2Cl_2 and PbCl_2
(d) HgCl_2 and PbCl_2

83. Ammonium salt gives brown colour with alkaline solution of Nessler's reagent to form iodide of Milon's base. The formula of this compound is:

- (a) $\text{NH}_2\text{—Hg—O—Hg—I}$
(b) $\text{NH}_2\text{—O—Hg—HgI}$
(c) K_2HgI_4
(d) $\text{NH}_2\text{—Hg—I}$

84. H_2S gas is passed into zinc acetate and zinc chloride solutions in separate test tubes(I) and (II). The ZnS is precipitated in:

- (a) (I) (b) (II)
(c) both (d) none of these

85. Which compound will not give positive chromyl chloride test?

- (a) Copper chloride, CuCl_2
(b) Mercuric chloride, HgCl_2
(c) Mercuric chloride, HgCl_2
(d) Zinc chloride, ZnCl_2

86. A metal salt solution forms a yellow precipitate with potassium chromate in acetic acid, a white precipitate with dil. Sulphuric acid, but gives no precipitate with sodium chloride or iodide. The white precipitate obtained when sodium carbonate is added to the metal salt solution consists of:

- (a) lead carbonate
(b) basic lead carbonate

- (c) barium carbonate
(d) strontium carbonate
87. A pale green crystalline inorganic salt (A) dissolves freely in water. It gives a brown precipitate on addition of aqueous NaOH. The solution of (A) also gives a black precipitate on bubbling H_2S in alkaline medium. An aqueous solution of (A) decolourizes the pink colour of the permanganate solution. The metal in the salt solution is:
(a) Copper
(b) Aluminium
(c) Lead
(d) Iron
88. $X + HNO_3 \rightarrow Y + NO_2 + H_2O + S$, Y + Ammonium molybdate $\rightarrow$ yellow ppt.
Identify X :
(a) As_2S_5 (b) Sb_2S_5
(c) SnS_2 (d) CdS
89. An organic compound (A) on heating produces two gases (B) and (C) and neutral oxide (D) which turns cobalt chloride paper pink. Gas (B) turns lime water milky and produces an acidic solution with water. Gas (C) produces a poisonous gas (E) with chlorine gas, this gas with ammonia gives an organic compound (F) which on further reaction with (D) gives NH_3 gas. Then, compounds (A) and (F) can be found as:
(a) $H_2C_2O_4$ and NH_2CONH_2
(b) CH_3COOH and NH_2CONH_2
(c) $CHCl_3$ and CH_3CONH_2
(d) CH_3Cl and NH_2COONH_4
90. Which of the following pairs of cations can be separated by adding NH_4Cl , NH_4OH and then $(NH_4)_2CO_3$ to the mixture?
(a) Ca^{2+} , Mg^{2+} (b) Ba^{2+} , Sr^{2+}
(c) Sr^{2+} , Ca^{2+} (d) Ba^{2+} , Ca^{2+}

COMPREHENSION BASED QUESTIONS

COMPREHENSION-I

Borax Bead Test is carried out when the original mixture is coloured. It is done with the help of a clean platinum wire on which a small loop is made at the end. When borax is heated on platinum wire loop a transparent glass like bead is obtained. The hot bead is brought in contact with salt till it reacts with fused borax and colour is imparted to the bead. Bead colour is noted.

Colour of the bead	Ion
1. Blue green	Cu^{2+}
2. Yellow	Fe^{3+}
3. Green	Cr^{3+}
4. Violet	Mn^{2+}
5. Dark blue	Co^{2+}
6. Brown	Ni^{2+}

1. Glassy bead is of:
(a) $B_2O_3 + NaBO_2$
(b) $NaBO_2 + Na_3BO_3$
(c) $Na_2B_4O_7 + B_2O_3$
(d) $SiO_2 + B_2O_3$

2. Blue bead can be of:

- (a) $Cu(BO_2)_2$ (b) $Co(BO_2)_2$
(c) Both (a) and (b) (d) None of these

3. The flame used in Borax Bead Test is:

- (a) Reducing
(b) Oxidizing
(c) Both (a) and (b)
(d) Neither (a) nor (b)

COMPREHENSION-II

- (i) A yellow precipitate of the compound A is formed on passing H_2S through a neutral solution of the salt B.
(ii) The compound A is soluble in hot dilute HNO_3 but insoluble in yellow ammonium sulphide.
(iii) The solution of B, on treatment with a small quantity of NH_3 , gives a white precipitate soluble in an excess of the reagent, forming a compound C.
(iv) The solution of B gives a white precipitate with a small concentration of KCN. The precipitate is soluble in an excess of the reagent, forming a compound D.

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- (v) The solution of D, on treatment with H_2S , gives A.
 (vi) The solution of B in dilute HCl , on treatment with a solution of BaCl_2 , gives a white precipitate of the compound E, which is insoluble in concentrated NO_3^- .

1. Which of the following is the cation present in B?

- (a) As^{3+} (b) Sb^{3+}
 (c) Zn^{2+} (d) Cd^{2+}

2. Which of the following anions is present in B?

- (a) SO_4^{2-} (b) CO_3^{2-}
 (c) SO_3^{2-} (d) S^{2-}

3. Which of the following are the white precipitate and the soluble substance formed by the excess of the NaOH reagent, respectively?

- (a) AsOCl and AsO_3^{3-}
 (b) SbOCl and SbO_3^{3-}
 (c) $\text{Zn}(\text{OH})_2$ and $[\text{Zn}(\text{NH}_3)_4]^{2+}$
 (d) $\text{Cd}(\text{OH})_2$ and $[\text{Cd}(\text{NH}_3)_4]^{2+}$

4. Which of the following are the white precipitate and the soluble substance formed by the excess of the KCN reagent, respectively, in (iv)?

- (a) $\text{As}(\text{CN})_3$ and $[\text{As}(\text{CN})_6]^{3-}$
 (b) $\text{Sb}(\text{CN})_3$ and $[\text{Sb}(\text{CN})_6]^{3-}$
 (c) $\text{Zn}(\text{CN})_2$ and $[\text{Zn}(\text{CN})_4]^{2-}$
 (d) $\text{Cd}(\text{CN})_2$ and $[\text{Cd}(\text{CN})_4]^{2-}$

ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

1. Which of the following carbonates do not give metal oxide on heating?

- (a) CuCO_3 (b) Ag_2CO_3
 (c) Na_2CO_3 (d) MgCO_3

2. The sulphide, which is insoluble in 50% HNO_3 is:

- (a) HgS (b) CuS
 (c) Bi_2S_3 (d) PbS

3. The ions which can be precipitated with both HCl and H_2S are:

- (a) Pb^{2+} (b) Cu^{2+}
 (c) Ag^+ (d) Sn^{2+}

4. Which of the following are soluble in water?

- (a) CaC_2O_4 (b) SrSO_4
 (c) BaCl_2 (d) $(\text{NH}_4)_2\text{C}_2\text{O}_4$

5. Salt of which of the following radicals will give brown fumes when treated with conc. HNO_3 ?

- (a) NO_3^- (b) Br^-
 (c) I^- (d) Cl^-

6. Which of the following halides are not soluble in water?

- (a) AgCl (b) AgBr
 (c) PbCl_2 (d) AgF

7. Aqueous solution of KI is used to detect:

- (a) Hg^{2+} (b) Pb^{2+}
 (c) Ag^+ (d) Be^{3+}

8. Potassium chromate (K_2CrO_4) is used to identify:

- (a) Pb^{2+} (b) Ba^{2+}
 (c) Ag^+ (d) Ca^{2+}

9. A solution of salt/salts in HCl when diluted with water turns milky. It indicates the presence of:

- (a) Sn salt (b) Bi salt
 (c) Sb salt (d) Cu salt

10. Which give/s a black precipitate when H_2S is passed through an acidic solution of?

- (a) Nickel acetate (b) Copper acetate
 (c) Cobalt acetate (d) Lead acetate

11. Which of the following do/does not give borax bead test?

- (a) Mg^{2+} (b) Mn^{2+}
 (c) Ca^{2+} (d) Mg^{2+}

12. Which of the following do/does not give borax bead test?

- (a) Mg^{2+} (b) Mn^{2+}
 (c) Cu^{2+} (d) Pb^{2+}

13. Which of the following statement(s) is/are correct when a mixture of NaCl and $\text{K}_2\text{Cr}_2\text{O}_7$ is gently warmed with concentrated H_2SO_4 ?

- (a) deep red vapour is evolved
 (b) the vapour when passed into NaOH solution gives a yellow solution of Na_2CrO_4
 (c) chlorine gas is evolved
 (d) chromyl chloride is formed

14. A solution of the mixture was prepared in conc. HCl. On diluting this solution with water a turbidity appeared. This indicates the presence of
 (a) As^{3+} (b) Sb^{3+}
 (c) Hg^{2+} (d) Bi^{3+}
15. Which of the following pairs of cations cannot be separate by using an NaOH solution?
 (a) Fe^{3+} , Al^{3+} (b) Cr^{3+} , Al^{3+}
 (c) Sn^{2+} , Pb^{2+} (d) Cu^{2+} , Pb^{2+}
16. Which statement is/are correct with reference to the ferrous and ferric ions?
 (a) Fe^{2+} gives brown colour with potassium ferricyanide
 (b) Fe^{2+} gives blue colour with potassium ferricyanide
 (c) Fe^{2+} gives red colour with potassium thiocyanate
 (d) Fe^{2+} gives brown colour with potassium thiocyanate
17. Which of the following reagents will not be useful in separating a mixture of Zn^{2+} and Cu^{2+} ?
 (a) H_2S in an acid medium
 (b) H_2S in an alkaline medium
 (c) Excess NaOH solution
 (d) NH_3 solution
18. A yellow precipitate is obtained when:
 (a) lead acetate solution is treated with K_2CrO_4
 (b) $\text{Pb}(\text{NO}_3)_2$ solution is treated with K_2CrO_4
 (c) AgNO_3 solution treated with KI
 (d) H_2S is passed through a solution of CdSO_4
19. Which of the following cations cannot be separated by passing H_2S through their solutions to which NH_4Cl and NH_4OH have been added?
 (a) Ca^{2+} and Ni^{2+}
 (b) Mg^{2+} and Mn^{2+}
 (c) Ni^{2+} and Mn^{2+}
 (d) Co^{2+} and Zn^{2+}
20. Which of the following mixtures of ions in solution can be separated by using NH_3 solution?
 (a) Hg_2^{2+} and Ag^+ (b) Bi^{2+} and Cu^{2+}
 (c) Ag^+ and Pb^{2+} (d) Cu^{2+} and Cd^{2+}

ASSERTION/REASON TYPE QUESTIONS

Each question contains Assertion and Reason. Examine the statements carefully and mark the correct answer according to the instructions given below:

- (a) A is correct but R is wrong.
 (b) A is wrong but R is correct.
 (c) Both A and R are correct and R is correct explanation of A.
 (d) Both A and R are correct but R is not correct explanation of A.

1. **Assertion:** A mixture on heating with dil. H_2SO_4 gives a gas with burning smell which turns lime water milky.

Reason: The lime water reacts with gas to give insoluble substance and the mixture may contain CO_3^{2-} or SO_3^{2-} .

2. **Assertion:** Sodium carbonate extract of salt containing sulphide gives a violet colour with appropriate reagent.

Reason: The appropriate reagent sodium nitro prusside gives violet colour due to the formation of sodium thionitroprusside.

3. **Assertion:** A salt on boiling with conc. H_2SO_4 gives brown vapours which on absorbing in water do not give brown solution.

Reason: The salt contains bromide and releases brown coloured Br_2 .

4. **Assertion:** Sodium carbonate extract of salt gives yellow precipitate with ammonium molybdate as well as yellow ppt if H_2S is passed in its solution in dil. HCl but the filtrate after removal of yellow ppt. does not give yellow ppt. with ammonium molybdate.

Reason: The salt contains As^{3+} or PO_4^{3-} .

5. **Assertion:** PO_4^{3-} if present in mixture interferes in the test of basic radicals after 2nd gp. in qualitative analysis.

Reason: Phosphates of all metals are insoluble in alkaline medium.

6. **Assertion:** MnS is pink in colour which on dissolution in dilute HCl and then heating with NaOH and Br_2 gives pink colour solution changing to brown on heating.

Reason: Mn^{2+} is oxidized to Mn^{4+} .

7. Assertion: Addition of KCNS to Fe^{3+} ions in solution gives blood red colour.

Reason: A complex $\text{Fe}(\text{CNS})\text{Cl}_2$ or $\text{Fe}(\text{CNS})_2\text{Cl}$ or $\text{Fe}(\text{CNS})_2$ is formed.

8. Statement: Heating effect on NH_4NO_3 , LiNO_3 , NaNO_3 and AgNO_3 is different.

Explanation: The decomposition always gives O_2 .

MATCH THE COLUMN TYPE QUESTIONS

Column-I and Column-II contains four entries each. Entries of column-I are to be matched with some entries of column-II. Each entry of column-I may have the matching with one or more than one entries of column-II.

1. Coloum-I

- (a) Colourless gas evolved on addition of dil. H_2SO_4

Coloum-II

- (p) $\text{S}_2\text{O}_3^{2-}$

- (b) White ppt. on addition of AgNO_3 (q) S^{2-}
 (c) Black ppt. obtained when HgCl_2 is added in little amount (r) NO_2^-
 (d) The ppt. obtained on addition of AgNO_3 followed by NH_3 solution (s) CH_3CO_2^-

RECENT YEARS' QUESTIONS OF IIT-JEE AND AIEEE

1. How do we differentiate between Fe^{3+} and Cr^{3+} in qualitative analysis gp. III: [AIEEE 2002]

- (a) By taking excess of NH_4OH
 (b) By increasing NH_4 ion concentration
 (c) By decreasing OH^- ion concentration
 (d) Both (b) and (c)

2. When H_2S is passed through Hg_2^{2+} , we get: [AIEEE 2002]

- (a) HgS (b) $\text{HgS} + \text{Hg}_2\text{S}$
 (c) $\text{HgS} + \text{Hg}$ (d) Hg_2S

3. A red solid is insoluble in water. However it becomes soluble if some KI is added to water. Heating red solid in a test tube produces violet coloured fumes and droplets of metal appear on the colour parts of test tube. The red solid is: [AIEEE 2003]

- (a) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ (b) HgI_2
 (c) HgO (d) Pb_3O_4

4. Which statement is correct? [AIEEE2003]

- (a) Fe^{3+} ions give deep green precipitate with $\text{K}_4\text{Fe}(\text{CN})_6$ solution.
 (b) On heating K^+ , Ca^{2+} and HCO_3^- ions we get a precipitate of $\text{K}_2\text{Ca}(\text{CO}_3)_2$.
 (c) Manganeses salts give a violet borax bead test in the reducing flame.
 (d) From a mixed precipitate of AgCl and AgI ammonia solution dissolves only AgCl .

5. An aqueous solution of a substance gives a white precipitate on treatment with dilute hydrochloric acid, which dissolves on heating. When hydrogen sulphide is passed through the hot acidic solution, a black precipitate is obtained. The substance is a: [IIT 2002]

- (a) Hg_2^{2+} salt (b) Cu^{2+} salt
 (c) Ag^+ salt (d) Pb^{2+} salt

6. Identify the correct order of solubility of Na_2S , CuS and ZnS in aqueous medium: [IIT 2002]

- (a) $\text{CuS} > \text{ZnS} > \text{Na}_2\text{S}$
 (b) $\text{ZnS} > \text{Na}_2\text{S} > \text{CuS}$
 (c) $\text{Na}_2\text{S} > \text{CuS} > \text{ZnS}$
 (d) $\text{Na}_2\text{S} > \text{ZnS} > \text{CuS}$

7. A gas 'X' is passed through water to form a saturated solution. The aqueous solution on treatment with silver nitrate gives a white precipitate. The saturated aqueous solution also dissolves magnesium ribbon with evolution of a colourless gas T. Identify 'X' and 'Y': [IIT2002]

- (a) $\text{X} = \text{CO}_2$, $\text{Y} = \text{Cl}_2$ (b) $\text{X} = \text{Cl}_2$, $\text{Y} = \text{CO}_2$
 (c) $\text{X} = \text{Cl}_2$, $\text{Y} = \text{H}_2$ (d) $\text{X} = \text{H}_2$, $\text{Y} = \text{Cl}_2$

8. $[\text{X}] + \text{H}_2\text{SO}_4 \rightarrow [\text{Y}]$ a colourless gas with irritating smell, $[\text{Y}] + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \rightarrow$ Green solution, $[\text{X}]$ and $[\text{Y}]$ is: [IIT 2003]

- (a) SO_3^{2-} , SO_2 (b) Cl^- , HCl
 (c) S^{2-} , H_2S (d) CO_3^{2-} , CO_2

9. A solution which is 10^{-3} M each in Mn^{2+} , Fe^{2+} , Zn^{2+} and Hg^{2+} is treated with 10^{-16} M sulphide ion. If K_{sp} of MnS , FeS , ZnS and HgS are 10^{-13} , 10^{-18} , 10^{-24} and 10^{-53} respectively, which one will precipitate first: [IIT2003]
- (a) FeS (b) MgS
(c) HgS (d) ZnS
10. A sodium salt on treatment with MgCl_2 gives white precipitate only on heating. The anion of the sodium salt is: [IIT 2004]
- (a) HCO_3^- (b) CO_3^{2-}
(c) NO_3^- (d) SO_4^{2-}
11. A metal nitrate reacts with KI to give a black precipitate which on addition of excess of KI converts into orange colour solution. The cation of metal nitrate is: [IIT 2005]
- (a) Hg^{2+} (b) Bi^{3+}
(c) Pb^{2+} (d) Cu^+
12. Which pair of compounds is expected to show similar colour in aqueous medium: [IIT 2005]
- (a) FeCl_2 and CuCl_2 (b) VOCl_2 and CuCl_2
(c) VOCl_2 and FeCl_2 (d) FeCl_2 and CuCl_2
13. A solution of a metal ion when treated with KI gives a red ppt. which dissolves in excess KI to give a colourless solution. Moreover, the solution of metal ion on treatment with a solution of cobalt (II) thiocyanate gives rise to a deep blue crystalline ppt. The metal ion is: [IIT 2007]
- (a) Pb^{2+} (b) Hg^{2+}
(c) Cu^{2+} (d) Co^{2+}
14. A solution when diluted with H_2O and boiled, gives a white precipitate. On addition of excess $\text{NH}_4\text{Cl}/\text{NH}_4\text{OH}$, the volume of precipitate decreases leaving behind a white gelatinous precipitate. Identify the precipitate which dissolves in $\text{NH}_4\text{OH}/\text{NH}_4\text{Cl}$
- (a) $\text{Al}(\text{OH})_3$ (b) $\text{Zn}(\text{OH})_2$
(c) $\text{Ca}(\text{OH})_2$ (d) $\text{Mg}(\text{OH})_2$

ANSWER KEYS

ONE CORRECT OPTION TYPE QUESTIONS

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (c) | 2. (b) | 3. (b) | 4. (c) | 5. (a) | 6. (a) | 7. (b) | 8. (a) | 9. (a) | 10. (c) |
| 11. (d) | 12. (a) | 13. (c) | 14. (a) | 15. (c) | 16. (c) | 17. (c) | 18. (c) | 19. (c) | 20. (b) |
| 21. (c) | 22. (a) | 23. (b) | 24. (a) | 25. (a) | 26. (a) | 27. (b) | 28. (b) | 29. (b) | 30. (b) |
| 31. (b) | 32. (c) | 33. (b) | 34. (b) | 35. (a) | 36. (b) | 37. (d) | 38. (c) | 39. (a) | 40. (b) |
| 41. (a) | 42. (a) | 43. (d) | 44. (b) | 45. (d) | 46. (d) | 47. (c) | 48. (b) | 49. (a) | 50. (c) |
| 51. (d) | 52. (c) | 53. (b) | 54. (d) | 55. (b) | 56. (a) | 57. (a) | 58. (a) | 59. (b) | 60. (c) |
| 61. (a) | 62. (a) | 63. (c) | 64. (a) | 65. (c) | 66. (c) | 67. (d) | 68. (c) | 69. (c) | 70. (b) |
| 71. (a) | 72. (c) | 73. (d) | 74. (a) | 75. (a) | 76. (c) | 77. (d) | 78. (b) | 79. (b) | 80. (b) |
| 81. (a) | 82. (c) | 83. (a) | 84. (a) | 85. (b) | 86. (c) | 87. (d) | 88. (a) | 89. (a) | 90. (a) |

COMPREHENSION BASED QUESTIONS

Comprehension-I

1. (a) 2. (c) 3. (c)

Comprehension-II

1. (d) 2. (a) 3. (d) 4. (d)

ONE OR MORE THAN ONE CORRECT OPTION TYPE QUESTIONS

1. (b, c) 2. (a) 3. (a, c) 4. (c, d) 5. (a, b) 6. (a, b, c) 7. (a, b, c) 8. (a, b, c) 9. (a, b, c)
10. (b, d) 11. (a, b, c) 12. (d) 13. (a, b, d) 14. (b, d) 15. (b, c) 16. (b, c) 17. (d) 18. (a, b, c, d)
19. (c, d) 20. (a, b, c)

ASSERTION/REASON TYPE QUESTIONS

1. (c) 2. (c) 3. (a) 4. (a) 5. (c) 6. (c) 7. (c) 8. (a)

MATCH THE COLUMN TYPE QUESTIONS

1. (a) $\rightarrow$ p, q, s; (b) $\rightarrow$ p, r, s; (c) $\rightarrow$ q; (d) $\rightarrow$ q

RECENT YEARS' QUESTIONS OF IIT-JEE AND AIEEE

1. (d) 2. (c) 3. (b) 4. (d) 5. (d) 6. (d) 7. (c) 8. (a) 9. (c) 10. (a)
11. (b) 12. (a) 13. (b) 14. (b)

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Question Bank

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Periodic Properties

STRAIGHT OBJECTIVE TYPE QUESTIONS

- Which of the following statement concerning lanthanides elements is false?
 - Lanthanides are separated from one another by ion exchange method.
 - Ionic radii of trivalent lanthanides steadily increases with increase in the atomic number.
 - All lanthanides are highly dense metals.
 - More characteristic oxidation state of lanthanide elements is +3.
- In the isoelectronic species, the ionic radii (Å) of N^{3-} , O^{2-} and F^{-} are respectively given by
 - 1.36, 1.40, 1.71
 - 1.36, 1.71, 1.40
 - 1.71, 1.40, 1.36
 - 1.71, 1.36, 1.40
- A trend common to both groups I and VII elements in the periodic table as atomic number increases is
 - Oxidizing power increases
 - Atomic radius increases
 - Maximum valency increases
 - Reactivity with water increases
- The incorrect statement among the following is
 - The first ionization potential of Al is less than the first ionization potential of Mg
 - The second ionization potential of is greater than the second ionization potential of
 - The first ionization potential of is less than the first ionization potential of
 - The third ionisation potential of is greater than the third ionization potential of
- Which of the following explanation is best for not placing hydrogen in either the group of alkali metals or halogens?
 - The ionization energy of hydrogen is too high for group of alkali metals, but too low of halogen group
 - Hydrogen can form compounds with all other elements
 - Hydrogen is much lighter element than the alkali metals or the halogens
 - None of the above
- The formation of the oxide ion $O^{2-}_{(g)}$ requires first an exothermic and then an endothermic step as shown below:

$$O_{(g)} + e^{-} = O^{-}_{(g)} \quad \Delta H^{\circ} = -142 \text{ kJmol}^{-1}$$

$$O^{-}_{(g)} + e^{-} = O^{2-}_{(g)} \quad \Delta H^{\circ} = -844 \text{ kJmol}^{-1}$$
 This is because
 - O^{-} ion will tend to resist the addition of another electron
 - Oxygen has high electron affinity
 - Oxygen is more electronegative
 - ion has comparatively larger size than oxygen atom
- The electron affinity values for the halogens show the following trend
 - $F < Cl > Br > I$
 - $F < Cl < Br < I$
 - $F > Cl > Br > I$
 - $F < Cl > Br < I$
- In which of the following arrangements, the order is NOT according to the property indicated against it.
 - $Al^{3+} < Mg^{2+} < Na^{+} < F^{-}$: Increasing ionic size
 - $B < C < N < O$: Increasing first ionization enthalpy
 - $I < Br < F < Cl$: Increasing electron gain enthalpy (with negative sign)
 - $Li < Na < K < Rb$: Increasing metallic radius
- There are four elements 'p', 'q', 'r' and 's' having atomic numbers $Z - 1$, Z , $Z + 1$ and $Z + 2$ respectively. If the element 'q' is an inert gas, select the correct answers from the following statements.
 - 'p' has most negative electron gain enthalpy in the respective period.
 - 'r' is an alkali metal.
 - 's' exists in +2 oxidation state.
 - (i) and (ii)
 - (ii) and (iii)
 - (i) and (iii)
 - (i), (ii) and (iii)

QB.4

10. Which of the following orders is correct for the size?
- $\text{Mg}^{2+} < \text{Na}^+ < \text{F}^- < \text{Al}$
 - $\text{Al}^{3+} < \text{Mg}^{2+} < \text{Li}^+ < \text{K}^+$
 - $\text{Fe}^{4+} < \text{Fe}^{3+} < \text{Fe}^{2+} < \text{Fe}$
 - $\text{Mg} > \text{Al} > \text{Si} > \text{P}$
- (1), (2) and (3)
 - (2), (3) and (4)
 - (1), (3) and (5)
 - (1), (2), (3) and (4)
11. Which of the following orders are correct for the ionization energies?
- $\text{Ba} < \text{Sr} < \text{Ca}$
 - $\text{S}^{2-} < \text{S}, \text{S}^{2+}$
 - $\text{C} < \text{O} < \text{N}$
 - $\text{Mg} < \text{Al} < \text{Si}$
- 1, 2 and 4
 - 1, 3 and 4
 - 1, 2 and 3
 - 1, 2, 3 and 4
12. The electron gain enthalpies of halogens in kJ mol^{-1} are as given below.
 $\text{F} = -332$, $\text{Cl} = -349$, $\text{Br} = -234$, $\text{I} = -295$.
 The least negative value for F as compared to that of Cl is due to:
- Strong electron-electron repulsions in the compact 2p-sub shell of F.
 - Weak electron-electron repulsions in the bigger 3s-sub shell of F.
 - Smaller electronegativity value of F than Cl.
 - (A) and (B) both
13. Which of the following statements is not correct?
- The first ionization energies (in KJ mol^{-1}) of carbon, silicon, germanium, tin, and lead are 1086, 786, 761, 708 and 715 respectively.
 - Down the group, electronegativity decreases from B to Tl in boron family.
 - Among oxides of the elements of carbon family, CO is neutral, GeO is acidic and SnO is amphoteric.
 - The 4f- and 5f-inner transition elements are placed separately at the bottom of the periodic table to maintain its structure.
14. The correct order of acidic strength is:
- $\text{Cl}_2\text{O}_7 > \text{SO}_2 > \text{P}_4\text{O}_{10}$
 - $\text{CO}_2 > \text{N}_2\text{O}_5 > \text{SO}_3$
 - $\text{Na}_2\text{O} > \text{MgO} > \text{Al}_2\text{O}_3$
 - $\text{K}_2\text{O} > \text{CaO} > \text{MgO}$
15. Which of the following orders is correct?
- $\text{F} > \text{N} > \text{C} > \text{Si} > \text{Ga}$ – non-metallic character
 - $\text{F} > \text{Cl} > \text{O} > \text{N}$ – oxidising property
 - $\text{C} < \text{Si} > \text{P} > \text{N}$ – electron affinity value
 - All of these

MULTIPLE CORRECT ANSWER TYPE QUESTIONS

16. The screening effect of inner electrons of an atom causes
- decrease in ionization energy
 - increase in atomic size
 - increase in ionization energy
 - decrease in atomic size
17. Which of the following atoms and ions is (are) iso-electronic with Br^- ?
- Se^{2+}
 - Se
 - As^-
 - Ga^{3+}
18. Mark out the correct options.
- First ionization energy $\text{Ca} > \text{K}$
 - Second ionization energy $\text{Mg} > \text{Al}$
 - Electron affinity $\text{S} > \text{O}$
 - Ionic radius $\text{Sc}^{3+} > \text{K}^+$
19. Which of the following statement(s) is/are true for IA elements?
- Na is regarded as a typical/representative element.
 - Basic character of their oxides increases down the group.
 - Among alkali metal ions degree of hydration is highest in Li.
 - In general electron affinity values decrease from top to bottom in the group.
20. Which of the following represent(s) the correct order of electron affinities?
- $\text{F} > \text{Cl} > \text{Br} > \text{I}$
 - $\text{C} < \text{N} < \text{Cl} < \text{F}$
 - $\text{N} < \text{C} < \text{O} < \text{F}$
 - $\text{C} < \text{Si} > \text{O} > \text{N}$
21. The process(s) requiring the absorption of energy is (are):
- $\text{Cl} - \text{Cl}$
 - $\text{O}^- \rightarrow \text{O}^{2-}$
 - $\text{Fe}^{3+} \rightarrow \text{Fe}^{2+}$
 - $\text{Ar} \rightarrow \text{Ar}^-$

22. Which of the following statement(s) is(are) true?

- (a) Ionization energy $\propto \frac{1}{\text{Screening effect}}$
- (b) The first ionization energies of Be and Mg are more than ionization energies of B and Al respectively.
- (c) Atomic and ionic radii of Niobium and Tantalum are almost same.
- (d) Metallic and covalent radii of potassium are 2.4Å and 2.03Å respectively.

23. Which of the following statement(s) is (are) correct?

- (a) The electron affinity for sulphur is more exothermic than that for oxygen.
- (b) Successive ionization energies of an atom always increase.
- (c) First ionization energy of As is greater than that of Se.
- (d) Chlorine has larger atomic size as well as electron affinity than that of fluorine.

ASSERTION/REASON TYPE QUESTIONS

24. **Assertion:** Noble gases have maximum electron affinity. [AIIMS 1995]

Reason: High electron affinity shows that the electron is loosely bonded to the atom.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
- (c) Statement-1 is True, Statement-2 is False.
- (d) Statement-1 is False, Statement-2 is True.

25. **Statement-1:** The 5th period of periodic table contains 18 elements and not 32.

Statement-2: $n = 5$, $-l = 0, 1, 2, 3$. The order in which the energy of available orbits 4d, 5s and 5p increases is $5s < 4d < 5p$ and the total number of orbitals available are 9 and thus 18 electrons can be accommodated.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
- (c) Statement-1 is True, Statement-2 is False.
- (d) Statement-1 is False, Statement-2 is True.

26. **Statement-1:** The first ionization energy of Be is greater than that of B.

Statement-2: 2p orbital is lower in energy than 2s.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

27. **Statement-1:** Manganese (atomic number 25) has a less favourable electron affinity than its neighbours on either side because.

Statement-2: (i) The Manganese has stable, $[\text{Ar}]^{18} 3d^5 4s^2$ electrons configuration.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
- (c) Statement-1 is True, Statement-2 is False.
- (d) Statement-1 is False, Statement-2 is True.

28. **Statement-1:** The electron gain enthalpies have large negative values to the upper right of the periodic table preceding the noble gases.

Statement-2: (i) The effective nuclear charge increases from left to right across a period and to consequently it will be easier to add an electron to a smaller atom since the added electron on an average would be closer to the positively charged nucleus.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
- (c) Statement-1 is True, Statement-2 is False.
- (d) Statement-1 is False, Statement-2 is True.

TRUE AND FALSE TYPE QUESTIONS

29. 1. In modern periodic table, each block contains a number of columns equal to the number of electrons that can occupy that sub-shell.
 2. The greatest increase in ionization enthalpy is experienced on removal of electron from core noble gas configuration.
 3. The size of the isoelectronic species is effected by electron-electron interaction in the outer orbitals.
4. Any thing influences the valence electrons will affect the chemistry of the element and the valence shell is not affected by nuclear mass.
30. 1. $\text{Na}_2\text{O}_2 < \text{MgO} < \text{ZnO} < \text{P}_4\text{O}_{10}$: Acidic property.
 2. $\text{Na} < \text{Si} > \text{Mg} < \text{Al}$: First ionization potential.
 3. $\text{F} > \text{Cl} > \text{Br}$: Electron affinity.
 4. $\text{Te}^{2-} > \text{I}^- > \text{Cs}^+ > \text{Ba}^{2+}$: Ionic size.
 (a) TTT T (b) TTFT
 (c) TFFT (d) TFFT

COMPREHENSION BASED QUESTIONS

Comprehension-1

The periodicity is related to the electronic configuration. That is, all chemical and physical properties are a manifestation of the electronic configuration of the elements.

The atomic and ionic radii generally decrease in a period from left to right. As a consequence, the ionisation enthalpies generally increase and electron gain enthalpies become more negative across a period. In other words, the ionization enthalpy of the extreme left element in a period is the least and the electron gain enthalpy of the element on the extreme right is the highest negative. This results into high chemical reactivity at the two extremes and the lowest in the centre. Similarly, down the group, the increase in atomic and ionic radii result in gradual decrease in ionization enthalpies and a regular decrease (with exception in some third period elements) in electron gain enthalpies in the case of main group elements.

These properties can be related with the:

- (i) reducing and oxidizing behaviour of the elements
- (ii) metallic and non-metallic character of element
- (iii) acidic, basic, amphoteric and neutral character of the oxides of the elements.

31. The correct order of the metallic character is:

- (a) $\text{B} > \text{Al} > \text{Mg} > \text{K}$ (b) $\text{Al} > \text{Mg} > \text{B} > \text{K}$
 (c) $\text{Mg} > \text{Al} > \text{K} > \text{B}$ (d) $\text{K} > \text{Mg} > \text{Al} > \text{B}$

32. The correct order of chemical reactivity in terms of oxidizing property is:

- (a) $\text{F} > \text{Cl} > \text{O} > \text{N}$
 (b) $\text{F} > \text{O} > \text{Cl} > \text{N}$
 (c) $\text{Cl} > \text{F} > \text{O} > \text{N}$
 (d) $\text{O} > \text{F} > \text{N} > \text{Cl}$

33. Which of the following statement is incorrect?

- (a) In general, metallic character increases down the group and decreases across a period.
 (b) In general, reducing property decreases down the group and increases across a period.
 (c) In general, the oxide formed by the element on extreme right is the most acidic.
 (d) Both (b) and (c)

34. Among Al_2O_3 , SiO_2 , P_2O_3 and SO_2 the correct order of acid strength is

- (a) $\text{Al}_2\text{O}_3 < \text{SiO}_2 < \text{SO}_2 < \text{P}_2\text{O}_3$
 (b) $\text{SiO}_2 < \text{SO}_2 < \text{Al}_2\text{O}_3 < \text{P}_2\text{O}_3$
 (c) $\text{SO}_2 < \text{P}_2\text{O}_3 < \text{SiO}_2 < \text{Al}_2\text{O}_3$
 (d) $\text{Al}_2\text{O}_3 < \text{SiO}_2 < \text{P}_2\text{O}_3 < \text{SO}_2$

Comprehension-2

It is not possible to measure the atomic radius precisely since the electron cloud surrounding the atom does not have a sharp boundary. One practical approach to estimate the size of an atom of a non-metallic element is to measure the distance between two atoms when they are bound together by a single bond in a covalent molecule and then dividing by two. For metals we define the term “metallic radius” which is taken as half the internuclear distance separating the metal cores in the metallic crystal. The van der Waal’s radius represents the over all size of the atoms which includes its valence shell in a non bonded situation. It is the half of the distance between two similar atoms in separate molecules in a solid. The atomic radius decreases across a period and increases down the group. Generally same trends are observed in case of ionic radius. Ionic radius of the species having same number of electrons depends on the number of protons in their nuclei.

35. The size of isoelectronic species – F^- , Na^+ and Mg^{2+} is effected by:
- nuclear charge
 - valence principal quantum number (n)
 - electron–electron interaction in the outer orbitals
 - atomic mass
36. Atomic radii of the noble gases are larger than the precedent elements of the same periods because:
- Atomic radii of the noble gases is expressed as van der Waal's radius
 - Valence shell electrons are completely filled so there is intereletronic repulsions.
 - Both (a) and (b)
 - None of these
37. Which of the following statements is correct?
- Metallic radius refer to metals only and is greater than covalent radius
 - Metallic radius refer to metals only and is smaller than covalent radius
 - Generally covalent radius refer to non-metals as well as metals in bonded state (Covalent bond).
 - Atomic radii of noble gases are expressed as van der Waal's radii which are smaller than metallic radii.

Comprehension-3

The first ($\Delta_i H_1$) and second ($\Delta_i H_2$) ionization enthalpies (in kJ mol^{-1}) and the ($\Delta_{eg} H$) electron gain enthalpy (in kJ mol^{-1}) of a few elements are given below:

Elements	$\Delta_i H_1$	$\Delta_i H_2$	$\Delta_{eg} H$
(a) P	520	7300	–60
(b) Q	419	3051	–48
(c) R	1681	3374	–328
(d) S	1008	1846	–295
(e) T	2372	5251	+48
(f) U	738	1451	–40

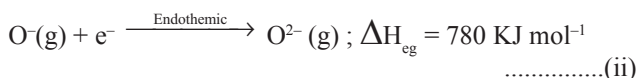
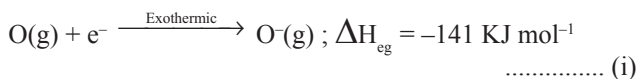
38. The least reactive element is:
- P
 - Q
 - R
 - T
39. The most reactive element is:
- P
 - Q
 - S
 - U
40. The most reactive element is:
- R
 - S
 - P
 - U

Comprehension-4

The amount of energy required to remove the most loosely bound electron from an isolated gaseous atom is called as first ionization energy (IE_1). Similarly, the amount of energies required to knock out second, third etc. electrons from the isolated gaseous cation are called successive ionization energies and $IE_3 > IE_2 > IE_1$.

- Nuclear charge
- atomic size
- penetration effect of the electrons
- shielding effect of the inner extra stable) affect the ionisation energies.

On the other hand, the amount of energy released when a neutral isolated gaseous atom accepts an extra electron to form gaseous anion is called electron affinity.



In (ii), the energy has to be supplied for the addition of second electron due to electrostatic repulsion between an anion and extra electron (same charged species). The electron affinity of an element depends upon (i) atomic size (ii) nuclear charge and (iii) electronic configuration. In general, ionization energy and electron affinity increases as the atomic radii decreases and nuclear charge increases across a period. In general, in a group, ionization energy and electron affinity decreases as the atomic size increases.

The members of third period of have some higher (e.g., S and Cl) electron affinity values than the members of the second period (e.g., O and F) because second period elements have very small atomic size. Hence there is a tendency of electron-electron repulsion, which results in less evolution of energy in the formation of corresponding anion.

41. The first ionization energies of Na, Mg, Al and Si are in the order of:
- $Na < Mg > Al < Si$
 - $Na > Mg > Al > Si$
 - $Na < Mg < Al < Si$
 - $Na > Mg > Al < Si$
42. Which of the following statements is correct?
- The element like F, Cl, Br etc, having high values of electron affinity act as strong oxidising agent.
 - The elements having low values of ionisation energies act as strong reducing agent.
 - The formation of $Be^- (g)$ from $Be(g)$ is an endothermic process
 - All of these

QB.8

43. Which of the following statements is correct?
- Ionization enthalpy increases for each successive valence shell electron.
 - The greatest increase in ionization enthalpy is experienced on removal of electron from core of noble gas configuration.
 - End of valence electrons is marked by a big jump in ionization enthalpy.
 - Removal of electron from orbitals bearing lower n value is easier than from orbital having higher n value.
44. Considering the elements F, Cl, O and N, the correct order of their electron affinity is:
- $F > Cl > O > S$
 - $F > O > Cl > S$
 - $Cl > F > S > O$
 - $O > F > S > Cl$

MATCH THE COLUMN TYPE QUESTIONS

45. Match the following:

Column-I

- $Fe(III) > Fe(II)$
- $Al > Na$
- $Cl > F$
- $N > C$

Column-II

- Electronegativity
- Basic character of their oxides
- Electron gain enthalpy ($-\Delta_{eg}H$)
- Degree of hydration

46. Match the following:

Column-I

- SO_2, NO_3^-, CO_3^{2-}
- B, Si, Ge, As, Sb
- He, Ne, Ar, Kr, Xe
- $M(g) + \text{energy} \rightarrow M^+(g) + e^-$
- $X(g) + e^- \rightarrow X^-(g) + \text{Energy}$

Column-II

- Semi-metals
- Electron affinity
- Isoelectronic species
- Ionization energy
- van der Waal's radii

47. Match the values of ionization energy and electron gain enthalpy listed in column-I with characteristic(s) of elements listed in column-II.

Column-I

$\Delta_f H_1, \Delta_f H_2, \Delta_{eg} H$ (in KJ mol^{-1})

- 2372 5251 +48
- 419 3051 -48
- 1681 3374 -333
- 1008 1846 -295

Column-II

- Elements which acts as a strong reducing agent
- Elements which exists as a monoatomic molecule.
- Least reactive non-metal
- Element which acts as strong oxidising agent
- Elements which oxide is a stronger basic in nature.

48. Match the increasing order given in column-I with the property(ies) given in column-II.

Column-I

- $Na^+ < F^- < O^{2-} < N^{3-}$
- $Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$
- $O < S < F < Cl$
- $Cl^- < K^+ < Ca^{2+} < Sc^{3+}$

Column-II

- Electronegativity
- Mobility of hydrated ions
- Ionic size
- Electron affinity

SUBJECTIVE TYPE QUESTIONS
Short Subjective Questions

49. A metal has electronic configuration $[Ar]18 3d^7 4s^2$. On the basis of this electronic configuration, find out the group member.

50. An ion having a 4+ charge and a mass of 51.99 amu has two electrons with $n = 1$, eight electrons with $n = 2$, and ten electrons with $n = 3$. Give the total number of protons present in the nucleus of the atom of metal.

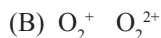
Chemical Bonding

STRAIGHT OBJECTIVE TYPE QUESTIONS

- Point out the incorrect statement about resonance.
 - Resonance structures should have equal energy
 - In resonance structures, the constituent atoms should be in the same position
 - In resonance structures, there should not be the same number of electron pairs
 - Resonance structures should differ only in the location of electrons around the constituent atoms
- Out of the following, which one has smallest bond length?
 - O_2
 - O_2^+
 - O_2^-
 - O_2^{2-}
- In which of the following pairs the two molecules have identical bond order
 - N_2, O_2^{2+}
 - N_2, O_2^-
 - N_2^-, O_2
 - O_2^+, N_2
- Select the most ionic and most covalent compounds respectively from the following.
 $CrO_5, Mn_2O_7, PbO, P_4O_{10}, SnO_2$
 - CrO_5, Mn_2O_7
 - PbO, Mn_2O_7
 - CrO_5, P_4O_{10}
 - SnO_2, CrO_5
- The hybridization of atomic orbitals of nitrogen in NO^{2+}, NO^{3-} and NH_4^+ are:
 - sp, sp^3 and sp^2 respectively
 - sp, sp^2 and sp^3 respectively
 - sp^2, sp and sp^3 respectively
 - sp^2, sp^3 and sp respectively
- In which of the following sets, the central atom of each member involves sp^3 hybridization?
 - IO_4^-, ICl_4^- and IF_4^+
 - X_3O_3, XeO_4, X_3F_4
 - $SO_3, SO_3^{2-}, SO_4^{2-}$
 - PCl_4^+, BF_4^-, ClO_4^-
- In which of the following groups, all the members have a linear shape?
 - NO_2, N_3^-, ICl_2^+
 - N_3^-, I_3^-, NO_2^+
 - XeF_2, C_2H_2, SO_2
 - $CO_2, BeCl_2, SnCl_2$
- Which reaction involves a change in the electro-pair geometry for the underlined geometry?
 - $\underline{BF}_3 + F^- \rightarrow BF_4^+$
 - $\underline{NH}_3 + H^+ \rightarrow \underline{NH}_4^+$
 - $2\underline{SO}_2 + O_2 \rightarrow 2\underline{SO}_3$
 - $H_2\underline{O} + H^+ \rightarrow H_3\underline{O}^+$
- Which of the species has the maximum number of lone pairs of electrons on the central atom?
 - $XeOF_4$
 - IF_4^+
 - XeF_2
 - BrF_3
- In the structure of H_2CSF_4 , to decide the plane in which $C = S$ is present, the following bond angle value are given
 Axial FSF angle (idealised = 180°) $\Rightarrow 170^\circ$
 Equatorial FSF angle (idealised = 120°) $\Rightarrow 97^\circ$
 After deciding the plan of double bond, which of the following statements is correct?
 - Two C-H bonds are in the same plane of axial S-F bonds.
 - Two C-H bonds are in the same plane of equatorial S-F bonds.
 - Total five atoms are in the same plane.
 - Equatorial S-F bonds are perpendicular to plane of π bond.
- Which of the following are isoelectronic and isostructural?
 - NO_2^+, CO_2
 - NO_3^-, BF_3
 - NH_3, CH^-
 - All of these
- In which of the following molecules/ions are all the bond angles not equal?
 - SiF_4
 - ICl_4^-
 - SF_4
 - PCl_4^+
- There is no S-S bond in
 - $S_2O_3^{2-}$
 - $S_2O_4^{2-}$
 - $S_2O_5^{2-}$
 - $S_2O_7^{2-}$

QB.10

15. Match the ionization processes listed in column-I with the changes observed as listed in column-II. For this, use the codes given below:

Column-I

Column-II

(P) Bond order increases and magnetic property is changed.

(Q) Bond order decreases and magnetic property is changed.

(R) Bond order increases and magnetic property is not changed.

(S) Bond order decreases and magnetic property is changed.

Note: Here change in magnetic property refers to change from diamagnetic to paramagnetic or paramagnetic to diamagnetic.

A B C D

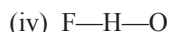
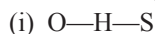
(a) s b c d

(b) s p r q

(c) r q s p

(d) p s q r

16. The increasing order of the strength of hydrogen bond in the following mentioned linkages is



(a) (i) < (ii) < (iv) < (iii)

(b) (i) < (i) < (iv) < (iii)

(c) (i) < (ii) < (iii) < (iv)

(d) (ii) < (i) < (iii) < (iv)

17. The boiling point of CCl_4 is higher than that of $CHCl_3$ because

(a) The dipole moment of CCl_4 is greater than that of $CHCl_3$

(b) CCl_4 forms hydrogen bonds.

(c) CCl_4 has higher molecular mass than that of $CHCl_3$.

(d) CCl_4 is more ionic than $CHCl_3$.

MULTIPLE CORRECT ANSWER TYPE QUESTIONS

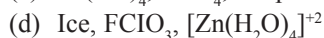
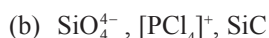
18. In which of the following pairs, hybridization of the central atom is same?



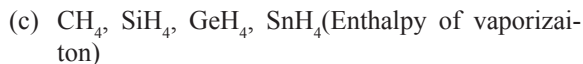
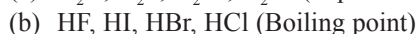
19. In O_2 molecule, which of the following molecular orbitals are lower in energy than $\sigma 2p_z$ orbital



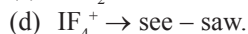
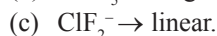
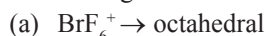
20. Which of the following set correctly represent sp^3 hybridization and tetrahedral geometry?



21. Mark out the incorrect decreasing order of physical properties given in bracket.



22. Which of the following species are correctly matched with their geometries according to VSEPR theory?



23. Which of the following statements is (are) true?

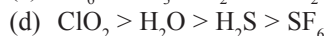
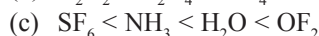
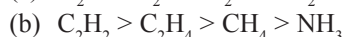
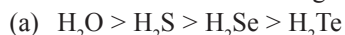
(a) It is impossible to satisfy the octet rule for all atoms in XeF_2 .

(b) As SF_4 exists, OF_4 should also exist because oxygen is in the same family.

(c) The bond in NO^+ should be stronger than the bond in NO^- .

(d) For ozone molecule, one oxygen-oxygen bond is stronger than the other oxygen-oxygen bond.

24. Correct order about bond angle is:



25. Which charge(s) for the N_2 molecule would give a bond order of 2.5?

- (a) +1 (b) +2
(c) -1 (d) -2
26. Which of the following are polar?
- (a) XeF_4 (b) XeF_6
(c) XeOF_4 (d) XeF_5^-

27. Hydrogen bonding is present in which of the following species?

- (a) CH_3NH_2 (b) CH_3
(c) CH_3COOH (d) $\text{CCl}_3\text{CH}(\text{OH})_2$

ASSERTION/REASON TYPE QUESTIONS

28. **Statement-1:** All $\text{F}-\text{S}-\text{F}$ angle in SF_4 greater than 90° but less than 180° .

Statement-2: The lone pair-bond pair repulsion is weaker than bond pair-bond pair repulsion.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

29. **Statement-1:** Compounds of Hg^{2+} ions having an ionic radius of 116 pm are more covalent in character than those of Ca^{2+} ions with almost identical size (114 pm) and the same charge.

Statement-2: For two ions of the same size and charge, one with an $(n-1)d^{10}ns^0$ electronic configuration will be more polarizing than a cation with an $(n-1)s^2(n-1)p^6ns^0$ electronic configuration.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

30. **Statement-1:** Carbon has unique ability to form $p\pi-p\pi$ multiple bonds with itself and with other atoms of small size and high electronegativity.

Statement-2: Heavier elements of group 14th do not form $p\pi-p\pi$ multiple bonds with itself because their atomic orbitals are too large and diffuse to have effective overlapping.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

31. **Statement-1:** NO^+ and CN^- both have same bond order and magnetism.

Statement-2: NO^+ and CN^- are isoelectronic species.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

32. **Statement-1:** Crystals of hydrated calcium sulphate (gypsum: $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) are soft and easily cleaved.

Statement-2: Crystals of anhydrous calcium sulphate (anhydrite: CaSO_4) are very hard and very difficult to cleave.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

TRUE AND FALSE TYPE QUESTIONS

Consider the following statements and arrange in the order true/false as given in the codes.

33. 1. In AB_2L_2 type, the BAB bond angle will always be greater than normal tetrahedral.
2. Maleic acid has higher K_a value than fumaric acid due to intramolecular hydrogen bonding.
3. In ClO_3^- , NH_3 , XeO_3 , the hybridization and the number of lone pair(s) on central atoms are same
- (a) T T T (b) F T F
- (c) F T T (d) F F T
34. 1. SbF_4^- and SF_4 are isostructural.
2. SeOCl_2 has a lone pair and a double bond.
3. In IOF_5 , the hybridization of central atom is sp^3d^2 .
4. In SO_4 and XeO_3F_2 , double bond(s) to oxygen are equatorial.

- (a) T T T T (b) F T F T
(c) T F F T (d) F F T T

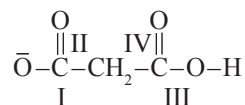
35. 1. N_2H_4 is pyramidal about each N atom.
2. NH_2OH is pyramidal about the N atom and bent about the O atom.
3. CH_3COCl is trigonal planar about the carbon atom (attached to O and Cl).
(a) F T T (b) F T F
(c) T T T (d) F F T
36. 1. The hydrogen bond in KHF_2 is stronger than that in gaseous HF.
2. The dipole moment of CH_3F is greater than that of CH_3Cl .
3. SP_2 hybrid orbitals have equal s and p character.
4. Two types of bonds present in B_2H_6 are covalent and three center-two electron bonds.
(a) T T F F (b) T F F T
(c) F F F T (d) F F T T

COMPREHENSION BASED QUESTIONS

A number of molecules and polyatomic ions cannot be described accurately by a single Lewis structure and a number of descriptions (representations) based on the same skeletal structure are written and these taken together represent the molecule or ion. This is a very important and useful concept called resonance. The contributing structures or canonical forms taken together constitute the resonance hybrid which represents the molecule or ion.

37. Which of the following is not considered as the canonical structure of N_3^- ion?
- (a) $:\ddot{\text{N}}=\text{N}=\ddot{\text{N}}:^-$ (b) $:\ddot{\text{N}}-\text{N}\equiv\text{N}:^-$
 (c) $:\text{N}\equiv\text{N}-\ddot{\text{N}}:^-$ (d) $[\text{N}=\text{N}:]^-$
 $\quad\quad\quad \diagdown \quad \diagup$
 $\quad\quad\quad :\text{N}:$
38. Choose the option having incorrectly indicated more stable canonical form.
- (a) $\text{CH}_2^+ - \text{F} \leftrightarrow \text{CH}_2 = \text{F}^+$
 (b) $\text{CH}_2 = \overset{\text{O}}{\parallel}{\text{CH}}^- \leftrightarrow \text{CH}_2 = \overset{\text{O}^-}{\mid}{\text{CH}}$
 (c) $\text{CH}_2 = \overset{+}{\text{CH}} \text{NR}_3^+ \leftrightarrow \text{CH}_2 = \overset{+}{\text{CH}} \text{NR}_3^+$
 (d) $\text{CH}_3 \text{O} \overset{+}{\text{CH}} \text{CH}_2^+ \leftrightarrow \text{CH}_3 \text{O} \overset{+}{\text{CH}} \text{CH}_2^+$

39. Which of the following comparison of bond length (C—O) is correct for the indicated bonds in the given molecule?



- (a) $I = II = III = IV$ (b) $I = II; III = IV$
(c) $IV < I = II < III$ (d) $II = IV < I < III$

Comprehension-1

When anions and cations, approach each other, the valence shell of anions are pulled towards a cation, is known as polarization and ability of the cation to polarize the anion is called as polarizing power of cation. Due to polarization, sharing of electrons occurs between two ions to some extent and bond shows some covalent character. Fajan's suggested following factors on which polarization depend.

- (i) As the charge on cation or anion increases, polarization increases
- (ii) Size of cation decreases or size of anion increases, polarization increases.

(iii) Cation with pseudo noble gas configuration shows higher polarization power than that with inert gas configuration.

40. Arrange the following compounds in the increasing order of covalent character.

- (I) SiCl_4 (II) MgCl_2
 (III) NaCl (IV) AlCl_3
 (a) $\text{NaCl} < \text{MgCl}_2 < \text{AlCl}_3 < \text{SiCl}_4$
 (b) $\text{SiCl}_4 < \text{AlCl}_3 < \text{MgCl}_2 < \text{NaCl}$
 (c) $\text{NaCl} < \text{MgCl}_2 < \text{SiCl}_4 < \text{AlCl}_3$
 (d) $\text{SiCl}_4 < \text{AlCl}_3 < \text{NaCl} < \text{MgCl}_2$

41. Which one is most ionic?

- (a) P_4O_{10} (b) TiO_2
 (c) SnO (d) SO_3

42. In which of the following compounds, highest polarization is observed?

- (a) MgCl_2 (b) BaCl_2
 (c) AgCl (d) AgI

Comprehension-2

The approximate shape of a molecule can often be predicted by using what is called the valence-shell electron-pair repulsion (VSEPR) model. Electron in bonds and in lone pairs can be thought of as a "Charge cloud" that repels one another and stay as far apart possible, thus causing molecules to assume specific shapes.

The repulsive interactions of electron pairs decrease in order:

Lone pair – lone pair > Lone pair – Bond pair > Bond pair – Bond pair

These repulsions effect result in deviations from idealised shapes and alteration in bond and molecules

43. Which of the following statement is correct with respect to bond angle?

- (a) The $\text{F} - \text{S} - \text{F}$ angle in SF_2 is more than $109^\circ 28'$
 (b) The $\text{H} - \text{N} - \text{N}$ angle in N_2H_2 is approximately 180°
 (c) The $\text{F} - \text{Kr} - \text{F}$ angle in KrF_4 is 90°
 (d) The $\text{Cl} - \text{N} - \text{O}$ angle in NOCl is more than 120° .

44. Molecular shape of XeF_3^+ , SF_3^+ and CF_3^+ are

- (a) the same with 2, 1 and 0 lone pairs of electrons respectively.
 (b) different with 2, 1 and 0 lone pairs of electrons respectively.
 (c) different with 0, 1 and 2 lone pairs of electrons respectively.
 (d) the same with 2, 1 and 0 lone pairs of electrons respectively.

45. Which of the following statements is incorrect?

- (a) In ClF_3 , the axial $\text{Cl} - \text{F}$ bond length is larger than equatorial $\text{Cl} - \text{F}$ bond length.
 (b) In SF_4 , $\text{F} - \text{S} - \text{F}$ equatorial bond angle is not 120° but 104° due to $\text{lp} - \text{bp}$ repulsions.
 (c) In $[\text{Cl}_4]^-$, $\text{Cl} - \text{I} - \text{Cl}$ bond angle is 90° .
 (d) In OBr_2 , the bond angle is less than OCl_2 .

Comprehension-3

Chemical bonding between two atoms is necessarily associated with an electrical moment arising out of the difference in electronegativity of two atoms. This means that every bond carries with it an electrical moment called the "bond moment". The dipole moment of a molecule is really the vectorial sum of the individual bond moment present in it. To compute the dipole moment it is necessary to find out the values of various bond moment in the following table dipole moment of different bonds are as given.

Bond	$\xrightarrow{\text{H}} \text{H}$	$\xrightarrow{\text{C}} \text{Cl}$	$\xrightarrow{\text{C}} \text{O}$
Bond moments	0.4D	1.5D	2.5D

The Group moments of few group as given

Group	NO_2	OH	CH	CH_3
direction of dipole	toward N	towards O	toward N	away from CH_3
dipole moment	4D	1.6 D	3.8 D	0.4 D

46. The bond angle in H_2S 97° and its dipole moment is 1.5 D. The $\text{S} - \text{H}$ bond distance is 0.15 nm. Therefore approximately percentage ionic character of $\text{S} - \text{H}$ bond is (neglect the effect of dipole moment of lone pair on sulphur atom in H_2S). (Given $[\cos 97^\circ = -0.12]$ and " $0.88 = 0 - 94$ ")

- (a) 32 % (b) 16 %
 (c) 84% (d) 10 %

47. In CH_3CCl_3 (I), CH_3Cl_3 (II) and CH_3Cl (III) the normal tetrahedral bond angle is maintained. Also given $\cos 70.5^\circ = \frac{1}{2}$. Therefore dipole moments of the given compounds are, (given due to -1 effect of Cl , the Bond moment of $\text{H} - \text{C}$ bond directed toward the H in CHCl_3).

- (a) I = 1.9 D, II = 1.9 D, III = 1.7 D
 (b) I = 1.9 D, II = 1.7 D, III = 1.9 D
 (c) I = 1.9 D, II = 1.7 D, III = 1.7 D
 (d) I = 1.9 D, II = 1.1 D, III = 1.9 D

48. In the acetone molecule considering the normal planer structure, the observed dipole moment of acetone molecule is –

- (a) 2.9 D (b) 2.75 D
 (c) 3 D (d) None of these

MATCH THE COLUMN TYPE QUESTIONS

49. Match the following:

Column-I (Moleculation)	Column-II (Hybridisation of Central)
(a) IO_2F_2^-	(P) sp^3d
(b) F_2SeO	(Q) sp^3
(c) ClOF_3	(R) sp^2
(d) XeF_5^+	(S) sp^3d^2

50. Match the following:

Column-I (Decreasing order)	Column-II (Physical Properties)
(A) $\text{NH}_3, \text{SbH}_3, \text{AsH}_3, \text{PH}_3$	(P) Dipole moment
(B) $\text{HF}, \text{HCl}, \text{HBr}, \text{HI}$	(Q) Melting point
(C) $\text{SnH}_4, \text{GeH}_4, \text{SiH}_4, \text{CH}_4$	(R) Enthalpies of vapourization
(D) $\text{H}_2\text{O}, \text{H}_2\text{Te}, \text{H}_2\text{Se}, \text{H}_2\text{S}$	(S) Boiling point

51. Match the compounds listed in column-I with characteristic(s) listed in column-II.

Column-I	Column-II
(A) SF_3	(P) One of the element has -1 oxidation state,
(B) KrF_4	(Q) Central atom has lone pair(s) of electrons.

- (C) NOCl (R) Central atom has sp^3 hybridization.
 (D) $\text{N}(\text{Me})_3$ (S) $p-d$ bonding.

52. Match the species listed in column-I with characteristic (s)/property (s) listed in column-II.

Column-I	Column-II
(A) C_2	(P) Diamagnetic
(B) N_2	(Q) Undergoes oxidation
(C) O_2^{2-}	(R) Undergoes reduction
(D) F_2	(S) Bond order ≥ 2
	(T) No mixing of 's' and 'p' orbitals

53. Match the compounds listed in column-I with characteristic(s) listed in column-II.

Column-I	Column-II
(A) Chloral hydrate	(P) Forms a zig zag chain in both the solid and also in the liquid.
(B) Hydrogen fluoride	(Q) Forms two dimensional sheet with almost hexagonal symmetry.
(C) Crystalline boric acid	(R) Have low volatility
(D) Sulphuric acid	(S) Intramolecular H-bond
	(T) Intermolecular H-bond

SUBJECTIVE TYPE QUESTIONS

54. How many bond pairs of electrons are around N in NO_3^- ?

55. How many hydrogen bonded water molecule(s) are associated in $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$.

56. If the dipole moment of AB molecule is given by 1.2 D and A-B the bond length is 1 Å then % covalent character of the bond is.

57. The number of vacant hybrid orbitals which participate in the formation of 3-centre 2 electron bonds i.e., banana bonds in diborane structure is:

Coordination Compounds

STRAIGHT OBJECTIVE TYPE QUESTIONS

- Which of the following pairs is not correctly matched?
 - Effective atomic number of Pt in $[\text{PtCl}_6]^{2-} = 84$
 - Absorption peak for $[\text{Cr}^{\text{III}}(\text{NH}_3)_6]^{+3} = 21680 \text{ cm}^{-1}$
 - Crystal field stabilization energy of d^2 in weak ligand field $= (-)0.8\Delta_0$
 - Example of weak ligand field for d^5 configuration $= [\text{Mn}^{\text{II}} \text{F}_6]^{-4}$
- The total number of possible isomers for the complex compound $[\text{Cu}^{\text{II}}(\text{NH}_3)_4][\text{Pt}^{\text{II}}\text{Cl}_4]$ are
 - 3
 - 4
 - 5
 - 6
- $[\text{CoF}_6]^{-3}$ is formed by hybridization
 - d^2sp^3
 - dsp^2
 - d^2sp^3
 - sp^3d^2
- Considering H_2O as a weak field ligand, the number of unpaired electrons in $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ will be (At. No. of Mn = 25)
 - Two
 - Four
 - Three
 - Five
- The EAN of metal atoms in $[\text{Fe}(\text{CO}_2)(\text{NO}^+)_2]$ and $\text{Co}_2(\text{CO})_8$ respectively are:
 - 34, 35
 - 34, 36
 - 36, 36
 - 36, 35
- Match the complex ions listed in column-I with the characteristics listed in column-II using the codes given below.

Column-I	Column-II
(A) $[\text{Ni}(\text{NH}_3)_6]^{2+}$	(P) d^2sp^3 and diamagnetic
(B) $[\text{Cr}(\text{NH}_3)_6]^{3+}$	(Q) sp^3d^2 and diamagnetic
(C) $[\text{Co}(\text{NH}_3)_6]^{3+}$	(R) sp^3d^2 and two unpaired electrons.
(D) $[\text{Zn}(\text{NH}_3)_6]^{2+}$	(S) d^2sp^3 and three unpaired electrons.

A	B	C	D
(a) R	S	P	Q
(b) S	R	Q	P
(c) R	S	Q	P
(d) P	Q	R	S
- Which of the following statements is correct?
 - $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}(\text{aq})$ and $[\text{Cr}(\text{CN})_6]^{4-}(\text{aq})$ are high spin and low spin complexes respectively, but both have same number of unpaired electrons.
 - $[\text{V}(\text{NH}_3)_6]^{3+}$ is inner orbital complex and paramagnetic with two unpaired electrons.
 - $[\text{Co}(\text{CN})_6]^{4-}$ is outer orbital complex and paramagnetic with three unpaired electrons.
 - $[\text{Mn}(\text{NO})(\text{CO})]$ is square planar and diamagnetic.

Note: NO is three electrons donor.
- Among $[\text{NiCl}_2(\text{PPh}_3)_2]$, $[\text{Ni}(\text{HDMG})_2]$ and $[\text{Ni}(\text{CO})_4]$ complexes the hybridization states of Ni atom are respectively.
 - sp^3, sp^3, dsp^2
 - sp^3, dsp^2, sp^3
 - dsp^2, dsp^2, sp^3
 - sp^3, dsp^2, dsp^2
- Amongst $[\text{Co}(\text{ox})_3]^{3-}$, $[\text{CoF}_6]^{3-}$ and $[\text{Co}(\text{NH}_3)_6]^{3+}$
 - $[\text{Co}(\text{ox})_3]^{3-}$ and $[\text{CoF}_6]^{3-}$ are paramagnetic and $[\text{Co}(\text{NH}_3)_6]^{3+}$ is diamagnetic.
 - $[\text{Co}(\text{ox})_3]^{3-}$ and $[\text{Co}(\text{NH}_3)_6]^{3+}$ are paramagnetic and $[\text{CoF}_6]^{3-}$ is diamagnetic.
 - $[\text{Co}(\text{ox})_3]^{3-}$ and $[\text{Co}(\text{NH}_3)_6]^{3+}$ are diamagnetic and $[\text{CoF}_6]^{3-}$ is paramagnetic.
 - $[\text{Co}(\text{NH}_3)_6]^{3+}$ and $[\text{CoF}_6]^{3-}$ are paramagnetic and $[\text{Co}(\text{ox})_3]^{3-}$ is diamagnetic.
- The 'brown ring' formed at the junction of two layers in the test of nitrate is due to the formation of a complex ion, $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]^{2+}$. Which of the following statements are correct for this complex. [$\mu = 3.87 \text{ B.M.}$]
 - Oxidation state of Fe is +1 and NO exists as NO^+ .
 - The complex ion is in octahedral geometry as attained by sp^3d^2 hybridization.
 - The complex is paramagnetic and has three unpaired electrons due to transfer of electron from NO to Fe^{2+} .
 - The complex is in octahedral geometry as attained by d^2sp^3 hybridization.
 - The brown colour of the complex is attributed to d-d transition of electron.

QB.16

16. Select the correct statement:
 - (a) Chelation effect is maximum for five and six-membered rings
 - (b) Greater the charge on the central metal cation, greater the value of $\Delta(\text{CFSE})$
 - (c) In complex ion $[\text{CoF}_6]^{3-}$, F^- is a weak field ligand, so that $\Delta_{\text{oct}} < P$ (pairing energy) and it is low spin complex
 - (d) $[\text{CoCl}_2(\text{NH}_3)_2(\text{en})]^+$ complex ion will have four different isomers
17. Wilkinson's catalyst reacts with H_2 to form an octahedral complex in which $\text{Rh}(Z = 45)$ has the following electronic configuration in the ligand field $t_{2g}^{2,2,2}, e_g^{0,0}$. Then which of the following is (are) correct about this new complex?
 - (a) It is diamagnetic
 - (b) Its IUPAC name is chlorodihydrido tris (triphenyl phosphine) rhodium (III)
 - (c) Hybridization of Rh (I) is d^2sp^3
 - (d) It is a paramagnetic complex
18. Which of the following statement is true about the complex $[\text{CrCl}_3(\text{OH})_2(\text{NH}_3)]^{2-}$ ion?
 - (a) It has three geometrical isomers
 - (b) Only one space isomer is optically active and remain inactive
19. A d-block element forms octahedral complex but its magnetic moment remains same either in strong field or in weak field ligand. Which of the following is/are correct?
 - (a) Element always forms colourless compound
 - (b) Number of electrons in t_{2g} orbitals are higher than in e_g orbitals
 - (c) It can have either d^3 or d^8 configuration
 - (d) It can have either d^7 or d^8 configuration
20. Which of the following is (are) correct statement(s) (assuming oxidation number of metal does not affect crystal field energy)?
 - (a) Considering H_2O to be a weak ligand then on the basis of CFSE only, we can say, that $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ is more stable than $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$
 - (b) On the basis of CFSE only $[\text{Fe}(\text{NH}_3)_6]^{2+}$ is more stable than $[\text{Fe}(\text{NH}_3)_6]^{3+}$
 - (c) All octahedral complexes of Ni(II) are bound to be outer d-complex
 - (d) The type of d-orbital involved in the hybridization for a square planar complex ($\text{CN} = 4$) is dz^2

MULTIPLE CORRECT ANSWER TYPE QUESTIONS

16. Select the correct statement:
 - (a) Chelation effect is maximum for five and six-membered rings
 - (b) Greater the charge on the central metal cation, greater the value of $\Delta(\text{CFSE})$
 - (c) In complex ion $[\text{CoF}_6]^{3-}$, F^- is a weak field ligand, so that $\Delta_{\text{oct}} < P$ (pairing energy) and it is low spin complex
 - (d) $[\text{CoCl}_2(\text{NH}_3)_2(\text{en})]^+$ complex ion will have four different isomers
17. Wilkinson's catalyst reacts with H_2 to form an octahedral complex in which $\text{Rh}(Z = 45)$ has the following electronic configuration in the ligand field $t_{2g}^{2,2,2}, e_g^{0,0}$. Then which of the following is (are) correct about this new complex?
 - (a) It is diamagnetic
 - (b) Its IUPAC name is chlorodihydrido tris (triphenyl phosphine) rhodium (III)
 - (c) Hybridization of Rh (I) is d^2sp^3
 - (d) It is a paramagnetic complex
18. Which of the following statement is true about the complex $[\text{CrCl}_3(\text{OH})_2(\text{NH}_3)]^{2-}$ ion?
 - (a) It has three geometrical isomers
 - (b) Only one space isomer is optically active and remain inactive
19. A d-block element forms octahedral complex but its magnetic moment remains same either in strong field or in weak field ligand. Which of the following is/are correct?
 - (a) Element always forms colourless compound
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 - (c) It can have either d^3 or d^8 configuration
 - (d) It can have either d^7 or d^8 configuration
20. Which of the following is (are) correct statement(s) (assuming oxidation number of metal does not affect crystal field energy)?
 - (a) Considering H_2O to be a weak ligand then on the basis of CFSE only, we can say, that $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ is more stable than $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$
 - (b) On the basis of CFSE only $[\text{Fe}(\text{NH}_3)_6]^{2+}$ is more stable than $[\text{Fe}(\text{NH}_3)_6]^{3+}$
 - (c) All octahedral complexes of Ni(II) are bound to be outer d-complex
 - (d) The type of d-orbital involved in the hybridization for a square planar complex ($\text{CN} = 4$) is dz^2

21. Which of the following molecules or ions may act as bidentate ligand?
 (a) CO_3^{2-} (b) NO_3^-
 (c) $\text{CH}_3\text{C}=\text{N}$ (d) $\text{C}_2\text{O}_4^{2-}$
22. Which of the following complexes is(are) correctly matched with their geometry?
 (a) $[\text{CoCl}_4]^{2-}$ – tetrahedral
 (b) $[\text{Co}(\text{Py})_4]^{2+}$ – square planar
 (c) $[\text{CuCN}_4]^{3-}$ – tetrahedral
 (b) $[\text{Fe}(\text{Co})_4]^{2-}$ – square planar
23. Which of the following complexes is (are) paramagnetic?
 (a) $[\text{Mn}(\text{CN})_6]^{3-}$ (b) $[\text{Cr}(\text{NH}_3)_6]^{3+}$
 (c) $[\text{Fe}(\text{CN})_6]^{4-}$ (d) $[\text{Co}(\text{CN})_6]^{3-}$
24. Which of the following compounds/complex ions would not show geometrical isomerism?
 (a) $[\text{Co}(\text{en})(\text{SCN})_4]^-$ (b) $[\text{Pt}(\text{en})_2\text{ClBr}]^{2+}$
 (c) $[\text{Cr}(\text{gly})_3]$ (d) $[\text{Pt}(\text{gly})_2\text{Cl}_2]$
25. In which of the following pair(s), both the complex show optical isomerism?
 (a) cis – $[\text{Cr}(\text{C}_2\text{O}_4)_2\text{Cl}_2]^{3-}$, mer – $[\text{Co}(\text{gly})_3]$
 (b) $[\text{Co}(\text{en})_3]\text{Cl}_3$, cis – $[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$
 (c) $[\text{PtCl}(\text{dien})]\text{Cl}$, $[\text{NiCl}_2(\text{PPh}_3)_2]$
 (d) $[\text{Co}(\text{NO}_3)_3(\text{NH}_3)_3]$, $[\text{Pt}(\text{en})_2\text{Cl}_2]\text{Cl}_2$

ASSERTION/REASON TYPE QUESTIONS

26. **Statement-1:** In complex $[\text{Cr}(\text{NH}_3)_4\text{BrCl}]\text{Cl}$, the 'spin only' magnetic moment is close to 1.73 B.M.
Statement-2: All known mononuclear complexes of chromium(III), irrespective of the strength of the ligand field, must have three unpaired electrons.
 (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
 (c) Statement-1 is True, Statement-2 is False.
 (d) Statement-1 is False, Statement-2 is True.
27. **Statement-1:** The magnitude for a given ligand increases as the charge on the metal ion increases.
Statement-2: NH_3 is a weak field ligand towards Co^{2+} but acts as a strong field ligand towards Co^{3+} .
 (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement 1.
 (c) Statement-1 is True, Statement-2 is False.
 (d) Statement-1 is False, Statement-2 is True.
28. **Statement-1:** $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ is green but a solution of $[\text{Ni}(\text{CN})_4]^{2-}$ is colourless.
Statement-2: Energy difference between d levels (i.e., Δ_o) for H_2O complex (paramagnetic) is in the visible region and that for the cyano complex (diamagnetic) is in the UV region
 (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement 1.
 (c) Statement-1 is True, Statement-2 is False.
 (d) Statement-1 is False, Statement-2 is True.
29. **Statement-1:** Bis(dimethylglyoximate)nickel(II) can show geometrical isomerism.
Statement-2: Tetrahedral complexes with chiral structure exhibit optical isomerism.
 (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement 1.
 (c) Statement-1 is True, Statement-2 is False.
 (d) Statement-1 is False, Statement-2 is True.
30. **Statement-1:** $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ can exist in cis-trans-forms but neither can display optical activity.
Statement-2: Each of these possesses at least one plane of symmetry.
 (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement 1.
 (c) Statement-1 is True, Statement-2 is False.
 (d) Statement-1 is False, Statement-2 is True.

TRUE AND FALSE TYPE QUESTIONS

31. 1. The order of the wave length of light absorbed in visible region is
 $[\text{Co}(\text{H}_2\text{O})(\text{NH}_3)_5]^{3+} > [\text{Co}(\text{NH}_3)_6]^{3+} > [\text{Co}(\text{CN})_6]^{3-}$
 2. The order of the stability of the complexes in aqueous solution is
 $[\text{Co}(\text{CN})_6]^{3-} > [\text{Cu}(\text{NH}_3)_4]^{2+} > [\text{Cu}(\text{H}_2\text{O})_6]^{2+}$
 3. Living Williams's order of stability is
 $\text{Mn}^{2+} < \text{Fe}^{2+} < \text{Co}^{2+} < \text{Ni}^{2+} < \text{Cu}^{2+} < \text{Zn}^{2+}$
 4. The order of magnetic moments of the complex is
 $[\text{Fe}(\text{H}_2\text{O})_6]^{3+} > [\text{CoF}_6]^{3-} > [\text{Cr}(\text{CN})_6]^{3-}$
 (a) F T T F (b) T T T T
 (c) T F T F (d) T T T F
32. 1. The $[\text{Co}(\text{ox})_3]^{3-}$ complex is diamagnetic and gains stability through chelation.
 2. The $[\text{Co}(\text{NO}_2)_6]^{4-}$ is inner orbital complex involving d^2sp^3 hybridization and is paramagnetic.
 3. The complex $[\text{PtCl}_4]^{2-}$ is square planar and is diamagnetic.
 (a) F T T (b) T T T
 (c) T F T (d) T T T
33. 1. $[\text{Co}(\text{EDTA})]^-$ has two optical isomers
 2. $[\text{Co}(\text{NH}_3)_4(\text{NO}_2)_2]^+$
 3. For $[\text{Pt Py}(\text{NH}_3)(\text{NO}_2)\text{Cl BrI}]$, theoretically fifteen different geometrical isomers are possible.
 4. For $\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2 \cdot 2\text{H}_2\text{O}$ can show hydrate as well as ionisation isomerism.
 (a) F T T F (b) F F T F
 (c) T F T F (d) T T T F
34. 1. The IUPAC name of the compound $[\text{Cr}(\text{NH}_3)_5(\text{NCS})][\text{ZnCl}_4]$ is pentaamminethiocyanato-N-chromate III tetrachlorozincate (II).
 2. CoSO_4 is paramagnetic and coloured.
 3. $[\text{Co}(\text{NH}_3)_4(\text{NO}_2)_2]\text{NO}_3$ shows ionization, linkage and geometrical isomerism.
 4. The C—O bond length in the complex, $[\text{Fe}(\text{CO})_5]$ is same as that found in carbon monoxide itself.
 (a) F T T F (b) T T T T
 (c) F T T F (d) T T T F

COMPREHENSION BASED QUESTIONS

Comprehension-1

Formula of complex is determined by many experimental methods. One of which is measurement of depression in freezing point. Which gives the number of ions present in the complex. In one of the experiment.....

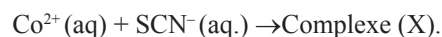
Compound $(\text{PdCl}_4 \cdot 6\text{H}_2\text{O})$ is a hydrated complex; 1 mole aqueous solution of it has freezing point of 269.28 K. Assuming 100% ionization of the complex, calculate the magnetic moment of the complex (given: K_f of water = 1.86 K kg mol⁻¹; and atomic number of Pd = 46).

35. The correct formula of complex is:
 (a) $[\text{Pd}(\text{H}_2\text{O})_3\text{Cl}_3]\text{Cl} \cdot 3\text{H}_2\text{O}$
 (b) $[\text{Pd}(\text{H}_2\text{O})_3\text{Cl}]\text{Cl}_3 \cdot 3\text{H}_2\text{O}$
 (c) $[\text{Pd}(\text{H}_2\text{O})_3\text{Cl}_2]\text{Cl}_4$
 (d) $[\text{Pd}(\text{H}_2\text{O})_3\text{Cl}_2]\text{Cl}_2 \cdot 3\text{H}_2\text{O}$
36. Magnetic moment of the complex can be:
 (a) 1.3 B.M. (b) Zero
 (c) 5.9 B.M. (d) 3.89 B.M.

37. The complex in the paragraph will have how many geometrical isomers?

- (a) 2
 (b) 3
 (c) 4
 (d) It will not exhibit geometrical isomerism.

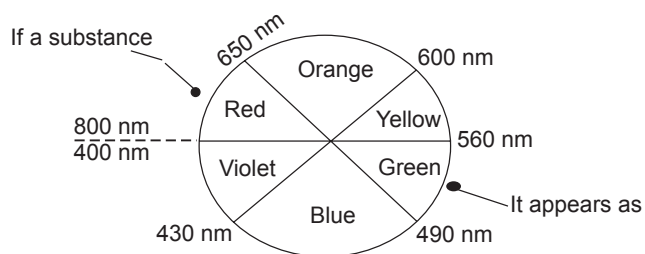
Comprehension-2



The coordination number of cobalt and nickel in complexes X and Y are four.

38. The IUPAC names of the complexes (X) and (Y) are respectively
 (a) Tetrathiocyanato-S-cobalt (II) and bis (dimethylglyoximate) nickel(II).

- (b) Tetrathiocyanato-S-cobaltate (II) and bis (dimethylglyoximato) nickel(II).
 (c) Tetrathiocyanato-S-cobaltate (II) and bis (dimethylglyoximato) nickelate(II).
 (d) Tetrathiocyanato-S-cobaltate (III) and bis (dimethylglyoximato) nickel(II).
39. The geometry of complexes (X) and (Y) are respectively:
 (a) tetrahedral and square planar
 (b) both tetrahedral
 (c) square planar and tetrahedral
 (d) both square planar
40. Select the correct statement the complexes (X) and (Y)
 (a) (X) is paramagnetic with two unpaired electrons.
 (b) (Y) is diamagnetic and shows intermolecular H-bonding.
 (c) (X) is paramagnetic with three unpaired electrons and (Y) is diamagnetic.
 (d) (X) and (Y) both are diamagnetic.



Different metal ions have different values of Δ , which explains why their complexes with the same ligand have different colour.

Similarly, the crystal field splitting also depends on the nature of ligands and as the ligand for the same metal varies from H_2O to NH_3 to ethylenediamine, Δ for complexes increases. Accordingly, the electronic transition shifts to higher energy (shorter wavelength) as the ligand varies from H_2O to NH_3 to en, thus accounting for the variation in colour.

Crystal field theory accounts for the magnetic properties of complexes in terms of the relative values of Δ and the spin pairing energy P . Small Δ values favour high spin complexes, and large Δ values favour low spin complexes.

41. The $[\text{Ti}(\text{NCS})_6]^{3-}$ ion exhibits a single absorption band at 544 nm. What will be the crystal field splitting energy (KJ mol^{-1}) of the complex? ($h = 6.626 \times 10^{-34}$ J.s; $c = 3.0 \times 10^8$ m/s; $N_A = 6.02 \times 10^{23}$ ions/mole.

- (a) 240 (b) 220
 (c) 270 (d) 250

42. Which of the following statements is incorrect?
 (a) The Ni^{2+} (aq) cation is coloured because Ni^{2+} ion can absorb light, which promotes electrons from the filled d-orbitals to the higher energy half filled d-orbitals.
 (b) The Zn^{2+} (aq) cation is coloured because the d-orbitals are completely filled and no electrons can be promoted, so no light is absorbed.
 (c) A complex which has just one absorption band at 455 nm, must be red coloured.
 (d) None of these
43. Which of the following complexes are diamagnetic?
 (i) $[\text{Pt}(\text{NH}_3)_4]^{2+}$ square planar
 (ii) $[\text{Co}(\text{SCN})_4]^{2-}$ tetrahedral
 (iii) $[\text{Cu}(\text{en})_2]^{2+}$ square planar
 (iv) $[\text{HgI}_4]^{2-}$ tetrahedral
 (a) (i) and (ii) (b) (ii) and (iii)
 (c) (i) and (iv) (d) (iii) and (iv)

Comprehension-3

In metal caronyls, there is synergic bonding interaction between metal and carbon monoxide. This leads to increase in strength of metal ligand bond and decrease in bond order of CO in carbonyl complex as compared to bond order in carbon monoxide.

Simple carbonyls are invariably spin-paired complexes except for vanadium metal.

44. The increase in bond length in CO as compared to carbon monoxide is due to
 (a) the donation of lone pair of electrons on the carbon into a vacant orbital of the metal atom
 (b) the donation of a pair of electrons from a filled d-orbital of metal into the vacant antibonding orbital of carbon monoxide.
 (c) both (a) and (b) both
 (d) None of these
45. Which amongst the following metal carbonyls are inner orbital complexes with diamagnetic property?
 (a) $\text{Ni}(\text{CO})_4$ (b) $\text{Fe}(\text{CO})_5$
 (c) $\text{V}(\text{CO})_6$ (d) $\text{Cr}(\text{CO})_6$
46. Which one of the following metal carbonyls involves the d^2sp^3 hybridization for the formation of metal-carbon?? bonds and is paramagnetic.
 (a) $\text{Cr}(\text{CO})_6$ (b) $[\text{V}(\text{CO})_6]$
 (c) $[\text{Mo}(\text{CO})_6]$ (d) $[\text{W}(\text{CO})_6]$

MATCH THE COLUMN TYPE QUESTIONS

47. Match the following

Column-I

- (A) $[\text{Fe}(\text{CN})_6]^{4-}$
 (B) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
 (C) $[\text{Cu}(\text{NH}_3)_6]^{2+}$
 (D) $[\text{Ni}(\text{CN})_6]^{4-}$

Column-II

- (P) Paramagnetic
 (Q) Diamagnetic
 (R) Inner orbital complex
 (S) Outer orbital complex

48. Match the following

Column-I

- (A) $[\text{Ni}(\text{CO})_4]$
 (B) $[\text{Fe}(\text{CO})_2(\text{NO})_2]$
 (C) $[\text{Ni}(\text{PF}_3)_4]$
 (D) $[\text{Ni}(\text{PPh}_3)_2\text{Br}_2]$

Column-II

- (P) Tetrahedral
 (Q) π back bonding
 (R) diamagnetic
 (S) One of the ligand is three electron donor

49. Match the following

Column-I

- (A) $[\text{Co}(\text{en})_3]^{3+}$
 (B) $[\text{Co}(\text{OX}_3)]^{3-}$
 (C) $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$
 (D) $[\text{Co}(\text{NO}_2)_6]^{4-}$

Column-II

- (P) sp^3d^2 hybridization
 (Q) Diamagnetic
 (R) d^2sp^3 hybridization
 (S) Paramagnetic

50. Match the complexes/complex ions given in column-I with the characteristic(s) given in column-II.

Column-I

- (A) $[\text{Co}(\text{H}_2\text{O})\text{BrCl}]$
 (B) $[\text{Co}(\text{gly})_3]$
 (C) $[\text{Co}(\text{NH}_3)_2\text{Cl}_2(\text{en})]\text{NO}_3$
 (D) $[\text{K}_3(\text{CoCl}_2)_2(\text{C}_2\text{O}_4)_2]$

Column-II

- (P) Show optical isomerism
 (Q) Show geometrical isomerism
 (R) Metal is in +3 oxidation state.
 (S) Non-electrolyte
 (T) Homoleptic complex

51. Match the complexes listed in column-I with type of hybridization listed in column-II.

Column-I

- (A) $[\text{AuF}_4]^-$
 (B) $[\text{Cu}(\text{CN})_4]^{3-}$
 (C) $[\text{Co}(\text{NH}_3)_6]^{2+}$
 (D) $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]^{2+}$

Column-II

- (P) dsp^2 hybridization
 (Q) sp^3 hybridization
 (R) sp^3d^2 hybridization
 (S) d^2sp^3 hybridization

SUBJECTIVE TYPE QUESTIONS

Short Subjective Questions

52. AlF_3 is soluble in KF forming a soluble complex. Calculate the EAN for the complex.
53. How many isomers of the complex, $\text{Pt}(\text{NH}_3)_2(\text{SCN})_2$ are possible?

Long Subjective Questions

54. Calculate the difference in net stabilization energies (in kJ mol^{-1}) of Fe^{2+} complexes with CN^- ligands ($\Delta = 25000 \text{ cm}^{-1}$) and with H_2O ligands ($\Delta_0 = 10000 \text{ cm}^{-1}$).

Given that the pairing energy, $P = 15000 \text{ cm}^{-1}$ for Fe^{2+} ion for both the complexes.

(Given that $h = 6 \times 10^{-34} \text{ J sec.}$, Avogadro Number = 6×10^{23} , Speed of light in vacuum = $3 \times 10^8 \text{ m sec}^{-1}$).

55. A metal complex having composition $\text{Cr}(\text{NH}_3)_4\text{Cl}_2\text{Br}$ has been isolated in two forms (A) and (B). The form A reacts with AgNO_3 to give a white precipitate readily soluble in dilute aqueous ammonia, whereas (B) gives a pale yellow precipitate partially soluble in concentrated ammonia. Calculate their magnetic moments (spin – only value).

Metallurgy

STRAIGHT OBJECTIVE TYPE QUESTIONS

- A mineral is called an ore if
 - Metal present in mineral is precious
 - Metal can be extracted from it
 - Metal can be extracted profitably from it
 - Metal cannot be extracted from it
- Which of the following statement is incorrect?
 - Silver glance mainly contains silver sulphide
 - Gold is found in native state
 - Zinc blende mainly contain zinc chloride
 - Copper pyrites also contain Fe_2S_3
- For which ore of the metal, froth floatation method is used for concentration?
 - Horn silver
 - Bauxite
 - Cinnabar
 - Haematite
- Froth-floatation method is successful in separating impurities from ores because
 - The pure ore is lighter than water containing additives like pine oil, cresylic acid etc.
 - The pure ore is soluble in water containing additives like pine oil, cresylic acid etc.
 - The impurities are soluble in water containing additives like pine oil, cresylic acid etc.
 - The pure ore is not as easily wetted by water as by pine oil, cresylic acid etc
- Which of the following processes involves smelting?
 - $\text{ZnCO}_3 \rightarrow \text{ZnO} + \text{CO}_2$
 - $\text{Fe}_2\text{O}_3 + 3\text{C} \rightarrow 2\text{Fe} + 3\text{CO}$
 - $2\text{PbS} + 3\text{O}_2 \rightarrow 2\text{PbO} + 2\text{SO}_2$
 - $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O} \rightarrow \text{Al}_2\text{O}_3 + 2\text{H}_2\text{O}$
- Flux added in the extraction of iron is
 - Silica
 - Felspar
 - Limestone
 - Flint
- Match column-I with column-II and select the correct answer using the codes given below.

Column-I	Column-II
(a) Argentite	(1) Halide ore
(b) Cuprite	(2) Carbonate ore
(c) Siderite	(3) Oxide ore
(d) Carnallite	(4) Sulphide ore

Codes:

A B C D	A B C D
(a) 4 3 2 1	(b) 1 2 3 4
(c) 2 3 4 1	(d) 3 4 1 2
- NaCN is sometimes added in the froth floatation process as a depressant when mineral contains ZnS and PbS because
 - $\text{Pb}(\text{CN})_2$ is precipitated while there is no effect of ZnS .
 - ZnS forms soluble complex $\text{Na}_2[\text{Zn}(\text{CN})_4]$ while PbS forms froth.
 - PbS forms soluble complex $\text{Na}_2[\text{Pb}(\text{CN})_4]$ while ZnS forms froth.
 - silicious impurities settle down on the bottom.
- Which does not represent correct method?
 - $\text{TiCl}_2 + 2\text{Mg} \rightarrow \text{Ti} + 2\text{MgCl}_2$: Kroll
 - $\text{Ni}(\text{CO})_4 \rightarrow \text{Ni} + 4\text{CO}$: Mond
 - $\text{Ag}_2\text{CO}_3 \rightarrow 2\text{Ag} + \text{CO}_2 + \frac{1}{2} \text{O}_2$: Van Arkel
 - $\text{ZrI}_4 \rightarrow \text{Zr} + 2\text{I}_2$: Van Arkel
- Main source of lead is galena (PbS). It is converted to Pb by

(A) PbS	$\xrightarrow[\Delta]{\text{air}}$	$\text{PbO} + \text{SO}_2$ $\text{Pb} + \text{CO}_2$
(B) PbS	$\xrightarrow[\Delta]{\text{air}}$	$\text{PbO} + \text{PbS}$ $\text{Pb} + \text{SO}_2$

 - A
 - B
 - Both (a) and (b)
 - None of these
- The chemical composition of "slag" formed during the smelting process in the extraction of copper is
 - $\text{Cu}_2 + \text{FeS}$
 - FeSiO
 - CuFeS
 - $\text{Cu}_2\text{S} + \text{FeO}$
- Which of the following statement is incorrect about the extractive metallurgy of copper?
 - Matte chiefly consists of cuprous sulphide and some ferrous sulphide.

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- (b) Most of the impurity of iron sulphide is removed as fusible slag during roasting.
 (c) The copper pyrites is concentrated by froth floatation process.
 (d) self reduction and electrolysis respectively.
- 13.** Which of the following statements is incorrect?
 (a) Cassiterite ore of tin contains the impurities of Wolframite which are separated by electromagnetic separator.
 (b) Tin metal is obtained by the carbon reduction of black tin.
- 14.** White bauxite is purified by
 (a) Baeyer's process
 (b) Hoop's process

- (c) Serpeck's process
 (d) Hall's process

- 15.** Match column-I with column-II and select the correct answer using codes given below in the lists.

Column-I

Column-II

- | | |
|--|----------------------|
| (i) Cyanide process | (a) Extraction of Al |
| (ii) Self reduction | (b) Extraction of Ag |
| (iii) Electrolytic reduction | (c) Extraction of Cu |
| (iv) Carbon reduction | (d) Extraction of Cu |
| (a) (i) – (b), (ii) – (c), (iii) – (a), (iv) – (d) | |
| (b) (i) – (b), (ii) – (d), (iii) – (a), (iv) – (c) | |
| (c) (i) – (d), (ii) – (a), (iii) – (c), (iv) – (b) | |
| (d) (i) – (c), (ii) – (b), (iii) – (d), (iv) – (a) | |

MULTIPLE CORRECT ANSWER TYPE QUESTIONS

- 16.** Addition of manganese in high concentration to steel makes it useful for making railways because
 (a) manganese removes oxygen and sulphur
 (b) manganese provides hardness to steel
 (c) manganese forms soft alloy with iron
 (d) manganese makes the iron coloured
- 17.** The correct statement(s) regarding aluminothermite process is/are
 (a) it uses aluminium as a reducing agent
 (b) it uses BaO_2 as a source of oxygen
 (c) thermite process is particularly used for Cr, Mn and Fe
 (d) thermite mixture gives highly exothermic reaction
- 18.** Metallurgical process of zinc involves roasting of zinc sulphide followed by reduction. Metallic zinc distils over as it is volatile and impurities like Cd, Pd and Fe gets condensed. The crude metal obtained is called spelter, which may be purified by
 (a) electrolysis process
 (b) fractional distillation
 (c) polling
 (d) heating with iodine
- 19.** Calcination and roasting processes of reduction of ores to their oxides are beneficial
 (a) to convert ores into porous form so that their reduction becomes easier.

- (b) as volatile impurities like P, As, Sb, S are removed.
 (c) as organic impurities are removed.
 (d) as the ores are converted into oxide form which makes the reduction easier.

- 20.** Which of the following common element(s) is/are present in the anode mud in electrolytic refining of copper?

- (a) Selenium (b) Tellurium
 (c) Silver (d) Gold

- 21.** Which of the following is(are) correctly matched?

- (a) Copper – Bessemer converter.
 (b) Iron – Blast furnace
 (c) Chromium – Aluminothermic process
 (d) Tin – Electrolytic reduction

- 22.** The reaction(s) which does (do) not occur in the reduction zone in the extraction of iron from haematite ore is(are):

- (a) $\text{Fe}_2\text{O}_3 + \text{CO} \rightarrow 2\text{FeO} + \text{CO}_2$
 (b) $\text{FeO} + \text{CO} \rightarrow \text{Fe} + \text{CO}_2$
 (c) $\text{Fe}_2\text{O}_3 + 3\text{C} \rightarrow 2\text{Fe} + 3\text{CO}$
 (d) $\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$

- 23.** Which of the following statement(s) is (are) true?

- (a) In the process of the precipitation of silver from sodium dicyanoargentate (I), the zinc acts as reducing agent.

- (b) In the process of roasting, the copper pyrites is converted into a mixture of Cu_2S and FeS which, in turn, are partially oxides.
- (c) Limonite, haematite and magnesite are ores of iron.
- (d) Tin and lead both are extracted from their ores by self-reduction.
24. Why lime stone is added in the extraction of lead from galena?
- (a) It stops the formation of PbSO_4
- (b) It remove the impurity of silica as fusible slag.
- (c) It converts lead silicate to lead oxide.
- (d) It remove the impurity of iron oxide as fusible slag.
25. The role of fluorspar (CaF_2) which is added in the electrolytic reduction of alumina dissolved in fused cryolite is(are):
- (a) To acts as a catalyst.
- (b) To make the fused mixture very conducting.
- (c) To increase the temperature of the melt.
- (d) To decrease the rate of oxidation of carbon at anode.

ASSERTION/REASON TYPE QUESTIONS

26. **Assertion:** Gold is recovered from its solution containing aurocyanide complex by adding zinc dust.
Reason: Zinc is more electropositive than gold.
- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
- (c) Statement-1 is True, Statement-2 is False.
- (d) Statement-1 is False, Statement-2 is True.
27. **Statement-1:** Extraction of gold from its native ore involves leaching the metal with dilute solution of NaCN in presence of air.
Statement-2: This is an oxidation reaction and leads to the formation of a soluble complex.
- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
- (c) Statement-1 is True, Statement-2 is False.
- (d) Statement-1 is False, Statement-2 is True.
28. **Statement-1:** Silica is added as a flux in reverberatory furnace, in the extraction of copper from copper pyrites.
Statement-2: Silica decreases the melting point of the ore and removes the impurity of lead sulphide as PbSiO_3 .
- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
- (c) Statement-1 is True, Statement-2 is False.
- (d) Statement-1 is False, Statement-2 is True.
29. **Statement-1:** Cast iron is different from pig iron.
Statement-2: Cast iron is made by melting pig iron with scrap iron and coke using hot air blast and has about 3% carbon content.
- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
- (c) Statement-1 is True, Statement-2 is False.
- (d) Statement-1 is False, Statement-2 is True.
30. **Statement-1:** Sodium chloride is added during electrolysis of fused anhydrous magnesium chloride.
Statement-2: Anhydrous magnesium chloride is obtained by heating hydrated magnesium chloride, $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$.
- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
- (c) Statement-1 is True, Statement-2 is False.
- (d) Statement-1 is False, Statement-2 is True.

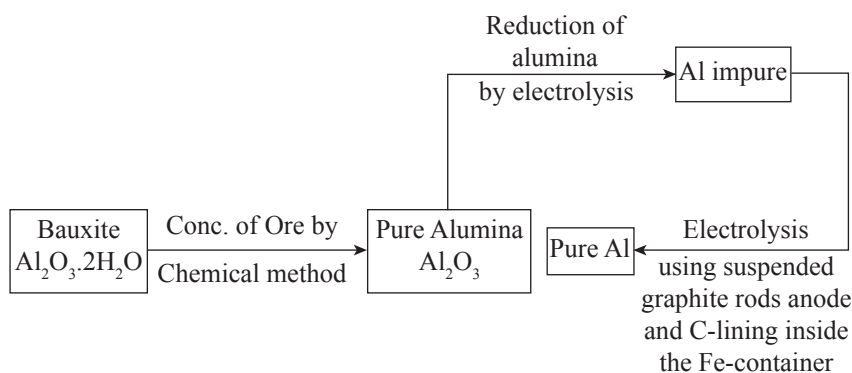
TRUE AND FALSE TYPE QUESTIONS

31. The slag is lighter and has higher M.P. than the metal.
32. Carbon cannot be used in the reduction of Al_2O_3
33. 1. In extraction of iron from haematite ore, the reduction reactions, take place in the lower temperature range and in the higher temperature range, in the blast furnace.
 2. Sphalerite is a carbonate ore of zinc.
 3. The principal ore of aluminium, bauxite, usually contains silica, iron oxides and titanium oxide as impurities.
 4. Solidified copper obtained from silica lined converter (Bessemer converter) has blistered appearance due to the evolution of SO_2 .
34. 1. In the aluminothermite process, aluminium acts as reducing agent.
 2. The process of extraction of gold involves the formation of $[\text{Au}(\text{CN})_4]^-$.
 3. In the extractive metallurgy of zinc, partial fusion of ZnO with coke is called sintering and reduction of ore to the molten metal is called smelting.
 4. Extractive metallurgy of silver from its ore argentite involves complex formation and displacement by more electropositive metal.
- (a) T F T T (b) F T F F
 (c) F F T T (d) T F F T
- (a) T F F T (b) T T T T
 (c) T T F T (d) T F T F

COMPREHENSION BASED QUESTIONS

Comprehension-1

Extraction of aluminium can be understand by:



Electrolytic reduction of Al_2O_3

Electrolyte: ($\text{Al}_2\text{O}_3 + \text{Cryolite}$)

Cathode: Graphite inside the Fe container

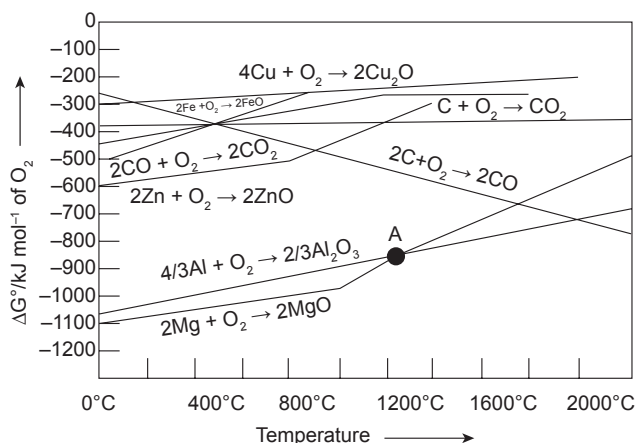
Anode: Graphite rods

35. The purpose of adding cryolite is:
- (a) to increase the electrical conductivity of pure aluminium
 (b) to lower the melting point of Al_2O_3
 (c) to remove the impurities as slag
 (d) to increase the Al% in the yield
36. Coke power is spreaded over the molten electrolyte due to:
- (a) prevent the heat radiation from the surface
 (b) prevent the corrosion of graphite anode
 (c) prevent oxidation of molten aluminium by air
 (d) both (b) and (c)

37. The function of fluorspar (CaF_2) is:
- to decrease the melting point of electrolyte
 - to increase electrolytic conductivity power
 - to remove the impurities as slag
 - all of the above

Comprehension-2

Go through the following graph and answer the following questions.



38. At what approximate temperature, zinc and carbon have equal affinity for oxygen?
- 1000°C
 - 1500°C
 - 500°C
 - 1200°C
39. To make the following reduction process spontaneous, temperature should be
- $$\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$$
- 1000°C
 - > 1000°C
 - < 500°C
 - > 500°C but < 1000°C
40. Which of the following statement is true?
- Reduction of calcined/roasted haematite ore to pig iron in blast furnace takes place in the lower temperature range and in the higher temperature range by CO and C respectively.
 - The reduction of zinc oxide using coke takes place at higher temperature than that in case of copper.

- It is quite easy to reduce oxide ores of copper directly to the metal by heating with coke after 500 – 600K.
- All of these

Comprehension 3

Amongst the various ores of a metal (M) (sulphide, carbonates, oxides, hydrated or hydroxides) two ores [X] and [Y] show the following reactivity.

- [X] On calcinations gives a black solid (S), carbon dioxide and water.
- [X] Dissolved in dil. HCl on reaction with KI gives a white precipitate (P) and iodine.
- [Y] On roasting gives metal (M) and a gas (G₁) which turns acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution green.
- [Y] On reaction with dil. HCl gives a white precipitate (Q) and another gas (G₂) which turns lead acetate solution black and also reacts with gas (G₁) to precipitate colloidal sulphur in presence of moisture. The M, S [X] and [Y] gives greenish blue flame.

41. The metal ores [X] and [Y] are respectively;
- Carbonate and sulphide ores
 - Sulphide and carbonate ores
 - Carbonate and hydroxide ores
 - Carbonate and oxide ores
42. Which of the following statements is correct about [Y]?
- [Y] is converted to metal (M) by self reduction.
 - Carbonate extract of [Y] gives yellow precipitate with suspension of CdCO_3 .
 - [Y] is chalcocites or chalcopyrites
 - All of the above
43. The gas (G₁) acts as
- Oxidizing agent
 - Reducing agent
 - Oxidizing and reducing agent
 - Fluxing agent
44. The white precipitate (P) is of:
- Cu_2I_2
 - Cu_2I_2
 - $\text{K}_2[\text{CuI}_4]$
 - None

MATCH THE COLUMN TYPE QUESTIONS

45. Match the following:

Column-I (Purification method)	Column-II (Metal preferred)
(A) Poling	(P) Titanium
(B) Cupellation	(Q) Cooper
(C) Electro-refining	(R) Electrolytic reduction
(D) Van Arkel method	(S) Silver
	(T) Germanium

46. Match the following:

Column-I	Column-II
(A) Argentite	(P) Halide ore
(B) Cuprite	(Q) Carbonate ore
(C) Siderite	(R) Oxide ore
(D) Carnallite	(S) Sulphide ore

47. Match the following:

Column-I	Column-II
(A) Acidic refractory material	(P) Quartz
(B) Acidic flux	(Q) Silica
(C) Basic refractory material	(R) Dolomite
(D) Neutral refractory material	(S) Bone ash

48. Match the ores listed in column-I with the appropriate process(es) listed in column-II.

Column-I	Column-II
(A) Haematite	(P) Slag formation during both roasting/smelting and bessemerisation.
(B) Copper pyrites	(Q) Reduction by carbon monoxide (mainly)
(C) Carnallite	(R) Electrolytic reduction
(D) Bauxite	(S) Calcination
	(T) Leaching

49. Match the reactions given in column-I with the appropriate method(s) listed in column-II.

Column-I	Column-II
(A) $4\text{Au} + 8\text{NaCN} + 2\text{H}_2\text{O} + \text{O}_2 (\text{air}) \rightarrow 4\text{Na}[\text{Au}(\text{CN})_2] + 4\text{NaOH}$	(P) Leaching
(B) $\text{CuFeS}_2 + 2\text{H}_2\text{SO}_4 \rightarrow \text{CuSO}_4 + \text{FeSO}_4 + 2\text{H}_2\text{S}$	(Q) Smelting
(C) $\text{Fe}_3\text{O}_4 + 4\text{CO} \xrightarrow{823\text{ K}} 3\text{Fe} + 4\text{CO}_2$	(R) Roasting
(D) $\text{MgCl}_2 \cdot 6\text{H}_2\text{O} \xrightarrow{\Delta} \text{MgCl}_2 + 6\text{H}_2\text{O}$	(S) Calcination

SUBJECTIVE TYPE QUESTIONS

Short Subjective Questions

50. What is the coordination number of aluminium in mineral cryolite?
51. The number of reducing agents involved in the extraction of iron (as pig iron) using blast furnace from ore haematite is(are).

52. Among the following metals, how many metals are extracted by self-reduction method from their respective ores? (Give total number). Hg, Zn, Cu, Al, Mg, Pb, Fe, Sn.

s-Block Elements and Compounds

STRAIGHT OBJECTIVE TYPE QUESTIONS

- During the electrolysis of fused sodium chloride, the anodic reaction is
 - Reduction of sodium ions
 - Oxidation of sodium ions
 - Reduction of chloride ions
 - Oxidation of chloride ions
- In the manufacture of metallic sodium by the fused salt electrolysis (Down's process), a small amount of calcium chloride is added to
 - Improve the electrical conduction
 - Increase the temperature of electrolysis
 - Bring down the melt temperature
 - Stabilize the metallic sodium
- Of the metals Be, Mg, Ca and Sr of group II A in the periodic table, the least ionic chloride would be formed by
 - Be
 - Mg
 - Ca
 - Sr
- Which of the following decreases on going gradually from Be to Ba (in periodic table)?
 - Basic character of hydroxides
 - Solubility of sulphates in water
 - Solubility of hydroxides in water
 - Strength of elements as reducing agent
- Which of the following alkaline-earth metal hydroxides is the strongest base.
 - Be(OH)₂
 - Mg(OH)₂
 - Ca(OH)₂
 - Ba(OH)₂
- Be(OH)₂ is insoluble in water while Ba(OH)₂ is highly soluble due to
 - Bond order
 - Lattice energy difference
 - Common ion effect
 - Hard acid
- Which of the following statements is true for all the alkali metals?
 - Their nitrate decomposes on heating to give the corresponding nitrites and oxygen.
 - Their chlorides are deliquescent and crystallize as hydrates.
 - They react with water to form hydroxide and hydrogen.
 - They really react with halogens to form ionic halides, M⁺X⁻.
- Sodium and potassium react with water much more vigorously than lithium because:
 - Sodium and potassium have high values of hydration energy as compared to that of lithium.
 - Sodium and potassium have higher melting point than that of lithium.
 - Sodium and potassium have lower melting point than that of lithium.
 - Sodium and potassium have lower hydration energy than that of lithium.
- Which of the following statements is not true about the dilute solutions of alkali metals in liquid ammonia?
 - They are deep blue coloured solutions.
 - They are highly conducting in nature.
 - They are diamagnetic in nature.
 - Ammoniated cations and ammoniated anions are formed in solution.
- Which of these is/are not correctly matched?
 - Basic strength of oxides Cs₂O < Rb₂O < K₂O < Na₂O < Li₂O
 - Stability of peroxides Na₂O < K₂O₂ < Rb₂O₂ < Cs₂O₂
 - Stability of bicarbonate LiHCO₃ < NaHCO₃ < KHCO₃ < RbHCO₃ < CsHCO₃
 - Melting Point NaF < NaCl < NaBr < NaI
 - 1 and 4
 - 1 and 3
 - 1 and 2
 - 2 and 3
- In manufacture of sodium carbonate from Solvay (or ammonia soda) process, the raw material used is

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- (a) NaOH (b) Na_2SO_4
(c) NaCl (d) NaHCO_3

12. Which of the following acts as an oxidizing as well as reducing agent?
(a) Na_2O (b) NaO_3
(c) NaNO (d) NaNO_2

13. Alkaline earth metal nitrates on heating decompose to give:
(a) $\text{M}(\text{NO}_2)_2$ and O_2 only
(b) MO, N_2 and O_2
(c) MO, NO_2 and O_2
(d) MO and NO_2 only

14. Consider the following statements and pick out the wrong one.
(1) The solubility, thermal stability and the basic character of the hydroxides of alkaline earth metals increases from $\text{Mg}(\text{OH})_2$ to $\text{Ba}(\text{OH})_2$.

(2) The dehydration of hydrated chlorides, bromides, iodides of Ca, Sr and Ba can be achieved on heating.

(3) The chlorides of both beryllium and aluminium are soluble in organic solvents and are strong Lewis acids.

(a) (1) and (2) only
(b) (1) and (3) only
(c) (1), (2) and (3)
(d) None of these

15. A metal [X] on heating on nitrogen gas gives [Y]. [Y] on treatment with H_2O gives a colourless gas which when passed through CuSO_4 solution gives a blue colour. [Y] is
(a) $\text{Mg}(\text{NO}_3)_2$
(b) Mg_3N_2
(c) NaN_3
(d) MgO

MULTIPLE CORRECT ANSWER TYPE QUESTIONS

16. Which of the following statements is correct for alkali metals?
 - (a) They can form covalent molecules such as Li_2 , Na_2 and K_2 . About 1% of these are in the vapour state.
 - (b) $\text{C}_6\text{H}_5\text{Li}$ is highly reactive.
 - (c) They form compounds with the liberation of heat.
 - (d) They all adopt ccp structures.
17. Alkali metals are characterized by
 - (a) Good conductor of heat and electricity
 - (b) High oxidation potentials
 - (c) High melting points
 - (d) Solubility in liquid ammonia
18. Which is/are true statements?
 - (a) The heats of hydration of the diapositive alkaline earth metal ions decrease with an increase in their ionic size.
 - (b) Hydration of alkali metal ion is less than that of II A.
 - (c) Alkaline earth metal ions, because of their much larger charge to size ratio, exert a much stronger electrostatic attraction on the oxygen of water molecule surrounding them.
 - (d) None of these
19. Which of the following statements are true for alkali metals?
 - (a) Lithium is the strongest reducing agent.
 - (b) Na is amphoteric in nature.
 - (c) Li^+ is exceptionally small.
 - (d) All alkali metals give blue solution in liquid ammonia.
20. Which of the following statement(s) is/are true for the solutions of alkali metals and alkaline earth metals in ammonia (*l*)?
 - (a) Concentrated solutions of alkali metals in ammonia are copper-bronzed coloured and have a metallic lusture.
 - (b) Dilute solutions of alkaline earth metals are bright blue in colour due to the spectrum from the sol-vated electron.
 - (c) Concentrated solutions of the alkaline earth metals in ammonia are bronze coloured.
 - (d) Evaporation of the ammonia from solutions of alkali metals yields the metal, but with alkaline earth metals, evaporation of ammonia gives hex-ammoniates of the metals.
21.
$$\text{M(alkali metal)} + (x + y) \text{NH}_3 \rightarrow [\text{M}(\text{NH}_3)_x]^+ + [\text{e}(\text{NH}_3)_y]^-$$

Solution	ammoniated metal cation
	+ $[\text{e}(\text{NH}_3)_y]^-$
	ammoniated electron

Mark the correct statement/s about the solution.

- (a) Solution has high electrical conductivity due to ammoniated electron.
 - (b) Solution imparts blue colour due to greater polarization of metal ion.
 - (c) Solution is quite stable, which is considered as a dilute metal.
 - (d) On the addition of substance like iron oxide, solution decomposes and releases H_2 gas.
22. Which of the following statements are true about the alkali metals?
- (a) All alkali-metal salts impart a characteristic colour to the Bunsen flame.
 - (b) Among $LiOH$, $CsOH$, KOH and $RbOH$, $CsOH$ has the highest solubility in water.
 - (c) Among the alkali metals, cesium is the most reactive.
 - (d) The reducing character of the alkali metal hydrides follow the order: $LiH > NaH > KH > RbH > CsH$.
23. Which of the following statement(s) is(are) false?
- (a) The halides of beryllium are electron deficient and are polymeric with halogen bridges.
 - (b) BeF_2 is very soluble in water owing to high salivation energy of Be but fluorides of other alkaline earth metals are almost insoluble.
 - (c) CaO , SrO and BaO react exothermically with water forming soluble hydroxides.
 - (d) The bicarbonates of alkaline earth metals are only stable in solutions.
24. Which of the following statement(s) is(are) true?
- (a) Both $LiCl$ and $MgCl_2$ are deliquescent.
 - (b) Lithium is least reactive but the strongest reducing agent amongst all the alkali metals.
 - (c) Liquid sodium metal is used as a coolant in fast breeder nuclear reactors.
 - (d) Lithium iodide is the most ionic in nature among alkali metal halides.
25. Select the correct statement with respect to sodium peroxide.
- (a) It decolourizes the acidified $KMnO_4$ solution.
 - (b) On heating with oxygen at $450^\circ C$ and 300 atm pressure, it becomes paramagnetic.
 - (c) It is obtained along with sodium metal, when sodium oxide is heated to a temperature more than $400^\circ C$.
 - (d) It gives both hydrogen peroxide and oxygen gas with water as well with sulphuric acid.

ASSERTION/REASON TYPE QUESTIONS

26. **Statement-1:** Na_2SO_4 is soluble while $BaSO_4$ is insoluble.
- Statement-2:** Lattice energy of exceeds its hydration energy.
- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 - (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
 - (c) Statement-1 is True, Statement-2 is False.
 - (d) Statement-1 is False, Statement-2 is True.
27. **Statement-1:** The lithium ion itself, and also its compounds, are more heavily hydrated than those of the rest of the group members.
- Statement-2:** The Li^+ is smallest among alkali metals and polarizes water molecules more easily than the other alkali metal ions.
- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 - (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
 - (c) Statement-1 is True, Statement-2 is False.
 - (d) Statement-1 is False, Statement-2 is True.
28. **Statement-1:** Beryllium resembles aluminium.
- Statement-2:** Charge/radius of Be^{2+} ion is nearly the same as that of the Al^{3+} ion.
- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 - (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
 - (c) Statement-1 is True, Statement-2 is False.
 - (d) Statement-1 is False, Statement-2 is True.

QB.30

29. Statement-1: As the electropositive character increases down the group, the stability of carbonates and hydrogen carbonates of alkali metals increases.

Statement-2: Lithium carbonates is not so stable to heat; lithium being very small in size polarizes a large CO_3^{2-} ion leading to the formation of more stable Li_2O and CO_2 .

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
- (c) Statement-1 is True, Statement-2 is False.
- (d) Statement-1 is False, Statement-2 is True.

30. Statement-1: Potassium and caesium are useful as electrodes in photoelectric cells.

Statement-2: Potassium and caesium, when irradiated with light, the light energy absorbed is sufficient to eject out the electron from an atom.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
- (c) Statement-1 is True, Statement-2 is False.
- (d) Statement-1 is False, Statement-2 is True.

TRUE AND FALSE TYPE QUESTIONS

- 31. 1.** Concentrated solution of an alkali metal in liquid ammonia is bronze coloured and diamagnetic.
 - 2.** Increasing stability of the peroxide and superoxide of alkaline earth metal down the group, is due to the stabilization of large anions by large cations through lattice energy effects.
 - 3.** In the vapour phase, BeCl_2 tends to form a chloro-bridged dimer.
 - 4.** Beryllium hydroxide reacts with acid and alkali both to form $[\text{Be}(\text{OH})_4]\text{Cl}_2$ and $[\text{Be}(\text{OH})_4]_2$ respectively.
- (a) T T T T
 - (b) F F F F
 - (c) T F T F
 - (d) F T F T
- 32. 1.** Sodium amalgam reacts with hot water and produces sodium hydroxide and hydrogen gas.
 - 2.** Sodium is more reactive than potassium.

- 3.** Alkaline earth metals dissolve in liquid ammonia to give deep blue black solutions which are conducting.
 - 4.** Alkaline earth metal salts have more number of water molecules as compared to those of alkali metal salts.
- (a) T T F F
 - (b) T F T T
 - (c) F T T F
 - (d) T F F T
- 33. 1.** $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ on heating gives Mg_2OCl_2 ($\text{MgO} \cdot \text{MgCl}_2$) because of hydrolysis.
 - 2.** The softness of group 1-A metals increases down the group with increasing atomic number.
 - 3.** Plaster of Paris is $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$.
 - 4.** Sodium carbonate is made by solvay process but the same process is not extended to the manufacture of potassium carbonate.
- (a) T T T F
 - (b) T F F T
 - (c) T T F T
 - (d) T T T T

COMPREHENSION BASED QUESTIONS

Comprehension-1

The solubility of most of the salts of group 1 elements in water decreases on descending the group. For a substance to dissolve, the energy evolved when the ions are hydrated is more than the energy required to break the crystal lattice. If the solid is insoluble, the hydration energy is less than the lattice energy. The lattice energy is an enthalpy ΔH° term and we should use ΔG° the standard free energy for converting the crystalline salt into gaseous ions. The

reason why the solubility of most group metals decreases on descending the group is that the lattice energy only changes slightly but the free energy of hydration changes rather more.

- 34.** Which of the following shows highest free energy of hydration?
- (a) Rb^+
 - (b) K^+
 - (c) Na^+
 - (d) Li^+

35. Which of the following shows highest lattice energy?
 (a) LiF (b) LiCl
 (c) RbF (d) RbI
36. Which of the following has highest solubility in aqueous solution?
 (a) NaF (b) NaCl
 (c) NaBr (d) NaI

Comprehension-2

Alkali metals oxides are obtained by combustion of the metals. Although Na normally gives Na_2O_2 , it will take up further oxygen at elevated pressure and temperatures to form NaO_2 . The per and superoxides of the heavier alkalis can also be prepared by passed stoichiometric amounts of oxygen into the solution in liquid ammonia.

The different alkali metal oxides can be distinguished by reaction with water. The superoxides reacts with CO_2 and give oxygen gas. The stability of per and superoxides is based upon that larger cation can stabilize larger anion, due to larger lattice energy.

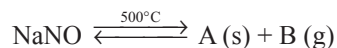
Alkali metals dissolve in liquid ammonia. Dilute solutions are dark blue in colour but as the concentration increases above 3M, the colour changes to copper bronze and the solution acquires the metallic lustre due to the formation of metal ions clusters. The solution of alkali metals in liquid ammonia is good conductors of electricity due to the presence of ammoniated cations and ammoniated electrons. However, the conductivity decreases as the concentrations increases, since ammoniated electrons and ammoniated cation associate.

37. Solution of sodium metals in liquid ammonia is strongly reducing due to the presence of:
 (a) Sodium hydride (b) Sodium atoms
 (c) Sodium amide (d) Solvated electrons.
38. KO_2 is used in oxygen cylinders in space and submarines because it
 (a) Eliminates moisture
 (b) Absorbs CO_2 only

- (c) Absorbs CO_2 and increases O_2 contents
 (d) produces ozone

39. Select the correct choice for alkali metal oxides.
 (a) Metal oxides react with water forming only metal hydroxides.
 (b) Metal peroxides react with water forming metal hydroxides and oxygen gas.
 (c) Metal superoxides react with water forming metal hydroxides, hydrogen peroxide and O_2 gas.
 (d) All of these.
40. On dissolving smaller (less than 3M) amount of sodium metal in liquid ammonia at low temperature, which one of the following does not occur?
 (a) Blue coloured solution is obtained
 (b) Ammoniated Na^+ ions are formed in the solution.
 (c) Liquid NH_3 becomes good conductor of electricity.
 (d) Liquid ammonia remains diamagnetic.

Comprehension-3



41. Compound A is produced by absorbing dinitrogen trioxide in Na_2CO_3 solution. Compound A is:
 (a) Na_2O (b) NaNO_2
 (c) N_2O (d) Na_2O_2
42. Gas B is paramagnetic and supports the combustion. Compound B is:
 (a) N_2 (b) N_2O
 (c) O_2 (d) Na_2O_2
43. D is a inert gas, which is also obtained by strongly heating the ammonium nitrite. Compound D is:
 (a) N_2O (b) NH_3
 (c) O_2 (d) N_2

MATCH THE COLUMN TYPE QUESTIONS

44. Match column-I with column-II

Column-I

- (A) $\text{Ca (OH)}_2 + \text{Cl}_2 \xrightarrow[\text{(slaked lime)}]{\text{below } 35^\circ\text{C}}$
 (B) $\text{Ca(OH)}_2 + \text{Cl}_2 \xrightarrow{\text{cold}} \text{(milk of lime)}$
 (C) $2\text{Ca(OH)}_2 + 2\text{Cl}_2 \xrightarrow{\text{heat}}$
 (D) $2\text{Ca(OH)}_2 + 2\text{Cl}_2 \xrightarrow[\text{(slaked lime)}]{\text{red heat}}$

Column-II

- (P) Calcium chloride
 (Q) Calcium chloride and cal. chlorate
 (R) Bleaching powder
 (S) Calcium chloride and calcium hypochlorite
 (T) Calcium chloride and calcium chlorite

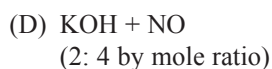
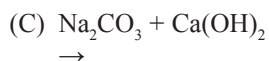
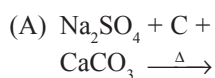
45. Match column-I with column-II

Column-I (Reaction)	Column-II (Product's Character)
(A) $2\text{Na}_2\text{O} \xrightarrow{400^\circ\text{C}}$	(P) Diamagnetic
(B) $\text{KOH} + \text{O}_3 \rightarrow$	(Q) Paramagnetic
(C) $\text{Na} + \text{O}_2 \xrightarrow{350^\circ\text{C}}$ (excess)	(R) Bond order 1
(D) $\text{K}(\text{dissolved in liquid NH}_3) \xrightarrow{302}$	(S) Bond order 1.5

46. Match column-I with column-II.

Column-I (Property)	Column-II (Order)
(A) Atomic radii	(P) $\text{Li} < \text{K} < \text{Na} < \text{Rb}$.
(B) Density	(Q) $\text{Li} < \text{Na} < \text{K} < \text{Rb}$.
(C) Electropositive character	(R) $\text{Li} > \text{Na} > \text{K} > \text{Rb}$.
(D) Hydration energy	(S) $\text{Rb} > \text{K} > \text{Na} > \text{Li}$

47. Match the products of the given reaction in column-I with the property(ies) in column-II

Column-I**Column-II**(P) One of the products has sp^2 hybridization of central atom.(Q) One of the products has sp^3 hybridization of central atom.

(R) One of the products is obtained as insoluble precipitate.

(S) One of the products is neutral oxide.

48. Match the particular(s) given in column-I with the compound(s) in column-II.

Column-I

(A) Solvay process

(B) Evolve CO_2 – on heating

(C) Aqueous solution is neutral towards litmus

(D) Oxone

Column-II(P) NaCl (Q) Na_2O_2 (R) NaHCO_3 (S) Na_2CO_3 **SUBJECTIVE TYPE QUESTIONS**

49. How many type of products are obtained on thermal decomposition of alkaline earth metal nitrates?

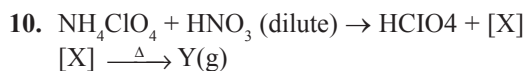
50. The decahydrate form of sodium carbonate i.e., washing soda on standing in air effloresces

and crumbles to powder. What is the number of water molecule(s) present in the compound formed?

p-Block Elements and Compounds

STRAIGHT OBJECTIVE TYPE QUESTIONS

- Aqueous solution of orthoboric acid can be titrated against sodium hydroxide using phenolphthalein indicator only in presence of:
 - trans-glycerol
 - catechol
 - cis-glycerol
 - both (b) and (c)
- Which one of the following is the correct statement?
 - Boric acid is a protonic acid.
 - Both Ti^{3+} and Al^{3+} ions act as oxidizing agent in aqueous solution.
 - Hydrogen bonding in H_3BO_3 gives it a layered structure.
 - $B(OEt)_3$ imparts blue colour to the burner flamer.
- $HCN + H_2O \rightarrow [X] + NH_3$
 $[X] \xrightarrow{\Delta} [Y] + H_2O$
 (Unbalanced equations)
 [Y] May be prepared by which one of the following methods?
 - By dehydration of malonic acid with P_2O_5
 - By thermal decomposition of carbon sub-oxide in air.
 - By heating potassium hexacyanoferrate (II) with conc. H_2SO_4 .
 - By the action of conc. HNO_3 on charcoal.
- Which of the following statements is correct about the water-gas shift reaction used for the commercial preparation of hydrogen gas?
 - It involves the selective oxidation of CO of water gas to CO_2 with the help of steam in presence of iron chromate as catalyst.
 - It involves the separation of CO and H_2 by solvent extraction method.
 - It involves the absorption of CO in aqueous copper (I) chloride solution.
 - It involves the removal of H_2 through occlusion with Pd.
- Select the correct statement about elements of group 15.
 - The order of stability of oxidation state for +3 is $Bi^{3+} > Sb^{3+} > As^{3+}$ and for +5 is $Bi^{5+} < Sb^{5+} < As^{5+}$
 - In the case of nitrogen, all oxidation states from +1 to +4 tend to disproportionate in acid solution.
 - There is considerable increase in covalent radius from N to P but from As to Bi only a small increase in covalent radius is observed.
 - All of these.
- Which of the following compounds does not give nitrogen on heating?
 - NaN_3
 - $(NH_4)_2SO_4$
 - NH_4NO_2
 - $(NH_4)_2Cr_2O_7$
- A brown coloured mixture of two gases is obtained by the reduction of 6N nitric acid with metallic copper. This mixture on cooling condenses to a blue liquid which on freezing ($-30^\circ C$) gives a blue solid. The correct choice for blue liquid or solid is.
 - It is referred to as the anhydride of nitrous acid.
 - It is an acidic oxide and hence dissolves in alkalis producing nitrites.
 - It can also be prepared by the action of 50% HNO_3 on arsenious oxide and then cooling to 250 K.
 - All of these.
- Select the incorrect statement.
 - Nitrous oxide supports combustion more vigorously than air.
 - Phosphorous pentoxide dehydrates nitric acid forming nitrogen pentoxide.
 - Reactivity order of various allotropic forms of phosphorous is white > red > black.
 - Red phosphorous changes to white phosphorus on heating in the atmosphere of CO_2 or coal gas at 573 K.
- For H_3PO_3 and H_3PO_4 the correct choice is
 - H_3PO_3 is dibasic and reducing agent.
 - H_3PO_3 is dibasic and non-reducing agent.
 - H_3PO_4 is tribasic and reducing agent.
 - H_3PO_4 is tribasic and non-reducing agent.

QB.34


$[\text{X}]$ and $[\text{Y}]$ are respectively.

- (a) NH_4NO_3 and N_2O (b) NH_4NO_2 and N_2
 (c) HNO_4 and O_2 (d) None of these

11. Which of the following orders is correct?

- (1) $\text{SbH}_3 > \text{NH}_3 > \text{AsH}_3 > \text{PH}_3$ – Boiling Point
 (2) $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3$ – Thermal Stability
 (3) $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3$ – Basic Character
 (4) $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3$ – Bond Angle

- (a) (1), (2) and (3) only
 (b) (2), (3) and (4) only
 (c) (1), (3) and (4) only
 (d) (1), (2), (3) and (4).

12. Which of the following is correct?

- (a) S_3O_9 – contains no S – S linkage.
 (b) $\text{S}_2\text{O}_6^{2-}$ – contains – O – O linkage.
 (c) $(\text{HPO}_3)_3$ – contains P – P linkage
 (c) $\text{S}_2\text{O}_8^{2-}$ contains S – S linkage.

13. Which of the following statements in incorrect for hydrogen peroxide?

- (a) It is stored in plastic bottles in dark
 (b) It acts as an oxidizing as well as a reducing agent.
 (c) It is used as a bleaching agent.
 (d) It has acidic as well as basic properties.

14. One gas bleaches the coloured matter by reduction temporary while the other one by oxidation permanently. The gases are respectively:

- (a) SO_2 and Cl_2 (b) O_3 and Cl_2
 (c) H_2S and CO_2 (d) Cl_2 and SO_2

15. H_2SO_4 cannot be used to prepare HI from KI as:

- (a) Its reaction with KI is very slow.
 (b) It oxidizes HI to iodine.
 (c) It disproportionates HI.
 (d) It reduces HI.

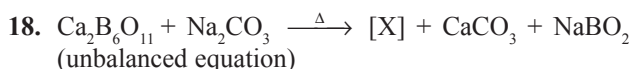
16. Which of the following statements is false?

- (a) H_3PO_2 , H_3PO_3 and H_3PO_4 all are tribasic and reducing in nature.
 (b) Amongst anions NO_3^- , SO_3^{2-} , CO_3^{2-} and BO_3^{3-} , only SO_3^{2-} have $p\pi - \pi$ d bonding.
 (c) Amongst anions SO_3^{2-} , SO_4^{2-} , $\text{S}_2\text{O}_4^{2-}$ and HSO_4^- , SO_3^{2-} is basic and reducing in nature.
 (d) Number of lone pair(s) of electrons on X_3 atoms in XeF_2 , XeF_4 and XeF_6 are 3, 2 and 1 respectively.

17. Which one of the following statements regarding helium is incorrect?

- (a) It is used to produce and sustain powerful superconducting magnets.
 (b) It is used as a cryogenic agent for carrying out experiments at low temperatures.
 (c) It is used in filling balloons for meteorological observations because it is lighter and non-inflammable.
 (d) It is used as dilute for oxygen in modern diving apparatus because of its high solubility in blood.

MULTIPLE CORRECT ANSWER TYPE QUESTIONS



Correct choice(s) for $[\text{X}] + \text{CaCO}_3 + \text{NaBO}_3$ (unbalanced equation)

- (a) Structure of anion of crystalline $[\text{X}]$ has one boron atom sp_3 hybridized and other three atoms sp_2 hybridized.
 (b) $[\text{X}]$ with NaOH (aq.) gives a compound which on reaction with hydrogen peroxide in alkaline medium yields a compound used as brighter in soaps.
 (c) Hydrolysis of $[\text{X}]$ with HCl or H_2SO_4 yields a compound which on reaction with HF gives fluoroboric acid.
 (d) $[\text{X}]$ on heating with chromium salts in oxidising flame gives green coloured bead.

19. On the basis of structure of graphite which of the following is (are) true for it?

- (a) It is diamagnetic substance.
 (b) It behaves like metallic as well as semiconductor.
 (c) C – C bond length in it is more than the diamond.
 (d) It is thermodynamically more stable as well as more dense than diamond.

20. Which of the following order(s) is (are) incorrect?

- (a) $\text{H}_3\text{PO}_4 > \text{H}_3\text{PO}_3 > \text{H}_3\text{PO}_2$ (Reducing character)
 (b) $\text{N}_2\text{O} < \text{NO}_2 < \text{N}_2\text{O}_3 < \text{N}_2\text{O}_5$ (Oxidation state on nitrogen atom.)
 (c) $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3$ (Basicity.)
 (d) $\text{SbH}_3 > \text{NH}_3 > \text{AsH}_3 > \text{PH}_3$ (Reducing character.)

21. Which of the following statement(s) is(are) incorrect?
- Ammonia is oxidized to NO_2 by oxygen at 800°C in presence of a catalyst platinum.
 - Nitric acid on standing slowly turns yellow.
 - Colloidal sulphur is formed when H_2S gas is passed through nitric acid solution.
 - N_2O_3 gas dissolves in water giving a pale blue solution.
22. Which of the following statement(s) is(are) false for sulphurdioxide?
- It reacts with dry chlorine in presence of charcoal to form sulphuryl chloride.
 - It reduces KIO_3 to iodine in acidic medium.
 - It when passed through a solution of sodium sulphide, produces Na_2SO_3 .
 - It oxidizes SnCl_2 to SnCl_4 in presence of HCl .
23. Which of the following statement(s) is(are) correct?
- Marshall's acid on partial hydrolysis gives caro's acid and sulphuric acid.
 - Conc. Sulphuric acid on reaction with potassium chlorate gives chlorine dioxide.
 - Iodine solution is decolourised by sodium thio-sulphate solution.
 - Aqueous solution of SO_2 liberates iodine from an iodate solution and decolourizes iodine solution.
24. Which of the following statements regarding hydrogen peroxide is(are) incorrect?
- The two hydroxyl groups in hydrogen peroxide lie in the same plane.
 - Aqueous solution of H_2O_2 turns blue litmus red.
 - When H_2O_2 behaves as a reducing agent, the $\text{O}-\text{O}$ bond in its molecules is not broken down.
 - Aqueous solution of H_2O_2 is stored in plastic bottles and some urea, phosphoric acid or glycerol is added to that solution.
25. The compound(s) which (have) peroxo linkage is (are):
- $\text{H}_2\text{S}_2\text{O}_3$
 - H_2SO_5
 - $\text{H}_2\text{S}_2\text{O}_7$
 - $\text{H}_2\text{S}_2\text{O}_8$
26. Sodium thiosulphate is prepared by
- Boiling Na_2SO_3 solution with elemental sulphur in alkaline medium.
 - Boiling Na_2SO_3 solution with elemental sulphur in acidic medium.
 - Oxidation of poly sulphides, Na_2S_3 with air.
 - Reduction of bisulphate solution plus SO_2 with zinc dust.
27. Choose the correct statement.
- Ferric bromide is obtained when chlorine gas is bubbled through a solution of ferrous bromide.
 - Sodium sulphate is obtained when $\text{SO}_2(\text{g})$, $\text{H}_2\text{O}(\text{g})$ and air are passed over heated sodium chloride.
 - H_2S gas is liberated when aluminium sulphide becomes damp.
28. Select the incorrect order(s) from the following.
- $\text{ClO}^- < \text{BrO}^- < \text{IO}^-$ – disproportionation
 - $\text{ClO}_4^- < \text{BrO}_4^- < \text{IO}_4^-$ – oxidizing power.
 - $\text{HClO} < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}_4$ – acid strength
 - $\text{HOCl} > \text{HOI} > \text{HOBr}$ – acid strength
29. $\text{XeF}_4 + \text{O}_2\text{F}_2 \rightarrow [\text{X}] + \text{O}_2$ (Unbalanced equation)
Select the correct statement(s) for [X].
- Partial hydrolysis of [X] gives XeOF_4 as one of the product.
 - [X] gives yellow liquid on melting.
 - [X] reacts with SiO_2 to form XeOF_4 and SiF_4 .
 - [X] reacts with XeO_3 to form XeOF_4 .

ASSERTION/REASON TYPE QUESTIONS

30. **Statement-1:** In water, orthoboric acid behaves as a weak monobasic acid.
Statement-2: In water, orthoboric acid behaves as a proton donor.
- Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 - Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 - Statement-1 is True, Statement-2 is False.
 - Statement-1 is False, Statement-2 is True.
31. **Statement-1:** Tetrachlorides of group 14th elements except CCl_4 are easily hydrolysed by water.

Statement-2: The central atoms of these chlorides can accommodate the lone pair of electrons from oxygen atom of water molecule in d – orbitals.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
- (c) Statement-1 is True, Statement-2 is False.
- (d) Statement-1 is False, Statement-2 is True.

32. Statement-1: P_4O_{10} is not used to dry NH_3 .

Statement-2: P_4O_{10} is acidic and NH_3 is basic.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
- (c) Statement-1 is True, Statement-2 is False.
- (d) Statement-1 is False, Statement-2 is True.

33. Statement-1: HNO_3 oxidizes sulphur to H_2SO_4 acid (S + VI) but only oxidizes selenium to H_2SeO_3 (Se + IV).

Statement-2: In Selenium atoms, the electrons are more tightly held by the nucleus on account of increased nuclear charge.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
- (c) Statement-1 is True, Statement-2 is False.
- (d) Statement-1 is False, Statement-2 is True.

34. Statement-1: Aqueous solution of hydrogen peroxide is kept in glass or metal container containing some urea or phosphoric acid.

Statement-2: Urea or phosphoric acid acts as a negative catalyst for the decomposition of hydrogen peroxide.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
- (c) Statement-1 is True, Statement-2 is False.
- (d) Statement-1 is False, Statement-2 is True.

35. Statement-1: Sulphur dioxide is more powerful reducing agent in alkaline medium than in acidic medium.

Statement-2: Sulphur dioxide reacts with hydrogen sulphide in presence of moisture and sulphur is precipitated.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
- (c) Statement-1 is True, Statement-2 is False.
- (d) Statement-1 is False, Statement-2 is True.

36. Statement-1: All halogens except F_2 are more reactive than interhalogens.

Statement-2: Bond in the interhalogens (X – Y) is weaker than X – X bond in the halogens.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
- (c) Statement-1 is True, Statement-2 is False.
- (d) Statement-1 is False, Statement-2 is True.

TRUE AND FALSE TYPE QUESTIONS

- 37. 1.** Oxides of boron (B_2O_3) and silicon (SiO_2) are acidic in nature.
- 2.** Oxides of aluminium (Al_2O_3) and gallium (Ga_2O_3) are amphoteric in nature.
- 3.** Oxides of indium (In_2O_3) and thallium (Tl_2O_3) are basic in nature.

4. Oxides of germanium (GeO_2) and tin (SnO_2) are acidic in nature.

- (a) T F T T (b) F F T T
- (c) T T T F (d) T F T F

38. 1. N_2O with sodium metal in liquid ammonia forms sodium azide and nitrogen gas is liberated.

2. Ammonium dichromate on heating decomposes to give nitrogen and a green coloured compound.
 3. CaNCN on hydrolysis produces a white precipitate and a gas which turns filter paper moistened with copper sulphate solution deep blue.
 - (a) T T T (b) T F T
 - (c) F F T (d) F T T
39. 1. $(\text{HPO}_3)_n$ can be prepared by heating phosphorous acid and bromine in a sealed tube.
2. Dry iodine reacts with ozone and forms a yellow solid, I_4O_9 .
 3. Oxides of indium (In_2O_3) and thallium (Tl_2O_3) are basic in nature.
 4. Oxides of germanium (GeO_2) and tin (SnO_2) are acidic in nature.
- (a) F T F (b) T T F
 - (c) T T T (d) T F F
40. 1. Argon is used in arc welding of metals or allows to provide and inert atmosphere.
2. XeF_2 , XeF_4 and X_3F_6 are colourless crystalline solids and sublime readily at 298K.
 3. XeF_2 , XeF_4 and X_3F_6 are readily hydrolysed.
 4. Xenon fluorides react with fluoride ion acceptor to form cationic species and fluoride ion donors to form fluoro anions.
- (a) F F F F
 - (b) T F T F
 - (c) T T T T
 - (d) F T F T

COMPREHENSION BASED QUESTIONS

Comprehension-1

The small size and high charge of Al^{3+} ion gives it a high charge density which is responsible for its tendency to show (a) covalency in its compounds in the gaseous state (b) high hydration energy which stabilizes its compounds in solution, and (c) high lattice energy of its compounds in the solid state. Thus aluminium can form both covalent and ionic bond.

Like halides of boron, halides of aluminium do not show backbonding because of increase in size of aluminium. Actually aluminium atoms complete their octets by forming dimers. Thus chloride and bromide of aluminium exist as dimers, both in the vapour state and in polar-solvents like benzene while the corresponding boron halides exist as monomer. In boron trihalides the extent of back bonding decreases with increase in size of halogens and thus lewis acid character increases. All BX_3 are hydrolysed by water but BF_3 shows a different behaviour.

41. The dimeric structure of aluminium chloride disappears when
 - (a) it dissolves in water
 - (b) it reacts with donor molecules like R_3N
 - (c) it dissolves in benzene
 - (d) both (a) and (b)
42. Which one of the following statements is correct?
 - (a) All boron trihalides are hydrolysed to boric acid.
 - (b) Anhydrous aluminium chloride is an ionic compound

- (c) Aluminium halides make up the electron deficiency by bridging with halide or alkyl groups
- (d) None of these

43. Which of the following statements about anhydrous aluminium chloride is correct?
 - (a) It exists as AlCl_3 molecule
 - (b) It is not easily hydrolysed
 - (c) It sublimes at 100°C under vacuum
 - (d) It is a strong lewis base.
44. Which of the following reaction is incorrect?
 - (a) $\text{BF}_3(\text{g}) + \text{F}^-(\text{aq}) \rightarrow \text{BF}_4^-$
 - (b) $\text{BF}_3(\text{g}) + 2\text{H}_2\text{O} \rightarrow [\text{BF}_3\text{OH}]^- + \text{H}_3\text{O}^+$
 - (c) $\text{BCl}_3(\text{g}) + 3\text{EtOH}(\text{l}) \rightarrow \text{B}(\text{OEt})_3(\text{l}) + 3\text{HCl}$
 - (d) $\text{BCl}_3(\text{g}) + 2\text{C}_5\text{H}_5\text{N}(\text{l}) \rightarrow \text{Cl}_3\text{B}(\text{C}_5\text{H}_5\text{N})_2(\text{s})$

Comprehension-2

Nitrogen forms the largest number of oxides as it is capable of forming stable multiple bonds with oxygen.

They range from N_2O (O. S of nitrogen +1) through NO , N_2O_3 , NO_2 , N_2O_4 to N_2O_5 (O. S of nitrogen +5). Following points are important regarding the study of oxides of nitrogen.

- (a) All oxides of nitrogen except N_2O_5 are endothermic as a large amount of energy is required to dissociate the stable molecule of oxygen and nitrogen.
- (b) The small electronegativity difference between oxygen and nitrogen make N – O bond easily breakable to

give oxygen and hence oxides of nitrogen are said to be better oxidising agents.

- (c) Except N_2O_5 , all are gases at ordinary temperature. N_2O_3 is stable only at lower temperature 9253K).
- (d) Except N_2O and NO which are neutral oxides, all are acidic oxides which dissolve in water forming corresponding oxy acids.
- (e) They are also good example for illustrating the concept of resonance.

45. The gas which is acidic in nature is:

- (a) NO (b) N_2O
- (c) NO_2 (d) both (a) and (c)

46. Which of the following statement(s) is correct for the oxides of nitrogen?

- (a) Dinitrogen trioxide dissolves in potassium hydroxide forming potassium nitrate.
- (b) Aqueous solution of nitrogen dioxide behaves both as a reducing agent and as an oxidising agent.
- (c) Nitrous oxide is fairly soluble in cold water and turns blue litmus red.
- (d) Nitrogen dioxide is an odd electron molecule and in the liquid phase, exists in equilibrium with the dimer N_2O_4 .

47. Identify the incorrect statement.

- (a) In N_2O_2 , the N-N bond length is longer than the usual N-N single bond distance.
- (b) NO_2 molecule is angular with N – O distance equal to the intermediate distance between a single and double bond.
- (c) N_2O is a linear molecule and has a small dipole moment.
- (d) None of these.

Comprehension-3

Ozone is an unstable, dark blue diamagnetic gas. It absorbs strongly in the UV region, thus protecting the people on the earth from the harmful UV radiation from the sun. The use of chlorofluorocarbon (CFC) in aerosols and refrigerators, and their subsequent escape into the atmosphere is blamed for making holes in the ozone layer over the Antarctic, and Arctic.

Ozone acts as a strong oxidizing agent in acidic and alkaline medium. For this property, ozone is used as a germicide and disinfectant for sterilizing water and improving the atmosphere of crowded places.

48. CFC damages ozone layer by reactions:

- (a) $\text{O}_3 + h\nu \rightarrow \text{O} + \text{O}_2$
- (b) $\text{Cl} + \text{O}_3 \rightarrow \text{ClO} + \text{O}_2$

- (c) $\text{ClO} + \text{O} \rightarrow \text{Cl} + \text{O}_2$
- (d) All of the above.

49. Identify the incorrect statement with respect to ozone.

- (a) Ozone is formed in the upper atmosphere by a photochemical reaction involving dioxygen.
- (b) Ozone protects the earth's inhabitants by absorbing UV radiations.
- (c) Ozone can also be made by heating O_2 over 2500°C and quenching.
- (d) Chlorine gas is preferred over ozone for the purification of drinking water and for water treatment in swimming pools.

50. Which of the following statements is correct?

- (a) The dark blue colour of ozone is due to intense absorption of green light.
- (b) Oxides of nitrogen and the halogen cannot damage the O_3 layer.
- (c) Ozone oxidizes dry iodine to I_2O_5 .
- (d) Ozone forms orange coloured compound KO_3 with potassium hydroxide.

Comprehension-4

All neutral halogens are diatomic and readily reduced to halide ions.

All combine with hydrogen to form gases which except for HF, are strong acids in aqueous solution. The chemistry of the halogens is governed in large part by their tendency to acquire an electron to attain a noble gas electron configuration.

Except, fluorine, all halogens disproportionate in sodium hydroxide (dilute) forming halides and hypohalites. All the hypohalite ions are unstable with respect to the disproportionation; $3 \text{OX}^- \rightarrow \text{XO}_3^- + 2\text{X}^-$.

51. Which of the following product is formed when one more periodate ion (IO_4^-) reacts with excess of iodide ions followed by the acidification of solution?

- (a) IO (b) IO_3^-
- (c) I_2 (d) I_2O_5

52. Which of the following statements is incorrect?

- (a) At ordinary temperature, the rate of disproportionation of hypohalites of chlorine, bromine and iodine follows the order $\text{IO}^- > \text{BrO}^- > \text{ClO}^-$
- (b) Fluorine can not be prepared in aqueous medium by electrolysis since it decomposes water with liberation of ozonized oxygen.
- (c) HI is a stronger acid than HBr because of the low dissociation energy of HI.
- (d) In aqueous solution chlorine is a strong oxidizing agent than fluorine.

53. Consider the following statements;

- (I) Amongst HCl, HBr, HI and HOCl, HCl is most stable to heat.
 - (II) Chlorine gas is evolved when potassium chlorate reacts with iodine.
 - (III) The basicity of F, Cl, Br, and I follows the order $F > Cl > Br > I$.
 - (IV) Sodium hypochlorite is used as a bleaching and sterilizing agent.
- (a) (I), (II) and (III) are correct
 (c) (I), (II) and (IV) are correct
 (b) (I), (III) and (IV) are correct
 (d) All of these.

Comprehension-5

A crystalline solid (A) reacts with hydrogen gas to form a monoatomic gas (B) and a highly associated liquid (C). (A) is soluble in water and undergoes hydrolysis slowly to form (B), (C) and a diatomic gas (D). When pure and dry (D) is subjected to a silent electric discharge another pale blue gas (E) is produced which like (A) also acts as a strong oxidizing agent. (E) dissolves in potassium hydroxide forming an orange compound. (C) forms an addition compound (F) with KF when electrolysed in molten state forms a most reactive gas (G). (B) and (G) (taken in 2: 1 ratio) when heated in nickel tube at 400°C and 6 atm pressure forms (A). The first ionization energy (IE_1) of (B) is fairly close to (D).

54. Gas (G) is produced by reacting K_2MnF_6 with:

- (a) SbF_5 (b) MnF_3
 (c) $KSbF_6$ (d) MnF_5

55. Which of the following statements is correct for (A)?

- (a) The central atom of (A) is sp_3d hybridized with three lone pairs of electrons.
 (b) (A) reacts with S_8 molecules in presence of (C) to form a compound in which the S atom is sp_3d_2 hybridized.
 (c) (A) reacts with SbF_5 as a Lewis base.
 (d) All of these.

56. Which is correct choice for (D) and (E)?

- (a) (E) reacts with dry iodine to form I_2O_5 .
 (b) A filter paper soaked in alcoholic benzidine becomes brown when brought in contact with (E).
 (c) (d) is prepared by Brin's process.
 (d) (b) and (c) both.

57. Which of the following statements is true?

- (a) Gas (G) reacts with conc NaOH to form OF_2 .
 (b) Pale blue gas (E) oxidizes alkaline KI to KIO_3 and KIO_4 .
 (c) Pale blue gas (E) in acidic medium reacts with $K_2Cr_2O_7$ to form a bright blue coloured compound (CrO_5).
 (d) (b) and (c) both.

MATCH THE COLUMN TYPE QUESTIONS

58. Match the reactions listed in column-I with the product(s) listed in column-II.

Column-I	Column-II
(A) $B_2O_3 + H_2O$	(P) H_3BO_3
(B) $B_2H_6 + H_2O$	(Q) H_2
(C) $B_3N_3H_6 + H_2O$	(R) HCl
(D) $BCl_3 + H_2O$	(S) NH_3
	(T) N_2

59. Match the acids in column-I with their properties/type of hybridised listed in column-II.

Column-I	Column-II
(A) Hypophosphorus acid.	(P) All hydrogen are ionizable in water.
(B) Orthoboric acid	(Q) Monobasic acid.
(C) Hypophosphoric acid	(R) sp^3 hybridized central atom(s).

(D) Pyrosulphurous acid. (S) Lewis acid.

60. Match the reactions mentioned in column-I with the nature of reaction/characteristic(s) of the products mentioned in column-II.

Column-I	Column-II
(A) $(NH_4)_2S_2O_8 + H_2O \xrightarrow{\text{distillation}}$	(P) Hydrolysis
(B) $NaBO_2 + H_2O + H_2O_2 \xrightarrow{OH^-}$	(Q) One of the product has peroxide linkage.
(C) $Na_2SO_3 \xrightarrow[\text{temperature}]{\text{fusion}}$	(R) Disproportionation
(D) 2-Ethylanthraquinol + air $\rightarrow$	(S) In one of the products the central atom has sp^3 hybridisation

QB.40

61. Match the reactions mentioned in column-I with the use of product/characteristic(s) of the products mentioned in column-II

Column-I
Column-II

- (A) $\text{HSO}_3^- + \text{SO}_2 \xrightarrow{\text{Zn}}$ (P) In one of the products the central atom(s) is(are) in +3 oxidation state.
- (B) $\text{Na}_2\text{Cr}_2\text{O}_7 + \text{C} \xrightarrow{\Delta}$ (Q) One of the products acts as a reducing agent.
- (C) $\text{P}_4 + \text{Cl}_2(\text{g}) + \text{H}_2\text{O} \rightarrow$ (R) One of the products or its anion has trigonal planar geometry.

- (D) $\text{H}_2\text{SO}_4 + \text{HI} \rightarrow$ (S) One of the products is a gas which is used in the purification of nickel metal.

62. Match the compounds listed in column-I with characteristic(s)/type of reaction(s) listed in column-II.

Column-I
Column-II

- (A) XeF_2 (P) Undergoes hydrolysis with water.
- (B) XeF_4 (Q) Acts as oxidising agent.
- (C) XeF_6 (R) Undergoes addition reaction.
- (D) XeO_3 (S) Has lone pair(s) of electrons.
- (T) Gives disproportionation reaction with H_2O or OH^- .

SUBJECTIVE TYPE QUESTIONS
Short Subjective Questions

63. The number of two center – two center – two electron bonds i.e., $2c - 2e$ in diborane is:
64. The number of hydrogen atom(s) attached to phosphorus atom in hypophosphorus acid is

65. XeF_4 disproportionates in water giving reduced and oxidized products. What is the ratio of oxidation states of Xe in the reduced and the oxidized products?

d-Block Elements and Compounds

STRAIGHT OBJECTIVE TYPE QUESTIONS

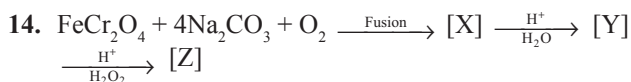
- Cerium ($z = 58$) is an important member of the lanthanoids. Which of the following statements about cerium is incorrect
 - The +4 oxidation state of cerium is not known in solutions
 - The +3 oxidation state of cerium is more stable than the +4 oxidation state
 - The common oxidation states of cerium are +3 and +4
 - Cerium (IV) acts as an oxidizing agent
- Silver nitrate is prepared by
 - The action of only conc. HNO_3 on silver
 - Heating silver oxide with NO^2
 - The action of hot dil. on silver
 - Dissolve in aqua-regia
- Which of the following is true for the species $3d^4$ configuration?
 - Cr^{2+} is reducing in nature
 - Mn^{3+} is oxidizing in nature.
 - Both (a) and (b)
 - None of these
- In which compound does vanadium have an oxidation number of +4?

(a) NH_4VO_2	(b) $\text{K}_4[\text{V}(\text{CN})_6]$
(c) VSO_4	(d) VOSO_4
- Among the following, the species that is both paramagnetic and coloured is:

(a) $[\text{MnO}_4]^{2-}$	(b) $\text{K}_4[\text{V}(\text{CN})_6]$
(c) $[\text{VO}_4]^{3-}$	(d) CrO_2Cl_2
- Which of the following increasing order of oxidizing power is correct for the following species?
 VO_2^+ , MnO_4^- , $\text{Cr}_2\text{O}_7^{2-}$
 - $\text{VO}_2^+ < \text{Cr}_2\text{O}_7^{2-} < \text{MnO}_4^-$
 - $\text{VO}_2^+ < \text{MnO}_4^- < \text{Cr}_2\text{O}_7^{2-}$
 - $\text{Cr}_2\text{O}_7^{2-} < \text{VO}_2^+ < \text{MnO}_4^-$
 - $\text{Cr}_2\text{O}_7^{2-} < \text{MnO}_4^- < \text{VO}_2^+$
- At 300°C , FeCl_3 :
 - Decomposes into FeCl_2 and Cl_2 .
 - Decomposes into Fe and Cl_2 .
 - Sublimes to give liquid FeCl_3 .
 - Sublimes to give gaseous dimer $(\text{FeCl}_3)_2$.
- Anhydrous ferric chloride is prepared by:
 - Heating hydrated ferric chloride at a high temperature in a stream of air.
 - Heating metallic iron in a stream of dry chlorine gas.
 - Reaction of ferric oxide with HCl (aq).
 - Reaction of metallic iron with HCl (aq).
- Anhydrous ferrous sulphate on heating gives:
 - Both SO_2 and SO_3
 - SO_2 only
 - SO_3 only
 - $\text{Fe}_2(\text{SO}_4)_3$ only
- $\text{FeCl}_3 \cdot 6\text{H}_2\text{O} + \text{C}(\text{CH}_3)_2(\text{CH}_3\text{O})_2 \rightarrow$ Products
 Reaction products are:
 - FeCl_3 , CH_3OH and CH_3COCH_3
 - $(\text{CH}_3\text{O})\text{Fe}$, HCl and H_2O
 - FeCl_2 , HCl and CH_3COCH_3
 - $\text{Fe}(\text{OH})_3$, FeCl_3 and CH_3COCH_3
- Excess of KI reactions with CuSO_4 solution and Na_2SO_4 solution is added to it. Which of the following statement is incorrect for this reaction?
 - Evolved I_2 is reduced
 - CuI_2 is formed
 - $\text{Na}_2\text{S}_2\text{O}_3$ is oxides
 - CuI_2 is formed.
- Select the correct statement:
 - PH_3 is reduces AgNO_3 to metallic Ag.
 - Organic tissues turn AgNO_3 black by reducing it to Ag.
 - AgCN is soluble in KCN .
 - All are correct statements.

QB.42

13. The yellow colour solution of Na_2CrO_4 changes to orange red on passing CO_2 gas due to the formation of:
- (a) CrO_5 (b) CrO_3
(c) $\text{Na}_2\text{Cr}_2\text{O}_7$ (d) Cr_2O_3



Which of the following statement is true for the compounds [X], [Y] and [Z]?

- (a) In all three compounds, the chromium is in +6 oxidation states.

- (b) [Z] is a deep blue-violet coloured compound which decomposes rapidly in aqueous solution into Cr^{3+} and dioxygen.
(c) Saturated solution of [Y] gives bright orange compound, chromic anhydride, with concentrated H_2SO_4 .
(d) All of these

15. KMnO_4 on treatment with conc. H_2SO_4 forms a compound (X) which decomposes explosively on heating forming (Y). The (X) and (Y) are respectively:

- (a) Mn_2O_7 , MnO_2 (b) Mn_2O_7 , Mn_2O_3
(c) MnSO_4 , MnO_2 (d) Mn_2O_3 , MnO_4^{2-}

MULTIPLE CORRECT ANSWER TYPE QUESTIONS

16. The hydrated cupric chloride is strongly heated. Which of the following statement(s) is(are) correct for this?

- (a) It is reduced to Cu_2Cl_2
(b) Cupric oxide is formed along with Cu_2Cl_2
(c) Only Cl_2 is liberated
(d) Cl_2 and HCl both are liberated

17. Explain giving reason:

- (a) Transition metals and many of their compounds show paramagnetic behaviour.
(b) The enthalpies of atomisation of the transition metals are high
(c) The transition metals generally form coloured compounds
(d) Transition metals and their many compounds act as good catalyst.

18. To an acidified dichromate solution, pinch of Na_2O_2 is added and shaken. What is observed.

- (a) Blue colour
(b) Orange colour changing to green
(c) Copious evolution of oxygen
(d) Bluish-green precipitate

19. When CO_2 is passed into aqueous:

- (a) Na_2CrO_4 solution, its yellow colour changes to orange.
(b) K_2MnO_4 solution, its disproportionates to KMnO_4 and MnO_2 .
(c) $\text{Na}_2\text{Cr}_2\text{O}_7$ solution, its orange colour changes to green.
(d) KMnO_4 solution, its pink colour changes to green.

20. Pyrolusite is MnO_2 , used to prepare KMnO_4 . Steps are $\text{MnO}_2 \xrightarrow{\text{I}} \text{MnO}_4^{2-} \xrightarrow{\text{II}} \text{MnO}_4^-$.

Steps I and II are respectively:

- (a) Fuse with KOH /air and electrolytic oxidation.
(b) Fuse with KOH/KNO_3 and electrolytic oxidation.
(c) Fuse with conc. HNO_3 and electrolytic reduction.
(d) Dissolve in H_2O and oxidation.

21. Which of the following chemical reaction(s) is(are) involved in the developing of the photographic plate?

- (a) $\text{C}_6\text{H}_4(\text{OH})_2 + 2\text{AgBr} \rightarrow 2\text{Ag} + \text{C}_6\text{H}_4\text{O}_2 + 2\text{HBr}$
(b) $\text{AgBr} + 2\text{Na}_2\text{S}_2\text{O}_3 \rightarrow \text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)_2] + \text{NaBr}$
(c) $\text{AgBr} + 2\text{NH}_3(\text{aq}) \rightarrow [\text{Ag}(\text{NH}_3)_2] \text{Br}$
(d) $2\text{AgBr} + \text{Na}_2\text{S}_2\text{O}_3 \rightarrow \text{Ag}_2\text{S}_2\text{O}_3 + 2\text{NaBr}$

22. The hydrated cupric chloride is strongly heated. Which of the following statement(s) is(are) correct for this?

- (a) It is reduced to Cu_2Cl_2 .
(b) Cupric oxide is formed along with Cu_2Cl_2 .
(c) Only Cl_2 is liberated.
(d) Cl_2 and HCl both are liberated.

23. Select the incorrect statement(s).

- (a) In K_2MnO_4 and CrO_2Cl_2 , the central transition metals/ions have the same oxidation state.
(b) Both sodium and potassium dichromate can be used as primary standard in volumetric estimations.
(c) Potassium dichromate on strong heating evolves oxygen gas and forms green powder.
(d) Potassium permanganate on fusion with potassium hydroxide liberates oxygen gas leaving behind green residue.

24. Cuprous chloride cannot be prepared:
- By passing SO_2 through the solution containing CuSO_4 and NaCl .
 - By heating excess of copper with conc. HCl in presence of a little KClO_3 .
 - By boiling copper sulphate solution with excess of copper turnings in presence of hydrochloric acid.
 - By dissolving cupric oxide or copper carbonated in conc. HCl .
25. Potassium manganate (K_2MnO_4) is formed. When:
- Cuprous oxide reacts with alkaline KMnO_4 solution.
 - Manganese dioxide is fused with K_2CO_3 in presence of KNO_3 .
 - Formaldehyde reacts with potassium permanganate in presence of concentrated potassium hydroxide solution.
 - Potassium permanganate is heated with potassium hydroxide.

ASSERTION AND REASON TYPE QUESTIONS

26. **Statement-1:** Dipositive zinc exhibits diamagnetism due to loss of two electrons from $3d$ – orbital of neutral atom.
Statement-2: The electronic configuration of zinc ($z = 30$) is $[\text{Ar}]^{18} 3d^{10} 4s^2$.
- Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 - Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
 - Statement-1 is True, Statement-2 is False.
 - Statement-1 is False, Statement-2 is True.
27. **Statement-1:** The number of unpaired electrons in the following gaseous ions:
 Mn^{3+} , Cr^{3+} , V^{3+} and Ti^{3+} are 4, 3, 2 and 1 respectively
Statement-2: Cr^{3+} is stable in aqueous solution.
- Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 - Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
 - Statement-1 is True, Statement-2 is False.
 - Statement-1 is False, Statement-2 is True.
28. **Statement-1:** E° for $\text{Mn}^{3+}/\text{Mn}^{2+}$ is more positive than $\text{Cr}^{3+}/\text{Cr}^{2+}$.
Statement-2: The third ionization energy of Mn is larger than that of Cr.
- Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 - Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
 - Statement-1 is True, Statement-2 is False.
 - Statement-1 is False, Statement-2 is True.
29. **Statement-1:** Cuprous ion (Cu^+) is colourless whereas cupric ion (Cu^{2+}) produces blue colour in solution.
Statement-2: Cuprous ion (Cu^+) has unpaired electrons while cupric ion (Cu^{2+}) does not.
- Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 - Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
 - Statement-1 is True, Statement-2 is False.
 - Statement-1 is False, Statement-2 is True.
30. **Statement-1:** To a solution of potassium chromate, if a strong acid is added, it changes its colour from yellow to orange.
Statement-2: The colour change is due to the oxidation of potassium chromate.
- Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 - Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
 - Statement-1 is True, Statement-2 is False.
 - Statement-1 is False, Statement-2 is True.

TRUE AND FALSE TYPE QUESTIONS

Consider the following statements and arrange in the order of true/false as given in the codes.

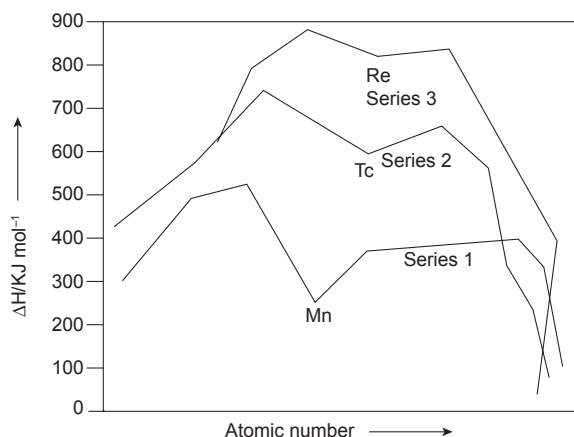
31. Copper (II) oxide is red powder. (True or False)
32. Mercury is the only metal which is liquid at 0°C. This is due to its high atomic weight. (True or False)
33.
 1. Mn^{2+} compounds more stable than Fe^{2+} towards oxidation to their +3 states.
 2. Titanium and copper both in the first series of transition metals exhibits +1 oxidation state most frequently
 3. Cu^+ ion is stable in aqueous solutions.

- | | |
|-----------|-----------|
| (a) T T F | (b) T F F |
| (c) T F T | (d) F F T |

34.
 1. Interstitial compounds have high melting points, higher than those of pure metals.
 2. Permanganate titrations in presence of hydrochloric acid are unsatisfactory.
 3. KMnO_4 does not act as an oxidizing agent in strong alkaline medium.
 4. KMnO_4 on heating in a current of H_2 gives MnO .
- | | |
|-------------|-------------|
| (a) T T F T | (b) T F F T |
| (c) T F T T | (d) F F T F |

COMPREHENSION BASED QUESTIONS

Comprehension-1



Trends in enthalpies of atomization of transition elements

35. The dip in the boiling point of manganese in the 3d series is due to
 - (a) less delocalisation of valence electrons owing to extra stable half filled configuration.
 - (b) smaller size of atom.
 - (c) less electronegativity of atom.
 - (d) all of these
36. Consider the following statements:
 - I. The transition elements (except the 12th group elements) are very much hard and have low volatility.

- II. Generally, in a given series the melting points of the transition elements rises to maximum and then fall as the atomic number increases.
- III. Transition metals have high thermal and electrical conductivity and metallic lustre.

Select the correct from the codes given.

- | | |
|--------------------|---------------------|
| (a) I and III only | (b) II and III only |
| (c) All of these | (d) None of these |

37. A: The value of enthalpy of atomisation is maximum at about the middle of each series. R: There is one unpaired electron per d-orbital and this results in stronger interatomic interaction.
 - (a) A is true, R is true and R is the correct explanation for A.
 - (b) A is true, R is also true but R is not the correct explanation of A.
 - (c) A is true, R is false.
 - (d) A is false, R is true.

Comprehension-2

Paramagnetism is properly due to the presence of unpaired electrons. In case of transition metals, as they contain unpaired electrons in the $(n-1)$ orbitals, most of the transition metal ions and their compounds are paramagnetic. Paramagnetism increases with increases in number of unpaired electrons. Magnetic moment is calculated from 'Spin only formula' viz.,

$$\mu = \sqrt{n(n+1)} \text{ B.M}$$

n = no. of unpaired electrons.

Similarly, the colour of the compounds of transition metals may be attributed to the presence of incomplete (n-1)d subshell. When an electron from a lower energy of d-orbital is excited to a higher energy d-orbital, the energy of excitation corresponds to the frequency of light absorbed. This frequency generally lies in the visible region. The colour observed corresponds to complementary colour of the light absorbed. The frequency of the light absorbed is determined by the nature of the ligand.

38. Which of the following pair of compounds is expected to exhibit same colour in aqueous solution?

- (a) FeCl_2 , CuCl_2 (b) VOCl_2 , CuCl_2
(c) VOCl_2 , FeCl_2 (d) FeCl_3 , MnCl_2

39. Titanium shows magnetic moment of 1.73 BM in its compound. What is the oxidation state of titanium in the compound?

- (a) +2 (b) +1
(c) +3 (d) +4

40. Identify the incorrect statement.

- (a) Mn^{2+} has the highest Paramagnetism amongst the bivalent cations of the 1st transition series.
(b) The coloured ions or compounds of transition elements are due to d-d transition, polarization of anion and charge transfer spectrum.
(c) In 3 d series the paramagnetic character first increase to maximum and then starts decreasing.
(d) None of these.

Comprehension- 3

- (i) A powdered substance (A) on treatment with fusion mixture gives a green coloured compound (B).
(ii) The solution of (B) in boiling water on acidification with dilute H_2SO_4 gives a pink coloured compound (C).
(iii) The aqueous solution of (A) on treatment with NaOH and Br^- -water gives a compound (D)

- (iv) A solution of (D) in conc. HNO_3 on treatment with lead peroxide at boiling temperature produced a compound (E) which was of the same colour as that of (C).
(v) A solution of (A) on treatment with a solution of barium chloride gave a white precipitate of compound (F) which was insoluble in conc. HNO_3 and conc. HCl .

41. Which of the following is true for compound (C)?

- (a) It oxidizes ammonia to nitrogen dioxide in neutral medium.
(b) It is coloured and colour is due to d-d transition.
(c) It can be oxidized by ozone.
(d) It is obtained by alkaline fusion of pyrolusite in presence of air followed by electrolytic oxidation.

42. The oxidation state of central metal ions of (A), (B) and (C) compounds are respectively.

- (a) +II, +VI and +VII
(b) +II, +VI and +VI
(c) +II, +VII and +VII
(d) +VI, +VII and +VII

43. Which of the following statements is correct?

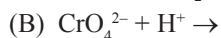
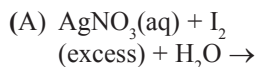
- (I) Anions of both (B) and (C) are diamagnetic and have tetrahedral geometry.
(II) Anions of both (B) and (C) are paramagnetic and have tetrahedral geometry.
(III) Anions of (B) is paramagnetic and that of (C) is diamagnetic but both have same tetrahedral geometry.
(IV) Green coloured compound (B) in a neutral or acidic medium disproportionates to give (C) and (D).

- (a) I and III only (b) II and III only
(c) II and IV only (d) III and IV only

MATCH THE COLUMN TYPE QUESTIONS

44. Match the reactions in column-I with nature of the reactions/type of the products in column-II.

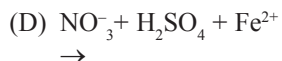
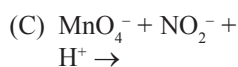
Column-I



Column-II

(P) Redox reaction

(Q) One of the product has trigonal planar structure



(R) Dimetric bridged tetrahedral metal ion

(S) Disproportionation

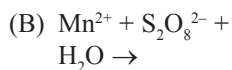
(T) Colour of one of the products is due to charge transfer spectrum.

QB.46

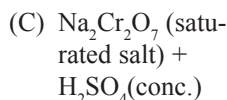
45. Match the reactions in column-I with nature of the reactions/type of the products in column-II.

Column-I
Column-II

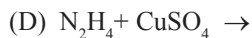

(P) One of the products is coloured due to charge transfer.



(Q) One of the product is in + VI oxidation state.

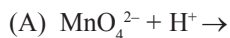


(R) Redox reaction

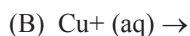


(S) One of the product is acidic oxide.

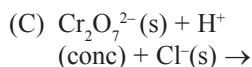
46. Match the reactions in column-I with nature of the reactions/type of the products in column-II.

Column-I
Column-II


(P) One of the products is paramagnetic.



(Q) Disproportionation

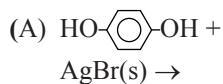


(R) One of the products is liberated as coloured vapours

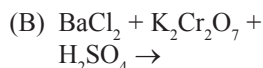


(S) In one of the products central atom exhibits its highest oxidation state.

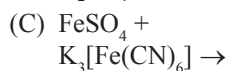
47. Match the reactions in column-I with type of the products/applications in column-II.

Column-I
Column-II


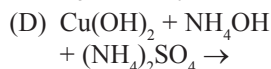
(P) Turnbull's blue pigment



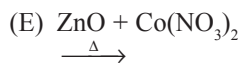
(Q) Schwoitzer's reagent



(R) Rinmann's green pigment



(S) Chromyl chloride test



(T) Photography

SUBJECTIVE TYPE QUESTIONS

Short Subjective Questions

48. The oxidation number of Cr in the product of alkaline oxidative fusion of FeCr_2O_4 is.

49. Find out the number of 3d electrons occupied in t_{2g} orbitals of hydrated Cr^{3+} ion (octahedral).

50. How many Cr – O bonds are equivalent in chromate dianion?

Qualitative Analysis

STRAIGHT OBJECTIVE TYPE QUESTIONS

- Which of the following anions would decolourize acidified KMnO_4 solution?
(a) S_4^{2-} (b) S^{2-}
(c) NO_3^- (d) CH_3COO^-
- AgCl dissolves in ammonia solution giving
(a) Ag^+ , NH_4^+ and Cl^-
(b) $\text{Ag}(\text{NH}_3)^+$ and Cl^-
(c) $\text{Ag}_2(\text{NH}_3)^4$ and Cl^-
(d) $\text{Ag}(\text{NH}_3)_2^+$ and Cl^-
- Which radicals are precipitated in $(\text{NH}_4)_2\text{CO}_3$ in presence of alkali
(a) Ca, Ba, Sr (b) Mg
(c) Both (a) and (b) (d) None of these
- The salt used for performing 'bead' test in qualitative inorganic analysis is
(a) $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
(b) $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$
(c) $\text{Na}(\text{NH}_4)\text{HPO}_4 \cdot 4\text{H}_2\text{O}$
(d) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
- On performing a borax-bead test with a given inorganic mixture for qualitative analysis, the colour of the bead was found to be emerald green both in oxidizing and reducing flame. It indicates the possibility of the presence of
(a) Co^{+2} (b) Ni^{+2}
(c) Cr^{+3} (d) Cu^{+2}
- Zinc pieces are added to acidified solution of SO_3^{2-} . Gas liberated can:
(a) Turn lead acetate paper black
(b) Turn lime water milky.
(c) Give both of the above tests
(d) Give none of the above tests
- Ammonium salts on heating with slaked lime liberates a colourless gas (X). Identify the correct statement for gas (X).
(a) (X) turns red litmus blue and produces dense white fumes in contact with dilute HCl.
(b) (X) turns filter paper moistened with mercurous nitrate black and gives intense blue coloured solution with CuSO_4 (aq).
(c) (X) when passed through Nessler's reagent produces a brown colour precipitate.
(d) All of these.
- An aqueous blue coloured solution of a transition metal sulphate reacts with H_2S to give a black precipitate. The black precipitate dissolves in 50% nitric acid forming a blue coloured solution. The blue solution on treatment with KI in weakly acidic medium turns yellow/brown and produces a white precipitate.
(a) Co^{2+} (b) Cu^{2+}
(c) Hg^{2+} (d) Pb^{2+}
- Which one of the following reagent can be used for differentiating Cu^{2+} and Bi^{3+} ?
(a) H_2S gas in presence of dil. HCl.
(b) NH_4OH (excess)
(c) $\text{K}_4[\text{Fe}(\text{CN})_6]$
(d) (b) and (c) both
- The ion that cannot be precipitated by both HCl and H_2S is
(a) Pb^{2+} (b) Cu^+
(c) Ag^+ (d) Sn^{2+}
- The only cations present in a slightly acidic solution are Fe^{3+} , Zn^{2+} and Cu^{2+} . The reagent that when added in excess to this solution would identify the separate Fe^{3+} in one step is:
(a) 2 M HCl (b) 6 M NH_3
(c) 6 M NaOH (d) H_2S gas
- $\text{Fe}(\text{OH})_3$ and $\text{Cr}(\text{OH})_3$ precipitates are completely separated by:
(a) Aqueous NH_3 (b) HCl
(c) $\text{NaOH}/\text{H}_2\text{O}$ (d) H_2SO_4
- A mixture of two white substance was dissolved in water. On passing Cl_2 gas through the solution a deep brown colour is developed. Addition of BaCl_2 solution to the original solution gives a white precipitate.

Addition of a large amount of NaOH solution to the original solution gives a white precipitate, whose suspension in water is used as an ant-acid, the mixture gives golden yellow colour flame. The mixture contains:

- (a) Na^+ , Mg^{2+} , Br^- and SO_4^{2-}
- (b) Na^+ , Ca^{2+} , Br^- and SO_4^{2-}
- (c) Na^+ , Al^{3+} , Br^- and SO_4^{2-}
- (d) Na^+ , Zn^{2+} , Br^- and SO_4^{2-}

14. $\text{K}_4[\text{Fe}(\text{CN})_6]$ can be used to detect one or more out of F^{2+} , Fe^{3+} , Zn^{2+} , Cu^{2+} , Ca^{2+}

- (a) Fe^{2+} , Fe^{3+} only
- (b) Fe^{3+} , Zn^{2+} , Cu^{2+} only
- (c) All but not Ca^{2+}
- (d) All of these

15. Intense blue precipitate of $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$ and sodium hydroxide solution when mixed gives:

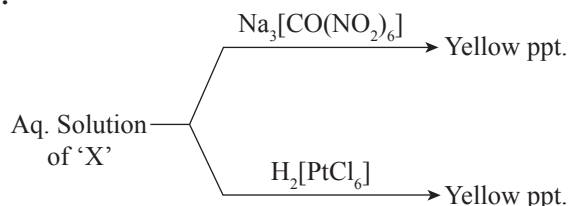
- (a) Soluble prussian salt
- (b) Reddish-brown precipitate
- (c) Deep-red colouration
- (d) Turnbull's blue

MULTIPLE CORRECT ANSWER TYPE QUESTIONS

16. In the dark brown ring test for nitrate ion:

- (a) the colour is due to charge transfer spectra
- (b) iron has a formal +1 oxidation state and NO has formal +1 charge ;
- (c) the complex species can be represented as $[\text{Fe}^{\text{I}}(\text{H}_2\text{O})_5\text{NO}^+]$
- (d) iron has +2 oxidation state and NO is neutral

17.



The cation (s) present in 'X' is/are:

- (a) NH_4^+
- (b) Na^+
- (c) Mg^{2+}
- (d) K^+

18. Which of the following mixtures cannot be separated by passing H_2S through their solutions containing dilute HCl?

- (a) Cu^{2+} and Sb^{3+}
- (b) Pb^{2+} and Cd^{2+}
- (c) Pb^{2+} and Al^{3+}
- (d) Zn^{2+} and Mn^{2+}

19. Which of the following statement(s) is/are correct when a solid mixture of NaCl and $\text{K}_2\text{Cr}_2\text{O}_7$ is gently warmed with cone. H_2SO_4 ?

- (a) A deep red vapour is evolved
- (b) The vapour when passed into NaOH solution gives a yellow solution of Na_2CrO_4
- (c) Chlorine gas is evolved
- (d) Chromyl chloride is formed

20. Choose the correct statements:

- (a) $\text{BaCl}_2 + \text{AcOH} + \text{K}_2\text{CrO}_4 \rightarrow$ yellow ppt
- (b) $\text{BaCO}_3(\text{s}) + \text{K}_2\text{CrO}_4 \rightarrow$ yellow ppt
- (c) $\text{BaCO}_3(\text{s}) + \text{K}_2\text{CrO}_4 + \text{AcOH} \rightarrow$ No ppt
- (d) $\text{SrCO}_3(\text{s}) + \text{K}_2\text{CrO}_4 + \text{AcOH} \rightarrow$ No-ppt

21. $\text{Al}_2(\text{SO}_4)_3 + \text{NH}_4\text{OH} \rightarrow \text{X}$

Select the correct statement(s) about compound X:

- (a) X is a white coloured compound
- (b) X is insoluble in excess of NH_4OH
- (c) X is soluble in NaOH
- (d) X can be used as an antacid

22. Which of the following statement(s) is/are correct?

- (a) Soluble sulphide gives black precipitate with AgNO_3 solution which is soluble in hot dilute nitric acid.
- (b) Soluble sulphide produces a yellow precipitate with a suspension of cadmium carbonate.
- (c) Sulphide ions reacts with sodium nitroprusside and gives a purple colouration.
- (d) Free H_2S gas forms white precipitate with tetrahydroxo plumbate (II) solution.

23. Each of these solution is added to a mixture of aqueous solutions of iodide and chloroform (CHCl_3), separately. Which will give a positive test for iodine when the solutions are vigorously mixed?

- (a) NaCl solution
- (b) NaBr solution
- (c) Chlorine water
- (d) Bromine water

24. Which of the following is(are) correctly matched?

- (a) $[Al(OH)_4]^- (aq) + NH_4^+ (aq) \rightarrow$ White precipitate and liberation of ammonia.
- (b) $Pb^{2+} (aq) + 2Br^- (aq) \rightarrow$ Red precipitate.
- (c) BiI_3 (black precipitate) + $H_2O (l) \rightarrow$ Orange precipitate
- (d) $Ca^{2+} (aq) + K_4 [Fe (CN)_6] (aq) \rightarrow$ White precipitate

25. Which of the following is/are correctly matched?

- (a) $Fe^{3+} + C_6H_4N(NO)ONH_4$ Reddish brown precipitate
- (b) $AsO_4^{3-} + Mg^{2+} + NH_4^+$ White crystalline precipitate
- (c) $Bi^{3+} + C_9H_7ON$ (5% + H^+ + I^-) Red precipitate
- (d) $Hg^{2+} + Co^{2+} + 5CN^-$ Deep blue crystalline precipitate.

ASSERTION/REASON TYPE QUESTIONS

26. **Assertion:** Sb(III) is not precipitated as sulphide when in its alkaline solution H_2S is passed.

Reason: The concentration of S^{2-} ion in alkaline medium is inadequate for precipitation.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
- (c) Statement-1 is True, Statement-2 is False.
- (d) Statement-1 is False, Statement-2 is True.

27. **Statement-1:** Baryta water becomes turbid on passing CO_2 gas through it but turbidity becomes clean on passing more CO_2 gas.

Statement-2: Carbonates give white precipitate with silver nitrate solution. The precipitate becomes yellow or brown if the mixture is boiled.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
- (c) Statement-1 is True, Statement-2 is False.
- (d) Statement-1 is False, Statement-2 is True.

28. **Statement-1:** No yellow precipitate is formed when an excess of a more concentrated (6M) solution of KI is added to a solution containing Pb^{2+} ions.

Statement-2: Yellow precipitate of PbI_2 does not dissolve in excess of dilute solution of KI.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

29. **Statement-1:** A solution of $BiCl_3$ in concentrated HCl when diluted with water gives white precipitate.

Statement-2: $BiCl_3$ forms insoluble BiO^+Cl when diluted with a large quantity of water.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
- (c) Statement-1 is True, Statement-2 is False.
- (d) Statement-1 is False, Statement-2 is True.

30. **Statement-1:** When H_2S gas is passed into an aqueous solution of $ZnCl_2$, Zn^{2+} ions are completely precipitated as zinc sulphide.

Statement-2: Zinc sulphide is insoluble in solutions of caustic alkali but dissolves in dilute HCl.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
- (c) Statement-1 is True, Statement-2 is False.
- (d) Statement-1 is False, Statement-2 is True.

TRUE AND FALSE TYPE QUESTIONS

31. 1. Like CO_3^{2-} , SO_3^{2-} also gives test with baryta water, $\text{Ba}(\text{OH})_2$.
 2. Ag_2SO_3 is insoluble in dilute HNO_3 .
 3. Triiodide ions (I_3^-) produced by the reaction of Cu^{2+} and KI solution are not reduced by excess of sodium thiosulphate solution.
 4. Black precipitate of $\text{Cu}(\text{SCN})_2$ turns into white precipitate when it reacts with saturated solution of SO_2 in water.
 (a) F T T F (b) T F F T
 (c) F T F T (d) T T F T
32. 1. The addition of chlorine water dropwise to a solution of a bromide liberates free bromine, which colours the solution orange red.
 2. Chlorine gas bleaches the filter paper moistened with yellow dyestuff, fluorescein.
 3. Nitrogen gas is evolved when a solution of nitrate is boiled with zinc dust and sodium hydroxide solution.
 (a) T T T (b) T F T
 (c) T F F (d) T T F
33. 1. When copper sulphate is added to $\text{K}_2[\text{HgI}_4]$ solution and SO_2 is passed into this solution, white precipitate is obtained.
 2. When copper sulphate solution is boiled with glucose in presence of caustic soda, a blue precipitate is obtained.
 3. Pure Cu_2Cl_2 is white, but moist air converts it into a dark blue material.
 (a) T T T (b) T F T
 (c) F F F (d) F T F
34. 1. Ammonium chloride on heating with an aqueous solution of $\text{K}_2\text{Cr}_2\text{O}_7$ and concentrated H_2SO_4 produces deep red vapours.
 2. Barium nitrate dissolved in water responds to brown ring test.
 3. Deep red solution of ferric acetate on boiling with water turns to a brownish red solution.
 4. NaNO_2 solution acidified with acetic acid produces blue colour with a starch paste contaminated with iodide ions.
 (a) T T T F (b) F F F T
 (c) T T F F (d) F F T T

COMPREHENSION BASED QUESTIONS

Comprehension-1

Aqueous solution of a salt 'A', when mixed with NaOH solution and warmed, a black precipitate is formed. Black ppt, is filtered and dissolved in dil. H_2SO_4 solution. The resulting solution gives a chocolate coloured precipitate with potassium ferrocyanide solution. The filtrate obtained after filtering off the black precipitate, upon warming with Zn and NaOH evolves a pungent smelling gas. The resulting solution also responds to the ring test. The filtrate does not evolve any gas when it is boiled with urea in the presence of H_2SO_4 .

35. Salt 'A' consists of:
 (a) Cu^{2+} (b) Hg^{2+}
 (c) Cu^{2+} (d) Pb^{2+}
36. The filtrate obtained after filtering off the black precipitate consists of:
 (a) NO_2^- (b) NO_3^-
 (c) CO_3^{2-} (d) Cl^-
37. The chocolate coloured precipitate is:
 (a) $\text{Fe}_2[\text{Fe}(\text{CN})_6]$ (b) $\text{Cu}_2[\text{Fe}(\text{CN})_6]$
 (c) HgSO_4 (d) $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]\text{SO}_4$

Comprehension-2

The following flow chart describes the scheme of classifications of anions on the basis of the nature of the products or nature of reaction with chemical reagents and acids.

Group-A Evolve the volatile products with acids (HCl or H_2SO_4)		Group-B Depend upon the reactions in solutions.	
Group-A ₁ Evolve gases or acid vapours with dilute HCl or H_2SO_4 , e.g., carbonate, sulphide, sulphate, nitrite, acetate, etc.	Group-A ₂ Evolve gases with H_2SO_4 , e.g., halides, nitrates, etc.	Group-B ₁ Give precipitation reaction, e.g., sulphate, phosphate, etc.	Group-B ₂ Give oxidation-reduction reaction in solution, e.g., manganate, chromate, etc.

38. A salt on reaction with dil. H_2SO_4 produces a colourless gas which turns lime water milky and an acidified orange coloured potassium dichromate solution green. The salt may be

- (a) A carbonate (b) A sulphite
(c) A sulphide (d) Acetate

39. A solution of a salt in concentrated H_2SO_4 turns white paste of starch containing potassium iodide blue. the salt may be a:

- (a) Chloride (b) Carbonate
(c) Bromide (d) Acetate

40. Which of the following statements is incorrect?

- (a) Metal iodide with excess chlorine water forms colourless iodic acid.
(b) Nitrates on gentle warming with aluminium powder and sodium hydroxide produces ammonia gas.
(c) Mercury nitrate solution produces yellow precipitate of basic mercury (II) sulphate with sodium sulphate solution.
(d) A deep red solution of complex ion $[\text{Fe}_3(\text{OH})_2(\text{CH}_3\text{COO})_6]^+$ on boiling with water produces a greenish-yellow precipitate.

Comprehension-3

The reagents like silver nitrate, potassium ferrocyanide, potassium thiocyanate, potassium iodide, potassium chromate. Nessler's reagent, etc., find extensive and very important applications in qualitative analysis because these reagents form different types of products with different cations. For example, potassium iodide forms yellow precipitate with Pb^{2+} but it forms red precipitate with Hg^{2+} . Hence, these reagents are widely used in the qualitative analysis of inorganic salts:

41. Which of the following is not correctly matched?

- (a) $[\text{A}(\text{OH})_4]^- (\text{aq}) + \text{NH}_4^+ (\text{aq}) \rightarrow$ White precipitate and liberation of ammonia
(b) $\text{Pb}^{2+} (\text{aq}) + 2\text{Br}^- (\text{aq}) \rightarrow$ Yellow precipitate
(c) BiI_3 (black precipitate) + $\text{H}_2\text{O} (\ell) \rightarrow$ Orange precipitate
(d) $\text{Ca}^{2+} (\text{aq}) + \text{K}_4[\text{Fe}(\text{CN})_6] (\text{aq}) \rightarrow$ White precipitate.

42. Which of the following cations (i.e., basic radical) forms coloured (not white) precipitates with aqueous solution of potassium iodide but precipitate does not dissolve in excess of reagent?

- (a) Hg^{2+} (b) Hg_2^{2+}
(c) Bi^{3+} (d) Cu^{2+}

43. Which of the following hydroxide does not dissolve in ammonia solution but dissolves in sodium hydroxide?

- (a) $\text{Zn}(\text{OH})_2$ (b) $\text{Cd}(\text{OH})_3$
(c) $\text{Cu}(\text{OH})_2$ (d) $\text{Al}(\text{OH})_3$

Comprehension-4

A chemist opened a cupboard and found four bottles containing water solutions, each of which had lost its label. Bottles 1, 2, 3 contained colourless solutions, while bottle 4 contained a blue solution. The labels from the bottles were lying scattered on the floor of the cupboard. They were:

Copper (II) sulphate, hydrochloric acid
Lead nitrate, sodium carbonate

By mixing samples of the contents of the bottles, in pairs, the chemist made the following observations:

- Bottle 1 + Bottle 2 $\rightarrow$ White precipitate is formed.
Bottle 1 + Bottle 3 $\rightarrow$ White precipitate is formed.
Bottle 1 + Bottle 4 $\rightarrow$ White precipitate is formed.
Bottle 2 + Bottle 3 $\rightarrow$ Colourless and odourless gas is evolved.
Bottle 2 + Bottle 4 $\rightarrow$ No visible reaction is observed.
Bottle 3 + Bottle 4 $\rightarrow$ Blue precipitate is formed.

With the help of the above observations, answer the following questions.

44. Bottle 3 contains:

- (a) Copper (II) sulphate
(b) Hydrochloric acid
(c) Lead nitrate
(d) Sodium carbonate.

45. Which of the following statements is correct for salts contained in bottle 1 and 4?

- (a) Bottle 4 gives white precipitate with excess of KI solution.
(b) Bottle 4 gives white precipitate with excess of $\text{K}_4[\text{Fe}(\text{CN})_6]$ solution.
(c) Bottle 1 and 4 both gives precipitate with excess of NaOH solution.
(d) Bottle 1 gives white precipitate with concentrated HCl solution.

46. Which of the following bottles develops intensive deep blue colour with aqueous ammonia?

- (a) Bottle 1 (b) Bottle 2
(c) Bottle 3 (d) Bottle 4

47. Which of the following will give white precipitate with bottle 1?

- (a) Bottle 2 (b) Bottle 3
(c) Bottle 4 (d) All of these

MATCH THE COLUMN TYPE QUESTIONS

48. Match the following:

Column-I

(Radicals)

- (A) Oxalate
(B) Acetate
(C) Sulphide
(D) Thiosulphate

Column-II

(Test reagent)

- (P) Sodium nitroprusside
(Q) Conc. H_2SO_4
(R) Neutral FeCl_3
(T) Dil. H_2SO_4

49. Column-I contains some reagents which when taken in excess produce precipitate with radical(s) given in column-II. Accordingly match the column-I with column-II.

Column-I

(Reagent)

- (A) KCN (aq)
(B) NaOH (aq)
(C) KI (aq)
(D) K_2CrO_4 (aq)

Column-II

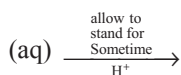
(Radical)

- (P) Pb^{+2}
(Q) Ag^+
(R) Hg_2^{2+}
(T) Cu^{2+}

50. Match the reactions listed in column-I with the colour of the precipitate(s) listed in column-II.

Column-I

- (A) $\text{Cu}^{2+}(\text{aq.}) + 2\text{SCN}$



- (B) $\text{Hot Bi}^{3+}(\text{aq.}) + \text{C}_6\text{H}_3(\text{OH}_3)^{\rightarrow}$

- (C) $\text{Ag}^+(\text{aq.}) \text{AsO}_4^{3-}(\text{aq.})^{\rightarrow}$

- (D) $\text{Mn}^{2+}(\text{aq.}) + 2\text{OH}^-(\text{aq.}) \xrightarrow{\text{air}}$

Column-II

- (P) Brown

- (Q) Brownish-red

- (R) Yellow

- (S) White

51. Match the particulars given in column-I with the appropriate chemical reagents given in column-II

Column-I

- (A) Fe^{3+} , Zn^{2+} and Cu^{2+} can be separated by
(B) PbS , CuS and CdS dissolve in
(C) Pb^{2+} gives yellow precipitate with
(D) $[\text{Ag}(\text{NH}_3)_2\text{Cl}]$ gives black precipitate with

Column-II

- (P) KI solution
(Q) Alkaline Na_2SnO_2 solution
(R) 50% HNO_3
(S) K_2CrO_4 solution
(T) Aqueous NH_3

52. Match the precipitates listed in of the compounds listed in-I with the solvent(s) listed in column-II.

Column-I

- (E) $\text{Zn}(\text{OH})_2$ precipitate dissolves in
(F) $\text{Cr}(\text{OH})_3$ precipitate dissolves in
(G) AgCl precipitate dissolves in
(H) CuS precipitate dissolves in

Column-II

- (P) Potassium cyanide
(Q) Ammonia
(R) Sodium hydroxide
(S) Sodium peroxide

SUBJECTIVE TYPE QUESTIONS

Short Subjective Questions

53. $\text{Fe}^{2+}(\text{aq}) + \text{NO}_3^-(\text{aq}) + \text{H}_2\text{SO}_4(\text{conc.}) \rightarrow$ Brown ring
The brown ring is due to the formation of complex, $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$. What is the oxidation state of iron in the complex?
54. A solution of Hg^{2+} ion on treatment with a solution of cobalt (II) thiocyanate gives rise to a deep blue

crystalline precipitate. The coordination number of mercury in the deep blue coloured compound is:

Long Subjective Questions

55. A mixture of two inorganic salts gives following chemical reactivity:
- (i) Mixture on reaction with dil. H_2SO_4 produces a colourless suffocating gas which turns acidified potassium dichromate solution green.

- (ii) Mixture on reaction with dil. H_2SO_4 gives reddish brown gas and on adding paper pellets the evolution of reddish brown vapours becomes brisk.
- (iii) Original solution of mixture gives white precipitate with dil. HCl which is soluble in ammonium hydroxide forming a soluble complex. The soluble complex gives white precipitate with HNO_3 and yellow precipitate with KI solution.
- (iv) The filtrate obtained after separating the white precipitate also gives white precipitate with H_2S in presence of NH_4OH and NH_4Cl . The solution of white precipitate in dil. HCl again gives white precipitate with excess of potassium ferrocyanide solution. Identify the cations and anions present in the mixture.

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Solutions

PERIODIC TABLE AND PERIODICITY IN PROPERTIES

1. (b) Ionic radius of trivalent lanthanide's almost remains constant with increase in the atomic number.
2. (c)
3. (b) Atomic radius increases as no. of shells increases.
4. (b) I.E.(II) of Na is higher than that of Mg because in case of, the second has to be remove from the noble gas core while in case of removal of second gives a noble gas core, has high first ionization potential than because of its stable configuration.
5. (a) The ionization energy of hydrogen is too high for group of alkali metals, but too low for halogen group.
6. (a) The addition of second electron in an atom or ion is always endothermic.
7. (a) Electron affinity value of Cl is greater than F and then decreases down the group.
8. (b) $B < C < N < O$; When we move from B to O in a periodic table the first ionization enthalpy increase due to the attraction of nucleus towards the outer most of electron.
9. (d) As 'Q' is noble gas, P, R and S having atomic number $Z - 1$, $Z + 1$ and $Z + 2$ should belong to halogen, alkali metal and alkaline earth metal respectively.
As halogen has one electron less than stable noble gas configuration it has greater tendency to accept an additional electron forming anion. Alkaline earth metal having valence shell configuration ns^2 exists in +2 oxidation state.
10. (d) (a) Mg^{2+} , Na^+ and F^- are isoelectronic and thus follows the order $_{12}Mg^{2+} < _{11}Na^+ < _9F^-$.
Al belongs to third period and has no charge so it is largest.
 $Na^+ = 102 \text{ pm}$; $Mg^{2+} = 72 \text{ pm}$; $Al = 143 \text{ pm}$, $F^- = 133 \text{ pm}$.
(b) K^+ has more number of shells than Mg^{2+} and Al^{3+} and Mg^{2+} are isoelectronic but Al^{3+} has higher nuclear charge so $Al^{3+} < Mg^{2+}$. Mg^{2+} and Li^+ has diagonal relationship. But due to +2 charge in Mg^{2+} , the Mg^{2+} is smaller than Li^+ . Hence Al^{3+} is the smallest one.
 $K^+ = 1.38 \text{ \AA}$, $Li^{2+} = 0.76 \text{ \AA}$, $Mg^{2+} = 0.72 \text{ \AA}$ and $Al^{3+} = 0.535 \text{ \AA}$.
(c) As the number of electrons are lost, the attraction between valence shell electrons and nucleus increases. As a consequence of this the electrons are pulled closer to the nucleus leading to the contraction in size of ions.
(d) Across the period the nuclear charge increases and thus the size of atoms decreases. $Mg = 160 \text{ pm}$; $Al = 143 \text{ pm}$; $Si = 118$; $P = 110 \text{ pm}$.
11. (c) (a) Down the group size increases and therefore, ionization energies decrease. Hence the order is correct.
(2) Cation is smaller but anion is bigger than its parent atom. As charge on cation increases the ionic radius decreases. Similarly as charge on anion increases the ionic radius increases.
$$IE \propto \frac{1}{\text{atomic/ionic radius}}$$
 Hence the order is correct.
12. (d) Due to small size of F atom, the electron-electron repulsions in compact 2-p sub shell are large and hence the incoming electron is not accepted with the same ease as is the case with Cl (less electron-electron repulsions)
13. (b) (a), (c) and (d) are correct statements
(b) First decreases from B to Al and then increases marginally owing to discrepancies in atomic size of the element.
14. (a) As non-metallic character of element attached to oxygen atom increases, the difference between the electronegativity values of element and oxygen decreases and the acid character of oxides increases and vice versa.
15. (d) (a) As electronegativity increases the non-metallic character increases as tendency to form anion increases.

- (b) It is based on their SRP values. (Oxidizing power may be cumulative effect of hydration energies, electronegativities, bond dissociation energies and electron gain enthalpies)
- (c) $C = -121$; $Si = -135$; $P = -60$; $N = +31$ (all values are in KJ/mole). It depends on various factors like size of atom, nuclear charge, partially filled, half filled and completely filled electronic configurations.
16. (a, b) Due to screening effect of inner electrons, the force of attraction between nucleus and valence electrons decreases, which results in an increase in atomic size and decrease in ionisation energy.
17. (a, c) Metallic hydroxide, basic CsOH, non-metallic hydroxide acidic HBr, sulphur is a good oxidising agent.
- $$S + 2e^- \rightarrow S^{2-}$$
18. (a, c) $IE_1(Al) < IE_1(Mg)$ [due to stable configuration of Mg]
After ionisation, configuration:
 $Al[Ne] 3s^2 3p^0$ $Mg[Ne] 3s^1$
Now, Al has stable configuration and higher effective nuclear charge, that's why
 $IE_{11}(Al) > IE_{11}(Mg)$
 Sc^{3+} and K^+ are isoelectronic species.
 $Sc^{3+} < K^+$
19. (a, b, c, d) All are obvious facts.
20. (c, d)
Generally, the electron affinity values of 3rd period elements are higher than the elements of 2nd period of the same group because of less interelectronic repulsions. Atomic size, nuclear charge and electronic configurations also affect the electron configurations also affect the electron affinity.
Electron gain enthalpies in kJ mol^{-1} are:
 $C = -121$; $Si = -134$; $P = -74$; $N = 0$; $O = -142$; $F = -328$
21. (b, d)
(b) Addition of second electron to O^- is opposed by electrostatic repulsion due to same charge. Hence energy is given for the addition of 2nd electron.
(d) Ar has stable electronic configuration ($ns^2 np^6$). Hence energy has to be given to add an extra electron to form Ar^-
22. (a, b, c, d)
(a) As screening effect increases, effective nuclear charge decreases thus valence shell electron is loosely bound. Hence I.E decreases.
(b) Be and Mg has ns^2 configuration (stable configuration).
- (c) Due to lanthanide contraction.
- (d) $r_{\text{metallic}} > r_{\text{covalent}}$ (covalent bond formation involves the overlapping of orbitals).
23. (a, b, c, d)
(a) Oxygen atom is smaller as compared to sulphur; so in oxygen there is more inter-electronic repulsion than that of sulphur.
(b) Charge increases so size decreases and number of electrons per proton also decreases; so valence shell electron gradually becomes more tightly held by the nucleus.
(c) As = valence shell electron configuration is $ns^2 np^3$ (half filled stable configuration). Se = valence shell electron configuration is $ns^2 np^4$ (partially filled less stable configuration).
As = 0.947 MJ/mole ; Se = 0.941 MJ/mole .
(d) Down the group atomic size increases. Fluorine experiences more inter electronic repulsion than that of chlorine; so fluorine has less electron affinity than that of chlorine.
24. (d) All noble gases have stable configuration. Therefore, they can not take any electron means that they have no affinity for electrons. High electron affinity shows that electron is strongly bonded to the atom. Here both assertion and reason are false.
25. (a)
26. (c) Be has completely filled stable $2s^2$ orbital and thus Be has higher ionisation energy than B. $2s$ orbital has less energy than orbital.
27. (a) Both statements are true and Statement-2 is the true explanation of the Statement-1. The magnitude of an element's electron affinity depends on the element's valence shell electrons configuration.
 ${}^{25}_{25}\text{Mn} = [\text{Ar}]^{18}3d^5 4s^2$ configuration,
 ${}^{25}_{25}\text{Cr} = [\text{Ar}]^{18}3d^5 4s^2$ configuration
 ${}^{25}_{25}\text{Fe} = [\text{Ar}]^{18}3d^6 4s^2$ configuration
28. (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
29. (b) 3. The size of the isoelectronic species is not affected by electron-electron interaction in the outer orbitals but is affected by the nuclear charge.
1, 2 and 4 are correct statements.
30. (c) 1. $\text{Na}_2\text{O}_2 < \text{MgO} < \text{ZnO} < \text{P}_4\text{O}_{10}$: as non-metallic character increases the acidic character increases. MgO , Na_2O are basic, ZnO amphoteric and P_4O_{10} acidic.

2. $\text{Na} < \text{Si} > \text{Al} < \text{Mg}$: Mg has higher than Na due to small size and higher nuclear charge. Mg has higher than Al because of ns^2 configuration (has extra stability and high electron penetration power of s-subshell electrons) and Si has higher than Al because of higher nuclear charge and small size. IE_1 : $\text{Na} = 496$, $\text{Al} = 577$, $\text{Mg} = 737$ and $\text{Si} = 786$ kJ/mole
3. There is more interelectronic repulsion in 2p-subshell of fluorine than chlorine (3p). So extra electron will be added easily in 3-subshell of chlorine as compared to 2p-subshell of fluorine. Down the group electron affinity values generally decreases with increasing atomic number due to increase in atomic size, So $\text{Cl} > \text{F} > \text{Br}$.
4. Isoelectronic series of ion; all have the xenon configuration.

$$\text{Ionic radius} = \frac{1}{\text{nuclear charge}}$$

Atomic number: $\text{Te} = 52$; $\text{I} = 53$; $\text{Cs} = 55$; $\text{Ba} = 56$.

31. (d) As ionization energy decreases electropositive character increases and thus the tendency to form cation increases. Therefore, metallic character increases.
32. (a) On the basis of their SRP values.
33. (b) As ionisation energies decrease down the group the reducing character increases and as across the period, ionization energies increase, so reducing character, decreases.
34. (d) As electronegativity difference between element and oxygen decreases the acidic character of oxides increases. The electronegativity also increases with increasing oxidation states. Hence, (d) is the correct
35. (a) Ionic size $\propto \frac{1}{\text{nuclear charge}}$
36. (c) Both (a) and (b) are correct statements.
37. (a) Covalent bond formation involves the overlapping of atomic orbitals and in doing this some part of the orbitals becomes common. So $r^{\text{covalent}} < r^{\text{metallic}}$
38. (d) Value corresponds to Noble gas. Hence, T is inert toward chemical reactivity.
39. (b) Metal having low IE_1 and higher IE_2 corresponds to alkali metal. Hence, Q is most reactive metal (easily forms cation).

40. (a) Non-metal having higher negative value of electron gain enthalpy corresponds to halogen. Hence R is more reactive non-metal (easily forms anion)
41. (a) In a period the IE increases due to decrease in atomic size and increase in nuclear charge. But IE_1 of Mg is greater than Al due to more penetration power of s-subshell electrons and stable $3s^2$ configuration (completely filled s-orbital)
42. (d) All statements are correct.
43. (d) Orbitals with lower value of n will be more closer to the nucleus and thus electrons will experience greater attraction from nucleus and so its removal will be difficult not easier.
44. (a) $\Delta H_{\text{eg}} \text{ kJ mol}^{-1} \rightarrow = -349$; $\text{F} = -328$; $\text{S} = -200$; $\text{O} = -141$
45. (a) P, R, S (b) P, R, S
(c) R (d) P, S
- (a) E.N. $\uparrow$ and O.S. $\uparrow$ because attraction of shared pair electron $\uparrow$ E.N. difference between $\text{E} - \text{O} \uparrow$ basic character $\uparrow$. Attraction for electrons $-\uparrow \Delta_{\text{eg}} H \uparrow$. Degree of hydration $\propto$ charge on cation and $\propto 1/\text{size of cation}$.
- (b) Size $\downarrow$ E.N. $\downarrow$; ΔEN of $\text{E} - \text{O} \downarrow$ basic character $\uparrow$ but Al_2O_3 is amphoteric, $-\Delta_{\text{eg}} H \uparrow$ with size $\downarrow$.
- (c) Same concepts are applied; acidity of their oxides seems to be nearly same so can not predict their relative strength.
- (d) —Do—; N has half-filled stable configuration So $-\Delta_{\text{eg}} H$ will be $\downarrow$ than C.
46. (a) R (b) P
(c) T (d) S
(e) Q
- (a) Have same number of electrons — So isoelectronic species.
- (b) Has metallic as well as non-metallic properties — semi metal
- (c) Exist as mono atomic molecules and are held together by weak van der Waal's forces — so radius is defined as van der Waal's radius.
- (d) Energy required to remove an electron from valence shell of a gaseous atom is called I.E.
- (e) Energy released when an extra electron is added to valence shell of an isolated gaseous atom to form an anion is called electron affinity.
47. (A-Q); (B-P, T); (C-S); (D-R)
- (a) inert gases exist as monoatomic molecule. They have highest IE_1 and IE_2 and positive electron gain enthalpies.

- (b) Alkali metal- IE_1 is low because of bigger size of atom and IE_2 is high due to noble gas configuration. As it has low IE_1 , it can lose one electron easily and thus more electropositive. So it acts as strong reducing agents and their oxides are basic in nature. They have low negative value of ΔH_{eg} because of large size.
- (c) IE_1 and IE_2 both high and negative value of electron gain enthalpy is very high because of high nuclear charge and small size of atoms, so it has a greater tendency to accept an additional electron. So it (F) acts as a strong oxidising agent.
- (d) As it has less negative value for ΔH_{eg} then (c) thus it will be least reactive non-metal (I).

48. (A-R); (B-Q, R); (C-S); (D-P, S)

- (a) For isoelectronic species, the ionic size decreases with increase in nuclear charge. Hydration $\propto$ charge on anion and heavier hydrated ions move slowly. Number of atomic shells increases, ionic size increases.

- (b) heavier hydrated ions move slowly. Number of atomic shells increases, ionic size increases.
- (c) correct order; as Cl has less inter electronic repulsions than F due to bigger size of 3p-sub-shell.
- (d) Oxidation state increases, the electronegativity increases. For isoelectronic species ionisation energy and electron affinity increases with increasing nuclear charge.

49. Last electron enters in d-subshell so it belongs to d-block elements.

For d-block elements group number

$$= \text{no of electrons in } (n-1) \text{ d subshell} + \text{number of electrons in valence shell (i.e., ns)} \\ = 7 + 2 = 9.$$

50. In M^{4+} ion total number of electrons = 20

$$\text{Electronic configuration} = \frac{1s^2}{2} \frac{2s^2 2p^6}{8} \frac{3s^2 3p^6 3d^2}{10}$$

$$\text{Number of electrons in metal} = \text{number of protons} \\ = 4 + 20 = 24$$

CHEMICAL BONDING

1. (c) C_2H_2 has linear structure because carbons are sp -hybridised and lies at 180° .

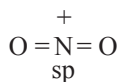
2. (b) Bond order of O_2^+ is highest so its bond length is smallest.

$$3. (a) B.O. = \frac{1}{2}[N_b - N_a] \quad N_2 = \frac{1}{2}[10 - 4] = \frac{6}{2} = 3;$$

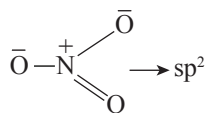
$$O_2^+ = \frac{1}{2}[10 - 4] = \frac{6}{2} = 3.$$

4. (b) As anion is same, the covalent character in ionic compound will depend upon the size and charge on the cation. According to fajan's rule, PbO (most ionic) = Pb^{2+} , Mn_2O_7 (most covalent) = Mn^{+7}

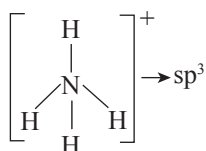
5. (b) NO_2^+ Number of electron pairs = 2
 Number of bond pairs = 2
 Number of lone pair = 0
 So, the species is linear with sp hybridisation.



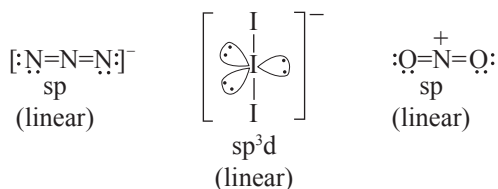
NO_3^- Number of electron pairs = 3
 Number of bond pairs = 3
 Number of lone pair = 0
 So, the species is trigonal planar with sp^2 hybridisation.



NH_4^+ Number of electron pairs = 4
 Number of bond pairs = 4
 Number of lone pair = 0
 So, the species is tetrahedral with sp^3 hybridisation.

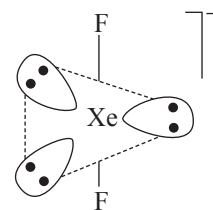


6. (b)

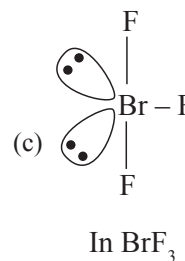
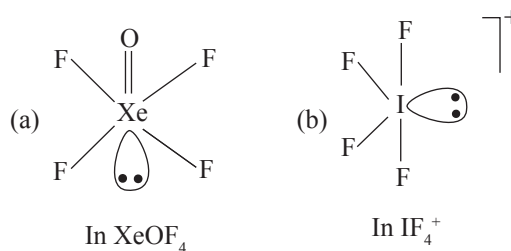


7. (a) $BF_3 (sp^2) + F \rightarrow BF_4^- (sp^3)$.

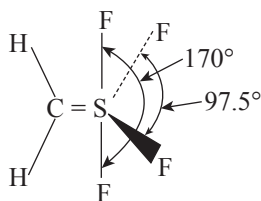
8. (c) According to VSEPR theory
 Number of valence electron pairs = 5
 Number of bond pairs = 2
 Number of lone pairs = 3
 Hence structure is linear



hybridization = sp^3d



9. (a) The hydrogen atoms are in a vertical plane with the axial fluorine atoms.



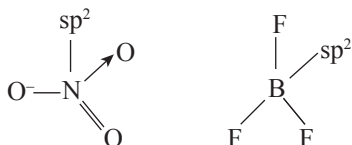
Hydrogen atoms lie in the CSF_2 axial plane. We know that the p bond involving a p-orbital on the carbon atom must lie in the equatorial plane of the molecule. And the resulting repulsion between the p electrons and the electron pair bonding the equatorial fluorine atoms is dramatic; The $\text{F}_{\text{eq}} - \text{F}_{\text{eq}}$ angle has been reduced to 97.5° .

10. (d) (a) $\text{NO}_2^+ = 7 + 16 - 1 = 22$; $\text{CO}_2 = 6 + 16 = 22$. So isoelectronic (both have 22 electrons in valence shells)

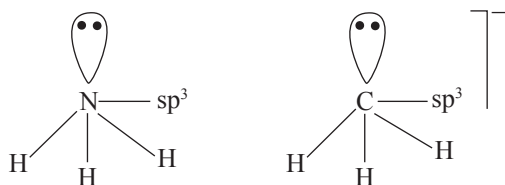
$\text{O} = \overset{+}{\text{N}}_{\text{sp}} = \text{O}$, $\text{O} = \text{C} = \text{O}$, both are linear so isostructural.

- (b) $\text{NO}_3^- = 7 + 24 + 1 = 32$; $\text{BF}_3 = 5 + 27 = 32$.

So isoelectronic and are isostructural as both have trigonal planar structures.

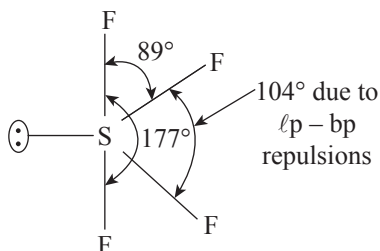


- (c) $\text{NH}_3 = 7 + 3 = 10$ $\text{CH}_3^- = 6 + 3 + 1 = 10$
So isoelectronic.



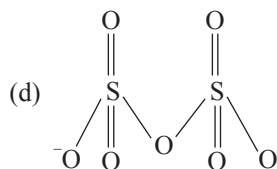
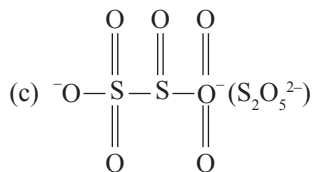
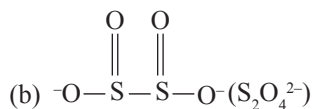
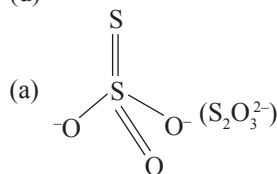
Both are pyramidal so isostructural.

11. (d)



Remaining molecules/ions have symmetrical structures; so have all bonds equal.

12. (d)



In structure $\text{OS S}_2\text{O}_7^{2-}$ there is one S-O-S bond but no S-S bond.

13. (b) (a) $\text{N}_2 = \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2$
 $= \pi 2p_y^2 \sigma 2p_z^2$

$$\text{B.O.} = \frac{10-4}{2} = 3; n = 0 \text{ (D)}$$

$$\text{N}_2^+ = \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2 = \pi 2p_y^2 \sigma 2p_z^1$$

$$\text{B.O.} = \frac{9-4}{2} = 2.5; n = 1 \text{ (P)}$$

- (b) $\text{O}_2^+ = \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 2p_z^2 \pi 2p_x^2$
 $= \pi 2p_y^2 \pi^* 2p_x^1 = \pi^* 2p_y^0$

$$\text{B.O.} = \frac{10-5}{2} = 2.5; n = 1 \text{ (P)}$$

$$\text{O}_2^{2+} = \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 2p_z^2 \pi 2p_x^2$$

$$= \pi 2p_y^2 \pi^* 2p_x^0 = \pi^* 2p_y^0$$

$$\text{B.O.} = \frac{10-4}{2} = 3; n = 0 \text{ (D)}$$

- (c) $\text{B}^2 = \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2 = \pi 2p_y^1$

$$\text{B.O.} = \frac{6-4}{2} = 1; n = 2 \text{ (P)}$$

$$\text{B}^{2+} = \frac{10-4}{2}$$

$$\text{B.O.} = \frac{5-4}{2} = 1/2; n = 1 \text{ (P)}$$

$$(d) \text{NO}^- = \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 2p_z^2 \pi 2p_x^2$$

$$= \pi 2p_y^2 \pi^* 2p_x^1 = \pi^* 2p_y^1$$

$$\text{B.O.} = \frac{10 - 6}{2} = 1.0; n = 2 \text{ (P)}$$

$$\text{NO} = \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 2p_z^2 \pi 2p_x^2$$

$$= \pi 2p_y^2 \pi^* 2p_x^1 = \pi^* 2p_y^0$$

$$\text{B.O.} = \frac{10 - 5}{2} = 2.5; n = 1 \text{ (P)}$$

14. (b) Strength of H-bonds on following factors.

- Electronegativity of element covalently bonded to hydrogen atom.
- Size of electronegative element.
- Ease of donation of lone pair of electrons by electronegative element.

15. (c) Number of polarisable electrons $\propto$ strength of vander waal's force $\propto$ boiling point. boiling point $\text{CHCl}_3 = 61^\circ\text{C}$ and $\text{CCl}_4 = 73^\circ\text{C}$.

16. (a, b, d)

$$\text{Steric number of } \text{ClF}_3 = 3 + 2 = 5;$$

$$\text{Steric number of } \text{ClF}_3\text{O} = 4 + 1 = 5;$$

$$\text{Steric number of } \text{ClF}_3\text{O}_2 = 5 + 0 = 5;$$

$$\text{Steric number of } \text{ClF}_2\text{O}^+ = 3 + 1 = 4;$$

$$\text{Steric number of } \text{ClF}_4\text{O}^- = 5 + 1 = 6;$$

$$\text{Steric number of } \text{XeOF}_4 = 5 + 1 = 6;$$

So answer are A, B, D as steric no. are same for these molecules and hybridisation is same in each pair.

17. (b, c)

In O_2 , MO electronic configuration in increasing energy order is

$$\sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \sigma 2p_z^2 < \pi 2p_y^2$$

$$< \pi^* 2p_x^1 = \pi^* 2p_y^1$$

18. (a, c)

For sp^3d^2 steric no. is 6

$$\text{Steric no. of } \text{IF}_5 = 5 + 1 = 6;$$

$$\text{Steric no. of } \text{XeF}_4 = 4 + 2 = 6;$$

$$\text{Steric no. of } \text{XeOF}_4 = 5 + 1 = 6;$$

$$\text{Steric no. of } \text{ICl}_4^- = 4 + 2 = 6$$

But XeF_4 and ICl_4^- dipole moment is zero and these are nonpolar.

19. (c, d)

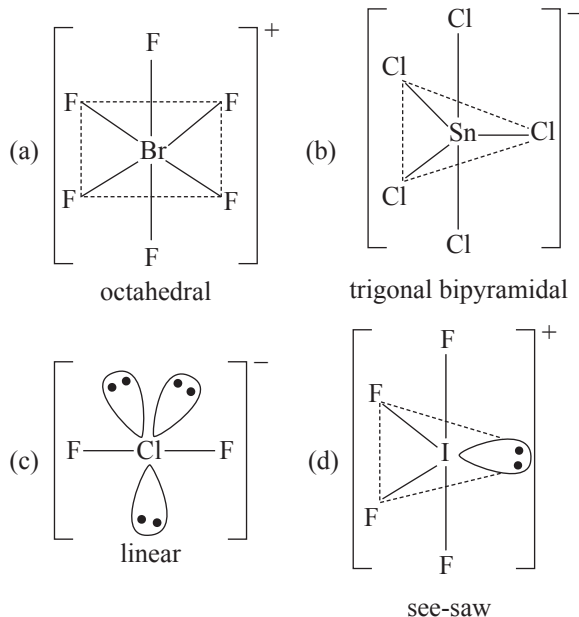
Enthalpy of vapourisation

SnH_4 , GeH_4 , SiH_4 , CH_4 (decreasing order)

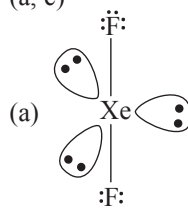
Melting point.

NH_4 , SbH_4 , AsH_3 , PH_3 (decreasing order)

20. (a, b, c, d)



21. (a, c)

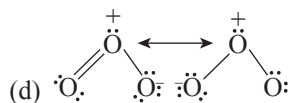


(b) Oxygen can not exceed its valence as it does not have vacant d-orbital.

$$(c) \text{NO}^+; \text{Bond order} = \frac{10 - 4}{2} = 3$$

$$\text{NO}^-; \text{Bond order} = \frac{10 - 5}{2} = 2.5$$

Bond order $\propto$ bond strength.



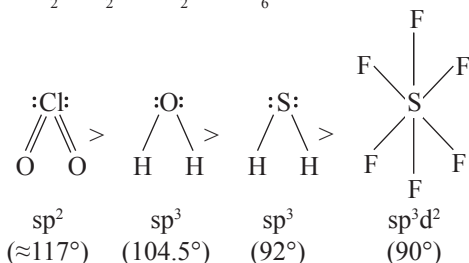
22. (a, b, d)

(a) According to VSEPR theory, as electronegativity of central atom decreases, bond angle decreases, so bond angle of $\text{H}_2\text{O} > \text{H}_2\text{S} > \text{H}_2\text{Se} > \text{H}_2\text{Te}$

(b) $\text{C}_2\text{H}_2 > \text{C}_2\text{H}_4 > \text{CH}_4 > \text{NH}_3$. In NH_3 there is bp-ip repulsion so bond angle decreases to 107° from 109.5° .

$\text{sp} \quad \text{sp}^2 \quad \text{sp}^3 \quad \text{sp}^3$
 $180^\circ \quad 120^\circ \quad 109^\circ \quad 107^\circ$

(c) $\text{SF}_6 < \text{NH}_3 < \text{H}_2\text{O} < \text{OF}_2$ in this case bond angle of NH_3 is highest because lp-lp repulsion is absent in it.

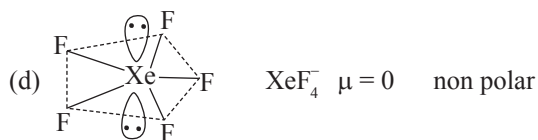
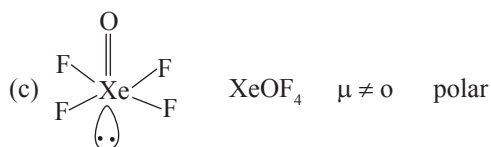
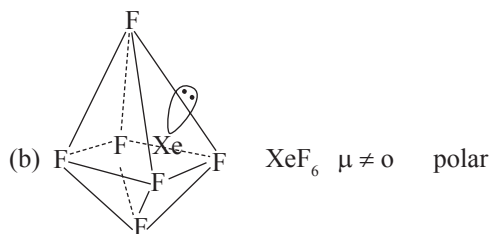
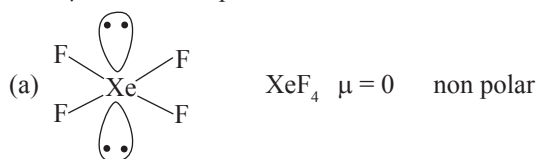


Note: It is supposed that in H_2S the hybrid orbitals do not participate in bonding but pure p-atomic orbitals participate in bonding.

23. (a, c) (a) $N_2 = \frac{10-4}{2} = 3.0$; $N_2^+ = \frac{9-4}{2} = 2.5$
 (c) $N_2 = \frac{10-4}{2} = 3.0$; $N_2^- = \frac{10-4}{2} = 2.5$

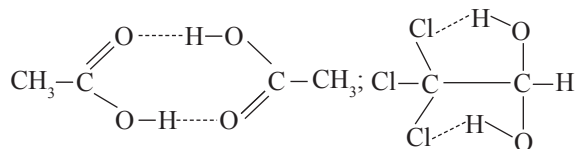
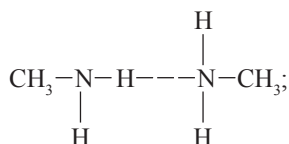
24. (b, c)

Polarity depends on net dipole moment. If dipole moment $\mu = 0$ it is nonpolar.

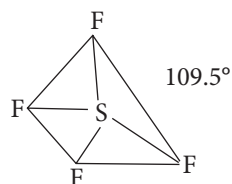


25. (a, c, d)

Hydrogen bonding is possible in CH_3NH_2 , CH_3COOH , $\text{CCl}_3\text{CH}(\text{OH})_2$



26. (c) SiF_4 has sp^3 hybridization and a shape of regular tetrahedral where the bond angle of $\text{F}-\text{Si}-\text{F}$ are found 109.5° which is greater than 90° but less than 180° . Repulsion sequence is $\text{Lp} - \text{Lp} > \text{Lp} - \text{Bp} > \text{Bp} - \text{Bp}$ so assertions are true but the reasons are false.



27. (a) Transition metal ions have pseudo inert gas configuration which have higher polarizing power due to poor shielding of d-electrons. Alkali metal and alkaline earth metal ions have less polarizing power due to inert gas configuration (more shielding effect of s- and p-electrons to d-electrons).

28. (b) Both S-1 and S-2 are true statements but are different and different and reason is not true explanation of assertion. High electronegativity of atoms attached to carbon atom indirectly results in the contraction of the size of p-orbitals. As carbon atomic size is smaller and thus it has effective overlapping.

29. (a) Number of electrons $\text{NO}^+ = 7 + 8 - 1 = 14$
 Number of electrons $\text{CN}^- = 6 + 7 + 1 = 14$

Bond order = $\frac{10-4}{2} = 3$
 Both are diamagnetic.

30. (b) Within the $\text{Ca}^{2+}/\text{SO}_4^{2-}$ layers the ions are held together by strong electrovalent bonds. But these separated $\text{Ca}^{2+}/\text{SO}_4^{2-}$ layers are linked by relatively weak H-bonds. The weaker H-bonds link SO_4^{2-} ions in the intermediate region. Consequently, the gypsum can be readily cleaved and stretched along the layer of water molecules.

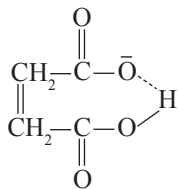
Anhydride has a completely ionic structure involving only Ca^{2+} and SO_4^{2-} ions.

31. (c) 1. $\text{OF}_2 < 109.5^\circ$ and $\text{OBr}_2 > 109.5^\circ$

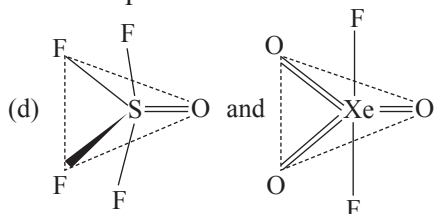
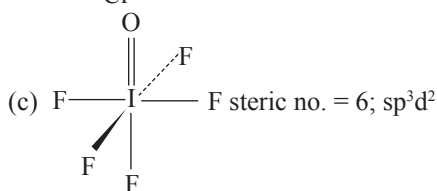
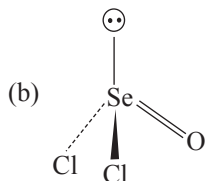
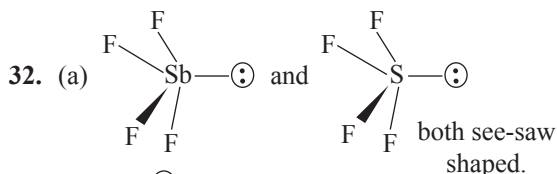
OF_2 : Fluorine being more electronegative attracts the bond pairs of electrons towards itself and thus there is more lp-lp repulsion decreasing the bond angle.

OBr₂: The bond pairs are close to more electronegative oxygen atom and thus there is repulsion between bp-bp increasing the bond angle. The bulkier Br atoms also create steric repulsion which also increase the bond angle.

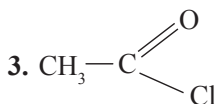
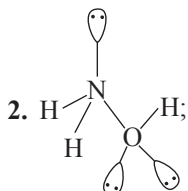
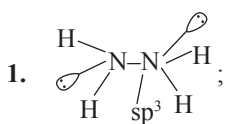
2. Anion of maleic acid (cis-form) gets stabilized through intramolecular hydrogen bonding after giving one protons as given in following structure. In fumaric acid (trans-form) no such intramolecular H-bonding is possible.



3. All have sp³ hybridization and one lone pair of electrons.



33. (c)



34. (b) 1. KHF₂, F—H—F; $\Delta H \approx -212 \text{ kJ mol}^{-1}$

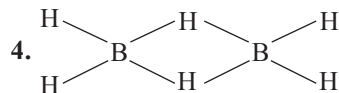
involves resonance $\text{H—F—H} \leftrightarrow \text{H—F—H}$.

HF, F—H—F; $\Delta H \approx -28.6 \text{ kJ mol}^{-1}$

2. CH₃Cl > CH₃F; due to large charge separation in CH₃Cl as compared to that in CH₃F.

CH₃F; $\mu = 1.85 \text{ D}$, CH₃Cl; $\mu = 1.87 \text{ D}$

3. In sp² hybrid orbitals character is 33.3% and p character is 66.6%



Terminal B—H bond are covalent while bridging B—H bonds are three centre-two electron bonds.

35. (d) Position of nuclei is changed in (d), that's why not considered as canonical structure.

36. (a) CH₂ = Fe⁺ is having more number of covalent bonds.

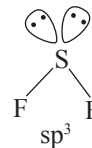
37. (c) I and II are fairly delocalized while IV and III are not localized fairly.

38. (a) As size of cation decreases and charge on cation increases, the polarization increases according to fajan's rule. Hence the covalent character increases.

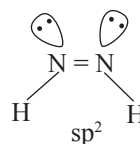
39. (c) As size of cation increases and charge on cation decreases, the polarization decreases according to Fajan's rule. Thus the ionic character increases. So SnO with +2 oxidation state will be most ionic.

40. (d) AgI shows highest polarization due to following two factors (a) Ag⁺ has pseudo inert gas configuration (a) I⁻ is biggest anion.

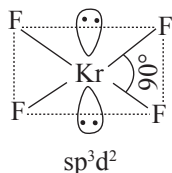
41. (c) Bond angle is less than 109°28' because of lp-lp repulsion as bond pairs are more close to fluorine atoms.



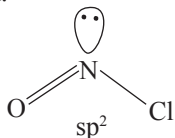
- (b) Bond angle is close to 120° because of sp² hybridization.



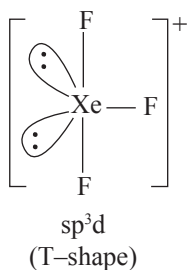
(c) KrF_4 has square planar shape.



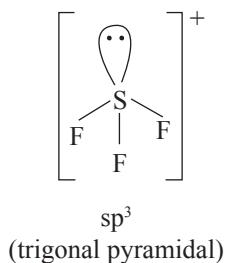
(d) Bond angle is less than 120° because of repulsion between double bond and lone pair. π -electrons of double bond create more repulsion than single congalent bond.



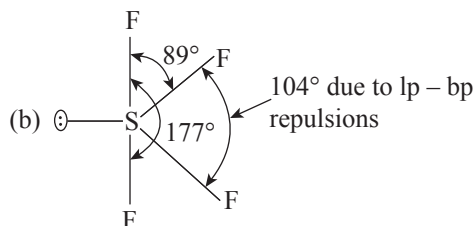
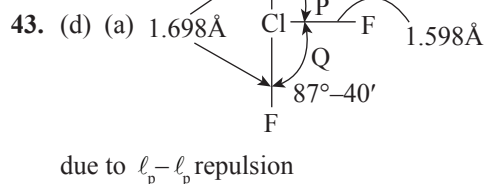
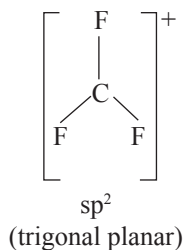
42. (b) (i)



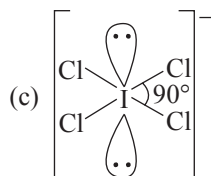
(ii)



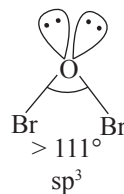
(iii)



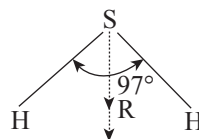
Adjacent bonds, thereby distorting the structure as donated oxygen atom require more space than a single bonding pair and repel.



(d) Bond angle in OBr_2 is more than OCl_2 due to lp-lp repulsion and steric hindrance of bulkier Br.



44. (b)



$$\mu_R = \sqrt{2\mu_{\text{S-H}}^2 + 2\mu_{\text{S-H}}^2 \times \cos 97^\circ}$$

$$\mu_R = \sqrt{2} \times \mu_{\text{S-H}} \times \sqrt{1 - 0.12}$$

$$\mu_R = \Delta$$

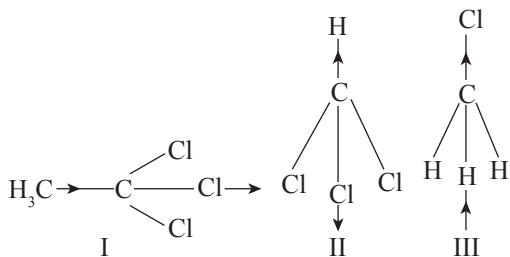
$$\mu_R = \frac{1.5}{\sqrt{2} \times 0.94} \text{ D}$$

$$(\mu_{\text{S-H}})_{\text{Cal}} = \frac{1.5 \times 10^{-9} \times 1.6 \times 10^{-19}}{\frac{10}{3} \times 10^{-30}}$$

$$(\mu_{\text{S-H}})_{\text{Cal}} = 3 \times 0.15 \times 1.6 \times 10 \text{ D}$$

$$\text{So \% Ionic charcter} = \frac{1.5}{\sqrt{2} \times 0.94} \times \left[\frac{1}{0.15 \times 16 \times 3} \right] \times 100 = 16 \%$$

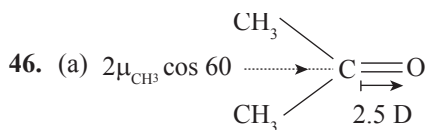
45. (d)



$$\mu_{\text{I}} = 3 \mu_{\text{C-Cl}} \times \cos 70.5 + \mu_{\text{CH}_3} = 1.5 + 0.4 = 1.9 \text{ D}$$

$$\mu_{\text{II}} = 3 \mu_{\text{C-Cl}} \times \cos 70.5 - \mu_{\text{C-H}} = 1.5 - 0.4 = 1.1 \text{ D}$$

$$\mu_{\text{III}} = 3 \mu_{\text{C-H}} \times \cos 70.5 + \mu_{\text{C-Cl}} = 1.9 \text{ D}$$



$$\mu_{\text{net}} = 2.5 + \mu_{\text{CH}_3} = 2.5 + 0.4 = 2.9 \text{ D.}$$

47. (A-P); (B-Q); (C-P); (D-S)

Steric no. of $\text{IOF}_2^- = 4 + 1 = 5$ sp^3d ;Steric no. of $\text{F}_2\text{SeO} = 3 + 1 = 4$, sp^3 ;Steric no. of $\text{ClOF}_3 = 4 + 1 = 5$, sp^3d ;Steric no. of $\text{XeF}_5^+ = 5 + 1 = 6$, sp^3d^2

48. (A-P, Q, R, S); (B-P); (C-S, R); (D-Q, S, R)

Dipole moment

	NH_3	SbH_3	AsH_3	PH_3
Electronegativity of central atom	3.0	1.9	2.0	2.1

Electronegativity of hydrogen	HF	HCl	HBr	HI
-2.1	$\text{max}^m \Delta E_n$		$\text{least } \Delta E_n$	
	most polar		least polar	
	$\text{CH}_4, \text{SiH}_4 \dots$			

Symmetrical tetrahedral structure.	H_2O	H_2S	H_2Se	H_2Te
		dipole moment decreasing		

Melting point:

$\text{NH}_3, \text{SbH}_3, \text{AsH}_3, \text{PH}_3$	HI, HF, HBr, HCl
$\text{SnH}_4, \text{GeH}_4, \text{CH}_4, \text{SiH}_4$	$\text{H}_2\text{O, H}_2\text{Te, H}_2\text{Se, H}_2\text{S}$

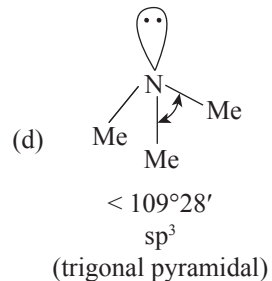
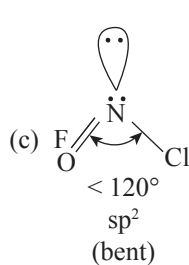
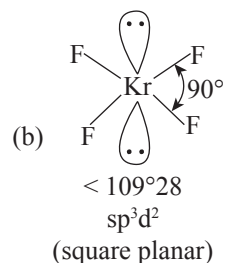
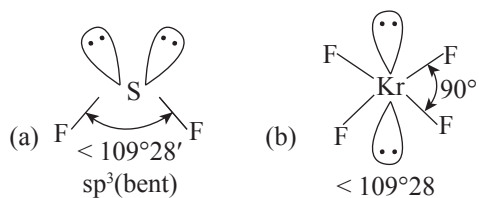
Boiling point:

$\text{NH}_3, \text{SbH}_3, \text{AsH}_3, \text{PH}_3$	HF, HI, HBr, HCl
$\text{SnH}_4, \text{GeH}_4, \text{SiH}_4, \text{CH}_4$	$\text{H}_2\text{O, H}_2\text{Te, H}_2\text{Se, H}_2\text{S}$

Enthalpy of vaporization

$\text{NH}_3, \text{SbH}_3, \text{AsH}_3, \text{PH}_3$	HF, HI, HBr, HCl
$\text{SnH}_4, \text{GeH}_4, \text{SiH}_4, \text{CH}_4$	$\text{H}_2\text{O, H}_2\text{Te, H}_2\text{Se, H}_2\text{S}$

49. (A-P, Q, R, S); (B-P, Q, R, T); (C-Q, S); (D-R)



50. (A-P, Q, R, S); (B-P, Q, R, S); (C-P, Q, R, T); (D-P, R, T)

(a) C_2 bond order = $\frac{8-4}{2} = 2$

All electrons are paired so diamagnetic.

Undergoes oxidation as well as reduction.

Mixing of 2s and 2p orbitals takes place.

(b) N_2 bond order = $\frac{10-4}{2} = 3$

All electrons are paired so diamagnetic.

Undergoes oxidation as well as reduction.

Mixing of 2s and 2p orbitals takes place.

(c) O_2^{2-} bond order = $\frac{10-8}{2} = 1$

All electrons are paired so diamagnetic.

Undergoes oxidation as well as reduction.

Mixing of 2s and 2p orbitals does not takes place.

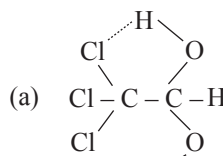
(d) F_2 bond order = $\frac{10-8}{2} = 1$

All electrons are paired so diamagnetic.

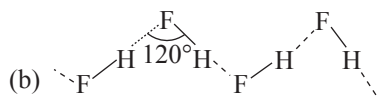
Undergoes reduction only.

Mixing of 2s and 2p orbitals does not takes place.

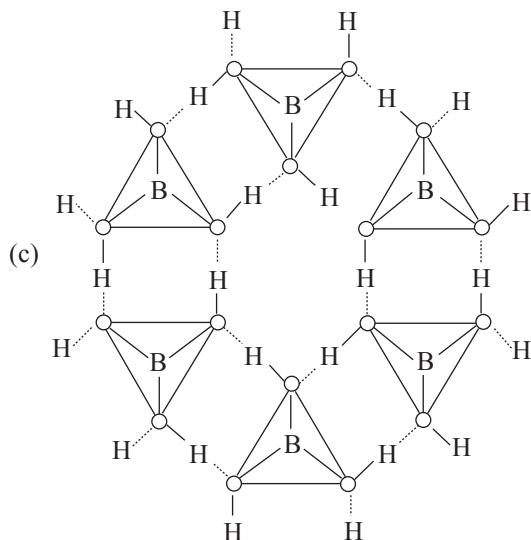
51. (A—S); (B-P, R, T); (C-Q, R, T); (D-R, T)



Chloral hydrate

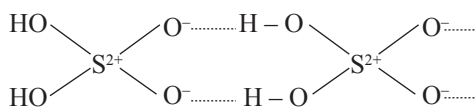


Hydrogen fluoride

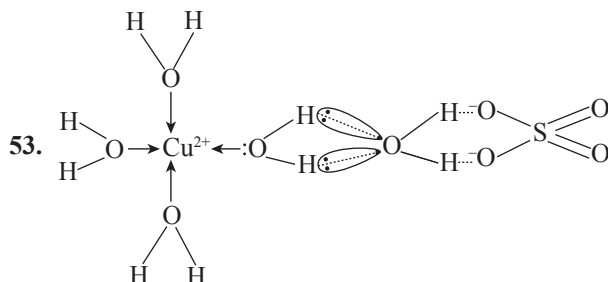
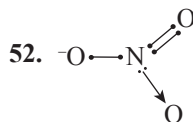


Boric acid

In the solid state, the B(OH)_3 units are hydrogen bonded together in two dimensional sheets with almost hexagonal symmetry. The layers are quite a large distance apart (3.18 Å).



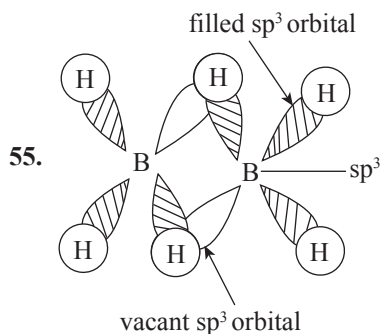
Note: HF, H_3BO_3 and H_2SO_4 have low volatility i.e., high boiling points due to intermolecular hydrogen bonding.



$$\begin{aligned} 54. N_{\text{cal.}} &= 4.8 \times 10^{-10} \text{ e.s.u.} \times 10^{-8} \text{ cm} \\ &= 4.8 \times 10^{-18} \text{ esu cm} \\ &= 4.8 \text{ D.} \end{aligned}$$

$$\therefore \% \text{ ionic character} = \frac{1.2}{4.8} \times 100 = 25\%$$

$$\therefore \% \text{ co-valent character} = 75\%$$



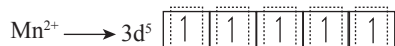
No of vacant sp^3 hybrid orbitals participating in the formation of banana bonds are 2.

COORDINATION COMPOUNDS

1. (a) EAN of Pt in $[\text{PtCl}_6]^{2-} = 80$
2. (c) EAN of a central metal ion = (atomic no. of central atom) – oxidation state + no. of ligands $\times 2$
 $= 26 - 3 + (6 \times 2) = 23 + 12 = 35$

3. (c) $[\text{CoF}_6]^{3-}$ is an outer complex having d^2sp^3 hybridization.

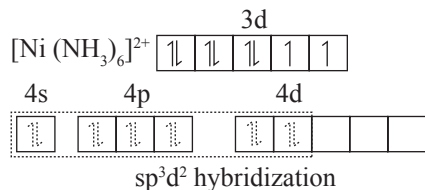
4. (a) $\text{Mn}^{25} \longrightarrow 3d^5 + 4s^2$



In presence of weak ligand field, there will be no pairing of electrons. So it will form a high spin complex. i.e., the number of unpaired electrons = 5.

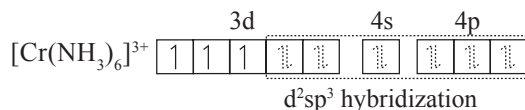
5. (c) Oxidation state of iron in $[\text{Fe}^{2+}(\text{CO})_2(\text{NO})_2] = x + 2(0) + 2 = 0$; $x = -2$
 So EAN = $28 = 8 + 36$
 Oxidation state of cobalt in $\text{Co}_2(\text{CO})_8 = 2x + 8(-0) = 0$ or $x = 0$
 So EAN = $27 + 1 + 8 = 36$.

6. (a) (a) 28Ni is in +2 oxidation state with electron configuration $[\text{Ar}]18 3d^8 4s^0$, so



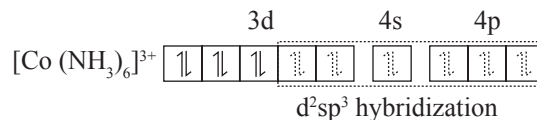
Number of unpaired electrons = 2

- (b) $_{24}\text{Cr}$ is in +3 oxidation state with electron state with electron configuration $[\text{Ar}]^{18} 3d^3 4s^0$; so



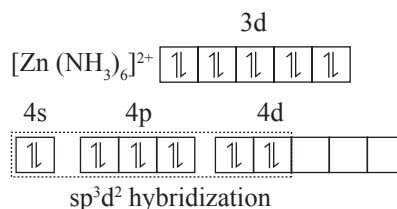
Number of unpaired electrons remains three, whether ligand is strong field or weak field ligand when electron configuration is $3d^3$.

- (c) $_{27}\text{Co}$ is in +3 oxidation state with electron configuration $[\text{Ar}]^{18} 3d^6 4s^0$, so



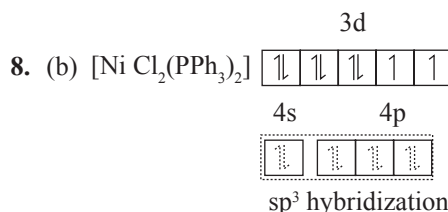
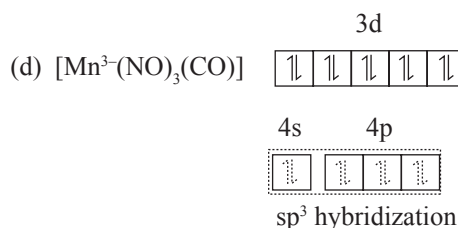
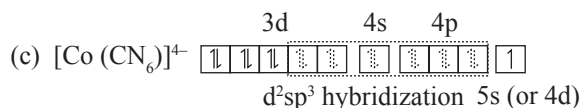
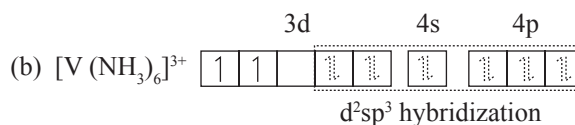
All electrons are paired, so complex ion is diamagnetic.

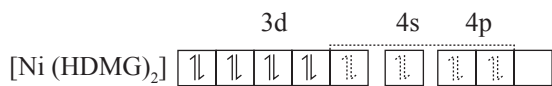
- (d) $_{30}\text{Zn}$ is in +2 oxidation state with electron configuration $[\text{Ar}]^{18} 3d^{10} 4s^0$, so



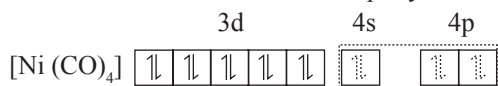
7. (b) (a) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ is high spin complex with four unpaired electrons.

$[\text{Cr}(\text{CN})_6]^{4-}$ is low spin complex with two unpaired electrons.



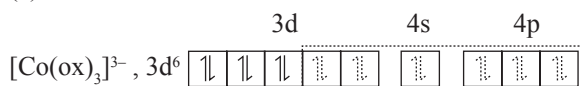


dsp² hybridization



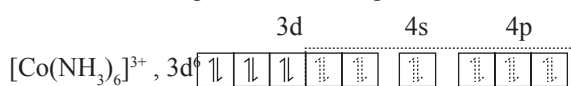
sp³ hybridization

9. (c)



d²sp³ hybridization

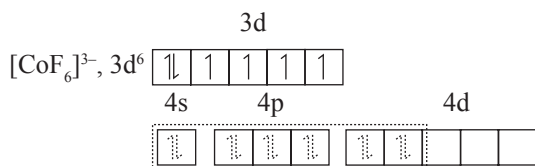
All electrons are paired, so diamagnetic.



d²sp³ hybridization

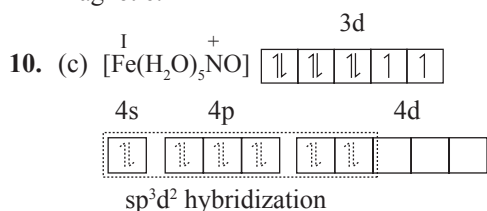
All electrons are paired, so diamagnetic.

F⁻ is a weak ligand



sp³d² hybridization

It contains four unpaired electrons, so it is paramagnetic.



Number of unpaired electrons = 3; So, $\mu = \sqrt{3(2+3)} = 3.87$ B.M.

Brown colour is due to charge transfer from ligand to metal.

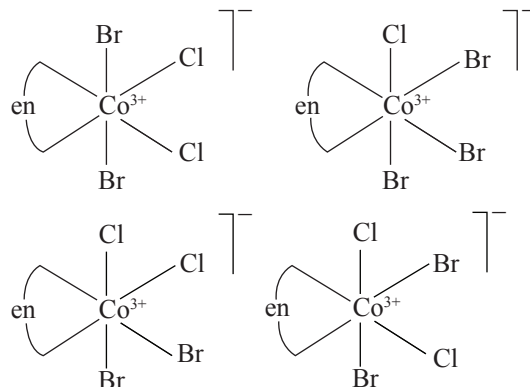
11. (a) Δ corresponds to the wave length of the visible light and $\Delta = \frac{hc}{\lambda}$

The order of ability to produce d-orbital splitting of various ligands;

CO > CN⁻ > NO₂⁻ > en > NH₃ > H₂O > OH⁻ > F⁻ > Cl⁻ > Br⁻ > I⁻

strong field	weak field
ligands	ligands
(larger Δ)	(smaller Δ)

12. (c)



13. (a) It shown geometrical, ionization and linkage isomers.

NO₂ – NO₂ cis-trans.

NO₂ – ONO cis-trans.

ONO – ONO cis-trans.

NO₂ – NO₃ cis-trans.

ONO – NO cis-trans.

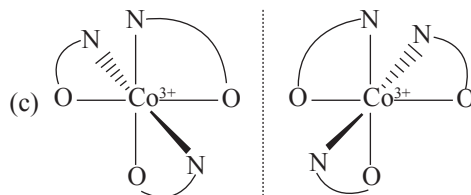
Total number of isomers = 10.

14. (a) (a) has three symmetrical bidentate ligands, which lack plane of symmetry.

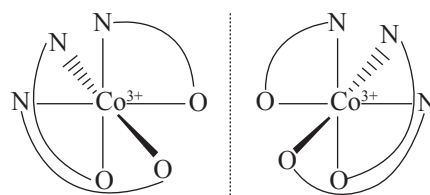
(b) and (c) has square planar geometry and has plane of symmetry. (d) has one of the plane of symmetry. So (b), (c) and (d) will not show chirality.

15. (c) (a) [Co (NH₃)₃ (NO₃)₃] shows fac-mer isomerism but does not show optical isomerism.

(b) [Co(ox)₃]³⁻ does not show fac-mer isomerism but shows optical isomerism.

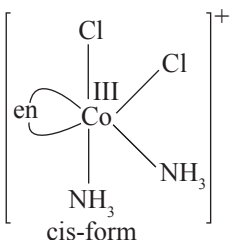


mer – one pair of enantiomers

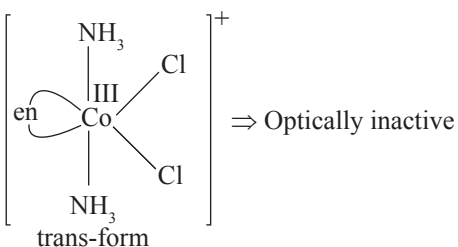
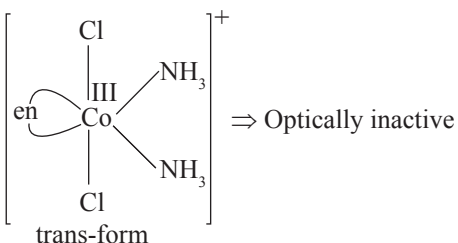


fac – one pair of natiomers

16. (a, b, d) $[\text{CoF}_6]^{3-}$: $\Delta_o < P$ and High spin complex.



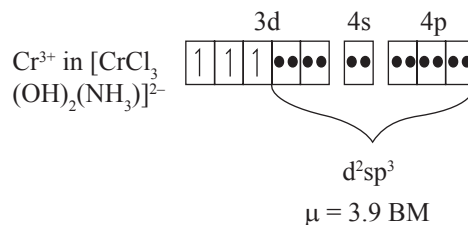
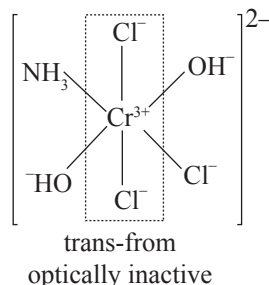
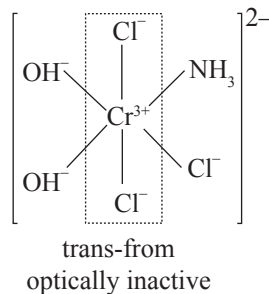
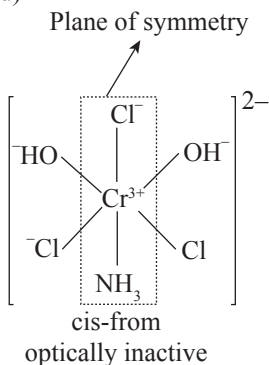
$\Rightarrow$ Asymmetric configuration hence exist in d and l forms



Total isomers = 4

17. (a, b) $[\text{Rh}^{\text{III}}\text{ClH}_2(\text{PPh}_3)_3]$ gives electronic configuration of $t_{2g}^{2,2,2}$, $e_g^{0,0}$, inner orbital complex and thus it has two d-orbitals empty, electrons will pair up and give d_2dp_3 hybridization. RhI with C.N = 4 has dsp^2 hybridization (4d series element)

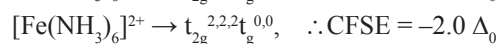
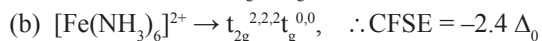
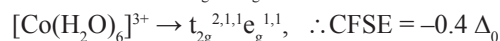
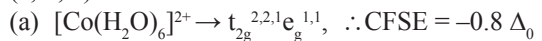
18. (a, d)



19. (b, c) Irrespective of nature of ligand field, magnetic moment of octahedral complexes having metal ions in d^1 , d^2d^3 , d^8 , d^9 configuration, remains same.

Element may or may not form colourless complex.

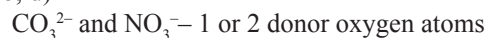
20. (a, b, c)



- (c) For co-ordination no six, two empty d-orbitals are not available for d^2sp^3 as Ni(II) has $3d^84s^0$ configuration.

- (d) d-orbital involved is $dx^2 - y^2$ not dz^2

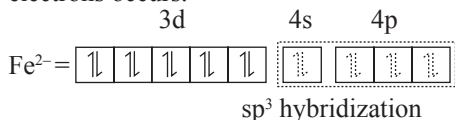
21. (a, b, d)



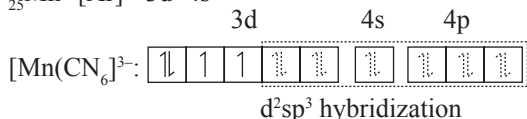
22. (a, b, c)

a, b and c are correctly matched.

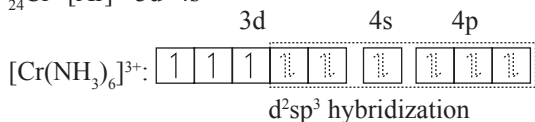
(d) – tetrahedral, CO stronger ligand so pairing of electrons occurs.



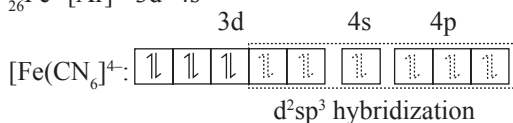
23. (a, b)

(a) ${}_{25}\text{Mn}^{3+}[\text{Ar}]^{18} 3d^4 4s^0$ 

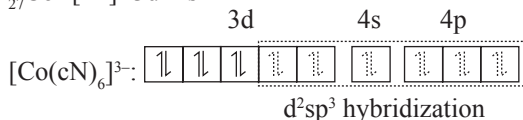
two unpaired electrons.

(b) ${}_{24}\text{Cr}^{3+}[\text{Ar}]^{18} 3d^3 4s^0$ 

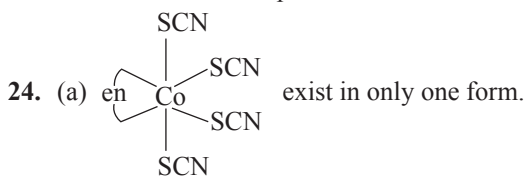
three unpaired electrons.

(c) ${}_{26}\text{Fe}^{2+}[\text{Ar}]^{18} 3d^6 4s^0$ 

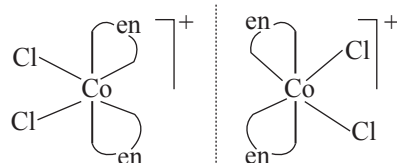
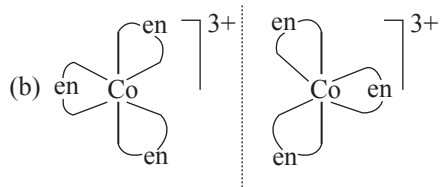
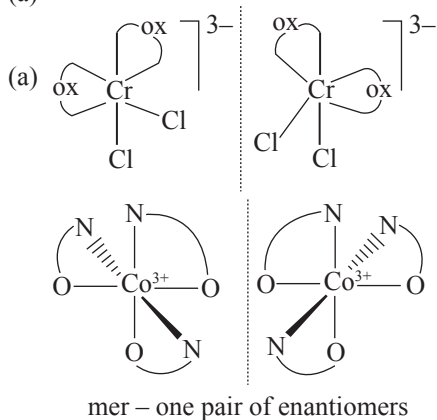
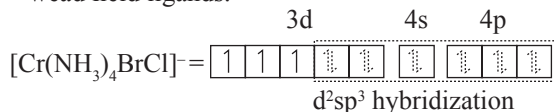
all electrons are paired.

(d) ${}_{27}\text{Co}^{3+}[\text{Ar}]^{18} 3d^6 4s^0$ 

all electrons are paired.



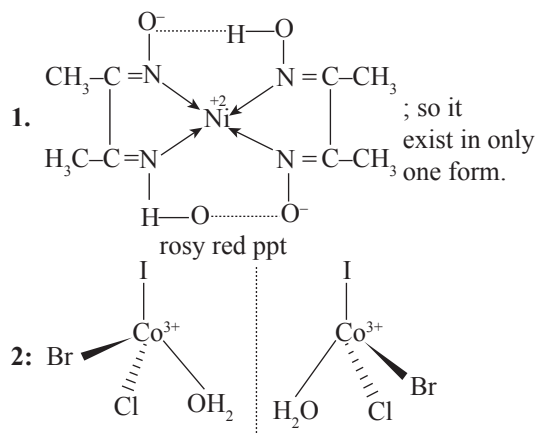
25. (a)

(c) [Pt Cl(dien)]Cl, [Ni Cl₂ (PPh₃)₂] complex is square planar whereas [Ni Cl₂ (PPh₃)₂] is tetrahedral.(d) [Co (NO₃)₃ (NH₃)₃] has plane of symmetry but cis-[Pt (en)₂ Cl₂] Cl₂ shows optical isomerism.26. (d) Cr³⁺ having 3d³ configuration always have 3 unpaired electrons with strong field as well as weak field ligands.

27. (b) As metal ion charge increases the ligands are drawn closer to the metal ion because of its increased charge density. As the ligands move closer, they cause greater splitting of the d-orbital, thereby producing a large Δ value.

28. (a)

29. (d)



30. (a)

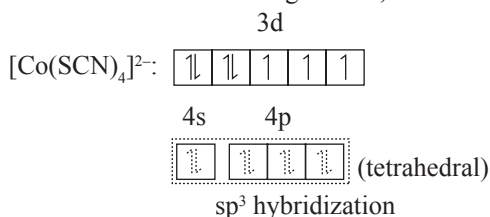
2. V.B.T. does not give any interpretation about the relative thermodynamic stabilities of various complexes. This is one of the limitations of V.B.T. S₃ is correct statement.

31. (b) All statements are correct.

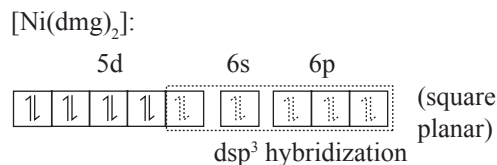
32. (a) 1. (ox) chelating ligand acts as stronger ligand. Therefore all electrons are paired so as to give d^2sp^3 hybridisation.
2. NO_2^- is a stronger ligand and with d^7 configuration to give d^2sp^3 hybridisation one of the 3d-electron jumps to higher energy orbital giving complex the paramagnetic character.
3. Pt(I) has 5d8 configuration. Hence, complex is square planar and diamagnetic.
33. (d) 1, 2 and 3 are correct statements.
4. It is an example of only hydrate isomerism not ionisation isomerism because such isomerism occurs owing to exchange of ions between coordination and ionisation spheres.
34. (c) 1. Pentaamminethiocyanato-N-chromium (III) tetrachlorozincate (II). or Pentaammineisothiocyanatochromium (III) tetrachlorozincate (II).
2. In $CoSO_4$, the oxidation state of Co is +2 i.e., $[Ar]^{18} 3d^7 4s^0$ electrons configuration. It has two unpaired electrons so it is paramagnetic and coloured.
3. $[Co(NH_3)_4(NO_2)_2]NO_3$
 $[Co(NH_3)_4(NO_2)(NO_3)]NO_2$ ionisation isomerism
 $[Co(NH_3)_4(NO_2)_2]NO_3$
 $[Co(NH_3)_4(NO_2)(ONO)]NO_3$ linkage isomerism
 This complex has cis- and trans isomers.
 Total isomers = 10.
4. The bond length C–O in CO itself is 1.128, while the bond length in $[Fe(CCO)_5]$ molecules are 1.15 Å on account of synergic interaction between Fe and Co.

Comprehension Type

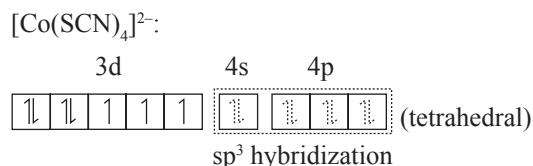
35. (a)
36. (b)
37. (a)
38. (b) (X) = $[Co(SCN)_4]^{2-}$; (Y) = $[Co(d, g)_2]$.
39. (a) X = $[Co(SCN)_4]^{2-}$, cobalt is in +2 oxidation state and SCN^- is weak field ligand. So,



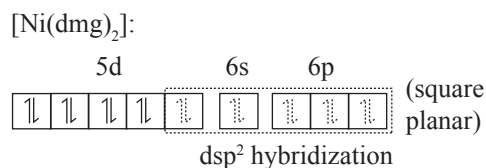
$[Ni(dmg)_2]$: Nickel is in +2 oxidation state and dmg is chelation ligand. So,



40. (c) X = $[Co(SCN)_4]^{2-}$, cobalt is in +2 oxidation state and SCN^- is weak field ligand. So,



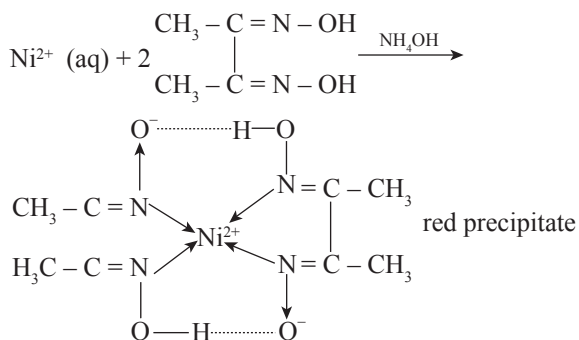
$[Ni(dmg)_2]$: Nickel is in +2 oxidation state and dmg is chelating ligand. So,



$[Ni(dmg)_2]$ shows intra molecular H-bonding as shown below.

Reactions involved:

$Co^{2+}(aq) + 4SCN^-(aq) \rightarrow [Co(SCN)_4]^{2-}(aq)$ blue colour complex.



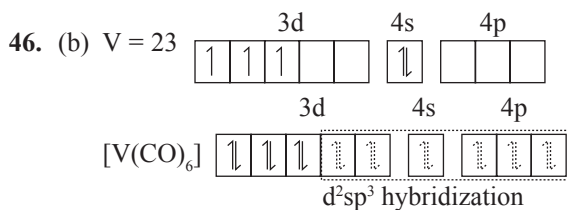
41. (b) $\Delta = h\nu = \frac{hc}{\lambda} = \frac{(6.626 \times 10^{-34} \text{ J.s}) \times (3.0 \times 10^8 \text{ m/s})}{544 \times 10^{-9} \text{ m}}$
- $3.65 \times 10^{-19} \text{ J/ion}$
- $= (3.65 \times 10^{-19} \text{ J/ion}) \times (6.02 \times 10^{23} \text{ ions/mole})$
- $= \frac{21.973 \times 10^4}{1000} \text{ kJ mol}^{-1} = 219.73 \text{ kJ mol}^{-1}$
42. (c) As it absorbs blue colour light ($\lambda = 455 \text{ nm}$), the colour of the complex must be orange.

43. (c) (i) $_{78}\text{Pt(II)}$ has $5d_8$ configuration, all electrons are paired; so diamagnetic.
 (ii) $_{27}\text{Co}^{2+}$ has $3d^7$ configuration; SCN^- is weak field ligand. So the complex is paramagnetic with three unpaired electrons.
 (iii) $_{29}\text{Cu}^{2+}$ has $3d^9$ configuration; complex is paramagnetic with one unpaired electron.
 (iv) $_{80}\text{Hg}^{2+}$ has $5d^{10}$ configuration; all electrons are paired so diamagnetic.

44. (b) Donation of electrons from a filled d-orbital of metal into the vacant antibonding π^* bonding orbital of CO decreases the bond order three to two and half or slightly more thus increasing the bond length between C–O.

$$\text{Bond order} \propto \frac{1}{\text{Bond length}}$$

45. (c) $\text{Ni}(\text{CO})_4$ sp^3 diamagnetic
 $\text{Fe}(\text{CO})_5$ dsp^3 diamagnetic
 $\text{V}(\text{CO})_6$ dsp^3 paramagnetic
 $\text{Cr}(\text{C}_6\text{O})_6$ dsp^3 diamagnetic



47.

A. Q, R
C. P, S

B. P, S
D. Q, S

Complex	Oxidation Type state	Magnetic property
$[\text{Fe}(\text{CN})_6]^{4-}$	+2	d^2sp^3 (inner-orbital) Diamagnetic complex)
$[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$	+2	sp^3d^2 (outer-orbital) Paramagnetic complex)
$[\text{Cu}(\text{NH}_3)_6]^{2+}$	+2	sp^3d^2 (outer complex) paramagnetic)
$[\text{Ni}(\text{CN})_6]^{4-}$	+2	sp^3d^2 (outer complex) Diamagnetic)

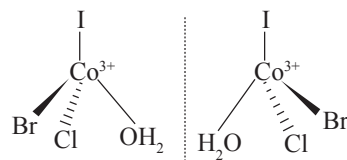
48. (A-P, Q, R); (B-P, Q, R, S); (C-P, Q, R); (D, P);
 (a) CO stronger ligand complex for pairing of electrons, sp^3 hybridization and diamagnetic, ligand is two electron donor,
 (b) When NO molecule coordinates with metal atom to form metallic nitrosyl, the single electron

present in π^* antibonding molecular orbital is transferred to metal atom M so that NO molecule is converted to NO^+ . Since NO^+ is isoelectronic with CO molecule, this ion coordinates with M^+ ion as a two electron donor in metal nitrosyl in the same way as CO coordinate to M atom in metal carbonyl. Note that NO molecule is a three electron donor. Now the empty π^* antibonding molecular orbital can overlap with the filled d-orbital of metal to form $\text{M} \rightarrow \text{NO}^+ \pi^*$ bond.

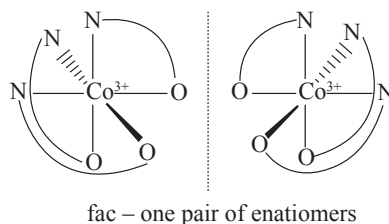
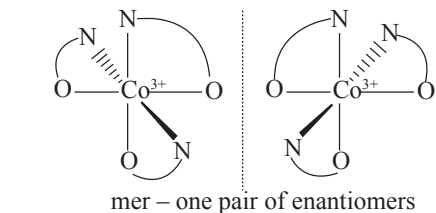
- (c) $[\text{Ni}(\text{PF}_3)_4]$ has bonding like that of $[\text{Ni}(\text{CO})_4]$

49. (A-Q, R); (B-Q, R); (C-P, S); (D-R, S)
 (a) d^6 -configuration, $\text{Co}^{3+} \rightarrow d^2sp^3$, diamagnetic. en multidentate ligand-chelation.
 (b) d^6 -configuration, Co^{3+} and diamagnetic $\rightarrow d^2sp^3$. Stabilization through chelation –OX multidentate ligand.
 (c) d^7 -configuration, Co^{2+} and paramagnetic with three unpaired electrons- sp^3d^2 (H_2O weaker ligand).
 (d) d^7 -configuration, Co^{2+} and paramagnetic, one electron jumps to 5s-orbital d^2sp^3 .

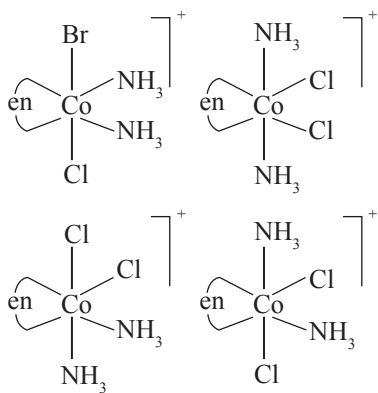
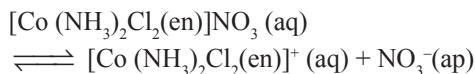
50. (A-P, R, S); (B-P, Q, R, S, T); (C-P, Q, R); (D-P, Q, R)
 (a) It has tetrahedral geometry; so it does not show geometrical isomerism. but it has chiral atom and therefore, have a pair of enantiomers. Oxidation state of cobalt in the complex is +3 and the compound is heteroleptic (having different types of ligands) and neutral.



- (b) Oxidation state of cobalt in the complex is +3 and the compound is homoleptic (having same type of ligands) and neutral.

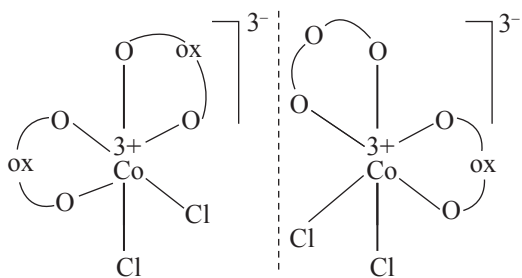
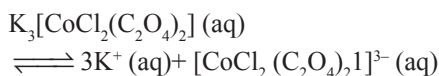


- (c) Oxidation state of cobalt in the complex is +3 and the compound is heteroleptic (having different types of ligands).

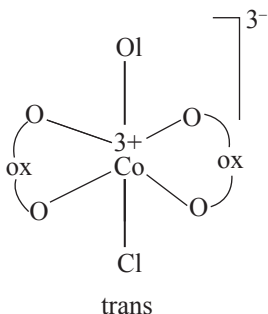


d and ℓ forms d and ℓ forms

- (d) Oxidation state of cobalt in the complex is +3 and the compound is heteroleptic (having different types of ligands).



cis-one pair of enantiomers

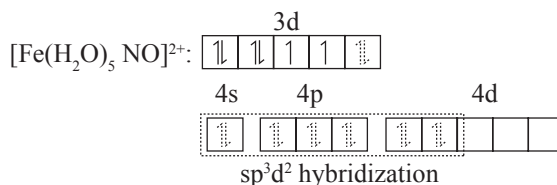


trans

51. (A-P); (B-Q); (C-P, R); (D-R).

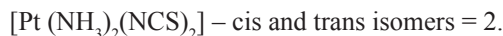
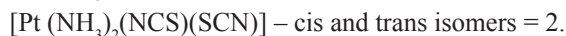
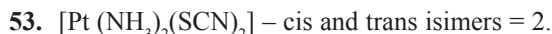
- (a) dsp^2 hybridization; Au is in +3 oxidation state and complex is diamagnetic. (5d^8 configuration)

- (b) sp^3 hybridization; Cu is in +1 oxidation state and complex is diamagnetic. CN^- is strong field ligand but no empty d-orbitals available for dsp^2 hybridization.
- (c) sp^3d^2 hybridization; Co is in +2 oxidation state and complex is paramagnetic with one unpaired electron.
- (d) sp^3d^2 hybridization with Fe in +1 oxidation state and paramagnetic with three unpaired electrons.

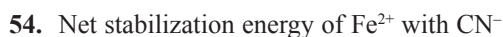


Electronic configuration of Al is $[\text{Ne}]^{10} 3\text{s}^2 3\text{p}^1$.

EAN for $\text{K}_3[\text{AlF}_6] = 13 - 3 + 2 \times 6 = 22$.



The complex is square planar; so it does not have a pair of enantiomers on account of the presence of plane of symmetry.



$$\begin{aligned} \text{ligands } t_2g^{2,2,2} e_g^{0,0} &= -2.4 \Delta_0 + 2P = m (-2.4 \times 25000 + 2 \times 15000) \text{ cm}^{-1} \\ &= -30000 \text{ cm}^{-1} \end{aligned}$$

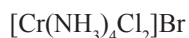
Net stabilisation energy of Fe^{2+} ions with H_2O ligands $t_2g^{2,1,1} e_g^{1,1} = -0.4\Delta_0 = 4000 \text{ cm}^{-1}$

$$\text{So required difference} = 26000 \text{ cm}^{-1} = 26 \times 10^5 \text{ m}^{-1}$$

$$= 6 \times 10^{-34} \times 3 \times 10^8 \times 26 \times 10^5 \times 6 \times 10^{23} \times 10^{-3} \text{ kJ/mole}$$

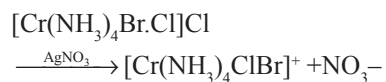
$$= 2808 \times 10^{-1} \text{ kJ/mole} = 281 \text{ kJ/mole} = 280.8 \text{ kJ/mol}$$

55. Complex $\text{Cr}(\text{NH}_3)_4\text{ClBr}$ has two isomers. Since coordination number of Cr is six and thus two forms may be



yellow ppt, soluble partially in conc. NH_3 i.e., $[\text{Ag}(\text{NH}_3)_2]\text{Br}$

QB.72



+ AgCl

white ppt soluble

in dil. NH_3 i.e., $[\text{Ag}(\text{NH}_3)_2\text{Cl}]$

NH_3 is strong field ligand so,



3d

4s

4p



d^2sp^3 hybridization

Hybridization of Cr in (a) and (b) is $d^2 sp^3$ having 3 unpaired electrons (3d3)

For d^3 configuration, the magnetic moment is always 3.87 with weak field or strong field ligand.

Magnetic moment = $\sqrt{n(n+2)}$

B. M. = $\sqrt{3(3+2)} = 3.87$ B.M.

METALLURGY

1. (c) It is fact
2. (c) Zinc blende is ZnS not
3. (c) Cinnabar (HgS) the ore of mercury is concentrated by froth floatation process.
4. (d) It is fact
5. (b) Reduction with carbon is called smelting

$$\text{Fe}_2\text{O}_3 + 3\text{C} \rightarrow 2\text{Fe} + 3\text{CO}$$
6. (c) Impurities of SiO_2 is present in the iron ore so basic flux CaCO_3 is added.

$$\underset{\text{Flux}}{\text{CaO}} + \underset{\text{Impurity}}{\text{SiO}_2} \rightarrow \underset{\text{Slag}}{\text{CaSiO}_3}$$
7. (a) Argentite $\rightarrow \text{Ag}_2\text{S}$; Cuprite $\rightarrow \text{Cu}_2\text{O}$; Sidirite $\rightarrow \text{FeCO}_3$; jCarnallite $\rightarrow \text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$
8. (b) $\text{ZnS} + 4\text{NCN} \rightarrow \text{Na}_2[\text{Zn}(\text{CN})_4]$ (soluble complex) + Na_2S .
9. (c) This process is si, ply calcination.
10. (b) PbS reduces PbO to Pb and SO_2 is liberaated. This is called as self reduction.
11. (b) Copper pyrites contain Fe and Cu. Iron oxide is removed during smelting, FeSiO_3 .

$$\text{FeO} + \text{SiO}_2 \rightarrow \text{FeSiO}_3$$
12. (b) Most of the impurity of iron sulphide is removed during smelting, $2\text{FeS} + 3\text{O}_2 \rightarrow 2\text{FeO} + 2\text{SO}_2$

$$\text{FeO} + \text{SiO}_2 \rightarrow \text{FeSiO}_3$$
 (3)
13. (d) (a) Cassiterite contains iron and manganese tungstate, called as Wol-framite. They have magnetic properties.
 (b) Purified ore that contains 70% SnO_2 is called black tin.

$$\text{SnO}_2 + 2\text{C} \rightarrow \text{Sn} + 2\text{CO}$$
 (carbon reduction)
 (c) Roasting is done at moderate tempe in presence of air in reverberatory furnace. Then air supply is stopped and temperature is increased to melt the mass when self-reduction takes place.
 (d) At 500 – 800K (lower timperature range in the upper part of blast furnace)

$$3\text{Fe}_2\text{O}_3 + \text{CO} \rightarrow 2\text{Fe}_3\text{O}_4 + \text{CO}_2$$

$$\text{Fe}_3\text{O}_4 + \text{CO} \rightarrow 3\text{Fe} + 4\text{CO}_2$$

$$\text{Fe}_2\text{O}_3 + \text{CO} \rightarrow 2\text{FeO} + \text{CO}_2$$

 At 900 – 1500K (higher timperature range in the lower part of blast furnace:

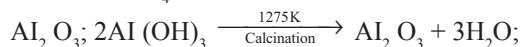
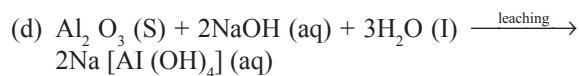
$$\text{C} + \text{CO}_2 \rightarrow 2\text{CO} ; \text{FeO} + \text{CO} \rightarrow \text{Fe} + \text{CO}_2$$
14. (c) White bauxite containing the impurity of silica only is removed by heating with coke in atmosphere of nitrogen.

$$\text{Al}_2\text{O}_3 + 3\text{C} + \text{N}_2 \rightarrow 2\text{AlN} + 3\text{CO}$$

$$\text{SiO}_2 + 2\text{C} \xrightarrow{\Delta} \uparrow \text{Si} + 2\text{CO} \uparrow ; \text{AlN} + \text{H}_2\text{O} \rightarrow \text{Al}(\text{OH})_3 + \text{NH}_3 ; 2\text{Al}(\text{OH})_3 \xrightarrow{\Delta} \text{Al}_2\text{O}_3 + \text{H}_2\text{O}$$
15. (a) Cyanide process $\rightarrow \text{Ag}$; Self reduction $\rightarrow \text{Cu}$; Electrolytic reduction $\rightarrow \text{Al}$; Carbon reduction $\rightarrow \text{Sn}$.
16. (a, b) Alloy manganese-steel is formed, which is very hard. Manganese also removes oxygen and sulphur from iron.
17. (a, b, c, d)
All are correct regarding aluminothermite process.
18. (a, b)
Zn (spelter) is purified by electrolysis process or fractional distillation.
19. (a, b, c, d)
All are the benefits of roasting and calcination.
20. (a, b, c, d)
Selenium, tellurium, silver, gold and platinum are the metals present in anode mud. This is becaue these are les reactive than copper.
21. (a, b, c)
 (a) Self reduction takes place in Bessemer converter.
 (b) Slag formation and reduction of haemalite to iron take place in blast furnance at different temperatures.
 (c) $\text{Cr}_2\text{O}_3 + \text{AL AL O}_3 + 2\text{Cr}$.
 This is called sluminothermic process.
 (d) Tin is Obtained by carbon reduction of SnO_2 (cassiterite ore)
22. (c, d)
 (a) It occurs in fusion zone at higher tempt. with Fe_2O_3 which escapes reduction zone.
 (d) It occurs in slag formation zone.
23. (a, b)
 (a) Magnesite (MgCO_3) is ore of magnesium.
 (b) Tin is extracted by carbon reduction. $\text{SnO}_2 + 2\text{C} \rightarrow \text{Sn} + 2\text{CO}$
 (c) and (a) are correct statements.
24. (a, b, c)
(a) True statement

- (b) $\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$ (slag); (a) $\text{PbO} + \text{SiO}_2 \rightarrow \text{PbSiO}_3$
 CaO Converts the PbSiO_3 to PbO , $\text{PbSiO}_3 + \text{CaO} \rightarrow \text{PbO} + \text{CaSiO}_3$
25. (b) CaF_2 -Ionic compound ionises to give ions and thus increases the number of ions in the electrolyte and as impurity decreases the melting point of Al_2O_3 .
26. (a) Au is recovered from the solution by the addition of electropositive metal.

$$2\text{NaAu}(\text{Cu})_2 + \text{Zn} \longrightarrow \underset{\text{Soluble complex}}{\text{Na}_2\text{Zn}(\text{CN})_4} + 2\text{Au} \downarrow$$
27. (a) $\text{Au}(\text{S}) \xrightarrow{\text{O}^2} \text{Au} + (\text{aq.})$
 $\text{Au}^+ + 2\text{CN}^- (\text{aq.}) \rightarrow [\text{Au}(\text{CN})_2]^-$ soluble complex.
28. (c) $2\text{CuFeS}_2 + \text{O}_2 \rightarrow \text{Cu}_2\text{S} + 2\text{FeS} + \text{SO}_2$;
 $2\text{FeS} + 3\text{O}_2 \rightarrow 2\text{FeO} + 2\text{SO}_2$
 $\text{FeO} + \text{SiO}_2 \rightarrow \text{FeSiO}_3$; thus remove the impurity of iron oxide.
29. (a) Cast iron contains 3% carbon.
30. (c) 1. It lowers the fusion temperature of MgCl_2 and provide conductivity to the electrolyte.
 2. $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ is calcined in a stream of dry HCl gas to have anhydrous MgCl_2 . In absence of dry HCl, It forms MgOCl_2 ($\text{MgO} \cdot \text{MgCl}_2$) because of hydrolysis.
31. **False** As log is lighter but it has lower M.P. than the metal.
32. **False** As the enthalpy of formation of Al_2O_3
33. (a) 1. In Upper part at lower temperature reduction of Fe_2O_3 takes place by carbon monoxide but unreduced FeO is reduced by carbon at higher temperature in lower part of blast furnace.
 2. Sphalerite = ZnS
 3. True statement.
 4. Evolution of SO_2 Produces blister like appearance on the surface of solidified copper.
34. (b) (a) $\text{Cr}_2\text{O}_3 + 2\text{AlO}_3 \rightarrow \text{Al}_2\text{O}_3 + 2\text{Cr} + 2\text{Cr}$.
 (b) $\text{Au} + 2\text{CN}^- + \text{H}_2\text{O} + \frac{1}{2}\text{O}_2 \rightarrow [\text{Au}(\text{CN})_2]^- + \text{Zn} = [\text{Au}(\text{CN})_2]^- + 2\text{Au}$.
 (b) True statement.
 (c) $\text{Ag}_2\text{S} + 4\text{NaCN} \xrightarrow{\text{O}^2} 2\text{Na}[\text{Ag}(\text{CN})_2] + \text{Na}_2\text{S}$
 $2\text{Na}[\text{Ag}(\text{CN})_2] + \text{Zn} \rightarrow \text{Na}_2[\text{Zn}(\text{CN})_4] + 2\text{Ag}$
35. (a) Na_3AlF_6 lowers the m.pt. and increases the conductance of melt.
36. (d) Code powder prevents oxidation of Al and graphite.
37. (b) Fluorspar increases the electrical conductance.
38. (a) At about 10000 C, lines cross and thus at this temperature zinc and carbon have equal affinity for oxygen.
39. (b) This will be above a temperature when two curves intersect that is > 10000 C.
40. (d) All statements are true.
41. (a) $[\text{X}] = \text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$ or $2\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$; $[\text{Y}] = \text{Cu}_2\text{S}$ or CuFeS_2
42. (d) It is sulphide ore (Cu_2S or CuFeS_2) and is called as chalcopyrites or chalcocites; S^{2-} gives yellow precipitate of CdS with CdCO_3 .
43. (c) $\text{G1} = \text{SO}_2$; sulphur can increase and decrease its oxidation state.
44. (a) $\text{CuCO}_3 + 2\text{HCl} \rightarrow \text{CuCl}_2 + \text{CO}_2 + \text{H}_2\text{O}$; $\text{CuCl}_2 + 2\text{KI} \rightarrow \text{CuI}_2 + 2\text{KCl}$; $2\text{CuI}_2 \rightarrow \text{Cu}_2\text{I}_2 (\text{P}) + \text{I}_2$
45. (A-Q); (B-R); (C-Q, R); (D-P)
 Poling-impure molten metal is stirred with green wood poles, oxide impurities are removed, mainly used for Cu and Sn
 Cupellation is used when impurities are other metals, mainly used for silver.
 Electrometallurgy-Cu, Ag, Au, Cr, Zn, Ni, etc., are mainly treated to remove insoluble impurities as anode mud.
 Van Arkel method vapour phase refining.
 Metals like titanium, zirconium, thorium and uranium are purified.
46. (A-S); (B-R); (C-Q); (D-P)
 Argentite $\rightarrow \text{Ag}_2\text{S}$; Cuprite $\rightarrow \text{Cu}_2\text{O}$;
 Siderite $\rightarrow \text{FeCO}_3$; Carnallite $\rightarrow \text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$.
47. (A-R); (B-Q); (C-P, S); (D-Q)
 Cuprite Cu_2O oxide Cerussite PbCO_3 carbonate Kainite $\text{KCl} \cdot \text{MgCO}_3 \cdot 3\text{H}_2\text{O}$ chloride and sulphate Calamine ZnCO_3 carbonate
48. (A-Q, S); (B-P); (C-R, S); (D-R, S, T)
 (a) Fe_2O_3 ; calcination; At lower part of furnace reduction of unreduced FeO by carbon at higher temperature and in upper part of furnace reduction by CO at lower temperature.
 (b) CuFeS_2 ; FeSiO_3 formation mainly occurs during roasting (now a day)/smelting and left FeO reacts with SiO_2 in Bessemer converter.
 (c) $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O} \rightarrow$ calcination in presence of dry HCl to remove 6 water molecules. Mg extraction by electrolytic reduction from molten mixture of anhydrous $\text{MgCl}_2 + \text{CaCl}_2$.



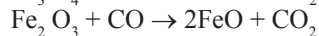
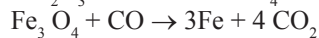
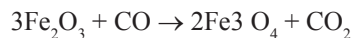
Extraction of Al from purified Al_2O_3 by electrolytic reduction.

49. (A-P); (B-P); (C-Q); (D-S)

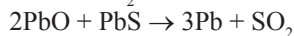
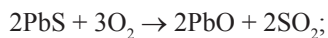
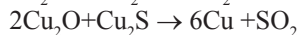
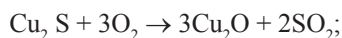
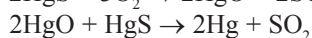
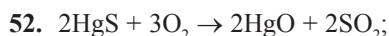
- Gold dissolves in NaCN solution forming a soluble complex leaving behind impurities is called leaching process. Hence the extraction of gold is called hydrometallurgy.
- Insoluble copper pyrites dissolves in H_2SO_4 forming soluble CuSO_4 is called leaching. Hence the extraction of copper is called hydrometallurgy
- Reduction of Fe_3O_4 by CO in blast furnace is smelting.
- Heating Mg in dry HCl gas to convert into anhydrous MgCl_2 is calcination.

50. Cryolite is $3\text{NaF} \cdot \text{AlF}_3$ or $\text{Na}_3[\text{AlF}_6]$. It gives that coordination number of aluminium is six. It can expand its covalency by using empty d-orbitals.

51. At 500-800 K (lower temperature range in the blast furnace)



At 900-1500 K (higher temperature range in the blast furnace):



S-BLOCK ELEMENTS AND COMPOUNDS

1. (d) $2\text{NaCl} \xrightarrow{\text{Electric current}} 2\text{Na}^+ + 2\text{Cl}^-$

Cation
Anion
2. (c) CaCl_2 bring down the melt temperature from 1075 K to 850 K
3. (a) Because of small atomic size and high I.E. Be forms covalent chloride.
4. (d) Because hydration energy decreases down the group.
5. (d) $\text{Be}(\text{OH})_2 < \text{Mg}(\text{OH})_2 < \text{Ca}(\text{OH})_2$
6. (b) On moving down the group; Lattice energy decreases with increase in size of cation.
7. (c) (a) $4\text{LiNO}_3 \longrightarrow 2\text{Li}_2\text{O} + 4\text{NO}_2 + \text{O}_2$
 $2\text{NaNO}_3 \longrightarrow 2\text{NaNO}_2 + \text{O}_2$ (similar decomposition with the nitrates of K, Rb and Cs)
 (b) Only LiCl is deliquescent and crystallises as a hydrate $\text{LiCl} \cdot 2\text{H}_2\text{O}$
 (c) $2\text{M} + 2\text{H}_2\text{O} \longrightarrow 2\text{M}^+ + 2\text{OH}^- + \text{H}_2$
 (M = an alkali metal)
 (d) Halides of Li are covalent in nature.
8. (c) When sodium and potassium react with water, the heat evolved causes them to melt, giving a larger area of contact with water, lithium on the other hand, does not melt under these condition and thus reacts more slowly.

	Li	Na	K
Melting point ($^{\circ}\text{C}$)	180	98	64.
9. (c) $\text{M} + (\text{x} + \text{y})\text{NH}_3 \rightarrow [\text{M}(\text{NH}_3)_x]^+ + [\text{e}(\text{NH}_3)_y]^-$
 Solutions are deep blue coloured due to the absorption of light in visible region by solvated electrons. It is paramagnetic due to the presence of the unpaired electrons.
10. (a) Basic strength of the oxides increases in the order $\text{Li}_2\text{O} < \text{Na}_2\text{O} < \text{K}_2\text{O} < \text{Rb}_2\text{O} < \text{Cs}_2\text{O}$. The increase in basic strength is due to the decrease in I.E. and increase in electropositive character.
 The melting points of the halides decrease in the order $\text{NaF} > \text{NaCl} > \text{NaBr} > \text{NaI}$, as the size of the halide ion increases. The decrease in melting point is due to increase the covalent character with increase in the size of anion according to Fajan's rule.
11. (c) $\text{NH}_3 + \text{CO}_2 + \text{H}_2\text{O} \longrightarrow \text{NH}_4\text{HCO}_3$
 $\text{NaCl} + \text{NH}_4\text{HCO}_3 \longrightarrow \text{NaHCO}_3 + \text{NH}_4\text{Cl}$
 $2\text{NaHCO}_3 \xrightarrow{150^{\circ}\text{C}} \text{Na}_2\text{CO}_3 + \text{CO}_2 + \text{H}_2\text{O}$
 $2\text{NH}_4\text{Cl} + \text{Ca}(\text{OH})_2 \longrightarrow 2\text{NH}_3 + \text{H}_2\text{O} + \text{CaCl}_2$
 (by product)
12. (d) NaNO_2 (sodium nitrite) acts both as oxidizing agent and reducing agent because the nitrogen atom in it is in +3 oxidation state (+3 intermediate oxidation state for nitrogen)
Oxidizing property:
 $2\text{NaNO}_2 + 2\text{KI} + 2\text{H}_2\text{SO}_4 \longrightarrow$
 $\text{Na}_2\text{SO}_4 + \text{K}_2\text{SO}_4 + 2\text{NO} + 2\text{H}_2\text{O} + \text{I}_2$
Reducing property:
 $\text{NaNO}_2 + \text{H}_2\text{O}_2 \longrightarrow \text{NaNO}_3 + \text{H}_2\text{O}.$
13. (c) $2\text{M}(\text{NO}_3)_2 \xrightarrow{\Delta} 2\text{MO} + 4\text{NO}_2 + \text{O}_2$
 (M = Be, Mg, Ca, Sr, Ba)
14. (c) (a) The solubility increases down the group because the change in lattice energy is more as compared to hydration energy. Thermal stability and the basic character both increase down the group as metallic character increases.
 (b) It is correct statement.
 (c) Beryllium and aluminium and diagonally related. Chlorides of both are covalent in nature and thus are soluble in organic solvents. Chlorides of both are electron deficient and thus act as strong Lewis acids.
15. (b) $3\text{Mg} + \text{N}_2 \xrightarrow{\Delta} \text{Mg}_3\text{N}_2$;
 $\text{Mg}_3\text{N}_2 + 6\text{H}_2\text{O} \longrightarrow 3\text{Mg}(\text{OH})_2 + 2\text{NH}_3$
 (X) (Y)
 $\text{CuSO}_4 \cdot 4\text{NH}_3 \rightarrow [\text{Cu}(\text{NH}_3)_4]\text{SO}_4$ (blue colour)
16. (a, b, c) The ions of alkali metals have small radii.
17. (a, b, d) Alkali metals has lower melting points due to larger size.
18. (a, b, c) Alkaline earth metals have larger charge to size ratio, so they shows high hydration.
19. (a, c, d) Solution of alkali metals in liquid ammonia is blue coloured due to ammoniated electrons.
20. (a, b, c, d)
 (a) Due to the formation of metal ion clusters

- (b) $M + (x + y) \text{NH}_3 \rightarrow M^+(\text{NH}_3)_x + e^-(\text{NH}_3)_y$
 (c) due to the formation of metal clusters.
 (d) $M(\text{NH}_3)_6 \rightarrow$ true statement

21. (a, b, d)

Solution consists of $[M(\text{NH}_3)_x]^+$ and $[e(\text{NH}_3)_y]$

It imparts blue colour not due to polarisation but due to excitation of free electrons to higher energy levels by the absorption of light in visible region. On the addition of iron oxide solution decomposes:



22. (a, b, c) (a) This is because the heat from the flame excites the outermost orbital electron to a higher energy level. When the excited electron comes back to the ground state, there is emission of radiation in the visible region as given below:

Metal	Li	Na	K	Rb	Cs
			Violet/ violet		
Colour	Crimson red	Yellow	Lilac	Red	Blue

- (b) Down the group the change in lattice energy is more than that of hydration energy.
 (c) Because of low ionization energy and melting point.
 (d) Reducing nature increases down the group as their stability decreases down the group
 $\text{CsH} > \text{RbH} > \text{KH} > \text{NaH} > \text{LiH}$.

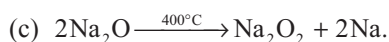
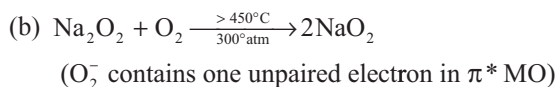
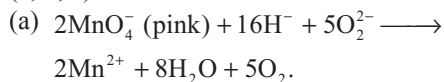
23. (a, b, c, d)

All statements are correct.

24. (a, b, c)

- (a) Li^+ and Mg^{2+} have higher polarizing power and thus attract more number of water molecules.
 (b) Li is least reactive because of its highest ionization energy on account of smaller size.
 (c) It is one of the uses of liquid sodium. It has low melting point.
 (d) LiI is the most covalent due to smaller cation and bigger anion. (According to Fajan's rule)

25. (a, b, c)



(d) gives only H_2O_2 .

26. (b) Both assertion and reason are true but reason is not the correct explanation of assertion.

Lattice energy of Na_2SO_4 is less than its hydration energy but the lattice energy of BaSO_4 exceeds its hydration energy.

27. (a) Statement-1 and Statement-2 are correct statements and statement-2 is the correct explanation of statement-1.

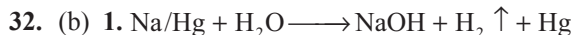
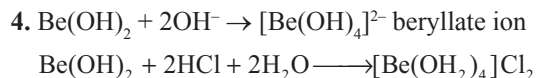
28. (a) Beryllium and aluminium have diagonal relationship.

29. (b) (i) Bigger cation stabilizes bigger anions and vice-versa.

(ii) Alkali metal carbonates and their bicarbonates stabilities depend on their metallic character. Stability of carbonates and bicarbonates $\propto$ Metallic character of their i.e., electropositive character.

30. (a) Potassium and caesium have low ionization energies and, therefore, their valence shell electrons can be easily ejected out by the energy of light.

31. (a) 1. In concentrated solution unpaired electrons pair up together and become diamagnetic
 2 and 3 are correct statements.



2. Potassium is more reactive than sodium than sodium due to its more negative value of reduction potential.

3. True statements.

4. Alkaline earth metal salts have more number of water molecules as compared to those of alkali metal salts, because of higher nuclear charge and smaller ionic radius.

33. (c) 1. True statement-salt of weak base and strong acid.
 2. As we move down the group from Li to Cs the strength of metallic bonding decreases as size of atom increases. Due to this there occurs a decrease in close packing of atoms in crystal lattice from Li to Cs and thus the softness increases from top to bottom.

3. It is hemihydrate of $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$.

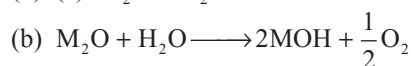
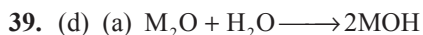
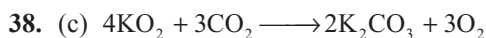
4. KHCO_3 is soluble in water and cannot be precipitated like NaHCO_3 .

34. (d)

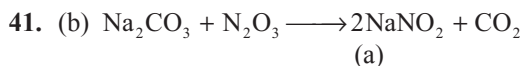
35. (a)

36. (d)

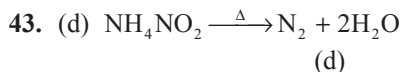
37. (d) Here solvated electrons acts as a reducing agent.



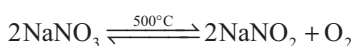
40. (d) Dilute solution of Na metal in liquid ammonia is deep blue coloured and highly conducting and paramagnetic in nature because of solvated cations and solvated electrons.



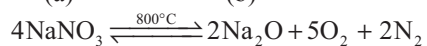
42. (c) O_2 is paramagnetic as it contains two unpaired electrons in anti-bonding $\pi 2p_x$ and $\pi 2p_y$ orbitals.



Reactions involved:



(a) (b)

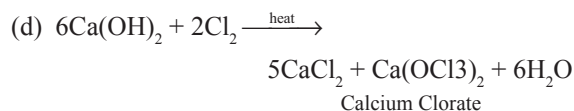
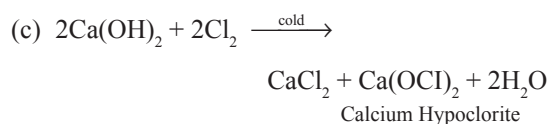
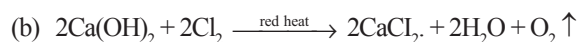
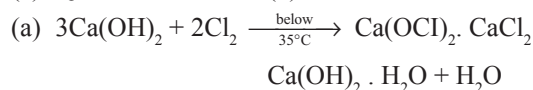


(c) (b) (d)

44.

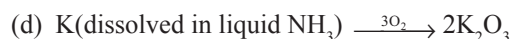
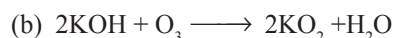
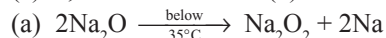
(a) R (b) S

(c) Q (d) P



46. (a) P, R (b) S

(c) P, R (d) P



46.

(a) Q, S (b) P

(c) Q, S (d) R

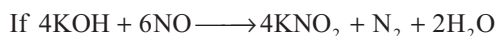
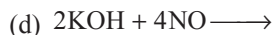
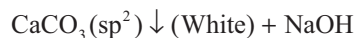
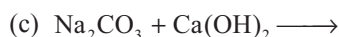
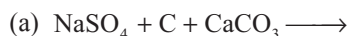
(a) Atomic radii increases down the group.

(b) Density increases down the group but density of Na > K

(c) As the atomic radii increases electropositive character increases

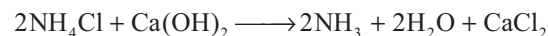
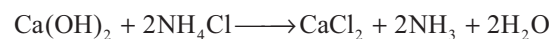
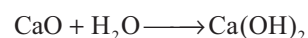
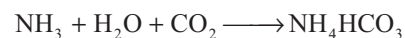
(d) Hydration energy increases as the size of cation decreases.

47. (A-P, Q, R); (B-P, Q, R); (C-P, R); (D-P, Q, S)



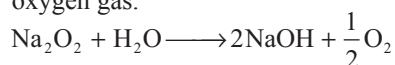
48. (A-R, S); (B-R); (C-P); (D-Q)

(a) and (b)

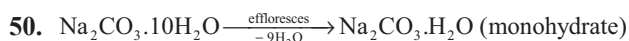
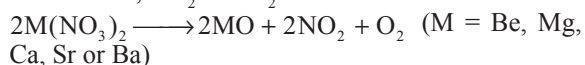


(c) The solutions of Na_2CO_3 and NaHCO_3 in water are alkaline in nature and thus turn the litmus paper blue.

(d) The commercial name of sodium peroxide is oxone which is used for the manufacture of oxygen gas.



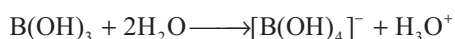
49. All alkaline earth metal nitrates decompose to form metal oxide, NO_2 and O_2 .



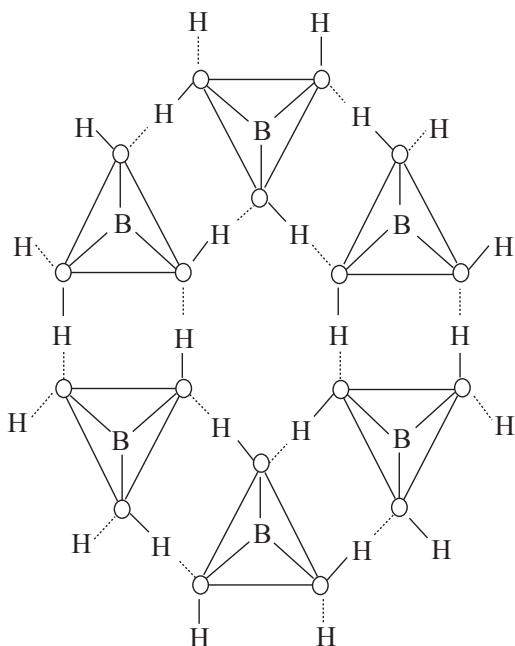
P-BLOCK ELEMENTS AND COMPOUNDS

1. (d) It forms a stable cyclic complex with polyhydroxy compounds like cis-glycerol, catechol, etc. This helps in the release of H^+ as H_3O^+ and, therefore, boric acid acts as a strong acid and hence can be titrated with sodium hydroxide using phenolphthalein.

2. (c) (a) It is a weak monobasic acid soluble in water and in aqueous solution in boron atom completes its octet by accepting OH^- from water molecules:

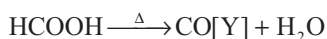


- (b) Only Tl^{3+} acts as an oxidising agent on account of inert pair effect.
(c) In the solid state, the $B(OH)_3$ units are hydrogen bonded together into two dimensional sheets with almost hexagonal symmetry.

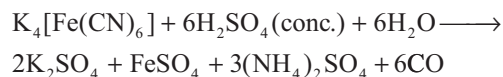


- (d) $B(OEt)_3$ imparts green colour to the burner flame.

3. (c) $HCN + 2H_2O \longrightarrow HCOOH(X) + 2NH_3$;



- (c) Only $K_4[Fe(CN)_6]$ gives CO with conc. H_2SO_4 according to the following reaction.



- (a) Malonic acid gives C_3O_2 ; (d) Action of conc. HNO_3 on charcoal gives $(CO)_2$. (b) C_3O_2 gives CO_2 .

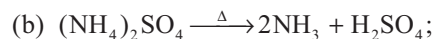
4. (a) $CO(g) + H_2(g) + H_2O(g) \xrightarrow[673K]{FeCrO_4} 2H_2(g) + CO_2(g)$

5. (d) (a) This is due to the inert pair effect in the heavier p-block elements.

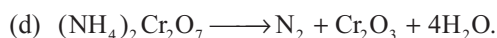


- (c) This is due to the presence of completely filled d and/or f orbitals in heavier elements. Covalent radius/pm; N = 70, P = 110, As = 120, Bi = 150.

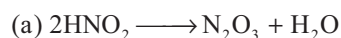
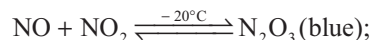
6. (b) (a) $2NaN_3 \xrightarrow{\Delta} 2Na + 3N_2$;



- (c) $NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O$.

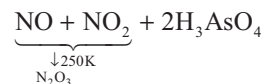


7. (d) $2Cu + 6HNO_3 \longrightarrow 2Cu(NO_3)_2 + NO + NO_2 + 3H_2O$



- (b) $N_2O_3 + 2KOH \longrightarrow 2KNO_2 + H_2O$

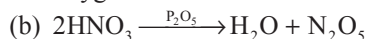
- (c) $2HNO_3 + As_2O_3 + 2H_2O \longrightarrow$



All options are correct.

8. (d) (a) $2N_2O \longrightarrow 2N_2 + O_2$.

It produces one third oxygen by volume of total gas produced while air has one fifth volume of oxygen.

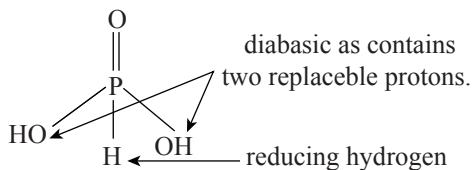


- (c) White phosphorus most reactive and black phosphorus is almost inert.

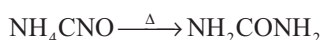
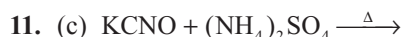
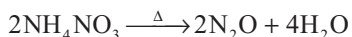
- (d) White phosphorus turns to red phosphorus on heating in the atmosphere of CO_2 or coal gas at 573 K.

So (a), (b), (c) are correct and (d) is correct.

9. (a)



Oxidation state of P is + 3 can be decrease to - 3 and increase to + 5. Hence it can act as both reducing as well as oxidising agent.



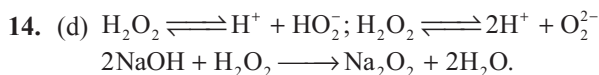
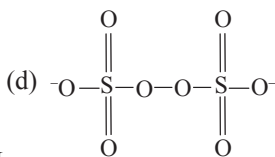
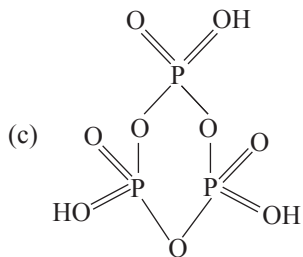
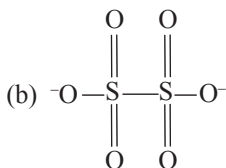
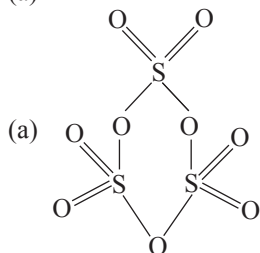
12. (c) (a) NH_3 has higher-boiling point than AsH_3 and PH_3 on account H-bonding. SbH_3 has highest boiling point on account of highest molecular weight.

(b) As the size of element increases, the strength of M-H bond decreases and thus tability decreases.

(c) Basicity decreases down the group. (Drago rule).

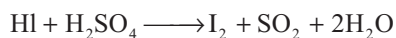
(d) HEH angle ($^\circ$): $\text{NH}_3 = 107.8$; $\text{PH}_3 = 93.6$, $\text{AsH}_3 = 91.8$; $\text{SbH}_3 = 91.3$.

13. (a)

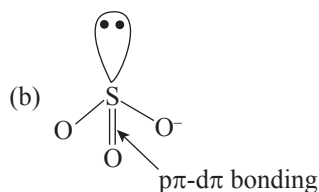
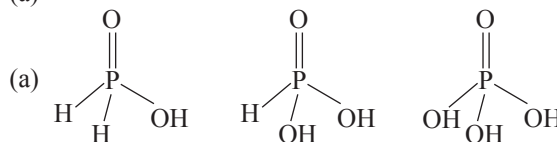


15. (a) (a) $\text{SO}_2 + 2\text{H}_2\text{O} \longrightarrow \text{H}_2\text{SO}_4 + 2\text{H}.$ Hence bleaching is through reducing (temporary) and $\text{Cl}_2 + \text{H}_2\text{O} \longrightarrow 2\text{HCl} + [\text{O}]$ bleaching is through oxidation (permanent). So (a) is correct and (b), (c) and (d) are incorrect.

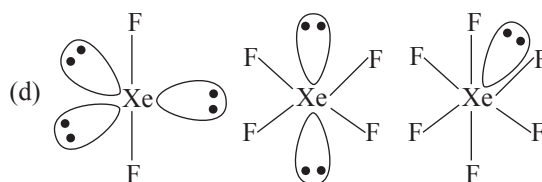
16. (b) H_2SO_4 is stronger oxidizing agent and HI is stronger reducing agent. So H_2SO_4 oxidizes HI to I_2 and is itself reduced to SO_2 .



17. (a)



(c) SO_3^{2-} contains one lone pair of electrons on sulphur atom.



18. (d) It is used because it has very low solubility in blood.

19. (b, c, d) Compound [X] is $\text{Na}_2\text{B}_4\text{O}_7$.

(a) two boron atoms are sp^2 while other two boron atoms are sp^3 hybridized.

(b) product is sodium peroxoborate which is used as brighter in soaps.

(c) H_3BO_3 is formed with HF gives HBF_4 .

(d) Green bead in oxidising as well as in reducing flame in cold.

20. (a, b) (a) (b) Correct statements.

(c) Graphite sp^2 hybridization bond length 1.42 Å, diamond sp^3 hybridization bond length 1.54 Å.

(d) Diamond more dense (3.51 g/ml) than graphite (2.25 g/ml)

21. (a, d) (a) As number of reducing hydrogen increases, the reducing character also increases. In H_3PO_4 there is no reducing hydrogen so it should be least reducing.

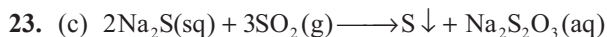
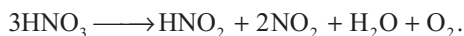
(b) $+1 < +2 < +3 < +5$

(c) Down the group the availability of lone pair of electrons decreases as they are present in more concentrated s-orbital.

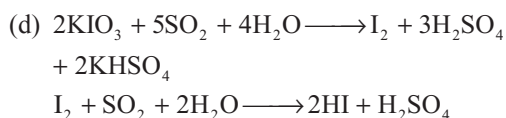
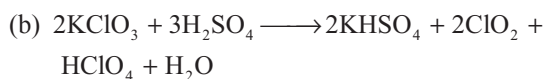
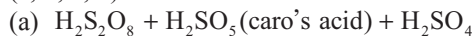
(d) As down the group bond energy decreases, the removal of H becomes easier.



(b) Due to the formation of NO_2 according to following reaction



24. (a, b, c, d)

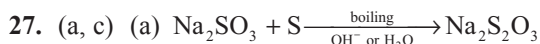
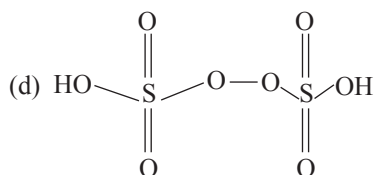
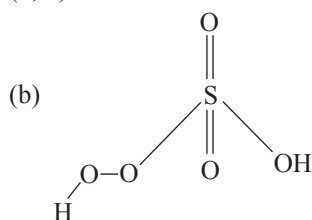


25. (a, b)

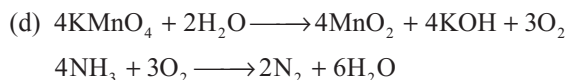
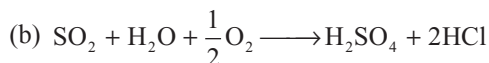
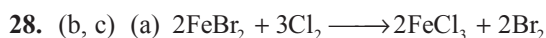
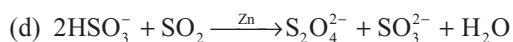
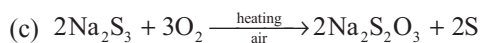
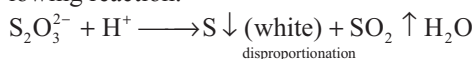
(a) Two hydroxyl groups lie in different plane (have open book like structure)

(b) It turns blue litmus red because of acidic character which is then bleached by bleaching action of H_2O_3 .

26. (b, d)

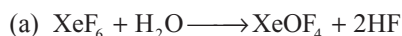
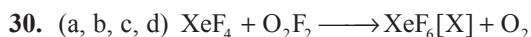


(b) SO_3^{2-} formed in presence of acid precipitates sulphur and liberates SO_2 gas according to the following reaction.

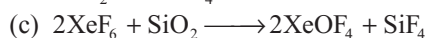


29. (b, d) (a) $\text{ClO}_4^- < \text{IO}_4^- < \text{BrO}_4^-$ according to their reduction potential.

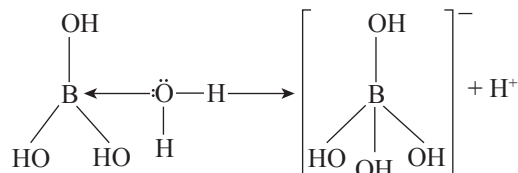
(a) $\text{HOCl} > \text{HOBr} > \text{HOI}$. As electronegativity of halogens increases the acidic character increases. Given (b) and (d) are correct order.



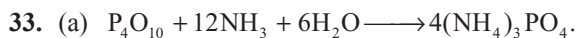
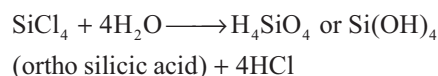
(b) XeF_6 turns to yellow liquid on melting whereas XeF_2 and XeF_4 turn white.



31. (c) Orthoboric acid acts as Lewis acid in water not as proton donor because it completes its octet by accepting OH^- from water.

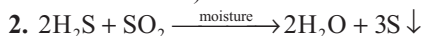
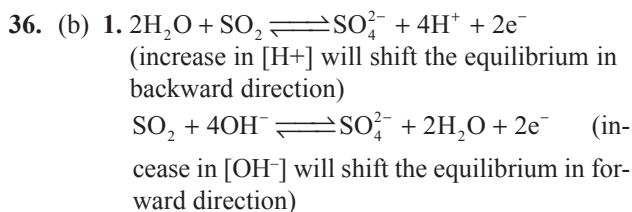


32. (a) Statement-1 is True, Statement-2 is True, Statement-2 is a correct explanation for Statement-1. For example



34. (a) Filling of 10 electrons in 3d-subshell enhances the effective nuclear charge on account of the poorer shielding and thus the electrons are held more tightly. This is the reason why selenium is reluctant to attain the highest oxidation state of (VI).

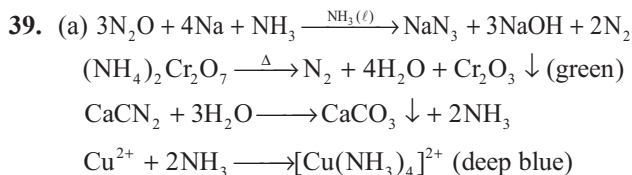
35. (d) Kept in plastic or wax-lined glass containers containing urea or phosphoric acid, because traces of alkali metal ions and metal ions from the glass and the metal surface respectively can catalyse the explosive decomposition of H_2O_2 .



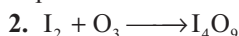
37. (d) X-Y is polar due to difference in their electronegativities where as X-X bond is non-polar. In X-Y, X is bigger halogen so X-Y bond length is bigger and thus weaker as compared to X-X bond.

38. (c) 1, 2 and 3 are correct statements.

4. GeO_2 is acidic while SnO_2 is amphoteric in nature.

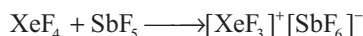
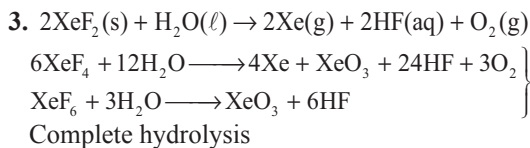


40. (b) 1. Metaphosphoric is prepared by heating phosphorus acid and Br_2 in a sealed tube, and



3. β -Sulphur is stable above 369K.

41. (c) 1 and 2 are correct statements.

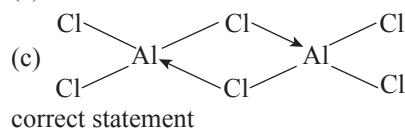


M = Na, K, Rb or Cs.

42. (d)

(a) due to high heat of hydration in water $[\text{Al}(\text{H}_2\text{O})_6]\text{Cl}_3$

43. (c)

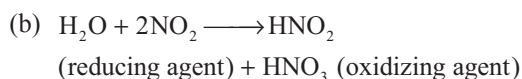
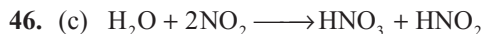


- (a) BF_3 forms HBF_4
(b) Anhydrous AlCl_3 is covalent

44. (c)

- (a) Covalent compound so it has lower melting point.
(b) Exists as Al_2Cl_6
(c) Being covalent easily hydrolysed
(d) It is Lewis acid.

45. (d) Only one electron pair pyridine can be accommodated not two electron pairs.

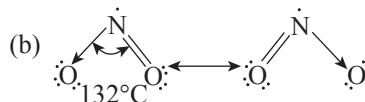
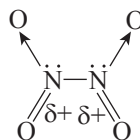


(c) Neutral to litmus as neutral oxide

(d) In the liquid state, N_2O_4 tends to ionise

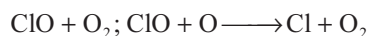
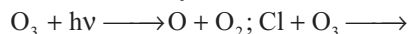


48. (d) (a) N-N bond length (1.75 Å) is longer than usual N-N single bond length which is believed to be due to δ^+ charge and lone pair of electrons on N atoms which causes repulsion.

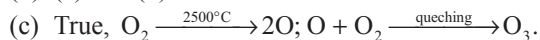


- (c) $\text{:N}\equiv\text{N:}$ and the structure suggests that it will have some dipole moment.

49. (d) Following reactions are responsible for the depletion of ozone layer



50. (d) (a) and (b) are correct statements.

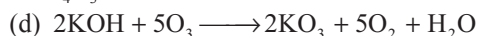


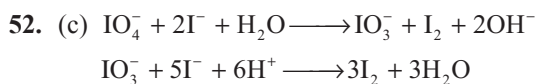
- (d) Its (i.e., ozone) advantage over chlorine is that it avoids the unpleasant smell and taste of chlorine.

51. (d) (a) Red light.

(b) Oxides of nitrogen and the halogen can damage the O_3 layer.

(c) I_4O_3 is formed.





53. (d) (a) Disproportionation is slow at the ordinary temperature for OCl^- , it is fast for OBr^- and very fast for OI^- .

(b) E° for the $1/2 \text{F}_2 | \text{F}^-$ electrode = + 2.9 V.

(c) HI is stronger acid than HBr because of its low bond dissociation enthalpy.

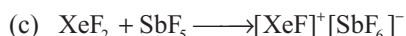
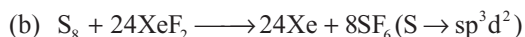
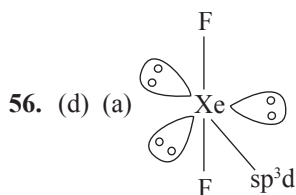
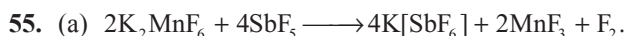
(d) Because of low bond dissociation energy and high hydration energy of F_2 , F_2 acts as strong oxidizing agent and oxidizing power decrease down the group. $E^\circ 1/2 \text{F}_2 | \text{F} = + 2.9 \text{ V}$.

54. (d) (i) HCl because of high bond enthalpy of it.

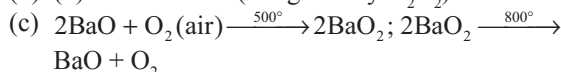
(ii) $2\text{KClO}_3 + \text{I}_2 \longrightarrow 2\text{KIO}_3 + \text{Cl}_2$.

(iii) Correct order, F^- because of high electron density on account of small size easily donate the electron pair, is strongest base.

(iv) It is used as bleaching as well as sterilizing agent (as it acts strong oxidizing agent).

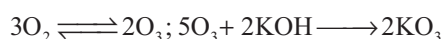
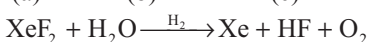
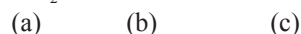
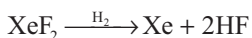
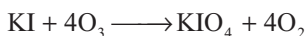
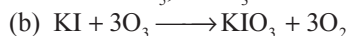


57. (d) (b) test of ozone (not given by H_2O_2)

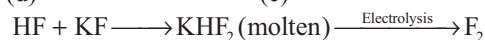


(a) O_3 forms I_4O_9 with dry iodine (incorrect statement)

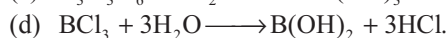
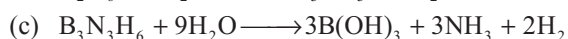
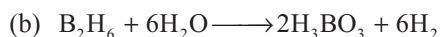
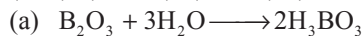
58. (b) (a) F_2 gives O_2 gas (c) H_2O_2 in acidic medium form CrO_5 , not O_3



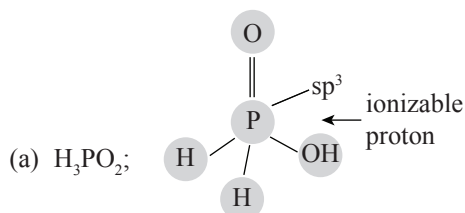
(orange solid) + $5\text{O}_2 + \text{H}_2\text{O}$



59. (A-P); (B-P, Q); (C-P, R); (D-P, R)

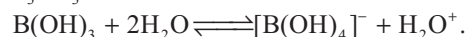


60. (A-Q, R); (B-Q, S); (C-P, R); (D-P, R)

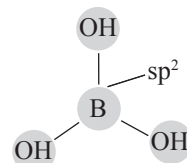


Hypophosphorus acid

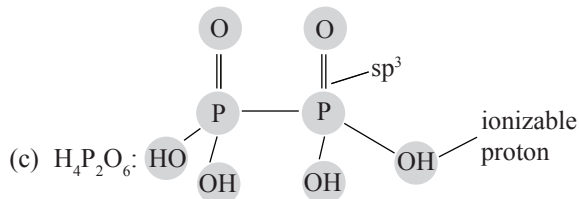
(b) H_3BO_3 ;



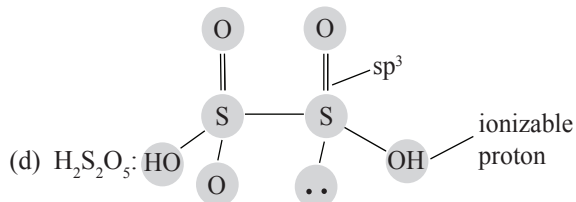
Monobasic Lewis acid as it completes its octet by accepting OH^- from water.



Orthoboric acid

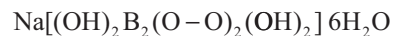
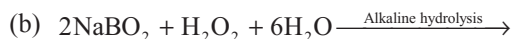
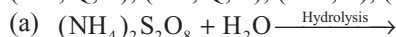


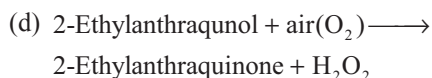
Hypophosphoric acid



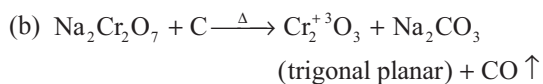
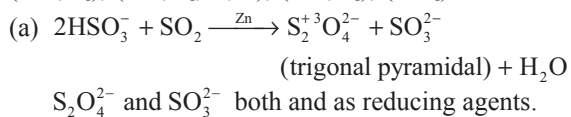
Pyrosulphurous acid

61. (A-P, Q, S); (B-P, Q, S); (C-R, S); (D-Q, S)

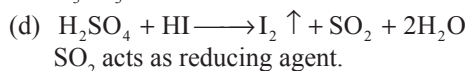
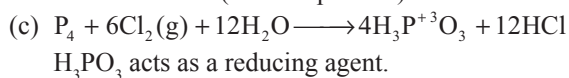




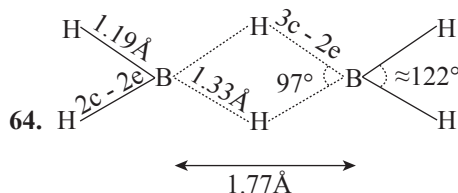
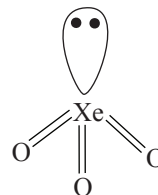
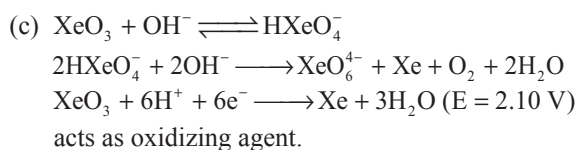
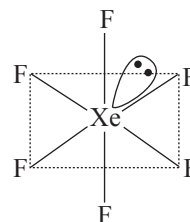
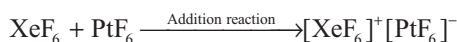
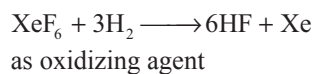
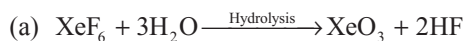
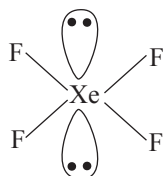
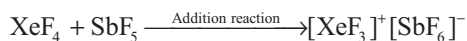
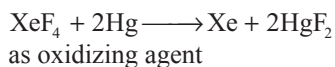
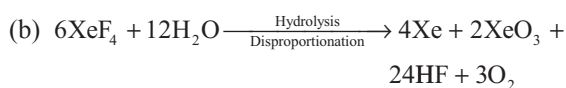
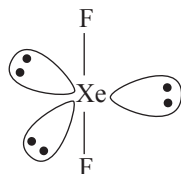
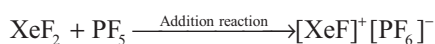
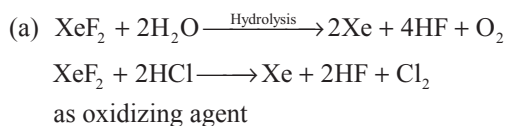
62. (A-P, Q); (B-P, Q, R, S); (C-P, Q); (D-Q)



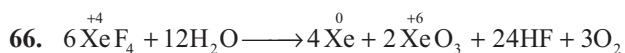
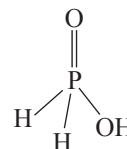
CO acts as a reducing agent and in the purification of nickel metal (Mond's process)



63. (A-P, Q, R, S); (B-P, Q, R, S); (C-P, Q, R, S); (D-Q, S, T)



65. Hypophosphorous acid is H_3PO_2 which has following structure.



D-BLOCK ELEMENTS AND COMPOUNDS

1. (a) + 4 oxidation state of cerium is also known in solution.
2. (c) $3\text{Ag} + 4\text{HNO}_3 \xrightarrow[\text{dilute}]{\text{heat}} 3\text{AgNO}_3 + \text{NO} + 2\text{H}_2\text{O}$
3. (c) Cr^{2+} is reducing as its configuration changes from d^4 and d^3 , the latter having a half-filled t_{2g} level. On the other hand, the change from Mn^{3+} to Mn^{2+} results in the half-filled (d^5) configuration which has extra stability.
4. (d) $\text{VO}_2^+ = \text{VO}^{2+} + \text{SO}_4^{2-}$; $\Delta x(-2) = +2 = +4$.
5. (a) $[\text{MO}_4]^{2-}$; Mn is in + 6 oxidation state. Electronic configuration is $[\text{Ar}]1s^2 2s^2 2p^6 3s^2 3p^4$. As it contains one unpaired electron it is paramagnetic and green coloured because of d-d transition in visible region. $[\text{TiCl}_6]^{3-}$, $[\text{VO}_4]^{3-}$ and CrO_2Cl_2 are diamagnetic as all electrons are paired.
6. (a) This is due to the increasing stability of the lower species to which they are reduced.
7. (d) True statement, $2\text{FeCl}_3 \xrightarrow{300^\circ\text{C}} (\text{FeCl}_3)_2(\text{g})$
8. (b) $2\text{Fe} + 3\text{Cl}_2 \longrightarrow 2\text{FeCl}_3$ (anhydrous); (a), (c), (d) do not give anhydrous FeCl_3 but hydrated FeCl_3 .
9. (a)

$$\begin{array}{c} \text{OCH}_3 \\ | \\ \text{CH} - \text{C} - \text{CH}_3 \\ | \\ \text{OCH}_3 \end{array}$$
10. (a) $\text{FeCl}_3 \cdot 6\text{H}_2\text{O} + \text{CH}_3\text{COOH} \longrightarrow \text{FeCl}_3 + 12\text{CH}_3\text{OH} + 6\text{CH}_3\text{COOH}_3$
11. (b) $2\text{CuSO}_4 + 4\text{KI} \longrightarrow 2\text{K}_2\text{SO}_4 + \text{Cu}_2\text{I}_2 + \text{I}_2$
 $2\text{Na}_2\text{S}_2\text{O}_3 + \text{I}_2 \longrightarrow \text{Na}_2\text{S}_4\text{O}_6 + 2\text{NaI}$; CuI_2 is not formed according to following reaction.
 2CuI_2 (unstable) $\longrightarrow \text{Cu}_2\text{I}_2 + \text{I}_2$.
12. (d) (a) $\text{AgNO}_3 + \text{PH}_3 \longrightarrow \text{Ag} \downarrow + \text{HNO}_3 + \text{H}_3\text{PO}_3$
 (b) $\text{AgNO}_3 \xrightarrow{\text{Organic tissue (proteins)}} \text{Ag} \downarrow$ (black)
 (c) $\text{Ag} + 2\text{KCN} \longrightarrow [\text{Ag}(\text{CN})_2]^+$
13. (c) $\text{CO}_2 + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{CO}_3 \rightleftharpoons 2\text{H}^+ + \text{CO}_3^{2-}$;
 $\text{CrO}_4^{2-} + 2\text{H}^+ \rightleftharpoons \text{Cr}_2\text{O}_7^{2-}$ (orange red) + H_2O
14. (d) $4\text{FeCr}_2\text{O}_4 + 8\text{Na}_2\text{CO}_3 + 7\text{O}_2 \xrightarrow{\text{Fusion}} \text{Na}_2^{+VI}\text{CrO}_4 \xrightarrow[\text{H}_2\text{O}]{\text{H}^+} \text{Na}_2^{+VI}\text{Cr}_2\text{O}_7 \xrightarrow[\text{H}_2\text{O}]{\text{H}^+/\text{H}_2\text{O}_2} \text{CrO}(\text{O}_2)_2$ (deep blue violet) $\xrightarrow{\text{H}_2\text{O}} \text{O}_2 + \text{H}_2\text{O} + \text{Cr}^{3+}$.
 $\text{Na}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{O}_4 \longrightarrow 2\text{CrO}_3$ (chronic anhydride) + $\text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$.
15. (a) $2\text{KMnO}_4 + 3\text{H}_2\text{SO}_4 \longrightarrow 2\text{KHSO}_4 + (\text{MnO}_3)_2\text{SO}_4 + 2\text{H}_2\text{O}$
 $(\text{MnO}_3)_2\text{SO}_4 + \text{H}_2\text{O} \longrightarrow \text{Mn}_2\text{O}_7 + \text{H}_2\text{SO}_4$
 $\text{Mn}_2\text{O}_7 \xrightarrow{\Delta} 2\text{MnO}_2 + 3\text{O}_2$
16. (a, b, d) $3\text{CuCl}_2 \cdot 2\text{H}_2\text{O} \xrightarrow{\Delta} \text{CuO} + \text{Cu}_2\text{Cl}_2 + 2\text{HCl} + \text{Cl}_2 + \text{H}_2\text{O}$; (a) only Cl_2 is liberated is wrong as evident from chemical equation.
17. (a, b, c, d)
 - (a) As metal ions generally contain one or more unpaired electrons in them and hence their complexes are generally paramagnetic.
 - (b) Because of having larger no. of unpaired electron in their atoms, they have stronger interatomic interaction and hence stronger bonding between the atoms.
 - (c) May be attributed to the presence of unpaired electrons (d-d transition in most of the compounds)
 - (d) This activity is ascribed to their ability to adopt multiple oxidation state and to form complexes.
18. (a, c)
 - (a) $\text{Cr}_2\text{O}_7^{2-} + 10\text{H}^+ + 4\text{Na}_2\text{O}_2 \longrightarrow$

$$\begin{array}{c} \text{O} \\ || \\ \text{O} \diagdown \text{Cr} \diagup \text{O} \\ | \quad | \\ \text{O} \quad \text{O} \end{array} + 8\text{Na}^+ + 5\text{H}_2\text{O}$$

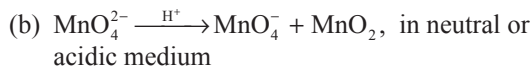
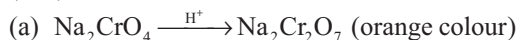
(blue colour)

In the absence of enter or amyl alcohol, CrO_5 in acidic medium decomposes to Cr^{3+} with evolution of oxygen.

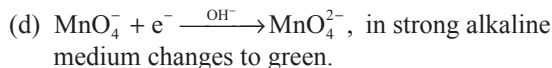
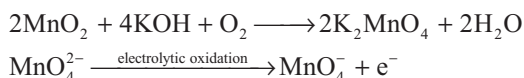
$$\begin{array}{c} \text{O} \\ || \\ \text{O} \diagdown \text{Cr} \diagup \text{O} \\ | \quad | \\ \text{O} \quad \text{O} \end{array} + 12\text{H}^+ \longrightarrow 4\text{Cr}^{3+} + 7\text{O}_2 + 6\text{H}_2\text{O}$$

(blue colour)

19. (a, b)



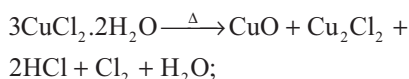
(c) False-In acidic medium no colour change.

20. (a, b) Fuse with KOH in presence of air or oxidizing agents like KNO_3 , KClO_3 etc.

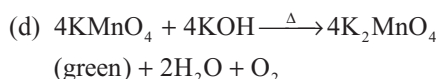
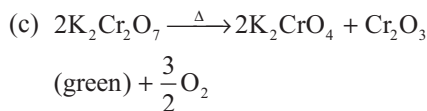
21. (a, b) (a) In making negative/positive of photographic film.

(b) In fixing of photographic film by removing unreduced AgBr as soluble complex.

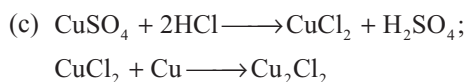
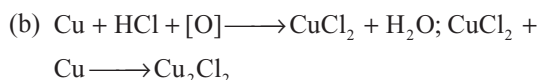
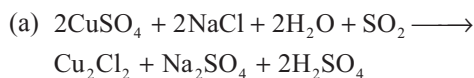
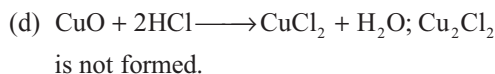
22. (a, b, d)

(c) only Cl_2 is liberated is wrong as evident from chemical equation.

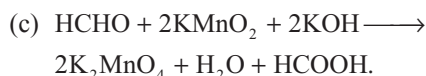
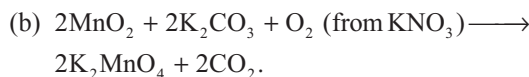
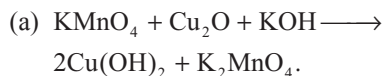
23. (b) (a) In both compounds central metal has + 6 oxidation state.

(b) Solution dichromate is $\text{Na}_2\text{Cr}_2\text{O}_7 \cdot 2\text{H}_2\text{O}$, so it can not be weighed correctly for the preparation of standard solution.

24. (d)



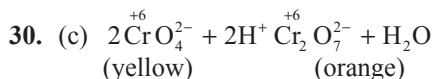
25. (a, b, c, d)

26. (d) The electrons are removed from outer most shell i.e., $4s^2$.27. (b) $\text{Mn}^{3+} = [\text{Ar}]^{18} 3d^4$, $\text{Cr}^{3+} = [\text{Ar}]^{18} 3d^3$, $\text{V}^{3+} = [\text{Ar}]^{18} 3d^2$, $\text{Ti}^{3+} = [\text{Ar}]^{18} 3d^1$ and Cr^{3+} is most stable due to greater CFSE of t_{2g}^3 configuration.28. (a, c) It is because of required change from d^5 (half filled) to d^4 (partially filled) in case of Mn.

$$E^\circ \text{Cr}^{3+}/\text{Cr}^{2+} = -0.41 \text{ V}$$

$$E^\circ \text{Mn}^{3+}/\text{Mn}^{2+} = +1.57 \text{ V}$$

29. (c) Cupric ion produces blue colour in solution due to the presence of unpaired electron (d-d transition) while cuprous ion does not have unpaired electron.



There is no change in oxidation state. So the explanation is wrong.

31. False

Copper (I) oxide is red powder

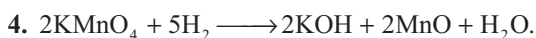
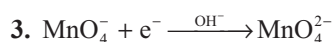
32. False

Correct reason is its very high I.E. and weak metallic bond.

33. (b) 1. It is because Mn^{2+} has $3d^5$ configuration which has extra stability.2. Note titanium but Copper, because with + 1 oxidation state an extra stable configuration, $3d^{10}$ results.3. Note stable as it undergoes disproportionation $2\text{Cu}^+(\text{aq}) \longrightarrow \text{Cu}^{2+}(\text{aq}) + \text{Cu(s)}$. The E° value for this is favourable.

34. (a) 1. Due to strong interatomic forces.

2. Hydrochloric acid is oxidised to chlorine.



35. (a) Due to stable half filled electronic configuration the delocalization of electrons is less and the metallic bond is much weaker than preceding elements.

36. (d) All statements are correct.

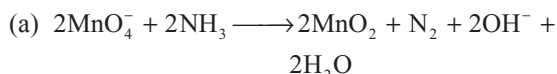
37. (a) Greater the number of valence electrons, stronger is the resultant bonding and higher the enthalpy of atomization.

38. (b) Both V^{+4} ($[Ar]^{18} d^1 s^0$) and Cu^{2+} ($[Ar]^{18} d^9 4s^0$) have one unpaired electron in the d-orbital and thus exhibit same colour.

39. (c) $\mu = 1.73$ i.e., one unpaired electron, Ti (E.C.) = $[Ar]^{18} 3d^2, 4s^2$, so it will lose three electrons to have one unpaired electron and thus the oxidation state of titanium is +3.

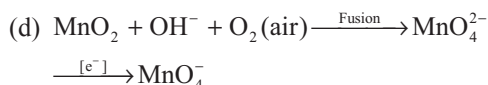
40. (d) (a) Has five unpaired electrons in 3d-sub shell. (b) is a obvious fact (c) number of unpaired electrons first increases to five and then pairing begins so number of unpaired electrons decreases.

41. (d)



(b) MnO_4^- is pink coloured and colour is due to charge transfer.

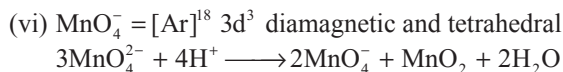
(c) MnO_4^- is in highest oxidation state i.e., +VII and hence cannot be oxidised.



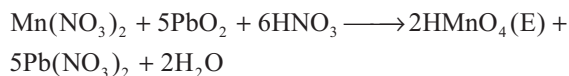
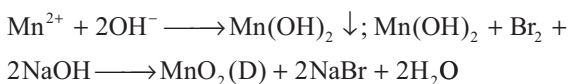
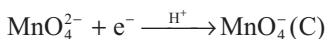
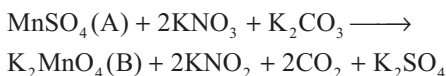
42. (c) (a) $MnSO_4 = +II$; (b) $MnO_4^{2-} = +VI$;

(c) $MnO_4^- = +VII$

43. (d) (iii) $MnO_4^{2-} = [Ar]^{18} 3d^1$ paramagnetic and tetrahedral

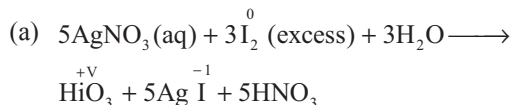


Solution of Passage

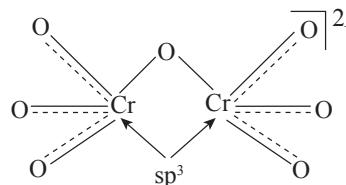
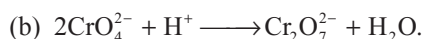


A = $MnSO_4$, B = K_2MnO_4 , C = $KMnO_4$, D = MnO_2 , E = $HMnO_4$, F = $BaSO_4$.

44. (A-P, Q, S); (B-R, T); (C-P, Q); (D-P, T)

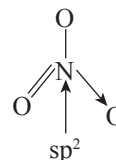


It is disproportionation and redox reaction. NO_3^- has planar structure.

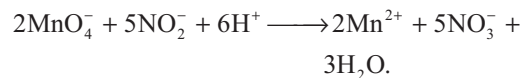


$Cr_2O_7^{2-}$ has dimeric bridged tetrahedral metal ion. The colour is due to charge transfer spectrum.

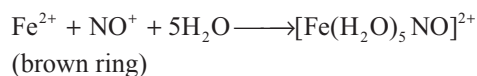
(c) NO_2^- is being oxidized and MnO_4^- being reduced according to the following reaction.



So it is redox reaction,

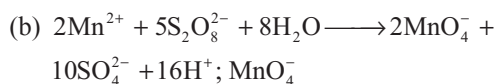
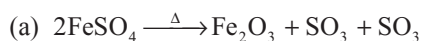


(d) Fe^{2+} is being oxidized and NO_3^- being reduced according to the following reaction. So it is redox reaction,

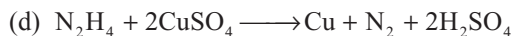
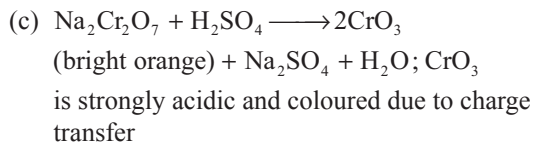


The colour is due to charge transfer spectrum.

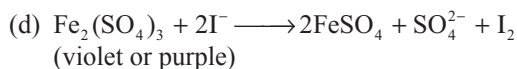
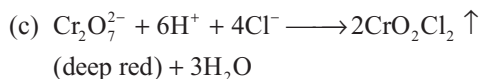
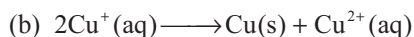
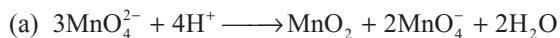
45. (A-Q, R, S); (B-P, Q, R); (C-P, Q, S); (D-Q, R)



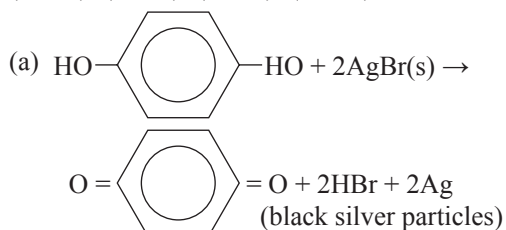
is coloured due to charge transfer.



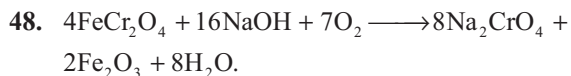
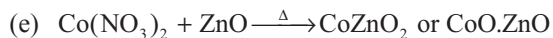
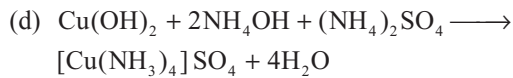
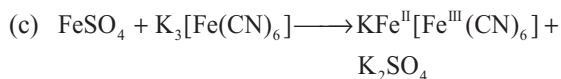
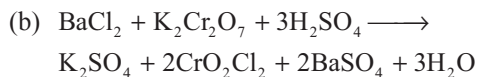
46. (A-Q, R, S); (B-P, Q, R); (C-P, Q, S); (D-Q, R)



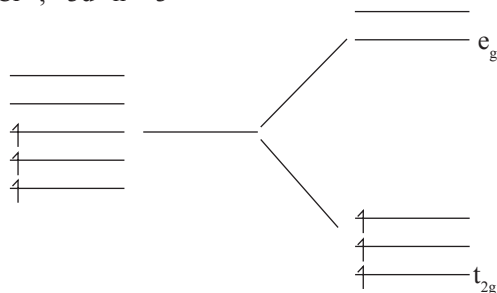
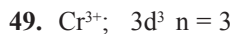
47. (A - T); (B - S); (C - P); (D - R)



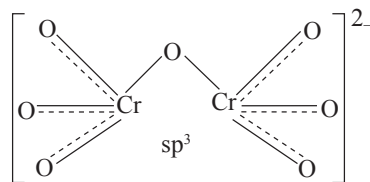
Hydroquinone acts as developer



Let the oxidation state of Cr in CrO_4^{2-} is x.
 So $x + 4(-2) = -2$ or $x = 6$



50.

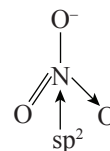


Six Cr-O bonds are equivalent due to resonance and two bridged Cr-O bonds are equivalent (no resonance).

QUALITATIVE ANALYSIS

1. (b) $2\text{KMnO}_4 + 3\text{H}_2\text{SO}_4 + 5\text{H}_2\text{S} \rightarrow \text{K}_2\text{SO}_4 + 2\text{MnSO}_4 + 8\text{H}_2\text{O} + 5\text{S}$
2. (d) $\text{AgCl} + 2\text{NH}_3 \rightarrow [\text{Ag}(\text{NH}_3)_2]\text{Cl}$
 $[\text{Ag}(\text{NH}_3)_2]^+ + \text{Cl}^-$
complex
3. (a) Ca, Ba, Sr these radicals are precipitated in basic medium.
4. (c) $\text{Na}(\text{NH}_4)\text{HPO}_4 \xrightarrow{\Delta} \text{NaPO}_3 + \text{NH}_3 + \text{H}_2\text{O}$
Micro cosmic salt Sodium meta phosphate (transparent)
5. (c) Chromium ion gives in hot and cold. Oxidizing and reducing flame green-colour.
6. (a) $\text{SO}_3^{2-} + \text{Zn} + 8\text{H}^+ \rightarrow \text{H}_2\text{S} \uparrow + 3\text{Zn}^{2+} + 3\text{H}_2\text{O}$
 $\text{Pb}^{2+} + \text{S}^{2-} \rightarrow \text{PbS} \downarrow \text{ (black)}$
7. (d) $\text{NH}_4^+ + \text{OH}^- \xrightarrow{\Delta} \text{NH}_3 \uparrow + \text{H}_2\text{O}$
 (a) NH_3 , alkaline in nature turns red litmus blue;
 $\text{NH}_3 + \text{HCl} \rightarrow \text{NH}_4\text{Cl}$ (white fumes)
 (b) $2\text{HgNO}_3 + 2\text{NH}_3 \rightarrow$
 $\text{Hg}(\text{NH}_2)\text{NO}_3 + \text{Hg} + \text{NH}_4\text{NO}_3$;
black
 $\text{CuSO}_4 + 4\text{NH}_3 \rightarrow [\text{Cu}(\text{NH}_3)_4]\text{SO}_4$ (intense blue)
 (c) $2\text{K}_2(\text{Hgt}_4) + \text{NH}_3 + 3\text{KOH} \rightarrow$
 $\text{HgOHgNH}_2\text{I} \downarrow \text{ (brown)} + 7\text{KI} + 2\text{H}_2\text{O}$
8. (b) $\text{CuSO}_4 + \text{H}_2\text{S} \rightarrow \text{CuS} \downarrow \text{ (black)} + \text{H}_2\text{SO}_4$
 $\text{CuS} + 2\text{HNO}_3 \rightarrow$
 $\text{Cu}(\text{NO}_3)_2$ (blue solution) + NO + S + H_2O
 $2\text{Cu}^{2+} + 4\text{I}^- \rightarrow \text{Cu}_2\text{I}_2 \downarrow \text{ (white)} +$
 $\text{I}_2 \uparrow$ (colour of solution is yellow/brown)
9. (d)
 (b) $\text{Cu}^{2+} + 2\text{NH}_4\text{OH} \rightarrow [\text{Cu}(\text{NH}_3)_4]^{+2}$
 (deep blue) + $4\text{H}_2\text{O}$
 $\text{Bi}^{3+} + 3\text{NH}_4\text{OH} \rightarrow \text{Bi}(\text{OH})_3 \downarrow \text{ (white)} +$
 3NH_4^+
 (c) $\text{Cu}^{2+} + [\text{Fe}(\text{CN})_6]^{4-} \rightarrow \text{Cu}_2[\text{Fe}(\text{CN})_6] \downarrow$
 (chocolate brown)
 $\text{Bi}^{3+} + [\text{Fe}(\text{CN})_6]^{4-} \rightarrow$ no change observed.

10. (d) (a) (b) (c) can be precipitated by HCl as well as H_2S as their insoluble chlorides and sulphides respectively.
 (d) $\text{Sn}^{2+} + 2\text{HCl} \rightarrow \text{SnCl}_2$ (soluble) + 2H^+ . It can be precipitated only by H_2S as its insoluble sulphide (brown).
 (b) $\text{Fe}^{3+} + 3\text{NH}_3 + 3\text{H}_2\text{O} \rightarrow \text{Fe}(\text{OH})_3 \downarrow$
 (reddish – brown) + 3NH_4^+
 $\text{Cu}^{2+} + 4\text{NH}_3 \rightarrow$
 $[\text{Cu}(\text{NH}_3)_4]^{2+}$ deep blue solution.
 $\text{Zn}^{2+} + 4\text{NH}_3 \rightarrow$
 $[\text{Zn}(\text{NH}_3)_4]^{2+}$ colourless solution.

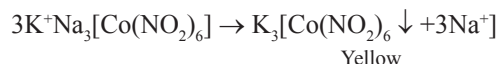


11. (c) $\text{Cr}(\text{OH})_3 + \text{OH}^- \rightleftharpoons [\text{Cr}(\text{OH})_4]^-$
 $2[\text{Cr}(\text{OH})_4]^- + 2\text{OH}^- + 3\text{H}_2\text{O}_2 \rightarrow$
 2CrO_4^{2-} (yellow colour solution) + $8\text{H}_2\text{O}$
 $\text{Fe}(\text{OH})_3$ remains as brown precipitate.
12. (a) $2\text{Br}^- + \text{Cl}_2 \rightarrow \text{Br}_2 \uparrow$ (reddish brown) + 2Cl^- ;
 $\text{SO}_4^{2-} + \text{BaCl}_2 \rightarrow \text{BaSO}_4 \downarrow$ (white) + 2Cl^-
 $\text{Mg}^{2+} + 2\text{NaOH} \rightarrow \text{Mg}(\text{OH})_2 \downarrow$ (white) + 2Na^+
 Suspension of $\text{Mg}(\text{OH})_2$ in water is used as ant-acid and sodium gives golden yellow colour flame.
13. (d) $\text{Fe}^{2+} + [\text{Fe}(\text{CN})_6]^{4-} \rightarrow \text{K}_2\text{Fe}[\text{Fe}(\text{CN})_6] \downarrow$
 [white] (in absence of air)
 $\text{Fe}^{3+} + [\text{Fe}(\text{CN})_6]^{4-} \rightarrow \text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
 (prussian blue precipitate or solution)
 $\text{Zn}^{2+} + [\text{Fe}(\text{CN})_6]^{4-} \text{ (excess)} \rightarrow$
 $\text{K}_2\text{Zn}_3[\text{Fe}(\text{CN})_6]_2 \downarrow$ [bluish white]
 $\text{Ca}^{2+} + [\text{Fe}(\text{CN})_6]^{4-} \rightarrow$
 $\text{K}_2\text{Ca}[\text{Fe}(\text{CN})_6] \downarrow$ [white]
 $\text{Cu}^+ + 2\text{K}^+ + [\text{Fe}(\text{CN})_6]^{4-} \rightarrow$
 $\text{Cu}_2[\text{Fe}(\text{CN})_6] \downarrow$ [chocolate brown]
14. (b) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3 + 2\text{OH}^- \rightarrow 4\text{Fe}(\text{OH})_3 \downarrow +$
 $3[\text{Fe}(\text{CN})_6]^{4-}$.

15. (a, b, c)

Nitrate gives a brown ring when reacts with conc. H_2SO_4 in presence of FeSO_4 due to formation of $\text{FeSO}_4 \cdot \text{NO}$ or $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$ complex compound. Brown colour is just due to charge transfer spectra.

16. (a, d)



Na^+ and Mg^{2+} do not give test.

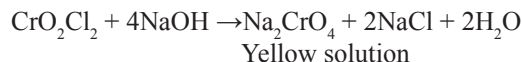
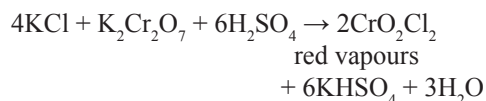
17. (a, b, d)

(i) Cations of the same group cannot be separated by the group reagent, and (ii) A cation, not treated with its own group reagent, is generally precipitated by the reagent of a later group.

H_2S in an acid medium is the reagent for group II. So the cations mentioned in (a) and (b) cannot be separated as they all belong to group II. Those mentioned in (c) can be separated because Pb^{2+} (group II) is precipitated as PbS and Al^{3+} (group III) is not. The cations mentioned in (d) remain unaffected by the reagent.

18. (a, b, d)

When a solid mixture of NaCl , $\text{K}_2\text{Cr}_2\text{O}_7$ and conc. H_2SO_4 is heated, the products obtained are:



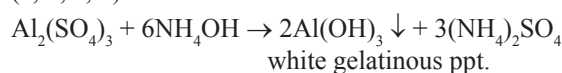
19. (a, d)

Ba^{2+} ions produce yellow ppt. with CrO_4^{2-} , which is not soluble in CH_3COOH

BaCO_3 dissolves in CH_3COOH and produces Ba^{2+} ion which gives yellow ppt. with CrO_4^{2-} which is insoluble in CH_3COOH

SrCO_3 is soluble in CH_3COOH and produces Sr^{2+} ion which does not give yellow ppt. of SrCrO_4 in CH_3COOH

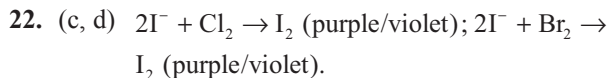
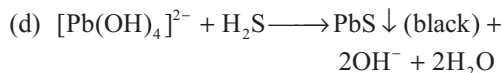
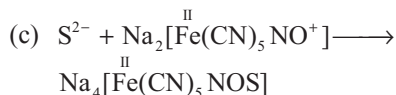
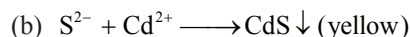
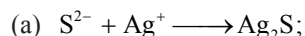
20. (a, b, c, d)



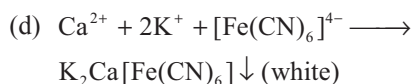
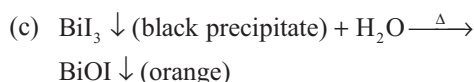
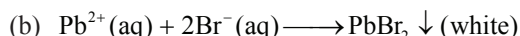
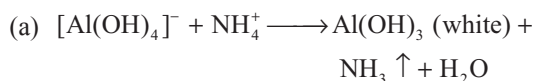
$\Rightarrow$ ppt. of $\text{Al}(\text{OH})_3$ is insoluble in NH_3 sol. but soluble in NaOH

$\Rightarrow \text{Al}(\text{OH})_3$ can be used as ant-acid.

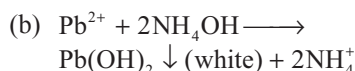
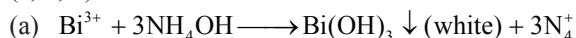
21. (a, b, c)



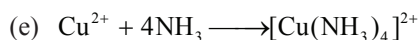
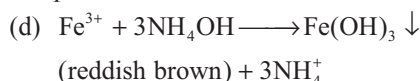
23. (a, c, d)



24. (a, b, d)

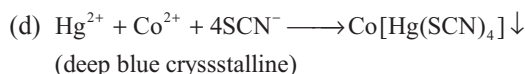
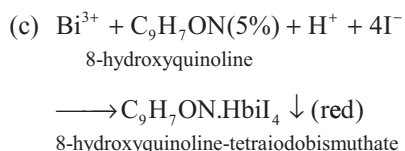
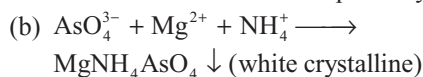
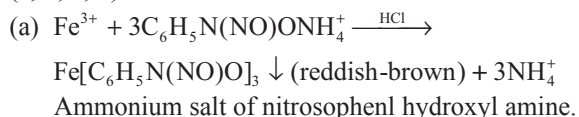


(c) Due to common ion effect of NH_4^+ , the OH^- ion concentration is not excess enough to precipitate Mg^{2+} as $\text{Mg}(\text{OH})_2$ because of its high solubility product.



In presence of ammonium salts, precipitation does not occur at all, but blue colour solution is formed right away.

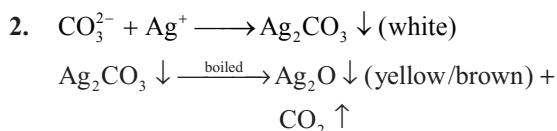
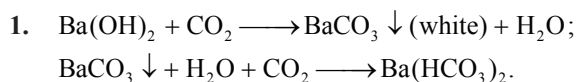
25. (a, b, c, d)



26. (c) Assertion is true but reason is false. Sb (III) is a basic radical of IIB group for which group reagent

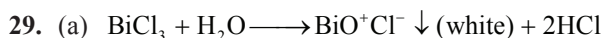
is H_2O is presence of dilute HCl . It is necessary to maintain the proper hydrogen ion concentration for the precipitation of IV group cations.

27. (b)

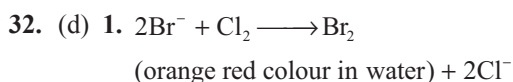
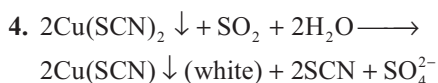
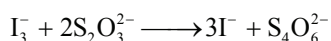
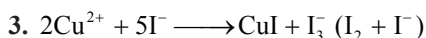
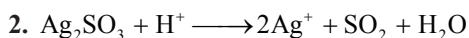
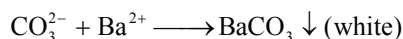
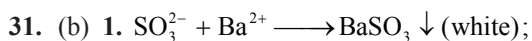


28. (b) $\text{PbI}_2 + 2\text{I}^- \rightleftharpoons [\text{PbI}_4]^{2-}$ so soluble complex.

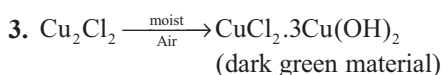
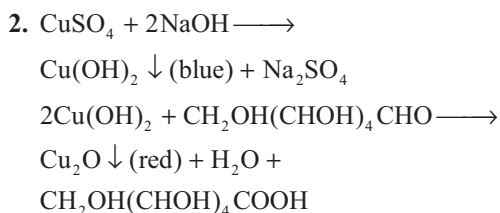
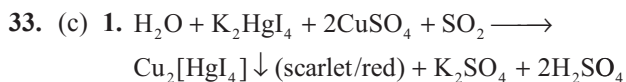
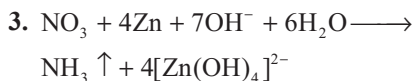
Reaction is reversible and shifts in backward direction on dilution with water.



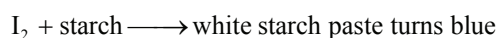
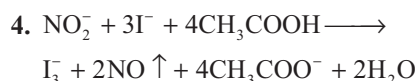
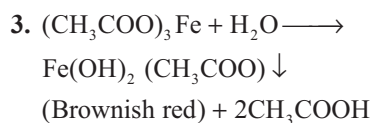
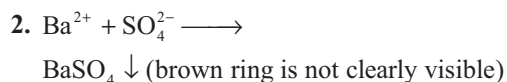
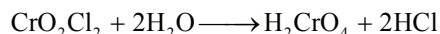
30. (d) K_{sp} of ZnS is large and thus is completely precipitated only in presence of ammonium hydroxide.



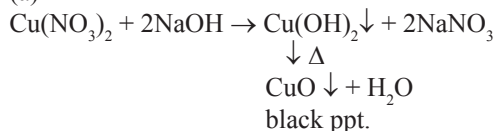
2. True statement.



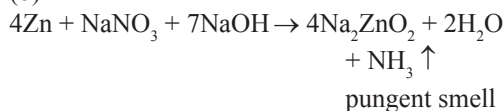
34. (b) 1. CrO_2Cl_2 reacts with water forming H_2CrO_4 and thus prevents the evolution of deep red vapour.



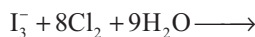
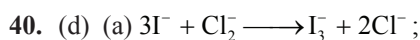
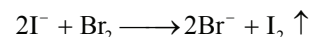
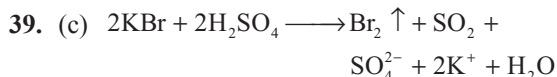
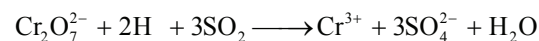
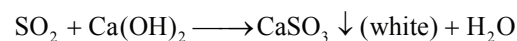
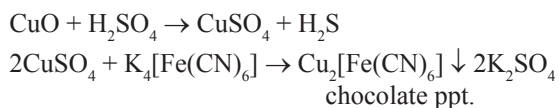
35. (a)

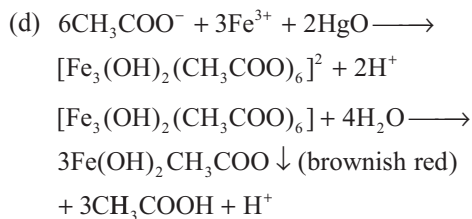
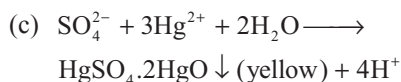


36. (b)

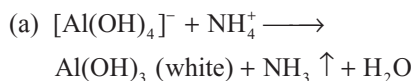


37. (b)

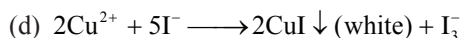
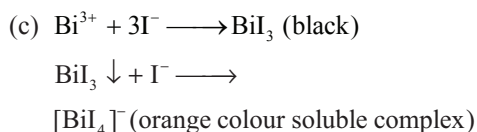
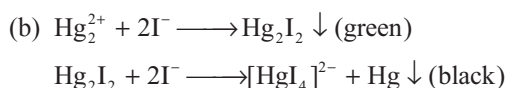
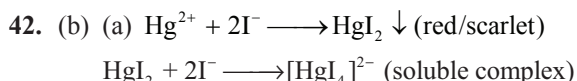
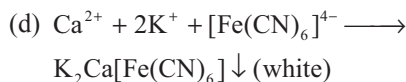
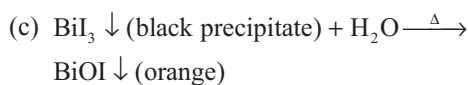
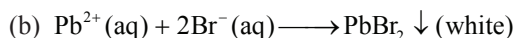




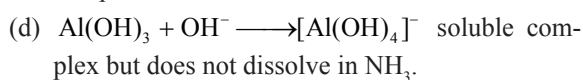
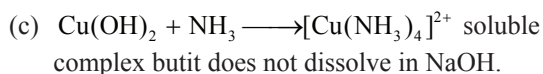
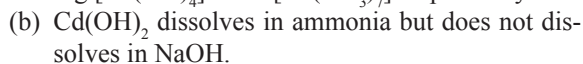
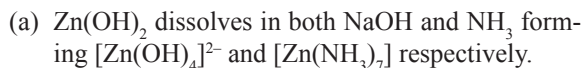
41. (b)



On addition of ammonium chloride solution of $[\text{Al}(\text{OH})_4]^-$ soluble complex, the hydroxyl ion concentration is reduced owing to the formation of the weak base ammonia, which can be readily removed as ammonia gas by heating. Thus, white precipitate of $\text{Al}(\text{OH})_3$ is obtained.



43. (d)



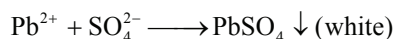
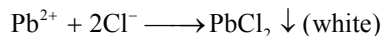
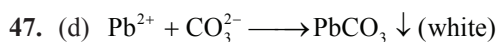
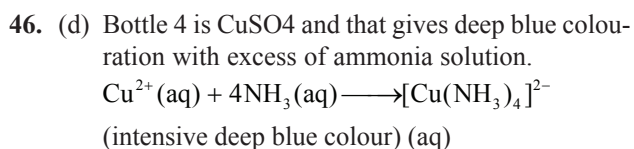
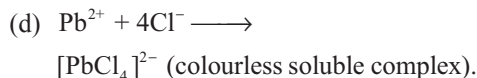
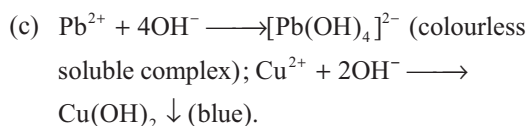
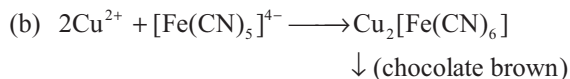
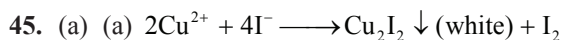
44. (d) As bottle 2 + bottle 3 gives colourless odourless gas, it may be carbon dioxide. Generally carbonates are decomposed by acids giving CO_2 gas. It suggests bottle 2 and 3 contain sodium carbonate and HCl .

Bottle 3 + 4 gives blue precipitate which confirms the Cu^{2+} in either of bottles. CuSO_4 , CuCl_2 and $\text{Cu}(\text{NO}_3)_2$ are soluble and CuCO_3 is insoluble in water as evident from the reaction.

$\text{Cu}^{2+} + \text{CO}_3^{2-} \rightarrow \text{CuCO}_3 \downarrow (\text{blue})$. Thus blue precipitate must be of copper carbonate.

Hence, bottle 4 is CuSO_4 , 3 is Na_2CO_3 , 2 is HCl (from above) and 1 is $\text{Pb}(\text{NO}_3)_2$ as it gives white precipitate of PbCl_2 with bottle (2).

Note: PbNO_3 , Na_2CO_3 and HCl solutions are colourless but CuSO_4 is blue colour solution.



48. (A-Q), (B-Q, R), (C-P, S), (D-S)

Oxalate on heating with conc. H_2OSO_4 gives a mixture of CO and CO_2 gases.

Acetate on heating with conc. H_2SO_4 , a vinegar smell of acetic acid is produced with FeCl_3 (neutral), gives a red solution of ferric acetate.

Sulphide-with dil. H_2SO_4 producing SO_2 gas which turns lead acetate paper black with sodium nitropruside-produce pink violet colour



Thiosulphate-with dil. H_2SO_4 producing SO_2 gas and colloidal sulphur.

49. (A-P, R), (B-Q, R, S), (C-P, Q, R, S), (D-P, Q, R)

(a) $\text{Pb}(\text{CN})_2 \downarrow$ white, not soluble in excess of reagent
 $\text{AgCN} \downarrow$ white, soluble in excess of reagent $\text{Hg} + \text{Hg}(\text{CN})_2 \downarrow$ black, not soluble in excess reagent.
 $\text{Cu}(\text{CN})_2 \downarrow$ yellow $\xrightarrow{\text{Excess}}$ $\text{K}_3[\text{Cu}(\text{CN})_4]$ soluble.

(b) $\text{Pb}(\text{OH})_2 \downarrow$ white $\xrightarrow{\text{Excess}}$

$\text{Na}_2[\text{Pb}(\text{OH})_4]$ soluble.

$\text{Ag}_2\text{O} \downarrow$ brown, $\text{Hg}_2\text{O} \downarrow$ black, $\text{Cu}(\text{OH})_2 \downarrow$ blue

(c) $\text{PbI}_2 \downarrow$ yellow $\xrightarrow{\text{Excess}}$ $[\text{PbI}_4]^{2-}$ soluble complex only with concentrated solution of KI (4M).

$\text{AgI} \downarrow$ yellow $\xrightarrow{\text{Excess}}$ No reaction

$\text{Hg}_2\text{I}_2 \downarrow$ Green $\xrightarrow{\text{Excess}}$ $[\text{HgI}_4]^- + \text{Hg} \downarrow$ black.

$\text{CuI} \downarrow$ white + $\text{I}_3^- \longrightarrow$ White ppt in brown coloured solution.

(d) $\text{PbCrO}_4 \downarrow$ yellow

$\text{Ag}_2\text{CrO}_4 \downarrow$ (brick red/brownish red)

$\text{Hg}_2\text{CrO}_4 \downarrow$ red (with hot solution of K_2CrO_4)

In cold, brown precipitate is obtained.

50. (A-S), (B-R), (C-Q), (D-P)

(a) $\text{Cu}^{2+} + 2\text{SCN}^- \longrightarrow \text{Cu}(\text{SCN})_2 \downarrow$ (black)

$2\text{Cu}(\text{SCN})_2 \downarrow \longrightarrow$

$2\text{CuSCN} \downarrow$ (white) + $(\text{SCN})_2 \uparrow$

(b) Hot $\text{Bi}^{3+}(\text{aq}) + \text{C}_6\text{H}_3(\text{OH})_3 \xrightarrow{\text{H}^+}$

$\text{Bi}(\text{C}_6\text{H}_3\text{O}_3) \downarrow$ (yellow) + 3H^+

(c) $2\text{Ag}^+(\text{aq}) + \text{AsO}_4^{3-}(\text{aq}) \longrightarrow$

$\text{Ag}_3\text{AsO}_4 \downarrow$ (brownish-red)

(d) $\text{Mn}^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq}) \longrightarrow$

$\text{Mn}(\text{OH})_2 \downarrow$ (white)

$\text{Mn}(\text{OH})_2 \downarrow + \text{O}_2 \longrightarrow 2\text{MnO}(\text{OH})_2 \downarrow$ (brown)

51. (A-T), (B-R), (C-P, S), (D-P, Q), (D-P, Q), (E-P, R, S)

(a) $\text{Fe}^{3+} \longrightarrow \text{Fe}(\text{OH})_3 \downarrow$ (reddish brown)

$\text{Cu}^{2+} \longrightarrow [\text{Cu}(\text{NH}_3)_4]^{2+}$ deep blue solution;

$\text{Zn}^{2+} \longrightarrow [\text{Zn}(\text{NH}_3)_4]^{2+}$ colourless soluble complex

(b) dissolve in 50% HNO_3 forming soluble nitrates

(c) $\text{Pb}^{2+} + 2\text{KI} \longrightarrow \text{PbI}_2 \downarrow$ (yellow) + 2K^+ ;

$\text{Pb}^{2+} + \text{K}_2\text{CrO}_4 \longrightarrow \text{PbCrO}_4 \downarrow$ (yellow) + 2K^+

(d) $\text{Bi}^{3+} + 3\text{KI} \longrightarrow \text{BiI}_3 \downarrow$ (black) + 3K^+

$\text{BiCl}_3 + 3\text{Na}_2\text{SnO}_2 + 6\text{NaOH} \longrightarrow$

$2\text{Bi} \downarrow$ (black) + $3\text{Na}_2\text{SnO}_3 + 6\text{NaCl} + 3\text{H}_2\text{O}$

(e) $[\text{Ag}(\text{NH}_3)_2]\text{Cl} + \text{KI} \longrightarrow \text{AgI} \downarrow$ (yellow) + $2\text{NH}_3 + \text{KCl}$

$[\text{Ag}(\text{NH}_3)_2]\text{Cl} + 2\text{HNO}_3 \longrightarrow$

$\text{AgCl} \downarrow$ (white) + $2\text{NH}_4\text{NO}_3$

$[\text{Ag}(\text{NH}_3)_2]\text{Cl} + \text{K}_2\text{CrO}_4 \longrightarrow$

$\text{Ag}_2\text{CrO}_4 \downarrow$ (brick red) + $4\text{NH}_3 + 2\text{KCl}$

52. (A-P, Q, R), (B-R, S), (C-P, Q), (D-P, S)

(a) $\text{Zn}(\text{OH})_2 \downarrow + 4\text{KCN} \longrightarrow \text{K}_2[\text{Zn}(\text{CN})_4] + 2\text{KOH}$
 colourless soluble complex

$\text{Zn}(\text{OH})_2 \downarrow + 4\text{NH}_3 \longrightarrow [\text{Zn}(\text{NH}_3)_4](\text{OH})_2$
 colourless soluble complex

$\text{Zn}(\text{OH})_2 \downarrow + 2\text{NaOH} \longrightarrow \text{Na}_2[\text{Zn}(\text{OH})_4]$
 colourless soluble complex

(b) It is slightly soluble in cold

$\text{Cr}(\text{OH})_3 \downarrow + 6\text{NH}_3 \longrightarrow [\text{Cr}(\text{NH}_3)_6]^{3+}$ (pink or violet soluble complex) + 3OH^-

$\text{Cr}(\text{OH})_3 \downarrow + \text{OH}^- \longrightarrow [\text{Cr}(\text{OH})_4]^-$
 green soluble complex

$2\text{Cr}(\text{OH})_3 \downarrow + 3\text{Na}_2\text{O}_2 \longrightarrow$

$2\text{Na}_2\text{CrO}_4 + 2\text{NaOH} + 2\text{H}_2\text{O}$
 yellow solution

(c) $\text{AgCl} \downarrow + 2\text{NH}_3 \longrightarrow [\text{Ag}(\text{NH}_3)_2]\text{Cl}$

$\text{AgCl} + 2\text{CN}^- \longrightarrow [\text{Ag}(\text{CN})_2]^- + \text{Cl}^-$

r, s $\longrightarrow \text{NO}$

(d) $2\text{CuS} \downarrow + 8\text{CN}^- \longrightarrow 2[\text{Cu}(\text{CN})_4]^{3-} + \text{S}_2^{2-}$
 colourless soluble complex (Redox reaction).

$\text{CuS} + \text{Na}_2\text{O}_2 \xrightarrow{\text{H}_2\text{O}} \text{CuSO}_4 + \text{NaOH}$

$\text{CuS}^- + \text{H}_2\text{O} \longrightarrow \text{CuSO}_4$ soluble compound

53. Here NO is NO^+ ; so in $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}^+]^{2+}$, the oxidation state of iron is +1.

54. $\text{Hg}^{2+} + \text{Co}(\text{SCN})_2 \longrightarrow \text{Co}[\text{Hg}(\text{SCN})_4] \downarrow$
 (deep blue)

55. $\text{SO}_3^{2-} + \text{H}_2\text{SO}_4 \longrightarrow \text{SO}_4^{2-} + \text{SO}_2 + \text{H}_2\text{O}$;
 $\text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 + 3\text{SO}_2 \longrightarrow \text{Cr}_2(\text{SO}_4)_3$ (green)
 (gjk) + $\text{K}_2\text{SO}_4 + \text{H}_2\text{O}$

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